The Knowledge of Failure

A Cross-Domain Manual of Structural Failure

24 mechanisms · 14 domains · 53 case cards

November 2026 · English edition v1.0

METHOD

Mechanisms first, cases second. Every case is told only from what the people in it could see at the time, and stops at the decision, so the reader judges before turning the page.

SCOPE

404,683 words · 24 mechanism chapters · 14 domain pages · 53 full case cards · 159 indexed cases

EDITION

v1.0 · November 2026

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Preface: The Reader’ s Contract

What this book is

Everything this book asks of a working manual on organizational failure fits in one sentence: after reading it, you can say, before you decide, which ways of dying you most resemble.

Three things follow from that sentence. Failure scattered across hundreds of pages of investigation reports becomes numbered case cards, twelve fields each, with the field definitions in Chapter 5. The cases are gathered into twenty-four mechanisms, one chapter each, with trigger criteria, generating conditions, weak-signal lists and countermeasures. Then comes a method of use: a situation table, a hundred-question review checklist, and a script for a ninety-minute rehearsal.

Every fact in the book comes from a primary document: investig - ation reports, court judgments, regulatory penalties, recall notices, audit reports, financial statements and prospectuses, and the parties’ own published postmortems. Each case carries a full citation and link at the end, and the numbering corresponds one to one with the master case table in Appendix B. The book conducts no new investigations. Investigation is a separate craft, practiced for decades by other people, among them the writers at the US National Transportation Safety Board (NTSB). What this book does is circulation: pulling the reusable parts out of those documents, putting an index on them, and arranging for them to come to mind at the moment a decision is made.

The reader’s contract

Four commitments, each of which every chapter has to honor. A chapter that fails one is a defective copy.

1. Three lines at the start of each chapter: what it covers, when to come back to it, how long it takes to read.
2. Each chapter ends with five self-checks rather than a summary. All five ask you to answer about work you currently hold. If you cannot answer them, the mechanism is running in you now, and you are not reading about someone else.
3. Cases stop at the decision point. Judge first, then turn the page. One marker throughout:

   Stop point: judge first, then turn the page.

   1. What information is in front of the people involved at this moment?
   2. If you were them, do you sign?
   3. What do you bet happens in the next three months?

   Outcome.

   The stop has a purpose. Postmortems are written after the result is known, and a reader of a postmortem holds the answer while hunting for the process, so everything looks obvious in hindsight. The stop withholds the answer, forces a choice where the information is incomplete, and then sets your choice beside theirs. Where your choice went wrong and where theirs went wrong are two separate accounts.
4. No moral instruction. The book writes structure, not “should”. Where a structural problem appeared, which signal came first, what change would alter the incentive or the information flow: that is what gets written. Whether the people involved were stupid or arrogant gets no verdict, because such

verdicts cannot be verified and they get in the way of seeing structure.

No assumptions about who you are

No industry, no rank. An aviation maintenance release case and a hospital disinfection case sit in the same chapter because the mechanism is the same, and when the mechanism is the same you can find the isomorphic instance in your own industry. Every term is explained the first time it appears, all terms are registered in the glossary in Chinese and English side by side, and no term drifts between chapters.

If a case detail is opaque, skip the detail and read the structure. Every case carries a decision point and a minimum counterfactual, and neither depends on industry knowledge.

Evidence grades

Each case carries a grade from A to E. A is the full report of an independent investigation body. B is an official regulator bulletin or communiqué. C is a judgment or penalty document. D is the parties' own published postmortem. E is press coverage, usable as a lead only, never as a source of fact. A case's minimum counterfactual and escalation point can be supported only by A-grade material and by part of B and C; cases below that threshold enter the appendix matrix and stay out of the body. The grades are defined and their use governed in Chapter 10 and Appendix C.

One rule belongs at the start: every number in the body of this book needs a four-part provenance, meaning who counted it, when, on what sample, under what definition. A number that cannot supply all four gets written as mechanism description in -

stead. Most paragraphs you read without a number are there because the discipline held.

What this book does not cover

Three things are out of scope. Psychological consolation after fail - ure, of which the market has plenty. Blame and compensation, which belong to lawyers. Prediction of which company will fold: the book classifies ways of dying, and the divination is left out.

One class of failure is deliberately confined to Part 6: exploratory failure. Without it there are no new drugs, no new routes, no new technology, and handling it together with preventable operational failure would amount to teaching people to refuse every attempt. Part 6 gives the test that separates the two.

One reading route

If you are short of time, read three chapters: Chapter 2, on how your sample of cases was filtered; Chapter 35, the situation table; Chapter 37, the review question list. If you have a project to review, add the two mechanism chapters that match your situation. If you write postmortems, read Chapters 5 and 38. Treat the rest as tools to consult.

Chapters stand on their own, and you can start anywhere. The one requirement: when you reach a stop point, actually stop, answer the three questions, then read the outcome. A book read with the stops skipped is half a book.

Three shortest paths

Three shortest paths, restated here to close the preface. Scenario one, your organization has just had an incident: go to Chapter 38

for the postmortem template, fill it in, then follow the mechanism chapters the postmortem raises back into Part 3 for close reading. Scenario two, you face a large decision: fill the table in Chapter 35, run the walkthrough in Chapter 36, rehearse in Chapter 39, three steps inside one day. Scenario three, you want to learn the subject properly: read straight from Chapter 1, write your answers at each stop-point card from Chapter 5 onward, and by the end of Part 5 install the five instruments in your own organization. All three routes end in the same place, the ledger. The ledger that remains after you finish the book is what the book actually delivers.

And a note for writers who cite this book: cite the number and the source ("av-034, NTSB AAR-02/01"). A retelling that drops the number loses checkability after two generations of citation, and checkability is the whole floor that separates this book from a collection of anecdotes.

May these numbers outlive the book in your hands.

Chapter 1: Why Failure Knowledge Is the Most Expensive, and the Least Organized

> What this chapter covers: the price of failure knowledge and the supply of it, plus the distance between having heard of a way to die and knowing one. When to come back: when you need to persuade someone, yourself included, to spend hours and budget on organizing failure; when a postmortem lands on your desk and you have to judge whether it deserves reading. Reading time: about 30 minutes.

1.1 Eliminative Knowledge

An organization makes the same kind of decision every day. Fund this project or not, release this batch or hold it, sign this build or send it back, fill this shift or run short. Each decision takes one item out of a set and abandons the rest. Knowledge has exactly one job in that act: it makes the set smaller.

Two kinds of evidence shrink a set. A success case says this path works, and validates one point. A failure case says this region contains a path that kills, and removes an area. For the same amount of information, the second removes more.

Organizations do not decide once. They decide daily, on structur - ally similar questions, which is what makes negative samples worth counting. The expected value of one failure record equals the number of wrong paths it removes, multiplied by the expected loss on each of them. Losses run in a long tail: nearly every error produces damage the organization absorbs, and a small number erase it. The large losses cluster on a handful of paths. Those are the paths the investigation agencies write about, year after year.

Failure knowledge therefore wins on two counts at once. It removes more area, and the area it removes carries the largest losses. Together the two support a claim about single records: a failure record carries more information than a success record.

The claim has a premise, and the premise is this chapter's question. Knowledge, here, means organized failure records. Failure that has not been organized does not qualify as knowledge at all. The rest of this section sets the threshold.

The eliminative arithmetic has a boundary, and it should be drawn before the argument goes further. A negative sample removes a region only if the region already has a name. One failure record eliminates a whole area when the area sits inside a pattern library; outside one, the same record eliminates a single point, that company's practice in that year, and the value decays to nothing as circumstances change. This explains why a good failure story told across a dinner table changes no decisions at the table. It is a point, and there is no library to multiply it. It also explains the order of this book: twenty-four chapters build the library before a single page spends itself on cases. The situation table in Chapter 35 is the interface between that library and your own circumstances.

There is a second boundary. Elimination value multiplies by expected loss, and expected losses in the tail depend on data. Rare large losses are exactly the ones nobody estimates well. Appendix C therefore fixes the evidence grades and the reconstruction method, and any figure that cannot be estimated is left blank instead of being handed to a decision-maker with false precision. The two boundaries make the chapter's first promise: every eliminations figure discussed here is backed by a pattern library and by an evidence grade.

One further compression of the opening is worth making. If a decision is an act of elimination, then the unit of knowledge value is the area it cuts away, not the quality of the telling. On that view the

twelve-field card is a measuring instrument. The fields are gradu‑ations, a card is a reading, a mechanism is the conversion table. The instrument view has a consequence for prose: this book does not chase narrative tension, because the reader should feel tension at the decision point and nowhere else. Its negative example is the narrativizing tradition in failure writing, where an accident is told as a mystery. Readers walk away remembering the protagonist and the reversal, and lose the conditions and the structure. That habit is the largest single drag on the circulation of failure knowledge, and it owns about half the cost counted in Section 1.5.

1.2 Is "I Know a Company That Died That Way" Knowledge?

The failure knowledge circulating at dinner and in meeting rooms usually sounds like this. I know a company that died exactly that way. So-and-so went hard at channel expansion and collapsed overnight. The person saying it often did watch a company die, and the information is not false. As knowledge it has three holes.

The first hole is in the fact layer. The sentence cannot be checked. Which company, which year, on the authority of what document, was there an investigation afterwards: all missing. The listener can believe it or not. Knowledge's first requirement is verifiability, meaning a third party can test the statement against source ma‑terial. Without that layer the statement drops to rumor, and rumor has no part in decisions. It only sets a mood.

The second hole is in the condition layer. The sentence has no boundaries. It says nothing about the company's size at the time, its regulatory density, its cash structure, its market stage, so you cannot tell whether this way of dying applies to your project. Every failure conclusion has a range of validity, and knowledge used outside its range manufactures accidents on its own. Taking "in‑

ternet companies must burn cash for scale" into a cash-flow business, or "manufacturing must be asset-heavy" into a contract-manufacturing model, are the same error twice. Failure knowledge with the condition layer missing sounds deeper the more damage it does, because it invites the listener to carry the conclusion straight across.

The third hole is in the structure layer. The sentence cannot be retrieved, accumulated or cited. Three months later you cannot reconstruct the full version; when a colleague needs it you cannot produce a source; it will not merge with similar records into a regularity. Knowledge enters organizational decisions only through a system: searchable, comparable, citable, contestable. All four verbs presume structure.

Set against that a failure record done seriously. A formal NTSB aviation accident report has four fixed parts: factual information, analysis, conclusions, and safety recommendations. The factual part carries a complete timeline, and every statement in it traces back through the investigation docket to physical evidence, recorded audio and interview transcripts. The conclusions part lists each finding and marks whether it is contributory or an observation. Safety recommendations are numbered one by one, and the recipient must answer in writing within 90 days. A reader may disagree with the analysis, but every step of checking has a route. That is what fielded means: fact, condition and structure, all three present.

A contrast makes the return on fielding concrete. For one and the same accident, the narrative version, in the register of a press release, has an information half-life measured in months: news is buried by news, and a source rots with its link. The fielded version, in the register of a formal report, has a half-life measured in decades: the document number enters an archive system, and paper and digital copies are held in parallel. The gap between the two

half-lives is the compound interest of fielding. Only knowledge that can still be cited in ten years is entitled to enter an organization's long-term decisions, and equipment selection, process design and staff development are all ten-year decisions. One practical instruction for decision-makers: before any piece of failure knowledge enters your decision library, ask for its half-life. Anything with a shorter half-life than your decision cycle is background, not evidence.

Chapter 5 compresses this fielding into a twelve-field case card. Here only the standard matters: before discussing failure, check whether the material passes the three gates of this section.

The three gates become a short set of questions. Handed any failure narrative, ask three in order. Can I follow the source to the original text, and does that text carry a time, a place, an institution and a document number? Did the narrator state the scale, the regulation, the cash position and the market stage of that moment, or jump straight from the story to the lesson? Can the record be searched, compared and cited, which means does it carry a number that lives inside a system? An answer that fails sends the item down a grade. Downgrading is not deletion, it changes the use: a rumor is a lead, you follow the lead to primary material, and you upgrade one record at a time.

The gates have another side. Material that passes them has not thereby become a conclusion. A primary report supplies facts and conditions, and the lesson layer still requires the reader to walk the pattern library. The gates sort material by grade. They do not think on the reader's behalf.

One objection comes up often enough to answer here: does all this make dinner-table stories worthless? No. They serve two purposes. A story is a topic lead, telling you where a primary document worth finding might exist. It is also a thermometer for organizational culture, since the way a team retells failure exposes its

habits of knowledge. What needs governing is the use of rumor as evidence, meaning a decision document with no source attached, not the story at the table. The gates route each narrative to the position that matches its evidence grade. Chapter 10 is the formal version of the three gates. Section 1.2 is the everyday one.

1.3 The Word "Replicate" Has Misled Everyone

The claim travels widely: success can be replicated, failure cannot. It points attention in the wrong direction. The accurate version runs like this. Success cannot be replicated. Failure can be avoided. Neither of them is replication.

People who learn from failure meet a third casualty of the replication myth. They get asked, in effect, whether failure cases are not all special, and whether their own project is not different. The objection is half right, and the halves need separating. The wrong half treats applicability as all or nothing; the value of the condition layer is precisely that it hands you a comparability checklist, and scoring that list item by item turns applicability into a measurement rather than a stance. The right half is a reminder that migration carries a tax, which the rest of this section and the domain pages in Part 4 pay. Once separated, the objection converts into a tool sentence: you say it is different, so name the condition that is different. A clause-level rebuttal returns the conversation to the field layer, where disagreement can be checked, instead of the position layer, where it can only be staked out. The sentence travels into review meetings, project approvals and postmortems, and it is the three gates in portable form.

Replicating a success requires the conditions to replay. The list runs: timing, team composition, competitive picture, regulatory window, funding environment, and a quantity of luck the people involved cannot articulate. Business school cases compress success

into a three-part story, vision, execution, result, and what the compression removes is exactly what replication needs. Following success cases therefore tends to raise the failure rate rather than lower it, and when you fail nobody writes your postmortem. That is the survivorship bias of Chapter 2.

The reusable part of failure knowledge is far more stable. A lethal path usually puts broader requirements on initial conditions than a success path does, and the reason is structural. Each link in a failure chain acts as a filter, so organizations that pass every filter and arrive at the same ending can start from wildly different states. Goal displacement appears with nearly identical structure in hospital waiting-time targets, school progression-rate targets and platform content targets. Success runs the other way. Every success path rests on a narrow set of conditions.

So failure knowledge exists naturally as patterns, not as cases. Patterns migrate across industries; cases are reference material. This is why Part 3 is built as twenty-four mechanism chapters with four cases each, drawn from four different industries: the definition of a mechanism requires it to hold across industries. Something that appears in only one industry is not a mechanism. It is an industry characteristic, and it belongs in the domain pages of Part 4.

Migrating a pattern carries a tax called adaptation cost. When a mechanism in the shape it takes in industry A enters industry B, every parameter has to be re-labeled. The decay rate in Chapter 16 is not the same number in an organization with two reporting layers as in one with six. There are two ways to pay. You can label the parameters yourself while reading cases from the home industry, which costs time and carries error, or you can read a book that has already labeled them, which is what Part 4 and the mechanism-by-domain matrix in Appendix A supply. The labeling has to be honest about its limits. Each cell of that matrix answers presence only,

meaning a reliable case exists for the mechanism in that domain. It says nothing about strength, which varies by organization and which this book does not pretend to deliver as an industry manual. The right use is three steps. Check the cell to confirm the mechanism is present, go to the mechanism chapter for the criteria, then plug in your own parameters. Those three steps pay most of the tax.

What remains sits in semantics. A mechanism's name carries the smell of its home industry when it crosses over: goal displacement is called compliance-target assessment in healthcare and exam-focused teaching in education, and the smell persuades readers they are looking at a different mechanism. The deodorant is the criteria sentence in each mechanism chapter. All three criteria per chapter are deliberately written in industry-neutral language, because criteria are a mechanism's fingerprint and fingerprints do not smell of an industry. When you migrate across industries, hold on to the fingerprint. If it matches, the industry nouns can be swapped freely. If it does not, a familiar name does not make it the same mechanism. That principle is the user manual for the Appendix A matrix, and it marks the line between this book and industry lore: lore collects smells, this book collects fingerprints.

1.4 Who Produces Failure Knowledge

If failure records were worth this much, the market should have produced them already. The producer list explains why it has not.

Statutory investigation bodies come first. Aviation has the NTSB, the UK AAIB, France's BEA and Japan's JTSB. Chemical and process industries have the US CSB. Nuclear has the IAEA incident reporting system. Maritime has the UK MAIB. Their output is steady, free and citable, which makes them the largest legitimate supply of failure knowledge. It arrives as legal documents: several

hundred pages, dense with technical terminology, written for hearings and litigation, with very little of it written for reading.

Insurance and reinsurance companies hold the largest failure datasets anywhere, spanning industries and years, with loss amounts attached. Those datasets are assets for pricing cover, and they are closed. You cannot buy them on an open market. You pay for them inside your premium.

Judicial and regulatory documents make up the third group: judgments, penalty decisions, recall notices, public and steadily growing, with China Judgments Online having once offered the largest disclosure surface in the world. Their weakness is fragmentation. One document covers one slice of one dispute, and nobody maintains a data structure across documents.

Academia comes fourth. Vaughan's normalization of deviance, Perrow's normal accidents, Snook's practical drift, Dekker's drift into failure: Part 3 takes its mechanism names from this line of work. But academic theory arrives wrapped in cases, the cases wrapped in second-hand retellings, and the citation chain back to the source frequently ends at a link that will not open. The theory is reusable. The cases cannot be checked.

The organizations involved come last, and produce least, for the hardest reasons. Writing failure in detail carries legal risk, hands intelligence to competitors, and costs employees and founders their reputations. All three point toward silence. Chapter 4 takes each of them apart.

Lay the five side by side and a mismatch appears. The largest producer of failure knowledge is a body of documents that its largest potential consumer group happens to be unable to read. Investigation agencies have written for decades. Business schools scarcely cite them. Corporate training does not use them. Decision-makers

do not recall them at the moment of choosing. Production is not the shortage. Circulation is.

The reverse face of the circulation gap is a misalignment of producer incentives. An investigation agency's best output is procedural completeness, which survives litigation, rather than a format that travels well. An insurer's best output is pricing precision, not industry education. A party to the event has one best output, which is not writing. Three optima add up to a structural blank, since every side is rationally declining to do the circulating. A structural blank admits only structural fixes: an independent identity for whoever circulates, belonging to none of the parties, which is where this book places itself; a stable unit of account for what gets circulated, which is the twelve-field card; and a checkable base, which is the links in Appendix B. None of the three depends on anyone's goodwill changing. They ask one thing: that somebody stand between upstream production and downstream consumption and translate between the two formats. The scarcity of translators is the field's central fact, and imitation is what this book would most like to provoke.

The entry cost of imitating it can be estimated for you. A first edition of one industry's card library, at this book's specification, is fifty cards, which is one knowledgeable part-timer for a year at a card a week. Fifty cards cover an industry's high-frequency ways of dying, following Pareto: most losses concentrate in a few patterns, and that coverage is enough to run the situation table in Chapter 35. The other end of the budget is maintenance, thirty percent refreshed each year plus ten cards from new reports. Maintenance cost is the real threshold, because the excitement of building is universal and the patience to renew is scarce. Chapter 39 therefore puts maintenance in its own line in the budget template and ties it to output, where each new card is matched against

the decisions it blocked that year. Paying for maintenance out of output is the only way a card library reaches its third year.

1.5 Circulation Stops at the Last Mile

Take a real NTSB formal report and watch where circulation breaks. It may be a three-hundred-page PDF in English, sitting on the fourth level of a database, written in airworthiness terminology, with three numbering systems running in parallel: accident number, recommendation number, docket number. The density is high. Every page carries timestamped communications, instrument readings, signed documents. For an engineer about to make a release decision, that report holds everything he needs except three things. He does not know the file exists. He does not know which file matches his situation. He cannot finish three hundred pages.

Three unknowns and one impossibility make up the whole of circulation cost: discovery, matching, reading. Once those costs pass a certain level, the effective supply of a public good falls to zero, however free it is on paper.

In the world as it stands, the circulation cost of success narratives has been driven down industrially. Media, biographies, podcasts, business school texts, trending feeds: somebody pays at every stage. Nobody pays for the circulation of failure records. An investigation agency publishes and stops there. Citing the report earns no academic citation, retelling it earns no traffic, and using it requires understanding it first. This is market failure in its standard shape, applied to an obscure commodity.

The obscurity has a secondary disaster worth naming. The blank in circulation gets filled by poor supply: bestseller narratives built on survivorship bias, success literature repackaged backwards, social media posts that turn an accident into a tragedy of character. The

crowding-out is real, and it works on appetite. An organization that reads three bad books on failure concludes that the field holds nothing serious, and its budget for the subject is spent before a serious title appears. This book's answer is not to criticize the neighbors, since criticism cannot be checked, but to demonstrate seriousness in public: twelve fields, four-part provenance, links, stated gaps. Each of those is verification waiting to happen. Verifiable seriousness obeys a harsh rule. It spreads ten times slower than the other kind, and every copy it leaves behind can be checked. Slow is the right speed here, because the object is a decision, not traffic.

Look at one piece of failure knowledge that has completed circulation. ValuJet 592, case av-035 in Appendix B. The NTSB formal report, AAR-97/06, spends one page pinning the physical origin: a maintenance contractor loaded chemical oxygen generators that had not been expended into the cargo hold as ordinary spares. It spends several hundred pages establishing the condition layer: the gap in dangerous-goods rules for the category "maintenance residue", the handover habits of the contractor system, the regulator's handling of earlier fires of the same kind. The pattern, the invisibility of maintenance, Chapter 29, turns up again in warehousing, in chemical processing, in software configuration management. The card's value is that it pins how a maintenance by-product becomes a dangerous good to a number you can cite. Circulation engineering is the work of repeating that operation across more reports.

The supply side also admits an incentive design. Price the identity of a primary-material reader, because reading a formal report is a skill: airworthiness terminology, the investigation framework, and a pattern library, three layers of training. The other side of pricing is certification. If an industry has a few dozen recognized report readers, examined by interpreting a new case and judged by peers,

then the translation capacity behind circulation has a human base. Certification costs the building of an exam system and returns compounding capacity, since each certified reader can produce several dozen qualified cards a year. The format in this book, twelve fields, evidence grades, the three stop-point questions, can serve directly as the syllabus: public syllabus, transparent stand - ard, verifiable work, and the profession grows on its own.

Repeating the operation across more reports requires deciding what the product is. Three forms are available, with a division of labor. Form one is the card library, which is the body of this book: twelve fields, numbers, links, built for retrieval and citation. Form two is the bulletin stream, a monthly digest of new reports in a field, which keeps an organization in contact with the material; contact is the precondition of use, and most organizations abstain not from unwillingness but from not knowing there is new stock. Form three is the morning case, ten minutes once a week on a single card, stopping at the decision point, letting the team judge, then releasing the outcome, which presses circulation into the room where decisions happen. The three differ in cost and return: the library is dearest and accumulates, the bulletin is cheapest and breaks easily, the morning case is closest to the floor and depends on a host who keeps at it. An organization picks what its budget allows. What this book can supply directly is form one, 158 cards in Appendix B, and the raw material for form two in the retrieval handbooks of the domain pages. Scripts for form three are ready in Chapters 35 and 39.

1.6 Where This Book Sits

This book does not investigate. It does circulation engineering, in three moves. Compression: several hundred pages become a twelve-field case card that keeps the fact layer, the condition layer and the structure layer. Matching: cases hang from twenty-four

mechanisms, so a reader starting from their own position, with the table in Chapter 35, can pull the list of ways of dying that applies. Delayed conclusion: the case stops at the decision point, the reader judges, and only then does the outcome arrive with the minimum counterfactual.

The third move is where this book separates from retelling. Retelling hands out conclusions: so-and-so failed through arrogance, so-and-so succeeded through humility. A conclusion of that kind cannot be reviewed and cannot enter a decision. What this book hands over is a slice of process. On a given date, in a given room, on a given document, these were the facts available, these were the options, and this is what the people did. The judgment stays with the reader, and the structural analysis is already written. That is the specific remedy for the attribution bias of Chapter 2.

Circulation engineering has edges. Compression loses information, so every case keeps its citation and its link and the reader can walk the number back to the original report. Compression can also introduce error, so every number needs a four-part source, and whatever cannot supply one is written as mechanism description. What this book submits is a traceable middle layer. Its value depends on how hard it is to check, not on how well it is written.

The last layer of checking is an admission that the book can be wrong. Any card in the appendices may differ from its source, since compression always loses, transcription always risks, and links always rot. The institutional reply to that is errata: somewhere to report the discrepancy, a record of the report, and that record published with the same weight as the text it corrects. A correction is credit rather than disgrace, because a long errata list means the material has been checked often. A system that welcomes correction accumulates trust faster than a declaration of zero defects, and that accumulated trust is what lets a body of knowledge like this one be finished at all.

The loop closes with a channel running the other way, for checking the book itself. The self-audit machinery is in the standards directory, and three entry points are open to any reader. Appendix B links each card, so the original text opens from the page and you can test the citation against the body yourself. Every number carries its four-part provenance, who measured it, when, on what sample, under what definition, and anything that cannot answer all four is marked pending or blocked. The gaps are stated plainly: Chapters 26, 28 and 34 each record what was not collected, why, and where collection should go next. With those three entry points, the right posture toward this book is neither belief nor disbelief. Pull three cards and check them. Only a book that can be checked deserves a place in a decision process, which is Section 1.2's standard applied to this book.

Five Self-Checks

1. Did your organization's last major loss leave a primary document anyone can retrieve? Ask three colleagues, write the three answers down, and compare them.
2. The last time you retold a case of failure, which source document had you read it in? If you cannot name the document, which category in Section 1.2 does that retelling fall into?
3. For the success cases your team consults before deciding, can you list the three narrow conditions that case required? If you cannot, what does that tell you?
4. Of the three circulation costs, discovery, matching and reading, which is highest in your organization? Is anyone working to lower it?
5. Given a budget for organizing your industry's failure knowledge, what would you build first? How does that answer connect to the situation table in Chapter 35?

Use the five together, once. The first checks inventory, whether the organization has any stock of failure knowledge at all. The second checks citation, whether the team's language carries sources. The third checks conditions, the real cost of the success cases you measure yourself against. The fourth checks circulation, which of the three costs is jammed. The fifth checks budget, what you would build if the money were real. Write the answers on one page, and that page is a map of where your organization stands in relation to failure knowledge. The next step from the map, what to build, how, and who maintains it, is the subject of Part 5, Chapters 35 to 39. The division of labor between this chapter and Part 5 follows from that. The meta-question chapters exist to convince you the work is worth doing, and the usage part exists to make it doable. The bridge between them is the situation table in Chapter 35, which pulls the cards you should read straight out of your own circum - stances. If you plan to jump from here to Part 5, remember the bridge is there: between the argument and the tool sits an afternoon in which you already started.

That afternoon can begin with the smallest possible item. Take the most instructive thing that happened to your team last year and write it as one page through the three gates of Section 1.2: three lines of fact, three lines of condition, three lines of structure, meaning source, number, retrievability. The page will most likely fail the gates, because most internal postmortems are missing the condition layer and the structure layer. Failing is a baseline rather than a failure. The distance between that baseline and a card in this book is the distance between your organization and a world that records failure seriously. Part 5 holds the plan for closing it. This chapter promises only one thing: the distance can be quanti - fied, in how many lines are missing, whether the source is missing, whether the number is missing. A quantified distance is a distance someone can shorten.

Chapter 2 Three Biases: Recording, Survivorship, Attribution

> What this chapter covers: how the failure sample you are handed has been filtered three times, by institutions, by human nature and by narrative, and the three specific things you will learn wrongly if you study from a filtered sample. When to turn back: when you cite a case to support a decision, when you design a review process, when you meet a number like "industry accident rate". Estimated reading time: 35 minutes.

Before you read any failure case, settle one question first: where did this sample come from? The sample in your hands was never drawn from the full population of failures. It passed through three filters, and each filter systematically favors a particular kind of failure. The first filter sits in the recording stage, the second in the telling stage, the third in the attribution stage. This chapter takes them apart one at a time, and after each one asks the same question: if you learn from this filtered sample, what exactly do you learn wrong?

2.1 The Recording Bias: Filing Rules Choose Your Sample for You

For a failure to leave a written trace, it has to clear a threshold. Institutions set that threshold. They decide how severe an event must be before it is investigated, written up and archived. Failures below the line, however many there are, do not exist in the written world.

Beyond the filing threshold stands a second gate that nobody advertises: the language gate. Records are written in professional documentary prose, while most first-hand descriptions of failure arrive in speech. Speech does not fit the format, so it is lost or de-

formed in transcription, and the casualties cluster at the front line. The veteran fitter's way of putting it, the nurse's hunch, the driver's foreboding: all unstructured language, none of it able to fill the boxes on an accident form. The remedy is dual-track recording. Fill the official track as required and store the spoken track separately, as an attachment layer to the file: interview transcripts, the raw wording from the floor, kept verbatim. The cost is storage and tidying. The value lands in future investigations, because a formal document answers compliance questions while a spoken track answers understanding questions, and the understanding layer is what most investigations lack. Dual-track recording compensates for the threshold and the language gate at once, and it is an advance line of defence for the handover rupture in Chapter 30: what the spoken track preserves is the part handovers lose first.

Aviation has the lowest recording threshold and the most complete coverage of any industry, which makes it a convenient ruler. The U.S. National Transportation Safety Board (NTSB) investigates every accident involving scheduled passenger airlines (carriers operating under 14 CFR Part 121). For general aviation accidents (private light aircraft, flying clubs, agricultural and forestry work) it sends an on-scene team to only a share of them, closing the rest in writing. 〔Verification pending: on-scene coverage rates would need checking against NTSB Annual Review volumes; this sentence is a qualitative statement〕 Even in the best-recorded domain on earth, then, the reports you read cluster around the gravest consequences, the most common aircraft types, or the events that happen to be instructive. All forty aviation cases in this book's case library earned a formal investigation. That is what a threshold looks like from the inside.

China's recording system supplies a second ruler. The Regulations on the Reporting, Investigation and Handling of Production

Safety Accidents (State Council Decree No. 493, 2007) sort accidents into four grades by deaths, injuries and direct economic loss: general, relatively serious, serious, particularly serious. Each grade goes to a different tier of government for investigation. Grading is a sensible way to allocate resources, and it doubles as a screen. Investigative capacity follows the grade, so the industry's large volume of sub-serious injuries, damaged equipment and near misses has almost no public written existence. Civil aviation is the exception: the Civil Aviation Administration of China publishes investigation reports and progress bulletins inside the International Civil Aviation Organization (ICAO) Annex 13 framework, which makes it one of the few domains with a full-text outlet. 〔Verification pending: the current text of the Civil Aircraft Incident Investigation Regulations (CCAR-395) needs checking〕

The shape of the threshold is legible from inside the records too. The U.S. Consumer Product Safety Commission recall notice for Cub Cadet commercial lawn mowers (Appendix B, mn-010) records that 106 fuel-leak reports had piled up, one of them a fire, before the recall came. The records were never missing; institutional pacing kept them behind the risk. Set that against a foodborne outbreak investigation (Appendix B, ag-001: 33 days from first illness to production halt and recall) and the difference in threshold height between two industries in the same period is plain.

There is a counter-intuitive corollary here worth spelling out. The threshold is the price tag on a learning opportunity. The mechanism is sample size. In a high-threshold industry, reports on a class of accident are sparse, so confirming a pattern means waiting for the pattern to repeat several times. In a low-threshold industry, the pattern is written into the textbook on its first appearance. Aviation's long run of progress has an underrated cause here: people credit the technology and the culture, while the density of the recording system is the foundation underneath both. For a policy

maker the corollary is direct. Every cut in the filing threshold, every near miss pulled into the record, is a trade of administrative cost for learning speed, and inside this book's frame that trade is almost always worth making: records are priced in thousands, accidents in tens of millions. So before you quote any accident rate, ask where the threshold sits.

A quick table helps fix an industry's position by its recording regime, in four tiers. Tier one, mandatory filing with full-text publication: civil aviation accidents, major hazardous-chemical accidents. Tier two, mandatory filing with restricted publication: most workplace injuries, routine food and drug enforcement. Tier three, voluntary reporting with aggregate publication: near-miss systems, consumer complaint databases. Tier four, no institutional record at all: accidents in small and mid-sized firms, the quiet death of software, accidents at home. The table's use is to calibrate your language. Quote the tier whenever you quote a rate, because a rate without a tier is just a word count. One further corollary: comparing rates across industries is meaningless, since a tier one number and a tier four number are not the same kind of object. What is comparable across industries is whether a mechanism is present or absent, which is precisely what the matrix in Appendix A sets out to show.

No record below the threshold does not mean no information. The information exists in another form: experience carried in practitioners' bodies. A senior mechanic knows which line has a temper. A nurse knows which combination of orders tends to go wrong. That knowledge leaves with staff turnover and resets to zero at retirement, because it never touched paper. The handover rupture in Chapter 30 is the account of how this sub-threshold knowledge disappears.

What you learn wrong, in order of the damage it does. One: you treat recording frequency as occurrence frequency. "We never

hear of that kind of accident here" usually means "we do not record that kind of accident here". Two: you treat recording quality as safety level. Aviation looks like the industry with the sharpest accident analysis because it has the best recording system, and those two facts are correlated while pointing in different directions. A field that suddenly produces a lot of failure analysis may be getting safer, or may simply have just cleared its filing threshold. Three: you ignore near misses by default. The statistics assembled after the fact show a crowd of near-miss precursors behind every major accident; a precursor that never crosses the threshold never enters the sample, so it never earns a place in the learning material.

Formal recovery of near misses is the half-answer aviation has produced to the recording bias. The form is the voluntary reporting system: front-line staff report errors and near misses anonymously or under protection, and the report enters the database without penalty. Aviation's confidential reporting systems and flight safety action programmes are the mature models, and healthcare and maritime transport have carried versions across. The voluntary system moved sub-threshold failure into the written world for the first time, shifting the sample from a grading by consequence to a grading by risk: a near miss with small consequences and high risk finally gets a file. The word "half" deserves honesty. Voluntary rates rest on trust, meaning the protection promise and its delivery; break the promise once and the rate falls off a cliff, with the price list in Chapter 19 applying directly. So before you read any near-miss data, check the protection clause and the record of how it has been honoured. The temperature of the data is the temperature of the trust.

A second route is active inspection: do not wait to be told, walk the site against a list. Safety inspections, process walkthroughs and shadow audits all belong here. Inspection pushes the defence from

voluntary to institutional, so it no longer depends on the courage of a reporter; its cost is labour and its coverage is always partial. Its bias is checklist-shaped, finding the hazards on the list and missing new forms that are not on it, which means the list itself has to be updated from the accident library. Together the two routes give an industry a recognisable profile. Voluntary systems supply depth, the fine grain from the floor, and inspection supplies breadth, the corners nobody volunteers. How well those two legs cover the ground measures an industry's recording maturity more truly than any claim about its safety culture.

2.2 The Survivorship Bias: The Storytellers Have Been Sifted

The threshold filters events. Survivorship bias filters storytellers. Failure leaves no teller behind as a matter of structure. The failed lose the position, the platform and the credibility that a story needs, and the degree of failure runs in inverse proportion to will - ingness to tell: the more total the collapse, the fewer people left to describe it. The failure narratives in circulation therefore come mostly from people who are doing reasonably well.

The filter has an occupational layer. People who stay in an industry after a failure work for someone, and what they can say is bounded by that employer: describing a former employer's failure carries career risk, because the trade is small and the circle is tight. The constraint pushes public telling toward two safe subjects, a former employer that has already folded and cannot retaliate, and epis - odes unconnected to the present employer that time has laundered. The overlap between the two sets is tiny, so total output is pressed very low. This is one micro-mechanism behind the question Chapter 4 takes up, why organizations seal off failure. The seal is a structure rather than a policy, and every potential teller is

running career arithmetic inside it. The arithmetic shows up as scarcity in public content. When a candid failure review does reach you, remember that an author produced it after running that arithmetic: the courage is real, and so is the cost.

Commercial publishing amplifies the filter. A failure chapter in a CEO memoir is written by the person who was on stage, recalling a peril that has since resolved; the executive who carried out the bad decision and lost the job over it usually has no book contract. Interview samples behave the same way, since founders who agree to sit for a failure interview are the ones with energy and social appetite left after failing. The U.S. data firm CB Insights assembles startup failure analyses from postmortems that founders submit voluntarily, a self-reported sample in which the more painful a case is, the likelier it is to be missing. (A compilation drawn from public postmortems and shutdown notices, self-selected sample; its pages have been checked)

Software offers a sharper observation window because it has a public postmortem tradition, and Chapter 4 treats that exception on its own terms. The relevant point here is who writes them: companies that were still alive after the incident and still had an engineering team to write the document. A company that died of its architecture never issues a postmortem of the death; its architecture documents go offline with the servers. Even in the most transparent industry, then, the set of public postmortems is a survivor set. It covers the cases that fell ill and recovered, never the cases that died of the illness. The mechanisms behind those two may have nothing in common. One is an immune-system problem, the other organ failure, and only the first is available to you.

Survivorship has a temporal form as well. Old cases that reach you have been through several rounds of selection, and the ones retold are the ones with drama, a teachable point, reach. The dull lethal cases drop out of circulation in silence. The countermeasure is to

go back to the archive: pull every filed report for a given year rather than reading the three that everybody cites. Uniform sampling from the archive gives you an unbiased sample. It costs reading volume, since a year's output may run to dozens of reports. What it returns is the true density of an industry's memory: which patterns are frequent, and which are merely amplified by transmission. The gap between the two is the distortion of narrative, and being able to see that gap is a basic skill of reading reports. The three biases in this chapter are its tutorial.

What you learn wrong, again in order of damage. One: the catalogue of ways to die systematically omits the most lethal entries. The fast, destructive endings, cash flow rupture, the core team leaving together, a single regulatory veto, are the endings with no time to write a review and the highest shame cost, so they appear least often in circulating knowledge and sit lowest in your own head. Two: you misjudge an industry's transparency. An industry with many postmortem articles may simply have many companies alive to write them; review culture and mortality are separate variables. Three: you learn risk avoidance from survivors. A survivor explaining what they got right is a lottery winner explaining how he picked numbers, methodologically speaking. Extract what the survivor says about what was visible at the time; discard the explanation of why they won.

A purchasing-level move is available against survivorship: buy death samples. Three categories of public source exist. Regulators publish cancellation and revocation records, courts and registries publish bankruptcy and liquidation filings, industry databases publish shutdown lists. None carries a postmortem and all carry facts: when the firm died, how large it was, what its last public act was. The right use of a death sample is not to learn the cause of death, since nobody tells you, but to calibrate distribution: how many firms in your industry die in a year, at what size, in what

pattern. The picture you build from survivor narratives is missing that entire block. Once the distribution is calibrated, every survivor lesson shows its conditions, and the first question becomes whether the dead ever tried the survivor's practice. Mostly there is no record either way, and that absence is itself a reading.

The archival view of death samples holds one more item that surprises people: a dead company's digital estate outlives it longer than expected. Domain registration records, the delisting date in an app store, archived job postings, the last post on a social account: any of these puts dates on a death. Assembling the timeline means cross-dating public traces, precise enough to answer half the questions of business history, such as when decline began and when the last expansion move was made. The methodological bar is low, public sources plus patience. What is scarce is that nobody does it, since digital archaeology produces neither traffic nor degrees. This chapter carries it as a public toolkit for death-sample collection.

2.3 The Attribution Bias: Every Review Is Written After the Ending

The first two biases decide which cases reach you. Attribution decides what shape they are told in. Every review is written after the outcome is known, and knowing the ending changes how the reteller rebuilds the causal chain. Chance gets woven into necessity. Forks get flattened. The fog that was actually there gets compressed into clear error or clear correctness.

The Challenger accident is the most thoroughly studied sample available. At 11:38 a.m. Eastern time on 28 January 1986, Challenger lifted off from Cape Canaveral, Florida, and broke up about 73 seconds into flight, killing all seven astronauts aboard. The Rogers Commission report opens Volume 1, Chapter 3, "The Acci-

dent”, with the sentence that the flight began at 11:38 a.m. EST on 28 January 1986, and records that it ended 73 seconds later in an explosive burn of hydrogen and oxygen propellants; the official text is at nasa.gov/history/rogersrep/v1ch3.htm, read and checked for this book on 2026-11-01. On the evening before, 27 January 1986, engineers at the contractor Morton Thiokol recommended in a teleconference against launching in the cold. Their grounds were that O-rings in the solid rocket booster field joints recovered from earlier flights had shown erosion at low temperature, and that evidence linked low temperature to seal failure. Management at the company then changed its position and approved the launch. One sentence from the meeting where the position changed is on the record: take off your engineering hat and put on your man‑agement hat (Rogers Commission report, Volume 1, Chapter 5, “Contributing Causes”, where Boisjoly’s testimony recounts Mason’s words to Lund as “take off your engineering hat and put on your management hat”; official text at nasa.gov/history/roger‑srep/v1ch5.htm, read and checked for this book on 2026-11-01).

With the ending known, attribution can be run at several depths. The physical layer is the first: cold robbed the O-rings of their re‑silience, and hot gas burned through the seal. That layer holds fully, and it is the physical mechanism the Rogers Commission it‑self established. The organizational layer is the second, and it is the explanation Diane Vaughan gave in The Challenger Launch Decision (University of Chicago Press, 1996). The company and the agency had launched before with erosion visible in the joints, and had succeeded each time; each success was written up as evidence that the deviation sat inside acceptable limits; the decision stand‑ard moved gradually under the weight of those successes, and the decision on the evening of 27 January was one routine slide along that track. This layer holds as well.

The two layers do not conflict, and they train entirely different readers. A reader who stops at the physical layer learns that cold is dangerous, and goes back to his own organization to put a gate on a single parameter called temperature. A reader who takes in the organizational layer goes back to his own ledger with a different question: do we have a deviation indicator that we have already run successfully many times, each time explained as still having margin? One case, and the depth of attribution determines what the learning produces. Which layer reaches you in the market de‑pends on who is telling. Accident bulletins gravitate to the physical layer, management scholars to the organizational layer, participants' memoirs to the luck layer.

Two further layers complete the set. The incentive layer reads the decision-maker's behaviour against the rewards and penalties in force at the time, what the appraisal system paid for and what it punished, which is the frame of Chapter 17. The structural layer asks whether another person in the same structure would have made the same choice, and the historical cases where the structure stayed put, the personnel changed and the accident repeated are its hard proof. The mnemonic for running all four in order: the physical layer answers how it broke, the organizational layer how it went wrong, the incentive layer why the pull toward proceeding was strong, the structural layer why it happened again. Only a reader who runs all four has earned the whole lesson from one accident; a reader who runs one gets the first page of the same syllabus every time.

Attribution bias also has an institutional form, and it deserves a straight look: an investigation report is itself written after the fact. The investigators know the ending. Procedure requires them to exhaust hypotheses before converging, yet the pull of narrative is always there. The counterweights are procedural: evidence num‑bering, a public case file, screening of multiple hypotheses, every

conclusion tagged with its evidence grade. NTSB reports attach an evidence chain to each conclusion, so a reader can disagree with a finding and still audit the reasoning, which is the fundamental reason such reports beat news retellings. Residual bias cannot be zeroed. It can be compressed.

Three steps correct for what remains. Read the structure: the table of contents and the numbering of conclusions. A report whose conclusions are individually numbered and evidence-tagged has had its bias pressed down by procedure; a report that runs as one continuous story lets bias grow freely. Read the dissent: did the board record any disagreement during the investigation? Dissent makes an investigation more credible, and a unanimous conclusion deserves a second look for narrative gravity. Read the follow-up: evidence that arrived after publication. A report whose conclusions were revised keeps its most honest material in the revision record, and an agency willing to revise has earned long-term trust. The three steps cost an extra hour, and they buy a defensible price for any investigative conclusion. A report is neither scripture nor waste paper: it is the best sample available, carrying residual bias.

What you learn wrong here, in order of damage. One: you overrate "they should have seen it". Played backwards from the ending, every fork is clearly visible; standing at the decision point, information arrives mixed with noise. Two: you fail to acquire field judgment. Reading finished attribution is watching someone else's finished game, and your own play does not improve, which is why every case in this book stops at the decision point, the third clause of the reader's contract in the preface. Three: you flatten complex failure into a single cause. Post hoc attribution tends to converge on one root cause, while a real failure chain usually spans the technical, organizational and incentive layers (the failure-chain model in Chapter 9). A flattened root cause is convenient for briefing and useless for prevention.

Rebuilding a decision point is a semi-technical drill you can teach yourself on any review document. Step one, sort by time: order the material's information by arrival, separate what came before the decision from what came after, and tag everything after as hindsight. Step two, assign weight: for each pre-decision item, record how reliable it looked, how urgent, and how much noise it was mixed into. Step three, find the forks: list every option that existed at the moment, including the ones killed off in a single sentence. What you hold is a desk reconstruction of the decision point, more accurate the fuller the material; where the material is thin, say so. The drill costs about an hour per document, and it trains the capability Chapter 6 formalizes as what was visible at the time. This chapter gives the household version; Chapter 6 gives the instrument version.

Against the single-cause briefing habit there is an institutional antidote: conclusions written in layers. The report states its physical-layer conclusions, its organizational-layer conclusions and its incentive-layer conclusions in separate sections, and the summary is not allowed to merge them into one sentence. This is the standard structure of a good investigation report, where findings are numbered one by one and each carries a layer tag and an evidence-strength tag. Move it into internal reviews and the rule becomes concrete: no root-cause field on the conclusions page. Use three columns instead, one each for technical, organizational and incentive, with at least one entry per column. The three-column review costs two extra minutes in the briefing and yields three times the handles in prevention, since one entry per layer is three things that can actually be changed. Between briefing efficiency and preventive effect, this chapter recommends the second every time.

2.4 What You Actually Hold Once the Three Biases Stack

A compressed answer first, then the unpacking. Once the three biases stack, what you hold is a collection of failures written for you by survivors, filtered by thresholds, and reassembled after the fact, and your brain will digest it as a full sample, a neutral account and causal truth. The compressed version sets the tone for the rest of this chapter. Every technique here, the threshold table, the three quality checks, the four layers, the desk reconstruction, is resistance to that digestion error. The posture behind the resistance is an admission: bias cannot be cleaned to zero. What can be done is to put a scale on each document so the reader carries it into the reading. Two readers, one with the scale and one without, will split their trust differently over the same document. That is the whole capability this chapter means to hand over.

All three biases act on every case at once. Run a full check-up on aviation, the lightest case in any industry: recording is mandatory, telling is done by a statutory body, attribution runs under procedural discipline. Even so it carries residuals. Near misses are largely absent, which is the recording bias. Public reviews are written by institutions that still exist, which is the survivorship bias. The report's structure converges on causes, which is the attribution bias. Other industries carry more, never less. Chemicals has a filing regime whose coverage concentrates on major accidents. Software has a review culture whose sample is all survivors. Healthcare event reporting runs at low voluntary rates in every country. Finance scatters its failure record across penalty documents and litigation, and almost nobody assembles it at industry level.

Here is a sample quality check you can run on any case before reading it, three questions long.

1. Check the threshold. How did this case enter the record: voluntary report, mandatory filing, media exposure? The threshold tells you which consequence tier it belongs to.
2. Count the autopsies. What is the teller's stake in the outcome? Does the institution still exist? Was the teller a decision-maker, an executor or a bystander at the time?
3. Take the attribution apart. Does the account mark the line between visible at the time and reconstructed afterwards? An account that marks nothing defaults to hindsight from beginning to end.

Run the three and your trust in a case has a scale on it. Fields 6 and 12 of the case card, visible at the time and evidence grade, are these three checks done in advance on the writing side.

The scale has one last place to go: into other people's hands. The fastest way to internalize the three biases is to explain them to your team, and one telling beats ten readings. The skeleton is in this chapter: one sentence per bias, the four-tier industry table, the three sample checks, the four-layer mnemonic. Four pieces of material fill ninety minutes without strain. For the live exercise, take the team's most recent internal review and run the three checks on it in the room: threshold, teller, layers. The effect shows at once, because the team reads reviews differently afterwards. The cost is ninety minutes, and what it buys is an upgrade in the team's literacy in failure knowledge, the most practical of the gifts the meta-question part gives an organization.

One more drill on the reader's side consolidates the writing-side work: spot-check the book at random. Pick a card, read fields 1 to 5 only, commit to your own judgment and conclusion, then compare against field 6 and the outcome. The reading works two ways. The

farther your judgment sits from the outcome, the stronger the hindsight habit in you. The closer it sits to field 6, the better you have learned to stand at the decision point. One card a month, twelve a year, and the bias becomes muscle memory, which forms no other way than in real judgments; the morning meeting script in Chapter 35 is the practice ground.

The organizational form of the same immunity is worth writing down: put the three biases into the review template. Every internal review carries three fixed columns, what was this report's recording threshold, what is the teller's stake, how many layers did the attribution take. The cost is one extra page, and the effect is that every review declares its own provenance. A review that declares its bias has a credibility scale; one that declares nothing is discounted to the worst case by default. The columns share their source with the record-keeping regime in Chapter 38, since the credibility of a review is part of the record, and the philosophy extends one step: even the bias gets recorded.

Five Self-Checks

1. What is the accident recording threshold in your industry? In the past year, how many events crossed it, and how many did you know about that did not?
2. Of the failure cases you have cited, how many trace back to a primary full text? If you can name the document and its number, it counts.
3. Of the reviews your organization ran last year, how many separated information visible at the time from information known only afterwards?
4. In your industry, who talks publicly about failure, the winners or the losers? Can you put an order of magnitude on the industry's autopsy rate?

5. Take your team's most recent review and rewrite one causal conclusion in its visible-at-the-time version. Does the conclusion still stand?

Beyond the five, there is an annual self-audit card for the three biases, kept personally. Question one: of the failure cases you cited in the past year, how many clear the threshold in 2.1, meaning they were formally filed and have a full text? Question two: of the failures you have told, how many were told with the people involved absent, since those present cannot tell it and you are telling it in their place? Question three: of the reviews you wrote, how many drew the line between visible at the time and reconstructed afterwards? The three answers make up your personal bias profile, and every gap in it is an improvement item for the coming year. The tools are all in this chapter: contact with voluntary systems, the purchase of death samples, the four-layer mnemonic, the three-step desk reconstruction. Profile and tools are both in place. What remains is the hours you put on the calendar.

Chapter 3 The Five Fields That Record Failure in Earnest

> What this chapter covers: how failure-record institutions are built in aviation, chemicals, maritime, medicine, and nuclear power; the four devices that make a record "in earnest"; and the channels a reader in China can actually reach. When to come back: when you need first-hand failure material for your own industry, or when you have to judge how much an outside accident report deserves to be trusted. Reading time: about 30 minutes.

Failure knowledge does not accumulate out of conscientiousness. It accumulates out of institutions. Worldwide there are only about five fields that keep failure records at scale, and although none of them ever compared notes with the others, the machinery they grew looks strikingly alike. This chapter takes that machinery apart into four parts, then checks how far each of the five fields has assembled them, and finishes on the channels a reader in China can reach today.

3.1 What "In Earnest" Means: Four Devices

A field's failure-record system earns the word "earnest" only if four devices are in place at the same time.

Device one: mandatory reporting. Statutes must state which events carry a reporting duty and what happens when a report is withheld. Where the duty is optional, the reporting rate tracks individual willingness from month to month, and the sample stops meaning anything.

Device two: independent investigation. The investigating body answers neither to the party under investigation nor to the enforcement arm that issues penalties. Its product is a statement of what

happened and why. Allocating blame is a separate trade. Merge the two and the party under investigation will buy safety with silence.

Device three: publication in full. Reports must be obtainable whole, numbered, and citable. A field that issues press releases and nothing else has headlines and no body text.

Device four: liability separation. What is written in a report must, as a matter of principle, stay out of proceedings against individu‑ als, and the statements of those involved must carry no threat of prosecution (gross negligence excepted). Take this one away and the other three collapse together, because nobody writes into a report the details that would put him in prison.

The four devices lock into one another. Without the second, the fourth has no institutional carrier. Without the third, the output of the first two cannot circulate. Without the first, the sample facing the other three is defective at the source.

One further criterion belongs in front of the inspection, and it is a criterion about criteria: redundancy. For each device, is a second body keeping a backup copy of the function? Beyond the NTSB there is the FAA’s continuing oversight, the insurance industry’s investigation teams, and the industry’s own voluntary reporting systems. An agency that is the only one doing its job is a candidate for Chapter 22 no matter how good it is. Redundancy varies enormously between fields. Aviation’s four devices each stand on several legs; most industries have one leg per device, sometimes one leg across all four. The number of legs is what decides how a record system survives a budget crisis, a political intervention, or the departure of the one person who cared. A reader scoring his own field should add a fifth column to the four: how many legs does each device stand on. The five-column total is the operational definition of the word in this chapter’s title.

The interlock also has a direction, and the direction is worth drawing as a dependency graph. Let liability separation fail and report content self-censors, a failure that propagates downstream from device four. Let publication fail and independent investigation works for nothing, propagating from three to two. Let independent investigation fail and mandatory reporting loses its point, propagating from two to one. The repair order is that graph read backwards: first the holes in mandatory reporting, then the independence of investigation, then publication, and liability separation last. The value of the order is diagnostic. When a field's record system has gone to seed, find the worst device and start there; repair anywhere else has no leverage. The lever is checked with one question: in this field's reports, does the person involved dare write the truth? On that question the first priority is always device four.

3.2 Aviation

Aviation is the prototype of this system, and it earned the role through accumulation. Aviation's record institutions were not designed in one sitting; they were bought layer by layer with disasters, and behind each reporting duty and each independent agency sits a famous investigation report. For a designer the stratified history is a free course: each founding document records the correspondence between the accident shape of its moment and the gap it exposed. The stratification continues, with record rules for unmanned aircraft and the airworthiness framework for electric vertical take-off and landing vehicles being written now, which makes new vehicle categories a live sample of institution-building. Anyone pushing for better records at home can treat aviation's history as the most complete set of construction drawings available. The drawings are aviation's; the mechanics of the structure are general.

At the international level, ICAO Annex 13, Aircraft Accident and Incident Investigation, sets three principles. The state of registry and the state of occurrence both carry an obligation to investigate. Investigation is separated from judicial accountability, and the report is not used to apportion blame. Reports should be published as promptly as possible. Major states have set up independent agencies under Annex 13 and produce a steady output: the NTSB in the United States, the AAIB in the United Kingdom, the BEA in France, the JTSB in Japan, the BFU in Germany, the ANSV in Italy, the AESA in Spain (with the CIAIAC under it), the ATSB in Australia, and the TSB in Canada. Reports are published in full and numbered, as with NTSB AAR-10/01 or AAIB AAR 1/2023, and major investigations open a public docket in which physical evidence, interview records, and transcripts of communications are itemized entry by entry.

China's civil aviation runs inside that international framework. The Civil Aviation Administration of China organizes accident investigations and publishes reports under the Annex 13 structure. Major accidents appear as progress bulletins: the March 21, 2022 crash of China Eastern flight MU5735, in which all 123 passengers and 9 crew members died, produced a progress bulletin from the CAAC on March 20, 2024. How fast the final report arrives and how complete it is are questions the reader can settle from the case library (see the av-038 entry in Appendix B).

Aviation produces a second thing that gets overlooked: follow-up. Every NTSB safety recommendation carries a number, the recipient must answer in writing, and whether the recommendation was adopted is a matter of public record. Failure knowledge is not only the analysis. It is also the ledger of what changed after the analysis.

The ledger has cross-industry value. The adoption-rate curve is a direct indicator of safety governance: sustained high adoption means a working channel from recommendation to practice, and a

slide means fatigue or financial strain. The data are public in the NTSB's recommendation status database and on the CSB's recommendation tracking pages, and the reading belongs to Chapter 19's landing-rate logic, where the landing rate is what an organization charges for the truth. The two chapters interlock exactly here. A reader outside aviation should still memorize the mechanism, because it is portable: any regulator's recommendation deserves a number, a named recipient, a response deadline, and a public status.

The cost of the aviation system is on the record too: investigations run long. Take the January 2, 2024 collision at Tokyo Haneda between a Japan Airlines A350 and a Japan Coast Guard aircraft. Five people died aboard the Coast Guard aircraft and its captain was seriously injured; all 379 people aboard the airliner were evacuated, with one passenger seriously injured and four lightly injured. The JTSB's second interim report, released December 25, 2025, still came without a final report, close to two years after the event (av-037, Appendix B). Deep investigation is priced in years. That is what "in earnest" costs.

The other half of the price list is that speed is tiered. Aviation's graduated publication is the mature pattern: a preliminary report within days, an interim within months, a final when it is ready, safety recommendations at any time. The effect is to break the information flow into timestamped statements, each declaring what is known and what is not at that moment, which serves public information and investigative rigour at once, two demands usually treated as opposites. The cost of transplanting it is publication discipline, since every interim must separate the confirmed from the pending in its wording. The return is credibility: public patience with progress bulletins comes from bulletins that never claim more than the evidence holds. The Haneda interim reports are a working sample, and a reader can study them as a template.

3.3 Chemicals and Process Industries

In the United States the chemical sector's investigative device is the Chemical Safety and Hazard Investigation Board (CSB), created under the Clean Air Act Amendments of 1990, independent of the enforcement agencies OSHA and EPA, and restricted to investigation with no power to penalize. (The CSB began actual operations in 1998; checked against the About page of its website.) Its output comes in two parts: full investigation reports, which include animated reconstructions of the event and are unusually public-facing for an investigation agency, and execution tracking of its safety recommendations.

The institutional difference from aviation is coverage. Aviation's investigator has jurisdiction over every major accident in the field. Chemical investigation resources concentrate on the worst events, while small and mid-sized releases and near misses are left to company management and local regulators. The UK's HSE system and the continental European arrangements are comparatively complete. In most developing countries, chemical accident records depend on press coverage and full texts are scarce. In China, chemical accident investigation reports are issued by the Ministry of Emergency Management and by provincial and municipal emergency management departments; major accidents get a public report, smaller ones get a bulletin, and completeness varies (Chapter 31 carries the channel details).

A further dimension of chemical recordkeeping sits inside the fence line: the internal records of a process safety management (PSM) system. A mature PSM system keeps three families of record alive year round, covering process hazard analysis, management of change, and mechanical integrity. Their quality sets the pace at which the company learns, and in an investigation they form a key evidence layer. Internal records and public investigation are complements: the report yields the industry's lesson, the internal re-

cord yields this plant's specifics. How well the layers are connected is a gauge of learning capacity, and at most companies they are not connected at all. The report is read and filed, the records stay locked in the system, and nobody does the translation. That post, mapping an industry report's lessons onto the facilities on your own site, has a ready tool in Chapter 35's locating table.

3.4 Maritime and Transport

Maritime failure records are built from the rule framework of the International Maritime Organization (IMO) plus the accident investigation bodies of individual states. The UK's MAIB is the model agency: reports published in full, independence from maritime justice, and an investigative stance aimed at safety improvement rather than fault-finding.

What makes maritime distinctive is fragmented jurisdiction. The state whose flag the ship flies leads the investigation, and the flag state, the coastal state, and the state of the crew's nationality may be three different agencies. Report quality therefore tracks the institutional level of whichever flag happens to be involved. The same accident under a different flag may leave no public full text at all, which is a warning worth carrying: in the maritime field, the absence of a report frequently says nothing about the absence of an accident.

The industry has a consensus model for this: a lead investigating state with assisting states, jurisdiction concentrated in one body while the others help as needed. How well it works depends on the good faith of the assistance, meaning how fast data are shared and whether access to witnesses is granted, and gaps in that good faith are the largest single reason maritime investigations are slow and incomplete. Hence a cheap reading habit: look twice at a report's acknowledgements. The list of assisting states and technical as-

sistance is the report's collaboration network, and the wider the net, the more complete the perspectives inside.

A second line of repair runs through the industry's own records. Safety management system rules require ship operators to keep maintenance and event records aboard, which gives a second layer of documentation beneath the flag state's paperwork; the cross-checking value of reading the two layers side by side was set out in Chapter 16. And fragmentation has a search implication. A single accident can leave several documents behind: the lead state's investigation report, the flag state's position paper, the coastal state's administrative record, the insurer's survey report. They fill in different parts of the picture, with the lead report giving the process, the insurance report giving the loss, and the administrative record giving the penalty. The complete search move is document genealogy, pulling the existence of the other documents out of one report's references, and A maritime reader should practise it until it is a habit.

One more public channel in this field deserves registration: port state control detention records. Which ships were detained and why are public, and the record works as a health check on the state of a hull, in a form that rewards batch analysis.

3.5 Medicine and Devices

Medicine's device takes a different shape. What it records is not the accident but the adverse event and the adverse reaction. The FDA's MAUDE database collects medical device adverse event reports, with mandatory reporting duties on manufacturers and on user facilities; the drug side runs on the FDA's Adverse Event Reporting System (FAERS) and on the European Union's pharmacovigilance system; China's counterpart is the National Medical Products Administration's (NMPA) bulletins and recall an-

nouncements together with the drug adverse reaction monitoring system. 〔Verification pending: the NMPA site loads dynamically in JavaScript; annual report figures to be checked once access is in place; this sentence is qualitative.〕

Score medicine against the four devices. Reporting is mandatory, so device one is present. The investigation side has no aviation-style independent accident body, so device two is missing, with post-market "investigation" replaced by signal detection and safety updates. Publication is half there, since the databases are queryable while individual records vary in quality. Liability separation is present, because the device adverse event reporting rules state explicitly that a report is not a determination of liability (see the language of 21 CFR Part 803, with the manufacturer duty at 21 CFR 803.50, checked against the current eCFR; user facility reporting has its own provision in the same part).

That structure produces a skew peculiar to medical data. Reporting rates move with how recent an event is and with how hard the causal link is to judge, and the same adverse event can arrive several times through several channels or never arrive at all.

Duplication and undercount together give the data user an operating rule: aggregate by event rather than by report count, merge the several channel reports of one event, and declare the gap left by under-reporting when you interpret anything. Aggregation needs event identification fields, a composite key of product, time, and symptom; records missing them, typically free text in older systems, must be read by hand, and that reading is the main cost of medical data analysis. Cross-border advice for readers from other fields: whenever you cite a medical figure, state that it counts reports, not cases. The distance between the two numbers is a function of the reporting system, and every loosening or tightening moves the ratio. One line of disclosure blocks nine tenths of the misuse. To Chapter 2's threshold checks add one rule: read med-

ical data as signals, and treat any use as evidence as requiring separate epidemiological verification.

There is a further direction worth adding, about the gap between signal and evidence: the absence of a signal is itself a signal. A device that has been on the market for years and shows zero reports in MAUDE may be safe, or may reflect an absent reporting habit, since compliance with user facility reporting duties is low worldwide. The correct reading of a zero is uncertainty, which is neither proof of safety nor evidence of a hidden problem, and the action it prompts is active checking: follow-up literature from clinical trials, comparable data from other jurisdictions. Misreading the zero is the common cold of medical data use, taking a quiet database for a clean product. This chapter files it under the medical edition of record skew. The phrase to keep on hand when citing medical data is: what grade of quiet is this quiet?

3.6 Nuclear Power and Energy

Nuclear power has the deepest international sharing of any field.

One detail of that sharing deserves study by other industries: the grading of events. The International Nuclear Event Scale (INES) puts events on a single 0 to 7 scale with explicit criteria at every level. Its governance value lies in the language, since a common scale lets insiders and outsiders talk: the public understands that a 6 outweighs a 5, and the press no longer invents its own explanation. The cost is coarsening, because differences within a level are flattened and two level 1 events can carry wildly different risk. Coarse and fine is a permanent trade-off; coarse serves public communication, fine serves professional analysis, and running both at once is the mature arrangement. Any field building event grading can copy the two-layer structure, coarse outside and fine

inside, with INES's public documents as a model for writing the criteria.

The International Atomic Energy Agency (IAEA) and the OECD Nuclear Energy Agency jointly operate the Incident Reporting System (IRS). Operating organizations in member states report operating events, and the database is shared among member states. The US Nuclear Regulatory Commission requires licensed operators to submit operating event reports (regulatory location: 10 CFR Part 50.73, the licensee event report system, checked against the current eCFR). The World Association of Nuclear Operators (WANO) runs a further reporting and assessment system inside the industry that is not open to outsiders. Nuclear events also carry an INES rating, published on the 0 to 7 scale.

The weak point in nuclear's assembly is publication. IRS detail is restricted to member states, WANO is entirely closed, and what the open world can obtain is whatever each national regulator chooses to publish. This is the first sample in the book of a system that is institutionally serious and externally sealed: how deep a reader can go depends on the openness of his own country's regulator.

The sealed side has an open substitute inside the industry. Some of WANO's technical material is published as ordinary documents, training material and summaries of operating experience analysis, shallower than the database but usable. The way to use it is to borrow across fields: a reader from another industry can read nuclear's public output as a textbook, because nuclear operating experience analysis is among the most rigorously written public material in process industries. The direction of the borrowing is worth naming plainly. Nuclear is missing device three and its internal rigour is the ceiling for the whole industrial world. The gap between inside and outside is a fact of the institution, and the spill of internal material into publications is a ride the reader can take for free.

3.7 Assembling the Four Devices

Field	Mandatory reporting	Independent investigation	Publication in full	Liability separation	Notes
Aviation	Full	Full	Full	Full (Annex 13 principles)	The prototype; the most complete tracking ledger
Chemicals	Medium (major accidents)	Independent in the US through the CSB; uneven elsewhere	Medium	Medium	Small and mid-sized accidents rest on internal records
Maritime	Full	Split (follows the flag state)	Medium	Full	Fragmented jurisdiction
Medicine and devices	Full (adverse events)	Missing (replaced by signal detection)	Medium (individual database records vary)	Full	Read the data as signals
Nuclear power	Full	Full	Split (international systems closed; national regulatory documents vary in openness)	Full	External publication is the weak point

The first use of the table: find the row for your own industry, or the nearest neighbouring row. The gaps in that row are the discount to apply when you read outside cases.

The second use: find the device your field is missing and borrow cases from a field that has it. Chemicals has no public sample for small accidents, so a chemical reader borrows maritime individual

reports; the mechanism is the same and the parameters are con - verted on your own desk. The borrowing rule is in Section 3.9 be - low and in Chapter 23, which requires extrapolation to be declared: name the source industry, state the conversion assumptions, keep the account visible.

The third use is for institution builders. If you get a seat at the design of a record system, this table is the parts list: copy avi - ation's full set, de-specify to the budget, and declare what you de-specced rather than hiding it. All three uses rest on one stance. The table grades no field; it tells each reader where he stands. In chemicals, major accidents give you CSB-grade full texts and small ones give you nothing public, so your study of small accidents runs on analogy from aviation or maritime, and how much to discount that analogy is a number you write down yourself.

3.8 Where China Stands

A reader in China has five channels to failure records, and they differ sharply in availability and in quality.

Channel one: accident investigation reports. Under the Regulations on the Reporting, Investigation and Handling of Production Safety Accidents (State Council Decree No. 493), serious and especially serious accidents are investigated at the provincial level or by the State Council. Reports on major chemical and industrial accidents are published by the Ministry of Emergency Management and by provincial departments, with completeness varying from case to case. Against the four devices: mandatory reporting is present; in - dependent investigation is weak, because the investigation team is government-led and staffed from several departments, leaving in - vestigators and the people who allocate blame heavily overlapping; publication is incomplete, with plenty of bulletins and few full texts; liability separation is weak, because reporting and account -

ability run as one process, and a sequence that assigns blame be‑fore it finishes investigating steers every statement toward self-protection. 〔Verification pending: the original State Council gazette, Article 19 onward, needs checking.〕

A secondary phenomenon of missing liability separation deserves its own entry: language drift. A report written under accountability pressure drifts toward the unfalsifiable. Timestamps lose preci‑sion, descriptions of actions turn into characterizations, names turn into job titles. The drift is measurable. Compare the reports of a single institution across years, and the annual change in lin‑guistic precision works as a thermometer of accountability pres‑sure. The measurement has governance value, since the year the language shifts is usually the year the accountability procedure changed, and that correlation points the repair at procedure design rather than at individuals. The use of linguistic evidence in this chapter is institutional diagnosis: take the language’s temperat‑ure first, then talk about amending the institution.

A reading exercise for this section: take one published major acci‑dent report and work through it section by section, asking which section corresponds to which device. Facts map to device one, analysis to device two, the manner of publication to device three, the accountability section to device four. What comes out is a credibility map of that report, showing which sections can be cited freely and which need a second look, because accountability and facts often carry different weight. Three reports are enough to learn the drawing, and the skill lasts a working lifetime, since any investigation report, Chinese or foreign, can be priced quickly with the same frame. Making that pricing ability common is this chapter’s central delivery to Chinese readers: the number of channels is a fact of where you were born, and pricing what they produce is trainable.

The comparison also gives a company reader a place to start on his own organization. Score your internal reporting system against the four devices. Is internal reporting a duty or a virtue? Where does the internal investigation team sit relative to the department it in - vestigates? How granularly are findings published to the whole staff? What written protection does a reporter have? Most com - panies come out short on devices two and four, with internal in - vestigation and HR running on one line and no written protection for the person who speaks. Chapter 19 covers external intake and exemption clauses and Chapter 30 covers handover protection, and both carry ready designs. This chapter's only job is the reminder: internal systems and national systems run on the same mechanics, and what a state can learn, a firm can learn.

Channel two: civil aviation. The CAAC works within the ICAO An - nex 13 framework, which makes this the channel that meshes best with the four devices of the five. The MU5735 progress bulletin came through it.

Channel three: drugs and devices. NMPA recall announcements, bulletins, and the annual report on drug adverse reactions are standardized public outlets. The depth of any single record is limited, but retrieval works.

Channel four: court judgments. China Judgements Online once offered the largest public body of judgments in the world, and tort, product liability, and labour dispute documents carry a great deal of detail about organizational failure. Availability has changed in recent years. 〔Verification pending: the Supreme People's Court's annual work reports and judicial statistics need checking; this sentence is qualitative.〕

Channel five: exchange and regulatory documents. CSRC admin - istrative penalty decisions, exchange inquiry letters and regulatory letters, and audit reports cover financial and governance failure at

listed companies, in full text, publicly available, in a uniform format.

Stack the five and the conclusion is this: readers in China have usable full-text channels in civil aviation, in drugs, and in listed-company compliance. In chemicals, public safety, and several other areas, full texts are scarce, bulletins and press paraphrase carry the load, and the grade of the learning material drops below B (grade definitions in Chapter 10).

One combined technique with the five channels deserves teaching on its own: cross-channel corroboration, where documents about the same matter check one another. A CSRC penalty decision and a court judgment corroborate the detail of financial fraud; an NMPA recall announcement and a judgment corroborate a product quality dispute; a CAAC bulletin and a judgment corroborate an operational dispute. The output is an upgrade in evidence grade: B-grade material from a single channel can support a reconstruction tag once its core facts are corroborated through a second channel, under the rules in Chapter 10 and Appendix C. The cost is one search. The return is material promoted a grade, and corroboration is the retrieval move a Chinese reader should practise until it is automatic. It is the dividend this five-channel structure pays to the careful. The same structure sets the distribution of evidence grades among this book’s China cases, and explains why most of them are tagged “reconstructed”.

3.9 Putting the Map to Work

Three moves.

Move one: find the nearest primary library for your own industry. That means a specific database and a specific report numbering scheme, not a search engine.

Move two: read three recent full-text reports from your industry and note, for each, which of the four devices is present and which is missing. The missing one is the extra caution to carry into every page.

Move three: if none of the five fields is close to your industry, then before you set the strength of any analogy from a neighbouring field, run Chapter 2's three sample checks.

A fourth move sits outside those three: build the table of Section 3.7 for your own industry, with a column per device, a mark for present or missing, and a count of legs. The criteria are this chapter's and can be copied straight across. Building it is itself a survey of your field's record institutions, with the question list already supplied above, and it costs an afternoon. What it produces is a map of your industry's learning infrastructure as it stands, and the gaps on that map are the agenda for a proposal to your trade association or regulator. An agenda beats a complaint: a proposal that arrives with an assembly table is the only kind taken seriously at an industry meeting.

The Five Self-Checks

1. Which of the four devices is missing from your field's accident reporting system? Last year, how much detail did that gap cost in one specific case?
2. Your industry's most recent major failure: at what address does the full investigation report sit? How many minutes do you need to find it?
3. Beyond regulatory inspection, does your organization keep its own failure records? How would you estimate the reporting rate?

4. If only one could survive, the regulatory report or the internal postmortem, which would your organization keep? What follows from that choice?
5. Of China's five channels, how many have you used? For the ones you have not: did you not know the entrance, or did you search and come away empty?

A weekend assignment goes with the five questions, one full tour of the channels in half a day. In the morning, run channels one and two: a full text of a major accident report, and an aviation bulletin. In the afternoon, run channels three through five: an NMPA recall announcement, a court judgment, and a CSRC penalty decision. Read every document with the same question in hand: how many of the four devices show in this text. The output is five sets of reading notes plus a quick-reference table of China's record in - stitutions drawn by you, closer to your own use than the table in Section 3.8. A reader who finishes the assignment has converted the map of Chinese failure records from something heard about into something walked. That sense of footing comes before any retrieval work: the map comes first, and the tools only bite once the map is there.

Chapter 4 Why This Method Did Not Spread

> What this chapter covers: five domains proved that failure can be turned into an institution, so why has that institution stayed inside them for sixty years, and what special conditions let software become the one exception. When to come back: when you are about to propose "let's build a postmortem system" to management, or when you need to judge whether an industry's failure knowledge can be expected to grow on its own. Reading time: about 25 minutes.

None of the four devices listed in Chapter 3 depends on advanced technology, and none is expensive to legislate. Aviation had the full set installed by the middle of the twentieth century. In the seventy years since, no industry outside safety regulation has copied the set whole. The reason is structural. It sits in the incentive structure, and no one's negligence is required. This chapter counts the five gates in the way, takes apart the software exception, then hands over a framework for judging your own industry: whether it could grow this system, and whether you should wait for it.

The chapter's place in the book is worth stating once. It works as a pressure valve. A reader who finishes Chapter 3 tends to ask why, if the arrangement is this good, their own industry does not have it. The answer this chapter gives is that the absence is not a verdict on your industry. It is the mechanics of five gates.

Relief is followed by a turn, from waiting for an institution to building one. The templates in Chapters 38 and 39 exist for people who will not wait. The psychological threshold on that turn is loneliness. An organization that builds its own system is a minority inside its industry, and a minority does not need encouragement. It needs proof that peers exist. Software and the five precedent do-

mains supply that proof: building your own was never naive, because it has precedents, tools, and an endpoint.

This chapter hands over all three. Precedents are in Chapter 3. Tools are in Part 5. The endpoint is in Chapter 40, which draws the boundary of a self-built methodology.

4.1 The Five Gates

Gate one: liability. Writing down the details of a failure amounts to organizing the evidence for a lawsuit nobody has filed yet.

The five gates do not weigh the same, and the weight distribution varies enormously across industries. Aviation carries a light liability gate, because statutory liability isolation exists, and a heavy cost gate, because the public does not read accident reports. Software carries the lightest liability gate of any industry and the heaviest institutional gate, since nothing is legislated. Manufacturing carries the heaviest competition gate, because process detail is intelligence. A reader can score the five gates for their own industry, one to five each, and treat the two highest scores as the principal obstacles. Resources belong there. Spreading effort evenly across all five is the standard waste of institution building.

The distribution also moves. Technology changes the weight of the competition gate, since digitized process information lowers the threshold for arming a rival. Changes in the legal environment move the liability gate. Re-scoring once a year is routine work for anyone who keeps an industry's records.

Where the liability gate sits for internal postmortems is a question every legal department knows cold, under the discovery rules that govern what must be produced in litigation. The clearer the document, the harder the defense later. The rational choice is to write the postmortem vaguely, or to leave it unwritten. Compliance

documents such as inspection records and training sign-in sheets have mandated formats and must exist. A causal postmortem carries no such obligation, and is better not written at all. That is the real legal position of postmortem culture in a great many organizations.

There is a mature line of remedy for the liability gate, and it lives inside the law. Privilege and protection regimes already exist in various forms across legal systems: privilege attaching to audit workpapers, statutory protection for quality assessments, immunity clauses covering investigation reports. The design point is to pin down what a document may be used for, safety improvement only, never liability or litigation. Pinning it down takes a written clause plus usage discipline, and the clause has to specify where each part of the document is allowed to travel.

Protected documents already exist in regulated industries, which shows the route works: operating event reports in nuclear power enjoy limited protection under the laws of several countries. Resistance to transplanting the idea comes from an instinct inside legal departments, since every newly protected document is one more object that can be challenged. The trade-off between resistance and benefit is best settled in one three-party legal meeting, with corporate counsel, industry association counsel, and the regulatory line arguing it out together. Loosening the liability gate has always been talked through by lawyers. It has never been shouted through by engineers.

A document management protocol is the second tool on this gate, and it complements protection. Most jurisdictions have well-developed means for handling internal documents: labeling the purpose, restricting distribution, applying retention rules. The purpose is not to destroy evidence, which is a serious offense, but to keep causal discussion and compliance records under separate

management during normal operations, before any litigation starts.

The protocol has to be signed jointly by legal and safety. Safety understands what the content is worth. Legal understands where exposure begins. A protocol designed by one side alone will be defective on the other side. The output is a single document: which class of postmortem text carries which purpose label, how long it is kept, who may read it. The cost is one cross-department meeting and half a page of text. What it buys is a downgrade of the liability gate from absolute barrier to manageable constraint.

The boundary clause goes in the first paragraph. The protocol covers causal analysis written voluntarily; it does not cover records under a statutory retention obligation. Once that boundary is clear, the protocol survives scrutiny from both legal and the regulator. 〔Verification pending: a cross-jurisdiction comparison would need legal databases; this passage is a qualitative statement〕

Gate two: competition. Failure detail is concentrated competitive intelligence. A crash investigation report names the component, the fuel line, the checklist that failed. A postmortem of a commercial failure names the supplier, the parameter, the channel that failed. The first is read by the whole industry. The second is read by your competitors, among others. The patent system protects technical solutions. It does not protect lessons. Lessons lie in the rubble of the loser, and only competitors go digging there.

One economic exit exists for this gate: the alliance scale of a shared pool. The competition gate is tight between two rivals and loose among forty, because competitive intensity is diluted by scale. The more members a pool has, the less any single firm exposes and the more it gains.

A pool needs two mechanisms. A contribution threshold, meaning you submit cards to get in. An access rule, meaning you see as

much as you contribute. Both exist to stop free-riding, since read-only members will kill the pool. Precedents are everywhere once you look: insurance loss pools, and shared engine fleet data in aviation, which is a mature model. The model rests on two conditions, member homogeneity and a trusted third-party custodian, usually an industry association or a reinsurer. If competition in your industry is too fierce to share anything, the first question is whether a custodian has been found.

Gate three: face. The organizational narrative is an asset.

The deepest layer of this gate is individual psychology, where admitting error collides with narrative identity. A manager's professional identity is built on the correctness of past decisions, and a postmortem that looks directly at error is a direct challenge to that identity. At the individual level, the face gate runs on the motivated reasoning of Chapter 13.

The individual remedy comes out of psychology as identity expansion: fold the ability to write postmortems into the professional identity itself, so that a manager's image moves from never wrong to good at correction. Redefinition turns the postmortem from a threat into a stage. The driver of that redefinition is almost always the top of the organization, because only modeling from the top gives it credibility. Chapter 19's endpoint criterion, senior people admitting error in public at no cost, functions here as a cultural switch again.

A listed company's prospectus carries a section headed "Risk Factors", and any reader can run an experiment on it. Pull ten prospectuses at random and compare how the risk factors are written. The language comes out heavily templated. Disclosure specific enough to say "this accident happened in our history, and it cost us this much" is rare, because that would be using a statutory document to characterize your own past. 〔Verification pending: systematic testing would require sampling full prospectus

text on SEC EDGAR; this is an operational suggestion the reader can check in an afternoon〕

The face gate reaches well past listed companies. Founders' public narratives, anniversary celebrations, government achievement reports, all of them run in the same direction. Failure has no column inches in that direction.

Inside the organization the gate has a human resources variant, since the space for failure in the internal narrative is just as scarce. The annual meeting tells success stories. The internal newsletter profiles excellent teams. Promotion defenses report achievements. Internal memory skews toward success exactly as the external narrative does. The consequence is quiet: failure knowledge can only circulate as gossip, passed by word of mouth and private aside. Gossip has no structure, fails all three tests of Chapter 1, and the more of it circulates the lower the quality of the organization's failure knowledge.

The remedy is to give failure a formal slot: a standing item on the monthly review meeting agenda, a lessons column in the internal newsletter, a failure section in the internal knowledge base. All three share one requirement, a named-credit system that makes writing about failure an achievement that can appear in a promotion file. The cost of the arrangement is opening an agenda item and a column. What changes is the ratio of the organizational narrative, and a ratio held for one year changes the failure literacy of a generation inside that organization.

A communications exit exists as well: turn failure into an honor narrative, finance it from the opposite direction. An industry award for the best postmortem makes writing about failure an act that earns standing. The design point is the judging standard, which should score by this book's card quality criteria, the completeness of fact layer, condition layer, and structural layer, and not by how good the story is. The supplier of such an award should be

an industry association or a foundation, a third party with credibility. The output is an annual selection, and that selection is the industry's first batch of qualified cards.

Loosening an honor gate has plenty of precedent in human experience. Medicine's morbidity and mortality conference turned public discussion of death cases into part of professional honor. The M&M conference is a ready template for industries outside healthcare to copy, and its charter, its blameless culture, and its conventions of frequency are all documented in public literature.

Gate four: institutions. The most expensive of the four devices is the independent investigation body, and it needs legislation. A body without investigative powers cannot write the full process. Reporters cannot obtain the docket, scholars cannot get onto the site, insiders cannot publish the whole text. Investigative power is a public power, and public power is generally paid for by a disaster large enough. Section 4.3 takes that up.

A lighter substitute sits in the middle: an industry joint investigation panel. No standing institution, no permanent staff. An association convenes it, member firms supply experts, and the team forms per event. Independence and expertise land between an internal postmortem and a national agency. The panel fits industry-level accidents of medium size, the ones too small for a national case and too murky for an internal review. Its institutional cost is very low, a charter plus an expert roster. Its credibility rests on two clauses: recusal, so that no investigator has a material interest in the company involved, and a commitment to publish the report in full. Prototypes of the joint panel can be found in industry bodies of the Flight Safety Foundation type. The distance from prototype to working model is one charter meeting and five live investigations.

Gate five: cost. Writing a postmortem consumes hours. Reading one consumes capability. Indexing and maintaining the collection consumes headcount.

One remedy on this gate is underestimated: let students and newcomers make the cards. Card production is unusually good entry training, since it drills reading a report, extracting fields, and mapping to mechanisms in a single pass. The channel is a course partnership with a school, with the accident analysis course using real cards as assignments and the best of them entering the industry library under the student's name. The design point is the quality gate. The inclusion standard does not drop because the author is a student. A single standard is what keeps a library alive.

The economics work out on both sides: teaching needs real material, the industry needs throughput, and cards are the medium of exchange. Of all the remedies on the cost gate this is the most elegant. The education system is the circulation partner this field should always have had and never did.

Circulation works for success because a media industry pays for it. Failure has no business model. Nobody subscribes to accident reports, nobody pays for a postmortem, and retelling one generates no traffic.

The cost gate has a hidden dimension, an asymmetry in capability. Writing can be made cheaper with a template. Reading cannot, because reading needs a pattern library: without the twenty-four mechanisms as a frame, what comes off the page is a story. That half of the capability is the reason this book exists. The routes to building it are laid out in Chapters 35 to 39, through the morning meeting, the positioning table, the playbook, and the budget. This chapter only needs to explain why the capability belongs on the cost ledger. Teaching one person to read postmortems well takes dozens of cards read closely, and those hours are a real cost. If the cost never enters a budget, the capability never gets built.

Writing reading capability into job descriptions and the training budget, which the budget template of Chapter 39 carries as a line item, is the enterprise-side remedy for this gate. An enterprise can treat exactly one of the five: cost. The other four sit at industry and legal level. Enterprise readers should therefore put all of their attention here.

Of the five gates, the first four are incentive problems. The fifth is plain market failure.

A technical exit on the cost gate is maturing: make the reports machine-readable. Regulatory documents are trending toward structured data, and XBRL in financial reporting proves that a document can be read by a person and by a machine at once. Machine-readable failure knowledge pushes reading cost down by an order of magnitude, since retrieval, comparison, and aggregation become automatic. Reading three hundred pages by hand and extracting fields by machine belong to two different eras.

The precondition is a field standard for the document. This book's twelve fields are one candidate: if regulatory reports were labeled to a comparable scheme, the reading cost of circulation would drop a tier at once. The promoter of such a standard can be a user alliance, with financial institutions, insurers, and research bodies jointly proposing structure to a regulator. Collective user voice is the strongest force behind document standardization, and the model is the history of financial data users pushing XBRL into place.

4.2 The Software Exception: How the Public Postmortem Survived

Over the past decade and a half, the software and internet services industry grew the only large-scale body of public failure knowledge in the world with no statutory basis behind it: the public

postmortem, and the methodology built around it. A landmark is the methods literature that Google's Site Reliability Engineering (SRE) team published openly in 2016, whose operating discipline for the blameless postmortem has been widely paraphrased since. Earlier practice took shape inside several leading internet companies, and the name itself is usually traced to the engineering blogs of a handful of those firms in the early 2010s. The representative public text is the SRE handbook at sre.google, verified as accessible. Earlier texts that carried the name as it spread have been lost or migrated, and are not cited here.

The exception deserves a device-by-device dissection, because it shows how four devices get assembled with no legislation at all.

The substitute for device one, mandatory reporting: changes and outages are high-frequency events, monitoring systems generate event records automatically, and the on-call regime requires a postmortem. The ticket does not close until the write-up is done. No statute exists, but a workflow does.

The substitute for device two, independent investigation: the postmortem is moderated by an engineer from a team that was not involved, on a fixed pattern. The people involved narrate, the moderator probes, the conclusions are anonymized. Independence comes from rotating the process, with no institutional headcount behind it.

The substitute for device three, publication of the full text: status page culture and incident notices to customers. Public postmortems cover only outages with a wide blast radius, yet the public libraries of the leading firms have accumulated several hundred entries, all of them searchable in full.

The substitute for device four, liability isolation: the blameless principle. The output of a postmortem is defined as a set of system improvements, never as a list of human errors. Whether the per-

formance system actually quarantines incident records varies from company to company. This is the device most often honored in name only across software's postmortem institutions: the institution is called blameless, and the incident history still counts in a promotion defense.

There is a blunt way to test whether the name matches the practice. Trace where one incident record flows after it is written. Who read it, whose file it entered, whose evaluation it affected. Tracing the flow is the measurement.

The measurement usually produces embarrassment. The readers turn out to be management rather than the improvement group, and the attachment turns out to be review material rather than a work ticket. That embarrassment is the starting point for governance, because unless the flow changes, blameless is decoration. The design work for flow governance is set out in full in Chapter 19, covering external custody, immunity clauses, and archive separation. This chapter sets one test: a postmortem that cannot answer where it goes means the postmortem culture around it sits somewhere between the name and the practice.

The substitutes hold together because of four conditions in the industry.

A portability test is a small tool worth carrying away: score your own industry against the four conditions, present or absent. An industry missing one or two cannot copy the software model whole, but it can copy its parts. The blameless moderation process, the change-linked method of attribution, the output definition inside a postmortem template. Parts do not need the full set of industry conditions, and the parts-level transplant list is laid out in the templates of Part 5. What this chapter owes the reader is the reason parts detach and still work: their conditions are an order of magnitude looser than those of the assembled system. Parts solve the technical side of a postmortem, how to moderate, how to at-

tribute, how to define the output. The industry conditions constrain the incentive side, why anyone writes at all and who reads the result. Technical parts carry no incentive precondition. Only the incentive side needs the conditions. Keep the two apart and the transplant budget goes to the right place.

Condition one: the object of failure is infrastructure. A service outage has no criminal dimension, the loss is capped by service level agreement (SLA) payouts, and the legal exposure of writing a postmortem is far smaller than in other industries.

Condition two: the industry has a written culture. Design documents, code review, change tickets. An engineer's ordinary work already lands on the page, and a postmortem is one more document type among familiar ones.

Condition three: failures are frequent and causally traceable. Canary releases and change management systems let the link between a given change and a given outage be drawn automatically. The sample size and the attributability are things other industries do not have.

Condition four: talent moves quickly. Engineers pass between companies and carry postmortem practice as a job skill, so the norm standardizes across the industry without any legislation.

Those same conditions mark the boundary of the exception. Software postmortems cover availability failures. Security incidents, data breaches and intrusions, hardly ever get a public postmortem, because the text itself is an attack manual and the legal liability is nowhere near capped. Product failures, meaning the wrong feature was built, and business failures, meaning the wrong market was chosen, sit outside this tradition altogether and fall back under the five gates of section 4.1.

A further boundary is a trend. In recent years the complete public postmortems from leading companies have grown fewer, status

page disclosure has grown briefer, and the public libraries have slowed their rate of update. 〔Verification pending: a quantitative trend would need year-by-year counts of the public libraries; this is a qualitative judgment〕 The exception has no legislative moat. It is a function of industry conditions, and when the conditions shift, the exception contracts.

The contraction can be read in the other direction too, and that reading should be stated plainly. Contraction is market behavior rather than decline. The volume of public postmortems rises and falls with the cost-benefit calculation of the industry publishing them. It follows the market.

That reading carries a warning for anyone who cites these libraries. Do not treat any industry's transparency as a permanent feature. The library you quote may stop updating in five years, so cited knowledge needs a local copy. The knowledge pack principle of Chapter 30 applies here: an external library will walk away, your own card library will not. The same discipline serves researchers and procurement people alike. Archive a snapshot whenever you cite an external postmortem, because the archiving rules in Appendix C are as necessary in software as anywhere else.

4.3 The Institution Is the Price a Disaster Paid

Look back at the legislative history of the four devices and one pattern repeats. They are almost always written after a major disaster.

In chemicals, the U.S. Chemical Safety and Hazard Investigation Board (CSB) was established under the Clean Air Act Amendments of 1990. The legislative backdrop included the pressure from events such as the December 1984 toxic gas release at the pesticide plant in Bhopal, India, where thousands died; the count of immediate deaths is a matter for later investigations and government statist-

ics. 〔Verification pending: the official figure would require Indian government archives; estimates vary widely, and the text keeps the qualitative "thousands"〕 In finance, the Sarbanes–Oxley Act of 2002 followed close behind the 2001 bankruptcies of Enron and WorldCom.

In aviation, the international investigation framework ratcheted upward through a run of accidents in the postwar jet age. After the collision of two aircraft at Tenerife on 27 March 1977, which killed 583 people and was investigated by the Dutch, Spanish, and American sides with the final report issued by the Spanish au‑thorities, the International Civil Aviation Organization (ICAO) moved revisions through on radio phraseology and cockpit com‑munication procedure. 〔Verification pending: the official archive page for the Spanish investigation report remains to be located; this sentence has been written as a qualitative statement〕 〔Veri‑fication pending: the original ICAO Doc series texts need check‑ing〕

Disaster pays first and the institution is written afterwards. That order tells you institutions are not created because someone re‑cognized their importance. They are forced into existence by the pricing of loss.

The order has an observable byproduct. An institution built after a disaster usually fixes only the class of problem that disaster be‑longed to, which is what targeted legislation looks like: chemicals after Bhopal, financial audit after Enron. Targeting makes legisla‑tion fast, since the problem is clear and the pressure is concen‑trated. It also fragments the record system. Each disaster yields one agency or one act, and the institutions share no infrastructure with each other. The five precedent industries cannot exchange anything.

The direction out of fragmentation is to make the institution's infrastructure. Shared field standards for reports, shared retrieval

infrastructure, mutually recognized evidence grading. One build at the infrastructure layer would serve every targeted institution at once. The promoter of that can only be a cross-industry user alliance, with insurance, academia, and media as natural members, and the object of the push is a shared tool of the kind this book is. If the twelve-field standard gets adopted by card libraries in several industries at once, infrastructure has begun to exist.

Three uses follow for the reader.

Use one: assess your industry. If your industry has no failure record system, the meaning is that losses have not yet grown large enough to force legislation, or that losses are absorbed by many small parties whose aggregation never becomes legislative pressure.

Use two: look for cheaper substitutes. Waiting for a disaster to buy the institution is the most expensive route there is. Industry self-regulatory alliances, shared postmortem libraries with liability immunity clauses, and insurer-led data pools have all been used as substitutes in the past, and section 4.4 gives the conditions under which each one stands.

Use three: the timing problem for whoever builds institutions. The disaster-buys-institution model has a corollary that runs against intuition. The window after a disaster is very short, since a legislative window usually closes before public memory fades. The operational answer is to pre-draft: an industry body keeps a finished legislative proposal on the shelf year round, so that when the disaster arrives, while everyone else is mourning, it has text.

Pre-drafting has precedent throughout legislative history. Most of the provisions that pass quickly after a disaster existed as proposals before it. The cost of keeping text ready is one drafting pass and an annual update. The return shows up in the catch rate at each window. The case format and the appendix system in this book are

half of that pre-drafted text for your industry. The other half has to be filled in with your industry's own parameters.

4.4 Will Your Industry Grow This System?

Three criteria. Score each one.

Criterion one: the externality of failure. Does the failure hurt you, or the public? A failure that hurts the public eventually draws regulation, and regulation will sooner or later be pushed into building a record system. All five precedent domains are high-externality industries. A failure that hurts only you will never grow an institution, and self-discipline is the ceiling.

Criterion two: the rate of technical iteration. When the rules update more slowly than the technology, a regulator's record system cannot keep up with its own object. Fast-iterating industries are more likely to spread the norm through practitioner mobility, which is the necessary condition of the software model. Slow-iterating industries get there through legislation, which is the nuclear model.

Criterion three: practitioner mobility. Talent flow is what turns a case lesson into industry common knowledge. In low-mobility industries, utilities and some monopolies among them, lessons stay trapped in the original organization until that organization disappears.

An industry scoring on all three can expect a software-style postmortem tradition to grow on its own. An industry missing two of the three should build internally and stop waiting for the industry, and Chapters 38 and 39 supply the templates. The judgment is itself a form of prevention: knowing which kind of disaster your industry is waiting to pay the bill means at least knowing the shape of the invoice.

One addition at the level of use: read the three criteria dynamically. All three parameters move with time. Mobility rises and falls with the industry cycle, externality heats up as a regulatory issue gains prominence, iteration speed shifts with each technology genera - tion. The operational form is to fold the three criteria into the an - nual industry scan, which the intelligence function of Chapter 33 can do in passing: rescore once a year.

The movement of the scores carries more information than the level of the scores. An industry whose mobility has turned from rising to falling is losing the conditions for the software model. The window is not only legislative. It is also cultural. The annual scan of windows is the last lesson this chapter has for an industry ob - server: institutions do not wait indefinitely, and they have a flowering season.

The shape of the invoice can be drawn as a prediction table from your industry’ s parameters. Score the three criteria, then lay the combination against the history of the five precedent industries and ask which way your industry’ s institution would arrive. The columns of the table are institutional forms: a national investiga - tion body, an industry joint panel, an insurance data pool, an en - terprise-built system, or nothing ever. The rows are combinations of scores.

The table is not fatalism. It gives a baseline, meaning the default trajectory with no intervention, and the value of a baseline is that it makes the need for intervention calculable. If the baseline says nothing ever appears, and you judge that the industry needs a re - cord system, then the material for your argument is the difference row.

The difference row is the strongest opening a policy discussion can have: by baseline our industry will never grow a record system, and our loss curve says we need one, which is why there is a proposal on the table.

A ready comparison is the Indian cryptocurrency exchange Koinex, case st-002 in Appendix B. The founders' shutdown postmortem put the cause of death on banks freezing payment channels. Regulatory uncertainty travelled through financial infrastructure, and the industry made no wrong product decision at any point. This is the startup-shaped sample of criterion one, with the externality landing outside the practitioner: the disaster your industry is waiting for is sometimes one you did not make.

Five Self-Checks

1. In your organization's most recent internal postmortem, did legal take part in reviewing the draft? What did the review change?
2. In your industry, how much is failure detail worth as competitive intelligence? Name one specific field where writing it down amounts to arming a rival.
3. In your industry's prospectuses, annual reports, or public materials, how templated is risk disclosure? Compare two of them and count the places where disclosure gets as far as a number.
4. Of the four software conditions, infrastructure as the object of failure, a written culture, frequent and attributable failures, and talent mobility, which does your industry lack? The missing conditions decide whether you copy the software model or the nuclear model.
5. The last time your industry's disaster bought an institution, what did it buy? If that invoice were issued again, where would your organization rank among the claimants?

Chapter 5 The Twelve Fields of the Case Card

> What this chapter covers: the twelve fields of the case card, the analytic unit this book uses everywhere, defined one at a time, each with one badly written example and one corrected version. When to come back: when you write a postmortem of your own, review someone else's, or need to compress an investigation report onto one page. Reading time: about 40 minutes.

A three-hundred-page investigation report holds almost everything and reads badly. A postmortem article reads well and holds fragments. The case card is the layer in between: one page, twelve fields, a hard requirement attached to each field, and no case admitted to the main text until the fields are filled. This chapter defines the fields one at a time and gives each one a bad example. The bad examples come from real postmortem and briefing prose. You will find the same species inside your own organization.

One term of use before the table. The twelve fields are an engineering format, not an academic one. Every field is on the card because something downstream consumes it: the ID feeds citation, what was visible then feeds training, the minimum counterfactual feeds a work order. A field with no downstream consumer has no claim on a line of the page.

Engineering formats are strict, and the strictness is the point. An engineering drawing carries tolerances because a dimension without a tolerance cannot be built to. The field requirements below are tolerances of the same kind, and "maybe good enough" quietly removes the use the field existed for.

Read with a pen in hand. Take your organization's most recent postmortem and fill each field as you reach its definition. By the

end you hold one completed card. A reader who only reads ends up with a list of format facts, which is not the same thing as the skill.

The twelve fields at a glance:

#	Field	Hard requirement
1	Case ID	domain code plus serial, unique across the book
2	One-line summary	30 characters or fewer, states only what happened
3	What and who	organization plus equipment or project, nameable down to model and registration number
4	When and where	year, month, day plus location
5	Domain and mechanism	domain code plus main mechanism, 1–3 attached, main mechanism first
6	What was visible then	information the actors held, with timestamps, at the moment they decided
7	Options ruled out	options tabled and rejected, and options that never reached the table
8	The decision point	who did what, in which meeting or on which document
9	Point of no return	the action after signature, go-live, takeoff, or pour
10	Scale of consequences	figures from primary material; where none exist, “no public quantification”
11	Minimum counterfactual	one action changed avoids the event; write one

#	Field	Hard requirement
12	Evidence grade and sources	grade A–E plus full citation plus link

5.1 Definitions and Examples, Field by Field

Field 1: Case ID

Definition: `domain code-serial`. Aviation av, chemicals ch, nuclear nu, healthcare me, pharmaceuticals ph, construction ce, maritime ma, software sw, finance fi, startups st, public policy po, military mi, manufacturing mn, agriculture ag.

The scheme travels. An organization building its own library may change the format and draw the prefix from its own industry vocabulary. Three governance rules do not travel, because they are what keeps any citation system alive: append only, keep state separate, reserve ranges. Formats can vary because an ID means something only inside its own system; your av-037 and this book's av-037 may point at different events, and carrying the title when you cite resolves it. The rules cannot be dropped because a broken link or a collided number kills a library outright. Take the three rules and the field definitions and use them as needed. Permission to take them is why this chapter exists.

Bad example: "Accident Case No. 3", or "Compilation of Typical Cases in Our City, Volume XI". The serial hangs off a table of contents. Revise the contents and the serial is void, and every citation into it snaps.

Good example: `av-037`. Three years later, after a contents revision, this ID still points to the Japan Airlines A350 record and its collision with the Japan Coast Guard aircraft. The ID is the anchor between this book and other material, and the primary key of the master case table in Appendix B.

Three engineering rules decide how long a scheme lives. Appending is the only permitted edit: when an ID turns out to sit under the wrong mechanism, add a correction entry and leave the number alone, because citation integrity outranks catalogue tidiness. State is a separate field, so revisions, v1 and v2, and errata never enter the ID. Ranges are reserved, with hand-collected, bulk-imported, and externally exchanged records given their own blocks of serials, so a merge cannot collide. The cost is one line of explanation in the register. The return arrives ten years out, in a system with no broken links, no duplicate numbers, and no silent renumbering, which is the credential a knowledge base needs before it can talk to outside material at all.

Field 2: One-line summary

Definition: 30 characters or fewer, stating only what happened. The test is negative: no cause, no judgment, no adjective.

Bad example: "Due to inadequate safety management and weak employee safety awareness, an avoidable serious accident occurred." Six information slots, four of them judgments (inadequate, weak, avoidable, serious) and none of them fact.

Good example: "737-9 door plug separates in climb; rapid depressurization; return." Fifteen characters in the Chinese, one action sequence: component, phase, consequence, outcome. The reader knows what happened and does not yet need to know why.

The method is to find the first moment on the timeline when something went wrong physically and write it as subject, verb, object. "Door plug separated." "Ball-nut thread worn to failure." "Oxygen generator caught fire." Evaluation gets its turn later.

The field has a second use as the catalogue index line. A catalogue page carries nothing but IDs and one-line summaries, and one

page then scans the whole stock of ways things die. Its standard is the scan test: fifty lines in thirty seconds, after which the reader can say roughly which families of death the library holds. A summary that fails is too long or too clever. Put the page on the wall of the decision room rather than the canteen, updated monthly as new cards arrive, since the readers of a death catalogue should be the people deciding.

Three sentence rules hold the field down. Verbs beat nominalizations: "door plug separated" beats "a door plug separation event", since nominalization hides the action inside an abstraction and is where bureaucratic prose starts. Quantifiers stay conservative: the 30-character limit forbids piling up, yet writers still squeeze in three parallel actions, so write the first physical event and leave the rest to the body. Names stay out: no personal names, no characterizations, because the summary is public-information layer while individual accountability belongs in field 8 and in the investigation documents. All three serve one purpose. The summary is a retrieval tag, a thing that exists to be found rather than read to the end.

Field 3: What and who

Definition: whose thing failed. Organization name, equipment model, any identifier that can be named: flight number, registration number, construction section.

The field carries a second layer. Write the subject on two levels, the operating level and the owning level. Ask whether the crew sailed for the owner or was hired with the hull, whether the plant runs its own staff or a contractor's. Responsibility and the landing point of remediation usually sit between the two, so write a subject chain of brand owner, operator, facility. Every name on the chain holds a decision point, and primary documents usually write the subject by level, which means the chain can be copied from the report's

subject section. The chain pays off in tracking: a requirement handed to one level has to be executed and verified at another, and a card with no chain gives tracking nothing to hang on.

Bad example: "an aircraft operated by a certain airline". One word, "certain", and the case drops to a fable. It cannot be checked, extended, or confirmed at the level of conditions.

Naming is also a weapon against survivorship bias. Narrated failures tend to lose their subject, and once it is gone the reader cannot check whether that company still exists, how it has done since, or whether the accident recurred. A named case allows the full verification pass, and the result is often more instructive than the story. The action is to look up what happened to the organization afterwards: its remediation, its later record, its events of the same kind. Doing this repeatedly is the advanced drill of card reading and the technical form of Chapter 2's instruction to go out and buy the death samples. Enough passes and a reader develops an instinct for the time depth of failure. An accident occupies a stretch of an organization's history; its single day is only the visible part.

Good example: "Alaska Airlines 1282, Boeing 737-9 (N704AL)." Checkable, citable, condition layer complete: the model fixes the airworthiness background, the airline fixes the maintenance system, the registration number lets anyone walk the aircraft's history in public databases.

Where naming carries legal risk, with a judgment not yet final or a penalty not yet published, drop one level: name the industry and the category, and state why the subject is not named.

One recurring argument deserves an answer. Are small subjects worth naming, the sole trader, the one-person company? Yes. Small operators account for most of the statistical mass, and anonymizing them teaches readers that risk belongs to large organizations;

naming restores the distribution to its real shape. The privacy boundary is prior publication: where the subject information is already public in a primary document, the card copies it, and where it is not, the card follows the document. Beyond that boundary there is an emotional cost, and it is real. A small operator's failure, written into a book, will be read by the people involved or their families. Write only the facts the documents contain, with no characterization of anyone. Characterization is the sugar of narrative and a second injury to the people in it.

Field 4: When and where

Definition: the event's year, month, day and location, precise to the day. The investigation's publication date belongs to field 12.

The field carries a geographic discipline. Write the location to the level where it can be found on a map: country, city, facility. Where the facility can be identified, name it; where it cannot, name the administrative area. The level decides whether the reader can place the case in its geographic and regulatory context, because the same mechanism in another jurisdiction carries a different condition layer, and location is the reader's first evidence about the regulator. Where a case cannot be located at all, for open sea or a facility that cannot be disclosed, say why. A stated reason distinguishes a limit from laziness.

Bad example: "early 2024", or "last summer". Early and late in a year are eleven months apart, enough room for a full maintenance cycle.

Good example: "2024-01-05, Portland, Oregon, USA, about 20 minutes after takeoff" (NTSB AIR-25-04). The date anchors the information environment: weather bulletins from before that day, maintenance records, meeting minutes all fall inside what was visible then; anything dated after is hindsight.

Precision has three more rules. Keep one time basis in a card, with event times in local time and the zone marked, and publication times in the issuing body's local time, because a mixed timeline steals the reader's sense of sequence. Distinguish an instant from a window: the minute where a minute is known, the window where only a window is known, with the source's precision marked, since false precision misleads as much as vagueness. Mark the genealogy of the timeline: where a preliminary bulletin and the fi‑nal report disagree, follow the final report and note the difference, which is itself a record of how the investigation moved.

Field 5: Domain and mechanism

Definition: the domain code plus the main mechanism. Up to three mechanisms may be attached, the first being the main one. Names come from the twenty-four-term vocabulary in Chapters 11–34. Do not invent one.

Bad example: "management problems", "human factors", "systemic risk". These three phrases turn up in the conclusion paragraph of most postmortems. They can hold anything, and so they explain nothing.

Good example: "av; The Invisibility of Maintenance; Practical Drift." Attaching a mechanism is an act of attribution, and an in‑accurate attachment is worse than a single accurate one.

The ratio is one main mechanism and at most two companions. Three or more says the attribution has not converged; go back to the mechanism chapters, compare against their criteria, and attach again. The ratio protects the index, since a card that attaches five mechanisms has voted in five rows of the matrix and thinned the signal-to-noise ratio of every search through it. It also prevents drift. The main mechanism declared here may not change in the body text: a case can be cited in several chapters, and every cita‑

tion presents it under the same main mechanism. Stable identity is part of what makes a case citable, and it is the specific defence against a case retold in each chapter as a story supporting that chapter's thesis.

Field 6: What was visible then

Definition: the timestamped information the actors held when they decided. Only records whose phase can be fixed from the material are admissible: meeting minutes, work orders, the issue time of a weather forecast, the previous shift's handover notes. Anything learned only afterwards stays out, without exception.

Bad example: "hidden hazards had long existed, and management cannot escape responsibility." The phrase "long existed" erases every piece of time information. Existing for how long? Who saw it, and when? Did the person who saw it have a duty to report it?

Good example: in the Alaska Airlines 261 case (NTSB AAR-02/01), what was visible then included the inspection records from before the lubrication and end-play check intervals were extended, the application for the extended intervals with the Federal Aviation Administration (FAA) approval, and the accident crew's reports of abnormal jackscrew position and unusual cabin noise. Each item carries the date it entered someone's field of view. The thread wear found when the nut was opened after the accident is post-event information and cannot enter this field.

This is one of the two fields with the highest value density in the book, field 11 being the other. Its purpose is to keep Chapter 2's attribution bias outside the reasoning: only what the actors could see is admitted. Chapter 6 is given over to operating it.

One implicit standard needs stating. Visible does not mean noticed. The field collects information that was reachable, meaning it sat on the desk, in the system, or in the minutes; whether the per-

son saw it is a separate question and belongs to the reader. Conflating the two lets the card quietly excuse or convict, since a thing unseen may be an attention problem or a presentation problem. Keep them apart by recording how the information was reachable: in the system, on paper, or by word of mouth. The three modes carry different attention costs. Once the mode is written, field 6 becomes the material from which a reader rebuilds the desk of that day, and every judgment made from it is the reader's own.

Field 7: Options ruled out

Definition: the options on the table at the decision point that were rejected, plus the options that should have appeared and did not. Record both and mark which is which.

Bad example: "had they been more careful, it would not have happened." More careful is an option that does not exist. No action, no cost, no decision-maker. Writing it is writing nothing.

Good example, with the format shown on Challenger (Rogers Commission report, 1986). On the teleconference of the evening of January 27, 1986, the options on the table were: launch on schedule on January 28, which was selected; delay until daytime temperatures recovered, rejected on schedule pressure; and honor the engineers' proposed threshold of no launch below 53 degrees Fahrenheit (about 12 degrees Celsius), rejected because launches had already succeeded several times at lower temperatures. The absent option was any reopening of why the launch temperature threshold had been sliding downward launch by launch. Volume 1, Chapter 5 of the Rogers Commission report records that Thiokol engineers recommended against launch below 53 degrees Fahrenheit, resting largely on blow-by data from flight 51-C; the official text is at nasa.gov/history/rogersrep/v1ch5.htm, and this book read it on 2026-11-01.

Ruled-out options are valuable because they expose the structure of the decision: how options were generated, how they were rejected, who was entitled to put them on the table. Most postmortems record what was chosen and omit what was choosable, and the same option-generating machinery then produces the same menu at the next decision.

Absent options need restraint. The evidence for one is a process document. An option that procedure required and the record never shows is a legitimate absence; an option the reviewer thought of afterwards is an invention. Restraint is needed because absent options are the easiest thing in the card to write with hindsight, turning what the actors never considered into what they should have considered, which is Chapter 2's attribution bias returning through the side door. The evidentiary standard is a comparison: the step exists in the procedure, its counterpart is missing from the minutes, and the gap is the evidence. Where the evidence is thin, write the absence weakly: "the procedure appears to have offered this path; no record of discussion appears; unresolved." A weak entry costs nothing. An invented one costs the card.

Field 8: The decision point

Definition: four elements, person, meeting, document, action. Who decided what, in which meeting or on which document. The date falls on a day.

A discipline is buried in that definition. A decision point has to be an identifiable human action: who, when, in what form, doing what. Automatic system actions and physical processes do not occupy this field; automated actions are recorded in the failure chain, and Chapters 21 and 22 cover the paths taken by things and by systems. The reason is what the field is for. Postmortems improve the environment in which people decide, which covers how options get generated, how information arrives, and how a signature

comes to be given; with no identifiable action there is nowhere for the improvement to land. The more automated the system, the fewer human decision points remain and the more each one mat - ters. The takeover moment in Chapter 26 is the scarce form of the decision point, and scarcity raises the standard of description, be - cause those points are everywhere a human being can still inter - vene.

Bad example: "company management made the wrong decision." No name, no meeting, no document. The sentence can be neither contested nor cited.

Good example: Alaska Airlines 261 (NTSB AAR-02/01, executive summary). Alaska Airlines extended the lubrication interval and the end-play check interval for the horizontal stabilizer jackscrew assembly, and the FAA approved both; the investigation found that the extended intervals let a lubrication shortfall the old schedule would have caught develop into excessive wear and then failure (text from the executive summary, in the portions I read myself). All four elements present: the company as who, the documents and approvals as the vehicle, the interval extension as the action, the FAA approval as the gatekeeping act.

Field 9: Point of no return

Definition: the moment after which the event's main con - sequences can no longer be undone by ordinary means. The test is in Chapter 7: money can undo it, time can undo it, a public an - nouncement can undo it; a signature and a go-live cannot.

The register links to the duty to notify. Information arriving after the point of no return, a weather update after takeoff or a test re - port after the pour, is post-event information, and what it does is open a new decision point: continue, turn back, stop work. At the point of no return the question changes from whether to do the

thing to whether to rescue it. Cards handle that phase as a separate entry, the post-irreversible decision, and it carries high value in review because it tests the ability to lose less in a bad position. Most postmortems only train the ability not to lose at all.

Bad example: "from the moment the project started, failure was already certain." Fatalism is not a definition of the point of no return. It folds every decision point into one and erases every position in between where intervention was still possible.

This chapter registers position only: which step, once passed, cannot be walked back. Chapter 7 owns the typology, where incremental, one-way valve, and compound types call for different interventions. The division of labour keeps the card's density manageable, since a list of positions already carries large engineering value; most organizations cannot state the irreversible points in their own key processes. Chapter 7 gives the method for compiling the list the first time, walking the process node by node and asking at each one whether the step can still be reversed. The cost is a single process walkthrough. The output is a register covering every key process, and every row should carry a named custodian, which is where Chapter 18's single-responsible-person clause comes in.

Good example: in the Japan Airlines 516 event (av-037), the point of no return formed when the Coast Guard aircraft entered the runway while the A350 crew held landing clearance. From that combination onward, the two aircraft's speeds and the remaining distance made the collision an event most operating paths could not avoid. Assembly-type cases are cleaner: the door closes, the concrete is poured, the reactor is synchronized to the grid, and every minute of work afterwards is paying the fare for that action.

Field 10: Scale of consequences

Definition: the figures in the primary material. Deaths, injuries, money lost, hours of downtime, units recalled; write whatever exists and name its source. With no figures, write "no public quantification".

Bad example: "caused enormous economic loss and severe social impact." Enormous is how much? Severe to what degree? The sentence's only function is to hold a place.

Good example: "0 deaths; 7 minor injuries (all passengers); 169 uninjured; aircraft substantially damaged" (av-030, NTSB AIR-25-04). Four figures, each with its own basis, all from one report.

"No public quantification" is a legitimate answer. It means the consequences never entered any primary document. It does not mean they were zero.

Casualty figures follow an official basis and carry its version. Preliminary and final counts often differ, since an initial bulletin gets revised as identities are confirmed, and the card takes the final adjudicated basis while preserving the revision trail. The reason is respect as much as law: every number in a casualty count corresponds to particular people and families, and a careless basis does harm to the people behind the digits. The operation is to date the number, as in "7 minor injuries (adjudicated, 2025-03)". Number and adjudication date travel together, and the reader knows which version is in hand. Version management looks fussy. It is the minimum courtesy this book extends to this field.

Field 11: Minimum counterfactual

Definition: the one action which, changed, would have avoided the event or sharply reduced its consequences. Work backward from

the latest action that did not depend on luck. Book-wide discipline: write exactly one.

Bad example: "had management been strengthened, systems perfected, awareness raised, and oversight tightened, the accident could have been avoided." Four actions amount to none. The reader cannot tell which comes first, and so does none of them.

Good example: Alaska Airlines 261 (AAR-02/01). Leave the end-play check interval unextended and inspect on the original schedule, and the wear had a chance of being found before it reached failure. One action, at the latest position on the chain of decisions actually taken, resting on no luck of the "happened to be spotted" kind. (AAR-02/01 executive summary p. 4, and the Safety Board's statistical study of end-play data in section 1.18.2; PDF verified.)

The field is a training device. Writing it forces the author to answer where the minimal intervention point lay along the whole chain. Chapter 10 gives the backward-walking method. Two counterfactuals amount to none, because the reader remembers the easy one, and the easy one is rarely the minimal intervention point.

One test is gold. Write the counterfactual as an instruction: who, by what date, does what, at what cost, approved by whom. Fill all five and it is usable; fail any and it is a wish. The test pulls the lesson down from the moral layer to the operating layer. "Tighten oversight" fails it; "change this check from annual to semiannual" passes, and the gap between the two is the gap between a lesson and a slogan. Organizations can use the test at scale by turning every passing counterfactual straight into a work order. A counterfactual that reaches a work order is closed. One that stays on the page is decoration.

Field 12: Evidence grade and sources

Definition: grade (A, full independent investigation report; B, official bulletin or communiqué; C, judgment or penalty document; D, public postmortem by a party involved; E, media lead), a full citation, and a link. Rule: E-grade material may never be the sole source for a case in the main text.

Bad example: "compiled from multiple media reports." When several outlets retell one wire story, "multiple" is a fake number.

Good example: "Grade A. NTSB, Aviation Investigation Final Report DCA21FA085, 2023; link at Appendix B, av-006."

A citation has to resolve to a document name and number, and the link has to have been opened by hand. No exceptions, because this is the collateral behind every credibility claim the book makes.

Readers need a conversion table. Read the grade first; it tells you how the card may be used. Grade A: all details quotable, down to signatory-level actions, and a postmortem can benchmark against it directly. Grade B: the event frame and official conclusions quotable, with detail discounted, since a bulletin carries limited resolution. Grade C: the facts legally established are quotable, and a judgment's fact section has been tested in adversarial proceedings, which is the distinctive value of the grade, though the disputed issues may have shaped which facts were selected. Grade D: frame only, no details, because a party's own postmortem has not been adversarially tested. Grade E: a lead only, to be followed toward primary or secondary material and not cited until something is found. The table also sets house rules: an organization can decide that only A through C cards enter the library used as a basis for decisions, and with that rule standing, the library has a gate.

Link rot needs a policy, because rot is the norm. Sites get rebuilt, agencies merge, PDFs come down, in batches every year. Three layers answer it. Snapshot at first reading, archiving the page or

keeping the local PDF. Register the dead link on the errata page with a substitute entry point. Downgrade where no substitute exists, following Appendix C, or convert the entry into a checkpoint. All three belong to the reading routine, and their cost is small: archive while collecting, click the batch once a year. They keep the book's citations checkable five years out, and checkability is the difference between a card library and a scrapbook of news.

5.2 Why Three Fields Carry the Value

Of the twelve, fields 2, 3, 4, 5, and 10 supply coordinates, fields 8 and 9 supply structure, and fields 6, 7, and 11 supply training. Strip 6, 7, and 11 from a card and what remains is indistinguishable from news. News tells you that something happened. Those three let you rehearse what you would have done.

Be honest about the cost. The three take more than sixty percent of the hours in one card: field 6 means checking phases along a timeline, field 7 means digging rejected options out of documents, field 11 means the full backward walk. Tooling answers it. Phase markers for field 6 come from a timeline template, the absence record in field 7 from a process comparison table, the backward walk in field 11 from the four questions in Chapter 10, and good tools compress the sixty percent to thirty. Tooling has a second payoff in consistency: cards by different authors reach the same level on the fields that matter, and a library that reads as though one person wrote it becomes possible. Templates sit in Part 5. This chapter's job is to say that the expensive part is expensive for a reason, and that it has tools.

The three relate in a fixed order. Field 6 fixes the information environment, since you can only decide with the cards on the table. Field 7 lists the real options that were on it, including the one never played. Field 11 points at the card whose replacement wins

the hand. Together they make one repeatable drill: read field 6, write your own field 7, make your own choice, then turn the page and compare against the actors' choice in field 8 and the minimal intervention point in field 11.

The drill grades by difficulty, and one card can be worked three times. In the first round the card is open: field 6 gives the full information set, and what gets trained is the imagination of options. In the second the card is half covered: fields 7 and 11 are hidden, field 6 and the situation are given, and what gets trained is the handling of information. In the third the card is closed: only time, place, and subject are given, and the reader reconstructs what was visible and what the options were from industry knowledge before comparing, which trains the actual stock of that knowledge and usually stings most. Space the rounds a month apart, since repeating a card too soon contaminates memory and a longer gap keeps the comparison honest. The card is a dumbbell, the weights come in steps, and different weights work different muscles. This is the specific remedy for Chapter 2's attribution bias, and the reason every case in this book stops at the decision point.

Two variants serve readers at different levels. Solo: read field 6 and the situation, write your option list and your choice on paper, then compare. Writing is the point, since an unwritten choice is an imagining while a written one can be tested. Group: split a morning meeting in two, hand out the front of the card, keep the back with the facilitator, discuss the options, then reveal. The group variant buys option diversity, because a room generates a fuller list than one person and absent options surface more easily in company. Both cost about ten minutes per card, and both are the basic forms of the morning meeting script in Chapter 35, which is nothing more than the variants fixed into a procedure.

5.3 How to Use the Fields

Three uses. Writing a postmortem: Chapter 38 turns the twelve fields into a template and attaches a paragraph-by-paragraph rewrite of a "strengthen management" style postmortem. Reviewing: the question lists in Chapter 37 are grouped by mechanism, and their answers backfill fields 6 and 7. Building a ledger: an organization indexes its industry's cards by mechanism under field 5, sorts them onto a timeline by field 4, and distils check items out of field 11; that ledger is the ammunition for the rehearsal meeting in Chapter 39.

Two more uses are worth adding. External communication: when an event description goes to a regulator, a customer, or an insurer, the twelve fields are a complete materials list, and a party that prepares by field always has a page number ready when a question comes. The value is that failure knowledge becomes a negotiating asset, and the posture of a party with complete materials in accident communication, claim negotiation, and regulatory inquiry is a different posture. Teaching: an industry induction course can be taught from that industry's cards, which carry three parts less drama and seven parts less bias than an accident story, and ten parts more structure. A newcomer learns how the industry dies faster from cards than from legend.

The five uses converge. The twelve fields are the working language of the field, and someone who has learned them saves a layer of translation in every conversation with this book, with their own organization, and with outside parties.

One caution on the other side of the standard: the twelve fields are a floor, never a ceiling. Additions should still be governed. Use the twelve for three years before considering a thirteenth, and put a candidate through three gates: it has a clear downstream consumer, it has a fill standard that can be executed, and the cost of

producing a card stays bearable with it added. The gates stop field inflation, the geriatric disease of every format, since each added field is a tax on every card already written. The exception to the tax is the optional field, marked optional, with no backfill of old cards and no obligation on new ones. Version the field set as Chapter 1 governs IDs, with v1 holding the twelve and v2 listing the optional additions. A format can stay alive, provided it stays disciplined.

Blank space on a card can hold any supplementary material. A case that cannot complete any one of the twelve is demoted to the appendix matrix, which accepts an ID, a domain, a mechanism, and consequences. The main text accepts nothing less.

Demotion runs both ways. A matrix card returns to the main text when the twelve fields are complete, the sources reach grade A through C, and the three key fields are written, and the return is reviewed on the same standard as first admission. Because the clause runs both ways, the matrix is a bench rather than a bin, and the bench matters because it keeps the route back open. Maintaining that route keeps a library alive: once a year, go through the matrix and ask which cards are close to promotion, which fields they lack, and what the path to filling them is. The review produces the work order for the next collection round. Chapter 28 makes the same argument about lists in general, that a list needs a destination. A bench means demotion is not an ending.

Five Self-Checks

1. Write a twelve-field card for your organization's last major loss. Note which fields the first card cannot fill.
2. In your postmortem documents, which passages are information known only afterwards, written as though it were known at the time? Rewrite one passage.

3. At your last significant decision, which options were on the table? For each rejected option, who supplied the reason, and where is it recorded?
4. Can your organization state the point of no return for each of its key processes? Do the people who can state it hold rank enough to call a halt?
5. Take your industry’s most recent investigation report and restate it using only the content of field 6. Has your conclusion changed?

One graduation standard sits beyond the five questions: complete one full card alone, from choosing the report to checking the citation, asking no one for help. Hand it to someone who has read this chapter and have them review it field by field against the requirements here. Their feedback is the final list of gaps between you and this chapter. A reader who clears that list can build a card library for any organization. What the skill is worth in the market, which Chapter 4 discussed as the scarcity of the circulating translator, and in governance, since the postmortem template in Chapter 38 takes this as its skeleton, is argued later in the book. Get the skill into your hands first. The occasions for using it will announce themselves.

Chapter 6 Visible at the Time

> What this chapter covers: how to drive the hindsight perspective out of your postmortems and out of your reading. Three operating disciplines, one test, three worked examples. When to turn back: when writing a postmortem, reading one, reviewing a design, or catching yourself saying "they should have seen it". Estimated reading time: 45 minutes.

The main enemy of the postmortem is a fact about memory: a person who knows how something ended can no longer think the way they thought before they knew.

The fact applies to the case card as well. Every card is written after the outcome is known, and no card can eliminate hindsight; the discipline has to be forced into the fields. Field 6, what was visible at the time, stands alone in the left column while the analytic fields are held back during composition. One rule follows: fill Field 6 first, and read the outcome section only after it is complete. An ordering discipline is imperfect and better than nothing. It rests on the writer's self-control, and self-control fails, so every card library needs a fallibility statement and an errata channel alongside it.

What follows compresses the contamination to a minimum, and states its ceiling in the same breath: these disciplines cannot return a postmortem to zero bias. Bias can be compressed, never zeroed. Memory and the narrative instinct reintroduce the outcome perspective at every layer of writing. Chapter 2 made that claim. This chapter is the compression manual.

Teach the three disciplines to your team. One person compressing their own bias still leaves a team postmortem with bias in it. Only when the whole team carries the disciplines does an organiza-

tion’s postmortem turn collectively. The teaching material is this chapter: forty minutes to read, an afternoon to practise.

The three rules: use only timestamped material; use only information the actors could obtain; list separately everything learned later. Executed together, they hold hindsight outside the door.

Whether the door is actually shut can be tested. The exchange test hands your left column to someone who does not know the outcome and asks them to infer what happened. The distance between their inference and the real ending is the purity of your column. A short distance means the column leaves the right blanks and leaks nothing from the future. The tester should if possible be a new colleague, who carries none of the organization’s existing narrative and so measures more accurately. The test returns a purity score and a recording of the tester’s reasoning, which shows what path your column led the reader down. Run it on a sample of postmortems once a quarter. Half a day of a newcomer buys an objective third-party reading of postmortem quality.

6.1 Why the Rearview Mirror Cannot Be Shut Off

Psychological experiments in the 1970s gave the phenomenon its name. Baruch Fischhoff’s classic study had two groups read the same historical situation, one told how it turned out and one not told. The group that knew judged the outcome to have been systematically more likely all along, and did not realize that knowing the ending was why (Fischhoff, “Hindsight Foresight: The Effect of Outcome Knowledge on Judgment under Uncertainty”, 1975; full citation: Fischhoff, ‘Hindsight-foresight’, Journal of Experimental Psychology: Human Perception and Performance 1(3): 288–299, 1975, bibliographic record verified via Crossref).

One laboratory detail carries a sharper implication for organizations. People who know the outcome do not merely raise their es-

timate for the outcome that happened. They lower their estimates for the others. The whole probability distribution is dragged toward the ending, and a distribution dragged that way is useless for risk work. The historical risk list written inside a postmortem has already been reshaped by the outcome, so using it to identify future risks is measuring new events with a bent ruler.

The correction is to write intervals instead of points. A risk retrospective records the several outcomes that seemed possible at the time, each with the basis it rested on. That spreads the bent distribution back out. The difficulty is evidential: probability judgments made at the time were almost never written down. That is the gap the decision record of Chapter 7 fills. Write the distribution at the moment of decision, and a future postmortem inherits a draft the ending has not touched.

The draft does not need to be elaborate. Line 2.5 of the three-line decision record covers it: the main alternative outcomes then believed, and roughly how much confidence each carried, in words rather than numbers. “About seventy percent confident” is enough for a later comparison. Verbal probability is unintimidating, since numerical probability demands calibration training while anyone can write down how confident they felt. Vagueness beats absence, because a postmortem with no draft can only be reshaped by the ending. File it in the same repository as the decision record. Half a line added to the Chapter 7 system saves a chapter’s worth of archaeology later.

Later research gave the effect a plainer name: creeping determinism. Once the ending is known, memory rewrites the process into a single road running to that ending.

Inside an organization the consequences are worse than in a laboratory, because postmortems are written by people who know the ending, for readers who will know it too. Three layers of con-

tamination stack up: the author knows, the reader knows, and both find it obvious that anyone at the time would have known.

Each layer has one counter. The author's is a writing-order discipline: left column first. The reader's is a reading-order discipline: left column alone, form a judgment, then read the right column and compare. The editor's is a layout discipline: columns physically separated, the entry point to the right column at the end of the left column, behind an explicit button. All three cost layout work and process discipline, and none is technically difficult. The difficulty is persistence, which comes from template fixation, Chapter 38's subject. The template hard-codes where the columns sit, so an editor who wants to interleave them has to fight the tooling.

Layout discipline sounds like housekeeping. It is the only one of the three that keeps working after the author leaves. Institutionalized layout outlives personal conscientiousness. Without it, postmortems train no judgment at all. Every fork in the road looks like open highway on paper, the reader concludes he would have seen it too, and back at his own desk he finds information arriving mixed with noise.

Accident investigation bodies maintain a whole procedural defence against this, and five of their devices can be lifted directly. Facts and analysis are published as separate parts, and the boundary between them is the boundary between the columns. Evidence is numbered item by item, so the analysis cites numbers and the citation chain can be checked. Competing hypotheses are worked through before any conclusion, and the report lists the hypotheses it excluded with the grounds. Conclusions are graded, each annotated with the strength of the evidence behind it, reusing the grade table in Chapter 10. Dissent is recorded: disagreements inside the team go into the report, because the presence of dissent is a mark of honesty.

All five have been verified as transferable, since they recur across industries in the better investigations. Resistance is measured in working hours. Together they add roughly thirty percent to the time a postmortem takes, and what that thirty percent buys is a generation of difference in credibility. Put a postmortem with all five beside one missing three, read both, and the gap needs no ar - gument.

The anti-forgery principle deserves one more step. A structured document' s verification path closes because every number it carries walks down to a physical object in the file. The report cites an investigation number, the number resolves to an interview re - cord number, and the interview record carries a signature page. Three levels down, and forgery would have to be committed at one of the three.

The organizational version of that landing is an evidence chain of numbers. Each fact number maps to a number in the source re - cord, which can be a work order number, a log ID, a page number in meeting minutes, anything that lands on a physical object. A postmortem whose numbering chain closes gains anti-forgery and anti-forgetting at once, because retrieval runs on the numbers. Three years later, the factual basis of a conclusion takes about five minutes to reach.

Reports of the NTSB type split facts and analysis into two parts. The facts part admits only material that carries a source and a phase. The analysis part annotates each statement with the fact it rests on. The structure' s deeper function is chronological sorting: for any statement, the first question is whether it happened before the accident or after.

For readers who work in investigation, that question has an up - graded form. Tag each statement with three moments: when the event occurred, when the information about it was recorded, when the author used it. Three-phase tagging makes recording delay

visible, and recording delay is itself an object of postmortem. A record system with long delays produces a left column that is in - complete by construction. One more column than two-phase tag - ging buys an evaluation dimension for the record system, and gives improvement a precise target under Chapter 38's trace-keeping requirements: the link with the longest delay gets rebuilt first.

This chapter's discipline is that whole procedure compressed into three rules a person or a team can execute on Monday.

The scope has to be drawn before the rules are applied. They con - strain the writing and reading of postmortems, whose purpose is to preserve the training value of judgment. They do not constrain the conduct of an accident investigation, which should exhaust every piece of evidence available after the fact. An investigation estab - lishes what happened; a postmortem re-enacts a decision. The two document types face different directions, so their disciplines differ.

The two meet at the reading of an investigation report: facts part under this chapter's discipline, analysis part as conclusions, the switch between the two readings being the section heading. With the scope drawn this way, no reader need fear that the discipline turns a postmortem into a crippled version of the truth. The right column, the post-hoc reconstruction, is where the truth is kept. The discipline asks only that the two worlds be labelled.

6.2 Discipline One: Timestamped Material Only

The operational definition: every item admitted to the visible-at-the-time set must answer when it entered whose field of view. Only items that can answer it may stand as inputs to the decision.

Timestamped material includes meeting minutes, with date, at - tendees and the order of the agenda; work orders and checklists, with completion time and signatory; the publication times of

weather and market data; the timelines of recordings and chat logs; the sequence of steps in an approval flow.

One digital-age caution belongs on that list. System timestamps are not automatically trustworthy. Server clocks can be reset, clocks across machines drift out of sync, log time zones can be a mess. Judgment about sequence across systems needs clock alignment first. Record the clock source and the time zone for each class of log, in the system configuration, so that at postmortem time the clocks are checked before the timeline is ordered. Cross-system accidents are where this trap bites hardest: two systems five minutes apart in their logs will invert the order of events, and an inverted order drives the analysis in a completely different direc‑tion. Synchronize the watches, then read the cards.

Material without timestamps includes the recollections of parti‑cipants, summary reports, retellings in the press, and undated statements like "hidden dangers had been found repeatedly before this".

An aviation case shows how hard the discipline bites. In the fact chain of Alaska Airlines 261 (NTSB AAR-02/01), the items that can carry timestamps include the application documents for extending the lubrication and end-play check intervals, with the dates on which the FAA approved them; the accident crew's reports to the ground about abnormal jackscrew position on the horizontal sta‑bilizer, with call times; and cabin reports of unusual noise, with times. The thread wear morphology of the nut found in the tear‑down after the accident, and the laboratory metallurgical analysis, have no moment at which they entered anyone's field of view. They go to the hindsight column whole.

Once the columns are separated, the reader sees what the decision-makers held: the interval had been extended by their own organ‑ization, and the aircraft had begun reporting abnormal position.

What they were not holding was any card showing the condition of the nut.

The method asks something cruel of the writer. Every clever in‑sight has to pass four questions, and most fail, because they are thoughts available only after the ending is known. The cruelty pays. The thinner the left column, the more real the reader's im‑mersion, and fewer cards on the table means the judgment being trained is closer to the judgment the field demanded.

The method extends outside postmortems. Split a history book's account into what was visible then and what was learned later, and you will see at once how much of what gets called foresight was assembled afterward. The demystification of historical reading comes free with this discipline.

The frequent error in practice is to treat the existence of informa‑tion somewhere inside the organization as visibility to the actor. Between existing and visible lie four gaps: channel, format, timing, language. A test report buried on the ninth directory level of a shared drive does not exist for the engineer making a release decision at two in the morning.

Judging visibility takes four questions. Did the item enter the channel the actor actually uses? The duty room reads the duty system, not the shared drive. Can the format be read on the spot? An original English report handed to a night-shift assembler may not be. Did it arrive before the action or after it? And can the mag‑nitude be grasped on the spot, when the one number that matters sits inside a hundred pages?

Institutionalized, those questions become an information access‑ibility audit: an annual sample of key decision points, drawn from every major decision the organization made that year, audited by postmortem-style walkthrough. It produces an annual accessibility report that lists the decision points where accessibility failed and

classifies the causes, channel missing, format wrong, timing late, magnitude hidden, with counts for each. The report goes to the annual budget for information infrastructure, Chapter 39's line item, where accessibility spending and its returns are reconciled. The audit costs about two working days a year, sampling plus walkthrough. An individual discipline governs the pen; an annual audit governs the system.

The four questions also compress into a pocket card, taped to the monitor while a postmortem is written: channel, format, timing, magnitude. Read the four words in order. If any word fails, the item moves to the right column.

From the pocket card the next step is organizational: an information accessibility matrix, key decision points as rows, the four questions as columns, the actual situation at the time in the cells, maintained annually by Chapter 33's intelligence function. Its output is an accessibility terrain map. The depressions on it, information that should have been visible at a key decision point and was not, form the construction list for information infrastructure. That list's budget line exists in Chapter 39. From terrain map to budget, the chain is complete except for a first move.

6.3 Discipline Two: Only What the Actors Could Obtain

The direct corollary: any "data at the time" cited in a postmortem must state whose hands it was in.

Where the same information sits in an organization determines what it is worth. Risk control knew that customer concentration had passed its limit, but the lending decision belonged to a branch manager whose system displayed only a ranking of customer contribution. A postmortem that writes "the bank knew concentration was over the limit and lent anyway" has placed the risk de-

partment's knowledge inside the branch manager's field of view. The accurate version reads: "the branch's lending system interface carries no customer-concentration field; that field appears only in the head office risk monthly report, dated the fifteenth of each month." Both sentences describe the same organization. The first invents a decision-maker who never existed.

A companion field belongs beside the distribution note: the people who should have known. Record which positions, by design, were supposed to have this information, then record who actually knew. The designed side comes from process documents, job descriptions and flow charts that specify the intended scope of knowledge. The actual side comes from systems and records, which show who could really see. Put the two columns side by side and the difference is a symptom of the information system.

A distribution table built this way serves two later chapters at once: information attenuation up the hierarchy (Chapter 16) and lateral information gaps (Chapter 21). Draw it once and both analyses have ground under them.

The Challenger case shows why the discipline is necessary, and shows it most completely. On the teleconference of the evening of January 27, 1986, the engineers' hand held this: O-rings recovered from booster joints on earlier flights had shown erosion and blow-by, with flight numbers and photographs on record; those signs correlated statistically with low temperature; the joint had performed worst at the lowest launch temperature reached so far. Their hand did not hold the on-site high-speed imagery, which came into existence only after the accident; nor the precise order in which the joint failed, reconstructed afterward; nor data on how the second seal behaved dynamically at low temperature, supplied by later experiments. The decision in that meeting was made with the first hand only. The Rogers Commission report can support attribution at the organizational layer precisely because its evid-

ence chain keeps the two hands strictly apart. (Each item checked: the flight-numbered records of erosion and blow-through, and the Morton Thiokol engineer's memorandum of July 31, 1985 warning of "a catastrophe of the highest order", both appear in Volume 1, Chapter 6, "An Accident Rooted in History"; the conduct of the meeting, with the engineers' photographs and diagrams entered as evidence, appears in Volume 1, Chapter 5. Official full text at nasa.gov/history/rogersrep/, read and verified for this book on 2026-11-01.)

The material leaves an advanced note for postmortem writers. Split the actors' hands by person. The engineers and the managers in that night's meeting held different cards: the engineers held cards leaning on data, the managers held cards about schedule and precedent. Only when the hands are split does the source of the disagreement become readable.

The technique is a seat-by-seat card chart, one column per attendee, recording what he brought into the room, what he received there, and who received what he sent out. The cost is a few more columns per person. The return is that a meeting postmortem stops being a blur and becomes a card game analyzable seat by seat: whose card was suppressed and never played, whose was played at the wrong moment. Chapter 19's price list for speaking and Chapter 16's attenuation analysis both run on that chart.

6.4 Discipline Three: What Was Learned Later Goes in Its Own Column

On the page, Discipline Three is two columns. The left column is visible at the time. The right column is reconstructed afterward, and every right-column entry states how it became known: teardown, interview, experiment, data analysis.

The right column carries a source discipline of its own. The means by which post-hoc information was obtained must be named as a tool: which instrument performed the teardown, whom the interview spoke to, what parameters the experiment ran. Tool information is what makes the right column's evidential strength assessable. Spoken post-hoc information and measured post-hoc information differ by an order of magnitude.

The execution standard is one line of source per entry, abbreviations acceptable: teardown report number, the interviewee's position, the experiment's parameter table number. The benefit is that the right column becomes checkable in its own right. A postmortem with both columns checkable is the highest definition of a postmortem this book offers.

The columns may not be interleaved. Any sentence that borrows from the right column to support reasoning in the left gets rewritten or deleted.

Visible at the time (left column)	Reconstructed afterward (right column)
Around 20:00 on January 27, contractor engineers on the teleconference recommended against launching below 12 degrees Celsius	Launch imagery shows gas leakage at the joint beginning around the 58th second after liftoff [pending: page number awaits the official Rogers report text]
Erosion records from O-rings recovered on earlier flights, flight numbers checkable	Laboratory low-temperature elasticity test data, run after the accident
Contractor management reversed its position after a recess of roughly 30 minutes and agreed to launch	Vaughan's 1996 quantitative reconstruction of the successive lowering of launch temperature thresholds

The value of the two columns lies in the existence of the right one. It admits on paper which content is post-hoc knowledge, and in doing so exposes an illusion the author and the reader share. A postmortem with interleaved columns never writes the word "afterward" and depends on hindsight in every sentence. The two-

column version drives those sentences into the annotation column, where anyone can see them.

One maintenance rule: keep the growth history of the right column. An early draft's right column is thin and thickens as the investigation goes deeper. Keeping the two versions side by side leaves a live record of how the postmortem's understanding evolved. That record has teaching value. Where the draft judgment and the final judgment differ is the trace of an author's cognition being corrected by evidence, and such a trace is unusually good material for training judgment.

The organizational application is a postmortem of the postmortems. Once a year, sample the differences between first draft and final version across everything the organization produced, and look for the pattern. Which class of problem is always misread in the first draft is the organization's systematic blind spot in judgment. That list is the target paper for the next training cycle, and Chapter 39's rehearsal meetings should pick their topics from it first.

6.5 The Test: Three Signal Classes and One Dissection

A quick test for readers. When a piece of narration contains any of the following three classes of signal words, the hindsight perspective is doing the talking.

The first class is the counterfactual inside a process account: "if only, at the time, they had". A counterfactual appearing there means the author is filtering the process through the outcome.

The list has to keep growing as the genre's style evolves. Three phrases have come into fashion and belong on it: "essentially", "in the final analysis", "when it comes down to it". They almost

always land where attribution is converging, and they work as rhetorical tools for compressing a complex failure, which is the flattening disease of Chapter 2, section 2.3. The mechanism for growing the list is a team blacklist of signal words, kept in a shared document and extended after each month's postmortem meeting. The longer the list, the stronger the team's immunity. The blacklist is for marking, not banning: writing does not have to avoid these words, but where one appears the author checks the perspective. Marking trains self-awareness. It does not censor vocabulary.

The second class is the vocabulary of surprise and retrospection: "had they known", "to everyone's surprise", "unexpectedly", "nobody imagined", "and yet". The subject of these words is a person who knows the ending, not an actor in the event.

The third class is "in fact, by then it had already". Behind every "in fact" sits a piece of information with no timestamp. Interrogate it: who saw it, when, through which channel.

For a deeper check, dissect. Break the narration into event cards, each card carrying two lines: the moment the event happened, and the moment the information reached the decision-maker's field of view. The difference between the two lines is the post-hoc content.

The tool version is a dissection table with three columns, event, event time, information arrival time, and a fourth that computes the difference. Every row gets filled. Where arrival time cannot be established, the row is marked N/A with a note. The N/A rows become a list of record gaps, which is also a map of where the trace-keeping system is missing pieces.

The payoff usually arrives about thirty minutes in, when the person filling the table first sees, physically, how many rows carry a large gap. Those rows are where hindsight is concentrated, and

they get rewritten or moved to the right column. The template belongs in the postmortem toolkit of Chapter 38, alongside the two-column template and the signal word list. Those three pieces are the complete set of implements for this chapter's discipline.

Do it once and you will find that in most of your own cards the two lines carry the same moment. As you wrote, you promoted the ending into the information environment of the time.

More phrases are worth collecting, and the acknowledgements and closing paragraphs are where they cluster: "if only it had been a little earlier", "thankfully", "a blessing in misfortune", "fortunately". Words of that kind in a closing paragraph mean the author's perspective was suspended over the outcome from beginning to end.

The correction for a closing paragraph is to return to the beginning. What the ending should answer is this: if we did it again, which items on the list of decision points would change? That question forces the author from the endpoint back to the start, and the round trip is itself a full self-check of the three disciplines across the document. An alternative honest ending states the limits of the judgment: "this postmortem's reconstruction of what was visible then is limited by missing records; the following three points cannot be restored." A limitation statement tells the reader where the thin ice is. Thin ice is not off limits; it has to be crossed with the uncertainty in view.

6.6 Three Walkthroughs

Walkthrough one: rewriting an Alaska Airlines 261 postmortem.

The hindsight version reads: "the threads of the jackscrew nut were already severely worn, the safety margin exhausted, and ditching was only a matter of time." Three post-hoc components

sit in that sentence. The degree of wear was known only from the teardown. “The safety margin exhausted” is an analytic conclusion. “Only a matter of time” is an inference running backward from the ending.

The visible-at-the-time version reads: “the crew of the accident flight reported abnormal stabilizer position and motor overload in succession, with call records of repeated attempts and exchanges over roughly ten-plus minutes; the component’s lubrication and check intervals had earlier been extended following the company’s application and FAA approval (AAR-02/01 executive summary p.4, verified).”

Both versions describe one event. The first teaches the reader to watch out for wear. The second teaches the reader to watch the decision chain that extended the interval, and the procedure for handling abnormal signals. The first cannot be acted on. The second can.

There is an extreme form of actionable writing, the decision-point re-enactment. The left column becomes a drill. The reader reaches the decision point, closes the material, writes down a judgment, then opens it again to compare. Re-enactment is the name of this book’s stop-point design, and it converts a postmortem from something read into something practised.

The organizational use is a rebuilt postmortem meeting, with the agenda changed from reading the report aloud to walking the decision points. At each point the meeting stops, everyone writes a judgment, and then the discussion starts. The meeting takes twice as long and trains ten times as much. Resistance comes from the time cost, and the honest comparison is that one rebuilt meeting is worth three accidents. The account is worked out in Chapter 39’s budget.

An actionable left column has a check standard: the density of executable instruction. Can the reader, after each sentence, list an action? Sentences that allow it are information. Sentences that do not are atmosphere. The strict form is one action per paragraph: every paragraph contains at least one item convertible into an action check, and paragraphs that fall short get cut or supplemented.

The organizational version is quantitative quality control. Count the executable instructions in each postmortem and send anything below threshold back for rewriting, with the threshold starting at three per page. This control and the Chapter 38 template requirements are two halves of one gate: the template governs structure, the count governs density.

Walkthrough two: the 737-9 door plug (av-030, NTSB AIR-25-04).

Among the safety issues listed in the report's executive summary is the state of work on the door area during final assembly at Boeing's Renton factory. Sorted under the two-column discipline, the visible side holds the operational fact that this door had been opened for riveting rework within the assembly flow, together with the states of the work-order records along the release chain. The hindsight side holds the laboratory confirmation made after the accident that four positioning bolts were missing and no reinstallation was recorded (AIR-25-04 PDF pp. 10, 22, 26, verified).

Standing at the release node, what the actors could see was that the records were incomplete. What they could not see was that the bolts were missing. A postmortem that writes "they knew the bolts were not installed and released the aircraft anyway" has pushed the right column's physical evidence into the left column's heads.

The accurate question sits in the left column: how did work on a critical area, with no record of reinstallation, pass down the release chain without any additional confirmation being required?

That question points at structure, specifically at the rule that in - complete records for critical-area work should by themselves trig - ger manual review. The remedies are in Chapter 29 (the invisibility of maintenance) and Chapter 15 (practical drift). The attribution of knowing violation points at nothing.

Structure-pointing questions come in a set of five patterns, all running backward from the conclusion. How was it allowed, meaning who did the structure authorize? How was it not stopped, meaning which gate was missing? How was it not seen, meaning which information chain was broken? How was it not written down, meaning which record requirement was absent? How did nobody own it, meaning which seat was empty?

The output is a list of five structural questions, each pointing at an object that can be modified. That list is the translation layer between postmortem and remediation, and the layer is what lets a postmortem end in a work order instead of a sigh. A sigh-style postmortem leaves the reader with a feeling. A postmortem built from the sentence patterns leaves a work order.

Attribution by knowing violation deserves a specific prohibition, because it is the most harmful pattern in the genre. It disguises a structural problem as a problem of character, and writes a repair - able organizational defect as an unrepairable human one.

Enforcement takes a substitution drill. Whenever the pen produces “knowingly” or “deliberately”, stop and ask why a rational person inside that structure would have done this. The answers that come back are almost always structural: the appraisal system, the process, the information available, the workload.

The psychological resistance is that the drill looks like excusing the actors. Chapter 2 covered the difference. Attribution places a be - haviour inside its conditions in order to understand it. Excusing wipes responsibility off the conditions. This chapter trains attri -

bution. When it is done, responsibility is still distributed through the structure, each piece where it belongs. Fixing the structure is more useful than cursing the people.

Walkthrough three: the R44 that took off at night in fog (av-021, NTSB CEN23FA125).

Visible at the time: the pilot's night qualification and instrument experience status, checkable in person and in the records; his own visual reading of the weather at the departure point; the night visual conditions before departure and in the direction of the destination. Reconstructed afterward: the flight path rebuilt from ADS-B data, which shows ground impact within a very short distance after liftoff.

A postmortem that writes "he did not know he had entered dense fog" breaks Discipline Two. Whether he knew cannot be established from the materials. What can be established is the climb and the turn he performed at that moment. Write actions, leave the psychology out: rule 2 of the case-writing method in Chapter 5, applied here.

The rule has a companion. Gaps between actions are left blank, honestly. If forty minutes pass between two calls with nothing recorded, write that nothing was recorded. A blank is honesty toward the recording system; filling it is a deception of the reader. Mark the blank with a gap row in the timeline: forty minutes with no record, no calls inbound or outbound and no system event log. The gap row reminds the reader that every judgment in that interval is a blind spot.

Gap density is diagnostic data in itself. Many gaps along the key decision chain mean the trace-keeping system is in poor condition, and Chapter 38's trace requirements exist to treat exactly that. The psychological threshold for leaving blanks is higher than it appears. A writer's instinct is to fill the gaps with reasonable in-

ference so the narrative runs smoothly. Smoothness is a virtue in narrative and a poison in postmortem.

6.7 The Limits of the Discipline

The three disciplines have an honest boundary. In most organizations, timestamped material does not cover every key decision. Meetings without minutes, decisions reached in a corridor, verbal instructions that left no trace: the organization's own recording gaps thin the visible-at-the-time column itself.

That is a diagnosis rather than a failure of the discipline. A key decision point in some failure carrying no timestamped record at all is itself a failure report on the organization's trace-keeping system. Chapter 38 lists the trace requirement as a standing clause of the postmortem system.

This chapter can supply a design standard for that system. The minimum requirement is three moments: when the decision was made, when it was recorded, when it was communicated. With all three present, a future postmortem has something to restore. The registration cost is tiny, since the three-line decision record already carries it under the Chapter 7 system. The maximum requirement is the probability draft described earlier: the alternative outcomes then believed, with confidence attached, in words. Between the minimum and the maximum, each organization sets its level by the risk grade of the decision. Low-risk decisions take the minimum. High-risk decisions, the ones before an irreversible point, take the maximum. Grading the requirement is what keeps a trace-keeping system from being resented, because a blanket requirement for full tracing dies in its first week.

Where the record is blank, the blank is marked as blank. Memory is not permitted to fill it.

Blank marking carries a second value. The distribution of blanks is a disease map of the organization's record system: which classes of decision carry the most blanks. Levels where verbal command dominates, cross-department coordination points, and night shifts are the three high-incidence zones.

Drawing the map takes a blank count across every postmortem of the past three years, classifying and counting the blank points in each left column. The output is a heat map, and its three hot zones correspond exactly to the handover failure of Chapter 28, the coordination problem of Chapter 18, and the night shift of Chapter 26. The remedies in those three chapters are the heat map's prescription. After the prescription comes a retest: the same blank count run again next year, against this year's. The speed at which blank density falls is the speed at which the record system improves.

The heat map is where this chapter's discipline scales. An individual's honesty, aggregated, becomes an organization's diagnosis.

Five Self-Checks

1. Take your most recent postmortem and scan it with the three signal classes from section 6.5. How many hits?
2. In your organization, what key information exists somewhere in a system but cannot be seen on the interface of the person who decides? Name the system and the field.
3. For your last important decision, can you reconstruct the timestamped sequence of information arrival? Where you cannot, is the record missing or is your memory?
4. Does your team's postmortem template have columns for visible at the time and reconstructed afterward? If not, whose resistance will adding them meet?

5. Rewrite one paragraph of your industry's most recent public postmortem in two columns, three items in each.

Beyond the five, the graduation exercise for this chapter: run the whole sequence on your organization's last major postmortem, all four steps, the signal scan, the two-column rewrite, the dissection table, and the exchange test. It takes about one working day. Its four products, the scan list, the two-column version, the dissection table, and the purity score, are the first physical examination report on your organization's postmortem capability.

The report's value shows over time, in the annual comparison. Run the same sequence next year, and the improvement across the four products is a growth curve for organizational postmortem capability. What that curve means gets said again in Chapter 38, where postmortem capability is treated as the cardiopulmonary function of organizational learning.

Chapter 7 Decision Points and Points of No Return

> What this chapter covers: how to recognise the two most valuable positions on a failure chain, the decision point where the track can still be switched and the point of no return where it cannot, with a reversibility test table you can apply directly. When to turn back: when setting checkpoints into a process, when defining signature authority, or when judging whether there is still time to brake. A fourth use, past the three openings of the chapter: training a newcomer. These two concepts are the first lesson of induction in any industry. A newcomer who lacks them treats every moment as equally important and spreads attention like pepper from a shaker. The teaching kit is this chapter's table plus three cases from the newcomer's own industry. Fill three tables and the newcomer has the feel for where the points sit in that trade. The same kit scales to an industry association. A published checklist of points of no return, drafted once from this chapter's table plus cases contributed by member firms, gives every member's release form a common skeleton. It is the highest-yield service an association can offer: one drafting, a whole sector served. Estimated reading time: 40 minutes.

Not every moment on a failure chain deserves your attention.

Attention is the scarcest resource in a postmortem. The session runs two hours, and where those two hours land decides what the session produces. The allocation rule is the two tools of this chapter: mark the decision points first, because a fork earns the time; mark the points of no return next, because a cliff edge earns the time; run every remaining advance moment in a single line each.

Put a number on the rule and you have decision-point density. A competent postmortem carries between five and ten marked decision points. Fewer than three means the failure chain was drawn too coarse. More than fifteen means advance moments have been promoted to forks. Calibrate against good published work: the count of findings in a formal investigation report is a usable yardstick for where the density should sit.

Most moments are the event advancing. At a few of them, someone stands at a fork with the untaken roads still in hand. This chapter supplies two tools for finding those moments: recognising the fork (the decision point), and recognising, past the fork, the step from which no road leads back (the point of no return). Both tools come with one test table.

The relationship between the two tools is worth drawing on a timeline. Decision points cluster before the point of no return. A failure chain usually carries several decision points and one or a few points of no return. Decision points are opportunities; points of no return are boundaries.

The timeline has a management consequence. The last decision point before the point of no return is the most valuable position on the chain, and the minimum counterfactual of Chapter 5 usually lands there. Staffing that position is a governance question: who stands there, and does that person hold enough authority and enough information to use what they see. Chapter 18's single-owner list carries the answer.

The timeline also reads backwards. Count decision points backward from the point of no return: first back, second back, third back. The smaller the backward index, the lower the intervention cost and the more indirect the effect. Choosing among the backward indices is the whole art of the postmortem.

7.1 The Decision Point: Three Necessary Conditions

A moment that satisfies three conditions at once is a decision point.

Condition one: at least two real options exist. Real means the option carries a concrete action, a concrete cost, and could have been chosen at the time. “Be more careful” is not an option. “Delay until the next day, when daytime temperature rises” is one, and it was the option rejected in the Challenger case (Rogers Commission report, 1986).

Condition two: the person with authority to choose sits within reach. In the room, on the other end of the line, at the next node of the approval flow.

Options laid on the wrong table have a standard shape. The one who could actually veto is not present, and the ones present believe they hold no veto.

An option set can also fail before anyone enters the room. The menu was fixed in advance. The real choice happened in informal contact the week before, and the meeting is a coronation. A ritualised meeting satisfies the three conditions of condition one on paper and none of them in substance.

Detection runs on the agenda timeline. Look at how mature each item was when it reached the table. If the pre-read already carries a conclusion and its endorsements, the decision point sits before the meeting, and the search has to move backward.

Moving backward is done by reconstructing the shadow meeting: the email, the phone call, the corridor. The table in the shadow meeting is the real table. Read the shadow record beside the formal minutes. Reading two meetings together is advanced postmortem work, and it is the only way to recover a decision that was made in a room with no minutes.

Shadow material is usually absent from the file. Words in a corridor leave no trace, and that is the decision-point version of the recording gap described in Chapter 6. Move a decision point into the untraced zone and the postmortem runs out of food. Chapter 18's diffusion of responsibility has this collapse as its core symptom: everyone present, nobody holding the authority to stop.

Prevention sits at the level of the criterion, and it is cheap. Run a power pre-check when the process is designed. Annotate each decision point with three answers: who may choose here, who generates the options, who confirms a veto. Write the three answers into the process document. The cost is three extra lines at design time. What it prevents is a collapse of authority during operation, and that collapse is far more expensive to repair. Learning after an accident that nobody at a given position held the authority to stop is the most expensive way to learn it.

Review the annotations annually, folded into the process review. Every major process change updates the authority annotation in the same pass. A stale annotation is as dangerous as a stale process.

Condition three: the consequence of the choice appears later. Time sits between the decision and its consequence, and that interval is both a window for correction and a bed for illusion.

Mid-course checkpoints are the governance tool for the interval. Place one or two mandatory review points between decision and consequence, and ask a single question at each: how far does the current actual diverge from what we expected. This converts the window from passive to active. Divergence is measurable before the consequence lands, and a measurable divergence triggers a correction rather than a blame round. The institutional footing for a correction culture sits exactly here.

Density scales with the consequence cycle: one checkpoint per third of the cycle. A decision whose consequence matures in three years gets review points at nine months and at eighteen months.

Route the output into the decision record. It is the counter-document to the probability draft of Chapter 6: the expectation then, the actual now. The comparison of two drafts is the learning itself.

Deferred appearance supports two readings. "We chose correctly, which is why nothing happened," and "our turn has not come yet." Telling them apart waits for Chapter 11 on the normalization of deviance. For now, hold one line: a deferred decision is free in the mind and never free on the ledger.

The psychological accounting has a forward-facing version. Deferral does more than make the current decision feel free; it teaches the current decider a wrong way to read feedback. Nothing bad happens in the short term, the decider reads nothing-bad as positive feedback, and the misreading accumulates into the confidence illusion of Chapter 14.

The correction device is the post-decision audit. Thirty days after a major decision, review once: what did we expect, what actually happened, how do we explain the difference.

Cost, half an hour. What it buys is speed in the feedback loop. Without it, feedback waits for the consequence to cash, and a decision with a three-year consequence cycle carries a three-year learning cycle.

Give the audit a slot in the decision record template: the three lines of Chapter 38 plus a fourth, the audit date. One line of text turns every major decision into a learning loop.

Moments satisfying all three conditions appear more than once on a failure chain, which is why the list matters more than any single point on it.

The list also feeds the book's stop points. Every decision point on it is raw material for a stop point. A stop point is not a literary flourish; it is a decision point made teachable, the real fork on a real chain carried to the reader intact.

One operation converts list to stop point: lay the information that was visible at the time in front of the fork. With the information assembled, the fork becomes the reader's problem.

The craft sits in the proportion. Information too complete and the problem is trivial. Too fragmentary and the problem is unfair. Calibrate by testing the distribution of reader judgments. A problem everyone answers the same way carries no value, and a problem that splits the room is a good one.

The principle is stated here so the reader knows a stop point is not decoration. Behind every one of them sits a list of decision points dug out of a real failure chain.

The discipline for writing and for postmortems is to take the latest one. The latest decision point on the chain is the anchor of the minimum counterfactual (field 11 of Chapter 5; Chapter 10 gives the full backward-walking method). The latest decision point carries the lowest intervention cost, sits closest to the consequence, and holds the most complete evidence.

The latest-one rule has a boundary case. At the latest decision point, every option is bad. Postponement is bad; degradation is bad. A dead corner is not a good anchor for a minimum counterfactual, because an anchor should be a position that held one good option not dependent on luck. Handle a dead corner by moving forward: find where a good option appeared, and anchor there. Record the dead corner as evidence of the organisation's helplessness.

Keep both marks. The anchor is marked for the counterfactual, the dead corner for structural repair. A dead corner shows that the

organisation needs a mechanism for generating new options, not better technique at choosing among the options it has.

A second boundary case: the latest decision point sometimes carries the least authority. The person who could actually stop the thing left the chain at an earlier position, so the latest position where intervention remained feasible sits with someone who holds no authority. Annotate both levels: the latest feasible position for training, the earliest position holding authority for institutional change.

Add a column to the decision point list for the authority grade of the decider at that position, one of full, partial, none. Statistics on that column expose the organisation's authorization disease. If the last feasible position at most key decision points concentrates in the hands of people with no authority, the authorization structure needs rebuilding, and the column is the proof.

One more distinction has to hold: a decision point and a point of accountability are different objects. A postmortem chases decision points, meaning where the track could be switched. Courts and disciplinary processes chase points of accountability, meaning who owes a legal duty for the consequence.

The two sit on different scales. Decision points are positions on the failure chain, distributed across the whole advance of the event. Points of accountability are legal anchors: a signature, a job title, a duty of care. They are far sparser. The sparsity follows from what law will connect, which is foreseeability and control. Law does not assign fault to a link you could neither foresee nor control.

The gap between sparse and dense is the division of labour between legal learning and engineering learning. Law teaches people to discharge duty at sparse anchors. Engineering teaches people to switch tracks at dense forks. An organisation needs both, and merging them into a single meeting loses both.

Coincidence, never necessity. The latest feasible switch point may sit with someone who holds no authority, while those holding au - thority narrowed the options to zero at several earlier positions. Take the accountability answer as the postmortem answer and the organisation learns "sign carefully." It does not learn who nar - rowed the options, in which meeting, on what grounds.

Resisting that gravity has an organisational form: an independent right to convene the postmortem. Convening does not pass through the accountability line. Safety or quality convenes directly, and the accountability line assembles its own material separately. Separ - ating the two lines is standard in industries with mature investig - ation systems (Device 2 and Device 4 of Chapter 3), and it works inside a single firm.

The resistance comes from the efficiency argument: two meetings for one accident is expensive. The cost comparison is the output quality of the two meetings. A merged meeting produces a finding of responsibility. A separated meeting produces an improvement list. The two documents do not sit on the same shelf.

The answer to the efficiency argument sits in the cost table. The half-day saved by merging buys no effective improvement item at all.

A technical form of resistance: the conclusion layer of the post - mortem and the conclusion layer of the legal document cite each other and never substitute for each other. The postmortem writes structural findings; the judgment's factual section is cited as an evidence source. Different readers, different purposes, different standards, and a document that mixes them serves neither reader.

The organisational form is two texts on two signature lines. Safety signs the postmortem, answering the improvement question. Legal signs the legal document, answering the responsibility question. The lines meet at the fact layer, which exists once, and diverge at

the analysis layer, where the purposes part: the postmortem asks how to prevent, the court asks who answers.

Cost, one more document. Return, both kinds of learning hold and prevention learning does not contaminate responsibility learning.

7.2 The Three Currencies of Reversibility

Irreversibility is a spectrum, never a switch. Decompose “undoing an action” into the capacity to pay in three currencies.

Currency one is money. After the action, can money buy it back? Rebuild a part, switch a supplier, recall a batch. Money works while the object is unused, unconsumed, not yet grafted onto something else. A bad supplier on the approved list comes off for a switching fee. Concrete poured into a wall does not come back for any sum.

Currency two is time. After the action, can waiting undo it? Post - pone a release, a departure, a construction window. Time works while the window stands open. Before it closes, postponement is a costless undo. After it closes, the same action jumps from the re - versible column into the irreversible one with nothing in between.

Currency three is credibility. After the action, can it be withdrawn in public? A correction notice, a recall statement, an apology with compensation. Credibility works while nobody has yet taken an irreversible action of their own on the strength of your statement. Correcting a notice nobody followed costs little. Correcting a standard the downstream has executed for six months costs in proportion to the breadth of that execution.

What none of the three currencies can buy back forms the fourth category: the signature and the go-live. The release engineer signs the airworthiness release, the software team presses deploy, the aircraft finishes its takeoff roll, the reactor synchronises to the

grid, the drug enters the formulary, the announcement goes out. The shared structure: a statement has been made about the world (“this thing is good”), the world has begun to act on the state - ment (passengers buy tickets, clients call the endpoint, physicians prescribe), and withdrawal no longer depends on your capacity to pay. It depends on what others have already sunk.

Split the fourth category in two tiers. The signature tier makes a single-point statement: one release, one approval. The go-live tier makes a systemic statement: a product launch, a regulation taking effect, a public announcement. The withdrawal cost structures differ. Withdrawing a signature faces the validity of a document. Withdrawing a go-live faces an ecosystem’ s adaptation, and that cost scales at network level with what others have built on the statement.

The tiering has a budget consequence. At the signature tier, a second confirmation slot can be a person. At the go-live tier it has to be a system, because human confirmation does not hold against launch momentum. Grey release and the canary are the engin - eered second confirmation slot at that tier.

That engineering is the software industry’ s best contribution to this chapter’ s table. For any go-live tier action, in any industry, one question is worth asking: where is my canary.

One-line test, expanding field 9 of Chapter 5: money can change it, time can change it, a public correction can change it; a signature and a go-live cannot.

The test carries a time dimension, because reversibility decays. Right after the action, money still works: the part is uninstalled, the stock is on hand. Three months on, money no longer saves it: the part sits in the airframe, the stock has dispersed. Reversibility has a half-life.

The management consequence is to pull the undo decision early. Every day of delay after the error is discovered burns reversibility. The escalation described in Chapter 27 burns a psychological account; delay burns physical reversibility.

Quantify the half-life by annotating each action with a reversible window: the installation cycle of the part, the turnover days of the stock. Supply chain and production systems already hold that data.

The annotation costs one comparison table, and it upgrades the test from a static instrument to a dynamic one. The test says whether the action can still be undone. The window table says for how much longer.

One misreading has to be blocked. Irreversibility is a spectrum, and the three currencies are usually paid in part. Money recovers one portion, time another. An assessment of partial reversibility must name which portion is reversible and which is not, and a half-reversible action is defended on its irreversible half.

The mechanism behind the misreading is the normalization of Chapter 11. Reading half-reversible as reversible is step one of normalizing, because the language game about margin begins with a vague currency.

The action for the reader: for any action assessed as partially re - versible, write the irreversible half into the notes column of the table. If you cannot name that half concretely, the assessment is not finished.

7.3 The Reversibility Test Table

Section 7.2 becomes a form you can hand to someone.

A digitised version embeds the test in the workflow system. The action node forces a result to be entered, the four answers go into the system, and an irreversible class action triggers additional

steps automatically: second confirmation, cooling period, higher authority.

The benefit is that the judgment stops depending on the individu - al’ s diligence, which drifts. By the thirtieth release of the year, the judgment has gone from deliberate to reflex.

The cost is mechanisation. The system asks fixed questions, and a new form of action may not fit a fixed test. Keep an exception channel, a manual judgment path, and let anything routed through it leave a trace. The trace is the safety valve on a digital test system.

For the action you are assessing, answer row by row.

Row	Question	“Yes” means	“No” means
1	Does undoing this cost within the normal budget?	Money-reversible	Move to row 2
2	Does an unclosed window allow postponement or a redo?	Time-reversible	Move to row 3
3	Can the external commitment be withdrawn, with nobody yet acting on it?	Credibility-reversible	Move to row 4
4	Is the bearer of the undo cost identified, and does that bearer consent to carry it?	Actionable	Irreversible, even if rows 1 to 3 answered “yes”
5	Does a second confirmation point stand between the action and its effect?	One intervention slot remains	You are at the point of no return

Row 4 carries a generational dimension. If the bearer of the undo cost appears years later, because the action belongs to this period and the cost to a successor or to the public, then that bearer's consent cannot be obtained at all. The row answers "no" by construction, and intergenerational costs are graded strictly. The reason is the billing logic of Chapter 28: an irreversible cost that crosses periods must be signed for across periods, and with no signature it is treated as irreversible. The exception is statutory authority, where law explicitly lets a decider of this period bind a future one. The boundary of that authority is written in the statute, and an intergenerational action that exceeds the boundary is graded strictly as well.

Reading the table: the count of "yes" answers in rows 1 to 3 gives the reversibility grade. All three present is reversible; two or more missing is irreversible. Row 4 carries a single-vote veto. Row 5 tells you whether one last intervention slot remains.

The grey zone needs a rule too. Two yes answers and one ambiguous answer: grade the ambiguous row strictly, which means counting it as a no. The reason is what the verdict is for. The verdict exists to set defences, and the two errors are not symmetric. Treating a half-reversible action as irreversible costs an extra confirmation. Treating an irreversible action as half-reversible costs a disaster. Strictness is the rational side of that asymmetry.

The exception to strictness is paralysis. Grade everything strictly and the organisation stops doing anything. Use a quota: a fixed number of lenient verdicts per year, with the list of lenient verdicts published. The quota lets strictness and action coexist, and publication keeps the leniency from becoming invisible.

The table has three uses, one per kind of person. The decider fills it in before acting; the moment row 5 answers "no" is the point of no return, and it goes into the decision record. The process designer places a second confirmation point ahead of every point of

no return and sets that point's minimum information standard, which is the visible-at-the-time discipline of Chapter 6. The post-mortem writer marks the point of no return on the failure chain and examines what happened at the last second-confirmation slot before it.

Examining that slot has a time dimension. The slot is designed to sit before the action, and in practice it may be moved after it: release first, signature later. A signature collected afterward is a pseudo-confirmation.

Detect it by comparing timestamps. If the confirmation record is timestamped later than the action, the confirmation is false.

When pseudo-confirmation is systematic, when a spot audit finds a backdating rate above twenty percent, the slot has ceased to exist in substance. Count a slot that exists only on paper as absent, which is the downgrade clause of Chapter 22: paper confirmations do not count toward the defence inventory.

Governing pseudo-confirmation requires a system lock. The confirmation action must be locked in the system before the action can complete, so a release cannot be generated at all until the lock has fired.

7.4 Points of No Return in the Cases

The 737-9 door plug (av-030, NTSB AIR-25-04) puts the point of no return on the final assembly line, at the moment the door closes and the aircraft passes into delivery and service. Before that position, one records check on work that opened a critical area, run by hand or forced by system, would have stopped everything downstream. After it, the risk entered the life ledger of the 176 people on board: 169 uninjured and 7 with minor injuries, figures taken from

AIR-25-04. The currency of withdrawal switched from person-hours to lives.

The design question is therefore concrete. In the factory's release chain, was the information "this critical area was opened" entitled to trigger a mandatory re-check? At the time it triggered nothing, and that failure is the generating condition taken up in Chapter 29 (the invisibility of maintenance) and Chapter 15 (practical drift).

Alaska Airlines 261 (av-034, AAR-02/01) contributes a finer type: the irreversibility of an institutional decision, realised one installment at a time.

The approval to extend the check intervals, company application and FAA approval together, was itself reversible. The intervals could have been set back. From the day the approval took effect, every flight released under the new intervals was a micro point of no return: an aircraft flying with components no longer checked to the original standard. The failure did not occur on the day of approval. It cashed on one of dozens of successive releases.

A second sample of the installment type sits in Chapter 12's first case, the quarterly reversals. Each quarter's reversal was reversible in itself, since the accounts could be adjusted back. Five consecutive quarters of reversal locked the disclosure basis into a new normal, and a decision that was reversible once became structure through repetition.

Read the two cases together and the identification list completes itself. The decision can be withdrawn individually. Repetition requires no additional approval. The conditions of each execution match the last. No feedback loop carries the consequence back. An institutional decision that hits all four is an installment machine, and the moment to notice it is the second item: where repetition needs no approval, the machine has started turning.

Counting is how you recognise this type. The more exposures an institutional decision accumulates, the less negligible the cumulative probability of a single deviation, which is the arithmetic footing of the normalization described in Chapter 11.

Give the count an institutional form: an exposure ledger, with every realisation of an installment-type irreversibility recorded, its date, its object, and the state of deviation at the time.

The ledger has two readings: the cumulative count, which says how close you are to the accident probability in the statistical sense, and the rate of increase, which says whether exposure through this institutional decision is accelerating. Chapter 25's margin logic sets the alarm line: where exposure count multiplied by single-event probability approaches one, that is the place to stop.

The ledger belongs to the person signing the release, who adds one line as they sign. The cost of recording matches the cost of releasing.

The template is one line with four columns: date, object released, summary of deviation state, signature. Keep the template deliberately crude. Crudeness is a survival strategy, because a complex ledger has a very high probability of being abandoned within three months.

Electronically, fold the ledger into the release system, so the release action generates its own row at zero extra cost. A merged ledger produces something for free: an automatic count of releases, which gives the installment count a live reading for the first time.

Put that live reading on the signature screen. A number the signer sees at every release is a number the signer will consider.

The Haneda runway collision (av-037, JTSB Interim Report No. 2, 2025) puts the point of no return at the moment a combination

forms: the Japan Coast Guard aircraft on the runway and the A350 holding a landing clearance.

This case's decision-point density is worth counting with this chapter's instrument. The decision points are the Coast Guard aircraft's confirmation before the stop line, the controller's issuance of the clearance, the A350 crew's acceptance of the clearance, and the tower's monitoring of runway occupancy. Four points strung on one chain, each of which held a chance to rewrite the ending.

The joint-failure structure, where a different action at any one point breaks the chain, is the series-dependency form of Chapter 33: decision points in series are as fatal as equipment in series.

The postmortem reading of a joint failure is to file the four failure modes separately, each point with its own visible-at-the-time and its own option set: four small cards rather than one large one, since a large card dilutes four responsibilities into a single sentence.

Past the point of no return, the option space left to the two aircraft was set by speed, distance and lighting. All the value of the postmortem sits before it: the confirmation at the stop line, the readback and comprehension of the control instruction, the cockpit's visual search for runway occupancy. The final report, unreleased as of this book's writing, should concentrate its hypothesis space on the same segment. Material at the interim-report stage carries grade A-minus, an independent investigation body's document that is not the final one, and the minus sign stays when citing it (on the use of evidence grades, Chapter 10).

This chapter is one of the load-bearing columns of the anatomy part. Chapter 6 supplied the discipline of the record; this chapter supplies the language of position. Every instrument in the chapters that follow installs onto these two words: a ledger mounts at a decision point, a checklist hangs in front of a point of no return, a

drill is scheduled at a second confirmation slot. Position language is the grammar of the anatomy part, and a reader who holds it can find the standing place for every device in the four chapters that follow. The auditor of this chapter is the annual output-ledger review of Chapter 20, where the exposure ledger and the catch ledger are audited side by side: one counts how much you exposed, the other counts how much your defences caught.

7.5 Designing Your Own Points of No Return

Identification is the smaller half of this chapter. The output should be design moves. Four of them.

Move one: place a second confirmation slot ahead of every point of no return, and set an information threshold for the slot.

Aviation's release order is the most mature template. The checks required before signing are listed item by item on the form, and a signature declares that every item on the form has been verified. Every go-live, loan disbursement, product launch and public announcement in your organisation deserves an equivalent.

A general template for the equivalent takes five columns: the action, the irreversibility type (the verdict from the four currencies), the checklist (each item verifiable), the confirmer (named, with a named alternate), and the visible-at-the-time summary (three to five key items).

Grade the form by the risk of the action. A low-risk action merges the five columns into three. A high-risk action fills all five and attaches a timestamp.

Electronic versions should enforce the fields: no completion, no release button. It is the same mechanism as the release validation of Chapter 29.

Review the template annually against that year's accidents and near misses. Which columns proved redundant in real events, and which were missing. The review is how a release form evolves with the organisation that uses it.

Move two: give irreversible decisions a cooling-off period.

Judging whether an action is reversible costs far less than judging whether it is correct, and the gap widens for the third currency. Public announcements and public commitments wait one night, without exception.

Credibility has a property the other currencies lack: other people cash it in. Once an announcement is out, the cost of withdrawing it depends on how many people acted on it, and the more irreversible their actions (a move, an investment, a surgery scheduled), the more expensive the withdrawal.

Assess that population before the announcement. Who will do something irreversible because of this notice? Build a forecast list, which turns credibility assessment from a feeling into arithmetic.

Make the forecast compulsory. Add one step to the signing flow for credibility-class announcements: an impact forecast table, with an empty cell meaning the document goes back.

The forecast table also serves another chapter. It is the same table as the commitment tiers of Chapter 13: grading an external commitment and forecasting its credibility impact are one piece of homework, and one piece of homework can feed two chapters.

What the overnight rule filters out is not error. It is impulse.

Set the cooling period by currency tier: overnight at the signature tier, a week at the go-live tier, and at the credibility tier by impact, with longer cooling for wider impact.

Tiered cooling meets institutional friction from the business calendar, since the business side reads a cooling period as an effi-

ciency tax. The justification for the tax is the cost of one withdrawal. A single withdrawal avoided by one night pays a year of the tax.

Enforce cooling in the system: build a time lock into the signing button for announcement-class documents, reusing the mechanism of Device 4 in Chapter 13 on the credibility currency. The virtue of a system lock is that cooling stops depending on memory, and impulse is exactly what memory fails to hold.

Move three: write the point of no return into the decision record.

Three lines: what moment the point of no return was, where the last second confirmation slot before it stood, and what information was visible at the time (the two columns of Chapter 6).

Three years on, this record is the only raw material a postmortem can still use.

Test the material for usability with a simulated postmortem. Take a decision record from three years ago, look at nothing else, and try to rebuild the table as it stood then from the record alone. The completeness of the reconstruction is the report card of the record-keeping system.

Run the simulation as an annual sample: three historical decision records a year, rebuilt as an exercise. The output is an improvement list for the record-keeping system, naming which class of missing information stalled the reconstruction.

Half a day a year buys the record system a real stress test. A record system with no stress test degrades quietly into form: the fields filled, the information gone.

Move four: the signer is the terminal of responsibility, and a chain of signers is the absence of responsibility.

This move carries a protection for the signer that has to be written alongside it. Information rights come with the halt right. If one

person is accountable at a point of no return, that person needs access to everything before it and unilateral authority to stop the action. A signature post with mismatched authority and duty is a timer set inside the organisation.

The institutional form is a rights list for the signature post: the scope of information access, the halt procedure, and the immunity clause that applies after a halt. Three items, written into the job description.

Detect a missing item with one question to a candidate for the post: why did the last signer sign? A post that answers vaguely will produce the next vague signer.

Then write it into contracts, processes and job descriptions: who signs for this action's point of no return, and whether that signer holds unilateral authority to halt. Chapter 18 shows the case structure in which seven signers equal zero signers.

Self-Check: Five Questions

1. Write down three points of no return in your project over the coming thirty days, and grade each one with the table in section 7.3.
2. What was your organisation's most recent case of money-reversibility that drifted into irreversibility? Was the drift done by the process or by a person?
3. Of the signatures you have signed, how many carried fields you did not verify item by item? Can you carry that ratio?
4. At the last second confirmation slot in your process, does the check examine the record or the physical thing? Do the two fail in the same way?
5. Does your organisation carry installment-type irreversibility, a reversible institutional decision accumulating exposure

through every routine release? Can you put a number on the installments?

Beyond the five questions, a pocket version of this chapter: the three questions of the reversibility table, sized for a phone. Can I afford to undo it (money)? Is there still time (time)? Can I take it back in public (credibility)?

Each question takes a yes or a no, and the combination gives four states: fully reversible, half reversible, critical, irreversible. The action a state calls for is one sentence spoken to yourself: which of the four is this.

A week of practice installs the habit. Ask the three questions of one action a day, and at the end of the week the table has been internalised.

The test of internalisation is a falling misjudgment rate. Compare past verdicts against what later happened, and keep the statistic for yourself. Above thirty percent, go back to the table and learn it again.

Chapter 8: Escalation

> What this chapter covers: the seven rungs between "something has gone wrong" and "there is no way back", the symptom, the test, and the cover story at each rung, plus a way to find out which rung you are standing on. When to turn back: when a project has already gone wrong and you are deciding whether to stop or keep funding it; when reviewing a proposal to push harder. Reading time: 40 minutes.

This chapter and Chapter 27 (Sunk Cost and Escalation) divide the work between them, and the line is worth drawing first. Chapter 27 is a mechanism chapter and answers why: why people and organizations keep pouring resources into a path that has already failed, the psychological ledger of sunk cost, the binding of a decision-maker to a project, the institutional pressure that makes exit read as an admission of error. This chapter is anatomy and answers how. Escalation is not one decision. It is a series of rungs, and each rung has its own test and its own vocabulary of cover. What the chapter tracks, rung by rung, is the top-up: a commitment enlarged, then dated to a meeting or a signature. That date is the point of escalation.

A second boundary runs against Chapter 13 (Narrative before Evidence). The rungs are an internal process, the way an organization locks itself in step by step. Narrative is the external presentation, the way it tells the world what is happening. The two meet at rung four: a larger public commitment is both a rung and a narrative, so cases at that junction, such as the Challenger story in the walkthroughs (8.4), need both chapters read together. The internal rung decides which way the external story runs; the external story then reinforces the internal rung, because words already spoken become the reason there is no way back. That two-way reinforce-

ment is the most complete dissection an escalation case offers, and one chapter alone leaves half the mechanics out.

The ladder metaphor needs one more piece: the elevator between rungs. Some organizations skip. A small team need not climb rung by rung, and an organization run by one strong person can go from rung one to rung six without touching the middle. Skipping saves the cost of a single voice, and the saving is the motive: the intermediate rungs go with their ceremonies. The identification is a gap in the vocabulary. Rung-one language is followed directly by rung-six language, with rungs three and four absent, and the gap is the evidence. A skipping organization is also harder to walk back. A skipped ladder has no intermediate platform, so it either turns at rung one or falls at rung six, and the intervention window is narrower rather than wider. Hence the countermeasure: where the single-voice signature is obvious, outside parties should set the warning line earlier.

One misuse has to be blocked before the model is handed over. The ladder is not an instrument for accusing colleagues. Pasting a rung number onto a person (“that is rung-two thinking”) turns anatomy into a weapon. Anatomy describes processes and structures, not personalities; the ladder describes a sequence of organizational behavior, and applying it to an individual converts it into a trial of motives. Hence the first rule of use. The ladder is for self-testing, for diagnosing an organization, and for analyzing published material. Judging the motives of a named person lies outside this book’ s tools, always.

The other half of the usable model is what to do once you have located yourself. Every rung has a step-down action, the cheapest one that still works at that level. Rungs one and two: register and restate, so the explanation is deposited and the original accounting restored. Rungs three and four: independent review, and a freeze on commitments. Rungs five and six: re-examine the accounting

basis, and freeze new investment. Rung seven has one action only: go outside, to an external adjudicator or a regulator. Stepping down is cheaper than descending a staircase, since you need not reach the bottom in one move. Two readers by design, the person inside (a self-rescue manual) and the governing layer (an intervention manual).

8.1 The Ladder at a Glance

Seven rungs, ordered by severity. The ordering principle takes one sentence: the higher the rung, the more expensive the currency of withdrawal. Chapter 7's three currencies project onto the ladder neatly. Rungs one to three are paid in paper and meetings, in announcements and registers, both reversible. Rung four begins paying credibility. Rungs five and six pay real money. Past rung seven even information, the raw material of every later decision, has been rebuilt. The price of the currency is the order of the rungs.

The order has a second reading, by people. Rungs one and two are mostly technical handling by the front line and middle management. From rung four upward senior management has to appear in person, because the signing authority for credibility sits with them. That hands an outside scout a useful detail: the density of senior executives personally delivering set phrases in public material is a high-precision indicator of ladder position.

Each rung also pays for the next in commitments. Explaining the anomaly away classifies the fact as an exception. Shrinking it books the cost at zero. Finding a scapegoat spares everyone inside from repenting. Promising more ties the public image to the failure. Changing the measure stops the ledger showing deterioration. Enlarging investment rewrites the cost of exit. Sealing information leaves nobody inside able to object. An outside observer sees "no

way back" only after all seven, while inside nobody has looked for a way back since rung three.

Two uses, for two audiences. Outside observers, whether investors, regulators or journalists, use the ladder as reconnaissance: count how often each of the seven vocabularies appears in the target's public material. An organization carrying all the rung-three-and-above phrases is a high-risk target, and the cost of looking is one read-through of the public file. Insiders use it as a self-test. You cannot stop senior management climbing, but you can judge your own rung, and after rung three the cost of speaking plainly rises steeply. The purpose is not martyrdom. It is deciding where your information should go, whether the direct reporting channel of Chapter 16 or the external channel of Chapter 19.

Both uses rest on one precondition: visibility. The linguistic signature of each rung has to be written out in full, which is what the cover-story lines in 8.2 are for. Learn those sentences by heart and the radar is installed.

8.2 Rung by Rung

Rung One: Explain It Away

Symptom. The anomaly is assigned a one-off cause. Weather, an individual case, one new employee's mistake, the coincidence of several very unlikely events. The attribution never enters the register and produces no process change.

Test. When the same class of anomaly appears a second time, is the explanation identical to the first? If it is, rung one is running.

The automated version is explanation comparison. The attribution field for anomalies sharing a keyword is displayed side by side, and a human judges the similarity; no algorithm is needed, the eye is

enough. A quarterly rhythm is suggested, with workload running linearly with the number of anomalies, so a busy quarter costs half an hour more. The output is an explanation-drift report: each year's explanations of the same anomaly in parallel, where the drift in wording is the running record of rung one. It doubles as knowledge capture, since a collapsed explanation becomes a counterexample, and the counterexample library is live teaching material for new staff. That is the hidden dividend of an anomaly register.

A detail belongs in the countermeasure against resetting to zero. The first-attribution column records an explanation; the same column on a later, similar entry records a test. When a repeat appears, the register pulls the earlier explanation up for comparison, triggered by a matching keyword rather than by anyone's memory. Three outcomes get three handlings: the explanation holds, so the attribution is confirmed and archived; it collapses, so the attribution is escalated and analysis triggered; it cannot be tested, so the attribution is marked open and held. Those branches go into the register's field logic, with a comparison-result field of holds, collapses, or open. The point is to make the register think: it stops being a warehouse and becomes a testing machine for attribution quality.

Cover story. "This one was an exception." "Everyone in the industry runs it this way." "It just happened to coincide."

Rung Two: Shrink It

Symptom. The scope of the impact assessment narrows after the fact. An event first reported as three hours of downtime becomes, in the management pack, "brief latency perceived by a small number of users". Costs outside the casualty count, such as handling hours, customer churn and premium increases, never reach the ledger.

Test. Put the two impact descriptions of the same event side by side. Has the scope moved? Is there an impact register opened at the original scope, or only one written at the final scope? No register means shrinkage is running.

The anchor row has an external version. The first report to a regulator is a legal anchor: the initial filing under a statutory reporting duty is protected by law, and any later change of scope has to be justified in writing to a regulator who reviews the reason. That makes retrospective shrinkage show up. If the internal report says the impact was contained while the initial filing says three hours of downtime, the gap between the two documents is a ready-made line of inquiry. Consistency between the internal register and external filings therefore has two motives, legal and governance, and the annual self-check on it belongs in the internal audit plan.

One discipline should be fixed hard. The initial scope, once registered, is the anchor of the ledger, and any later change may only be added as a new row citing the anchor row. Change the scope, never the history. The target is retrospective shrinkage: three hours of downtime at the time, "a brief fluctuation" in the quarterly report. Enforcement is row-level locking, where a registered row cannot be edited and corrections are new rows. That is mature in financial systems, where it is the irreversible journal entry; porting it to an impact register is a configuration job.

Cover story. "The impact is contained." "Actual losses are far below the book figure." "The vast majority of users noticed nothing."

Rung Three: Find a Scapegoat

Symptom. Responsibility is placed where it cannot answer back. A departed employee, an outside supplier, the previous management team, the macroeconomy, user misuse. The attribution has one

property in common: every serving member of the organization is safe.

The precondition deserves naming first: the candidate pool. Whoever has already left, whoever is most junior, the outsourced, the deceased, anyone unable to defend themselves. The pool's shared property is that the cost of resisting is infinite. Its contents are a reading of the culture: an organization whose first thought after a failure is who to blame shows that systems analysis never had a place in its accountability culture. The countermeasure is proceduralized accountability, in which accident analysis is driven by a template whose question list carries the names of stages and never the names of people. That leaves a scapegoat no grammar to stand in.

Test. Does the attribution contain no serving, named party at all? A scapegoat is the cheap substitute for accountability rather than its opposite: one head buys the system an exemption from inspection.

Who gets chosen follows a pattern worth a footnote. Outsourcing and agency labor are the high-yield pool: the relationship is a contract rather than employment, and the legal procedure for disposing of them is the simplest. The scapegoating rate of contract workers is a product of labor structure and accountability culture together. The public observers here are labor inspectors and the press. The construction shows up in official statements as "the temporary worker", "the outsourced contractor", "the agency staff", and the appearance of those words is the fingerprint, which is why inspectors can put them on the list of flagged terms for reviewing a public notice.

A second fingerprint is grammatical. Is the subject of the attribution sentence a singular person or a plural system? "An employee violated procedure" is a suspect sentence. "The procedure did not cover this case" is a systems sentence. The deciding point is where the remedy lands. A remedy that punishes a person leaves

the scapegoat standing. A remedy that revises a procedure serves accountability and improvement at once. The two readings are not mutually exclusive, and the test is proportion: only the person handled and no system action taken, and confidence is very high. An outside observer can then rewrite the scapegoat sentence into a systems sentence, and that rewriting right, held by journalists and researchers, is the public tool against a scapegoating culture.

Cover story. "The employee concerned has been dealt with under company policy." "We were let down mainly by our supplier." "This is a legacy issue."

Rung Four: Promise More

Symptom. An existing failure is met with a larger target. A bigger feature set announced after a delay, a higher revenue target announced after widening losses, "a higher safety standard" announced after an accident. The public commitment is defensive: occupy the narrative first, so that "we have failed" has no place left to stand in the language.

Defensive commitment has a more concealed form: borrowing someone else's mouth, in a vision speech at an industry conference or a long-range plan in a media interview. Commitments through third-party channels never enter the formal disclosure process and escape internal review. The identifying point is the separation between the channel and the formal one. The vision appears only in the interview, never in the annual report, and the separation is the signal. The consequences match a formal commitment, since the market prices the stock on what it hears, but the company keeps a right of denial: that was only an outlook offered in an interview. That right is the clearest feature of the type.

Two market indicators turn defensive commitment into a score rather than a feeling. The ratio of commitment to resource comes first: a jump in the ratio, with the target doubling while the budget stays flat, is high confidence. The data are public, since a listed company's targets and budgets both sit in its filings, and for a private organization you read the supplementary budget requests. The interval between announcement and exposure comes second, where a major new commitment published within two weeks of the exposure is probably defensive. Together they move identification from impression to grading: a ratio jump plus a coinciding window is two points, and a total over threshold flags red.

Test. Where does the resource for the new commitment come from? If the resources behind the raised target come from cutting testing, cutting validation or cutting redundancy, which is ground held by Chapter 20 (the slack of success) and Chapter 25 (margin erosion), then this rung has started structurally.

Resources are not only money, either. Headcount, hours and management attention are resources, and the common form of rung four is pressing the same people into two jobs at once, closing out the old project while sprinting at the new target. Borrowing from one line to pay another is invisible on the budget sheet, which records only money; it shows in rosters and timesheets. The tracking tool is a quarterly comparison of hours distribution, where the drift in a key team's share of hours is the invisible progress bar of rung four. Its readers are the PMO and the govern - ing layer, because the war the budget cannot see is visible in the roster.

Cover story. "Turning a challenge into an opportunity." "Double down and burn the boats." "Hold ourselves to a higher standard."

Rung Five: Change the Measure

Symptom. The indicator is swapped, the basis adjusted, the baseline reset. This is a change of ruler rather than a falsification. The indicator measuring the quantity that deteriorated most visibly is quietly replaced by one that performs better.

Test. Was the statistical basis changed in the same period, without independent review? Can the reason for the reset baseline be independently verified? What separates rung five from Chapter 12 (goal displacement) is timing. Chapter 12 describes an indicator optimized over years until it no longer measures what it names. Rung five is an emergency ruler change at the moment failure surfaces, and the discriminator is the coincidence in time.

The coincidence can be quantified. Measure the interval between the ruler-change date and the bad-news date: look back from the announcement for the nearest bad news, a loss warning, an accident notice or a regulatory inquiry, and record a change within thirty days as an emergency suspect. Suspects go into the register described in Chapter 12 with one added column: days since the nearest bad news. The annual statistics on that column are a thermometer for the organization's ruler-changing culture. A high share of emergency suspects means ruler changes are stress responses rather than science, and stress-driven changes also carry a high later correction rate, which is the subsequent evidence of stress.

Cover story. "To measure it more scientifically." "To align with industry practice." "Adjusting the statistical method."

Rung Six: Enlarge the Commitment

Symptom. Resources are added to a path already shown to fail. More budget, more people, more schedule, more leverage. The reasoning no longer depends on new evidence; the argument is the

size of what has been spent. We have put in this much, so stopping now loses all of it. This is the home ground of Chapter 27, which supplies the classic experimental basis of the sunk-cost effect, Staw's 1976 design, along with the institutional conditions. (Staw, "Knee-deep in the big muddy", Organizational Behavior and Human Performance 16: 27–44, 1976; the Crossref record has been verified.)

Test. In the minutes of the incremental decision meeting, what share of the arguments is "we have already invested"? Once is rhetoric. Where it becomes the main argument, that is rung six.

A measurement basis for that share: the ratio of already-invested arguments to new-return arguments in one set of minutes, where a ratio above three to one marks the documentary form of rung six. What makes the measure executable is that both classes are countable, and annotating one set of minutes is half an hour of reading. The training material is this chapter's list of cover stor‑ ies, which maps phrases to argument types. The output is an es‑ calation-concentration curve for budget meetings, quarter by quarter. The budget season in which concentration rises is the season in which the ladder is being climbed.

Cover story. "Stop now and everything we have put in is wasted." "One more doubling and we turn it around." "Our competitors are escalating too."

Rung Seven: Seal Off Information

Symptom. The circle of the informed contracts, the external line is unified, the channels for challenge stop working. Classification levels rise, minutes move from written to verbal, the person who raised an objection is transferred, which is Chapter 19 (why the truth-teller goes first) cashing out, and data stop reaching the board or the regulator in full.

Test. Do access tightening, a demand for one line, and the movement of dissenters coincide in time with the exposure of failure? Coincidence means rung seven.

In a digital workplace the coincidence test has a new fingerprint: the change log of the permission system. Bulk adjustments to access rights, bulk upgrades of classification, temporary rerouting of the approval chain. The log records every tightening faithfully, and timeline analysis against the exposure date is basic digital forensics; for a management audit it takes one query. Its advantage is that it asks nobody's cooperation. In the paper era, investigating a seal took interviews and courage; in the log era the seal leaves its own trace. The precondition is that the log itself is not sealed, so its access control must be independent of the business system under audit. If the auditor can reach the log, the fingerprint means something. Past this rung the organization holds no route by which "turn back" can be raised, and the problem worsens until outside parties see it.

One variable sets the speed of that worsening: cash flow. How fast a rung-seven organization deteriorates is finally decided by cash, since a problem can be carried as long as cash flow holds, and cash flow is the fuel depot of delay. The reading is available from outside: payables stretching, abnormal inventory turnover, the rolling frequency of short-term borrowing. Creditors and suppliers watch all three daily, and their abnormal combination, stretched payables plus rolling borrowings plus inventory piling up, is the financial portrait of a rung-seven organization. That portrait is a direct input to a lender's credit decision, and the money that keeps such an organization alive is a prepayment on the next bill.

Outside observers have an intervention point too. A rung-seven organization still answers hard external signals: a regulatory inquiry, a qualified audit opinion, a core supplier shortening credit terms, an unusual rhythm of executive departures. What rung

seven seals is internal dissent; statutory and contractual pressure still has to be taken, usually by reassurance and delay. The check-list is therefore which external actions penetrate the reassurance layer: a qualified audit opinion, a regulator's on-site inspection, a major customer's contractual right to audit. Those three carry legal or contractual force, and the layer cannot hold them. The list is valuable in the hands of creditors, large customers and regulators, the external braking system of a rung-seven organization. Where the brake bites is this chapter's offering to readers in finance and procurement.

Cover story. "To prevent the release of unverified information." "One message, to avoid panic." "Exceptional measures for an exceptional period."

8.3 The Mechanics: Why Nobody Stops Halfway

Before the seventh rung there are many positions where the climb could stop, and it rarely does. Three layers explain the failure to stop.

Layer one: every rung delivers a real local gain, and the list deserves to be complete. Rung one buys output without stopping the line. Rung two buys a presentable report. Rung three buys a temporarily steady morale. Rung four buys the share price and the financing. Rung five buys a passed appraisal. Rung six buys a diluted unit cost. Rung seven buys surface quiet inside. The gains are real, and a defense built on denying them gets nowhere. The design principle is substitution: give the organization a way to obtain the same gain without escalating. Rung two's gain, for instance, can be bought with disciplined disclosure of the impact register, which also buys a credible report and a reassured regulator. Substitution rather than confrontation is the core technique, finding a clean source for the benefit. The accounts of escalation

are global and intertemporal, while decisions are local and current, which is the ground Chapter 28 (horizon mismatch) covers.

Layer two: irreversibility rises rung by rung, and the early warning indicator is the lead time of reversible actions. At rung one the explanation is written after the fact, the next day. By rung four it is pre-written before the meeting, with the commitment text finalized ahead of the event's conclusion. That inversion, conclusion before evidence, is the precursor of an irreversible jump. Detect it by comparing document timestamps, where a commitment text finalized earlier than the record confirming failure is an alarm. The response is an audit, since an inverted sequence means the decision process has been turned upside down and every document it produced needs re-reading. In Chapter 7's three currencies, the actions of rungs one to three (explaining, booking, attributing) are almost entirely reversible, undoable with an announcement and an edit to the register. Rung four pays credibility, rung five pays the credibility of the ledger, rung six pays cash and contractual obligation, and past rung seven even information has been rebuilt. The further down, the dearer the way back: that is the physical shape of a slide.

Layer three: each rung shrinks the option space of the next. A quantifiable expression exists: topic survival rate, the frequency with which challenge-the-status-quo items appear on meeting agendas and how many rounds they survive. In a rung-seven organization's agenda they are extinct. The method is an annual sample: pull the year's agendas, count the challenge items, and follow what happened to each, whether discussed, shelved or vanished. The year-on-year comparison is a trend in openness, and an organization on a downward trend has rung seven at the door. The signal belongs to the governing layer, since topic survival is one of the few cultural indicators visible without a survey, needing only the agenda archive. Rung three's scapegoat deletes "the

system has a problem" from the agenda. Rung five's ruler change hides the real data from rung six's incremental decision. Rung seven's seal removes the input that would have triggered a stop-loss after rung six. The rungs are not seven parallel behaviors on a menu. They are a one-way street, each later rung built on the foundation of an earlier one.

The one-way street can be walked backwards inside a specific window. A triggering event that changes level, a new chief executive, regulatory intervention, pressure from a major customer, a financial alarm, can sometimes knock an organization from rung five back to rung two, with the new ruler scrapped and the impact restated. An external shock resets the internal cost of the cover story. For outside parties the lesson is timing: the first three months after a shock are the best window for demanding a restatement, because the cost of self-justification is highest and the cost of compliance lowest. The window closes fast, with a new vocabulary rebuilt inside a quarter.

8.4 Two Walkthroughs

Walkthrough one: locating Challenger on the ladder, drawing on the Rogers Commission report (1986) and Vaughan's study (1996). The O-ring erosion found after earlier flights was explained as "normal within acceptable limits", which is rung one running flight after flight. The cold-temperature risk on launch day was pressed in the evening teleconference of January 27, 1986, and that pressure has the same structure as rungs one and two. Contractor management reversed itself after the meeting was paused and came out for launch, which is the meeting form of rung four, a commitment set against a doubt. (The official text of the Rogers report was read and verified on November 1, 2026. The original-source counterpart of the rung-one phrase appears in Volume 1, Chapter 6, where Thiokol's conclusion after flight 51-C described

the condition as not ideal, though acceptable, and the launch constraint established in July 1985 was waived by the project manager flight after flight. The rung-four meeting form appears in the Volume 1, Chapter 5 record of the afternoon caucus. Vaughan (1996) remains available only on paper, pending interlibrary loan.) The point of the walkthrough is to aim an abstract rung at a specific meeting, so that each rung lands on a date, a meeting name and a signatory.

Walkthrough two: the judgment made in flight on Columbia. At the launch on January 16, 2003, foam shed from the external tank and struck the left wing. In the project assessments before the accident, foam shedding was classified as "in-family", meaning the same class as shedding already observed on earlier flights and therefore not a new safety concern. The Columbia Accident Investigation Board gave that classification a dedicated review: the standard itself absorbed "a larger piece of foam" into an existing category, and the new information, the location and energy of the strike, was never entitled to trigger new analysis. (CAIB Vol. 1, pp. 122–123, verified.) This is rungs one and two working together, explaining it away as a familiar phenomenon and shrinking it by grading against historical experience, while the information visible at the time did contain an option that would have forced reassessment: requesting external imagery. That request was cancelled partway along the decision chain, and the CAIB report carries a section on it. (CAIB Vol. 1, p. 39, verified.) The conclusion lands on structure again. The absence of a rule for updating the classification standard let rung one run automatically.

Both walkthroughs share one teaching point: locatability. An abstract escalation concept, once landed on a date, a meeting name and a signatory, becomes a concrete object for a corrective action list. The output is coordinates, not a verdict. The suggested format is a ladder-location table, the seven rungs as rows and the case's

evidence in the columns: which document, which meeting, which sentence as spoken. A row that can be filled in is a rung that has already happened. The table has a second use in postmortem training: give trainees a historical case and have them fill it in, and the filling is how the tests get internalized. The check on internalization is how fast a trainee locates a new case.

8.5 Devices for Stepping Down

Four devices, ordered by installation cost, cheapest first.

Device one: the anomaly register. Three columns per entry: initial impact scope, attribution, and whether the same class recurred. Its function is to remove rung one's hiding place, since an explanation has to be deposited, and when the same anomaly returns, the earlier explanation becomes the object under test.

A threshold has to be designed into its maintenance: the bar for registering an anomaly must be low, with the front line defining what counts. A high bar, admitting only anomalies whose consequences exceed some limit, leaves rung one a permanent hiding place, because small-consequence anomalies are never registered and explaining them away costs nothing. The price of a low bar is noise, and noise is handled by quarterly folding, which merges anomalies of one pattern into a single line so the register stays readable. The ratio between threshold and folding is the core operating parameter: low enough for the front line to register without burden, folded enough that management is not drowned.

Device two: a register of accounting changes. Any adjustment to a statistical basis, a baseline or a measurement method is registered separately, stating the proposer, the reason and the effect, and it needs an independent reviewer's signature. Under this register, rung five's change of ruler turns from a covert operation into an explicit act.

The step after an explicit act is an explicit trail: an annual compilation of the change register, one year's ruler changes in one view, which ruler, proposed by whom, what the reviewer said. The compilation should be visible to the whole governing layer, since the list is their first-hand material for judging whether reporting is credible. A governing layer that reads the reports but never the list is reading a world someone else has already adjusted. The second reader is the audit, because basis drift is a high-value lead and the register cuts the evidence-gathering cost by an order of magnitude.

Device three: exit conditions up front. At approval, write down which signals start an exit review, who adjudicates, and that at the review the investment already made no longer counts as an argument. Chapter 27 gives the full design. The after-the-fact defense of rungs four and six fails against an ex-ante clause.

Timing of the signature matters for the clause's force: it has to be signed at approval. Exit conditions added after a project runs into trouble have no teeth, because by then the drafter is already standing on the ladder. Being ex-ante is psychological as well as political, since nobody knows whether their own project will fail, and that makes the clause fair to everyone. A fair clause signed at the start, cited in a crisis three years later, leaves no party able to call it unjust.

Device four: rewards for termination. Reward, as policy, the person who stops a project, so that whoever halts a sunk project receives the same level of recognition as whoever rescues one.

One technical point sits in how the award is judged: post-hoc verification of the stop. What became of the stopped project, and how does the reason it died compare with the reason for stopping it. Where the comparison supports the decision, the award stands. Where it overturns the decision, the case is recorded anyway, because a stop that was a misjudgment is also learning material. Termination rewards do not only reward the right calls; they record

every call, and the completeness keeps the institution from de‑forming into a correctly-guessing contest. The PMO compiles the aftermath of all stopped projects each year into the review mater‑ial. This is the reverse-incentive design that answers Chapter 27's mechanism. Without it, the people who run the first three devices are systematically eliminated by the performance system (Chapter 19).

Publicity is an execution detail. Recognition has to be organiza‑tion-wide and public: an all-hands meeting, an internal newsletter, a promotion announcement. Private recognition carries no cul‑tural effect. The worry is that the stopper gets labelled, and nobody dares let them run a project again. The hedge is to classify the stop: stopped because the market moved versus stopped because exe‑cution failed, the first an award for judgment, the second an award for correction, with both framings in the citation. The long-run effect is desensitization. As stopping becomes ordinary, the or‑ganization's cost of failure falls, and the end point is a stop that is no longer news. On that day the termination reward has done its work and can sit quietly inside the system.

A self-test to close the section. Take the project you are responsible for now and answer three questions. Did the most recent anomaly get an attribution in the register? Is the scope of the latest impact assessment consistent with the previous one? If all investment went to zero tomorrow, is your first reaction to check the data or to check who leaked? If you answer the third with the second, you are at rung six at least.

The third question deserves expansion into a graded set: it is the detector for rung seven, the second for rung five, the first for rung one. The complete five run like this: did the attribution reach the register (rung one), is the accounting scope consistent (rung five), are the signatures complete (rung four), has anyone been moved (rung three), has information been sealed (rung seven). Five ques‑

tions together locate the rung. The organizational use is an annual cultural check, answered anonymously by everyone, where the distribution of positive answers by rung is the organization's escalation climate map and the year-on-year comparison is the trend. The reader is the governing layer. The map is more honest than any cultural slogan, because it measures behavior rather than statements.

One note on reading order. Read this chapter before Chapter 27. The ladder gives the skeleton: locate the rung with the tests, then go to Chapter 27 for why the stopping is so hard. Locating before explaining matches the natural rhythm of handling a crisis, which is to stop the bleeding before looking for the cause. The pair is finished when you can say two things about any escalation case without pausing: which rung it stands on now, and why it cannot stop. Those two sentences are everything the two chapters have to give.

Self-Check: Five Questions

1. Which of the seven rungs does the most recent anomaly on your project belong to? Test it against the criteria in 8.2 one by one, not by feel.
2. When did your organization last change its performance metric? Was the change registered? Who reviewed it?
3. How many times have you heard "we have already put in so much" as the main argument for a budget increase? Did the project turn around on any of those occasions?
4. In your organization, who was most recently rewarded for stopping a project? If you cannot remember, what does that mean?

5. Across the seven rungs, which rung' s cover story appears most often in your organization? Write the sentence down and pin it to the agenda of the next review meeting.

Chapter 9 The Failure Chain: Three Layers and the Breaks at Their Boundaries

> What this chapter covers: how the technical, organizational and incentive layers are divided, how to recognize the breaks where they meet, and how to use the test that says an event is a failure only when all three layers break, while a single broken layer is a fault. When to turn back: when you are reviewing an accident, reviewing a system's defence-in-depth design, or working out which layer your own improvement measures land on. Estimated reading time: 45 minutes.

Any failure with weight to it can be taken apart into three layers: what the physics and the system were doing, what the people and the procedures were doing, and what the structure of interest was rewarding. Each layer can break alone. Where the break sits decides what kind of event you have.

The classification also decides who gets told. A fault, meaning one layer broken, can be handled inside the maintenance department. A failure, meaning several layers broken, has to be escalated to governance. Mix the two paths and governance drowns in faults while the real failures inside the flood get diluted. The criterion for choosing the path is the test in this chapter: two or more layers broken, report it as a failure. Filing a failure as a fault has its own cost. Deliberate downgrading is Chapter 18's subject; uninten‑tional downgrading usually comes from people who cannot layer an event at all, and the criterion below is written for them.

The walkthrough in section 9.4 carries a precondition. At least four of the nine cells must be fillable with evidence. Below four, the exercise degrades into three-layer guessing, and its output is a

collection list rather than an analysis: which evidence columns are empty, what can be recovered, what cannot and should be logged as a checkpoint. Walk the case when collection is done. The threshold comes from Chapter 10’s evidence grades, at least two dated pieces at grade B or above. Cases that fall short go into the matrix first, under Chapter 5’s downgrade rule.

Fill the table as a small team. Three people, one from a technical background, one from process, one from management, each filling a copy alone before the copies are merged. A single filler carries a single-layer bias: the technically trained one leaves the organizational row thin, the manager writes the incentive row in the language of intentions. The disagreements at the merge are the most valuable cells. Put them in the notes row instead of resolving them by seniority; disagreement means the organization holds no single model of what happened, and that is a diagnosis.

One stance before the method. The three layers are an instrument, not an ontology. The claim is not that the world is built in three storeys, but that sorting causes by the currency you would have to pay to remove them gives analysis a handle. A tool is judged by whether it works, meaning whether it locates a place to intervene, not by whether it is true. The stance removes a burden: you need not settle which layer a phenomenon philosophically belongs to. Where attribution is disputed, run the hypothetical intervention both ways and keep whichever reading changes the outcome.

9.1 How the Three Layers Are Divided

The dividing line is not the object of study. It is the intervention currency: the variable whose change would eliminate this class of event. Whichever layer holds that variable is the layer the event belongs to.

Technical layer. The currency is physical: swap a part, change code, add a sensor, move a parameter. Events here are failures of components, materials, algorithms and physical processes. A hydraulic line punched through, thread wear, a battery in thermal runaway, a pitot tube with its intake blocked. The list has a second use: the search radius of your currency is the radius of your authority. A frontline engineer can reach parts and parameters, never assessment schemes or contracts. So write the currency list into the job description, in a column of its own beside the duties. That column feeds Chapter 35's positioning table, whose first parameter reads it.

Organizational layer. The currency is process and information: change the checklist, change what a handover contains, change the reporting route, change the shape of the meeting, change the training. Events here are information that did not arrive, responsibilities nobody claimed, process forked away from practice, training detached from the real scenario.

Incentive layer. The currency is contracts and assessment: change the KPI, change the pay structure, change the tenure, change the liability rules. Events here are local rational choices summing into collective loss (Chapter 17's territory), present benefit set against distant cost (Chapter 28), and assessed indicators diverging from the real goal (Chapter 12).

A technical break needs one further question before you write it down: primary or induced? Primary means an unknown failure mode in a new material or process. Induced means brought on by deferred maintenance, service past life, or a wrong modification. Induced technical failure does not count as an independent technical break on the nine-cell table, because its root sits in the organizational or incentive layer. The evidence is the maintenance record and the usage history; the criterion is Chapter 29's. The

distinction sets the level of the fix: for an induced failure you do not change the part, you change the maintenance budget.

The incentive layer has a break that spans organizations: buyer and supplier incentives pulling apart. The buyer wants cheap and on time, the supplier wants margin and capacity. Both sets lock at contract signature and meet years later as a quality event. The non-OEM bearing in ma-001 is that meeting point, buyer price pressure crossing supplier margin protection. Detection happens at contract review, where you look at the weight ratio between price clauses and quality clauses. A contract with that ratio badly skewed is a break waiting for its year.

The incentive layer also has a terminal form: an entire industry measured the same way. Pharma runs on sales targets, aviation on on-time rates, construction on schedule. When every firm is scored alike, the break grows in the same place across the industry, and what follows has the character of an industry design rather than a scandal at one firm. Only regulation can break that isomorphism. One firm changing its assessment alone buys a competitive disadvantage; the first to change is the first to die. A regulator forcing the change industry-wide is the executable route, whether that means legislating hours-of-service limits or retiring on-time-rate scoring. The argument for such a move is the break-point register: this industry's break cases in a list, which is page one of any lobbying document.

There is a fourth currency, less visible: the default option. The incentive layer is strongest when it changes a default without touching a single assessment. The scoring stays, the submission workflow opens on a different screen, and behavior follows the default. Governing defaults is behavioral economics inside an organization, and it is cheaper than rewriting an assessment, so pick a default that moves no currency at all. These moves appear in Chapter 12's register of settings, because a change of default must be

logged like any other: which default changed, who benefits, who will not notice.

Before using the criterion, guard against its commonest misuse. The three layers are not a ranking of blame. A technical break is not the shallow one and an incentive break is not the deep one; the layers classify currencies, they do not rank importance. The misuse comes from reading the model as an onion, the outer layers assumed more fundamental. That reading drags every postmortem toward the incentive layer, because digging always arrives there, and it teaches the organization to despise the technical fix in front of it, which is also part of learning. Read the layers as parallel. All three are candidate roots at once.

The test is the hypothetical intervention, run layer by layer. "If we fitted a better sensor, would this still have happened?" Yes means the sensor is not the root layer. "If we changed the handover process?" "If we took the downtime loss out of the department's assessment?" Keep asking down the layers. The one that answers "it would not have happened" is the event's main resident layer. Most major events answer "it would not have happened" at more than one layer, which is the next section's subject.

An event may be tagged to more than one layer, but the tags need rank. The main layer is the one a hypothetical intervention eliminates; the secondary layer is the one that deepened the consequences. Write the ranks in the table header. The standard for ranking is intervention sensitivity: run the intervention on each layer, order by effect size, take the strongest as main. The output is intervention priority, because budget and attention go to the main layer first. Pick it wrong and the secondary repairs, however well executed, only postpone one failure by one season.

The walkthrough is finished when one judgement sentence exists: which layer is this case's main resident layer, and in which cell is the anchor. Write it on the last row. That sentence feeds two

downstream users, the improvement list ordered by resident layer, and the case card, which fills field 11 from the anchor cell. Its presence or absence is the only completion test. A walkthrough without it is a table with boxes ticked.

Keep a scoring habit alongside. Build the hypothetical-intervention version of the table: in each cell, a line predicting what the intervention would change. Write predictions before the fix and results after it. Prediction against outcome is the raw material of an intervention knowledge base, and it pays off in Chapter 39, where the rehearsal scenario library needs calibration data from historical predictions. No prediction made in advance, nothing to calibrate. Predictions must carry a date stamp: only one written before the fact has standing to be compared against it afterwards. One written in afterwards is narrative, not prediction, and Chapter 6's discipline applies.

9.2 One Broken Layer Is a Fault

An event becomes a failure when all three layers break. When one layer breaks and the others catch it, the event is a fault. This is the operational core of the chapter.

Take the technical layer broken alone, with the organizational layer in place. In place has a standard: the defensive action actually ran. The checklist was signed, the backup actually switched, the training actually happened. Presence on paper does not count, and Chapter 22's real-rehearsal standard governs. The judgement that a technical layer broke alone presupposes a gate verified by rehearsal, because unverified presence is statistically indistinguishable from absence, which is why near misses must separate what the design caught from what luck stopped. If redundancy, checklists, backups and training were genuinely there, the event becomes a near miss: the part failed and the system caught it.

Caught events do not enter accident statistics. They enter the maintenance log.

Near misses deserve rescuing for a second reason: they are the free teaching material of the model. A fault that got caught proves some gate worked. Written up as a case, it gives the model's abstract criteria a concrete face. Production cost is trivial, one maintenance log plus one interview, and the return is high. A newcomer learns inter-layer cooperation more completely from a successful interception than from a failure, since a failure case shows only the layers that were penetrated while a success case shows all of them acting. The production system is Chapter 20's capture ledger, the same mechanism under another name.

Now break the organizational layer alone: one handover omission, one checklist item skipped. Technical redundancy is still in place and the consequence is usually absorbed as well.

Absorption has another name, and it is luck. An absorbed organizational break is usually stopped by the incidental conditions of the day: someone happened to look twice, the day happened not to be busy. Luck absorption and design absorption must be counted separately, and Chapter 20's ledger keeps them apart. What the design caught is a credit to the defence; what luck stopped is an IOU written by the defence, and the list of IOUs feeds Chapter 25's margin ledger. The distinction matters for risk measurement, because an event stopped by luck will not keep being stopped under repeated exposure. An organization that counts luck as defence has a margin ledger reading zero.

Break the incentive layer alone, with nothing yet broken technically or in process, and the consequence arrives as chronic attrition: maintenance deferred, skills draining. Years pass before it meets the technical layer.

That rendezvous has warning value, because attrition is monitorable. Three quantities will do: the balance of deferred maintenance, training hours as a share of total hours, and the duration of vacancies in key posts. Where those curves cross the technical accident rate, you have the precursor. The two or three years before the crossing is the last window for intervention. Put the monitoring on the governance dashboard beside Chapter 25's defence concentration, and when both indicators appear in the same year, treat it as a yellow alert.

In aviation's language this structure is defence in depth, and the academic statement is the Swiss cheese model: each layer of defence is a slice with holes, holes are normal, and an accident is the rare penetration where the holes line up. The formulation is James Reason's, in Human Error (Cambridge University Press, 1990; the Crossref record has been checked). The arithmetic deserves a sentence. Any single layer fails with low probability, and several failing at once is a product, which is why failure is rare. But the layers do not fail independently. Organizational and incentive failures move to align with the technical holes: cut maintenance and the technical slice gains holes, raise assessment pressure and the organizational layer stops watching for anomalies. That active alignment is what Part III's twenty-four mechanism chapters describe.

The holes can be managed instead of awaited. Waiting for alignment is fatalism; active management means a census per layer, a written list of that layer's known blind spots, and cross-comparison of the lists as a forecast of future failures. A frame for the forecast: rows are combinations of three-layer blind spots (technical A × organizational B × incentive C), columns are similarity to real cases (has a close combination happened before), and similar cases link into Appendix B. The frame turns forecasting from intuition into a lookup table. Hit rates will be imperfect, and responsibility still shifts from genius to method; method can be

taught, copied and audited. The tools are already distributed: Chapter 22 for real rehearsal of redundancy, Chapter 20 for the capture ledger, Chapter 19 for dissent records; the annual joint inspection of the three is the census. Its output is a hole-alignment risk table, one row per potential penetration with a management action attached. Update it twice a year, because holes are dynamic: new equipment, new processes and new faces all cut new holes.

The direction of alignment fits one line: holes in the lower layer are made by disease in the upper. Maintenance debt drills holes, training cuts drill holes, configuration decisions drill holes. Holes in the upper layer are mostly absorbed below: the process gate catches technical badness, the redundancy gate catches process leakage. The governing implication is to repair from the top down. When a technical hole census turns up a problem, the repair usually lands in the organizational or incentive layer, and charging the hole itself treats the symptom. The nine-cell time-span column shows the year each hole was drilled, an incentive decision ten years back against a technical hole last month, which makes the drilling relation visible.

The three-layer model also gives the mechanism chapters their coordinate system. Part II's tools work inside a layer and across a boundary; Part III's mechanisms explain how the layers pull each other off true.

9.3 Three Kinds of Break at the Boundaries

Boundaries between layers are where breaks concentrate, because a boundary needs a translation act and translation has no job title.

The prescription follows from the diagnosis. Create the translator post at the boundary: between the technical layer's anomaly report and the organizational layer's risk register, between the organizational layer's process change and the incentive layer's

assessment adjustment. The post need not be full time. An existing quality or safety role can hold it concurrently, provided the job description states the translation duty in writing, including which boundary information must pass through that desk. Once written down, translation stops being something a conscientious individual volunteers and becomes something the organization runs.

The workload of the post can be estimated. Count the boundaries: after a nine-cell walkthrough, count the cross-cell links, technical to organizational, organizational to incentive. The number of links measures translator workload, and the denser the cross-layer collaboration the more necessary the post. Compare year on year, and you learn whether the density is natural to the organization or manufactured by management. Density is not a disease. Density without translation is.

The mechanism is one sentence. Every cross-cell link is an information interchange point, and with no translator there information arrives in the sender's language. The receiver, unable to parse it, processes it by his own understanding, and the probability of distortion multiplies with each transfer. After three transfers, fidelity is under fifty percent. That is why dense cross-layer work needs translators more, not less.

Each break gets a shape.

Break one: a technical signal the organizational layer never translated. The front line sees an anomaly, but the recording and reporting system holds no category word that can receive it. The information passes through in the form "not an anomaly".

Category words also fail by being too coarse. The register exists, but the category is a generic "equipment anomaly" and everything goes into it. Coarse categories stop like signals from aggregating, and a signal that never aggregates never becomes a pattern. Design fine categories by failure mechanism, not by

equipment, since equipment is replaced and mechanisms persist. The list of mechanism categories is Part III's twenty-four-chapter vocabulary, and aligning the category table with it is the translation bridge between recording and analysis. Chapter 20's capture ledger already files by mechanism.

The 737-9 door plug (av-030) gives the shape concretely. That door had been opened for work during final assembly, and the form the fact took on the process record was "no reinstallation record". The absence of a record is a technical-layer fact. The release chain's organizational layer never translated "no record on a critical area" into the trigger "stop and verify by hand", so the information passed through the inspection in compliant form.

That compliant passage is open to any checklist-driven process. A checklist asks about clauses, and an anomaly no clause mentions walks through as compliant. The defence is scenario-based clauses, and Chapter 32's scenario-driven inspection is the specific remedy for break one. The division of labour: this chapter diagnoses that information passed through in compliant form, Chapter 32 prescribes by putting the scenario into the inspection system. Diagnosis in this part, prescription in that one, is the book's arrangement.

The Atlas Air 747 freighter at Narita (av-007, JTSB report, 2025) supplies a second instance. The abnormal state of the tires during taxi was a technical-layer signal. The crew's felt sense of "running over something" never met an organizational handling procedure, meaning a procedure stating which taxi-phase anomalies require a stop, and the signal went into the air with the aircraft. The JTSB report records it at page 2: the crew perceived a vibration "like running over something" but heard no abnormal noise, and decided to continue the taxi (checked).

Break two: an organizational requirement in conflict with an incentive assessment. The process says A, the assessment rewards

B, and behavior follows B for years. On release and dispatch desks the tension between on-time rate and safety margin is the standard sample: the process says do not release a doubtful aircraft, the as - sessment says a delayed flight has a cost. You need no site visit to identify this break. Read two documents, the process file and the assessment scheme, and put them side by side. Chapter 17 gives cases from four industries.

Read specific clauses, not summaries. Summary language is uni - form everywhere (“safety first”), and the conflict lives at clause level, in the clause stating release conditions and the line of the assessment that deducts for delay. Only clause-level reading reaches the numbers and the trigger conditions. The output is a conflict register: rows are conflicting clause pairs, process clause against assessment item, columns are the behavioral consequence, meaning what the employee does when condition X arises. The register goes to governance, because only the incentive layer has authority to resolve it. A frontline employee cannot rewrite an as - sessment.

Break three: the incentive layer shaping the technical layer’ s selection. Configuration decisions are made under incentive con - straint and repaid years later as technical failure. In the same Atlas 747 event, the poor visibility of the tires from inside the cabin re - moved the information channel that could have turned a taxi an - omaly into a reason to stop. Set that against me-002, where safety needles left the supply catalogue and changed the boundary of clinical practice. Supply and configuration decisions are made under incentive constraint and repaid years later as failure. The technical layer’ s “invisibility” is not a law of physics. It is the product of one configuration meeting years earlier, and the option list at that meeting had the price column in a prominent place. This break has the longest lookback and is the one postmortems drop most often. The accountability report stops at the organiza -

tional layer, but the ledger of the configuration decision sits in the incentive layer.

Monitoring can be made visual. Draw the three-layer timeline: years across, three swimlanes for the density of technical events, the density of organizational changes, and the incentive budget curve. A year where the three converge is a candidate for break resonance, and the list of such years goes to governance, because each deserves a nine-cell walkthrough in retrospect. All the data comes from ledgers you already keep, maintenance logs, change orders, budget tables. The chart costs half a day and carries more insight than ten special meetings.

Incentive lookback needs a method of its own. Configuration archaeology runs on year spans, since break three's lookback is often measured in decades: the configuration meeting, the material selection, the supplier contract from fifteen years ago. How deep you dig depends on how much archive survives. Are the minutes and the option lists still there? Where they are not, the substitute is comparison with contemporaneous projects of the same kind: what did others select? The comparison restores the sense of what the decision's boundaries were at the time. The output goes into the incentive row, with the evidence column stating the archive's condition and the method of comparison. Mark the limits of the archaeology honestly.

9.4 The Walkthrough: Fitting a Case into Three Layers

Here is the reusable checklist. For any case, answer three questions per layer, nine cells in all:

	Where this layer broke in the event	Dated evidence for the break	Corresponding intervention
Technical layer	The chain of physical and system behavior	Physical evidence, logs, data (the fact sections of the investigation report)	Replace parts, add sensors, change the design
Organizational layer	Information flow and the structure of responsibility	Meetings, work orders, handover records, training records	Change the process, change the checkpoint, change what the handover contains
Incentive layer	Contract and assessment structure	Assessment schemes, contract clauses, pricing and configuration decisions	Change the assessment, change the contract, change the payment structure

The easiest column is the first, because everyone can tell a story. The one most often missing is the second, the dated evidence.

Column two has a width rule. Dated evidence need not be perfect: a frontline original record, an email from the time, a photograph of the duty log will do. Where the date is fuzzy, state the fuzziness, "roughly that week". Honest width beats precise fabrication, because a fabricated date collapses under cross-verification, and one collapse takes the credibility of the whole walkthrough to zero. An empty cell may not be filled with a story. It may hold "no record", and under Chapter 6's discipline "no record" is a diagnostic result.

Complex cases should use the extended version, with a column for time span: the year and month each layer's break formed. Technical breaks usually form days before the event, organizational breaks months before, incentive breaks years before. The column makes the layers' temporal depth visible, and once visible the timescale of the improvement follows: an incentive repair is

measured in years, so do not expect quarterly results. Fill it from the dates of each layer's evidence, the manufacture date of the physical item, the date on the minutes, the signature date of the contract. Spans are inferred, so write the basis.

A worked example: Alaska Airlines 261 (av-034, NTSB AAR-02/01, executive summary checked). Technical layer: the Acme nut threads in the horizontal stabilizer jackscrew assembly were excessively worn, the assembly failed in flight, the aircraft lost pitch control and went into the sea, and all 88 people on board died, in the report's own wording. Organizational layer: the company extended the lubrication interval and the end-play check interval for that assembly, the FAA approved the extension, and the investigation found that the extension let inadequate lubrication develop into failure-level wear. Incentive layer: the report's conclusions place the maintenance program and regulatory oversight on the list of issues. The specific economic trade-off behind the extension, downtime cost and scheduling pressure, is discussed but not quantified to a citable degree, so this cell is marked "insufficient evidence; reconstruction requires the maintenance engineering record". No story is inserted.

The discipline of not inserting stories meets a psychological obstacle: the social pressure to fill the blank. In a review meeting an empty cell invites the question of why you did not find out, and the correct answer is to display the search process: which archives you checked, whom you asked, what came back. Displaying the process turns the blank from an embarrassment into proof of work. Auditing calls this the trace, and the trace of the search matters as much as the trace of the conclusion.

The last intervenable position in this case falls on the organizational layer, at the node where the extension was approved. That is where field 11 of Chapter 5, the minimum counterfactual, takes its value. The two routes meet at that node, which tells you something

about the book's architecture: the twelve-field card and the three-layer walkthrough are two tables for one object.

One detail of the join matters. A counterfactual action must be tagged with its layer, which layer it falls on and which currency it moves. A tagged counterfactual enters the improvement list for that layer. An untagged one is an orphan: after the review meeting it has nowhere to live and the list will not take it in. One line of tagging saves one lesson.

Where the incentive row is thin, hand it to Chapter 17's principal-agent framework. The row usually gets no further than assessments and contracts, and the framework asks one level down: who is the principal, whose objective is the agent optimizing, where is the monitoring cost spent. The deeper look often rewrites the row, because what read as an assessment problem turns out to be an absent principal. Not required every time: clear contract evidence closes the case at nine cells, and the framework is for rows where evidence is vague.

Merge the two tables and you have the script for Chapter 39's rehearsal. The nine cells fix the scenario, the twelve-field card lays out the material. The join is a mapping: field 8, the decision point, corresponds to the organizational row; field 9 to the time-span column; field 6 to the evidence column. The mapping table sits in the user notes to Appendix A. Because the mapping exists, a reader who learns either table gets the other for free. The two are one instrument, one cut from the training angle, one from the structural angle.

9.5 What "All Three Broken" Means for Three Kinds of Reader

For the person writing the postmortem. A postmortem that stops at the technical layer is a fault report.

When the walkthrough is done, read the nine-cell table beside the body of the postmortem and work out which cells the narrative paragraphs correspond to. The cells the text never touches are the blind spots, and the list goes back to the author as material for the next version, without reproach. The procedure also runs in reverse: walk first, then write, with the table as the outline. Such an outline covers the three layers automatically and carries the evidence column with it.

Whether your conclusion crossed the technical layer takes one question: is the subject of the conclusion sentence a part or a structure? "Nut wear caused the crash" has a part as its subject, so it is a fault report. "The approval of the extended inspection interval let wear develop into failure" has a structure as its subject, and only now are you approaching failure. The upgraded form is the subject-chain test. Write the causal chain in full: who approved what in which year, letting whose work under what conditions develop into what result. Written in full, every link is an intervenable position and the list of links is a rich seam of measures. Chain length is postmortem depth: three links or fewer is shallow, five or more begins to touch the incentive layer.

For the person designing defences. Leave one gate on each layer.

Acceptance of the three gates should be done together. The technical gate is accepted by real rehearsal, the organizational gate by a drill, the incentive gate by a case replay. The institutional carrier is Chapter 22's annual real rehearsal plus this chapter's walkthrough. Cost is amortized by the combination and completeness rises, because single-gate acceptance cannot see problems of mutual exclusion and three gates in one room expose them.

The point of three-layer defence is independence. The organizational gate, meaning a mandatory re-check, must not take its signal from the technical gate, meaning a sensor. The incentive gate, meaning no penalty for calling a stop, must guarantee that the or-

ganizational gate dares to be used. Chapter 22 carries a case where all three gates were shared.

One failure mode of the combination deserves flagging: gates that exclude each other. When the incentive gate, no penalty for stop - ping, fights the organizational assessment of schedule bonuses and penalties, the organizational gate prefers not to be used, because the person who uses it loses the schedule bonus. Detection is the two-document reading of break two in section 9.3. Once detected, the repair is in the incentive layer: put gate use into positive scor - ing, count uses as a plus and not a minus. That reversal is the de - cisive switch for whether three gates work together at all.

For the person deciding on improvement measures. Which layer you can move decides which chapters of Part III you read.

Lay the measure list out in three columns, incentive, organiza - tional, technical. The columns force every measure to show its currency, and a measure that fits none, "pay more attention" or "strengthen management", is deleted on the spot. The layout costs a table with two extra columns and its filter value shows on first use: in most organizations the first version of the list has an empty incentive column, and the empty column is itself a finding.

Frontline engineers can move the technical and organizational layers (Chapters 15, 21, 30). Middle managers can move the organ - izational and incentive layers (Chapters 16, 17, 18). Only those who can change contracts and assessments can move the incentive layer (Chapters 12, 17, 28). An improvement plan cannot exceed your intervention currency. Work out which currency you hold before writing the plan; this is the first parameter of Chapter 35's positioning table.

The second parameter is layer span, meaning how many layers you can coordinate across. The frontline acts within its own layer, the middle manager crosses two, the executive moves all three. An

honest assessment of span sets the size of the plan. A middle manager who cannot cross into the incentive layer should not write incentive actions into the plan, or it will stay on paper. The correct way to cross is to find a bridge, an ally at executive level, and Chapter 35 describes how to choose one.

Recognizing your currency also applies at organization level: tiered sign-off on the improvement list. The list sits in three columns by currency, and each has a different signer. An engineer signs the technical column, a manager the organizational, and whoever holds assessment authority signs the incentive column. Every measure then carries a signature from someone able to execute it. A measure with no corresponding signer is deleted or transferred to someone with authority. Ownerless measures are the dead stock of an improvement list.

Self-Check: Five Questions

1. Run the nine-cell walkthrough of section 9.4 on your organization's most recent significant loss. Which cells cannot be filled with dated evidence?
2. In your organization's latest improvement list, how many measures fall on the incentive layer? If none do, how many years will the technical repairs hold?
3. Find one concrete place in your processes where a technical signal has no organizational category word: what anomaly can occur for which the system holds no registration category?
4. What is your post's intervention currency? Can you change a process, change an assessment, or only change a parameter? The answer decides which chapters of Part III you need.
5. In your organization's postmortem conclusions, is the subject usually a part or a structure? Who trained that habit?

Beyond the five questions, here is the end-of-part test for the anatomy section. Take a real investigation report from your own industry at grade A and complete the full cycle from zero: draw the failure chain, mark decision points and points of no return, fill the nine-cell table, judge reversibility, walk the three signal words, write the minimum counterfactual, assign the evidence grade. The first run takes about a day, half a day with practice, and the output is one complete case card, your first grade-A card, at once the graduation certificate of Part II and the entry ticket to Part III.

One reminder above the model. The three layers are an analytical framework, not an organization chart. Departments such as engineering, operations and HR do not map cleanly onto the layers. HR is not the incentive layer. The incentive currencies, assessments, contracts and pricing, are scattered across several departments. Fill the table by currency ownership, not by department ownership, and ask what change would eliminate this event, never whose department it is.

Chapter 10 The Minimum Counterfactual and Evidence Grades

> What this chapter covers: the five-step backward method for the minimum counterfactual, with the three kinds that fail; and the A to E evidence grades, with the rules for moving a case up or down and for labelling it "reconstructed". When to turn back: when writing fields 11 and 12 of a case card, reviewing someone else's postmortem conclusion, or deciding whether a case may enter the main text. Estimated reading time: 35 minutes.

Chapter 5 defined the twelve fields. Chapter 6 set the discipline for one of them, what was visible at the time. Chapter 7 gave the test for the decision point. Chapter 9 supplied the framework for combining the parts. This chapter finishes the two most valuable fields on the card: field 11, the minimum counterfactual, and field 12, the evidence grade.

The recommended way to learn the craft is by doing, one chapter at a time. Build a single card and let it grow. Fill the fields after Chapter 5; add "what was visible then" after Chapter 6; add the point of no return after Chapter 7; run the nine cells after Chapter 9; do the backcast and the grading after this chapter. Five passes, and the same card is complete in every cell. The card that grew this way is your graduation piece.

Beyond the card, the rest of the book has a route. A reader who finishes Part II should treat every case in Part III as practice material: take one case per chapter, run the five-step counterfactual on it, set its grade from the table, walk it through the nine cells. That exercise runs across the twenty-four chapters of Part III, and by the end the domain pages of Part IV read as revision, with the tools unchanged and only the parameters swapped for an industry. The learning curve reaches its first plateau at the end of Part II: the

reader standing there holds every tool, and every later part stacks parameters and situations on top of them.

One pace suggestion for the climb: a full run every four chapters (11 to 14, 15 to 18, and so on), six groups in all, one case per group taken through the whole set of card, counterfactual, nine cells and grade. After six groups the tools sit in the hands rather than in the notes. The test is speed. The first full run takes half a day; the sixth takes an hour.

For teachers and trainers, the five-step method and the grade table work as course material without alteration, because a training programme is a sequence of exercises. Session one: the five-step method and one demonstration. Session two: learners critique each other's counterfactuals. Session three: grade judgement, with cross-verification. Session four: a full run, one card, start to finish. The package costs almost nothing, since the material is this book's chapters plus public cases, and it produces the organization's first cohort of people who can write a card. Cards are the bottleneck on the industry's circulation of failure knowledge (Chapter 4), and a four-session course is one of the ways through that bottleneck.

The class should be mixed: one third line engineers, one third safety staff, one third middle managers. A single-composition class trains a homogeneous skill. A mixed class produces cross-layer conversation, with engineers and managers finding holes in each other's counterfactuals, and that argument is where cross-layer understanding forms. Chapter 9's theory of intervention currencies gets its first field trial in that room: each side learns what the other side's currency looks like in the hand.

The extension of the course is organizational certification. Four sessions plus a graduation card earns an internal certificate, and the roster and the card work go into the knowledge base. A certificate is a capacity marker, not a title: projects that need cards draw

names from the roster, and the skill becomes visible for the first time, which is the precondition for deploying and rewarding it. Chapter 39's budget template carries the two lines that close the loop, certification training and card-writing hours.

10.1 The discipline of the minimum counterfactual

Definition, expanding field 11 of Chapter 5: the one action whose change would have avoided the event or greatly reduced its consequences.

One word in that definition needs slow reading: greatly. A counterfactual does not have to erase every consequence. Turning 110 deaths into 110 injuries is a qualified counterfactual. What it must satisfy is proportionality: the scale of the intervention has to match the scale of the consequence. Set the counterfactual for a 97-death accident at a reminder issued in a pre-shift meeting, and the reminder cannot stop a consequence of that size. The test for proportionality is magnitude coverage: does the intervention carry enough leverage to move this consequence? A counterfactual without that leverage is a toy. The way to identify a toy is to run it for real and ask whether the consequence would actually have shrunk to the magnitude claimed.

The least-cost rehearsal is a variant of that exercise. Assume the counterfactual action is executed once, and answer three questions: what does it cost, who objects, and does the process allow it. The three answers score the counterfactual's realism. A low-realism counterfactual can still be qualified, but it is only theoretically qualified. Theoretical counterfactuals mark the position; realistic ones can be executed. Keep both in the working draft, and put only the one that passes realism into the signature line.

Reduction is measured by what can be verified. State the magnitude: half the consequence, or nine tenths of it. Declaring the

magnitude gives the assessment a scale. An unverifiable claim that the consequences would have been much better does not count.

Three disciplines follow, and all three have to pass. Write one counterfactual only. Test it for dependence on luck. Test it for overreach.

Why only one, argued from the author's side rather than the reader's (Chapter 5 made the reader's argument): the impulse to write several comes from a fear of omission, the worry that a better candidate is being left out. The working draft answers that fear. Put every candidate in the draft, attach the draft behind the card, publish nothing from it. The card carries one conclusion, and every candidate leaves a trace. Separating draft from conclusion is routine in academic review, and postmortem writing can borrow it as is. The serious form of the discipline: two counterfactuals sitting side by side on the table mean the backcast is not finished. Compare their intervention positions and their evidence strength, choose one, and move the loser into the draft. A reference book owes its reader exactly one conclusion.

There is a game-theoretic reason as well. When several counterfactuals stand side by side, the reader takes the cheapest as the lesson, and the cheapest is often not the most effective. The choice of lesson slides from the author to the reader's laziness, and the result is that one case yields different lessons in different hands. Randomised lessons amount to no lesson at all. A single-counterfactual discipline returns the choice to the author, whose comparison sits in the draft while one conclusion is delivered. The two layers, delivery and draft, get rigour and readability at once.

10.2 The five-step backward method

Step one: reverse the failure chain. Start from the consequence and list the nodes backwards: the consequence, the point of no

return (Chapter 7), the last decision point, the earlier structural conditions. Go down to the organizational and incentive layers; the nine-cell table of Chapter 9 can be used directly for the sorting. Every node carries a date and an actor. "Management was lax" is not a node. "The 14 March review meeting put this item at position 7 on the agenda and spent 4 minutes on it" is.

Timeline software or paper will do, and the tool is not what matters. What matters is that each node is written on a card. One node, one card. A backcast is trial and error, and a wrong order gets redone; carding drives the cost of re-sorting to near zero. The physical act (writing the cards, laying out the order, drawing the lines) looks crude and beats most thinking software. The book draws one practical conclusion from this: do the postmortem and the backcast on paper.

A team version is worth adding. Tape the node cards to a wall and let the whole group sort them. The sorting disagreements mark where the group's understanding diverges, so mark the disputed cards in a different colour. The colour wall is the advanced form of the review meeting: the meeting stops being a report-reading session and becomes a sorting session, and its output is a failure chain calibrated across the group. A shared organizational picture of a failure survives handover better than any individual's picture (Chapter 30).

That picture also needs a documentary exit. Photograph the sorted wall and archive the photograph alongside a digital version of the final order, both into the knowledge base and the knowledge pack (Chapter 30). The archived wall is the permanent draft of that failure chain. A newcomer rebuilding the chain three years later compares against it, and the deviation becomes measurable. The trail of deviations is the record of how the organization's understanding evolved.

The starting level deserves its own warning. Choose which consequence to backcast from: the death, the injury, or the near miss. A different start gives a different chain. A major consequence gives a longer chain; a near-miss chain stops at the defence gate. Choose by the purpose of the exercise: to study defences, start from a near miss; to study penetration, start from an accident. Purpose comes before backcast, and this is the first appearance of Chapter 35's positioning table inside the method.

One more node criterion: a node has to be an action, not a state. A meeting being held is an action; an item's position on the agenda is an action. Lax management is a state, and so is a poor culture. Actions can be swapped, and states cannot be swapped directly, since nobody can swap in a different culture. States change through the actions that generate them, so the work is to find the decision that produced the state. States are regular guests on the incentive layer of the nine-cell table, and writing down which action generated a given state completes the reduction from state back to action.

Step two: test each node. For every node ask whether a different action there would have changed the consequence. Mark the ones that would as effective, the rest as ineffective. Use only what was visible at the time (the discipline of Chapter 6). For an action to count as a qualified intervention point, its necessity has to hold inside the actor's own information environment.

The information environment itself is a boundary case worth naming. Sometimes the actor could not see something because the system did not display it. In that situation the qualified intervention point moves up a layer, to the information system: making the information visible is the counterfactual, and making the actor cleverer is not. The logic matches the nine-cell table, where redesigning an information environment is an organizational-layer intervention. Both levels can hold a qualified counterfactual at

once, each labelled with its layer, following Chapter 9's labelling discipline.

Step three: strike the luck types. Any candidate that depends on coincidence goes out. "If the inspector had looked once more that day." "If the person on site had happened to be a senior engineer." The identifying feature of a luck-type counterfactual is that the action it demands has no position in the system. Whether it happens depends on staffing and on coincidence.

Step four: strike the overreach types. A candidate that requires the actor to break procedure, defy a superior, or exercise authority he did not hold gets removed, or moved up a layer. "The shop-floor operator should have refused to release the batch" is an overreach counterfactual unless the procedure actually grants him an independent veto. Where the veto does not exist, the qualified intervention point sits in the authorisation design, and the node moves up to the meeting that wrote the procedure. Overreach is the most common defective item in postmortems, and its function is to dress a structural problem as a courage problem.

The overreach check needs a companion step. A node that has been moved up must go through the luck test again. If the repair at the authorisation layer depends on an enlightened leader, on some executive happening to care, the candidate is a luck type once more. Luck does not discriminate by layer, and it can enter a candidate at any layer. The mnemonic is one action, one question: does this action have a position in the system? If it does, carry on; if it does not, move up or strike it.

Step five: take the latest one and sign it. Among the qualified candidates that remain, take the one latest in time and write its three elements: date, location (whose ground it sits on), action. The signature format reads: "Minimum counterfactual: on (date), on the (carrier), (actor) performs (action)." If the three elements

cannot be filled in, the backcast is not finished, and the answer is to go back to step one.

The signature has a quantitative use. Collect a year's qualified counterfactuals and classify each by layer, position and action type. The reading is a distribution of the organization's improvement opportunities: which layer produces the most counterfactuals. That distribution tells next year's improvement money where to go, since the layer with the most counterfactuals is the layer the budget belongs to. This path from tally to budget is the interface between the chapter's method and the budget template of Chapter 39.

It has an organizational use too: conversion into a work order. A qualified counterfactual generates a work-order draft, with the fields mapping one to one, date to deadline, location to owner, action to task. Conversion is what turns a lesson into a standing to-do item. A counterfactual that is not converted dies when the review meeting ends. The trigger for conversion is a pass at review, and the review asks the five-step questions, so passing the five steps makes the work order legitimate on its own.

10.3 Three disqualifying counterfactuals

Type	Example	Defect
All-purpose	"strengthen management, improve the system, raise awareness"	No action, no actor, no date; cannot be executed and cannot be refuted
Luck	"if the inspector had happened to look one more time"	The action demanded has no position in the system
Overreach	"the operator should have held out against the pressure and refused release"	Requires authority the person did not hold; the qualified intervention point sits in the authorisation design

All-purpose is the most common. The test is to compress the sentence into one verb and one object. If it will not compress, it fails.

A hybrid form needs recording: the disguised specific. "Management should have recognised the fragility of the supply chain earlier" looks as though it carries an action and an actor. Take it apart. Management is a collective noun, so there is no named actor; "recognised earlier" is a state, so there is no action. A disguised specific is more dangerous than an obvious void because it passes the two-second intuition check. Only the five steps, run item by item, bring it back into view. The mnemonic for the disguised version: name a person, land a verb, give a date; missing any one of the three sends it back.

10.4 Evidence grades A to E

Every case in the book carries one of five grades, and the grade decides what that case is entitled to support.

Grade A: the full report of an independent investigating body. Independent of the party investigated and of the prosecuting function, published in full, with a factual section and an analytical section. Supports the decision sequence, the options ruled out, and the minimum counterfactual. This is the grade of the cases in the main text.

Grade B: official notices, communiqués and regulatory announcements. Authoritative, without a complete evidence chain. Supports the magnitude of the consequence and conclusive findings. Does not support the detail of a decision point. B-grade cases go into the appendix matrix; a B case that enters the main text must be labelled "reconstructed". A process report, such as the JTSB's 経過報告, is graded A− to match Appendix C.

Grade C: judgments, administrative penalty decisions, arbitration awards. The facts have gone through adversarial testing, but the selection of facts serves the legal issue. Supports amounts, responsible parties, and the timeline of acts. Does not support technical processes that never entered the dispute.

One directional hint for using C-grade material: prefer the findings that went against your own side. A fact confirmed by the court against the party that contested it has been tested hardest, so an adverse finding is the harder finding. When a postmortem cites C-grade documents, leading with the facts the party admitted is a technique for raising credibility.

Grade D: the party's own public postmortem (software industry postmortems, company statements, participants' memoirs). First-hand detail, inherently self-interested. Supports the shape of an organization's internal process. Supports a finding of fact only after cross-verification against A, B and C material.

Cross-verification of D-grade material has an efficiency trick: an order of work. Verify first the details that appear nowhere else, since only the unique ones need testing. Where the material overlaps a public report, the report has already vouched for that part, and re-verifying it wastes effort. Verify the unique, skip the overlap. This order cuts the cross-verification workload by 70%.

Grade E: media reporting. Lead generation only. Use it to find a case, then go and find A to D material on that case. The main text may never rest on E-grade material alone.

A search tip belongs here as well: media reporting has a time structure. The incident report is fast and wrong often. The investigative piece is slow and corrects more. The anniversary retrospective is slowest and adds the long view. Reading all three layers in sequence is the complete move in E-grade lead mining; stop at the first layer and the direction you take is often the wrong one.

The reading list for those layers sits in’ s search manual, which names the deep outlets for each domain and the search strings for anniversary pieces.

Two rules govern movement between grades. Combination up - grade: a core fact corroborated by B and C together can support a main-text case labelled “reconstructed”, while a single B-grade document cannot. Downgrade: a dead link with no archive drops the case to pending verification, and a quoted passage that does not match the original is removed outright. the link check and the number-to-source check carry these out.

Corroboration has one further detail: what gets corroborated is the fact, not the conclusion. Two B-grade documents recording the same fact identically establish corroboration. Two documents agreeing on a conclusion while describing the facts differently do not, because agreement on a conclusion can be a loop of retelling. This is Chapter 2’ s circulation skew in its evidentiary form. A cites B, B cites C, and three sources turn out to be one. The test that breaks the loop is to find the earliest source: is the first of the three an independent interview, or a retelling of the others? The mne - monic: trace it back, find the earliest, and ask whether that source saw the thing itself.

Two further states exist beyond up and down.

A frozen case: evidence grade undetermined, investigation run - ning, litigation unresolved. Mark it frozen. A frozen card enters neither the main text nor the matrix; it waits in the knowledge base, with the reason written down and the thaw condition stated, which is publication of the final report. On thaw it is graded afresh. Freezing prevents two errors, concluding from an out-of-date pro - cess report and forgetting to come back once the final report ap - pears.

A partial downgrade: within one card the facts carry different grades, the core fact at A, circumstantial material at C. The card does not drop as a whole. What drops is the support range of one fact, and the labelling switches to per-fact grades, so that a user of the card knows precisely which line is hard and which is soft. A coarse whole-card grade misleads on a complex case, and per-fact labelling is the mark of a mature one.

Disagreement about a grade can be arbitrated. Grading has a subjective element, and two people can grade the same document differently. The rule is to take the lower of the two, since the margin of safety in that rule belongs to the reader, who uses the material at the lower grade and loses nothing. Record the dispute in the case's errata line. The dispute is itself material.

10.5 Labelling "reconstructed"

Cases on medium evidence are still worth writing, on one condition: the inference is marked. Most organizations' internal material is B, C or D by nature, a front-line review being D, an internal investigation B, legal paperwork C. A grade system gives those materials a grammar of mutual recognition for the first time, so that an internal review can be used correctly, neither overstated nor thrown away. Teaching that grammar takes a day, and afterwards the organization's stock of old reviews turns from a pile of paper into a graded knowledge base.

The labelling convention: narrative text carries verifiable factual sentences only, and an inference opens with "Reconstructed" and states its basis.

Inference has a tone discipline of its own. Keep the modals restrained, and never write "obviously", "undoubtedly" or "of course" inside an inference. An inference is an inference, and an author who writes it as an assertion is using tone to make up for

evidence. Tone runs in inverse proportion to evidence, which is a hard rule of postmortem writing: the stronger a sentence sounds, the more the reader should ask for its evidence number.

The format, by example:

"Minutes of the 14 March review meeting show the risk listed as agenda item 7 (factual sentence, grade C: meeting minutes). Reconstructed: from the agenda order and the meeting's length, the risk received less than 5 minutes of discussion and no resolution item was recorded (inference)."

The discipline for a reconstructed sentence: a reader must be able to re-check the inference along the basis given. An inference with no re-check path is not written.

Density needs self-control as well, since a case reconstructed on every page reads as a page of footnotes. When reconstruction exceeds 40% of the body text, the evidence is not yet sufficient to write the piece, and the material goes back to the matrix. The quantitative standard is a ratio of factual sentences to inferences of at least three to one. Below that ratio, the title must state that this is a reconstruction, not a report of fact.

Grades are not for life. New evidence moves a case up or down: publication of the investigation report lifts a B case to A, and a quoted original that gets overturned drops the case or removes it. The institutional carrier for that maintenance is the case library's annual review, which reads each card's evidence status in turn, asking whether a new report exists and whether the links still resolve. The output is a list of grade changes, published to readers, so that what a reader holds is always the current grade. Appendix B's case table keeps a grade column for every case. A reader's trust should follow the grade.

10.6 Numbers back to source, and citation convention

Every number in the main text carries the four-part record set out in Chapter 5: who counted it, when, on what sample, under what definition. Two execution rules are added here.

Rule one: the set check. Before a chapter is delivered, extract every number in the body and take the difference against the set of numbers in the source cards that chapter cites. An empty difference passes.

The difference can be audited in the other direction as well. Numbers in the cards that the text does not use are not a problem; numbers used in the text that the cards do not contain are. A reverse difference shows that the source cards are richer than the chapter, which is a good sign, since it means the author made selections rather than copies. Write the reason for each omission into the card's note, so the selection stays traceable. The practical meaning of the rule: no approximate numbers written on the fly. Every number in the body comes from a source card, or it is deleted.

The cost and the return of that discipline are worth stating. The cost is slower writing, since every number goes back to its card for a check. The return is that nobody has to check you. A reader can trust each number because the system guarantees them, and the trust compounds at the point of citation, where another author cites your number without re-verifying it, because you already did. Being citable is the long-run return on numeric discipline.

Rule two: a quotation keeps the language of the original, with a translation in brackets. Institution names, document titles and regulation names follow the glossary's bilingual table, and no chapter drifts from it.

The bibliographic form is fixed: (institution, document number or title, year, link at the Appendix B entry). Page numbers are completed at the checking stage. Where a page cannot be obtained, widen the cited range, writing "the report's executive summary" rather than "page 41 of the report". That is how a style handles uncertainty honestly.

The honesty principle extends to telling the reader the grade. A conclusion resting on A-grade evidence may be written as a flat statement. One resting on B-grade evidence is written as a probable statement. A reconstructed sentence always carries its marker. Grade disclosure lets the reader read at the right level of confidence, and an uncertain sentence never appears in a certain tone. The disclosure goes into the postmortem template of Chapter 38, where the template governs the format and the discipline governs the substance.

One more application: what to quote. Quote only sentences you have checked word for word. A reported quotation is marked as reported ("according to a paraphrase"). The distinction between paraphrase and direct quotation is the foundation of citation ethics. On that foundation sits one more check: every directly quoted sentence must be findable on a page. A direct quotation that cannot be found on a page is downgraded to a paraphrase. The check costs a minute a sentence and buys the book's zero-tolerance commitment, which is to quote one sentence fewer rather than one sentence that cannot be verified.

The five self-checks

1. Write the minimum counterfactual for your organization's most recent loss, then run the five steps to see whether it is a luck type or an overreach type.

2. Do your project's decision records support grade A, or only grade D? Where is the gap?
3. Of the industry cases you have cited, how many survive the grade test in 10.4? How many conclusions still stand after the downgrade?
4. When did an all-purpose counterfactual last appear in one of your postmortems, and how did it pass review at the time?
5. Set a citation rule for your team: material below which grade may be cited as fact in your meeting room?

The graduation exercise sits outside the five. Write a complete case card for your organization's most recent loss: twelve fields, five-step backcast, grade labels, reconstructed sentences, the whole sequence. The work takes about half a day, and the acceptance standard is a colleague who does not know the event being able to re-check it end to end from your card and its links alone. Pass that, and you hold every component of the book's method. Writing a card is the last tool this book hands over, and every later part uses it again, the domain pages organising cases by card, the usage chapters expanding templates from cards, the boundary chapters checking against cards.

Your first card will not be good, and it is not meant to be. Its job is to expose the distance between where you are and the standard. A list of that distance is more useful than a perfect card, because the list is your study plan for the next stage. Take the finished card and check it against the field table of Chapter 5, cell by cell: which cell is empty, which cell has a story impersonating a fact, which cell's counterfactual is all-purpose. Write the result as a one-page memo. Run the memo, and your second card passes about half of it. All of Part II exists to get you to a passing second card, a fluent fifth, and a tenth you can teach from.

Before the tools are put away, run the evidence disciplines of the five chapters into one route. A reader facing a new piece of failure material passes five gates. Gate one: what grade is it (this chapter). Gate two: has it cleared the recording threshold (Chapter 2). Gate three: are the visible-at-the-time and the reconstructed-afterward kept apart (Chapter 6). Gate four: how many layers did its failure chain break (Chapter 9). Gate five: is its minimum counterfactual qualified (this chapter's five steps). Once the five are passed, the material's use is settled: cite it, refer to it, keep it as a lead, or discard it. The route fits on one sheet of paper, and that sheet is Part II's final compression.

Not every reading needs all five. Skimming the news needs gates one and two. Citing needs all five. The gates are used at levels, the way Chapter 23's releases are tiered: the use decides the depth. Material mentioned in passing needs none of them; material written into the basis of a decision passes all five.

Closing Part II

This chapter is the last of Part II, and the anatomy section hands in its paper here. The chapters it collects (the fields, what was visible, the decision point and the point of no return, the escalation ladder, the failure chain, the counterfactual and the grades, six in all) form one evidence workflow. Find material, at the threshold of Chapter 2. Read it, under the discipline of Chapter 6. Take its structure apart, with the nine cells of Chapter 9. Locate the positions that matter, with Chapter 7. Give the conclusion, with the fields of Chapter 5 and the methods of this chapter. Every station on that route has a tool, a discipline and an exercise, and the route ends at one capacity: turning any failure into a checkable card. That capacity is the foundation under all of the book's analysis. Each of Part III's twenty-four chapters was built with it, and a reader who has practised it can test every stone of the foundation by hand.

The six lessons, named once more with their equipment: Chapter 5, 12 fields. Chapter 6, 3 disciplines. Chapter 7, 2 tools and 1 judgement table. Chapter 8, 7 steps and 4 devices. Chapter 9, 3 layers, 3 breakpoints and a 9-cell table. Chapter 10, a 5-step method and a 5-grade table. The arithmetic of the six comes to 12+3+3+11+15+10, which is 54 operable items. That number is Part II's entire output and the craft cost behind every case in Part III. Large, and every item usable: this is the line between a reference book and a book of opinions, and the reason the book can hand 158 cards to its readers for checking.

The whole of Part II in plain words: treat failure as something that matters, work it with method, leave a paper trail. Treating it as something that matters is the attitude, and Part II exists to show that an attitude can be converted into method. Working with method is the tools from Chapter 5 to this chapter, and method can be learned. Leaving a paper trail is the institution of Chapter 38, and a trail can be checked. None of the three is mystical, and each can start tomorrow afternoon.

The twenty-four chapters behind the door use every tool practised in these six lessons. The case cards change shape from chapter to chapter, some full, some condensed, but the skeleton is always the twelve fields, the evidence always carries a grade, the counterfac - tual always passes five steps. A reader who has finished this chapter meets no new tool in Part III, only the same tools in twenty-four uses. That variety is evidence the tools were designed well, and it is why the book puts them all up front. The cost of front-loading is two parts of abstraction; the return is four parts of fluency.

The other argument for front-loading is time. Learning the tools costs once: six chapters, then usable for life. Learning the mech - anisms late costs continuously, with the method re-explained in every chapter. A one-time cost beats a continuing one in any long

project, and Chapter 28's logic of horizons here sides, unusually, with the investment. Money spent on tools is the highest-return money in this book.

The next part visits five industries, and each has an accent. Aviation's accent is procedure, where every sentence has a number. Chemical processing's accent is layers, physical chemistry and management wound into a double helix. Maritime's accent is the international, since every case sits in several jurisdictions at once. Drugs and devices speak statistics, converting a single case into a signal. Nuclear speaks conservatively, logging even one millisievert. Each accent gets a page in Part IV, and a reader who has heard all five can pick up a report from any industry and hear where it comes from. Hearing the origin is the prerequisite skill for evaluating a piece of material.

The closing inventory of this part, in its fullest form: Chapter 5, 12 fields. Chapter 6, 3 disciplines, 1 criterion and 3 signal words. Chapter 7, 3 conditions, 3 currencies and 1 judgement table. Chapter 8, 7 steps, 4 devices and 1 self-test. Chapter 9, 3 layers, 3 breakpoints and a 9-cell table. Chapter 10, 3 disciplines, a 5-step method, 5 grades and 1 labelling convention. More than fifty operable items between them, each usable before the end of today. Part I gave the reader a view of failure, Part II gave the scalpel, and Part III takes both on rounds through twenty-four organizational failures.

For the reader who wants to skip Part II and go straight to Part III: you can. Every case card in Part III stands on its own, and the story reads without the anatomy. What you will miss is the second half of each chapter, the failure-chain analysis and the mapping from each defence to an intervention currency, and that missing half is exactly what separates this book from a collection of stories. If you reach Part III, find you need the tools and come back, Chapters 5 to

10 are here. A reference book is best opened with a question in hand; reading it in order is only its most respectable form.

Part II is also worth a second reading once you have made a card by hand. On the second pass, every counterexample in Chapter 5 is one you committed, and every discipline in Chapter 6 is one you nearly broke. All the detail invisible the first time lights up.

What marks the tools as yours is not remembering the definitions but muscle memory. Your hand splits the two columns when you write a postmortem. Your hand fills the judgement table when you decide. Your eye looks for the grade when you read a report. Muscle memory has no fast route; each tool wants ten uses by hand, and the list of ten uses is each chapter’ s five self-checks. Five questions plus practice is the whole training plan, and the starting point is one card.

Chapter 11 Normalization of Deviance (Vaughan, 1996)

> What this chapter covers: how a deviation turns into standard practice on the strength of "nothing happened last time"; three trigger criteria, the shape the mechanism takes in four domains, eight weak signals and the devices that block it. When to come back to it: when someone answers a challenge with "we have always done it this way"; when you are about to approve a request to relax a standard, extend an interval or waive an inspection. Reading time: about 55 minutes.

11.1 What It Is

Definition: the first appearance of a deviation is handled as an anomaly. As it recurs and produces no consequence, it is reinterpreted as acceptable normal; the explanation stops entering the register; the inspection and release criteria move quietly. Diane Vaughan named the mechanism while studying the Challenger launch decision (Vaughan, The Challenger Launch Decision, University of Chicago Press, 1996). The coldest part of her account is this. Normalization does not happen in the explanation. It happens in the filing. Every deviant event is explained by the organization, which says why this one did no harm, and then filed, which says which category it belongs to. Explanations are public and arguable. Filings are quiet and not arguable. The substantive move of normalization is to shift a deviation out of the drawer marked "to be eliminated" and into the drawer marked "known, acceptable", and the basis for the filing is the outcome: nothing happened.

Three trigger criteria. If any one of them is present, the mechanism is already running.

① A deviation of the same class has appeared twice or more, and none of the occurrences produced a consequence. ② Each occurrence was explained as "there is still margin", "a one-off", "contained", and none of the explanations generated a register entry. ③ An inspection standard, a limit or an interval was lowered or relaxed and written into the process, or practice took the place of process.

Normalization runs through three phases, each with its own archival signature. In the anomaly phase the deviation appears for the first time, enters the register, the entry stays open, and the handling work is organized around removing the cause. In the case phase the deviation appears a second and a third time, each occurrence is explained on its own terms (margin, coincidence, special conditions), the register entries are closed one by one, and no combined statistic across entries is ever produced. In the routine phase the appearance of the deviation no longer triggers a register action at all. Handling is fixed in the process or in habit, and "we did it this way before" becomes the oral register. What separates the three phases is attention. The anomaly phase consumes attention, the case phase sustains it, the routine phase releases it. The mechanism's end product is a reallocation of attention: the attention that used to watch the deviation is moved elsewhere by the organization, while the deviation keeps running where it was.

People assume normalization needs an explicit decision to tolerate something. In most cases no such decision can be found. The "in family" clause in section 11.3, the alarms that close in seconds, the noise explanation: each is a structure that only becomes visible when a large number of small handling actions are strung together and read backwards. Standing at the moment any one of those actions was taken, the person saw a reasonable piece of work. That is the argument for the three-state register in section 11.5. Reason-

ableness seen from one seat cannot be re-checked by addition. Only the register can do that.

Why "no consequence" carries almost no evidential weight needs saying once, plainly. A run with a deviation that lands safely proves very little. It proves this run did not hit the tail, and nobody knows where the tail sits. Stack fifty safe landings and the evidence grows far more slowly than intuition suggests, because the runs are correlated with one another (the same people, the same practice, the same tempo) and the information they carry overlaps heavily. An organization that reads a consequence-free period as proof of safety is reading an intuitive count of independent trials. The real trials are neither independent nor repeated often enough to separate hypothesis A from hypothesis B. That is also why weak signal 8 in section 11.4 is so hard to refute. The challenger cannot produce more data, because the data only exists after the consequence. Normalization wins arguments in a peculiar way. It does not need to win. It needs to last until the data arrives.

The arithmetic core is worth setting out, because it explains why "we have always done it this way" is an understandable error rather than a stupid one. The deviation has appeared n times with no consequence, and two hypotheses are on the table. Under A the margin really is sufficient, and the absence of consequence is the structure's doing. Under B the margin is not sufficient, and the absence of consequence is luck. The organization usually holds no data that discriminates between them. Discrimination requires knowing the tail of the distribution, and the tail is precisely what has not happened yet. Normalization is the default to A when the data cannot discriminate. The cost of that default is set by the compounding of exposures. Every run with the deviation is one exposure. Under B the accident probability rises with the number of exposures; under A it stays constant. The installment-type irre-

versibility of Chapter 7 is this arithmetic in the form it takes at a release decision.

11.2 How It Grows

Four generating conditions, corresponding to structure, incentive, time scale and information.

Structural: the deviation is produced at the front line, and the au - thority to remove it sits far away. Stopping a line, changing a sup - plier, redesigning one detail: high cost, long process. Writing an explanation: low cost, finished on the spot. When the gap between the cost of removing and the cost of explaining is wide enough, the organization chooses explaining again and again.

Incentive: the person who explains the deviation does not carry the deviation’ s tail risk. The manager who signs the relaxation is usually transferred before the consequence lands (Chapter 28), or the deviation itself yields a current-period benefit in output, on- time rate or labor hours. Where the person who sees the deviation is also the person who benefits from it (Chapter 17), the explana - tion drifts toward hypothesis A on its own.

Time scale: the consequence is delayed. Delay breaks the learning loop. Months or years sit between the deviation and the con - sequence, and by the time the consequence arrives, the record of the deviation has already been overwritten with “explained”. Delay has a side effect. A stretch of no consequences is itself evid - ence, to the organization, that hypothesis A holds. Success be - comes the fuel of deviation; the slack of success in Chapter 20 is the sister mechanism.

Information: no cumulative register. Each deviation is assessed on its own, from the material of that one event plus the memory of the people involved. A list of historical deviations does not exist, or

exists with every entry marked "explained". With no independent review, the hypothesis-A explanation has nothing to be confronted with.

There is one amplifier: years in the trade. Personal experience is not built like an organizational register. "I did this for twenty years and nothing happened" treats a personal exposure count as data, and its sampling bias is the microscopic version of this chapter's mechanism. It also sits on the border with Chapter 14: intuition is most confident inside its own failure zone.

One archival property of the mechanism needs adding after the generating conditions. The status of a register entry decides who reads it later, and with how much attention. An entry marked "explained" presents itself in every later search, every handover, every audit as a settled matter. The reader assumes it no longer needs judgment, only awareness. Chapter 30, handover rupture, meets this chapter exactly here. The entries on a handover sheet marked "legacy item, dealt with" are how normalization travels between departments.

The consequence-free period has a second contaminating effect: it rewrites the baseline for future assessment. Every run with the deviation produces a record that says the run was normal. Those records enter the experience base, and later risk assessments use them as their template. The sample the assessor sees has already been screened of bad outcomes. This is the survivorship bias of Chapter 2 reproduced inside the organization. The filing action rewrites the organization's definition of normal every day, and the rewriting raises no alarm.

Among the four generating conditions, the structural one is the most often misread, as "the front line is a long way from headquarters". Distance is the shell. The core is a difference in standing between two kinds of knowledge. Design knowledge is written into documents; operating knowledge is scattered among

people. Documents can be searched, cited and audited. Knowledge held in people cannot, so every collision between the two ends with the document winning, and the real operating practice survives outside the documents under the name of custom. Normalization is that survival, filed. Before custom produces a consequence, the organization's official language has no word for it. The front-line representative signature clause in the device list of section 11.5 is aimed at this gap in standing. Its job is to give operating knowledge the same right to be cited as a document.

11.3 The Pattern in Four Domains

The six cases below come from four domains. Following the book's convention, each stops at the decision point: you decide first, then turn the page for the outcome. Case numbers match Appendix B. Cases one, two and five are given as full cards, each backed by a grade-A full report from an independent investigation. The other three stay condensed.

Case 1 (Spaceflight): The "In-Family" Classification of a Foam Strike

mi-001.

When and where: 16 January 2003, launch at 10:39 a.m. local time; 1 February 2003, entry interface at 8:44:09 a.m., loss of communication with the ground at 8:59 a.m. Locations: Kennedy Space Center, Florida (launch and imagery interpretation); Marshall Space Flight Center, Huntsville, Alabama (external tank project, flight readiness review); Johnson Space Center, Houston, Texas (debris assessment team, mission evaluation room, mission management team).

What and who: The NASA Space Shuttle Program office (SSP); the Marshall external tank project (ET Project); the Johnson engineer-

ing directorate and structures division; engineers from United Space Alliance and Boeing; the Debris Assessment Team (DAT, led by two co-chairs, one from NASA and one from United Space Alliance); the Mission Evaluation Room (MER, the project office's engineering support group); the Mission Management Team (MMT, chaired by an appointee of Johnson); and the Program Requirements Control Board (PRCB), the body that signs the flight readiness review. The STS-107 crew numbered 7 and spent 16 days on orbit.

What they were doing: On 7 October 2002, STS-112 launched and a piece of foam roughly 4×5×12 inches came off the left bipod ramp of the external tank, tearing the insulation on the solid rocket booster attach ring (CAIB p.126, figure 6.1-5). On 31 October, at Marshall, the PRCB heard the external tank project manager, Jerry Smelser, at the STS-113 flight readiness review. His briefing was titled "STS-112/ET-115 Bipod Ramp Foam Loss" and ran four pages (CAIB p.126). That loss had occurred three months before Columbia's last launch (CAIB p.124, figure 6.1-3). The briefing gave three flight rationale: the foam came off the -Y bipod ramp and exposed the bipod fitting; the current fitting had not changed since STS-54 (ET-51); and across more than a hundred external tanks only 3 significant bipod foam losses had been recorded (CAIB p.126). At the same review the NASA headquarters safety office submitted a report estimating a 99% probability that no further foam would come off that area (CAIB p.127). The review's conclusion characterized the foam loss as an "acceptable risk" rather than as "not a flight safety issue", and the STS-113 certificate of flight readiness was signed (CAIB p.127). STS-113 launched at night in November; no imagery was obtained, so whether foam came off again could not be determined (CAIB p.127). The judgment rested on a long ledger. After the first flight in 1981, 300 tiles were replaced. Of the 79 missions with imagery, 65 captured foam loss, and the orbiter's lower surface averaged 143 dings after re-

turn (CAIB p.123). STS-107, meanwhile, flew as a 16-day science mission with no robotic arm, and RCC panels 1 through 11 on the left wing leading edge could not be seen from inside the cabin (CAIB p.147).

The decision point: ① 31 October 2002, Marshall, STS-113 flight readiness review. On the table: the foam dimensions and damage photographs from STS-112, the three flight rationale from the external tank project, and the 99% estimate from the headquarters safety office. The Image Working Group had earlier recommended that the STS-112 loss be recorded as an In-Flight Anomaly; the recommendation was not adopted at the review (CAIB p.126). CAIB's re-check found the first and third rationale incorrect. The bipod ramp design had been changed at ET-76, and the 99% calculation pooled the left and right bipods and set the start of its sample after STS-112, which left out STS-7, STS-32R and STS-50 (CAIB p.127). The meeting's conclusion was written as "acceptable risk", and the certificate was signed (CAIB p.127).

② 22 January 2003, in the morning, Houston: the imaging request is withdrawn. At 8:09 the MER log records that "Engelauf wanted to know who was having the Air Force look at the vehicle" (CAIB p.154). At 8:30 NASA's Department of Defense liaison called US Strategic Command to withdraw the imaging request; less than 90 minutes separated initiation from withdrawal (CAIB p.154). After the withdrawal, a 9:33 email from Ham to Austin and Roe was still asking the same question in another form: could anyone state that no external tank foam loss could cause a flight safety problem, on grounds of density (CAIB p.155). At 10:15 Dittemore replied that someone first had to confirm that foam density could not strike a tile hard enough to affect the orbiter (CAIB p.156). At 10:37 the log records the conclusion: "Shriver, Ham, Roe have indicated they do not need Air Force look at vehicle" (CAIB p.157).

③ 24 January 2003, 7 a.m., Houston: the debris assessment team briefed the MER. That day the MER manager gave the MMT an oral summary with no data. The MMT classified the impact as a "turnaround issue" and stopped pursuing imagery (CAIB p.39).

What was visible then: On 17 January at 9:30 EST, image analysts at Marshall saw the impact at 81.9 seconds on film, and within an hour Kennedy confirmed it on higher-resolution film (CAIB p.141). Only three ground cameras could see the impact at all: one lost tracking during ascent, one was out of focus from improper lens maintenance, and one captured only the upper surface of the left wing (CAIB p.141). At 10:58 CST the same day, the MER log recorded debris "from the external tank forward bipod area, traveling down the left side, striking the left wing leading edge near the fuselage" (CAIB p.142). At 15:15 Bob Page, chair of the Image Working Group, telephoned Wayne Hale and Lambert Austin: Boeing was computing trajectory, velocity, angle and energy (CAIB p.142). At 16:28 the log records that "USA and Boeing are not working the debris problem over the weekend, waiting for film on Monday" (CAIB p.143). On 18 January Boeing produced two size estimates, 20×20×2 inches and 20×16×6 inches, with a velocity of about 750 feet per second and an incidence angle under 20 degrees (CAIB p.144). At 11:39 on 20 January the log records debris "about 20 inches, plus or minus 10 inches, similar in size to the STS-112 event" (CAIB p.147). At 9:26 on 21 January, Schomburg wrote by email: "TPS took a hit, probably not a problem" (CAIB p.150). The debris assessment team held its first formal meeting the same day (CAIB p.152). At 9:01 on 22 January, Air Force Lieutenant Colonel Lee emailed McCormack: "Austin called yesterday, requesting DoD high-resolution photos of the shuttle on orbit" (CAIB p.152). That same day Rocha wrote an email that was never sent: "in my poor technical judgment, SSP and the orbiter project not requesting additional imaging is wrong, close to irresponsible" (CAIB p.158). At 23:13 on 23 January, Stich emailed the

crew: “the experts have looked at the high-speed photography, there is no concern for RCC or tile, and absolutely no concern for entry” (CAIB p.160). At 12:15 on 28 January, the safety and quality assurance seat’s log mentioned debris for the second time, asking whether anyone from SR&QA had been involved in the decision that the ascent debris strike was safe (CAIB p.167). Information that became available only afterwards: the foam struck RCC panels 5 through 9, panel 8 was breached, hot gas entered the wing leading edge spar, and the aluminum structure melted. None of this appeared before the debris reconstruction that followed 1 February.

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not “pay more attention”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 8:44:09 on 1 February 2003 the orbiter entered the atmosphere. At 8:59 communication with the ground was lost, and the vehicle broke up over Texas and Louisiana. All seven astronauts died. During the search a Bell 407 helicopter crashed in Angelina National Forest, killing the contract pilot and a Texas Forest Service employee and injuring 3 (CAIB p.47). Ground searchers passed 3,000 within two weeks and 4,000 within a month (CAIB p.46). Back to the three questions. The most common answer to the first is “ask for the imagery again”. CAIB lists it as the 4th of eight missed opportunities: the request was withdrawn within 90 minutes, and the person who withdrew it was not an engineer (CAIB p.154, p.168). The item most often missed on the second

question is the four slides of 31 October. They described a design that had already been changed as unchanged, and that sentence was the pivot of the entire flight rationale (CAIB p.127). The third question lands on 17 January. The Image Working Group had classified the strike as "out of family" that day, and under project procedure an event of that class should have been handled as a Tiger Team. What was formed was a DAT, which is not bound by the Tiger Team checklist, so the event belonged nowhere in the procedure (CAIB p.143).

Options ruled out: Have the crew photograph the left wing with a handheld camera. Rocha asked on 17 January whether the crew could inspect the wing; no response is recorded (CAIB p.168). Columbia had no robotic arm, RCC panels 1 through 11 were not visible from the cabin, and a visual check required a spacewalk (CAIB p.147). Request Department of Defense imaging. It was initiated on 22 January and withdrawn within 90 minutes (CAIB p. 154). Change the entry trajectory to reduce heating. The engineering analysis concluded that no alternative trajectory would significantly reduce the heat load on that area (CAIB p.161). Upgrade the DAT to a Tiger Team. The event had been classified out of family, but that procedure was never run; the team was left "organizationally adrift" with only a date to report: 24 January (CAIB p.143). Extend the mission and wait for on-orbit imagery. Not found in any meeting record. Declare the strike a flight safety issue outright. CAIB records the actual requirement as the reverse: engineers had to prove it unsafe before management would move, while the standing expectation was to prove it safe (CAIB p.170).

Early signals: ① 18 June 1983, STS-7: the left bipod ramp lost a 19×12 inch piece of foam, leaving 25 shallow divots forward of the bipod strut and 40 divots on the lower external tank. The post-flight PRCB recorded it as an in-flight anomaly and closed it at the next review on the grounds that the orbiter thermal protection system

had been repaired. Neither the closure document nor the flight readiness certificate mentioned the cause of the foam loss (CAIB p. 124). Suppression reason: it was logged under procedure, so it counted as handled. ② 9 January 1990, STS-32R: 5 divots in inter‐tank foam with diameters from 6 to 28 inches, 111 strikes on the orbiter's lower surface with 13 over 1 inch. The anomaly was closed at the STS-36 review as a "local foam void", vent holes were subsequently drilled in the foam, and the mechanism has never been confirmed (CAIB p.125). ③ 2 December 1988, STS-27R: 707 divots, 298 of them over 1 inch, one tile knocked off and struc‐ture exposed. After the ablator was replaced, the problem counted as closed before the next review (CAIB p.128). The same investiga‐tion recommended that the project "actively solicit design im‐provements to eliminate the source of debris or reduce the poten‐tial for damage"; 14 years later there was still no corresponding action (CAIB p.129). ④ 25 June 1992, STS-50: the third bipod foam loss, characterized in Hazard Report 37 as an "accepted risk", with the external tank project writing at the same time that it was "not a flight safety issue" (CAIB p.125). ⑤ 19 November 1997, STS-87: thermal protection damage prompted NASA to run 9 flight tests to settle the foam loss question. The fix did not work, and the in-flight anomaly was closed after STS-101 as an "accepted risk" (CAIB p.129). ⑥ 7 October 2002, STS-112: the fourth, with foam of 4×5×12 inches tearing the booster attach ring insulation. The review of 31 October passed it with two lines, "design un‐changed" and "99% no loss" (CAIB pp. 126–127). The suppres‐sion reasons collapse into one sentence: this is a class of event already logged, already explained, with no consequence on record. CAIB's count is 7 left bipod losses in 113 flights, about 10% of the 72 flights with usable imagery (CAIB p.124).

The point of escalation: 31 October 2002, Marshall, where the STS-113 flight readiness review wrote "acceptable risk" into its conclusion and the certificate of flight readiness was signed shortly

after (CAIB p.127). CAIB's judgment is that this decision "made it, and possibly encouraged it, easier" for members of the MMT to use the same reasoning when they assessed the STS-107 foam strike (CAIB p.126). STS-113 then launched at night with no imagery, and in November the same rationale was forwarded by Paul Shack to Rocha by email as a reference (CAIB p.151). By 17 January, an engineer arguing "this one is different" first had to overturn the conclusion of the project's most rigorous review, held a month earlier.

Failure chain: Technical layer: foam separated from the left bipod ramp at 81.7 seconds and struck RCC panels 5 through 9 at 625 to 840 feet per second at 81.9 seconds; panel 8 was penetrated and hot gas entered the wing leading edge spar. There is a design background underneath this. The thermal protection system was designed to withstand impacts below 0.006 foot-pounds, and the external tank requirement was that it produce no debris; both requirements were breached on the first flight in 1981, when 300 tiles were replaced (CAIB p.123). Organizational layer: the strike was filed as "in family", meaning within known experience, considered understood, and not a flight safety issue. The classification rested on the absence of past consequences, not on the energy and location of this strike (CAIB p.124). The same filing broke once, on 17 January, when the strike was reclassified as out of family, but the Tiger Team procedure that out-of-family should trigger was never started (CAIB p.143). Foam damage monitoring was judged inadequate by the project itself, a fuller record would have helped identify trends, and the improvement never landed (CAIB p.129). Incentive layer: STS-107 was a long-scheduled 16-day science mission, imaging would consume science time and require interagency channels, and the trade on 22 January was to stop a 90-minute request (CAIB pp. 154–155). The break sits between the organizational and incentive layers. The filing clause exists precisely so that anomaly signals can be dissolved quickly under schedule

and resource pressure, and what it saves management is the cost of one more look.

Minimum counterfactual: At 8:30 on 22 January 2003, do not withdraw the Department of Defense imaging request that had already been initiated. Let it run through the procedure Strategic Command was already following. One action, ten days before the breakup. The cost was a few hours of science observing time, and it required no luck and no new technical judgment, because that morning the request had already reached the stage of identifying available imaging assets (CAIB p.153). CAIB lists it as the 4th of eight missed opportunities (CAIB p.168).

Outcome and sources: Seven astronauts died. Two search personnel died in a helicopter crash and 3 were injured. The orbiter broke up, and debris was scattered across Texas and Louisiana. CAIB found that the accident "was not a random event" and listed the way the foam loss was filed as an organizational cause (CAIB p.10, p.124). Primary source: Columbia Accident Investigation Board, Report Volume I, August 2003, NASA official PDF, link at Appendix B mi-001.

Case 2 (Chemicals): Thirty-Two Alarms

ch-001.

When and where: 10 November 2014, a Monday, a mis-dilution shutdown; 12 November, an attempted restart; the morning of 14 November, the hydrate plug confirmed and a handling plan settled; 15 November, high-pressure alarm inside the building at 2:51 a.m., valves opened from 3:01 to 3:13 a.m., the last victim found between 11:15 and 11:55 a.m. Location: La Porte, Texas, the Lannate insecticide unit of the DuPont La Porte plant. The point of failure was the second-floor drain area and the third-floor vent header collection level of a multi-storey enclosed building; the control room stood outside the building.

What and who: E. I. du Pont de Nemours and Company, La Porte plant, insecticides business unit (IBU), Lannate (methomyl) unit. The shift was nights: control board operator, shift supervisor (who also served as Process Coordinator, responsible in an emergency for giving the rescue team process information), and field operator. Three plants on the site shared one emergency response team (ERT) of about 60 people, managed by DuPont (CSB p.20, footnote). The same building also handled chlorine and methyl isocyanate (MIC), both more toxic than methyl mercaptan under site standards (CSB p.13, footnote; p.28). Methyl mercaptan was detected with borrowed hydrogen sulfide detectors, with the alarm set at 25 ppm for methyl mercaptan (CSB p.14, footnote).

What they were doing: On 10 November a tank truck of 50% acetaldoxime (AAO) was being unloaded when a water dilution valve that should have been interlocked shut stayed fully open, because a jumper installed during the spring 2014 turnaround had not been removed. Water entered the tank, the AAO was diluted, and the unit shut down (CSB p.9, footnote; p.13). Water entered the methyl mercaptan feed line during the shutdown. Houston temperatures held below 55 Fahrenheit from 11 November, averaging about 40 Fahrenheit over the first 24 hours, hydrates formed in the line, and the restart on 12 November failed (CSB p.13, footnote; p.102). On the morning of 14 November the unit technical team met the day-shift supervisor, concluded that water had entered and formed hydrates, and decided to heat the line under its insulation with hot water hoses until the hydrates decomposed. The team also recognized that heating would expand the methyl mercaptan and that it needed somewhere to relieve, so operators opened three valves between the methyl mercaptan feed line and the vent header; at the time of the incident one was fully open and one was slightly open (CSB pp. 99–100, pp. 102–103). None of this work had a process hazard analysis, a written procedure or a management-of-change review (CSB p.100). Execution began at the tank and feed

pump section and moved along the feed line toward the reactor system, with pressure gauges on the three connecting valves used to locate the plug and track progress (CSB p.103). In the early hours of 15 November the hot water melted the hydrate, and liquid methyl mercaptan passed through those three valves into the vent header, a line built for vent gas and designed never to carry liquid (CSB p.104). At 2:51 a high-pressure alarm came from equipment inside the building. The board operator treated it as condensate accumulation, as custom dictated, and sent the shift supervisor and operator 1 to drain the third floor (CSB p.15). Third-floor low-point draining had been written into the daily work instructions since 14 January 2014, once per shift (CSB p.108).

The decision point: ① 2011: the $20 million NRS incinerator pro - ject went into service. The vent header running to the incinerator had not been designed with liquid accumulation properly in mind. The line has low points, there is no engineering measure such as a knockout drum, and accumulated liquid could only be removed by hand from the low points (CSB p.106, p.108). CSB records that this was not considered in design; no individual signature for project approval or acceptance appears in the report. Over the following three years, several pieces of equipment connected to that header had repeated high-pressure events, and the fixed response was manual draining (CSB p.107).

② March 2010: a corrective action closed. A second-party PSM audit in 2009 found that the site held no documentation showing the building ventilation system had ever been tested for supply ef - fectiveness, and recommended a periodic test procedure. The site configured an automatic work-order generator in the computerized maintenance system and closed the corrective action on that basis in March 2010, without assigning an owner (CSB pp. 97–98). The first test should have run in November 2010 and was actually delayed to 2012. The dry end was measured for airflow but never

analyzed and had no acceptance criteria. The wet end, the side that failed in 2014, was never tested at all (CSB p.98).

③ 14 January 2014: the daily work instruction. Area management required operators, in the daily instruction book, to drain accumulated liquid from the third-floor vent header "low point" once per shift. A paragraph was added later: attach a hose and drain the liquid under an open floor safety shower until it stops flowing (CSB p.108). The instruction came with no management of change, no dedicated procedure, no line-breaking permit, and no requirement for additional protection (CSB p.108, p.111). The instruction book had no per-item signature block, and CSB names the approving source only as "area management" (CSB p.108).

④ The morning of 14 November 2014: the meeting of the unit technical team and the day-shift supervisor. The meeting settled on external hot water heating with three valves connected to the vent header, with no management of change and no written procedure. The hot water heating also exceeded the heat input design value of the system's thermal expansion relief valve (CSB p.104).

What was visible then: 32 methyl mercaptan alarms at the control panel over 17 hours (CSB p.14). The detector range was 0 to 25 ppm and the alarm was set at 25 ppm, which is 2.5 times the OSHA limit of 10 ppm and 50 times the NIOSH recommended value of 0.5 ppm; readings appeared only on the DCS and never entered the historian (CSB p.15, footnote). The high-pressure alarm at 2:51 on 15 November arrived with the "condensate" custom attached to it (CSB p.15). The odor threshold of methyl mercaptan is about 2 ppb, one five-thousandth of the OSHA permissible exposure limit of 10 ppm, so smelling it says nothing about a low concentration, and prolonged exposure brings olfactory fatigue (CSB p.14, footnote). Two shift suggestion meetings, in February and October 2014, both recorded the low-point accumulation and both proposed changing the piping to eliminate the low point (CSB p.112). A management-

of-change review started weeks before the incident stated in its own text that third-floor draining "was not a good practice, be - cause the line is sometimes nitrogen-purged, which could release nitrogen inside the building" (CSB pp. 108–109). The only written note on the shift turnover record was one line: "Unplugging M[e]SH Header." (CSB p.113, footnote). An "urgent" work order had been opened on 20 October after another roof fan stopped with an abnormal noise, and a second fan had been down since June (CSB p.85). The ventilation system had no scrubber or incin - erator of its own; exhaust went straight to atmosphere (CSB p.97, footnote). The 2009 audit finding and the March 2010 closure were on file (CSB pp. 97–98). Information that became available only af - terwards: the concentration gradient on each floor, the release rate and the dispersion, all reconstructed by CSB modeling, with a total of about 24,000 pounds (CSB p.7, p.24).

> **Stop here. Decide before you turn the page.**
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> You are in that seat now, with the materials above in front of you.
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> 1. What would you do? Write the action, not "pay more atten - tion".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: Four people died: 3 operators and 1 shift super - visor, from the combined effect of asphyxiation and acute methyl mercaptan inhalation (CSB p.7, p.24). At 3:30 operator 1 radioed for help, and different listeners understood the wording as "we need help" or as "I need help on the fourth floor" (CSB p.17). Operat - ors 2, 3 and 4 went in without respiratory protection. Operator 2 died on the third floor. Operator 4 advanced about 10 feet, hit what he later called "a wall" of methyl mercaptan, and fell back (CSB

p.18). Operator 6 entered with three 5-minute escape respirators, put one on operator 4, went up to the third floor for his brother, and was overcome before he could connect the mask to the air cylinder (CSB pp. 19–20). The ERT was called at 3:50, and the other party understood "rescue" as high-angle and confined-space work, so it brought harnesses and rope (CSB pp. 20–21). At 4:05 the incident commander asked for blast suits and SCBA. The small pumper truck carrying the SCBA would not start, and it stood more than 2,000 feet from the control room (CSB p.21). The first team wearing SCBA entered the building at 4:25 (CSB p.24). About 40 minutes passed before the control room understood how serious things were inside the building, and full control took more than six hours (CSB p.27). The pump was not stopped until 6:02, and the tank and the leaking line were fully isolated at 10:07 (CSB pp. 25–26). On the three questions: the most common answer is "go into the building and look", and operator 4's ten feet is that answer. The item most easily missed is the fans, two "PSM critical" machines with one down five months and one down nearly a month (CSB p.85). The third question does not land on alarm handling. It lands on why that header had a low point in 2011 at all.

Options ruled out: Stop draining and evacuate. The emergency plan required that a release be reported to the shift supervisor, that the site toxic gas alarm be sounded, and that SCBA be worn during response; what happened was confirmation followed by continuation (CSB p.29). Start the ventilation. Neither roof fan would turn (CSB p.85). Confirm at the valve mouth with a portable detector. Personal methyl mercaptan monitors existed on site and were never issued to field operators to wear (CSB p.28). Work under a line-breaking permit. By DuPont's own standard, opening a valve to atmosphere is line breaking and requires a written work plan; the people on site did not regard a drain hose as line breaking (CSB p.111). Remove the low point by engineering. Raised at the shift suggestion meetings of February and October 2014, never opened

as a project (CSB p.112). Wait for the management of change to finish before draining. That review had started only weeks earlier, and draining went on (CSB p.108). Add technical support to the night shift. The handling plan gave the night shift no adequate technical resource (CSB p.95).

Early signals: ① The 2009 second-party PSM audit: ventilation never verified; the corrective action closed in March 2010, the first test slipped to 2012, the wet end never tested (CSB pp. 97–98). Suppression reason: the entry's status already read "closed", and CSB calls this kind of closure a paper closure (CSB p.97). ② The 2011 process hazard analysis: hydrate plugging was ranked consequence C-1 and frequency F-1, the lowest rung of both scales, risk level IV, "no mitigation required", so there was neither heat tracing nor any procedure for decomposing hydrates (CSB p.94). Suppression reason: the matrix had already answered, unlikely and minor. ③ Text left behind by the 2012 safety perception survey and the July 2014 front-line supervisor interviews: "our safety passion has slipped over the past ten years"; "we are just not as good as we thought we were... a long slow decline, and we may not have noticed how far we had dropped" (CSB p.84). Suppression reason: a culture survey is not a deficiency entry and does not enter the corrective action register. ④ The daily drain instruction from 14 January 2014, which turned a known design defect into a per-shift action (CSB p.108). Suppression reason: ten months in writing, nothing had happened. ⑤ Both shift suggestion meetings in 2014 identified the low-point accumulation and proposed changing the piping (CSB p.112). Suppression reason: it had been discussed in a meeting, which counted as handled. ⑥ 32 alarms in 17 hours, at an alarm value of 25 ppm (CSB pp. 14–15). Suppression reason: they coincided with maintenance, so they were maintenance-related.

The point of escalation: The morning of 14 November 2014, when the unit technical team and the day-shift supervisor settled on hot

water heating with three valves connected to the vent header. From that point liquid methyl mercaptan had a route into the building (CSB pp. 102–104). CSB's own words: an effective management-of-change review of the heating and of this abnormal line configuration "could have identified and controlled the hazards that led to the incident" (CSB p.104). After that step the only interception point left was confirming what the line contained before the valves were opened, and sampling was written into the daily instruction and never carried out (CSB p.108).

Failure chain: Technical layer: the vent header to the 2011 incinerator has low points and no knockout drum, and liquid was removed by hand (CSB p.108); building ventilation was designed to 25% of the lower explosive limit for combustible gas with no allowance for toxicity, while 25% of the LEL for methyl mercaptan is 65 times the immediately dangerous to life or health concentration of 150 ppm (CSB p.8, footnote); both roof fans were unavailable at once (CSB p.85); the detector alarm sat at 25 ppm and displayed only in the control room (CSB p.15, footnote). Organizational layer: the 2009 audit finding was closed by building a procedure, with no test performed (CSB p.98); the 2011 hazard analysis rated hydrates as the lowest risk (CSB p.94); draining had no management of change, no procedure and no line-breaking permit (CSB p.108, p. 111); turnover was oral, with a single written line, "Unplugging M[e]SH Header." (CSB p.113); the definition of line breaking did not match field understanding, and a drain hose was not treated as line breaking (CSB p.111). Incentive layer: the $20 million project of 2011 was for more output and lower emissions, and Lannate was a full-rate product (CSB p.106); the 2014 bonus plan, LPBC, was tied only to OSHA recordable injuries, deducting 5% per injury up to a cap of 50%, with no process safety metric in it (CSB pp. 115–116). The break sits between the organizational and incentive layers: making the accumulation into an engineering fix requires a project

and a shutdown, while writing it into a daily instruction requires one line.

Minimum counterfactual: At the meeting of 14 November 2014, run a management-of-change review on "hot water heating plus three valves connected to the vent header". One action, available the same day, needing no new data. CSB found that this step "could have identified and controlled" the hazards that led to the incident (CSB p.104).

Outcome and sources: Four dead and about 24,000 pounds of methyl mercaptan released. In May 2015 OSHA issued 11 citations for this incident; on 9 July it issued 8 more for the adjacent unit and placed DuPont in the Severe Violator Enforcement Program; a settlement was reached on 17 July 2017 (CSB p.62, footnote). EPA proposed 22 risk management program violations after its inspection, the Department of Justice filed a complaint with a consent decree on 23 July 2018, and DuPont paid a $3.1 million civil penalty without admitting liability (CSB p.65). DuPont announced in spring 2016 that the Lannate unit would not restart, and it was demolished in 2017 (CSB p.10). Primary source: CSB, Investigation Report No. 2015-01-I-TX, June 2019, link at Appendix B ch-001.

One detail in this case deserves a second look from the chapter's angle. The maintenance work order for the failed exhaust fan sat in the system for nearly a month with the status "urgent", and CSB's text uses that word in English, where it means handling within two to three days. The work-order system was not wrong, and the person who set the priority had a reason: ventilation plays no part in the reaction, and a stopped fan does not affect output. The gap sits between the status language and the handling. "Urgent" reads, in this chapter's archival sense, as the maintenance edition of "explained": a known defect formally logged as urgent, with nobody asking why it had not been handled within two or three days. What arrived was that defect becoming the condition

that turned a release into a poisoning on the night of the incident. Device 1's three-state register, ported to a work-order system, is a one-line patch. A defect in equipment that carries a safety function may only be given the status "remediation by date", and the date is set by a party whose appraisal does not depend on output. An "urgent" with no time standard is an adjective.

Case 3 (Aviation Maintenance): Interval Extensions, Approved One Signature at a Time (condensed)

av-034. Alaska Airlines 261, an MD-83 (N963AS), 31 January 2000, 2.7 miles north of Anacapa Island in the Pacific. The NTSB final report (AAR-02/01; this book downloaded and read the PDF): the lubrication interval and the end-play check interval for the horizontal stabilizer jackscrew assembly had been extended at the company's request with FAA approval (executive summary p.4, verified). The report found that the extensions allowed inadequate lubrication to develop into excessive wear, that the jackscrew nut threads failed in flight, and that the 2 pilots, 3 cabin crew and 83 passengers on board all died. The decision point is the moment the FAA signed the extension. Minimum counterfactual: do not extend the end-play check interval, and check at the original interval. Primary source: NTSB, AAR-02/01, 2002, link at Appendix B av-034.

The condensed version is worth a second look because it shows normalization's most ordinary carrier: a relaxation document with unimpeachable wording. The application came from the operator, the approval from the regulator, both acting inside their own authority, and every step is compliant when read alone. The mechanism leaves one trace. Almost all the information the approval rested on came from the applicant's own records. NTSB's executive summary sets out the path between the extension and the development of wear plainly; in hindsight, the separate reasonableness of each extension request adds up to putting the wear process of the jackscrew nut inside a sparser inspection window.

Devices 2 and 4 in section 11.5 are built to this shape: numbers must be measured, and a signature must have a second pen.

Case 4 (Finance): Alarms Closed in Seconds (condensed)

fi-013. Moors & Cabot, a broker-dealer, anti-money-laundering (AML) monitoring. FINRA disciplinary decision of August 2026 (this book downloaded and read the monthly summary PDF): a large number of AML alerts were closed within seconds with no corresponding review, after an earlier independent AML test had flagged the alert backlog and no remediation followed. The decision point is the moment the independent test report reached management and named the backlog. Minimum counterfactual: when the report arrived, add staff and systems as it recommended. Primary source: FINRA, Disciplinary and Other FINRA Actions – October 2026, 2026, link at Appendix B fi-013.

The condensed version hides the standard form this mechanism takes in finance. The handling action for an alert is closing it, the reason field is a structured dropdown, and the menu has no option reading "needs more time". The choice architecture has paid for the normalization in advance: each closure taking seconds is, on its own, a non-compliant act, but the interface makes non-compliance the default path. Once the independent test had flagged the backlog, the entry should have moved into remediation. It moved into the next audit calendar instead. A reader in financial services can carry Device 1 back unchanged: alert entries get three states only, and a closure measured in seconds must be filed under "running at risk" with a signature attached.

Case 5 (Aero Engine): The Low-Amplitude Indications of 2016 (full card)

av-006.

When and where: 20 February 2021, departure from Denver International at 13:04 local time, failure in the climb at 12,500 feet at 13:09, return and landing at 13:28, about 24 minutes from takeoff to touchdown (the flight was bound for Honolulu); the failure occurred above Broomfield, Colorado. The inspection that matters happened in April 2016, at the Pratt & Whitney overhaul shop (NTSB pp. 19–20).

What and who: United Airlines flight 328, a Boeing 777-222 (N772UA, manufactured September 1995, 400 seats, maximum takeoff weight 545,000 pounds), with two Pratt & Whitney PW4077 engines rated at 77,000 pounds of thrust each. The right engine was manufactured in 1995 and installed on the aircraft in August 2016; its most recent inspection before the event was a continuing airworthiness inspection on 16 January 2021, with airframe total time 96,814.41 hours (NTSB p.9). The maintenance program was United's FAA-approved engine maintenance program, built on the Pratt & Whitney PW4000 112-inch maintenance planning document (NTSB p.19). Fan blade overhaul was done at the manufacturer's overhaul shop and included fluorescent penetrant inspection, visual inspection and thermal acoustic imaging (TAI) (NTSB p.19).

What they were doing: The aircraft was climbing at 12,500 feet at about 280 knots, with moderate turbulence forecast between 13,000 and 23,000 feet, and the crew increased thrust to shorten the time in it. Five to seven seconds after the thrust increase, the cockpit voice recorder captured a thump; the flight data recorder showed an uncommanded shutdown of the right engine, followed by a fire warning (NTSB p.6). The engine had accumulated 2,979 cycles and 12,384 hours since overhaul, and 15,262 cycles and 81,768 hours since new (NTSB p.9). The blade that failed was number 19, fractured transversely across the full span at the root platform plane (NTSB p.14). That blade had been through two

overhauls, one in 2014 and one in 2016; the April 2016 TAI left two low-amplitude indications on the record (NTSB pp. 19–20). The overhaul interval in force at the time was 6,500 cycles (NTSB p.19). The blade failed at less than half the interval, and the next TAI was scheduled at 6,500 cycles (NTSB p.3). TAI works like this. The blade airfoil is coated with a thermally radiative paint, a transducer at the root vibrates the whole blade, the faying surfaces on either side of a crack rub at high frequency and generate heat, an infrared camera reads the temperature signature at the surface, and soft - ware amplifies it for an inspector to interpret (NTSB p.19). In - spectors do meet suspicious indications and have to judge whether a real crack or a benign cause produced them, the benign ones being paint debonding at the surface or sand trapped in the in - ternal cavity, neither of which justifies removing the blade (NTSB pp. 19–20).

The decision point: ① April 2016, the overhaul shop, the TAI in - terpretation. The software flagged two low-amplitude indications on the convex side, in the mid-chord flowpath region, and both fell inside the fatigue origin area that later fractured. The inspector classified them as extraneous, judging them to come from camera sensor noise or from loose particles inside the cavity (NTSB p.20). Under nondestructive inspection procedure NDIP 1065, then in force, an indication of that amplitude required one of two actions: strip the coating, re-spray and run a second TAI, or escalate the ambiguous indication to a team review. The report states that "there was no evidence that either of these reviews occurred", and the blade was released for continued service (NTSB p.3, p.20). The interpretation left one word behind, extraneous, and no second opinion (NTSB p.20).

② 13 February 2018: another PW4077 on a United 777-222, over the Pacific, released a fan blade in the same way, NTSB case DCA18I - A092. NDIP 1065 then went through revisions B, C and D, which

added process controls and sharpened the illustrations of acceptable and unacceptable indications (NTSB p.20). The same passage records that after the 2018 event the April 2016 thermal data were pulled for review, that "the reviewers accepted the previous interpretation", and that the blade stayed in service (NTSB p.20).

③ The 6,500-cycle overhaul interval itself. The figure comes from the manufacturer's maintenance planning document, applies uniformly to PW4000 112-inch stage 1 fan blades, and was executed through United's maintenance program with FAA approval; NTSB lists the manufacturer's insufficient inspection interval frequency among the contributing causes (NTSB p.4, p.19). After the event the TAI interval went from 6,500 cycles to 1,000, one sixth of what it had been (NTSB p.4).

What was visible then: The hierarchy of the maintenance program: United's engine maintenance program approved by the FAA, resting on the PW4000 112-inch maintenance planning document, under which a blade continues in service as long as it passes the prescribed inspections (NTSB p.19); this set of blades had been overhauled in 2014 and again in 2016, both at the manufacturer's shop (NTSB p.19). The April 2016 TAI images, and the two low-amplitude indications the software flagged on its own (NTSB p.20). The NDIP 1065 criteria then in force, under which an ambiguous indication is either redone or escalated to a team review (NTSB p.20). The cycle counts for that blade since new and since overhaul, and the 6,500-cycle interval set by the maintenance planning document. The work cards from the 2014 and 2016 overhauls, covering fluorescent penetrant, visual and TAI, with April 2016 the second shop visit at which this set of blades received TAI. The procedure requiring that an indication an inspector cannot clearly interpret be referred, with the images and the blade, to a process engineer (NTSB p.20). Information that became available later: the fracture originated on the internal cavity radius surface

about 6.6 inches above the blade root and grew from 0.063 inch to 0.199 inch over about 3,150 cycles of stable low-cycle fatigue; the two low-amplitude indications sat "very close" to that origin, and the report concludes they "most likely" are the images of the initiating fatigue crack (NTSB p.18, p.20). Two manufacturing reasons for the short blade life also only hold after metallurgical analysis: carbon contamination on the cavity surface accounts for about two thirds of the total difference, and a contour discontinuity left by machining for about one third, the latter raising local steady stress by 30% and cutting fatigue life by 50% (NTSB p.4, pp. 18–19).

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What happened: At 13:09 the fan blade fractured across its full span. The inlet cowl lip, the fan cowl support beam and thrust reverser components separated, and debris fell over about 40 acres that included a park and a residential area; one vehicle and one house were damaged, and the 239 people on board were unhurt, as was everyone on the ground (NTSB p.1, p.7). Blade 18, next to 19, fractured about 25 inches above the platform with overload features; the remaining blades were intact but their tips were bent against the direction of rotation (NTSB p.14). The fire warning came seconds after the fracture, and both fire bottles discharged without putting the fire out. K-flange bolts failed under the high dynamic loads, hot compressed air escaped and ignited leaking

high-pressure fuel, and the lower thrust reverser strut area burned through about 6 to 9 minutes after the fracture, while the airworthiness requirement for that area is at least 15 minutes of fire resistance (NTSB pp. 2–3). The fire had a second, independent path: the engine-driven hydraulic pump supply shutoff valve did not close as commanded because of silicone grease contamination on the motor brushes and commutator, and hydraulic fluid leaked into the engine compartment and fed the fire (NTSB p.3). Certification standards do not allow the inlet cowl and fan cowl to separate when a blade is released. Boeing changed the inlet cowl structure, adding metal to raise strength and ductility, and the FAA mandated the change by airworthiness directives 2022-06-10 and 2022-06-11 (NTSB pp. 1–2). Back to the three questions. The most common answer is "re-test the signal", but the problem here was not a reading that drifted. It was two indications that nobody chased. The item most easily missed is the number 6,500, because it decided that between April 2016 and the failure there would be no mandatory second look. The third question lands on the filing. The noise explanation could stand, but the entry belonged in "remediation by date" with a review deadline, and what it got was a one-time closure (NTSB p.3, p.20).

Options ruled out: Strip the coating and re-spray for a second TAI. NDIP 1065 says so; it was not done (NTSB p.20). Escalate the ambiguous indication to a team review. NDIP 1065 says so; it was not done (NTSB p.20). Treat the indication as a suspected crack and remove the blade. No record shows it was considered. Register the indication as a trackable item with a re-check required before the next overhaul. The NDIP of the time offered two paths only, redo and escalate, with no tracking tier (NTSB p.20). Shorten the inspection interval for this blade type. After the 2018 event only the procedure document changed and the interval stayed where it was, until airworthiness directive 2022-06-09 on 15 April 2022 moved TAI from 6,500 cycles to 1,000 (NTSB p.4, p.20). Screen the fleet for the

manufacturing anomaly. Pratt & Whitney moved to a dedicated clean argon system in 1997, the failed blade was hot-formed in 1994, and no screening aimed at pre-1997 blades appears (NTSB p. 19).

Early signals: ① April 2016: the software flagged two low-amplitude indications, both inside the crack origin area (NTSB p.20). Suppression reason: judged to be camera noise or cavity particles, a "known benign cause". ② 13 February 2018: a full-span fracture on an engine of the same type (NTSB p.20). Suppression reason: handled as a procedure revision, which changed the interpretation text and not this engine's record; the 2016 images were pulled and the original call was upheld. ③ The first PW4000 fan blade separation came in 2010, when about 50% of the airfoil came away; 2018 brought one full-span fracture; this event is the fourth of its type (NTSB p.2). Suppression reason: each handled as its own case, and the four never merged into a trend. ④ From the first confirmed cracked blade in December 2004 to 31 January 2023, 17 blades were confirmed cracked, 7 of them found by TAI, 1 visually and 9 by the new ultrasonic inspection (NTSB p.21). Suppression reason: found meant replaced, and the count never entered the interval assessment. ⑤ Two on the manufacturing side, cavity surface carbon contamination and contour discontinuity (NTSB pp. 18–19). Suppression reason: part of the manufacturer's process history, outside the operator's inspection scope.

The point of escalation: After the fracture of another PW4077 on 13 February 2018, the thermal data from April 2016 were pulled out for review, the reviewers accepted the original interpretation, and the blade stayed in service (NTSB p.20; the report gives no date for the re-review itself). The engine that fractured in 2018 was of the same type, also a PW4077, also on a United 777-222 (NTSB p.20). After that step the same images had been read twice with the same result. The only chance left to open the question was the 6,500-

cycle overhaul, and the blade broke at 2,979 cycles before reaching it (NTSB p.3, p.19).

Failure chain: Technical layer: a fatigue crack initiated on the internal cavity radius of blade 19 and grew stably from 0.063 inch to 0.199 inch over about 3,150 cycles, then the blade separated across its full span; the blade's own life was short because of carbon contamination and the contour discontinuity (NTSB p.18). The growth rate from detectable size to through-wall thickness was relatively stable and predictable across this fracture and the earlier ones, because the initiation sites were all at the surface (NTSB p.4). The fan case did contain the blade fragments, but the displacement wave from the impact deflected the case into the fan cowl hinge, and the damage moved out into the nacelle (NTSB p.1). Damage also reached the nacelle by a second route: the inlet cowl and aft bulkhead of the event engine were carbon-fiber honeycomb sandwich, and simulation showed it could not dissipate a blade release the way the aluminum cowl used in certification testing could. Predicted displacement for the 2018 event was 0.55 inch, past the sandwich's compressive rupture strain, while the aluminum certification article was predicted at 0.47 inch with only local yielding (NTSB pp. 20–21). Organizational layer: two low-amplitude indications filed as extraneous, with the second TAI or team review that the procedure required never happening and no record of why (NTSB p.3, p.20); a 6,500-cycle interval that did not match the blade's actual life, named by NTSB in its probable cause as insufficient inspection interval frequency (NTSB p.4). Incentive layer: a second TAI means stripping and re-spraying and consumes overhaul hours, and a shorter interval means more inspections across the fleet. Both costs land in the current period, and the crack risk lands five years out (NTSB pp. 20–21). NTSB's findings read "scheduled maintenance inspections: inspection inadequate" and "organizational issues: manufacturer equipment design" (NTSB p.5). The break sits in the organizational layer,

where a filing action turned an entry the procedure said needed a re-check into a one-time closure.

Minimum counterfactual: April 2016: run the second TAI that NDIP 1065 specifies on those two low-amplitude indications, after stripping and re-spraying. One action. The procedure existed, the equipment existed, the criteria existed, and NTSB's words are that the blade "should have" received a second TAI or a team review (NTSB p.3, p.20). At the stable growth rate of 3,150 cycles, there was at least one chance to see it within five years.

Outcome and sources: Nobody died. The inlet cowl and fan cowl support beam separated, the nacelle burned, and property on the ground was damaged. The FAA issued emergency airworthiness directive 2021-05-51, grounding every 777-200 and 777-300 powered by PW4000 112-inch engines; from 15 April 2022 TAI runs every 1,000 cycles and ultrasonic inspection every 275 cycles (NTSB p.4, p.21); Pratt & Whitney's service bulletin also required an immediate fleet-wide TAI, with the new ultrasonic inspection covering three high-risk areas (NTSB p.21). NTSB's probable cause: fatigue failure of the right engine fan blade, with the blade inspection's failure to identify low-level crack indications and the manufacturer's insufficient inspection interval frequency as contributing factors (NTSB p.4). Primary source: NTSB, Aviation Investigation Final Report DCA21FA085, published 7 September 2023, link at Appendix B av-006.

Where this sits against the mechanism: No relaxation in this case was pushed in bad faith, and no explanation in it was absurd. The whole work of the mechanism was done by one default action, explain and close. The opening against it is not a better class of explanation. It is Device 1's rule that every entry carry one of three states.

Case 6 (Nuclear Power): Tagging With No Verification (condensed)

nu-002. Unit 3 of the Gravelines nuclear plant, France. The ASNR event notification (October 2026, this book read the page): a tagging error disabled an existing compensatory measure, the affected equipment was unavailable for 7 hours 45 minutes, and the event was rated INES 1. Tagging is one of the most frequent routine actions on a site: performed several times a day, with no consequences for years, its execution carried by custom. From this chapter's angle the decision point is the verification step after the tag is applied. No second pair of eyes checks the correspondence between the tag and the compensatory measure, and the error walks into an undefended interval. Minimum counterfactual: make independent verification mandatory once tagging is complete. Primary source: ASNR, 2026, link at Appendix B nu-002.

Set the six cases side by side and the common structure appears. None of them lacked a procedure. Spaceflight had a classification clause, chemicals had an SCBA response clause, aviation had established inspection intervals, finance had an alert handling process, nuclear power had a tagging system. What all six lacked is the same thing: a mechanism that forces the procedure and reality to meet. Foam came off, alarms sounded, signals arrived, alerts piled up, tags were hung, and none of those meetings produced a review action, because the entry's status, the alarm's calibration and the signal's handling had already supplied the answer "no further thought needed". Normalization is that "no further thought needed" accumulating item by item in the archive. The signal list and the defenses in the two sections that follow are aimed entirely at that accumulation.

One more observation belongs at the end of the case set. In all six cases the procedure text was in place before the event, and a paper compliance check would have looked at it and seen nothing. A

checklist verifies that a procedure exists. The three-state register verifies that the procedure met reality. Separating those two jobs is the first operational distinction this chapter offers.

11.4 What Can Be Seen Earliest

Weak signals, each with the reason usually used to push it aside. The list runs from earliest to latest.

1. The register of same-class deviations is missing, or every entry in it carries the status "explained". The reason it is pushed aside: "we have already dealt with that."
2. The number of anomaly reports falls while the number of problems raised at the front line stays the same. Reason: "safety is improving." (A falling report count is more likely to mean the reporters gave up.)
3. The wording of an inspection standard or an acceptance clause is rewritten from "must" to "in principle". Reason: "for workability."
4. "We did it this way before" enters meeting minutes for the first time, as a basis for a decision. Reason: "respect for practical experience."
5. A request to relax a limit, an interval or a threshold comes from the user, and the review is carried out by the user's usual collaborator. Reason: "standards should fit reality."
6. A newcomer's question about existing practice is answered with a statement of custom rather than a statement of grounds. Reason: "old hands training newcomers is the best training there is."
7. The same problem has two or more formal explanations in circulation, and they do not cite each other. Reason: "a difference of wording, nothing more."

8. “No consequence so far” is used as safety evidence for the first time, rather than as a statement that the data are insuffi‑cient. Reason: “let the results speak.”

Signal 8 deserves its own paragraph. It marks the fully formed stage. By the time it appears, the mechanism’s output is a system of argument with “nothing happened” as its evidence, and any later challenge has to refute that system before it reaches the facts. Most misuse of black swan rhetoric (Chapter 41) grows on this foundation.

Put the eight signals back against the cases in this chapter and their landing points show. Signal 1, a missing register or one where everything reads “explained”, lands on fi-013: the independent test had already written the alert backlog up as a finding, and the entry never moved into a remediation status. Signal 4, “we did it this way before” written into minutes, lands on mi-001: the STS-113 review formally recorded the foam loss flight rationale as safe to fly (CAIB p.126). Signal 5, an application by the user re‑viewed by a collaborator, lands on av-034: the lubrication and end-play check intervals were extended on the company’s application with FAA approval, on a single information source (NTSB AAR-02/01, executive summary p.4). Signal 8, absence of con‑sequence used as safety evidence, lands in two places at once. In mi-001 it is twenty years of foam loss without a catastrophe read as the system working, and the per-flight table at CAIB p.124 is the ledger of that argument. In av-006 it is every normal inspection result read as support for the explanation, once the anomaly had been explained and closed.

There is an order of use worth writing down. Check the register first, then talk to people. Half the eight signals (1, 3, 5 and 8) can be confirmed from the register alone; the other half need interviews and site observation. Run it the other way round and the inter‑viewee’s “explained” narrative arrives first and explains away

the fork that was sitting in plain sight on the register. The order of the audit is itself part of the defense.

Who runs the audit matters too. The best candidates often sit outside the organization chart: internal audit, a third-party assessor, or a newcomer less than six months in post. What those three share is the absence of stock memory about the "explained" archive. They ask the questions that a stock reader has stopped asking. That also explains why the same register gets quieter the older the organization is: every reader has already accepted the "explained" entries. The list of questions a newcomer raises in the first year is usually the best index of what the register actually contains.

11.5 How to Prevent It

The four devices share one principle: take the filing action out of the hands of the explanation and give it to the register and the signature.

Device 1, the three-state deviation register. An entry may carry one of three statuses only: eliminated (with evidence of elimination), remediation by date (with a date and an owner), running at risk (with a risk assessment and a signature). A fourth status, "explained", is prohibited. Explanations may go in the remarks field, but the status must land in one of the three. Device 1 in Chapter 8 is the same register used in an escalation setting.

Device 2, margin quantified. Any statement that "there is still margin" must be written as a number together with its data source, and the source may not be the consequence-free history itself. Evidence for hypothesis B cannot be used to prove hypothesis A. A margin statement with no numeric source counts as no margin.

Device 3, the exposure audit. Every running-at-risk item carries two numbers: the current exposure count, and the exposure increment added by each release. The audit has one question. Is the operation of this deviation a finite exposure that we authorized, or an infinite one? This is the register form of the installment-type irreversibility in Chapter 7.

Device 4, two signatures for any relaxation. Approval to relax a standard, extend an interval or waive an inspection requires two signatures: the head of the requesting party, and a representative of the risk-bearing party who has no interest in the deviation and whose appraisal is not affected by current output. The FAA approval in the Alaska 261 case could not act as independent gatekeeping because its information source was single. The two-signature rule pins the information source down: the material required for approval must include inspection or test records from outside the user.

The weak signals of normalization share one cross-industry property. None of them needs technical expertise to see. All of them need a register. An auditor with no industry experience who is handed a deviation register can count the share of entries marked "explained" and make a reliable judgment about how far the mechanism has run. Defending against normalization therefore does not require industry experts. It requires an honest register and someone willing to read it.

The signature requirements across the four devices add up like this. The register is maintained by a safety engineer, not a production engineer. Margin figures are written by the user's engineer and reviewed by the safety engineer. Exposure increment reports go directly to the technical director. Relaxation approvals carry the signature of both the user and the risk-bearing party. What the four roles share is the absence of any interest in current output, and that is the organizational base the whole defense stands on.

Each device has one most direct application in the cases of this chapter. Device 1 on fi-013: on the day the independent test report arrives, the alert backlog entry must land in one of the three states; on the facts it can only go to "remediation by date", with a date and an owner attached, and the quarterly audit after that looks at one thing only, whether this entry has closed. Device 2 on av-034: a statement in an extension request that lubrication remains adequate must carry a measured jackscrew lubrication condition and measured end play; without the measurement the request returns automatically. Device 3 on av-006: had the 2016 anomaly been registered as "remediation by date", the entry would have rolled into every inspection with a review deadline attached, and the blade that broke in 2021 would have had several chances to be looked at again. Device 4 on mi-001: had the STS-113 review run on two signatures, the second pen signing the safe-to-fly rationale would have had to come from a party with no interest in the launch schedule. CAIB calls that review a "key decision" (p.126), which shows how completely the second pen was missing.

None of the four devices can be installed once the drift has matured. Registers, numbers, audits and second signatures are slow variables, and they work only if the organization configures them during the consequence-free period. Install them as the consequence approaches and the mechanism has already passed into the routine phase; what the defenses then face is the whole "explained" archive. That is why these items belong in a standing budget rather than in a special remediation project. Slow variables need a standing carrier.

The failure modes of the four devices are worth naming, because each one has a dignified way to die. The three-state register dies when a "remediation by date" comes due and nobody asks: the entry rolls over, and after a few rolls it is indistinguishable from "explained". Margin quantification dies in the degradation of its

source: the first figure is measured, the second is copied from the first, the third is an experienced guess, and nobody notices that the wording in the source column has moved from “measured” to “estimated”. The exposure audit dies at the manual step: logging the increment depends on the releaser jotting it down, and when things get busy only the total keeps rising. The two-signature rule dies in the relationalization of the second pen: the risk-bearing representative sits with the user long enough that the independent check before signing shrinks into a courtesy initial. One antidote covers all four. Each quarter, draw one closed entry at random and re-audit it backwards, publish the draw, and put the result on the first page of the next quarterly report. Devices do not stay honest by themselves. Someone has to poke them.

11.6 Boundaries with Adjacent Mechanisms

Against practical drift (Chapter 15). Drift describes the fork between practice and rule, which is the question of how the work is done. Normalization describes the fork being redefined as acceptable, which is the question of how it is ruled on. Drift can run without normalization: drift that a register is watching stays in “remediation by date”. Normalization usually takes drift as its raw material, but it needs the second step, explain and file. Where both are present, find the fork point first, then find out who filed it.

Against margin erosion (Chapter 25). Eating into margin is an active reduction, a cut, and it leaves a register to check. Normalization is an interpretive slide, carried out in speech, and it usually leaves no register. A single approval to extend an interval is both a cut and a normalization, and the carrier decides which chapter owns it. What lands on paper belongs to Chapter 25. What lives in the tone of a meeting and in custom belongs here.

Against narrative before evidence (Chapter 13). Narrative operates after the conclusion is public: commitments made outward, sides taken inward. Normalization operates in the internal archive layer: filing, classification. The two usually appear as a pair, normalization inside first and narrative outside afterwards. The discriminator is audience. An action filed inward belongs to this chapter. An action promised outward belongs to Chapter 13.

11.7 Five Self-Checks

1. In your process's deviation register, how many entries carry the status "explained"? Pick one. Is its deviation still running now?
2. Where were you the last time you said "we have always done it this way"? Was there anything in the other person's objection that deserved a register entry?
3. Of the relaxations you have signed (intervals, limits, waivers), how many rested on material that included records from outside the user?
4. In your organization, who can state the current cumulative exposure count for a given running-at-risk item? If nobody can, in whose hands is the exposure accumulating?
5. Take a customary practice you have run for more than two years with no consequence, and walk it through criteria ①②③. If all three hit, what does this chapter call it?

Chapter 12 Goal Displacement; Goodhart's Law

> What this chapter covers: how a metric quietly takes the place of the goal it stands for; three criteria for telling that the mechanism has started; the shape it takes on the ground in four domains; eight weak signals; and four lines of defence for whoever sets a metric. When to turn back: when you are about to set a new KPI, when a polished metrics report lands on your desk, or when something "looks good in the numbers but feels wrong". Estimated reading time: 50 minutes.

12.1 What It Is

A metric is always a stand-in. The goals that matter, safety, health, whether anyone actually learned the material, whether trust is real, whether the assets are worth what the books say, are for the most part not measurable in themselves. So the organization measures something countable and installs it where the goal should be. Goal displacement names the moment the substitute takes over: management attention settles on the countable quantity, the organization starts optimizing the quantity itself, and the optimizing stops improving the thing the quantity stood for. Goodhart's law in its common form: when a measure becomes a target, it ceases to be a good measure. Charles Goodhart said something narrower in a 1975 lecture at the Bank of England; the familiar sentence was written by Marilyn Strathern ('Improving ratings', European Review 5(3): 305–321, 1997, bibliographic record verified via Crossref). The causal direction is the part people get wrong. The measure did not degrade. The behaviour around it changed.

Three criteria tell you the mechanism is running. Any one of them is enough:

1. Optimizing actions appear that are specific to the assessed quantity, and their cost is spent on the number rather than on the work.
2. Those actions do not improve the goal behind the number, and may consume it.
3. Someone who says so pays for it, in the appraisal or in the meeting.

The three are not parallel. Criterion 1 is the starting point. Any assessed metric attracts optimizing actions; that is rational behaviour, not a disease. Criterion 2 is the watershed, the point where the optimizing action comes off the goal it was supposed to serve. Criterion 3 is the freeze: the separation has been noticed, the noticer has been punished, and the organization's priorities are now on the record. How fast the three arrive depends on how quickly the distance between metric and goal becomes visible. The greater the distance and the later the consequence, the longer the window in which nothing appears to be wrong.

The occasion of Goodhart's observation is worth recording. In 1975, discussing monetary aggregate targets at the Bank of England, he put the point as a piece of regulatory shop talk: any observed statistical regularity will tend to collapse once pressure is placed on it for control purposes. Strathern later rewrote it into the sentence everyone cites. The difference between the two versions usually gets skipped. The original describes a game between the controller and the controlled: the controlled party knows what is being watched and moves real behaviour outside the observation point. The familiar version describes a divergence between measure and goal. This book uses both, because both are happening inside organizations. The first is gameplay, what the assessed do to the metric. The second is drift, what the metric's meaning does to

itself over time. Of the four defences in section 12.5, the first two watch gameplay and the last two watch drift.

Donald Campbell, a sociologist, said the same thing a few years before Goodhart, from the angle of educational testing: the more any quantitative indicator is used for social decision-making, the more it will be subject to corruption pressures and the more it will distort and corrupt the social processes it is meant to monitor. Read together, the two sentences show the mechanism ignoring systems and industries alike. Wherever measurement is used to decide people or resources, pressure appears. You do not need to remember who said it first. What you need is the conditional both sentences share: used for decisions. A metric on a wall that nobody looks at provokes no behaviour at all. The entire force of goal dis‑placement comes from the metric's wiring to appraisal, budget and personnel.

Take a metric apart and it has four parts: the definition (what counts as one overdue loan), the sample (which projects enter the rating), the frequency (how often it is measured), and the present‑ation (what type size, which page). Each part has room for play, and the professionalism of the play rises through that list. Changing the presentation is the crudest move: bad numbers set in small type. Changing the frequency sits in the middle, measuring before the review and stopping after it. Changing the sample is subtler, moving the projects that are about to fail out of the sample. Chan‑ging the definition is the professional's move, redefining the start date of "overdue". Goodhart effects do not arrive in step across the four. Play with the presentation and the goal may be un‑harmed. Play with the definition and the link between metric and goal is usually already severed. Defences can be laid out to match: presentation and frequency get caught by paired indicators (Device 2), sample and definition get stopped by the register of definitions (Device 4). Case 1 in this chapter works on the definition layer, on

whether a rescinded loan counts as a loss, and that is why it is misleading rather than merely aggressive accounting. A cut at the definition layer carries a disclosure obligation with it.

Readers of a metrics report come in three kinds, and each has a different failure to guard against. The first reads the trend: the curve is up, so things are fine. Trend is the reading most easily polluted, because gameplay manufactures trends by nature. The second reads rank, its position against peers. Rank moves an or‑ganization's attention from the goal to the competition, and goal displacement evolves fastest in a ranking context, because once gameplay spreads among peers, not playing becomes the way to fall behind. The third reads anomalies, caring only about moments of significant deviation. The anomaly reader is the least sensitive to mature gameplay, since mature gameplay keeps numbers inside the normal band and misses every anomaly detector by design. No one reading is sufficient. The workable combination is to have the three readers exchange question lists on a schedule: the trend reader asking the anomaly reader "what has felt wrong lately" usually gets further than any audit report.

One step down the same four parts yields three conditions a metric needs at the moment it is set. It must be reconcilable: the met‑ric's data has to reconcile against an independent source, system logs against reports, an external list against an internal ledger. A metric that will not reconcile has no standing to enter an appraisal. It needs an outside comparison: a comparison series, not collected by the assessed party, designated in advance as the reference for when the metric distorts. And its definition history must be on file: changes to definition, sample, frequency and presentation kept in one place, so the metric can be replayed through time. A metric failing all three can still be used, though only for self-reference, never for appraisal or resource allocation. These three are the four defences projected onto the moment of setting: reconcilability is

Device 2, the outside comparison is Device 3, the definition history is Device 4.

12.2 How It Grows

Three conditions generate it.

Condition one: the goal cannot be measured directly. A goal that can be counted needs no proxy. "How many conforming parts did we build this month" can be counted. "How good were this month's decisions" needs a stand-in, say the share of decisions revisited six months later. The further the proxy sits from the real goal, the more room the optimization has to skew.

Condition two: the levels above read only numbers. Management's bandwidth is finite, and reporting across levels compresses into numbers by default (Chapter 16 gives that attenuation its quantitative form). Once the basis of a decision converges on a single number, optimizing that number is the rational choice at every desk in the organization. Nobody breaks a rule. Every level completes what it is assessed on. That is Chapter 17's misaligned incentives arriving as a technical implementation.

Condition three: the assessment cycle is shorter than the real cycle. Loan quality runs in years and the report in quarters. Student growth runs in years and the exam in semesters. Where the two cycles differ, the metric and the goal come apart, because inside the gap the return on optimizing the proxy is certain while the cost of exposure is deferred.

A fourth condition is implicit and usually skipped: the neutral appearance of a number. Numbers do not look like interests. They look like facts. When two departments fight over a metric's definition, the scene presents as a technical discussion, how the sample is drawn, what sits in the denominator, and nobody has to

say out loud that this decides their year-end rating. The neutral surface keeps optimizing actions out of moral view. Nobody believes they are cheating; everyone is "getting the data right". This is why goal displacement is more common than open fraud. Fraud needs someone willing to lie. Metric optimization needs ordinary people doing their jobs carefully.

The spread of gameplay is the fastest step in the whole mechanism. Once an optimizing move has been invented, it needs no training to travel. The month-end push, the selective filing, the trick of moving an ugly quantity out of the denominator: these move with rotation, with meetings where one department watches another, with the annual reinterpretation of the assessment. Gameplay is the fastest-travelling knowledge in organizational memory, because it is the only knowledge that pays immediately on acquisition. Design the defence on the assumption that gameplay exists. A metrics system designed in the hope that nobody finds the game fails on the day it goes live.

Condition three has a seasonal form that deserves its own paragraph. Gameplay rises and falls with the assessment calendar. The month-end push, the quarter-end surge, the fiscal-year-end adjustment of a definition are direct projections of the cycle gap. That shape hands an auditor a time window. Check at the trough of the calendar and you see normal operations; check at the peak and you see the game. Sampling points for a metric audit should therefore be secret and random, with the days around the assessment date always included. The practice is cheap. What it encodes is the insight that gameplay is periodic, and only a periodic method will see it.

Composite metrics need a warning of their own. Weighting several sub-metrics into one score is the standard answer to thin management bandwidth, and it widens both entrances to this mechanism at once. Weights are the top layer of the definition: whoever sets

them sets the cheapest path for gameplay, and weight changes usually go unlogged, with a lower bar than changing a definition. Composition also hides compensation. A deteriorating sub-metric can be fully cancelled by a flattering one, so an improving safety score may come precisely from a falling count of reported incidents. Three minimum requirements apply to any composite. Weight changes go through the definition register. Every sub-metric is displayed separately. Any jump in the composite can be traced to a named sub-metric. A composite that fails all three is worse to read than its parts.

12.3 The Shape in Four Domains

Case 1 (Finance): The Delinquent Loans That Were Moved Off the Books

fi-001.

When and where: May 2019 (the special handling programme launched), August 2019 (the delinquency programme launched), through February 2021 (the end of the period the SEC treats as relevant), with rescissions in practice continuing to the end of August 2020. Orlando, Florida, at the headquarters of Travel + Leisure Co. The complaint was filed on 30 September 2026 in the United States District Court for the Southern District of Florida, case No. 26-cv-62760, with Litigation Release No. 26657 issued the same day.

What and who: Travel + Leisure Co., formerly Wyndham Destinations, Inc., renamed after its spin-off from Wyndham Worldwide on 31 May 2018 and renamed again to its present name on 17 February 2021, the largest timeshare company in the world, with 797,000 owners and more than 280 resort clubs; since 2014 the company has financed more than half of its timeshare sales itself every year (complaint p.6). The posts involved: senior management

responsible for finance and investor relations, who confirmed the rescission target count every quarter (p.10); the sales rescission committee, which reviewed rescission requests beyond the statutory rescission period; the operations departments that executed the rescission coding; and the financial planning staff who built the monthly forecast.

What they were doing: Through late 2018, third-party exit companies grew active, charging owners to terminate contracts after the statutory deadline and often advising them to stop paying outright (complaint p.7). The company tightened rescission approvals and rejected any customer suspected of working with such a firm (p.7). The outcome ran against the tightening. In the first quarter of 2019 delinquency and the loan loss provision rate jumped together, the provision rate rising from 20.4% in the same quarter a year earlier to 22.5% (p.8). On the 1 May 2019 first-quarter earnings call, five analysts pressed on that metric in succession, and management promised on the spot to stop the year-over-year increase and hold the 2019 provision rate at its 2018 level (p.8). That same month the company started a special handling programme that pulled delinquent or defaulted timeshare loans out of the portfolio and rescinded them in bulk, known internally by three names: "late-stage delinquency" rescission, the "late-stage rescission programme", and the "owner care programme" (pp. 9–10). In August it added a further delinquency programme, covering loans tied to customer complaints (p.2). The special handling programme screened the longest delinquency first, ahead of foreclosure (p.10). New rescissions stopped at the end of August 2020 because timeshare sales were falling under the pandemic (p.11). Every loan rescinded in the relevant period fell outside the statutory rescission period, five to seven days in the states concerned, which the company called "late-stage rescission" internally (p.3, p.9).

The decision point:

1. 1 May 2019, the first-quarter earnings call. Management turned the loan loss provision rate into a public commitment: stop the year-over-year increase, hold the 2019 guidance (complaint p.8). One quarter earlier the metric had appeared in the results at 22.5% against 20.4% (p.8). That sentence converted an internal metric into a public target, and every quarterly number after it had to answer to the sentence.

2. From May 2019, the monthly forecast. The company wrote a target count into the monthly forecast for how many delinquent or defaulted loans had to be rescinded, calculated backwards: to reach the provision guidance, a given number of defaults had to leave the portfolio (p.10). Senior management confirmed the target count at least once a quarter and then directed the relevant departments to execute it (p.10). The calculation had two inputs, the 2019 guidance holding level with 2018 and the internal 2020 target the company had set for itself (p.10). The carrier was a forecast spreadsheet rather than a set of minutes, and the spreadsheet ran from May 2019 to the end of 2020 (p. 10). It carried counts and balances. It carried no loan identities, and no comparison column for what the rate would have been without the rescissions.

3. August 2019, the weekly meeting of the sales rescission com - mittee. The delinquency programme compiled delinquent and defaulted loans into a "delinquency agenda" spreadsheet circulated before the meeting; every line on it was approved, with no review loan by loan (p.11). The committee's original function was to judge, one loan at a time, whether a request to rescind past the deadline should stand. After this step the judgement was a signature on a whole table. The committee still met and the agenda still went out.

4. 30 October 2019 to February 2021, seven periodic reports and five calls. The carriers the complaint lists: the MD&A of the 10-Q for the five quarters from 30 September 2019 to 30 September 2020; the quarterly results releases issued with 8-Ks; the MD&A of the 10-K for fiscal 2019 and fiscal 2020; and the calls for the third and fourth quarters of 2019, the third and fourth quarters of 2020, and the full year 2019 (p.12). The improvement was attributed to the customer care phone line, to getting owners back on vacation, to tighter underwriting, and to improvements in the booking system (p.13).

What was visible then: the loan-by-loan list of rescinded loans, with days delinquent and default status; the thresholds were the company's own, 30 days delinquent and 120 days in default (complaint p.7). The list held one particular class of entry: loans coded for rescission within a few days of the 120-day mark (p.10). The delinquency programme's table was grouped by customer complaint, with rescission decoupled from whether a complaint had merit and tied only to delinquency status (p.2, p.11). The "delinquency agenda" spreadsheet and the committee's weekly agenda (p.11). The rescission target count inside the monthly forecast (p.10). The record of the five analysts' questions on 1 May 2019 (p.8). Analyst reports forwarded by investor relations, one stating that the provision rate had a material impact on the stock price (p.8). 10-Q and 8-K drafts, the provisioning worksheets, the call scripts (pp. 12–13). The company's self-set 2020 internal target sat in the same forecast set (p.10). One piece of background was also of record: investors and analysts treated the metric as a barometer of whether the company could hold off the third-party exit firms (pp. 11–12), and the company knew it (p.8). At the end of every quarter these materials sat on the same desk at the same time, a sheet saying how many loans had to go and a draft saying how much the metric had improved. One thing was already blank. The company forecast the effect of rescissions on the metric and

never tracked the programme's actual effect on the financial statements (p.2, p.10), so nothing on that desk could answer how much of the improvement came from the rescissions. What became knowable only afterwards: the totals of more than 2,900 loans, roughly $77 million in balance, of which about $34 million was in default (p.3); the age profile, 44% over one year and 11% over two years (p.9); and the fact that in several quarters the programme moved both metrics by more than 5% (p.12).

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: on 30 September 2026 the SEC sued and settled on the same day. The company neither admitted nor denied the allegations, accepted a permanent injunction (Securities Act of 1933, 17(a)(2) and (a)(3); Exchange Act of 1934, 13(a) and 12b-20, 13a-1, 13a-11, 13a-13), and paid a civil penalty of $975,000 (Litigation Release 26657). The relevant period, October 2019 to February 2021, is not the same stretch as the rescission operations, May 2019 to December 2020, and the complaint writes the two apart (p.1, p.3). During the relevant period the company raised about $1.65 billion through three private note issuances, and issued $13.5 million of common stock for a 2019 acquisition and $96.8 million of common stock for an employee stock purchase plan (complaint p.14). Against the three questions: the most common answer to the first is "have legal confirm whether the right to rescind is available". The right was available. The complaint alleges non-disclosure, and

the rescissions themselves are not among the charges (pp. 1–3), so that answer is a layer off the mark: an available right and a disclosable fact are two different things in the complaint (p.3). The item most easily missed under the second question is the monthly forecast, the only sheet in the case that carries the target and the action in one place. The checkpoint for the third belongs at the first monthly forecast review in May 2019: check whether that line of numbers has a disclosure treatment attached to it, and who signs the sheet.

Options ruled out: disclose both programmes and their quantitative effect on the provision metric (not done; every allegation in the complaint rests on this omission, complaint pp. 1–3). Leave the rescinded loans in the portfolio and provision for them on the normal basis (not chosen; rescission and charge-off happened in the same place, leaving no trace in the accounts, p.3). Have the sales rescission committee review the delinquency agenda loan by loan (the process existed; the delinquency programme routed around it, p.11). Track the programme's actual effect on the financial statements (not done, p.2, p.10). Use the absolute balance of delinquent loans as the external pair for the provision rate (not done; no disclosure in the complaint sets balance beside rate). Confine rescissions to the statutory five-to-seven-day window in each state (not done; every loan rescinded in the relevant period was past it, p.3). Obtain a written opinion from an outside auditor on the accounting for the rescissions (the complaint does not mention any review opinion from an outside auditor). The complaint records behaviour, not the reasons these options were turned down. What can be checked is that none of them happened, and that each carried a cost legible on paper.

Early signals:

1. 1 May 2019: five analysts pressed on the provision rate in the same call (complaint p.8). The promise to stop the rise and hold

guidance absorbed the pressure, and nobody asked how the company meant to stop it.

2. First quarter 2019: the provision rate at 22.5%, up two percentage points year over year, a move the company itself acknowledged as meaningful to investors (p.8). It was treated as an operating problem and handled by operating means: take loans out of the portfolio, rather than say in the disclosure that the definition had changed.

3. An analyst report forwarded by investor relations, stating that the provision rate had a material impact on the stock price (p. 8). The sentence entered the internal channel and never entered a filed document.

4. August 2019: the "delinquency agenda" went live and rescission moved from loan-by-loan approval to approval of a whole table (p.11). The simplification looked like an efficiency gain. It is the only written procedural change in the case, and after it no document records why any individual loan was approved.

5. The age of the rescinded loans, 44% over one year and 11% over two (p.9). These were old accounts. Rescinding them did not improve the quality of new sales; it changed the numerator and the denominator of the current period.

Of the five, the earliest predates the programme's launch and the latest appears in the results release of 30 October 2019 (p.8, p.13).

The point of escalation: the "delinquency agenda" mechanism of August 2019. Judgement loan by loan was replaced by a signature on the whole table, and after that no rescission carried a record of whether it should have been approved (complaint p.11). In the same quarter the first rescissions entered the 10-Q and 8-K treatment. By 26 February 2020 the company described the full-year provision rate as "flat" and claimed it had met guidance (p. 13), and the fiscal 2019 10-K carried the same treatment into the

annual report (p.12). On 6 May 2020 it said the metric "would have been" the fourth consecutive quarter of year-over-year improvement (p.13). By then the rescission had moved from an operating action to a component of the compliance story. From May 2019 to at least December 2020 the cumulative count passed 2,900 (p.3), with the target reset on the forecast every quarter (p.10): each quarter stacked new rescissions on the last, and backing out would have meant unwinding a stack of spreadsheets. After new rescissions stopped in August 2020, the disclosure did not change its story; the fiscal 2020 10-K and the full-year call still described improvement (p.12). Reversing course would have required restating three years of external statements at once. That is what "cannot turn back" looks like in practice.

Failure chain:

Technical layer: the definition. Whether a loan counted as on the books, and whether it stayed in the portfolio after rescission, was decided by the company's own accounting treatment. Once coded as rescinded, revenue, provision already taken and default balance reversed together, and the accounts read as though the loan had never existed (complaint p.3). The delinquency and default thresholds were the company's own as well, 30 days and 120 days (p.7), and rescissions clustered just short of 120 days (p.10).

Organizational layer: the target count went from the monthly forecast to the executing departments; the delinquency programme bypassed the sales rescission committee's review loan by loan; and the company did not track the programme's actual effect, so no internal document could answer how much of the improvement came from rescissions (p.2, p.10). Of the three layers, only the accounting entries kept a complete record, and what they record is "never happened". What can reconstruct the history is the rescission coding table and the monthly forecast, and neither enters the disclosure process.

Incentive layer: the provision rate sat in external guidance, analysts called it material to the share price, and the May 2019 call turned it into a commitment (p.8). In the same period the third-party exit firms were pushing defaults up, and the company itself used the metric as a barometer of whether it was holding them off (p.7).

At the boundary: the definition layer supplied the action, move it out; the incentive layer supplied the number, how many. The two met in one row of the monthly forecast. The break is that row. It was an operating target and a disclosure treatment at once, and only the first was written into a process.

Minimum counterfactual: on the first monthly forecast of May 2019, add a line reading "provision rate if this month's rescinded loans had stayed in the portfolio", and carry that line unchanged into the MD&A of that quarter's 10-Q. One action, in the month the programme started, needing no luck: the company was already forecasting the effect (complaint p.2), and the data sat on the same sheet. The 10-Q is prepared every quarter and the MD&A written every quarter, so the line has a landing place already built (p.12). The cost is one line of text and one more signature. The line does not stop the rescissions. It makes "what is the metric after the rescissions" into a question somebody has to answer. Answer it and a disclosure obligation exists; leave it blank and the sheet carries a gap. The gatekeeper is the existing quarterly reporting sequence, and no new committee is needed. Fixed afterwards instead, the job becomes a comparison series across five quarters (p. 12): one line turns into a restatement, and the restatement then has to explain why the first five quarters said nothing.

Outcome and sources: civil penalty of $975,000, permanent injunction, neither admit nor deny (Litigation Release No. 26657 / AAER-4603, 30 September 2026). Two primary documents, issued the same day, checking each other: SEC v. Travel + Leisure Co.,

No. 26-cv-62760 (S.D. Fla.), Complaint, 17 pages, paragraphs 1–50, giving fact and page number paragraph by paragraph; and Litigation Release No. 26657, giving the penalties and the list of statutes, a web page without pagination, cited by release number. The two agree on the penalty amount, the list of statutes and the period of conduct (LR text; complaint ¶1, ¶9). The enforcement release page carries a download link to the complaint PDF, which closes the citation chain. Link at Appendix B, fi-001.

Case 1 also repays a separate look at the width of five reporting quarters. At the end of the first quarter the rescission could still be explained as an accounting judgement. Repeated at the end of the second, the judgement was anchored by precedent. By the third quarter it was beginning to look like a process. By the fifth it was a routine component of the financial statements, and apart from the few people who knew all the facts, nobody in the organization still treated it as a decision. That is the temporal shape of this mechanism: the sense of doing something decays with repetition while the undisclosed amount accumulates. The point where the two curves open away from each other is where outside intervention is worth most and an internal brake costs most. A hard rule follows from it. Any accounting choice that affects external disclosure, executed for two consecutive reporting periods, triggers a mandatory outside review, with reviewers including a party from outside the auditors and the disclosure committee. Five quarters of width could have been compressed into half a quarter of argument.

Case 2 (Public policy): Ratings That Drifted from Actual Risk

po-003.

When and where: March 2025 (the baseline month for the risk documents) and April 2025 (the month of the rating snapshot), with the report published on 7 October 2026. Washington, D.C. Policy

and definitions were set by the Office of Management and Budget (OMB) Federal CIO office, the dashboard at itdashboard.gov was operated by the General Services Administration (GSA), and the ratings were submitted by the CIOs of 12 agencies.

What and who: OMB's Federal CIO office, which set the rating definitions and the update frequency; GSA's Office of Government-Wide Policy (OGP), which took over dashboard operations in March 2022 (GAO-27-108416 p.5) and handled user feedback, prioritization and releases on three-week iterations (p.25); the CIOs of 12 agencies, together with the CPIC teams, investment managers, project health assessment groups and chief technology officer offices that drafted ratings for them (pp. 43–48); and GAO, which ran the comparison at the request of the House Committee on Oversight and Government Reform (p.1).

What they were doing: OMB launched the federal IT dashboard in June 2009. In December 2014 FITARA wrote into law that agency CIOs classify major IT investments by risk, and required that an investment rated high risk for four consecutive quarters trigger a review (GAO-27-108416 p.1, p.9). The dashboard covered nearly 600 investments across 26 agencies (p.5). The page carried three classes of rating. Cost and schedule were computed by formula, with a variance of 30% or more red, 10% or more but under 30% yellow, and under 10% green (p.6). The CIO rating was filled in by a person on a 1 to 5 scale, 1 the highest risk and 5 the lowest, then mapped onto five colours (p.6). OMB supplied six factors: risk management, requirements management, contractor oversight, historical performance, human capital, and other (p.7, table 1). Update frequency was not specified in A-11 Section 55 of 25 July 2024; only the version of 29 August 2025 says "at least quarterly" (p.7, footnotes 16 and 17). GAO began the audit in April 2025, took the April snapshot, and used the March risk registers for comparison, on the reasoning that ratings in the month an audit opens are least dis-

turbed by the audit (pp. 3–4). The report summary offers an example: a system already behind schedule that still displays a stale "low risk" on the page may never receive the review it needs (report summary page). In 2015 GAO had already placed IT acquisition and operations management on the federal high-risk list, partly on the reliability of dashboard data (p.2).

The decision point:

1. April 2025, 53 ratings posted to the dashboard. How each was calculated differed by agency. Treasury added three scores (the operational risk rating, the bureau CIO's integer from 1 to 5, and the CPIC chair's judgement), passed the sum through an overall rating formula and rounded to an integer (p.46, table 17). Education scored the six factors, combined them into a raw score from 1 to 5, and had the CIO review it; where the CIO believed other factors applied, the "CIO discretionary rating" became the final rating outright (p.44). HHS used a self-assessment followed by a "discretionary risk rating and rationale" from the CIO (p.45). GSA ignored risk scores and set a level when any three of eight criteria were met (p.47). VA used a quantified score, 1 point per category out of 5 (p.47). In that same month GSA rated Login.gov "medium" and GAO rated it "high": three risks in the investment's risk register averaged "elevated", and technical issues plus an incomplete data protection implementation remained open as of March 2025 (p. 20).

2. April 2025, all six Department of Veterans Affairs investments rated optimistically: none of the six agreed with GAO, and GAO judged all six riskier (p.16, table 4). The health management platform was rated "low" by the agency and "high" by GAO; benefits appeals and benefits payment were rated "low", with GAO at "medium" and "elevated" respectively; customer relationship management and enterprise data services were

rated "low", with GAO at "medium" for both (pp. 18–19). Education had a single investment in the sample and was off by one level: the loan servicing system was rated "medium" by the agency and "elevated" by GAO, on the ground that 16 of the project's 20 applicable risks were "elevated" or "high" (p.20).

3. April 2025, 21 ratings older than the agency's own update cycle (p.22, table 9). By agency: Commerce 5 of 5 overdue, HHS 9 of 9, DHS 4 of 4, Transportation 1 of 8, GSA and Education 1 each (p.22). The three classes are not the same case. Commerce, Transportation and GSA had set monthly updates and had not updated in April. HHS and DHS had set quarterly and had not updated in the three months before April. Education had set semi-annual and had not updated in the preceding six months (p.22).

4. April 2026, OMB announced on the dashboard that the system would be retired and replaced, with agencies moving to "a streamlined state focused on statutorily required data". It gave no timetable and did not answer GAO's inquiries (p.2, p.4). GSA said the page was still open to the public and to authorized users as of May 2026 (p.2).

What was visible then: each investment's risk register in its March 2025 version, listing likelihood and impact item by item; executive review board briefings; GAO and inspector general reports from March 2024 to March 2025 (p.40); the investment's prior ratings and their generation dates on the dashboard; OMB's A-11 Section 55 and the six-factor list; the agency's own CPIC guidance and rating criteria documents (pp. 43–48); and the FITARA clause on four consecutive quarters of high risk (p.9). The dates were public: when a rating was generated and whether the agency's cycle was monthly, quarterly or semi-annual sat on the same page (p.13, table 3; p.22). The FITARA text was in hand too,

including the requirement to review four consecutive quarters of high risk (p.9). GSA had run a user survey in October 2024, and 26 of 37 responses were satisfied with the overall experience (p.26), a result equally within the public record. One exception is worth noting: State said it could not obtain its own agency's historical ratings and so could not do trend analysis (p.24). For that agency, prior ratings were not on the desk. An older archive was also open: between 2010 and 2016 GAO issued six reports on the dashboard with 47 recommendations, 45 implemented and 2 closed unimplemented (p.10), with both the reports and the implementation status public. What became knowable only afterwards: GAO's weighting scheme, bands of 0, 1, 3, 9 and 27 (p.39, table 14); GAO's downgrading of nine assessments by one level on reports since March 2024 and one by two levels on review briefings (pp. 40–41); and the full-sample result of 27 consistent, 24 higher and 2 lower (p.16).

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: on 7 October 2026 GAO published the report. Of 53 investments, 27 matched the CIO rating, 24 were judged riskier by GAO and 2 less risky; 21 ratings had not been updated within the agency's own cycle; and 17 recommendations went to 9 agencies (p.16, p.22, p.30). The responses: 5 agencies agreed, DHS agreed with one recommendation and disagreed with one, Commerce and Defense disagreed, Treasury took no position, and OMB submitted

no comments (p.31); VA committed to reassess its six investments by 31 December 2026 (p.32). Defense objected that GAO had given no written description of its rating method, and GAO's answer was that the risk register for the Army's personnel and pay system Increment 2 held three "elevated" risks, which is what the "elevated" rating rested on (pp. 34–35); in the same exchange GAO accepted Defense's supplementary material on update timeliness and withdrew that recommendation (p.35). Against the three questions: the most common answer to the first is "rate honestly". All 24 deviations recorded in the report lean toward lower risk, and all six VA ratings lean the same way (p.16). That distribution will not yield to an explanation in terms of individual honesty. The item most easily missed under the second question is the rating date. The colour is on the page every day and nobody reads the date, so 21 overdue ratings look exactly like ratings generated that morning. The checkpoint for the third belongs at the step where a rating is finalized: check whether any discretionary adjustment left a written rationale, and whether the rating's date still falls inside the agency's own cycle.

Options ruled out: move rating submission from agency CIOs to an independent assessor (not implemented; the 2014 statute assigns the function to the CIO directly, p.9). Set the level purely from the risk register score (not adopted; Commerce, GSA and DOD all said the rating combined multiple factors, p.19, pp. 33–34). Raise every agency to monthly updates (seven had already chosen monthly and five quarterly or semi-annual, and OMB's August 2025 revision asked only for quarterly, p.13). Have the federal CIO carry out the FITARA advisory role for investments rated high risk four consecutive quarters (the law already requires it; in November 2024 GAO found 27 review records and the federal CIO at none of them, p.9). Display the rating beside the risk register score it corresponds to (not implemented; p.6 records converting a 1 to 5 score into a colour, with the conversion invisible on the page). Lower the

four-quarter trigger to two (not discussed; FITARA writes four into the statute, p.9). Have GAO fill in the ratings directly (not implemented; GAO states that its assessment offers a standardized cross-agency view and is not a prescription to agencies, p.41). All seven options are recorded in the report, and none was executed.

Early signals:

1. GAO-16-494 in June 2016 ran the same comparison: of 95 investments, GAO judged 60 riskier, 22 consistent and 13 less risky (p.10). Ten years on the direction had not changed, and the share had fallen from 63% to 45% (p.10, p.16). What overrode it was written into every response: the methods differ, so the results are not comparable.
2. GAO-25-107041 in November 2024: 17 investments across 8 agencies had been rated high risk for four consecutive quarters, and no agency fully met FITARA's review requirement; the federal CIO attended none of the 27 review records (p.9). What overrode it was procedural. OMB neither accepted nor rejected it, and by October 2026 nothing had moved (p.10, p.28).
3. The April 2025 rating dates: 21 past the agency's own cycle (p. 22). What overrode it was the interface. An overdue rating is still a normal block of colour on the page.
4. OMB's A-11 Section 55 of July 2024 did not specify update frequency (p.7). What overrode it is that it reads as blank space rather than as a gap. No frequency means the rhythm belongs to the agency, and once the rhythm belongs to the agency there is no such thing as overdue, until August 2025 closed the hole.
5. The agencies' own explanation, attributing deviation to having considered more factors (p.19). That sentence is itself a signal: it concedes that the rating is not the risk score, and the page shows only the rating.

The point of escalation: April 2026, when OMB announced retirement and replacement on the dashboard page (p.2). The announcement turned the addressee of 17 recommendations into a system awaiting decommissioning. It gave no specification for the new system, no timetable and no route for agency participation; GSA said all of that sat with OMB (p.6), and GAO records OMB's non-response in three places in the report (p.2, p.4, p.6). At that point the only route to better ratings runs through rebuilding them inside the new system, whose definitions are set by the same office. The 21 overdue ratings will not be fixed before retirement. They will disappear with the system. GAO drew the boundary at the same time: retirement is not immunity, and public reporting of investment risk is a FITARA requirement rather than a preference of that particular website (p.29). The recipients of the 17 recommendations are still in place. The carrier is about to go. Two classes of requirement in those recommendations, timely updates and ratings that reflect the ability to meet goals, will then have to be executed against a system that does not yet exist.

Failure chain:

Technical layer: of the three rating classes, two have a formula (cost, schedule) and one does not (risk). The CIO rating is a human judgement from 1 to 5, and the page shows a colour and a score with no components, no confidence and no mapping to the risk register (p.6). Given the same register, an agency combines it by its own weights and GAO combines it as 0, 1, 3, 9, 27; the two results can differ by two levels while both claim compliance (p.39, p.46).

Organizational layer: the rating is drafted by the agency being rated. Between draft and final sits a discretionary step, with Education's CIO able to override the raw score and HHS's CIO supplying a "discretionary rating and rationale" (pp. 44–45). The update cycle is set by the agency, and 21 are overdue (p.22). Two agencies' processes take longer than a quarter (p.23).

Incentive layer: the rating enters congressional view, and FITARA ties four consecutive quarters of high risk to a mandatory review (p.9). The cost of a low rating is an explanation this quarter. The cost of a high one arrives when the project fails, and project cycles run in years. A rating sitting at "medium" never triggers the FITARA review (p.9).

At the boundary: the discretionary opening left by the technical layer is filled by the self-assessment process of the organizational layer, with its direction set by the incentive layer. The break sits at that opening. One field is at once the agency's report upward and the public and congressional route to oversight, and the two uses demand opposite kinds of close reading.

Minimum counterfactual: beside the rating on every investment page, display the date the rating was generated and the agency's declared update cycle, and grey out the whole cell once the cycle is exceeded. One action, before the April 2025 snapshot, and available at the time: both fields already existed in the system (p.13, table 3; p.22). It changes no rating method and adds no new metric. It lets the 21 overdue ratings show themselves on the page. Feasibility rests on changing the front end rather than agency process: GSA makes front-end changes on OMB's instruction, and user-feedback improvements already shipped on three-week iterations (pp. 25–26), so no agency has to agree to change its rating method first. The cost is one front-end change and one note on definitions.

Outcome and sources: GAO's conclusion is that the selected agencies' ratings do not consistently provide an accurate assessment of investment risk (p.28), and it made 17 recommendations, all falling under two requirements: that ratings reflect the investment's ability to meet its goals, and that they be updated in line with agency process (pp. 30–31). Eight of the nine agencies receiving recommendations submitted written comments, carried in appendices 4 through 11 (p.32). Primary source: GAO, IT Dash-

board: Selected Agencies' Investment Ratings Fail to Fully Consider Risks, GAO-27-108416, 7 October 2026, 30 pages of body text, appendices 1 through 12 running to printed page 66, with front matter at i–iv; page numbers in this card are printed page numbers, matching the footers. Link at Appendix B, po-003.

Case 2 carries a number that is easy to read backwards: more than half of the 53 ratings agreed. Does a majority agreement mean the system is broadly sound? The GAO report will not support that comfort. Which projects the agreeing ratings cluster in, and which the deviating ones cluster in, is the test of whether the system has failed. If the deviations concentrate on the highest-risk projects, precisely where a rating most needs to tell the truth, then "more than half agree" says only that low-risk projects are easy to agree on, while the high-risk projects carry the credibility of the whole system. That is a general rule for metric audits: the overall agreement rate is a placebo indicator, and the agreement rate by stratum carries the information. Apply it inside your own organization, and the first question to ask of any "compliance rate" report is where the non-compliant cases sit.

Two more numbers deserve a second look through time. The fact that 21 ratings went unrevised means that even a rating once honest rots as time passes. The fact that the previous report's recommendation to OMB sat for more than a year with no action means the comparison mechanism had already sent a signal and the system did not digest it. These two numbers join this chapter to the archival view of Chapter 11. An unrevised rating is the "already explained" form of time. An unimplemented recommendation is the public-sector version of a remediation deadline that exists only on paper. The health check for a metrics system therefore needs two time items: when was this last updated, and how many of the previous round's recommendations were im-

plemented. Those two answers belong in the first line of any report. They matter more than any current number.

Case 3 (Software): The Gatekeeper That Watched the Replica (condensed)

sw-006. The GitHub platform, September 2026. The official monthly availability report, a page read and verified for this book: a background data cleanup job overwhelmed the primary database, and within about ninety minutes 96% of web requests to create issues failed. The protection mechanism monitored replica lag only, with the primary's own load outside its scope: the cleanup ran on the judgement that every replica metric was normal and pushed the primary past its limit. The decision point is the design review of the protection mechanism. Minimum counterfactual: add a primary load threshold to the protection, pausing the cleanup job when the threshold is crossed. Primary source: GitHub, 2026, link at Appendix B, sw-006.

Case 4 (Construction): The Broken Bolts That Never Entered the Ledger (condensed)

ce-002. Audenshaw, Manchester, the longitudinal sleeper system on a railway bridge. RAIB report, Report 10/2025, a page read and verified for this book: fatigue fracture of a base fixing bolt derailed a freight train, 9 wagons left the track, the line was closed for about 8 weeks, and nobody was injured. The investigation found the fatigue fracture of the base bolt to be the cause of the derailment, and found that earlier bolt failures had been neither recorded by the maintenance team nor reported: Network Rail's safety assurance system had not noticed for years. The metric read zero while the goal deteriorated. Minimum counterfactual: the maintenance team registers the failure under procedure and reports it the first time a bolt is found broken. Primary source: RAIB, Report 10/2025, 2025, link at Appendix B, ce-002.

Expansion note (sw-006): GitHub's post-incident review, published by the party and verified for this book, gives more detail than the condensed version, and the place worth stopping at is the design definition of the protection. The cleanup job carried an automatic safeguard that paused it under high load, and the safeguard's monitored object was replica lag alone, with the primary's own load outside the scope. The review confirms that the cleanup kept running on the judgement that the replica metrics were normal, and pushed the primary past its limit. Restated in this chapter's language: the goal of database health was broken into a set of metrics, and the one written into the protection logic, replica lag, took the place of the goal. The pressure the cleanup put on the primary was never put to the question of whether the goal was being damaged, because it was always put to the question of whether the metric was normal, and the metric was normal. The event ran about ninety minutes and about 96% of web requests to create issues failed. The minimum counterfactual is written on the source card: add a primary load threshold to the protection, pausing the cleanup job when it is crossed. That counterfactual costs almost nothing, and it works as a mirror for anyone designing metrics. Check whether your automatic protection watches the goal, or one shadow of the goal.

The two expansion notes together carry one more warning. In both cases the party added the missing metric afterwards, one adding a primary load threshold, the other being required to add failure registration. Adding a metric is the standard output of a post-incident review, and it is also the moment this chapter's mechanism watches most closely: a new metric goes live and the next round of gameplay begins to evolve. A reader of such reports should attach one sentence automatically to "monitoring has been added": was its gameplay rehearsed first.

Expansion note (ce-002): the broken bolts at Audenshaw can be read again under Chapter 11's three-state register. The coldest finding in the RAIB investigation is that earlier bolt failures were neither recorded by the maintenance team nor reported. The metric, the count of registered failures, read zero. The goal, real visibility of the joint's condition, deteriorated. The path here runs negative: the goal's information channel was never connected to a metric, so failures never entered the count. This chapter's eight weak signals cannot catch that kind of negative substitution, because they watch for visible optimizing actions, and here there was no action at all. Only Chapter 11's Device 1 catches it. The status of every defect on a physical system is allowed three states, and "never registered" reads at audit as "the three-state register is not running", which is itself a finding of the highest order.

Four cases side by side show the two directions of goal displacement. Positive substitution (fi-001, po-003): metric pressure drives visible optimizing actions, and rescissions and ratings are both actions; actions leave traces, and afterwards there is something to investigate. Negative substitution (ce-002, sw-006): nobody optimizes a metric, because the goal's information channel was never wired into one, with failures never entering a register and primary load never entering monitoring. Positive substitution needs a defence against gameplay. Negative substitution needs a defence against blind spots. The defences differ: gameplay against rehearsal and pairing (Devices 1 and 2), blind spots against an unassessed channel and a definition register (Devices 3 and 4). When you assess your own organization, start with the negative question: which of your key goals has an information channel that was never connected to any metric. That question reaches the weak point earlier than "is this metric being gamed".

Case 5 (Finance): The Codes Nobody Reconciled (condensed)

fi-010. Webull Financial, a broker-dealer. FINRA decision of August 2026, in the monthly summary PDF downloaded and verified for this book: the books and records for option order origin codes were inaccurate over a long period, drawing a $115,000 fine and a censure. The origin code is the basis of measurement for a broker's internal routing statistics and its regulatory reporting: a miscoded field distorts every downstream metric about which channel contributes how many orders. From this chapter's angle, what the case displaces is record accuracy itself. When a record has no reconciliation arrangement, its distortion has no exposure mechanism, and an error can persist for a long time without triggering any signal. The decision point is the reconciliation design of the data pipeline. Minimum counterfactual: reconcile origin codes against routing logs on a schedule, logging discrepancies separately. Primary source: FINRA, Disciplinary and Other FINRA Actions, 2026, link at Appendix B, fi-010.

Case 5 pushes this chapter's boundary out one more step. fi-001 is a metric actively optimized. fi-010 is a metric's raw material going wrong with nobody reconciling it. po-003 is a rating decoupled from real risk. ce-002 and sw-006 are goals whose information never reached a metric at all. Five forms sit under one mechanism name, and each needs its own defence, which is why the signal list in section 12.4 has to be used together with the four devices in section 12.5. The list locates the form. The devices do the fitting.

12.4 What Can Be Seen Earliest

Eight weak signals, each with the reason normally used to override it.

1. The footnotes on a metric's definition keep growing, until someone has to be assigned to explain them. The reason given: "Refining the definition is a mark of management maturity."
2. The same metric means different things in different departments. The reason given: "The businesses are different."
3. A full-time "data governance" post appears, and the problem list does not get any shorter. The reason given: "Data quality needs an owner."
4. The metric improves while the front line reports otherwise, and the gap lasts more than two assessment cycles. The reason given: "The front line only sees its own corner."
5. Optimizing actions on the assessed quantity are visible, month-end pushes, batch back-filling, selective filing, and nobody asks how those actions relate to the goal. The reason given: "Meeting a target takes some flexibility."
6. Someone who says the metric is distorted loses points in the appraisal or gets moved to another post. The reason given: "Think of the whole."
7. External comparison data, customer complaints, regulatory sampling, third-party assessment, diverges from the internal metric over a long period. The reason given: "Different definitions, so not comparable."
8. The metric's historical base gets restated. The reason given: "For comparability."

Put the eight back against the cases in this chapter. Signal 3, a data governance post and an unchanged problem list, matches po-003: the rating system had complete guidance and process, and what it

lacked was a third-party comparison between rating and real risk. Signal 4, improvement against what the front line reports, matches fi-001: the provision rate improved while the absolute balance of delinquent loans was piling up. Signal 5, visible optimizing actions nobody queries, matches fi-001 as well: the rescissions were batch operations, repeated, across five quarters. Signal 8, a restated base, is the latent form all four share: once a definition change goes unlogged, restatement meets no barrier.

The eight also have a practical reading by cost of discovery. Signals 1 through 3 can be found from a desk, by reading the metric documentation and looking at the organization chart. Signals 4 and 7 need external comparison data. Signals 5 and 6 need observation on site. Signal 8 needs an archive. An auditor with only a desk view catches about a third of them, and that is the real coverage of most metric audits. It is also why the right to obtain external comparison data, complaints, sampling results, third-party assessments, belongs in the rules rather than in the auditor's ingenuity.

A list of external comparison data by industry, restricted to items that are usually public: on the customer side, complaints and returns, with billing disputes and regulatory complaints in finance; on the quality side, rework and recalls; on the people side, voluntary turnover and the list of posts that are hard to fill; on the public side, regulatory penalties and third-party assessments. There is one principle for choosing a comparison. It should be mechanistically related to the metric under suspicion, and it should carry no shared optimization incentive. A shared incentive is the main cause of comparison failure: if the complaints unit and the sales unit sit inside the same assessment unit, complaint data stops being a comparison and becomes an accomplice.

There is a class of earlier signal buried in why the metric was created, and it is worth writing out. When a new metric appears, ask where it was first proposed. A metric born from a problem, an in-

cident rate added after an incident, is usually healthy. A metric born from a reporting need, a dashboard built to show something upward, carries an optimization mandate from birth. The only cure for the second kind is to give it an outside comparison, and to put the comparison into the same report.

12.5 How to Prevent It

Device 1: rehearse the gameplay. When a metric is created, write down how it will be gamed: at least three optimizing moves, and a probe for each move that would reveal it. The Goodhart effect cannot be removed, only observed. A metric that goes live without a gameplay rehearsal goes live unmonitored. Who signs: the pro - posal is drafted by whoever puts the metric forward, supplemented by whoever will be assessed on it, and reviewed by an independent party.

Device 2: pairing. An output metric needs a quality metric, volume needs a rework rate, speed needs an incident rate. The pair is col - lected by a different department, and the assessed party does not control the paired metric's definition or sample. The lesson from Travel + Leisure: had the provision rate's counter-metric, the change in the absolute balance of delinquent loans, been submit - ted by a risk function independent of finance and made public, the effect of the rescission programme could not have been hidden.

Device 3: keep one unassessed channel. Set up an information channel that produces no metric: frontline interviews, sampling of raw complaints, managers walking the floor. It does not enter ap - praisal, only decisions, and it must have a named reader. When its output diverges from the metric, the divergence is the most expensive signal available.

Device 4: register definition changes. Every adjustment to a met - ric's definition, sample or baseline is logged separately and

countersigned by an independent party. Travel + Leisure's rescission programme was an accounting definition change, GitHub's monitoring scope a technical definition change, and Audenshaw's failure registration categories a records definition change. All three failed in unregistered definition territory.

Set the four devices against the cases in this chapter and each has one direct use. Device 1 on po-003: at the design of the rating system, rehearse three moves, a rating that lags, a description updated while the rating does not move, and an answer of "the methods differ" to any query, each with a probe, an automatic reminder against the update deadline, an audit flag when description and rating change out of step, and outlier detection on the cross-agency rating distribution. Device 2 on fi-001: the paired metrics for the provision rate are the absolute balance of delinquent loans and the count of rescissions, collected by a risk function independent of finance and carried in the notes to the financial statements. Device 3 on ce-002: track walkers' verbal findings and informal reports from site staff, sampled periodically into the decision layer, unassessed. Device 4 on sw-006: the coverage of a monitoring system is a definition, and every change to it, what was added and what was removed, goes through the register.

The failure mode of each device needs writing down, because it determines where the maintenance cost lands. Rehearsed gameplay dies when rehearsal and reality separate: the rehearsal document is written and filed, and gameplay is already three generations on. Pairing dies when the paired metric's definition gets absorbed: sit the collecting department and the assessed department in the same room often enough and the definitions converge. The unassessed channel dies on a vacant "named reader": its output lands in an inbox with no decision authority. The definition register dies when the register and the use of it separate: the form is being filled in, the metric is being changed, and nothing forces

the two to meet. The maintenance method is one thing only. Once a quarter, replay a real event through all four devices, and repair on the spot whichever one the replay cannot pass.

Here is a five-step minimum for a reader who needs to start work, run in order.

1. Draw the proxy chain: the metric's definition, its data source, who enters it, who may change it, and its definition change history, on half a page. A metric you cannot draw goes to step two anyway.
2. Find the comparison: a piece of outside data the metric should relate to, complaints, chargebacks, rework, regulatory sampling, plotted with the metric across the last eight assessment cycles. A divergence lasting more than two cycles is a flag.
3. Catch the season: re-sort the metric by the assessment calendar and look at the peak-trough gap. An abnormal gap flags gameplay risk.
4. Read where closures go: count where all the exception items end up, alarms, tickets, negative reviews. A single destination for all closures, everything "no action needed" or everything "completed", is the fingerprint of negative substitution.
5. Ask who paid: who in the past year lost points or was moved for questioning the metric. An organization that cannot produce one example should go back and re-read criterion 3 at the start of the chapter.

Run those five and a metric health sheet exists. Nothing in them needs special access, and most of the material sits in public reports.

Turned around, the same five steps become the checklist for setting a new metric. At setup, run the first three, proxy chain, comparison, seasonal forecast, and add the two actions specific to setup: write the gameplay rehearsal (Device 1) and register the

baseline definition (Device 4). A metric that goes through all five on the day it is set costs about half a day. Reformed after three cycles of assessment, the same metric costs quarters. Metrics resemble machines in one respect: the running-in cost is lowest before de - livery.

12.6 Boundaries with Adjacent Mechanisms

Against normalization of deviance (Chapter 11): normalization re- files a deviation, which moves the archive; goal displacement re - defines the goal, which moves the ruler. In cases the two often share a bed, with an optimized metric concealing a deviation while the deviation's success reinforces the metric's credibility. Tell them apart by the object of the action. Archival moves belong to Chapter 11, measurement moves to this chapter.

Against misaligned incentives (Chapter 17): misaligned incentives is the structural disease, with interests at each level pointing dif - ferent ways, and goal displacement is its most common technical implementation. The division of judgement: if what needs chan - ging is the interest structure, who bears the consequence, go to Chapter 17; if the interest structure is sound and the measurement is skewed, stay here.

Against the fifth step of the ladder (Chapter 8): the ruler swap at step five is an emergency move in a crisis, with a meeting and one decision on record. The definitional drift in this chapter is gradual and belongs to normal times, with no single moment where the ruler changed. The two share the definition register as a defence, and their triggers are different.

12.7 Five Self-Checks

1. Write down your team' s three most important assessment metrics and, for each, the real goal it stands in for. A metric whose proxy relationship you cannot state is not ready for a conversation about optimizing it.
2. In your organization, when did someone last say a metric was distorted? What happened to that person afterwards?
3. Pick a metric that has improved for three consecutive cycles and test it against one piece of external comparison data, complaints, sampling, or what the front line feels. How wide is the gap?
4. Which metric in your system has a definition change with no register entry? How much of that metric can you still believe?
5. If your core KPI were abolished tomorrow, what is the first data you would want to see? Why is that data not in the metric today?

Chapter 13 Narrative before Evidence; Motivated Reasoning

> What this chapter covers: how the conclusion gets ahead of the investigation. An explanation that is not yet mature gets said out loud in full, and from that point it filters the evidence, raises the bar for its opponents, and ties the organization to its own first version of events. Three criteria, four domains on the ground, four defences. When to turn back: before roadshow material is finalised, halfway through an accident-attribution meeting, when a project proposal enters the signature chain, and when you no - tice you are waiting for "supporting data". Estimated reading time: 50 minutes.

There is a moment in the prehistory of almost every major failure, and it never appears in the causal chain of the accident report. It is the moment an immature explanation is spoken aloud in full for the first time. Said to investors, said to a superior, written into the summary field of a project proposal, dropped into a work chat. The nature of the explanation changes as it is said. Before that moment it is a hypothesis awaiting a test, and being wrong costs a revision. After it, it is a position, where being wrong costs credibility and testing it reads as provocation. This chapter is about what the organization does after that moment.

13.1 What It Is

Definition: narrative before evidence is an inversion in the order of publication. The conclusion goes public, gets committed to, and is written into documents before the evidence is assembled, and from then on every new piece of evidence is filed under the ex - planation already in place, while opponents carry a heavier burden of proof. The psychological engine is motivated reasoning. A large

body of experiments in cognitive psychology shows that people form a position first and select evidence afterwards, and that this process is transparent to the person it happens in, invisible to introspection. Kunda's 1990 review is the standard citation in the field: Kunda, "The Case for Motivated Reasoning", Psychological Bulletin 108(3), 480–498, 1990 (Crossref record checked, DOI 10.1037/0033-2909.108.3.480). Motivated reasoning in a private head is harmless. Public commitment by an organization upgrades it into an institution: an individual who changes his mind has no audience, and an organization that changes its mind does.

The experimental base is worth a paragraph, because it explains why "stay objective" is useless as a defence. Kunda's review concludes that direction enters at the stage of evaluating evidence: the same data set reads strong to one position and weak to another, and the whole operation presents itself subjectively as "I am simply analysing this rationally". Direction is transparent to the person holding it, which means any defence that relies on individual self-awareness is breached from the inside. That leaves one purpose for an organizational defence: turn the test into an external action that the process cannot skip, which is what Device 1 and Device 2 in this chapter do.

Commitment has a second layer. Once a narrative enters a written document it acquires an independent life. Documents get forwarded, cited, and used as input to the next document, and each handover wears one layer off the uncertainty qualifiers, the "preliminary estimate", the "to be verified". By the third citation the hypothesis has become a fact. Nobody has to lie for this to happen; the wear occurs in retelling. The defence is document-level grade marking, which Device 1 writes into the page header: the reader can see at any moment which grade the conclusion in his hand belongs to.

Narratives travel on three carriers, and each needs its own defence. The first is people: the founder's roadshow, the executive's statement, the expert's endorsement, defended by Device 2 (a disproof owner) and Device 3 (symmetric burden). The second is documents: offering documents, project proposals, approval cases, defended by Device 1 (grade marking) and Device 4 (the time gap). The third is metrics: the handful of numbers the narrative has selected will speak for it, as Chapter 12's presentation-layer move describes, and the defence is to pair every selected metric with a control that was not selected. Of the three, the metric carrier is the one most often left undefended. Numbers look more objective than people and more neutral than documents, and they are the most durable form the narrative takes: people go quiet when the meeting ends, documents get archived, and the metric refreshes on the first day of every month.

The moment named in criterion ① can be seen in advance with one exercise. Draw a timeline for a major conclusion: time on the horizontal axis, four markers on it, being when the conclusion took shape, when it was said outward, when the verification plan started, and when verification finished. A healthy project puts the four points in that order from left to right. A sick one puts the second point ahead of the third, sometimes ahead of the first. The line costs nothing and can be drawn by hand in any meeting. Its value is that it turns narrative before evidence from a retrospective mood into an order-of-events fact that can be pointed at in the room. Being pointed at does not stop a commitment. A commitment that has been pointed at at least knows it owes a verification line.

Three activating criteria. The mechanism has started if any one of them appears.

① The conclusion has already been said outward: to customers, investors, superiors, the public, a regulator, in any form of public commitment. The operative word is outward. What you say to your

own people can be taken back; what has gone out becomes an event when it is taken back.

② When new evidence arrives, the first question in the discussion is "how do we explain this", not "does it shake the premise". Filing replacing testing is the operational signature of a narrative that has taken over.

③ Opposing views are told to "bring more complete evidence", while the proposing side never cleared an equivalent bar when it published its conclusion. An asymmetric burden of proof is the institutional evidence that the narrative now holds status.

13.2 How It Grows

Three generating conditions and one amplifier.

Condition one: a public commitment has already been made. The form does not matter, whether a roadshow, a project proposal, a press release, a statement to a regulator, or a launch date promised to customers. Once the commitment is out, the cost of retreat changes from "revise a conclusion" into "admit the earlier judgment was wrong", and the second is an order of magnitude higher (the mechanism of Chapter 27 plugs in here).

Condition two: information overload. When the evidence stream is too large, the existing explanation is the only filter available. What reaches a project team in a day (data, rumour, competitor moves, internal complaints) far exceeds anyone's processing bandwidth, and fitting material into a frame that already exists is the cognitively cheap path. An overloaded environment turns the cheap path into the default one.

Condition three: no independent analytical route. Nowhere in the organization is one person or one post authorised to set the mainstream explanation aside and analyse separately. When that lane is

missing, every objection has to ride in the mainstream explana - tion’ s car, and the objection is disarmed at the level of wording.

The amplifier is time pressure. The more an occasion demands an answer immediately, whether the first hours of an accident, a fin - ancing window or a policy launch, the more easily the narrative outruns the evidence.

Two structural observations sit on top of the three conditions and the amplifier. The first is that commitment is a stock. Explanations said outward accumulate: said once at project approval, cited once in the interim report, repeated once in a press release, each repe - tition adding one to the stock. The larger the stock, the higher the unit cost of changing the story and the weaker the motive to test it. A successful test pays the organization; a failed test is charged to every person who signed along the decision chain. That is why narrative before evidence almost never corrects itself: its direction in time points at the continued inflation of committed stock.

The second is that the audience sets the hardness of the narrative. A narrative told to internal colleagues can be corrected the same day. One told to investors has to survive until the next earnings season. One told to a regulator and the public makes the correction procedure itself an event. The same explanation will diverge into versions for different audiences, and the divergence becomes a risk of its own: the organization starts managing internal expectations with the external version, and the gap between the two becomes a cost that has to be covered with more narrative. The replacement of customer contract counts by “market potential” in case one below is the moment those two versions merged at the interface of the offering document.

13.3 The Shape in Four Domains

Case 1 (Finance): The AI Narrative and Four Thousand Investors

Case ID: fi-002.

When and where: offering period December 2019 to December 2024. On 26 August 2026 the SEC filed in the United States District Court for the Southern District of Florida, case number 1:26-cv-25837-RKA, complaint of 15 pages. The company was incorporated in Delaware in July 2018, headquartered in Miami during the offering period, and had its principal place of business in Sunnyvale, California, at the time of filing.

What and who: GenesisAI Corp., running an AI services marketplace where model developers list and buyers pay a monthly subscription, with the platform taking 30% of transactions concluded directly between buyer and seller (complaint §16, p.4). Founder and then-CEO Archil Cheishvili, recorded in the complaint as 33 years old, CEO, executive and director from August 2018 to January 2025 (§11, p.3). Paragraph 15 of the complaint: every statement made or published by the company was created or approved by Cheishvili. Four offering channels: a first Wefunder round in December 2019; then three Netcapital rounds; a StartEngine round launched in September 2021 and cancelled by the platform; a Reg A offering through the broker Dalmore Group in 2022 (§23, p.7). The instruments were simple agreements for future equity (SAFEs) and stock, and the four crowdfunding rounds ran under provisions of the 2012 JOBS Act that open them to non-accredited investors (§21, p.6).

What they were doing: in December 2019 the Wefunder offering page carried a chart titled "Revenue Forecast". The revenue line starts at zero in mid-2020 and rises to $250 million by 2024 (§27, p. 8). That month Cheishvili put two numbers in an email to a pro-

spective investor: $1 million in 2020 and $8 million in 2021 (§ 26, p. 8). The email attached no customer contract and no letter of intent.

In February 2020, answering a question in the comment section of the Wefunder page: "We estimate to reach revenue of around $25M in 3 years!" (§ 29, p.9). The comment section, like the offering page, could be read without registering. In April 2020 another email said the company had "lots of companies and users in the waitlist", with a target of $500,000 that year and $5 million the next (§ 29, p.9). Three carriers, three dates, three versions of one set of numbers: $25 million in three years, $500,000 that year, $5 million the next, $250 million by 2024. No conversion relates them to one another, and no document ever reconciled them.

The state underneath, in the same period: the marketplace was still in beta, launched publicly in May 2022 with about 35 models (§ 17, p.5). No subscription fee was charged before launch, and there was no user agreement or commitment; the complaint's words are "there was no 'waitlist'" (§ 30, p.9). In October 2021 the company's own StartEngine page said: "[w]e have a beta version of the marketplace up and running...[with] 23 models deployed..." (§ 17, p.5). That sentence sits in the same year as two upward revisions of the valuation, in April and September 2021.

The valuation moved once a year. The first round in 2019 implied $7.5 million; the April 2021 second round documents about $47 million; the September 2021 third round about $89 million; the Netcapital page in July 2022 $204 million, more than double the year before (§ 33, § 34, p.10). The solicitation material carried a separate partnership list: slides posted on the website and the crowdfunding page in 2020 and 2021 read "Over 25 partnerships!", and what had actually been signed were non-binding memoranda of understanding, some of them expired by 2021 (§ 39, § 40, p.12). The 2021 offering page also said "over 15 companies willing to join [its] marketplace", and the 2022 offering circular

rewrote it as "over 15 agreements with AI companies and individuals who are interested in deploying AI tools" (§ 40, p.12). Two words for one fact, moving from "willing to join" to "agreements".

On the promotion side, from 2020 to 2024 the company paid at least $780,000 in advertising to X, Google and Meta, plus one sidewalk lightbox (§ 20, p.6). In 2023 it built a bulk-email server tool and used it to reach new prospective investors (§ 19, p.5). In the second half of 2020 the company stopped publishing revenue forecasts (§ 30, p.9). The $250 million chart stayed up.

The decision point: four of them.

The first in December 2019, when the first-round offering page and slides were finalised. The carriers were the "Revenue Forecast" chart on the Wefunder page and the investor email sent that month (§ 26, § 27). The signatory was Cheishvili, and paragraph 15 attributes the creation and approval of every company statement to him alone. On the desk: a marketplace in beta, no customer agreements of any kind, and $250 million as an extrapolated curve. The chart carried a disclaimer, "that this forecast is not guaranteed" (§ 28, p.9). The disclaimer sat under the chart, the chart stayed on the page, and neither changed for four years.

The second in April 2020, the email that used the word waitlist. It went out after the first round had closed, its purpose was continued solicitation, the carrier was private email, and the recipients were individual prospective investors (§ 29, p.9). The same month had a second carrier in the platform's comment section, publicly readable and open to anyone with a screenshot function. What made this step irreversible was the recipient: once the email is sent, the company owes that particular person a particular number.

The third in April and September 2021, two upward revisions of the valuation. The carriers were two sets of offering documents, and the number went from $47 million to $89 million (§33, p.10). Company revenue that year was zero (§31, p.9). The complaint records the method: the valuation was set by Cheishvili alone; when an investor asked for the basis he answered "comparables"; the comparables he chose had large customer bases, more mature products, or backing from well-known venture firms, and GenesisAI had none of the three (§37, p.12). Paragraph 5 of the complaint says the company had no robust comparable analysis and should have known, before making valuation statements to prospective investors, that a more objective analysis was needed.

The fourth in July 2022, when the number became $204 million and the carrier moved from an offering document to a public Net-capital page (§34, p.10). Before that the figure had lived only in material behind a registration wall; afterwards it sat on a page anyone could screenshot. Over the same period, February to June 2022, Cheishvili sold 12,000 shares through Reg A for about $51,000 (§23, p.7). The peculiarity of the Reg A channel is that it lets a founder sell shares he already holds, not only new issuance by the company.

What was visible then: the desk held a checklist that could verify itself. How long the marketplace had been in beta was the company's own development schedule. Whether customer agreements existed took one question to sales. Whether a waitlist existed took one lookup in the system that day. The February 2022 offering circular said "over 1,400 individual registered users and companies", and the complaint defines a registered user as someone who filled in a form with a name, a company and an email as an expression of potential interest, with no payment commitment required, in a system the company did not even track for active use (§36, p.11). The same metric was 1,000 in 2021, 2,000 in a 2022

YouTube solicitation video, and 2,600 in 2023 (§35, §36). Four numbers from four carriers, and the definition was never written down.

The valuation line was on the desk too: $7.5 million in 2019, $204 million in 2022, with zero revenue in between (§31, §33).

One more item. In September 2021 StartEngine cancelled the GenesisAI offering (§23, p.7). That was the only interception in the record made by an outside institution. After it, the company did not stop to check a single document.

Information that only became available afterwards: actual revenue. Zero for 2018 through 2021; about $8,000 in 2022, most of it from an AI conference held in Miami; about $9,000 in 2023; about $40,000 in 2024, mainly from the email server tool rather than the AI marketplace (§31, pp. 9–10). And on 2 January 2025 the company announced it would stop developing the product (§7, p.3).

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: on 2 January 2025 the company announced it would stop developing the product, and Cheishvili stepped down as CEO and from his other roles (§7, p.3). On 26 August 2026, the day the SEC filed, the settlement terms were published: the defendants did not admit the allegations and consented to a final judgment subject to court approval, a permanent injunction against violating Section 17(a)(2) of the Securities Act of 1933, dis-

gorgement by Cheishvili of $50,000 plus $9,184.53 in prejudgment interest, and a civil penalty of $50,000 (LR 26619). The complaint had sought a three-year officer and director bar (§ 9, Relief D); the settlement announcement does not list it.

Back to the three questions. The most common answer to the first is "make the forecast conservative", and the forecast in this case was in fact changed once: in the second half of 2020 the company stopped publishing revenue forecasts, while the chart already published stayed on the offering page for four years. The most commonly missed item under the second question is the September 2021 StartEngine cancellation, the only interception from outside, after which every subsequent offering ran through the same channel structure and the company treated the event as a channel problem rather than a narrative problem. The third question lands in April 2020: had someone been required, before that email went out, to replace "waitlist" with a checkable number and produce the list on the spot, the sentence could not have been written. One threshold does the work, which is being able to produce the list. If the list cannot be produced, the adjective has to go.

Options ruled out: six.

1. Replace the revenue forecast curve with signed contract value. Not done. What stayed on the page was the disclaimer "not guaranteed" (§ 28), treated as proof that the uncertainty had been disclosed. A disclaimer protects the tone and leaves the curve alone.
2. State publicly in 2020 that no waitlist existed. Not done. The April email that year still used the word, and no correction followed for two years (§ 29).
3. Commission an independent review of the valuation method. Not done. The complaint records no robust comparable analysis, a valuation set by the CEO alone, and no written criteria for choosing the comparables (§ 5, § 37).

4. Disclose the definition of "registered user" to investors. Not done. The definition was written down for the first time in the complaint, having existed inside the company all along (§ 36).
5. After the September 2021 channel cancellation, pause the offering and work on the product. Not done. The company kept offering on Netcapital and opened a Reg A channel as well (§ 23).
6. Rewrite "partnership" as "memorandum of understanding". Not done. The 2022 offering circular upgraded "willing to join" into "agreements", moving the words against the facts (§ 40).

Early signals: seven.

① December 2019: the forecast chart starts in mid-2020, and the company held no customer agreements that year (§ 26, § 27). Reason it was pressed past: the note under the chart says forecasts are not guaranteed. ② April 2020: "waitlist" coexists with beta status (§ 29, § 30). Pressed past because interest received during beta was treated as a list. ③ April 2021: the valuation jumps from $7.5 million to $47 million with revenue at zero in the same period (§ 31, § 33). Pressed past because registered users were growing. ④ October 2021: the company's own offering page still says beta and 23 models, in a year when the valuation was raised twice (§ 17, § 33). Pressed past because beta is a technical stage while valuation looks at the market. ⑤ February 2022: the offering circular presents "1,400 registered users" as demand-side evidence (§ 35). Pressed past because a registration count is a number that can be refreshed every month. ⑥ September 2021: StartEngine cancels the offering (§ 23). Pressed past by switching platform. ⑦ July 2022: the valuation doubles in a year to $204 million (§ 34). Pressed past with "comparables".

Four of the seven are visible at the document layer, two need the version history of platform pages, and one needs a letter from an

outside institution. The excuses share a shape: each swaps a checkable quantity for an interpretable one. Registered users is a number that can be refreshed; what registered users mean is an interpretation. A valuation is a signature; the basis of a valuation is an interpretation. A partnership is an intention; the nature of a partnership is an interpretation.

The point of escalation: the step after the September 2021 channel cancellation. The company kept offering on Netcapital, wrote $204 million onto Netcapital's public page in July 2022, and ran a Reg A through Dalmore the same year (§23, §34). $204 million was not a new fact; it was the same valuation method restated on a new carrier. The carrier changed, and with it the cost of retreat: withdrawing a number from an offering document is an edit, and withdrawing a number from a public page is a retraction. In the same year the founder sold 12,000 shares through Reg A (§23), a sale inside the same offering channel and at the same time as the $204 million pricing. By then the valuation was no longer a hypothesis open to internal discussion. It was a price already sold to more than four thousand people, and every additional investor added one more listener to any retreat.

The Reg A round had a second consequence in the shareholder register. The complaint describes the investor base as "predominantly retail investors" (§21, p.6), thousands of them scattered where they cannot check each other. A concentrated board demands an explanation once. Thousands of dispersed investors will not ask at the same time, and any one who does can be answered with the same paragraph. Part of why escalation works is that the audience is dispersed.

Failure chain: technical layer. The marketplace only launched in May 2022, and the company's eventual revenue came from an email server tool built in 2023, about $40,000 in 2024 (§19, §31). There is no transition band between what the product could do and

what the narrative said: the narrative described a marketplace and the revenue came from a bulk-mailing tool. Nothing failed in the technical layer; what failed was the exchange rate between the technical layer and the narrative.

Organizational layer: every outward statement was created or approved by one person (§15). No second reviewer touched the valuation, the registered user had no definition, and the nature of the cooperation agreements had no legal standard distinguishing an MOU from a partnership. Paragraph 40 of the complaint notes that some of the companies that signed MOUs did put their tools on the platform, which gives the word "partnership" a handhold in fact without giving it one in contract. The handhold's use is that when the question comes, a real page can be pointed at.

Incentive layer: Reg CF opened the rounds to non-accredited investors, four crowdfunding rounds plus one Reg A; from 2020 to 2024 at least $780,000 went into advertising, and the traffic it bought landed directly on offering pages (§20); each round's valuation sat above the last and supplied the reference for the next one's pricing. Cheap issuance, a dispersed audience and small tickets together slowed the feedback loop that asks whether anyone is challenging this.

The break at the joint sits between the organizational and incentive layers: checking the phrase "registered user" costs an afternoon, and raising a valuation costs one signature. The first appears in nobody's appraisal. The second does.

Minimum counterfactual: before the April 2020 email went out, a person who does not report to the CEO replaces "waitlist" in the outward text with two checkable numbers, beta registrations and paying customers, and requires the underlying list to be produced on the spot. One action, landing after the first round and before the second, with no luck required: the list is either in the system or it is not. If it is not, the sentence cannot be written that day, and the

valuation narrative that follows loses one of its supports. The cost is one email and one query. The return is an adjective removed permanently from the outward text.

Outcome and sources: announcement of 2 January 2025 that product development would stop; settlement of 26 August 2026, a permanent injunction plus $50,000 in disgorgement, $9,184.53 in interest and a $50,000 penalty, subject to court approval. Primary sources: SEC, Litigation Release No. 26619, 2026-08-26; SEC v. Gen‑esisAI Corp. and Archil Cheishvili, No. 1:26-cv-25837-RKA (S.D. Fla.), Complaint (15 pages), link in Appendix B, fi-002.

One detail in this case deserves to stop on the enforcement lan‑guage. The SEC characterised the conduct as "negligently mis‑representing". The choice cuts both ways for a reader of this book. Read lightly, it means the regulator did not need to prove an intention to lie, only that the gap between statement and fact was reachable, so narrative optimism carries a price by itself. Read heavily, it means motivated reasoning is no excuse: a founder who believed his own narrative still met the legal consequence. Both readings arrive at the same lesson. The truth of a narrative lives inside the person; the liability of a narrative lives outside him; the distance between those two positions is the width at which this mechanism requires an organization to dig in beforehand.

Case 2 (Public): The Replication Narrative and Its Precedent

Case ID: po-006.

When and where: in January 2021 the UK government and EDF opened formal negotiations on Sizewell C; construction started on site in April 2024; the final investment decision (FID) came in July 2025; financial close in November 2025. The audit report was pub‑lished on 20 May 2026, signed by the Comptroller and Auditor General on 12 May and laid before the House of Commons on 18

May, numbered HC 33, session 2026-27. The site is at Sizewell in Suffolk, England; the decision-makers sat in Whitehall, London.

What and who: the Department for Energy Security and Net Zero (DESNZ, including its predecessors the Department of Energy and Climate Change and the Department for Business, Energy and In - dustrial Strategy) as sponsor department; the project company Sizewell C Ltd (SZC Co.). Equity: DESNZ 44.9%, EDF 12.5%, La Caisse 20%, Centrica 15%, Amber Infrastructure 7.6%, with total equity capped at £8.5 billion in nominal terms; debt capped at £36.6 billion from the National Wealth Fund plus £5 billion of commercial loans (NAO p.5). The auditor is the National Audit Of - fice, signed by the Comptroller and Auditor General Gareth Davies. The accountable bodies are the department that approved FID and the Treasury, not the construction site.

What they were doing: the original text of the replication argu - ment is at paragraph 1.11 of the NAO report. By the time SZC was approved, HPC had completed more than 99% of its detailed civil design and 85% of its detailed mechanical design; for comparison, when EDF took FID on HPC in 2016 it had only outline design and about 10% of detailed design, and the Finnish and French projects were no more mature at their investors' decisions (p.24). About 85% of the above-ground design could be copied straight from HPC. The reactor design already met UK nuclear safety regulatory requirements. Key components continued to come from inside the EDF group: Framatome reactors, Arabelle turbines (§ 1.12, p.26).

The shape of the argument is worth seeing clearly. The NAO sum - marises the position of DESNZ and the company as follows: copy - ing HPC's design and using the teams that built HPC reduces schedule and cost risk; SZC's cost and schedule estimates are far more mature than those of the HPC, French and Finnish sponsors at the time they decided; the company considers about 60% of its cost estimate to have "high or very high" certainty (§ 8, p.7). All

three statements are true. "Risk is reduced" is derived from them, and sits outside them.

Three cost and schedule lines, in 2024-25 prices: the company's project baseline £38.2 billion with all units in service by July 2039; the DESNZ low regulatory threshold £40.5 billion by October 2039; the high threshold £47.7 billion by August 2043, with government and investors committed to fund up to that cap (NAO p.5, § 1.15, p. 26). Three lines correspond to three probability ranges, and once written into the company's licence they become contractual parameters.

Returns and bills: investor post-tax equity return of 10.8% to 13.0% in nominal terms, which the NAO judges above the level common in other utility sectors, including a "nuclear premium" (p.13, § 18). On the consumer side, a typical household bill rises by £4 in 2025-26 and climbs to a peak of £19 to £21 a year in the first decade of operation, in 2024-25 prices; consumers began paying through their bills from November 2025 (p.5, Summary § 3).

Money in before the decision: from 2017 to FID, DESNZ injected £5 billion, taking its holding to 90%; external consultancy fees over the same period came close to £77 million (§ 1.8, p.21). In January 2022 it paid EDF a £100 million option fee (Figure 2 note 1, p.23). In November 2022 it injected £679 million to become a 50% shareholder and arranged CGN's exit (§ 1.8).

The decision point: four of them.

The first in January 2021, when formal negotiations opened with an initial target of financial close by the end of March 2023 (§ 1.7, p. 21). The carriers were the negotiating mandate and the timetable. On the desk: the overruns and delays of the four precedent EPRs were all public, with Finland and France each more than ten years late at four times the original estimate (§ 1.10, p.24). Starting ne - gotiations altered every option after it. Once a party is in exclusive

negotiation, whether the project should be built leaves the table and whether the price can be closed takes its place.

The second in 2022, when the regulated asset base (RAB) model was legislated (§1.6, p.19). This moved the location of the risk: construction-period costs would be carried by consumer bills rather than by the developer alone. The carrier was the Nuclear Energy (Financing) Act. Once the Act passed, the audience for the approval argument widened from investors to billpayers, and bill-payers have no procedure for voting down an individual expenditure.

The third in April 2024, when construction started on site, fifteen months before FID. The NAO records that continued injections during the negotiations "helped keep the start of construction on schedule for April 2024", preserving the overlap with HPC's schedule so that people, supply chain and equipment were not dispersed; the report's own words are that DESNZ "funded the project at risk accepting it could lose its investment" (§1.9, p.22). This is escalation in this chapter's sense, and the card takes it apart separately below.

The fourth in July 2025, when DESNZ and the Treasury took FID together, 4.5 years after negotiations opened and at least 28 months later than planned (§1.7, p.21). That month DESNZ wrote the two regulatory thresholds into the company's licence (§1.15, p.26). Financial close came in November 2025, and in the same month £1.6 billion went to HPC for access to its cost, schedule and experience information (§1.13, p.26). The payment was designed to offset exactly the £3 per megawatt hour price reduction agreed in the 2016 HPC deal (§1.14, p.26). The name of the official who signed off FID is not stated in public records.

The order of the four nodes is worth holding on to. Opening negotiations decided who to talk to. Legislation decided who carries construction risk. The start of construction decided the direction

of the evidence. FID decided which numbers went into the con‑tract. By FID the first three had narrowed the options to price terms, and the only question left on the approval table was where to set the three cost lines.

What was visible then: Figure 3 in the NAO report is a table that fits on a negotiating table. Olkiluoto 3, construction started 2005, originally due in service 2009, actual 2023, 14 years late, cost from €2.3 billion to €9.6 billion in 2024 prices, an overrun of about 320%. Flamanville 3, started 2007, originally 2012, no earlier than 2026, 14 years late, cost from €3.2–3.3 billion to €13.2 billion in 2015 prices, an overrun of 300% to 310%. Taishan 1 and 2, five years late each. HPC unit 1, started 2018, originally in service 2025, revised to 2030–2031, five to seven years late, cost from £17.9 billion to £35–36 bil‑lion in 2015 prices, an overrun of about 100% (p.25; EDF annual report figures of February 2026). The same passage mentions Vo‑gtle and VC Summer in the United States, which shows overruns are not confined to the EPR design (§ 1.10).

The second item on the desk was the NAO’s own June 2017 report on HPC (HC 40). Same audit institution, same subject, previous conclusion, and already public when negotiations opened in 2021.

The third was the company’s self-assessment of cost certainty. Figure 5 of the NAO report, dated August 2025, splits the £38.2 bil‑lion estimate into seven cost areas. The three marked “very high” certainty total 48%, and the least certain 15% is non-replicable site works (pp. 29–30). The strength of the outward argument came from the 85% that can be replicated, and the risk is concentrated in the remaining 15%.

Information available only afterwards: SZC’s own cost and schedule. The project expects to generate from 2038, the three cost lines correspond to in-service dates of July 2039, October 2039 and August 2043, and the actual figures cannot be checked until after those dates. The NAO’s concluding sentence reads: “no nuclear

power station of its type has so far been built without delays or cost increases." (Summary § 21, p.15)

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the project reached financial close in November 2025 and is proceeding as planned, with consumers paying through their bills from November 2025. The NAO did not overturn the project. It wrote the gap between the narrative and the precedents up as an audit finding, in wording that says the path "depends on some significant assumptions", that investor returns "look high", and that the extent to which investors are incentivised to control costs is "unclear" (§ 19, § 22). The report makes two recommendations: any department considering a similar delivery model should be able to explain why so much risk has to be borne by taxpayers and consumers, and demonstrate that investor returns are commensurate with the risk actually borne; and DESNZ and the company should improve oversight of cost, schedule and private investors (Summary § 23, § 24, pp. 14–16).

Back to the three questions. The most common answer to the first is "wait until the HPC data is more complete", and the NAO records the other side of that: the delay did give the company time to improve its estimates and its supply chain arrangements (§ 1.9). The reason holds, and it is a different operation from comparing precedent overruns item by item. The first improves an estimate. The second tests an argument. The most commonly missed item

under the second question is one row of Figure 5: the least certain 15% is exactly the non-replicable site works, while the strength of the outward argument came from the 85% that replicates. The third question lands before the April 2024 start of construction, and the checkpoint is one table: take each precedent in Figure 3, break its overrun down by cause, and state item by item which arrangement SZC uses to avoid it.

Options ruled out: five.

1. Submit an item-by-item comparison of the causes of overrun at comparable projects before FID. Not done. The approval argument is built with similarities first and cost estimates second, and the overrun history is covered in one sentence in paragraph 7 of the report, that other nuclear stations have also suffered large overruns, and not treated as evidence requiring a response item by item (§ 7, § 8).
2. Prepare a credible alternative. Not done. The NAO's words are that DESNZ "did not develop a credible 'plan B' to its negotiations, despite being asked to do so by the major projects review group and the Infrastructure and Projects Authority recommending this" (§ 2.4, p.38).
3. Bring the £1.6 billion HPC information access payment into the project baseline. Not done. The baseline explicitly excludes it, and it is one of the main differences between the £40.5 billion threshold and the £38.2 billion baseline (§ 1.15, p.26).
4. Raise investors' risk exposure in exchange for a lower return. Not done. The NAO judges investors' cost-control incentive at no more than about 1.6 percentage points when costs rise, and writes that "whether these incentives are sufficient cannot be judged from the transaction structure alone" (§ 19, p.13).
5. Push the start of consumer payment until after commissioning. Not done. Consumers have been paying since November 2025 (p.5).

Early signals: six.

① January 2021, when negotiations opened: the overrun and delay record of the four precedent EPRs was already fully public (Figure 3). Pressed past because SZC replicates HPC and is not a first-of-a-kind build. ② June 2017: the NAO's HPC report had already identified problems in the way risk was transferred. Pressed past because the new financing model had replaced the old structure. ③ Late 2022: large nuclear projects under development had fallen from four in early 2017 to SZC alone (Figure 1, pp. 18–20). Pressed past because with one option left the negotiating leverage was gone, and the NAO's formulation is that taxpayers and consumers had to accept more risk (§10). ④ April 2024: construction starts fifteen months before FID, recorded by the NAO as "at risk". Pressed past to preserve the schedule overlap with HPC and avoid a break in skills and supply chain. ⑤ July 2025, at FID: the alternative required by the Major Projects Review Group and the Infrastructure and Projects Authority still did not exist. Pressed past because resetting the project would cause significant delay and delay would erode the benefits of replication, a causal chain the NAO records (§2.4). ⑥ Investor return of 10.8% to 13.0%, above the normal utility level. Pressed past because private capital had to be attracted.

Three of the six come from the NAO's own earlier report and the front half of the same report. Signals were not in short supply. What was missing was a column in the approval table to write them into.

The point of escalation: the April 2024 start of construction. At that point the object of the argument changes from a document to a site: fifteen months, £5 billion, a supply chain already scheduled, a reactor building with its base slab poured. Any later evidence against the replication narrative has to pass through the fact of construction already under way. The carrier of this approval was

DESNZ's continuous injection of money during the negotiations, £679 million for equity in November 2022 and £5 billion cumulative by November 2025; the name of the signatory is not stated in public records. The NAO's assessment of this stretch runs both ways: the money kept the start date at April 2024 and preserved the overlap with HPC, and the same money went out with no guarantee that a deal would be struck.

A second consequence of starting is that the direction of the evidence got locked. After the start, the project team had to prove that the station could be built to this schedule, and no longer whether it should be built at all. The two need different evidence. The first needs progress photographs, supply contracts and labour curves. The second needs a list of the causes of precedent overruns. The first batch generates itself every day. The second was never requested.

Failure chain: technical layer. The construction history of the EPR points the same way at four sites (Figure 3), and HPC itself overran by about 100%. SZC's response was to take the replicable part of the design to completion. Figure 5 shows the most certain 48% made up of three items, spend already incurred, nuclear equipment under signed fixed-price contracts, and bulk materials replicable from HPC, while the least certain 15% is non-replicable site works. The technical break sits on the site, and the site is the piece that cannot be copied.

Organizational layer: the approval argument is built around replication, and the overrun history was never required to answer item by item; the alternative was absent, though both opinions requiring one are written into the report; two regulatory thresholds went into the licence, fixing uncertainty as three numbers. Once in the licence, the thresholds are contractual parameters rather than assumptions awaiting discussion. The failure at this layer is one of argumentative form: a conclusion that could only be supported by

similarity was written into a contract that can only be tested by those three numbers.

Incentive layer: political commitment (energy security, jobs, net zero) and consumers already paying turn the cost of retreat from a fiscal question into a question of political credibility; an investor return of 10.8% to 13.0% coexists with a cost-control incentive of about 1.6 percentage points (§18, §19). High return and weak incentive together make cost control a secondary objective for investors, with getting the contract as the primary one.

The break at the joint sits between the technical and organizational layers: precedent overrun data never entered the evidence column of the approval table, because it was filed under "first-of-a-kind problems" and SZC described itself as not first-of-a-kind. Once the classification is made, the same data moves from evidence to background.

Minimum counterfactual: before the April 2024 start of construction, require the project side to submit a table that breaks down each precedent EPR in Figure 3 by cause of overrun and delay, states item by item which design, contract or organizational arrangement SZC uses to avoid it, and marks in red any item with no corresponding arrangement. One action, before the start, with no luck required: every cause of overrun is in public audit reports, and the person filling the table needs no new data, only existing data placed in a table that has to be completed. The value is in the columns rather than the answers. Columns that demand item-by-item correspondence force a correspondence to be written. Where none exists the row stays empty, and an empty row is easier to see than any argument.

Outcome and sources: financial close in November 2025, project proceeding; consumers paying from November 2025; the NAO wrote the gap between the replication narrative and the precedents into an audit finding, with two recommendations on oversight and

on the proportionality of returns. Primary source: National Audit Office, Sizewell C, HC 33, Session 2026-27, 20 May 2026 (signed by the Comptroller and Auditor General on 12 May 2026), link in Appendix B, po-006.

This case is worth one dissection of the replication narrative as a form. The bookkeeping is asymmetric: similarity (same reactor type, reusable design) gets written up as the conclusion, and difference (site geology, supply-chain generation, contractor mix, regulatory version) stays in invisible default values. The NAO sentence about no such station having been built without delay or cost increase is sharp because it tests the other side of similarity. If replication really holds, the overruns should replicate too; the argument replicated the success assumptions and left the failure data out. That gives a general probe for any narrative of this kind. Ask the proponent which parameters of the precedent are being replicated, and which parts of the precedent's full data set, including the bad data, fall inside the similarity claim. Similarity that cannot be answered for is narrative similarity. Similarity that can be checked item by item is engineering similarity.

The same dissection has a cross-generation sample in case one of Chapter 11 (mi-001). The shuttle era's "in-family" classification amounted to organizing twenty years of foam-shedding history into a narrative that supported flight: every shedding with no consequence was narrated as evidence of system reliability, and the opposite reading, that shedding had never been eliminated and only never cashed in, waited until after STS-107 to be written into an official conclusion. Same facts, two narratives, and the difference is only who holds the organization's attention and its power to classify. The lesson for a reader is that the ground here is the organization of facts rather than their quantity, and what an auditor can do is periodically re-check the alignment between the

narrative and a raw list, an event-by-event table of the kind at CAIB p.124.

Case 3 (Drugs and Devices): Into Distribution before Clearance (condensed)

ph-005. Adept Medical's ear ventilation T-tube, 9mm. FDA device recall record Z-0966-2025, whose API record has been read for this book: no 510(k) clearance, 6,280 units recalled (2,500 in the United States, 3,780 outside it), distributed to Canada, Australia and New Zealand. The decision point is the internal node at which distribution began while clearance was still pending. Minimum counterfactual: before the first shipment abroad, the quality and regulatory team confirms the 510(k) status and records it in the release record. Primary source: FDA, Z-0966-2025, 2025, link in Appendix B, ph-005.

Cases three and four read together as two process forms of assumption used in place of evidence. The assumption in case three is procedural: clearance in progress, shipping first, the distribution step running ahead of the regulatory status. The assumption in case four is technical: the safety of a raw-milk process rests on induction from batches that went well, until a state laboratory's finished-product test breaks it. In both, the assumption performs the function of evidence, and no line on the release sheet says that the basis here is an assumption. The action asked of the reader is cheap. In the release, shipping and go-live checklists you handle, find which line rests on an assumption, attach one measurement or one wait to it, and that line moves from the narrative list back to the evidence list.

Case 4 (Agriculture): A Process Assumption Standing in for a Safety Conclusion (condensed)

ag-003. Twin Sisters Creamery's Whatcom Blue raw-milk cheese. FDA enforcement report event 97880 / H-0193-2026, whose API re-

cord has been read for this book: a Class I recall of 3,438.2 pounds for potential E. coli O103:H2 contamination. Raw-milk cheese is not pasteurised, and its safety rests on a process assumption; the state agriculture department's finished-product testing broke the assumption. Minimum counterfactual: before the first commercial distribution, run one E. coli O103:H2 test on finished product and bring the result into the release standard. Primary source: FDA, event 97880, 2026, link in Appendix B, ag-003.

Case 5 (Startups): The $103 Million Automation Narrative (full card)

Case ID: st-003.

When and where: founded 2014, wind-down announced June 2020. Headquartered in Austin, Texas, with an outsourced office in the Philippines. The exact date of the wind-down, the date of the board resolution, and the signing dates of the funding rounds are not stated on the Failory page.

What and who: ScaleFactor, a bookkeeping and finance software company serving small and medium enterprises. One founder, Kurt Rathmann, also CEO; between 100 and 250 employees; 7 funding rounds totalling $103 million from 17 investors (Failory names Bessemer, Coatue, and Austin's Techstars, written "Tech Stars" in the original). This card is evidence grade D: a single media account of a party's own retrospective, with no primary documents, no round-by-round dates and no breakdown of amounts. Every node involving an internal meeting, resolution or signatory is marked "not stated in public records" and is not filled in by inference.

What they were doing: in 2017 the company launched its first software product, built on top of QuickBooks and Xero, doing bookkeeping workflow orchestration. Failory's words are "built on top of QuickBooks and Xero". The front end was an automated

dashboard covering five services: bookkeeping, tax filing, financial forecasting, bill payment and payroll. The delivery cadence promised outward was real-time reporting, set against the monthly close of a traditional bookkeeping firm. Pricing was about $6,000 a year, with the top tier at $30,000.

Behind the interface sat another structure. The bookkeeping was done by accounting staff in the Austin office plus an outsourced team in the Philippines. Reports still went out monthly. Their job title was not accountant but "customer service officer". Their cost did not sit in cost of goods sold. Failory's description of this: "creative accounting on their own numbers to boost the perception of growth and to give themselves a more tech-like sales margin."

What customers saw was the interface, and not who produced the reports behind it. Five services were packaged into one dashboard at $6,000 to $30,000 a year with "real time" at the centre of the sales pitch. At delivery the promise was downgraded to monthly, and monthly is the rhythm of a bookkeeping firm rather than a software product. That distance between promise and delivery showed up on the customer side as late reports, and inside the company as an unwritten definition that entered no document.

The customer side left specific evidence. Accounts handed to customers contained errors they had to correct themselves. Lindsey Reinders' business lost $17,000 to an error that went unnoticed for months; when she asked for compensation she was offered a partial refund conditional on not making her customer experience public. Failory quotes her: "If you're one of the investors that gave these clowns $100 million...You should know they've flushed it down the toilet with poor product and poor service."

Then 2020, and the pandemic. Failory's formulation is that the pandemic "allegedly halved" the company's $7 million in ARR. In June of that year the company announced it was winding down.

The decision point: four of them.

The first in 2017, when the first software product launched. The carriers were the product itself and the pricing page, the date is precise only to the year, and the page text no longer exists. On the desk: the product stood on two third-party accounting systems, delivery was still monthly, and people were in the loop. This step changed the company's outward identity from a bookkeeping service into software, and pricing, hiring and fundraising all followed software-company terms afterwards. The exact launch date and the person who approved it are not stated in public records.

The second was calling the accounting staff customer service and moving that labour out of cost of goods sold. The carriers were the organization chart and the P&L line items, both internal and both printable. The consequence is measurable: gross margin looked better, and the share of labour vanished from the statements. The date of the change, the person who approved it, and whether a finance lead objected are not stated in public records.

The third was the pivot to a matching platform. Failory gives one sentence as the reason, and the sentence is material for this chapter: "raised money like a tech startup, so they needed to develop a scalable tech product rather than to refine and grow their service business." Product form was decided by the terms of the fundraising, and the order is identifiable here: valuation logic first, product form second. The specific timing, the decision meeting and the approver are not stated in public records.

The fourth came in June 2020, with the announcement of the wind-down. Failory records that the investors decided to stop funding and that the pandemic was the outward attribution; the CEO's quote is: "Business owners went into fight or flight mode. You don't necessarily need all the planning tools, high end gadgets. You just get back to the simple 'pen and paper.'" The resolution documents and the signatories are not stated in public records.

Two of the four nodes have dates, one to the year and one to the month. That is the hard constraint of a D-grade source, and it sets how to read this card: the actions and their form can be checked, the moment and the signature cannot.

What was visible then: four things on the desk, all inside the company's own systems, with no outside adviser required.

1. The actual date the monthly reports went out. The gap against the "real time" promise is timestamped in the system.
2. The labour hours each customer account consumed. They sit in the rota and the timesheets.
3. The composition of errors and the rework. It sits in the tickets, the $17,000 one included.
4. The absence of customer-service labour from the cost of goods sold line. The P&L shows it without any audit.

A set of ratios could also have been computed on the spot, and the card does one division here, with every figure taken from numbers Failory has already published. Pre-pandemic ARR about $7 million; 100 to 250 employees; annual fee $6,000 to $30,000. Put the three together: ARR per head about $28,000 to $70,000; a customer count implied by the annual fee of about 233 to 1,167; and therefore about 1 to 12 customer accounts per employee. A product that genuinely automated bookkeeping would serve an order of magnitude more accounts per head. The division is a check that could have been run at the time rather than an accusation: divide two numbers from your own statements and see which business the answer looks like.

Information available only afterwards: after the wind-down the company returned remaining capital to investors, in an amount not disclosed, and employees received 12 weeks of severance, with a small team retained to serve existing customers and help them migrate (Failory). Also the overall judgment of former customers

and former employees, that the promised automation was “almost non-existent” and delivery depended on people.

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not “pay more attention”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the company wound down in June 2020, having raised $103 million, with pre-pandemic ARR of about $7 million. Remaining capital was returned to investors in an amount not disclosed, and 100 to 250 employees received 12 weeks of severance (Failory).

Back to the three questions. The most common answer to the first is “take a down round and extend the runway”, and that road was in fact walked here: first the pivot to a matching platform, then a platform that never produced traffic, then investors stopping funding. A down round solves cash and leaves the division standing. The most commonly missed item under the second question is the job title. Calling accountants customer service is the cheapest rewrite in this narrative and the hardest for an outsider to see, and it changed two tables at once, the people on the organization chart and the cost on the P&L. The third question lands in the first quarter after the 2017 launch: had someone measured the share of labour hours once at that point, the pricing basis for the following rounds would have been different. The checkpoint costs one week of counting.

Options ruled out: six.

1. Treat the service business as a service business, refine it and grow it. Failory records this path as rejected, with the reason given in the fundraising terms: money raised as a tech company demanded a scalable tech product.
2. Disclose to investors that the "customer service" staff behind the curtain were accountants. This was tested once, and Failory writes that "some investors were turned away after revealing the 'customer service staff' behind the curtains"; once the price of disclosure had been seen, the remaining 17 invested.
3. Compensate the customer affected by the error in full and handle it openly. Not done. What was offered was a partial refund conditional on confidentiality.
4. Price by account or by hour, and disclose the gross margin structure. Not stated in public records whether this was discussed.
5. Make the service business's unit economics positive before pivoting to a platform. Not stated in public records whether this was discussed. The checkable fact is that the pivot happened while the service business had not produced a profitable structure.
6. Cut the annual fee and narrow the customer base into a small, stable bookkeeping firm. Not stated in public records whether this was discussed. The checkable fact is pricing from $6,000 up to $30,000, a band that corresponds to full-service outsourcing for an SME rather than to self-serve software.

Of the six, the first two have source support, and the last four have only the fact side without the decision side. That line is the boundary of a D-grade source: it can tell you what the company did, and rarely what it considered.

Early signals: six.

① 2017: the product was built on QuickBooks and Xero, orchestrated the workflow, and replaced very little. Pressed past to get the workflow running on existing systems first. ② Reports went out monthly, against the "real time" promise. Pressed past for a reason not stated in public records. ③ The accounting staff were titled customer service officer on the organization chart. Pressed past because what is presented outward is a software company. ④ Customer-service labour was left out of cost of goods sold. Pressed past to bring gross margin close to a software company's. ⑤ A $17,000 customer loss, plus a refund conditional on confidentiality. Pressed past by handling it case by case. ⑥ A team of 100 to 250 against $7 million in ARR. Pressed past with the observation that SaaS companies are all labour-heavy early on.

The first five of the six are visible at the document layer, and the sixth needs one division. Of the two hardest signals, one was renamed and one was reclassified. Neither requires a lie, only a different option chosen while filling in a form. A signal does not predict the outcome; it asks the people in the room to answer one question while they are still there, which is how this number was calculated.

The point of escalation: the pivot to a matching platform. This restated a labour-intensive service as a platform and kept fundraising priced at tech-company multiples. 7 rounds and 17 investors, each one endorsing the classification that this was software. By the time the platform also failed to produce traffic, the company had no service business to retreat to, because customer relationships, pricing and staffing had all been rewritten in the previous step. Failory's judgment of this stretch is that the company "needed to develop a scalable tech product rather than to refine and grow their service business", a sentence that served as the reason at the time and describes the mechanism of escalation. The specific date and the approver are not stated in public records.

This step also changed what was measured. As a service business the company could count accounts, hours per account and retention; as a platform it could count registered sellers, transaction volume and retention. Once the metrics changed, the earlier set stopped appearing in the monthly material. No number was falsified. The company changed which numbers it reported, and at the moment of the change the quantities that could have falsified the original narrative left the field of view. This is the same machine as the presentation-layer move of Chapter 12.

Failure chain: technical layer. The automation was not complete enough to support a claim of no humans. Failory records the software as "very glitchy and mistake-prone", which added to the accountants' workload; delivery still depended on people and the cadence was still monthly. The technical problem sits in the role the software played: it orchestrated the workflow in this business, and the share of the work it replaced was small.

Organizational layer: two definitions were rewritten at once, the job roster and the cost of goods sold line, and the result is that the share of labour, the number best able to falsify the narrative, appears in neither the internal statements nor the outward material. The company had no measurement post independent of the growth narrative, and no second person put a written opinion on record about either definition (not stated in public records whether anyone raised one). The failure here is the removal of evidence from the statements rather than its concealment: the line items remained and the people remained, and they stopped sitting next to each other.

Incentive layer: tech-company multiples reward the classification "scalable", and a service company's gross margin structure depresses a valuation. The price of disclosure was measured once, when an investor left because of it, and non-disclosure became the

default afterwards. 7 rounds of fundraising hardened that default into process.

The break at the joint sits between the organizational and incentive layers: rewriting a definition costs nothing, funding arrives quickly, and the truth has to wait until customer complaints accumulate enough to be visible from outside. By the time it was visible, the company was already a platform.

Minimum counterfactual: in the first full quarter after the 2017 launch, draw a random sample of customer accounts, and have someone who does not report to the sales line measure the share of labour hours from receipt of documents to issued report, then write it into that quarter's investor update. One action, between two funding rounds, with no luck required: the hours are in the rota and the documents are in the system. Once the number sits in an investor update it becomes a monthly comparable series, and once the series exists every later claim about automation has to align with it.

The action lands in the investor update for a reason that concerns readers. The readers of a weekly report rise and fall with the narrative. The readers of an investor update set the price of the next round. The same number read by the first group is read as normal for an early stage; read by the second it enters the valuation discussion. Where the checkpoint sits decides whether it does anything, more than what it checks.

Outcome and sources: wind-down in June 2020; $103 million raised; pre-pandemic ARR of about $7 million; remaining capital returned to investors in an amount not disclosed; 12 weeks of severance. Source: Failory Cemetery, "Why Did ScaleFactor Fail?", link in Appendix B, st-003. Evidence grade D: a single media account of a party's own retrospective, with no primary documents and no round-by-round dates; every internal decision node on this card is marked "not stated in public records"; ARR per head and

accounts per employee are divisions this card performed on dis - closed numbers, and are labelled as such in the text.

Case 6 (Finance): A Final Judgment on Narrative Detached from Assets (condensed)

fi-005. American Patriot Brands and two of its executives. SEC lit - igation (Litigation Release No. 26587, whose text has been read for this book): on 10 July 2026 the Central District of California entered summary judgment finding liability for securities offering fraud; joint and several disgorgement of $17,786,703 plus $6,202,777 in interest, a company penalty of $4,729,004, $6,399,792 in total against CEO Robert Y. Lee, and officer and director bars against two executives. Unlike the settlement in case one, this is a litigated final judgment: the court made a court-level finding on the dis - connect between the offering narrative and the underlying assets. The angle for this chapter is that the scale of the narrative, a grand picture of agricultural assets, was disconnected from the real state of those assets, and the disconnect left checkable space in the of - fering documents. Minimum counterfactual: reconcile the de - scription of asset status in the offering documents against the un - derlying contracts, item by item. Primary source: SEC, LR-26587, 2026, link in Appendix B, fi-005.

Six cases side by side complete the exit list for narrative before evidence: crowdfunding offering documents (fi-002), an approval argument (po-006), a device distribution release (ph-005), a process assumption (ag-003), a funding-round narrative (st-003), and an asset narrative as found by a court (fi-005). What the six exits share is that once a commitment is complete, evidence is demoted from the basis of a decision to material for its defence. Every organiza - tion has a narrative, so the reader’s stocktake is a hunt for com - mitment exits: list every channel through which your organization makes an outward commitment, attach a grade gate to each, and this chapter’s defences have somewhere to stand.

13.4 What Shows Up Earliest

Eight weak signals, each with the reason it is usually pressed past.

1. The conclusion is published before a verification plan exists: the launch date comes first and the experiment schedule second. Pressed past with "the market window will not wait".
2. "We have already confirmed that" starts appearing in the minutes, and the carrier of the confirmation cannot be found. Reason: "it was said out loud."
3. When new data arrives the first question is "how do we explain it", and nobody asks whether it shakes the premise. Reason: "keep the discussion efficient."
4. The person questioning is asked for ten times the evidence, and the person supporting only has to restate the position. Reason: "don't re-litigate."
5. The wording of diligence, test and pilot reports begins aligning with the established conclusion: one fact gets characterised more and more consistently across reports. Reason: "the team has reached consensus."
6. "The person playing devil's advocate doesn't understand the business" becomes the standard classification in the room, and the person classifying never tests his own business judgment. Reason: "division of expertise."
7. Appendices thicken while the body thins: evidence is moved out of sight rather than into the argument. Reason: "leadership only reads the conclusion."
8. The original wording of an earlier version's conclusion is edited silently, with no revision record. Reason: "improving the phrasing." (Compare the register of definition changes in Chapter 38: revisions to a narrative need a trail as well.)

Put the eight back against this chapter's cases. Signal one, publication before a verification plan, is fi-002: when the offering documents were finalised, the revenue expectation had no verification arrangement attached. Signal three, new data explained and never tested, is po-006: the overrun history of comparable projects reached the approval table as a footnote and entered the discussion as "we adopted a new model". Signal four, an asymmetric burden, takes this form in po-006: people questioning the replication strategy were asked first to prove that replication did not hold, while the strategy itself was never asked to prove anything. Signal five, report wording drifting toward the conclusion, is st-003: the growth narrative pulled the characterisation of ticket composition and retention data toward "market education in progress". Signal eight, silent edits, is most common in fundraising, where the forecast numbers in one round's documents get rewritten in the next round's with no revision record.

Re-sorted by cost of discovery: signals two, five, seven and eight are findable at the document layer; one and three need meeting observation; four and six need a record of symmetry across meetings. Half can be caught at the document layer, which is why Device 1 builds grade marking into the document template. Make the defence that depends most on documents the most mechanical action, because a mechanical action does not depend on the actor's alertness.

The eight share a blind spot worth writing down. All of them watch externally visible acts, whether publication, minutes, reports or classification, while the earliest work of narrative before evidence happens inside cognition and leaves no external trace. By the time any of the eight appears, the mechanism has reached the commitment stage. The only way to move the detection window earlier is to watch the moment in criterion ① of section 13.1, when the explanation is first said outward. The only position in an organization

that can watch that moment is process: put a verification plan field into the template for outward-commitment documents, so that a blank field blocks signature. A template is more reliable than vigilance, for the same reason Device 1 is mechanical.

13.5 How to Defend Against It

Device 1: graded release of conclusions. Split outward expression into three grades, hypothesis (marked explicitly as unverified), working conclusion (with an evidence list and disconfirming conditions), and finding (with an independent review record). Only the third may support a commitment. Implementation: the three labels go into the header of the document template rather than into the author's conscientiousness. Signature: a finding-grade document needs an independent reviewer, and the reviewer may not be the person who first proposed the conclusion.

Device 2: appoint a disproof owner. Every major conclusion gets one person whose job description includes overturning it, with budget, data access and a no-blame guarantee. The comparison question from the GenesisAI case is this: who was authorised to check the gap between the company's actual revenue and the revenue the narrative implied? The disproof owner's annual review is not "I checked and found nothing wrong". It is a report of the disproofs found during the year and which of them were adopted.

Device 3: a symmetric burden rule. Write the meeting rule in advance: proponents and challengers face the same evidentiary threshold. The chair enforces it, and violations go into the minutes. The effect is that every triggering of criterion ③ leaves a trace.

Device 4: the time-gap clause. Early accident attribution, finalised offering documents and project commitments all wait overnight.

The clause is procedural in form: the sign button on an outward-commitment document must be more than twelve hours from the last edit, enforced by the system.

Each of the four has one direct use on this chapter's cases. Device 1 on fi-002: a crowdfunding offering document is an outward commitment and may only be supported by finding-grade material, and finding grade requires a reviewer's signature from someone other than the original proponent, which closes the path where a founder believes his own narrative. Device 2 on st-003: the disproof owner's annual review topic is the real composition of the automation completion rate, with data access and a no-blame guarantee; had that post reported once, about a year before the wind-down, the share of tickets would have reached the board pack. Device 3 on po-006: the approval hearing rules are written in advance so that the side arguing for replication and the side questioning it face the same threshold, with the precedent overrun comparison table submitted by the proposing side. Device 4 applies to the shared setting of cases one and five: the sign button on offering documents and commitment documents is forced overnight. The value of that mechanical action is the overnight window it gives motivated reasoning. No new evidence arrives by the next morning, and the only thing that has changed is the distance between the signer and the narrative.

The failure modes of the four devices need writing down as well. Graded release dies in the formalisation of the labels: the header says hypothesis while the prose runs in finding-grade tone, and readers are more sensitive to tone than to labels. The disproof owner dies of an orphaned post: budget without data access, a review without an audience, the annual counter-argument written and archived. Symmetric burden dies when the chair stops enforcing: the rule is in the meeting policy, the chair breaks it once with no consequence, and from then on the rule is symmetric only on

paper. The time-gap clause dies in signature outside the system: the button in the workflow waits overnight while the real commitment goes out immediately in the work chat. All four failure modes share one feature, in that the act of bypassing the device is small, local and unrecorded. The countermeasure is one meta-rule rather than more devices: every exception to a device is registered, and the exception list is reviewed quarterly.

The meta-rule has its own way of dying. The exception list becomes an exception budget: when "three exceptions this quarter" gets read as "nine left in this quarter's budget", registration turns from a defence into a quota. The fix is to make sure the list is read by someone with no stake in the commitment, and to compile it per commitment rather than per organization. The number of exceptions under one conclusion grade carries more information than the organization's total. The first points at a narrative that is currently out of control. The second only shows that the organization is busy.

The joints to other chapters can be gathered here. Section 13.1 describes how commitment rewrites the standing of evidence; Chapter 11 describes how the absence of consequence rewrites the standing of a deviation; Chapter 12 describes how a metric rewrites the standing of a goal. All three run on one engine: organizations need certainty, and certainty can be manufactured. Manufactured certainty is cheap, immediate and allocatable, and its bill arrives at the tail. The one action the book gives the reader is this: whenever certainty arrives too easily, find out where its bill is being sent.

One cross-chapter addition beyond the four devices: give every narrative a control metric. For each key quantitative claim in an outward commitment, whether a revenue expectation, a schedule commitment, a cost commitment or a capacity commitment, register at the time of setting an independently measurable control quantity and the party responsible for measuring it, and put the

commitment’s due date and the control measurement date on the same line. This moves Chapter 12’s pairing discipline onto the narrative: on the day the narrative comes due, the control number appears on the same table without being asked for. The register is an early-warning instrument in itself. A commitment whose control quantity has never been measured is automatically downgraded to unverified hypothesis, on the symmetry principle of criterion ③.

13.6 Boundaries with Neighbouring Mechanisms

Against normalization of deviance (Chapter 11): normalization works at the archival layer, where a deviation gets filed as acceptable, and this chapter works at the expressive layer, where a conclusion gets published early. A case can show both, with the deviation first archived internally as acceptable and then described outward as safe. Both are present in Sizewell’s replication narrative: precedent overruns were archived as manageable, and the replication hypothesis was published as low risk.

Against sunk cost and escalation (Chapter 27): escalation explains input, meaning why the organization continues, and this chapter explains expectation, meaning why it is optimistic. Narrative first is often the preparatory step for escalation, because once the story is out, exit becomes broken faith. Division of criteria: if the review table shows an alignment problem in benefit estimates, go to Chapter 27; if it shows a problem in the order in which conclusions were published, stay here.

Against the slack of success (Chapter 20): slack is the shrinkage of defensive investment in a period without accidents, and this chapter is the inflation of belief in a period without counter-evidence. They share a time condition, which is the absence of bad

news, and a remedy, which is one independent sceptical post with a supply line attached.

13.7 Five Self-Checks

1. The last outward commitment you made about a project's expectations: is the verification plan finished? If it is not, which of the three grades does the commitment belong to?
2. Who on your team is authorised to overturn the mainstream conclusion? When did that person last use the authority, and who paid the cost of using it?
3. The last time you questioned a superior's conclusion, what evidentiary threshold were you held to? Did the proposing side pass the same threshold at the time?
4. The judgment you are most confident of right now: can you write it in the form "this evidence would make me abandon it"? If you cannot, what does this chapter call it?
5. Look back at the first communiqué after the most recent accident in your industry: how long passed between the attribution and the investigation report, how many times did the narrative change in between, and where is the record of the changes?

Chapter 14 Where Intuitive Expertise Breaks Down (Kahneman & Klein)

> What this chapter covers: the conditions under which expert intuition is reliable and the conditions under which it fails; the three criteria; what the failure zone looks like on the ground in four domains; how to test "experience" when it is offered as an argument, and how to make intuition explicit. When to turn back: when a veteran is put into a new environment; when someone answers data with "I have done this for twenty years"; when your own sense that a new situation "feels right" arrives. Estimated reading time: 50 minutes.

Kahneman, an experimental psychologist who spent a career doubting intuition, and Klein, a naturalistic decision researcher who spent a career defending it, once wrote a paper together whose title is the answer: a failure to disagree. What they found is that under particular conditions the two schools agree completely on when intuition can be trusted. There are two conditions: the environment is regular enough, and the feedback is fast enough. This chapter is that agreement turned into engineering, moving "can this be done on intuition" out of folklore and into three criteria a practitioner can check against a situation.

14.1 What It Is

Definition: intuition is a recognition capacity trained jointly by the regularity of an environment and the feedback of practice. Kahneman and Klein give the two preconditions for it to hold. The environment must be regular enough for the regularities to be learned, and practice must carry timely, high-quality feedback, so that errors are seen and corrected (Kahneman & Klein, "Conditions for Intuitive Expertise: A Failure to Disagree", American

Psychologist 64(6), 515–526, 2009 (Crossref record checked, DOI 10.1037/a0016755)). Take away either precondition and intuition enters its failure zone. It still answers quickly, its confidence does not move, and its accuracy drops to the level of a beginner.

Three activating criteria. Any one of them appearing puts intuition out of service as a basis for decision:

1. The environment is unfamiliar, or the rules have just changed (new industry, new equipment, new population, new regulatory framework).
2. Feedback is delayed by more than a few months, or there is no feedback at all (the consequences of the decision cannot be seen and cannot be measured).
3. There is no opportunity for repeated practice (a once-in-a-lifetime situation).

What matters is admitting where the real danger of the failure zone sits. Inside it, intuition does not weaken; it decouples from its own confidence. The certainty the judge feels and the actual hit rate of the judgment part company, and the judge can feel only the first. That is why the most dangerous person in a failure zone is the old expert whose environment changed and whose confidence did not. He is not bluffing. He genuinely feels certain, and that certainty was once calibrated, in the old environment.

Klein's half of the school deserves a paragraph of its own, because it says what "a regular environment" concretely means. Klein studied fire commanders, nurses and military commanders, experts who grew up inside real environments, and the model he produced is the recognition-primed decision: a veteran reads a scene by pattern matching, the matched pattern arrives with a workable course of action attached, and the whole process runs fast enough that the person feels only "intuition". The joint between that model and the criteria is the phrase pattern library.

The environment feeds the library, so the environment's regularity sets the library's valid range. Regularity can be measured: the same class of situation recurs, and the same cues point to the same outcome. Change the environment and the library is still there. Matching still happens, options still come out, and every match is wrong. The recognition-primed decision model never says intuition fails. It says precisely where intuition fails: on the first day after the library is swapped.

Calibration needs writing out in full, because every test in this chapter rests on it. Calibration is the correspondence between confidence and hit rate: say you are nine tenths certain, and you are right nine times in ten. There is one route to it, a large volume of feedback. Weather forecasters are the old specimen, because the forecast is made daily, the outcome is checkable every time, and the check cannot be avoided. The same person, well calibrated in a feedback-dense domain, returns to ordinary performance the moment he moves into a feedback-sparse one. The corollary is hard: calibration is a property of the domain, and an expert who believes he carries it with him is making a mistake. So the question an organization should ask about intuition is never "how accurate are you". It is "what is the feedback density of your domain". The first question measures temperament. The second measures structure.

One organizational consequence of the confidence-accuracy split is worth writing down. A judgment in the failure zone cannot be repaired by thinking it through again. The second pass draws on the same pattern library and returns the same answer, and confidence may rise, because familiarity gets misread as confirmation. That is why all four defences in this chapter are external: an admission form, a checklist, a retraining trigger, an instrument. What they share is that the checking action is placed outside the judge's

pattern library. An organization that expects its judges to doubt themselves ends up with confidence squared.

14.2 How It Grows

There are three entry paths into the failure zone.

Path one: cross-domain transplant. Intuition accumulated in domain A gets carried into domain B (an old CEO dropped into a new industry, a senior mechanical engineer taking over a software system, a successful clinician moved into management). Environmental regularity changed; the recognition template did not. Transplanting happens often because promotion and secondment reward exactly the past record of intuition: the more successful the person, the more likely he is to be moved.

Path two: one success treated as a regularity. A single survival is distilled into reusable "experience". A rule with a sample size of one is unusually confident inside the failure zone, because it never meets a counterexample. A counterexample needs repetition, and this rule has one instance.

Path three: rule change erodes the old regularity. The environment did not change, the rules did (new regulation, new aircraft type, new version, new procedure), and the old intuition is already off target at the level of detail while the subject perceives no change. Not perceiving the rule change is the quietest entrance to the failure zone. Paths one and two at least come with a signal that the environment moved. Path three comes with no signal at all.

The amplifier is hierarchy. The more senior the judgment, the fewer people there are to check it. A front-line error is found the same day (a cross-threaded bolt will not go back in), while an executive's strategic judgment shows itself three to five years later, and the showing can be re-explained by narrative (Chapter 13). The

feedback loop is weakest at the top of an organization, which is where the failure zone stays exposed longest.

Beyond the three paths and the one amplifier, add an observation about time. The failure zone has an age structure. A young person's intuition library is thin, so the failure zone is small: there is not much intuition to fail. In mid-career the library is at its thickest and the environment at its most stable, and the hit rate is at its lifetime peak. Late in a career the library is still at its thickest, while the environment has turned over several times, and the library holds layered memory left by several generations of it, without the person necessarily knowing which layer he is drawing on. This explains a portrait that recurs in failure-zone cases: the judgment that goes wrong was made by the most senior person in the room, and that judgment was accurate earlier in that person's career. Organizational ageing therefore has a dimension that is usually left out of the count. Equipment and procedure age (Chapter 29), and judgment assets age as well: the thickest library corresponds to the oldest version of the environment.

The three paths are not exclusive. In real cases they feed each other. Someone who has crossed domains is especially likely to distil a single success into a rule (path two), because his old intuition occasionally lands in the new environment at a rate above a newcomer's chance level, and those few hits get read as proof that the old intuition holds in the new setting. Rule erosion (path three) gives the other two their best cover: when failure arrives, attribution goes anywhere except to the rule change, because nobody saw the rule change. In an audit, check the three paths separately. Path one checks the CV, the history of matching between person and environment. Path two checks the sample, the number of cases the basis rests on. Path three checks versions, the distance between the version of the rules and the version of the judging habit.

14.3 The Shape in Four Domains

Case 1 (Aviation): The Narrow-Runway Illusion

Case ID: av-011.

When and where: 19 July 2025, 10:53 local time, Lansing, Michigan, Capital Region International Airport (KLAN), runway 28L. The earlier leg that day had departed Coldwater, MI (KOEB) at 10:24, 29 minutes en route, under visual flight rules (VFR) throughout, with no flight plan filed and both departure and arrival inside Class C airspace (CEN25LA260 pp. 1, 4).

What and who: a Cessna A185F, registration N888DK, built in 1982, single piston engine (Continental IO-550-D027B, 300 hp), tailwheel landing gear, four seats; the annual inspection was current at the time of the accident (14 April 2025), and airframe total time was 1,652.4 hours. The operation was Part 91 private flying with no certificate holder. One pilot in the left seat, three passengers, no second pilot, four-point harnesses (pp. 3, 4).

What they were doing: the pilot held a private certificate with single-engine land and single-engine sea ratings, no instrument rating and no instructor rating, and a third-class medical certific‑ate with limitations. Total flight time 295 hours, 100 hours on type, 173 hours as pilot in command, 57 hours in the previous 90 days, 16 in the previous 30, 7 in the previous 24 hours; the last flight review recorded in the report was 16 August 2022, 35 months before the accident (p. 3). The recurring flight review period required by the FAA is 24 months (14 CFR 61.56(a)). The flight was a short cross-country. The destination field’s 10:53 observation read few clouds at 3,000 feet, overcast at 4,400 feet, visibility 10 miles, wind 200° at 8 knots, temperature 24°C, dew point 19°C, altimeter 30.02 inches of mercury, no turbulence; 28L was dry asphalt, 8,506 feet long and 150 feet wide (pp. 4, 5). With the wind from 200° and the runway’s magnetic bearing at 280°, the angle between them is 80°, the

crosswind component about 7.9 knots and the headwind component about 1.4 knots (converted from the wind data at p. 1 and the runway data at p. 5).

The decision point: three moments, each with a carrier that can be checked.

Moment one, before the 10:24 departure. Runway width is printed in the pre-flight material: both the airport diagram and the Airport/Facility Directory (A/FD) list runway dimensions as a separate item. The file holds no record of any pre-flight preparation. Report p. 6 states that the NTSB did not travel to the scene, that the investigation was Class 4, and that the report rests on the pilot's statement and material supplied by the FAA; the type and this operation were not required to carry a cockpit voice recorder or a flight data recorder, so what was said on approach and where the pilot looked left no trace in the file.

Moment two, the flare shortly before 10:53. The report records that the pilot stated he flared too high and suspected the runway width illusion had affected him (p. 1). Three things were in use in the cockpit at once: the visual angle the runway edge lines subtended, the altimeter reading, and the flare rhythm built up over 100 hours on type. The illusion acts on the first of them. Runway width changes the angle the edge lines subtend, that angle gets read as the remaining height, and the reading has a different source from the altimeter.

Moment three, after the touchdown bounce. The report's order is: bounced on contact with the runway, the nose began to weathervane into the crosswind, the pilot made control inputs to stop the ground loop, did not succeed, the right wing struck the runway (p. 1). The probable cause has two clauses: improper flare, and failure to maintain directional control while landing in crosswind conditions (p. 1). The findings run to three lines: aircraft

control (pilot); landing flare (unable to attain or maintain); crosswind (response, compensation) (p. 2).

The direction of the width illusion has a standard formulation in flight training material, and both directions are given there. A runway narrower than the one a pilot is used to makes the aircraft seem higher than it is, the approach goes low and the touchdown comes early. A runway wider than the familiar one makes the air‐craft seem lower, the approach goes high, and the touchdown is hard or runs off the end (FAA, Pilot's Handbook of Aeronautical Knowledge, Ch.17 "Runway Width Illusion"). Runway 28L is 150 feet wide. The report records neither the width of the pilot's usual runway nor the direction of the illusion, only his account of flaring too high (p. 1). This chapter keeps "narrow-runway illusion" as shorthand for the class.

What was visible then: what could be had in the cockpit, ordered by how checkable it is.

Pre-flight material. The runway is 8,506 by 150 feet, asphalt, dry, printed on the airport diagram and in the A/FD (p. 5); the same documents carry the field elevation of 860 feet and the approach type column for 28L, which reads None (p. 5). These numbers were in the flight bag before departure and needed no equipment to obtain.

The 10:53 local observation. Few clouds at 3,000 feet, overcast at 4,400 feet, visibility 10 miles, wind 200° at 8 knots, temperature 24°C, dew point 19°C, altimeter 30.02 inches of mercury, turbulence column None/None (p. 4). Field pressure altitude 860 feet with a temperature of 24°C is roughly 11°C above the standard atmosphere at that altitude, so density altitude sat above pressure altitude (converted from the data at pp. 4 and 5); this could be worked in the cockpit, and the report does not record whether it was.

Cockpit instruments. The report records no reading from the altimeter, the airspeed indicator or the vertical speed indicator. Aircraft of this class are not required to carry a flight data recorder, so the readings existed only in the pilot's memory, and memory appears in the report in a single phrase: flared too high (p. 1).

The runway itself. A 150-foot asphalt strip, and the angle its edge lines subtended was the only real-time height reference on this approach. It is more immediate than the altimeter and easier to mislead, since the angle changes with viewpoint, cockpit height and seating posture, while the altimeter does not (p. 5 gives the width; the mechanism is at FAA handbook Ch.17).

The pilot's own records. Licence, medical, and five lines of hours in the logbook (p. 3), plus his judgment of the aircraft's condition, which the report records as his statement that there had been no mechanical malfunction or failure beforehand (p. 1). That sentence removed a whole class of hypotheses at the time.

What was learned only afterwards: damage to the right wing and aileron was classified as substantial (p. 1); the defining event was coded as loss of control in flight (p. 1); the findings put the landing flare on its own line as unable to attain or maintain (p. 2); and the NTSB never went to the scene at all (p. 6). All four are in the report. None of them was in the cockpit.

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: all four people aboard were uninjured, the right wing and aileron were substantially damaged, and there was no fire and no explosion (pp. 1, 5). The NTSB published its final report on 5 September 2025 without travelling to the scene; the investig‑ ator in charge was Bradley Miller, and the FAA's Grand Rapids office participated (p. 6). Back to the three questions. The most common answer to the first is "go around". The sequence in this case put the cancellable moment before touchdown: by the bounce the aircraft was on the runway, the speed was decaying, the 7.9 knots of crosswind were still there, and the energy and attitude window a go-around needs had already closed, while the action's cost had been 30 seconds with a table before departure. The item most often missed under the second question is the five lines of experience figures on p. 3. All of them were valid and none of them was about width; no line records how many different runway widths he had landed on. The checkpoint for the third question belongs in the approach briefing before departure: one item only, how far this runway's width is from the usual one, and whether the flare reference is the edge lines or the touchdown markings.

Options ruled out: a go-around. The report does not record it be‑ ing considered. The structural fact that can be checked is that the cancellable window lay before touchdown, and that at the bounce the aircraft was already on the runway with the speed decaying and about 7.9 knots of crosswind still present (pp. 1, 4). Diverting to another runway or another airport. KLAN was the destination, the flight plan field reads None, and no alternate is recorded (p. 4); with 10 miles visibility the weather offered no obstacle, and the obstacle was that a diversion needs to be decided in advance, a decision that is not in any document. Taking an instructor or a second pilot along. The report records Second Pilot Present: No (p. 3); Part 91 private flying requires neither a second pilot nor an instructor aboard, and the cost of this option is one seat and one scheduling action. Moving the flare reference from the runway

edge lines to the touchdown markings. The report records no approach briefing at all, and the type and operation were not required to carry a voice or data recorder, so what was looked at and said on approach left no trace (p. 6). A fifth option was flying the trip under instrument flight rules (IFR). The pilot held no instrument rating (p. 3), which put this out of reach at the time; it never entered the option list.

Early signals: ① the flight review stopped at 16 August 2022, 35 months before the accident, past the 24-month cycle (p. 3; 14 CFR 61.56(a)); the reason it was waved through: licence valid, medical current, 57 hours in 90 days, three items all in order, and none of them triggers an interception. ② no instrument rating (p. 3), which means no training in the cross-check that treats instruments as primary and vision as secondary; the reason it was waved through: under visual flight rules vision as primary is the prescribed action anyway. ③ 100 hours on type and 295 in total (p. 3), ample for private flying but covering one type and one class of runway; the reason it was waved through: hours get used as a substitute for scenario coverage. ④ the destination runway is 150 feet wide (p. 5), with the usual runway width unrecorded; the reason it was waved through: 8,506 feet of length is enough, and width is on no checklist. ⑤ a crosswind component of about 7.9 knots (converted, pp. 1 and 5); the reason it was waved through: 8 knots is a routine value in private flying, and the report records the turbulence as None. ⑥ four seats, three passengers, a single pilot in the left seat (pp. 3, 4); the reason it was waved through: carrying passengers is not a regulatory problem, and Part 91 adds no review for it.

The point of escalation: the bounce. Before it, a high flare was a height error that could be covered with back pressure at the moment of touchdown. After it, the aircraft had been thrown back into the air, the speed was decaying, the nose was swinging into the crosswind, and the correction available had shrunk to rudder and

aileron, with ground contact already having happened once. A tailwheel aircraft yaws quickly at that moment: once the main wheels touch, the nose turns into the crosswind, the fuselage axis separates from the direction of travel, the angle of attack becomes asymmetric across the wings, and one wing drops. The report's order is flare too high, bounce, weathervane, control inputs, unsuccessful, right wing strikes the runway (p. 1); it records neither the height of the bounce nor the time from bounce to wingtip contact. No signature, no approval, no second person present. In Part 91 private flying this step has no documentary carrier at all, which is exactly why it cannot be audited afterwards. The pilot's retelling is all there is to check.

Failure chain: technical layer: runway width changed the visual height reference and the flare went high; touchdown bounce; the nose weathervaned in the crosswind; correction insufficient; right wing and aileron struck the ground, damage substantial (p. 1). Organizational layer: a Part 91 private operation has no operator safety management system, no dispatch and no second pilot (pp. 3, 4); regulation sets the flight review at 24 months and this case ran 35 (p. 3); the training and checking columns record review dates and not scenario subjects such as runway width. Incentive layer: empty in this case, and the emptiness is itself a fact. There was no scheduling pressure, no hourly target, no carrier rule, and on a 29-minute leg nothing became more expensive because of one more table lookup on runway width. All three layers broke at once. The interface sits between the organizational and technical layers: a perceptual error needs an external check to be caught, and nothing in the Part 91 structure holds a post responsible for supplying that check.

Minimum counterfactual: before the approach, read the runway dimensions line in the airport material (28L: 150 feet), put the width difference into the briefing, and move the flare reference

from the edge lines to the touchdown markings. One action, taken before the 10:24 departure, needing no luck: the data is in a public document and the report already carries it at p. 5, and the cost is 30 seconds and one sentence. The action changes no flying skill. It changes one thing, moving the "how much height is left" judgment off the visual angle and onto a ground marking that does not vary with runway width. Nor does it require the pilot to know the word illusion: look up the width, compare it, change the reference point, three steps that can be executed without understanding the perceptual mechanism behind them.

Outcome and sources: four people aboard uninjured; right wing and aileron substantially damaged; no fire, no explosion (pp. 1, 5). The NTSB concluded improper flare followed by failure to maintain directional control in a crosswind landing (p. 1). The investigation was Class 4 with no travel to the scene, and the report rests on the pilot's statement and FAA material (p. 6), so half of what this card can say depends on what the pilot said himself. He said he flared too high, and he said the runway width illusion may have affected him: the two most informative sentences in the case and the two that cannot be re-checked. Primary source: NTSB, Aviation Investigation Final Report, CEN25LA260, Cessna A185F, N888DK, Lansing, Michigan, July 19, 2025, adopted 2025-09-05 (PDF 6 pages), link at Appendix B av-011; both directions of the runway width illusion at FAA, Pilot's Handbook of Aeronautical Knowledge, Ch. 17 "Runway Width Illusion".

Case 2 (Chemicals): A Newcomer on a Screen with Three Valves

Case ID: ch-004.

When and where: 4 June 2025 at 14:14:28, Monaca, Pennsylvania, furnace 5 of the ethane cracker (ECU) at Shell Polymers Monaca; at 14:20:45 the gas in the firebox was ignited by the pilot lights and the furnace exploded (CSB 2025-05-I-PA p. 61, Appendix A

timeline). The site sits about 30 miles northwest of Pittsburgh, covers 386 acres, and started production in November 2022 (p. 10).

What and who: Shell Polymers Monaca, an ethane cracker of Shell Chemical Appalachia under Shell plc, with seven cracking furnaces in parallel and a site output of 3.3 billion pounds of polyethylene pellets a year. On the day of the incident the site held 351 employees, 529 contractors and 2 visitors (p. 10). Two people were in the posts at issue: one process control engineer (a PACO engineer in Shell's internal usage, meaning process, automation, controls and optimization) and one ethylene control room operator. The responsible parties are written as posts: the operations production specialist (P-Spec) who assigned the valve-opening task, the PACO engineer who executed it, and the operator who acknowledged the alarm (pp. 26, 28).

What they were doing: a planned return to service. The coke catchers on all seven furnaces had never been cleaned since production began in November 2022; in late March 2025 furnace 1 was opened for inspection and the coke had reached the manway (p. 24). In early April the unit shut down for maintenance, the catchers on furnaces 1, 2, 4 and 7 were cleaned for the first time, and furnace 5 was next in the queue (p. 24). Furnace 5 shut down on 30 May, and on the night shift of 31 May the energy isolation was completed: two 36-inch motor-operated valves (MOVs), one on the furnace side and one on the column side, were closed together, steam at a pressure above process pressure was introduced into the cavity between them, and the vent between them was opened (pp. 24, 16). At that point furnaces 1, 2, 3 and 4 were cracking; furnace 6 shut down on 1 June; furnace 7 moved to hot steam standby (HSSB) on 4 June (pp. 24, 59–60).

Catcher cleaning requires no entry into the furnace, so the work was not isolated as a confined-space entry. The furnace-side valve was closed, the decoking valve open, and the pilot lights left

burning, the reason being to keep the refractory from dropping below the dew point and condensing; on the night shift of 18 October 2021 water from a hydrostatic test had damaged the refractory brick at the bottom of a furnace (p. 25 and note c). That decision was inconsistent with the site's Furnace Isolation Plan, which requires isolating the furnace and cutting the gas to the pilot lights, and the change to lighting them did not go through the deviation review of the Safe Isolation Policy (p. 38). While a furnace sits in pilot-light-only mode, three alarm groups, furnace pressure high, carbon monoxide high and methane high, are suppressed by design to prevent alarm flooding and nuisance alarms, and the suppression appears on screen as a grey bell icon (p. 25 and note d). At 09:20 on 3 June, because the fan motor control firmware was updated during the cleaning, the operator on shift switched the induced draft fan from pressure control to speed control, at about 163 rpm (pp. 30, 60). At 13:40 that day the two MOVs were re-energized (pp. 26, 60). The 08:00 meeting on 4 June was to move furnace 5 out of pilot-light-only and into HSSB; the furnace start-up procedure does not cover the step of leaving pilot-light-only, so the operator had no procedure to follow (p. 26).

The decision point: four irreversible moments.

Moment one, the 2023 ECU process hazard analysis (PHA) revalidation. The document is on file, and the revalidation record identifies the case of "the furnace-side MOV not closed while the column-side MOV is open, causing cross-flow or backflow" as a scenario that could kill several people (p. 42, Figure 20). The same revalidation judged two safeguards, the operator's response to the methane high alarm and the start-up procedure, to be sufficient, and wrote that adding an engineering control was grossly disproportional to the risk reduction (p. 43). The same document carries a separate recommendation, to remove a backflow path created by the routing of the differential pressure transmitter tubing, and

management concluded that the risk was tolerable and did not adopt it (p. 43). The carriers are the PHA revalidation record and the human factors checklist; the report names no signatories.

Moment two, the 2016 and 2017 HMI design specifications. Shell commissioned a process automation contractor to write the HMI Detailed Design Specification, with a first edition in 2016 and a second in 2017 (p. 46). The specification required control logic in the same area or unit to be merged onto a single SIS logic drawing wherever possible, and limited annotation text on those drawings to 40 characters (pp. 47, 49). The three MOVs therefore share one screen, ordered top to bottom as furnace side 511, decoking 512 and column side 513, switched by scrolling, with labels that differ only in the last digit (pp. 27, 28).

Moment three, the 08:00 meeting on 4 June 2025. The operations production specialist assigned the task of opening the column-side valve to the PACO engineer, and the minutes are on file (p. 26, Fig - ure 13). The engineer had never performed that task before (pp. 7, 26, 40). The meeting did not discuss whether an SIS bypass permit was needed (p. 26). The path itself was not new: close both MOVs to isolate, have the PACO engineer open the column-side valve from the engineering station, then use the furnace-side valve to bring the furnace back to cracking. It had been executed successfully 19 times (pp. 20, 41).

Moment four, 14:14:28. The SIS lock was released at 14:04 (p. 61). At 14:13:47 the engineer entered configuration (debug) mode, the screen refreshed and jumped back to the top, and his gaze moved from the column-side valve logic in the lower third of the screen to the furnace-side valve logic in the upper third (p. 28). At 14:14:28 the open command went to 511 rather than 513 (p. 61).

What was visible then: the screen in front of the PACO engineer held three logic blocks laid out identically and numbered 511, 512 and 513, with nothing else describing furnace side, decoking or

column side (pp. 27, 28, 50); the scrollbar returned to the top on refresh (p. 28); there was one written job aid, in which the two steps "select SIS mode" and "issue the open command" are in the wrong order (pp. 36, 40); and there was an indication that the SIS was in read-only mode, which he did not recognize at first and which a process control engineer called over to help spotted (p. 27). On the control room side there was an alarm reading "MOV in unexpected position", its description text differing from the column-side valve alarm only in the last digit, and a grey bell showing the three safety alarm groups still suppressed (pp. 35, 25). On the process side, the induced draft fan was in speed control at about 163 rpm, and the furnace's structural pressure limit was 0.01 psig (p. 30).

About the engineer's experience the report supports only three sentences: he had never performed this task before (pp. 7, 26, 40); he was assigned at the daily production meeting and had to follow the written job aid (p. 40); his process knowledge was limited, and he could not see the process consequence of opening the column-side valve while the furnace-side valve was open (p. 40). The report records neither his years in the trade nor how many SIS bypasses he had worked before, only that this particular task was his first. Another person was in the same room that day. The process control engineer called in to help had opened valves before, saw at the door that the SIS was in read-only mode, and said it had to be switched to debug before it would accept commands (p. 27). One screen, and one man saw the mode first while the other saw the valve first. The difference is not intelligence; it is whether the pattern library has that cell in it.

What was learned only afterwards: two MOVs open at once made a path from the high-pressure header to the low-pressure firebox; within 6 minutes about 641 pounds of cracked gas accumulated in

the firebox; and the explosion tore the furnace wall open (pp. 29, 30).

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the explosion came at 14:20:45 and fire followed, with the fire fought until 16:00 (pp. 31, 61). Nobody was injured. Fifteen employees were evacuated from the ethane cracking area, and one contractor trapped in the unit elevator beside furnace 5 was rescued (p. 32). Furnaces 4 and 6 shut down for inspection and returned to service on 18 June and 20 June respectively (p. 32). Shell estimated property damage at $95 million, and the release was about 5,100 pounds (515 pounds of unburned cracked gas and about 4,600 pounds of combustion products) (pp. 7, 31). Furnace 5 was rebuilt over six months and restarted on 22 January 2026 (p. 32). OSHA issued three citations on 1 December 2025 with proposed penalties of $26,480, and the case closed on 2 January 2026 (p. 32). The CSB listed two major safety issues, reliance on administrative controls and the human-machine interface, and issued two recommendations to Shell (p. 9). Back to the three questions. The most common answer to the first is "colour-code the three valves". This screen already had numbers, colours and zones, and what it lacked was task analysis and a confirmation step; colour would only have added another layer for a person to look at. The item most often missed under the second question is the number 19: it converted an unreviewed work path into site routine. The

checkpoint for the third question belongs on the signature page of the PHA revalidation: one item only, for any scenario judged capable of killing several people, is at least one safeguard independent of human recognition.

Options ruled out: an engineering control. Linde's original design provided two, a key interlock used in sequence and a SIL 2 interlock that closes the furnace-side valve automatically when low backflow pressure is detected (p. 44). Both live on the field manual panel, and the panel was never programmed, at commissioning, to support removing the double isolation, so both interlocks were equivalent to absent whenever the furnace sat in double isolation (p. 44). Opening the valve in the field. Design and policy both require a field operator to work the panel and visually confirm valve position (p. 39), but the column-side valve can only be opened by the PACO engineer from the engineering station (p. 19). An SIS bypass permit. Shell policy requires a bypass permit naming the equipment tag to be monitored; staff understood the requirement to apply only when operations opened a valve, and the CSB could not find that exemption in the policy text (pp. 37, 41). The emergency stop (ESD). It was not pressed, and Shell judged afterwards that pressing it would not have changed the result, since an ESD takes longer to close the column-side valve than a close command issued inside the SIS (p. 37, note 2). Deferring the isolation removal to a period with someone watching. The report does not record the option; the site does not count moving in and out of maintenance status as start-up, shutdown or abnormal operation, so no isolation zone was set (p. 32 note a, p. 37).

Early signals: ① the 2023 PHA revalidation stated that the scenario could kill several people, and both safeguards were administrative controls (pp. 42, 43); the reason it was waved through: engineering controls were judged grossly disproportional to the risk reduction. ② the human factors checklist in the same revalidation did not

record opening and closing MOVs from the engineering station as a human factors problem, although it had been done a dozen or more times (p. 48); the reason it was waved through: this was routine work, not abnormal. ③ 19 successful restarts (pp. 20, 41); the reason it was waved through: nothing went wrong on any of them, which counts as having been validated, and the CCPS names this phenomenon outright in bypass work practice as normaliza - tion of deviance (p. 41). ④ the job aid has "select mode" and "issue command" in the wrong order (pp. 36, 40); the reason it was waved through: the aid was written during commissioning and has always been used this way. ⑤ three safety alarm groups sup - pressed by design in pilot-light-only mode, among them a carbon monoxide alarm that could have acted about 90 seconds before ig - nition (pp. 25, 35); the reason it was waved through: suppression exists to hold down alarm flooding and nuisance alarms. ⑥ the induced draft fan in speed control at 163 rpm rather than the nor - mal pressure control (pp. 30, 60); the reason it was waved through: raising speed by hand while the furnace is being placed in service is site routine (p. 30).

The point of escalation: 14:16:46. The column-side valve had not moved for two minutes, and the engineer concluded that the fur - nace-side valve was already open and the command had done nothing, so he withdrew the 511 open command from the SIS logic (pp. 28, 61). Withdrawing it deleted the only record showing that 511 had just been opened; what he saw with his own eyes was a valve that had been open all along. At 14:17:50 he scrolled back to the bottom of the screen and sent the open command to 513 (p. 61). From that moment both valves were open, the path existed, and a valve needs about 4 minutes to travel from closed to fully open (p. 30). The backflow clock had started at 14:14:28 and had already run 3 minutes 22 seconds.

There were two further chances in the six minutes between 14:14:28 and 14:20:45. At about 14:15:45, roughly five minutes before the explosion, the "MOV in unexpected position" alarm reached the ethylene control room; the operator on shift acknowledged it, read it as the column-side valve alarm because the column-side valve was the one under discussion that day, and took no corrective action (pp. 35, 39). Had he handled it the approved way, a field operator would have been sent to check the furnace-side valve, and the CSB adds a note: such a field person might have been standing near the explosion point, and the result could have been serious injury or death (p. 28, note a). At 14:20 the same operator found the furnace-side valve open and the column-side valve also open, walked to the engineering station and told the engineer to close the column-side valve; by then it was 78% open and still opening (p. 28). The close command was issued at 14:20:39, and the explosion came 6 seconds later (p. 61). Neither chance corresponds to an approval action, and neither had a second reviewer. The SIS lock had been released, configuration mode had been entered, and both the withdrawal and the reissue sit in the logic log (p. 61 A13.6, A13.7), yet no action required anyone's approval.

Failure chain: technical layer: two MOVs in series open at once, creating a path from the header to the furnace 5 firebox; a MOV's full stroke takes about 4 minutes and the backflow lasted about 2.5 minutes; the induced draft fan's capacity was below the backflow rate; the firebox first contained the gas within its 0.01 psig structural limit, and the furnace was not designed to withstand an internal explosion; about 641 pounds of cracked gas accumulated in 6 minutes and met the pilot lights (pp. 29, 30). Organizational layer: all 11 safeguards were policy, procedure or human response to an alarm, delivered as paper and speech (pp. 35–37, 8); the PHA knew the scenario could kill several people and still chose purely administrative controls (p. 43); the start-up procedure does not cover leaving pilot-light-only (p. 26); the job aid's steps are out of order

(p. 40); no SIS bypass permit was raised (p. 41); the HMI specification merged three valves onto one screen, capped labels at 40 characters and required no task analysis (pp. 47–49); the human factors checklist did not cover the task (p. 48). Incentive layer: the work plan for 4 June was to move furnace 5 from pilot-light-only to HSSB (p. 26), and the restart schedule competed with other site modification projects for the same money; reprogramming the manual panel was listed as a future project while other projects took priority and were approved (p. 44); OSHA's proposed penalty of $26,480 is not on the same scale as $95 million in damage (p. 32). All three layers broke at once. The interface: the organizational layer handed the step of identifying the right valve to the operator's pattern library, and the incentive layer put the modification that could have replaced that library, the reprogramming of the panel, at the back of the queue.

Minimum counterfactual: after the 2023 PHA revalidation and before the next removal of the double isolation, program the field manual panel so that the double isolation can be removed, which puts Linde's two interlocks, the key sequence and the SIL 2 low-pressure auto-close, into service in the double-isolated state, and retire the path of the PACO engineer opening the column-side valve from the SIS. One action, in the modification window after commissioning, needing no luck: the hardware is already on the panel and what is missing is programming, and report p. 44 records that this path sat on the future projects list, so what was missing was taking it off that list. The cost is one programming job and one training session; the return comes in two layers. The key interlock makes opening the wrong valve physically unreachable, and the SIL 2 interlock corrects two-valves-open automatically the instant low pressure appears. It also requires nobody to admit he can be wrong, because an interlock does not ask an operator for his years of service.

Outcome and sources: no injuries or deaths, 15 people evacuated, one contractor rescued from the elevator; property damage $95 million; furnace 5 out of service for six months (pp. 7, 31, 32). The CSB concluded that the cause was the simultaneous, inadvertent opening of two MOVs used for isolation, with contributing factors including the assignment of a process control engineer with lim - ited process knowledge to operate the valves and reliance on inef - fective administrative controls to prevent backflow (p. 8). The same report carries a harder finding: all 11 controls Shell identified failed, because employees had not been adequately trained be - forehand, did not understand the controls, or were not capable of following them (p. 37). That sentence rounds out the definition of an administrative control. Its effectiveness is not a property of a document; it is a property of people. Primary source: U.S. Chem - ical Safety and Hazard Investigation Board, Furnace Explosion and Fire at Shell Polymers, Monaca, Pennsylvania, Incident Date June 4, 2025, Report No. 2025-05-I-PA, September 2026 (PDF 70 pages), link at Appendix B ch-004. Page numbers follow the PDF' s physical page order.

Case 3 (Maritime): The Chain of Reaching In to Rescue (Condensed)

ma-010. The emergency response rescue vessel Viking Islay, September 2007, off the East Yorkshire coast, England. The MAIB report (Report 12/2008, pages read for this book) records that cor - rosion of steel structure inside the anchor chain locker consumed oxygen and left an oxygen-deficient atmosphere. The first crew member entered and collapsed. His shipmates went in after him one after another without gas detection equipment, and 3 seamen died. The report' s own words confirm it: "three seamen on board the ERRV Viking Islay lost their lives as a consequence of entering an enclosed space" . The rescue instinct is a virtue in open air; in an oxygen-deficient compartment it was the second

and third cause of death. The decision point is the moment the second crew member stood at the hatch. Minimum counterfactual: after the first man collapses, the second does not enter, and calls for support first, carrying a gas detector. Primary source: MAIB, Report 12/2008, 2008, link at Appendix B ma-010.

Case 5 (Healthcare): Pain That Never Entered the Triage Differential

Case ID: me-015.

When and where: the reference event took place in the emergency department of a non-specialist acute hospital in England, anonymised in the report as AH1, and the report records no specific date. HSIB opened its investigation on 20 July 2017 and published the final report on 23 January 2020. The clinical timeline runs on the clock: onset at 18:00, death at 03:15 the following day (HSIB 2020 § 2.1–2.67, Appendix 1).

What and who: a national investigation by the Healthcare Safety Investigation Branch (HSIB) into the recognition pathway for acute aortic dissection (AD). The bodies involved were the emergency department at AH1 and the regional cardiothoracic centre (RCC). The posts, in the order the patient met them: a triage nurse (EDS, 15 years in that emergency department), a nurse in the low-dependency area (EDSN, more than 7 years of nursing experience), an advanced care practitioner (EDACP, a registered nurse, 22 years in practice, less than 1 year in emergency care), a foundation year 2 doctor in emergency medicine (FY2, in the second year since qualifying, less than 1 week in the department), an emergency medicine consultant (EMC, 18 years in the department), and a medical specialty registrar (MSpR, in the sixth year since qualifying, in the first year of specialty training in medicine, 8 months at the hospital) (§ 2.14, 2.18, 2.21, 2.27, 2.30, 2.35).

What they were doing: the most ordinary class of work an emergency department does, sorting non-specific chest pain, investigating it, and deciding who stays. The report supplies the denominators. England sees about 2,500 acute dissections a year, and about 2,000 of those patients reach hospital alive; on that basis an average emergency department meets about 15 cases a year, which is about 1 in 400 of the patients who present with chest pain (§ 4.6.1.2). Chest pain accounts for about 5% of emergency attendances, and about 45% of those are eventually diagnosed as cardiac (§ 4.9.3.1). Mean time from arrival to diagnosis runs between 4 and 5 hours (§ 4.8.7.4), and diagnostic delay occurs in 16% to 40% of cases (§ 5.2.4). The mortality denominators are harder still: about 20% die before reaching any hospital, and about 50% die before reaching a specialist centre (§ 5.2.2).

The patient's path ran alongside those numbers. A 54-year-old man, ordinarily fit; at 18:00, while weightlifting, sudden retrosternal pain with nausea and a sense of impending death, the pain moving to between the shoulder blades; at 18:40 he rang NHS 111, the call handler followed the script and asked about the character of the pain (tearing, severe), he answered that it was severe and lasted no more than 15 to 20 seconds, then agreed to the description crushing, aching, band-like, and took 300 mg of aspirin (§ 2.2–2.8). At 19:04 an ambulance arrived, crewed principally by a first-year student paramedic (PM2); blood pressure in the two arms measured 127/61 and 114/84 mmHg, 19 minutes apart, with "no signs of abdominal aortic aneurysm" recorded; the ECG showed occasional ectopics and an enlarged cardiac shadow (§ 2.10–2.11). The crew agreed that everything pointed to muscle pain and conveyed him anyway, to rule out a myocardial infarction (§ 2.12). He reached AH1 at 20:04, was booked in at 20:12, and sat in the waiting area (§ 2.13).

The decision point: four irreversible moments.

Moment one, 20:46 to 20:49, triage. EDS entered “chest pain” into the electronic Manchester Triage System (MTS) with observations of pulse 128, blood pressure 108/47 and respiratory rate 17; the system assigned category 2 on a five-level scale, where category 2 requires assessment within 10 minutes, and she signed at 20:49 (§ 2.17, Appendix 1). After signing he was sent back to the waiting area and entered a cubicle only at about 21:00 (Appendix 1). The MTS chest pain flowchart contains no “aortic pain” discriminator, which is precisely what HSIB’s first recommendation set out to add (recommendation R/2020/066). The carrier is the electronic triage record, and the signatory is EDS.

Moment two, 21:21, the first assessment entry. EDACP picked the case up from the electronic board, and triage gave no verbal handover (§ 2.22). After examining the patient he wrote “looks well”, with a differential of musculoskeletal chest pain, chest infection and cardiac chest pain; a Wells score of 0, not suggestive of pulmonary embolism; neither palpation nor movement reproduced the pain, and he was unhappy with his own diagnosis, because the pain would not reproduce and the patient said he felt unwell (§ 2.23–2.26). The carrier is the 21:21 electronic record. He then handed the patient to FY2, verbally, and ended his shift.

Moment three, 22:48, FY2’s entry. The chest X-ray was read as no obvious abnormality, troponin T was 30 ng/l against a normal range of 0–14, and D-dimer was above 3000 ng/ml against a normal range of 0–230; the record notes only negative findings for venous thromboembolism, and the abnormal values were written down without being taken anywhere (§ 2.28–2.29). At 22:58 an addendum records discomfort, back pain and vomiting (§ 2.34). Radiology reported the chest X-ray the following day as an enlarged cardiac shadow with a widened mediastinum (§ 2.28).

Moment four, 23:27, the imaging request. Once MSpR took over she said she had seen a similar case before, suspected dissection, asked

for blood pressure in both arms (123 and 100 mmHg, a difference of 23) and requested an emergency CTA at 23:27 (§2.35–2.38); the scan ran at 00:01 and was formally reported at 01:06 as Stanford type A, with the tear extending to the right common iliac artery and the right brachiocephalic artery (§2.42–2.44).

What was visible then: there is no cockpit here, only consulting rooms, and what could be seen was just as limited. The list follows the order of contact.

At the triage desk, 20:46 to 20:49, EDS had the MTS electronic flowchart in front of her: select the presenting complaint "chest pain", click through the items, and the system assigns a category by algorithm. The evidence she could enter was four observations, pulse 128, blood pressure 108/47, respiratory rate 17 and a pain score of 1/10, together with the patient's colour and posture as he sat in the waiting area. She remembered afterwards that she had looked once and thought he was in a bad way (§2.15). The system returned category 2, to be seen within 10 minutes (§2.17). The flowchart has no aortic pain line, and no line anywhere asks for a comparison of the two arms.

At the assessment desk, 21:20 to 21:26, EDACP had no handover, only the triage note and the electronic board (§2.22). What he could see was a complete history (weightlifting, sudden onset, 10 to 15 seconds, left arm and back, a sense of impending death), one examination (neither palpation nor movement reproduced the pain), a Wells score table calculated at 0, and the patient's appearance: looks well, no pain face, no sign of circulatory failure (§2.23–2.24). Of those four, the last was the most persuasive and the least reliable, because it fitted muscle strain after exercise exactly.

At the handover desk, 22:48, FY2 had three new items, the chest X-ray she read as no obvious abnormality, troponin T of 30 and a D-dimer above 3000 (§2.28). She could also see how the patient had

changed over three hours: walking about, pale, sweating, vomiting, back pain (§ 2.33–2.34). Her seniority was second year since qualifying and less than 1 week in the department (§ 2.27). Of the three new items, two were abnormal and one had been judged normal, and the report records that her entry carried only the negative findings that excluded venous thromboembolism (§ 2.29).

At the consultant's end, around 22:50, EMC was the most senior person in the department with 18 years behind him. His information on this case amounted to one telephone call: he went to the bedside, the patient was in the toilet, and the examination did not happen (§ 2.30). His advice to FY2 was that abnormal blood results were worth worrying about, that she should speak to the medical registrar, and that admission might be needed (§ 2.31).

At the diagnostic end, 23:00 to 23:27, MSpR held the most unusual information of the five: one precedent, a similar case seen before, plus systolic pressures of 123 and 100, a difference of 23 mmHg (§ 2.35–2.37). In nurse training that figure is taught as significant once it passes 20, and it did not reach the record until 23:21.

What was learned only afterwards: the chest X-ray had in fact shown an enlarged cardiac shadow and a widened mediastinum (§ 2.28); the CTA showed the dissection extending along most of the aorta to the right common iliac artery and the right brachiocephalic artery (§ 2.44); scored retrospectively on the ADD-RS he would have had 1 point (§ 4.9.5.3); and his four hours in the department were shorter than the national mean of 4 to 5 hours (§ 4.8.7.4).

Stop point:

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".

> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 01:57 the transfer vehicle left. About 12 minutes later breathing stopped and the heart stopped soon after; the vehicle diverted to another hospital to collect blood, and on arrival received a message from the regional centre to come immediately, so it turned around without the patient ever leaving the vehicle. The regional centre said that after 32 minutes of CPR nothing more could be done, and death was declared at 03:15 (§2.62–2.67). HSIB made two recommendations: that the Manchester Triage international reference group add "aortic pain" to MTS as a chest pain discriminator, and that the Royal College of Emergency Medicine and the Royal College of Radiologists jointly develop, deploy and evaluate a national recognition and management pathway (R/2020/066, R/2020/067). There were also three safety observations and one safety action, under which NHS Pathways amended the telephone triage algorithm in version 18 (§5.2). Back to the three questions. The most common answer to the first is "more training for emergency doctors". Six people were involved in this case and three of them had more than ten years of experience, while the one who recognized the dissection was in his sixth year since qualifying with a single precedent for this disease to draw on. Training cannot change an exposure of 15 cases a year per person. The item most often missed under the second question is the denominator of 1 in 400 chest pain patients: it says that practice does not make perfect with this disease, and every judgment that the patient looked well was made on that premise. The checkpoint for the third question belongs within 10 minutes of triage: calculate the ADD-RS once, and arrange imaging at 1 point or above.

Options ruled out: adding "aortic pain" to the triage table. The MTS chest pain table had no such item at the time; HSIB made the recommendation in January 2020, and the Manchester Triage international reference group replied on 17 April 2020 that it had been added to four flowcharts, chest pain, back pain, abdominal pain and collapse, with a definition attached ("sudden onset, presenting with severe abdominal or chest pain, which may be sharp, stabbing or tearing..."). A national diagnostic pathway. None existed at the time; the report recommended that the two colleges develop one, and the reply said the working group planned to publish guidance in 2021. Routine CTA for all chest and back pain. The report discusses the cost of that route, which requires balancing detection rate against CTA capacity, and it therefore argues for screening first with a score and D-dimer (§ 4.9.5.2, 4.9.6.3). The ADD-RS itself. The tool was published and available as an online calculator at the time, and the report's retrospective calculation for this case comes to 1 point (§ 4.9.5.3); it was never outside the options, it simply went unused. Controlling blood pressure and heart rate before transfer. The report records that this was considered and given up, to save time and to avoid needing a doctor in the vehicle (§ 5.1.13), while the ICU consultant argued against delay on the grounds that the disease carries a 10% hourly risk of death (§ 2.50).

Early signals: ① about 2,500 cases a year nationally, with about half dying before reaching a specialist centre (§ 5.2.2); the reason it was waved through: the disease is intrinsically lethal, and the deaths were booked to the disease. ② neither palpation nor movement reproduced the pain (§ 2.24); the reason it was waved through: musculoskeletal pain stayed in the differential, and the report describes a tendency not to abandon that diagnosis while exploring more serious cardiopulmonary problems (§ 5.1.5). ③ troponin of 30 and a D-dimer above 3000 (§ 2.28); the reason it was waved through: the record said only that venous thromboembol-

ism was excluded, and the abnormal values were never converted into a next action (§ 2.29). ④ the chest X-ray judged as no obvious abnormality, then read the next day as an enlarged cardiac shadow and a widened mediastinum (§ 2.28); the reason it was waved through: the on-shift read is taken as the conclusion, and a chest X-ray is a poor test for dissection in any case (§ 5.1.8). ⑤ the patient walking about the department, saying he felt unwell, vomiting, with back pain (§ 2.33–2.34); the reason it was waved through: observations and oxygen saturation were normal and he looked well (§ 2.23, 2.29). ⑥ triaged as category 2 yet placed in the low-dependency area (§ 5.1.4); the reason it was waved through: category 2 sets a 10-minute assessment time and says nothing about the level of monitoring.

The point of escalation: 22:48. Three lines lay in the record at the same moment, pain that would not reproduce, a D-dimer above 3000 and a troponin of 30, and what FY2 wrote was a direction, chase the bloods and possibly home, resting on a chest X-ray reported as no obvious abnormality (§ 2.27–2.28). Before that moment, anyone taking the case over who laid the three side by side would have moved toward imaging; after it, the record carried a direction written as exclusion, and later people followed the exclusion.

From that moment to the CTA request at 23:27 lie 39 minutes, and the report records what filled them: EMC advised referral to medicine, FY2 called the medical registrar, went back to the patient and found him vomiting, and the patient's partner also agreed that something was wrong (§ 2.31–2.32). The force that overturned the direction came from the eye at the bedside, not from the record. The carrier is the 22:48 electronic entry; the step needed nobody's approval and got none. EDACP had gone off shift, and EMC was consulted by telephone only after this point (§ 2.26, 2.31).

Failure chain: technical layer: the symptoms are non-specific, since 85% of patients have sudden severe chest or back pain that can be described as tearing or as crushing (§ 1.9); diagnosis rests on imaging alone (§ 5.2.6); a chest X-ray is unsuited to detecting dissection (§ 5.1.8); and a dissection can extend from the ascending aorta to the iliac arteries within hours (§ 2.44). Organizational layer: the triage table lacks an aortic pain item, and the existence of recommendation R/2020/066 is the evidence; triage to assessment had no verbal handover (§ 2.22); for three hours the patient was carried by EDACP and FY2 while the consultant never examined him (§ 5.1.6); between diagnosis and transfer there is a further wait of more than an hour for the formal report (§ 5.1.11); and nationally there is no usable incidence and outcome data, which the report makes a safety observation in its own right (O/2020/053). Incentive layer: chest pain is about 5% of emergency attendances and about 45% of it is cardiac, so throughput and priority sit on that side; the report records that junior doctors may be reluctant to request a CTA out of concern about its availability (§ 4.9.4.5); and transfer time pressure overrode treatment before transfer (§ 2.50, § 5.1.13). All three layers broke at once. The interface: the epidemiological data is in the report and the checklist line is not, and no post existed to translate data into a line. The two recommendations were meant to create that post, and at the time it had no name.

Minimum counterfactual: before 22:48, calculate the ADD-RS for this patient. The score covers three classes of high-risk feature, predisposing conditions, pain features and examination features, on a 0 to 3 scale; here it comes to 1 point retrospectively, which in the published literature means considering a CTA, even without a D-dimer result at the time (§ 4.9.5.3). One action, taken in any minute after triage, needing no luck: the score is published with an online calculator, and it measures three facts that can be asked for, a family history of Marfan syndrome or aortic disease, whether the

pain was sudden and severe, and an absent pulse or an inter-arm blood pressure difference. Nor does it require the person calculating to trust his own intuition, because the score moves "should I suspect this" from feeling to counting.

Outcome and sources: the patient died at 03:15 the following day, and HSIB issued 2 recommendations, 3 safety observations and 1 safety action (§5.2). One of the report's central findings is blunt: during his time in the department the patient never saw a consultant, and was seen by an advanced care practitioner and an FY2 (§5.1.6). Another is equally blunt: a four-hour delay from arrival to diagnosis is not unusual with this disease (§5.1.10). The inability to produce national incidence and outcome data is a gap the investigation admits to itself (O/2020/053), and the gap is the physical evidence for criterion 2, since a system that cannot count its own numerator is in no position to give its judges feedback. Primary source: HSIB, Delayed recognition of acute aortic dissection, 23 January 2020, link at Appendix B me-015 (archived link, grade A–); this card cites by paragraph number (§) because the report is web-only and sets no page numbers.

The ending of case one is worth a second look, specifically the pilot's self-report that he suspected the runway width illusion had affected him. That report is rare twice over. Survivors of an illusion usually attribute the incident to the wind, the light or how they were feeling that day, and blaming an illusion amounts to admitting that perception itself was wrong, an admission that needs the help of evidence afterwards. Rarer still is blaming the illusion rather than the state of the day, because it shows that the man completed the upgrade from "I made an error" to "human perception errs on narrow runways". After that upgrade the experience remains usable, carrying a note on its range of validity from then on. That is what Device 2 is meant to produce in quantity:

experts who know where their confidence comes from, with the confidence intact and the annotation attached.

Case 6 (Healthcare): Judgment Alone on Non-Specific Symptoms (Condensed)

me-013. Diagnosis of pulmonary embolism (PE) in an English emergency department. The HSIB investigation report (March 2022, archive snapshot read for this book) drew on 14 serious incident reports, concluded that individual judgment under non-specific symptoms lacked decision support, and made 3 recommendations. The symptom spectrum of PE (breathlessness, chest pain, a fast heart rate) overlaps several common diagnoses, and clinical intuition carried far too much of the sorting load: the judge's experience holds few PE samples (criterion 3), the symptom environment is irregular (criterion 1), and the outcome feedback is broken (criterion 2). The decision point is the moment of individual judgment with no decision support tool present. Minimum counterfactual: embed a standardised scoring tool in the emergency pathway and make its use mandatory. Primary source: HSIB, Clinical decision making: diagnosis of pulmonary embolism in emergency departments, 2022, link at Appendix B me-013 (grade A–).

A number in case two deserves a paragraph of its own: 11 administrative controls. The figure looks generous, and taken apart item by item it is all the same material, procedures, training and prompts, every one of them resting on people having stable attention and stable recognition. Dense stacking of administrative controls has a name in this chapter's frame: transferring the safety load of the system onto people, and then treating human reliability as an input assumption. When the CSB lists reliance on administrative controls as one of its two major safety issues, an investigating body is confirming that the transfer failed, and confirming it at its own level of authority. A conversion rule for the reader: for each

administrative control you meet, ask once what guarantees its effectiveness. If the answer comes back to human attention, the control's real weight in that system is close to zero, unless the system has also provided a backup for attention, in the form of interlocks, fault tolerance or a forced wait.

Case 7 (Aviation): Three Minutes in Cloud for a VFR Pilot (Condensed)

av-014. A Cessna FR 182 on a private ferry flight, February 2021, Germany. The BFU investigation report (2022, read for this book): the flight entered cloud under visual flight rules, spatial disorientation led to loss of control, the aircraft struck the ground, and the pilot was killed. The environmental premise of visual flight is continuous visual reference to the ground, and cloud removes that premise entirely; with no ground reference the pilot's felt sense of attitude returns systematically wrong answers. This is criterion 1 in its most complete form, since the environment changes wholesale the instant the aircraft enters cloud. The decision point is the decision to keep flying visually in front of a gap in the cloud. Minimum counterfactual: check the route forecast during pre-flight planning, and set a personal line of "no going visual". Primary source: BFU, 2022, link at Appendix B av-014.

Cases five to seven push each of the three criteria to an extreme. Case five is the textbook for criteria 2 and 3, broken feedback plus a low-frequency scenario. Case six is a triple failure zone, with all three criteria satisfied at once. Case seven is the instantaneous, full version of criterion 1, the environment replaced wholesale inside the second it takes to enter cloud. The value of the criteria lies in how they combine. One criterion on its own can be negotiated, through training, experience or remediation; when two or more hold together, any confidence in intuition should be counted as zero until an instrument or a checklist is in place.

Combining them has a second use, which is profiling posts. Run the three criteria over each key judgment post in your organization and most will land in a stable combination. Scheduling and triage posts, with high judgment frequency and fast feedback, usually lack only the environmental upkeep of criterion 1. Purchasing and investment posts, with slow feedback, lack criterion 2. Emergency and safety posts, with once-in-a-lifetime events, lack criterion 3. The combination type decides the combination of defences: a post short of criterion 1 gets a change trigger, short of criterion 2 gets records and review, short of criterion 3 gets instruments and checklists. Draw that map across the whole organization and the budget for intuition defences has an ordering for the first time, with the posts carrying the heaviest combination going first.

The chain rescue in case three deserves a second look as a domain general rule. Serial deaths during rescue from an oxygen-deficient confined space are a recurring category in marine investigation: the first man goes down to the environment, and the second and third go down to the rescue instinct. The mechanism is at its purest in this category. Saving people is one of the strongest social in - stincts humans have, it is almost always right in open air, and it is fed by dense feedback in a stable environment; a confined space turns the environment' s single critical parameter, oxygen, into something that cannot be sensed, and the whole premise of the intuition is pulled away while the surface scene does not change at all. The MAIB report attributes the three deaths to the act of en - tering the enclosed space itself, which amounts to a ruling: in this setting the correct action is a veto on intuition, and the equipment (a gas detector) and the procedure (call first, then enter) exist to carry out that veto.

The under-deduction in case four reads differently under each criterion. The old calculation table is the specimen for path two, a conversion convention carried through years of tender offers. The

new instrument form, short call options appearing in the consid - eration of an offer, is the rule change of path three, and the in - ternal procedure does not define how the new form converts, which is the system admitting at its own level that judgment is still running on the old library. FINRA's penalty, $80,000 of which $40,000 went to FINRA, is small in scale; the value of the case is its generality. Financial instruments are innovated at a pace that runs steadily ahead of the pace at which internal definitions are up - dated, and the gap between the two paces is the interval in which this class of case is generated. Device 3's new column on the change form, the list of existing experience affected, is designed for exactly that interval: the day a new position type enters the system, the review of the conversion definition should be triggered.

The risk that went into the PHA with no independent guarantee mechanism deserves one further step along the administrative-control audit. After all 11 controls have been asked what guaran - tees their effectiveness, most organizations' lists shrink to two or three lines, and what survives is the set of items that carry a genu - ine independent guarantee. The shrinking is not alarming. It makes the real water level of risk control visible for the first time. What follows visibility is the watershed: add engineering controls, which are expensive, slow and effective, or accept the level and write it into the risk assessment, which is cheap and honest. The worst choice is to leave the list untouched, because the gap between the paper water level and the real one is the stock of future accidents.

One organizational detail in case two is worth enlarging: an oper - ator without experience was placed in an independent post on the grounds that he had completed standard training. That sentence quietly merges two concepts, training hours and intuition built. Standard training delivers explicit knowledge, the rules, the se -

quence, how to read a label; intuition needs repeated exposure in the environment plus feedback, and the two take construction periods an order of magnitude apart. Scheduling pressure looks only at the first, because the first has a certificate attached that can be checked. A corresponding check for the reader's own organization: in the roster for a key post, is the basis for working independently a certificate or a count of exposures? The first is the system's assumption, and the second is the system's reality.

Case 4 (Finance): The Short Options Left Out of the Deduction (Condensed)

fi-012. Cowen and Company. The FINRA penalty of August 2026 (monthly summary PDF read for this book): in a partial tender offer the internal procedure did not define how 100,000 short call options should be converted into a net long position, the calculation followed the old convention, the firm over-reported by 150,000 shares, and the penalty was $80,000, of which $40,000 went to FINRA. The decision point is the moment the net long calculation met a new instrument form. Minimum counterfactual: the internal procedure triggers a review of the conversion definition automatically when a new position type is added. Primary source: FINRA, Disciplinary and Other FINRA Actions – October 2026, 2026, link at Appendix B fi-012.

What the four cases share is this: every judge was fully confident at the moment of judging, and the source of that confidence, visual experience, post common sense, the rescue instinct, the old calculation table, lay inside the range of criterion 1 or criterion 3. Recognizing a failure zone does not require doubting the judge's intelligence. It requires checking which environment the experience behind the judgment was formed in.

14.4 What Can Be Seen Earliest

Eight weak signals, each with the reason that usually waves it through.

1. The reasoning behind a decision contains analogy rather than measurement (“this is much like that one back then”). Reason given: “experience is the fastest theory there is.”
2. The environment’ s rules, equipment or population changed recently, and the people in the key posts did not. Reason: “core competence is transferable.”
3. A veteran answers data in a meeting with seniority (“I have done this for twenty years”). Reason: “respect for practice.”
4. A class of decision has never had feedback: it was made, and nobody has ever checked whether it was right. Reason: “it has always run fine.”
5. Training hours get compressed on the grounds that the system will handle it. Reason: “efficiency.” (Chapter 26 develops this.)
6. A newcomer asks a basic question and the experts present answer vaguely. Reason: “the detail has gone; the direction I am certain of.”
7. The induction period for someone crossing domains is an order of magnitude shorter than the period their old domain took to build. Reason: “he was brought in to step straight in.”
8. The organization keeps no record of its expert judgment: the judgment was made, the basis was not written down, and afterwards nothing can be calibrated. Reason: “the business moves fast.”

Items 4 and 8 together are the definition of calibration. No record means no feedback, no feedback means no calibration, and confidence without calibration is only temperament.

Put the eight signals back against this chapter's cases. Signal 1, analogy instead of measurement, maps onto case one, where the sense of when to flare is analogy in its fastest form. Signal 2, environment changed while the post did not, maps onto case two, where the new three-valve screen met an operator population that had not changed. Signal 3, seniority answering data, appears in case five in a variant: experience answering checklisting, with the checklist read as distrust of clinical judgment. Signal 4, decisions without feedback, maps onto case five, where the missed diagnosis killed the patient after transfer and the emergency department never heard about it. Signal 5, compressed training, maps onto case two, where administrative controls replaced depth of training. Signal 7, induction too short across domains, maps onto case two's new operator. Signal 8, unrecorded judgment, maps onto all of them, since the basis of an intuitive judgment cannot be reconstructed after the event.

Read by discovery cost, the eight signals have a reading peculiar to this chapter. Signals 1, 3 and 6 come by ear, in meetings and conversations. Signals 2, 5 and 7 come by eye, from the organization chart and the roster. Signals 4 and 8 come by checking records. A failure-zone audit is therefore a division of labour among three senses: hear the analogies and the seniority talk, see the mismatch between environment and people, check for the absence of feedback and records. The three correspond to a conversation audit, a post audit and an archive audit, and any one of them done alone covers a third of the ground.

The conversation audit has three opening questions for a reader who wants to start on Monday. Ask the expert: "if this judgment is wrong, when is the earliest you would know?" A judgment that cannot name a time has no feedback loop, and criterion 2 lights up. Ask the expert: "when did you last meet a case of this kind?" An answer of never puts the judgment under criterion 3, to be handled

as a hypothesis awaiting verification. Ask the team: "if someone more junior made this judgment, how would he get it wrong?" The list of errors in the answer is the list of experience the judgment actually rests on, and the longer that list, the more exposed the judgment is to the failure zone. None of the three needs a tool. All three need a questioner who is not under seniority pressure.

The standard move in an archive audit is to sample judgments, not events. Draw a number of significant judgments at random from the past year and ask one thing about each: how much of the basis was recorded. Most organizations doing this for the first time come out with a glaring empty set, which is not a scandal but a starting point, since the empty set says the organization has no recording system for its judgment assets, and weak signal 8 has something to remediate at last. Sampling judgments reads cleaner than sampling events, because an event arrives with its own demand for attribution, while a judgment sits quietly inside the process and reflects the organization's real habits best.

14.5 How to Defend Against It

Device 1: an intuition admission form. Add one mandatory line to the review form for significant judgments: "Does the experience this judgment rests on hold in the current environment (rules, equipment, people)? What is the basis?" A judgment that cannot produce a basis is downgraded to a hypothesis awaiting verification, under the tiered release of Chapter 13. The cost is very low, one line. What it stops is the whole class of failure zones.

Device 2: convert intuition into a checklist. A veteran's intuition is an asset, so bring it out and write it as explicit check items, which lets newcomers benefit and lets the veteran be checked. The method is interview extraction: have the expert talk through what he is looking at while he works, and the fixation points recorded

become the candidates for check items. Whatever will not go down on paper is marked as the non-reliable part and signed by the expert himself. The signature is the step that matters, because it turns "only tacitly known" from a disclaimer into a risk statement.

Device 3: environmental change triggers retraining. For every change of rules, equipment or interface, list the intuitions that depend on the old environment and assess item by item whether retraining is needed. The interface change in the Shell Monaca case and the instrument innovation in the Cowen case should both have triggered this step. The trigger's carrier is the change form: add a column reading "list of existing experience affected", to be filled in by whoever raises the change.

Device 4: give the failure zone an instrument. A four-gas detector for a confined space, approach aids for a narrow runway, automatic conversion for a net long position: inside a failure zone, replace recognition with measurement. The budget ordering rule is to start with scenes where criterion 1 holds, since failure there is the most complete.

Now put the four devices onto this chapter's cases. Device 1 on case one: add one mandatory line to the approach briefing, "how far is this runway's width from the usual one, and how does the flare reference change". One line, twenty seconds, and what it stops is the entire class of perceptual illusion. Device 2 on case two: ask an experienced operator to say aloud what he is distinguishing while he works the three-valve screen, and write the fixation points down as check items; whatever will not go down gets his signature, "not reliable here", and after that signature the system must be fitted with a physical interlock. Device 3 on case five: every revision of a differential checklist lists the existing judgment habits it affects, and the triage group's habits are compared against it item by item. Device 4 on case six: embed the

scoring tool in the emergency system and make its use mandatory, moving the sorting from recognition to measurement.

The failure modes of the devices need writing down as well. The admission form dies of inflation in the answer to "what is the basis", since if anything counts as a basis then nothing was asked. The checklist dies when nobody maintains it after extraction, with the environment moving again and the list standing still, which is where Chapter 15's fork grows. Change-triggered retraining dies when the "experience affected" column stays empty all year. Instrumentation dies when the instrument's alarm gets turned down for the sake of good relations. The four deaths have a single common antidote: put device use into the audit, and have the auditor ask what the device last stopped, leaving aside the question of whether it exists at all. A device that has never stopped anything means either that the environment really is safe or that the device is already dead, and the audit's job is to tell those two apart.

Together the four devices point to one organizational document worth establishing, the intuition asset register. Each row records one significant intuition held by one expert: the environment in which it holds, the check items extracted from it, its failure conditions, and the date it was last reviewed. Maintenance costs little, one pass against the change forms each quarter, and the return is that the organization holds an inventory view of its judgment assets for the first time: whose intuition has been validated in which environment, whose intuition is owed an extraction, and which posts' judgments have no register at all. The intuition asset register belongs to the same family as Chapter 11's deviation ledger and Chapter 12's definition register, since all three turn key states that were scattered across individuals into entries that can be searched, reviewed and handed over.

The order in which the register gets built matters. Start with the posts that hold few people and heavy judgments, such as emer-

gency command, release signatures and investment decisions, then cover the posts that hold many people and scattered judgments, such as front-line operation and review. Every row in the first group can correspond to a major consequence, while the value of the second group is the statistical view that coverage gives. Most organizations start with the second group because people are easier to organize, and the register comes out thick with the critical judgments still unrecorded. The first iron rule of the register is to order by magnitude of consequence, not by headcount.

14.6 Boundaries with Adjacent Mechanisms

Against extrapolating past the validated domain (Chapter 23): extrapolation carries a conclusion into a new environment, at the data layer, while this chapter carries the judge into a new environment, at the person layer. When a scene occupies both, check the person first, meaning his experience premises, and the conclusion second, meaning its validation boundary. Fixing the person is cheap and fixing the conclusion is expensive.

Against practical drift (Chapter 15): drift is the fork between practice and text, an organizational phenomenon, while the failure zone is one person's recognition capacity crossing its edge, an individual phenomenon. A whole crew can drift together; a failure zone shows up only in specific people. The reverse holds too, since individual intuition inside a drifting organization can stay perfectly precise, precisely executing a practice that has already left the text behind.

Against all the talk that surrounds expert confidence: this chapter has nothing against experience; what it opposes is experience overstepping. The test is always the three criteria rather than an attitude toward experts, which means neither deferring to them nor dismissing what they say because of who says it.

14.7 The Five Self-Checks

1. Do the two premises, environmental regularity and feedback quality, still hold for the judgment you rely on most? What was the last change to either?
2. Whose data have you answered with "I have done this for twenty years"? Looking back, was that intuition or a failure zone?
3. After your team's last environmental change (new system, new rules, new population), which pieces of old experience were reassessed? If there is no list at all, where did the reassessment happen?
4. Which decisions in your post are once in a lifetime? What are they made with now, measurement or recognition?
5. Can you write one veteran's intuition as a checklist? Pick one and try it. Will he sign "not reliable here" over the parts that will not go down?

Chapter 15 Practical Drift (Snook)

> What this chapter covers: how formal procedure and actual practice can diverge for years without producing a casualty; Scott Snook's drift model, its three activating criteria, four settings in the field, and the two-end alignment defence. When to turn back: when what training teaches no longer matches what the site does; before you write a new rule that "must be followed"; after an audit finds that practice and manual disagree. Estimated reading time: 50 minutes.

The hardest thing about spotting practical drift is that it produces no victims while it runs. Every crew works by a method it has worked out for itself, and the work gets done well. Every checklist is ticked as the procedure requires, and the ticks are all there. Studying the 1994 shootdown of a US Army Black Hawk by a friendly fighter, Scott Snook found the frightening shape of the structure: every link in that accident had been running in exactly the same way, every day, before it. He named the phenomenon practical drift. Organizations do not slide into chaos. They drift, each unit toward what is convenient for that unit.

15.1 What it is

Definition: practical drift is the gradual departure of actual practice from formal procedure, where the departure persists over a long period, nobody corrects it, and members of the organization come to regard the fork as normal. The departure is no sudden impulse toward risk. It is a local adaptation to real working conditions. The workload changed, the staffing changed, the tools changed, the manual did not, and reality grew a second method.

Three activating criteria. The mechanism has started if any one of them appears:

1. Formal procedure and actual practice have diverged for a long time, and the divergence has a substantial history;
2. The divergence is read as expertise: departing from the manual is treated as proof of professionalism;
3. Nobody believes he is violating anything, because the rule is written in a form no one could follow.

The third criterion is the dividing line between drift and ordinary violation, and the reason drift's consequences run heavier. A violator knows what he is risking. A drifter does not, because the procedure he bypassed is missing from his risk model altogether.

Snook's book deserves two paragraphs of background, since they explain why a model drawn from military research is not confined to the military. In April 1994, in the northern Iraq no-fly zone, two US Air Force F-15 fighters shot down two US Army Black Hawk helicopters, and the United Nations personnel aboard were all killed. Snook's finding was that every unit taking part in that sortie, the fighter wing, the helicopter detachment, the air traffic control agency and the joint operations centre, was running on its own actual version of the mission; each version departed from the joint task regulation, and each departure had been happening daily for months without consequence. He pictured the organization as two ends of a drift: at one end the baseline practice written in the regulation, at the other the actual practice each unit executed. During a consequence-free stretch, nobody looks at the distance between the two ends.

That background also supplies a key property. Drift happens unit by unit. Each crew, each post, each station drifts on its own, at a speed and in a direction set by local working conditions; the organization holds no single drift quantity that can be read off, be-

cause drift keeps no central record. What an accident needs is only to string already-drifted units onto one event chain, and on the day they are strung together every unit is doing what it does on any other day. Snook called this the latent structure of drift: every site on the eve of a disaster is as calm as routine.

Three words need separating under this chapter's usage: deviation, violation, drift. A deviation is a one-off departure, archivable afterwards, the raw material of Chapter 11. A violation is a deliberate bypass of a rule the actor knows exists. Drift is the process by which the fork becomes the default, and the actor gradually loses any sense that the rule exists. The relation is progressive. A deviation is tolerated repeatedly and hardens into a convention; the convention settles into the site's normal way of working; the gap between the normal way and the rule is forgotten, and drift is complete. The best position for a defence line is between the first level and the second, which is where the three-state register for deviation ledgers does its work (Chapter 11, Device 1). The worst position is to start checking for violations after the third level, because by then you face a complete parallel rulebook, and punishing anyone you pick is a misjudgement.

The two-end alignment mentioned at the head of the chapter can now be stated properly. Snook's remedy is not a return to the regulation, since most regulations genuinely lag reality. His remedy is to align baseline and actual at both ends on a regular schedule: either amend the regulation until it can be executed in reality, or resource reality until it can execute the regulation as written. The operative word is regular. Alignment is a calibration act; drift is a continuous process; a single alignment does not stop the next drift. Alignment has to be mounted on the organizational calendar, twice a year and on any change, and become a standing operation the way instrument calibration is. Device 1 in section

15.5, the manual-reality comparison, is that alignment in institutional form.

15.2 How it grows

Four generating conditions, in structure, incentive, time and in - formation.

The structural condition: rules are written by people who do not work the line, and they are written from the design state rather than the operating state. The incentive condition: following the book carries an immediate cost, while the cost of departing is delayed and probabilistic. The time condition: drift needs time, and every "this trip is special, simplify it once" leaves no record behind. The information condition: no comparison mechanism exists, so nobody measures the difference between manual and reality.

Then there is drift's arithmetic. A single departure is usually harmless, which is Chapter 9's point that one broken layer is a fault. The disaster form of drift is several fork points stacking in the same direction on one event chain.

Three features of the dynamics deserve adding.

Drift has a directional bias. It always drifts toward lower immedi - ate cost. Pausing, cross-checking, waiting, reading back: each of these actions costs time now and pays off somewhere else, and the first cut of drift always lands here. To observe an organization's drift direction is to observe how it prices time.

Drift is contagious. A new member meets two demonstrations, the manual version and the site version, and the trainer will teach the site version, because that is the one that gets the job done. Drift therefore copies itself automatically through staff turnover, and

every handover is an inoculation. Chapter 30's handover rupture connects to this chapter at exactly that point.

Drift self-organizes a shared understanding of when drifting is allowed. Crews form default terms among themselves: which steps may be dropped when it is busy, which may never be dropped. The understanding is never written down and is executed with fair consistency, and its existence is itself proof that drift is no error; it is the organization revising its rules through action. This is also where prevention gets hard. What you face is not a lapse of memory. It is a parallel rule system, running in fact.

Push the arithmetic one step further and the disaster form appears. Each fork point strung on an event chain passed every safety review on its own, because the object of review is the regulation, not the practice. A drifted organization therefore shows a particular compliance face: the paper system is complete, and the paper system has nothing to do with reality. Three of this chapter's four cases carry that face.

The relation between drift and audit needs its own warning. Routine audit is drift's incubator. The benchmark an audit compares against is a document, and the carrier it inspects is a document as well: records, signatures, ledgers. Drift grows precisely on the layer outside documents. A fully drifted practice can generate a complete set of passing audit evidence, because the drifted organization learned long ago to complete the paperwork. Case 9's working papers and Case 3's surgical checklist are both conforming output from that production line. Two conclusions follow. An organization whose audit finds zero differences is the one whose drift level most deserves suspicion. And audit method has to include site observation and shadow comparison, described in section 15.4; an audit that reads only documents is structurally blind to drift.

15.3 The pattern in four domains

This section held four cases when it was first drafted (aviation, maritime, healthcare, construction) and holds nine after expansion; Cases 5 to 9 (aviation, maritime, nuclear, finance) were added. Nine cases cover the hosts drift commonly lives in: the seam between training and design, the seam between scheduling and the body, the seam between tagging and handover, the seam between checklist and pause, the seam between template and object.

Case 1 (Aviation): The Logbook in Flight

Case ID: av-020.

When and where: 18:57 local time, 9 March 2023, mountainous terrain near the town of Franklin, Macon County, North Carolina, United States. The helicopter departed Murphy (5NC4) at 18:44 bound for Asheville (NC95), 13 minutes en route (NTSB ERA23LA147 p. 1, p. 4). The flight was at night, in Class G airspace, on a company VFR flight plan, with no air traffic control clearance (p. 4).

What and who: an EC135 P2+ helicopter, registration N558MT, manufactured in 2012, operated by Med-Trans Corporation (MED-TRANS CORPORATION, trading as LifeForce) under Part 135 on-demand air taxi, air medical, holding a Part 135 operating certificate (p. 3). Four people were aboard: one pilot and three passengers, one of whom the report mentions was a flight nurse. The pilot was 51, held a commercial certificate with airplane and helicopter ratings, 4,723 total flight hours, 1,867 on type; his last proficiency check was on 1 March 2023, and his second-class medical carried no limitations (p. 3). He flew single-pilot from the right seat (p. 3). Airframe total time was 6,067.7 hours at the AAIP inspection of 23 February 2023; the emergency locator transmitter was installed and did not activate (pp. 3, 4).

What they were doing: Before departure the pilot worked out the highest obstacle on the route himself, 6,100 ft, with several 5,000 to 5,500 ft peaks along it; he also calculated that the flight would run into the end of evening civil twilight (p. 1). Both calculations were his, and the report contains no second route analysis.

After takeoff he set the autopilot at 5,500 ft and levelled off (p. 1). He then reached for his night vision goggles and found them on the co-pilot's seat, lying under an aircraft logbook. He moved the logbook to the pilot's door pocket, "where it normally was kept" in the report's words, and inside that same movement decided to check the logbook's flight hours against the maintenance hours due (p. 1).

Once the check was under way he noticed the cloud base descending, reset the autopilot to 5,000 ft, and then "put his head back down" into the logbook (p. 1). The flight nurse in the rear asked for an updated arrival time. He looked up and saw the helicopter heading for a tree-covered peak. He pulled aft on the cyclic to climb, the tail boom struck several trees, the vertical stabiliser separated from the tail boom, and he put the helicopter down on a road (p. 1).

Thirteen minutes passed between takeoff and ground contact. In those thirteen minutes the cockpit held one person, and no second pair of eyes was watching the relative position of the altitude, the cloud base and that peak.

A word on the nature of this material. The report runs five pages, its factual section cites no recorded data from the aircraft, and the entire sequence comes from the pilot's statement and the site examination (pp. 1, 5). Whether the helicopter carried recording equipment, and of which kinds, is not stated in the report. Every moment on this card, the level-off, the moving of the logbook, the altitude change, the look up, has a single source. That is not thin reporting. It is the standing condition of this class of investigation

when there is no recorder and no second crew member: the sequence can only be retold by the person who lived it, and what he retells is exactly how his attention was allocated at the time.

The decision point: three moments, and not one of them left a document.

Before 18:44 on 9 March, in cockpit preparation, the cruise altitude was set at 5,500 ft. The grounds were his own route data: highest obstacle 6,100 ft, several peaks between 5,000 and 5,500 ft (p. 1). That single choice fixed every margin that followed, because 5,500 ft is level with the highest of those peaks, which leaves zero tolerance. The choice had no paper carrier of the kind a weight and balance sheet provides; the report holds only the pilot's statement.

After takeoff, in the stowage that preceded donning the goggles, the logbook was on the co-pilot's seat with the goggles on top of it, instead of in the pilot's door pocket (p. 1). He moved it to its usual place and then decided to check it there and then. No meeting, no approval, no second person.

After he saw the cloud base descending, he took the autopilot from 5,500 to 5,000 ft and went back head-down into the logbook (p. 1). This is the last reversible moment in the sequence. Changing altitude took a few actions; stopping the check would have taken a few actions. He took the first.

What all three share is the absence of any paper carrier. This is not the author's inference. The investigation was graded Class 4, the NTSB did not travel to the site (p. 5), and the report cites no company operations manual, no training record, no duty regulation. Most cards in this book can name a meeting date and a signatory at the decision point. This one cannot, because on that day the cockpit held one person.

There is a fourth moment of a different kind: the aft cyclic input after he looked up and saw the peak. It lasted on the order of a second, and the report gives it one sentence, that he pulled aft to climb and the tail boom struck several trees (p. 1). It does not be - long to drift. It is the compensating action after drift. When a per - son finds himself at the wrong altitude and on the wrong heading, the action space left is a few seconds wide.

What was visible then: Before the flight on 9 March, what he had in hand was his own computed obstacle data, highest 6,100 ft with several peaks at 5,000 to 5,500 ft, and his own conclusion that the flight would meet the end of evening civil twilight (p. 1).

In flight, what was actually in the cockpit: the autopilot setting at 5,500 ft and later 5,000 ft; the night vision goggles on the co-pi - lot's seat; the logbook under the goggles; the pilot's door pocket, the logbook's usual place (p. 1). Outside the window, the cloud base, which he himself described as descending (p. 1). On the in - tercom, the rear medical crew, and the flight nurse's request for an arrival time, the only external input that pulled him out of the head-down state (p. 1).

What was learned afterwards: the distance between the helicopter and the tree-covered peak at the moment he looked up; the separ - ation of the vertical stabiliser from the tail boom after the tail struck trees; the extent of airframe damage from the landing on the road (p. 1, pilot statement and site examination). The K1A5 weather station, 4 nautical miles from the accident site, recorded at 19:15, eighteen minutes after the accident, a cloud base of 5,000 ft, visibility 10 miles, temperature 12 Celsius, dew point 0 Celsius, no precipitation and no obscuration, at a station elevation of 2,020 ft (p. 4). That observation was not on his desk, and the report does not say he consulted that station before departure. It establishes something else: visibility that night was not the obstacle. The obstacle was terrain and a lowered head, present at the same time.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the tail boom struck trees and the vertical stabiliser separated, the helicopter landed on a road, three aboard were slightly injured and one was unhurt, and the airframe was substantially damaged (pp. 1, 4). The NTSB's probable cause, in the report's words, was the pilot's "improper decision to review the aircraft logbook in flight, which resulted in a controlled flight into terrain" (p. 2). The report carries five findings: aircraft control (pilot), mountainous terrain (decision related to conditions), task allocation (pilot), decision-making and judgement (pilot), and task monitoring and vigilance (pilot) (p. 2).

The three questions, looking back. The commonest answer to the first is "check the logbook after landing", and that action costs nothing: the logbook has its place in the door pocket and can be checked on the ground just as well. What the case spent instead was the whole margin of a 13-minute leg. The item most often missed under the second question is the number 5,000 ft. When the cloud base fell he changed altitude, and the route carried peaks of 5,000 to 5,500 ft; that information was in the pre-flight calculation, written by him. The third question lands in pre-flight preparation: check which tasks on this trip must be done in the air, and if you cannot name one, do not carry the object for it into the cockpit.

Options ruled out: the report records neither the alternatives the pilot considered nor any company rule on when the logbook is to

be checked. The four items below are built from what was on his desk and from conditions in the report that make them unavail - able.

Check the logbook after landing. Zero cost, since the pocket is where it normally lived (p. 1). The report records no rule either forbidding or requiring an in-flight check: at the level where doc - uments exist, the act was neither prohibited nor required, only habitually scheduled.

Have a second person read the figures. This was a Part 135 single-pilot operation, the right seat was empty, and the goggles were ly - ing on it (pp. 1, 3). The option did not exist on this leg.

Hold 5,500 ft and leave the altitude alone. The action the pilot de - scribed was a 500 ft descent (p. 1). The report records no consid - eration of routing around, climbing, or turning back; when the cloud base fell, he did not treat changing altitude and stopping the current task as one act.

Switch to instrument flight. The leg was on a company VFR plan with no clearance in Class G airspace (p. 4). IFR was not part of that day' s mode of operation, and changing mode is itself an abort of the mission.

Early signals: no meeting record exists to cite, so the four signals below can only be described by how they were handled, not by what reason dismissed them. This is drift landing on a single-pilot post: there was no second person who needed persuading.

Signal one: the cruise altitude of 5,500 ft, level with peaks of 5,000 to 5,500 ft (p. 1). The signal sat on the pre-flight calculation sheet, and the handling was to fly it as computed.

Signal two: before flight he had already calculated that evening civil twilight would end during the flight (p. 1). Daylight disap -

pearing inside the leg was his own result, and cabin paperwork was then scheduled into the same leg.

Signal three: the pre-flight stowage order had already changed. The logbook had left the door pocket and was lying on the co-pilot's seat, with the goggles on top (p. 1). The signal was an object in the wrong place, and the handling was to move the logbook back and start checking it on the spot.

Signal four: the descending cloud base was seen, and the handling was 500 ft down and the head down again (p. 1). The signal was processed into an altitude number. It was never processed into an action stop.

The point of escalation: in cockpit preparation before 18:44 on 9 March, the logbook was set down on the co-pilot's seat with the night vision goggles on top of it, and was later fetched for checking in flight (p. 1). The report does not record the day this practice began, nor how many times it had been executed before. What can be checked is that one day: it happened, no second person was present, and no one signed.

Failure chain: Technical layer. The autopilot held 5,000 ft; the route carried peaks of 5,000 to 5,500 ft; there was no visual reference while his head was down; at look-up the helicopter was close to a tree-covered peak; aft cyclic, tail boom striking trees, vertical stabiliser separating; a landing on a road and substantial damage (p. 1). Organizational layer. Part 135 single-pilot operation with no second pair of eyes in the cockpit; a company VFR plan with no clearance, so no outside opportunity to warn (pp. 3, 4); no company document in the report governing when the logbook is checked (pp. 1, 5). Incentive layer. En-route time was treated as time available for other work, a point the report does not support with any document and which can only be read from the sequence: he carried the paperwork aboard, and after seeing the cloud base fall he went back to it (p. 1). All three layers broke at once. The

junction: the falling cloud base was a technical-layer signal, and the one channel that could translate it into "stop looking down" was another person in the cockpit, a channel that single-pilot operation removes by design.

Minimum counterfactual: put the logbook back in the pilot's door pocket before departure and check it after landing. One action, taken in cockpit preparation before 18:44, with no luck required: the pocket is where the logbook normally lived, and the report says so in those words (p. 1).

Outcome and sources: controlled flight into terrain at 18:57 on 9 March 2023; three people slightly injured, one unhurt, airframe substantially damaged; investigation graded Class 4, the NTSB did not attend the site, and the report was first issued on 6 June 2023 (pp. 1, 5). Primary source: NTSB, Aviation Investigation Final Report, ERA23LA147, 2023; link in Appendix B, av-020.

One layer in this case is worth naming. The act of a crew member going head-down into a logbook has a respectable name in aviation: task management, the business of putting en-route time to use. The material of drift is often exactly this kind of respectable practice wisdom. It makes sense, colleagues vouch for it, it has a record of efficiency, and its only problem is that it was never checked against another rule. Preventing drift does not mean abolishing practice wisdom. It means giving practice wisdom an interface for that check: every clever local practice deserves one question, which accident was the rule you bypassed written for. Answer it, and the practice at least knows what it is betting. Fail to answer it, and the practice is betting against a history nobody has read.

Case 2 (Maritime): The Watch Alarm the Music Covered

Case ID: ma-002.

When and where: about 02:38 on 14 October 2025, off Lochinver in Sutherland, Scotland, on the rocky shore at Rubha Rodha. The trawler left Stornoway at 21:30 on 13 October for the fishing grounds (MAIB Report 17/2026 p. 1, p. 2). At the time of the accident it was dark, with slight seas and moderate to good visibility (p. 8).

What and who: the prawn trawler Silver Cloud II (fishing number WK 80), wooden, built in 1972, registered length 15.58 m (overall length 16.55 m), gross tonnage 42, bought in 2024 by Silver Cloud Shellfish Ltd and normally based at Ullapool and Gairloch (pp. 2, 8). Three people were aboard: the skipper, the mate and one deckhand. The skipper and the mate each had about 25 years in fishing, and the deckhand had been aboard about 12 months; all held the certificates Seafish requires, and safety drills were held on board regularly (p. 2). The accident was investigated as a very serious marine casualty (p. 8).

What they were doing: On 10 October the vessel finished a repair period. The skipper helped on site and went home to rest around midnight (p. 2).

At 11:00 on 11 October the vessel sailed from Gairloch to start fishing. The skipper already had influenza-like symptoms at sailing, headache, sore throat, blocked nose and muscle aches, and was taking paracetamol; he judged himself fit to sail (p. 2). Fishing ran until the evening, the vessel entered Ullapool at 23:00, and he went to bed at midnight.

He rose at 03:30 on 12 October and sailed immediately. The mate took the first watch until 05:30, the skipper took over until shooting the gear at 06:30; hauling at 11:30, then fishing through the day; that evening the vessel steamed to an old submarine mooring buoy in Loch Ewe, arriving at 20:30, and he went to bed at 23:00.

At 04:15 on 13 October the vessel left the buoy for the grounds and shot at 06:45; hauling at 11:00 found the net badly damaged, and the decision was taken to go to Stornoway to land the catch and repair, berthing at 15:00. The skipper went to sleep while the mate and deckhand repaired the net, landed the catch and ate. He rose at 21:00, the crew decided to add a night tow to make up for the shot lost that day, and the vessel left port at 21:30 (p. 2).

The mate stood the first two hours in the wheelhouse, and at 23:30 the skipper took over and was left alone. He set a waypoint near the grounds, expected to arrive at 01:15, and engaged the autopilot (p. 2). Between 23:30 and 01:15 he fell asleep in the wheelhouse. The vessel missed the waypoint, held 092 degrees and continued toward the Scottish mainland, and at 02:38 it struck the rocks at Rubha Rodha (p. 2).

The decision point: three moments, none of them signed.

At 11:00 on 11 October, at the Gairloch quay before sailing, he sailed with symptoms. The report's words are that the skipper had influenza-like symptoms at sailing but judged himself fit to sail (p. 2). That judgement involved no medical examination, no review and no second signature. It locked the next three days of rota into a body already running at a discount.

At 21:00 on 13 October, aboard at Stornoway, the crew added a night tow. The background was the badly damaged net found at the 11:00 haul and a shot lost that day (p. 2). The crew made the de‑cision jointly; the report's phrase is "the crew decided". The immediate consequence is written in Table 1: on 13 October, 16 hours 30 minutes of work, a longest unbroken work period of 15 hours, and a longest unbroken rest of 4 hours (p. 4).

At 23:30 on 13 October, in the wheelhouse, the mate handed over and the skipper kept the watch alone. At that point 19 hours 15 minutes had passed since he rose at 04:15 that day (p. 2; the inter‑

val follows from the report's narrative). Three people were aboard, the mate had just finished two hours on watch, and the deckhand had been aboard 12 months (p. 2). The decision at that moment was not to call anyone.

None of the three moments has a paper carrier, and the report is blunt about why. MSN 1884 (F) amendment 1 requires work and rest arrangements to be planned and recorded (p. 5), and "a complete record for the four days prior to the vessel's departure could not be obtained" (p. 6). No record means nothing to sign, and without anything to sign, all three decisions are absent from the paper.

What was visible then: Before sailing on 11 October: his own symptoms, four of them, being held down with paracetamol (p. 2); the schedule, a cycle of 10 days at sea and 4 to 5 ashore (p. 4).

Aboard on 13 October: three people available, of whom the mate had about 25 years and the deckhand about 12 months (p. 2); the fact that the net had been badly damaged that day (p. 2); the fact that a shot had been lost (p. 2); the vessel's equipment fit, a watch alarm mounted in the wheelhouse that armed when the autopilot was engaged and needed a button pressed every 4 minutes, with the volume rising if no one intervened for 1 minute after the first alarm, and no visual alarm or audible repeater in the crew accommodation (p. 4).

In the wheelhouse at 23:30 on 13 October: the waypoint and the expected arrival time, the autopilot, and the music that was playing. Also present was a division of labour already customary on that vessel: during trawling the skipper or mate would usually keep the wheelhouse for long stretches, while watches on passage were taken by the mate and the deckhand (p. 4). That night the passage watch fell to the skipper himself. The report records that music was often played on this vessel when hauling and often in the

wheelhouse, and that music was playing in the wheelhouse on the night of the accident (p. 4).

What was learned afterwards: the skipper's total rest across three days (6, 6.5 and 7.5 hours) and his longest unbroken rest (6, 3.5 and 4 hours), in Table 1, which the MAIB reconstructed from interviews (p. 4); the missing records for the four days before departure (p. 6); and the judgement that the alarm was highly likely to have been masked by music in the wheelhouse, the report's phrase being highly likely (p. 6). One more item: the period when human alertness falls naturally runs from about 02:00 to 06:00 (p. 5), and the grounding at 02:38 falls inside that window (p. 6).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the vessel grounded at 02:38. The impact and the movement that followed woke the skipper and the crew; an inspection found the fish room flooding to about 1 m and still rising, and the skipper decided to abandon. The three donned life jackets, the deckhand stepped straight onto the rocks, and the mate entered the water on the starboard side and was pulled ashore by the deckhand. The skipper put the engine in neutral, activated the emergency position indicating radio beacon, sent a distress alert by digital selective calling, took the VHF radio, and entered the water on a line to be pulled ashore. At about 03:00 the RNLI lifeboat arrived and recovered all three (pp. 2, 3). There were no injuries; the hull was breached, flooded and partly submerged, then

broke up on the rocks and was declared a constructive total loss (pp. 1, 8).

The MAIB made no new recommendations and issued no safety flyer, on the grounds that findings of this kind had recurred repeatedly in earlier investigations: Sustain (16 November 2023, off the entrance to Loch Broom, skipper asleep on watch) and Crystal Stream (6 June 2025) (pp. 5, 7).

The three questions, looking back. The commonest answer to the first is "fit a watch alarm". This vessel had one, and it was not a legal requirement (p. 6, original: although not a requirement on board Silver Cloud II, a watch alarm was fitted). It still did nothing, and the cost of that answer is turning a four-minute button press into an action that may or may not be taken. The item most often missed under the second question is the absence of a repeater in the crew accommodation. It reads like an equipment detail. In fact it decides that this alarm can only wake someone already in the wheelhouse, which makes it structurally useless against the watchkeeper himself falling asleep. The third question lands before 21:00 on 13 October: count the seven-day rest total, and cancel the night tow if it falls short.

Options ruled out: the report records no discussion of alternative arrangements among the crew. The five items below are built from conditions aboard at the time and from facts the report can verify.

Two-person watches. Three people were aboard, and the mate had already stood the first two hours before 23:30 (p. 2), so he was capable of taking over. Single-person watches are not prohibited at this vessel size, though the arrangement runs against MGN 313 (F) guidance on the risk of lone lookout (p. 6). The real cost of this option is the mate's rest, and the mate's rest that day was fragmented as well (p. 4).

Cancel the night tow. The direct cost is one shot of gear, which is exactly what the 21:00 decision set out to recover (p. 2). The report records no one proposing cancellation.

Turn the wheelhouse music off or down. The vessel had no ban on music; it was played routinely when hauling and in the wheel‑house, part of the work (p. 4). MGN 313 (F) lists distractions in the wheelhouse, music systems among them, as a contributory factor in collisions and groundings (p. 5). The guidance was written and the vessel's practice had not followed it: a verifiable fork between text and practice.

Have the deckhand keep him company or take the watch on pas‑sage. The third person was in his bunk at the time; the report re‑cords that the deckhand had been aboard about 12 months, was building experience, and did not stand trawl watches independ‑ently (pp. 2, 4). Passage watches were normally shared by the mate and the deckhand (p. 4), and that night they were not arranged that way.

Fit a repeater for the alarm in the accommodation. That is an equipment modification requiring the owner to spend money (p. 4 records that the vessel had no repeater). The report records no discussion of it.

Early signals: five, each with the account available at the time.

Signal one: sailing with symptoms at 11:00 on 11 October (p. 2). The account available: the symptoms were influenza-like, held down with paracetamol, and he judged himself fit to sail.

Signal two: three days of rest at 6, 6.5 and 7.5 hours, with longest unbroken rest of 6, 3.5 and 4 hours (p. 4, Table 1). The rules require no less than 10 hours in any 24 and no less than 77 hours in any 7 days, with daily rest split into no more than two periods, one of them at least 6 hours (p. 5). Rest on 12 and 13 October was split into three periods each, with longest periods of 3.5 and 4 hours, so both

conditions failed at once. The account available: MSN 1884 carries its own exception, allowing an authorised departure for operational reasons provided safety is not harmed (p. 5). That exception had become the default channel.

Signal three: 16.5 hours of work on 13 October, with a longest unbroken stretch of 15 hours (p. 4). The account available: the net broke during the day, a shot was lost, and it had to be made up.

Signal four: the alarm sounded only in the wheelhouse, with no repeater in the accommodation (p. 4). The account available: the equipment was not a legal requirement, and having it at all was a bonus (p. 6).

Signal five: music and alarm sharing one space (p. 4). The account available: music was the normal background of the work, played when hauling and in the wheelhouse.

The point of escalation: two consecutive actions between 21:00 and 23:30 on 13 October. The crew decided on a night tow and the vessel left at 21:30; two hours later the mate handed over and the skipper stayed alone (p. 2). Once the vessel had left, it had no physical means of relieving the watchkeeper before the 01:15 waypoint, and two of the three remaining people were in their bunks. Nothing was signed at this step and nothing was recorded: the work and rest arrangements for that day were never written down (p. 6).

One time gap deserves recording separately. The vessel missed the waypoint at 01:15 and grounded at 02:38, 83 minutes later (p. 2). On the alarm's setting, with the autopilot engaged, a button press was required every 4 minutes, the alarm sounding if it was not pressed and rising in volume after 1 minute without intervention (p. 4). Eighty-three minutes is room for more than twenty alarms. The report does not record whether the alarm sounded that night.

It records a conclusion: the crew did not know the skipper had lost the ability to keep watch until the vessel struck (p. 6).

Failure chain: Technical layer. Grounding at 02:38, hull breach, fish room flooding to about 1 m, subsequent break-up (pp. 1, 2). Organizational layer. A single watchkeeper and no structured rota (p. 4, original: an informal watchkeeping arrangement and had no structured rota); no work and rest records, with the four days be - fore departure missing (p. 6); an alarm with no repeater in the ac - commodation, designed on the assumption that someone in the wheelhouse could hear it (pp. 4, 6). Incentive layer. Catch and trip rhythm: a shot was lost to net damage on 13 October and a tow was added that night to recover it (p. 2); a cycle of 10 days at sea and 4 to 5 ashore, landing catch every two days (p. 4). All three layers broke at once. The junction: the one channel that could carry the fact of the skipper asleep to the accommodation was the alarm, and the alarm was designed to sound only in the wheelhouse, where it was highly likely to be masked by music that night (p. 6). The channel broke at the level of equipment fit and volume, and that level sits on nobody' s checklist: the alarm was not broken and the music was not a violation.

Minimum counterfactual: before deciding on the night tow at 21:00 on 13 October, check the rest total against 77 hours in 7 days, and cancel the tow if it falls short. One action, taken two hours before departure, with no luck required: the three days' figures in Table 1 were already in his own routine, and the floor set by the rules is written in MSN 1884 (p. 5).

Outcome and sources: all three rescued, unhurt; the hull breached, flooded, broken up and declared a constructive total loss; the MAIB made no recommendations and issued no safety flyer (pp. 1, 7, 8). Primary source: MAIB, Report 17/2026, Silver Cloud II, August 2026; link in Appendix B, ma-002.

There is a quieter layer in Case 2: the fact that the alarm really was fitted to that vessel. The equipment was real, the function was real, the defence line existed, even though no requirement demanded it (MAIB Report 17/2026 p. 6). What drift does is not dismantle de‑fences. It makes them mismatched. The alarm was kept as its design scenario specifies, a quiet wheelhouse, while the wheel‑house's real scenario, music, had long since stopped being the design scenario. That sentence generalizes into a test question for drift: is this defence line's design scenario still the same scenario as the site today. The difference between the two is the part drift has eaten, and that part appears on no defect list, because the alarm was not broken and the music was not a violation.

Case 3 (Healthcare): The Checklist That Drifted (condensed)

me-001. An NHS orthopaedic treatment centre in the United King‑dom. HSIB investigation report (Administering a wrong site nerve block, 2018; the archived snapshot was read and verified for this book): the nerve block for a 69-year-old patient's right ankle op‑eration was injected into the left leg, and the patient was not harmed. The Stop Before You Block checklist was in use and was being filled in, but the way it was executed had evolved from pausing at each item and confirming aloud in both directions to ticking items off as you went. National figure: about 192 wrong-site operations a year. The decision point sits at the moment the checklist was simplified from an item-by-item pause and check to a running tick. Minimum counterfactual: execute the checklist as designed, with a pause and a cross-check at every item. Primary source: HSIB, 2018; link in Appendix B, me-001 (archive link, grade A−).

This condensed case holds the sharpest phenomenon in the chapter. After drift, the checklist still produces complete compli‑ance records. The ticks are real, since every form was signed. The check did not happen, and that is real too. A paper record proves

that the form was filled in; it cannot prove that the check took place. The difference between the two is the measure of the pause that got eaten. Device 2, which registers pause-type actions on a separate list, is aimed at exactly that difference: a checklist protects the pauses between the ticks, not the ticks on the paper.

Case 4 (Construction): The Safety Space Filled with Cable (condensed)

ce-003. Ickenham station, London Underground. RAIB report (Report 02/2026; the pages were read and verified for this book): a passenger fell onto the track and was struck and killed by a train. The recess beneath the platform edge is the provision that gives a person who falls a space to get into; the investigation found the recess filled with communications cable, so the refuge was too small. Each cable installation permit was one small occupation, and no occupation was ever approved under the heading of safety space. The decision point sits at one of the sign-offs that approved cable being run under the platform. Minimum counterfactual: keep a register of protected space, and route every occupation through a separate approval. Primary source: RAIB, Report 02/2026, 2026; link in Appendix B, ce-003.

One sentence in Case 4 deserves taking apart: no occupation was ever approved under the heading of safety space. Every occupation went through approval, with engineering permits and works permits complete. What was missing was any approver being required to answer how much refuge space would remain after this occupation. The absence of that heading made each occupation compliant at every level of approval, while the accumulated result was the loss of the safety function in full. Device 3’s register turns space into a quantity with a number, the remaining clear width, updated at every occupation, so that the heading finally has an account behind it. This is a general rule for drift defences: drift

happens in domains that have no dimension, and the first step of the defence is to give that domain a quantity.

Case 5 (Aviation): The Full Rudder the Training Taught

Case ID: av-033.

When and where: 09:16:15 Eastern Standard Time on 12 November 2001, over the residential area of Belle Harbor, Queens, New York, United States. American Airlines Flight 587 began its takeoff roll at 09:13:51 and became airborne at 09:14:29, departing runway 31L at Kennedy airport (NTSB AAR-04/04 p. 1, p. 3).

What and who: American Airlines Flight 587, an Airbus A300-605R (N14053), operating under 14 CFR Part 121 on instruments, the first leg of the crew's one-day round trip (p. 1). Aboard were 2 flight crew, 7 cabin crew and 251 passengers, 5 of them under the age of two on laps (p. 1). On the training side the subject is the American Airlines Flight Training Academy in Fort Worth, Texas, and its Advanced Aircraft Maneuvering Program (AAMP), the upset recovery module within the company's selected event training, taught in initial, recurrent, transition and upgrade training (pp. 79, 80). Under the company's General Flight Training Manual, edition of 1 September 2001, maneuver training was delivered by professional simulator instructors who did not hold airline pilot qualifications (p. 79).

What they were doing: AAMP was developed from 1996 onward, after a review of worldwide large multi-engine transport accidents between 1987 and 1996 identified loss of control as the leading cause (p. 80). The course took shape in 1997: 6.5 hours of ground school and six simulator items, one of which was an uncommanded roll to at least 90 degrees of bank (pp. 81, 82).

How that item was run is set out in the report. The instructor pressed a touchscreen button and the simulator first rolled about

10 degrees one way, then rolled past 90 degrees the other way; before that, the simulator suppressed aileron, spoiler and rudder response, and kept suppressing them until bank reached 50 degrees or a full 10 seconds had passed; after 50 degrees, yaw and roll control were restored progressively within 1.3 seconds (p. 83). During the item the instructor told the pilots that a heavy aircraft had just passed ahead, some instructors naming a Boeing 747, and issued a wake turbulence warning (p. 82). Trainees were told to recover the way ground school had taught them.

The first officer's training record: AAMP ground school in March 1997, his first AAMP simulator session in November 1997, then recurrent sessions in November 1998, November 1999 and December 2000, with the next scheduled for late November 2001 (pp. 81, 82). The captain attended ground school in May 1997 and his first simulator session in December 1997 (pp. 81, 82). The course had run in full on the first officer for four years, and he passed every cycle.

The decision point: three irreversible moments. The first two carry signatures. The third is only an action.

On 27 July 1995 the airline's managing director of flight training and standards wrote to the Principal Operations Inspector (POI) requesting approval of a training manual revision, on the grounds of "recent NTSB concerns and recommendations", noting that the company was "proactively" running this training in simulator periods, and listing maneuvers that included upset recovery from more than 90 degrees of bank and pitch attitudes beyond 35 degrees (p. 80). On 1 August 1995 the POI approved the manual revision (p. 80). This is the only moment in the case signed by both the company and the regulator, and it is where the drift begins: the training scope was expanded voluntarily by the company, and the expanded part carried no statement of its structural boundary.

On 6 October 1997 the company's chief pilot and vice president of flight replied to a joint letter from the Boeing Commercial Airplane

Group, Boeing Douglas Products Division, Airbus and the FAA (p. 88). That joint letter had gone out on 20 August 1997, observing that "overemphasis on rudder's superior roll authority relative to aileron and spoiler at high angle of attack is a concern", and recommending full aileron first and rudder only after no response (p. 87). The reply refused that order, on the grounds that the company did "not advocate rudder first or rudder only", and stating that if pitch rate after unload was insufficient, company pilots "would not hesitate to roll the lift vector off the vertical axis" (p. 88). Airbus later told the NTSB that it did not receive the reply until 20 January 1998 (p. 88, note 143). The signatory was the chief pilot and vice president of flight.

The third moment is the seven seconds from 09:15:51 on 12 November 2001. The first wake encounter came at 09:15:36, and the first officer answered it with wheel inputs of 37 degrees right and 34 degrees left, while rudder pedal travel stayed on the order of 0.1 inch, inside the authority of the yaw damper (pp. 3, 4). The second encounter came at 09:15:51, with the wheel at 64 degrees right and the pedal at 1.7 inches right; between then and 09:15:58.5 the pedal alternated through 1.7 inches right, 1.7 left, 1.7 right, 2.0 right, 2.4 left and 1.3 right (pp. 4, 5). At 09:15:58.4 the right rear main attachment fitting of the vertical stabiliser failed and the whole fin came off, at an airspeed of about 251 knots (p. 4).

The third moment carries no signature and needed none. The first officer was executing a sequence he had practised and passed for four years.

What was visible then: On the course side, what was in hand, in date order.

The AAMP training workbook, edition of 1 January 1997, one copy per pilot. It covered wake turbulence response, and one line read "High AOA maneuvering = RUDDER"; another passage read that the rudder's effectiveness as a roll control increases with angle of

attack, and that at high angle of attack the rudder becomes the most effective roll control (pp. 84, 85). The workbook was revised later, and the edition in use at the time of the accident was dated 1 May 2000 (p. 84).

The company's own A300 operating manual, Flight Controls page 8, edition of 8 September 1999: two independent rudder travel limiting systems, reducing maximum rudder deflection from plus or minus 30 degrees below 165 knots, in steps, to plus or minus 3.5 degrees above 310 knots (p. 93). Two years before the accident, that manual already said this rudder had very little travel left at high speed.

Also on the desk was a body of external and internal feedback, every item dated. On 22 May 1997 the company's A300 technical pilot wrote to the Airbus chief test pilot, noting that the handout described the rudder as the primary roll control at high angle of attack, and that instructors were already teaching pilots to use rudder to control roll in wake encounters (p. 87); on 23 May the Airbus chief test pilot faxed back agreeing to a telephone conference (p. 87). On 27 May an internal memorandum from the general manager of flight operations technical to the chief pilot and vice president of flight recorded "serious concern" about "certain flawed aerodynamic theory and flying techniques" in AAMP, called the teaching of rudder as the primary means of roll control in high angle of attack recovery "extremely dangerous", and asked for "immediate corrective action" (pp. 88, 89). On 13 June an internal note from the Airbus chief test pilot observed that the simulator was "particularly inaccurate at large sideslip angles", so pilots might draw the wrong conclusions from it (p. 87). On 20 August the four-party joint letter went out (p. 87).

Public statistics were available at the time. The NTSB searched the ASRS database afterwards, and among reports from 1988 to 1999 in which large transport aircraft encountered wake turbulence as the

trailing aircraft, the maximum bank pilots estimated was usually within 30 degrees and never above 60 degrees (p. 83). The data sat in the database throughout; the search was done after the fact.

What was learned afterwards: the load at the moment the vertical stabiliser failed was about twice the certified limit load and above the certified ultimate load (p. 159); on the real aircraft at 240 knots pedal travel should have been limited to 7.9 degrees, while full rudder input in the simulator measured 10.3 to 18.9 degrees, and some trainees said they "could not feel the rudder stop" (p. 90); and the first officer's two wake responses on the 727 in 1997, which come from interviews with pilots who flew with him (pp. 137, 138).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 09:15:58.4 the right rear main attachment fitting of the vertical stabiliser failed and the whole fin separated; both engines then separated from the airframe, and the aircraft fell into the Belle Harbor residential area, killing all 260 people aboard and 5 on the ground (pp. 1, 4). The NTSB's probable cause was the in-flight separation of the vertical stabiliser, the loads on it having exceeded ultimate design values, caused by the first officer's "unnecessary and excessive" rudder pedal inputs, with those inputs contributed to by characteristics of the A300-600 rudder system design and by elements of the American Airlines Advanced Aircraft Maneuvering Program (p. 160). Two board members dissented from changes made to the ordering of the contributing

factors; their dissent is printed separately after the main text, and one line of it reads that the investigators could find no instance of abnormal rudder use by this first officer before his AAMP training, and that when a captain asked him about an earlier event, he "insisted that AAMP instructed him to use the rudder pedals that way" (p. 165).

The three questions, looking back. The commonest answer to the first is "revise the course", and revision presumes that someone knows there is a gap between what the course does and what the structure can take. That knowledge was scattered, before the accident, across four dated letters and memoranda, and not one of them entered a course approval form. The item most often missed under the second question is page 8 of the company operating manual, Flight Controls: the line that runs from plus or minus 30 degrees down to plus or minus 3.5 degrees. It reads like a system description. It is in fact the entire travel of that rudder at high speed, and the pilots who had been trained did not know the line existed (p. 159, finding 10). The third question lands at the manual approval of 1 August 1995: had the approval form carried one line asking what structural boundary corresponds to the full rudder input this course teaches, two departments would have had to meet.

Options ruled out: four, each of them on the desk at the time, and each with an outcome the report can verify.

Change the order of actions as the joint letter recommended, full aileron first and rudder only after no response. The reply of 6 October 1997 refused (p. 88). The refusal argued the point item by item and changed wording in the text; it did not change the simulator item.

Cancel the 90-degree bank item and replace it with a low-fidelity demonstration. The NTSB wrote the feasibility of this path into the report: without a motion platform or with a simple screen display,

the exercise might be better suited to awareness training and less likely to introduce wrong information (p. 160, finding 18). It was not done at the time. After the accident the company changed the item: freeze the simulator first, brief, then hand control back, and stop describing the setup as following a heavy aircraft (p. 83).

Model the real aircraft's pedal travel in the simulator. The simulator's elastic cable stiffness was calculated mathematically rather than built from real cable, and that is where the difference comes from (p. 90, note 146); static measurements on the real aircraft were not made until 31 January to 2 February 2002, in Tulsa (p. 91). It was not done at the time, and nobody proposed doing it.

Have the structures and airworthiness side co-sign the training course. On the certification side there was one POI approval (p. 80), and the training instructors did not have to be airline pilots (p. 79). This path had no institutional entry point at the time.

Early signals: five, each with a date, and each answered with a document.

On 22 May 1997 the company's own A300 technical pilot wrote to the Airbus chief test pilot that the handout made rudder the primary roll control at high angle of attack and that instructors were already teaching it that way (p. 87). The answer was the fax of 23 May agreeing to a meeting.

On 27 May 1997 the internal memorandum from the general manager of flight operations technical called part of the aerodynamic theory "flawed", called teaching pilots to use rudder as the primary roll control "extremely dangerous", and demanded immediate correction (pp. 88, 89). The report records no course change following that memorandum; the next verifiable textual change is the workbook rewording of 1 September 1997 (p. 84, note 138).

On 20 August 1997 the joint letter from Boeing, Boeing Douglas, Airbus and the FAA went straight at the "overemphasis on the superiority of rudder" (p. 87). The answer was the reply of 6 October, which rejected it point by point (p. 88).

On a 727 flight in 1997 a captain watched this first officer, levelling off between 1,000 and 1,500 ft, answer a momentary wake encounter with repeated full-rudder alternating inputs; the captain said he had never seen any pilot react that way (p. 137). He spoke to the first officer afterwards, and the first officer said AAMP had taught him to use the rudder that way; the captain advised him to use less of it, and did not report it as a course problem (p. 138 and note 181). On the next flight together, the first officer still went to the rudder quickly when he met wake (p. 138).

Signal five: Airbus says it did not receive the reply of 6 October 1997 until 20 January 1998 (p. 88, note 143). The last of the four external opinions arrived three and a half months after it was sent.

The point of escalation: on 1 September 1997 the AAMP workbook was revised, deleting the word "maneuver" and adding "co-ordinated" before "rudder" (p. 84, note 138). The report records that the change answered the manufacturer's concern about the company's "emphasis on the use of rudder". The change happened at the level of wording only: the simulator's 90-degree item stayed as it was, the instructor's "following a heavy aircraft" setup stayed as it was, the 10-second suppression and 1.3-second restoration stayed as they were (p. 83). Once the text was fixed, the objections lost their target: the company's reply could say it did "not advocate rudder first or rudder only" (p. 88), because the workbook no longer said so. Nobody signed this step; the report records only the revision date and the content of the change.

Failure chain: Technical layer. After the second wake encounter the pedals swung through six large alternating inputs in seven seconds (p. 5); at 09:15:58.4 the right rear main attachment fitting

failed under a load about twice the certified limit load and above the certified ultimate load (p. 159); the fin separated and the aircraft went out of control. Organizational layer. Ground school taught pilots to use rudder to help roll (p. 159, finding 8); the 90-degree item produced bank angles beyond anything in the real wake statistics, so trainees linked wake encounters to the need for a violent recovery (p. 83; p. 159, finding 9); pilots were not adequately trained before the accident on the effects of pedal input at high speed or on how the travel limiter works (p. 159, finding 10); the simulator's pedal travel did not match the aircraft (p. 90). Incentive layer. The course was the company's safety calling card, certification rested on a single POI approval (p. 80), training instructors did not have to be airline pilots (p. 79), and the training system and the airworthiness system each ran on its own line. All three layers broke at once. The junction broke in the cell marked text changed, action unchanged: the rewording of 1 September 1997 brought the text into line with the external objections, while the simulator item and the instructor's spoken setup did not move; four years later the first officer executed an action, not a text.

Minimum counterfactual: in the second half of 1997, calibrate the simulator's pedal travel against one static measurement on the real aircraft, with the aircraft parked and actual pedal travel recorded across airspeed bands, then re-run the course. One action, taken four years before the accident, with no luck required: the NTSB performed that measurement in January 2002, at American's Tulsa maintenance base, on an A300-600 (p. 91).

Outcome and sources: 265 people killed, 260 aboard and 5 on the ground; the NTSB issued eight new recommendations, A-04-56 through A-04-63, in addition to A-02-01, issued earlier on this accident (pp. 161, 162). Primary source: NTSB, Aircraft Accident Report

AAR-04/04, In-Flight Separation of Vertical Stabilizer, American Airlines Flight 587, 2004; link in Appendix B, av-033.

Case 6 (Maritime): A Watch Rota with No Ledger

Case ID: ma-003.

When and where: about 00:06 on 6 June 2025, at Barmore Island in Loch Fyne, Argyll and Bute, Scotland. The trawler left Tarbert at 03:30 on 5 June, the fourth and last fishing day of that week (MAIB Report 11/2026 p. 1, p. 2).

What and who: the prawn trawler Crystal Stream (fishing number LH 147), wooden, registered length 16.43 m, overall length 17.78 m, gross tonnage 112.34, under the UK flag, owned by M & I MacKin - non Ltd, bought in 2023, based at Tarbert, fishing for prawns in Loch Fyne and the Clyde with the catch going to the owner' s transport business (pp. 4, 12). Four people were aboard: the skip - per, the mate and two deckhands, one of whom was clearly unwell that day (p. 2). The skipper held a near-coastal fishing certificate issued in 1998, and all hands had completed Seafish' s one-day course in sea survival, firefighting, first aid and safety awareness (p. 5).

What they were doing: The vessel left at 03:30 on 5 June. The skipper steered south, handed the watch to the mate at 03:50 and went below to sleep (p. 2).

The vessel' s routine ran Monday to Friday, leaving in the early hours and returning each night or every other night depending on the catch; weekends were nominally for rest, though the skipper and crew sometimes used them for maintenance (p. 5). Before this departure the vessel had fished prawns three days running (p. 5).

At 05:00 a drop in engine revolutions woke him, the vessel being on the grounds. The first tow went out at 05:30 and was hauled at 10:30, with the second shot immediately after. He stayed in the

wheelhouse through this stretch with the windows open, drinking coffee often (p. 2). At 11:45 a coaster passed ahead of the vessel, he took avoiding action, and then went back to trawling. At 18:30 the fourth and last tow of the day began, and he set a north-west course for Tarbert (p. 2).

At 19:30 the whole crew ate a heavy dinner near the Inchmarnock channel. At 22:00, with the vessel passing Inchmarnock Island, the first deckhand offered to take the watch, and he refused (p. 2). At 23:15 the last tow was finished and the gear recovered. The crew sorted catch on deck while he stayed in the wheelhouse and brought the speed up to about 7 knots (p. 2). From that moment until the vessel reached the water off Tarbert, he was asleep at the helm (p. 2).

At 23:45 the first deckhand saw the vessel passing Tarbert and took it that the skipper had decided to put in another tow in Loch Fyne before returning to port. He carried on sorting catch (p. 2). Shortly after midnight the vessel struck a submerged obstruction, the im‑pact threw the first deckhand across the sorting table, the others lost their footing, and the skipper was woken by the blow (p. 2).

The decision point: three moments. The first two have documents. The third has none.

On 12 September 2023 a Maritime and Coastguard Agency (MCA) surveyor boarded the vessel after its change of owner, and the survey report set out three requirements: that the safety folder and risk assessments be reviewed and updated; that evidence be provided that the crew had completed the required safety training and held the appropriate qualifications and medical certificates; and that new crew induction records and rest time records be es‑tablished (pp. 4, 5). The owner and skipper submitted partial close-out evidence by email and submitted nothing specific on the crew’s medical certificates. On 25 September 2023 the vessel was

issued a UK Fishing Vessel Certificate (UKFVC) (p. 5). The signatory was the MCA.

The renewal survey of 14 May 2024 listed seven deficiencies, one of them that two crew members held no medical certificates; the report records that "no documentation was available to confirm that the crew held valid medical certificates to close the deficiency" (p. 5). On 21 June 2024 a new UKFVC was issued, valid until 25 June 2029 (p. 5). The signatory was the MCA. Thirty-eight days separate the two documents, and the report records no supplementary check in those thirty-eight days.

At 22:00 on 5 June 2025, in the wheelhouse, the deckhand offered to take over and the skipper refused (p. 2). This was the last moment at which the watch could have been handed over, and the only decision in the case he made on the spot, with no second person reviewing it. It has no document; the report gives one sentence of narrative and records no reason.

Put the three moments in a row and the shape of the gap appears. The rules require rest time to be planned and recorded (p. 7). The survey of September 2023 already stated that rest records had to be established. By June 2025 the vessel still showed "no evidence of any system for managing crew rest time" (p. 7). Between the two sits a renewal survey and a certificate valid to 2029.

What was visible then: Before departure: three consecutive days of prawn fishing that week; his own sense of fatigue (p. 5); a catch that had begun to worsen two days before the accident, and operating difficulties on board that were putting him under stress (p. 5). The rota held only himself: the report's wording is that the skipper "was managing his own rest" (p. 9), and no second person on board was counting the hours.

On that day he handed over at 03:50, was woken by the change in engine revolutions at 05:00, and between then and 23:15 his sleep amounted to that one period of 1 hour 10 minutes (p. 2).

Through the daylight of 5 June: one deckhand was clearly unwell (p. 2); and his own condition, handled by keeping the windows open, drinking coffee often and getting up to move about, all three of which were his own response to his state. The report notes that the open windows and the moving about match the advice in MGN 313 (F) on maintaining alertness, and that coffee is not on that list; it then records that these measures "were not sufficient to offset the effects of accumulated fatigue" (p. 9).

At 22:00 on 5 June: the person available, since the first deckhand offered (p. 2); the custom of the vessel, the report recording that he "was satisfied with the crew's ability to keep a navigational watch, but preferred to steer himself during fishing operations and when entering or leaving port" (p. 5); the instruments, an autopilot that followed a magnetic compass heading, had no link to the plotter and could not follow a preset track (p. 7), and a plotter he "generally did not use" (p. 7); and the watch alarm in the wheelhouse, which was fitted but had no interface with the autopilot and was not working at the time (p. 4).

What was learned afterwards: rest within 24 hours of "slightly more than 4 hours", taken in two periods; rest across six days "just over 24 hours" (p. 9, Figure 3 reconstructed by the MAIB from interviews); a regulatory floor of 10 hours in any 24 and 77 hours in any 7 days (p. 7); and the refusal of the deckhand's offer, which the MAIB wrote up directly as "a missed opportunity to reduce sleep debt" (p. 9).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".

> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the hull struck the submerged obstruction and breached, and the fish room flooded fast. The crew rigged the vessel's emergency pump and tried to keep up with the water; at 00:09 the skipper reported the impact to the coastguard on VHF; he then deliberately put the vessel ashore on rocks on the south-west side of Barmore Island. The liferaft inflated upside down and took time to right; the second deckhand was too weak to evacuate and was helped into the raft; the crew did not know the raft's equipment well and could not find the knife for the painter, so the skipper went back aboard to cast it off. At 00:49 the four reached the shore. At 01:16 the RNLI inshore lifeboat from Tighnabruaich arrived and took them off (pp. 2, 3). Injuries were minor and pollution was slight; the vessel took a list, partly submerged, was declared a constructive total loss and was later broken up and removed (pp. 1, 12).

The MAIB made one recommendation to the owner (2026/136): establish a safety management system under the Fishing Safety Management Code, establish watchkeeping procedures under MGN 313 (F) and the Fishermen's Safety Guide, and establish a fleet-wide fatigue management system (p. 11). The owner had watch alarms fitted across its vessels (p. 11), and the MAIB issued a safety flyer to the fishing industry on this case (p. 11).

The three questions, looking back. The commonest answer to the first is "fit a watch alarm". This vessel had one, with no interface to the autopilot, and it was not working (p. 4); what the owner fitted fleet-wide after the accident was the same thing, the same generation of equipment. The item most often missed under the second question is one line in the survey report of 12 September

2023, the requirement to establish rest time records. It reads like a paperwork requirement. It is in fact the only action in that system that ever counts hours; without it, rest is nothing but the person's own sense of it. The third question lands at the certificate issued on 21 June 2024: the survey listed seven deficiencies, one of them had no close-out evidence, and the certificate was issued anyway, valid for five years.

Options ruled out: four.

Accept the deckhand's offer at 22:00. The first deckhand raised it himself (p. 2), and he was capable of keeping watch, the report recording that the skipper "was satisfied with the crew's ability to keep a navigational watch" (p. 5). The cost is the deckhand's own rest that day, and one pair of hands fewer on the sorting table.

Cut the day's work short or stop early. The catch had been worsening for two days before the accident (p. 5), and the fourth day was the last fishing day of the week; stopping early costs that week's income directly.

Assess the risk of lone lookout in the wheelhouse and adjust on the basis of the assessment. MGN 137 (M+F) strongly recommends that a vessel in darkness not rely on the navigational watchkeeper as the sole lookout (p. 8); no such assessment had been done on board, and the report states plainly that "the risks of single-handed watchkeeping were not identified in the vessel's risk assessment" (p. 1).

Connect the watch alarm to the autopilot. The alarm had no interface with it (p. 4); the modification is equipment work, and the owner fitted alarms fleet-wide only after the accident (p. 11).

Early signals: five.

Signal one: the survey report of 12 September 2023, requiring the risk assessment to be reviewed and updated, evidence of training

and medical certificates, and induction and rest time records to be established (pp. 4, 5). Handling: partial evidence by email, and the certificate issued on 25 September as before.

Signal two: the seven deficiencies at the renewal survey of 14 May 2024, including two crew without medical certificates (p. 5). Handling: no close-out evidence, a new certificate issued on 21 June 2024, valid until 2029.

Signal three: the risk assessments in the safety folder. The online copy held one, covering boarding and disembarking; the one filed at survey covered a person overboard and had last been reviewed on 30 March 2022 (p. 7). Handling: at the 2024 survey the surveyor reminded the owner to review risk assessments regularly, and nothing changed after the reminder (p. 10).

Signal four: six-day rest of "just over 24 hours", and "slightly more than 4 hours" in 24 (p. 9). The account available is written in the report: the vessel had "no system for managing rest time", and rest was arranged by the skipper himself (pp. 7, 9).

Signal five: on 5 June one deckhand was clearly unwell and later unable to evacuate unaided (p. 2). Handling: sailed as normal, worked the fourth day as normal.

The point of escalation: two consecutive actions between 22:00 and 23:15 on 5 June 2025, the refusal of the offer and then bringing the speed up to about 7 knots alone (p. 2). After 23:15 three people were on deck sorting catch and he was alone in the wheelhouse; the alarm had no connection to the autopilot (p. 4); the autopilot would hold only a magnetic heading and would not follow a track (p. 7); and he did not use the plotter (p. 7). Nothing was signed at this step and nothing was recorded: the work and rest arrangements for that day were never written down (p. 7).

Failure chain: Technical layer. At 00:06 the hull struck a submerged obstruction, the fish room flooded, and the vessel listed

and partly submerged after going aground (pp. 1, 2). Organizational layer. A single watchkeeper with no risk assessment (pp. 1, 9); no written safety management system, with one safety folder aboard and one maintained online by an external provider, the online copy holding a single risk assessment, on boarding and disembarking (p. 7); no rest time management system and no records (p. 7); a watch alarm with no interface to the autopilot, not working at the time (p. 4); crew without valid medical certificates (p. 5); no induction procedure and no regular drills, with the liferaft handled inexpertly (pp. 7, 10). Incentive layer. A worsening catch that week, a fourth and final fishing day, and a catch delivered straight into the owner's transport business (pp. 4, 5). All three layers broke at once. The junction: fatigue was known to the man himself (p. 9 states that the risk "was not identified by those responsible for implementing the requirements", with operational demand overriding the rest requirement), and two entry points could have turned it into "not on watch tonight". Handover was one, and the deckhand offered it at 22:00 and was refused. Recording was the other, and that entry point had never been built.

Minimum counterfactual: accept the deckhand's offer to take the watch at 22:00 on 5 June 2025. One action, taken two hours before the accident, with no luck required: the man was already in the wheelhouse, and the words only had to be said (p. 2).

Outcome and sources: all four survived, with minor injuries and slight pollution; the vessel was a constructive total loss and was broken up and removed; the MAIB made one recommendation to the owner, who had watch alarms fitted across the fleet (pp. 1, 11, 12). Primary source: MAIB, Report 11/2026, Crystal Stream, May 2026; link in Appendix B, ma-003.

Cases 5 and 6 sit side by side at the end of this section to complete the two far ends of drift. Case 5 shows drift growing in the seam

between training and design, the drift furthest from the line. Case 6 shows drift growing in the seam between scheduling and the body, the drift furthest from paperwork. Put those two ends together with the intermediate forms in Cases 1 and 2, and the distribution map is complete: drift lives in no department. It lives in the seam between each department and the rule.

Case 7 (Nuclear): Two Tagging Events at One Plant in a Fortnight (condensed)

nu-001. Unit 3 at the Gravelines nuclear power station, France. ASNR event notification (October 2026; the notification was read and verified for this book): during an outage an electrical tagging error left a standby emergency diesel generator de-energised for longer than the 24-hour ceiling allowed by the operating rules, and the event was rated INES 1; within two weeks a second tagging-type event occurred at the same plant (nu-002, equipment unavailable for 7 hours 45 minutes). An outage is the period when procedure density at a nuclear plant is at its highest, and tagging and state restoration change hands hundreds of times: every handover is an action carried by convention, the 24-hour ceiling is written in the operating rules, and the fork in execution is written in habit. Two weeks separated the events, and the second was executed no differently from the first. Minimum counterfactual: build an item-by-item handover sheet for outage electrical tagging and state restoration, with mandatory double verification. Primary source: ASNR, 2026; link in Appendix B, nu-001.

Case 8 (Maritime): A Deck with No Separation (condensed)

ma-011. The fishing vessel Weston Bay (GY123). MAIB Report 9/2026 (pages read and verified for this book): two men went overboard while crab pots were being deployed, and one died. The investigation found no physical separation between people and gear, no man-overboard risk assessment, no man-overboard drill, and

rescue equipment aboard that was unsuitable for recovering an unconscious person. Work on a fishing deck is carried by genera - tions of experience: how the pots are stacked, where a man stands, how the ropes are flaked, all of it governed by conventions that work, except that the possibility of a person being taken overboard by pots and ropes had never entered any assessment as a risk. Minimum counterfactual: run a man-overboard risk assessment before the operation, and configure drills and rescue equipment to what it finds. Primary source: MAIB, Report 9/2026, 2026; link in Appendix B, ma-011.

Case 9 (Finance): After Changing Audit Firm (condensed)

fi-009. Zhong Shen Ya Tai Accounting Firm. Administrative penalty decision of the China Securities Regulatory Commission (No. 54 of 2026; the original text was read and verified for this book): the firm failed to exercise due diligence in auditing the annual report of the same entity, with confiscation and fine each of 1,226,415.10 yuan, and the two signing accountants fined 200,000 yuan each; the finding concerned revenue inflated through fictitious powertrain sales. This case forms a pair with another firm in Chapter 14’ s materials library (fi-008): two firms in succession failed to intercept the same fraud, which indicates that the failure point is not the personal quality of one group of people but the executed form of the industry’ s standard audit procedures. The procedure is writ - ten into the working-paper template, execution follows the tem - plate, and a filled template means the procedure is complete. The template and the fraud technique never had a mandatory meeting point. Minimum counterfactual: upgrade confirmation and site-visit procedures for revenue accounts as fraud techniques evolve, instead of reusing the working-paper template. Primary source: China Securities Regulatory Commission, No. 54 of 2026, 2026; link in Appendix B, fi-009.

Set the nine cases side by side and three regularities of drift are complete. The host of drift is the high-frequency action: a logbook check in flight (every trip), music in the wheelhouse (every night), a tag handover (hundreds in one outage), a tick on a checklist (every operation), a filled working paper (every engagement). Drift never parasitizes a low-frequency decision, because a low-frequency action never gets the chance to become a convention. Drift's camouflage is efficiency and professionalism: every site version is faster, smoother and more experienced-looking than the manual version, so drift gets celebrated as improvement until the day it is reclassified as an accident. Drift's reconciliation cost runs three orders of magnitude below the accident cost, and the organization pays only the latter, because the former arrives without an invoice. Those three together give the budget logic of prevention: every penny allocated to a reconciliation interface, Devices 1 through 4, saves a thousand times its value at the tail.

15.4 The earliest you can see

Eight weak signals, each with the reason it is usually waved through.

1. What training teaches differs from what the site does, and both sides know it. The reason: "training covers principles, the site covers what actually happens."
2. New people are brought up to speed by older staff, not by the manual. The reason: "learning the job from someone experienced is the most effective way."
3. A step in the manual has never once been executed in practice and still appears on every compliance checklist. The reason: "better to have it written than not."
4. The site's verdict on the manual is that it was written for the auditors. The reason: "everyone understands that."

5. When workload rises, the first thing simplified is the pause in the process: the cross-check, the read-back, the wait. The reason: "efficiency."
6. An audit finds that practice differs from the manual, and the outcome is a change to the site rather than the manual, or a change to the manual without asking the site. The reason: "one end has to move."
7. A key action is demonstrated in two versions, the factory version and the site version, and new people are told to watch the site version. The reason: "the manual is out of date, don't learn it wrong."
8. The person who signed the procedure cannot say which accident or risk the procedure was written against. The reason: "compliance requires it."

Put those signals back against the cases. Signal one, training and site differ and both sides know it, matches Case 1: handling paperwork in flight was a default the site taught. Signal two, new people taught by colleagues rather than by the manual, matches Case 8: every practice on that deck came from being shown. Signal three, a manual step never executed but still on the compliance form, matches Case 9: the procedure lines in the audit template were filled in as usual, with the real intensity of confirmations and site visits accounted for elsewhere. Signal five, the pause simplified first, matches Cases 1 and 6: a head down in a logbook and a watch stood alone both traded a pause for output. Signal six, the audit finds the fork and only one end moves, matches Case 2: the alarm was fitted, which is the manual end, and nobody touched the music in the wheelhouse, which is the site end. Signal eight, the signatory cannot say what accident the procedure addresses, is the hidden thread of the chapter: in Case 5's training actions and Case 8's deck conventions, neither the signatories nor the trainers knew which stretch of history they had routed around.

The method for finding drift deserves a section of its own, because its difficulty differs from that of normalization. Normalization is audited against ledgers, which have archives. Drift is audited against practice, which has none. The workable method is shadow comparison: take one key process, send one group through a full walkthrough following only the manual and another group through following site habit, and have the two write up a difference list item by item. Every entry on that list then gets classified. Site practice better: amend the manual. Site lacking resources: supply the resources. Site taking the easy road: put it on a ledger. Shadow comparison costs one crew's time, and it measures the one thing in the organization that nothing else measures: the distance between text and practice.

A cheaper high-frequency probe sits alongside shadow comparison. Register the difference between "the manual says" and "we actually do" at the crew meeting, ten minutes at a fixed slot each time, on one question only: which steps did we do differently from the manual this week. Nothing registered needs handling straight away; let it accumulate. A quarter of registrations is that crew's drift map. Most crews are startled the first time they draw it: the entries do not number two or three, they number dozens, and more than half of them are differences nobody can remember starting. The value of the map is not the correction, which runs through Chapter 11's three-state register. The value is making the fork visible: an invisible fork is climate, and a visible fork is an agenda item.

15.5 How to prevent it

Device 1: manual-reality comparison. Twice a year, write the manual's steps for a key process alongside actual practice and produce a difference list. Each difference is either an amendment to the manual, conceding that reality is better, or a change to the

process, resourcing reality to follow the manual. Only one thing is forbidden: failing to compare. Signatories: the process owner and one representative from the line.

Device 2: list protection for pause actions. Every pause-type action in a process, the cross-check, the read-back, the visual confirmation, goes on a separate register, and any request to simplify one has to be heard item by item, with the default answer being no. Pause actions are the first thing drift eats.

Device 3: a spatial dimension for occupation approval. Wherever a physical space in a facility carries a safety function, keep a register of protected space, route every occupation through a separate approval, and separate the approver from the user.

Device 4: a resource clause in every rule you write. Every mandatory procedure carries a statement of its execution cost, the time, the people and the tools required, and whenever the procedure changes, that cost is checked against the real workload on the site. Writing a rule nobody can execute is prepaying for one drift.

Each of the four devices has one direct use on this chapter's cases. Device 1 on Case 1: write the site practice of using en-route time alongside the manual's see-and-avoid clause, and classify the difference as a site lacking resources, since only a document mount or a two-person crew would let the practice stand. Device 2 on Case 6: register the handover confirmation as a pause-type action and hear any request to simplify the watch rota item by item, because simplifying the rota is exactly the entry point for watchkeeping drift. Device 3 on Case 4: re-examine the cable occupation of the platform recess against the protected-space register, and either approve each existing occupation retroactively or remove it. Device 4 on Case 9: attach execution cost and conditions of applicability to each line of the audit programme template, and require any engagement outside those conditions to escalate to a different procedure rather than filling in the template.

Four failure modes, one per device. Manual-reality comparison dies when the comparison becomes a formality: the difference list is written and filed, and the revision schedule is always next quarter. Pause-action protection dies when the register goes stale: a new process goes live without the register being updated, and the shield protects an action that no longer exists. The space register dies when its scope is too narrow: only physical space is registered, and nobody watches the occupation of process space, meaning authority, approval slots and handover slots. The resource clause dies when the cost statement is wrong: the department writing the rule underestimates what execution costs, and the clause drifts the day it goes live. All four share one patch. Each device's output, the difference list, the register of pause actions, the space ledger, the cost statement, carries a mandatory annual review date, and the review is done by someone other than the device's owner.

One organizational precondition is worth naming. The reconcili‑ation interface needs a third party both sides accept. The owner of the manual believes the site does not follow the rules; the site be‑lieves the manual has nothing to do with the ground; without a third party the reconciliation talks break down every time. The line representative's signature in Device 1 and the cost statement in Device 4 are both ways of institutionalizing that third party. An organization without the role can only convene the conversation between manual and reality in the form of an accident.

The reconciliation meeting has one output that is easy to overlook: drift contains improvement proposals. When site practice departs from the manual, sometimes the site is wrong, sometimes the manual is wrong because the rule has not kept up with changes in equipment, process or users, and most of the time each side holds part of the truth. The meeting's decisions therefore have only three legitimate exits: amend the manual, conceding that reality is better; supply resources so reality returns to the manual; or put the

difference on a ledger and run with the risk. An organization that presumes site practice is guilty by definition will miss the first exit, and the first exit is often the most valuable of the three, since the site wisdom absorbed into a manual revision is exactly what lets the next edition still be executed. An organization that bans every fork does not buy compliance. It buys better-kept false accounts.

The last design detail is who attends. The manual side, the people who write the rule; the site side, the people who execute it; the safety side, the people accountable for the consequences. All three in the room is the minimum for the meeting to be valid, and with any one missing the alignment degrades into a one-sided decision. The chair is best held by the safety side, since safety is the one function among the three whose performance is measured by finding forks. Frequency follows risk: twice a year for high-risk processes, once a year for low-risk ones, plus ad hoc meetings triggered by any change. An organization that writes this meeting into its quality system has fitted drift with a standing brake.

How well the brake works depends on where the minutes go. The direct readers of the reconciliation minutes should be the manual's owner and the training department, as input to revision, with the safety committee copied for the risk view. Minutes that end up in any other drawer turn the brake into one more document ritual. A fixed three-column format is worth adopting: difference described, cause classified as site error, manual error or missing resource, and decision with a deadline. Only minutes with all three columns can be checked against at the next meeting six months later, and the completion rate of the previous round's decisions is the only proof that the mechanism is alive.

15.6 Borders with neighbouring mechanisms

Against normalization of deviance (Chapter 11): drift describes the existence of the fork, that how the work is done has changed; normalization describes the fork being refiled, that how it is classified has changed. Drift can occur without normalization, since a fork held on a ledger stays in the state of correction by a deadline. Normalization normally uses drift as its raw material but needs that second step, explanation and filing. The order of treatment: align first, which is this chapter, then defend against refiling, which is Chapter 11.

Against compliant and unsafe (Chapter 32): compliant and unsafe means every inspection passes and the whole is still dangerous, because the inspection system and the real risk are misaligned. Practical drift is text and practice forking, because documents and behaviour are misaligned. The Ickenham recess sits on both: the occupation went through engineering approval, which is compliant, and the safety function was gone, which is unsafe. When filing a case, classify by the mechanism that set it off and attach the residual mechanisms as secondary.

Against automation and skill decay (Chapter 26): drift is a person's practice departing from the text; decay is a person's ability to execute an existing action diminishing. Rising automation accelerates both at once, since the text updates more slowly than the system, which is drift, and the person stops practising as the system takes over, which is decay. Drift in an automated environment has one further form: the system's own configuration forks from the text, and no old hand is left who would notice.

15.7 The five self-checks

1. Write down three differences between what the manual says and what is actually done in a process you own. If you cannot, is that because you do not know, or because there are none?
2. Which item on your organization's checklists do you know has never actually been executed? Is it still on every form?
3. At the last process simplification, what kind of action was cut: output-producing or pause-type? When a pause action is cut, is there a ledger entry?
4. Which rules on your site are written in a form nobody could follow? Name three, and ask yourself whether you dare propose a revision.
5. Does protected space, physical or procedural, have a register in your organization? When was it last occupied?

Chapter 16 Information Attenuation up the Hierarchy | MUM Effect

> What this chapter covers: how bad news is softened on its way up; the psychology and the structure behind the MUM effect, three start criteria, scenes from four domains, and the design points of a channel that skips the hierarchy. When to turn back: when you notice that the top's read of a situation plainly disagrees with the front line; before you write a report that says "overall under control"; when you are designing a reporting system. Estimated reading time: 50 minutes.

Bad news travels through an organization in a way that one sentence can hold: every floor it climbs, it puts on a more presentable coat. "Both units have a major safety hazard" arrives at the department manager as "there is some risk", at the director as "worth watching", at the executive vice president as "overall under control", and by the time it reaches the person who can release the money, the thing no longer looks like itself. Organizational psychology has a sour and accurate name for this: the MUM effect, mutual unwillingness to uphold bad news upward. The sourness is in "mutual". The sender does not want to report, the receiver does not want to hear, and the two of them complete every step of the softening together. No step in that sequence requires anyone to lie.

The experimental origin is worth stating. Sidney Rosen and Abraham Tesser named the phenomenon in a series of studies around 1970, and the core fact they measured was this: the discomfort of delivering bad news is large enough to change delivery behaviour even when the truth of the message has nothing to do with the messenger's interests. Unwillingness to report sits close to the default setting. What an organization has to do is build channels

against that default, not ask its members to overcome discomfort. That placement decides the shape of every defence in this chapter: they are structural devices, and not one of them depends on cour - age.

The moral asymmetry between attenuation and lying deserves to be spelled out, because it is the reason this mechanism is so hard to govern. A lie has an author, a moment and a content, and can be traced to someone. Attenuation has none of that. Each layer's wording survives inspection ("I reported what the department knew"), and no layer has lied. Moral judgement fails here. In - formational judgement still works: put the versions from each level side by side and the points where intensity drops are plain to see. The defences in this chapter therefore point at material that can be checked (minutes, originals, numbers), and not at anyone's atti - tude.

The receiving end's unwillingness to hear needs a paragraph of its own, because it is usually reduced to a matter of temperament. Three structural sources feed it. The first is asymmetric burden: the person who hears bad news is expected to solve it, and plenty of bad news (market shifts, policy changes, technology routes) lies outside the listener's remit. Every hearing loads a pack onto his back, and the appetite drains. The second is attribution inertia: the person who brings bad news is easily read as the person who cre - ates problems, and the stronger the association in organizational memory between bearer and trouble, the less the receiving end wants to hear. The third is agenda pressure: bad news has no seat. Agendas are organized around the period's targets, so bad news can always be deferred to next time, and by next time its wording is one layer lighter. The devices aimed at the receiving end sit in Section 16.5, Devices 1 and 2: side-by-side reading turns hearing into an obligation, and a named reader turns hearing into a post.

16.1 What It Is

Definition: information attenuation up the hierarchy is the systematic dilution of bad news as it climbs. Each level makes a small wording choice, for self-protection or out of consideration for the person above, and the choices compound until the decision-maker's version and the front-line version have drifted apart. Nobody has to lie for this to happen. That is the distance between attenuation and falsification, and the reason it resists governance: you cannot discipline a behaviour assembled entirely from compliant wording.

Three start criteria. Any one of them means the mechanism is running.

1. The same piece of bad news reaches different levels at different intensity. Lay the minutes side by side and check.
2. The decision layer's judgement of the situation has visibly diverged from the front-line version. The front line laughs when it hears what the top thinks.
3. Skipping a level is defined in the organization's language as disloyalty, as not knowing how things work, or at minimum as "talk to your manager first".

16.2 How It Grows

Three generation conditions. Condition one: the reporter's interests are bound to what he reports. When the severity of the news decides his appraisal, his bonus and his prospects, the wording tilts light. This is Chapter 17's misaligned incentives projected onto the information layer: the message takes the colour of the reporter's interests rather than the colour of the facts. Condition two: many layers, narrow bandwidth. Each layer is one compressed retelling, compression is lossy, and the direction of the loss is set

by the reporter's interests. A layer is not a relay station. It is a filter, and each filter takes out a little, little enough that nobody is answerable for it, enough that the accumulation is lethal. Condition three: no receiving mechanism for bad news. The organization runs channels that reward good news (wins, highlights, targets beaten) and no channel through which a problem can be reported without cost. Where the receiving end sets no place, the sending end trims the message to fit.

The concealment is worth repeating: every step is small. From "serious accident hazard" to "some risk" to "overall under control", each choice survives examination inside a single meeting. Only five sets of minutes read side by side make the slide visible. Side-by-side reading is the core action of this chapter's defence.

The arithmetic behaves like compound interest. Suppose each layer passes on eighty per cent of the intensity, which is a conscientious layer by the standards of most organizations. After five layers the decision layer receives a little more than one fifth of the original. Every one of those five layers believes it merely distilled the message. This arithmetic explains why training people to "report more faithfully" does nothing: no layer is skimming anything beyond its eighty per cent, and the problem lives in the multiplication. Multiplication admits two remedies. Cut the number of layers, which is the direct channel. Or keep a raw value in a path that does not pass through them, which is the original document delivered unedited. Devices 1 and 2 in Section 16.5 are those two remedies.

Good news and bad news do not climb at the same speed, and the gap is one of the steadier findings in this literature. Good news moves fast, because every layer is glad to pass it on. Bad news moves slowly, because every layer wants to digest it, dress it and pick a moment. The speed gap is itself a probe: the arrival-time

difference between the two measures how deep the attenuation runs. The practical measurement: take one group of good-news items and one group of bad-news items of the same class, count the average days from origin to decision-layer minutes, and the ratio of the two is the organization's attenuation index. The index does not need precision, only trend. Two consecutive quarters in which bad news slows down, and criterion three is very likely met.

Trust is the moderator that keeps turning up in the evidence. Every notch down in a reporter's trust in the receiver adds a layer of self-protection to his wording. In an organization short on trust, attenuation is not a discipline problem. It is the individually rational choice of every reporter. Trust builds slowly and is spent in one transaction: a single instance of being held to account for bad news wipes out ten years of speaking freely, and the count restarts at zero. That asymmetry is why Device 3, the immunity clause, has to be carried by the institution and not by a leader's verbal promise. A verbal promise is honoured at the moment of accountability, and the discretion at that moment is precisely what the promise cannot bind. Institutional immunity goes into two documents, the performance one and the accountability one, and any blank left in either will be cited in the first real case.

16.3 Shapes in Four Domains

Case 1 (Chemical & Process): The Information Chain after the Report Came In

Case ID: ch-007.

When and where: 22 November 2013, 02:12 to 10:25; the junction of Qinhuangdao Road and Zhaitangdao Street, Qingdao Economic and Technological Development Zone, Shandong, where the Donghuang crude oil pipeline crosses the municipal drainage cul-

vert; the information chain reaches the Weifang oil transport division dispatch centre, the zone main duty office and the Qingdao city main duty office.

What and who: The pipeline storage and transportation branch of Sinopec Corp, with its Weifang oil transport division, Qingdao oil transport station and Huangdao tank farm; the Qingdao Economic and Technological Development Zone management committee with its emergency office, safety supervision bureau and municipal bureau; the Qingdao city main duty office and the municipal economy and information technology commission. Responsibility attaches to posts: the duty dispatcher in the dispatch centre, the marketing and sales section, the division head and his deputy, the Qingdao station master, the zone emergency office director and deputy director, and the committee's deputy director in charge.

What they were doing: The Donghuang pipeline went into service in July 1986, 248.5 km long, 711 mm in diameter, design throughput 20 million tonnes a year, reversed to opposite flow in October 1998 at 10 million tonnes a year (report p. 2). The accident section passed overhead under the south half of the Qinhuangdao Road bridge culvert roof, 110 cm from the top of the pipe to the culvert roof slab and 148 cm from the bottom of the pipe to the culvert floor (p. 2). At the time of the accident the line carried light crude with a closed-cup flash point of −16 °C, running at full load, outlet pressure 4.67 MPa, outlet temperature 27.8 °C (p. 2). The Weifang division tested the external coating and local pipe body in 2009, 2011 and 2013, and none of the three found serious corrosion in the accident section (p. 6). A coating overhaul was scheduled from 2011, and by October 2013 the 15 km containing the leak point had still not been overhauled (p. 6). Over the same period the zone carried out the Qinhuangdao Road comprehensive improvement project without consulting the pipeline company and without calculating the project's effect on pipeline safety (p. 7). The Qingdao

station’s patrol system was incomplete and the patrol staff lacked the professional knowledge, so no pipeline protection requirement was raised when the municipal open channel was covered and slabbed or when the road was widened and resurfaced; the emergency plan had no drills and the rescue staff did not know their own duties (p. 6). At the Sinopec Group and listed company level the safety inspections had dead corners and blind zones, found no hazard in the accident section and took no protective measure (p. 6). In 2013 the Qingdao city and zone pipeline protection departments ran 6 special rectification inspections, and none of them found the effect of the road works on the pipeline (p. 7).

The decision point: Four moments.

One. At 02:12 the supervisory control and data acquisition system in the Weifang division dispatch centre showed the Huangdao tank farm outlet pressure falling from 4.56 MPa to 4.52 MPa. Two telephone calls to the tank farm ruled out any operating cause, and the drop was judged a leak; at 02:25 the pumps were stopped and the line shut down (p. 3). At 02:35 the Qingdao station was told to close the block valve at the Yanghe valve chamber, which was shut around 03:20; the chamber sits 24.5 km from the Huangdao tank farm and was the nearest downstream chamber to the leak (p. 3). At that moment the desk held a pressure drop of 0.04 MPa and one distance to a valve chamber.

Two. At 02:57 the repair centre was called out. At 03:40 Qingdao station staff were on site and confirmed the leak about 1.5 km from the Huangdao outlet (p. 3). Around 04:00 excavation of the leak point and pipeline repair began, alongside clearing materials and cleaning up oil spilled at sea (p. 3). The excavation came first. Combustible gas detection does not appear in the record (p. 6).

Three. At 04:47 the marketing and sales section reported to the head of the Weifang division, at 05:07 reported the overall situation to the pipeline company dispatch centre, and at 05:30 the division

head sent his deputy to the site (p. 3). At 07:40 a work pit of 2 m by 2 m by 1.5 m was dug out, and at 08:20 the leak point was found and reported (p. 4). At 09:15 the pipeline company gave the order to form an on-site command under the emergency plan (p. 4). From the shutdown to that order, 6 hours and 50 minutes.

Four. The government ran a second chain. At 02:31 the zone public security sub-bureau's 110 line took a call reporting crude oil near the south gate of Lidong Chemical; at 03:10 the call was passed to the zone main duty office (p. 4). By 04:17 the emergency office, the municipal bureau, the safety supervision bureau, the environmental protection sub-bureau and the Huangdao street office had received it in batches; at 04:51, 07:46 and 07:48 the committee's deputy director, director and party working committee secretary received it (p. 4). At 07:49 it reached the Qingdao city main duty office. Between 08:18 and 08:27 the city duty office telephoned three departments "requiring further verification of the information"; between 08:34 and 08:40 text messages went to the secretary-general and deputy secretary-general; between 09:01 and 09:06 to the mayor and 4 deputy mayors (p. 4). The zone management committee graded the event a general emergency on the basis of the enterprise's report (p. 8), and the report does not state when the grading was made or who signed it.

What was visible then: On the enterprise side, the pressure drop at 02:12, the shutdown at 02:25, the valve closed at 03:20, the position of the leak at 03:40, the dimensions of the work pit at 07:40, the leak point confirmed at 08:20 (pp. 3–4). On the government side, the content of the 02:31 call, the 03:10 report to the duty office, the units that had received it by 04:17, three zone leaders receiving it between 04:51 and 07:48, the city's demand for verification at 08:18 (p. 4). The units that arrived between 04:10 and 05:00 were the emergency office, the municipal bureau, the safety supervision bureau, the environmental protection sub-bureau and

the Huangdao street office, and their work was cleaning oil off the water (p. 4); the report records no department as responsible for the culvert under the road surface. One report stands apart: Lidong Chemical told the zone safety supervision bureau that crude oil had been found in an open channel inside its plant (p. 8), and that information stopped at the bureau, never passed to the government or the other departments. The text messages that went up between 08:34 and 09:06 carried "basic situation of the leak accident" (p. 4); by the time of the explosion no upward text contained a leak volume. The crews on site had excavators and hydraulic breakers, and the record holds no combustible gas reading (p. 6). Every report on the enterprise side carried a time and an action, and none carried a quantity. At 09:01, 1 hour and 24 minutes before the explosion, the mayor was still receiving that basic situation (p. 4). Known only afterwards: the leaked volume of about 2,000 tonnes, established by calculation (p. 5); the 8 hours and more between the leak and the explosion (p. 6); the drainage culvert's total length of about 1,945 m (p. 3); the mechanism of the corrosion thinning (p. 5); the temporary shed Lidong Chemical had put up over the culvert without authorisation (p. 8).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 10:25 the crew on site perforated the culvert cover slab with a hydraulic breaker, and the impact spark ignited the oil vapour inside. The explosion affected more than 5,000 m of the culvert and damaged water, power, heating and gas lines (p. 5). 62 people died, 136 were injured, and direct economic losses came

to 751.72 million yuan (p. 1). By 2 December all 62 of the dead had been identified (p. 5). 15 people were subject to judicial measures (pp. 8–9) and 48 were recommended for Party and administrative discipline (pp. 9–15); the Shandong provincial safety supervision bureau was ordered to fine Sinopec Pipeline Company at the stat‑utory maximum and to fine its party secretary and general manager 80% of their 2012 income each (p. 15).

Back to the three questions. The commonest answer to the first is "raise the response level". This case did grade the event, as a general emergency, and the grading rested on the enterprise report alone, which carried no volume and no direction of flow. That ac‑tion costs one telephone call and one signature. What was missing is the line of numbers before the signature. To the second ques‑tion, the item most easily missed is the 02:31 call to 110: it came 5 hours and 18 minutes before the enterprise report reached the city at 07:49, it came from outside the system, and it was never merged into the same chain. The third question lands before the 04:00 excavation, on one gas test.

Options ruled out: Activating the emergency plan (the report finds no activation order was issued in time, p. 6); raising the response level (not raised after the general grading, p. 8); marking a cordon, closing roads, notifying residents to evacuate (the report finds none of it was done in time, p. 8); combustible gas detection and explosion-proof equipment (the report finds neither was done and non-explosion-proof equipment was used blindly, p. 6); passing Lidong Chemical's report to the government (the report finds it was not passed in time, p. 8); calling in the repair centre (first raised by the deputy division head at 09:30, 55 minutes before the explosion, p. 4). All six carry "not" in the report's wording. Not one carries "considered and rejected": no record shows anyone proposing, anyone vetoing, anyone signing a refusal. The excluded option here takes the form of never reaching the agenda. In its

prevention section the report turns these into positive require‑ments: measure the oil and gas concentration on site, identify and assess the hazards and risks, and mark off a cordon and evacuate the surrounding population in time according to conditions on site (p. 17).

Early signals: Six items, each with the excuse that carried it past.

1. Three tests in 2009, 2011 and 2013 found no serious corrosion in the accident section (p. 6). Each closed as “not found”, so the hazard ledger stayed empty, and the reason it was pushed past was “three tests in three years”.
2. The coating overhaul scheduled from 2011 had not started on the 15 km section by October 2013 (p. 6). It sat in the annual plan and counted as in hand, and the reason it was pushed past was “it is on the plan”.
3. The Qinhuangdao Road improvement project ran without con‑sulting the pipeline company and without calculating its effect on the pipeline (p. 7). The works completed the municipal ap‑proval route on their own, and the reason it was pushed past was “the paperwork is complete”.
4. 6 special rectification inspections in 2013 found no effect from the road works (p. 7). The inspections were completed by tick‑ing a list, and the reason it was pushed past was “not in this round’s scope”.
5. The patrol system was incomplete, the patrol staff lacked pro‑fessional knowledge, the emergency plan had no drills (p. 6). All three were booked under training, and the reason it was pushed past was “training pending”.
6. The pipe ran suspended inside the culvert in an area subject to seawater backflow (p. 7). It was a 1986 design, compliant when handed over, and the reason it was pushed past was “the design was approved”.

None of the six was hidden. They sat in a test report, an annual plan, a works approval, an inspection checklist, a training plan and a design file (pp. 3–7).

The point of escalation: After the valve closed at 03:20 the source was cut, and the oil was still in the culvert. The excavator came in at 07:00, the pit took shape at 07:40, the leak point was found at 08:20; from that moment the object of the work changed from plugging a leak to breaking up a structure above an oil-bearing culvert. Only at 09:15 did the pipeline company order the on-site command formed, and at 09:30 the deputy division head reported that the repair could not be finished independently and asked for support (p. 4), 55 minutes before the explosion. The decision to open the culvert cover and switch to the hydraulic breaker is recorded only as "decided on site" (p. 5), with no name and no time. The shape of this escalation: every step advanced the repair, no step needed approval, and no step needed a magnitude written upward. Across the two hours and five minutes from 08:20 to 10:25 the site held actions and no numbers. At 09:55 the deputy director of the municipal economy and information technology commission arrived, at 10:21 reported sea-surface pollution to the city's deputy secretary-general, and at 10:27 reported the explosion and fire on site (p. 4): the word "explosion" enters the government chain two minutes after the explosion. The report's prevention section answers with one sentence: "text messages shall not be used in place of telephone reports of accident information" (p. 17).

Failure chain: Technical layer: the pipe wall thinned by corrosion in alternating wet and dry seawater and salt spray, with road load and vibration on top; the rupture released about 2,000 tonnes of crude into the culvert; vapour accumulated in a confined space; the spark from the breaker ignited it (pp. 5–6). Organizational layer: the leak volume was never assessed under the plan, the risk

assessment was seriously wrong, no activation order was issued, no cordon was set, no combustible gas was measured (p. 6); at group and listed company level the division of pipeline protection and safety duties was unclear and responsibility unnamed (p. 6); the zone graded the event a general emergency, did not raise the level, did not close roads, did not evacuate, and the emergency office suppressed and delayed reports of the accident and misreported whether the responsible leader had attended (p. 8). Incentive layer: the line ran at full load (p. 2), the overhaul scheduled for three years never reached the 15 km section (p. 6), and six inspections and three tests all produced "not found" (pp. 6–7); the outputs of inspection and testing pointed the same way, and nobody had to answer for failing to find anything. All three layers broke between 04:00 and 10:25. The magnitude gap has an exact address: the figure of 2,000 tonnes first appears in the investigation report's calculation (p. 5) and in no upward document. The junction broke between the organizational and the incentive layer: the arrangement that made "not found" a qualified output also made "unquantified" a qualified report, and neither needed a signature.

Minimum counterfactual: Before the excavation around 04:00, one combustible gas measurement at the work face and at the culvert opening, with the reading entered in the site record, and no non-explosion-proof equipment used while it was over the limit. One action, 1 hour and 35 minutes after the shutdown. The instrument is standard kit for a repair crew, the crew was already on site, no new budget was needed, no approval had to be waited for, and no judgement was required: over the limit, change the method; within it, the demolition may proceed. The timing is 04:00 because the leak was not yet opened, the road surface was not yet broken and the culvert cover was untouched; one step later and the action falls inside demolition already under way. The report lists "no combustible gas detection" among the indirect causes (p. 6),

which says the step was available on the day and was not taken. It is also an information action: once the reading is in the record it has to travel up with the site report, and the receiving chain then has something to pass at that cell.

Outcome and sources: 62 dead, 136 injured, direct economic losses of 751.72 million yuan; 15 people subject to judicial meas - ures, 48 recommended for Party and administrative discipline; the company fined at the maximum, the party secretary and general manager each fined 80% of 2012 income; the Shandong provincial government and the Sinopec Group ordered to report to the State Council with a deep self-examination (p. 15). Primary source: State Council investigation team for the "11·22" Sinopec Donghuang crude oil pipeline leak and explosion, Investigation report, 10 January 2014, published on the website of the State Administration of Work Safety, link in Appendix B, ch-007. The official release is a full web page with no official PDF; the card's page numbers fol - low an A4 print of the full text, 18 pages, and the page-to-section map is in _evidence/ch-007.md.

Case one carries a reading that deserves a paragraph of its own: in Qingdao the horizontal failure to assemble and the vertical delay existed side by side. The vertical one is pinned by the report's timestamps. The leak was found at 02:12 and reported up the company chain only at 05:07; the 110 call came at 02:31 and the city duty office heard of it at 07:49. The horizontal one is this chapter's subject. The company, the local government and the site have no line of authority between them, and the mechanism at work is that each holds one facet. Every party's information is true, local, and sufficient for its own duties. The act of assembly has no owner. Reading therefore has to widen: the minutes of each level side by side, and then the information versions of each party side by side. The check has one question. Where on which desk

does the full picture of this event exist at this moment. If you cannot answer, you are looking at the Qingdao shape.

Case 2 (Maritime & Transport): The New Procedure That Stopped at the Paper

ma-007. The ro-ro cargo ship Laureline, 13 July 2024, Purfleet terminal, United Kingdom. MAIB report (Report 7/2026, page verified for this book): a sailor was crushed to death by a trailer during vehicle loading. The investigation found that the new vehicle-deck loading safety procedure was not understood by the crew and was being carried out in distorted form; supervision was assessed as ineffective; the trailer driver did not know where the crew were. The procedure had been signed off at the management end and had never landed at the execution end. Between issuance and accident, "the procedure has been understood and is being followed" was management's default information, and that information had never once been verified.

Stop at the decision point: the moment after the new procedure was issued and before the first shift worked to it. On the desk: the procedure text distributed, the training records on file, the next shift's work plan, and one missing item: no mechanism anywhere answers the question "did the people on board understand it". **Outcome:** one death. Primary source: MAIB, Report 7/2026, 2026, link in Appendix B, ma-007.

The phrase "supervision was assessed as ineffective" deserves magnification. Between the supervisor and the supervised lies the distance from the deck to the wheelhouse, and the supervisor's version came from the ship's routine reports: procedure trained, operations normal. In MAIB's language that version was wrong. Upward information completed its attenuation on one clause, "the procedure has been understood": the training record is true, and the absence of understanding is also true. This is the same phe-

nomenon as the checklist ticking in Chapter 15, seen from the other side: paperwork proves delivery, not receipt. The equivalent of Device 2 on board is management reading one deck record or one interview a month and asking "how do you understand the new procedure". One question, and the depth of attenuation is measured once.

Case 3 (Public Policy & Administration): The Conceptual Schedule

po-002. The US Department of Veterans Affairs Electronic Health Record Modernization programme (VA EHRM). GAO report (GAO-26-109393, September 2026, page verified for this book): the programme plans to complete deployment at about 170 sites by 2031; deployment was paused in 2023 after poor user reviews and restarted at the end of 2024; GAO noted that what VA gave Congress was only a conceptual schedule, with no cost estimate and no detailed schedule document; of 18 recommendations, 4 had been implemented. This is attenuation in its public-administration form. Information passed to the overseer is deliberately set at a granularity that can be reported. Nobody concealed anything. A form of presentation was chosen that cannot be interrogated.

Stop at the decision point: the reporting season in which the conceptual schedule was chosen over a detailed plan in answer to congressional oversight. On the desk: the real risks and setbacks of deployment, the expectation that a detailed plan would be interrogated item by item once written, and the complete procedural legality of the word "conceptual". **Outcome:** the oversight body recorded the gap in information baseline in a dedicated report. Primary source: GAO, GAO-26-109393, 2026, link in Appendix B, po-002.

The conceptual schedule pushes attenuation into its public-administration form: information is deliberately parked at a granu-

larity that cannot be interrogated. It is the same coin as the gran - ularity choice in case four, the single incident. Whoever chooses the granularity decides whether interrogation is possible. GAO's countermeasure is worth recording: the report asks directly for a cost estimate and detailed schedule documents, which forces the granularity down. A reader can borrow the move inside his own organization. Whenever the words conceptual, framework or dir - ectional appear in material submitted to an overseer, a board, a regulator, a superior, ask one question: can you give me a version I can check line by line. What is being demanded is not honesty. It is checkability, and checkability by itself changes how the upstream writes.

Case 4 (Manufacturing & Supply Chain): Six Hundred Incident Reports

mn-001. The HAIARA power bank (Mitrytech, Amazon channel). CPSC recall notice (page verified for this book): about 11,000 power banks recalled; the company knew of 600 reports of fire and other thermal incidents, among them 3 burns and 9 cases of property damage. Six hundred incidents are not one piece of bad news. They are six hundred. Each looked small on its own: one overheating, one smoking; each travelled its own reporting path to the end; and the act of adding them up happened at no level at all until the regulator did it.

Stop at the decision point: the window in which incident report N arrived and the sales decision was not touched. On the desk: the severity of each single case, the channel pressure from units still on sale, the cost of a recall, and a number nobody was responsible for watching, the cumulative count. **Outcome:** CPSC intervened and a recall followed. Primary source: CPSC, 2026, link in Appendix B, mn-001.

Case four has a layer that needs saying plainly. Every one of the 600 reports completed the compliance flow. The company logged it, filed it, assessed it as a single case, and the assessment concluded that each one fell below the reporting threshold. Attenuation here works through a choice of granularity rather than softened word - ing: at the granularity of the single incident, six hundred add up to zero; at the granularity of the cumulative count, six hundred is a number that demands action. The regulator did one thing, and it changed the granularity. The conversion for a reader: who set the granularity of your safety ledger, and what colour does it show at that granularity, and what colour at another.

Case 5 (Healthcare): Twelve Days and One Unclaimed Chest X-ray

Case ID: me-005.

When and where: 15 July 2017, 07:40, to 31 January 2018. The emergency department, the assessment ward and cardiology ward 24 of an NHS trust in England, plus a coronary ward at another hospital in the same trust. The report refers throughout to "the Trust" and gives no hospital or place names (HSIB pp. 12–16). The clock times: the X-ray taken at 10:18 on 15 July, discharge on the evening of 21 July, the report issued on 27 July, the handover on 28 July, the GP retrieving it on 20 October.

What and who: the national investigation team of the Healthcare Safety Investigation Branch (HSIB; the report cover carries HSSIB), published 21 October 2022. The trust departments under investig - ation were three: the emergency department, which had a con - sultant on results management duty; radiology, with its consultant radiologists, radiology administrative team, the Return to Accident Room tray and the bouncer register; and cardiology, with ward 24, the coronary ward and the consultant who performed angioplasty. Outside the trust: the GP practice, the national serious incident

database StEIS, the Royal College of Radiologists (RCR) and the Care Quality Commission (CQC) for England. Responsibility attaches to posts: the junior doctor who wrote "lung fields clear", the consultant radiologist who issued the report, the emergency consultant on results management that day, the cardiology consultant who received the patient, the radiology administrative courier, and the receptionist who signed for the report.

What they were doing: At 07:40 on 15 July the patient arrived at the emergency department with chest pain and shortness of breath; the workup included a chest X-ray and an ECG; symptoms and results pointed to a myocardial infarction and she was moved to the assessment ward (p. 12). The film was taken at 10:18. At 13:50 the assessment ward junior doctor wrote "lung fields clear" in the notes. At 14:00 the consultant ward round recorded only the blood results and the ECG and did not mention the film (p. 12). At 14:30 cardiology saw her, and she was transferred to the cardiology ward that day, where she stayed five days waiting for transfer for angiography (p. 12). On 20 July she moved to another hospital in the same trust, had angiography and a stent the next morning, and went home that evening with a cardiology outpatient appointment six weeks out (p. 13).

The action on the day of discharge was the most ordinary of the lot: the discharge summary went to the GP practice and did not mention the emergency chest X-ray still waiting for a report (p. 13). That same week radiology was running a manual process the staff called the bouncer. The reporting doctor emailed the administrative team, or put the paper report in the RTAR tray in the radiology office; the administrators printed it and entered the patient's details in the bouncer book; they delivered twice a day, at 09:00 and 16:00, to the emergency department reception; reception put the report in the results tray and signed the bouncer book (p. 26). Interviewees described it as long-standing customary practice, and

no policy document set it out (p. 25). Radiology had just changed its information system, and stability and speed had been a problem for months before it was resolved, with the backlog taking months more to clear (p. 24). The clinical director said emergency chest X-rays were reported "broadly" within 48 hours (p. 24).

The decision point: Four moments.

One. At 13:50 on 15 July the assessment ward junior doctor wrote "lung fields clear" in the record system. Ten minutes later the consultant did the ward round, and the 14:00 note records blood results and ECG supporting the diagnosis of infarction, with no mention of imaging (pp. 12, 21). At the moment that line was written the formal report did not exist. On the desk was a film nobody had formally read.

Two. Discharge on the evening of 21 July. The discharge summary went to the GP practice and its body carried no item for "emergency chest X-ray awaiting report"; the summary arranged a cardiology clinic appointment six weeks out (p. 13). The discharge was handled by the coronary ward team, while the requesting department for the film was the emergency department. Six days had passed since the X-ray, and no report existed.

Three. On 27 July, the twelfth day after the examination, the consultant radiologist completed the report. The text: "There is a 45mm opacity projected over the liver which may represent a right lung base mass." (p. 13). It carried no recommendation for the next step and no time frame (p. 23). Under radiology policy the radiologist emailed the administrative team asking that the report go back to the emergency department; that day the report was printed, delivered to the department and signed for at reception (p. 13). The comparison sits on paper. RCEM best practice guidance from 2016 requires that emergency imaging be reviewed by a radiologist or a reporting radiographer within 48 hours of the request (p. 23). The trust's own policy set no time limit on reports at all, requir-

ing only that clinically urgent cases be telephoned to the requester promptly and that suspected cancer be notified within one day (p. 24). This report travelled on day twelve. The notification travelled by letter.

Four. On 28 July the emergency consultant on results management reviewed the report, looked the patient up in the system, found her under cardiology, wrote to the cardiology consultant and emailed at the same time, copying the GP (p. 13). The letter was generated by the system, and the report says it "would normally be printed and posted"; nobody could establish whether this one had been printed (p. 14). The cardiology consultant opened the email but could not recall reading it (p. 14). He had just come back from three weeks of leave, estimated 50 to 100 emails a day and up to 700 a week, and was on the emergency cardiac surgery rota in that first week back (p. 26). Both the letter and the email were accurate: 45 mm, right lung base, possible mass.

What was visible then: The image from 10:18 on 15 July; the "lung fields clear" in the notes at 13:50 that day; the 14:00 ward round note with no imaging item; the discharge summary of 21 July with no pending-imaging line; the report text of 27 July and the figure of 45 mm; the emergency reception signature in the bouncer book that same day; the transfer letter in the system on 28 July, the email that had gone out, and the email's opened status. Five further texts. The trust's Radiology Reporting Arrangements of November 2015 contains the sentence "We are...unable to con‑firm that results have been received by the referrer, or have been acted upon." (p. 25). The same policy sets no reporting time limit (p. 24). The department did not use the critical, urgent and unex‑pected significant three-tier classification (p. 22). The report con‑tains no "what next" (p. 23). The bouncer process exists only in custom and not in policy, and it has four manual steps, any one of which can fail (pp. 25–26). Two external numbers can be checked.

Between 1 April 2017 and 14 May 2018 StEIS held 41 serious incidents of this class, 13 of them from emergency departments (p. 11). CQC records emergency reporting time limits across trusts ranging from 1 hour to 2 working days (p. 30).

Known only afterwards: whether the letter was printed, posted, or where in the chain it was lost (p. 14); why the cardiology consultant opened the email without reading it (p. 14); that the twelve-day delay came from the backlog after the system change (p. 24); the composition of the 41, of which 27 were cases where serious harm might have been avoided had the patient been informed (p. 11); the trust's internal investigation conclusion, that with her cardiac and respiratory condition as it was, radical treatment may not have been appropriate, and earlier palliative care "might have brought slightly better symptom relief or a small improvement in survival" (p. 17); and that the patient's husband learned of the film from a hospital letter five months after his wife's death (p. 16).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: The cancer diagnosis was delayed and the patient died. HSIB made 4 safety recommendations, all aimed at the reporting and follow-up closure of unexpected significant findings. Back to the three questions. The commonest answer to the first is "chase the report", but chasing only compresses twelve days to five, and with no ownership mechanism the finding still sits. To the second question, the item most often missed is the national figure of 41 cases: it says the accident belongs to no single hospital. It

marks an ownership vacuum in the standard path by which reports come back. The third question lands on closure design: every report with an abnormal finding needs a single named owner and an escalation on timeout, and an empty owner field is the system's red state. Primary source: HSIB, Failures in communication or follow-up of unexpected significant radiological findings, 2019, link in Appendix B, me-005 (archived link, grade A–).

Options ruled out: The report records, item by item, the routes not taken at the time. Hot reporting, with a report issued as the emergency film is taken: five radiology consultants agreed it was the ideal, but radiologists and reporting radiographers were short, CQC recorded a vacancy rate averaging 14%, and most imaging departments did not consider it realistic (pp. 29, 30 note 24). Telephone handover for every unexpected significant finding: the telephone "definitely gets heard" but leaves no auditable record, and no document required it at the time (p. 32). Radiology arranging the next step directly: it would duplicate CT the emergency department may already have ordered, and coordination costs more while requests are still on paper (p. 35). Writing "what next" into the report: three radiology consultants said reports suggesting lung cancer usually carry a recommendation, and they also considered this report clearly written, the point being whether risk control was triggered, and it was (p. 23). A dedicated results coordinator watching unclaimed reports: the report concludes the post is needed, states that it has a resource impact, and notes that most trusts' IT could not support it (p. 56). The NHS telling patients directly: this became recommendation 2019/041, the NHS App could not do it at the time, the time frame under discussion was two to three weeks (p. 60), and the recommendation text is at p. 61. The trust itself took a different route and took it halfway: an electronic acknowledgement system, expected live in August 2019 (p. 28).

Early signals: Five, each with the reason it was carried past.

1. The 2015 RCR national audit: 154 radiology departments replied, 88% had a written policy, 17% had an electronic acknowledgement system, and of those with a system only 42% had anyone reviewing the read data regularly; the audit concluded that most departments did not meet the published national guidance (pp. 52–53). The signal sat in a public audit, and the reason it was pushed past was "we have a policy, and we have the bouncer book".
2. The trust's own Radiology Reporting Arrangements, November 2015 edition, says in its own words that it cannot confirm results were received by the requester or acted upon (pp. 22, 25). A self-declared gap went into an in-force document, and the reason it was pushed past was that it stood under notes rather than defects.
3. 41 serious incidents, 13 of them from emergency departments (p. 11). Anyone could pull that number with database keywords, 58 retrieved and 17 excluded (p. 12 note 8), and the reason it was pushed past was that each case closed on its own terms, with different causes.
4. Reporting times stretched markedly after the system change, with a backlog running for months (p. 24). The clinical director called it a system problem being resolved, and the department line stayed at "broadly 48 hours".
5. After the 2007 NPSA safety practice notice the RCR issued four more standards documents, and the same harm kept happening in England; NHS Resolution recorded 662 radiology claims within two years, 24 of them coded as failure to act on abnormal results, with more than £2.5 million already paid (p. 8, p. 11). These numbers were public, and the reason they were pushed past was that they were a national trend and not a local to-do.

The point of escalation: The moment the letter went out on 28 July. Until then the report sat with the only department that had a results-handling process: the emergency department had the bouncer book, two deliveries a day at 09:00 and 16:00, a duty results consultant, and an audit trail at the point where reception signed. Once the letter went out, the finding moved to a department with no results-handling process, and the step complied with the guidance then in force. RCEM guidance requires that when a patient is admitted under a non-emergency team, responsibility for review and follow-up be handed explicitly to that team (p. 25), and the report quotes the BMA's judgement on clinical handover, one of the most dangerous procedures in medicine (p. 33). The handover was carried out on 28 July by the emergency consultant on results management, under that guidance and trust policy; the report records no approval step, because the step needed no approval and no signature. The chain runs longer than that: emergency to cardiology, cardiology to CT booking, then respiratory (p. 32). For the next 85 days no status in the system changed because this report had not been handled. Two of the trust's four internal remediation items land on this handover: writing the process for how unexpected significant findings reach the emergency department, and creating a Cancer Imaging and Pathway Co-ordinator (p. 28).

Failure chain: Technical layer: the mass sat behind the diaphragm, where a non-specialist easily misses it (p. 21); the report was completed on the twelfth day after the examination (p. 13); it states findings only, with no next step and no time frame (p. 23); the system's read status means opened, while the RCR standard requires seen and acknowledged as two separate steps (p. 50); the report travelled as a paper print carried by hand, and any link could drop it (p. 26). Organizational layer: trust policy never says how an unexpected significant finding goes back to the emergency department, and the process lived only in custom (p. 25); receipt had an

audit trail and handling did not, so the bouncer book proves the report reached reception and not that anyone took it on (p. 26); the results management consultant's duty stopped at delivery to the receiving team (p. 25); the receiving team had no acknowledgement field and nobody needed to acknowledge (p. 27); the discharge summary carried no pending imaging (p. 13); nationally, 17% of departments had an electronic acknowledgement system (p. 52). Incentive layer: no national standard set a reporting time limit and the trust's policy set none either (pp. 23–24); CQC inspection did not check whether acknowledgement systems were running (p. 4); radiology capacity was held back by radiographer shortage, with a CQC-recorded vacancy rate of 14% (p. 30 note 24), and output was counted in reports completed; the cardiology consultant's inbox ran 50 to 100 items a day (p. 26); the cost of the harm arrived later as claims, 24 in two years and more than £2.5 million (p. 11), and entered no post's current-period metrics.

The junction broke between the organizational and the incentive layer: the receipt chain closed completely on 27 July, and the handling chain broke at its first hop on 28 July. The arrangement that made "delivered" a qualified output also kept "unhandled" out of anyone's metrics, and neither needed a signature.

Minimum counterfactual: On 21 July, writing the discharge summary, add one line to it: "Emergency chest X-ray (15/07): report not returned, responsibility passing to cardiology outpatient review." One action, on a document that had to be written that day, at a cost of one line. It needs no new system (the trust's electronic acknowledgement system arrives only in August 2019, p. 28), no budget, no reliance on anyone remembering, and no change in anyone's judgement. The timing is 21 July because the discharge summary is the one document from this admission that crosses both institutions and teams: it goes to the GP practice and to the cardiology clinic, and the clinic is six weeks away. The report re-

cords this gap itself (p. 13), and records the clinical consensus that had the report been issued while the patient was an inpatient, the follow-up failure might have been avoided (p. 24). It does not guarantee a cure, since the trust's internal investigation considered she may not have been suitable for radical treatment (p. 17). It guarantees the film enters a list that someone has to open.

Outcome and sources: On 31 January 2018 the patient died at home (p. 16). The delay was 12 days, the report went unclaimed for 85 days, and the diagnosis was pushed back by about three months; CT on 10 November confirmed bilateral lung cancer, she was told the same day, and treatment was set as palliative (p. 16). Four safety recommendations: 2019/039, that the RCR set out the principles for judging unexpected significant, critical and urgent findings, an alert coding framework and a mandatory trigger list; 2019/040, that the NHSE/I patient safety team ensure organisations implement monitored acknowledgement systems; 2019/041, that NHSX develop a digital way of telling patients results; and 2018/042, that CQC write risk control into the core service framework. There is one further safety observation, that radiology opening CT directly for chest X-rays suspicious of lung cancer is worth evaluating (pp. 5–6, 66–67). The national figure is 41 serious incidents, 13 of them from emergency departments (p. 11). Primary source: HSIB, Investigation report: Failures in communication or follow-up of unexpected significant radiological findings, published 21 October 2022 (recommendation numbers follow the case numbering of 2019/039 to 2019/041 and 2018/042), link in Appendix B, me-005 (archived link, grade A−). The card's page numbers are report PDF pages, 95 in total, and the item-by-item sources are in _evidence/me-005-加厚.md.

Case five shares a detail with Chapter 30 that is worth setting down. Across the twelve days, nobody broke a rule. Radiology completed

the report in queue order, slowly but compliantly; the system delivered it to the requesting department, and that link worked as designed; after the review and the handover nobody owned the follow-up, because the system had no acknowledgement field and nothing flagged the abnormal. Every link ran as designed, and the design left the ownership of the finding with no landing place. Remediation in a case like this points at fields and not at people: owner, timeout, escalation. Add those three fields and the next finding in the same sequence lands. HSIB's four recommendations sit largely on those three fields, and a reader can borrow them directly.

Case 6 (Public Policy & Administration): The Public-Administration Edition of Side-by-Side Reading (condensed)

po-008. The National Audit Office of China, audit of central budget execution for 2025 (reported to the Standing Committee of the National People's Congress in June 2026, PDF text verified for this book): the audit found 40.427 billion yuan of corporate income tax, VAT and other taxes levied excessively or collected in advance from 1,954 enterprises; 59.651 billion yuan of inflated fiscal revenue; and 10.935 billion yuan of tax evaded. The case is this chapter's mechanism read backwards. Levying excessive taxes and fees is a workaround at the execution level, and it is invisible in routine upward reporting, where every level's revenue task shows as met. Only the audit channel, which does not pass through the hierarchy (Device 2, national edition), added it up. The decision point is the moment a basic collection unit, facing a revenue task, chooses an enforcement standard. Minimum counterfactual: give tax execution data a sampled direct channel that does not go through hierarchical aggregation. Primary source: National Audit Office, 2026, link in Appendix B, po-008.

Case six has a second reading. Audit is the attenuation detector that the modern state installs in advance. Its technical features are

this chapter's entirely: no hierarchy, in an independent office reporting to the legislature; originals, in extended audit pulling the working papers; side-by-side reading, in final accounts set against the books. Publication is itself a governance act, since it turns the measured attenuation into the starting point for accountability and remediation. A reader in a company can borrow one design principle from the public edition: the detector must have no personnel overlap with the reporting system it tests. The day audit independence is written into the governance documents, one more line appears on attenuation's price list.

There is a layer beyond the numbers. The audit report lists the three figures separately, excess and advance collection, inflated revenue, evaded tax, and the separation is itself an anti-attenuation way of writing. Merging them into one sentence about collection and management problems involving over a hundred billion yuan would be grammatical and would be informational suicide: three classes of problem map to three remediation paths, and merging blurs all three at once. That is a public lesson for anyone who writes reports. Remediability is the only legitimate reason to keep figures apart, and before any impulse to merge, ask whether the object of remediation can still be found afterwards.

Case 7 (Aviation): The Engineer's Unease (condensed)

av-017. Eurocopter EC135P2+, registration JA824H, August 2024, Japan. JTSB serious incident investigation report (2025, verified for this book): the helicopter was cleared to land at a helipad and landed on the runway instead. The detail recorded in the material card is a textbook sample for this chapter. The maintenance engineer riding on board felt something wrong about the landing position and did not say so. The unease was a piece of bad news in perfect condition, and it stopped at the step where saying it would disturb the flying. The upward half of the MUM effect, the sender's unwillingness to report, completed itself inside the

cabin. The decision point is the moment of re-confirming the cleared landing point before final approach. Minimum counter-factual: put a fixed mutual confirmation into the approach brief, that the expected landing point matches the clearance, and that unease from any seat is spoken. Primary source: JTSB, AI2025-8-1, 2025, link in Appendix B, av-017.

The engineer's silence has a context that deserves a word in his favour. The role order in a cabin, who flies and who fixes, is itself a piece of safety design, and what it prevents is interference from outside the role. That is the hard part of designing a channel for bad news: protect the role order and still give every seat's unease an exit. Aviation's mature answer is worth copying. Use an explicit callout phrase, so that anyone in any seat who sees a mismatch says it in a fixed form. Make the consequence of speaking institutional: always recorded, always thanked, never questioned for motive. The phrase lowers the cost of the exit and the institution removes the price of using it. Another industry copying the combination can drop the phraseology and cannot drop the two principles. The exit has to be cheap, and the exit has to be blameless.

The counter-examples sit in this chapter. In case four the 600 reports all had a registration channel, so an exit existed, and the consequence of registering was an archive nobody cared about, so the exit had no value. In case seven the unease had no fixed phrase, so the exit was expensive, and the consequence of speaking was unknown, so the exit carried liability. That gives the three elements of exit design: it exists, it is cheap, it is blameless. Add a fourth, an echo, which is the feedback ledger in Device 2. A channel missing any one of the four will idle at the first real test.

The line at the head of this chapter, that every floor up changes the coat, can carry one more note: the coats are changed according to a grammar. The first layer changes the quantifier, so a major hazard becomes some risk. The second changes the verb, so will hap-

pen becomes may exist. The third changes the subject, so our unit becomes some units in the industry. The fourth changes the time, so now becomes in the long run. Four steps and the sentence is grammatical, morally spotless and empty. The reader’ s quick check is to watch those four substitutions. When any of them ap - pears on the way up, put the original back and read the sentence again. The restored sentence makes the receiver uncomfortable, and that discomfort is the first arrival of the real intensity.

Beyond grammar, the layout of a report takes part in the attenu - ation. Where the bad news sits, which page, which paragraph, whether bold; how the heading is worded, risk note or serious in - cident; how it is ordered against the good news, in the sandwich of good, bad, good. All of it is the layout layer. Layout is the hardest to defend because even the wording can be called accurate: it merely placed the bad news where the reading would not be interrupted. The side-by-side table in Device 1 catches layout as well, because the same item is a heading in the original and a footnote in the quarterly report, and the difference in position is the difference in intensity.

The eight cases run across five domains, chemical, maritime, public policy, manufacturing and healthcare, and produce three host shapes for attenuation. The chain type, cases one, five and eight, where information passes through many links, ownership evaporates and nobody assembles the whole. The paper type, cases two and three, where a document proves delivery while under - standing and execution never happened, so the upward word complete is wrong. The granularity type, cases four and six, where everything looks normal at a grain convenient to oneself and problems pile up at another grain. Each host takes a different de - tection action: for the chain type, find the assembly table and ask where the whole picture sits; for the paper type, test understanding by spot-reading and asking for a playback; for the granularity type,

change the grain and look again on a cumulative or an alternative basis. The six signals in Section 16.4 are the common exit for all three, and the four devices in Section 16.5 are matched to the host type.

Case 8 (Construction & Transport): The Hand the Door Test Missed (condensed)

ce-006. Ealing Broadway station, London. RAIB report 01/2026 (page verified for this book): a passenger's hand was trapped in a train door and the passenger was dragged along, suffering minor injuries. The investigation found that the door obstacle detection did not identify the trapped hand and the anti-drag measures did not work. The state of the equipment in this case has the form of locally knowable, globally unreachable: the people on the platform saw it, the train system did not know, and the driver departed on the signal. The piece of information that existed stopped at the platform's position and never entered the departure decision chain. The decision point is the visual confirmation before depar‑ture. Minimum counterfactual: a mandatory voice confirmation between platform and driver on any obstruction anomaly, with no departure until it is confirmed. Primary source: RAIB, Report 01/2026, 2026, link in Appendix B, ce-006.

Cases seven and eight side by side are the same mechanism in two industries. In both, the bad news was produced intact at the front line: an engineer's unease in one case, a hand seen in a door in the other. In both it stopped at the last gap before the decision chain. The position of the gap is the same too: between the person closest to departure or landing and the person who makes the de‑cision. So the defence lands in the same place. A forced confirma‑tion at the final gap costs an order of magnitude less than detection installed along the whole chain.

16.4 What Shows Up Earliest

Six weak signals, each with the excuse that usually carries it past.

1. The share of adjectives in reporting material rises and the share of numbers falls. Excuse: "leadership wants the key points."
2. "Overall under control" and "the trend is improving" appear more often. Excuse: "to steady expectations."
3. The same event carries different intensity in the minutes of different levels. Excuse: "each level has its own focus."
4. The front-line originals (work orders, recordings, ledgers) do not match the version that went up. Excuse: "the report was distilled."
5. A bypass channel exists and has never been used, with zero users. Excuse: "that shows there is no problem."
6. The top's judgement of conditions at the front line makes the front line laugh. Excuse: "leadership sees the bigger picture."

Signal three is the operational starting point. It needs no new system, only someone willing to photocopy two sets of minutes onto one sheet of paper.

Put the six signals back against the cases. Signal one, adjectives up and numbers down, matches case one, where the magnitude of the leak was cut into facets across the parties describing it. Signal two, the rising frequency of "overall under control", matches case three, where the conceptual schedule is the documentary form of control. Signal three, intensity differing by level, matches case six, where the execution-level workaround is invisible as a whole in routine reporting. Signal four, originals not matching the upward version, matches case two, where the training record and the crew's understanding are two versions. Signal five, a bypass channel at zero use, is the negative footnote to case seven, where the engineer's unease never even reached the channel's door-

way. Signal six takes this form in case five: the system's belief that follow-up was running normally, set against the national fact of 41 cases.

The six sort by cost of collection. Signals three and four are cheapest, a photocopy of minutes and a comparison against originals, half a day's work. Signals one and two come next, needing a quarter's material for a word-frequency observation. Signals five and six are dearest, needing interviews and site time. The starting advice is the usual one: begin with the cheapest. Do signal three this quarter, and it alone decides whether the mechanism is running.

The intensity table deserves a template. Four columns: time, level, verbatim excerpt with no rewriting, and a self-rating of intensity from one to five, where the rating answers "how much more would I worry if I read only this". Each level's version of the same topic takes one row, in time order. There is one discipline in making it: the excerpt must be verbatim, because rewriting re-does the attenuation you are auditing. There is one rule in reading it: do not look for who was wrong, since that leads into Chapter 19, and mark only the change points, the intensity dropping from what to what, at which level. The level where change points cluster is the first site for attenuation governance, and whether that level's receiving end lacks a seat or its sending end lacks protection decides which device to add.

16.5 How to Defend

Device 1: reading the minutes side by side. Once a quarter, take one topic's reporting minutes from different levels, lay them out in time order, and mark the points where intensity changes. This is the only method that makes "every step small, the total large" visible. Execution: the comparison is done by the person who re-

ports directly to the top, since he stands outside the attenuation chain, and the result is the first item on that period's meeting agenda, with no pre-screening.

Device 2: a direct channel for originals. Front-line work orders, incident flash reports and raw customer complaints get a channel straight to a senior reader, unpolished, with a named reader each month who signs the reading log. What the channel gives is exist - ence rather than throughput: writers who know the originals will be read moderate their wording. Capacity design: a senior read - er's real monthly capacity is small, so sample rather than pile up, 20 items drawn at random each month, because piling everything up is the same as having none.

Device 3: an immunity clause for reporting. The system states in writing that a person who reports bad news factually is not held to account for the news itself, and that accountability applies only to concealment and fabrication. Without this clause, what Devices 1 and 2 collect dries up at the source: side-by-side reading will show that every level rewrote, and the first level that rewrote is the one that did not dare write it straight.

Device 4: quantifier discipline. Internal reporting bans unquanti - fied modifiers, "certain", "rather many", "overall", and every modifier must carry a number or the words "not quantified". The lesson of Qingdao is not that some level said the wrong thing. It is that the response decision was made on an assembly that was never complete. Quantifier discipline cannot abolish that, and it can expose where the problem sits: the paragraph where unquan - tified modifiers begin is where the attenuation begins.

The four devices on this chapter's cases. Device 1 on case six: set collection execution data beside the level-by-level aggregates, and the difference is the forty odd billion the audit office measured. Device 2 on case two: management reads one front-line work re - cord a month. Device 3 on case seven: an unease spoken in the

cabin goes into the flight record whether it turns out right or wrong, and nobody asks about motive. Device 4 on case one: every status update after the leak carries a magnitude, leak volume, affected area, progress of the work, and where no number exists the words "not quantified" go in its place.

The failure modes of the four. Side-by-side reading dies of sample bias: only the minutes of material events get read, and the worst attenuation sits in items that looked small at the time; the patch is to draw the topic at random each quarter. The originals channel dies of reading as performance: the signature is there and the reading is not; the patch is for the named reader to tell the front line periodically what he read, since that feedback is the only evidence that reading happened. The immunity clause dies of exceptions: accountability shifts from the message to an improper manner or a wrong moment, and immunity is bypassed; the patch is a written explanation for every bypass. Quantifier discipline dies of number inflation: the modifiers go and the numbers get inflated; the patch is for Device 1's side-by-side reading to check where the numbers came from at the same time.

The weighting of the four is worth stating. If only one can be installed, install Device 3, immunity, because goods in the channel precede everything else. If two, add Device 1, because once there are goods you need to see the deformation. Three and four follow in that order. The order runs against intuition, since most organizations install side-by-side reading first, and the result is a distortion with nowhere to be reported, which leaves the mechanism where it started. Channel first, detection after.

One common mistake in staffing a direct channel is worth walking around in advance: making it a full-time inspection post. A dedicated inspector turns finding bad news into a departmental KPI, the department reports noise as signal to meet its volume, and the channel loses credibility fast. A workable design is part-time plus

rotation. The duty is carried by front-line staff on rotation, quarterly, whose day job stays on site and who report what they themselves see; rotation keeps the channel from being bound to anyone's career over the long run. The far end, the senior reader, keeps a record too: who read, what was read, and who was given feedback, one page a month. With a ledger at both ends, the channel survives an assessment cycle.

16.6 Boundaries with Neighbouring Mechanisms

Against handover rupture (Chapter 30): attenuation is vertical loss between levels, and rupture is horizontal loss between the parties on either side of a handover. Laureline's new procedure, case two here, and the two-ship assumption in ma-008 look like the same class, and the criterion that separates them is direction. What was not passed downward is attenuation; what was not passed sideways is rupture.

Against why the truth-teller goes first (Chapter 19): this chapter is the channel question, how the message deforms; Chapter 19 is the price question, what happens to the person who speaks. Which to check first. If the channel carries plenty of traffic and it is distorted, fix the channel. If the channel carries nothing, check the price first.

Against goal displacement (Chapter 12): metric distortion is attenuation that has been structured, since compressing risk into a rating is one institutionalised wording choice. GAO's cross-check of dashboards, case two in Chapter 12, is the public edition of Device 1 here.

16.7 Five Self-Checks

1. Take the three sets of minutes from your organization's most recent material event and read them side by side. At which level does the intensity start to change?
2. How many unquantified modifiers are in the last report you signed?
3. In your organization, who has read the original complaint or the incident flash report? When did he last verify anything with the front line?
4. If tomorrow you wrote a piece of bad news into the weekly report exactly as it stands, what would happen? What does that answer tell you?
5. How old is your judgement of the front line? When did you last spend half a day on site?

Chapter 17 Misaligned Incentives | Principal-Agent

> What this chapter covers: how each layer's rational choice adds up into a loss for the whole; the failure structure of the principal-agent chain, four scenes on the ground, and three ways to write externalities back into assessment. When to turn back to it: when you are designing an assessment scheme, signing an outsourcing or agency contract, or have just found that everyone did the right thing and the result was still bad. Reading time: about 50 minutes.

Misaligned incentives is the most counterintuitive mechanism in this book, because it asks you to accept an uncomfortable fact: the organization that failed may contain no bad person, and possibly no one who made an error at all. The auditor maintains the client relationship by commercial logic. The tax authority completes its task by target logic. The regional carrier presses cost down by market structure. The seller races to list by platform rule. Each layer's choice is correct on its own ledger, and the loss their sum produces is paid by someone outside the chain. This chapter supplies no villains, only structure, which is why every defence in it is structural as well: change the assessment, change the contract, change who owns the verification. Changing the people does nothing.

17.1 What It Is

Definition: misaligned incentives is a structural disease of the principal-agent chain. Every party optimizes the quantity it is assessed on, and the sum of the parties' choices damages the whole.

The identification signature is clean enough to replace criteria with three questions:

1. Take each layer's choice on its own. Is it rational inside that layer's own objective?
2. Has anyone broken a written rule?
3. Add the choices up. On whose head does the loss land, and does that person hold a seat at the decision table?

The third question carries the diagnosis. When the loss lands outside the chain, on passengers, investors, taxpayers or users, no link inside the chain has a reason to change its behavior on that account. The people in a position to find the problem are the people who benefit from it, as Chapter 37's question set puts it, and where the beneficiary runs the verification the finding persists for years in the form "verified, does not exist".

The academic frame takes one paragraph. An agency problem appears when one party's interest depends on another party's action and the quality of that action cannot be verified cheaply. Jensen and Meckling's 1976 paper splits the cost into monitoring costs, bonding costs and residual loss, and the sum of the three is the price of the misalignment. This book skips the formal model and keeps one corollary: verification cost sets the depth of the misalignment. Auditing is the textbook industry because the quality of an audit is extremely expensive to verify (readers of the statements cannot observe whether audit procedures were compressed). Listing on a platform is the consumer edition, because the authenticity of a compliance document is just as expensive for a buyer to check. Wherever verification is expensive, misalignment runs deep. The economic content of every defence in this chapter is one sentence: make verification cheap, or move it to someone with a reason to look.

One distinction has to be made early: misalignment against corruption. Corruption is the agent selling the layer's objective, taking money to let something pass. Misalignment is the agent faithfully executing the objective he was assessed on: collecting tax to hit a target, pressing cost under a per-departure fee, loading goods to hold a listing window. The line matters because the remedies belong to different trades. Corruption removes people. Misalignment changes structure. An organization that confuses the two chases a misalignment as corruption, replaces the people, issues warnings, makes an example, then finds the structure intact and the next outbreak under a new name. The two firms that failed one after the other in case 1 are the ready-made footnote on how little that achieves.

17.2 How It Grows

Three generating conditions.

Condition one: the externality cannot be priced at its own layer. Your decision damages the ledger of another link (audit distortion damages investors, over-collection damages enterprises, fatigued flying damages passengers), and your own ledger cannot see it. The invisibility is not a management lapse. It is what the accounting categories were designed to do.

Condition two: the assessment measures only what the layer controls, and cross-layer consequences have no owner. No post signs for the outcome of the chain as a whole, so every layer's optimization contributes its own error to the whole. Chapter 18 picks that up from here.

Condition three: tenure is shorter than the consequence cycle (Chapter 28). When the cost is paid by a successor or by a third party outside the system, the current-period rational choice keeps tilting, and the successor inherits the tilt.

Misaligned incentives is the main disease of the incentive layer in Chapter 9's three-layer model. Almost every technical implementation of it runs through a metric, which makes Chapter 12 its standing complication. Almost every defence of it needs someone to call a stop, which makes Chapter 18 its standing precondition.

One length variable sits outside the three conditions: the longer the chain, the further the sum drifts from the overall objective. Two reasons. Each subcontracting layer attenuates the principal's visibility of end quality by one notch, which is Chapter 16's mechanism replayed at contract level, and the agent's room for interpretation expands to fill it. Each layer's assessment cycle is also shorter than the level above it: headquarters reads the year, the supplier reads the order, the order's upstream reads the delivery week. The shorter the cycle, the more completely the externality leaves the field of view. The key position in defending a long chain is therefore the contract layer. Verifiable end-quality clauses, and not process-compliance clauses, written into every level of the contract are the only thing that lets every link see the end of the chain.

Price is the resource the incentive layer most often leaves lying. The cheapest form of writing an externality back into assessment is to let payment terms that already exist carry it: an audit fee paid in installments with the final payment tied to the regulator's inspection result, platform settlement to a seller delayed in step with compliance spot checks. None of these creates a post or a report. Each changes the trigger condition on money that was going to move anyway. Two of the three write-ups in section 5 belong to this class. What can be written with money does not need a person watching.

The material deserves separating, because different materials take different tools. Money (commission, audit fee, pay per unit) is fixed by changing the payment trigger: fastest to work, fastest to regress,

since restoring the clause brings the behavior back with it. Assessment (KPI, ranking, performance review) is fixed by changing the definitions and the weights, which is the whole toolkit of Chapter 12: half a beat slower, and it holds. Identity (“we are a safe company” , “professionalism”) is the cheapest and the most fragile. In quiet years it substitutes for a great deal of money; on the night something happens it is gone. Chapter 13’s narrative assets meet the incentive layer at this point. The three usually appear together. An auditor draws an audit fee, carries a partner assessment and believes in professionalism, and the misalignment runs along the resultant of the three. Diagnose them separately, then repair in the order money, assessment, identity: change the payment trigger, then the metric definition, then rebuild the identity. Run that backwards, starting with the slogan, and the identity you shout gives the misalignment cover.

17.3 The Shape in Four Domains

Case 1 (Finance): Paying the Auditor to Be Strict

fi-008. Zhongshen Zhonghuan Accounting Firm, audits of Yuebo Power’s 2019-2021 annual reports. The China Securities Regulatory Commission’s administrative penalty decision (〔2026〕 No. 56, page checked for this book): the audit was not performed with due diligence and the reports issued contained false records; the CSRC confiscated the audit fee income of 3,679,245.18 yuan and imposed a fine of the same amount, and the three signing certified public accountants were fined 400,000, 300,000 and 200,000 yuan. The market structure of auditing is this case’s entire structure. The auditor’s fee is paid by the party being audited, while the real consumers of the audit product, the readers of the statements, neither set the price nor inspect the goods. When keeping the cli-

ent conflicts with standing up to the client, the direction of the tilt was settled on the day the engagement letter was signed.

Stop at the decision point: the moment a check in the audit programme is compressed, or an anomaly is waved through. On the desk: the cost of the procedure, the survival of the client relationship, the accumulated experience that "it went through before and nothing happened". 〔What happened〕 Confiscation plus a fine of the same amount. The same issuer later failed the same way under another firm (Zhongshen Yatai, fi-009, Appendix B): the structural disease relapsed on the same chain, with one firm swapped out between the first occurrence and the second, which rules out the explanation from individual quality. Primary source: CSRC 〔2026〕 No. 56, 2026, link at Appendix B fi-008.

One account in case 1 is worth working out for the reader. Confiscation of 3,679,245.18 yuan plus an equal fine, plus 900,000 yuan across the three signing accountants: that is the price tag of the structure. The comparison is the audit fee itself, roughly 3.68 million yuan of annual report audit income across the three years, around 1.2 million a year. When the expected cost of waving one anomaly through (probability of being caught times the penalty) sits below the certain cost of holding the line against a client (the fee income and the renewal), the tilt is a matter of arithmetic rather than morals. The function of a regulatory penalty is to change the arithmetic, lifting the expected cost above the certain one. The same view explains why penalty severity is a core parameter of incentive governance: a rule that fines too lightly is not a rule; it is a price list.

Case 2 (Public): Over-collection under a Revenue Target (full card)

po-008.

When and where: 23 June 2026, the 23rd session of the Standing Committee of the 14th National People's Congress (Beijing). The audited collection behavior took place in the 2025 budget year, on the tax and customs collection front lines and in the treasuries of 21 provinces and municipalities.

What and who: the National Audit Office, Auditor General Hou Kai reporting to the NPC Standing Committee on behalf of the State Council. The implementing bodies named are the tax and fee collection posts of the tax and customs departments, and local finance and treasuries in 21 provinces and municipalities. The report runs 25 pages in seven parts; this item sits in Part One, audit of central fiscal management, as the first item and the first sub-item, which makes it the opening of the entire problem list. The report discloses at aggregate level and names no individual enterprise.

What they were doing: in March 2025 the third session of the 14th NPC approved that year's budget, with "organizing fiscal revenue" ranked first among the five items of this central fiscal management audit (p.3). The approved frame was a central general public budget with revenue of 9.737253 trillion yuan, expenditure of 14.597253 trillion yuan and a deficit of 4.86 trillion yuan (p.3). Once execution began, the institutional calendar laid this out as a sequence of actions: corporate income tax prepaid by quarter and settled by the end of May the following year; VAT declared and paid monthly; import-stage taxes collected by customs on behalf of the tax authority; local finance booking expenditure and making transfers at the treasury end. The audit entered after the budget year closed and aggregated the year's collection records by enterprise. The aggregation produced this: tax and customs departments over-collected or collected in advance corporate income tax, VAT and other levies from 1,954 enterprises and others, 40.427 billion yuan (p.4). The same account holds two further sets of actions: 21 provinces and municipalities inflated fiscal revenue by 59.651

billion yuan by booking treasury funds as expenditure and then routing them back (p.4), and 4,976 enterprises evaded 10.935 billion yuan of tax by issuing invoices irregularly for deduction (p.4). The item also lists separately 20 enterprises that evaded 41.4702 million yuan of domestic VAT on export goods (p.4). Four groups of numbers close the item, and the audit gives them one common description: over-collection of "excess taxes and fees", inflation of fiscal revenue and other unhealthy tendencies (p.4).

The decision point: three irreversible moments, each with a documentary carrier.

The first is the budget approval of March 2025. The carrier is the budget resolution the NPC approved, which the report records as "strictly implementing the relevant resolutions of the third session of the 14th NPC" (p.1). That moment fixes the twelve-month baseline: revenue of 9.737253 trillion yuan, checked against the table quarter by quarter and month by month.

The second is a single-enterprise operation inside some filing period of 2025. The carrier is the tax return and the treasury receipt: the prepayment ratio, the supplementary collection inside the filing period, the cross-period adjustment, each action with legal parameters in the collection system. The report discloses no single-enterprise documents and no signing post, only aggregate figures (p.4). Both places read "from 1,954 enterprises and others", and the "and others" leaves collection targets that never entered the enterprise count (p.4; 2024 report p.4). The opacity at this level is a matter of scope: the audit report's problem list stops at enterprises times amount and breaks down nothing below that. The technique in this position differs from one year to the next. The 2024 report had cultivated-land occupation tax, VAT and land transfer revenue (2024 report p.4); the 2025 report has corporate income tax and VAT (p.4). The technique changes. The position does not.

The third is the rectification closure of 22 December 2025. The carrier is item (1)1 of the State Council report on rectification of the problems found in the 2024 audit of central budget execution and other fiscal revenue and expenditure: tax departments in 28 provinces and municipalities completed rectification of 33.189 billion yuan of over-collected or prematurely collected taxes and late fees by processing refunds and offsets and coaching enterprises to file corrected returns. In the same month the State Taxation Administration's working mechanism against excess taxes and fees, revenue quality monitoring and accountability notices among it, was written into the rectification results. The person reporting the rectification is the Auditor General on behalf of the State Council; the parties executing it are 28 provincial tax bureaus.

The order of the three is this: the first fixes the number, the second produces it, the third returns it and then stands the mechanism up.

What was visible then: the approved budget figures of March 2025 (p.3); monthly and quarterly treasury progress in the collection system; individual enterprises' returns, prepayment records and tax source ledgers; the full text of Article 55 of the Budget Law, which forbids collection departments to over-collect or collect in advance budget revenue that is due, and whose other paragraph provides that governments at all levels may not issue revenue targets to budget revenue collection departments and units, in force since 1 January 2015; the previous year's audit report published on 24 June 2025, which put over-collection of excess taxes and fees and similar at 88.997 billion yuan, of which tax departments in 28 provinces and municipalities over-collected or collected in advance 33.189 billion yuan in taxes and late fees from 631 enterprises (2024 report p.4); the recommendation on p.24 of that report, in its own words, to "vigorously rectify the collection of excess taxes and fees, arbitrary charges and arbitrary fines" (2024

report p.24); the rectification ledger of December 2025, with the refund amounts and the number of enterprises coached into corrected returns reported by 28 provincial tax bureaus (rectification report, item (1)1); the correction and refund menus in the collection system, the route the December 2025 rectification actually took; the deficit of 4.86 trillion yuan and the conclusion that the budget was on target (p.3); and the draft for comment and problem list the audited units received before the report was published, recorded on the report's last page, where the National Audit Office "solicited the views of the audited units in accordance with law, issued the audit report and put forward rectification requirements" (p.24). What became known only afterwards: the list of the 1,954 enterprises and the audit's sampling basis; the full chain of the 59.651 billion yuan on the treasury books; the total of 111.013 billion yuan across the three items, which the book arrives at by adding the three figures on p.4. This item also carries no example anywhere in the report's format, being composed entirely of enterprise counts and amounts, where the "for instance, in one province" that appears elsewhere in the report is absent (p.4).

You are in the seat now, with the material above on the desk.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: for 2025, 1,954 enterprises were over-collected or collected in advance to the tune of 40.427 billion yuan; 21 provinces and municipalities inflated fiscal revenue by 59.651 billion yuan; 4,976 enterprises evaded 10.935 billion yuan of tax; 20

enterprises evaded 41.4702 million yuan of domestic VAT on export goods (p.4). On 11 June 2026 a State Council executive meeting took up rectification of the problems found in the 2025 audit and set specific requirements (p.24), and the National Audit Office undertook to report full rectification before the end of the year (p.25). The same item in the previous year had been closed with 33.189 billion yuan returned in full through refunds and offsets (rectification report, 22 December 2025). One year later the new year's figure was 40.427 billion yuan. Back to the three questions. The commonest answer to the first is "refund the over-collection", an action already performed once, a full refund, at the cost of a year and a sum of late fees, and the second year's number did not get smaller. The item most often missed in the second question is the phrase "treasury funds booked as expenditure and then routed back", which readers take for an accounting error, while it takes three parties' vouchers aligned, finance, treasury and the spending unit, before it can be done at all. The landing point for the third question is the December rectification closure, and the check is whether those enterprises' prepayment parameters returned to tax source levels after the refund.

Options ruled out: four, each with a procedure ready at the time, none with a record of use in the report or the public material.

The first is the budget adjustment procedure, lowering the revenue expectation. The Budget Law requires a budget adjustment to be approved by the standing committee of the people's congress at the same level; the procedure exists, and the cost is waiting for a session.

The second is lawful approval of deferred payment. The Tax Collection and Administration Law gives provincial tax bureaus authority to approve deferral of tax payment, for up to three months, and the reduced revenue lands on that year's treasury receipts.

The third is refund and offset inside the filing period. This route was used later, by the rectification: the 33.189 billion yuan was returned through processing refunds and offsets and coaching enterprises into corrected returns (rectification report, 22 December 2025). The same channel was in the system during the collection year.

The fourth is a written report to the superior level that the jurisdiction's tax source was insufficient, with a request to adjust the task. The report holds no record of this kind.

All four share one cost: each reduces the current-period treasury receipts. No record exists of the reasons for ruling them out. The only cost readable from the institutions is that one, and that one is visible on the monthly progress table.

Early signals: ① On 24 June 2025 the previous year's audit report named over-collection of excess taxes and fees and similar at 88.997 billion yuan, of which 33.189 billion yuan fell on tax departments in 28 provinces and municipalities and 631 enterprises, with a further 55.808 billion yuan of land transfer revenue over-collected in 14 provinces and municipalities (2024 report p.4). The finding was classified as irregular organization of fiscal revenue and rectified as a single item. ② The same page carries two numbers pointing the other way, 21,000 eligible enterprises not receiving tax and fee reductions of 280.631 billion yuan in time, and 72.423 billion yuan of taxes and fees under-collected since 2021 because data were not shared (2024 report p.4). Over-collection and under-collection are listed as two items, rectified separately, never put in one place. ③ The recommendation of 24 June 2025 was to rectify vigorously; the recommendation of 23 June 2026 was to clean up vigorously, one character changed, the same recommendation in the same first sentence under fiscal revenue both years, read as last year's problem already under rectification (2024 report p.24; p.22). ④ The rectification report of 22 December

2025 listed the 33.189 billion yuan as rectified and the mechanism as standing, with completion judged by refund amounts and institutional text and with no relation to whether the audited year's behavior had stopped. ⑤ The 59.651 billion yuan technique of booking expenditure and then routing back requires three parties' vouchers aligned, and any one party seen alone shows a compliant entry (p.4). It was read as an accounts technique problem and handled apart from the collection behavior.

The point of escalation: 22 December 2025, the rectification closed on refunds and offsets, coaching into corrected returns and a new mechanism, with 33.189 billion yuan returned in full and revenue quality monitoring and accountability notices online (rectification report, 22 December 2025). Six months later, the report of 23 June 2026 lists 40.427 billion yuan again in the same subsection at the same ordinal position, with the enterprise count up from 631 to 1,954 (p.4; 2024 report p.4). The two numbers cannot be subtracted from each other. Audit scope and sampling targets differ year by year, and a count that tripled more likely means smaller enterprises were sampled, the average per enterprise falling from 52.6 million to 20.69 million yuan by the book's calculation; what is comparable is the position, not the size. The two reports also place the question in the same position, under "(1) irregular organization of fiscal revenue, policy implementation not in place" (2024 report p.4; p.4). The escalation has no single signatory. Its landing point is the correspondence between the two reports: the definition of rectification stops at money returned and mechanism standing, while the audited behavior ran its same course inside that same year. The State Council executive meeting of 11 June 2026 deployed the same kind of rectification again (p.24), which is the second turn of this loop, and the body that certifies rectification complete is the body that finds the relapse.

Failure chain: technical layer. Every action in the collection system carries legal parameters, prepayment ratio, filing period, receipt account, corrected return, all of them on the menu; the booking and the routing back at the treasury end use different vouchers, each complete on its own; the data sit in segments across tax, finance, treasury and the People's Bank of China, and every segment adds up. Organizational layer. The audit is ex-post external sampling whose scope stops at enterprises times amount, and the 1,954 enterprises have no list; collection is ex-ante and enterprise by enterprise, decided at the console inside the filing period; rectification acceptance is self-reported by the party being rectified, with refund amounts compiled and reported by the tax departments themselves. Incentive layer. Revenue is a monthly measurable output of the layer; tax source authenticity, enterprise cash flow and business environment are cross-period unmeasurable externalities that enter no monthly table; once the approved 9.737253 trillion yuan becomes the progress baseline, every period's shortfall has to be made up in the next. All three layers broke at once. The break at the boundary sits between the organizational and the incentive layer: Article 55 of the Budget Law closed the door on issuing revenue targets in 2015, while approved budget figures and execution progress were not prohibited, so the number stayed on the table and only the door plate changed its name. By the end of April 2026, rectification of the 2024 audit findings had reached 1.41 trillion yuan in amounts, more than 1,150 institutional documents drawn up or improved, and more than 3,690 people disciplined or handled (p.3), with the scale of rectification and the recurrence of the same problem present at the same time. The three units of measurement are money, documents and people (p.3), and none of them measures onto the accounts of the enterprises that were over-collected.

Minimum counterfactual: before the rectification closed in December 2025, draw 100 enterprises from the 33.189 billion yuan

refund list and check their prepayment parameters in the next filing period. One action, in December 2025. Refund data, filing data and tax source ledgers already sit in the same system, so the step needs no new authority and no new body, and the cost is one sample. It also needs nobody to admit a problem: prepayment parameters are numbers, and one comparison with the previous year's actual payment produces them. The 100 enterprises are a workload figure, one review team finishing in two weeks, and the result has two forms, parameters back at actual payment or parameters not back. The second form, arriving in December, lands just before the next year's budget is issued. If the parameters are still above actual payment, the 40.427 billion yuan gets seen as it happens instead of a year later, when the audit aggregates it enterprise by enterprise.

Outcome and sources: across the three items of 2025, 111.013 billion yuan (40.427 plus 59.651 plus 10.935, p.4). The report publishes at the aggregate level of 21 provinces and municipalities, with no province and no enterprise named. Rectification enters the next budget year's procedure, with a report to the NPC Standing Committee before the end of the year (p.25). The page references cited in this item are concentrated on pp. 3, 4, 22, 24 and 25. Primary source: National Audit Office, Report of the State Council on the Audit of Central Budget Execution and Other Fiscal Revenue and Expenditure for 2025, delivered on 23 June 2026 at the 23rd session of the 14th NPC Standing Committee, PDF of 25 pages, all page numbers in this card being pages of that PDF, link at Appendix B po-008. The comparison documents are the same-named report for 2024 (24 June 2025, 27 pages) and the report on rectification of the problems found in the 2024 audit (22 December 2025).

Case 2 has one layer left to take apart. The 59.651 billion yuan of inflated fiscal revenue was produced by techniques of the kind described as booking treasury funds as expenditure and routing

them back. The existence of such a technique is itself a diagnosis of the incentive layer: it needs several links to cooperate, booking, routing back, reporting, which tells you the inflation was not an individual collector improvising but an organizational craft for completing a target. Past a certain intensity of target pressure an organization invents its own craft for meeting the target, and the complexity of the craft is proportional to the pressure. The lesson for whoever designs targets is written in the arithmetic: every unit by which a target exceeds the real tax source makes the craft one notch more precise, and the 59.651 billion yuan the audit found is the full width of the gap between target and source.

Case 3 (Aviation): The Chain Where Every Layer Saved a Little

av-032. Colgan Air 3407 (Continental Connection), 12 February 2009, Clarence Center, New York. NTSB final report (AAR-10/01, PDF downloaded and checked for this book): during a night approach, control error after the stall warning, the aircraft stalling into a house, 49 people on board and 1 on the ground dead. An economic structure sits inside the report's factor chain. The regional carrier operated for the mainline airline under a per-departure fee structure, with the mainline brand and the regional capacity on two separate ledgers. Pilot pay levels, commuting-style scheduling and their interaction with fatigue are discussed explicitly in the report: the captain had commuted a long distance to reach base that day, and fatigue management and the scheduling rules were compliant on their face. The company optimized capacity cost, the pilots optimized their personal living arrangements, the regulator set its defences by carrier class. Every layer's choice made sense on its own books.

Stop at the decision point: the operating day on which the scheduling and fatigue management rules took effect for a com-

muting crew. On the desk: the roster, the commuting legs, rest rules met on paper, and a question no party was required to answer, which is how much safety margin this combination was worth in that night's approach. 〔What happened〕 50 dead; the accident drove revisions to the fatigue rules and to captain experience requirements. Primary source: NTSB, AAR-10/01, 2010, link at Appendix B av-032.

The coldest layer in case 3 is compliance on paper, since the fatigue management and scheduling rules were met on their face. The compliance measure tests duty and rest duration for a single segment, and it cannot measure what a cross-city commute actually costs; the parameters are written in an office, the cost is incurred on the red-eye commute. In a chain where every layer is compliant, the safety margin leaks out at the seams between layers: the seam between company and pilot, where whether commuting counts as duty time is undefined; the seam between mainline and regional, where the per-departure fee turns speed into revenue; the seam between regulator and carrier, where tiered defence assumes regional capacity carries lower risk. The rule revisions after the accident, fatigue rules and captain experience requirements, are in essence an assignment of the seam's account to whoever holds signature authority. The reader's self-check follows from that: at the seams of your chain, whose margin is leaking now.

Case 4 (Manufacturing): The Cost of Compliance on a Platform Shelf

mn-004. The SOOWERY eight-drawer fabric dresser. CPSC recall notice, page checked for this book: about 138 units recalled for violating the mandatory safety standard for clothing storage units (the requirement of the STURDY Act), sold through the Amazon channel for about nine months, with no injury reports. On a third-

party platform, listing speed is the seller's lifeline and compliance verification is a cost the seller pays: the gap between the platform's spot checks and the mandatory standard gets filled in by each seller's own diligence. The other end of the same structure is the consumer, who cannot see the absence of a test report, exactly as the reader of an audit report cannot see the compression of audit procedures.

Stop at the decision point: the operating window before the batch goes up, with the compliance documents not yet complete. On the desk: the timing of the sales window, the turnaround and cost of testing, the actual state of similar goods already on the platform. 〔What happened〕138 units recalled; the structural risk of small-batch imported goods of the same kind remains in the channel. Primary source: CPSC, 2026, link at Appendix B mn-004.

One detail on the consumer side of case 4 needs filling in. The recall notice is standard. The consumer's perception is probabilistic. The 138 units spread across nine months of sales, and a buyer's exposure on either side of the recall is decided by the efficiency of the platform's push. The incentive layer gets a symmetry test here: the platform's motive to push a recall is tied to its own liability exposure, while for small-batch goods the recall liability falls in law mainly on the seller, and often ends when the seller deregisters. Every link acts on its own ledger, and the last link, the consumer, receives protection as a probability. The hedge designed for platforms is Device 3 in section 5: spot checks funded by the platform and run by an independent party, with the cost booked to the seller's violation reserve instead of the platform's cost. Change where the money flows and the sampling density acquires its own funding source.

Case 5 (Platform and Data): A Structural Conflict That Survived Two Redesigns (full card)

sw-007.

When and where: 15 December 2020, when the consent mechanism went live, to 1 September 2025, when the decision issued; France. The inspections ran online and in Paris: a remote inspection from 20 October to 4 November 2022; three on-site inspections at Google France on 29 November 2022, 6 December 2022 and 5 September 2023, at 8 rue de Londres in the 9th arrondissement; an online check of the cookies on google.fr, YouTube, Google Maps and Google Search on 20 April 2023 (§ 10, § 11, § 12, § 13, § 42). The restricted committee heard the case on 26 June 2025, issued the decision on 1 September and published it on 3 September.

What and who: Google LLC (California, a wholly owned subsidiary of Alphabet Inc.) and Google Ireland Limited, which handles Google products for the European Economic Area and Switzerland. Both were found to be joint controllers within one economic entity (§ 2, § 262). Google France is the French establishment of both, which is the basis on which CNIL claimed territorial jurisdiction and excluded the GDPR one-stop-shop mechanism from the case: cookies and electronic prospecting fall under Article 82 of the Data Protection Act and Article L. 34-5 of the Postal and Electronic Communications Code, not under the GDPR (§ 29, § 30, § 38). The restricted committee sat with five members under its chair, Philippe-Pierre Cabourdin (cover page).

What they were doing: on 15 December 2020 the account creation flow went live with two paths. The first, called "quick personalisation (1 step)", took two clicks in all: click confirm and the personalised advertising cookies drop. The second, "manual personalisation (5 steps)", took five screens before personalisation could be switched off, six clicks in all (§ 113, § 117, § 133). Both paths arrive at the same endpoint: a choice between personalised

and generic advertising, with no step at which advertising cookies can be refused (§125). In the same period Gmail's smart features split the inbox into Primary, Promotions and Social tabs, and an account with the feature on saw advertising entries between the messages of the Promotions and Social tabs (§158, §159). An entry is the same size as a message, has a white background and a sender name in the same format, differs only in the date position, which carries the word Ad in green, and the web version lacks a checkbox (§187, §188). Inspectors walked this combination themselves during the remote inspection of 20 October to 4 November 2022: choose generic advertising, click continue using smart features, the inbox splits into three tabs and advertising appears between two of them (§158). Gmail has been a free service since 2004, requires a Google account before use, and the account is at once the entry to advertising, YouTube and other products (§6). On the company's statements in the proceedings these advertising entries enter no mail storage, cannot be searched, moved or replied to, are allocated by auction and change with the conversation (§194). Two earlier penalties on the same subject hang on this same path, of 7 December 2020 and 31 December 2021 (§84, §244).

The decision point: three dates, each with a carrier.

The first is 15 December 2020, when the account creation consent mechanism went live. The carrier is the account creation path on google.fr itself; the decision records the date and the two-step and five-step structure, with no signing post appearing (§113, §117). The first path's page is titled "Confirm your personalisation settings and cookies", and it holds two buttons, "Confirm" and "Back" (§118). The asymmetry fixed at that moment is two clicks to accept and six to refuse (§133).

The second is April 2023, when the Gmail advertising entries were restyled. The carrier is the format change on the delivery side, and

during the proceedings the company submitted an IPSOS format study it had commissioned, in which 76 percent of respondents found the redesigned advertising not hard to tell apart from messages (§ 164). The decision records the change as intended to reduce the risk of confusion with messages, and the restricted committee held it insufficient to overturn the legal characterization (§ 160, § 249). The same period holds the second on-site inspection of 5 September 2023 and company submissions of 26 September and 14 November 2023 (§ 12, § 14).

The third is October 2023, when the path was redesigned. Quick personalisation (1 step) was renamed "quick choice (1 step)", two buttons were added, "Refuse all" and "Accept all", and clicking through led to a "Confirm your settings" page that still offered "Confirm" or "Back" (§ 121). The manual path was renamed "manual choice (4 steps)", with the content of pages 3 and 5 untouched and page 3 still titled "Enable the 'customised ads' parameters to see more relevant ads" (§ 122, § 123). The decision's conclusion splits in two: from that date the freedom limb of consent was no longer missing (§ 136), and the flow as a whole satisfied the informed limb neither before nor after the redesign, because nowhere from beginning to end did it state that access to the service is conditional on advertising cookies being dropped (§ 138, § 141).

The notice sentence at step 5 of the manual path still comes after the user has picked an advertising type (§ 124). The order of the three moments is structure first, appearance second, buttons last. The round that added buttons changed the click count and left the disclosure alone, and the choice on page 3 still happens before the explanation on page 5, so a user has to pick an advertising type before learning that what he picked is a whole set of cookies (§ 143).

What was visible then: the NOYB complaint filed on 24 August 2022 on behalf of three complainants (§ 7); the full text of resolutions 2020-091 and 2020-092 of 17 September 2020, requiring refusal to be as easy as acceptance and the interface not to highlight one option visually (§ 97, § 101); the Court of Justice's judgment of 25 November 2021 in C-102/20, holding that advertising in an inbox, of the same shape and in the same position as messages, constitutes direct marketing by electronic mail and needs specific, informed consent in advance (§ 157), with paragraphs 41 to 43 of that judgment giving as grounds the occupation of a line meant for private mail and the resemblance to a message that makes a mistaken click likely (§ 192), the judgment given nearly a year before the inspection began (§ 242); paragraph 150 of the Court's judgment of 4 July 2023 in Meta, C-252/21, which writes the equivalent alternative into the test of free consent (§ 102); its own account scale in the backend, meaning the number of consumer Gmail accounts in France active at least once in the 28 days to 11 December 2022 whose inbox carried a Promotions or Social tab, redacted as [...] in the decision (§ 241); two prior cases, SAN-2020-012 of 7 December 2020 and SAN-2021-023 of 31 December 2021, same defendant and same subject (§ 84, § 244); the notice on step 5 of account creation, which opens with "if you agree" while the path holds no refuse-all step (§ 145); its own path's click counts, two to accept and six to refuse, countable on any walkthrough (§ 133); the path and buttons after the October 2023 redesign (§ 121); EDPB Opinion 08/2024 of 17 April 2024, which lists free service with generic advertising only as an equivalent alternative and which the company cited in its defence (§ 101, § 114); the report served on 25 July 2024, written submissions of 4 October 2024, 29 January and 28 April 2025, and oral submissions of 26 June 2025 (§ 16, § 17, § 20, § 23). What became known only afterwards: more than 74 million accounts in France, of which 53 million people were shown the advertising at issue (CNIL press release, 3 September 2025; the

corresponding figures are redacted as [...] in the decision, § 239, § 241); the fine and the daily penalty (§ 264, § 265, § 271); and the informed limb still unrepaired after October 2023 (§ 138).

You are in the seat now, with the material above on the desk.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: on 1 September 2025 the restricted committee fined Google LLC 200 million euros and Google Ireland Limited 125 million euros, 325 million euros in total, the two amounts split by company with each covering both violations and no split by violation (§ 264, § 265 and the operative part). It also ordered both to remedy within six months of notification: French users must be made to understand that creating an account necessarily means advertising cookies, and prior consent must be obtained before advertising is inserted between messages; after the deadline the penalty is 100,000 euros a day (§ 269, § 270, § 271). The decision was published, with the names removed after two years, and an appeal to the Conseil d'État lies within four months (last page of the operative part). Back to the three questions. The commonest answer to the first is another round of remediation, though two rounds have already shown that the same structure regenerates the same violation, so the third round's output will most likely be a third remediation document. The item most often missed in the second question is the revenue function itself: whether the consent mechanism is bolted onto the architecture or built into it decides whether remediation fixes the boundary or the engine. The correct

landing point for the third question is the interface design review, whose mandatory question is "what is the revenue state of this architecture with all consent switched off", a question that forces revenue dependence and compliance boundary into the same room.

Options ruled out: four, each with ready practice at the time.

The first is an explicit cookie wall. The restricted committee says plainly that tying access to the service to cookies is not in itself unlawful, provided consent is free and refusal carries no loss (§ 127); the decision even gives two sample lines, one for creating a free account and accepting cookies for personalised advertising and content, one for creating a free account and accepting only cookies for generic advertising (§ 146). The route was there and was not taken, and the decision's assessment is that where the company could have chosen a cookie wall that is unambiguous to users, it chose a path that was complex and ambiguous (§ 147).

The second is making refusal one step. The "Refuse all" button of October 2023 proves it can be done (§ 121); it did not appear before the redesign, and after the redesign it solved only the click count (§ 136).

The third is stating the condition on the first screen. The remedi‑ation order requires exactly that sentence (§ 269); the decision notes that neither path carried it at any time (§ 141), and that the manual path split it across steps 3 and 5, leaving the user to join the two (§ 143).

The fourth is making advertising entries plainly unlike messages. The change of April 2023 did part of it (§ 160); the entries in the Court's case had a grey background and showed no sender (§ 190), while Gmail's have a white background and a sender name in the same format as messages (§ 193).

All four leave the revenue model untouched and move disclosure and clicks instead. The decision holds no record of ruling them out.

Early signals: ① The two penalties of 7 December 2020 and 31 December 2021, same defendant and same subject, both on cookie consent under Article 82, the 2021 one naming the inconvenience of refusal on the home pages of google.fr and youtube.com (§ 84, § 244). They were read as a fine on the website banner and not on the account creation flow, a distinction the company itself raised in defence (§ 28, § 34, § 35). ② The Court's judgment of 25 Novem - ber 2021 (§ 157), given nearly a year before the inspection (§ 242), was read as a civil judgment about advertising that is not mail, the company arguing against applying it by analogy (§ 162), and the appearance change was pushed back to April 2023 on that reading (§ 160). ③ The guidelines of 17 September 2020 required refusal to be as easy as acceptance (§ 101), and the penalty of 31 December 2021 had already stated as a principle that the two options must take an equal number of operations (§ 135), while the company's own path still ran two against six (§ 133). It was read as already offering a refusal path; the decision acknowledges the choice ex - isted and finds the manner of expression distorted it (§ 134). ④ After the buttons went in during October 2023 the informed limb was still unrepaired (§ 138), and that was read as remediation complete, with the company arguing that the order no longer served a purpose under Article 82 because the redesign had been finished (§ 267). ⑤ From September 2024 the company stopped using some cookies connected with personalised advertising (§ 115, § 249), which is a different matter from opting out of all advertising cookies; the decision records it as a mitigating cir - cumstance (§ 249, § 247) without changing the finding, and the account scale kept rising through the same period, with the meas - urements of 13 May 2023 and 10 September 2024 each higher than the one before (§ 241).

The point of escalation: the October 2023 redesign is that step. The date is in the decision, the action was adding two buttons and renaming the paths (§ 121), and no signatory appears. The timing sits after the inspections closed and nine months before the report was served on 25 July 2024 (§ 16). In the year before, both inspection procedures had run their course (§ 8, § 11, § 12), and the April 2023 format change had already been made once (§ 160). The escalation holds because the redesign closed the freedom limb while fixing the informed limb in place: the path set from then on, and every later claim of compliance landed on "the user can choose", which is the defence's own wording at § 267, while the gap the decision finds is exactly that the user does not know he has no choice (§ 141, § 142). Adding the sentence about access being conditional on cookies after the redesign would mean overturning the whole path design of December 2020; leaving it out, every step of the path can be called a choice. The acceptance standard of that round shows here too. The deliverable list is buttons and renaming, and the decision, comparing page by page, records that the content of pages 3 and 5 of the manual path was unchanged (§ 122), so the deliverable is two buttons live while the running state is whether the user knows what he is choosing. A deliverable can be accepted. A running state can only be sampled.

Failure chain: technical layer. The advertising entries share the size, the white background and the sender format of messages and differ only in a green Ad and the date position, the web version lacks a checkbox, and clicking opens the whole advertisement in the form of a message (§ 187, § 188); account creation holds two paths, two clicks to accept and six to refuse (§ 133); the October 2023 redesign added buttons and left disclosure split across steps 3 and 5 (§ 143). Organizational layer. Both changes land on the product path, and the decision records dates and structures only, with no signing post (§ 113, § 121); the joint-controller finding rests on the parent's full ownership and the decision chain and

has nothing to do with who signed (§ 262); the approval column is empty in the decision; remediation was delivered by the party be - ing remediated, out of its own path and its own explanation (§ 267); after the report was served on 25 July 2024 the company's four rounds of written submissions argued jurisdiction and ne bis in idem (§ 28, § 33, § 37), not facts. Incentive layer. Gmail is provided free, and its maintenance and development are funded by the group's advertising revenue, a layer the decision's own text describes as therefore intrinsically connected with the dropping of cookies (§ 130), and it provides that a party adopting a cookie wall must show its purpose is to keep an economic balance between providing the service and its cost (§ 129, § 131). The difference between the personalised and the generic inventory is recorded as degree of intrusiveness (§ 109, § 110), with the unit price absent from the file; group revenue in 2024 was 350 billion dollars, about 312 billion euros (§ 263), and the advertising line's own revenue figures are redacted in the published version (§ 252). All three lay - ers broke at once. The break at the boundary runs between the technical and the incentive layer: every click on the path is at the same time a pricing of inventory, and the ease of the click is decided by the technical implementation.

Minimum counterfactual: when the path went live on 15 Decem - ber 2020, put the sentence "creating an account requires accept - ing advertising cookies" on the first screen, beside the two op - tions. One action, on the day the mechanism went live, changing no button and adding one line of copy. The cost is a screen of copy and one copy review, with the two sample lines at paragraph 146 of the decision serving as template; no architecture change, no change to the revenue model, and the six-click asymmetry can be left as it is. The step was doable at the time and did not depend on luck: the guidelines of September 2020 already required clear dis - closure (§ 101), and the remediation order at paragraph 269 re - quires exactly that sentence. Acceptance takes one person and one

pass: open google.fr, walk the first path, see whether the first screen carries the sentence. It does not erase all liability, since the freedom problem before the redesign still stands, and it erases the informed limb, which is the one thing this Article 82 violation lacked from 2020 to 2025 (§ 138, § 153, § 154).

Outcome and sources: 325 million euros in total (200 plus 125), two violations found, a six-month remediation period and 100,000 euros a day after it (§ 264, § 265, § 269 to § 271). Primary source: CNIL restricted committee decision SAN-2025-004 of 1 September 2025, English courtesy translation, PDF of 54 pages; citations in this card are paragraph numbers (§) and page numbers are PDF pages. The decision runs from a cover page with the subject matter and the service record, through Part One on facts and procedure (§ 1 to § 23), Part Two on reasoning, jurisdiction, procedural objections, controller determination, Article 82 and L. 34-5, and Part Three on the fine, the remediation order and publication (§ 225 to § 275), to the operative part on the last page. The translation's header states that the French original prevails where the two differ. Accompanying press release of 3 September 2025; prior cases SAN-2020-012 of 7 December 2020 and SAN-2021-023 of 31 December 2021; Court of Justice C-102/20 of 25 November 2021. The published version carries [...] redactions, and where a figure is redacted this card fills in none (§ 239, § 241, § 252). Link at Appendix B sw-007.

Case 5 is the standard portrait of misaligned incentives in the platform economy: the form of the violation upgrades with each remediation while the structure of the conflict persists unchanged. Its sister case (sw-008, Appendix B: FNAC DARTY fined 7 million euros for an app sending browsing data to third parties through an SDK without consent) shows the small and medium version of the same structure, where privacy compliance never covered the third-party SDK chain. As in case 5, the gap sits in the chain rather than

in awareness: the compliance team sees its own code and not the code integrated into it.

Case 6 (Manufacturing): Fifty Mattresses and Eleven Thousand Bottles of Lotion (condensed)

mn-007 and mn-009. Two CPSC recalls of October 2026, pages checked for this book: about 50 Longjili Trading mattresses failing the mandatory flammability standard, and about 11,000 units of RootStim minoxidil hair regrowth liquid missing child-resistant packaging, a PPPA mandatory standard. Both are isomorphic with case 4 in this chapter, the SOOWERY dresser: small-batch import, violation of a mandatory standard, no injury reports, delisting from the platform channel. The reading of a recall sequence three in a row at the incentive layer is this. The compliance cost for a small-batch importer, testing, certification, packaging change, is badly out of proportion to the batch size, and list first and withdraw if caught is for many small sellers a strategy that holds up mathematically. A mandatory standard settles whether a thing may be sold; it does not settle whether anyone verified anything before it went up, and with the verification step empty the standard degrades into a recall catalogue. Minimum counterfactual: mandatory compliance spot testing by the platform on a small seller's first order. Primary source: CPSC, 2026, links at Appendix B mn-007 and mn-009.

Case 7 (Finance): Even the Identity Was Borrowed (condensed)

fi-003. Ichcoin Tech Corp. SEC enforcement of August 2026, text checked for this book: an investment adviser practised on forged ADV filing information, using the CRD number of an unrelated 87-year-old individual. The principal-agent chain is pushed to its absurd edge here, since the compliance identity itself, the investment adviser registration, was forged, so the agent the principal faces, regulator and client both, is already broken at the identity layer. It

gives the chapter's three questions an extreme test. Is each lay - er's choice rational? Borrowing an identity is rational inside a fraud frame. Did nobody break a rule? Someone did, and this case enters the territory of intent. On whose head does the summed loss land? On investors and on the 87-year-old whose number was used. The answers push the case to the corruption and fraud side, which is exactly what calibrates the line drawn at the start of the chapter: misalignment changes structure, fraud moves the courts. Primary source: SEC, 2026, link at Appendix B fi-003.

Laid out by industry, the seven cases run audit, tax collection, re - gional aviation, platform retail, platform data, small-batch manu - facturing and adviser identity, and the distribution follows one rule: the more expensive verification is in an industry, the heavier the case. Audit and securities, cases 1 and 7, verify quality in units of years, and their cases end in criminal exposure and large pen - alties. Platform retail, cases 4 and 6, verifies in units of spot checks, and its cases end in a recall catalogue. Data platforms, case 5, sit between the two and end in a seven-figure euro fine. Verification cost grades the failure style of an industry: expensive verification produces big cases, cheap verification produces catalogues. Inver - ted, the rule is the budget table for defences. Whichever grade your industry's verification cost sits in is the grade your incentive-layer governance budget should sit in.

One diagnostic tool belongs beside that budget table: principal identification. Every agency relation has a nominal principal, the one who pays, and a real principal, the one who carries the con - sequence, and the separation between the two is the depth of the misalignment. The nominal principal of an audit is the audited company, the real principal is the reader of the statements. The nominal principal of goods on a platform is the seller, the real principal is the buyer. The nominal principal of regional capacity is the mainline carrier, the real principal is the passenger. The tool

is one question: when this thing breaks, who loses most, and who has no seat today? Find that person and you have the legal starting point of every defence in this chapter, since Device 2's whole-result signatory acts, nominally, for this absent party. The absent party's seat is the origin of the whole corporate governance literature, which this chapter does not re-expand; it only notes that any proposal claiming to solve misaligned incentives while giving the absent party no place to speak is circling inside agency upon agency.

17.4 What Shows Up Earliest

Seven weak signals, each with the reason that usually presses it down.

1. The phrases "we followed the contract" and "we followed the rules" appear at abnormal density in meeting minutes. Reason given: following the rules is not a fault.
2. Every piece of bad news reaching a given post comes off another link's ledger. Reason: that is the division of labor.
3. The beneficiary is the verifier of some problem, and the verification reads "no issues" year after year. Reason: it is running normally.
4. Decisions with large externalities are never made by anyone who meets the people who carry them. Reason: hierarchical division of labor.
5. The assessment metrics fight each other, and every side in the fight is winning. Reason: multi-dimensional assessment.
6. The same structural disease recurs at different suppliers or divisions, and every response is to replace the people. Reason: an execution problem. If replacing people worked, the second occurrence would not have happened.

7. In contract negotiation, risk clauses get far less time than price clauses. Reason: price is the business.

Put the seven back against this chapter's cases. Signal 2, bad news arriving from another link's ledger, is case 1: the first victim of an audit quality problem is the statement reader, who appears on no ledger of the firm. Signal 3, the beneficiary verifying and finding nothing year after year, covers cases 1 and 4: the party maintaining the client relationship shares a table with the party guarding audit quality, and on the platform the spot check sits on the same side as the listing. Signal 4, deciders who never meet the bearers, is case 3: the designer of the capacity cost structure does not sit in the cockpit on a night approach. Signal 5, metrics fighting while every side wins, is case 2: the revenue metric wins and tax source reality loses. Signal 6, a recurring structural disease handled by replacing people, covers the two firms of case 1 and applies as well to the three recalls of case 6. Signal 7, risk clauses getting short negotiation time, is the native form of the listing terms in case 4.

One indicator of the layer's own deserves writing down: the whole-chain checkup, an extension of question 5 in section 7. Take the most recent loss that crossed a link boundary, write out each layer's choice and its rationality in that layer, and put the summed result and the party carrying it on the last line. The layer-by-layer rationality table is the incentive layer's examination sheet: every row, read alone, is innocent, and the last row is the lesion. It is also the strongest tool for talking to outsiders, since any party shown the table can find on it the evidence of his own correctness and, on the last line, the cost he took part in. The persuasion problem of misaligned incentives is hard exactly there, and that is also where it is solved.

The three questions need operating instructions. For the first, on whether each layer is rational, the material to check is each layer's assessment documents and contracts, never interviews: an

interview will say of course we put safety first, an assessment document tells the truth. For the second, on whether anyone broke a rule, the material is the compliance audit's conclusion: an all-green compliance record next to a bad outcome is the signature of the second question holding. For the third, on whose head the loss lands, the material is the consequences chapter of the accident report set against the seating chart of the decision table, and the seat gap shows up in the first comparison of the two lists. All of it is available inside the organization, which makes this one of the lowest-threshold diagnoses in the book: it needs no industry knowledge, only the one act of laying the documents side by side.

17.5 How to Prevent It

Device 1: externality accounting. Write the foreseeable effect of this layer's decisions on the other parties of the chain into this layer's own assessment. Audit firms get assessed retrospectively on later penalties and litigation records, and the CSRC penalty is an instance of the externality returning to the account, since what it fines is the consideration the audit fee bought. The quality of lawful collection enters the collection agency's assessment weights. What will not fit a quantitative clause goes in as a qualitative item with mandatory reporting at every review.

Device 2: a signatory for the whole result. Every multi-link chain names one role answerable only to the whole: the engagement partner answerable to the readers of the statements, the collection quality officer answerable to the taxpayer's experience, the operations control chief answerable to passenger safety. Its assessment is tied to no output of the layer. An empty title achieves nothing. The person must hold the power to stop (Chapter 18), and the stop must not be subject to performance assessment by any party inside the layer (Chapter 19's protection interface).

Device 3: beneficiary recusal from verification. Whoever benefits from a problem may not act as its verifier, and an arrangement already in place is replaced by external sampling. The whole institutional history of audit independence is the implementation record of this one clause, and its failures always begin with an exception: "this one is special, let the in-house team review it first".

Device 4: a structure-review trigger. When the same failure mode appears at two or more suppliers or divisions, escalation to a structure review is mandatory: examine the contracts and the assessments, not the people. Recurrence is the incentive layer speaking. A problem that yields to a personnel change does not recur, and a problem that recurs never yields to one.

The four devices against this chapter's cases. Device 1 on case 1: add regulatory penalties and litigation lookback to partner assessment, with the 3,679,245.18 yuan of this case as a ready-made lookback entry. Device 2 on case 3: add a co-signing role answerable for passenger safety to the capacity clauses of the mainline and regional contract. Device 3 on case 5: move the verifier of the consent mechanism's remediation from the advertising line to outside it. Device 4 on case 6: the third recall of the same kind triggers a structure review of the platform's small-batch import channel automatically.

The three write-ups for putting an externality back into assessment are worth giving in full. Write-up one is money: final payment, revenue share and settlement cycle tied to verifiable end quality, as described above. Write-up two is a post: the whole-result signatory, assessed on the end result alone, which is Device 2. Write-up three is an account: convert the externality into an internal cost and book it, as a compliance cost line for collection quality or a fatigue risk provision inside the capacity structure. The common enemy of all three is the excuse that it cannot be quantified: the accounting version of write-up three allows interval estimates (Chapter 11's

Device 2, the digitizing of margin), and what will not fit a quantitative clause goes in as a qualitative item reported at each review. None of the three is expensive. What is expensive is the reason for refusing to write them, which is that the layer does not want to see its own externality. Seeing it is the first step of every defence in this chapter.

The four failure modes deserve equal space, since each has a respectable way to die. Externality accounting dies of account inflation: the lookback line takes everything, and in the end it is indistinguishable from an ordinary cost line. The whole-result signatory dies of being an empty title: signatures get signed and the stop power is never wired, with no budget veto and no direct reporting line. Whether a signatory is real takes two facts, whether he has ever stopped something and what happened afterwards. Beneficiary recusal dies of spreading exceptions: the list of special cases grows longer every year until recusal is a principle on paper, so every exception needs written approval and a quarterly review in aggregate. The structure-review trigger dies of threshold decay: two or more becomes three, then four, and the trigger becomes a souvenir. All four share one examination question: when did this device last actually fire? A device that cannot produce a date is dead.

17.6 Boundaries with Adjacent Mechanisms

Against goal displacement (Chapter 12): misaligned incentives is the interest structure, goal displacement is its technical implementation. The criteria divide the work. If what needs changing is the interest structure, meaning who carries the consequence, go to Chapter 17. If the interest structure is sound and the measurement is crooked, stay in Chapter 12. The two often need work at the same time, and the order is to plug the structure first, by beneficiary recusal from verification, and repair the measurement after.

Against diffusion of responsibility (Chapter 18): dilution is nobody being responsible, with the signature chain ownerless; misalign - ment is someone being responsible in the wrong direction, with the assessment pointing the wrong way. The criterion is one ques - tion. Give the responsibility to one named person: does the situ - ation change? If it changes, the disease is dilution. If giving it changes nothing because his assessment forces him to act against it, the disease is misalignment.

Against horizon mismatch (Chapter 28): when the current-period tilt is amplified by short tenure, the horizon dimension stacks on top. The criterion: extend the decision-maker’ s tenure three times over, would the choice change? If it would, the main disease is in Chapter 28. If it would not, the main disease is in this chapter.

17.7 Self-Check: Five Questions

1. Write out the assessment objective of every link in your chain and see where the sum points. Does it point at the objective of the whole?
2. Which problems in your organization have the beneficiary for a verifier? Can you list three?
3. Has one of your decisions damaged another link’ s ledger? Did that damage appear in any assessment document of yours?
4. If your role were given one more line, answerable only for the result of the whole, which line of your present assessment would have to be deleted to make room?
5. Take an old incident in which every layer was blameless, write out each layer’ s rational choice in turn, and put the summed result on the last line. That line is the incentive layer’ s examination report.

(This chapter: 14,553 characters.)

Chapter 18 Diffusion of Responsibility

> What this chapter covers: how responsibility thins as the number of signatures rises; three criteria, four field sites, and the design of unilateral stop authority together with a ceiling on signature counts. When to turn back: when the organization runs a coun - tersigning workflow, stands up a committee, or hands one matter to three departments. Estimated reading time: 50 minutes.

Responsibility dilutes quietly. Nobody refuses it. It is averaged away in the act of dividing the work. A fraudulent annual report carries four signature lines. A collision in a shipping lane has four classes of party. A level crossing’ s warning equipment spans two owners. A helicopter’ s landing rests on two sets of assumptions. Seen afterwards, every party signed, every party covered its own segment, and every party can state exactly what it did. No link in the chain is in charge, and every link sincerely believes that somebody else is watching. This chapter is about that somebody else.

18.1 What It Is

Definition: diffusion of responsibility is the psychological and structural phenomenon in which responsibility thins as the num - ber of participants rises. The structural face is dispersal: staged approval, multi-party signature, committee decision. Each gate added is meant to add a line of defence. What it adds is one more contributor to the assumption that someone else will check. The psychological face is the office version of the bystander effect: the more parties involved, the higher the cost of unilateral interven - tion and the greater the social risk of making a fuss over nothing. Lay the two together and no link in the chain is in charge.

Three trigger criteria. Any one of them means the mechanism has already started.

1. Many participants, and the responsibility interfaces between them are not drawn in writing.
2. Every party can state exactly what it did; no party can state who owns the whole.
3. The organization holds no unilateral stop authority anywhere.

Criterion 2 has a self-test that is harsh and does work. After an incident the organization's first reflex is to find out who signed. The signature chain carries seven names. Seven signatures mean the step was confirmed seven times, which means not one of the seven confirmations was necessary.

The psychological face has a dated experimental origin. John Darley and Bibb Latané ran a series of experiments in 1968 in which subjects waited alone or in company while smoke came into the room next door, or a person fell next door. Subjects on their own reported the anomaly at a far higher rate than subjects in company. Company did not add up to more vigilance. It diluted each person's inclination to step in. The central variable is structurally identical to this chapter's criteria: the presence of others supplies the assumption that somebody has already dealt with it, and that assumption holds in every head at once. The office version differs in one respect. The others there carry documented roles, so the assumption looks better founded. The experiment also supplies the usable converse. Name one person, even if the only change is that one more person is nominally responsible for raising the alarm, and intervention rates return to the solo level. The antidote has sat in the literature for decades. Replace everybody with a name.

The structural face needs one more definition: what it means to be in charge. The test here is three observable conditions satisfied at once. You know what you are looking at, which is the information

state. You know what you are allowed to do, which is the authority state. You know what follows if you do nothing, which is the con - sequence state. Take away any one and you are not in charge, even where the paperwork carries your name. The VTS operator in Case 1 (ma-008) lacked the second: the boundaries of authority were unclear. The two parties in Case 3 (ce-008) lacked the third and the first: nobody assessed what the equipment actually did to pedes - trians. Auditing the three states beats auditing the name of the re - sponsible person, because the paper name is usually written in a clear hand and the three states are usually all empty.

18.2 How It Grows

Three generating conditions.

Condition one is an institutional tradition of multi-party signature and staged approval. The intention is another line of defence; the cost structure of signing decides the behaviour. Checking every prior signature benefits the chain and costs the signer, so a rational signer checks his own cell. Defences stack on paper and run in parallel in practice.

Condition two is layered outsourcing and subcontracting. Each tier down drops the visibility of risk a grade and weakens the respons - ibility link a grade. Where a subcontract carries service clauses and no risk clauses, nothing but a commercial relationship remains between the tiers.

Condition three is committee decision. The resolution belongs to the collective, execution needs an assignee, and supervision has no post. What a collective passes has no individual claimant by con - struction, and the credential of having taken part makes each member feel the duty is already discharged.

The tragedy of dilution is that it is asymptomatic in ordinary times. Every link signs, approves and meets. The system looks deeply defended, which is Chapter 22's illusion of redundancy projected onto the responsibility layer. The symptom shows up only at the moment somebody has to call a stop. At that moment seven signatories equal none.

The arithmetic fits in a sentence. Responsibility divides by headcount. The drive to intervene declines by headcount. The two curves run in opposite directions. Let the cost of dealing with a risk be C. With n parties and no written interface, each party perceives its own share of "I ought to act" as roughly C/n. The larger n gets, the smaller each share, and the cheaper the waiting strategy of letting somebody else move first. The equilibrium of waiting strategies is the classic dilution scene: everybody waits and the exposure runs its full length. The design implication is as plain. Either squeeze n to one, a single accountable person, or put each party's perceived share back at C by writing the whole consequence into each party's assessment. Those two moves are Device 1 and Device 2.

n also has a time dimension. Dilution is not static. The participant count grows with the age of the process. A new process has one owner. Every small incident adds a countersignature. Every cross-departmental collaboration adds a party. n climbs monotonically with age, the perceived share falls monotonically, until some unpleasant event drags the chain onto an audit table. That is why old organizations carry the most signatures on their gates and not necessarily the strongest defence. The growth of n records the organization's history of frights; each signature is a souvenir of the last accident, the organizational-design counterpart of Chapter 11's deviation ledger. Device 4, the ceiling on signature counts, is aimed at exactly this monotonic growth. Souvenirs can stay in the minutes. They do not have to stay in the workflow.

Signing has one further behavioural layer: how far a signer rationally checks. Making every signature verify all preceding content grows the cost linearly with chain length while the benefit falls with the marginal contribution of each added signature. A rational signer checks his own cell. Chapter 1 set out that structure; its industry version is the signature chain in Case 4. The patch is not to ask signers to be more conscientious. It is to write the verification protocol into the system. Signatures come in two kinds. A verifying signature means substantive checking of the content in that cell. A passing signature means confirming only that the paper moved through. Mark the two in separate columns, and rule that a passing signature may never be cited as quality evidence. Once the columns exist, the defensive illusion of seven signatures has no raw material left.

The outsourcing chain has a contract-layer patch worth spelling out: vertical continuity of risk clauses. A subcontract usually carries service levels and nothing else. The risk clauses, meaning the duty to notify when something goes wrong, the duty to cooperate with an investigation, and the cap on liability, are signed only between main contractor and client. At the second and third tiers the obligations to risk fall to zero tier by tier. The continuous form is the flow-down clause: the main contractor writes the core risk clauses verbatim into every tier of subcontract, and no tier may override them with looser terms. Flow-down is established practice in aviation maintenance and maritime contracting. It treats the disease in condition two, the tier where nothing but a commercial relationship survives. Checking the penetration rate of flow-down clauses is the first move in auditing an outsourced chain.

18.3 The Shape in Four Domains

Case 1 (Maritime): A VTS Whose Staff Did Not Know Their Own Authority

Case ID: ma-008.

When and where: 24 October 2023, 0455 (UTC+2). Eastern German Bight traffic separation scheme, collision point 54°01.5N 007°38.7E, 0.2 nautical miles inside international waters. The watch station was the German Bight vessel traffic service centre at Wilhelmshaven. Wind force 5 to 6 with gusts to 7, visibility between 10 and 31 km, dark (pp. 10, 24, 65).

What and who: the Bahamian bulk carrier Polesie (24,055 gross tons, built 2009, managed by Polska Żegluga Morska P.P., 20 crew, 32,997 tonnes of grain) and the Isle of Man general cargo ship Verity (2,601 gross tons, built 2001, managed by Faversham Ships Ltd, 7 crew, 3,262 tonnes of steel coils). The monitoring party was German Bight Traffic, operated by the Federal Waterways and Shipping Administration under the direction of the Federal Waterways and Shipping Agency (GDWS). Responsibility is recorded by post: the officers of the watch on both ships, the VTS nautical assistant, the VTS nautical supervisor (pp. 10, 32–33).

What they were doing: at 1940 on 23 October Polesie sailed from Hamburg for A Coruña with 32,997 tonnes of grain, draught 10 m, after seven days loading in port. At 2008 the same day Verity sailed from Bremen for Immingham with 3,262 tonnes of steel coils, draught 5.4 m, after five days loading (p. 12).

At 0310 on the 24th the pilots left both ships. On the Polesie bridge were the master, the second officer and a helmsman. The master handed control to the second officer and left. Course 273°, speed 11 knots, time in hand (p. 12). At 0410 the chief officer took the 0400 to 0800 watch. The starboard radar was on 6 nautical mile range with

offset display. Verity was already on AIS at CPA 0.15 nautical miles and TCPA 45 minutes. Neither radar had its CPA alarm switched on, and the ARPA had not acquired the target (p. 15).

At 0423 Verity entered the northbound lane on course 335° at 8.3 knots, keeping to the eastern side. At 0444:30 Polesie passed the planned waypoint, where the plan was to alter to 256° onto the westbound lane; she was still making 273°. At 0446:03 she crossed the 256° planned track with no change of course (p. 16). Between 0448:48 and 0449:59 the chief officer entered nine successive course changes on the autopilot, taking the heading from 264° to 245°. The 28° port turn went in instalments, and from another ship or from the VTS screen it did not look like steering (pp. 17, 58).

Over the same period the VTS assistant was monitoring both ships, judged that there was a risk of collision, and noticed that Verity was not acting as the give-way vessel. In the preceding three minutes the assistant had also handled two routine ship reports; the rest of the watch's tasks are unknown. The Channel 79 west sector assistant was absent that night and one supervisor covered two sectors (pp. 17, 36).

The decision point: the first moment sits at the document layer. On 28 October 2013 Coral Ivory and Siderfly collided in the Kiel Canal, and the Federal Bureau of Maritime Casualty Investigation (BSU) recommended in report 330/13 that GDWS study changing the VTS ship-shore duplex channel to simplex (p. 53). Nobody died in that collision. Siderfly was at risk of sinking from damage below the waterline and the canal was shut for three days (p. 53). The report records no formal reply from GDWS. By the time the 2026 report was published, GDWS said a next-generation VTS system was in development with the simplex requirement included, over 2028 to 2034 (p. 69). For the ten years between, the two ships could reach each other only through a shore station relaying words.

The second moment is 0450:09. The assistant called Verity on Channel 80 with the ships less than 1.3 nautical miles apart, CPA 160 m, TCPA 8 minutes 11 seconds. Three sentences: the CPA is very small; what is your intention; under the rules you must take avoiding action (pp. 17, 60). The report notes that the VTS's own post-event report does not explain why the intervention waited until TCPA was down to 7 minutes 28 seconds (p. 60). When the call ended the ships were a nautical mile apart, CPA under 0.2 nautical miles, TCPA 5 minutes 16 seconds (p. 17). The call carried no IALA WARNING marker and the tone was the tone of ordinary conversation (pp. 17, 61).

The third moment runs from 0453:43 to 0454:00. The supervisor called Polesie on Channel 80, confirmed radio contact, then in a raised, urgent voice said "do not go left, do not go left, turn right, Polesie, alter course to starboard" (p. 20). The chief officer answered "yes, starboard", and nine seconds later went to hand steering and hard starboard (p. 21). The VTS post-event report states that the supervisor was acting under Rule 17(c) of the International Regulations for Preventing Collisions at Sea (p. 61). The report's analysis holds that the chief officer was more likely acting under Rule 17(b), and that the restriction in 17(c) did not apply in that situation (p. 62). At the moment the instruction went out the ships were under half a nautical mile apart, and the instruction finished within 90 seconds of the collision (pp. 20, 62). Polesie was still inside German territorial waters, where the VTS had authority to instruct her. Half a mile away Verity was already in international waters, where the VTS could only advise (p. 65). Two ships half a mile apart, two sets of authority.

None of the three had a second person check it. The document was issued by GDWS. Both calls were decided by the watchkeeper on duty. Annex A holds only time, call sign and the words as spoken (p. 72).

What was visible then: on the Polesie bridge, both pilots were off the ship and the master was aboard but not on the bridge (p. 12); two tracks on radar and ECDIS, with the CPA and TCPA figures from AIS; the territorial sea baseline drawn on the ECDIS at 0433, the moment Verity passed from German territorial waters into international waters (p. 16); at 0443 a readout of CPA 0.11 nautical miles and TCPA 14 minutes 42 seconds (p. 16); at 0452:37 the Verity AIS icon selected, reading 0.83 nautical miles, CPA 0.2 nautical miles, TCPA 3 minutes 40 seconds (p. 19); four written instructions from the master, one of them "call the master in any doubt", specifying a call at 5 nautical miles or more (p. 30).

At the VTS station: a chart terminal fusing radar and AIS, on whose reconstruction of 0450:09 the two ships show a CPA under 0.1 nautical miles with 8 minutes left (p. 18); the automatic alarm for CPA under 0.5 nautical miles muted because there were too many small craft, with the indication still on screen (p. 35); no German territorial sea boundary on the display (p. 65); Channel 80 duplex, so the shore station could hear the ships but the ships could not hear each other (p. 37); one assistant short on the watch (p. 36); a three-shift rotation with the night shift from 22:00 to 06:00 (p. 35); no written guidance on what to do at which CPA (p. 35).

On the Verity bridge: two X-band radars with ARPA, AIS, ECDIS, BNWAS (p. 30); a steady 8.3 knots along the eastern side of the lane since 0423, nearly 8 nautical miles without a change of course (p. 57); no voyage data recorder, not then required on a ship of 2,601 gross tons (pp. 11, 70).

What became known only afterwards: the 28° port turn by Polesie was hard to make out on radar or by eye (p. 58); Verity was in fact turning into the track of Polesie (p. 59); the CPA between the two ships never exceeded 0.53 nautical miles from entry into the traffic separation scheme to the collision (p. 56); Polesie entered international waters at 0454:18, 1 minute 10 seconds before the collision

(pp. 21, 65); had the port turn continued there would have been no collision (p. 61); the investigation found that the chief officer "felt obliged to comply" with the supervisor's instruction (p. 62); the investigation found no evidence that either the operator or the supervisor knew the geographic limits of their authority (p. 65).

You are in the seat now, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 0455:28 the bulbous bow of Polesie struck Verity on the starboard side at a relative speed of about 12 knots, some 11 m forward of the accommodation (p. 22). Verity sank in 36 m of water, about 5 minutes after the collision. Five of the seven crew died; two were rescued, the second officer and one seaman (pp. 9, 23, 54). The search ran into the night. Four crew were unaccounted for, and two of the bodies were found inside the vessel during the recovery (p. 64).

Go back to the three questions. The commonest answer to the first is to intervene earlier and to use the WARNING marker. In this case the cost of that action does not sit in speaking. It sits in what follows speaking. After the first call at 0450:09 the VTS internal report records that "at 1 nautical mile VTS had no further opportunity to intervene" and that it was "satisfied that the earlier interaction had produced a collision avoidance solution" (p. 36). Three and a half minutes later it spoke again, and that second sentence pulled the chief officer out of a port turn that had already been working. The item most easily missed under the second question is the ab-

sent territorial sea boundary on the display. It looks like a charting detail. It was the shared blind spot of all three people on watch that night. The answer to the third question lands before anybody speaks: establish which side of the water the ship is in, then decide whether the sentence is advice or instruction (p. 65).

Options ruled out: a broadcast on Channel 16 so both ships hear the same words. The report states there is no indication that Channel 16 was considered, and concedes that a broadcast has drawbacks, but it does offer a way to clarify what is happening (p. 63). Having the two ships move to a working channel and talk directly: not possible on a duplex Channel 80, which needs a separately agreed channel; the VTS would not be a party to that call, and once advice had been given it would not know what the ships had agreed (pp. 62–63).

A single instruction from the VTS to both ships: on the Verity side the VTS had advisory authority only (p. 65). Intervening before 0450: the VTS internal report sets out as experience that small craft in the area often give way late, need little swinging room and are reluctant to leave their course (p. 36). The experience held, and the price of it was a threshold pushed out to 1.3 nautical miles. Continuing to intervene inside 1 nautical mile: the internal report sets its threshold at "at 1 nautical mile VTS had no further opportunity to intervene" (p. 36). The chief officer carrying the port turn began at 0452:55 through to the end: the reconstruction shows no collision, but he went to hand steering nine seconds after the supervisor's instruction (pp. 21, 61). The master taking the bridge: his written instructions required a call at 5 nautical miles or more in any doubt (p. 30), and the master was called after the collision (p. 22).

Early signals: 1. On 28 October 2013 Coral Ivory and Siderfly collided in the Kiel Canal, and BSU report 330/13 recommended that GDWS study changing the duplex channel to simplex (p. 53). Reas-

on it was overridden: the canal was closed three days, nobody died, and the recommendation stopped at "study". 2. On 11 February 2015 Ever Smart and Alexandra 1 collided inside the Jebel Ali port limits, and the container ship's bridge sighted the other vessel seconds after the port control call (p. 52). Reason it was overridden: a one-off inside port limits where both pilotage and port control had failed. 3. On 11 December 2013 Paula C and Darya Gayatri collided in the Dover Strait, and the investigation found that a timely and well-meant VHF intervention by the coastguard almost certainly pushed the Paula C officer of the watch into the wrong action (p. 53). Reason it was overridden: blame landed on the officer of the watch for losing situational awareness, and the intervention itself was never questioned. 4. The VTS compulsory refresher scenario VK02-K-GBT-De trains exactly the case of a risk of collision where the VTS does not intervene (p. 36). Reason it was overridden: the training supplies the right answer, and the boundary was only ever exercised in one direction. 5. In 2022 the IMO member state audit of Germany raised no finding or observation on the VTS (p. 36). Reason it was overridden: the external audit passed, so the system was taken as confirmed. 6. A shipowner inspection record of 19 October states that the bridge personnel knew all navigational safety procedures; the master's written instructions were a verbatim copy of the Liverpool to Hamburg version of 12 October, signed by the master and three officers (p. 30). Reason it was overridden: complete paperwork was read as execution.

The point of escalation: 0451:36 to 0451:50. The assistant called Polesie with the sentence: "the Verity on your port side will pass astern of you, he tells me he will turn to starboard and pass astern of you, keep a check on the situation" (p. 19).

Before that sentence each ship held its own picture. Verity held course and speed, expecting Polesie to give way to the lane and let

her pass ahead. Polesie expected Verity to pass astern (pp. 57, 19). The two pictures contradicted each other and neither ship had spoken to the other. After that sentence, the Polesie picture had been confirmed by the shore station. The chief officer answered "yes, I will keep a watch, a little to starboard" (p. 19). The report notes that this answer did not meet the closed-loop readback standard of IALA 1132, and that had the assistant used the WARN - ING marker before the call, the ambiguous reply would not have been let pass (p. 61).

The escalation took the form of a relayed sentence. It sent each ship a confirmation that the other would give way, and the two confirmations pointed opposite ways. It also put the VTS itself in the position of a problem already handed over: for the next four minutes nobody spoke, until 0453:43. In those four minutes Verity only began to turn to starboard at 0452:26, while Polesie at 0451:02 set the autopilot from 245° to 252° and stopped the port turn that was under way (pp. 17, 19). Who approved it: the assistant issued it, the supervisor sat at the same watch station, and there is no record of a review. The words are in Annex A (p. 72).

Failure chain: technical layer. The duplex Channel 80 kept ship from speaking to ship, leaving the VTS as the only channel and re - ducing it to relaying what was meant (pp. 60, 62). The automatic alarm for CPA under 0.5 nautical miles was muted (p. 35). The ter - minal did not show the territorial sea boundary (p. 65). Both ships made small incremental alterations on autopilot that did not read as steering at a distance (pp. 56–57). Verity had no voyage data re - corder (p. 11). Neither ship had its CPA alarm on. Polesie's sea - men were on a lookout rota and by custom did not keep lookout (pp. 15, 29). Whether Verity had an extra lookout that night, the report records as impossible to establish (p. 55).

Organizational layer. Authority was cut into two grades of post, supervisor and assistant, and three grades of measure, traffic in -

formation, traffic assistance and instruction, with the assistant empowered to issue instructions only on the supervisor's authority (p. 33). The three sentences at 0450 belong in the files to information and assistance, yet they went past plain notification ("under the rules you must act"), and the authority to instruct was not the assistant's (pp. 17, 33). The ordering rule that traffic assistance is provided when traffic information has failed to achieve its purpose came with no CPA or TCPA threshold and no statement of when to stand back (pp. 34, 62). The supervisor issued the instruction on the basis of Rule 17(c), and the report holds that 17(b) was the rule that applied (pp. 61–62).

Incentive layer. The VTS was not assessed on CPA at all; assessment fell on whether interaction followed procedure and whether records were kept. Four posts of thirty were vacant and the watch ran anyway, with cover arranged "opportunistically" (p. 36). The two ships were assessed on schedule reliability and surface compliance. Both had time in hand that night (p. 55) and both kept the CPA inside 0.5 nautical miles (p. 56).

The seam broke between the document and the experience. The document left blank who must speak at what distance. The watchkeeper filled the blank from experience. In that experience small craft always give way late, so silence read as a judgement.

Minimum counterfactual: add one line on standing back to the intervention clause of VV-WSV 2408. When TCPA falls below 5 minutes, the VTS issues warnings only and no instructions. One action, dated before October 2023, needing no new equipment and no extra staff. The report states plainly that neither VV-WSV 2408 nor VV-GDWS 24-2 gives guidance on the CPA or TCPA at which intervention becomes counterproductive (p. 62). With that line in place, the "turn to starboard" at 0454:00 would not have gone out. The report's reconstruction shows that had the chief officer continued his port turn the ships would not have collided (p. 61).

The figure of 5 minutes has support in the report: when the supervisor spoke at 0453:43 the TCPA was under 3 minutes, and the report describes that band as counterproductive for intervention (p. 62). The report did not put the threshold into a recommendation. 2026/117 asks GDWS only to "provide guidance", with no number attached (p. 70). The same line would hold for next time: the decision to stand back becomes citable, and therefore answerable.

Outcome and sources: Verity was a total loss, five people died, Polesie suffered slight damage to bow and starboard side with minor oil pollution (p. 10). The report makes seven recommendations, numbered 2026/115 to 2026/121. The IMO should amend SOLAS Chapter V/20 to bring internationally trading cargo ships of 500 to 3,000 gross tons within the voyage data recorder requirement. GDWS should review the use of duplex channels and provide guidance to intervention staff covering when to intervene, the use of information markers, and when to stand back (p. 70). IALA has revised Guideline 1141 to require shore relay wherever a duplex channel is used (p. 69). The limits of the investigation are also recorded: Verity carried no recorder, key witnesses were not interviewed, and the internal decision process at the VTS could not be reconstructed (pp. 9, 11).

Primary source: MAIB, Report on the investigation of the collision between the bulk carrier Polesie and the general cargo ship Verity, Report No 5/2026, February 2026, 82 pages, with Annex A a translation of the German Bight VTS report and Annex B extracts from the Collision Regulations. Link in Appendix B, ma-008.

The four-party shape of Case 1 is worth reading again against the three states, because it is the complete specimen of this chapter's criteria. Information state: each of the four parties held a fragment (the two ships each knew their own constraints, the VTS could see both tracks), and the whole picture existed only on radar, never in

a judgement. Authority state: the authority clauses for operator and supervisor were vague on what could be required, whether it could be enforced, and who carried the failure if enforcement failed, which was the fatal gap on the monitoring side. Consequence state: each bridge knew the physical consequence of a collision, and the assumption that the other would act thinned the urgency of it. Each state has a patch: a common picture table for information, an authority list for authority, and making assumptions explicit for consequence, where the watch rule requires "I think the other will move" to be said aloud, at which point it becomes something that must be verified. The MAIB recommendations fall broadly inside that frame, and the reader can check them against the original.

Case 2 (Healthcare): The Incompatibility Only the Registry Knew

me-008. Joint replacement surgery in the English NHS. HSIB report (Implantation of wrong prostheses during joint replacement surgery, 2018; the archived snapshot has been checked against the original for this book). During surgery a 62-year-old patient was implanted with prostheses from different manufacturers that were not compatible with each other, and the defect surfaced days later when data at the national joint registry were matched. Surgical team, instrument supply, pre-operative check, registry entry: each person at each link could explain how his own step met that link's process. The theatre took implants against the implant list, the supplier shipped against the order, the check ran against the checklist. No link carried full responsibility for whether the two components matched each other, because matchability straddled the seam between four links, and seams do not sign.

The stop point is the instant on the operating table between the implantation of the first component and of the second. On the

table: implants each carrying its own label, process records showing every link verified, and a registry that nobody thought to query on the spot. **What happened**: the patient underwent a second operation; the report names the registry's detect-afterwards pattern, and detection afterwards means nobody was responsible beforehand. Primary source: HSIB, 2018, link in Appendix B, me-008 (archived link, grade A–).

"Seams do not sign" can be pushed a step further. The seam went unclaimed because matchability straddles the catalogue systems of two manufacturers, and neither catalogue contains the other. The gap in the information structure comes before the gap in responsibility: even a person who wanted to own it would have no data to check against. The order of repair for seams of this kind is therefore data first, responsibility second. Until a compatibility tool exists, one that queries both catalogues together, naming a responsible person only creates a post with a duty and no instrument. The HSIB report makes the registry's after-the-fact detection a principal finding, which is an acknowledgement that the data layer was missing, and the direction of its recommendations agrees with this chapter's order. The generalisation for the reader: at a check point spanning two systems, ask first whether the tool exists, then whether the responsibility exists. Responsibility written where the tool is absent is written in the air.

Case 3 (Construction): The Governance Interface at a Level Crossing

ce-008. The Staniforth Road level crossing of the Supertram network in Sheffield, 22 June 2025. RAIB report (Report 10/2026, pages checked against the original for this book): a tram struck two pedestrians, one seriously injured and one slightly injured, both taken to hospital. The pedestrians did not notice the tram coming, the audible and visual warnings had no effect, and risk management

between the infrastructure owner and the operator at the crossing was weak. The crossing's warning equipment straddles the gov - ernance interface of two bodies. Ownership of the equipment, its maintenance, its assessment for effectiveness and its actual effect on pedestrians sit on different responsibility lists, and the un - worded seam between those lists falls exactly on the question of whether a pedestrian can cross safely.

The stop point is the last moment at which the crossing's warning equipment was assessed on the measure of actual effect on pedes - trians. On the table: the equipment status record, which reads normal, the owner's and the operator's respective duty docu - ments, both complete, and the seam between them that neither party has ever signed. **What happened**: two people went to hos - pital; the report lists the unclear risk interface among its investig - ation findings. Primary source: RAIB, Report 10/2026, 2026, link in Appendix B, ce-008.

The governance interface here has one feature worth marking. The equipment status record reads normal. Normal means the equip - ment has power and is working. What failed was the measure of actual effect on pedestrians, meaning whether the warning is enough for a pedestrian to notice the tram. The split between the two measures falls along the split between the two bodies: the owner manages equipment condition, the operator manages op - erational safety, and the measure of effect sits in neither assess - ment. This is the responsibility version of the negative substitution in Chapter 12, where the information channel for the real goal was never wired into either party. The repair copies Chapter 12's general rule: build a quantity for the effect, here a frontline ob - servation of whether pedestrians notice, then assign it an owner under this chapter's device. On governance interfaces like a level crossing the two chapters' defences always arrive as a pair.

Case 4 (Finance): The Ex-Post Section Through a Signature Chain

fi-007. The decision against the responsible persons in the Qingyue Technology fraudulent issuance case (〔2026〕No. 59, pages checked against the original for this book): chairman Gao Yudi, warning, a fine of 9 million yuan, an 8-year market ban; executive deputy general manager Mu Xinju, a fine of 9 million yuan, a 6-year ban; chief financial officer and board secretary Zhang Xiaobo, a fine of 6 million yuan, a 5-year ban; supervisory board chairman Wu Lei, a fine of 4 million yuan, a 3-year ban (the same case as fi-006). The penalty decision is a section cut through responsibility diluted in advance. The fraudulent offering document and annual report carried four names, and the four signatures made up the organizational narrative of layer upon layer of checking. Afterwards the regulator took each signature back to its true degree of knowledge, one by one: the layering of signatures was the shape of dilution beforehand and the order of accountability afterwards.

The stop point is the stretch of process in which the fraudulent content was finalised and the four signatures went down in sequence. On the table: what each signatory actually knew, the route by which each document arrived already signed above, and the half step by which each signatory pushed the duty to check onto the signature before or after his own. **What happened**: fines totalling 28 million yuan, with market bans of 3 to 8 years. Primary source: China Securities Regulatory Commission, 〔2026〕No. 59, 2026, link in Appendix B, fi-007.

The layering in Case 4 has one more thing to read. The fines and ban lengths fall away in strict order down the four signatures, because the regulator prices by degree of knowledge and by the influence of the post. The counterpart of that pricing beforehand is that knowledge and influence ought to weigh a signature. Dilution flattens them into the same undifferentiated “signed”. The lesson

for a board is written in the pricing. The further down the chain a signature sits, the greater the signatory's information disadvantage, so the designer of the chain owes the lower signatories an independent information route, whether an outside adviser with direct access or a channel for dissent that goes straight up. Without one, the lower signature is only a guarantee given by the upper, and dilution runs from the first signature to the last.

Case 4 and Case 1 of Chapter 17 (fi-008, the audit firm's side of the same issuer) are a mirror pair for this chapter. The firm is the outsourced end of the chain failing, the four company signatures are the core of the chain failing, and the same fraudulent annual report sits between them. The regulator's handling of the two ends, put together, is the full section through dilution: the structural disease at the outsourced end, which is paying for strictness, meets the structural disease at the core, which is the flattening of the signature chain, on one issuer. The two cases are placed side by side so the reader can see the whole-chain shape of dilution. It does not stop at the boundary of an organization, at committees and countersignatures. It also runs between organizations, through commissioning, outsourcing and subcontracting. The instruments that defend against it are the same set. The governance interface register, Device 3, carries across organizations unchanged, and the requirement to name four parties holds across an organizational boundary as well.

Case 5 (Public Policy and Administration): Thirteen Years, Four Options, Two (full card)

Case ID: po-005.

When and where: 2000 to June 2026. The Palace of Westminster, London, a Grade I listed building inside a UNESCO World Heritage Site. The report was signed by the Comptroller and Auditor General

on 8 June 2026, ordered to be printed on 11 June and published on 15 June, numbered HC 32, session 2026-27 (pp. 1, 3).

What and who: the Restoration and Renewal Programme. The parties involved: the House of Commons and the House of Lords, each of which must approve the scope and funding of Phase 2; the House Commissions, one per House, responsible for that House's administration and services and setting the spending limit for Phase 1; the Clerk of the House of Commons and the Clerk of the Parliaments, corporate officers who serve as accounting officers for the programme; the Client Team, which acts for the Clerks and manages the Delivery Authority day to day and replaced the Sponsor Body in 2023; the Client Board, a joint body of the two House Commissions; the Programme Board, made up of members of both Houses, the two Clerks and external independent experts; the Delivery Authority, responsible for day-to-day delivery; and Strategic Estates, which manages the parliamentary estate and reports through the Director of Operations to the Commissions. Oversight sits with the National Audit Office, and the report is signed by the Comptroller and Auditor General, Gareth Davies (pp. 3, 7–8, 30).

What they were doing: in 2000 the first survey of the plant rooms raised concerns about the building and its services (pp. 6, 16). In 2013 the two Houses established the programme (pp. 7, 17). In 2014 the Commons Commission agreed to proceed with refurbishment of the Northern Estate; in 2016 that Commission decided to give Richmond House to the programme as decant space (p. 28). In January 2018 the Commons voted to approve construction and to vacate the Palace entirely during the works, with a commitment to return as soon as possible; in February the Lords endorsed the approach (p. 17). In 2019 Parliament passed the Parliamentary Buildings (Restoration and Renewal) Act, establishing the Delivery Authority and the Parliamentary Works Sponsor Body and writing in nine scope considerations: value for money, safety and security,

environmental protection, accessibility, visitor access, education and other facilities, public engagement, archaeology and historic value, and benefits across the whole of the UK (pp. 17–18). In December 2020 the House Commissions agreed to ask the Sponsor Body to study the implications of continued presence (p. 18). Through 2022 and 2023, over concerns about cost and schedule, the Sponsor Body was replaced by the Client Board and the Client Team (pp. 18, 30). In 2023 the Client Board added two options for construction inside the Palace, EMI and EMI+, and asked Strategic Estates to develop them; the same year the two Clerks said it was uncertain how much longer the Palace could be used as it stands, because the building has no reliable system for stopping the spread of fire (pp. 16, 18). In March 2024 the Client Board published a strategic case that fixed scope at outcome level 4, "reasonably ambitious" (pp. 18–19). In 2025 the Commons Commission required the Northern Estate refurbishment to meet a "Parliamentary grade" standard, raising the refurbishment standard above the minimum intervention originally set (p. 28). In February 2026 the Client Board published the costed options report, the first time four delivery options had been set side by side with complete cost and schedule ranges (p. 8).

The decision point: the first moment is January to February 2018. The vehicle was two votes, the Commons in January and the Lords in February, and the report records only the month. Each House voted separately and neither could stand in for the other. That vote turned "move out" from an option into a premise, and every option since has had to find room between that premise and the commitment to return as soon as possible (p. 17).

The second moment is December 2020. The vehicle was a decision of the House Commissions to ask the Sponsor Body to study the implications of continued presence. The report records the decision in a single line: ask the Sponsor Body to carry out work so as

to understand fully the implications of retaining a continued presence in the Palace during construction (p. 18). It sets no dead-line for the study's conclusions, and it does not say that the 2018 decision stands if the conclusions come back unfavourable. From that year on, the 2018 premise was open to discussion, and the discussion carried no expiry date.

The third moment is 2022 to 2023. The vehicle was a change of governance: the Sponsor Body was abolished and the Client Board and Client Team took over. The report records the range of years, with no months and no document numbers. The stated reason was to increase members' engagement with the programme, and the remit handed to the successor was "to develop options for the Houses to consider and vote on" (pp. 18, 30). The responsible body changed its name, and the verb in the remit changed from over-seeing delivery to developing options. The direction of accounting responsibility appears in the same section: the two Clerks answer to their own House and Commission, not to a minister or to the Treasury (p. 30).

The fourth moment runs from February to June 2026 and is not complete. The vehicle is the costed options report plus a parliamentary debate that has yet to be scheduled. The Client Board asks the Houses to approve cutting the options from four to two, full decant and EMI+, and to authorise a seven-year Phase 1 with a £3bn cap; the target date for the final choice is 2030 (pp. 6, 8–9). The request goes to the Houses, and the debate required for approval has to be scheduled by the government (p. 9). The membership of the Houses keeps changing. The report records that since 2024 the Client Team has met around 542 sitting members, of whom around 283 are sitting Commons members, roughly 43 per cent, and that many of them took no part in the earlier decisions and met the programme at one point only in its development (pp. 10, 19). The date of 2030 was set by the Client Board itself, and

the report calls it a target rather than a statutory deadline (pp. 6, 31).

What was visible then: in January 2018, the conclusions of every review since 2000, the current condition of the building and its services, and the option set of full decant with a return as soon as possible. The costs of the four options did not appear until February 2026 (p. 8).

In 2019 the nine scope considerations of the Act sat side by side on one page. The report's comment is that these broad and potentially competing considerations needed clear trade-offs (p. 18).

In December 2020 the table held the NAO's 2020 report on early value-for-money risk (p. 9). The cost and schedule of continued presence had not been estimated, and its range of £14.4bn to £22.0bn over 33 to 45 years did not stand beside the other three options until 2026 (p. 8).

In February 2026 the four ranges were on the record together: full decant £11.1bn to £15.6bn over 19 to 24 years; continued presence £14.4bn to £22.0bn over 33 to 45 years; enhanced maintenance and improvement, or EMI, with the Palace divided into 14 zones and no more than 30 per cent vacated at any one time, £23.3bn to £56.3bn over 52 to 84 years; and EMI+, with the Lords vacating for 8 to 13 years and the Commons borrowing the Lords chamber for no more than 2 years, £19.5bn to £39.2bn over 38 to 61 years (p. 8). Phase 1 in the same month was divided into five work packages: temporary accommodation, at £417m for the Queen Elizabeth II Centre with the acquisition price counted separately and £328m for Richmond House; enabling works at about £691m; early works at about £160m; and pre-construction activity at about £860m (p. 25). Design of the conference centre stood at RIBA stage 2 and the Northern Estate group containing Richmond House at stage 1, and the former had under one month of buffer between completion and handover to operations (pp. 27–28).

What became known only afterwards: the technical feasibility of the two EMI options cannot be confirmed until the next design stage (p. 20); the precise location and quantity of asbestos are unknown (p. 23); the Northern Estate cannot provide temporary accommodation before 2034, pushing completion to 2039 (p. 28); and if the Commons decant were phased, some members and staff would remain on site for about half the duration of a full decant (p. 28).

You are in the seat now, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: as of the NAO's June 2026 update, none of the four delivery options had been selected, and both requests, cutting the options to two and authorising a seven-year Phase 1 capped at £3bn, were waiting on debates in the two Houses, which the government schedules (pp. 8–9). The spread between the highest and lowest option was £45.2bn in cost and 65 years in schedule (p. 8). Each year of delay adds £320m to £420m to the programme's overall cost (p. 10). Maintenance of the Palace averaged about £1.5m a week between 2017-18 and 2025-26 and is expected to rise to about £2m a week from 2026-27 to 2029-30; the programme itself is expected to bring that weekly figure down by about £700,000 (p. 17). Go back to the three questions. The commonest answer to the first is to create a new lead body, yet the machinery has already been reorganised several times in thirteen years, and the nth reshuffle of the machinery will not by itself produce the decision

mechanism that the n+1th reshuffle lacks. The item most easily missed under the second question is the state of "four cut to two" itself: the completeness of the comparison material makes "still choosing" look like progress, when in fact no document carries the duty to decide. The right landing point for the third question is the deadline clause: give the final choice a statutory deadline and a default option, so that failure to choose by the date triggers one of them, and the duty to decide acquires an expiry date.

Options ruled out: continued presence. Its figures are £14.4bn to £22.0bn over 33 to 45 years, at least £3.3bn more than full decant and at least 14 years longer (p. 8). It did not enter the two options recommended for further development, and the report records no reason for leaving it out, only the sentence that "two rather than one was recommended in order to reduce continuing cost while still giving the Houses a choice" (p. 9).

EMI. £23.3bn to £56.3bn over 52 to 84 years. Its underground design and delivery strategy differ from the other options, its technical feasibility cannot be confirmed until the next design stage, and Strategic Estates only began designing the two EMI options in 2023 (pp. 10, 20).

Reducing scope to level 3. The analysis shows that core requirements drive most of the cost, and moving one level down or up changes it by only about 5 per cent, so narrowing scope buys no room on cost (pp. 11, 23).

Keeping all four options on the table. The Client Team estimates the additional design and planning cost of developing more options at £115m to £530m, depending on how many are kept (p. 17).

Delegating authority to a fully empowered body. The report records this under what Phase 1 may need, notes that the Client Team is reviewing governance, and adds that any change may require legislation (p. 31).

Early signals: 1. The first survey in 2000 had already raised concerns about the building and its services, and the programme was not established until 2013 (pp. 6, 16). Reason it was overridden: survey first, project second. 2. The vote in January 2018 gave a commitment to return as soon as possible (p. 17), while the four options run from 19 years at the shortest to 84 at the longest (p. 8). Reason it was overridden: the commitment lived in a political promise and the durations lived in the options; the two sentences were never set side by side. 3. The December 2020 request described reopening the 2018 decision as understanding the implications fully (p. 18). Reason it was overridden: the document gave the work a name and gave the direction no name. 4. In 2014 and 2016 the Commons Commission agreed the Northern Estate refurbishment and designated Richmond House as decant space, two to four years before the 2018 decant vote (p. 28). By April 2026 Strategic Estates took the view that the Northern Estate could not provide temporary accommodation before 2034, and the report writes that if phasing follows available accommodation, the pace of construction in the Palace may be set by the availability of decant space rather than by the needs of programme delivery and parliamentary business (p. 28). Reason it was overridden: decant space was treated as enabling works, never as the clock of the option. 5. The report itself records that successive reports explained the objectives and recommendations of each stage but gave no simple account of how those decisions fitted together (p. 10). Reason it was overridden: every report answered the question of its moment.

The point of escalation: in 2023 the Client Board decided to add EMI and EMI+, taking the option set from two back out to four; the same year Phase 1 was designed to be usable under whichever option was finally chosen (pp. 18, 24). The report states that the more options left on the table beyond the two recommended, the more

complex Phase 1 becomes and the more unnecessary cost it carries (p. 24).

In February 2026 the four options and a Phase 1 package capped at £3bn over seven years were bound into one costed options report (p. 8). The structure of the binding is that money is spent on the basis of being usable under any option while the option is kept on the basis of being chosen later. The first phase includes pre-construction procurement, which the report estimates at 12 to 18 months and places on the critical path (p. 26). Of the five work packages, the Commons decant and staff accommodation have to happen whichever option is finally chosen (p. 28). That money does not wait for the option.

By this point the options on the table are no longer four items on paper. They are four sets of construction commitments that have already begun to spend. Who approved it: the addition in 2023 sat with the Client Board, and the report records only the year; the request of February 2026 still awaits approval by the two Houses, and the debate has to be scheduled by the government (p. 9).

Failure chain: technical layer. The Palace is Grade I listed, its electrical, heating and drainage services run through the walls, and asbestos is everywhere with precise location and quantity unknown (pp. 19, 23). Eighty-five per cent of the estimated construction cost goes to priority works, meaning building services renewal, stone conservation, fire protection and asbestos removal (p. 11). The judgement of the two Clerks in 2023 was that the building has no reliable system for stopping the spread of fire, and that under a major fire it would be hard to save (p. 16). Across the four options the lowest and highest estimates differ by £45.2bn and the durations by 65 years (p. 8). Contingency runs from 32 per cent to 148 per cent of base cost (p. 22). The technical feasibility of the EMI options cannot be confirmed until the next design stage (p. 20).

Organizational layer. Governance is spread over seven bodies (pp. 7–8). The remits and membership of the Client Board and the Programme Board overlap, and both take part in strategic decisions and trade-offs (p. 30). The two Clerks are accounting officers in law, and in practice lack the authority an accounting officer normally holds on a major government project (p. 12). The programme has no single accounting officer carrying personal statutory responsibility (p. 30). The remit of the Client Board is "to develop options for the Houses to consider and vote on" (p. 30). The funding model in the Act has no flex: Phase 2 cannot start until Parliament has approved the scope and the total funding for the whole works (p. 31). Key activities run outside the Client Team and the Delivery Authority, and the report writes that it is not clear who takes decisions, or who is responsible for delays or trade-offs on those projects (p. 13).

Incentive layer. The report's judgement on the current governance is that it is designed for parliamentary engagement and the development of options, not for clear accountability and delivery decisions (p. 12). The five work packages of Phase 1 sit with different funders and different delivery parties (p. 25). In the remits of the various bodies, producing options is stated plainly and choosing between them rests with the two Houses (p. 30). Every party is delivering its own cell, and every party can produce evidence of it.

The seam broke between producing options and choosing one. Producing has an owner, a budget and deliverables. Choosing needs approval from each House separately, and the power to put that approval into the order paper sits with the government (p. 9). There is one due date, and it is a target, not a deadline (p. 6).

Minimum counterfactual: in December 2020 the House Commissions, in the same decision that asked the Sponsor Body to study continued presence, write in one lapse clause: the study must

conclude within 12 months, and if it does not, the full decant approved by both Houses in 2018 stands as the default. One action, dated December 2020, needing no legislation and no budget, only a sentence in the Commissions' own decision. The conditions were all in place at the time: the default had already been voted through by each House separately in 2018 (p. 17), and the scope and duration of the study were for the requesting party to set (p. 18). With that line, the premise for adding EMI and EMI+ in 2023 does not hold, since the addition was possible only because continued presence was still an open option. The side-by-side ranges and the 65-year schedule spread would not appear in the 2026 report. The report does not put this into a recommendation; its recommendation (a) asks for a published account of how decisions evolved at each key stage (p. 14), and an account lets a reader understand the past without letting the next decision fall due.

Outcome and sources: no delivery option selected, Phase 1 awaiting approval by the two Houses, the target date for the final choice 2030 (pp. 6, 8–9). The four cost ranges run from £11.1bn to £56.3bn in total, over 19 to 84 years (p. 8). The amounts shown on the Phase 1 diagram total about £2.46bn against the £3bn cap; note 3 to the exhibit states that Phase 1 costs do not yet include Cloister Court, system segregation and decommissioning, or riverbank security (p. 25). Primary source: NAO, Restoration and Renewal of the Palace of Westminster: 2026 Update, Report by the Comptroller and Auditor General, Session 2026-27, HC 32, 15 June 2026, 40 pages; the evidence base is a document review from January to June 2026, including board minutes and papers from January 2022 to March 2026, eight interviews and four teach-ins (pp. 30–32). The report states that it offers no opinion on policy or on which option should be chosen (p. 32). Link in Appendix B, po-005.

Case 6 (Public Policy and Administration): The Railway That Kept Redrawing Its Frame (condensed)

po-007. Northern Powerhouse Rail in England. NAO report (March 2026, the pages checked against the original for this book; the NAO rates its own evidence at the weak-judgement grade). A regional rail programme was re-scoped substantially again and again over more than ten years, and the realisation of benefits slipped accordingly. In this chapter's reading, the repeated changes of scope mean the programme had no single owner of its direction: at each change of strategic leadership or fiscal cycle the scope was recut to the new preference. Each cut had its own reasons, and the sum of them is that the benefits are always still to come. Minimum counterfactual: once the programme's scope is set, any substantial change must be confirmed by an independent assessment that outlasts a single administration, with a record kept. Primary source: NAO, Northern Powerhouse Rail, 2026, link in Appendix B, po-007.

The contrast between Case 6 and Case 5 deserves a sentence. One programme has gone thirteen years without a decision (Case 5) and the other has redrawn its scope for more than a decade (Case 6). The symptoms look opposite, one frozen and one churning, and the structural disease is the same: neither deciding nor changing has a document to carry it. Freezing means nobody is obliged to choose; churning means nobody is obliged to hold still. The state between them, where a choice once made is neither overturned nor shelved, is what a carrying document buys. A reader who governs public projects can turn that into a self-check: for your long-horizon programme, who made the last major decision, in which document, and under which procedure. A programme that cannot answer all three is on the road between Case 5 and Case 6.

Case 7 (Agriculture and Food): The Relay of a Recall Along the Distribution Chain (condensed)

ag-004. The retail end at Whole Foods, Whatcom Blue bulk sliced cheese. FDA enforcement report (event 97931 / H-0210-2026, the API record checked for this book). Two days after the upstream recall, the retail end followed with a recall of 411 pounds, Class I. Responsibility passing along a distribution chain takes the form of a relay here: the manufacturer starts the recall and distribution and retail each follow at their own pace. Two days is not slow in the context of food recalls, and no link in the chain was required to answer when every downstream location would be clear. Recall effectiveness is measured at the end of the chain while recall or - ganisation is measured by link, and the gap between the two measures is the shape of Case 7. Minimum counterfactual: gener - ate the list of downstream locations and a clearance deadline when the recall is opened, and close it out location by location. Primary source: FDA, event 97931, 2026, link in Appendix B, ag-004.

The two-day relay is worth reading against the time frame of food safety. The exposure window for STEC contamination, Shiga toxin producing E. coli, is counted in days, and every day the product sits in the chain is another day of exposure at the consumer end. The reason this case belongs in a chapter on dilution is that every link’ s notion of “as soon as possible” was sincere, since every link’ s own procedure requires prompt action, and promptness was defined separately by each link: the manufacturer counted from issuing the notice, distribution from receiving it, retail from the internal transfer order. Nobody owned the total elapsed time of the three promptnesses strung together. The counterfactual of closing out by location works as an antidote because it moves the definition of prompt from each link’ s own measure to the meas - ure at the end of the chain: the clearance deadline is worked back

from the endpoint, and each link is allotted not the time it considers reasonable but the time the endpoint needs.

Working back from the endpoint has a further organizational meaning. It turns the chain's responsibility from participation into delivery. Under participation, every link's action counts as a contribution. Under delivery, only the state of the endpoint counts, meaning downstream locations cleared, and the actions of the links are process. Switching between the two measures is not difficult. The first week is difficult, because every link will object that it does not control the next link. The objection is true and the answer is structural: the part you do not control is exactly the part to write into the interface register, meaning the duty of the upstream party to notify the downstream party and the duty of the downstream party to confirm back. Once the register is written, what you do not control becomes traceable. That conversion, from objection to register, is the most everyday use of Device 3.

Case 8 (Nuclear): Inadequate Supervision Among Four Violations (condensed)

nu-004. Acuren Inspection Inc., Alaska site. NRC press release (October 2026, checked for this book): a proposed civil penalty of $36,000 for four violations, namely failure to perform radiation surveys, violation of operating procedures, inadequate supervision, and inadequate instructions for responding to alarms. The one worth stopping on is inadequate supervision. The supervisory post existed, since a violation list that names it shows the post is in the establishment, and the act of supervision did not happen: nobody did the surveys and nobody watched the procedures. A supervisory post is the classic hiding place for diluted responsibility, because its duty points by nature at whether others are doing things right while its assessment usually points at its own output of paperwork. Penalising the four violations together prices supervision and work the same: a failure of supervision is not an aggrav-

ated case of management failure, it is a violation in its own right. Minimum counterfactual: keep a record of the supervisory post's survey checks, who checked, when and what, and treat a missing record as a supervision violation. Primary source: NRC, 2026, link in Appendix B, nu-004.

Inadequate supervision also yields three lines of design advice for whoever creates a supervisory post. The output of the post is findings, meaning what it caught, not the number of inspections, meaning how many it ran, so the assessment should point at the findings register. The reporting line should be independent of the party being supervised, at least one level across, or supervision becomes an internal function of the supervised. The post's establishment budget should be carried separately, with no shared cost centre with the supervised party's output targets. The logic behind all three is to make the supervisory post itself a single point that is in charge: information in the findings register, authority in the independent reporting line, consequence in its own budget and assessment. Supervisory posts are a disaster area for dilution precisely because they span several parties by nature. Make the post a single point first under Device 1, and only then is it fit to supervise anybody else.

18.4 What Shows Earliest

Six weak signals, each with the reason that usually overrides it.

1. Every box on the process chart has a person and the whole has no owner. Override: "the division of labour is clear."
2. After a problem surfaces, the first meeting called is a meeting to establish who is responsible, not a meeting to fix it. Override: "we have to learn the lesson."

3. A committee resolution has no named executor, or the executors are not the people who passed it. Override: "collective decisions are steadier."
4. The outsourcing contract carries service clauses and no risk clauses and no stop clauses. Override: "commercial is commercial."
5. Three departments could all take charge of one matter, nobody does before an incident and everybody does after it. Override: "joint management."
6. The frequency of "that is not mine" rises in internal communication. Override: "follow the process."

18.5 How to Defend

Device 1, the single accountable person clause. For every critical process and every risk point, the system names one person who carries the responsibility, with the person replaceable and the seat not allowed to stand empty. Every other signatory carries a verification duty rather than a gatekeeping duty, and the verification duty is defined in writing: your signature states only that the items you checked are true. Had the four signature lines in the Qingyue case said beforehand what each one verified, the regulator would not have had to divide it up for them afterwards.

Device 2, unilateral stop authority. At least one post holds independent authority to stop a critical irreversible action, written into the job description and into the contract (Chapter 7), with the cost of stopping carried by the organization, which is the interface with the termination reward in Chapter 27. Stop authority that does not land on a named person is not stop authority, and stop authority that lands on a person with no budget behind it is authority nobody dares use.

Device 3, the governance interface register. For any object maintained jointly by several parties, whether crossing equipment, shared data, an interface system or a matchability check, build an interface register that names for each item the owner, the operator, the assessor and the party that accepts the effect, with no two of the four allowed to be the same party. Once a year run a drill against the register: if this thing breaks, who finds out first. Write the drill's answer into the register in place of the assumption.

Device 4, the ceiling on signature counts. Any process that carries more than three gatekeeping signatures on one action is restructured by force, because gatekeeping past three signatures equals no gatekeeping at all. The extra signatures either convert to informed copy, which does not constitute gatekeeping, or are deleted, and every signature kept must state the specific items it verifies. The resistance will be heavy, since every deleted line will claim to be critical. The way through is to have each line write down what it verifies, and to delete automatically any line that cannot name its items.

The four devices on this chapter's cases. Device 1 applied to Case 1: the VTS authority list states the trigger conditions for a mandatory instruction and where the responsibility falls, turning the operator's "what must I do" from a vague clause into three lines. Device 2 applied to Case 3: either party at the crossing gains authority to suspend use of the crossing unilaterally, with the cost of suspension paid from a shared fund, because authority nobody can afford to use goes unused. Device 3 applied to Case 2: prosthesis matchability is entered as an interface item, with the owner, the supplier, the assessor and the accepting party all named and none of them the same. Device 4 applied to Case 4: each of the four signature lines on the offering document states what it verifies, and any that cannot is converted to informed copy.

The failure modes of the four devices. The single accountable person dies of departure: a change of person carries no handover list, the name changes and the three states reset to zero. The patch is to write the three states into an appendix of the job description, so they travel with the post and not with the person. Stop authority dies of nobody daring to use it: the authority is on paper and the psychological cost of using it sits in the culture. The patch is to publish the record of use at intervals, since stop authority used three times is authority that is alive. The governance interface register dies of staying static: a change to the object does not trigger an update. The patch is to link the register to the change note. The ceiling on signatures dies of column drift: informed copy quietly gets promoted into a source cited by gatekeeping signatures. The patch is citation review, returning automatically any document that cites copied circulation as quality evidence. All four patches share one principle. The state of a device has to be observable, and a device that cannot be observed is treated as absent.

18.6 Boundaries with Adjacent Mechanisms

Against misaligned incentives (Chapter 17): misalignment is a question of direction, where the assessment points at the wrong target, and dilution is a question of existence, where no carrier of the direction is there at all. The test question: if you hand the responsibility to one named person, does the situation change. If it does, the case is dilution. If handing it over changes nothing because his assessment forces him to act against it, the case is misalignment. Colgan 3407 (Case 3 of Chapter 17) has both layers: the economic structure of the regional carrier was misalignment and the absent stop authority was dilution.

Against the invisibility of maintenance (Chapter 29): unclaimed maintenance debt often has dilution as its cause. The Westminster restoration (po-005) shows both faces, with responsibility scattered

across two Houses and several bodies as the dilution and the debt itself as invisible maintenance. Classify by the principal contradiction: repair the responsibility structure first, and the debt will reach the agenda.

Against information attenuation up the hierarchy (Chapter 16): attenuation is vertical distortion and dilution is horizontal dissipation. In an organization that has both, bad news neither travels up nor gets claimed sideways, and Chapter 16's direct channel and this chapter's single accountable person are a pair of parts that have to be fitted together.

18.7 Five Self-Checks

1. Which of your organization's recent processes ran a gatekeeping signature chain longer than three? Can you write down the specific items each signature verified?
2. For the matter you are responsible for, is unilateral stop authority in your hands? Have you used it? Who paid the cost when it was used?
3. Take one shared object that spans departments, whether equipment, data or process, and write its four-party register. If you cannot write down the assessor and the party that accepts the effect, who owns it now?
4. In the last committee resolution you took part in, was the executor named in the text of the resolution?
5. After an incident, does your organization call the repair meeting or the responsibility meeting first? Where is that order written down?

Chapter 19 Why the Truth-Teller Goes First | Whistleblower Suppression and Retaliation

> What this chapter covers: how the alarms inside an organization get muted one at a time; three shapes of muting, three start criteria, four cases, and the design rule that protection has to be carried from outside. When to turn back: before someone brings you a problem after first scanning the meeting room; when you design a reporting or dissent channel; when you want to speak and hesitate. Estimated reading time: 50 minutes.

No organization announces that truth is unwelcome here. It needs to do one thing: make the last person who spoke up pay a visible price. The price travels through the organization in a few weeks. After that the alarms go quiet one by one, and nothing has been removed. The quiet is the sound of nobody pulling them. When the next accident arrives, the investigation report will carry a finding that has become boilerplate: the risk had previously been raised by staff. This chapter is about the years between that sentence and the accident.

19.1 What It Is

Definition: when the cost of raising a problem is higher than the cost of silence, an organization's alarms get switched off one at a time. The problem is ignored, then the person who raised it is pushed to the edge, then the person is gone. Three shapes belong in one chapter. In the first, the raiser is retaliated against, which is whistleblower retaliation in the narrow sense. In the second, the truth is procedurally shelved: raised, minuted, never handled. In

the third, the truth is censored before it leaves the mouth, the commonest shape, with a record of zero.

The third shape explains why evidence here is hard to gather by nature. A muted alarm leaves no record that it ever rang.

Three start criteria. Any one of them means the mechanism is already running.

1. The last person in the organization who raised a sharp question in public is worse off: reassigned, marginalized, departed, or with a performance record that slipped.
2. Internal reporting and dissent channels exist, and have gone unused or opened no case for a long stretch. Zero does not mean no problems. It means nobody speaks.
3. A class of problem is tied to one leader's reputation, and that class of problem has become undiscussable here.

19.2 How It Grows

Three generation conditions. Condition one: performance is rated by the superior alone. When the cost of speaking is priced by the person spoken about, the price rises. The power to price is the mute button. Condition two: reporting has no outside receiver. When the intake officer and the accused sit inside the same system, the channel becomes plumbing that returns the problem to its source. What the intake officer has to balance is not the right and wrong of one problem but his own position between two parties. Condition three: the problem is bound to the decision-maker. The existence of the problem equals the failure of the decision, and the return on concealment rises with the depth of the binding. Chapter 13, narrative before evidence, gains enforcement power here: the narrative is already published, so the truth becomes testimony for the crime of lying.

The evidence base of this chapter needs saying plainly. Formal investigation reports almost never record the retaliation process, because recording it requires the person to speak, and speaking is the act being punished. This is the deepest layer of Chapter 2's record bias. Retaliation leaves no text. What it leaves is the outcome of personnel movements.

The micro-structure of condition one, the pricing power, deserves one more cut. When the superior alone rates performance, every utterance by a subordinate draws on an invisible deposit: the superior's judgement of whether this person is easy to work with. The withdrawal rules are tacit knowledge in the organization. Which sentence costs how much is learned by every member in the first year. Tacit knowledge of this kind spreads fast, and the spread of workarounds in Chapter 15 repeats here, so the whole organization shares one price list without anyone stating it. The list can be read from two external indicators: the dissent rate in meetings, lower the higher you go, and the wording used for bad news, which is Chapter 16's attenuation curve. In an organization where both are deteriorating, the list is already priced high, and any slogan encouraging candour prints more currency without changing the price. Behaviour follows price.

Condition three, the reputation binding, comes with an amplifier: compound interest on time. The longer a problem stays bound to a decision, the higher the cost of opening it, which is Chapter 27's escalation in its reputational form, and the higher the marginal return on silence. Case four shows the terminal state. A CSB recommendation was refused procedurally, and years later the same class of recommendation went out again because of an explosion. The remediation cost saved by the first refusal and the accident cost after the second fall into two tenures and two assessment systems. The personal ledger of the person who did the binding stays positive the whole way. The organization settles its ledger

once, at the tail. That is the entire secret of the compound interest on binding.

Procedural shelving needs a layer of its own, because it is the standard craft of the second shape. The tool kit holds four pieces. Referral to study passes the matter to some department for study, with no deadline. Merger establishes that the problem is already under consideration in another programme. Conditional start waits for a precondition that may never hold. Filing keeps the record and thanks the person for raising it. All four are defensible. Study is prudence, merger is coordination, conditions are robustness, filing is respect. The tell is time. None of the four carries a due date, so the problem enters a state with no expiry.

Between the second shape and the third sits a transitional state worth naming: public thanks. The person who raised the problem is praised in public and the problem is filed in public. The next time the same person raises something, the answer is that the matter raised last time has already been handled. Praise completes the muting at that point, because it marks the person as having been heard. The cost of later utterances goes up instead of down, since a second raising now reads as a failure to understand the bigger picture.

Beyond the three main shapes, retaliation has a set of sub-shapes worth listing, because they leave personnel records entirely clean. Sub-shape one is loss of opportunity: a key project, a promotion nomination, a posting abroad, lost with no document produced, and only frequency can be counted. Sub-shape two is information isolation: the person stops being pulled into key groups and meetings, always for a reason that holds, that he is too busy, that the topic does not concern him. Sub-shape three is social cooling: the density of corridor and lunch encounters drops, with nobody organizing it and nobody announcing it. The three together do the work of a reassignment, at an order of magnitude more difficulty to

prove. The only detection instruments they share are frequency and time series: the allocation table for opportunities, the meeting attendee lists, the density of communication. That is why signal four in Section 19.4 says only that the name disappears. The name usually goes before the job title does. The marker of this transitional state is the separation of thanks from handling: the thanks are public, the handling has no follow-up, and a second utterance from the same source is turned back with a reference to the first round of thanks.

The defence is the one used throughout this book, deadlines and a ledger. Thanks stay thanks, the ledger opens an account per problem, and a second utterance in the same account automatically raises priority. The hook on the craft is a deadline and nothing else: any shelving action carries a review date, and a review that is overdue counts as unhandled. The hook already appeared in Chapter 11, in the three-state register's rectify-within-a-deadline state. Here it is recast for the oversight recommendation.

The return-flow dynamics of the intake channel are the core of condition two and deserve the same treatment. The intake officer faces two relationship networks rather than one question of right and wrong. The reporter and the accused are both inside his network, and he is asked to rule between them. The rational officer's optimum is usually to return the problem to the accused for self-inspection. That completes the procedure, since the channel holds a record, and it starts no feud with either side. The net effect is a channel that pipes problems to the people who make them. The organizational version in case four is structurally identical: the party that answers a recommendation is the party the recommendation is about, and the rating power sits with the object. The defences are identical too. Separating intake authority from the authority under review, which is Device 1, is called external intake in the personal case and a review mechanism in the organizational

one. This is also why the four cases in this chapter record the silencing of alarms rather than the retaliation process itself. Thin evidence is a measurement of how well the mechanism is running.

The three shapes get harder to prove in ascending order, and the order decides the audit method. The first, retaliation, leaves personnel records: reassignment papers, rating outcomes, departure dates. The reason column always says something else, but the timelines can be laid side by side. The second, procedural shelving, leaves intake records: cases opened, implementation rates, the phrase held for further study in the minutes. These are produced on a schedule, and the only difficulty is that nobody counts them. The third, self-censorship, leaves nothing. It happens inside the speaker, at the moment a sentence is half spoken and then withdrawn. It is the only one of the three that can be measured solely by proxy: a channel at zero use, zero dissent in meetings, word frequencies in exit interviews. The detection tools therefore sit on three different desks, personnel data, ledger data, and word-frequency and usage data. The six signals in Section 19.4 are the merged list from those three desks.

The cost-benefit structure of truth-telling is worth calculating once, because it drives all three shapes. The cost is immediate, certain, and carried by the individual: social price, career risk, damaged relationships. The benefit is distant, probabilistic, and enjoyed by the organization: a risk removed, an accident that did not happen. Individual rationality requires the benefit to land on the individual. The organization books it collectively. That mismatch is the first mover of this chapter's mechanism, and Chapter 17's frame goes back to work here: writing the externality back into the assessment is the economics behind every form of protection. The evidence from the other direction supports the same frame. In organizations where truth carries a price, with rewards for finding problems, public thanks, and promotion open to

dissenters, all three shapes recede. Change the price and behaviour changes. The prevention in this chapter is, in substance, a price list.

The price list needs its storage media named, because muting spreads through cases rather than through policy. Organizational memory has three storage points. The first is the person: his experience travels outward through exit interviews, private warnings and industry reputation, and one retaliation case travels further than the organization expects, especially in a small industry. The second is the witness: watching a colleague being handled produces a more conservative inference than the colleague's own account, because witnesses price the worst case. The third is the handover from old staff to new: the part of induction that covers what not to say is the oral edition of the price list. Three storage points mean the organizational cost of one retaliation is not one person's job. It is the price of speech across the organization for several years. A protected instance of candour has the same three storage points, and the price is booked in both directions, so the cost and benefit of this chapter's devices read at compound interest.

19.3 Four Domains, Four Shapes

Case One (Aviation): A Sense of Wrongness, Left Unsaid

av-017.

When and where: around 13:45 on Thursday, 8 August 2024 (Japan time), Kobe Airport (神戸空港), Kobe City, Hyogo Prefecture, Japan, on runway 27, slightly before the centreline marking. The aircraft had taken off at 13:28 from the W helipad at the same airport (p. 1, p. 3).

What and who: a Eurocopter EC135P2+ rotorcraft belonging to Hirata Gakuen (学校法人ヒラタ学園), registered JA824H, manufac‑turer's serial number 0941 (p. 1). The captain in the right seat was 68, held a commercial rotorcraft licence, and had 11,749 hours 51 minutes total flight time. The left seat held a maintenance engineer flying along, filling in a checklist and not flying the aircraft (p. 1, p. 3). The aircraft carried no airworthiness certificate that day and flew under the proviso to Article 11, paragraph 1 of the Civil Aero‑nautics Act as a pre-inspection test flight (p. 1). The landing clear‑ance was issued by Kobe Airport control (p. 1, p. 6).

What they were doing: the flight was an equipment check. It left the W pad at 13:28 and flew to a point about 15 kilometres (8 naut‑ical miles) northwest of the airport for a radio transmit and receive test (p. 2). Around 13:38 the captain asked for landing instructions and was told to report on the right downwind leg for runway 27. Around 13:41, as the aircraft was about to enter the right down‑wind, the controller redirected it to the C helipad and passed the position of another helicopter, the preceding aircraft, on final. The captain read the instruction back and reported the preceding aircraft in sight (p. 2).

Once the preceding aircraft had landed and vacated the C pad, the controller issued "CLEARED TO LAND AT C-HELIPAD 27". The captain read it back and turned base (pp. 2–3). Flying the circuit, he remembered that on the previous day's flight the downwind heading specified in the school's Subject Implementation Guid‑ance had been slightly off, and decided to fly the route once to es‑tablish the correct heading. In his statement he said he then treated the flight as a good chance to sharpen his skill, and decided to fly precisely to the figures in the guidance (pp. 2–3). The engin‑eer filled in the checklist and listened to the radio (p. 3).

For roughly two months up to a week before the event the captain had barely flown. After returning to flying he had made 27 landings

by the previous day, and only one of them was flown by him onto the runway. That one approach was undershot, and the circuit was not flown to the specified figures (p. 3). Whether helicopters used the pad or the runway was the controller’s decision, taken on the state of scheduled passenger traffic (p. 3).

The decision point: this case has no meeting room and no signature block. The traceable moments hang on radio calls, one checklist, one internal school document, and one manoeuvre.

Moment one, 13:38, the captain asks for landing instructions. He expected the short final marked (1) in Figure 2, straight back to the pad, which is the airport’s usual practice outside training flights. Because the preceding aircraft was already on final, the controller gave the circuit approach marked (2) (p. 3). After that call the aircraft had one extra circuit over the airport, and with it a stretch of time in which to think about something else.

Moment two, 13:41, the controller redirects the aircraft to the C pad and the captain reads it back. The read-back was correct (p. 2). The captain stated that he never heard “CLEARED TO LAND AT C-HELIPAD 27” as a runway clearance, and that what he understood while reading back was the C pad (p. 4). The Kobe convention appends 09 or 27 to a pad name to indicate the approach direction, and the 27 in that phrase is not the touchdown point (p. 4). The touchdown error did not happen at this second (p. 4).

Moment three, mid-circuit, the captain decides to fly faithfully to the figures in the Subject Implementation Guidance. In his statement he placed that decision alongside the good chance to sharpen his skill, and said that landing precisely to figures is mostly used for touch-and-go training on a runway (p. 3). No second person took part and no document was produced. On the return leg of a test flight, the figures in a training document took priority.

Moment four, turning final, the aircraft did not line up with the parallel taxiway where the C pad sits. It made its final turn toward runway 27 (p. 3). The report's analysis: while flying to training figures, the approach's sense of target became the runway, and touchdown awareness shifted from the helipad to the runway (p. 4).

Moment five, touchdown around 13:45, slightly before the centreline marking of runway 27 (p. 3). Between 13:41 and touch‑down the report records no further control instruction (p. 3).

Not one of the five moments left a signature. Two of them happened with only two people in the cabin, and the second of those had a checklist in his hands (pp. 2–3).

What was visible then: from 13:28, the W pad as the departure point, the two people on board, the checklist in the engineer's hands (p. 2). Known before takeoff: no airworthiness certificate, this was a pre-inspection test flight (p. 1). At 13:38, the instruction to report on the right downwind for runway 27 (p. 2), and the fact that the aircraft had been sequenced behind the preceding aircraft and would fly a circuit rather than a short final (pp. 2–3). At 13:41, the redirect to the C pad and the preceding aircraft's position (p. 2). Then "CLEARED TO LAND AT C-HELIPAD 27" and the read-back (p. 2). The preceding aircraft landed at the C pad and vacated, so the pad was free (p. 2). Mid-circuit, the previous day's finding came back: the downwind heading in the guidance was off (p. 2). Outside the window, five usable touchdown surfaces: run‑way, W pad, C pad, E pad, grass area (p. 3). Weather fine, wind southwest at 10 knots, visibility good (p. 1). Both men could hear the exchange and understood it differently (p. 3). The engineer heard the clearance as the C pad, and therefore assumed the touchdown point was the C pad (p. 3). The two men did not hold the same information. On the captain's side: the C pad sitting on the parallel taxiway, the threshold markings of runway 27, the T5

taxiway intersection, the gap the preceding aircraft left when it vacated (p. 3, Figure 2). No third person in the cockpit, and no rule at the time requiring a touchdown-point cross-check (p. 4). On the controller's side: three visual acquisitions, when the clearance was issued, when the aircraft turned final, and when it crossed over taxiway T5 (p. 3).

Information that arrived only afterwards. The captain learned of the wrong touchdown after landing, back at the office, from a telephone call from the aviation bureau (p. 3). Touchdown awareness being replaced is the report's inference (p. 4). The claim that correction was possible only in the stretch after the final turn, while descending to the touchdown point, and only by the captain himself, is likewise a retrospective judgement (p. 5). The count of 14 cases over 18 years in which a correct read-back still departed from the instruction was also compiled after the fact (p. 4). The controller's judgement that at three visual points he could not see that the aircraft meant to land on the runway also comes from a retrospective interview (p. 3).

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 13:45 the aircraft landed on runway 27. The report records no injuries and no damage to the airframe (p. 1, p. 3). After confirming the state of the area around the C pad it had cleared, the controller looked at the aircraft and found it about to touch down on runway 27, too late to issue a correction (p. 3, p. 5).

The captain himself learned that the touchdown point differed from the clearance only back at the office, on the telephone with the aviation bureau (p. 3). The report gives one cause: the captain's attention was on flying accurately to the figures, the approach's sense of target was the same as in touch-and-go training, touchdown awareness moved from the helipad to the runway, and the aircraft landed on the runway (p. 5). Back to the three questions. The commonest answer to the first is to confirm the touchdown point once more before landing. In this case that action costs one word: saying the C pad while reading back. The item most likely missed under the second question is the heading error in the Subject Implementation Guidance. It looks like a small error in a document, and the handling was to have the captain fly the route himself to confirm it. The document was never amended. The checkpoint for the third question falls after the read-back at 13:41 and before the final turn: check whether the touchdown point named in the clearance matches where the nose is pointed, and if it does not, do not continue the approach.

Options ruled out: the report records no discussion of options in the crew. The four items below were available at the time, with the reason each did not happen. One: go around or break off the approach. There was no trigger condition, the pad change had been read back correctly, and the rules did not require a touchdown-point cross-check before the final descent (p. 4, p. 5). A go-around would also have required noticing the wrong heading, and noticing is the missing link. Two: the engineer speaks up on the spot. His statement says he had no confidence in his knowledge of control communications and thought he had misheard, so he said nothing to the captain (p. 3). His only basis was the gap between the clearance he heard and the heading he saw. Three: the controller issues a correction. At none of the three visual points did he see that the aircraft meant to land on the runway, and by the time he could confirm it the aircraft was about to touch down (p. 3, p. 5). Four:

the captain's final-approach call. He stated that when flying aeroplanes he calls the landing runway together with the flap setting and the gear down, and that flying this school's helicopter requires no gear confirmation, so the final-approach call was gone (p. 4). The trigger for the call disappeared along with the item, and nobody moved it to the touchdown point.

Early signals: (1) Almost no flying for about two months up to a week before the event, then 27 landings by the previous day, of which one was a runway landing flown by the captain himself (p. 3). This was buried by the total, 27 flights already made, and the 27 were never broken out by touchdown surface. (2) That single runway landing came in undershot, with the circuit not flown to figures (p. 3). It was read as one deviation inside a recovery. (3) The previous day's finding that the downwind heading in the Subject Implementation Guidance was off (p. 2). It was buried by the decision to fly the route once to confirm, and the document was not amended, so the correction moved from the document into one person's memory. (4) The flight was a test flight, and the school's habit was pads for test flights and the runway for training, with both touchdown surfaces sharing one piece of airspace (p. 3). The label test flight outweighed where the nose was actually pointed. (5) The clearance wording "CLEARED TO LAND AT CHELIPAD 27" ends in 27, which by Kobe convention only marks the approach direction (p. 4). The phrase runs the pad name and the direction together, and heard quickly it leaves only the 27. Convention outweighed the literal text, and nobody read it as touchdown information. (6) In 18 years, 14 cases of a correct readback followed by a departure from the instruction, 5 of them arrivals (p. 4). They were treated as a collection of individual cases, never converted into a procedure, and never entered any post-readback check.

The point of escalation: mid-circuit, when the captain decides to fly to figures, and then, without lining up on the parallel taxiway, makes the final turn toward runway 27 (p. 3). The step from something is slightly off to there is no way back is this one. In Figure 2 the approach line to the C pad and the approach line to runway 27 separate out of the same piece of airspace, and no procedure divided flying to the pad from flying to the runway into two different tasks (p. 3). By this point height and distance had entered the band where only the captain could notice and correct, and the report states that the correction window lay after the final turn, in the descent toward the touchdown point (p. 5). Inside the aircraft, this step had no signatory and no second person was consulted. Outside it, the report's record of all three visual acquisitions is that the controller could not see the aircraft meant to land on the runway (p. 3), and by the time he could confirm it the aircraft was about to touch down (p. 5). The escalation took the form of an autonomous decision after a read-back, leaving no trace in any document, recoverable afterwards only through interviews. The school later wrote into its rules that an approach is not continued when the touchdown point cannot be established accurately (p. 5), which supplies exactly the gate this step lacked. The rule puts the checkpoint before the final descent begins.

Failure chain: technical layer. On this aircraft type at this school there was no touchdown call on final approach: the captain stated that flying aeroplanes he calls the landing runway with the flap setting and gear down, and that this school's helicopter needs no gear confirmation, so the call was dropped (p. 4). The left seat held a maintenance engineer whose duty was the checklist (p. 3). The C pad sits on the parallel taxiway, running alongside the runway (p. 3). Organizational layer. The airport divides five touchdown surfaces by use: runway for training flights, pads for test flights, E pad for visiting aircraft, grass area for hovering (p. 3). The division is held by habit, and no document pins down which surface this

particular flight should use. The school's Subject Implementation Guidance carried a known heading error, found the day before, handled by having the captain fly the route himself (p. 2). The control side treated a correct read-back as proof the instruction had been conveyed, and the report lists 14 such cases over 18 years for that reason (p. 4). Incentive layer. Two people sat in the front, a captain and a maintenance engineer. The one who might have spoken was not a flight crew member, and the one to be corrected was the captain. How a clearance is read was the captain's to interpret. The engineer's statement is one sentence of self-doubt: no confidence in control communications, a belief that he had misheard (p. 3). On the control side attention went to aircraft that read back incorrectly, and the aircraft that read back correctly was seen at three visual points without being warned (p. 3, p. 5). The boundary break sits between the organizational and incentive layers. The procedure put the entire weight of confirmation on the single act of read-back, and a read-back proves the words were heard, not that the touchdown point was seen. No step then required a second look out of the window.

Minimum counterfactual: at 13:41, while reading back "CLEARED TO LAND AT C-HELIPAD 27", the captain says the pad name in full and lines up on the parallel taxiway accordingly. One action, one word, placed before the touchdown error occurs. It needs no luck and does not depend on the engineer overcoming his self-doubt. It changes nobody's position and asks nobody to admit fault first. The cost comparison: after this case the school amended its rules, ran TEM and CRM training for pilots and accompanying maintenance engineers, and added mutual confirmation for critical clearances (p. 5). The three together are an institutional expense. The counterfactual costs one extra word in a read-back. The rule the school wrote runs in that direction: before the final descent for landing begins, confirm that the control instruction, including the landing surface, matches the surface

about to be used, and do not continue the approach when this cannot be established accurately (p. 5).

Outcome and sources: no injuries, no airframe damage recorded in the report, and the report records no other aircraft on the runway at the time, so the consequence stops at landing somewhere other than the cleared place (p. 1, p. 3). The school amended its rules, ran TEM and CRM training for pilots and accompanying maintenance engineers, and introduced mutual confirmation of three critical clearances whenever a communicable person sits in the co-pilot's seat. Kobe Airport control changed to monitoring through the final stage of landing and asking for an instruction to be repeated whenever doubt arises (pp. 5–6). Primary source: Japan Transport Safety Board (Aviation Division), Aircraft Serious Incident Investigation Report AI2025-8-1-JA824H, decided 11 September 2025 (Reiwa 7), Hirata Gakuen, Eurocopter EC135P2+ JA824H, 8 August 2024 (Reiwa 6) around 13:45, landing on the runway at Kobe Airport. Six pages, downloaded and read in full for this book. Link at Appendix B, av-017 (grade A).

The counter-condition in case one deserves half a step further, because it is the most valuable sample type in the study of self-censorship. Silence in a setting with no conflict of interest rules out retaliation as an explanation and pushes the mechanism down to the stratum of pure social calculation. What happened in the engineer's half second can be recorded only by its result, that he said nothing, but the calculation can be reconstructed. The expected cost of speaking was a step down in the captain's estimation if he had misread the situation. The expected cost of silence was a wrong touchdown point if he had read it correctly. The difference between the two lies in visibility: the cost of being wrong shows up at once, as embarrassment, and the benefit of being right shows up late, as an accident that did not happen. Social calculation systematically overweights the cost that is visible now. That is

the cognitive floor under the first shape, self-censorship. The investigation finding the report lists, no reconfirmation before the approach, is a patch at the structural layer. The cognitive layer has exactly one patch: bring the price of speaking to zero, where being wrong costs nothing and speaking itself is recorded. Device 4 in Section 19.5, the chair's silence rule, governs that price.

Case Two (Public): Seventeen Deferrals of an Outside Truth

po-002. The electronic health record modernization programme at the United States Department of Veterans Affairs. The GAO report, GAO-26-109393, has been read and checked page by page for this book. GAO, as the statutory oversight body, issued 18 recommendations, and as of August 2026 only 4 had been fully implemented. The case widens the definition of a truth-teller. GAO sits outside the organization, needs no shelter from it, has congressional backing for its recommendations, and publishes its reports in full. Even so, the landing rate stayed at one fifth for years. If truth configured like that can be shelved procedurally, the fate of internal truth needs no thought experiment.

The decision point is the budget moment each year when the recommendation list is sorted by priority. On the desk: the recommendation list, the deployment pressure of that budget cycle, and the resource cost of implementing recommendations. Shelving needs no decision here, only an ordering: put implementing the recommendations after deployment progress, year after year.

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?

> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the gap between oversight and remediation was recorded year by year and became input to the next round of oversight. Primary source: GAO, GAO-26-109393, 2026. Link at Appendix B, po-002.

The sorting mechanism of case two can be written as a general rule: shelving needs no decision, only an ordering. The rule applies widely: to budget ordering, with implementation behind deployment; to agenda ordering, with risk items behind business items; to hiring ordering, with safety posts behind capacity posts. None of the three produces an accountable decision, since nobody votes against implementing a recommendation, and the mechanism therefore escapes ordinary governance. You cannot oppose a decision that never happened. The observable handles are the outputs of the ordering: landing rate, share of agenda, ratio of posts. Put the three numbers side by side each quarter and the real priority becomes visible. GAO runs that comparison once a year; a reader should run it inside his own organization at least once a quarter.

In public administration the sorting rule has a second, interagency reading: the relay step applied to oversight reports. GAO reports go direct to Congress and are published in full, so the truth of oversight bypasses relay up the line. NAO reports go direct to parliament and are public in the same way. The effectiveness of both oversight systems rests on one institutional variable, direct delivery and publication of what oversight produces. Against the internal sorting table in case two, where deployment pressure outweighs implementation, directness and publication move that table into daylight. Sorting can continue, but it happens in front of the principal. The same design applies to corporate oversight functions, internal audit and the supervisory board: the report

goes straight to governance and is available in full, with no summary step in between. The summary step is the commercial edition of the relay step.

Case Three (Healthcare): A Report Twelve Days Late

me-005. Emergency imaging in the English NHS. The HSIB investigation, Failures in communication or follow-up of unexpected significant radiological findings, 2019, read from an archived snapshot checked for this book: a chest X-ray report raising the possibility of a lung mass was issued 12 days late, follow-up of the unexpected finding had no owner, the patient was diagnosed about three months later and died in January 2018. Nationally, 41 serious events of the same kind were recorded that year. Truth met no retaliation in this case. It met a queue. A report flagging possible malignancy queued in the system for 12 days, and each link in the chain handled something more urgent by its own standard. Muting an alarm does not require punishment. Priority ordering is enough. This is the mildest and commonest shape of the mechanism in this chapter.

The decision point is any one of the twelve days in which the report moved back a day at a time in the queue. On the desk: the length of the queue, the priority rules of each link, and the default ordering that treats important but not urgent as last.

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not “pay more attention”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the diagnosis was delayed, and 4 safety recommendations were issued. Primary source: HSIB, 2019. Link at Appendix B, me-005 (archive link, grade A–).

Chapter 16 reads the queue in case three as information attenuation, where priority ordering dilutes the urgency of a finding. This chapter reads its silent side: nobody in the queue shouts. In the imaging department's queue the reports are not the only thing waiting. The people who know the report is in there are waiting too, since the referring doctor remembers writing the request, and in twelve days not one chasing call was made. That tells you the cost of chasing, which reads as looking amateurish or disturbing a colleague, outweighed the value of the finding. The system does not punish chasing. The queue culture prices it. The mechanism that makes chasing free sits on the extension of Device 4: for any overdue item, a single enquiry from any role generates a record and raises priority automatically, and the enquirer's name appears in no assessment. Ask once and the price list thins by one entry.

An operational endpoint tells you when the price list has grown thin enough: the highest-paid level in the organization has publicly admitted one of its own errors of judgement, and paid no visible price for it. That criterion is harder than any policy clause because it tests the top of the list. What a senior admission costs sets the ceiling for the whole organization, and while the ceiling holds, every discount below it is temporary. The way to use the criterion is plain. It needs no measurement, only an event to wait for. After each senior admission, watch the organization for a month, at the dissent rate and the wording of bad news, and the movement is the true position of the price list.

Beyond reading the number, senior people have one move only they can make: explicit repair of costs already paid. For the most recent person damaged for speaking, whether reassigned, mar-

ginalized or gone, the senior move is to correct the assessment of him in public, restore his participation, or at minimum cite his contribution in a policy review. Two requirements attach. It has to be explicit, visible to the organization rather than a private reassurance, and it has to attribute, saying clearly what the cost was paid for instead of pushing it onto process. A full refund is not needed. The organization acknowledging that the earlier pricing was wrong is enough, because the price list is an expectation, and changing an expectation needs no full refund.

Case Four (Chemical): Five Recommendations Refused

ch-009. The OPC Polymers resin plant in Columbus, Ohio, 8 April 2021. The CSB investigation page has been checked for this book. It records a leak from a low-pressure vessel that produced a vapour cloud explosion, 1 employee dead and 8 taken to hospital. The CSB noted in the investigation that low-pressure vessels below 15 psig fall outside the main ASME and API codes, which leaves a blank in the regulatory field of view. One further fact carries more weight. Before the accident the CSB had issued a safety recommendation about hazards at the same plant, and the company's response was rated Unacceptable Response. Refusing an external recommendation is the most formal silence an organization can give to truth. There is no rebuttal and no argument, only a procedural entry that says unacceptable.

The decision point is the moment the CSB recommendation arrives and the company writes its response letter. On the desk: the cost of remediation, the boundary of a CSB recommendation's legal force, which is none, and the legal distance between being named and being obliged to change.

> **Stop here. Decide before you turn the page.**

> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the vapour cloud exploded on 8 April 2021, 1 employee died and 8 were taken to hospital, and after the accident the CSB issued recommendations on the same class of problem again. Primary source: CSB, Yenkin-Majestic Resin Plant Vapor Cloud Explosion and Fire, 2021. Link at Appendix B, ch-009.

Case Four, Full Card: The Most Formal Silence

ch-009.

When and where: 8 April 2021, Columbus, Ohio, United States, the OPC Polymers resin plant, part of Yenkin-Majestic.

What and who: a resin production facility, with the US Chemical Safety Board (CSB) as the investigating and recommending body.

What they were doing: a leak from a low-pressure vessel produced a vapour cloud explosion, killing 1 employee and sending 8 to hospital. Two structural facts hold the case up. Low-pressure vessels below 15 psig fall outside the main ASME and API codes, which leaves a blank in the regulatory field of view, and equipment inside that blank sits where the compliance system cannot see it well. Before the accident, the CSB had issued a safety recommendation about hazards at the same plant, and the company's written response was rated Unacceptable Response, with no rebuttal of the technical content and a procedural door closed in the response.

The decision point: the moment the CSB recommendation arrives and the company writes its response letter. On the desk: the estimated cost of remediation, the boundary of legal force behind a CSB recommendation, which carries no enforcement power, how peer firms have generally handled earlier recommendations, and the experienced probability that regulatory resources will not return.

What was visible then: the full recommendation, the mapping onto hazards at this plant, the cost of remediation, and the coverage blank in the codes as a technical fact. Information that arrived only afterwards: the full chain from leak path to ignition source, and the overpressure footprint of the explosion, both of which appear only in the CSB's reconstruction.

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the vapour cloud exploded on 8 April 2021, 1 employee died and 8 were taken to hospital, and afterwards the CSB issued recommendations on the same class of problem again. Back to the three questions. The commonest answer to the first is to adopt the recommendation in part, though partial adoption leaves the same procedural record as refusal, the response-letter rating, and the structural difference is only the length of the remediation list. The item most often missed under the second question is the coverage blank in the codes. The 15 psig line sorts equipment into two regulatory worlds, and risk accumulates on the

blank side with no external alarm. The right place for the check-point under the third question is the rating system for responses to recommendations: make unacceptable a state with a remediation deadline and a review date, rather than the end of a response letter. Primary source: CSB, Yenkin-Majestic Resin Plant Vapor Cloud Explosion and Fire, 2021. Link at Appendix B, ch-009.

Options ruled out: adopt the recommendation in full and remediate on schedule, which the cost basis does not allow; mount a technical rebuttal, which consumes technical and legal resources and would be recorded; bring the low-pressure vessels under an internal inspection programme stricter than the codes require, which is voluntary, with no external pressure behind it.

Early signals: (1) The equipment list inside the code's coverage blank was never drawn up separately, buried under the reason that the codes do not require it. (2) The CSB had already named hazards at this plant, buried under the reason that recommendations carry no force. (3) Internal inspection covered the low-pressure side less thoroughly than the code-covered side, buried under the reason that following the code is enough.

The point of escalation: after the response letter was rated unacceptable, no external mechanism compelled a review, and the hazard list stayed where it was. The escalation took the form of a refusal with no consequence, entered into the organization's library of experience with regulators.

Failure chain: technical layer. A leak from a low-pressure vessel formed a vapour cloud that ignited. Organizational layer. Equipment inside the regulatory blank carried risk with no external alarm, and internal inspection was lighter than on the code-covered side. Incentive layer. Remediation cost was immediate and certain, explosion risk was delayed and probabilistic, and refusing a recommendation carried no legal consequence. All three layers broke at once. At the boundary: the unacceptable rating, an or-

ganizational-layer signal, triggered no review, and the incentive layer carried no cost, so the alarm was muted in the most formal way available.

Minimum counterfactual: complete the remediation the CSB recommended, or at least bring the equipment on the blank side into the internal inspection programme with a record of review. One action, years before the event, at a cost far below the direct loss, 1 dead and 8 injured plus a plant shutdown.

Outcome and sources: 1 person died and 8 were taken to hospital, and the CSB issued recommendations again on the same blank. Primary source: CSB, Yenkin-Majestic Resin Plant Vapor Cloud Explosion and Fire, 2021. Link at Appendix B, ch-009.

Where it sits against this chapter's mechanism: this case pushes muting up to the organizational level, where a company can mute a regulator's recommendation as easily as a person can be muted. Personal muting runs on the price list; organizational muting runs on the boundary of enforcement power. The defences are isomorphic. Give the truth an enforcement mechanism, protection in the personal case and a review mechanism in the organizational one, and the price changes.

The four cases occupy one notch each on the scale of muting, and together they are the full spectrum of the mechanism. Case one is self-censorship, the third shape, cheapest and least recorded: the engineer's half second. Case three is queue muting, the mild edition of the second shape: twelve days and nobody chasing. Case two is procedural shelving, the standard edition of the second shape: external recommendations sorted and shelved year after year. Case four is formal refusal, the heavy edition of the second shape, closing the door on truth from outside the organization: the response-letter rating. From the lightest half second to the heaviest letter, the instrument of muting grows more formal and the cost of the problem grows more severe. The self-check path runs the other

way. Start with the heaviest, the recommendation-response ledger; then the middle, channels and landing rates; then measure the lightest, the dissent gradient. Formal muting leaves documents. Half-second muting leaves only a gradient. Heavy to light is also easy to hard.

19.4 What Shows Up Earliest

Six weak signals, each with the reason that usually buries it.

1. Nobody objects in the meeting, and the corridor afterwards is full of objections. Reason given: internal harmony.
2. The reporting channel has existed for three years with zero use and zero cases opened. Reason given: the culture is good.
3. The word culture appears at high frequency in exit interviews. Reason given: personal choice.
4. Names that once raised sharp questions stop appearing on the key meeting lists. Reason given: work assignments.
5. A class of problem has its own set of euphemisms in internal documents. Reason given: communication efficiency.
6. A leader's oral remarks get compiled into written discipline. Reason given: understanding the spirit.

Put the six signals back against the cases. Signal one, agreement in the room and objection in the corridor, is case one's self-check at meeting scale: the sense of wrongness in av-017 that never entered the cockpit conversation. Signal two, a channel at zero use and zero cases, is the mirror of case two. If external oversight lands one recommendation in five, a zero at the internal channel is the normal state in most organizations rather than an anomaly, so read the trend and the comparison, using other organizations' case baselines. Signal four, the disappearing name, corresponds to case four: after the CSB recommendation was rated unacceptable, the

hazards at that plant appeared less often in later documents. The problem was not solved. The vocabulary for raising it was.

Signal five, the euphemism, corresponds to case three, where an overdue report sits in the system under the label pending.

The six signals sort onto three desks by evidential difficulty, the same three desks as Section 19.1. The personnel desk holds signal four, and its reading is timelines laid side by side, the month a question was raised against the month a personnel change fell. One case proves nothing; a time correlation across three or more cases is a finding. The ledger desk holds signals two and five, read against a baseline: zero use against the baseline for similar organ - izations, euphemism against the language of public documents. The word-frequency desk holds signals one, three and six, read as trends: the gap between corridor topics and meeting topics, the drift in exit-interview vocabulary. Data from all three desks is available inside the permission boundaries of most organizations, which is the precondition for auditing this mechanism at all.

19.5 How to Prevent It

Device 1: protection carried from outside. Intake and investigation of reports are handled by a party independent of the organization, whether a regulator, an industry inspectorate or outside counsel, and the internal channel only passes things on. It does not adju - dicate. The criterion is fixed in writing: the party reported has no assessment, personnel or budget authority over the intake officer. The lesson of po-002 runs both ways. External intake guarantees that truth gets recorded, and does not guarantee that it gets im - plemented. Recording and landing are two gates, each with its own device, and the landing rate belongs to Device 3.

Device 2: put the questioning into a post. Chapter 13’ s counter-witness role, Chapter 11’ s ledger auditor, and this chapter’ s

dissent post all do the same thing: they write saying no into a job description, pay for it, give it authority, and bring it inside the protected range. The point of the post is that the cost of truth moves off personal reputation and onto role behaviour. A role that fails is a failure of judgement and can be reviewed. A person who speaks and fails has committed a political error, and a political error gets replayed rather than reviewed.

Device 3: an observable landing rate. Keep a line-by-line ledger of oversight recommendations, from internal audit, external audit and regulators, and publish it to governance every quarter. po-002 shows that a landing rate can be recorded year by year, which also shows it can be held to account. The landing rate is what the or‑ganization actually charges for truth, and if that number is not published, any published welcome for oversight is decoration.

The technical rules for publishing the rate have to be fixed as well, or the publication itself gets muted. Three rules, none optional. Rule one is line by line: one row per recommendation, no aggreg‑ation, because an aggregate hides the distribution and the items that fail to land are usually the important ones. Rule two is four states: closed, in remediation, replied, not started. Replied stand‑ing as its own state is the key, since it leaves the responded-but-not-done item with nowhere to hide. Rule three is by name: the owner of each recommendation is written to a post, not a department, because a department is a dilution unit, Chapter 18’s subject, and a post is the smallest carrier of responsibility. The publication cost of the three rules is one page a quarter, and that page is the whole of the organization’s good faith toward truth.

Device 4: the chair’s silence rule. This is the meeting-level micro-defence, the second sentence of the Chapter 39 script: announce that every point raised is recorded and not rebutted. The earliest rebuttal in a rehearsal or a postmortem is the smallest unit of re‑taliation, and it is the one most worth blocking first, because the

habit of silence it trains gets carried all the way into the reporting system.

Place the four devices on the cases. Device 1 on case four: review of the response goes to a third-party technical body, and unacceptable stops being a terminal state. Device 2 on case one: cross-checking in the crew was made a post long ago, which is aviation's counter-condition, and what is missing is extending post-based speaking rules to low-confidence signals such as a sense of wrongness. Device 3 on case two: the recommendation landing rate published quarterly to Congress, with the name of the owner written down. Device 4 on case three: any enquiry about an overdue queue item exempt from rebuttal, with priority raised automatically.

The four devices also have failure modes. External intake dies in an intake vacuum, where an outside body takes the channel and has no budget to investigate, and a complaint that goes into a black hole damages trust more than one that goes into a pipe. The patch is contractual intake deadlines plus investigation resources. The post dies in isolation, where the dissent post exists but sits far from the decision table and the last metre of the conveyor is cut. The patch is a standing agenda seat for the post. The landing rate dies in definition drift, where completed relaxes from remediation closed to replied. The patch is publishing the three states. The silence rule dies when the chair makes one exception, since the first rebuttal with no consequence ends the rule on the spot. The patch is Chapter 13's, with the chair's enforcement record entered in the minutes. All four patches do the same work: they turn the device from a promise into an observable state, and an observable state leaves retaliation nowhere to hide.

The scope of protection needs one boundary drawn explicitly: what is protected is the act of speaking, not the content of what is said. The device set does not guarantee that everything said is correct,

since false alarms are common, and guarantees only that raising a problem is not punished. Judging whether the alarm was right is later work for the ledger and the investigation. The boundary carries two meanings. For the person raising the problem, being wrong carries no liability, which removes the largest single source of self-censorship, and the engineer in case one needed exactly that sentence. For the organization, the duty to protect does not cancel the duty to verify, and verification runs alongside protection without conflict. With the boundary drawn, the devices cannot be dragged down by the two abuses: spreading false accusations under the name of protection, and retaliating under the name of verification. Every instance of either abuse becomes a new entry in the price list, which makes the boundary a survival condition for the devices rather than a legal detail.

Beyond the boundary, the depth of protection needs grading: question, challenge, report. The question level, case one's sense of wrongness, needs only meeting rules. The challenge level, opposing an established plan, needs Device 4 plus a ledger. The report level, alleging a specific violation, is where Device 1's external intake comes in. The reason for grading is cost matching. External intake is expensive, and applying it to everything floods the channel with trivia, which damages the protection that is genuinely needed. The sequence is simple to state: ask in the room first, then the ledger, then outside. A problem that has been through all three carries stacked protection. A problem that jumps levels falls back to an individual exception once the three-level system matures, and exceptions themselves run through Device 1's standard process.

19.6 Boundaries with Neighbouring Mechanisms

Against information attenuation up the hierarchy, Chapter 16: attenuation is the channel deforming, with the message distorted in

transit, and this chapter is the source running dry, with the message no longer produced. Which to check first? A channel carrying plenty of messages that arrive distorted calls for fixing the channel. A channel carrying no messages calls for checking the price first.

Against narrative before evidence, Chapter 13: where the narrative goes first, opposition is required to carry an overload of proof, and this chapter fills in who enforces that asymmetry and what it costs. Chapter 13's symmetric-evidence rule is where Device 2 lands in a meeting.

Against normalization of deviance, Chapter 11: the filing moves of normalization run smoothly largely because nobody stands up to demand the to-be-eliminated state. With the alarm fitted and ringing, normalization is slow. With the alarm muted, normalization runs at full speed. A ledger of the kind Chapter 11 describes will itself be muted, in an organization without the protection this chapter describes.

19.7 The Five Self-Checks

1. What is the position now of the person who last raised a problem publicly in your organization? If you do not know, go and find out. The act of finding out is itself a signal.
2. How many cases did your reporting channel open in the past three years? If the number is zero, whose assessment system does the intake officer sit inside?
3. The last time someone opposed you in a meeting, did you record the objection and follow it up?
4. What is the landing rate on the recommendations external oversight has given you? When did you last show that number to governance?
5. The piece of truth you are holding back, what price are you betting against? Who sets that price?

(This chapter: 12,407 characters.)

Chapter 20 The Slack of Success (Failure of Success, Sitkin)

> What this chapter covers: how a long accident-free run makes defensive spending shrink; three criteria that show the mechanism is running, four carriers that let it shrink without anyone deciding, and the four devices that keep a defence alive through the years when nobody needs it. When to turn back: at the budget review where someone proposes cutting the backup, the drill or the inspection that "has not been used in years"; or when you hear yourself call one of your own processes a formality. Estimated reading time: 50 minutes.

Every line of defence fights an unwinnable campaign for its own existence. The standby generator has not started in ten years. The drill has found nothing in three. The inspection process has never stopped anything from leaving the gate. On the budget sheet these items carry a single name: no output for years. Their real output is the events that did not happen, and a non-event enters no ledger anywhere.

Lyle Sitkin's argument in the organizational learning literature sits exactly there. Success weakens learning. What an organization learns during a good run is "this way of working is fine", not "this way of working has produced nothing because the defences are in place". The two lessons differ by a clause. Some years later they differ by an entire accident.

20.1 What It Is

Definition: the slack of success (failure of success) is the structural tendency of success to weaken learning. During a good run the organization treats defensive spending, meaning backups, drills,

inspections and redundancy, as cost with no output, and trims it or turns it into ritual. The cost of the shrink arrives late, and by the time it arrives the cut can no longer be traced to the decision that made it. The mechanism runs on the time axis. It is the organizational-learning version of margin erosion (Chapter 25). That chapter eats the reserve. This one eats the willingness to maintain the reserve.

Three criteria for onset. Any one of them means the mechanism is already running.

1. A defensive item has produced no visible value for a long time. There is no record of it ever catching anything.
2. It starts to be treated as surplus. A cutting proposal cites it as a reason, or in practice its execution has become a walk-through.
3. The first accident after the cut or the drift lands more than two years out. The causal chain has already broken in time, and nobody will join the ends.

The number in criterion 3 deserves a pause. Why two years? Budget cycles are counted in years, staff turn over on a two-year rhythm, and memory decays by the quarter. By the time the consequence of a cut appears in year three, the person who decided is gone, and the person who inherited the arrangement faces a condition that, as far as anyone there can tell, has always been the condition. Criterion 3 does not describe the size of the risk. It describes a structural guarantee that accountability will never land.

Sitkin's argument has two lines of context that matter here. What he describes is an asymmetry in organizational learning. Learning from failure is easy, because failure is concrete, vivid and has a party to attribute it to. Learning from success is hard, because success is diffuse, multiply caused, and offers no attribution point. The consequence of that difficulty is not that nothing gets learned. The wrong thing gets learned. The absence of consequence is at-

tributed to current practice rather than to the defences and to luck. The output of that bad learning is a stable false belief: our way of working is safe. Once the belief sets, a defensive item's identity slides from necessary cost to historical leftover, and every cutting proposal at the budget table becomes a market test of the belief. A belief that is never contradicted keeps appreciating.

The attribution problem around "nothing happened" needs one finer note. The output of a defence and the output of luck cannot be told apart in advance, and are hard to tell apart afterwards, because they share one form of evidence: nothing occurred. The only window for telling them apart is the defence's catch record, the drill that found a fault, the sample check that stopped a batch, the review that corrected an error. An organization whose window has closed can keep its record going only on luck, and nobody meters the consumption of luck. That is the deep logic of Device 1 in this chapter, the ledger for non-events. The ledger does not exist so a defence can claim credit. It exists so the organization knows how much of its safety is design and how much is coincidence. Both proportions have value. Mixed together they are worth nothing, because the design share gets cut by mistake and the coincidence share gets trusted by mistake.

20.2 How It Grows

Three generating conditions.

The good run has to be long enough. A defence is treated as necessary only while the memory of the accident is fresh, and memory decays faster than accidents recur. The gap between those two rates is the growing season.

The cause of the good run has to be unattributable. The organization cannot say how much of "nothing happened" was the defence and how much was luck (Chapter 11's two hypotheses:

margin to spare, or a good draw). If it cannot say, it cannot book the defence. An item that cannot be booked is an item waiting to be cut in any budget review.

Budget pressure has to recur. Something has to go every year, and the safest thing to cut is the line with no accident attached to it, because the argument is unanswerable in the room. It is true, and it cannot be refuted, unless the organization runs a system that books the non-events.

The consequence of slack has one precise property. It lands outside the decision-maker’ s tenure (Chapter 28), and the party it lands on is usually not the party who decided. That is what lets the mechanism repeat generation after generation inside one organ - ization, with each generation convinced it is the first.

Slack travels in four carriers, and budget is only the most visible.

The first carrier is budget: line items cut, easiest to observe. The second is attention: inspection slides from measurement to signing in, and the four years in case 1 are a history of attention slacken - ing. The third is skill: a drill slides from rehearsal to performance, which is the defensive side of Chapter 26’ s skill decay, and what is lost is the ability to do the task under pressure, not the ability to do it at all. The fourth is upkeep: standby equipment stops being powered up, switched over, turned over, which is the pump in case 3.

All four carriers share one property. They can shrink without a decision. Nobody has to vote for the cut. Busyness and a clean re - cord only have to act together for long enough. That property also sets the difficulty of the fix: fighting a shrink that was never de - cided requires mechanisms that are decided, a ledger, a test peri - od, a lookback date, and the whole function of those mechanisms is to turn unconscious drift back into conscious action.

The generational repeat has a further mechanism inside organizational memory. Accident memory is stored in people, and people move faster than accidents recur. The old hand leaves with the story of that time, and the story did not enter the file, or entered as one line. The newcomer faces a defence system that has process but no stories. Process can drift, because it specifies actions. Only a story commands respect, because it specifies the reason. The point of Device 4, institutionalized memory, is therefore not to write the story down. It is to write the story as a causal account a newcomer can use: what this defends against, what happened here, what happens if you cut it. One page against one resignation cycle is a poor ratio. It is also the only archive that can speak at a budget table.

The odds on that page can be narrowed further. Its real battlefield is somewhere else: the first ninety days of each new management team. A new manager has no memory of the defence system, and a first budget instinct runs toward cutting, which is the general preference of a new broom facing old accounts. Whether the page reaches the hand inside that ninety-day window decides whether the defence system survives each change of guard. The delivery mechanics are copied from Chapter 17: put the page in the standard attachments of the handover checklist, as a fixed column of the handover form. The incoming manager who signs for the handover signs for the page, and a signed receipt removes the excuse of not having seen it. A change of guard is a high-risk moment for a defence system (Chapter 28's tenure mechanism). A page in the handover pack is this chapter's seatbelt for that moment, at the cost of one procedure revision.

Beyond procedure revision, the page gets one gift of an update opportunity: after every relevant accident or industry event. An event makes memory briefly fresh, which is the gap described in condition one, and at that moment rewriting the page is cheapest

and most accurate, because what the defence prevents has just been demonstrated. Write the trigger into Device 4's terms: within one month of a comparable accident anywhere in the industry, the relevant page is reviewed by compulsion, and the old version is archived once the reviewer signs. The month of an accident is when memory is most expensive and most honest. What gets written into "what it prevents" in that month is the highest fidelity the page will ever hold. An industry has few such months in a year. Catch them and the page stays accurate across two changes of guard.

The industry-level tuition in case 4 has one last layer: industry memory and company memory are stored in different places. A company's "never had an accident" rests on its own operating record, which in that case was the operator's past safety record. An industry's "never had an accident" needs a failure-mode library built across companies. The first is narrow by construction, since one company's luck is not the industry's rule. The second is the evidence base a certification decision should stand on. Building the library is the job of an industry body, which most sectors already have in their safety organizations. The company-side duty runs both ways: submit, meaning report your own abnormal events to the library even when they had no consequence, and read, meaning consult the library before certification and before a design change. Two-way flow is the circulation system of industry-level defensive memory. A blockage in either direction, and most companies read without writing, distorts the library's coverage, and a distorted library then misleads the next company's certification judgement.

The last link in that circulation is a compulsory lookback after an industry accident, and the grounding in case 4 was one industry-wide lookback. Once a new failure mode is demonstrated, the library is updated and every company's certification assumptions

and inspection procedures are refreshed against it. The efficiency of the lookback depends on one quantity: the lag from event to procedure update. In an industry where the lag is measured in months, the industry layer compresses the slack window. In one where the lag runs to years, the second company is still working from the old assumptions, and the several similar events after case 4 are what that lag costs. The lag is a measure of industry gov - ernance, and shortening it needs no new technology. It needs the update process registered in advance: which procedures a library update triggers, who owns the rewrite, how many days are allowed for it, all written into the industry's own procedures so that when the event comes the update runs on the registered route.

The bill for the five cases can be settled side by side here. Case 1, four years of idling for one dangerous occurrence, with no injuries. Case 2, five days of retention and a second injury risk. Case 3, one vessel. Case 4, a fleet grounded for months and one injured fire - fighter. Case 5, two hundred million in deposits and a returned li - cence. The spread in scale is the point. Slack does not choose an industry and does not choose a size. It chooses time. Any organiz - ation whose good run is long enough and whose defensive ac - counts are unclear sits on the same spectrum, whether it runs a fishing boat or certifies a new aircraft type. The spectrum has no terminus, only the start of the next good run, and the four devices and three inserts in this chapter are the whole toolkit for making that run cost something different from the last one.

The procurement list is worth pricing for the finance reader. Device 1 is a ledger, costed in person-days. Device 2 is the live-testing regime, costed in equipment hours and labour, scaled to the length of the list. Device 3 is a lookback register, costed in pa - perwork. Device 4 is a page, costed in writing time and induction time. The four together usually cost less than one exercise. On the benefit side sit the five bills: one dangerous occurrence, one re -

tention, one vessel, one grounding, one licence. This is not a precise cost-benefit analysis. It is an order-of-magnitude sanity check. Between the annual cost of defensive spending and the single-event cost of a defence failing lie two or three orders of magnitude. What is genuinely worth arguing about at the budget table is never whether to buy these tools. It is who maintains their ledgers and their dates once they are bought, and that post is written into the second sentence of Device 1 in section 5: a ledger needs an owner, and the owner needs to be assessed.

Assess the owner on this chapter's own measures: the ledger's update rate, the on-time rate of live tests, the execution rate of lookbacks, the timeliness rate of page reviews. All four are process measures, and the defect of process measures, that done is not the same as done right, is covered by sampling, with Chapter 11's quarterly spot check auditing the same post once a year. The owner brings a second benefit, which is that responsibility for the defence system lands inside this chapter's criteria. Every device has a name, the name has a second line, a deputy, and the organization's defensive ledger stops being a set of dates with no owners.

Once the dates have owners, the defence system stands on the same management table as every other asset: equipment depreciates, software renews, defences carry dates. The managerial meaning of that table needs one closing note. It turns safety from an atmosphere into a set of manageable state variables. An atmosphere slackens in a good run, necessarily, because it is a function of attention. A state variable does not slacken, because it is a function of the ledger. All of this chapter's effort comes down to that one translation: moving defence from atmosphere to variable. In an organization where the translation is done, the slackening still happens during a good run, since that is what people are like, but the floor it reaches is held by the ledger. Attention can leave. Dates do not move themselves.

The successor is the other beneficiary of dates that do not move. A new management team facing a dated defensive table starts the cutting discussion from a different question: whether these dates can be afforded, rather than whether the items are worth keeping. The ledger converts the argument about a defence's existence into an accounting of maintenance cost, and an accounting of maintenance cost is always cheap for a rational manager. That also fills in one column of Chapter 28's bill handover list, the column headed which defences I cut, since the entries can be exported straight from the ledger by date. This kind of mutual parts supply between mechanisms is normal in the later chapters of Part III. As the slack chapter, this one supplies more parts than most: the ledger to Chapter 11, the lookback to Chapter 28, live testing to Chapter 22, the page to Chapter 30. That is why it is one of the most cited nodes in the book's network of countermeasures. The time property of a defence is needed by nearly every chapter.

The most cited node is also the one most easily read as a universal remedy, so draw its boundary clearly. The ledger, live testing, the lookback and the page defend against a defence dying slowly. They do not defend against a defence that was wrong from the start. A defence designed against the wrong risk, sited in the wrong place, or run at a frequency too low to cover the exposure will show up on the ledger as a long run of zero catches, and under Device 1's convention a zero has two readings, genuine calm or nothing being caught. This book offers no automatic way to detect the third reading, that the defence itself is misdesigned. What that reading needs is an industry comparison, a catch baseline from comparable organizations, and the boundary check in Chapter 23. With the four tools plus an industry comparison, the correctness of a defence is fully monitored. Install the tools without the comparison and an organization will sit contentedly guarding a lighthouse that turns to no purpose.

The lighthouse image can light the whole of Part III. The twenty-four mechanisms are twenty-four ways for a lighthouse to fail, and each chapter's four countermeasures are the materials for rebuilding it. This chapter's lighthouse is the strangest, because its failure is the quietest. The others are broken by waves, which is what an accident is. This one goes out slowly on its own, eroded by the absence of accidents. Quiet is its most dangerous property, and the entire justification for the four tools. A quiet thing has to be made to make a noise on a schedule. Drills, live tests and lookbacks are the bells rung for a defence at fixed intervals. An organization that rings them is not necessarily safe. An organization that has been silent for years is certainly going slack.

There is one piece of good news for the slackening organization, and it makes the real ending of the chapter. Slack is reversible. The ledger can start being kept today. Live testing can be scheduled from the next cycle. The lookback can be logged from the next cutting proposal. The page can start with one hour tomorrow. None of the four tools waits for an accident. That is the largest difference between this chapter and most of the others. The countermeasures in a disaster chapter are easiest to accept after the disaster. The tools in this chapter are worth most before it. Every accident-free day the reader is having right now is the best installation window the four tools will ever get. Not acting while the window is open is the mechanism this chapter defines.

20.3 The Shape It Takes in Four Domains

Case 1 (construction): Four Years in Which Nobody Noticed the Correct Ramp Was Not There

ce-007. Norwood Junction station, London Overground. RAIB report 09/2026, read in full for this book. The mobile ramp used by

wheelchair passengers to board and alight had not been available since September 2021. On 1 July 2025 a train left the station carrying a ramp that should not have been on it, which is a dangerous occurrence, and by good fortune nobody was hurt. The report's key sentence is worth quoting whole: "The correct ramp had not been available since September 2021 and none of the station checks in the period up to the accident had identified this." Four years of absence, and every platform check in the period missed it. Over a long accident-free stretch, the check had become a signing-in. Its output was a record that somebody had looked, not a chance that somebody had found something.

Stop at the decision point: the platform check after the PGI of 14 January 2022. On the table: the ramp photographs in the check record, the missing model information in those photographs, and a question nobody asked, which is what a photograph of this can prove.

What happened: a dangerous occurrence, no injuries; the absence and the failure of the checks both enter the report. Primary source: RAIB, report 09/2026, 2026; link at appendix B, ce-007.

Case 1, full card: Four Years and One Tick

Case ID: ce-007.

When and where: September 2021, the month the platform 1 ramps were replaced and the month from which the report dates the absence, through 09:52 on 1 July 2025. Platform 1, Norwood Junction station, London Borough of Croydon, the station standing 8 miles 55 chains south of London Bridge (RAIB p. 10). The report was published in June 2026 and runs to 52 pages.

What and who: Arriva Rail London (ARL), operating the London Overground concession for TfL since November 2016 and the station operator at Norwood Junction, employing both the driver of

9D18 and the station staff who deploy the ramp (RAIB p. 11). TfL is the concessioning authority and the rolling stock owner (p. 11). The mobile platform ramps are provided by Govia Thameslink Railway (GTR) under contract (p. 26), and the manufacturer is Portaramp UK Limited. Carlisle Support Services supplies contract station staff at the station (p. 11). The train was class 378 five-car electric multiple unit 378148 (p. 11). Posts involved: station manager, who carries out the PGI, station assistant dispatcher (SAD), train dispatcher (TD), driver.

What they were doing: Norwood Junction has six platforms, and platform 1 is the only one reachable without stairs and a subway. There is no level boarding between the platform surface and the floor of a class 378, so a wheelchair user can board and alight only by the mobile ramp (RAIB p. 18). The ramp is kept locked on the platform and opened by station staff with a key. It is used a few times a day, with no pattern, and a driver would not meet it daily (p. 18).

On the morning of 1 July 2025 a passenger who lives in Norwood arrived by mobility scooter about 10 minutes before 9D18 was due, passed through the gateline and asked the SAD for assistance under TfL's turn up and go service (p. 13). The passenger travelled this line regularly and had boarded at Norwood Junction and at other stations before (p. 13). The SAD telephoned the ARL passenger assistance team, gave the destination and the service, then took the ramp from its store with the key and waited with the passenger towards the rear of the platform (p. 13). 9D18 was the 09:46 West Croydon to Highbury & Islington, and it entered platform 1 at 09:51:35 (p. 9, p. 13). Once the alighting passengers were clear, the SAD went to the accessible door of the third car, put the ramp's locking feet into the holes in the door threshold, then stood on the ramp to test that it was secure (p. 13, paragraph 22 and figure 5). For the driver this was the fourth trip of the day: booked on at New

Cross Gate depot at 06:44, then 07:15, 07:45 and 08:47 back to back, with 9D18 the last turn round (p. 13).

The decision point: four moments.

1. September 2021, platform 1, GTR replacing the platform ramps. Before the replacement the station held two ramps, one com‑patible with class 377 and one compatible with class 378 (RAIB p. 26, paragraph 57). After it, both were a type compatible with class 377 only. The report's word for this is unintentionally (p. 26, paragraph 57). The scope was the two ramps on platform 1. The report does not record the replacement as an equipment change, and records no work order, no approving post and no accepting officer. What remains is the result: the class 378 ramp with the extension arm was no longer on the station. Where the extension arm came from is stated. In 2014 ARL's predecessor, London Overground Rail Operations Limited, specified a ramp with an extension arm for class 378 trains (p. 26, paragraph 54). The report's conclusion states the step as a finding, "The correct portable access ramp for class 378 trains had been re‑moved", and lists it as the first causal factor (p. 43, paragraph 122a).
2. 14 January 2022, a PGI, the station manager's planned general inspection, and the first record after the replacement to leave images of the ramps. Two photographs were taken. Neither ramp in them had a door-blocking arm, and the model was not recorded (p. 27). The same record set on 16 July 2020 had re‑corded three types, one Overground, one Southern, one Thameslink, with the arms on the latter two visible in the pho‑tographs (p. 26). The criteria did not change between the two occasions. What the record contained did.
3. 10 December 2024, the platform risk assessment for Norwood Junction, prepared against SQE 04.11-A, Train Dispatch and Platform Train Interface Risk Assessment, issue 2, January 2024

(p. 39, paragraph 108). The report states that the assessment did not consider the risk arising from ramp use, and that no other assessment covering the task was found.

4. From February 2025, ARL issued a new step-by-step ramp procedure: hold the ramp before the train arrives, stand where the driver can see you (p. 28, paragraph 68). By the time of the accident, 261 of the 291 directly employed station staff had been briefed. The SAD on duty that morning was not among them (p. 28, paragraph 68). The report records neither the post that approved the procedure nor any technical check of it against ramp type.

What was visible then: the two ramps on the station looked almost identical, the locking feet and the threshold holes were the same size, and a class 377 ramp could be fitted to a class 378, take a passenger's weight and work (RAIB p. 25, paragraph 51). The ramp label carried a serial number and a model, and the model denoted size, not the trains it suited (p. 27, paragraph 59 and figure 17). The PGI criteria came from RIS-3703-TOM appendix B, and the two questions touching the ramp asked whether it was secured correctly and whether it was in good condition and compatible with the trains calling at the platform. The standard does not explain what compatible means (p. 40, paragraph 110).

Three PGI records carry ramp entries and photographs: 16 July 2020, 14 January 2022 and 16 February 2023 (pp. 26–27). In 2020 a train had left Edmonton Green with a ramp still on it, and ARL had issued a Safe Every Time bulletin afterwards requiring a high-visibility vest and a position where the driver could see the staff member (p. 28, paragraph 67). On the driver's side the document in force was SQE 20.6, the dispatch action list, which includes a z-scan of the monitors, a check that the timestamp is running, and the words "never rely only on the interlock light" (p. 20, p. 22, paragraph 39). Before February 2025 the only formal document on

the ramp was that 2020 bulletin, and the report records no refresher training in the interval (p. 28, paragraph 68, p. 38, paragraph 102). Station staff had no training in matching a ramp to a train type (p. 27, paragraph 61, p. 38, paragraph 102). The platform was not busy that morning, and the SAD was the only person in the dispatch corridor (p. 33, paragraph 87).

What was learned afterwards. September 2021 is the start of the absence. Class 377 doors are of the plug type, with the leaves closing inwards, so a ramp lies inside that sweep and needs no extension arm, while class 378 doors slide into pockets in the bodyside and a ramp without an arm does not block them (pp. 23–25). The driver had used a personal phone on duty, with five calls lasting from 33 seconds to 25 minutes (p. 31). The monitor on the driver's side cropped the staff member's raised arms out of frame (p. 37). A reconstruction measured the ramp's vertical image height on the screen at 2 to 3 millimetres, below the lower limit of 2.6 to 3.6 millimetres, while the report also judged the ramp clearly visible in the reconstructed image (p. 36, paragraphs 91 to 92).

Stop point

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 09:52:05 the driver pressed door close, with the SAD still standing on the ramp carrying out the weight test and the scooter passenger still on the platform. At 09:52:13 the train started away with the ramp still on it (RAIB p. 14). The ramp was dragged to the platform end, struck the end stop and fell into the side drain

between the running line and the railway boundary (p. 16). Nobody was injured. The SAD was beside the ramp as it began to move, and a passenger sitting on a platform bench was warned by the TD's whistle and pulled her legs back before the ramp reached her (p. 7, p. 14). One door and a section of bodyside panelling on 9D18 were lightly damaged, and the ramp was written off (p. 7). Platform 1 was closed for about 25 minutes and reopened around 10:15 (p. 16). The passenger was assisted onto a later train, and told the RAIB that she now travels by train less often and experiences severe anxiety when she does (p. 17). The driver was screened for alcohol and drugs at Silwood sidings with a clear result (p. 17), and has not driven a train for ARL since (p. 46). The report classifies the event as a dangerous occurrence and makes six recommendations and two learning points (p. 1, p. 7). Recommendations 3 and 4 sit at the level of regulation and industry standard: the accessibility technical specification requires a ramp to be secured on the train but does not require door closing to be prevented while a ramp is in place (p. 25, paragraph 52, p. 48).

Back to the three questions. The commonest answer to the first is to add photographic evidence to the checklist, but a photograph prevents a false tick and does not prevent a look that never happens. The action has to be defined first, as walk, look, compare, and then a photograph has something to be a photograph of. The most overlooked item in the second question is the four years. They show that the idling was structural rather than personal: the system had never defined what the check was checking. The third question lands on the definition of the checking action. Every tick should correspond to one physical action and one comparison standard, and any check item whose definition is missing should come off the list, since such items produce nothing but a feeling of safety. Primary source: RAIB, report 09/2026, 2026; link at appendix B, ce-007.

Options ruled out: adding a functional check to the PGI, fitting the ramp, closing the doors and seeing whether the interlock is obtained, never reached the table. The PGI worked to the RIS-3703-TOM criteria, and the standard asks about securing and condition only, not about door closing (p. 40, paragraphs 110 to 111). Marking ramps by train type was also never tabled, since the label held a serial number and a size model (p. 27, paragraph 59). What ARL did after the accident was exactly that, adding the trains a ramp suits on ramps at multi-type stations and hanging a tag on the handle of any ramp incompatible with class 378 (p. 46, paragraph 130). Station staff telling the driver when a passenger was boarding had been ARL practice at its stations and faded over time, and the report's most likely explanation is the lengthening of class 378 units from three cars to five (p. 28, paragraph 65). The February 2025 procedure required only that staff stand in the driver's sightline, and did not restore the verbal notice (p. 28, paragraph 68). Making ramp deployment a safety-critical task never reached the table either. ROGS guidance does not clearly list ramp deployment among safety-critical activities, so ARL kept it out of its competence system, and the practice on the station was that anyone who had been shown the job could do it (p. 39, paragraphs 104 to 105). Driving on the interlock light alone was ruled out on paper: SQE 20.6 states that an object under 30 millimetres trapped in a door still allows the interlock to be obtained, and adds "never rely only on the interlock light" (p. 22, paragraph 39).

Early signals:

1. 14 January 2022: neither ramp in the PGI photographs had a door-blocking arm, while the photographs of 16 July 2020 showed the arms on the Southern and Thameslink ramps (pp. 26–27). The two sets sit side by side in figure 16 of the report (p. 27). Reason it was pushed past: the criteria asked about

securing and condition, both ramps fitted and took weight, and "in good condition" was satisfied.

2. 16 February 2023: the PGI checked the ramp labels, and the label detail went unrecorded (p. 27). Reason it was pushed past: the act of checking the label entered the record, the model column was blank, and the record looked complete.
3. A class 377 ramp on a class 378 worked perfectly, except that it did not block the door (p. 25, paragraph 51, p. 27, paragraph 61). Reason it was pushed past: fitting was taken as compatible, and the report records that staff therefore had no reason to raise it (p. 27, paragraph 61).
4. The Safe Every Time bulletin issued after the Edmonton Green event of 2020 reached the mailbox of the SAD involved, and the report found no evidence that it was read, with no refresher training between then and February 2025 (p. 28, paragraphs 67 to 68). Reason it was pushed past: the bulletin had been issued, and the covering was booked.
5. The depth of the PGI had been moving since 2020. The 2020 record listed three ramps and their uses, the 2022 record listed no model, and the 2023 record noted photographs and a label check without the label's detail (p. 26, paragraph 58, p. 27). Reason it was pushed past: the depth of a check is the station manager's call, and either depth counts as done.

The point of escalation: February 2025, when ARL began issuing the new step-by-step ramp procedure (RAIB p. 28, paragraph 68). This was the last attempt before the accident to put ramp knowledge back out onto the platform, and it reached 261 of 291 directly employed station staff. Thirty were not covered. The SAD on duty was one of the thirty (p. 28, paragraph 68). The illustration in the procedure shows the staff member holding the ramp, standing where the driver can see, which is the same instruction as the 2020 bulletin, and neither mentions train type (p. 28, paragraph 68, fig-

ure 21). The report records no approving post for the procedure and no technical review by anyone. On the day, the SAD's position was screened by passengers and by the platform canopy, so the instruction to stand in the driver's sightline did not hold (p. 28, paragraph 66). Over the four months that followed, the only ramp information circulating on the platform was that it fitted. The difference between fits and matches was carried by no document. Go back one step further, to September 2021: once only one model remained on the station, the PGI photographs of 2022 and of 2023 recorded it as this platform's standard fit (p. 27).

Failure chain: technical layer. Class 378 doors slide into a pocket in the bodyside and need an arm reaching out from the doorway to block them. A class 377 ramp has no arm, but its locking feet match the threshold holes, so it fits, takes weight and still lets the doors close and the interlock be obtained (RAIB pp. 23–25). The door system detects only an obstruction thicker than 30 millimetres (p. 23, paragraph 47). The ramp hangs outside the vehicle, beyond the gauge, and sweeps people and objects off the platform as soon as the train moves (p. 23, paragraph 44).

Organizational layer: the ramps are provided by GTR under contract and used by ARL staff (p. 26, paragraph 57), and the ARL PGI did not require anyone to check which ramp the station should hold (p. 40, paragraph 111). The PGI criteria came from appendix B of RIS-3703-TOM, compatible was never defined, and the standard does not ask for door closing to be checked (p. 40, paragraphs 110 to 111). Staff had no training in matching ramp to train type (p. 27, paragraph 61). Ramp deployment was outside ARL's competence system (p. 39, paragraphs 104 to 105). The platform risk assessment of 10 December 2024 did not touch the ramp (p. 39, paragraph 108).

Incentive layer: the ramp is used a few times a day with no pattern (p. 18, paragraph 31). Under DOO the driver alone is responsible for dispatch, and station staff have no role in the dispatch process

(p. 18, paragraph 35). Identifying the train type produced nothing in the period, since the ramp still took weight, while the cost fell in the period, because reading a label, comparing types and reporting all had to fit inside the few minutes of one arrival.

Three PGI records across three years treated "the station has a ramp" as a verified fact. Not one of them was required to answer which ramp it was. The boundary breaks in the organizational layer: the word compatible in the criteria never landed on any document, so the check passed every time.

Minimum counterfactual: in September 2021, after GTR replaced the platform 1 ramps, the station manager carries out one comparison of the ramp against the trains that call, and writes the result into that month's PGI record. One action, in the month of the replacement, and no luck required. The ramp label carries the model (p. 27, paragraph 59), the three train types calling at the station are known (p. 23, paragraph 48), and the PGI record of 16 July 2020 already listed what each of the three ramps was for (p. 26). Every PGI in the four years after had a chance to see the same thing (pp. 26–27). The report does not put this action forward. It is reconstructed from what the PGI could already do: the station manager was taking photographs and writing records, and in 2020 wrote down the purpose of each ramp (p. 26, paragraph 58). The report's own landing point runs the same way. Recommendation 1 asks the operator to review its management processes to confirm that ramps fully compatible with the trains that call are available (p. 47).

Outcome and sources: no injuries. One door and a section of bodyside panelling lightly damaged, one ramp written off, platform 1 closed for about 25 minutes. The report lists a direct cause, three causal factors and three underlying factors (p. 43). Six recommendations go to First Rail London, items 1 and 2, the Department for Transport, item 3, the RSSB, items 4 and 5, and the ORR, item 6,

with two learning points besides (pp. 47–49). After the accident ARL audited the ramps at every London Overground station, an - notated the trains each ramp suits at multi-type stations, and tagged the ramps incompatible with class 378 (p. 46, paragraph 130). It also carried out a ramp use risk assessment and agreed a proactive maintenance and inspection plan with the manufacturer (p. 46, paragraph 129). Primary source: RAIB, report 09/2026, Dan - gerous occurrence involving a portable access ramp at Norwood Junction station, 1 July 2025, June 2026, 52 pages, Department for Transport, full PDF read.

Case 1 carries an extension that runs across industries. Every check item of the confirm-it-is-there kind, whether the ramp is there, the extinguisher is there, the spare part is there, the per - mission is there, shares one failure mode. Presence is the state most easily assumed, because presence is not consumed by use. Items that measure consumption, oil level, pressure, remaining margin, have physical decay to remind people. Presence items have none. Their state changes at the moment the object is taken, and the moment of taking is exactly when the organization is least likely to be watching, since that is a transfer, a loan or a redeploy - ment. The design requirement for a presence item is therefore higher than for any other: a physical action definition, meaning walk to it and look at it, plus a change trigger, meaning any move - ment of the object must be logged. With both in place, four years of idling has a chance of being stopped by a sample check in year four.

Case 2 (healthcare): Counting Drifts in an Accident-Free Period

me-004. UK obstetrics. HSIB investigation, Detection of retained vaginal swabs and tampons following childbirth, 2019, archived snapshot read for this book. A vaginal tampon was left in place

after delivery and was not discovered until five days after discharge, in the report's words: "It was left in and not discovered until five days after leaving hospital." The swab count and tracking process exists to prevent exactly this, and its rigour varies naturally during a period with no accidents. No decision is needed. Only busyness. During the five days the patient attended twice outside the hospital, and neither attendance thought to check something that ought to have been cleared already.

Stop at the decision point: the moment during a delivery when a counting action was skipped or simplified. On the table: the pace of the labour, the count sheet present, and the belief that this has never happened here.

What happened: five days of retention; the report makes the counting and tracking system an investigation focus. Primary source: HSIB, 2019; link at appendix B, me-004 (archived link, grade A–).

The phrase from case 2, no decision needed, only busyness, deserves one further note. Busyness is slack's best disguise. A count simplified on a busy day differs from a count skipped on a quiet one, because the busy simplification has a wholly legitimate reason, an urgent labour, and the person doing it believes the reason. A justified simplification is the hardest kind to defend against, because it bypasses every control designed for laziness. Nobody in the room thinks they are breaking the process. Everyone thinks they are saving the patient. The defence therefore belongs in structure rather than supervision: design the count as a physical sequence that cannot be simplified, with the swabs on a physical stub carrying a count strip, so that simplification is physically impossible and busyness cannot compress it. An engineered defence does not depend on anyone's conscientiousness, which is Chapter 9's three-layer model speaking to this chapter again. Any action

guarded only by conscientiousness is missing an engineering con‑straint.

Case 3 (maritime): A Standby Pump Nobody Had Verified

ma-006. Stern trawler Freedom II (CN 111), 21 February 2024, 11 nautical miles south-west of Oban, Scotland. MAIB report 8/2026, read in full for this book. The suction pipe of the deck wash pump lacked support, fractured and let water in, and the vessel sank. The crew was rescued by liferaft and nobody was injured. The invest‑igation also found that not every pump fitted on board was ser‑viceable. Paper and steel had parted company over the accident-free years. The list said the pump was there. The deck said it was not. The only way to make those two lines agree, which is to start the thing once, had not been scheduled for years.

Stop at the decision point: the most recent moment at which the standby pump’s availability was assumed rather than verified. On the table: the entry in the equipment list, the distant record of the last live test, and the day’s work priorities.

What happened: the vessel was lost, with no injuries. Primary source: MAIB, report 8/2026, 2026; link at appendix B, ma-006.

The split between list and steel in case 3 has an intermediate state worth naming: serviceable on paper. An equipment list is an ac‑count, and reconciling an account against objects has a mature institution in fixed assets, which is the stock check. In safety equipment that institution is often missing. Who confirms that a standby pump, a life-saving appliance or an emergency supply is available, when, and by what action, is a question most lists do not answer. The gap in case 3, a reconciliation action that was to start the machine once and was not carried out for years, is on the same scale as the four years in case 1. A small fishing vessel without a stock-check regime can be forgiven, but the same structural gap

exists in systems far larger than a trawler, as case 4's certification system shows. A stock-check regime for safety equipment is the institutional form of Device 2 in this chapter: every claim of availability carries the date of a live test, and availability without a date is treated as unverified.

The automatic downgrade of unverified status needs a recovery channel beside it: a retest restores the status. Downgrading is a prompt, not a penalty. Its organizational meaning is that this line of defence is currently guaranteed by the ledger rather than by a test, and once the guarantor has changed, a decision-maker citing the defence sees unverified instead of available, which makes the citation honest by itself. Honest list states get at the root of this chapter's mechanism. Most of slack's harm travels through lists, since both case 1 and case 3 are lists telling lies. Once lists stop lying, the remaining harm is the shrink itself, which is visible, measurable and repairable.

Case 5 (finance): Cost Slack in a Cheap-Money Era (Condensed)

st-004. Xinja, an Australian neobank. Failory Cemetery archive, quoting a customer email from CEO Eric Wilson, read for this book. The bank raised A$204.7 million, took about US$200 million in deposits in its first month, and then, as capital conditions tightened, stopped taking business and returned its licence. The CEO's email names the model's dependence on its environment: "increasingly difficult capital-raising environment". This chapter's reading: the success of the loose-capital years, measured in the speed of deposit gathering, hid a slackening cost structure. A customer acquisition model built on high deposit rates works only in years when funding is cheap. The longer the good run, the looser cost discipline became, and the deeper the model's hidden dependence on its environment. A tightening environment is not a black swan. It is the normal state of an interest rate cycle, and the model held no margin for the normal state. The de-

cision point is the moment high-rate acquisition was written into the growth model without an environmental sensitivity test. Min - imum counterfactual: require an environmental sensitivity test as an attachment to the growth model, covering the effect of rising funding costs on the model' s survival. Primary source: Failory Cemetery, quoting the CEO' s customer email, 2020; link at appendix B, st-004 (evidence grade E).

Case 4 (aviation): Assumptions Not Verified at Certification

Case ID: av-002.

When and where: November 2006, the key test, to 26 August 2011, when the type certificate was issued. The event occurred at 10:21 on 7 January 2013, US Eastern time. Seattle, Washington, for the certification review at the FAA aircraft certification office, and gate 8A, terminal E, Edward Lawrence Logan International Airport, Boston, Massachusetts, for the event itself.

What and who: the Boeing Company' s 787-8 programme and its electrical power system (EPS) safety assessment team; Thales Avionics Electrical Systems, designer of the electrical power con - version subsystem; GS Yuasa, designer and manufacturer of the main and APU batteries; the FAA' s Seattle Aircraft Certification Office and the Boeing authorized representatives acting under its organization designation authorization (ODA). The aircraft was operated by Japan Airlines (JAL).

What they were doing: the 787 programme was launched in April 2004, first flew in December 2009, was certified in August 2011 and was first delivered in September 2011 (AIR-14/01 p. 17). In Septem - ber 2004 Boeing told the Seattle certification office that the main and APU batteries would be lithium-ion. This would be the first transport category aeroplane with permanently installed re - chargeable lithium-ion batteries (p. 60). The FAA reviewed 14 CFR

25.1353 and concluded that the existing rules covered several failure, use and maintenance characteristics of lithium-ion batteries insufficiently, and went to special conditions under 14 CFR 21.16 (p. 60). Issue paper SE-9 was issued on 31 March 2006, draft special conditions on 23 April 2007, and the final special conditions 25-359-SC on 28 September 2007, nine in number and effective 13 November 2007 (p. 61).

Three strands ran in parallel through certification. A means of compliance was proposed for each of the nine special conditions, by analysis or by test, with the deliverables list submitted alongside the certification plan. GS Yuasa and Thales carried out battery-level analysis and qualification testing, with Boeing reviewing and approving the results. Boeing then folded those results into the EPS safety assessment (p. 62). The report's later judgement is that traceability between the special conditions and the deliverables was never established (p. 86). Battery qualification evidence came in two classes, abuse testing (external short circuit, overcharge, overdischarge, high-temperature storage) and the safety assessment, and the compliance conclusion for each of the nine conditions rested on one class or the other (p. 62, p. 83). The FAA approved the initial certification plan on 22 December 2005 and revision C on 8 January 2007 (p. 62). On 16 September 2009 the EPS safety assessment was finalized, containing a failure modes and effects analysis (FMEA), a functional hazard assessment (FHA) and a fault tree (p. 62).

The decision point: three irreversible moments, each attached to a document, five years apart.

The first came in November 2006. GS Yuasa ran a research and development nail penetration test: a steel nail driven through the electrodes of a fully charged cell, with the battery fully charged, ungrounded and of pre-production build, at an ambient temperature of 50–59°F, the temperature of an electrical and electronic

equipment bay on a typical flight (p. 64, footnote 108). The penetrated cell's surface heated, the cell and the battery case emitted smoke, and adjacent cells heated without venting (p. 65). The test was run once (p. 64, p. 65). It was not a certification requirement but an R&D test (p. 64, footnote 107), and its result went into Boeing's EPS safety assessment as the measured basis for the conclusion that the effects of an internal short were limited (p. 65). The reviewers were a Boeing project engineer, a safety, reliability and maintainability engineer, and an authorized representative acting for the FAA (p. 64, footnote 109). This is the moment a single-point reading was promoted to a general conclusion. Boeing, Thales and GS Yuasa later said further battery-level R&D nail penetration tests had been carried out, but no records survived that the NTSB could examine (p. 81, footnote 135).

The second came on 16 September 2009, when the EPS safety assessment was finalized (p. 62). The FHA classified venting with smoke and fire as catastrophic, and venting with smoke but no fire as hazardous (p. 63). Three probability requirements followed: venting below 1×10^{-7}, smoke below 1×10^{-7}, and flammable liquid spill below 1×10^{-9}. The assessment concluded that the only known failure mode leading to venting with smoke and fire was overcharge, so overcharge protection was designed to 1×10^{-9}, and internal short circuit was grouped with the venting without fire case (p. 63). In the same assessment the battery-level FMEA states that no single-point failure of the main or APU battery leads to catastrophic consequences (p. 64).

The third came on 26 August 2011, when the FAA issued the type certificate, with a certification basis of Part 25 plus 25-359-SC (p. 61). The issuer was the FAA, and the compliance conclusions were made by Boeing's authorized representatives, Boeing having received ODA approval in August 2009, which allows its represent-

atives to make FAA compliance findings on specified reports and tests (p. 65).

All three occasions carry a date and named hands. What they lack is one document on the same table. None of them required the assumption to be written down, that thermal runaway in a single cell does not propagate to adjacent cells, and none required a test of it (p. 12, p. 86).

What was visible then: on 31 March 2006 the FAA wrote the danger itself into SE-9, the issue paper produced for this aircraft. Lithium-ion batteries are "significantly more susceptible to internal failures that can result in self-sustaining increases in temperature and pressure", and manufacturers of mobile phones and electric vehicles had already had problems (p. 61). GS Yuasa's service data showed more than 14,000 cells of the same design in industrial use, none of which had vented or gone into thermal runaway, and from that record a venting rate for the 787 battery of about once per ten million flight hours was extrapolated (p. 64). No analysis of how closely industrial duty and environment resembled airborne duty and environment accompanied the data (p. 64).

The text of special condition 2 reads "the design of the lithium-ion battery must prevent self-sustaining, uncontrolled increases in temperature or pressure" (p. 86, footnote 147). None of the nine conditions required battery-level test data under single-cell thermal runaway (pp. 99–100). In July 2009, during a whole-aircraft test at the UTC Aerospace Systems APSIF, a battery was inadvertently overcharged, one cell vented, and nothing propagated (p. 82). In September 2009 two engineering venting tests observed electrolyte leakage, the battery case was given additional sealing, and the pre-production battery's vent disk arrangement differed from the final design (p. 82, footnote 138). Four qualification abuse tests were run, two external short circuits, one overcharge, one over-

discharge, all at 158°F, none of which produced thermal runaway (p. 83).

What was learned afterwards: the NTSB and UL repeated the nail penetration at 158°F on an ungrounded battery, and single-cell thermal runaway propagated to every cell in the case (p. 82, footnote 137). Two of the three heat transfer tests run by the Japan Transport Safety Board propagated, and both were on grounded batteries (p. 82). The failure in the accident battery began with an internal short in cell 5 or cell 6 (p. 91).

Stop point

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: at 10:21 on 7 January 2013 the APU battery failed and the APU shut down automatically. Maintenance found heavy smoke coming from the battery case lid and two flames about 3 inches long at the connector. The battery was removed at 11:57, about 80 minutes after the first report, and the event was under control at 12:19 (p. 14, p. 17). The 183 passengers and 11 crew were already off the aircraft, and nobody on board was hurt. One firefighter suffered a minor injury, and a firefighting captain suffered minor burns to the neck when the battery "exploded" (p. 14, p. 17).

Grounding and modification: after the Takamatsu event of 16 January, the FAA issued emergency airworthiness directive 2013-02-51 the same day, Japan's Ministry of Land, Infrastructure,

Transport and Tourism issued an equivalent the next day, and other national authorities required grounding in turn (p. 65). On 19 April the FAA approved a redesign of the battery system, on 22 April AD 2013-08-12 replaced the emergency directive, effective 26 April, and the fleet returned to service once modified (p. 66).

Back to the three questions. The commonest answer to the first is to repeat the test at 158°F, on a grounded battery, of final config - uration. In 2009 that cost weeks of rig time and one battery. The saving from skipping it was paid in grounding, redesign and re - certification. The most easily missed item in the second question is the three conditions buried in the 2006 test report: the temperat - ure, the absence of grounding, and the pre-production battery. They sit in brackets and footnotes, they read like lab notes, and they are in fact the scope within which the conclusion holds. The third question lands on the certification plan. Every special con - dition needs to trace backwards to a measurable performance re - quirement and to a test, and any item that cannot be traced is a gap.

Options ruled out: a thermal runaway test at the 158°F maximum operating temperature was not on the qualification test list, be - cause the assessment held the effect of that failure mode to be no worse than a single cell venting (p. 82). A grounded battery to re - produce the installation on the aircraft was not used; the batteries in the 2006 test and in the abuse tests were ungrounded (p. 83). Repeated tests to establish repeatability were not run, only the single test (p. 82). A retest on a final-design battery was not done, the test battery being of pre-production build with a different vent disk arrangement (p. 82, footnote 138). Writing "battery perform - ance under single-cell thermal runaway" into the battery spe - cification control document (SCD), with a test hung off it, was not done, and because the requirement was absent, no such item ap - peared on the qualification test list. The report notes that the ab -

sence is exactly why the need for a thermal runaway test was "less visible" to Boeing's authorized representatives and to FAA engineers (p. 86). A structured comparison of industrial cell service against 787 duty was not done, and the report found no such document (p. 83). The reason given for turning down each of these comes back to the same sentence: the assessment had already shown the effect to be limited, so no further test was needed.

Early signals:

1. 31 March 2006: the FAA wrote in SE-9, prepared specifically for the 787, that lithium-ion was "significantly more susceptible" to internal failure leading to thermal runaway, and listed the mobile phone and electric vehicle precedents (p. 61). Reason it was pushed past: 14,000 industrial cells with no failures.
2. November 2006: three departures in the test conditions, 50–59°F rather than the 158°F maximum operating temperature, no grounding, and a pre-production battery. The report's words are that the test "did not adequately represent the battery installation on the 787" (p. 64, p. 82). Reason it was pushed past: this was an R&D test and the conditions were representative.
3. July 2009: the APSIF test, with the battery inadvertently overcharged, one cell venting and nothing propagating (p. 82). Reason it was pushed past: another confirmation that nothing propagates. The report later notes that the test was not designed to assess propagation and cannot serve as confirmation of the 2006 test (p. 82).
4. September 2009: two venting tests showed electrolyte leakage and the case gained additional sealing (p. 82). Reason it was pushed past: the change touched sealing only, not the propagation judgement.
5. The failure mode list made overcharge the only entry into venting with fire, put internal short circuit under venting without fire, and the whole protection budget therefore went to

overcharge monitoring (p. 63). Reason it was pushed past: a list ought to be classified that way, and classification needs no verification.

The point of escalation: 26 August 2011, when the type certificate was issued (p. 61). The assessment of 16 September 2009 was the last document that could still have been changed. Once the certificate was out, "single-cell thermal runaway does not propagate" stopped being a reading from a 2006 R&D test and became an established conclusion in the airworthiness file. The certification basis recorded on the certificate is Part 25 plus nine special conditions, checkable one by one. Condition 2 requires prevention of self-sustaining rises in temperature or pressure, and does not say by what test (p. 61, p. 86). Any later change has to go through the change process, that process is triggered by an objection, and an objection must be aimed at an assumption written on paper. This assumption was never written. The escalation had no separate signing ceremony, since issuing a certificate is routine FAA work and making a compliance finding is routine work for a Boeing authorized representative, and the two together fixed an unverified inference in place until an event. Less than eighteen months after the certificate, with the fleet under 52,000 flight hours, two battery venting events occurred within 9 days (p. 10, p. 25).

Failure chain: technical layer. An internal short in cell 5 or cell 6, thermal runaway propagating cell by cell inside the case, venting, electrolyte spill, fire (p. 91). Nothing propagated under the 2006 test conditions; the nail penetration repeated at 158°F propagated to every cell (p. 82, footnote 137).

Organizational layer. The key assumption never entered the safety assessment (p. 12). An assumption that is not written has nothing for a reviewer to challenge, and the several independent rounds of review by Boeing authorized representatives and FAA certification engineers found no gap (p. 85). The gap then ran down the docu -

ment chain: no requirement in the SCD for battery behaviour under single-cell thermal runaway (p. 86), so no thermal runaway test on the qualification list, so nothing of the kind in view when the certification plan was approved. The regulation says to prevent self-sustaining rises in temperature or pressure (p. 86, footnote 147) and does not say what proves it.

Incentive layer. The zero-failure record of industrial cells is the cheapest reliability evidence there is, needing no rig and no time. Verifying the assumption needs 158°F, grounding, final configuration and repetition, which is the most expensive class of evidence available. Someone had to ask for the expensive evidence. The place to ask was the SCD, and the SCD's source was the assumption nobody had written.

All three layers break. The boundary breaks between the organizational and the incentive layer: no document required that test to take place, and the operating record of nothing having happened signed for it instead.

Minimum counterfactual: before the EPS safety assessment was finalized in September 2009, write "thermal runaway caused by an internal short in one cell does not propagate to adjacent cells" as an explicit assumption, and hang a verification on it: repeated nail penetration at 158°F on a grounded battery of final configuration. One action, in the week the assessment was finalized. It was doable at the time. 158°F is the battery's own stated maximum operating temperature (p. 82), grounding is the installation on the aircraft (p. 86), and nail penetration was a method GS Yuasa had already used (p. 64). No luck is required. The cost is weeks of rig time and one battery. The output is a test report that can be filed beside the 2006 report, and the difference between the two sets of conditions is itself the basis for deciding whether the design has to change. The report's causal judgement runs the same way: had the assessment included the possibility of propagation, the authorized

representatives and FAA engineers would very likely have required additional testing, or a design able to accommodate full-case cascading thermal runaway (p. 84).

Outcome and sources: the NTSB's probable cause finding is that an internal short circuit in a single cell of the APU lithium-ion battery initiated thermal runaway that cascaded to adjacent cells, releasing smoke and fire; Boeing had not incorporated into the design a requirement to mitigate the most serious consequences of a single-cell internal short circuit, and the FAA had not identified that deficiency during type design certification (p. 92). The report also states that GS Yuasa did not test the battery under the most severe conditions likely in service, and that the test batteries differed from the final certified design (p. 11). The report was adopted on 21 November 2014. Safety recommendations A-14-113 to A-14-130 go to the FAA, Boeing and GS Yuasa, of which A-14-119 to A-14-123 address the identification, verification and traceability of assumptions, and A-14-129 requires Boeing to revise its safety assessment process for new technology designs (pp. 93–95). Primary source: NTSB, Auxiliary Power Unit Battery Fire, Japan Airlines Boeing 787-8, JA829J, Boston, Massachusetts, January 7, 2013, NTSB/AIR-14/01, adopted November 21, 2014; link at appendix B, av-002.

The industry-level tuition in case 4 has one layer worth separating out. In a certification setting, substituting analysis for measurement takes the form of analysis coverage. A safety assessment covers a failure mode by analysis, meaning models, analogy and prior data, and measurement is carried out only where the analysis shows it to be necessary. Whether it is necessary is itself a judgement about the failure mode, so where understanding is thin the analysis will show measurement to be unnecessary. That is a closed loop referring to itself: a gap in understanding decides the absence of testing, and the absence of testing preserves the gap in

understanding. There is one way out, which is to require measurement for failure modes that are new, and the lithium thermal runaway in this case was new, whatever the analysis says. The test for novelty has to be independent of the analyst, which means an industry-level list of failure modes kept current. That is institutionalized memory for a certification system: the industry stops letting one company's safety record vouch for the whole industry's assumptions.

20.4 What Shows Up Earliest

Six weak signals, each with the reason that usually pushes it aside.

1. The argument for keeping a defensive line item at the budget meeting starts to be "nothing has ever happened". Reason given: "that is a fact".
2. The actual execution of a drill, a sample check or a review slides from measurement to signing in. Reason given: "the process is being followed".
3. Standby equipment has not been powered up or tested for a long time, while the list says serviceable throughout. Reason given: "the equipment is new". (The pump on Freedom II.)
4. A new starter describes a defensive process as "the way we do things here" and cannot say what it prevents. Reason given: "just follow it".
5. The last record of a defensive action catching a problem is more than two years old. Reason given: "there are fewer problems now".
6. The year counts quoted in a cutting proposal leave out the events the defence caught and nobody totalled. Reason given: "there is no data". (The absence of data is itself criterion 2.)

Put the six signals back against the cases in this chapter. Signal 2, measurement becoming signing in, maps to case 1, four years of idling platform checks. Signal 3, a list saying serviceable while nothing has been verified, maps to case 3, the paper state of the standby pump. Signal 4, a new starter unable to say what the process prevents, maps to case 2, where the count process in a new nurse's hands means nothing more than a form. Signal 5, the last catch older than two years, maps to case 4, a certification system with no catch record for a new failure mode. Signal 6, the data gap, maps to all of them. None of the four cuts in those cases went through a budget meeting, and none of the shrink passed a decision point. That is the fine difference between this chapter and the one before it. The cuts in Chapter 25 have signatures. Most of the shrink in this chapter has none.

Ordering the six signals by cost of discovery gives a reading specific to this chapter. Signals 3 and 5 are cheapest, since they need only the dates in a ledger. Signals 1 and 6 are middling, since they need the budget papers read. Signals 2 and 4 are dearest, since they need site observation and interviews. Start with the cheapest, as usual. Pull the organization's list of defensive equipment and processes, mark against each item the date of its last live test or last catch, and mark in red anything older than two years. The length of the red list is the operating level of this chapter's mechanism. For most organizations doing this inventory for the first time, the number of red items comes as a surprise, and the surprise is itself evidence that criterion 1 holds.

20.5 How to Defend Against It

Device 1: keep a ledger for the non-events. A defensive line gets an output ledger: every problem a drill found, every batch a sample check stopped, every error a review corrected, all entered. The ledger has to separate two kinds of empty. Nothing to catch, which

is genuine calm, and nothing being caught, which is ritual. The first is evidence that the spending works. The second is evidence that the spending is dead. Mixed together they can never be told apart.

Device 2: live testing of defences. Backups and standby processes are verified for real, on a cycle: the standby pump is started and run, the standby line is actually switched over, the backup staff actually take the post. Chapter 22's principle that redundancy must be rehearsed for real is repeated here because it is the line most often crossed in this chapter. Results go into the ledger and are bound to the claim of availability: an item not verified in the current cycle drops automatically to unverified.

Device 3: a two-year lookback on every cut. Any defensive cut, when approved, is logged with a lookback date two years out, and at the lookback the domain's actual risk indicators are compared against the pre-cut baseline. A lookback recovers no losses. It does give the next cutting meeting one more piece of history to be asked about, and each time the institution remembers, the next defence has to be a paragraph longer.

Device 4: institutionalize the memory. Write the reason a defence exists onto one page headed what this prevents, and keep it with the process, the same instrument as the why document in Chapter 30. A process whose purpose a new starter can read is a process that survives. A line item whose accident the organization still remembers is a line item that cannot be cut. The page's format is fixed: what it prevents, what happened before (an accident ID or an industry case), and the minimum cost of cutting it.

Set the four devices against this chapter's cases. Device 1 on case 1: the platform check ledger splits findings from checks, and four years of zero findings against continuous checks is written evidence of idling. Device 2 on case 3: the standby pump is started quarterly and the run recorded, with serviceable bound to the date

of the last test. Device 3 on case 5: the cut to the high-rate acquisi‑tion model, the rate reduction, is logged with a lookback date, and the lookback compares deposit retention. Device 4 on case 2: the one page for the count process carries an industry retention case, this report being the material, and new starters read it as part of induction.

The four devices have four ways to fail. The output ledger dies of record inflation, as drills begin manufacturing findings to make the ledger look good and the record detaches from reality. The patch is to grade findings, with real findings and improvement suggestions in separate columns, and to put the ledger itself under audit. Live testing dies of a distorted test environment, as the standby pump passes an unloaded run and fails under working load. The patch is to design test conditions around the worst case, in the sense of Chapter 25’s margin. The two-year lookback dies of formality, as the comparison indicator is chosen because the cut does not affect it. The patch is to lock the indicator at the moment the cut is approved, with no substitution afterwards. Institution‑alized memory dies by turning into a leaflet: written, distributed, unread. The patch is to bind the page to training assessment and to require a rewrite after every relevant accident, since an accident is the moment that tests whether the page is still accurate. The four failures share one checkup question. Did this device last change a decision? A device that has never changed a decision is decoration.

20.6 Boundaries With Neighbouring Mechanisms

Against margin erosion (Chapter 25): eating margin is the cutting action itself, a fall in reserve that can be read off the accounts, while slack is the growth of the willingness to cut, the reason someone believed a cut was affordable. In ma-004 the seawater pipe is margin (Chapter 29), while “nobody since 1998 has felt the need to look at it” is slack. Keep the two ledgers on the same page.

Against normalization of deviance (Chapter 11): normalization files a deviation away, with the words the margin is still sufficient, while slack downgrades a defence, with the words we no longer need it. Both work from the same false evidence, that nothing happened, and they run in opposite directions: one loosens the defence against the abnormal, the other loosens the guard on the normal.

Against horizon mismatch (Chapter 28): the cost of slack is cashed late, and by the time it is cashed the decision-maker has left the building. That is the structural reason it repeats generation after generation, and it is the same time condition as Chapter 28. The bill handover list in Chapter 28 should carry a column headed which defences I cut.

20.7 Five Self-Checks

1. Name three defensive items in your organization that have not had an accident in years. When was each last tested, or last known to catch something?
2. At your last budget meeting, what did you approve or cut on the argument that nothing has ever happened?
3. When did your standby system last switch over for real, and was the switch made at peak load?
4. Take one defensive process and ask a new colleague what it prevents. Would you be comfortable with the answer?
5. If the defensive ledger from Device 1 were opened today, how many caught problems do you estimate it would record in its first year? The estimate is itself the answer.

(The Chinese text of this chapter runs to 15,248 characters.)

Chapter 21: Hidden Coupling | Interactive Complexity and Tight Coupling (Perrow)

> What this chapter covers: how two systems that "should have nothing to do with each other" come to share one inconspicuous thing; Perrow's interactive complexity, the three criteria, four field sites, and an annual dependency-mapping regime. When to turn back: after a merger, an architecture rebuild or a supplier change; when a fault report starts saying "unknown cause". Reading time: 50 minutes.

Hidden coupling has one property that separates it from every other mechanism in this book. It cannot be managed before it is seen, and once it is managed it stops being hidden. That sounds like a tautology. It is the hardest problem statement in the field, because you cannot open a project, allocate a budget or set a date for something you do not know exists. Perrow, in Normal Accidents (1984), put interactive complexity among the structural sources of accidents: the connections between the parts of a modern system exceed anyone's full understanding, and failures travel along paths their designers never drew. Forty years on, the systems are only more complex. What this chapter can supply is a mapping discipline that turns the unseen into the seen, one pass at a time.

21.1 What It Is

Definition: hidden coupling is the engineering face of interactive complexity. Two systems share a connection that neither their designers nor their users are aware of, and a small fault crosses between them along a path nobody has drawn. Nothing is hidden at the physical layer, where the path is real and has always been

there. The hiding is at the cognitive layer. Nobody put it on paper, so nobody managed it.

Three criteria start the mechanism. Any one of them is enough.

1. Two systems believed to be independent turn out to share something trivial: one account, one pipeline, one permission, one battery, one bearing supply chain.
2. Nobody can answer the question "who is affected if this goes down".
3. Fault reports start containing "unknown cause" or "unexpected propagation path".

21.2 How It Grows

Four generative conditions.

Condition one is merger and system integration. The link between two systems is struck from the merged drawings while remaining connected in the physical world. The merger document records "functional migration complete" and omits "old connection physically retained".

Condition two is legacy. A workaround installed twenty years ago outlives everyone who remembers it, and the half-life of an expedient is the tenure of the person who installed it.

Condition three is outsourcing and supply chain depth. What you buy is a component made of other people's components (a near relative of Chapter 33, cascading failure: that chapter covers the concentration of dependency, this one covers the invisibility of the connection).

Condition four is configuration drift. Permissions, parameters and switches quietly link into new paths across a series of changes. Every change passes its own review. The review looks at the

change, never at the change combined with the rest of the network.

One asymmetry has to be learned by heart. Coupling that is seen gets managed, and managed coupling stops being hidden. The stock of this mechanism therefore always equals the part nobody has drawn yet. The defence is an unending mapping exercise, never a one-off cleanup.

Perrow's framework has two axes. This chapter uses the first and owes the reader the second, because mixing them up is the standard misreading. Axis one is interactive complexity, the way parts are connected: in a linear system failures propagate along known paths, in an interactively complex system they can combine along unexpected ones. Axis two is tight coupling, the time available to work the problem after a fault: a tightly coupled system has almost no slack, and the chain completes before anyone intervenes. The axes take their values independently. An interactively complex but loosely coupled system (a university) can produce strange failures and digest them slowly. A linear but tightly coupled one (a conveyor factory) fails fast and predictably. Disasters cluster where both axes are at maximum: nuclear power, chemicals, aviation, global networks. The mapping discipline in this chapter works on axis one, making interaction visible, while the priority order for mapping comes from axis two. Dark lines in tightly coupled areas get drawn first, because once energised they leave no time for a human response. All four main cases sit in the double-high region. The helicopter's control interaction happens at the scale of seconds, Cloudflare's file propagation at the scale of minutes, the Dow relief path had no waiting time at all, and on Kommandor Susan the interval from loss of power to loss of the anchor was a few minutes.

The cognitive layer of "hidden" has one more source: the language wall between specialisms. Every subsystem has its own

vocabulary (process, network, mechanical, permissions), and coupling points tend to fall exactly on the border between two of them. No specialist on either side can describe the line completely in their own words. A network engineer says "authorisation", a process engineer says "pipeline", and on their own drawings neither word points at the other's system. The language wall explains why dependency mapping has to be run by a questioner and an answerer from opposite sides (the organisational note under Device 1). It also explains why the artefact has to be written in a cross-system common language (resource name, blast radius, owner) and not in either side's dialect.

21.3 Four Domains, Four Shapes

Case One (Aviation): A Hidden Link Between Control Channels

av-040.

When and where: 27 October 2018, 1937 (UTC, night local time), King Power Stadium, Leicester, England. The helicopter lifted from the centre circle of the pitch, intending to reposition to London Stansted. After loss of right yaw control it came down on the concrete of the staff car park on the south-east side of the stadium. That surface has a 0.5 metre step, and the airframe was still rotating fast when it touched (AAIB p. 71).

What and who: type certificate holder Leonardo S.p.A. (AgustaWestland at the time the certificate was issued), accident aircraft serial number 69018; the tail rotor double-row ball bearing was manufactured in France (the BEA took part in the investigation), the tail rotor actuator in the United States (the NTSB took part); operator Starspeed Ltd, owner Foxborough Ltd (Isle of Man), single-pilot operations under Part-NCC; airworthiness authority

EASA, type certificate data sheet EASA.R.5098, certified to CS-29 Amendment 2 (17 November 2008) (p. 26). One pilot (53, ATPL(H), 12,947 hours total, 177 on type), four passengers (p. 21, p. 22, p. 38).

What they were doing: the airframe was built on 11 July 2016 and had 330.9 hours and 1,034 landings at the accident; the last bi-weekly check was on 22 October (p. 24). The bearing is a hybrid, steel races with silicon nitride ceramic balls (p. 22). It carried over the AW139's on-condition status: no scheduled removal time, a rotational smoothness check by hand every 600 flight hours or once a year, removal for visual inspection at 2,400 hours (p. 35). The load spectrum and discard life for the AW169 and AW189 were completed at design acceptance with the AW139 as the reference, producing the same recommended life and the same inspection interval (p. 37). Certification flight tests measured tail rotor axial and bending loads, and the data were never passed to the bearing manufacturer for review (p. 214). The same test data later showed that a few classes of manoeuvre inside the normal envelope generate contact pressure high enough to cause damage (p. 213). The accident bearing was the most damaged of the same-type bearings submitted for examination, and had the shortest time in service (p. 213). Takeoff was at 1937, rearward climb at 600 to 700 feet per minute, at 250 feet a 15 degree nose-down input held for 6 seconds with gear retraction, and a right turn entered at about 300 feet (p. 14, p. 60). The departure was flown as a Cat A procedure under a congested-area approval: reach the decision point after lift-off, hold a climb margin of 130 feet, accelerate to VY, then transition. The aircraft's average rate of climb exceeded the procedure's parameters and the turn began before the acceleration segment was complete. The investigation found that neither deviation materially affected controllability after the failure (p. 165, p. 197).

The decision point:

One. 2012. An AW139 in Qatar seized the same bearing and lost yaw control. There were rotation marks on the control shaft. The AW139 thread is left-hand, so the nut seat tightened instead of backing out, torque went back into the bearing until it failed com - pletely, the blades stopped at about +10 degrees, and the helicopter ran on and landed with nobody injured (p. 36). The event did not start an independent Annex 13 investigation; the manufacturer and the local airworthiness authority judged the bearing too badly damaged for a meaningful failure analysis (p. 36). The response was two maintenance requirements: no refitting after removal, and a discard life of 3,000 hours. Both went out as a service bulletin with no airworthiness directive behind them, so the life never be - came an airworthiness requirement (p. 36). The 3,000 hours from the bulletin sat alongside the 2,400-hour inspection point, and the evidence shows operators commonly refitted bearings after the 2,400-hour check in order to run them to 3,000 (p. 36 footnote). In the same period the load spectrum and life for the AW169 and AW189 were going through design acceptance (p. 37).

Two. July 2015, EASA issued the AW169 type certificate. The system safety assessment placed the bearing in the tail rotor assembly (CS 29.547, 29.602) and the control shaft with its actuator-end lock nut in the tail rotor actuator (CS 29.1309, 29.602). Two reports, two failure analyses (p. 38). The actuator manufacturer' s analysis covered its own parts only, with a written statement that it had not been told the effects of failure in connected parts, "for example shaft rotation driven by gearbox torque". The thread handedness of the nut was not in the design specification either (p. 38). Both failures were classified catastrophic. The actuator-end lock nut was listed as a catastrophic single-point failure, but only two failure modes were considered, nut fracture and loosening under vibra - tion. On the accident aircraft that nut was right-hand thread (p. 28, p. 214). The authority confirmed that once a failure is classified

catastrophic the manufacturer need not consider its knock-on effects at all (p. 39).

Three. After 1937 that night, the start of the right turn came with a right pedal input. The control shaft moved right under hydraulic pressure and reduced tail rotor pitch through the pitch link. AAIB's analysis of the recorded data says plainly that the failure sequence was triggered by this pedal input, and that inputs of this kind are common at this stage and could be connected to anything, a crosswind correction among them (p. 165).

What was visible then: at the certification table in July 2015, the bearing was never certified separately; it was assessed as part of the system by compliance method (p. 37); the two AW139 part numbers had accumulated 3,699,826 hours (p. 35); the 2012 wreckage had been judged "too damaged for a meaningful failure analysis" (p. 36); CS-29 held no design or test requirement for rolling contact fatigue in bearings (p. 37); bearings were not stripped and inspected after the certification tests (p. 214); operator records in the UK and Italy showed several bearings rejected for roughness and axial play (p. 35). In the cockpit, after 1937: the flight manual held one combined tail rotor failure procedure (LTE), split into hover and forward flight, both opening with lowering the collective, both stating that this will produce rapid right yaw and loss of yaw control, and both noting that a high yaw rate brings pitch and roll with it (pp. 56 to 57); there was no procedure for tail rotor pitch runaway, catastrophic by certification and therefore not required (p. 57); the aircraft was moving from rearward to forward flight and was already turning, so neither case fitted, and no simulator exercise matched either (p. 165); yaw rate reached 128 degrees per second, the ADAHRS reached the limit of its output to the autopilot, data went invalid, the autopilot disengaged and annunciated "ap ahrs 1 fail" and "ap ahrs 2 fail" (p. 170); a low rotor RPM warning and three oil system pressure warnings followed,

later analysed as effects of the yaw rate while the systems themselves were sound (p. 170); radio altitude peaked at about 430 feet (p. 62); yaw rate reached close to 10 per cent of main rotor RPM (p. 170). Information that arrived only afterwards: subsurface-origin rolling contact fatigue on the raceway, fracture of the cage, and debris ground to powder and re-plated into a new raceway surface (p. 199); the rotating control shaft screwed the lock nut out, friction heat locally melted nut and seat into a weld, and both were pushed off the end of the shaft (p. 171); with mechanical feedback gone the control shaft travelled to the end of its range, and the yaw primary stop sat on the pilot's side of the disconnect, so the stop failed (p. 172).

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What happened: within 1 second of the yaw starting, left pedal was fully down. After about 5 seconds of full left pedal with no effect the pilot began lowering the collective, reaching the bottom in 2 seconds (p. 166). Yaw rate peaked at 209 degrees per second, altitude peaked near 430 feet, and the aircraft then descended at about 4,000 feet per minute, with full collective applied at about 75 feet (pp. 62 to 63). The airframe rolled over on impact and the fuel caught fire. All five on board died, one crew and four passengers, and the vertical deceleration exceeded the 30g energy-absorption limit of the seats; four firefighters who arrived first suffered heat injuries (p. 21, p. 198). Back to the three questions. The most com-

mon answer to the first is to step on the opposite pedal earlier and harder. The cost of that action in this case was zero to one second: the pilot had full left pedal inside 1 second, and the pedals lost their mechanical connection to the tail rotor the moment the con - trol shaft disconnected. Simulator trials showed that no rate of collective movement restored heading control (p. 167). The item most likely missed under the second question is the word cata - strophic. It reads as a classification, and in the document chain it also works as an order to stop analysing. The third question lands at the type certification review board: for any failure classed cata - strophic, ask what happens if that failure occurs in a part belonging to the neighbouring system.

Options ruled out: a rolling contact fatigue requirement for crit - ical bearings in CS-29 (no clause existed before 2023, and the ex - isting acceptable means were enough to pass; AAIB raised it after - wards as Recommendation 2023-019, p. 37, p. 218); giving the cer - tification load data to the bearing manufacturer for review (not required, p. 214); a system-level mitigation analysis after a cata - strophic classification, meaning warning and fault-tolerant design (explicitly not required, p. 39, raised afterwards as Recommenda - tion 2023-025); a flight manual procedure for tail rotor pitch run - away (catastrophic by certification, not required, p. 57); turning the bearing into a life-limited part with mandatory removal (the AW139 already had the 2,400-hour inspection, but the life came from a service bulletin rather than an airworthiness requirement, p. 36); adding a third locking feature (the two required by CS 29.607, split pin and lockwire, were already fitted, p. 39); using the existing vibration monitoring to catch degradation (manufacturer tests judged the existing sensor set would not help with this bear - ing, and an accelerometer was added at the lever position only afterwards, p. 155).

Early signals: (1) The 2012 AW139 event: control shaft already rotating, blades stopped at +10 degrees, forced landing. No independent investigation was opened, the wreckage was judged too damaged for a meaningful failure analysis, and the handling consisted of two maintenance requirements in a service bulletin. The case was closed as an individual occurrence with the mechanism unresolved (p. 36). (2) Ceramic balls produce contact pressure about 12 per cent above steel, and CS-29 had no rolling contact fatigue clause; the bearing was not certified separately, and passing the compliance method was compliance (p. 22, p. 37). (3) Bearings were not stripped after the certification flight tests. The rules did not require it, the tests produced no abnormal readings, and a strip was therefore treated as unnecessary (p. 214). (4) Operator records in the UK and Italy showed bearings rejected for roughness and axial play, while a 3,000-hour life coexisted with a 2,400-hour inspection point and operators commonly refitted. Rejections and refits were both read as normal wear on an on-condition part and never rolled up into a fleet-wide trend (p. 35, p. 36 footnote). (5) The AW139's opposite thread went into effect in 2012 while the AW169 and AW189 actuators were being designed in the same period, and the handedness never entered the specification. The type family was assumed mature (p. 38, p. 214).

The point of escalation: after the 2012 service bulletin, the AW169 and AW189 load spectra were referenced to the AW139 and produced the same recommended life and the same inspection interval (p. 37); the actuator design specification had no line for thread handedness (p. 38); certification came in July 2015, the two failures were each classified catastrophic, split across two reports, and the nut's failure modes were limited to fracture and loosening under vibration (p. 38, p. 214). What 2012 left behind was two maintenance numbers. Where the torque went after the seizure has no sequel in the report, so the design input for the AW169 and AW189 remained a theoretical load spectrum (pp. 36 to 37). Those three

actions together made the repetition of one failure mode on a second type into a question nobody had to answer. The July 2015 certification analysis never addressed how this shaft would rotate once the bearing seized (p. 214). There is no single signing moment. The signatures are scattered across three kinds of document: a service bulletin, a design specification, a type certification review.

Failure chain: technical layer. The hybrid bearing ran with a lubricant film that broke down under the contact pressures of ordinary manoeuvres, the ceramic balls went from rolling to sliding, and the raceway developed surface-origin rolling contact fatigue. The cage wore and fractured, the released material was ground to powder and re-plated into a new raceway surface, clearance disappeared and the bearing seized. After the seizure, tail rotor drive torque passed through the bearing into a control shaft that should have been stationary. The shaft rotated at speed, broke the lockwire and sheared the split pin. Hydraulic pressure held nut and seat in contact, the thread wound the whole way out, and friction heat welded the two together and pushed them off the end of the shaft. Mechanical feedback failed, the control shaft travelled to the end of its range under hydraulic pressure, and the blades reached the physical limit at negative pitch. With main rotor torque no longer opposed, right yaw accelerated, and the yaw rate coupled through the airframe into a change of 43 degrees pitch and 25 degrees roll inside half a second (p. 62). Organizational layer. The bearing belonged to the tail rotor assembly and the control shaft and nut belonged to the actuator: two systems, two reports, two failure analyses. The actuator manufacturer stated it had not been told the effects of failure in connected parts. Both failures were classified catastrophic, which by regulation removed the requirement to analyse knock-on effects (pp. 38 to 39). Incentive layer. Certification is a reconciliation of compliance methods, and an item with no clause passes on the existing acceptable means. Once

a failure is classified catastrophic the risk is carried by the statistical basis, and system-side mitigation never enters the review's scope (p. 218). On the in-service side the cost sits on the discard life and the 600-hour hand check, both from a service bulletin and neither from an airworthiness directive (pp. 35 to 36). The boundary break sits between the organizational layer's system boundary and the technical layer's torque path: the drawings put the bearing and the actuator in two systems, and physically they are the two ends of one shaft.

Minimum counterfactual: the design specification for the AW169 and AW189 actuator states that the actuator-end lock nut is left-hand thread, opposite to the direction the control shaft turns after a bearing seizure. One action, in the design specification, costing one line of text and one annotation on a drawing. The AW139 design proved itself in a real seizure in 2012 (p. 214). The manufacturer's early 2020 modification did exactly this, a left-hand thread plus a washer, mandated on 6 March 2020 by airworthiness directive 2020-0048 (p. 100). The action needed no new analysis. The AW139 drawing was on file and the 2012 result was on file; what was missing was copying one design conclusion from the type family into the specification. It does not prevent a bearing from seizing and does not guarantee that the 2018 night departure could have been landed. It changes the direction of the consequences: the nut tightens instead of backing out, the control path stays connected, and the blades stop at +10 degrees instead of running to the negative pitch limit.

Outcome and sources: AAIB issued 8 recommendations, numbered 2023-018 to 2023-025, all to EASA, aimed at critical-part bearings, load spectrum validation, the airworthiness status of lives, and system-level mitigation for catastrophic failures (p. 15, p. 218); the manufacturer issued 16 service bulletins and EASA issued 9 airworthiness directives (p. 15); steel ball bearings came in

during September 2020 with an initial life of 400 hours (p. 100). The investigation also noted that other bearings removed from aircraft may have carried damage of varying degree, and that they were neither returned to the manufacturer nor given the same strip-down examination (p. 214). In June 2022 a UK AW139 was found with the same bearing failed at 2,750 hours during a ground check, and AAIB opened a separate investigation; the AW139 fleet had not embodied the modification, and the manufacturer considered it unaffected (p. 37, p. 100). Primary source: AAIB, Aircraft Accident Report 1/2023 (AAIB-25398), published 6 September 2023, Volume 1 of 224 pages with Volume 2 as appendices; link at Appendix B, av-040.

Case One leaves the certification system with a structural problem. Two control functions were each certified and each found compliant, and their combined behaviour was on no checklist anywhere. The general name for the problem is the combined certification gap: the parts of a system are verified against single-part standards, each with its own limits and failure modes, and the interaction between parts is carried by a line of engineering judgement that no significant risk exists, usually an intuitive extrapolation from the single-part results (both passed, so the combination should pass). Criterion one of this chapter takes this form here: the certification list is the exact inverse of the shared-resource list. It registers every part and registers nothing between parts. Closing the gap does not require testing every combination, whose number explodes geometrically. It requires a combination list: the scenarios in which two functions may act at the same time, one analysis or one test attached to each entry, and everything off the list declared an unverified area. The list turns a geometric explosion into a manageable register, which is this chapter's mapping idea applied directly to certification.

Case Two (Software): One Permission Doubles a File

sw-002. Cloudflare’s global edge network, 18 November 2025. The official postmortem, read and checked for this book: a database permission change doubled the size of a feature file managed by Bot Management and pushed it past its limit. The bad file spread across the network through the deployment mechanism, and core traffic was down over a wide area for about three hours, 11:20 to 14:30 UTC, with full recovery at 17:06. The permission system and the feature distribution system are two governance regimes, two monitoring stacks, two ledgers. Their connection is one trivial database grant. At the moment the change review passed, nobody was asked the question: once this permission is granted, whose file grows?

The decision point is the moment the permission change request is approved. On the desk: the rationale for the grant, the completeness of the change process, and one field that appears on no form, the downstream physical consequence of the grant.

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> 1. What would you do? Write the action, not “pay more attention”.
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> 3. If you may set one checkpoint, when is it and what does it check?

What happened: core traffic was interrupted for about three hours. The handling was to stop the bad file spreading and roll back to the last-known-good configuration. Primary source: Cloudflare, 2025. Link at Appendix B, sw-002.

Case Two has a second lesson worth taking separately, in the propagation mechanism itself. The bad file was not pushed by hand. It travelled to the whole network on the deployment mechanism, on its own. A deployment pipeline is a genuine advance in efficiency, and it is also the fastest path any coupling can travel (the prefix withdrawal in Case Five spread through deployment and synchronisation machinery too). The coupling properties of a pipeline deserve a paragraph in its own design document: it trusts by default whatever enters it, and the oversized bad file was admitted, and it synchronises everywhere by default, so the bad file travelled at network speed. Two gates belong on the pipe. Artifact validation, meaning size, signature and schema, as a precondition for propagation. Staged propagation, meaning batches across the network with a health check between them. Both gates aim at the path this case used, and both are the general answer to the class of incident in which one bad file takes the whole network down.

Case Three (Chemical): The Relief Line Became a Propagation Path

ch-005. Dow's Louisiana Glycol II ethylene oxide plant, 14 July 2023, Plaquemine, Louisiana. The CSB report (No. 2023-03-I-LA), PDF read and checked for this book: a fragment of a portable work light left behind during maintenance punctured a rupture disk, causing an explosion and fire, the release of more than 31,000 pounds of ethylene oxide, and shelter-in-place for hundreds of nearby residents. The designed coupling sits in the relief routing. The relief line was piped back to the ethylene oxide reflux drum, so the system meant to send an abnormal release to a safe location became the path by which explosion energy entered the process. The vessel's closure procedure had no step for confirming the drum was empty, and nothing on the paperwork knew the work light was inside it.

The decision point has two moments. The first is the design and change review that tied the relief system back to the reflux drum: the review passed on process grounds, and nobody examined the tie as a disaster propagation path. The second is the closeout of the maintenance and the sealing of the vessel: the checklist had no emptiness confirmation, and the light stayed inside. On the desk at the first: the relief capacity calculation, and the way the existing drawings showed the routing. On the desk at the second: the com - pletion status of the job, and the tick boxes on the checklist.

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> 1. What would you do? Write the action, not "pay more atten - tion".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: explosion, fire, and a large release. Primary source: CSB, No. 2023-03-I-LA, 2026. Link at Appendix B, ch-005.

The relief line in Case Three also carries a design philosophy question. Why was the relief system tied back into the process at all? The answer is capacity and environmental protection, recov - ering the relieved material instead of venting it, and every one of those reasons holds under the process-rationality terms of the change review. What was missing is a review under a different set of terms: how this connection behaves in an accident scenario, with explosion energy running back along it. The safety review and the process review answer their own questions on the same change form, and the coupling point falls in the gap between the two an - swers. The three classes of query under Device 2, downstream system, shared resource, emergency path, aim at exactly that gap:

the emergency path question forces someone to answer where the relief line sends energy in an explosion. Chemical readers can keep this case as a sentence pattern. Any path meant to lead to safety gets one further question: what is at the other end of it.

Case Four (Maritime): Non-OEM Bearings and a Ship That Could Not Anchor

ma-001. The survey vessel Kommandor Susan, 25 January 2025, Firth of Forth, Scotland. The MAIB report (Report 10/2026), PDF read and checked for this book: the main generator was destroyed and caught fire and the ship lost all electrical power; one engineer suffered serious lung, kidney and eye injuries from smoke during the evacuation. The chain ran on from the blackout, since the anchor windlass could not work, and the vessel could not anchor in coastal waters. The root cause points to non-OEM bearings fitted during an out-of-yard overhaul in 2019, which wore out early, and to the fact that the swap was made without the owner's knowledge (the report's wording: non-OEM components, fitted without Hays Ships' knowledge). One supply chain decision, taken to save money and time, connects to an emergency capability six years later: the windlass depends on the main power plant.

The decision point is the approval of the overhaul plan with non-OEM parts in it, and the failure to disclose that to the owner. On the desk: the price and lead time of the spares, the equivalence statement for the OEM part, and one question outside the scope of the overhaul contract review: on some night six years from now, what will this generator be unable to carry away?

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> 3. If you may set one checkpoint, when is it and what does it check?

What happened: one person seriously injured, the main generator destroyed. Primary source: MAIB, Report 10/2026, 2026. Link at Appendix B, ma-001.

The non-OEM bearing in Case Four leaves one contract-layer patch worth writing down: who verifies the equivalence statement in an overhaul contract. The supplier states equivalence, meaning com - parable performance, and the duty to verify is assigned in most contracts to nobody. In this case the swap happened without the owner even knowing (MAIB' s wording), so the duty to verify landed on a party that had no knowledge. The credibility of an equivalence statement is a core subject in supply chain coupling management. The price gap on an OEM part is the market price of the statement' s risk. Whether to buy OEM is a fair argument. Once a non-OEM part is bought, converting the statement into verifiable material, meaning material certificates, life test data and service records from the same type fleet, has to be nailed into the contract. Equipment readers can generalise the line: every word of an equivalence claim is owed a verification document.

Case Five (Software): An Empty Parameter Means All (full card)

sw-003.

When and where: 20 February 2026, 17:56 (UTC), at the prefix an - nouncement layer of Cloudflare' s global edge network; impact ended at 23:03 (UTC) the same day. The body of the postmortem gives the start as 17:48 (UTC), the timeline gives the start of impact as 17:56 (UTC). Both markers stand, and the report does not say where the difference comes from; the total duration in the report is

6 hours 7 minutes. The postmortem was published on blog.cloud‑flare.com, and every time in it is UTC.

What and who: the Cloudflare network team and the Addressing API team. The business side is BYOIP (Bring Your Own IP), where customers bring their own address space onto the platform; the affected downstream products were Core CDN and Security Ser‑vices, Spectrum, Dedicated Egress, and Magic Transit. Scale: that BGP neighbour carried 6,500 announcements, of which 4,306 were BYOIP prefixes; 1,100 were withdrawn in error, 25 per cent of the BYOIP total. Responsibility is recorded only to the level of role: the person who made the change, and the person who approved the deployment; the postmortem names teams and subtasks and no individuals. Its wording uses "our addressing system" and "our own software", pointing at systems and software rather than at one operator's mistake. Two classes of object were hit: BYOIP prefixes, and service bindings.

What they were doing: on 5 February 2026 at 21:53 (UTC) a line of code was merged to main. The background is Code Orange: Fail Small, a programme whose aim is to make every change pushed to the network as controlled as a software release, mediated by health signals and quick to revert. Removing a BYOIP prefix at a custom‑er's request had been a manual operation, and it sat on the list of manual actions the programme existed to eliminate. The change turned it into a cleanup subtask that runs on a schedule: scan for BYOIP prefixes due for deletion, then delete them. A prefix can carry many associated objects, so batch processing was the inevit‑able way to build it. The subtask does more than the prefix. BYOIP prefixes hold service bindings on the platform, each product with its own (CDN, Spectrum, Magic Transit), and when the prefix is removed the product configuration hanging off it loses its host. In the manual version a person worked through those in order. Re‑written as a subtask, one function deleted the whole set at once.

The request the subtask issues is `GET /v1/prefixes?pending_delete`: the parameter name is there, the value is not. The server implementation reads `if v := req.URL.Query().Get("pending_delete"); v != ""`, so only a non-empty value enters the branch that returns pending-delete prefixes alone. The client sends nothing, the server reads an empty string, the condition is false, the branch is skipped, and execution drops into the default path that returns every prefix. The subtask treats each returned item as already queued for deletion and be‑gins systematically deleting prefixes together with every dependent object, service bindings included.

At 17:46 (UTC) on 20 February the Address API version carrying this subtask finished deploying. At 17:56 (UTC) the subtask ran for the first time and prefix announcements began to disappear. Fif‑teen days separate merge from deployment; ten minutes separate deployment from effect. The consequence of a withdrawn an‑nouncement is not slowness. The prefix is gone from BGP, so the internet holds no route to it, and end connections enter BGP Path Hunting, trying alternative paths until they time out. What a cus‑tomer sees is failure after failure to connect, never an error page.

The decision point:

One. 5 February 2026, 21:53 (UTC), the merge. The vehicle is the code review and merge record. The review’s frame was “turn manual removal into an automated flow”, and the object of review was the customer journey, meaning what happens after a customer clicks remove in the console. Nobody asked what the server returns when this parameter is empty. In the postmortem’s own words, initial testing and code review focused on the BYOIP self-service API journey and completed successfully, and the testing did not cover the scenario where “the task runner service independently executes a change to user data with no explicit input”. Those two paths are not the same path in test design. A customer journey

needs a human to press a button; a task runner needs the clock to reach a time. A passing review proves only that the first is safe. Three things sit behind the failure to stop this. In framing, the review asked whether the flow was correct and never what the server returns for an empty parameter, which cannot occur on the customer journey because a customer pressing the button always carries a specific prefix. In path, the subject of the test case is a person and the subject of the path that caused the incident is a clock. In motive, the change belonged to a risk-reduction programme whose direction needed no defending, so review attention went to whether the automation faithfully reproduced the manual flow, and under non-empty parameters it did.

Two. 20 February 2026, 17:46 (UTC), the deployment. The vehicle is the Address API release record. Before deployment the health-mediating configuration rollback system was not yet in production (the postmortem's phrasing: "that system is not in Production today"), so this step had no gate that stops on error by itself. The change went out iteratively rather than in one pass over all BYOIP customers. That existing practice left 75 per cent of prefixes untouched, but it is a rollout method and not a safeguard built for this change, and all 1,100 were withdrawn before anything stopped. The deployment step looks reversible, on the logic that a mistake can be changed back. The postmortem says that assumption did not hold at the time: deleting a prefix took its service bindings with it, so reverting had to run three paths for three damaged states, and the longest of them required pushing configuration to machines worldwide one at a time.

Three. 20 February 2026, 18:21 to 18:46 (UTC), locating and stopping. At 18:13 someone engaged because one.one.one.one was failing; at 18:18 an internal incident was declared; at 18:21 the Addressing API team was paged; at 18:46 an engineer identified the subtask, terminated it and disabled the schedule. Fifty minutes

passed between the start of impact and the stop, and the withdrawals also completed inside those fifty minutes.

What was visible then: on the desk on 5 February, the request line and the parameter definition; the subtask's functional description; the existing house style of calling the API with a parameter name and no value in comparable tasks; and the server's handling of that parameter, readable in the repository. On the desk between 5 and 20 February: a staging environment with data as close to production as the team could make it; test records showing the BYOIP self-service customer journey passing; simulated data. On the desk on 20 February: the 17:46 deployment completion record; from 17:56 one.one.one.one returning 403 with "Edge IP Restricted", while resolution of 1.1.1.1 including DNS over HTTPS was unaffected; on the BGP side, the prefix count on that neighbour falling continuously between 17:56 and 18:46, with individual customers' services dropping off the network. In the same period the progress of alternatives was visible to the team: staged test mediation and correctness validation for the Addressing API were already under way in parallel, while health-mediating configuration rollout and snapshot rollback were not yet in production (the postmortem's words: "critical work was already ongoing", "preventative measures weren't fully deployed before the outage"). Information that arrived only afterwards: empty was read on the server as no filter; the deletions included service bindings, not prefixes alone; and the full list of withdrawn prefixes and the three damaged states only came together during the recovery.

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?

> 3. If you may set one checkpoint, when is it and what does it check?

What happened: between 17:56 and 18:46 (UTC), 1,100 prefixes were withdrawn, 17 per cent of the 6,500 announcements on that neighbour and 25 per cent of the 4,306 BYOIP prefixes. Around 20:20 (UTC) the rollback was in place and 800 prefixes were announcing again. At 19:19 (UTC) guidance went out so that customers could re-announce from the console themselves. The remaining roughly 300 had lost their service configuration at the edge, with no service binding left in the console to switch to, so they could only come back once engineers completed a global machine configuration push at 23:03 (UTC). The count is of BGP announcements: on a curve of the 6,500 announcements on one neighbour over time, the 1,100 are the segment that vanishes. It is not a count of customer reports and not a count of affected domains. The report states that most of the 6 hours 7 minutes went into restoring prefix configuration to its pre-change state. Back to the three questions. The most common answer to the first is to add a confirmation dialog to the cleanup task, but a confirmation guards against misoperation and cannot guard against the automatic path through a semantic gap, since with an empty parameter no human is in the loop. The item most often missed under the second question is the empty value itself: too inconspicuous, with review attention on the main functional path. The third question lands on making parameter semantics explicit: empty is either forbidden by a required-field check or explicitly defined as no operation, one of the two, written into the code and into the review checklist. Primary source: Cloudflare, 2026. Link at Appendix B, sw-003.

Options ruled out: automatic rollback through the health-mediating system (a Code Orange build, not in production at the time; rollback could only run on database snapshots plus per-machine configuration push, the first 800 took that route and the last 300

took a longer one; the postmortem's wording is that engineers had to use database snapshots rather than reconcile desired state with actual state); customer self-service from the console (guidance from 19:19, effective for the 800 and useless for the remaining 300, whose service bindings were already deleted so the console had nothing to switch to, and that failure is a signal in itself); running the task runner once in staging with an empty parameter (the postmortem judged the simulated data insufficient, and the scenario was not in the test design either, which covered the customer-presses-a-button path); a global cap or circuit breaker on the withdrawal action (the follow-up item is to monitor for withdrawals that are too fast and too broad and stop the rollout, and nothing of the kind existed at the time); verification at a per-minute rate (1,100 in 50 minutes averages 22 a minute, derived from the two timeline points and the count, and entry-by-entry checking could not keep up with that); a health check between batches (the batching at the time was iterative rollout carrying no checks).

Early signals: (1) pending_delete was interpreted as a string, and client and server had trouble aligning on the value of that flag, which is the postmortem's own attribution. It was buried because the parameter was optional, the code was compact, and the customer journey test passed. (2) The cleanup subtask ran on a schedule and changed user data on its own, sitting on no customer path at all. It was buried because eliminating manual action was the point of the change, so automation read as reduced risk, and when the program misbehaved nobody was there to stop it. (3) Manual removal was a routine low-frequency customer request, one prefix at a time, with an operator who knew exactly which prefix was going. Automation turned it into a batch that could reach every prefix. It was buried because batching answered the fact that a prefix may have many associated objects, and nobody matched the step from one at a time to all at once with a limit. (4) The configuration change had no production-side snapshot roll-

back. It was buried because Code Orange was building one and it was nearly ready; before the outage the item was a line on a schedule, never a line in the risk register. (5) The subtask executed ten minutes after going live, with no observation window for the first run. It was buried because this was a routine task that had run countless times, and the fifteen days between merge and deployment spanned two iterations, so the context of the review was no longer in the room on deployment day.

The point of escalation: the merge at 21:53 (UTC) on 5 February 2026 is that step. The line turned an empty parameter from a harmless input into a trigger for deleting everything, and it handed the deletion to a scheduled execution that asks no one. Deployment came at 17:46 on 20 February and execution at 17:56; from then on every deleted prefix took its service bindings with it, and the recovery path degraded from change it back to push configuration to every machine again. The postmortem does not list the approvers of the merge or of the deployment, so the signatures of this escalation are scattered across two kinds of record. Reading the merge as the escalation holds because it changed two quantities at once: the width of the trigger, from a named prefix to every prefix, and the executing subject, from a person to a clock. With the width changed, the cost of an error moved from one object to a set. With the subject changed, nobody was present to see the error. The line sat on main for fifteen days between 5 and 20 February, and nobody blocked it on the grounds of what an empty value returns. The only possible interception point in that window was the merge review; after it, every step was automatic until someone called a halt at 18:46. The escalation has no minutes, only a merge timestamp and a deployment timestamp.

Failure chain: technical layer. The client treated the query parameter as a presence switch, sending the name without a value, and the server judged it by its value, where an empty string is false. The

empty string carries two meanings on the server at once: this parameter is absent, and this parameter is present with no value. The second was read as the first, the filter disappeared, and everything came back; the subtask read all as all pending deletion; deleting took the service bindings; BGP announcements were withdrawn; end connections entered BGP Path Hunting, changing paths until they timed out. Organizational layer. The review fol - lowed the customer journey and never covered the task runner acting on its own; the first run of a scheduled task had no obser - vation window, and the first signal of the fault came from the cus - tomer side, with one.one.one.one returning 403, which is what brought someone in at 18:13, while resolution through 1.1.1.1 was unaffected, so that signal pointed neither at the root cause nor at the affected object; the staging data was insufficient; there was no production-side snapshot system for rollback, and recovery split into three paths by three damaged states, the longest requiring a global machine-by-machine push. Incentive layer. The change was justified as reducing the risk of manual work, and the direction it - self, that more automation is safer, needed no defence in review; iterative rollout saved 75 per cent, but it was standard practice and not a measure aimed at this change. The boundary break sits between two parameter conventions: in HTTP a parameter name present means true, in Go a retrieved empty string means false, and the value empty has a reading on each side while the deletion fell exactly between them, in a stretch where no human was in the loop.

Minimum counterfactual: the server parses pending_delete as a boolean, accepts only true, and returns 400 for a missing or empty value. One action, taken in the code review before the merge, costing three lines of validation and one test case. Empty then has exactly one reading, and a subtask that gets a 400 deletes nothing. Cloudflare’ s own follow-up item does this in substance (API schema standardization, so that client and server align on the

value of the flag), and the implementation at the time treated it as a string with each side interpreting for itself. That line is visible to the eye in the 5 February diff. It needs no new system, no schedul - ing, and no judgement made on the day of the outage. It adds one question to the review checklist: when this parameter is empty, what does the server return. The action does not stop anyone writing a wrong request. It stops a wrong request from being ex - ecuted as all of them, and it does not stop a prefix from being de - leted either, only requires that the answer all be stated once, ex - plicitly.

Outcome and sources: 1,100 prefixes withdrawn, 25 per cent of BYOIP; 800 came back with the rollback; about 300 came back at 23:03 (UTC) through a global configuration push, for a total of 6 hours 7 minutes. The postmortem explicitly excludes network at - tack and malicious behaviour. It carries three follow-up items: standardising the API schema; separating operational state from configuration state with snapshot rollback; and a circuit breaker for withdrawals that are too fast and too broad. It places this change in the first category of Code Orange: Fail Small, in which any configuration change pushed to the network gets a controlled rollout on the same terms as a software binary release. The page notes a 12-minute read and carries a table of affected products and an itemised timeline. The report does not state how many cus - tomers were affected. Primary source: Cloudflare, "Cloudflare outage on February 20, 2026", David Tuber, February 2026, all times UTC; the postmortem is a web report, so the figures here are cited to timeline entries rather than to page numbers. Link at Appendix B, sw-003.

Case Six (Software): A Regex Outside the Load Test (condensed)

sw-001. Cloudflare's edge WAF, July 2019. The official post - mortem, read and checked for this book: one regular expression triggered catastrophic backtracking in the global environment,

exhausted CPU, took the network down worldwide for 27 minutes, and about 80 per cent of traffic was lost. The handling decision recorded in the postmortem is this chapter's best sample of stopping well: at 14:02 the proposal to disable the WAF globally, at 14:07 the execution, five minutes to stop the bleeding instead of working the problem point by point. The coupling sits in the environment. The same regex was safe in the load-test environment, where input volume is small, and in the global environment, at the scale of real traffic, its backtracking complexity exploded; a rule that takes effect across environments does not carry its performance characteristics across them, which is the standard shape of software coupling. Minimum counterfactual: performance-test each rule at production input scale before it goes live, and give every global rule its own kill switch. Primary source: Cloudflare, 2019. Link at Appendix B, sw-001.

Case Seven (Software): Customer Retries Became the Attack (condensed)

sw-004. Cloudflare's control plane, September 2025. The official postmortem, read and checked for this book: after a tenant service failed, retry mechanisms on the customer side all fired at once, the retry storm came back and crushed the control plane, and the control plane was unavailable or partly available for about an hour. The coupling crosses a system boundary. Customer systems, each with its own retry policy, and the platform system, with its control plane capacity, are joined into one thing by collective behaviour at the moment of failure. No review on either side alone sees that line, because it only becomes live during a failure. The standard patch after the handling combination of rate limiting, capacity and rollback is to write backoff and jitter requirements for retries into the client contract. Minimum counterfactual: put the client retry policy into the platform's public contract, with backoff mandatory. Primary source: Cloudflare, 2025. Link at Appendix B, sw-004.

Case Eight (Aviation): An Unlatched Door and a Lying Pitot (condensed)

av-026. A 747 freighter, February 2024, United Kingdom. The AAIB report (2026), read and checked for this book: the L1 door had not been properly closed in rain, water came through the drip shield into the electronics bay, the airspeed data went wrong, the aircraft pitched down 7.6 degrees at one point with a descent rate of 6,240 feet per minute, and the crew recovered and landed safely. The coupling chain: the open state of a cargo door, a ground handling item; the sealing of the drip shield, a design assumption about water ingress; and the pitot, an air data item. Three parts belonging to three specialisms were joined into one path by rainwater. The decision point is the dispatch after working with the cargo door open in rain, with no requirement to re-check the shield seal. Minimum counterfactual: add a visual seal check after any work with the door open in rain. Primary source: AAIB, 2026. Link at Appendix B, av-026.

Case Nine (Agriculture and Food): The Invisible End Product (condensed)

ag-008. A stabilizer ingredient carrying bacteria reached food plants in several states. The FDA enforcement report (F-1333-2024), API record read and checked for this book: the contaminated stabilizer went to downstream bakeries in 23 states plus Canada and Mexico, and 7,257.6 kilograms were recalled (Class I). The end product of a business-to-business ingredient is invisible: the ingredient supplier knows what was shipped and not which bag of flour became which item on which shelf. Supply chain coupling adds one layer of invisibility at every packaging conversion. Minimum counterfactual: keep a batch-level flow ledger for B2B ingredients, with conversion into end product mapped finely enough to recall. Primary source: FDA, F-1333-2024, 2024. Link at Appendix B, ag-008.

21.4 What Shows Up Earliest

Six weak signals, each with the reason normally used to bury it.

1. The question "who is affected if this goes down" needs a meeting before it can be answered. The reason: "the system is complex, that is normal."
2. Fault reports repeatedly carry "propagation path unclear" or "recovered for unknown reasons". The reason: "it is in the knowledge base now."
3. The maintenance windows and change windows of two systems overlap almost exactly. The reason: "one shutdown is less trouble."
4. There is no list of shared resources (accounts, certificates, piping, the grid, people), or the list has not been updated in years. The reason: "it is all in people's heads."
5. After a merger or an outsourcing deal, the old system's ledger was never merged in. The reason: "the function has already moved over."
6. On the review record of some change, the impact field says "none". The reason: "it was a small change."

Put the six back against the nine cases. Signal one, the impact question that needs a meeting, belongs to Case Five: before the incident nobody could have said who the 1,100 prefixes would hit. Signal two, the unknown cause that goes into the knowledge base, belongs to Case Six: the performance of that regex failed across environments, and the first attribution in the fault report was difficult. Signal four, the missing shared-resource list, belongs to Case Two: the connection between the permission system and the feature distribution system sat on no list at all. Signal five, the unmerged ledger, belongs to Cases Three and Four: the closure checklist in the maintenance procedure, and the information

boundary in the overhaul contract. Signal six, impact written as "none", is the change-form signature of every software case here.

The nine cases sort again by what carries the coupling, and this second sorting is the more practical one. Resources carry some of it: a permission, piping, a bearing, water. Behaviour carries some: retry, propagation, rollback. Semantics carry the rest: how an empty parameter is read, how a rule fails when it crosses environments. Each carrier has its own detection method. Resource carriers are checked against a register (Device 4). Behaviour carriers are checked by looking at collective behaviour during a failure, and by making retry policy contractual. Semantic carriers are checked with a review checklist (Device 2 plus a line for explicit semantics). The trivial thing that criterion one names has three aliases on these carriers: shared resource, collective behaviour, default semantics. Running all three classes during a mapping pass finds dark lines faster than working system by system.

There is a fourth carrier, the most hidden of the four and worth its own heading: people. A shared person, meaning the one person who understands a system, the duty officer holding three posts, the specialist who only appears in the key meetings, is the commonest shared resource in any organisation and almost never appears on a resource list. In the HR system he is a headcount in a department. In the technical system he is a critical path. Human coupling behaves at failure time exactly as any resource coupling does. The person is absent, through leave, departure, or being occupied by two incidents at once, and both lines break together. The case counterpart is the engineer on Kommandor Susan, seriously injured during the evacuation, after which nobody aboard could deal with the engine room; the vessel's emergency plan assumed that a person able to get back into the engine room would exist. Registering people, meaning a list of the sole skilled holder for each key post plus a proxy matrix, is the first chapter the register should be

given. Chapter 18's two-line instruction, a name and a deputy, joins this chapter here: the proxy matrix is the redundancy design for human resources.

True rehearsal, the Chapter 22 principle, has one form reserved for people: the deputy actually stands the watch. A name on the matrix does not mean the post can be covered. The deputy of a sole skilled holder has to have done that person's work at least once, stood the watch, run the operation, chaired the key meeting, with the date entered in the matrix. The crew's emergency training in Case Four, evacuation and abandon ship, is human proxy rehearsal in its purest form: a trained movement is the only one that has somewhere to come from once the smoke is thick. Rehearsing people is harder to schedule than testing equipment, since a person cannot be turned over once a quarter the way a pump can. The substitute is rehearsal at the level of the task: every six months the deputy actually performs three of the key person's most frequent tasks, and the key person writes the list of three. Six months is the right sampling rate for the decay of a human skill (Chapter 26).

Beyond sampling rate, human coupling has an organisational symptom that belongs on the weak-signal list: the meeting cancellation rate when one person is on leave. Move a key person's calendar and a string of meetings reschedules or cancels, which says the line runs through him alone. Measuring it needs no instrument, only the calendar. The names with the highest cancellation rates are the names the register most needs a deputy for. The usual misreading in the organisation is "this person is important", taken as credit. The correct reading is that there is no second line where this person sits, taken as risk. The two readings are settled at failure time: importance is counted in days, exposure is counted in occurrences, and one simultaneous occupation, with him caught between two incident scenes, is enough to break both lines at once.

Simultaneous occupation has a dedicated amplifier in a crisis: every scene wants the same capable person. The first reflex of a command post is to call for the most familiar name; he is sent to the first scene, the second loses him at the same moment, and the third call finds him already on the road. The staffing principle for a crisis, the crisis version at the end of Chapter 18, therefore needs one line: the commander's call list is ordered by availability and not by familiarity alone, and the call list and the proxy matrix come from one source.

Sharing the source has a second use, in drills: the drill's sampling rule deliberately skips the first name on the call list and calls the deputy up. One drill then tests two things, the deputy's ability to hold the post (Chapter 22's true rehearsal) and whether the call list is ordered sensibly. Two fixed debrief questions follow: did the deputy hold (human redundancy), and how far did the scene de‑grade once the first name was taken away (the price of the coup‑ling). Both answers go into the annual revision of the matrix and the plans. The sampling rule meets resistance in small organisa‑tions, where the mainstay idles on drill day. The answer to the resistance is to cast the mainstay as the failure party, since the person who knows the system best plays failure most convincingly; no labour is wasted and the drill gains realism.

Past realism, this chapter owes the reader one more line of exten‑sion across industries: data. Two datasets with nothing to do with each other can be joined, through an ID number, a device identifi‑er, a latitude and longitude, a timestamp, and the join can turn apparently anonymous records back into a personal profile after the fact. The carrier here is the key, and the object of mapping is key sharing between data flows. The consent conflicts in Case Five of Chapter 17 are at bottom keys moving across systems beyond the coverage of the compliance map. A data dependency graph, re‑cording which datasets share which keys, at what granularity, go‑

ing where, is the register mirrored into data governance. Privacy engineers can port the method as it stands: draw the key graph, register the keys, and ask about a key's blast radius at change review.

The last stop is finance. In trading systems the carriers are the clock and the market data feed. Every strategy shares one feed and one time reference, so a jitter in the feed revalues every strategy at once, and a small failure at a market data vendor is a regular industry-wide stress test. The register translated for finance: a shared list of feed sources, time sources, risk parameter sources and clearing channels, each entry carrying its users and its failure impact. Regulatory technology has already written this chapter into several trading rules, in the requirements for backup market data feeds and redundant timing. The register's incremental value is turning the requirement from a clause into a checkable graph. Nine cases plus three extensions, data, finance and people, close the chapter's reach: wherever there is sharing there is coupling, wherever there is coupling there is a mapping debt, and wherever there is a mapping debt there is a bill that arrives when the line is energised.

One clause in how that bill is delivered is worth knowing in advance: a dark line's bill is almost always larger than a lit line's. A lit failure, where a part the design depends on breaks, comes with a plan, a spare and a drill. A dark failure, where a line nobody knew about goes down, has none of the three: the plans were written from the drawn diagrams, the spares were stocked for the known single points, the drills were scheduled around the known scenarios. With none of the three, handling falls back on improvisation, and its duration is set by how fast the line is found. A good part of Case Two's three hours went into locating the propagation path, and the customer self-service portion of Case Five's six hours was guided only after location was finished. Loc-

ation speed is the only lever on a dark failure, and the training material for it has exactly one source: archived propagation lines from past failures, the three debrief questions above. The thicker the archive, the more improvisation converges on table lookup, and table lookup is an order of magnitude faster than a divergent search.

The gain is delivered only if the archive can be queried: debriefs indexed by propagation line, as a triple of origin, carrier and destination, and looked up by symptom when the next failure starts. Index maintenance is another view of the register. The two views, the graph for prevention and the index for response, share one underlying dataset, so the cost is paid once. Four devices, two views, one staffing recommendation and one review checklist are now delivered. What remains is a last reminder: a dark line does not respect an organisational boundary. It runs through the outsourced function, through the supplier, through the customer's system, so the mapping boundary cannot stop at the organisation's edge either. Register entries run out to the external system, with the counterparty's contact and the counterparty's failure impact, because the lines that cross a boundary are usually the ones that bill the most.

The evidence sits in the four main cases. A helicopter's set of control interactions crossed the boundary between supplier and airworthiness authority. One permission grant crossed the boundary between platform and customer. One relief line crossed the boundary between contractor maintenance and owner process. One bearing swap crossed the boundary between repair yard and shipowner. All four lines cross a seam between organisations, and all four bills are split between the two sides of the seam. Cross-boundary entries get registered down to the other party's name, and the annual mapping asks one layer beyond the boundary. The last line of the chapter belongs to everyone who works at a seam:

the line under your feet has probably never been drawn by the person on the other side of it, and drawing it is the only map the two of you will ever share. The cost is one more column in the plan. The benefit is that a crisis scene stops being a contest over the same person.

21.5 How to Prevent It

Device 1: the annual dependency map. Once a year, draw the real dependency graph, working from the bottom up. Start with the least conspicuous shared resources, the accounts, the piping, the certificates, the people, and ask of each one who uses it and who is affected if it breaks. Whatever cannot be drawn is the risk area, and it is registered on its own, at which point criterion two be - comes a ledger entry by itself. One organisational note decides whether the map is real: the questions are asked by someone who does not know the system and answered by someone who does. A person who knows the system and draws it alone will draw the system in his memory, which is not the system as built.

Device 2: a mandatory impact field in change review. The impact field on a change form may not say "none". It has to name at least three classes of party that were consulted, downstream sys - tem, shared resource, emergency path, and the consultation has to be recorded. The permission form at Cloudflare and the piping change form at Dow both need that question on them. The purpose of a mandatory field is not to make anyone smarter. It is to make "nobody asked" a matter of record.

Device 3: independent validation of emergency paths. Once a year, take each critical emergency capability, anchoring, relief, evacu - ation, rollback, and run an exercise on the assumption that the main system is unavailable, to check what the capability actually depends on. The windlass on Kommandor Susan and the failover

drills of Chapter 22 meet here. An emergency capability's dependence on the main system is the most expensive line on the coupling graph, because it only becomes live at the worst moment.

Device 4: the shared-resource register. Every resource used by two or more systems gets its own entry: name, users, failure impact, owner. The register is the skeleton of the dependency graph. Without it, the graph starts from zero every year.

Set the four devices against this chapter's cases. Device 1 on Case Four: start from the least conspicuous shared resource, the bearing, feeding the main engine, the standby and the windlass chain, and draw the dependency lines between generation and emergency capability. Device 2 on Case Five: add a parameter-semantics question to the three consultation classes on the change form. Device 3 on Case Three: the annual exercise on the relief system carries a mandatory answer to what is at the other end of the path. Device 4 on Case Two: the permission entry records the downstream files and distribution chain the grant feeds.

The four devices also have four ways of dying. The annual map dies when the drawing is finished and nobody uses it, with the graph on the wall and the change review never citing it; the patch is to make the change form cite the relevant node by number, which binds the graph to the change process. The mandatory impact field dies of perfunctory consultation, with every party returning the templated answer no impact; the patch is to sample-audit the consultation records and treat a perfunctory record as no consultation. The emergency validation dies of being done on paper, with the annual exercise turned into a slide review; the patch is to require one physical action in every exercise, a real cutover or a real anchor drop, scheduled alongside the true rehearsal of Chapter 22. The register dies of not being updated, when a resource is removed, merged or swapped and no edit follows; the patch is to link register entries to change forms, so a change touching a resource auto-

matically raises the entry for update. The shared health check for all four: when was the device's output last cited, and by whom.

One acceleration rule on cadence deserves its own line: tighten the interval during change-dense periods. The annual map assumes a system changes once a year, and a software system changes ten times a month. Where change density is high, the mapping follows the change rhythm, with a dependency recheck after each large change instead of a wait for the annual pass. The empty-parameter subtask in Case Five and the permission change in Case Two both sit in change-dense territory. In such places the annual pass is deadline management rather than a discovery mechanism, and discovery comes from mapping triggered by change.

21.6 Boundaries with Neighbouring Mechanisms

Against the illusion of redundancy (Chapter 22): the illusion is redundancy contaminated from a common source, with the backup dying alongside the primary. Hidden coupling is independent systems tied by a dark line, with two systems that should not die together dying together. The test: if the systems involved were supposed to back each other up, go to Chapter 22. If they were supposed to have nothing to do with each other, stay here.

Against cascading failure (Chapter 33): that chapter covers the concentration of dependency and the scale of propagation, through suppliers, cloud providers and logistics networks. This chapter covers the cognition of connection, where nobody knows the line exists. The dependency register of Chapter 33 is this chapter's mapping restricted to a supply chain subgraph.

Against crossing a scale threshold (Chapter 24): scale multiplies the search space of coupling, since the number of unmapped connections grows faster than the number of nodes. An organisation in

rapid expansion should tighten the annual map to quarterly. The interval follows geometry, not choice.

21.7 The Five Self-Checks

1. Name three things in your system that are shared by two or more subsystems. If you cannot name three, how long has the mapping been owed?
2. Where is the debrief for the most recent "unknown cause" failure filed? Dark lines usually hide in the second debrief.
3. Which item on your emergency capability list has never been validated on the assumption that the main system is unavail - able?
4. When did your organisation last go through a merger, an out - sourcing deal or a large restructuring, and was the old system's ledger merged in at the time?
5. Take a recent change form and read the impact field. If it says "none", ask the three consultation questions today, and see what comes back.

(This chapter: 15,585 characters.)

Chapter 22 The Illusion of Redundancy

> What this chapter covers: why two backups fail together, the three tests for common-cause failure, its shape in four domains, the arithmetic of redundancy, and the validation standard of systems that are different and have actually been rehearsed. When to turn back: when reviewing a redundant design, buying a backup system, or when someone answers a single-point challenge with "we have two". Estimated reading time: 50 minutes.

"We have two" is the sentence in an engineering conversation that most needs a follow-up. The follow-up is not "where is the second one". It is three more expensive questions. How does it differ from the first? Who has verified that it can work on its own? Has the switch to it ever been made under real pressure? Ask all three and a good number of "two" shrink to one and a half, or to one with a manual. This chapter is about that shrinkage: how redundancy loses its independence at each stage, in procurement, in design, in rehearsal, until the day the common cause arrives and every set fails at once.

22.1 What it is

Definition: the illusion of redundancy describes several lines of defence that stand side by side on paper and are in fact one line. They share a power supply, a vendor, a drawing, a crew, or an assumption. When the common cause arrives, all of them fail at once. Engineering calls this common-cause failure. The word illusion marks the cognitive side: an auditor counts the number of lines, an attacker counts the kinds.

Three tests for activation. Any one of them means the mechanism is already running.

1. Two "independent" lines share a common item: power, vendor, algorithm, drawing, people, location, or one shared assumption.
2. The redundancy has never been verified on its own. Nobody has ever shut the primary down and tried the backup.
3. The switchover has never been rehearsed under real pressure. Every rehearsal has been run in a planned window, at low load, with a way back.

22.2 How it grows

Three conditions generate it.

Condition one, the scale logic of procurement. A second system from the same supplier is cheaper, integrates more easily, and the operations team does not have to learn a second product. Hetero - geneity means twice the learning and twice the maintenance cost, so the budget moves rationally toward the same source. Nobody decides "we want a single source". Each local optimisation points there.

Condition two, design reuse. The backup inherits the primary's drawings and assumptions, and the errors in the design are copied faithfully into the backup. Even the errors are redundant.

Condition three, rehearsal cost avoidance. A real switchover car - ries risk, and real pressure carries business impact, so the switchover is scheduled as a ceremony both sides know: planned window, low load, a retreat available at any moment. The cere - mony completes, and the rehearsal record says successful.

The arithmetic of common-cause failure is the centre of this chapter. Let p be the annual failure probability of one line. Two

independent lines fail together with probability about p^2, so the return on redundancy comes from multiplication. If the two lines share a common cause with probability c, the joint probability is about $c + p^2$, and the order of magnitude drops back from multiplication to addition. The return evaporates. Until the common cause is removed, the arithmetic of redundancy is fiction, which is why test 1 asks about shared items rather than about the number of sets.

The family tree of common causes is worth listing in full, because the register (Device 1) is searched by family. The environmental family: one power supply, one server room, one flood, one electromagnetic environment. Arkema's flood is the specimen for this family. The design family: one drawing, one algorithm, one assumption. The assumption in the map case is that the screen carries all the traffic. The operational family: one maintenance team, one configuration change, one monitoring stack. The common item in the monitoring cases is people. The behavioural family: one habit of use, one allocation of trust. The sanding case rests on an assumption about how automatic and manual sanding divide the work.

Beyond those four sits a fifth and the most concealed: certification. Two systems passed the same certification to the same standard, and the blind spot in the certification method becomes the shared blind spot of both. The evidential assumption in case four of this chapter has the same origin as case four in Chapter 20. Register by family and the result is more reliable than scanning by intuition, because intuition misses environment and certification most often: neither is in view on an ordinary day.

One step further into the arithmetic gives the qualitative version of the beta factor, which reliability engineering knows well. Split the failure probability of the pair into an independent part and a common-cause part. The share held by the common-cause part,

beta, decides what the redundancy is worth. Beta near zero and the pair behaves almost multiplicatively. Beta near one and the pair behaves almost like a single line. Beta comes from a list rather than from theory: count the shared items under test 1, and the contribution of each is roughly visible. This chapter keeps no formulas, only one conversion for managers. The discount rate on the redundancy you bought is beta. More shared items means a deeper discount. The heterogeneity you saved on in the purchase contract is the price of a higher beta.

22.3 The shape in four domains

Case one (aviation): A map treated as a collision-avoidance system

av-022. A Bell 206L-4 (N193AL) and a float-equipped DHC-2 (N97EL), 4 September 2023, about a thousand feet over King Salmon, Alaska. NTSB final report (ANC23LA071, downloaded and read by this book): the two aircraft collided in flight. One person in the helicopter was slightly injured; seven people in the aeroplane were unhurt. The helicopter crew was using an ADS-B electronic map as its reference for traffic and collision avoidance, and the DHC-2 it struck carried no ADS-B at all, so the traffic picture on the helicopter's screen could not contain that class of target. The paper redundancy for collision avoidance, eyes outside plus electronic map, was down to one leg in reality: the crew's visual scan had been partly ceded to the screen, and the screen stayed silent about the traffic it could not see.

Stop at the decision point: the stretch of flight in which the helicopter crew treated the map as its traffic picture. What was on the table: the target list on the screen, the operating habit of seeing and avoiding, and a question nobody had checked, which is how

much the screen does not show. (What happened: mid-air collision.) Primary source: NTSB, ANC23LA071, 2024, link in Appendix B av-022.

What case one adds to a redundancy list is perception as a line of defence. Looking out and an electronic map should be two independent sensing lines, since eyes and screen fail in different ways. After the cession, the presence of the screen weakens the scan: the screen says there is no traffic, and the eyes stop working at it. Branches of a redundant system have a behavioural dependence as well as a problem of physical independence. The existence of one branch changes how hard the other is used. Human factors calls the aviation form perceptual complacency. The list needs a line for use-dependence between branches: does the presence of A reduce how much B is used. Automatic against manual sanding in the sanding case, and the purple syringe against the double check in the healthcare case, are recorded on that same line.

Case two (construction & infrastructure / rail): Two sanders fall silent together

Case ID: ce-001

When and where: 21 October 2024 at 19:26; the single line west of Talerddig passing loop, Powys, Wales, United Kingdom. The collision point lay about 890 m beyond the exit points at the west end of the loop, and 1,080 m beyond block marker MH1078. The train had been braking into the east end of the loop since 19:22.

What and who: Transport for Wales Rail Limited (TfWRL), operator of passenger services on the Cambrian lines. The unit was Class 158 diesel multiple unit 158841 (leading car 57841, rear car 52841). The driver qualified on that line and type in March 2022, after 13 years in other railway roles. The infrastructure, the Talerddig loop points and the Machynlleth signal control centre belong

to Network Rail's Wales and Borders route; the signaller on duty had 8 years of experience. The rolling stock is owned by Angel Trains. ERTMS signalling on the Cambrian lines was supplied by Ansaldo STS, now Compagnie des Signaux. Vehicle maintenance, fuelling point checks and sanding tests were carried out by TfWRL at Machynlleth depot.

What they were doing: In the 24 hours before the collision the unit was working its seventh diagram of the day, and its sanding system had just passed three "serviceable" confirmations in a row. In the early hours of 18 October, 158841 spent the night at Holyhead depot and three 5 kg bags of sand went into the under-floor sand boxes (recorded on CCTV). Overnight on 19 to 20 October the unit had a fuelling point check at Machynlleth depot, which falls about every 1,500 miles: the technician pressed the sanding test button on the sand box, saw sand come out, and recorded a pass. On the morning of 20 October the driver instructor and a trainee carried out the preparation, pressed the test button again, and sand came out again (p. 19). Through 20 October the unit ran four routes; disruption meant the planned overnight at Crewe became a night at a platform at Chester. On the morning of 21 October at Chester the driver on duty could not reach the underfloor equipment, so the sanding test was not done, and the procedure did not require it (p. 19). On 25 September the annual sanding rate test had been recorded as completed, though TfWRL's maintenance planning system had no field for the weight, only a tick box (p. 90). On 12 October a B exam, about every 7,500 miles, passed its wheelslide protection test (pp. 50, 90). On the evening of 21 October, 158841 coupled to 158828, was uncoupled at Shrewsbury and went forward as 1J25, departing at 18:33 two minutes late with 31 people aboard. The opposing service 1S71 (158824, 6 people aboard) left Machynlleth at 19:08. Both ran on time and were booked to pass at Talerddig: the signaller put 1J25 into the Up Loop to stop, and 1S71 through the Down Loop at

95 km/h (p. 20). At 19:22 the train approached the loop at 125 km/h; the driver closed the power controller, made two applications of brake step 1 within 39 seconds, and the speed fell to 94 km/h. At 19:23 he selected step 2, and three to four seconds later applied full service braking 640 m from the stopping point, with ETCS applying as well; the driver states that he applied too. About a second later the WSP intervened. Six seconds after that, ETCS commanded emergency braking, at about 500 m from the stopping point (p. 21).

The decision point: In January 2008, Overrun Risk Assessment ERTMS Level 2 Movement Authorities Cambrian, EDSCL-DES-004000, version 0.2 issue 10, 7 pages, covered the whole Cambrian line and was not location-specific. The document cited GI/RT7006 clause 4.4 b i, on the grounds that the standard for lineside signalling does not apply to a cab signalling system and that Cambrian ERTMS carries no AWS, and on that basis dispensed with a detailed overrun risk assessment. It set an overrun rate of 2×10^{-9} per train per hour, which is to say that a train under ETCS supervision will not pass the overlap (p. 72). In September 2009 the ERTMS project obtained a derogation on that basis: beyond MH1078 there is only 81 m of overlap where the standard requires 180 m (p. 71). Once the overlap was shortened, the margin against overrun rested on the braking performance of the train, and braking performance at low adhesion depends on sanding. RAIB found no evidence that any party to the project recognised that dependency (p. 89). The document made "ETCS supervises continuously, so no train passes the overlap" a premise for the whole line, and because the detailed assessment was waived, the combination beyond MH1078 of a low-adhesion high-risk area, a 1:56 descent and the entry to a single line was never assessed at all (p. 72).

On 25 September 2024, at Machynlleth depot, a technician carried out the annual sanding rate test. A bag went over the sand delivery

hose, the test button was pressed for 30 seconds, and the procedure required 0.75 to 1.0 kg from each hose. The system records only a tick for completed, with no weight (p. 90).

The fuelling point check in the early hours of 20 October, on the night shift of the 19th, and the departure preparation on the morning of 21 October confirmed "sanding serviceable" twice more, both times using the button on the sand box. That button has its own supply and bypasses the sanding isolation switch (SIS) and the low-speed relay (LSR4) (p. 90). The same bypass circuit had been used in the B exam of 12 October.

Between 19:23 and 19:26 on 21 October, in the cab of 57841: ETCS applied full service braking and then emergency braking. The driver called the signaller on the ordinary GSM-R and reported that he was slipping through Talerddig and expected to pass the block marker (p. 55). At 19:24 the train passed MH1078 at 54 km/h, and 190 m later it ran through the points at the west end of the loop at 33 km/h into the 1:56 descending single line. The emergency sanding button was never pressed (pp. 55, 56). Emergency sanding takes effect whenever emergency braking is set, and that condition already existed (p. 51).

What was visible then: The cab before 19:22. The planning area of the DMI showed a four-step speed profile of 130, 115, 95 and 50 km/h, with the 50 km/h step for entering the Up Loop falling at the loop entrance (pp. 56, 61). A lineside sign at MH1078 read low ad - hesion (p. 23). It was dark, and the driver had seen no rain on the approach (p. 36). The TfWRL Autumn Seasonal Risk Guide, version 2 of September 2023, marks Talerddig as a low-adhesion high-risk area (p. 67). The fixed items of departure preparation include two health indicator lights for emergency sanding, and both were lit that day (p. 52).

From 19:23 onward. The 95 km/h advisory sequence had not fin - ished when the DMI switched to 50 km/h; the pre-indication was

masked by the previous step and the display jumped straight to yellow (p. 62). The driver’ s statement is that he registered the need to slow further from the yellow speed hook pressing toward the needle, and that he does not recall an audible warning (p. 63). The overspeed message and the ETCS brake symbol followed. The WSP intervened about a second after the first full service brake application, and this appears only in the data record, with no separate cab display.

On the desk, the only sanding-related items were the emergency sanding button and the two health lights, which at night sit behind a shroud and were visible lit beneath it. Nothing told the driver that automatic sanding was delivering sand (pp. 29, 52).

The paper record for that day: 15 kg of sand added on 18 October, two tests with sand flow on 20 October, a B exam passed on 12 Oc - tober, the annual measurement completed on 25 September, and no test on the morning of 21 October. What was known only after - wards: the SIS contact showed high resistance in the on position, the LSR4 relay coil was intermittently open inside its base, and both sand delivery hoses on the leading car were blocked with leaf-like organic matter (pp. 43–48). The adhesion peak derived from WSPER data fell from 0.06 before MH1078 to 0.035 on the des - cending section (pp. 39, 67). Had automatic sanding been deliver - ing 1.5 to 2.0 kg per minute, the train would have stopped before MH1078 or just beyond it (p. 50).

You are sitting in that position, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not “pay more atten - tion”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?

> 3. If you may set one checkpoint, when is it and what does it check?

What happened: One passenger died (Tudor Evans). On 1J25, 3 people were seriously injured, including the conductor, and 18 received minor injuries; the driver of 1S71 was seriously injured and the other 5 people on board received minor injuries (pp. 7, 12). Neither train derailed. The leading car was severely deformed, and the cab of 1S71 was the worst deformed. 1J25 ran through the points at the west end of the loop, which were not set for it, and damaged them. The line reopened on 28 October, after 7 days closed (p. 12). RAIB made 9 recommendations, 3 safety observations and 1 learning point (p. 9).

The three questions, looking back. The most common answer to the first is "press the emergency sanding the moment slip is detected", and its cost is one shot of sand and a refill at the depot; the report calculated the effect: pressed at the moment ETCS commanded emergency braking, the train stops just beyond MH1078 (p. 54). The item most easily missed under the second question is the one that reads "test button passed": on paper it looks most like safety evidence, and its circuit happens to bypass the two failure points. The third question lands on the design of the sanding test. What has to be proven is that the circuit is live, the valve opens and sand reaches the rail, not that there is sand in the box.

Options ruled out: Trap points at the west end of the loop. They were installed in 1968, removed in 1986 when the line changed to RETB, and not restored when ERTMS came in; Network Rail limits their use to depots, yards and siding exits, because a trap point works by derailing a train deliberately (pp. 70, 71). A location-specific detailed overrun risk assessment: the 7-page document of 2008 had covered the whole line with a simple assessment (p. 72). Restoring the MH1078 overlap to the standard 180 m: derogated in

September 2009 on the grounds of continuous ETCS supervision (p. 71). Applying an adhesion modifier after the RHTT wash: Wales and Borders route policy is not to, because the effect is uncertain and because the modifier may glue leaves to rail that has just been cleaned (pp. 77, 95). An automatic stop command to other trains in the event of an overrun: ERTMS has no such function, it is a customer requirement item, and Network Rail did not ask for it; the signal box has a regional emergency stop, but procedure puts a GSM-R call first (p. 73). A cab indication that automatic sanding is delivering sand: this needs design and certification work, and no modification had been proposed before the accident (p. 108). Monitoring of the automatic sanding delivery rate: Angel Trains began a technical review only after the accident (p. 108).

Early signals: ① The Stonegate accident of 2010 (RAIB report 18/2011), in which another train's automatic sanding was not working, with the sand box almost empty, and the train ran 4 km past its planned stop; on a section treated with adhesion modifier, deceleration was better by about 0.5%g than on untreated track. Why it was pressed past: single data point, not quantifiable (p. 77). ② Braking Using ERTMS for the Cambrian Project, version 5.0, September 2010, section 4.1, which states that very low adhesion requires the driver to adjust driving to keep the speed safe. Why it was pressed past: the applicable standard places low adhesion outside its scope (pp. 87, 88). ③ GI/RT7006 of 2000, which explicitly excludes overruns caused by low adhesion from the overrun protection standard, with three supporting reasons: low adhesion is hard to predict, it will disappear over the years, and other measures can deal with it, namely sanding, rail treatment and vegetation management. Why it was pressed past: if other measures cope, overrun protection need not (p. 88). ④ During early ERTMS testing, ORR raised the point that closely spaced cascading speed restrictions let one indication mask another, and recommended avoiding the arrangement; RAIB could find no record of a formal response

from the project (p. 64). ⑤ A driver may meet an ETCS intervention about once a week, and 158824 had 17 service brake interventions in the 21 days before the accident. TfWRL had no software to read the JRU and could only look for traces in the OTDR, so it identified far fewer than occurred (p. 92). Why it was pressed past: interven - tions were treated as the system working normally. ⑥ Talerddig had been listed as a low-adhesion high-risk area for years, over a section 5.9 km long, and the original reason was that uphill trains could not climb it. Why it was pressed past: being registered as high risk was taken as being handled (p. 67).

The point of escalation: The fuelling point check in the early hours of 20 October, on the night shift of the 19th. The item sits in TfWRL's maintenance plan at about every 1,500 miles, is carried out by a technician at Machynlleth, and consists of pressing the test button on the sand box and watching the hose. The test passed and the box was ticked. After that day, the next ground check that could possibly find a fault in automatic sanding was the next B exam, up to two weeks away (p. 90). The two-week window comes from the B exam interval of about 7,500 miles: a fault can appear on any day inside the window, and checks happen only at its two ends. The bypass circuit let this check pass with the system already failed. The departure preparation on the morning of 21 October was the second line, and it was not done because the platform po - sition put the underfloor equipment out of reach, which the pro - cess permitted. After two confirmations on paper, the train went into its seventh diagram of 21 October carrying an automatic sanding system that delivered nothing. From that moment to the collision, no ground check would touch the system, and no cab display would expose it.

Failure chain: Technical layer. The delivery command for auto - matic sanding needs four conditions to hold at once: SIS closed, the LSR4 contact closed, the SWR contact closed, and the direction

and brake step selection relays in position (p. 45). After the acci‑dent the first two were found not to hold: the SIS contact showed high resistance in the on position, and the LSR4 coil was intermit‑tently open between coil and base terminal, either of which would stop the sand valve energising (pp. 43–47). Both delivery hoses on the leading car were blocked with leaf-like organic matter at the same time, and normal working back pressure would be enough to clear a blockage of that kind (pp. 44, 48). The cab had no delivery indication (p. 29). Wheelslide protection itself worked normally, and for that very reason the fact that the WSP operated cannot be used to argue that sanding was working (p. 42).

Organizational layer. The test button circuit has its own supply and bypasses SIS and LSR4 (p. 90), and the same button is the main evidence at the fuelling point check, at departure preparation and at the annual measurement, so every test succeeded. The annual measurement procedure required weighing, and the maintenance system had no field for weight (p. 90). The JRU recorded every ETCS intervention; TfWRL had no software to read it and had to route data requests through Network Rail (p. 92). On the overrun pro‑tection side: 81 m of overlap, no trap points at the west end, and the simple whole-line assessment of 2008 (pp. 71, 72).

Incentive layer. The RHTT only washes and applies no adhesion modifier, on the grounds that the effect is uncertain and leaves may stick to the rail (p. 95). Monitoring of the sanding delivery rate was classed as a technical review whose starting point came after the accident (p. 108). Trading 180 m of overlap for 81 m bought line capacity and left the existing signal layout untouched.

The break sits at the seam between the organizational and incent‑ive layers. The test circuit was designed to prove that sand comes out, not that the circuit is live. That cheap choice kept two silent technical faults permanently invisible to a check that ran once a fortnight.

Minimum counterfactual: At 19:23, at the moment ETCS commanded emergency braking, press the emergency sanding button on the desk. One action, and the conditions held at that moment: emergency sanding is available only with emergency braking set, and it was set; the button is on the desk, both health lights were lit, both pressure bottles were in the green, and the system delivered sand normally when tested after the accident (pp. 52, 53). No luck is involved: the button is fixed equipment in the cab, the sanding pressure and quantity had been inspected, and RAIB's simulation calculated the action: pressed at that moment, the train stops just beyond MH1078 and does not enter the single line (p. 54). The cost is one shot of sand, refilled at the depot. The action was available every second after 19:23, and its effect decays with distance: pressed while passing MH1078, the train stops beyond the loop exit points; pressed about 18 seconds after leaving the loop, it stops near MH1081; pressed 75 seconds after passing MH1078, the collision still happens, at about 14 km/h (p. 54).

Outcome and sources: One passenger died, 4 people were seriously injured (3 on 1J25 and the driver of 1S71), and 23 received minor injuries (18 on 1J25, 5 on 1S71). Neither train derailed. The Talerddig loop points were damaged and the line reopened on 28 October (pp. 7, 12). RAIB published the report on 18 June 2026, with 9 recommendations addressed to RSSB, Angel Trains, Network Rail and TfWRL, plus 3 safety observations and 1 learning point on safety-critical communications (p. 9). RAIB also wrote that had 1S71 been stopped by a railway emergency call (REC), the collision would not have been avoided but the collision energy would have been lower (p. 8). The report leaves room in its wording on the key judgements: the cause of the automatic sanding failure is written as possible, and the blocked hoses would probably have been cleared by a working system (pp. 7, 48). TfWRL issued two national incident notices after the accident, one on the hose blockage and one on the electrical fault, and changed the sanding test button

circuit on the Class 158 so that it passes through SIS and LSR4 (p. 108). Primary source: RAIB, Report 08/2026, Collision between two passenger trains near Talerddig, Powys, 21 October 2024, June 2026, 140 pages, link in Appendix B ce-001.

The division of labour between automatic and manual in this case deserves one more cut. An arrangement where the automatic system is primary and the manual system catches it is reasonable while warnings work. Its gap is the scenario in which the automatic system fails quietly: the system is broken and does not warn, so the signal the driver reads, which is silence, contradicts the true state. Silence read as normal is the general trap of automation, the silent failure mode. A common-item register for redundancy therefore needs an entry for silent failure: mark each system's failure as loud, meaning it warns, or quiet, meaning it does not. A quiet system needs periodic active verification, a check of state rather than a wait for a warning, because a monitoring strategy that waits for alarms has no purchase on silent failure at all.

Monitoring has a common cause of its own, and the recursion is worth stating. The monitoring of two systems can share a source: one collection script, one alerting platform, one on-call judgement process. A common cause in monitoring makes the failure of the backup itself invisible, because the alarm paths for primary and backup die together and you never get the chance to learn that the backup is dead. Verifying monitoring redundancy is the meta level of verifying redundancy: first prove the alarm path is alive by injecting a test alarm, then prove the redundancy works by a real switchover. The order cannot be reversed, since a switchover rehearsal on a dead path is a photograph of nothing. The meta level is verified more often than the object level, monthly injection against annual switchover, and costs far less, so it should be the first routine action in any redundant system.

After the routine action, one piece of the redundancy picture is still missing: retirement management. A superseded system, the old primary after a cutover or the previous generation of equip - ment, tends to stay on the register for a while, and readers of the register keep counting it as a line of defence. An old system that is not maintained, not upgraded and not rehearsed is a line of de - fence on paper only. The rule has to be written hard: from the date of formal retirement the system comes off the defence count and the register is updated with it. For the transition period when old and new run in parallel, state which one counts as defence and which is a shadow. The reader of a register is always someone in the future, an auditor, a successor, a regulator, and honesty toward that reader is the last exit for every device in this chapter. The ledger has to be true before redundancy counts.

Redundancy that counts still faces one last opponent: time eroding heterogeneity. Heterogeneity is not a property acquired at pur - chase, it is a property that has to be managed. The second supplier gets acquired and becomes the same source. The second team is rotated and merged into the same team. The second algorithm converges as its framework is upgraded and becomes the same stack. This decay is a slow variable, observed on a five-year cycle: each year, when the common-cause register is updated, compare the heterogeneity dimensions against last year and mark the entries that lost a dimension as heterogeneous erosion. The rem - edy depends on the dimension, either re-heterogenise by changing supplier or accept the downgrade and discount that pair in the defence count. Slow variables tend to reach zero in silence, and the annual comparison is how this chapter gives heterogeneity a place where it cannot go quiet.

The chapter now belongs in the book' s larger picture. Most mechanisms in Part III are many causes producing one effect; this one is the mirror image, one cause producing many effects. The

common cause arrives, several lines of defence fail together, and several mechanisms get their stage at once: once redundancy collapses, normalization of deviance, invisible maintenance and the slack of success each pay their bill inside the same accident. That mirrored structure makes this chapter a key to many accident reports. Where a report speaks of multiple defences failing, most of the time the single cause is found here. The method for finding it is the five-family list: pull out every failed line of defence in the report, look for their intersection, and the intersection is the common cause. Intersection is the tool this chapter gives readers of accidents, and the mirror it holds up to designers: every line of defence you design will one day stand in front of a common cause you did not think of, and whether it is ready has been written in the ledger for a long time.

One page of that ledger belongs to the buyer: acceptance criteria for redundant procurement. Acceptance does not count deliverables, meaning two systems arriving, it counts three things. A difference statement, with the list of differences between the two systems archived alongside the delivery. An independent validation report, in which the supplier tests the second system alone under agreed conditions. A switchover script, including the worst-condition version. Put all three into the acceptance clause of the purchase contract and the supplier's quote rises a little, which is the true cost of testing and documentation. That is exactly the price signal this chapter wants: the real cost of redundancy is validation, and copying is the cheap part. Redundancy sold without validation is cheap because what is being sold is the appearance of redundancy.

The last hiding place of appearance is insurance and compliance paperwork: redundancy clauses in policies, defence lists in compliance reports. Both are concentrations of appearance, and both have one good use. Take the defence list in the insurance and

compliance documents and reconcile it against this chapter's ledger; the entries that differ are the complete roster of paper defences. The roster is usually embarrassing, since the insurer believes you have things that measurement shows you do not, and the value of the embarrassment is that it prices the gap between two sets of books. A policy is priced on paper, an accident is paid out in reality, and the difference is shared on the day it happens between shareholders and the families of the dead. Reconcile once a year by taking the insurance declaration line by line against the rehearsal records. The other name for that action is repricing the policy against reality.

With pricing done, the redundancy story closes its loop inside this chapter. It starts as a verbal promise, "we have two", and ends as a ledger: difference statements, independent validation, rehearsal records, common-cause registration, heterogeneity maintenance, retirement management, policy reconciliation. No single link is expensive, and once the links are joined the verbal promise acquires an engineering meaning for the first time. The keeper of the ledger has to be written into a post, following the owner clause of Chapter 20: the ledger has an owner, and the owner has a performance measure, which is the on-time rate of each link. The redundancy ledger, combined with the dependency map of Chapter 21 and the output ledger of Chapter 20, makes three books for an organization's defence system: where the defences are (dependency map), whether they are alive (redundancy ledger), and whether they do any work (output ledger). Once a year the three books consolidate into a defence annual report, whose first page follows the Chapter 20 format: what we caught this year, what we validated this year, what we downgraded this year.

The annual report is also where the attacker's view gets its last page, because that view is where counting kinds comes from. An attacker here means any force looking for the weakest path

through a system: fire, flood, a market shock, a hostile intrusion. Such a force never defeats defences one at a time. It looks for their greatest common denominator, the one condition that kills several at once. Safe design mirrors the attacker: write down the greatest common denominator for each group of defences, which is the five-family list, then decide either to eliminate it through heterogeneity, segregation and dispersal, or to price it as a budget line for residual risk. An organization that has done both is helped by thinking like an attacker, since every condition capable of killing several defences at once is on the register, and the thickness of the register is the real safety of the system. Where the register is thin, the likely reason is that nobody has counted yet.

The feel of the thing changes once you have counted. In a review meeting, a reader trained by this chapter hears "double assurance" or "multiple layers of protection" and no longer relaxes; the reflex is to take the phrase apart. The standard move is three questions, difference, independent validation, worst condition, as in section 7. After three questions, double assurance usually shows one of two colours: real redundancy, with difference and validation both present, or paper redundancy, same source and never rehearsed. The two colours lead to different places, real redundancy into the asset column of the annual report, paper redundancy into the downgrade list and the gap line of the policy reconciliation. What this chapter wants to leave in the reader's hands is the craft of turning an adjective into a ledger entry. The word safety is worth exactly as much in this trade as the number of times it has been taken apart. If two sanding systems can both fail in silence, the shared item is the same one: waiting.

Case five (aviation): One rupture and two hydraulic systems (full card)

Case ID: av-007

When and where: Pushback at 21:10 on 12 August 2024, takeoff at 21:39, landing at 01:12 on 13 August. Narita International Airport, Chiba Prefecture, Japan, near 35°44′25″N 140°23′34″E. The tyre rupture happened near taxiway A between W6 and W5. Takeoff and landing were both on runway 16R, and the destination had been Los Angeles International. The 21:40 ground observation was a south-southeast wind at 3 m/s, visibility 10 km, clear, temperature 27°C, dew point 26°C, sea level pressure 1,005 hPa (pp. 1, 2, 3, 5).

What and who: Atlas Air, Inc., scheduled flight 7106, Boeing 747-400F, registration N404KZ, United States registry, manufacturer serial number 34283, built 5 March 2007, total flight time 64,170 hours 56 minutes. Seven people aboard. The captain, 47, was PF in the left seat, with 9,140 hours total and 1,181 hours 14 minutes on type. The first officer, 55, was PM in the right seat, with 3,830 hours total and 161 hours 14 minutes on type. Five supernumeraries were on board. The report abbreviates the left body main landing gear as the same main gear leg, and both hydraulic systems discussed below are mounted on it. The investigating authority was the aviation division of the Japan Transport Safety Board, which appointed one investigator-in-charge on 13 August 2024 and added another on 28 August; the United States participated as state of design and manufacture and state of registry (pp. 1, 4, 5).

What they were doing: The aircraft arrived that day on the Incheon to Narita sector, and four inspections found nothing abnormal in the body landing gear. Before departure from Incheon, maintenance measured each tyre pressure against the daily check and the crew did an external walkaround; after landing at Narita maintenance walked around the aircraft once, and the crew walked around it again before departure (p. 5). All four inspections touched the tyres, and they touched them by looking, with one pressure reading added to the fourth. The report does not record

the time of any of the four inspections, and it records no reading that departed from the norm (p. 5). The only numbers were tyre pressures at Incheon; after that, until the event, there was no second reading.

The history of the two problem tyres is on file. Position No. 7 on the body gear carried a second retread, fitted on 21 June 2024, which had run 89 sectors by the time of the event. Position No. 8 carried a first retread, fitted on 9 August 2024, which had run 2 sectors (p. 6). Both wheel assemblies had been inspected and prepared under the company maintenance procedure (p. 5).

At 21:10 the aircraft pushed back and taxied toward 16R, becoming airborne 29 minutes later. Near taxiway A between W6 and W5 the crew felt a vibration like running over something, heard no sound, and the taxi continued. Nothing abnormal occurred on the rest of the taxi to the runway. The takeoff roll produced no vibration and no abnormal instrument indication (p. 2).

The decision point: ① Between pushback at 21:10 and 21:39, on taxiway A between W6 and W5. One vibration, no second one, no sound, no warning. The captain was PF in the left seat and the first officer was PM in the right seat. The taxi continued. This segment is carried by the crew's account and by the 21:39 takeoff time. There is no document.

② Shortly after takeoff, EICAS showed "HYD QTY LOW 4", an Advisory. The report describes the No. 4 system as the main brake hydraulic system, and the message concerns quantity; the crew confirmed on EICAS that the quantity was falling rapidly, ran the non-normal checklist for loss of the No. 4 system, and found nothing else abnormal (p. 2). The captain decided to continue to Los Angeles, and the report gives two reasons: the checklist had been completed with no other abnormality, and a return would first require reducing gross weight below maximum landing weight, which meant burning fuel (p. 2). During the same period

the crew judged that hydraulic fluid might have leaked onto the runway during the takeoff roll, and asked Narita tower to inspect the surface of 16R (p. 2). The report gives no time for this decision, placing it only after the handling of the No. 4 warning and before the cabin warning at 21:52 (p. 2).

③ At 21:52, near 12,000 ft, EICAS showed "CABIN ALTITUDE", a Warning, and the cabin altitude gauge read 10,300 ft. The crew stopped the climb near FL150, declared an emergency to ATC, ran the cabin decompression checklist, descended to 10,000 ft as the checklist directs, and decided to return to Narita (p. 2). After this point the heading had no second possibility, and the only open question was weight.

④ From 00:06 on 13 August the crew dumped fuel over the sea. During the dump EICAS first showed "HYD QTY LOW 1", an Advisory, then "HYD PRESS SYS 1", a Caution; the report describes the No. 1 system as the backup brake hydraulic system (pp. 2, 3). The crew ran the checklist for loss of both No. 1 and No. 4, and after confirming that gross weight was below maximum landing weight they lowered the gear by the alternate method required by that checklist rather than normally (p. 3). On landing, nosewheel steering and autobrake were unavailable, and the crew used manual braking (p. 3).

What was visible then: One vibration during taxi described as running over something, no sound, nothing abnormal afterwards (p. 2). No vibration during the takeoff roll, no abnormal display (p. 2). After takeoff the Advisory "HYD QTY LOW 4", with the No. 4 quantity falling rapidly on the readout (p. 2). At 21:52 the Warning "CABIN ALTITUDE", cabin altitude 10,300 ft, aircraft altitude 12,000 ft (p. 2). During the fuel dump the Advisory "HYD QTY LOW 1" followed by the Caution "HYD PRESS SYS 1" (p. 3). After landing, no steering and no autobrake (p. 3). The night was

clear with 10 km visibility, temperature and dew point one degree apart, and no precipitation on the surface (p. 5).

There was no cockpit display relating to the tyres at all, because the aircraft was not fitted with a tyre pressure indication system, TPIS (p. 7). On paper: four inspection records with nothing abnormal (p. 5); installation records of 89 sectors at No. 7 and 2 sectors at No. 8 (p. 6); both wheel assemblies maintained to procedure (p. 5). The report does not record the crew speaking to the tower about the vibration during the taxi (p. 2).

What was known only afterwards: the tread on the No. 7 tyre had already separated between W6 and W5, and the separated tread did not fly off but turned with the wheel, striking the same shock strut again and again (pp. 7, 8). The thin white mist seen during the takeoff roll was the signature of a ruptured brake hydraulic line (pp. 7, 8). As the gear retracted, hub debris punctured the pressure bulkhead at the top of the main wheel well, opening a hole about 25 cm fore and aft and about 38 cm across (pp. 3, 8). The cause of the pressure loss in the tyre could not be determined (p. 8). The tyre manufacturer's investigation found no manufacturing defect in the retreaded article and judged the bond at the mating surface adequate (p. 6). The flight recorder captured no vibration or impact connected with the event, and the cockpit voice recording had been overwritten after landing (p. 7). The key video came from a photographer near the airport and reached the investigators only later (p. 7).

You are sitting in that position, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".

> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

Options ruled out: Stopping on the taxiway and asking maintenance to look at the wheels from the ground. The report does not record the option being discussed, and its cost is a delay and taxiway occupancy. Returning immediately after takeoff: the report states that weight had to come below maximum landing weight first, so the aircraft flew toward Los Angeles (p. 2). Landing overweight: the report records no evaluation of the option, only the constraint that weight had to come down (p. 2). Fitting TPIS to the type: the 747-400 series offers it as an option and this aircraft did not have it (p. 7); the report's analysis says that with it fitted the crew might have noticed abnormal tyre pressure during the taxi (p. 8). Physically separating the two brake hydraulic lines on the same strut: this is a design change, the report does not mention it being considered, and the operator's 8 post-accident corrective actions all sit on the maintenance procedure side (p. 9). Measuring daily tyre pressures after cooling: written into the work instruction only after the accident (p. 9). Lowering the gear normally: the checklist required the alternate method, and the crew followed it (p. 3).

Early signals: ① On 21 June 2024 a second retread went onto position No. 7 of the body gear, and it ran 89 sectors to the event (p. 6). Why it was pressed past: preparation followed the company maintenance procedure (p. 5), and the procedure at the time did not prohibit retreads at positions No. 7, 8, 11 and 12; it prohibited them only after the accident (p. 9). ② Tyre pressure check results had not been kept as records in the maintenance management system, and recording them in a traceable form became a requirement only after the accident (p. 9). Why it was pressed past: paper records

existed and could be read, but could not be assembled into a trend. ③ The daily check had no requirement on when to measure, and only after the accident was a cooling condition added: in principle after the brakes have cooled and the temperature has stabilised, more than 2 hours after block-in (p. 9). Why it was pressed past: following the job card was compliance, and the fact that a hot tyre reads high was not on the card. ④ The daily work instruction con - tained a clause allowing a tyre or brake that needed replacement to be deferred where replacement might delay the service (p. 9, de - leted after the accident). Why it was pressed past: the clause au - thorised it. The report does not say the clause was used in this case. ⑤ No cockpit visibility of tyre pressure (p. 7). Why it was pressed past: TPIS is optional and airworthiness does not require it.

The point of escalation: 21:39, runway 16R, rotation. Before rota - tion the only place that could see the wheels was off the aircraft: the photographer beside the taxiway recorded the bang and the tread block turning with the wheel (p. 7). After rotation nothing on board could see the tyres, and the aircraft lifted off with a wheel already stripped. This escalation has no document and no signa - ture; its carrier is one push of the thrust levers. The report can give only two time anchors: the 21:39 takeoff and the second warning 13 minutes later (p. 2). Inside those 13 minutes the record holds two things, the No. 4 checklist and the request for a runway inspection (p. 2); the aircraft climbed to 12,000 ft, and the presumed track is drawn on figure 2 of the report (pp. 2, 3). The warning level did not match the damage. Two brake hydraulic systems damaged at the same location were reported at the lowest level, Advisory, and the only Warning pointed at another system, cabin pressurisation. The third came back as an Advisory, two and a half hours later (pp. 2, 3).

Failure chain: Technical layer. The pressure loss in the No. 7 tyre, cause undetermined (p. 8), collapsed the contact patch, loaded the junction of tread and sidewall, and the tread cracked and separated (p. 8). The separated tread stayed on the wheel and turned with it, slapping the same shock strut repeatedly during the taxi, and the No. 1 and No. 4 brake hydraulic lines on that strut were ruptured as a result (p. 8). The white mist during the takeoff roll marks the point at which the piping is presumed to have failed (p. 8). The bare wheel kept rolling, hub components fractured and flew, and the debris punctured the pressure bulkhead as the gear retracted (p. 8). One source, three damages: No. 4 hydraulics, No. 1 hydraulics, cabin pressurisation.

Organizational layer. The two brake hydraulic systems run along the same shock strut. The report's own wording is that the No. 1 and No. 4 brake hydraulic systems on the same main gear shock strut were damaged (p. 8); their functions are primary and backup (pp. 2, 3), and their physical location is shared. The cockpit had no tyre pressure display and TPIS was not fitted (p. 7). All four inspections happened before the damage, and between the damage and liftoff there was no ground inspection at all (pp. 2, 5). Pressure records never entered the system, so no trend was visible (p. 9). The sensation of running over something during taxi had no corresponding handling clause, and the report contains no quantified trigger for one.

Incentive layer. The daily work instruction said in writing that a replacement which might cause a delay could be put off (p. 9). Tyre pressure was measured hot, so the reading ran high and the check passed (p. 9).

The break sits at the seam between the organizational and design layers. Design and maintenance procedure each complied. One put two lines on a single strut; the other left tyre pressure to a daily

reading and a walkaround. Neither was answerable for the pos - sibility that one physical event could reach both.

Minimum counterfactual: After the one vibration, stop the air - craft on the taxiway and ask maintenance to look at positions No. 7 and No. 8 from the ground. One action, taken before 21:39, at the cost of a delay and some taxiway occupancy. No luck is involved: the separated tread had not come off at that point and was still turning with the wheel, and the video shows it (p. 7); the place where the crew felt the vibration and the place where the video records the bang are the same stretch (pp. 2, 7). The action needs no luck and no new basis for judgement. It needs the sensation of running over something to be treated as real damage rather than a bump in the surface. The report' s only judgement about early detection is the possibility it raises about TPIS (p. 8); it does not evaluate a stop and a look. The second half of the action is also in the report: once tread separation is confirmed, the response is a wheel change, and the post-accident corrective action made it ex - plicit that a radial tyre whose tread has separated must be replaced (p. 9). The job card at the time did not carry that line.

What happened: The aircraft landed on 16R at 01:12 on 13 August, stopped after the runway centreline marker on manual braking, and the runway stayed closed for about 7 hours until the aircraft was moved to a stand (p. 3). The airframe damage was substantial: the pressure bulkhead at the top of the main wheel well had an opening about 25 cm (10 in) fore and aft and about 38 cm (15 in) across, within the scope of a major repair; the No. 7 and No. 8 wheel assemblies were damaged; hydraulic components on the shock strut were fractured and buckled; parts of the gear door were missing; and there were about 80 cracks and dents on the lower left fuselage skin and the left horizontal stabiliser (pp. 3, 4). The 7 people aboard were uninjured (p. 1). On the ground, the runway surface carried straight and curved scrape marks and

damaged light fittings; door fragments and tyre pieces were found on taxiway A, and multiple pieces including tyre and brake parts on the runway (p. 5). The report was decided on 11 July 2025 and runs to 9 pages (p. 1).

The three questions, looking back. The most common answer to the first is "monitor the hydraulics closely after takeoff", and monitoring cannot change a physical fact that has already happened; the stop and look is the action that would have caught the event on the taxiway. The item most easily missed under the second is the scope of the confidence carried by "if one fails there is another": it holds for independent failure and does not hold for common-cause failure, and the wheel well is precisely where the two systems physically coexist. The third question lands on the handling standard for a taxi anomaly. Abnormal vibration plus a suspicion about a tyre should mean stopping to look.

Outcome and sources: JTSB, Aircraft Accident Investigation Report, Atlas Air Boeing 747-400F (N404KZ), accident category airframe damage during takeoff, Narita International Airport, 12 August 2024, decided by the aviation division of the Japan Transport Safety Board on 11 July 2025 (Reiwa 7), 9 pages, PDF downloaded and read by this book. The cause section uses presumed throughout, and the cause of the No. 7 pressure loss is stated as undeterminable (p. 8). Preventive measures come in two parts. Section 5.1 recommends capturing tyre pressure trends and names TPIS as an effective means (p. 9). Section 5.2 lists the operator's 8 changes to the daily work instruction, among them measuring after cooling, printing hot and cold permissible pressures directly on the job card, deleting the clause that let a delay take precedence, prohibiting retreads at certain positions, entering records into the system, and making gauge numbers and calibration dates traceable (p. 9). Primary source: JTSB, AA2025-6-2, 2025, link in Appendix B av-007.

Case three (healthcare): The purple syringe and the second pair of eyes

me-003. Paediatric care in the United Kingdom, sedation before an elective renal biopsy. HSIB report (Inadvertent administration of an oral liquid medicine into a vein, 2019, archived snapshot read by this book): oral midazolam was drawn into an intravenous syringe and given intravenously by mistake. Two lines of defence existed. A purple syringe reserved for oral medicines, which is device-level segregation, since the colour makes oral and intravenous equip - ment physically hard to confuse. And a double check before ad - ministration, which is process-level segregation. Both failed in the same event: the syringe was substituted, and the double check de - graded into a formality. The child came to no obvious harm. The report cites a national figure of 61 wrong-route administrations and makes four safety recommendations (2019/028–031).

Stop at the decision point: the preparation room, just before ad - ministration, with the double check about to begin. What was on the table: the purple syringe out of stock or not to hand, two busy colleagues, and two signature boxes on the check sheet. The shared item behind both lines of defence was normal conditions, meaning a purple syringe available and a second person free. Break that premise and both fail together. (What happened: wrong-route ad - ministration, no harm.) Primary source: HSIB, 2019, link in Appendix B me-003 (archived link, grade A−).

The shared item in case three, normal conditions, has a general - isation worth keeping: a line of defence depends on resources. The purple syringe is a consumable and can be out of stock; the second person is labour and can be busy; both lines rest on an unstated premise of adequate resourcing. Adequate resourcing holds on most days, and the longer it holds the less visible the dependency becomes, which is the mechanism of Chapter 20 projected onto redundancy. The remedy is a stress register for the resource de -

pendency of each line: how much of this line remains when consumables run out, when staff are short, when an outsourced provider changes hands. The output of that register is usually brutal, since most multi-line designs have a real defence count of one when resources are tight. The brutality has one use, which is to pull the language of double insurance back to arithmetic. The premium for insurance, meaning heterogeneity and redundant resources, has to be affordable on the tightest day, or the policy is a weekend edition.

Case four (chemical & process): A fallback on the same power supply

ch-003. Arkema's organic peroxide plant at Crosby, Texas, late August 2017. CSB report (pages read by this book): the flooding from Hurricane Harvey cut power to the site's refrigeration, the stored organic peroxides warmed, the first trailers self-ignited on 31 August, and the rest were deliberately burned on 3 September. The emergency plan made transfer to refrigerated trailers the fallback after a power loss. The fallback sat under the same common cause as the primary system: the flood took out refrigeration in the buildings and also the refrigeration in the trailers and the site roads. On paper the backup was independent. In the topology of a flood it shared one life with the primary.

Stop at the decision point: the moment when the emergency plan named the refrigerated trailers as the fallback for a power loss, and was approved. What was on the table: a list of trailer refrigeration capacity, and an assumption that the site had never flooded. A common-cause review has one question: will the cause that takes out the primary act on you as well. The flood answered yes. (What happened: serial fires and explosions, with evacuation of the surrounding area.) Primary source: CSB, Arkema Inc. Chemical Plant Fire, 2018, link in Appendix B ch-003.

The flood in case four is worth a second reading from the angle of how plans are written. Plans are written by scenario, with the power-loss scenario and the flood scenario kept apart, and when the fallback was approved inside the power-loss scenario its au‑thors stayed inside that scenario, asking what to do when the power goes. The common-cause question, which is why the power went and what else that cause will touch, belongs to the territory between scenarios. The remedy in plan writing is a cross-scenario reading step: when a fallback is finished, go back to the physical cause that triggered it and read the other scenarios along that cause. If the flood both causes the power loss and floods the trail‑ers, then the power-loss fallback has to be reread inside the flood scenario. The step costs half a day per plan, and what it stops is the common cause that lives in the seams between scenarios. Most common causes do not live inside a scenario. They live on its boundaries.

Case six (maritime): Two emergency items fail together (condensed)

ma-009. The stern trawler Ocean Way, November 2014, United Kingdom waters. MAIB report (2015, pages read by this book): running before a following sea, the deck flooded and the vessel capsized, killing 3 and leaving 2 survivors. The redundancy in the emergency equipment failed in pairs. The liferaft was trapped by its cradle and never released. The EPIRB, the emergency position indicating radio beacon, had no GNSS positioning, so the search and rescue datum took 47 minutes to establish. Two emergency lines, one for escape and one for location, were each present on paper and each failed in the real scenario. Their shared item is that neither had ever been validated in that scenario, neither the cradle release nor the beacon’ s positioning. Minimum counterfactual: annual checks of emergency equipment under real conditions, one

real release and one real positioning test. Primary source: MAIB, 2015, link in Appendix B ma-009.

Case seven (aviation): Recurrence after the required actions were done (condensed)

av-027. A 767 freighter, October 2025, United Kingdom. AAIB report (2026, pages read by this book): the aileron control jammed during climb and the aircraft landed safely. The type has a history of ice-induced jamming, and both the mandated modification and the mandated inspection had been completed. The failure came back after the defence, modification plus inspection, was complete on paper, which shows a gap between the scope in which a defence is effective and the scope in which it is exposed: the scenarios the modification covers do not fully overlap the icing conditions actually met. The illusion takes this form here, the assumption that doing the required actions is safety. The actions were done, so the defence was recorded as present. Minimum counterfactual: bring the recurrence conditions into the control set used to validate the modification. Primary source: AAIB, 2026, link in Appendix B av-027.

Case eight (aviation): A performance error that no defence caught (condensed)

av-028. A Boeing 737-8 (G-CRUX), April 2025, London Luton. AAIB correspondence investigation (2026, pages read by this book): the crew computed performance for the full runway and took off from an intersection, the climb was extremely shallow, and the aircraft crossed the runway end at 13 ft. The airport's dispatch defences were several, a performance sheet recheck, a runway threshold confirmation and post-hoc FDM monitoring, and none of them caught the error on the spot. Only the FDM data found it afterwards. The shared blind spot of several defences is the assumption that the airport reference point used in the calculation matches the

one used in the takeoff, and the independence between the de‑fences did not cover that assumption. Minimum counterfactual: a consistency check between the performance sheet and the actual takeoff point, made a fixed dispatch item. Primary source: AAIB, 2026, link in Appendix B av-028.

22.4 What shows up earliest

Six weak signals, each with the reason usually used to press past it.

1. The backup and the primary share a supplier, an implementer or an operations team. Reason: “it integrates well and they respond fast.”
2. Every switchover rehearsal has been done at night in a low-load window, never at peak. Reason: “we can’t take the risk with the business.”
3. The alarms from both systems feed one duty desk and one person. Reason: “centralised monitoring is more efficient.”
4. The backup’s specification was adapted from the primary’s drawings. Reason: “to guarantee consistency.”
5. The fallback in the emergency plan never lists the failure con‑ditions it shares with the primary. Reason: “the plan is written by scenario.” (Arkema’s flood is the common variable between scenarios.)
6. The full version of “we have two” has never been questioned: two of what, different in what way, validated by whom. Reason: “why go into that much detail.”

Put the six signals back against the eight cases. Signal 1, the same supplier, matches the trailers in case four: fallback and primary sit inside one procurement radius. Signal 2, rehearsals in low-load windows only, matches all eight cases, since not one of them went through a real switchover or a real-scenario validation. Signal 3,

one duty desk, matches the sanding in case two: automatic and manual state feed a single judgement in one cab. Signal 4, an adapted specification, matches the hydraulics in case five: two lines routed under the constraint of one general-arrangement drawing. Signal 5, a common variable between scenarios, matches the flood in case four. Signal 6, the missing follow-up question, is the problem in case one: the "two" of ADS-B map and eyes was never questioned as to whether the other aircraft carried the equipment at all.

Sort the eight cases by the five families. Environment: the flood in case four. Design: the line routing in case five and the reference point assumption in case eight. Operational: the coverage gap between modification and inspection in case seven. Behavioural: the cession in case one, the division of labour in case two, normal conditions in case three. Certification: the paper validation in case six, and the dual certification of device and process in case three. The distribution suggests an order for registering. Environment and certification are the hardest to audit from inside, because they sit outside the organization's field of view, so bring in an outside view for those two first. Then self-audit the behavioural and design families. Operational comes last, since daily familiarity covers it most easily.

22.5 How to defend

Device 1: the common-cause register. Every redundant pair is made to register its shared items: power, network, vendor, algorithm, site, people, assumption. Each item carries the note that says whether both halves fail at once when this common cause arrives. The register is updated every year, and it and the dependency map of Chapter 21 index each other. The review item in the Arkema case comes from here: the register for a fallback must

contain a line for the failure conditions it shares with the primary system.

Device 2: the heterogeneity standard. Critical redundancy has to be heterogeneous in at least one dimension: a different principle, a different vendor, a different team. Automatic and manual sanding, the syringe colour and the double check, the map and the lookout: in every one of them the failure happened where the two sides were alike. The procurement cost of heterogeneity is real, and its consideration belongs on the purchase order. What the extra money buys is that the common cause is not shared.

Device 3: the true switchover. The validation standard is fixed in advance. A switchover must shut the primary down and be carried out under real load, at least once a year. Redundancy that cannot be switched for real is relabelled from redundancy to paper backup and dropped from the defence count. What that downgrade does is put the arithmetic back from multiplication to addition. The numbers get worse, and they are the true ones.

Apply the devices to this chapter's cases. Device 1 on case four: the register line for shared failure conditions gets flood level, site roads and the offsite grid. Device 2 on case three: the substitute procedure for when purple syringes are out of stock, temporary independent preparation by two people with a verbal read-back, is written in advance as the heterogeneous alternative rather than invented under pressure. Device 3 on case two: the real rehearsal of manual sanding is staged on wet rail, a genuine simulation of abnormal adhesion, not a button press on dry rail. Device 4, whose statement comes at the end of this section, on case six: the annual checks of liferaft release and EPIRB positioning are run under a simulated capsize rather than on a calm deck.

The four devices have four ways of dying. The common-cause register dies on granularity: it records same vendor and misses the same product line of the same vendor, where the common cause

actually sits; the patch is to take entries down to model and batch. The heterogeneity standard dies on formal heterogeneity: the vendor changed, but both vendors buy the same upstream chips or the same steel, which is second-order same source; the patch is to ask about second-tier suppliers for critical components. The true switchover dies on scope shrinkage: the core function is switched and the edge functions stay on the primary; the patch is to tick the switchover scope item by item against the function list and keep the evidence. The fallback review dies on being a one-off: it was reviewed at approval and later revisions never were; the patch is to bind the plan version number to the review record. All four deaths answer to one health question: when did this redundancy last work under bad conditions. Where the answer is never, the status drops automatically to paper backup, under the downgrade clause of Device 3.

Redundant procurement and design carry one economic point worth writing down. The heterogeneity premium is real: the learning cost of a second supplier, the integration cost, the double line of spares. Every decision that suppresses it raises beta. The honest way to build the budget is to put two columns side by side, whole-life cost for the same-source option and for the heterogen - eous option, and a third column for the difference in risk, meaning the difference in common-cause exposure. Most organizations compare only the first two columns, and with the third missing, saving money wins on paper every time. The third column does not need precision, a range is enough. What it needs is someone re - quired to fill it in, booking the externality to the account as writing three in Chapter 17 directs, and from then on common-cause risk has a price.

Once the price exists, one psychological obstacle remains: the comfort of homogeneity. An operations team’ s familiarity with same-source systems is real efficiency, with faster fault-finding,

common spares and shared on-call training, and heterogeneity is worse on every one of those counts. The true benefit of heterogeneity, that the common cause is not shared, is probabilistic, delayed and invisible. Trade certain efficiency for probabilistic safety and an organization will pick efficiency again and again, unless the probabilistic side is put in front of it often. The carrier for that reminder is the quarterly review of the common-cause register, where each shared item is read aloud to the operations team itself. The cost is trivial. What it maintains is the presence of the heterogeneity benefit in the mind of the decision-maker, and the rate at which that presence decays is the rate at which beta climbs back. The quarterly review is best understood as the frequency at which probabilistic safety is renewed.

The history of the idea is worth a paragraph, because it explains where the tests come from. The golden age of redundancy design was aviation and nuclear power, with fault trees, dual channels and diverse design, and the engineers of that era were close to obsessive about common cause, since Three Mile Island and Chernobyl were industry memory. Software took the practice over later, with multiple availability zones and multi-active architectures, and its sensitivity to common cause lagged behind. Cloud availability zones that share one control plane, with cases two and five of Chapter 21 both sitting in the control plane, show that a zone in a multi-zone architecture delivers physical redundancy while leaving logical redundancy untouched. The handover between the two generations has one clear point: the physical era looked for common causes in the environment, the logical era looks for them in the control plane and in the assumptions. Today's reader has to look in both, because the system holds cables and server rooms alongside algorithms and default semantics, and the five families were assembled from the failure history of both generations.

Device 4: same-cause review of the fallback. Every fallback in an emergency plan answers one question: will the cause that takes out the primary act on you as well. The question costs one line of text. What it buys is an Arkema-type scenario stopped on paper.

22.6 Boundaries with neighbouring mechanisms

Against hidden coupling (Chapter 21): coupling is the hidden line between independent systems, where nobody knows they are connected. The illusion is same-source inside a redundant pair, where everyone knows they are connected and nobody takes it seriously. The hidden lines that Chapter 21's mapping draws become this chapter's common-cause entries the moment they enter a redundancy register.

Against the slack of success (Chapter 20): slack lets a backup go unvalidated, because nobody feels the need to rehearse. The illusion lets a backup go un-heterogenised, because it was never independent at the point of purchase. Slack is a disease of time and the illusion is a disease of structure, and an organization that has been slack for long enough will see its redundancy register drift toward same-source on its own.

Against compliant and unsafe (Chapter 32): a compliance check counts the number of defences. This chapter counts the kinds. Domains where regulation requires redundancy and acceptance looks only at existence are where the illusion has the highest incidence.

22.7 The five self-checks

1. Write down three redundancies your organization is proud of. Where is the shared-item list for each one? Without a list, put quotation marks around the word redundancy.

2. When did your backup system last run independently of the primary? Who validated that it was independent?
3. At the last switchover rehearsal, was the primary actually shut down? Was the traffic real?
4. In your emergency plan, are the failure conditions shared by the fallback and the primary written down? Run one flood or one power loss through it.
5. How many defences exist on a screen, in a register or on a drawing but have never been validated on their own? Count them now.

(This chapter: 15,261 characters.)

Chapter 23 Extrapolating Past the Validated Domain

> What this chapter covers: the mechanism that carries "validated at A" straight across to B; three criteria, four field sites, the mandatory format of the difference list, and graded release. When to turn back: when a pilot becomes a rollout, when a method is reused across models, when an old instrument is put to a new use, when "we have always done it this way" is being written into a new setting. Reading time: about 50 minutes.

Every piece of knowledge has an edge. What varies is whether anyone has written down where it falls. A validation report that says a method works is, strictly read, a broken sentence. The whole version always reads: works under these conditions, within this range, on these objects. Extrapolating past the validated domain is the name for the default repair. Whoever cites the conclusion supplies the missing clause himself, silently, and reads "valid here" as "valid everywhere". Cognition does this by default. What this chapter asks for is a declaration fee charged on each act of carrying a conclusion across an edge. Abolishing extrapolation would abolish action, and this book is not going there.

23.1 What It Is

Definition: using a conclusion that holds inside a boundary as though it were a boundary-free law, while the boundary itself is never listed and never knocked down item by item.

Machine learning calls the same problem out-of-distribution use. Engineering and medicine have older names for it: off-label use, and use past the stated limit.

Three criteria open the mechanism. Any one of them is enough.

① The environment of use differs from the environment of validation, and the difference has never been drawn up as a list. ② "We validated it at A" appears in the argument as a reason to skip inspection. ③ Someone once pointed the difference out, and it never reached the decision record.

Extrapolation is not always wrong. An organization that bans it outright cannot move, and the exploratory failure in Chapter 40 of Part Six is extrapolation under control. What this chapter puts on trial is the missing list. Carry a conclusion across an edge without declaring the difference and you have the disease. Declare the differences, work through the effect of each, run the verification they demand, and you have a rollout. Do none of it and you have haulage: the box left the depot, and nobody checked the contents against the new address.

Out-of-distribution, off-label, past the limit: three names, one structure. The structure is worth spelling out before the cases arrive. Validated knowledge lives in a box held up by five dimensions: object (whom it works for), conditions (under what), duration (validated for how long), scale (at what magnitude), population (who applies it). Extrapolating is the reach from inside the box to outside it. Accidents happen when the reaching hand still believes it is inside.

Of the five dimensions, duration and population are the ones most often left unreported. Differences in conditions can be seen (dry ground against wet ground). Differences in population hide inside the statistics, as the share of children did in the raw-milk case. Differences in duration hide in the slow variables, as cumulative damage did across reprocessing cycles. The five fields on this chapter's declaration form, object, conditions, duration, scale, population, are cut from that box. Filling the form in is not a paperwork ritual. It is how the box gets drawn.

The cost of the declaration deserves its own paragraph, because the tuning decides whether the mechanism lands anywhere. What is required is a friction added to each act, since prohibition amounts to banning action. The size of the friction has to be chosen with care: too small (a verbal declaration) and it is not there; too large (full re-validation) and it is a ban. This chapter's form sits deliberately between half a day and a day. Expensive enough that a casual extrapolation stops being casual. Cheap enough that one worth doing can still be afforded. The tuning rule is general across the book's defences: friction has to sit in the narrow band where it changes behaviour without forbidding it. A defence placed outside that band gets walked around, or repealed.

Calibration inside the band leaves one more problem, which is who reviews the form. The intuitive answer, a specialist from the same field, reproduces condition two exactly. His analogical intuition and the declarant's come from the same source, and neither will catch a difference that sits underneath both of them. The review structure therefore has to cross domains. A specialist in the field signs the technical rows (conditions, scale). A risk function from outside the field signs the cognitive rows (population, duration, object). Both signatures, or the form does not clear. The reason is the cognitive shape of the mechanism: the blind spot in an extrapolation is not technical, it is the cell nobody believes needs looking at, and the second pair of eyes exists to force that cell open.

A cell that has been looked at needs somewhere to go. Archived forms go into a searchable library, indexed by object and by dimension, and the next declarant of a similar extrapolation checks the library first. The earlier form and its outcome, whether the thing worked and whether anything went wrong, are the cheapest prior on offer. A dimension with no entry in the library is itself a red mark on the new form: this organization has never validated anything on that axis. The library compounds in the way the dark

archive of Chapter 21 does. An organization's extrapolation intelligence accumulates through declarations rather than being paid for in accidents. Upkeep costs one indexing rule, and the library gives "we have done it this way before" an evidence form that can be checked for the first time.

Beyond its evidence form, extrapolation has a public exit worth gathering in, which is the citation chain of regulations and standards. An industry standard cites the standard above it, that one cites a research report, and the report's conclusion carries its own box. Every hop is an extrapolation, and most citations never re-read the boundary of what they cite. Cases 2 and 6 both run a chain of this kind, one protocol citing a device-class framework, one relief design citing an estimated heat release rate. The chain has a single self-discipline: at each citation, copy the cited conclusion's scope of validity into your own document, and the boundary travels down the chain with the citation (Device 3). The cost is one line of text. What it buys is that the boundary is not dropped at every hop.

The end of the chain is the person doing the work, which is where a boundary is most useful and hardest to deliver: the syringe in the nurse's hand, the exhaust on the tractor, the wind limit table in the tug pilot's cockpit. The field form of a boundary is an action criterion. Do not draw a dose when the graduations do not match. Check the exhaust after working wet ground. Stop when crosswind and turbulence together pass the limit. The difficulty sits in the parameterisation. Box dimensions are continuous (concentration, wind speed, cycle count); field criteria are discrete (do it, or do not). The method is to round toward strictness: cut the continuous variable into whole steps a person on site can read, and put the cut on the safe side of the true value, which is the margin convention of Chapter 25. The field version is this mechanism's last mile.

Forms and conclusion sentences are paper. Only a field criterion acts in the second before the event.

That second needs calibrating in the field, repeatedly. Once the field boundary is issued, every event where a criterion fires (a job stopped, a dose refused) earns ten minutes of review, with two questions: did it fire at the right point, and is the boundary set tightly enough. The answer is written back into the criterion. Write-back keeps the field version a live document. Criteria that are too strict get loosened gradually, so the alarm keeps its credibility. Criteria that are too loose get tightened in time. Calibration records merge with the form archive, and the organization's extrapolation ledger ends up with three levels: the conclusion sentence (knowledge), the declaration form (decision), the field criterion (action). All three are cut from the same box, and each serves a different reader.

23.2 How It Grows

Three generating conditions.

Condition one: the pilot and the rollout sit in different organizations. The pilot team knows where the boundary is (which users, which city, what weather, who was watching). The rollout team is handed the conclusion with the boundary stripped off. Boundary knowledge is tacit, and it stays behind with the team that earned it.

Condition two: the analogical pull of transplanting across domains. The similarity is on the surface (all insulin, all jet display flying, all dairy). The difference is underneath (concentration scales, aircraft performance, microbial ecology). Surface similarity carries the intuition, deep difference decides whether the conclusion lives, and intuition is what speaks in the decision meeting.

Condition three: the cost of validation is carried on one side. Writing the boundary down and re-validating item by item is paid for by the user. The speed is enjoyed by the decision-maker. Nobody writes the list.

23.3 The Shape in Four Domains

Case 1 (Aviation): An Old Jet Inside a Display Framework

av-042.

When and where: 22 August 2015, 12:22 UTC (13:22 local time, British Summer Time); north of Shoreham Airport, West Sussex, England, at the junction of the A27 Brighton bypass and the Old Shoreham Road. The impact point lay outside the airfield bound - ary, in ground the display organizer did not control (AAR 1/2017 p. 17, p.211).

What and who: owner and operator Canfield Hunter Ltd; a Hawker Hunter T7, registration G-BXFI, built in 1955, converted from single seat to two seats in 1959, airframe at 5,976 hours at the time of the accident (p.38). The display was staged by the company that ran the Shoreham RAFA Airshow, formed for the event and leasing the airfield (p.99); the flying display director (FDD) ran the flying and applied for the permission, and a four-person flying control committee (FCC) helped monitor it (p.38, p.100). The regulator was the General Aviation Department (GAU) of the UK Civil Aviation Authority (p.192). The pilot was alone on board, 51 years old, 14,249 hours total, 43 hours on type, 115 hours in the previous 90 days (p.28).

What they were doing: this was his fifth Hunter display of the 2015 season (p.187). After displaying at Bruntingthorpe on 8 August he flew 9 more displays inside two weeks, 7 in an RV-8 and 2 in a Jet Provost (p.35). The Shoreham sequence was a pass, a left-hand

Derry Turn, then a bent loop, a loop with roll in it so that the exit track is offset from the entry track. He had flown the same manoeuvre at the same venue in 2014, and to reduce the roll the bent loop needed he had then taken the left turn west of Lancing College; for 2015 he planned to turn between the airfield and the college (p.23). A hand-drawn map with the sequence numbered was found in his pocket afterwards (p.23). An engineer did the daily inspection on the afternoon of 21 August; the pilot did his own preflight check that day and asked for full tanks (p.24). Take-off was on runway 02, which slopes downhill and had a tailwind; with the high temperature and the tailwind the ground run was longer than usual and he rotated at 112 KIAS against a normal 120 (p.24). He was airborne at 12:04 (p.24). Before the display he held over the sea near Brighton at 1,800 to 2,500 ft; at 12:20 Shoreham reported wind 120° at 12 kt, temperature 24°C, QNH 1013 (p.25). He descended to a 100 ft pass on the display line, took the Derry Turn to an apex of 1,800 ft, then rolled into a descending left turn down to about 185 ft above ground, approaching the display line at about 25° (p.25). By the accident he had 19 hours 25 minutes of display and transit flying on the Hunter, the last sortie on 8 August (p.34, p.35). On 30 August 2014 he had flown the same manoeuvre at the same venue, and the exit track that day ran along the display line, parallel to the crowd line (p.118).

The decision point: four moments.

① 3 June 2011, the fifth Hunter conversion sortie, which was the final handling test; that day's log was endorsed "solo 'Standard' aerobatics DA" (p.31). The five training sorties before it came to 4 hours 5 minutes, three of them formation, the fourth a transit, only the fifth solo (pp. 30–31); follow-on training up to September 2011 added 2 hours 40 minutes (p.32). The test sortie's log lists handling, slow flight and "emergencies"; low-level recovery is not on it, and the report says elsewhere that he never

practised a recovery on the Hunter (p.31, p.37). The type exemption (ATRE) was renewed on 28 June 2012 and again on 27 August 2014 on the basis of "holds a valid licence", with no training or check requirement attached (p.32).

② 12 September 2014, the company's chief pilot, acting as display authorisation examiner (DAE), renewed his DA. The assessment used two Jet Provosts in formation and he flew No. 2 throughout (p. 32). The assessment words were about formation quality: "smooth accurate… crisp, safe and correct". The report then notes that, following a leader, he had no opportunity to show the safety gate for solo manoeuvring, the speeds, the handling or the minimum height, and no opportunity to show his positioning against the display line (p.33).

③ 24 June 2015, the DA was renewed again, this time in the RV-8, the assessment reading "leading and wingman in a two-ship RV-8 display" (p.33). Under the rules then in force, renewal in one cat‑egory renewed them all, and the Hunter and Jet Provost authorities came along with it (p.34).

④ Before the 2015 show the organizer obtained a permission issued by the CAA under Article 162 of the Air Navigation Order, with a display area map issued by the CAA. The area was drawn beyond the airfield, and a 1 km radius covered the A27 junction (p.100, p. 211 item 6). The FDD's risk assessment listed "aircraft impacting outside the airfield boundary" as hazard 8, severity catastrophic, probability "extremely unlikely", the judgement "unaccept‑able"; the mitigation read: the pilot complies with the Rules of the Air, the FDD and the FCC monitor by radio and stop the display if needed (p.102). The matrix wants the risk reduced before an "un‑acceptable" rating can stand, and that column was filled with those three measures. The report adds that the FDD considered the whole assessment compliant with CAP 403 (p.102, p.214 item 36).

What was visible then: the night before, he read his own ground notes, which covered type handling and the limits to be observed (p.23). He had firm numbers for the Jet Provost safety gate: 220 kt minimum entry for a low-level loop, 2,200 ft at the apex, plus the height the manoeuvre needs to finish (p.29). CAP 403, 13th edition, February 2015, paragraph 5.58: all aerobatic manoeuvres, inverted passes and manoeuvres pulled past the vertical included, must be flown above the approved display height; the DA currency requirement was three displays or practices in 90 days, one of them on the display type (p.36, p.37). His own DA form gave a Hunter minimum pass height of 100 ft, aerobatic category S, minimum aerobatic height 500 ft (p.34). His own written manoeuvre parameters: entry not below 350 KIAS, no entry below that; apex gate 4,000 ft, made up of 3,000 ft for the second half of the loop, plus 500 ft for the authorised minimum height, plus 500 ft of margin (p. 24). The organizer's briefing document for participants, section 4: fly only manoeuvres known and practised, and keep the aircraft positioned at all times so that, if the engine or the structure fails, the impact point is outside the crowd area (p.100). The ten-hazard list of the risk assessment; the Article 162 permission and its map (pp. 100–102). Weather: Stansted 150° at 14 kt, 28°C, dew point 16°C, QNH 1014; Shoreham at 12:20, 120° at 12 kt, 24°C, QNH 1013 (pp. 24–25). Weight: full fuel plus a 180 lb pilot, 18,814 lb before take-off against a maximum of 25,000 lb, centre of gravity inside limits (p.50). Technical state: a test flight after maintenance on 20 March, on which he reported a problem with the right altimeter verbally (p.51); the test flight record of 26 June 2011 shows engine speed at 8,250 rpm three times and 8,350 rpm once, above the 8,100±50 limit, with no technical log entry, no inspection, and the endorsement "airworthy and serviceable" (p.50). Known only afterwards: the fuel regulator diaphragm was aged past the point of airworthiness (p.200, p.212); the left altimeter read about 100 ft low in test and lagged, and at a 2,700 ft apex its pointer partly covered

the thousands counter (p.180, p.212); at the day's weight and density altitude the second half of the loop needed at least 2,600 to 2,950 ft (p.212); the bent loop rolled about 60° and the exit track pointed straight at the A27 (p.26, p.112).

Stop point. You are sitting in this seat with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened. 11 people on the ground died and 13 were injured (the pilot seriously, 1 seriously, 11 slightly), and the aircraft was destroyed (p.27). Impact speed was about 225 kt (p.213). The pilot survived; canopy jettison and the ejection seat were triggered by the impact, and he did not initiate ejection himself (p.216 items 57 and 59). Daily traffic on the A27 runs at about 58,500 vehicles (p. 215). The report made 11 safety recommendations, on top of 21 from three earlier special bulletins (p.21).

Back to the three questions. The commonest answer to the first is "roll out of the loop if the apex is short". In this case that action cost four seconds: flight test showed the recovery could still be made within four seconds of passing the apex, he had never practised it on the Hunter, and he had said he was unsure whether he could roll at 105 KIAS (p.37, p.181). The item most often missed in answer to the second is the four loops flown in the Jet Provost the weekend before: apexes at 2,750, 3,150, 2,800 and 2,900 ft, speeds of 105 to 130 KIAS, almost coincident with the 2,700 ft and 105 KIAS of the accident day (p.67, p.213). On the Jet Provost those are normal

figures. On the Hunter they are zero margin. The third question lands on the renewal before the season began: assess by type, and write the entry speed and the apex gate into the sequence list handed to the FDD. The report adds one sentence of its own: of the 13 fatal accidents at UK airshows between 1996 and 2015, this was the only one that killed a member of the public (p.123 table 8 note 38), and a history with no third-party deaths does not establish that the activity is safe (p.200).

Options ruled out: ejection, since the cockpit sat outside the ejection envelope for most of the segment after the apex, and the seat cartridge's two-year installed life was four and a half years over and its six-year total life a year over (p.198, p.216); a stop called by the FCC, since at the apex the aircraft was plainly below the gate, but the FCC did not know what the gate was and had no means of measuring height and speed accurately, and the 2014 video of the aircraft overflying the Lancing residential area had been neither noticed nor acted on at the time (p.190, p.215); sending the manoeuvre sequence into the approval process, since nothing then required the organizer or the FDD to hold a per-display sequence list, on the grounds that it was unnecessary (p.106, p.214), and the same report cites the Canadian practice of submitting distance, entry and exit points, speeds and minimum heights for each manoeuvre 10 days before the show (p.106); renewal by type, since AAIB only recommended renewal by category or type in a 2016 special bulletin and the CAA reply put the study out to April 2018 (pp. 188–189); separating aerobatic airspace from people on the ground, since AAIB recommended designated aerobatic airspace with no non-essential personnel, vehicles or vessels beneath it (2016-039) and the CAA did not accept it (p.194), after which the report records another line of thinking, that deviation from the Rules of the Air be allowed only where participants control the area and a sufficient risk assessment has been done (p.195); flying the 2014 sequence again as it stood, since that year the apex margin was suf-

ficient and the exit track ran along the display line (p.118), a precedent that made "fly it again" look like something needing no fresh validation.

Early signals: ① the Shoreham display of 30 August 2014, aerobatics over a residential area, on video, noticed and acted on by nobody at the time (p.190, p.194); the report later found such overflights were "not confined to one pilot, one aircraft or one venue" (p.192); it was pressed down because nothing had gone wrong. ② the renewal of 12 September 2014, flown as No. 2 in formation, the assessment text about formation quality only and the instruction column carrying a single line, "discussed the importance of instruction and continued supervision" (pp. 32–33); it was pressed down because formation flying is part of display flying too. ③ two renewals in 20 months, neither of them on the Hunter, Jet Provost formation in September 2014 and RV-8 in June 2015 (pp. 32–34); they were pressed down because the rules let one category cover all of them. ④ the test flight record of 26 June 2011, three readings at 8,250 and one at 8,350 rpm over the limit, no entry, no inspection, signed "airworthy" (p.50); it was pressed down because the flight itself was normal and the overspeed only momentary. ⑤ the previous weekend's four Jet Provost loops, apexes of 2,750 to 3,150 ft at 105 to 130 KIAS, close to the accident day (p.67), a point the report wrote into the causal analysis on its own (p.189); it was pressed down because that was another aircraft, and nobody had put the two sets of numbers side by side.

The point of escalation: the RV-8 renewal of 24 June 2015 locked the whole season's Hunter authority into place. From that day to 22 August no Hunter assessment stood behind it (pp. 33–34), and the six Hunter displays inside 90 days satisfied the currency requirement exactly, so the continuity of the authority showed no gap (p.37). On the day he had only the one public display (p.211 item 2), the tanks were full, and on paper the permission ran to 10

March 2016 (p.38, p.50). On the signing side, the 2014 signature came from the operator’s own chief pilot acting as DAE (p.32); the 2015 renewal needed assessment in one category only, and the as - sessor is not named (p.34). After that, full fuel on 21 August, the tailwind take-off on 22 August at 112 KIAS instead of 120, 24°C and an airfield density altitude of 1,310 ft each shaved the energy budget thinner (p.24, p.50, p.317). After the 20 March test flight he had reported the right altimeter verbally; the instrument went for repair, and the repairer said there was no component history card and its history could not be established (p.51, p.53). By 12:22, pulling at 310 KIAS and 185 ft, the only switch left was a rolling recovery within four seconds of the apex (p.178, p.181). The escal - ation took the form of continuity of authority and continuity of energy stacked on one timeline.

Failure chain: technical layer: entry at 310 KIAS, 40 below his own gate; engine speed falling from 7,530 to between 6,500 and 6,800 in the climb, 7,010 at the apex, all below maximum thrust (p.119); about 60° of roll, and tests show a loop with 90° of track change in the climb reaching an apex 300 to 400 ft lower than a straight loop (p.211); apex 2,700 ft against a requirement of 2,600 to 2,950 ft (p. 212); 105 KIAS at the apex, with a rolling recovery never practised on the type (p.37); the sorties that produced the gate data had apex density altitudes between 2,400 and 3,400 ft against 4,000 ft on the day, a difference of 1,500 ft (p.321).

Organizational layer: DA renewal in one category could cover them all (p.34); neither renewal was flown on the Hunter (p.213 item 30); the manoeuvre sequence did not have to go to the FDD (p.214 item 38); the risk assessment used the 500 ft rule in the Rules of the Air as its mitigation while the Article 162 permission waived that rule within 1 km of the crowd, and the FDD did not know about the waiver (p.214 items 40 and 41); the CAA inspected 7% of the dis -

plays it permitted in 2015 (p.215 item 55) and had never set an ac‑ceptable level of safety for display flying (p.216 item 56).

Incentive layer: the Hunter is expensive to run, its engine life is finite and spares are scarce, so training was arranged by riding along with other flying tasks (p.186); the cost of writing the boundary down and re-validating by type fell on the operator and the regulator, while a complete display season and an available venue paid out on the spot.

The break sits between the organizational layer and the incentive layer. The only two documents that could have stopped this extra‑polation before the event, the per-type renewal record and the manoeuvre sequence list, did not exist at the time, and they did not exist because neither carried any current return for any signatory.

Minimum counterfactual: turn the renewal of 24 June 2015 into a solo assessment on the Hunter, one sortie containing one bent loop and one rolling recovery past the apex, marked by the examiner against 350 KIAS and 4,000 ft. The action falls two months before the accident, the aircraft was at North Weald, the DAE was the company’s own chief pilot, and no luck is required (pp. 32–33, p. 186). The cost is calculable: his solo final handling test in 2011 was one sortie of 1 hour 05 minutes (p.31). No new regulation and no separate CAA approval are needed, because the rules then allowed renewal by category and did not forbid doing it by type (p.34, p. 188). Had that assessment been marked against the 4,000 ft gate, the “sufficient margin” success of 2014 would have stopped being a reason to skip the next check (p.118). The step changes no part of the display sequence and no clause of the framework. It puts one assessment on the aircraft being assessed. The report itself records that both renewals “complied with the rules of the time” (p.188, p.213 item 30), and inside those rules this step was available.

Outcome and sources: 11 dead, 13 injured; 11 recommendations, plus 21 from three special bulletins; the report also asked the De‑

partment for Transport to commission an independent review of UK airshow governance (2017-011), the CAA's own review having produced 29 actions (p.210); the answer to 2016-041, renewal by type, was a review of category structures and a study of behaviour transfer between types, to be completed by April 2018 (p.189). The CAA has since withdrawn the general minimum-height exemption for airshows, permissions now state the display area boundary, and a report is required within seven days of the display (p.193, p.195); 2016-039, clearing the ground beneath aerobatic airspace, was not accepted (p.194). Primary source: AAIB, Aircraft Accident Report 1/2017, Report on the accident to Hawker Hunter T7, G-BXFI, near Shoreham Airport, 22 August 2015, published 3 March 2017, 452 pp., including a flight test data appendix and a human factors study (p.317, p.423); the report finds two causes, apex height in‑sufficient to complete the manoeuvre (low entry speed, insufficient thrust in the climb) and the recovery not carried out, with six contributory factors (pp. 20–21). Page numbers on this card are PDF pages (printed page +16). Link: https://as‑sets.publishing.service.gov.uk/media/58b9247740f0b67ec80000fc/AAR_1-2017_G-BXFI.pdf (Appendix B av-042).

Case 1 leaves the word "framework" with a permanent problem. A safety framework is designed around the energy characteristics of the known fleet: display line distance and exclusion zone size are scaled to the speeds and turn radii of the common types. For every new type, the framework is therefore performing an implicit extrapolation, on the assumption that the new type's energy characteristics fall inside the existing envelope. The declarant in that implicit form is the framework's maintainer, and what the report pushed for after this case was, in substance, the writing of "a new type must submit an energy declaration on entry" as an explicit step. A reader can run the same check on his own frame‑work: which class of object are your isolation distances, authority thresholds and approval levels calibrated to, and whether a step

exists where a new object declares its energy, scale or risk characteristics. The generality of a framework is an assumption nobody has validated. Making it explicit costs one table.

Case 2 (Drugs and Devices): A Reprocessing Protocol That Covered a Model It Should Not Have

ph-007. An Ethicon XCEL trocar (5 mm), circulated after reprocessing by Stryker Sustainability Solutions. FDA device recall record Z-3183-2026, the API record checked for this book: the reprocessed device may crack, and 2,838 units were recalled. Reprocessing and reuse of single-use devices has mature protocols and a regulatory framework behind it. The framework's applicability to this model's structure, meaning the integrity of the bladeless trocar's seal and hub connections after reprocessing cycles, was covered over by the protocol's generality. The protocol had validated that devices of this class can be processed this way. It had never validated this one.

Stop at the decision point: the review moment when the reprocessing protocol added the new model to its list and the batch was released. On the table: the protocol's prior safety record, the structural difference between the new model and its predecessors, and the release schedule. **Outcome.** Recall, batch withdrawn. Primary source: FDA, Enforcement Report Z-3183-2026, 2026, link at Appendix B ph-007.

Case 2 has an institutional counterpart in devices. The FDA regulates reprocessed single-use devices by device class, and the seam between a class-level generic protocol and a model-level structural difference is exactly where this case sits. The management answer is to align the granularity of the two lists: the validity statement goes by model, which one was validated, and the release decision goes by class, any of which may pass. Alignment costs documentation work, and it buys a release step that can answer "was this

one validated" every time it is asked. Any reader can run the same alignment check: ask whether the validity statement behind your generic protocol, template or SOP is written at the granularity of a class or of an individual. One step off in granularity puts the boundary one step off, and a boundary one step off has already produced one accident in Case 2 and another in Case 3.

Case 6 (Chemical): A Scale-Up with No Reactivity Data (condensed)

ch-006. A reactor ran away into a decomposition explosion in November 2024, in the United States. CSB report page (2026, page checked for this book): 2 dead and 2 seriously injured, with vessel fragments thrown beyond the site boundary; the investigation records three findings, no reactivity test data, an emergency relief system inadequately designed, and no process safety management system. The chapter's criteria light on the first finding. Process scale-up from bench to production is the classic extrapolation, and reactivity data is the first line of a scale-up declaration. Without data, the relief system can only be sized against an extrapolated heat release rate, and the error in that estimate gets amplified, in a runaway, into vessel fragments. Minimum counterfactual: complete reactivity testing (adiabatic calorimetry, for instance) before scale-up, and size the relief system on measured data. Primary source: CSB, 2026, link at Appendix B ch-006.

Case 7 (Aviation): An Old Manoeuvre After a Safety Notice (condensed)

av-005. An R44 helicopter in October 2021, in Germany. BFU report (2022, checked for this book): during a low-g manoeuvre the rotor hub struck and cut through the mast, the aircraft broke up in flight, and 3 people died; the manufacturer's safety notice had stated the low-g risk long before. Transplanting display manoeuvre conventions between types is the aviation form of condition two: a

manoeuvre that holds on one type gets carried to another as part of the common display vocabulary. The safety notice wrote the boundary down, and the boundary statement never entered the operating convention. Minimum counterfactual: check the display fleet's manoeuvre list against manufacturer safety notices, type by type, and mark any manoeuvre a notice covers as barred. Primary source: BFU, 2022, link at Appendix B av-005.

Case 8 (Aviation): A Tug Take-Off in Crosswind (condensed)

av-025. A glider tug in December 2025, in the United Kingdom. AAIB report (2026, page checked for this book): after take-off in strong crosswind and turbulence the aircraft lost control and hit the ground; the tug pilot was seriously injured and nobody in the glider was hurt. The tug's operating envelope had been established in everyday conditions, and strong crosswind plus turbulence stacked two variables from outside it at the same moment. The decision to continue the take-off rested on a success baseline from everyday conditions, which is the weather version of criterion ②, "we validated it at A". Minimum counterfactual: write the combination limit on wind speed and turbulence into the operating standard, and stop when it is exceeded. Primary source: AAIB, 2026, link at Appendix B av-025.

Case 3 (Healthcare): The Boundary of a Syringe Scale

me-002. Medical wards of an English general hospital. HSIB report (Administering high-strength insulin from a pen device in hospital, 2022, archived snapshot checked for this book): U-500 high-strength insulin, at 500 units per millilitre. The dedicated safety needle for the pen device disappeared from the supply catalogue after a contract change, and nurses switched to drawing the dose with a U-100 syringe: a syringe graduated at 100 units per millilitre used on fluid at 500 units per millilitre, every graduation on the

barrel five times the intended dose, and the patient had two hypoglycaemic events. The U-100 syringe's validated boundary was the world of U-100 concentration. The instrument did not change. The world changed five-fold, and a qualified tool inside the boundary became an overdose generator outside it.

Stop at the decision point: the safety needle disappearing from the catalogue, and the nurse choosing a substitute instrument before drawing the dose. On the table: the syringe at hand, the concentration on the label, and the seam along which the catalogue change notice never reached the ward. **Outcome.** Two hypoglycaemic events; the report made recommendations along the chain from supply to bedside. Primary source: HSIB, 2022, link at Appendix B me-002 (archived link, grade A–).

The five-fold scale in Case 3 is worth one piece of risk arithmetic. The instrument's qualification did not change, since a U-100 syringe is still the right tool for U-100 fluid. What changed was the fluid. The matching relation between tool and fluid was implicit, held by catalogue pairing and by habit, and it had never been written as an explicit check item. When a distant action such as a contract change alters the matching relation, the clinical end knows nothing, because the change's blast radius on the supply chain (Chapter 21, Device 2) has no mechanism for reaching the end user. HSIB's recommendations along that chain repair the notification line. The cross-industry generalisation is plain: wherever a tool and a material sit in an implicit matching relation (instrument and fluid, blade and material, code and dependency version), write the relation as an explicit check item, and give changes to the relation a notification line that reaches the point of use. Making the relation explicit is one-off documentation work. Its return arrives at every supplier change and every catalogue adjustment.

Case 4 (Agriculture): Raw Milk's Safety Assumption and 171 Cases

ag-001. Commercially distributed raw milk in California, from September 2023. CDC MMWR report (Vol. 74, page checked for this book): 171 confirmed cases in a Salmonella Typhimurium outbreak, 22 hospitalised, of whom 120 (70%) were under 18; the timeline runs from first illness on 21 September, to the county health department's report on 18 October, to detection in samples on 23 October, to the production halt and recall on 24 October, to resumption of production on 31 October. Raw milk is unpasteurised, and its saleability rests on assumptions about process and testing. The outbreak showed the boundary of those assumptions, batches, microbial flora, sampling rate, consuming population, failing at commercial scale.

Stop at the decision point: the approval of resumption on 31 October, with the testing regime carried over unchanged. On the table: the post-recall remediation list, the sampling scheme, and the epidemiological fact of the share of children in the consuming population. **Outcome.** Further cases appeared inside the resumption period (MMWR coverage runs to March 2024). Primary source: CDC, MMWR Vol. 74, 2025, link at Appendix B ag-001.

The resumption approval deserves taking apart. The remediation list focused on testing, on sampling rate and batch management, and the population difference the outbreak had exposed, seven cases in ten being minors, was not among the conditions to be rechecked before production restarted. The testing question is whether this batch may leave the plant. The population question is who this batch is more dangerous to. The point where the two meet, risk weighted by population, was absent from the resumption decision, and the extrapolation ran on from the plant gate to the shelf. Further cases inside the coverage period turn that absence into an outcome. A general checklist for resumption de-

cisions therefore gains a line: review the structure of the consuming population, who eats the product and who is most vulnerable, with demographic difference entering the declaration form at the specification of its first line.

The wording of a declaration form has one more job to do, which is anti-narrative. Keep narrative words such as "natural", "pure" and "traditional process" out of validation documents. A narrative word need not be false. Raw milk is natural. What makes the word useful is that it underwrites the boundary: a reader who sees "natural" fills in "safe", and the basis of the fill-in is the feel of the word rather than data. The language discipline for validation documents, Device 3's boundary in the conclusion sentence, therefore gains a banned-word list: conclusions and boundaries may use data words only, concentration, dose, detection rate, temperature; narrative words belong in marketing documents, and the two document types are signed off by different owners carrying different liability. The list's value is not censorship, since nobody can police language. It is a reminder at the moment of writing. The moment you put a narrative word on the page, you should know that the sentence has left validated ground.

Case 5 (Manufacturing): An Exhaust Validated on Dry Ground (full card)

mn-011.

When and where: a recall in January 2013, after several years on sale; the United States. CPSC recall notice, page checked for this book.

What and who: about 25,000 commercial all-terrain vehicles of one model; the defect was that in wet conditions leaf debris accumulated and the exhaust ignited it.

What they were doing: performance validation was completed on dry ground. Exhaust temperature, insulation distance and clearance to combustibles were all designed and passed against dry-ground operation. Real use contained a variable dry-ground validation had not covered, which was vegetation debris from working wet ground. Wet debris clings to the exhaust surface, bakes on, accumulates, carbonises and ignites at exhaust temperature; on dry ground the debris either does not cling or gets blown away, so validation had no reason to doubt the surface temperature. After 18 fire reports and one hand burn, the manufacturer recalled about 25,000 units with a repair kit. The validation box (dry ground, typical vegetation, normal work rhythm) and the use box (wet ground, heavy debris, continuous low-speed work) shared one machine and not one set of physics.

The decision point: the review of the product definition and the validation plan, which is to say the approval of the validation condition list. On the table: the description of the target market and work scenarios, the cost and schedule constraints on validation, the validation conventions of comparable products, and one question that was not on the list, which is the environment we never simulated in which the target user will drive this machine.

What was visible then: the validation condition list, the scope-of-use statement in the product manual, the climate and work types of the target market, all of them discoverable. Known only afterwards: the form of the debris accumulation and the ignition path, which appeared only in the post-accident engineering analysis.

Stop point. You are sitting in this seat with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".

> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened. 18 fire reports and one hand burn, with about 25,000 units recalled for a repair kit. Back to the three questions. The commonest answer to the first is "add a heat shield", and a heat shield is a retrospective patch, while the prospective move is putting wet-ground operation into the validation matrix, even as a single extreme case. The item most often missed in answer to the second is the validation condition list itself, which reads as a technical document and is in fact the product's boundary declaration: the conditions absent from it are the world the company has not promised. The third question lands on the validation matrix review, which must answer with the list of extreme work scenarios in the target market; the gap between that list and the matrix is the difference list for the declaration form. Primary source: CPSC, 2013, link at Appendix B mn-011.

Options ruled out: putting wet-ground conditions into the validation matrix, squeezed out by cost and schedule; restricting wet use in the manual, commercially unworkable and unenforceable in practice; a dedicated post-launch complaint category for debris, since complaints went through the routine quality process and were never aggregated by scenario.

Early signals: ① wet-ground work appears in the marketing material and not in the validation matrix, pressed down because "marketing speaks in scenarios and validation works to standards"; ② occasional "the exhaust runs hot" feedback through the repair channel, pressed down as "normal"; ③ competitors' validation conventions also excluded wet ground, pressed down because "that is how the industry does it".

The point of escalation: every year of sales without a fire vouched for the dry-ground validation, and the sales share in wet regions was not tracked, so exposure had no ledger. The escalation took the form of exposure made invisible.

Failure chain: technical layer, wet debris accumulating, carbonising and igniting at exhaust temperature. Organizational layer, no declaration mechanism for the gap between validation conditions and real use. Incentive layer, validation cost immediate, fire risk delayed and probabilistic. All three broke at once. The break sits where marketing and validation use two scenario languages, marketing working by scenario and validation working by standard, and the fork in language is why the difference was never declared.

Minimum counterfactual: at the validation matrix review, force the marketing scenario list and the validation condition list side by side and declare each difference item. One action, in product definition, costing half a day of review.

Outcome and sources: 18 fires and one burn, with about 25,000 units recalled. Primary source: CPSC, 2013, link at Appendix B mn-011.

23.4 What Shows Earliest

Six weak signals, each with the reason usually used to press it down.

1. The rollout plan has no section on what changed against the pilot. Reason: "the pilot already proved it."
2. A difference is acknowledged out loud ("the users here are different") and never appears in the decision record. Reason: "everyone knows."
3. The conclusion sentence of a validation report carries no scope of validity. Reason: "who reads that closely."

4. Reuse across models, populations or regions triggers no re-validation. Reason: "same principle."
5. A difference someone raised disappears from the minutes. Reason: "meeting time was short."
6. A boundary event, some minor anomaly, is explained as normal fluctuation outside the pilot. Reason: "the sample is still small."

Put the six signals back against the eight cases. Signal 1, no difference section in the rollout plan, matches Case 5, where the validation matrix and the marketing scenarios were never placed side by side. Signal 2, difference acknowledged verbally, matches Case 3, where the concentration difference lived in the nurse's common sense (the graduations differ) and not in a process check item. Signal 3, no scope in the conclusion sentence, matches Case 2, where the reprocessing protocol stated its validity by class. Signal 4, reuse without re-validation, matches Case 7, where a manoeuvre vocabulary crossed between types. Signal 5, the raised difference vanishing from the minutes, matches Case 8, where the wind limit judgement sat in the crew's conversation and not in the operating standard. Signal 6, the boundary event explained as fluctuation, matches Case 4, where early signals before the outbreak were never aggregated by scenario.

Re-sort the six by what it costs to find them. Signal 3 is cheapest, a matter of reading conclusion sentences. Signals 1 and 2 come next, reading documents and minutes. Signals 4 and 5 need cross-department comparison. Signal 6 is the most expensive, because it needs anomaly data aggregated by scenario. The starting advice is the usual one: do the cheap thing first. run every conclusion sentence in your organization's validation documents through a single pass, once, and rewrite the ones without a scope of validity to Device 3's specification. That rewrite makes every downstream citation carry the boundary automatically. The document-layer

work is one-off, and its return is set by the number of times the document is cited.

23.5 How to Defend

Device 1: the extrapolation declaration form. Any act that carries a conclusion outside its validation environment files one form: the original validation's boundary (sample, conditions, duration), the target environment's difference list, and an item-by-item statement of why each difference does not affect the conclusion. The items whose effect cannot be stated are exactly the items that need supplementary verification, so the form's value is that it exposes what cannot be explained. The form is archived. It is the raw voucher of "what was known at the time" for any later post-mortem.

Device 2: independent supplementation of the difference list. The user drafts the list and the pilot team supplements it, the two sides listing separately and not permitted to merge their discussion before writing. Listed separately the differences come out complete, since what the pilot team knows is often in the user's blind spot and the reverse also holds. The union of the two lists is the real difference surface.

Device 3: the boundary written into the conclusion sentence. Every validation report's conclusion sentence carries its scope of validity ("holds under conditions X"). Downstream citations that strip the boundary count as citation errors and enter the post-mortem ledger. What this changes is a writing habit: once the boundary is promoted from footnote to body text, its probability of being cited stops being zero.

Device 4: graded release. Extrapolations are graded by the item count of the difference list. Zero differences, released directly. With differences, verification intensity and monitoring density rise with

the count. Where a difference touches a safety boundary, direct extrapolation is barred. The three criteria for exploratory failure in Chapter 40 connect here: controlled extrapolation needs a declaration, a budget ceiling and a stop-loss point. Grading turns extrapolation from "do it if you can" into "do it by the price list".

Run the four devices against this chapter's cases. Device 1 on Case 5: declare the validation condition list and the target market scenario list side by side, with the wet-ground row filled into the difference list. Device 2 on Case 3: the clinical side and the supply side each write a list of what this catalogue change altered, and "the safety needle is gone" is a row the clinical list carries and the supply list does not. Device 3 on Case 2: rewrite the reprocessing protocol's validity statement to run by model ("this protocol is validated for model X over N cycles"). Device 4 on Case 6: grade the scale-up project by its count of difference items, and bar direct release where reactivity data is missing.

The four devices have four failure modes. The declaration form dies when declaration becomes a formality: the difference list reads "no significant difference" and no reviewer asks a follow-up. The patch is that a "no significant difference" statement must name the dimensions checked, five checked and each written as no difference, and a declaration that cannot name its dimensions is treated as undeclared. Independent supplementation dies when the pilot side is absent: the two-column table gets drafted by the user, and the pilot column stays blank year after year as the team disperses and people move. The patch is that the pilot's boundary knowledge is written into the closing report when the pilot ends, with the boundary clauses a mandatory deliverable of closure. Boundary-in-the-conclusion dies of format compliance: a scope of validity is present and empty ("wide applicability"). The patch is that a scope must carry specific dimension values, conditions, sample size, duration. Graded release dies of standard drift: the

definition of a safety boundary gets loosened a step at a time, "touches a safety boundary" becoming "touches a major safety boundary". The patch is that the grading definitions go into the rulebook, with revision requiring a governance signature. All four deaths share one health question: did the last declaration form change a decision. A form that has never changed a decision is the new paperwork.

The relation between the declaration form and insurance or regulatory filings is worth a sentence. An archived form has a second use as an insurance submission, the difference list and the supplementary verification records going in with the application, and its value at claim time is that it is the raw voucher of what was known. A decision with a declaration record and a decision without one are treated very differently in claims and in litigation. The document exists to make the decision-maker's state of knowledge reconstructable, not to absolve him, and Chapter 10's evidence grading extends here to the decision document itself.

23.6 Boundaries with Adjacent Mechanisms

Against where intuitive expertise breaks down (Chapter 14): extrapolation is the transport of a conclusion, on the data layer; the failure zone is the transport of the judge, on the human layer. Where one scene occupies both, check the person first, his premises from experience, and the conclusion second, its validation boundary. Repairing the person is cheap and repairing the conclusion is expensive.

Against narrative before evidence (Chapter 13): narrative-first means the conclusion has run ahead of the evidence, while boundary extrapolation means the evidence has run past its range of validity. The first lacks evidence and the second lacks a bound-

ary declaration. Sizewell's "replication" narrative (Chapter 13, Case 2) and this chapter's difference list are two faces of one coin.

Against crossing a scale threshold (Chapter 24): a change of scale is the commonest difference item and the most underrated, since a pilot of ten becoming a rollout of a hundred thousand changes the distributions of users, load and anomalies all at once. A change of scale belongs on the first line of the difference list.

23.7 Five Self-Checks

1. Your organization's most recent pilot-to-rollout: does a difference list exist? Who drafted it? Has the pilot side read it?
2. The validation conclusions you have cited: does the conclusion sentence carry a scope of validity? If not, what boundary did you supply yourself?
3. The most recent reuse across models, populations or regions in your field: who was asked to re-validate? Nobody, or nobody?
4. Which steps of your product or process are done "the way we have always done them", with the original validation conditions no longer findable?
5. Take an old validation report of yours and rewrite its conclusion sentence with the boundary attached. Once rewritten, does it still support the way the thing is being used now?

Chapter 24: Crossing a Scale Threshold

> What this chapter covers: why the same method fails at a larger scale; three criteria, four field sites, the rhythm of re-validation at each magnitude step, and coordination cost put on the books. When to turn back: when growth accelerates, when a single site turns into a platform, when a point turns into a network, when self-operation turns into openness. Estimated reading time: 50 minutes.

Organizations in growth run on an illusion: what worked at a hundred people will work at a thousand if everyone tries harder. The arithmetic behind that illusion is wrong. Coordination cost grows superlinearly with the number of nodes. Management bandwidth decays with every layer added. Validation's sample assumptions were set at the old magnitude. What makes the crossing dangerous is the way it fails. The system does not collapse. It changes in kind. The same method is still running, everyone is still working hard, and the quality of the output has quietly taken on a different nature, while every current indicator still looks healthy, because the deterioration grows in a dimension the indicators do not measure.

24.1 What It Is

Definition: an organization's capability function is not linear. Coordination cost, information loss, the count of anomaly exposures and the count of interfaces all grow superlinearly with scale, while the governance structure (process, systems, headcount, validation rhythm) was designed at the size the organization last had. Crossing a scale threshold means the same method quietly stops

working at some magnitude, and the failure arrives before any alarm does.

Three trigger criteria. Any one of them means the mechanism has already started.

1. The same method has changed in kind at twice the size: it holds together only if someone coordinates a little more.
2. A management layer has been added, or the meeting count grows faster than headcount.
3. Local optima begin to cancel each other, every department optimizing while the total indicator does not move or moves down.

24.2 How It Grows

Three generating conditions.

Condition one. Growth runs faster than governance's renewal cycle. The governance structure was built at the size the organization had when it was designed, and growth leaves that boundary of validity behind. The boundary raises no alarm, because the old method's record of success at the old magnitude keeps being produced. Only the sample behind that record already belongs to an earlier scale.

Condition two. Validation's sample size was never reset with scale. A defect rate validated on a thousand users scales its exposure count in front of a million: the probability unchanged, the absolute count multiplied by a thousand. A rare defect becomes a daily report, and the capacity to process daily reports was not multiplied by a thousand.

Condition three. Coordination cost is invisible. Meeting time does not enter product cost, layer overhead does not enter the accounts, interface staffing does not enter gross margin. This is the sibling of

Chapter 29. Maintenance is invisible, and coordination is equally invisible. An invisible cost defaults to zero in a growth decision.

The threshold's most insidious property is that it is retrospective. It becomes visible only after it has been crossed, and the sign is usually the veteran's sentence: "it didn't use to be like this." In an organization that sentence is not nostalgia. It is data.

Three curves of nonlinear growth are worth keeping in the head, because each maps to a symptom. The coordination curve: fully connected coordination demand grows with the square of head-count, the organization answers with layers and division of labour, breaking one square into several small squares, and each added layer costs one more round of information attenuation (Chapter 16) and one more interface. The interface curve: connections between subsystems grow as combinations, interfaces are the breeding ground of hidden coupling (Chapter 21), and scale gives the dark lines geometric room to grow. The exposure curve: users, devices and transactions multiply the absolute count of anomalies, and validation's low probability enters the zone of the inevitable once volume arrives (Case 1). What the three share is that none of them enters the current set of accounts. Accounts record the linear items, revenue, output, order count. The nonlinear items, coordination, interfaces, exposure, are reconstructed backwards only after a symptom appears.

The difference between changing in kind and collapsing needs its own passage, because the two are defended differently. Collapse is a hard failure: the system goes down, the service stops, and there is an alarm, a review, a repair. A change in kind is a soft failure: the process still runs, the indicators still climb, the quality has changed its nature, and there is no alarm, no review, no repair. Detection therefore runs on comparison, Device 4's teardown question in this chapter, the external comparison of Chapter 12. Monitoring the indicators themselves will not find it, because a

change in kind is very good at keeping indicators looking well. One question for the reader: the last time you discovered that a process had changed in kind, did an indicator tell you, or did a new em - ployee's question? Most people answer the second. An indicator system goes collectively blind in front of a change in kind, because what it measures are the linear items.

24.3 How It Looks in Four Domains

Case 1 (Manufacturing): 338,800 Units Recalled in the First Year

mn-003. The DREO DR-HSH015 electric heater. CPSC recall notice, page checked for this book: about 338,800 units recalled, with about 36,300 more in Canada, 17 ignition reports, 1 hand burn and 3 reports of property damage, a wide recall triggered within one year of launch. With the per-unit defect rate unchanged, channel volume multiplied exposure, the summed hours of operation, by several orders of magnitude. The low probability that validation had found acceptable entered the zone of the inevitable in the first sales season at volume. The threshold sat in the channel, not in the factory.

Stop at the decision point: the moment the first production run and the channel distribution plan were fixed. On the table: the sample size of the validation batch, the channel's distribution multiple, and a relation nobody had written as an equation, since validation covered the usage intensity of a thousand units while the channel promised the exposure of a million. **What happened.** A large re - call. Primary source: CPSC, 2026; link in Appendix B, mn-003.

Case 1 carries a second subject worth a paragraph of its own, an exchange rate between marketing and validation. The validation sample size and the distribution multiple come from two different

departments, engineering setting the sample and marketing setting the distribution, and no exchange rate exists between the two numbers. Neither department has to answer what a thousand-unit validation means under a million units of exposure. That missing rate is the common blind spot of cross-department scale: each department optimizes locally at its own magnitude, and the summed risk has no owner, which is Chapter 18's dilution replayed on Chapter 24's ground. The fix is to write the exchange rate into the release process. The approval pack for a first release carries two columns side by side, exposure covered by validation and exposure planned for distribution, and a gap of more than one order of magnitude between the columns triggers additional validation or a staged release. Staged release is the engineering answer to the exchange rate problem: let exposure grow with validation rather than with the channel's ambition.

Case 6 (Manufacturing): Half a Century of Compound Exposure (condensed)

mn-006. A French press with a glass carafe. CPSC recall notice, October 2026, page checked for this book: 5.9 million units recalled, 70 breakage reports, 48 scald injuries. Half a century of cumulative sales amplified a per-unit small probability into a certainty for the user population. Micro-damage in glass cannot be seen before it fractures, and thermal shock and knocks in use push the exposure of a defect past the recall line across a user base built over decades. This is the time-axis version of the scale threshold. No volume ramp is needed, only time multiplied by installed base. Minimum counterfactual: monitor the ratio of cumulative sales to incident reports, an exposure-calibrated incident rate, and let a shift in the ratio trigger a design review. Primary source: CPSC, 2026; link in Appendix B, mn-006.

Case 7 (Fintech): The Scale Disease of Growth-Fed Spending (condensed)

st-001. A legal technology startup, about 200 employees and about 450 customers. Failory Cemetery archive, quoting the founder Justin Kan's public statements, checked for this book: $75.5 million raised, then closed. The founder's retrospective names the problem in his own words, continued expansion and further investment before product-market fit, so that scale only amplified the cost of an unvalidated model. This is the self-imposed version of the scale threshold. The organization applies resource scale to a business function that has not been validated, the function's error is amplified in proportion, and the amplification runs until the capital stops. Minimum counterfactual: bind each increment of expansion to a validation milestone, with no milestone meaning no next increment. Primary source: Failory Cemetery, 2020; link in Appendix B, st-001 (evidence grade E).

Case 2 (Public): Eleven Thousand Schedules

Case ID. po-001.

When and where. Report issued 14 September 2026, with an audit period from December 2024 to September 2026; field data runs to May 2026 and cost figures to June 2026 (GAO-26-107992 pp. 1, 4, 26). Location: Washington, DC, for FAA and the Department of Transportation, plus the Airspace Modernization Office created in January 2026; installation sites are 409 towers, 119 towers with terminal approach control facilities, 25 stand-alone terminal approach control facilities and 21 en route centres (pp. 5, 22).

What and who. The FAA's Brand New Air Traffic Control System programme, BNATCS. Execution sits with the Airspace Modernization Office, set up in January 2026; oversight sits with the BNATCS Executive Steering Committee, established in August 2025 (p. 22).

The prime systems integrator is Peraton, with the contract awarded on 5 December 2025 (p. 3, footnote 4). Accountable posts: the FAA Administrator, the programme managers of the 13 programmes and the integrator's project manager (pp. 29, 30, 32).

What they were doing. In May 2025 the Transportation Secretary announced BNATCS, compressing the modernization of 138 legacy air traffic control systems into two phases, phase one covering 13 programmes, all to be complete by December 2028 (pp. 2, 14, 15). Phase one was not a new start. It accelerated investment already running: voice switches from September 2037 to November 2028, 106 months earlier; radio sites from September 2037 to December 2028, 105 months earlier; information display systems from September 2034 to September 2028, 72 months earlier (p. 17, table 2). The workload is physical: 450 voice switches, 1,581 radio sites, 5,170 copper cables, close to 30,000 telecommunications services (pp. 13, 17). In July 2025 Public Law 119-21 appropriated $12.5 billion, $9 billion for systems modernization and $3.5 billion for facilities (p. 15, footnote 24). On 5 December 2025 the integrator contract was awarded at a ceiling of $1.5 billion and settled at about $876 million (pp. 22, 23 footnote 36). In January 2026 the FAA told the White House that the $9 billion would cover 10 of 12 programmes in full and that the other 2 were short by $450 million (p. 19). By March 2026, $1.8 billion was obligated, 20% of the appropriation (p. 23). By May 2026, 2,560 of the 5,170 copper cables were replaced and radio sites were 17% complete (pp. i, 31); 9 of the 13 programmes had started, and the remaining 4 were due to start between June 2026 and September 2027 (p. 24). In June 2026, 30 months remained to the December 2028 deadline (p. 26). Installation is scheduled by project, and what has not reached site is 2,610 fibre locations, 435 enterprise information display system sites, 72 terminal flight data manager units, 398 voice switches, 1,311 radio sites and 166 ground situational awareness systems (p. 30).

The decision point. Four moments.

May 2025. The Transportation Secretary announced BNATCS on presidential direction and wrote "substantially accelerate" into the programme's objectives, compressing phase one to 3 years (pp. 2, 14). The vehicle was a public announcement with an accompanying programme document, signed by the Transportation Secretary.

July 2025. Public Law 119-21 completed the appropriation (p. 15, footnote 24). The vehicle was a statute, signed by the President. This step fixed more than an amount. It fixed the shape of the accounts. The same law parcelled the $9 billion out by programme, so a shortfall could only be declared by programme. In June 2026 the FAA put the additional need at $10.8 billion, $574 million for phase one and $10.2 billion for phase two, while the fiscal 2027 budget request asked for $565 million and $860 million (pp. 27, 21). Table 3 sets appropriation against estimate row by programme: information display $300 million, terminal flight data manager $300 million, telecommunications $4.75 billion, radar $3 billion, runway safety and surface surveillance $500 million, weather $80 million, advanced training $100 million (p. 15 footnote 24, p. 20 table 3). No row in that table is the programme as a whole.

August 2025. The FAA Administrator established the BNATCS Executive Steering Committee, applied new accelerated acquisition rules to every BNATCS project, and signed a blanket waiver releasing all projects from nearly all requirements of the Acquisition Management System, AMS (p. 22). Page 28 records this waiver as one of the key reasons no complete, documented life-cycle cost estimate exists: after it, producing that estimate stopped being a written obligation (p. 28).

5 December 2025. The FAA awarded the prime systems integrator contract to Peraton at a ceiling of $1.5 billion (p. 3 footnote 4, p. 22). Existing programme contracts were modified in the same

step, giving the integrator authority to manage those contractors on the FAA's behalf (p. 23). The contract states that the integrator is responsible for an integrated master schedule covering phase one (p. 29). Ninety days of negotiation followed the award, and the deployment schedule was settled inside that negotiation (p. 4 footnote 7, p. 29). The money saved in the negotiation went elsewhere: the gap between ceiling and final price was more than $600 million, oversight of radar installation moved from the integrator to the radar manufacturer, and the traffic flow management data and services system was deferred (p. 23 footnote 36). Scope, responsibility and deadline locked at this step, and the integrated master schedule was written as a duty, with the delivery condition left empty.

What was visible then. On 11 January 2023 the notice to air missions system went down, aircraft nationwide were held on the ground for about 2 hours, more than 1,300 flights were cancelled that day and close to 10,000 were delayed; the same system failed again in February and in March 2025 (p. 6). On 23 September 2024 an earlier GAO report had already found 51 of the 138 systems (37%) unsustainable and 54 (39%) possibly unsustainable, with 58 of them carrying significant operational impact on airspace safety and efficiency; 17 were of particular concern, and 4 had no investment under way at all (pp. 1, 6, 7, 11). That same material carried the terminal flight data manager record: nearly 3 years late, cost up 20%, deployment sites cut from 89 to 49 (p. 12). In late April 2025 the old communications lines at the Philadelphia terminal approach control facility failed, controllers at Newark Liberty International lost radar and radio contact for about 90 seconds, and the FAA ordered a ground stop; the same facility failed again in May (pp. 2, 8). In May 2025 the FAA terminated the outside contract for the enterprise information display system and moved the work in-house (p. 26). In September 2025 a telecommunications failure at the Dallas terminal approach control facility sent controllers back

to manual clearances and delayed or cancelled more than 1,800 flights (p. 8). In January 2026 the comparison table sent to the Office of Management and Budget was in hand, $9 billion covering 10 programmes with 2 short by $450 million (p. 19), and in March 2026 the obligation figures were in hand, $1.8 billion, 20% (p. 23). In April 2026 the integrator delivered the first version of the integrated master schedule (p. 29). In May 2026 the counts for six categories of installation site were in hand (p. 30) and the executive dashboard was updating monthly (p. 22); GAO could not confirm the dashboard's completeness and reliability, because the FAA did not explain where the data came from or what validated the updates (p. 3). What only became known afterwards: the overlap of the 11,389 schedules at the same installation site, where the report can offer only the St. Louis example and writes of the rest that "numerous overlaps may exist" (p. 30); and the exact size of the full life-cycle cost gap, where the report establishes that the $10.6 billion figure is an underestimate and does not say by how much (p. 27).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened. On 14 September 2026 the report framed the programme as needing improvement in cost and schedule planning and made two recommendations to the FAA: prepare a life-cycle cost estimate following GAO cost estimating best practice, and prepare a phase-one integrated master schedule following GAO schedule assessment best practice (p. 32). The Department of Transportation partially concurred with both and did not list the

actions it would take (p. 32). No casualties, no penalty, no grounding. The outcome is a risk characterization and two recommendations not yet accepted. The numbers stay on the table: a phase-one estimate of $10.6 billion against an appropriation of $9.03 billion, a gap of $1.585 billion (p. 20 table 3); a phase two needing another $10.2 billion with no start date and no completion date (pp. 15, 21); and, from June 2026, 30 months of phase-one work left (p. 26). Back to the three questions. The commonest answer to the first is to make the integrator deliver the integrated master schedule before work starts, and that action costs a full stop on installation in the 9 programmes already started, with 30 months left, which pushes completion past 2029. The cheaper place to stand is the 90-day negotiation window, writing the integrated master schedule in as a payment condition, when the cost of stopping was still zero. The item most often missed in the second question is the dashboard, which updates monthly and looks like something that already exists, while nobody can explain its data sources or its update parameters (p. 3). The right point for the third question is the 90 days after contract award, and the object of the check is whether the integrated master schedule covers repeat visits to the same installation site; whether the document exists is not a check item.

Options ruled out. Five.

Merging all 11,389 schedules into the scheduling tool. The FAA officials' reason is in the report: there are too many projects, and merging everything into the tool "would not be helpful" (p. 29). Ruling the option out amounts to a declaration that the tool's magnitude has been overtaken by the project count. The occasion was an oral briefing to GAO in May 2026, with no document.

Rescheduling by facility rather than by project. This would require each of the 13 programmes to hand over the authority to decide installation timing. The report records no programme-level ar-

rangement of that kind, only manual identification in "limited cases" (pp. 29, 30).

Delaying installation until the integrated master schedule is finished. The December 2028 deadline and the 30-month remainder (p. 26) make this infeasible on budget and on politics.

Cutting the number of projects and merging similar installations. The contract was awarded on 5 December 2025 with scope divided by 13 programmes (pp. 22, 23), and changing scope means changing the contract.

Using an AI tool to identify conflicts automatically. This is the route the FAA chose in May 2026. The tool is in testing, and neither rules for its use nor a new process exists (p. 30).

Early signals. Five, each with the reason it was overridden.

The 1982 estimate. The FAA's first comprehensive modernization plan, in 1982, estimated $10 billion and expected full benefits by the late 1990s; in 1995 GAO put the effort on its high-risk list, the estimate rose to $36 billion, and the largest component, the advanced automation system, had to be substantially redesigned (p. 9). Reason: every round was treated as a new starting point.

NextGen. Launched in 2004 with completion planned for 2025; a 2023 GAO report put completion at 2030 or later; fiscal 2007 to fiscal 2022 spending passed $14 billion, and completion needs at least $35 billion from federal and industry sources combined (p. 10). The old ledger carries one more entry: in 2023 GAO recommended that NextGen establish a life-cycle cost estimate, and the FAA did not do it (p. 28, box). Reason: progress was presented by milestone, and the missing piece was the cross-project total.

The sustainability rating. The ratings of 23 September 2024 found 105 systems unsustainable or possibly unsustainable (pp. 6, 7).

Reason: it was read as fleet ageing, with hardware replacement and more money as the answer.

The OMB table. The January 2026 table reconciled 12 programmes line by line, and one of them, Alaska automation capability, received none of the appropriation at all (p. 19, p. 20 footnote b). This table was the only occasion in the case to put 12 programmes on one page, and the page carries no total row. Reason: every row was compliant, and no row was over budget.

The first integrated master schedule. Delivered in April 2026 and still not connected to the 11,389 schedules in May 2026 (p. 29). Reason: the deliverable exists, and the work is described as in progress.

The point of escalation. In August 2025 the FAA Administrator signed the blanket waiver that took every BNATCS project out of nearly all AMS requirements (p. 22). The waiver text states its grounds: AMS life-cycle process steps are numerous and document-heavy and may cause substantial delay, and financial, legal, technical, safety and programme review requirements may add months or years to new system deployment (p. 28). The same text struck the life-cycle cost estimate off the list of obligations (pp. 28, 33). After that stroke the integrated master schedule went from a missing output to a missing obligation, and no post was answerable for its absence. The Department of Transportation later commented that the Administrator had not waived the cost estimate requirement; GAO answered that the waiver text explicitly includes that requirement and that the high-level documents the FAA supplied do not constitute a reliable estimate (pp. 32, 33). In May 2026 FAA officials said that integrating all the projects "would not be helpful" and switched to waiting for an AI tool still under test (pp. 29, 30). By then the room left to retreat was short: 9 phase-one programmes had started, the contract scope was cut by 13 pro-

grammes, and the only lever remaining was a contract modification.

Failure chain. Technical layer. The scheduling tool's magnitude is a single-project, single-programme network. The 11,389 schedules, at an average of 30 activities per project task on the Department of Transportation's own figures (p. 33), put the activity count in the 340,000 range. At one installation site, projects from different programmes schedule themselves separately. The St. Louis example in the report has the voice switch doing site preparation in April 2027 and the information display system back at the same site in January 2028, about 8 months apart (p. 30). Two visits, two disturbances to controller positions. The voice switch carries one further constraint: old and new equipment must run in parallel for every controller (p. 25), and the length of the parallel window depends on how densely the site is scheduled, which is the quantity nobody computes without a total. Organizational layer. The unit of management is the project, not the facility. Each project's schedule can be defended inside its own programme; cross-project conflicts are identified by hand in "limited cases" and have not reached scale (pp. 29, 30). Each of the 13 programmes holds a detailed master schedule covering three years (p. 33, Department of Transportation comments). Every layer has one, which is exactly the shape of a missing total. Incentive layer. The deadline is December 2028, speed is written into the waiver text (p. 28), and money is appropriated programme by programme (p. 20 table 3). The cost of stopping to integrate is carried by the programme that stops, and the benefit of integration is shared by 13 programmes and the controllers. All three layers break at once. The break at the boundary sits between the organizational and the incentive layer. The only carrier that could make someone answerable for cross-project conflict is an integrated master schedule with a signature and a payment condition; the waiver removed the requirement, the contract wrote the duty onto the integrator, the delivery condition

column stayed empty, and so no layer had to answer for the absence of a total (pp. 28, 29).

Minimum counterfactual. In the 90-day negotiation after the 5 December 2025 award, running to March 2026, write "an integrated master schedule covering all installation sites" as the first payment condition of the contract, with the FAA spot-checking the overlapping schedules at three installation sites before payment. One action, placed between award and start of work, with no luck required: the negotiation window exists in the contract (p. 4 footnote 7, p. 29), and with a ceiling of $1.5 billion against a settled price near $876 million, the headroom was enough to hold the clause (p. 23 footnote 36). If the negotiation fails there is a second moment, at the April 2026 delivery of the first integrated master schedule, sent back for rework with "same-site overlaps eliminated" written as the acceptance criterion (p. 29). Neither moment depends on a change of personnel or on extra appropriation, and neither asks the FAA to build a new system first: the integrated master schedule needs the activities and dependencies on one network, and the contract already names the integrator as its author (p. 29).

Outcome and sources. A phase-one estimate of $10.6 billion, an appropriation of $9.03 billion, and a phase two needing another $10.2 billion with no timetable; both recommendations are addressed to the FAA Administrator, and the Department of Transportation partially concurred with both (pp. 20, 21, 32). Primary source: U.S. Government Accountability Office, Air Traffic Control Systems: Ambitious New Modernization Effort Needs to Improve Cost and Schedule Planning, GAO-26-107992, September 14, 2026. Key pages: p. 1 for the system count and sustainability, p. 15 for two-phase funding and the 13 programmes, p. 17 table 2 for the acceleration, pp. 19–20 for table 3, p. 26 for the cost estimate deficiency, pp. 29–30 for the eleven thousand schedules and the

St. Louis example, pp. 32–34 for the recommendations and the Department of Transportation's comments. Body text runs to p. 34; appendices with the methodology and the department's comments run to p. 47. Link in Appendix B, po-001.

The 11,389 schedules are worth a second reading from the angle of tool evolution. Project-by-project tracking is the previous magnitude form of the tool, the aggregation need of 138 systems is the next one, and the distance between the two is GAO's structural finding. Changing the tool is not buying new software. It is changing the unit: a full life-cycle cost estimate, a single milestone framework, a cross-system dependency view, which is Chapter 21's mapping applied directly to a very large programme. Resistance sits locally, since every system's team has to hand over part of its autonomy, and the size of the resistance is proportional to the political standing of the system. External oversight earns its place here by giving the change of unit an institutional reason, in the shape of congressional questioning. The lesson for anyone running a large programme: the coordination tool has to lead the engineering magnitude by one step. Leading costs redundant tooling; following costs what Case 2 cost.

Case 3 (Software): The Day of Record Load

sw-005. The GitHub platform, 17 August 2026. Official statement, page checked for this book: record load broke through components with insufficient headroom, and the outage lasted 7 hours 47 minutes. The core assumption of capacity planning was calibrated on historical peaks, and the record day reset that assumption's historical extrapolability to zero. The scaling coefficient was not wrong. The object it extrapolated from, the previous generation's load shape, was no longer this load. Scale arrived as a double growth, in traffic and in task shape, and the old peak curve could not measure the new workload.

Stop at the decision point: the last time the capacity model was parameterized on historical peaks and left no unjustified headroom for a record day. On the table: the growth curve, the architecture at the last peak, and a set of new behaviors that could not be verified. **What happened.** Close to eight hours unavailable, with recovery in stages and the follow-up work named "the work ahead". Primary source: GitHub, 2026; link in Appendix B, sw-005.

Case 3's capacity headroom has a further connection, to the illusion of redundancy in Chapter 22. Redundancy is a form of capacity, standby processing power, and its magnitude calibration faces the same record-day problem: backup capacity is sized on historical peaks, and a record day fills primary and standby together. The three cloud-era cases, Case 3 here with Cases 2 and 5 of Chapter 21, add up to a capacity governance checklist for platform readers: parameterize on historical peaks, keep unjustified headroom for the record day (Device 3), monitor load shape as well as load volume, and re-validate at each magnitude step, which is the exam re-sat on the ladder. What the four share is a philosophy, a humility coefficient, believing by default that this time differs from last time. Humility is not cheap, since headroom costs money, and its price is written on the bill for every outage.

Case 4 (Finance): Thirty-Eight Shells Registered in Bulk

fi-004. The SEC's action against 38 fictitious investment adviser entities, Litigation Release No. 26622, page checked for this book: through 2025 and 2026 the defendants made material false statements in Forms ADV filed with the Commission, presenting themselves as legitimate advisory firms, and the cases were consolidated in the federal district court in Colorado. The registration system's design capacity assumed firm-by-firm human review. Bulk, automated, template-driven fabrication voided that scale assumption wholesale: the threshold did not move, and the way of crossing it

did. This is the adversary's version of the scale threshold, where growth happens in the attack rather than in the organization, while the defence's granularity stays one firm at a time.

Stop at the decision point: every day the registration system's structure, its bulk submission interfaces and its formal review rules, stays as it is in front of the new automated use. On the table: the anomalous growth in submission volume and the compliant appearance of each firm's materials. **What happened.** The 38 cases consolidated in one action; systematic registration anti-fraud entered the agenda. Primary source: SEC, Litigation Release No. 26622, 2026; link in Appendix B, fi-004.

The adversary's version has a symmetry worth adding. Defence calibrates its magnitude to the history of attacks, designing review throughput against past submission volume, and attack calibrates its magnitude to the holes in the defence, with automation using the gaps in firm-by-firm review in bulk. The two calibrations chase each other, and the defence is always half a step behind for a structural reason: changing the defence requires procedure, and a rule change needs a process, while changing the attack requires a script and one person. What compensates for the half step is anomaly monitoring. Shifts in the distribution of submissions, bulk volume, template similarity, machine timing signatures, are the fingerprints scaled attack leaves behind, and fingerprint monitoring costs far less than scaling firm-by-firm review. Readers in regulatory technology can file this case as a pattern: when defence granularity and attack granularity stop matching, put fingerprint monitoring in first and discuss rule revision second.

Case 5 (Healthcare): The Oxygen Peak in the Pipes (full card)

Case ID. me-011.

When and where. 19 May 2021, report published (HSIB p. 1), official PDF of 61 pages. The event happened in the autumn of 2020,

during the second wave; the report redacts the trust's name and the specific dates, leaving only the weekday sequence "Thursday evening, late Friday night, Saturday, Sunday" (HSIB pp. 25–26). Location: two sites of one acute hospital trust in England. The points involved are the high dependency unit (HDU) and the operating theatres; the HDU had been built at the end of an existing Nightingale ward and used that ward's existing pipework (HSIB p. 42).

What and who. The trust's medical gas pipeline system (MGPS) and the estates and facilities team. Inside the hospital, HTM 02-01 sets four posts: the medical gas committee (MGC, a multidisciplinary forum), the authorising engineer (external reviewer, producing an annual report, the only external assurance mechanism in the HTM), the authorised person (the engineer running day-to-day management, appointed on the authorising engineer's recommendation and re-examined every three years), and the designated nursing or medical officer (the ward-level clinical contact who clears work affecting supply) (HSIB pp. 19, 35, 36). Outside: NHS England and NHS Improvement (NHSE/I) issue guidance and alerts, the CQC regulates site safety, and HSIB ran this investigation. The report does not disclose the trust's name or its bed count; each of the two sites has its own radial MGPS (HSIB p. 44).

What they were doing. In 2012 the trust recorded "lack of understanding of hospital-wide oxygen flow" on the estates risk register, and because of its severity the same entry went onto the trust-level risk register (HSIB p. 23). In April 2020 the estates team and a senior nurse with surgical and critical care experience produced the "red dot plan": red dots on the site floor plans marking where COVID patients could be placed, so that oxygen pickup points would be spread out (HSIB p. 23). The plan was laid out by where oxygen enters the hospital from the MGPS, and the trust at the time believed this also covered the flow limits of each ward

(HSIB p. 23). The flow basis was the understanding of treatment then current: patients on ventilators, about 15 litres per minute each (HSIB p. 23). On that basis the HDU was calculated to have 145 litres per minute available (HSIB p. 23). In March and April 2020 two CAS alerts reached the estates team by email, and the team replied to the trust's CAS officer confirming the measures had been taken (HSIB p. 24). The first wave hit this trust lightly and the red dot plan was never tested against real demand; operational need also ruled out simulation or dedicated testing, which would have affected patient care (HSIB pp. 24, 31). In the summer of 2020 attention moved back to outpatient clinics and elective surgery, non-COVID patients returned, and hospital-wide activity ran higher than in the first wave (HSIB p. 24). The autumn brought the second wave, with the red dot plan still in place (HSIB p. 24). Treatment had changed: CPAP and nasal high flow had become mainstream, and both consume more oxygen than a ventilator (HSIB p. 25). Typical flows for the three devices: ventilator 15 litres per minute, CPAP about 30, nasal high flow more than 60 (HSIB p. 11).

The decision point. Four moments.

1. The oxygen flow knowledge gap entered two levels of risk register (HSIB p. 23). The vehicle was the register itself, with "identified mitigations" in the action column. For the next eight years the risk sat on the register and the capacity was never recalculated; the authorising engineer's reports for 2019 and 2020 were still recording too few authorised persons and designated officers not appointed (HSIB pp. 36, 39).

April 2020. The red dot plan was finalized (HSIB p. 23). The vehicle was the site floor plans and the plan document, handled by the estates team and that senior nurse. The HDU's available flow went into the calculation as 145 litres per minute (HSIB p. 23). That single number decided how many patients the HDU could take and

which oxygen therapy they could receive (HSIB p. 24). The report's record of this step is direct: the trust initially considered that the HDU should be able to deliver the specified flow, and calculated patient numbers and therapy types accordingly (HSIB p. 5). The plan defined "balance" between areas on the floor plan, never down to pipe sections.

July 2020. The MGC held its scheduled meeting. The two NHSE/I alerts on piped oxygen supply, of 31 March and 6 April, were in the meeting pack, not listed as agenda items, and not raised under other business (HSIB p. 24). The alerts' requirement was specific: engineers, clinicians and managers should plan peak capacity together (HSIB p. 20). Nationally, some trusts held only 2 of 12 planned MGC meetings in three years (HSIB p. 33); this trust held the scheduled July meeting, with the alerts in the pack and not on the agenda.

Early September 2020. The authorising engineer emailed the NHSE/I guidance on piped oxygen, published in August, to the trust's two authorised persons (HSIB pp. 24, 49). One was on leave awaiting planned surgery, the other was in hospital with COVID-19, and the trust did not register the email (HSIB p. 24). This route was standard practice at the time: the guidance did not go through CAS and was delivered directly by the authorising engineer (HSIB p. 49); because it was not published on CAS, it carried no CAS target date and no completion return (HSIB pp. 49, 52). The four moments have four different vehicles, a register, a floor plan, a meeting pack, an email, and none of them required an answer to the question of how much flow a given pipe section can deliver.

What was visible then. The 2006 edition of HTM 02-01 sets minimum flows: general wards at 6 litres per minute for 1 bed in 4 plus 10 litres per minute for 1 further bed, giving 50.5 litres per minute for a 28-bed ward; the HDU at 10 litres per minute for 1 bed and 6 litres per minute for three quarters of the remaining beds, giving

37 litres per minute for a 7-bed HDU (HSIB p. 19). Beyond that, system redundancy is used first, and at the pipe limit there is a pressure drop and a loss of flow (HSIB p. 19). The HTM runs to 336 pages (HSIB p. 19). The report distinguishes the terms: capacity is the system's maximum oxygen production, flow is the volume delivered in a given time (HSIB p. 14). Network shape: most NHS hospitals are radial, so take-off early in the network lowers the pressure available downstream (HSIB p. 18).

The estates alert of 31 March 2020: when demand at several wall outlets exceeds the VIE delivery limit, supply pipes lose pressure quickly and hospital-wide failure becomes possible, and VIEs can be exhausted quickly and unpredictably (HSIB p. 20). The alert of 6 April 2020: some hospitals were already drawing more than the MGPS design maximum flow, and icing in the pipes can reduce flow unexpectedly (HSIB p. 20). Clinical guidance from NHSE/I on 9 April 2020: the adult oxygen saturation target was lowered from 94%–98% to 92%–96%, with 90%–94% to be considered when ne‑cessary (HSIB p. 22). Guidance in August 2020: the concerns were weaknesses in equipment, in system configuration and in opera‑tional management, and none of them concerned liquid oxygen running out (HSIB p. 21). The alert of 19 November 2020: winter demand could run higher than in spring, because elective surgery and recovery areas had restarted, and both new wards and old estate would add load (HSIB p. 21).

The trust's own reading: total hospital demand never exceeded 56% of total capacity, and oxygen production had headroom (HSIB p. 5). One more document is on file, the estates team's written confirmation to the CAS officer, in March and again in April, that measures were in place (HSIB p. 24). The division of labour is in the material too: estates understood pipes and pressure and not changes in clinical treatment, while clinicians understood patients

and saturation targets and not the oxygen consumption of different devices (HSIB p. 31).

What only became known afterwards: in a pressure test the HDU pressure fell from 4.2 bar to 3.1 bar in "about 10 seconds" and the alarm sounded (HSIB p. 25); the shortfall came from existing pipework upstream of the HDU being drawn on first (HSIB p. 25); and a causal link between the two patients' desaturation and the pressure drop was never established (HSIB p. 26).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
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What happened. A major incident was declared on the Saturday afternoon, and every patient arriving at the hospital was diverted to other trusts (HSIB p. 26). The diversion was later narrowed: one site cancelled it, and the other diverted only patients needing more than 6 litres per minute (HSIB p. 26). At least 40 elective operations were cancelled (HSIB p. 26). The major incident was stood down seven days later (HSIB p. 26). Two patients recovered with no adverse outcome, and the report records no deaths (HSIB p. 26). The MGC met in emergency session and thereafter weekly (HSIB p. 26). HSIB made four recommendations: define the duties and competences of the posts listed in the HTM (R/2021/132), establish continuing assurance of authorising engineer competence (R/2021/133), revise and reissue the HTM (R/2021/134), and have the CQC adjust its site assessment model (R/2021/135); an interim bulletin carried one further recommendation on the MGC (R/2021/120), with four observations and three actions, one of them

the trust procuring pipeline flow meters and electronic telemetry (A/2021/041) (HSIB pp. 7–9). Back to the three questions. The commonest answer to the first is to increase oxygen production capacity, and that action buys no flow here: the report states that increasing the oxygen available cannot overcome the limits of the hospital's internal pipeline infrastructure (HSIB p. 43). The item most often missed in the second question is the two lines of HTM minimum flow, 50.5 and 37 litres per minute, which look like a compliance floor and are in fact a design ceiling. The point for the third question is the flow basis in the April 2020 red dot plan: whether the basis is 15 or 60 decides how many patients the HDU can take.

Options ruled out. Six.

Simulating the peak before real demand arrived. The report states that operational need made it impossible to simulate MGPS demand or run more specific tests, because that would have affected patient care (HSIB p. 31).

Expanding the production side. NHSE/I had a capital programme replacing VIE components, and the report states in the same place that it does not solve the pipework (HSIB p. 43).

Converting to a ring main. Both trust sites are radial, the HTM does not require ring mains in new work, and it assumes most work will still be built radial (HSIB p. 44); the report lists ring main conversion as a mitigation other trusts used (HSIB p. 45), and this trust's two sites were still radial at the time of the event (HSIB p. 44).

Spreading patients across the hospital. This is the ideal from the estates side, since it balances pressure across the network (HSIB p. 32). On the clinical side, infection control requires COVID patients to be cohorted, and rota efficiency has to be weighed as well (HSIB p. 32). Two local optima collide directly over bed allocation.

Manifold cylinder packs as backup. The report lists this as mitigation engineering at other trusts; it tops up gas during a pressure drop and does not change the delivery capacity of the pipes (HSIB p. 45).

Waiting for the HTM revision. The revision work started in late 2019 and was shelved in 2020 because of the pandemic (HSIB p. 18), so it was unreachable inside the event window.

Early signals. Five.

The 2012 register entry (HSIB p. 23). Reason: registration was taken as disposal.

The authorising engineer's annual reports for 2019 and 2020, naming three things not done in two consecutive years: appointing designated officers, adding authorised persons, and cover out of hours (HSIB pp. 36, 39). The designated officer vacancy has a history: a cohort was trained about six years earlier, restructuring and staff loss left nobody to succeed them, and the DMO post stayed vacant (HSIB p. 39). Reason: the annual recommendation is copied from year to year and does not constitute a condition for starting work.

The two alerts of 31 March and 6 April 2020 (HSIB p. 20). Reason: they were issued as estates alerts, not as national patient safety alerts, and carried no completion return, so the trust did not treat them at the highest priority (HSIB p. 52); the 31 March one went to a wider list than comparable estates alerts and its level was still estates (HSIB p. 52). Nationally, CAS alert volume was up 177% year on year (HSIB p. 51).

The July 2020 MGC meeting, with the alerts in the pack (HSIB p. 24). Reason: not on the agenda, so nobody read them out; the MGC had long been treated as "the estates meeting", and clinical attendance was already unstable (HSIB p. 33).

Thursday evening, the theatre pressure alarm sounding intermit - tently (HSIB p. 25). Reason: the alarm stopped once patients were moved under the red dot plan, and that was taken as resolution. The HDU sounded no pressure alarm that night (HSIB p. 25).

The point of escalation. When the new HDU was completed, no additional money went into raising pipeline capacity. It was de - signed to HTM minimum flow and connected to the end of existing pipework, and that pipe passes the Nightingale ward beds before it reaches the HDU, so redundancy in the system may already have been drawn off upstream (HSIB p. 42). This step turned a pipework problem that could still have been changed into poured concrete, and raising capacity afterwards means building work. In the same period infection control concentrated COVID patients into a few wards, and the HDU sat at the end of the network (HSIB pp. 32, 42). The new ward got new equipment and new beds; the pipework stayed old (HSIB p. 42). The report records no signatory for this step, because every new bed and every new device was approved separately and complied separately; the two written replies to the CAS officer in March and April were already the latest conclusion in the file about this system (HSIB p. 24).

Failure chain. Technical layer. The report states that the radial network showed no problem before the pandemic, because the expected oxygen use sat inside the capacity of a radial system (HSIB p. 45); upstream draw lowers downstream pressure (HSIB p. 18); the design basis is bed count multiplied by per-bed flow, the actual use is device flow multiplied by devices running at once, and there is no conversion between the two measures (HSIB pp. 19, 11); and once redundancy is used up, pressure drops and flow is lost (HSIB p. 19). Organizational layer. The unit of capacity review is the site and the ward, and pipe sections are not in it; the unit of the red dot plan is a dot on a floor plan, and pipe routing is not in it; the MGC agenda carries no capacity item (HSIB pp. 23, 24).

Training is missing too: the NHS electronic staff record contains a subject called medical gases but no usable course module (HSIB p. 41). External assurance is missing as well: before April 2020 PAM was completed voluntarily, about three in ten trusts did so, and no data submission was required (HSIB p. 47); CQC inspection looks at the ward environment, and site-level pipeline assurance is not a routine check item (HSIB p. 48). Incentive layer. Capital goes to new clinical areas, and existing infrastructure does not get investment at the same priority (HSIB p. 42); the maintenance backlog for sites in England rose from £6.46 billion to £9 billion within a year, a 40% increase (HSIB p. 42); and the level of an alert decides the resources that answer it (HSIB p. 52). All three layers break at once. The break at the boundary is in the unit of evidence: technical readings arrive per outlet and per ward, organizational action starts per site and per specialty, and no post in between is responsible for turning a pipe section into a network.

Minimum counterfactual. When the red dot plan was finalized in April 2020, change the flow basis from the ventilator's 15 litres per minute to the CPAP and nasal high flow settings, 30 litres per minute and above 60, recalculate available flow at the most upstream pipe section on each site, and write the simultaneous draw limit for a single pipe section as a hard constraint on bed allocation. One action, in the window between the first and the second wave, April to September 2020. No luck required: the mainstream devices in November were CPAP and high flow (HSIB p. 25), and both flow figures could be obtained from the equipment side in April (HSIB p. 11); the authorising engineer's annual report mechanism was already running (HSIB p. 36). What gets checked is specific: how many beds draw on the most upstream section at peak, multiplied by the device flow per bed, against the available flow of that section. This is paper work, with no new equipment, no building work, and no dependence on a change of personnel. It

costs one calculation and one MGC agenda item. Not doing it cost the seven days in November and the 40 operations.

Outcome and sources. A major incident, diversion at both sites, at least 40 elective operations cancelled, two patients recovered, and no deaths recorded. Primary source: Healthcare Safety Investigation Branch, Investigation report: Oxygen issues during the COVID-19 pandemic, 19 May 2021, official PDF of 61 pages, downloaded locally and read section by section (the archived hsib.org.uk page returned 429, so the current PDF at hssib.org.uk was used; archived link in Appendix B, me-011). Evidence base: all MGPS documents and interviews at the trust concerned, comparable material from four other trusts, and interviews with nine organizations including the DHSC, NHSE/I, the CQC, the MHRA and the British Thoracic Society (HSIB p. 29). The report frames the case as a national safety risk: at least a dozen NHS trusts had raised the same problem, each of them had enough liquid oxygen, and what was limited was the section of pipe that carries gas to the bedside (HSIB p. 28). Key pages: p. 5 for the 56% total demand, p. 11 for device flows, p. 19 for HTM minimum flows, pp. 20–22 for alerts and guidance, pp. 23–26 for the event sequence, pp. 42–45 for infrastructure and network shape, pp. 51–52 for alert levels.

24.4 What Shows Earliest

Six weak signals, each with the reason that usually overrides it.

1. A management layer is added, on the grounds of strengthening coordination. Override: "we have reached that scale."
2. The number and length of meetings grow faster than headcount, and decisions get slower. Override: "people want to align."
3. The veterans' phrase becomes "it didn't use to be like this." Override: "nostalgia."

4. Every local indicator improves while the total stalls or worsens. Override: "every department is working hard."
5. Rare events rise in frequency in proportion to scale, with absolute counts growing, and get read as quality deterioration. Override: "a quality problem." (The numerator grows with the denominator, which is not a quality signal.)
6. The staffing share of interfaces, reconciliation and cross-department scheduling quietly passes the staffing share of core output. Override: "big companies are all like this."

Take the six back to this chapter's seven cases. The added-layer signal maps to Case 2, where coordination layers and system count inflated together across a multi-phase programme. The meeting-growth signal maps to Case 5, where coordination cost stayed off the books and cross-department peak planning at the hospital ran entirely on ad hoc meetings. The local-optima signal maps to Case 1, where engineering's conservative validation and the channel's aggressive distribution were each locally optimal. The assumption-reset signal maps to Cases 3 and 6, where capacity coefficients and exposure assumptions both ran at the old magnitude. The granularity signal maps to Case 4. The veteran's phrase is the spoken signature of all of them: an old hand senses the change in kind before any data does.

Reorder the six by cost of discovery. The local-optima and assumption-reset signals are cheapest, both being a matter of comparing numbers, coordination hours as a share and the date an assumption was last reset. The added-layer and meeting-growth signals come next, since they need the organization chart and the calendar. The granularity and veteran signals are dearest, needing external comparison and interviews. The starting suggestion is to make the coordination-hours signal a standing action of Device 2: publish coordination hours as a share of core hours once a quarter. That number is the thermometer of the scale disease. A single rise

is not necessarily illness; a sustained rise combined with local optima cancelling is the diagnosis.

24.5 How to Defend

Device 1: re-validation at each magnitude step. Divide organizational scale into steps, ten people, a hundred, a thousand, ten thousand, and force a re-validation of the core assumptions at every crossing: coordination method, validation sample size, capacity coefficient, review throughput. Not linear extrapolation. The problem is worked again from the beginning. GitHub's capacity coefficient, DREO's validation sample and the FAA's schedule tool should all re-sit the exam at the step, and the exam has one question: does this method still hold at the new magnitude, and on what evidence.

Device 2: coordination cost on the books. Hours spent in cross-department meetings, on reconciliation and on interface maintenance are counted quarterly and published. When the share passes a threshold, self-set, a fifth of core hours say, a structural review is triggered. Invisible costs are the ones that need booking, since a visible cost announces itself. Booking has a second use: it gives expansion justified by "we hire in order to coordinate" a real unit price.

Device 3: unjustified headroom in capacity assumptions. For record-day scenarios the capacity model keeps one tier of headroom that cannot be argued from history, two tenths on top of the peak coefficient for instance, and the document states plainly that the basis for this tier is that we do not know. The basis of headroom may be luck, provided someone remembers it is luck. Write it down and luck can be audited.

Device 4: the annual teardown question about structure. Once a year, run the thought experiment: if we designed from zero today,

how would the coordination structure differ from this one? The difference list does not serve an immediate reorganization. It serves early warning, and the speed at which the list lengthens is the speed at which the threshold approaches, which is Chapter 34's path-lock contrast question used here in reverse.

Put the four devices against this chapter's cases. Device 1 on Case 1: when a distribution plan crosses an order of magnitude, force the exposure model to be rerun, with the validation sample reset to the target exposure. Device 2 on Case 5: count the hospital's cross-department coordination hours quarterly, and when the share rises look for a capacity gap first, since more coordination usually means someone is compensating for missing capacity. Device 3 on Case 3: write the record-day tier's basis as not know‑ing, and audit headroom with a single question: what is this tier's basis? Device 4 on Case 2: the annual exercise for a multi-phase programme asks how the governance structure for 138 systems would be designed if the programme were started today, and the difference list goes to the external overseer.

Each device has a characteristic way of dying. Re-validation dies at the step that is never defined: re-validation is postponed after every small increase, on the grounds that the step has not been reached, and the definition of a step is relaxed by personal favour. The patch is to write the steps into the rules, round-number thresholds for headcount, users and volume, so that the crossing triggers by itself with no human review. Coordination accounting dies by definition drift: meeting hours go from all staff to a sample, and numerator and denominator each quietly shrink. The patch is to write the definition into the rules and put it in the annual internal audit. Unjustified headroom dies by being borrowed, with the record-day tier scheduled away in ordinary times; the patch is to make the tier unschedulable in the scheduling system, a hardware logic lock rather than discipline. The teardown question dies when the exer‑

cise turns into complaint, the difference list becoming a grievance collection with no entry reaching the early-warning ledger; the patch is that the exercise must produce measurable warning indicators, each entry in the difference list paired with a monitor. All four deaths share one health-check question: when did this device last change how resources were allocated?

The time structure of the scale problem deserves a passage of its own. Threshold crossings usually happen in the fastest-growing quarter, and the fastest-growing quarter is exactly the quarter when the organization has the least bandwidth for re-validation, with everyone chasing growth. That mismatch decides the design: re-validation has to be scheduled ahead, as an automatic process in which crossing the line queues the work into next quarter's validation calendar, rather than as an ad hoc decision. Scheduling ahead costs a rigid validation calendar, since people have to be freed during growth. Not doing it costs what Cases 1, 3 and 5 cost, and all three crossings happened in growth quarters with no re-validation.

24.6 Boundaries with Adjacent Mechanisms

Against extrapolating past the validated domain, Chapter 23. A change of scale is the item most often left off an extrapolation list. Chapter 23 governs whether the differences got listed; this chapter governs how they are re-validated once listed. Pilot to rollout goes through Chapter 23's declaration form, and every crossing of a magnitude step goes through this chapter's re-validation.

Against cascading failure, Chapter 33. Dependency concentration at networked scale is cascading's ground. This chapter governs the nonlinearity of coordination and validation; cascading governs a single point amplified through the network. A platform in growth should read the two together.

Against diffusion of responsibility, Chapter 18. The habitual answer to scale is more people and more signatures, and a longer signature chain is exactly what manufactures dilution. Treating this chapter's disease with Chapter 18's disease is the commonest self-injury of a growing organization. Responsibility design during expansion uses a single accountable person, not added signatures.

24.7 Five Self-Checks

1. Which magnitude step is your organization on now? At the last crossing, which core assumptions were re-validated?
2. What is coordination's share of core hours? Is there a number? Having no number is itself the answer.
3. The most recent reset of a capacity or review-throughput assumption: did data trigger it, or an incident?
4. What process did a veteran last describe with "it didn't use to be like this"? When did that process last run as designed?
5. If scale doubled tomorrow, what breaks first? Does what you are betting on match the headroom you have set?

Chapter 25: Margin Erosion (Dekker)

> What this chapter covers: how margin is eaten one bite at a time with nobody seeing the whole; Dekker's drift view, three criteria, four domains, the margin ledger and the cumulative-view review. When to come back: at every reduction review, at every "this one is special" exception request, and on the day a system crosses a boundary with no warning at all. Reading time: about 50 minutes.

No system that fails was pushed off the cliff by a single decision. Sidney Dekker called the real motion drift into failure: each departure legal under the rules of its day, each cut cleared by its review, each "just this once" survived in fact. Margin is not decided away. It is forgotten away. It has no ledger, no guardian, and no meeting at which anyone looks at it whole. On the day the boundary arrives, everyone in the room faces a thing that has never happened before.

25.1 What It Is

Definition: margin erosion is the design margin of a system, meaning time, strength, distance, redundancy, room to act, eaten back cut by cut. Each cut looks harmless on its own, and nobody holds the running total.

Three criteria for activation. Any one of them, and the mechanism is already running.

1. No single cut is enough to cause the event on its own. The defence made for each cut is usually true.
2. Nobody holds the cumulative quantity. There is no ledger, only the assessment made at the time of that one cut.

3. “Still enough” becomes the basis for the decision, and “enough” is measured against the state left by the previous cut.

Criterion (iii) is the self-locking part and deserves separate treatment. Every cut resets the baseline; the baseline then serves as the starting point for the next cut. Before the first cut the margin is the design value. After the tenth, the reference point has slid to wherever the sixth left it, and nobody remembers the origin. That is the difference between a margin being reduced and a margin being eaten: reduction is an action, erosion is a failure of memory.

Dekker’s angle needs two more sentences. Writing after normal accidents (Perrow) and the normalization of deviance (Vaughan), he turned the lens on where a system travels during the years in which nothing happens. His observation: every adjustment inside a system is made against current practice, and current practice has already absorbed all the earlier adjustments, so the reference frame itself is drifting. That observation gives criterion (iii) its floor. A sliding baseline is not managerial negligence; it is what reference-to-practice produces by construction. The only way to stop the slide is to move the reference from practice to paper, which means the original design value. That is the principle behind Device 3, the minimum waterline: the number is taken from the design, never from any “current state”.

Memory loss has an organizational face worth naming. Margin has no natural guardian. The designer created it, and when the designer changes jobs the knowledge disperses. The operating team maintains function (it still works), not margin (how much bad news it can still take). Management watches output and cost. None of the three treats margin as its own asset. It is a public good, and the fate of public goods is written in the economics textbook. The margin ledger is the device that turns the public good into owned property: every margin line carries an owner, with the clause from

Chapter 20 copied across, and the owner's appraisal carries one line for the completeness rate of that year's ledger. Giving margin a person is the second thing that has to happen, after giving it a book.

25.2 How It Grows

Three generating conditions.

Condition one: cost pressure that never lets up. The damage comes from medium and small cuts in every budget cycle, not from one crisis-sized cut. Once the pressure is routine the cuts are routine, and routine is precisely what strips the signal value out of each one.

Condition two: the invisibility of margin. Design margin does not appear in any report and produces no revenue. Its existence shows only afterwards, in the event. Anything invisible has a settled fate under cost pressure. The invisibility of maintenance in Chapter 29 is the asset-side version of the same story.

Condition three: the institution of the single assessment. Each change is assessed for its own effect. No post, no form and no standing meeting owns the sum of the history. Every one of those conclusions can be honest, and the total is systematic self-deception.

25.3 How It Looks in Four Domains

Case 1 (Maritime): Freeing Ports Cut Down One at a Time

ma-009. The twin-rigged stern trawler Ocean Way, November 2014, west coast of Scotland. MAIB report 23/2015 (page checked for this book): the vessel capsized and sank, three dead, two survived. The investigation pointed to the weight of water on deck. Water trapped

there capsized the boat, and the report's own term is entrapped water. Progressive earlier changes to the freeing ports made that water more likely to stay. The liferaft cradle jammed and could not be released, and the EPIRB had no GNSS fix, so search and rescue had a position 47 minutes later. Each freeing-port change had its own reason. Together they ate the righting moment the boat needed in a bad sea, which is the margin her design had promised.

Stop at the decision point: the moment one of those changes was approved. On the table: the functional reason for this change, normal running afterwards, and a deck drainage drawing with no cumulative record. Outcome: capsize and sinking, three dead. Primary source: MAIB, Report 23/2015, 2015; link at Appendix B, ma-009.

An engineering debt is still owed on this case. Why the loss of righting moment stayed invisible. A trawler's stability is designed with a full set of margins (freeboard, initial metacentric height, limiting line), and the effective value of all of them moves with every change to deck arrangement. Stability review is triggered only by specific events, a modification registered or a class survey, and successive small freeing-port changes never reached the trig‑ger line. The professional calculation existed; naval architecture can produce the number. The trigger mechanism was absent, since nobody was required to ask for it. So the lesson is not that the arithmetic was beyond them. Nobody did the arithmetic. Tying the stability review trigger to small modifications, so that accumulated change past a threshold forces a review, is the trawler version of the cumulative-view review. Another name for a trigger line: the arrangement that calls the professional calculation at the right time.

Case 2 (Chemical): An Undersized Relief Path and an Absent System

Case ID: ch-006.

When and where: 14:39 on 12 November 2024 (the top relief valve moving closed with no command) to 14:57:41 (reactor 6 rupturing); the central manufacturing area of the Givaudan Sense Colour caramel colour plant, Louisville, Kentucky; reactor 6 and the combined control room and laboratory 40 feet to its east (CSB report p. 28, p.31, p.69).

What and who: the Givaudan group, Switzerland; the wholly owned subsidiary D.D. Williamson & Co., LLC owns and operates the plant, acquired by Givaudan on 19 December 2021 (p.16, p.88). Batch reactors 1 through 6 on site; No. 6 built in 1978, run at the Modesto, California plant, stored here after that plant closed in 2008, modified and installed in 2021 (p.19). About 55 people on the register, with 30 employees plus at least four contractors and visitors on site at the time (p.16). The relevant posts: the plant engineer who took over the reactor 5 and 6 project in September 2020; the engineering manager who led the 2012 reactivity testing and left in April 2017; the operating crews and the maintenance technicians (p.25, p.88).

What they were doing: two things at once. One was restoring capacity. Reactor 6 finished installation in August 2021, received its first batch on 28 December and passed its pre-startup safety review (PSSR) on 29 December (p.88). The unit came from the closed Modesto plant together with reactor 5, was modified to meet this plant's design requirements, and went into the existing piping of the central manufacturing area. By design the two were to run a range of caramel colour products, not only 484 (p.27). The other was the day's routine batch. At 10:22 on 12 November the crew charged to the 484 batch record: water, invert syrup, sodium hy-

droxide, phosphoric acid, antifoam (p.28). Product 484 is a Class I caramel colour, made here at least since 2012, with the batch record issued on 27 January 2012; reactor 6 had run at least four 484 batches in the month before, without abnormality (p.28 footnote). At 14:17 the pressure setpoint went to 12 psig and the temperature setpoint to 300°F per the batch record; the PLC closed the top relief valve to build pressure and steam came on to heat (p.28). Pressure reached 12 psig at 14:24 and the valve modulated to hold it over the next 16 minutes. At 14:39 the valve moved toward the closed position with no command from the PLC; at 14:40 the PLC commanded full open and the valve did not move (p.28). The technical fault landed on that one valve. The plant collected no valve position data, and the report reconstructed position from surveillance video together with scrubber temperature and pressure (p.28 footnote).

The decision point: the report withholds individual names, so signatories can be recorded only to post and document. Four cuts, each with a date, a size and a carrier.

Cut one, 15 October 2007: the site took its ammonia inventory below the threshold of the EPA Risk Management Program (RMP) (p. 88). The size of the cut was an entire obligation disappearing. After a reactor at this plant ruptured on 11 April 2003, killing one person, releasing 26,000 pounds of ammonia and sending 1,500 people to shelter in place, the plant sat under RMP and had to run a hazard review at least every five years; leaving coverage took that obligation to zero (pp. 23–25, p.57). The carrier was an inventory change, and the report records no post responsible for it (p.22). The recommendation CSB issued after the 2003 accident, to establish a hazard evaluation procedure, stayed open until August 2009, when it was closed as “acceptable action taken” (p.57, p.88). That barrier was patched on paper. The 2007 exit from coverage removed the external beat that kept it alive.

Cut two, 30 November 2012: an outside consultant used reactivity test data from product 034 to size relief, and recommended enlarging the orifice on the reactor that would make 034, from 4.186 square inches to 10.304 square inches, a factor of 2.46 (p.26 footnote, p.88). It was implemented in 2013 on reactors 3 and 4 only; reactors 1 and 2 were not making 034 at the time and stayed at 4.186 (p.27). In the same period the management of change (MOC) for reactor 3 carried an action item to "update the HAZOP", with the completion and date columns blank; the reactor 4 modification went through no MOC at all, against the plant's own policy of June 2006 (p.57, Figure 35 p.58).

Cut three, 14 April 2021: the plant engineer who had inherited the project ordered relief valves for reactors 5 and 6, orifice 4.186 square inches (p.88). The basis was the existing configuration of reactor 2, on the grounds that the two new reactors were close to it in size and made the same products. The site held no design basis document for the reactor 2 relief system (p.27, p.47). That order took the 10.304 validated for 034 nine years earlier and put 4.186 back in its place, which is 40 percent of the validated figure.

Cut four, 19 October 2021: the MOC for reactors 5 and 6 was closed. The MOC itself listed a process hazard analysis (PHA) as required, then concluded it was unnecessary because the design was "a copy of No. 2" (p.60, p.88). The PHA was never done. Two months later the PSSR of 29 December checked that the equipment had been built to the drawing, not where the drawing's basis came from (p.88).

What was visible then: the 034 reactivity curve of 15 October 2012, showing a self-sustaining temperature rise under closed heating and a pressure rise driven by non-condensable gas (pp. 25–26); the consultant's relief sizing recommendation of 30 November 2012, requiring an orifice larger than 7.4 square inches (p.26 footnote); the 2013 modification records for reactors 3 and 4, and the un-

closed “update the HAZOP” item on the reactor 3 MOC (p.58); the HAZOP report of 21 February 2008, whose high-pressure deviation list included relief valve failure, blocked relief line and check valve failure, with sugar decomposition absent from it (p.25 footnote); the MOC policy text of June 2006, “design review completed be‑fore the change” (p.57); the missing design basis document for reactor 2 (p.47); the Safe Operating Limits document established in 2011, which set pressure limits at each vessel’s maximum allow‑able working pressure and temperature limits at maximum allow‑able working temperature, giving 75 psig and 355°F for reactor 6, and which also said that if a reaction could not be controlled and pressure approached the safe limit, the fire alarm should be pulled and the plant evacuated (p.65); the control room and laboratory combined in 2012, 40 feet from the reactors and not blast resistant (p.69); the plant’s 19 October 2015 statement closing out the EPA consent agreement, saying that all recommendations in the final work plan had been implemented by 17 September 2013 (p.88); the personnel record of the engineering manager’s departure in April 2017 (p.88); the production schedule around 2021, on which reactor 2 had begun making 034 without an MOC (p.88); section 10 of the supplier’s safety data sheet for invert syrup, which carried no decomposition warning, while a second supplier’s sheet for the same material, which the report also found, did state the decomposition hazard (pp. 53–54).

What was learned only afterwards: invert sugar begins to decom‑pose above 267°F, inside the normal operating range for 484 (p.54); in CSB’s adiabatic calorimetry the third exotherm began at 243°F, with a 284°F temperature rise and a peak pressure of 1,509 psig (p. 37); the relief area needed to stop this rupture was 16.74 square inches, requiring a 6R10 valve or a single 4-inch rupture disc (pp. 49–50); and why the relief valve failed cannot be determined, because the positioner was never recovered (p.75).

Stop point. You are sitting in this seat with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened. The relief device lifted at 14:54 and pressure dropped briefly, then rose again; pressure passed 75 psig at 14:55; temperature passed 355°F at 14:56; at 14:57:41 the reactor ruptured, with a recorded peak of 237 psig and 385°F (p.31, p.89). Six people were in the control room. Two died, a maintenance technician and a railroad operator, both crushed when the control room collapsed; the railroad operator was not found until about 10 hours after the event (p.32, p.34). Three were seriously injured: the reactor operator with fractured ribs, a maintenance technician with lacerations, and an administrative assistant in the neighbouring office with a concussion (p.32). Damage came to $30 million on site and $10 million off site; debris travelled up to 400 feet beyond the fence line, and a shell section of about 2,000 pounds landed 245 feet away in the yard of a private house (p.32). The plant has been shut down and demolished (p.9). Back to the three questions. The most common answer to the first is "pull the test data from that year and redo the relief sizing". In this case that action costs one archive pull and one calculation, and what it blocks is 237 psig. The most easily missed item under the second question is the number in the Safe Operating Limits document itself: 75 psig looks like the strictest line on the sheet, and it is in fact the vessel's rupture line, so a limit set at the boundary leaves no room to act. CSB calculated that pulling the fire alarm at 14:54:04 would still have left

about 3.5 minutes before the rupture (p.66). The third question lands on the MOC closure of 19 October 2021: test the evidence behind that sentence, meaning whether the drawing being copied from reactor 2 has a design basis at all.

Options ruled out: reactivity testing across other products and individual raw materials, which was done only to the minimum required by the 2008 HAZOP recommendation and the EPA consent agreement, since 034 had already been identified as the most haz‑ardous product for heat output and the rest were filed as lower risk (pp. 51–52). Enlarging relief on reactors 1 and 2 as well, which were not making 034, different products carrying different risk (p.27). A PHA for reactors 5 and 6, which the MOC identified and then judged unnecessary as "a copy of No. 2" (p.60). A separate relief sizing study for reactors 5 and 6, since the method and conclusions of the 2012 consultant study were in the plant and copying an ex‑isting design needs no new analysis; the report records no new analysis (p.27, p.47). An overpressure scenario list per vessel under the ASME code, which requires the owner to "identify all potential overpressure scenarios and the corresponding pressure relief means", while reactors 1, 2, 5 and 6 left no record of a relief design basis (p.47, p.59 footnote). Relocating or blast-hardening the control room, where the public reason for combining control room and laboratory in 2012 was to shorten the walk between the oper‑ator and the equipment and improve communication; the plant had left RMP by then, so no siting analysis was required by regu‑lation, and the report confirms the plant never did one (p.69).

Early signals: 1. On the 2013 reactor 3 MOC, the "update the HAZOP" action item had both columns blank, and the responsible person column blank as well (Figure 35, p.58). Why it was pushed through: the modification was installed and running normally, and all that was missing was a text update; the 2015 closure statement then stamped "all recommendations implemented" over it (p.88).

2. In April 2017 the engineering manager, the only person who had led the 2012 testing, left, and the knowledge never reached a document (p.88). The reason given: the record "should be in the MOC", and that MOC was precisely the one never finished; a departure is routine in a personnel system and triggered no document review. 3. Before 2021, reactor 2 began making 034, with an orifice of 4.186 against a validated 10.304 for 034, and the change went through no MOC (p.88). The reason given: a product switch counts as scheduling, not as an equipment change, and the schedule has no change column. 4. On the order of 14 April 2021, 4.186 and the missing design basis for reactor 2 sat on the same piece of paper, which carried only valve model and orifice size, no overpressure scenario (p.47, p.88). The reason given: same size, same product, copying is compliance; a purchasing process does not ask for a design basis. 5. The 2011 limits document set the ceiling at 75 psig and 355°F with no automatic alarm attached (p. 65, p.66). The reason given: the document already contained the instruction to pull the fire alarm and evacuate on loss of control, which was treated as a barrier that existed; when the control room and laboratory were combined in 2012 they stood 40 feet from the reactors (p.69).

The point of escalation: the MOC closure of 19 October 2021 put "no PHA" into the file, and that became the conclusion every later reader would see (p.60, p.88). The first batch went in on 28 December and the PSSR passed on 29 December. The PSSR was the last gate that could have stopped this, and it checked equipment against drawing, while the drawing's basis was already missing from the file, so the gate had nothing to say about the central question. The acquisition closed just before that batch: Givaudan completed the transaction on 19 December, and the archive it took over was in exactly this state, with a test curve and no design basis, an MOC and no PHA (p.88). From the day the MOC closed, "no PHA needed" stopped being one person's judgment and became

the plant's documented conclusion; later readers found that sentence. For the 34 months that followed, reactor 6 ran on the normal schedule and never once called on the relief system. Normal running kept underwriting the cut, because a reactor that has run three years without trouble does not look like equipment short of margin.

Failure chain: technical layer. At 14:39 the relief valve closed with no command. Sugar decomposition began above 267°F, and the exotherm plus carbon dioxide drove pressure up on its own. Cooling water came on 9 minutes before the rupture, and heat removal ran below heat generation. The relief system lifted as designed, and the 4.186 square inch orifice fell about four times short of the 16.74 square inches required, so pressure kept climbing to 237 psig and the shell failed.

Organizational layer. The 2012 test results were never extrapolated to other products or to individual raw materials. The 2013 modification covered two reactors, left one action item unclosed and skipped MOC on one unit. Sixteen years passed after 2008 with no further hazard analysis. No post was accountable for executing process safety policy. Safe operating limits sat at the vessel's design limit, with no automatic alarm and no record of refresher training. The control room stood 40 feet from the reactors, was not blast resistant, and had never been through a siting analysis.

Incentive layer. Syrup is not on the OSHA process safety management list of named chemicals or on the EPA RMP list, so neither regulation covers a process of this kind (pp. 71–74). After the 2007 exit from RMP, the five-year hazard review stopped being a legal duty. Capital projects moved as "copy the existing reactor", the fastest and cheapest path available.

The break sits at the junction of organizational and incentive layers. The incentive layer supplied a legal path called "don't", and the organizational layer turned "don't" into documents one

at a time: an unfilled action item, a product switch on a schedule, an order copied from reactor 2. The regulatory compliance list has no line for reactivity, so every "don't" passed. And the organizational layer had already accepted compliance as a substitute for hazard review. All three layers of margin were therefore absent at once: no data to set the waterline, no orifice to catch the flow, no post to keep the record.

Minimum counterfactual: before the MOC closed on 19 October 2021, run the PHA that the MOC itself listed, instead of closing it with "a copy of No. 2". One action, 37 months before the event, costing one cross-discipline review meeting, with no new equipment, no new budget and no new consultant. By the report's own definition of that PHA, it had four things to examine: earlier hazard analyses, MOC files over the years, reactivity test data, and the protection design about to be adopted (p.60). Any one of the four would have brought the 2012 curve and the 2013 modification to the table. The meeting would have had to answer one question: which document is the relief design basis for reactor 2. The answer was that no such document exists (p.47), and that empty answer would have put the 10.304 for 034 and the 4.186 on the order onto the same table.

Outcome and sources: 2 dead, 3 seriously injured; $30 million in on-site damage and $10 million off site; the plant demolished (p.9, p.32). CSB issued 12 new recommendations and reiterated 2 earlier ones (pp. 79–83), among them that Givaudan calorimetry-test all caramel colour products (R1), that a third party redesign emergency relief on every reactor (R5), that automatic alarms and training be added against the safe operating limits (R6), and that a siting study precede any new plant (R7). Primary source: U.S. Chemical Safety and Hazard Investigation Board, Fatal Runaway Reaction and Explosion at Givaudan Sense Colour / D.D. Williamson, Louisville, Kentucky, November 12, 2024, Investigation Report

No. 2024-06-I-KY, released 27 May 2026, 96 pages; Givaudan has said it is considering rebuilding the plant elsewhere, and CSB attached a comment requiring the new site to be kept away from residential areas (p.11 footnote). The report and the test data sit on the same case page; link at Appendix B, ch-006.

The three layers of margin in this case are worth restating in the ledger format of this chapter, because the case is the standard specimen of what a margin ledger should look like. Row one, process safety information margin, meaning whether reactivity data exists. Original design requirement: do it. Current value: none. Change history: never listed. Row two, physical margin, meaning the relief capacity's coverage of the runaway rate. Original design value: estimated from existing parameters. Current value: insufficient. Change history: no review after the design was fixed. Row three, organizational margin, meaning a process safety management system in place. Original requirement: recommended by industry practice. Current value: written policy, no accountable person, and outside both OSHA PSM and EPA RMP. Change history: October 2007, ammonia inventory taken below the RMP threshold and coverage exited, with no item raised since. All three rows share one signature, an empty change history. A margin with no change history has not necessarily stood still; nobody looked either way. The ledger's columns follow from these three rows: item, original value, current value, change history, owner. With all five in place, a blank in any row has nowhere to hide.

Case 3 (Software): Three Unpaid Debts in Security Control

sw-011. France Travail. The CNIL closing decision (page checked for this book): a €5 million fine for inadequate security of jobseeker data, with the findings including adviser accounts without multi-factor authentication (MFA), access rights wider than the duties required, and no active monitoring of tool activity logs; the case

closed in September 2026 after remediation. All three are margins. Authentication margin, where a single factor was enough to log in and the second factor was skipped. Permission margin, where broad access was handed out for convenience and least privilege was skipped. Logging margin, where records were kept and nobody read them, so active monitoring was skipped. Each of the three was "running normally" before anything happened, and each omission had a defensible reason at the time.

Stop at the decision point: the configuration decision, taken while the account system was being built, that made MFA optional rather than default. On the table: the usability request, the deployment cost, and a tiering assumption that only high-risk posts needed it. Outcome: a €5 million fine, closed after remediation. Primary source: CNIL, 2026; link at Appendix B, sw-011.

Those three omissions share a scale property worth naming. Each was a one-time configuration decision, an option chosen while the account system was being built, and the margin each consumed is continuously exposed, since every day of operation accumulates risk in an unprotected state. Configuration margin has a distinctive shape: the decision costs almost nothing and the regret costs a great deal. Ticking MFA is one click; remediating after a leak is five million euros. Decisions of that shape need their own review rule, and the rule is that every security-relevant default starts on, with opting out requiring a stated reason. The philosophy of defaults is this chapter's mechanism translated into a configuration screen: the freedom to eat margin should be off by default, and anyone who wants it should have to apply.

Case 6 (Maritime): The Seam the Inspection Plan Did Not Have (condensed)

ma-013. A sling failure: a deck crew member was struck by a towed block and killed, October 2023, United Kingdom. MAIB report

(2026, page checked for this book): a quick release link lost strength under long-term high bending stress and parted, and the vessel's inspection plan did not cover that defect form. The coverage of an inspection plan is the degree to which margin has been itemised: a check dimension the plan does not have, here link fatigue under bending stress, is where margin drains away in silence. Minimum counterfactual: add the stress-form dimension to the inspection plan for load-bearing links, with periodic strip-down and non-destructive testing. Primary source: MAIB, 2026; link at Appendix B, ma-013.

Case 7 (Aviation): A Fuel Margin That Was Calculated Low (condensed)

av-013. An instructional flight, July 2023, France. BEA report (2026, page checked for this book): the instructor's fuel consumption estimate was low, usable fuel ran out and the engine stopped, the aircraft ditched at sea and was destroyed, and the instructor and student were injured. The fuel plan is the most explicit margin in flying, the legal reserve, and here the margin was eaten at the planning layer: the consumption figure came from an optimistic reading, so the reserve met the book on paper and evaporated in the actual burn. The cheapest place to defend this margin is calibration of the planning parameter, backfilling estimates with real consumption, and one calibration costs one spreadsheet. Primary source: BEA, 2026; link at Appendix B, av-013.

Case 4 (Aviation): A Margin Whose Going Cannot Be Rebuilt

av-015. Reims/Cessna FA150L (I-FFSQ), 1 November 2019, Santa Severa, Italy. ANSV investigation report (Relazione d'inchiesta, PDF checked for this book): on a sightseeing flight the aircraft struck a power line on approach, crashed and caught fire, and both people on board, the pilot and the passenger, died (the original

reads “Le due persone a bordo (pilota e passeggero) perdevano la vita”). The report states plainly what it could not establish: why the aircraft came in below the height it should have flown. The mechanism of margin erosion ran to completion, with height margin, track margin and visibility margin failing one after another, and where each one went cannot be reconstructed from the evidence.

Stop at the decision point: each segment of the approach in which height margin was progressively spent. On the table: the visual references of the moment and the judgment of the two people on board, precisely the part the evidence cannot fully restore. Outcome: two dead. The case is a proof by contradiction for the margin ledger: when even the investigator cannot rebuild where the margin went, the ledger kept beforehand is the only memory there is. Primary source: ANSV, Relazione d’inchiesta I-FFSQ, 2023; link at Appendix B, av-015.

The unreconstructability carries an evidentiary corollary worth writing down. Margin is spent in real time during flight, with height falling and track drifting, but if nobody is monitoring the live data, with no altitude alert triggered and no track deviation watch, the spending happens only in the physical world and in no record at all. The investigation can use only what was recorded (radar data, wreckage position), and the granularity of the record sets the ceiling on the reconstruction. Here even the investigator could not confirm why the pilot was below the height to be flown, and the limit of the granularity is the limit of the reconstruction. The corollary for any real-time system: consumption of a critical margin should be recorded as it happens, even if the record is automatic and unmonitored, because the record is the raw material of every future reconstruction. The direct channel to original records in Chapter 16 gains a technical argument here.

Case 5 (Maritime): The Pipe from 1998 (full card)

Case ID: ma-004.

When and where: about 05:30 UTC on 24 March 2024, when the steel stern trawler Opportune sank about 36 nautical miles east of Lerwick, Shetland (60°16.6N 000°02.3E); the bilge level alarm had sounded on the bridge 45 minutes before the sinking. The water at the site is more than 100 metres deep and the wreck was not examined. The trip sailed from Scalloway at 01:00 on 20 March with Peterhead as destination (p.7, p.8, p.11).

What and who: the owner Quiet Waters Fishing Company Limited, which had held and managed the vessel since October 2020; UK flag, built in Spain in 1998, length overall 27.85 metres, registered length 23.95 metres, 331 gross tonnes, no class society (p.7, p.11). Eight people on board: the skipper held a Deck Officer Class 2 (Fishing) certificate of competency, the mate also served as engineer, and six deckhands made up the crew (p.11). Inspection and certification fell to the Maritime and Coastguard Agency (MCA) under MSN 1872 (F) Amendment 1, treating the vessel as existing; a local shipyard and a piping contractor hired by the owner also appear in the record (p.11, p.15).

What they were doing: two timelines were running at once. The first was a trip spent managing the stern shaft. The vessel sailed at 01:00 on 20 March and fished grounds to the east for three days; at 18:00 on 23 March, hauling about 100 nautical miles northeast of Lerwick, a stern shaft high-temperature alarm came up on the bridge, the mate measured the stern seal with an infrared thermometer and found it above normal running temperature, the stern seal oil was partly emulsified, and the crew changed the oil. The skipper spoke to the owner ashore and decided to stop fishing and steam for Peterhead to have the shaft examined out of the water. For the first hour they read stern seal and shaft temperature

every 10 minutes, then every 30 minutes for two hours, then hourly once the temperature held below 40°C (p.8, p.9).

The second timeline belonged to the machinery and had not moved in twenty-six years. The engine room sits under the main deck at the stern, mostly below the waterline, with a watertight door forward and one aft, and the door to the steering gear space behind it usually open (p.12). The main seawater intake manifold is 200 mm galvanised steel pipe, with cast iron sea valves either side of the main engine below the turn of the bilge, piping and valves under the floor plates about 4 metres below the waterline, installed as originally fitted at build (p.12). The same report carries numbers for the shell: 12 mm at the keel and 8 to 10 mm at the waterline when built, sacrificial anodes renewed in November 2022, impressed current cathodic protection also fitted (p.12). Somebody measured the shell. Nobody measured the pipe.

The decision point: the report names no individuals, so signatures can be recorded only to post and document. Four cuts, each with a date, a size and a carrier.

Cut one, 1998, when the build was fixed. Clause 4.1.11 of MSN 1872 (F) sets the default for seawater piping on a new vessel at aluminium bronze, cupronickel or a similar corrosion-resistant material, and the same clause leaves a gap: heavy-wall mild steel may be used for cross-ship intake mains, provided the bore is not less than 100 mm and the interior is galvanised after assembly (p.15). This main is 200 mm and satisfied both conditions. The size of the cut can be measured against SAN 29 of 2016: galvanising has a limited life, and within a 20-year installation life galvanised steel piping will almost certainly be replaced more than once (p.18). In 2024 the vessel was 26 years old (p.6), 6 years past that assumption. The carrier was the build drawing and the build survey, and the report records only "as originally fitted at build" with no approving post (p.12).

Cut two, October 2020, a one-day intermediate survey with the vessel lying at Scalloway. The MCA surveyor raised 38 deficiencies, three of which were written down: piping leaking, repair required; impressed current cathodic protection not working; lifejackets not stowed where readily accessible (p.14). The size of the cut lay in how the items were closed. The owner emailed photographs of the rectified defects to the MCA and the MCA closed them (p.14). The leak was treated as a local repair, with no pressure test and no thickness measurement. The carrier was the deficiency list and the email with photographs.

Cut three, November 2022, the five-year renewal survey completed at Fraserburgh, with one day out of the water and, six days later, one day in the water plus an inclining test (p.14). The carrier was the Fishing Vessel Hull Condition Report, and this was the largest and most quantifiable cut of the four. Shell plating inside and out was judged "good for age", with thickness measurement done by a local contractor six months before the survey and found compliant; shaft and rudder clearances independently checked by the yard and within tolerance; sea valves stripped, examined, found satisfactory and refitted; inclining test confirming the stability book still valid (p.14, p.15). Five items reached a conclusion one by one, hull, thickness, shaft and rudder, sea valves, stability, and the seawater piping item reached none; the certificate's own words are that no comment was made on the condition of the vessel's seawater piping (p.15). Over the same period, MCA internal guidance SAN 29 required the revised hull condition report to add a seawater piping section (p.18). The signatory was the MCA surveyor.

Cut four, 21 August 2023, when the vessel grounded while shifting berth in Fraserburgh harbour and a car tyre wrapped around the propeller; she was lifted out, the shaft removed, examined and repaired (p.15). The owner engaged a local contractor to examine the

engine room piping: several pipe runs above the fuel tanks were removed, cleaned, examined and refitted, while the main manifold ring feeding the whole system stayed where it was, limited by the arrangement under the floor plates, and received a high-pressure water flush and rodding to clear mussels (p.15). The size: only the removable part could be inspected, and the 200 mm main ring was neither removed nor measured. The carrier was the contractor's examination and cleaning work, whose content the report records but whose work order number it does not. Approval sat with the owner, as the employer and the party responsible for seaworthiness (p.14).

What was visible then: the October 2020 list of 38 deficiencies, piping leakage and failed cathodic protection among them; the November 2022 hull condition report with five conclusions and one blank column; the record of the 21 August 2023 grounding, lifting out and shaft removal, together with the contractor's scope cov‑ering only the runs above the fuel tanks (p.14, p.15). The safety folder: 52 risk assessments, two of which name flooding as the hazard ("seawater intake valve failure", "stern damage while trawling"); three monthly drill subjects, man overboard, abandon ship, fire; no seawater piping failure assessment and no flooding drill (p.12, p.24). The engine room itself: sea valve handwheels sit below the floor plates and are worked with an extended bar kept beside the generator through a small hinged flap in the plating, with no valve position markings on the plating and no arrange‑ment drawing at the entrance (p.13); both 25 m³/h electric bilge pumps could be started and stopped only inside the engine room (p.14). Public material: MGN 165 (July 2001) records that 18 of the 33 fishing vessel losses in 1999 came from flooding, and requires weekly pump trials, monthly sea valve exercise and sea valve in‑spection when the vessel is out of the water (p.17); of 230 fishing vessel flooding reports between 2013 and 2022, 78 (34 percent) led to the loss of the vessel, and 68 of those (87 percent) were more

than 15 years old (p.19); in August and September 2012 the Span - ish-built twin trawlers Audacious and Chloe T, built to the same design, sank one after the other, the combined investigation found seawater piping failure, neither crew could close the flooded intake valve, and that case produced SAN 29 (p.19). Another vessel of the same design was altered instead: fresh water cooling with box coolers removed the need for the two sea valves, and a fixed emer - gency bilge suction was fitted that could be worked from the upper deck with an independent power pump, without opening the en - gine room door (p.19). RNLI material from 2009 gives about 8 tonnes per minute through a ruptured 177.8 mm pipe (p.18). Total pump capacity on this vessel was about 66 m^3/h (p.25). The MCA had held SAN 29 internally since October 2016; it was never merged into MSIS 27 and never published to owners (p.18, p.22).

What was learned only afterwards: the origin of the flooding could not be determined (p.6); the position, wall thickness and corrosion distribution of the failed section are unknown, since the wreck lies in over 100 metres of water and was not examined (p.11, p.21); SAN 29 had still not been merged into MSIS 27 as of September 2025 (p. 18); the MCA never asked the owner for a piping diagram, and no piping test record existed before the renewal survey (p.23).

Stop point. You are sitting in this seat with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not “pay more atten - tion”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened. At 04:45 the bilge level alarm sounded. The mate went down to the engine room and found water at gearbox height, with the turning shaft throwing spray and water rising from under the floor plates; he went back for the skipper and minutes later the two went down together, and when the door opened there was white odourless steam, so they judged the space could not be entered (p.9, p.21). The skipper decided everyone should go to the bridge for lifejackets and immersion suits and prepare to abandon; at 0458 the mate sent a DSC distress alert on VHF; shortly after, propulsion and electrical power were lost (p.9, p.21). At 0510 the eight mustered on the bridge; the 10-person liferaft went over the port side and six deckhands jumped into it; the skipper activated the EPIRB and passed flares into the raft; the painter was torn away and parted, and the raft drifted off (p.9). At 0530 the stern went under and the bow floated for another 10 to 15 minutes; at 0556 a helicopter sighted two rafts; at 0623 all eight were recovered; at 0700 they landed at Sumburgh (p.11). No one was hurt, the vessel was a total loss, and pollution was minor (p.6, p.7). Back to the three questions. The most common answer to the first is "go over the pipes again", and in this case that action costs one pressure test or a set of thickness readings; the report records that before the renewal survey there was no record of any piping test (p.23). The item most easily missed under the second question is the phrase "rapid and practical means" in clause 2.2.6 of the MSN for existing vessels: a portable extension bar was on the register and formally reached the valve, and it stopped reaching it as soon as the floor plates flooded (p.16, p.24). The third question lands on the blank column in the November 2022 report, and the report supplies the time account: 45 minutes from alarm to sinking, about 15 minutes from alarm to loss of power (p.21).

Options ruled out: replacing the seawater piping as a whole, where the report records no section replacement at any point; the reasons can only have been cost and dock time, and the sentence

about replacement more than once in 20 years was something the owner could not see (p.22, p.23). Upgrading to the new-build standard, since the corrosion-resistant material requirement in clause 4.1.11 applies to an existing vessel only when the seawater piping is replaced, and no replacement means no trigger (p.15). A pressure test or thickness measurement at the 2022 renewal, where SAN 29 itself required vessels over 10 years old to have internal examination and pressure testing or gauging on sections not replaced within 5 years; the report records no such requirement from the MCA and no piping diagram supplied by the owner (p.18, p.23). Removing the main ring during the August 2023 lift-out, where the piping lay under the floor plates and the contractor could take out only a few runs above the fuel tanks (p.15, p.23). Converting to fresh water cooling with box coolers, which a vessel of the same design had done, with conversion cost and survey route unrecorded in the report (p.19). Fitting a permanent extension bar or remotely operable valves, where the MSN asks existing vessels only for "rapid and practical means" and makes permanent access mandatory only on new ones (p.16, p.24). Making the bilge pumps startable from outside the engine room, which the report records as a current state and does not discuss (p.25). Adding a piping risk assessment and a flooding drill, where chapter 8 of the MSN requires monthly drills including flooding and MGN 570 has a flooding drill, and the folder held neither (p.12, p.16).

Early signals: 1. In October 2020 "piping leaking, repair required" sat in a list of 38 deficiencies alongside lifejackets and cathodic protection, and the evidence of closure was a set of exterior photographs; "rectified" pushed it through (p.14). 2. In November 2022 the hull condition report concluded five items and left the piping item without a conclusion, and that column was exactly what SAN 29 asked for; "certificate signed, items cleared" pushed it through (p.15, p.18). 3. On 21 August 2023 the main ring was flushed and not examined; "cleaned, mussels cleared, no leak

seen” pushed it through (p.15, p.23). 4. Of 52 risk assessments none covered piping, and of three monthly drill subjects none covered flooding; “we already have the seawater intake valve fail‑ure one” pushed it through (p.12, p.24). 5. The 2013 combined in‑vestigation into Audacious and Chloe T had already written that neither crew could close the flooded intake valve, and had pro‑duced SAN 29 directly; “that was another vessel” pushed it through. SAN 29 was issued in October 2016, this owner took the vessel over in October 2020, and the public guidance available at handover did not include it (p.18, p.19, p.22). 6. The sea valves were “stripped, examined, found satisfactory” and the handwheels were 4 metres below the floor plates at the same time, and the first sentence pushed the second out of view (p.13, p.15). Five of these six have “satisfactory” as their keyword and appear as certific‑ates and photographs. The one thing actually missing appears as a blank, and a blank enters no checklist.

The point of escalation: the moment the renewal certificate was signed in November 2022. Before that the piping had merely gone unrecorded; after it the piping was an absence inside a valid certi‑ficate. For the next 16 months (p.23) any reader saw five items each concluded, an inclining test confirming stability, and a certificate within its validity; in the same round the sea valves were stripped, passed and refitted, which stamped formal compliance on the ar‑rangement (p.15, p.16). Through those 16 months the vessel worked the fishing season as usual, sailing, fishing for three days, hauling, raising an alarm, making for port, every step in the report inside procedure. A second chance came 7 months later, on 21 August 2023, when the grounding put the vessel out of the water, took the shaft out and brought the contractor aboard, and the main ring stayed in place and got a wash (p.15, p.23). The owner hired the contractor to examine the engine room piping, and what could actually be examined was a few runs above the fuel tanks. After that lift-out and before the sinking, the vessel never had another

opportunity to show the pipes the air. Seven months later a 200 mm pipe breached 4 metres below the waterline, the breach position could not be confirmed, and the wreck sits in more than 100 metres of water (p.11, p.21).

Failure chain: technical layer. The 200 mm galvanised mild steel pipe thinned by corrosion in seawater, unmeasured here, with no wall thickness figure anywhere in the report, and breached under the floor plates 4 metres below the waterline. On the RNLI basis a hole of that size admits about 480 t/hour, and the vessel's total pump capacity was about 66 m^3/h (two 25 m^3/h electric pumps plus an 8 m^3/h manual pump and an 8 to 10 m^3/h portable one), roughly one seventh of the demand, with the two electric pumps together at 50 m^3/h covering about one tenth (p.14, p.18, p.25). The report judged that even with the pumps running, meaningful dewatering before the loss of power would have been difficult (p.25). Closing the valve was the only means available, the handwheel sat under the floor, and reaching it took a hinged flap and an extension bar, while by 04:45, when the water was found, the floor plates were already submerged, so the valve was out of reach at the moment it was discovered (p.24). About 15 minutes separated the alarm from the loss of power (p.21).

Organizational layer. The hull condition report had no column for seawater piping (p.15). SAN 29 had existed since October 2016, was never merged into MSIS 27, and remained an internal document as of September 2025 (p.18). Piping shared a cramped space under the floor plates with the fuel lines, so visual inspection and non-destructive testing could cover only part of it, and the harder a section was to reach the easier it was to miss (p.23). There was no piping maintenance and inspection plan, no piping failure risk assessment and no flooding drill (p.12, p.24). The owner's records went down with the vessel (p.11).

Incentive layer. Replacement cost falls due at once and corrosion risk arrives later; about 75 percent of UK registered fishing vessels are over 15 years old (p.19); the rules drop a grade for the existing fleet, asking only for "rapid and practical means" and reserving permanent access for new vessels (p.16, p.24).

The break sits between the organizational and incentive layers: the rule downgraded "accessible" to "temporarily accessible" and the survey turned "piping" into a non-item, each move legal on its own, and together they left a 200 mm hole 4 metres below the waterline under the floor plates.

Minimum counterfactual: at the November 2022 renewal survey, do what SAN 29 required for vessels over 10 years old, a pressure test or thickness measurement on seawater piping sections not replaced within 5 years, and write the result into the piping section newly added to the hull condition report. One action, 16 months before the event, with the vessel already out of the water on her cycle, no extra dock time needed, and the cost of one pressure test or one set of thickness readings (p.14, p.18). The report confirms that the requirement already existed and that no record shows it was ever carried out (p.18, p.23). The same SAN 29 also required the owner to submit a piping diagram marking ferrous sections and their ages; the MCA did not ask and the owner did not submit, so neither side took that step (p.18, p.23). The report places this gap in the inspection framework, since SAN 29 was never merged into MSIS 27 and a surveyor might not know the document existed (p. 23). Had a section failed, a 200 mm section could be replaced within the same docking; had it passed, that column would be the first set of pipe wall numbers any later reader could see.

Outcome and sources: the vessel was a total loss, all eight were rescued unhurt, and pollution was minor; the wreck lies in more than 100 metres of water and was not examined (p.6, p.11, p.21). MAIB issued five recommendations: 2026/107, merge SAN 29 into

MSIS 27; 2026/108, require owners of vessels over 10 years old to provide piping diagrams marking the age of each ferrous section, and revise the hull condition report to add a seawater piping sec - tion; 2026/109, revise the safety guide for fishermen to require a valve position diagram posted outside the engine room entrance and a record of when each section was replaced; 2026/110, require owners to add flooding drills and a piping failure risk assessment and to implement MGN 165; 2026/111, require owners to compile piping diagrams and record the replacement date of each section (p.28). The investigation also issued a safety leaflet to the fishing industry (p.27, Annex A at pp. 30–31). These conclusions rest on crew statements, survey records and comparison with vessels of the same design (p.11, p.21, p.23). Primary source: MAIB, Report on the investigation of the flooding and foundering of the stern trawler Opportune (LK 209), approximately 36 nautical miles east of Lerwick, Shetland Islands, Scotland, on 24 March 2024, Very Serious Marine Casualty Report No. 3/2026, January 2026, 32 pages; link at Appendix B, ma-004.

25.4 What Shows Up Earliest

Six weak signals, each with the reason normally used to push it aside.

1. Design margin has no written account and lives only in the original designer’ s memory. Reason given: “it’ s all on the drawings.”
2. Each assessment looks only at the change in hand, and nobody builds the cumulative view. Reason given: “we assessed every one of them.”
3. “Still enough” has no quantified basis, or its basis is the state after the last cut. Reason given: “practice proves it.”

4. The list of exceptions and exemptions is getting longer, and exemptions have started citing other exemptions. Reason given: "it's a special case."
5. Recovery from disturbance is slowing, so the same small fault now takes longer to clear. Reason given: "coincidence."
6. Near-miss reports are falling while operating intensity holds steady. Reason given: "safety is improving." (The cognate signal of the Chapter 11 criterion.)

Put the six back against the seven cases in this chapter. Signal one, margin only in memory, maps to Case 5, where knowledge of the seawater pipe's wall thickness lived on a 1998 drawing. Signal two, assessment of the change only, maps to Case 1, where each freeing-port change was assessed by itself. Signal three, "enough" measured from the last cut, maps to Case 2, in the re-use logic that sized relief from existing parameters. Signal four, exemption citing exemption, is most visible in finance and compliance, and the tiering assumption in Case 3, that only high-risk posts need the extra factor, is its configuration-screen version. Signal five, slower recovery, maps to every case, and it is the only living signal that runs across all seven. Signal six, fewer near-miss reports, maps to Case 7, where a low consumption figure with no near-miss backfill will come out low again next time.

Rank the six by cost of detection. Signal five is cheapest, since most systems already hold recovery time data. Signals two and three come next, costing a read of the review material. Signals one and four are most expensive, needing archive and ledger comparison. Start with the cheap ones as usual: pull your system's fault recovery times and plot five years of them, and the year the curve turns up is the year margin started going; every "running normally" after that turn deserves one more question.

25.5 How to Defend

Device 1, the margin ledger. Write the system's critical margins as numbers that can be added up: structural margin, time margin, staffing margin, financial margin, each with its original design value, its current value and its change history. The person who keeps the ledger is separate from the person who applies to cut it (the review-independence principle of Chapter 17). I-FFSQ demonstrates the necessity: with no ledger, even reconstruction afterwards is impossible.

Device 2, the cumulative-view review. When a reduction application reaches the table, the pre-read must include the reduction history of that margin, and the object of review shifts from "this cut" to "this line". The test shifts from "is there enough left after this cut" to "how many more cuts does this line allow". The curve itself is the deterrent, because anyone who cuts on a curve signs a cut with a name on it.

Device 3, the minimum waterline for margin. For margins that cannot be restored (structure, relief capacity, emergency capacity), set a level that may not be crossed and put it in the rules: any reduction below the waterline needs an external independent signature (the second confirmation seat for points of no return in Chapter 7). The waterline's design basis is the original design value, never the value from the last review, and that is what separates it from "still enough".

Device 4, recovery time monitoring. Treat the time a system takes to recover from an equivalent disturbance as the live indicator of margin: slower recovery is an early reading that margin is being eaten. It does not need to know which cut did the damage, only that the boat is sitting lower. The baseline needs periodic remeasurement and archiving, or even "slower" becomes unanswerable.

The four devices on this chapter's cases. Device 1, ledger, on Case 5: build a pipe wall thickness ledger by measurement point, with three columns, original design thickness, each measured value, and minimum waterline. Device 2, cumulative review, on Case 1: any change to deck arrangement arrives with the stability margin history curve attached. Device 3, waterline, on Case 2: set a water - line on the coverage factor of relief capacity against runaway rate, with startup prohibited below it, and derive the waterline from reactivity test data, which makes testing a precondition (no test, no waterline). Device 4, recovery monitoring, on Case 3: use the time to close a security gap, from discovery to fix, as the live indicator of security margin, and raise an alarm when the stock of open gaps grows.

The four failure modes. The ledger dies when it drifts from the physical object, with old numbers against new steel; the patch is to bind it to thickness measurement and survey readings, writing measured data straight into the ledger. The cumulative view dies when the curve misses cuts, recording only approved reductions and losing the ones that happened in fact, drift in execution and unregistered changes; the patch is a two-source curve, approved plus measured, with an alarm whenever measurement departs from approval. The waterline dies when its basis drifts, with someone proposing to reset it against current state; the patch is that any revision of the waterline needs an external independent signature on the same terms as a cut below it, making a drifting basis the same offence as the cut. Recovery monitoring dies when the baseline goes stale, with a five-year-old baseline used to judge today's recovery; the patch is periodic remeasurement written into the device's own operating instructions. The four patches share one health question: when was the ledger's number last calibrated against the physical object.

Margin has an accounting side worth one more passage, because it decides whether the ledger can enter the budget process at all: record margin as a liability. Seen from accounting, margin is insurance already paid for and not yet used, since the strength, distance and redundancy bought at design time were real money. Eating margin is selling that insurance for cash. Selling insurance has a defined bookkeeping form (a liability falls); in most organizations the consumption of margin has no account at all. Give it one, a safety margin liability statement, and every reduction automatically generates an entry (debit cash, credit margin), so the budget process sees a price for margin for the first time. The account does not need precise valuation, a range will do. It needs to exist: once it exists, consumption leaves a trace, and with a trace the defence concentration indicator of Chapter 20 finally has a denominator.

25.6 Boundaries with Adjacent Mechanisms

Against the invisibility of maintenance (Chapter 29): a maintenance backlog is work that should have been done and was not, the asset-side version, while eating margin is quantity that should have been kept and was not, the design-side version. The two often share a source and an outcome (Surfside takes both chapters), and the test is the object: time and actions go to Chapter 29, margin and strength stay here.

Against practical drift (Chapter 15): drift moves the distance between practice and text, while margin erosion moves the distance between the system and its boundary. Drift normally comes first, with practice loosening before margin is cut, and the two ledgers are worth reading together, since the line that loosened first is usually the one cut next.

Against the normalization of deviance (Chapter 11): normalization supplies the psychological permission to eat margin (“it was fine

last time"), and this chapter supplies the physical result ("this time it was not enough"). Criterion (iii) in Chapter 11, the stand - ard quietly lowered, lands in engineering terms exactly on the ledger in this chapter.

25.7 Five Self-Checks

1. Do you have a record of the current value and the original design value for your system' s three critical margins? What are the ratios?
2. At the last reduction review, did you see the effect of this cut or the history of the line?
3. How long is your exception list now? How many entries were added last year? Have any ever been removed?
4. Is your system' s recovery time from an equivalent small fault getting longer? Do you have the data?
5. If you had to explain tomorrow how much safety margin this system has left to whoever takes it over, how many pages could you put in their hands?

Chapter 26 Automation and Skill Decay | Automation-Induced Skill Degradation; Out-of-the-Loop

> What this chapter covers: how human skill and takeover ability decay once automation moves people out of the control loop; the criteria, two deep field cases, and an honest account of the evid - ence gap. When to come back: when an automation project starts, when simulator and hands-on training hours get cut, or when "the system will handle it" becomes the reason not to practise. Reading time: about 50 minutes.

Automation and skill decay form a closed loop. The more auto - mated a system becomes, the more a person needs to hold the skill for the rare takeover moment; and automation itself supplies the reason to cut the training that would keep it: "the system will handle it." Each turn of the loop leaves the person in the room more out of practice when the takeover moment arrives, and the takeover moment is designed to get harder, because the system absorbs every routine and leaves the entire residue of exceptions to that one person. This chapter's case library has an honest weak spot on this mechanism: the grade A cases that record a takeover failure caused by skill decay cluster in aviation. The cause of that weakness is part of the mechanism, since other domains do not have a long enough automation history and takeover failures have not yet entered their investigation systems. The gap is marked as a gap rather than filled with two or three thin cases.

26.1 What It Is

Definition: once automation has taken over most of the system's running time and most of its routine actions, the human's un -

derstanding of the system and the human's ability to run it by hand decay together. The moments when automation leaves the person in the loop are precisely the hardest ones. The operator degrades into a monitor, the monitor becomes a taker-over after a long quiet stretch, and takeover becomes the action with the fewest repetitions in a working life.

Three criteria start the mechanism. Any one of them and it is running:

1. The person has not performed an action for a long time, counted in months and years, and that action is still required in an accident.
2. The takeover moment is the hardest moment: automation covers the routine and leaves the whole non-routine residue to the person.
3. The manual and the training assume the person can do it, and no mechanism verifies that the person can.

Where the mechanism sits in the research literature is worth setting down. Lisanne Bainbridge's 1983 paper "Ironies of Automation" put the paradox at its root: what automation does best is the part the human least needs to do, what is left to the human is the part the human finds hardest, and the human spends the years of waiting for the hardest moment losing practice on all the others. Research forty years later added two pieces. The first is the out-of-the-loop problem: work on situation awareness shows that a person's grasp of system state on a monitoring post decays systematically with time, and re-entry at takeover needs a substantial stretch of cognitive rebuilding. The second is automation complacency: reliability research shows that the higher an automation's historical reliability, the looser the calibration of trust in it and the later the response to its alarms. Together the two give criterion 3 a measurement basis. Takeover ability is a stock you can measure,

through quota, refresher and simulation. Measuring it is cheap. Not measuring it is expensive.

Two research lines meet at the workstation with a combined effect. Someone out of the loop, already decayed, meets a trust calibration leaning toward complacency, where fewer alarms mean more trust, and the real start of the takeover moment comes later than either factor alone would put it. Case One shows the combined effect most clearly: the crew did not fail to know where the flight mode annunciator sat. A calibrated trust had made "throttle in automatic" the default background, and a change in the background does not become a target for attention. Defence against the combination therefore runs on two tracks. The cure for out-of-the-loop is to pull the person back into the loop, with scheduled manual cross-checks of key parameters. The cure for complacency is to build trust calibration into the ordinary week, with periodic automation-anomaly drills that let people watch the system be wrong on a schedule. Neither cure is expensive. What is expensive is their absence, which costs nothing on any day before the accident.

26.2 How It Grows

Three generating conditions. Condition one: the level of automation rises. Nothing here is pathological. Higher automation is a choice, and usually the right one. The disease is in condition two. Condition two: training hours get compressed, and the compression is justified by the claim that the system handles it. Automation creates the training requirement, which is takeover skill, and supplies the reason to cut the training. The loop closes here, and every turn of it saves money. Condition three: rare scenarios. Takeover scenarios come up in real operations on a scale of years. Without institutionalised practice, proficiency in them only falls, and every

system upgrade changes the takeover interface and the takeover sequence, depreciating whatever stock the last practice left.

Depreciation deserves its own arithmetic. Take the annual upgrade rate of the system and the annual decay rate of takeover skill; the two together give the total depreciation rate of takeover ability. Both are percentages, and added up they usually exceed the growth rate of the organization's training spend. The reason depreciation stays hidden is that no single item trips an alarm. Upgrades get tested. Training gets recorded. The composite, the present value of takeover ability, has nobody accounting for it. The report that would account for it is sketched in this chapter's Device 1, the quota, and Device 2, the simulation. The quota measures the stock, meaning whether the person can still do the thing. The simulation measures the increment, meaning whether the person can do it on the new interface. Draw both numbers as a curve by year and the slope is the depreciation reading. An organization whose reading is not negative, whose stock is rising, hardly exists. The management target is not zero decay. It is to make the slope of training spend catch the slope of depreciation.

One asymmetry in the economics of the loop has to be written down. The training cost saved is current, certain and small. The cost of a failed takeover is distant, probabilistic and large, and the bill often goes to people outside the system: passengers, users, patients. That asymmetry makes the loop survive every budget table, because each cut looks correct on its own. Put together they are the lesson aviation bought with decades of accidents: hand flying is a skill that costs money to maintain. Money spent on sim - ulators is billed by the hour. Money spent on sea walls is billed in lives. The gap between the two prices is the budget argument for every defence in this chapter.

26.3 In the Field

Case One (Aviation): The Autothrottle on a Visual Approach

Case ID: av-031.

When and where: 6 July 2013, about 11:27:50 Pacific Daylight Time; the sea wall outside the threshold of runway 28L at San Francisco International Airport, California, USA. The accident happened in the last 3.5 nautical miles of the descent, with the cockpit about 1,500 feet above the ground.

What and who: Asiana Airlines flight 214, a Boeing 777-200ER, registration HL7742, Incheon to San Francisco, operating under 14 CFR Part 129. The left seat held a trainee captain (PF, the pilot flying). The right seat held a captain acting as instructor and pilot-in-command (PIC), who was also the pilot monitoring (PM). The jump seat held a relief first officer acting as observer. The flight was the trainee captain's operating experience (OE) training flight and the first leg of his two-day schedule (p.20). On board were 291 passengers, 12 flight attendants and 4 flight crew, 307 people in total (p. 31).

What they were doing: The trainee captain's 777 transition training began on 25 March 2013. He passed a simulator proficiency check on 18 May and a line-oriented flight training check on 30 May (p.32). From 14 June he flew observation legs, logging 9 hours 35 minutes. From 16 June he flew OE with an instructor (p. 32). Of the 20 segments and 60 hours Korean regulation required, he had completed 8 segments and 33 hours 31 minutes by the day of the accident (p.33). Every earlier OE approach had been an ILS approach, and the syllabus did not require a visual approach during OE (p.33).

The day's actions, in cockpit voice recorder order. 09:38, back in the cockpit (p.21). 10:42:28, approach briefing: the 28L glideslope

was out of service, they would hold lateral track on the localizer and manage the vertical profile with the automatic flight system (p. 21). 10:47:28, descent checklist, Vref 132 knots (p.22). The 28L glideslope was out for construction, a notice to airmen had gone out, and the crew knew it from the briefing (p.95). 11:12:33, the relief first officer came back to the cockpit and sat in the jump seat to observe (p.23). 11:21:49, clearance for the visual approach, with the aircraft at 6,300 feet and 211 knots, the autothrottle in HOLD, and the automatic flight guidance system in FLCH SPD pitch and HDG SEL roll (p.23). 11:23:17, the controller asked them to slow to 180 knots and hold that to 5 nautical miles from the runway; the PM acknowledged (p.23). 11:23:57, pitch mode changed to vertical speed and the autothrottle to speed mode, with -900 fpm selected and changed to -1,000 fpm a second later (p.24). 11:25:55, crossing 5 nautical miles at 2,085 feet and 174 knots, 400 feet above the 3-degree profile (p.25).

The decision point: 11:21:49, on the radio, the PM answered "runway in sight" and the crew accepted the visual approach clearance (p.23). Accepting gave up the vertical guidance of an instrument approach. Only the localizer was available that day, and the localizer approach's minimum descent altitude was 460 feet (p.21, p.95).

11:23:17, on the same frequency, the crew accepted the restriction of 180 knots to 5 nautical miles (p.23). Under FLCH SPD the selected speed is held by the elevator, and selecting 180 knots from 212 pitches the nose up and drops the sink rate to about 300 fpm (p.24, p.96). 11:25:43, the PF set the MCP altitude window to 3,000 feet, the go-around altitude (p.24). 11:26:25.7, the pitch mode changed to FLCH SPD and the autothrottle went to THR (p.25). With 3,000 feet sitting above the aircraft, the system began to accelerate and climb; the thrust levers moved forward and the nose came up slightly (p.25). Working from the flight data recorder, the PF's in-

terview and the actions that followed, the report judged that the person who pressed FLCH was most likely the PF (p.99), and that the selection was not called out as Asiana's 777 pilot operating handbook required (p.99).

11:26:28.8, the PF disconnected the autopilot with a double click of the yoke switch, and the aural warning was suppressed (p.100). 11:26:33, he pulled the thrust levers back to idle by hand, cancel - ling the climb the system had just begun, and the autothrottle went to HOLD (p.25, p.99). 11:26:40 he called "flight directors off", and at 11:26:43 turned off his own side's switch (p.26). 11:27:14.8, passing 500 feet with a sink rate of about 1,200 fpm and thrust at idle, N1 about 24%, he had called neither "500" nor "stable" and had not gone around (p.28, p.101).

The organizational layer holds an earlier decision point. Asiana's pilot operating handbook said to use the autopilot and autothrottle to the maximum extent, and to use the autothrottle even when hand flying (p.120). In 2012, 77.7% of the company's 777 landings were manual landings, but that figure does not include the altitude at which the autopilot was disconnected (p.122). The design layer holds another. In 2010, during 787 certification flight testing, the FAA and EASA raised that this logic provides no minimum speed protection in FLCH or VNAV SPD combined with HOLD; Boeing substituted a note in the 787 airplane flight manual for a change to the logic, and the FAA accepted that outcome (p.114).

What was visible then: The flight mode annunciator at the top of both primary flight displays, showing autothrottle mode, roll mode and pitch mode from left to right. When a mode has just activated, its column carries a green box for 10 seconds (p.43). After 11:26:33 the autothrottle column read HOLD, with the green box (p.100). At 11:26:38 the MCP speed window was set to 137 knots (p.26). Between about 1,500 feet and 500 feet the thrust levers did not move once (p.106). The navigation display carried a green altitude

arc showing where the aircraft would reach the MCP-selected altitude (p.96); the localizer 28L procedure had computed a profile, and the vertical path deviation indicator was on the screen as well (p.117).

Precision approach path indicator readings, in time order: 11:26:36 four white (p.26); 11:27:14.8 three white one red (p.28); 11:27:19.8 two white two red (p.28); 11:27:24.1 one white three red (p.28); 11:27:31 four red (p.28). The speed tape and the vertical speed indicator were equally in front of them: at 500 feet, 137 knots and 1,200 fpm (p.28); 11:27:23.3, below Vref 132 knots (p.28); 11:27:31, 122 knots (p.28). In the seconds where speed held briefly at 134 knots, the trend arrow still pointed down (p.106).

The observer called "sink rate sir" twice, at 11:26:58 and 11:27:05 (p.27). He said afterwards that his basis was the company operations manual, which set a sink rate not greater than 1,000 fpm below 1,000 feet (p.101). The stable approach criteria and the 500-foot callout requirement were on the table too: the PM was to call "500" and the PF to answer "stable" or "go around" (p.101).

Information that only came out afterwards. None of the three crew remembered seeing HOLD displayed (p.25). In his interview the PF's account of the autothrottle mode logic did not match the manual: he said that after a manual override the autothrottle resumes control of speed (p.111). If both flight director switches are in the off position, the modes are cleared and the autothrottle goes to SPD and holds 137 knots (p.100, p.120). The autothrottle engage function is not available in FLCH SPD, and low speed protection did not exist at that moment (p.45, p.112). On performance modelling, beginning the go-around 11 seconds (124 feet) earlier would likely have avoided ground impact (p.108).

Stop point

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: 3 passengers died, one of them after being taken to hospital, and 49 people were seriously injured: 40 passengers, 8 flight attendants and 1 flight crew member. The remaining 248 passengers, 4 flight attendants and 3 flight crew members were lightly injured or uninjured (p.15, p.31). Of the 307 people on board, 99% survived (p.15). The aircraft was destroyed by impact and fire, and the sea wall and the indicator lights were damaged (p.14, p.32). The NTSB determined the probable cause as: the crew's mismanagement of the descent profile during a visual approach, the PF's inadvertent deactivation of automatic speed control, the crew's inadequate monitoring of airspeed, and the delayed execution of a go-around after the deviation was detected (p.147).

Going back to the three questions. The most common answer to the first is "go around at 500 feet". That action's cost in this case lay in the record rather than in the flying: four red on an OE flight would very likely have been reported to the company's training supervisor (p.29, footnote 23), and industry statistics show 97% of unstable approaches continued to a landing (p.101). The item most often missed in the second question is the word HOLD at 11:26:33. It appeared less than a second after a thrust reduction the crew performed themselves, it lit for 10 seconds, and the fact that the thrust levers then stopped moving matched the expectation that speed had been set to 137 knots and the aircraft was decelerating (p.106). The right place to put the checkpoint, for the third question, is the gate at 500 feet. The manual already carried the callouts

and the go-around criteria. What was missing was execution and an independent cross-check.

Options ruled out: Speedbrakes to increase the sink rate; the report records that he did not use them (p.96). Flaps 20 at 8.5 nautical miles: speed was 188 knots, 7 knots below the flaps 20 limit, and the report records that he could have done it but did not; selecting flaps earlier would have allowed a steeper descent and saved deceleration time later (p.96). Vertical navigation: the flight management computer had already computed a 3-degree profile on the localizer 28L procedure and displayed vertical path deviation on both navigation displays, and the PF said in interview that he never used it (p.117). Flying the localizer approach: a minimum descent altitude of 460 feet was available, steadier than working out the whole profile by eye (p.21). The PM taking control: he had 3,208 hours on the 777 (p.34), but the role confusion created by two captains in the same cockpit kept that path from starting (p.106). Turning off both flight directors: Asiana's informal practice required turning both off and then turning the PM's one back on; in fact only the PF's was turned off (p.26, p.120). The PF saying out loud that he was uneasy about the missing glideslope: that is not on the cockpit voice recorder, and the report judged that if he had said it, the PM might have suggested flying vertical navigation (p. 105).

Early signals: 11:23:50, sink rate fell to about 300 fpm (p.24). This was explained away as complying with the controller's speed restriction. 11:25:46, crossing DUYET at 2,250 feet, 450 feet above the 1,800-foot reference for that point (p.25); this was overridden by the note that a visual approach does not require a crossing altitude (p.23, footnote 10). 11:24:32, the observer reminded them "180 knots to 5", the PF answered "huh?", the observer repeated, and only then came "okay" (p.24). 11:24:53, the PM said "this looks a little high", the PF said "yes", the PM said "should be a bit

high", the PF asked "you mean too high", then said "I'll descend a bit more" and changed the vertical speed to -1,500 fpm (p. 24). The conversation turned a deviation into an adjustable parameter. 11:25:31, vertical speed went from -1,500 back to -1,000, a change the report judged inappropriate because the aircraft was still several hundred feet high on the profile (p.97). The observer called sink rate at 11:26:58 and 11:27:05, the PF answered "yes sir", the sink rate briefly decreased, and the observer stopped calling (p.101). 11:27:14.8, passing 500 feet at 1,200 fpm and N1 24%, with speed sitting exactly at 137 knots (p.28); compliance on speed masked thrust not being in place. On the training side: in a 2012 joint survey by Boeing, the International Air Transport Association and the International Federation of Air Line Pilots' Associations, 42% of the transitioning pilots surveyed said training on automatic flight systems was minimal or insufficient, 23% were only marginally comfortable on their first line flight, and 62% needed more than 3 months to become comfortable (p.113). On the design side: in 2010, at 787 certification, the FAA and EASA had already raised the hazard in the same logic (p.114), and a note in a manual overrode it.

The point of escalation: 11:26:33. The thrust levers went to idle and the autothrottle went to HOLD. From then on it no longer moved the thrust levers, and speed was managed by the human alone. The only function in the system that restores speed by itself is autothrottle engagement, and it is explicitly excluded in FLCH SPD pitch mode (p.45). HOLD has five exit conditions: reaching the MCP target altitude, selecting a new pitch or autothrottle mode, turning off the autothrottle switch, manually advancing the levers past the thrust limit, and disconnecting the autopilot with both flight directors off (p.45). At 11:26:43 only the PF's was turned off, so the fifth condition was not met (p.26, p.100). At 11:26:36 the PM said "speed", the PF answered "target speed 137", and at 11:26:38 set it in the speed window (p.26), which confirmed once

more that speed still had a target. From the autopilot disconnect to impact was 81 seconds (p.121); HOLD lasted until 11:27:44.7, when movement of the thrust levers cleared it, 71 seconds in all (p.30). By 11:27:14.8, passing 500 feet, the chain had no way back: idle thrust, N1 24%, 500 feet above the ground, 1,200 fpm (p.28). Nobody approved that step. The report attributes the action to the PF's hand (p.99). What approved it was an operations manual that wrote "maximize the use of automation" as policy (p.120).

Failure chain: Technical layer: the target altitude sat above the aircraft and FLCH SPD climbs by its own logic; the manual override of the thrust levers triggered HOLD; HOLD provides no minimum speed protection (p.114); HOLD triggers no aural warning, because the warning only sounds when the autothrottle is disconnected (p.115); the interface cue is a 10-second green box (p.43); speed at 500 feet happened to be back at 137 knots and held there for about 4 seconds (p.101, p.106).

Organizational layer: the training material came from Boeing's flight crew training manual and did not explain the cases in which the autothrottle will not re-engage by itself (p.112); a stall protection demonstration in the simulator left the impression that the autothrottle wakes up on its own, and the report writes that this very likely gave him false confidence (p.112); a ground instructor had met "anomalies" three times and mentioned them only verbally in class, never in the syllabus (p.112); the callout discipline was not followed at several points on this leg (p.105, p.106); two captains in the same cockpit produced role confusion (p.106); the right seat was flying as an instructor for the first time (p.35); the PF believed only the captain could call a go-around, while the manual says the first officer may also call it when flying (p.110); all three were on duty in a circadian low, and the PF had slept 5.5 hours in the 24 before the accident (p.103, p.104).

Incentive layer: OE is the check before release to the line, and four red would be reported to the training supervisor (p.29, footnote 23); the manual required maximum use of automation (p.120); 77.7% of manual landings were concentrated below 1,000 feet (p. 122), so the quota number looked good while the scenarios it covered were the wrong ones; 97% of unstable approaches were continued (p.101).

The junction breaks between the organizational layer and the in‑centive layer. Policy wrote "use automation" as the correct practice. The check wrote "traces of instability" as a deduction. Both incentives pointed the same way, so no layer had a reason to go and look at the silent change called HOLD.

Minimum counterfactual: at 11:26:43, before turning off the PF's flight director, the PM turns off his own first, or both at the same time. The report's own words: if both flight director switches are in the off position, the automatic flight guidance system's modes are cleared and the autothrottle goes to SPD and holds the 137 knots selected on the MCP (p.100, p.120). One action, 14 seconds after the autopilot disconnect, needing no extra information and no luck: both switches sit on the MCP, the PM's hand was already there, and this was Asiana's own informal practice, merely ex‑ecuted in the wrong order (p.120). From 11:26:43 to impact there were 67 seconds left, enough for the autothrottle to hold 137 knots down to 500 feet. The report issued recommendation A-14-54 on that basis (p.150). The cost is zero and the division of labour does not change: the action falls to the same hand. Set a different counterfactual beside it: "go around earlier" would have to overturn the check, and "more manual flying" would have to change the syllabus, and neither of those is a single action.

Outcome and sources: 3 passengers died, 49 were seriously in‑jured, 99% of the 307 on board survived; the aircraft was destroyed (p.14, p.15, p.31). The NTSB issued 21 new recommendations

(A-14-37 to A-14-57), addressed to the FAA, Asiana Airlines, Boeing, the airport firefighting working group and the City of San Francisco; 4 went to Asiana and 2 to Boeing. The four directly relevant here are: enhance 777 autothrottle training (A-14-37), a special certification design review of the 777 automatic flight control system (A-14-42), guidance on shutting down flight directors after disconnecting the autopilot on a visual approach (A-14-54), and a revision of automation policy to increase manual flying (A-14-55) (pp. 148–150). After the accident Asiana added straight-in approaches and speed-constrained approaches to its 777 transition syllabus (p.118). Primary source: NTSB, Aircraft Accident Report NTSB/AAR-14/01, July 2014, official PDF, 207 pages, link at Appendix B, av-031. Page numbers in this card follow the PDF page order.

One general lesson about mode awareness translates for readers outside aviation. Every automated system has a board that says which mode it is in, and the characters on that board have to pass three checks: whether they can be noticed, whether they can be read correctly, and whether someone outside the system can verify them independently. The three checks are the routine checkup for automation, and most systems only do the second, which is writing the characters correctly. The FMA on Asiana 214 lost points on all three, each for its own reason: the change was too quiet, the characters too small, and there was no external verification. Turning the three checks into a list gives the monthly sheet for an automated workstation, and the cost is one walkthrough of the workstation.

One detail of a decision point in Case One deserves its own paragraph: the delay before the go-around decision. Between the obvious anomaly of a sinking glideslope and the execution of the go-around sits a stretch of crew consultation, and that delay is the readout of how long the last link of the takeover chain took, the

chain being recognise the anomaly, reach agreement, execute the action. The go-around itself is well covered in training, since it is a standard manoeuvre. What jammed the chain was the middle step, the social cost of saying "go around" from an unequal seat. The skill problem meets the problem in Chapter 19 here: however fluent the technical links of the takeover chain, a chain that jams at the speaking link still misses.

In Case One the miss cost three lives; in an ordinary organization the same miss costs a delayed process. The size of the cost does not change the conclusion: speaking up is part of the takeover chain, and practising takeover means practising speech. The drill design for it is cheap. Fix one role in every exercise, the most junior member present, and make that person responsible for calling the anomaly out loud; the consequence of a wrong call is one line in the exercise log. After ten repetitions the muscle for speaking grows alongside the muscle for taking over. In organizations where it does not grow, the problem is usually in the incentive rather than the muscle, because the person who spoke up was once answered with a question, and the repair of incentives belongs to another chapter (Chapter 19). This chapter's job is to put speaking on the list.

Beyond the list, one everyday device for calibrating trust in automation is worth writing down: let the system fail on purpose where people can see it. It sounds like a prank and is a mature practice in reliability engineering, the source of the idea behind chaos engineering. Inject real failure shapes into the exercise environment on a schedule, and real means shapes copied from postmortems rather than textbook failures, so that users see periodically what automation failure looks like. Someone who has watched it fail moves the anchor of trust from historical reliability to the possibility of failure, and the start of takeover moves earlier with it. The cost of injection is exercise environment resources. The return is

written on Case One's bill: the price of one silent mode change would fund ten years of injection.

One discipline in the design of injection deserves emphasis: the faults injected should come from real postmortems, your own industry's or an adjacent one's, and should not be invented. An invented fault tests the system's theoretical weaknesses. A real fault tests its actual weaknesses, and the two distributions rarely overlap, since faults in postmortems are always stranger than designers imagined. The dark-line archive of Chapter 21 supplies material here directly, because a dark-line fault carries both realism and surprise. The two chapters' devices mesh at this point: a mapped dark line is a topic in the injection script library.

A second supply route for the library comes from the user side. Automatic client behaviour such as retries, caching and local queueing is a new failure source in the automated era, and the retry storm in Case Two is user automation fighting platform automation. Injection of client behaviour, simulating thousands of users retrying at once, has mature tooling in the software industry, and its equivalent outside software is simulation of collective demand-side behaviour, with passengers rebooking at once or patients arriving at once. Adversarial drills with automation on both sides, the platform injecting faults and the client injecting a storm, are the full-stack rehearsal of the automated era. The era of rehearsing the server side alone is over.

Alongside that, one new regression has to be guarded against: automating the drill itself, with one-click exercises, generated faults and automatically scored results. A fully automated drill saves labour and quietly reproduces this chapter's mechanism, because the hand feel of the drill decays too and the people who design drills stop taking faults apart by hand. The right place for drill automation is the execution layer, where scripts run the routine injection. The design layer has to stay human, with each

quarter's script written by hand once, because the writing is the designer's own takeover practice. Execution can be outsourced to a script. Understanding cannot. That is the general discipline of the automated era, and the general demand this chapter's mechanism makes of every tool.

That discipline has another face, which is an opportunity. The spread of automation has pushed the marginal cost of drills to a historic low, with simulation environments billed by the hour and fault injection copied by script, so the bottleneck in training supply has moved from resources to willingness. The organizational answer on willingness sits in budgets and appraisals, which are this chapter's Device 1 and the audit list. The cultural answer sits in narrative: tell takeover drills as capability building rather than as a cost centre. The fastest way to start that shift is to write up every successful takeover as a case, who held what under what conditions with which combination of actions, and put the case in the internal knowledge base and on the meeting agenda. The visibility of successful takeover is the fuel of a takeover culture: an organization that can see its heroes trains heroes.

A second half of the hero story has to follow at once, or the story spoils. Visibility of heroes does not mean dependence on heroes. The healthy shape of a takeover culture is that everyone has practised takeover, with the quota covering the whole staff, and the unhealthy shape is a few heroes covering the field while the rest watch. An organization built that way resets to zero on the day the hero leaves (the mobility mechanism of Chapter 28), while an organization where everyone practises has a substitute when any one person is absent (the cover matrix of Chapter 18). The watershed between the two cultures is visible in the drill roster: does the list rotate, with everyone on it, or is it fixed, with the specialists on it? Distributed takeover ability is part of organizational redundancy, and the principle of real rehearsal in Chapter 22 applies to re-

dundancy in people as well: the real rehearsal of redundancy is a rotating drill roster.

Rotation carries a second function, across skill generations: veterans and newcomers drilling on the same floor. A mixed roster lets a newcomer learn sequence from a veteran's takeover, who throttles first and who rolls back first, and lets a veteran re-examine habits under a newcomer's questions about why that step comes first. Transmission between generations is the original channel by which skill survives, which is apprenticeship, and the drill institutionalises it into a transmission with a ledger. The ledger does not record who mentored whom, which is a personnel matter. It records the content transmitted, as in which three sequencing judgements this drill passed on, and the list goes into the knowledge base as teaching material for the next generation of newcomers. Organizational archiving of skill closes the loop at that point: individuals practise through quota, teams rehearse through rotation, the organization stores through the ledger, newcomers read through the material. Only when all four turn does takeover ability become an organizational asset rather than an individual gift.

Superstition about gifts is this mechanism's last stronghold at the cultural layer, and it deserves one deliberate demolition. A person who takes over well is often described as naturally calm, or quick to react. Temperament contributes, certainly, and close reading of postmortems gives a plainer answer: people who take over well are people fluent in sequence, who have rehearsed the order of the combination in their heads, and fluency in sequence comes from the density of practice and the quality of the review. Temperament sets how far performance decays under pressure. Practice sets the height of the baseline, and an organization with a high enough baseline still clears the pass mark after decay. An organization that attributes takeover to gift therefore skips the most important task,

which is rostering the practice, and does the most useless one, which is recruiting for gift. The position here matches Chapter 14: skill can be measured and practised, gift can be neither, and the budget goes to the half that can be practised.

Within the practicable half, one principle borrowed from sports training belongs in the design of practice: layer the scenarios by realism. The base layer drills decomposed actions, the correctness and speed of a single step. The middle layer drills combined se - quence, the order and trade-offs of several steps. The top layer drills the whole scenario under pressure, with incomplete in - formation, time pressure and concurrent alarms. The recommen - ded time split is an inverted pyramid, with the top layer most ex - pensive and least frequent, since one top-layer exercise returns more than ten base-layer ones. The reason for layering is the transfer rule of skill: fluency in decomposed actions does not transfer by itself to combination, and fluency in combination does not transfer by itself to pressure. Transfer is bought one layer at a time, and the currency at every layer is exposures. The whole eco - nomics of this mechanism is buying enough exposures at each of the three layers.

One invisible opponent sits in the budget for those exposures: the training package sold with the automation. Procurement of an automation system often comes with training, a vendor engineer walking through the operation, and that training is positioned as an introduction to operation rather than takeover training, since the vendor has no postmortem library for your industry and can - not design scenarios at the takeover level. An organization that counts it in the training budget gets a falsely good reading on the first item of the audit list, the training hours curve. The definition of the training budget therefore has to be written into the rules: takeover training hours count only exercise hours built on scen - arios from your own industry’ s postmortems, and vendor training

is listed separately and not counted. One line of definition guards against substitution in the budget and forms the foundation of audit credibility.

On that foundation, an annual report on takeover ability is worth establishing as a fixed document. Four pages, one thing on each: quota completion, drill coverage, red and amber audit items, and the closure rate of the jam ledger, one page one chart one table, ten minutes for the governance layer to read. The design of the read - ership has a hidden use: the report is ammunition for budget ne - gotiation, since red items map straight onto budget requests, and it is evidence of where responsibility was discharged, since an or - ganization that reported a red item and had its budget cut stands in a very different position at the next accident review. Combined with the margin ledger of Chapter 25, the annual report becomes a joint document, a double annual report on defence and skill, and the organization that produces it gives the governance layer its first complete view of defence: what the system can do, what the people can do, and how much each can still absorb.

The double report’ s final page is left blank for the reader, as the book’ s convention requires. Write down three numbers: the automation coverage of your system, as the proportion of actions completed automatically; your own count of takeover practice this year; and your subjective rating of familiarity with manual mode. The combination reads in two lines. High coverage with little practice means every risk in this chapter is waiting for you. High coverage with plenty of practice means you are already the person this chapter set out to shape: automation works for me, and I stand watch on its boundaries.

The watch roster can be refined down to the four links of the takeover chain: recognise the anomaly, exit automation, execute by hand, restore automation. Each link has its own way of being practised. Recognition drills alarm triage, picking the true signal

out of noise. Exit drills mode operation, the button sequence that hands the system back to human control. Manual execution drills psychomotor skill, the old subject the quota governs. Restoration drills handback, the correct posture and confirmation actions for returning the system to automation. Practice on the four links is often merged into one run, a single exercise from recognition to restoration. A merged run measures the whole; a layered exercise trains the links. An annual plan needs both, merged runs to measure the composite and layered exercises to fix the weak link, and the information about which link is weak comes from the jam ledger of each merged run: whichever link jammed most often becomes next quarter's layered priority.

The jam ledger carries one recording discipline worth writing down. Classify the cause of the jam when you record it, as not knowing what to do, knowing what to do but not being able to do it, or being able to do it but not daring to. The prescriptions for the three differ completely: the first is a problem of understanding boundaries and goes to Device 4, the second is psychomotor skill and goes to the quota, the third is a problem of speaking and au‑thorization and goes to Chapter 19. A ledger that mixes them keeps every drill review at the surface, as in "it jammed for five minutes here". A ledger that classifies them takes the review straight to the prescription, as in "this is the third kind, go and look at the in‑centive". The label costs one field at recording time and buys a direct channel from drill output to training action.

The channel has one more station at its end: verification that the training worked, meaning whether the point that jammed last time still jams at the next drill. The logic is as plain as any training evaluation, since practice should produce a change, and the action is a closure rate statistic over the jam ledger, the share of entries that no longer jam on retest. Low closure means either the method is wrong, with the layering badly chosen, or the dose is too small,

with too few exposures, or the target is wrong, with the wrong people trained. Each of the three has its own diagnostic path, change the layer, add volume, change the people, and the finding goes back into the ledger. The ledger then becomes the training system's own feedback loop, and this mechanism's ironic loop, in which automation eats training, is offset by a loop running the other way, in which training eats review. The long-run result of the two offsetting is a dynamic equilibrium of takeover ability: automation coverage rises, training supply rises with it, and the balance point is set jointly by budget and will.

Turnover is the final variable at the balance point, and it links to Chapter 28. The faster people move, the higher the training supply has to be, since the cost of rebuilding the stock in newcomers is continuous. High-turnover organizations, in consulting, outsourcing and rotation-heavy industries, therefore have to design takeover training as a production line: practise from the day of joining, quota by month, and a recorded handover drill at departure. The line costs more than a stable organization's annual plan, and its return is written in the handover checklist of Chapter 28: the people who leave do not take the organization's takeover floor with them.

Documentation standing in for people is the final line of that floor. Once the key knowledge of takeover is archived organizationally, meaning the scripts, the jam ledger, the four-link list and the one-pager, the organization's dependence on any individual drops to a substitutable level. The test of substitutability is a random name drill: assign any member to the takeover seat, once a quarter, and whoever is named takes it regardless of seniority. The deterrent value of the random name drill exceeds its training value, because it keeps everyone in a state of being nameable, and the by-product of the deterrence is that the three levels of practice become normal for everyone: knowing, being able to do, daring to step in. A team

with all three is the best finished product this chapter's defences can buy. The price is a few hours of drill time per quarter, and the warranty is written in the result of every real takeover.

At this point the chapter's two main cards, Asiana in aviation and Cloudflare in software, have shown both faces of the mechanism: one is the heavy tuition bought by decades of automation history, and the other is a bill from the new generation of automation that has only started arriving. The blank between them, the other domains, is held by three extrapolations, and the honesty of the placeholder is written in the note on the evidence gap. The hope attached to the placeholder is that a later edition can put three real cards in this section. The precondition is that those domains start accumulating postmortem libraries, and the first entry in such a library can start with the reader's organization's next takeover drill.

Beyond drills, the chapter owes readers a word to the designers of automation, because half the responsibility for keeping takeover ability fresh sits on the design side. Three points for designers. Treat manual mode as a product, a formal function rather than a debugging entry point, tested for usability to the same standard as the automation. Treat state changes as events, presented conspicuously, verifiable, traceable, with the three checks run before the factory gate. Write the takeover plan into the deliverable, with every function carrying the manual takeover path for its failure and one page for it in the delivery checklist. All three costs land in the design phase, which is the cheapest moment. All three absences land at the takeover moment, which is the most expensive. The faster automation iterates, the more these three matter: the system changes every year, and the human takeover path cannot break every year.

The closing word goes back to the human side. The two words in this chapter's title, automation and skill decay, are two halves of

one multiple-choice question. Automation is the compulsory part, since the gains in efficiency and safety are real and going back to manual is not an option. Skill decay is the elective part, since how far it decays, at which link, and what replaces it are all designable. An organization that answers the question well can summarise its relation with automation in one sentence: machines do what machines are good at, people do what people cannot be replaced at, and the handover between them has a set of actions both sides have practised. That set of actions already has a name in this chapter: takeover.

One footnote remains, for the time dimension of takeover. Takeover moments are unevenly distributed across a system's life, frequent early after launch, rare in the stable period, frequent again at replacement, and the practice roster should follow that distribution: drill exit and manual control at launch, recognition and alarms in the stable period, plan review and full-chain drills at replacement. The reading of the distribution comes from the change calendar, the schedule of upgrades, migrations and refactors, and binding the roster to the calendar makes training supply follow the rhythm of change automatically: the system changes, the practice changes with it, and takeover ability stays pointed at the current version of the system. This is the only mechanism among this chapter's defences that needs continuous maintenance, since the rest run for a long time once installed, and the maintenance is one action: after each change, look at the roster and change two lines.

Authority to change those two lines should sit with the shift leader, the person who knows the current version's hand feel best, and the record of the change goes into the attachment to the change ticket, under the same clause as the takeover plan review. Devolution is justified because the hand feel of a version is known only on the line. The tightening is justified because a devolved change

must leave a trace, and an untraced change is new drift (Chapter 15). Between the loosening and the tightening, takeover training stays in step with the system's evolution. This is the least conspicuous and most durable of the chapter's devices, and its name can simply be: change two lines.

Among the two lines changed, one is best left unchanged forever: the first person responsible for takeover ability is always the person on shift, not the system, not the procedure, not the score of the last drill. What the system can do is supply the conditions that keep that person's hand feel fresh, which are quota, drills, documentation and budget. The suppliers of those conditions and their consumers are two different groups, management and the line, and every conversation between them is a translation of this chapter, with management reading audits and budgets and the line reading quotas and scripts. The quality of the translation decides the quality of implementation, and implementation is inspected in the five questions of Section 7. All five ask the line, because takeover finally happens in the hands on the line. However good the upstream design, unfamiliar hands leave everything as it was; with familiar hands, takeover arrives in time even under a rough procedure.

Redundancy of hands on duty, at least two pairs, is what guarantees that takeover arrives in time. The worst shape of a takeover moment is the only experienced person being absent, on leave, departed, or already inside another fault, and two experienced people on shift is the staffing floor for takeover, with the minimum headcount on the roster computed from the division of labour in the takeover chain rather than from normal operations. The line that writes that floor into the rostering rules is this chapter's last line: staffing in the automated era is set by the worst shape of the takeover moment, not by the ordinary shape of the day. The extra pairs kept on in quiet times look like idle posts and read as lives at

the takeover moment. The defence against it sits in Chapter 19's armoury: the right of any seat to call upward without going through the chain, where calling a go-around needs no justification, and blameless recording afterwards. The training list for takeover ability therefore gains one soft subject: practice at speaking up.

Case Two (Software): Takeover in a Retry Storm

Case ID: sw-004.

When and where: 12 September 2025, 16:32 to 19:12 UTC. The impact window ran from 17:57 to 19:12, 75 minutes in all, in three segments: a first segment of 20 minutes, a middle segment of 41 minutes and a last segment of 14 minutes. The location is Cloudflare's global control plane: the dashboard and a set of APIs. Tenant Service is a Go process running on Kubernetes in some of the data centres (postmortem, Timeline, How we responded).

What and who: Cloudflare's Tenant Service. It sits on the authorization path of API requests: every request's authorization is decided by it, and a request it cannot decide returns a 5xx (postmortem, What Happened). The parties involved were the dashboard frontend team, the API platform team, the on-call SRE, and the automated alerting service that pulled responders into a call. The affected surface was the dashboard and a set of APIs. Isolation between the control plane and the data plane is strict, and services on the edge network were unaffected (postmortem, How we responded).

What they were doing: at 16:32 a new version of the dashboard went live (postmortem, Timeline). This version contained logic that used a React useEffect to manage calls to the /organizations endpoint. useEffect decides whether to run again from its dependency array: when a value in the array changes, the function executes

once more. In this version the dependency array held an object that was recreated on every state or prop change, so React treated it as new every time, reran the function on every render, issued several calls within one render, and retried when a call failed (postmortem, What Happened).

From 16:32 to 17:57 that version ran in production for 85 minutes and produced no visible failure on its own. At 17:50 a new Tenant API Service was deployed, and 7 minutes later the service was overwhelmed (postmortem, Timeline). The postmortem's word - ing for that stretch is that the Tenant API Service was overwhelmed during the deployment of the new version.

On release tooling, Cloudflare was already migrating services to Argo Rollouts, which watches deployment errors and rolls back automatically. Tenant Service had not yet been migrated, and the migration was scheduled (postmortem, Reducing impact). On ca - pacity, the Tenant Service quota was set for normal conditions and not for a spike of this kind (same section). On monitoring, API us - age was visible at the time, but whether a request was a retry or a new request could not be seen (postmortem, Improving visibility).

The decision point: 16:32, the dashboard version released, carried on a release ticket (postmortem, Timeline). The defect took the form above. The postmortem records neither the reviewers, the approvers nor the sign-off for that release, and does not record whether it went through a canary.

17:50, the Tenant API Service deployed, carried on a deployment ticket (postmortem, Timeline). The postmortem does not state what that version changed, only that the service was overwhelmed during the deployment. The two releases were 78 minutes apart and each went through its own process independently.

After the overload at 17:57, the first step in handling it was to re - duce load and add capacity. The postmortem writes the basis for

that step as a principle: the first goal in an incident is to restore service, and fixing the root cause is the common path but need not be the first step (postmortem, How we responded, original text "Our first goal in an incident is to restore service. Often that involves fixing the underlying issue directly, but not always."). At 18:17, after scaling, the API was back to 98% availability and the dashboard had not recovered (postmortem, Timeline).

At 18:58 they removed some erroring code paths and released a new Tenant Service (postmortem, Timeline). The basis was inference: the service reported healthy and the API had largely recovered, but large numbers of errors were still being logged, and the team inferred that those errors were the cause of the dashboard's continuing unavailability (postmortem, How we responded, original text "We theorized that these were responsible for the ongoing Dashboard availability issues"). That release affected the API a second time, and the timeline marks IMPACT INCREASE at that point.

At 19:01 a temporary rate limiting rule was published, and at 19:12 the problematic Tenant Service change was rolled back and the dashboard returned to 100% (postmortem, Timeline). One inconsistency in the account should be noted: the narrative places global rate limiting alongside scaling as load reduction in the same round, while the timeline records the rate limiting rule at 19:01. This card follows the timeline.

The design side holds an earlier decision: Tenant Service was not enrolled in automatic rollback. Its time carrier is the migration plan, and the postmortem writes only that the migration was scheduled, with no date (postmortem, Reducing impact). The external postmortem records no approver for any change, so that column has to stay empty.

What was visible then: at 17:57 the dashboard availability curve began to fall (postmortem, Timeline). The Tenant Service usage

curve rose abnormally (postmortem, How we responded). The API's 5xx ratio, and the 98% reading at 18:17 (postmortem, Timeline). "Substantial" errors in the service logs (postmortem, How we responded). The pod count and throughput before scaling. The automated alerting brought the right people into the call early in the incident (same section).

One phenomenon that showed up during the handling: after Tenant Service was restarted, every user's dashboard re-authenticated to the API and the API went unstable again (postmortem, Reducing impact). One gap was visible in the same way: monitoring could not distinguish retries from new requests (postmortem, Improving visibility). The postmortem records no action between 17:57 and 18:17 other than scaling, so the only verifiable readings in those 20 minutes are the curves. In the 3 minutes after 18:58, the 19:01 rate limiting rule landed, and at that moment the curve was still heading down (postmortem, Timeline).

Information that only came out afterwards: the root cause lay in the dependency array of that useEffect in the dashboard (postmortem, What Happened); the amplification once repeated calls and retries on failure were compounded (the postmortem gives no figure); and that if Argo Rollouts had been in place, it would have rolled back the second Tenant Service update automatically, "limiting the second segment" (postmortem, Reducing impact).

Stop point

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: the dashboard and the API were affected for 75 minutes, and the API was severely affected in two of the segments, corresponding to the two periods when Tenant Service was unavailable (postmortem, Timeline). Taken apart, 17:57 to 18:17 is 20 minutes, 18:17 to 18:58 is 41 minutes, and 18:58 to 19:12 is 14 minutes. The last segment was manufactured by the response itself: the only change in those 14 minutes was the 18:58 patch. At 18:17, when the API returned to 98%, the dashboard still showed zero recovery; after the 19:12 rollback the dashboard returned to 100%. The data plane was unaffected, and users who did not change configuration or use the dashboard were unaffected (postmortem, How we responded). The postmortem reports no data loss.

Going back to the three questions. The most common answer to the first is "scale first". That is what the case did, and the postmortem's assessment of it is that it helped availability but was not enough to restore normal service (postmortem, How we responded). The cost of the action was that the first 20 minutes of the response went into adding resources, with root cause work queued behind it. The item most often missed in the second question is that retries could not be distinguished from new requests. It reads like technical debt in a log field, and it is in fact the only entry point to the root cause: the postmortem says that had they known they were looking at a sustained flood of new requests, they would have identified the loop in the dashboard sooner (postmortem, Improving visibility). The right place to put the checkpoint, for the third question, is 18:17, and the thing to check is the divergence between the API curve and the dashboard curve. Authorization is decided by Tenant Service across the board, so the server side as a whole returning to 98% while the dashboard shows zero recovery means the source pressing on it is still there, and is not on the server side.

Options ruled out: locating the root cause before restoring service; the postmortem writes that fixing the root cause is the common path but need not be the first step (postmortem, How we responded). At 17:57 the root cause had not been located, so this path needs a conclusion that did not exist at the time. Rolling back the dashboard version from 16:32: that step is not in the timeline, and the 19:12 rollback was of the Tenant Service change (postmortem, Timeline). Relying on automatic rollback to catch it: Tenant Service was not enrolled in Argo Rollouts (postmortem, Reducing impact). Scaling without rate limiting: scaling was done, and the postmortem judges it insufficient (postmortem, How we responded). Treating a service restart as the recovery method: it was done, it triggered a thundering herd of re-authentication and pressed the API again (postmortem, Reducing impact). Locating the fault from logs: with retries indistinguishable from new requests, the logs cannot show the shape of the loop; the postmortem says knowing they were seeing a sustained flood of new requests would have made the dashboard loop easier to identify, and that sentence is exactly the conclusion the logs of the time could not supply (postmortem, Improving visibility).

Early signals: the 16:32 version would "trigger far more calls", a sentence written in the timeline (postmortem, Timeline). It was overridden by "new version traffic ramping up", and the 85 minutes that followed with no consequence let the signal suppress itself. The two changes were 78 minutes apart, at 16:32 and 17:50, which was overridden by "each reviewed independently". The deployment at 17:50 to the overload at 17:57 is an interval of 7 minutes (postmortem, Timeline), overridden by "jitter during a deployment". Automatic rollback was not yet migrated and the migration was scheduled, overridden by "the migration is in progress". Retries were written into the release content, and the backoff and jitter that belong with retries were absent (postmortem, Timeline, original text "including retries in the event of

failure"). Capacity was not configured for a spike and there was no capacity alert (the postmortem writes that monitoring was being improved so as to "proactively alert before reaching capacity limits", which means none existed then), overridden by "it has never happened". The divergence at 18:17, with the API at 98% and the dashboard at zero recovery, was overridden by the good news that the API had recovered.

The point of escalation: 18:58. Removing the erroring code paths and releasing a new version rested on inference, not verification (postmortem, How we responded). In the 61 minutes before it, the curve had one direction, backwards, and after it the API fell a second time (postmortem, Timeline, IMPACT INCREASE). The action acted on errors, while the input to the fault was call volume: removing erroring code paths did not reduce call volume at all. The step turned a single rollbackable point into two impacts, and the remedy for the second impact was the same thing: rollback. Only after the 19:12 rollback did the dashboard return to 100% (postmortem, Timeline). From 18:58 to 19:12 is 14 minutes, and those 14 minutes produced no new information, only new failure surface. The postmortem's characterization of that update is "a bad change" (postmortem, Timeline, original text "This was ultimately a bad change"), and it shares a timestamp with IMPACT INCREASE. The approver is not recorded.

Failure chain: Technical layer: the dependency array contained an object rebuilt on every render, producing multiple calls to /organizations within one render; a failed call was retried, with no backoff and no jitter; Tenant Service was overloaded and authorization could not complete, so the API returned 5xx across the board; the 5xx responses triggered more retries, a positive feedback loop; and the restart made every user re-authenticate, producing a second impact (postmortem, What Happened, Reducing impact).

Organizational layer: two changes, frontend and backend, went live separately with no joint review; a service sitting on every API call path was configured at the capacity of an ordinary service; it was not enrolled in automatic rollback; monitoring could not distinguish retries from new requests; and the patch went out before the root cause was verified (postmortem, How we responded, Improving visibility).

Incentive layer: the benefit of release frequency is current and the risk of coupling is deferred; investment in capacity and automatic rollback had no bill before the incident; and "the service reported healthy" was treated as evidence that the job could be closed (postmortem, How we responded).

The junction breaks between the organizational layer and the incentive layer. Tenant Service's position on the request path means its capacity, release tooling and monitoring should have been rated as a core component. It was rated as an ordinary internal service, and that rating cell has no owner.

Minimum counterfactual: before 17:50, enrol Tenant Service in Argo Rollouts automatic rollback. The postmortem's own words: if that mechanism had been in place, it would have rolled back the second Tenant Service update automatically, "limiting the second segment" (postmortem, Reducing impact). One action, with no luck involved: the tool was already in use, only one service was missing from it, and the migration was already scheduled at the time. It shortens the incident rather than eliminating it: the first segment still happens and the second does not, the 14 minutes of the last segment come off the bill, and the responders do not have to change production again at the end of their most tired hour. The cost is migration hours, and the bill before the incident was zero. The same fault with a dependency-array fix as the counterfactual covers a wider scope, but it demands distinguishing retries from

new requests within 75 minutes, and monitoring could not make that distinction.

Outcome and sources: 75 minutes, two segments of API impact, the dashboard severely affected throughout, the data plane unaffected (postmortem, Timeline, How we responded). The postmortem divides the improvements into two kinds: reducing the impact of similar changes, and improving observability. It also records two things that limited the impact: automated alerting found the right people quickly, and control plane and data plane isolation kept edge services unaffected (postmortem, How we responded). Five remediation items: migrate Tenant Service into Argo Rollouts and raise its priority; add randomized delay to dashboard retries so that retries spread out (the hotfix went out after the impact ended); scale up and add capacity alerting; tag requests as retry or new; deduplicate dashboard calls (postmortem, Reducing impact, Improving visibility). Primary source: Cloudflare Blog, "A deep dive into Cloudflare's September 12, 2025 dashboard and API outage", published 13 September 2025, author Tom Lianza, link at Appendix B, sw-004. That postmortem is an external account and attaches no change tickets, approvers or request volume figures; this card writes only the verifiable timestamps and the sections in which they appear.

One organizational observation about the combination of actions in Case Two belongs here. The decision chain in the response, the order of rate limiting, scaling and rollback, is assembled on the spot in a real incident by the individual on call. That assembly depends on experience, meaning what was done last time, and the full set of combinations, meaning which orders work, has never been institutionalised as a script. Scripting costs the work of sorting the order and effect of past responses into a one-page decision tree, and it buys a takeover moment that starts from a lookup rather than a recollection, the same move as the triple index in

Chapter 21. Script plus game day is the complete training loop for takeover ability in the software industry: the script supplies the lookup, the game day supplies the muscle, and the outputs of the two calibrate each other.

Note on the Evidence Gap

Verification cases for criterion 1 outside aviation, meaning marine autopilots, medical device default parameters, and operator takeover in industrial control, are missing from this book's collection scope in full grade A form. The cause of the gap is the mechanism itself: events of this kind have not yet entered the investigation systems of domains where automation penetration is shallow in large numbers. Directions for further collection are listed in, which covers regulatory reports and accident libraries on automation by domain. The gap is marked as it stands and does not affect the generality of the criteria or the defences, which come directly from the long literature on human factors in automation and from the cases above.

Beside the gap, the chapter offers extrapolations in three domains, marked explicitly: what follows is mechanism extrapolation, not case cards, and will be replaced by real cards once collection catches up. Extrapolation one, marine autopilot: after the autopilot takes over routine steering, the helmsman's manual steering practice decays by the month, while narrow channels and collision avoidance demand immediate hand feel; the check form for criteria 1 and 3 here is a quarterly quota of manual helm and simulated takeover drills. Extrapolation two, medical device default parameters: after infusion pumps and monitors handle routine configuration through their default parameters, nurses' familiarity with what the parameters mean and how to adjust them decays, and the manual parameter change in an anomaly happens at the busiest moment of the shift; the check form for criterion 3 is a

drill on changing parameters outside the defaults, written into training assessment. Extrapolation three, industrial control: the automation coverage of DCS operators rises year by year, and the execution frequency of high-load manual sequences such as plant start-up drops to once in several years under long-cycle operation; the check form for criterion 1 is simulator refresher training on the start-up sequence and archived operating video of veteran operators. The three extrapolations share one chain of criteria, and turning them into real cards is an open item on this book's collection list.

Beyond those extrapolations, one kind of automation now in progress deserves early monitoring: AI-assisted judgement automation, in aided diagnosis, aided writing and aided coding. It differs from classical execution automation in one key respect: what it takes over is judgement rather than hands, so the decay happens in the texture of judgement, in the feel for the right answer and the smell of a wrong one. Takeover ability under this kind of automation is harder to measure than under execution automation, since the question is how to set a quota for judgement, and a workable start is to keep a control sample for manual review, taking a number of cases each month and running them entirely by hand beside the AI output, where the difference between the two is the stock reading for judgement skill. Measurement methods will be forced into existence by the spread of AI automation, and the framework of this chapter's criteria, a measurable stock and a loop that can be broken, is directly usable as their first draft.

26.4 What Shows Up Earliest

Six weak signals, each with the reason it usually gets overridden.

1. Simulator and hands-on training hours fall year after year, and the stated reason tracks automation coverage. The reason: "training follows the job."
2. "The system will handle it" appears where a training document leaves something out. The reason: "focus on the increment."
3. A minimum requirement for manual execution, such as a pilot's manual landing quota, exists but carries no assessment record. The reason: "we have a procedure."
4. The handling of a takeover scenario is rated in a postmortem as correct but slow. The reason: "the outcome was good." The slowness is itself the decay reading.
5. New employees find the system's manual mode unfamiliar, or do not know it exists. The reason: "they will not need it."
6. After an automation change, a version upgrade or a logic adjustment, the takeover plan is not updated with it. The reason: "functionally equivalent."

Put the six back against this chapter's two cases and three extrapolations. Signal 1, falling training hours, matches Case One: what the training side of Asiana 214 shows is an industry-wide footnote to this signal. Signal 2, omissions in training material, matches extrapolation two: training in the era of default parameters is left with nothing but introductions to new features, and the manual path for a parameter has disappeared from the teaching material. Signal 3, a minimum requirement with no assessment record, matches Case One: organizations that hold a manual landing quota and keep no record of assessing it are not few, and they are more common outside aviation. Signal 4, correct but slow, matches Case Two: slowness is the measurable reading of decay, and a post-

mortem should record the time each step of the response took and align it against history. Signal 5, unfamiliarity with manual mode, matches extrapolation three: on plants running long cycles, the DCS manual mode is close to being lost. Signal 6, a plan not updated after a change, matches all of them: it is the high-frequency companion symptom of automation, and every upgrade should trigger a review of the takeover plan.

Order the six by the cost of finding them. Signals 2 and 3 are cheapest, since they mean reading training material and quota records. Signals 4 and 6 are middling, since they mean reading postmortems and change tickets. Signals 1 and 5 are most expensive, since they need training ledgers and interviews on the floor. The recommended start is signal 3: pull out the list of minimum manual requirements, look for an assessment record against each one, and treat a requirement with no record as a requirement that does not exist, which is criterion 3 carried out in plain words.

26.5 How to Defend

Device 1: a manual quota. Set a minimum frequency quota for manual operation of key systems, with aviation's manual flying quota as the mature model, and make quota completion an assessable metric: whether a person can still do the thing is a quantity to be measured, not an assumption. The quota's scenarios have to cover the whole takeover chain, recognising the anomaly, exiting automation, executing by hand and restoring automation, and all four are required.

Device 2: simulation of takeover scenarios. Rewrite takeover scenarios from postmortems into simulated exercises, with the software industry's game day and chaos engineering as the same class of thing, and cover the takeover list once a year. The exercises use the real material of a postmortem, at a cost far below the real

accident, and the output of an exercise, where it jammed and who did not know which button, goes into the ledger and becomes input to the next round of training.

Device 3: thresholds and cues for takeover. An automated system should give a clear entry signal for the state in which a human is needed, through a mode change, a fall in confidence or a combination of concurrent alarms, and the takeover plan is updated in step with the signal. After every change to the system, the takeover plan is reviewed under a mandatory clause, which is the cure for weak signal 6. The lesson of Asiana 214 takes this shape here: a state change has to be presented in a form human attention can catch, rather than as a line of annunciator characters that can be reasonably read past.

Device 4: write automation knowledge back into people. Every automation function carries a one-pager on what it does, what it does not do, and what you do when it fails, and training and refresher training follow that page. A person's understanding of the system is the base of takeover ability. Without an understanding of where automation stops, takeover starts after the correct moment. With it, takeover can start early, at the system's limit.

Set the four devices against this chapter's cases and extrapolations. Device 1 on Case One: quotas for visual approaches and manual flying assessed separately for captains and first officers, with the quota starting at the first officer's first line release. Device 2 on Case Two: the combined handling of the retry storm rewritten as a game day script, with the decision tree for rate limiting, scaling and rollback entered in the register. Device 3 on extrapolation two: mode changes on the infusion pump presented with a conspicuous cue, and the training on the parameter change path sourced from the same place as the cue. Device 4 on extrapolation three: a one-pager for the start-up sequence, covering

what it does, what it does not do and what you do on failure, stored with the operating procedures and entering refresher training.

Now the failure modes of the four devices. The quota dies on watered quota, with manual operations completed in unstressed periods, the easiest landings and the calmest sea states, so the quota is met and the difficulty of the takeover scenario has never been practised. The patch is to grade the quota by scenario, with difficulty tiers and a minimum share for the high tier. Simulation dies on the gap between script and reality, with game day faults that are too clean and too single-threaded. The patch is to rewrite scripts from real postmortems into compound faults, with concurrent alarms and incomplete information. Thresholds and cues die on alert fatigue, since a conspicuous cue that appears every day becomes background, the cry wolf effect. The patch is to grade the cues, keeping the ones that genuinely need takeover rare and denoising the routine ones. Writing knowledge back dies on a one-pager that is never updated, so that after an upgrade the page still describes the previous version. The patch is to bind the one-pager to the change ticket, with the ticket forcibly triggering a review of the page under the same clause as the takeover plan review. The four patches share one check sentence: in the most recent takeover practice, were the steps that jammed fed back into training?

An order among the devices is also worth fixing: Device 4 first, which is understanding the boundary, then Device 1, which is practising the action. The reason sits at the cognitive layer. Manual practice without an understanding of the boundary is blind practice, and a person who does not know when to take over practises perfect actions at the wrong moment. Once the boundary is understood, quota and simulation have the correct target, which is starting the takeover at the correct moment. Organizations whose training budget is growing should hold this order most strictly: buy the one-pager first, since it is cheap, and simulator hours after,

since they are expensive. The inverted order spends half the budget on blind practice.

26.6 Boundaries with Neighbouring Mechanisms

Against practical drift (Chapter 15): drift is practice departing from the text, an active divergence, while decay is capability departing from the requirement, a passive fading. In an automated setting the two cause each other, since the text updates more slowly than the system, which is drift, and people leave practice as the system moves on, which is decay. Once decay has set in, the distance between practice on the floor and the text is wider than any deliberate deviation would produce.

Against the illusion of redundancy (Chapter 22): automation is often counted as redundancy in people, on the reasoning that the system will handle it and people need not learn it, and the moment a person is crossed off the defence list this mechanism has every condition it needs. Whether people count as redundancy is a question for Chapter 22's verification standard: has a real switchover ever been made, and has takeover ever been rehearsed for real?

Against where intuitive expertise breaks down (Chapter 14): decay is the time version of the failure zone. Once feedback is cut off by automation, the preconditions of intuition, which are regularity and feedback, are systematically dismantled. Chapter 14's admission table should trigger on every row by default in an automated environment.

26.7 Five Self-Checks

1. When was the last composite fault in your system that a human took over? When was that person's previous practice?

2. Is your organization's training hours curve rising or falling? Which reason is written next to the fall?
3. Does a takeover scenario list exist? When did you last practise using real postmortem material?
4. In your automation documentation, is "what it does not do" written down? Pick a function at random and ask one of its users.
5. If every automation stopped tomorrow morning, who in your team could start the plant? How many names are on the list, and what is their average age?

(This chapter: 16,514 characters.)

Chapter 27: Sunk Cost and Escalation (Staw)

> What this chapter covers: how money already spent takes the next decision hostage; Staw's classic experiment, the criteria, three field sites, and the design of exit conditions up front. When to turn back: at a project renewal review, on a motion to top up the budget, or when you hear "just a little longer". Reading time: 50 minutes.

The experimental evidence for the sunk cost effect is fifty years old. Barry Staw's 1976 experiment in Organizational Behavior and Human Performance ("Knee-deep in the big muddy", volume 16, pages 27–44, Crossref record verified) put subjects in the role of head of a corporate R&D project. Half were told the project had failed and the responsibility was theirs. The other half were told responsibility lay with other people or with circumstances. Everyone was then given the chance to decide whether to commit more. The result held, and it is uncomfortable: the group carrying the responsibility chose to commit significantly more often. Admitting the earlier misjudgement cost too much, so future money went on keeping past standing alive. Fifty years on, the conditions of that experiment can be reproduced as they stand in any boardroom, command post or project review.

27.1 What It Is

Definition: the sunk cost effect means a decision driven by past investment instead of future return, where the size of what has already gone in becomes itself the argument for putting more in. Chapter 8 split escalation into seven rungs (the process view: explain it away, shrink it, find a scapegoat, promise more, change the

measure, enlarge the commitment, seal off information). This chapter returns to the decision point: why exit is so expensive in the mind and in the institution, and how to make it cheaper.

The criteria, all three present and the escalation structure stands:

① The investment already made is publicly visible, on the books, in buildings, in promises already published.

② The main argument for new money is "we have put in this much already", rather than a calculation of the return on the addition.

③ Exit is treated as an admission of error, and there is an identifiable person who pays for the admission.

Criterion ③ has an identity layer worth peeling back. Escalation is usually read as a question of face, and its actual structure is harder than face. Part of a sponsor's exchangeable value inside an organization is stored in the project's narrative: he is the man who built X, or the man building X. Exit is an asset write-down on that narrative, and what gets written down is not standing but the quotation on the job market. The structure explains why a transfer or a promotion so often fails to break escalation, since a new seat leaves the narrative where it was. Real unbinding waits for the narrative to depreciate by itself, as a newer project covers the older one, or it comes from Device 4 in this chapter, a reward for termination that makes exit a story worth telling. The two routes differ in cost by an order of magnitude, and which one an organization chooses is its true price for escalation.

Prospect theory, Kahneman and Tversky's loss aversion, supplies the last piece on the psychological side. Money already spent sits in the loss region of the mental account, and the decision curve in that region is not the curve in the gain region: sensitivity to further loss diminishes, so the larger the loss, the less each additional unit of it hurts. That is the shape of the escalation slope. The first top-

up hurts most, and each one after it goes down more easily, because an account already that deep in loss makes each further unit of loss less noticeable. The implication for defense is exact. The exit window is widest before the first top-up, and once the first top-up is made the resistance to exit doubles. The trigger point for an exit clause (Device 1) therefore belongs at a node before topping up starts, at the project's original milestones rather than the middle of the budget cycle.

27.2 How It Grows

Three generating conditions.

Condition one: the decision-maker is bound to the project. The project is his record, his body of work or his career wager, so failure at the level of identity equals personal failure, and no accounting treatment settles a question of identity. The "responsibility is yours" manipulation in Staw's experiment corresponds inside an organization to the answer to one question: whose project is this.

Condition two: the exit mechanism is missing. The approval document carries no exit conditions, or carries them with an interested party as the arbiter. Exit can then only arrive as a coup, by changing the person, changing the accounting line or waiting out a term, and the organizational cost of a coup makes everyone prefer to wait.

Condition three: external oversight is weak. No party is both able and willing to ask the same question at every review: if we were starting from zero today, would we still choose this project. The question costs nothing and its frequency of appearance is close to zero.

The structural side needs one more observation. Missing exit has a plausible stand-in, and the stand-in is called adjustment. What separates adjustment from exit is the continuity of the account. Adjustment continues the same mental account, same project, different method. Exit closes the account, which means saying that this money stops here. Adjustment is therefore escalation's most respectable form: every adjustment opens a new story, the account never returns to zero, and the snowball of investment keeps rolling without ever getting smaller. The probe for respectable escalation is the age of the account. How long has this project's mental account been open, counted from the approval date, and how many times has it been adjusted, counted from the number of re-justifications. A project with a long account and many adjustments is probably feeding the substance of escalation on the respectability of adjustment. That reading goes straight into the zero-based re-ask (Device 3), where the age of the account is itself the first argument for re-asking.

The oversight side has a subtler point. In an organization with weak external oversight, the opponent of escalation faces the project's coalition rather than the project: the sponsor, the contractors, the incumbent suppliers, the regional interests. The coalition makes the exit review table unequal by construction, one voice against many. Device 2, the external adjudicator, is then more than a procedural arrangement, since it rebalances the table. An adjudicator with no binding lets the coalition's narrative advantage expire, because how many times you have said the project is good has no bearing on whether it should continue today. The deeper reading is that governing escalation is a question of seats rather than of persuasion, which is Chapter 17's seating logic on its third appearance in this book.

27.3 Field Sites

Case 1 (Public): After Two Resets

Case ID: po-004.

When and where: February 2020 to May 2026, a timeline joined by three resets: February 2020, October 2023, January 2025. London, the Department for Transport and the HS2 Ltd head office; NAO fieldwork covered review, assurance and contract meetings at both bodies between March and May 2026 (NAO p.8).

What and who: the Department for Transport is the sponsor, holding the funding and the oversight; HS2 Ltd is the department's executive public body, responsible for design, construction and opening. In April 2026 a new company, Euston Delivery Company Limited, was set up to take over Euston station and the related works (NAO p.16, p.41). The auditor is the National Audit Office; the report was signed by the Comptroller and Auditor General, Gareth Davies, on 22 June 2026 and laid before the House of Commons (HC 52, NAO p.1, p.3). Responsibility is recorded by post: the Permanent Secretary at the Department for Transport, the Secretary of State for Transport, the chief executive of HS2 Ltd.

What they were doing: the third reset, launched in January 2025, was under way, with HS2 Ltd expecting to finish it in spring 2027. The reset has its own budget of £153 million, of which £101 million had been spent by the end of March 2026 (NAO p.18). Construction never stopped. Reset and construction ran in parallel, and four workstreams handled cost and schedule, supply chain and commercial, organizational capability and culture, and governance (NAO p.18, p.19). The cost and schedule workstream broke the project into 21 separate projects, fixed scope, schedule, cost and risk for each, merged them into a range, and had the result assured by an independent panel that included the Infrastructure and

Projects Authority (NAO p.25). Five stages each carry a readiness review, and a stage that fails does not open the next one (NAO p.20, figure 3). The renegotiation of four civils joint venture contracts, begun in spring 2024, failed that year; 2025 brought a different approach, settling historic changes and actual costs first and renegotiating future spend afterwards, with about £500 million of non-allowable cost identified and a commercial reset aimed at about £2 billion of savings (NAO p.10, p.29). In January 2026 the Secretary of State asked HS2 Ltd to assess what a lower speed would save; in May 2026 the department agreed to cut the maximum operating speed from 360 km/h to 320 km/h, cancel automatic train operation and downgrade the signalling specification, an estimated saving of £1 billion to £2.5 billion against about £1.3 billion of long-term benefits lost (NAO p.9, p.25, p.26). In April 2026 Euston was separated from the main line; Euston works had cost £3.8 billion by the end of March and needed £4.1 billion more in public money (NAO p.44).

The decision point:

① February 2020, around the notice to proceed: the full business case for Phase One was submitted with a benefit-cost ratio of 1.2, sunk spending before the end of December 2019 excluded on Treasury rules, a cost estimate of £35 billion to £45 billion (2019 prices) and a funding envelope of £44.6 billion (2019 prices) (NAO p.22, p.27). The assessment also recorded that Phase One would make the rest of the network, and other rail projects, possible (NAO p.27). Signed by: the government, through the notice to proceed; on file: the accounting officer assessment of the Permanent Secretary at the Department for Transport.

② October 2023: the previous government cancelled Phase Two, ending the line north of Birmingham, including the western and eastern legs of 2a and 2b; the project that emerged was smaller in scope, more expensive and later (NAO p.16, p.17). The accounting

officer assessment gave a benefit-cost ratio of 1.1 to 1.8, excluded £24.6 billion (2019 prices) of sunk spending to the end of September 2023, and deducted about £11 billion as the cost of undoing Phase One (NAO p.27); one month before the cancellation, the civils cost forecast had just risen by £6 billion (NAO p.29). Approved by: the government, announced to the Commons by the Secretary of State.

③ January 2025: the third reset began. In March 2025 the new chief executive finished a project assessment whose conclusions were that construction had started too early, that the contracts had not delivered the expected performance, and that the organization had not been set up to manage delivery effectively; the assessment listed four priorities and named capability gaps in contract management and assurance, with the delivery team short of people (NAO p.18, p.11). Approved by: the new chief executive of HS2 Ltd and the Permanent Secretary.

④ April to May 2026: Euston passed to the new company as a separate item; the department published a revised estimate of £87.7 billion to £102.7 billion and agreed the lower speed and the redesigned specification; the Permanent Secretary issued an accounting officer assessment with a benefit-cost ratio of 1.5 to 6.4, excluding £40.9 billion (nominal) of sunk spending to the end of September 2025 (NAO p.26, p.41, p.27). The same document produced a figure in the other direction for the first time: had the costs of today been known in 2020, the benefit-cost ratio would have been 0.3 to 0.4, which is poor value (NAO p.27). The formal baseline for the revised estimate still waits for a first version in autumn 2026 and a final one in spring 2027 (NAO p.25).

What was visible then:

In January 2020 the NAO report High Speed Two: A progress update (HC 40) had already warned about cost and governance, and stated that schedule and cost had both drifted (NAO p.16, footnote 3).

When the notice to proceed was issued, that report and the £35 billion to £45 billion range were in hand at the same time (NAO p. 16, p.22). In September 2023 HS2 Ltd's own estimate showed main civils works alone up £6 billion (2019 prices) on 2020 (NAO p.29). In April 2024 the department formally notified HS2 Ltd under the Development Agreement that it was in breach of its delivery obligations, a contract-level document rather than an internal judgement (NAO p.20, figure 3, note 2). That same year, with the four civils joint ventures in renegotiation, HS2 Ltd had no reliable total cost estimate to negotiate with (NAO p.29). In June 2025 the Stewart Review reported that the governance structure was not fit for purpose and that a different arrangement was needed (NAO p. 21, footnote 5). In February 2026, 45 of the 124 recommendations from four external reviews had been implemented, 36 per cent (NAO p.21). In May 2026 the cost of cancelling Phase One was estimated at £33 billion to £58 billion, comparable to completing it, and HS2 Ltd itself described this as a highly uncertain high-level estimate with no comparable project as a benchmark; in the same assessment the case for the government to carry on was written as a benefit-cost ratio above 1.5 rather than the range (NAO p.27, p.28, p.26).

What was learned afterwards: opening dates fall between May 2036 and October 2039 (Old Oak Common to Birmingham) and between May 2040 and December 2043 (Euston to Birmingham and the connection north to the West Coast Main Line), three to 13 years later than the 2020 plan; the cost increase decomposes into underestimation 32 per cent, inefficient delivery 23 per cent, scope change 11 per cent and inflation 34 per cent (NAO p.23, p.24, p.23 figure 4). Both items come from the body of the June 2026 report.

> **Stop here. Decide before you turn the page.**

> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: by the end of March 2026 the department and HS2 Ltd had spent £46.8 billion (nominal), of which £2.6 billion went on the cancelled Phase Two (NAO p.22). The revised cost to complete is £87.7 billion to £102.7 billion, with HS2 Ltd's own target at £93.2 billion; against the 2020 estimate of £35 billion to £45 billion, the figure is still up £35.9 billion to £37.2 billion after inflation, a rise of more than 100 per cent at the low end (NAO p. 22). Full opening is three to 13 years later than planned in 2020 (NAO p.23). The Phase Two shutdown carries a bill of its own: about 560 properties to be disposed of, a disposal programme costing about £9 million, with about £170 million expected back (NAO p.47). The audit conclusion is that both bodies are avoiding further cost increases and delay, but that the reset will not be complete until spring 2027, and this time it has to be done right (NAO p.14). Back to the three questions. The most common answer to the first is another reset review with a fresh baseline; in this case that action is priced at the £153 million reset budget plus the time to spring 2027 (NAO p.18). The item most easily missed in the second is the £33 billion to £58 billion cancellation estimate, which looks like a technical parameter and is in fact the body of the argument for continuing. The third lands on the October 2023 cancellation: had the from-zero benefit-cost ratio been put on the table beside the continuation option, 0.3 to 0.4 would have appeared two and a half years earlier.

Options ruled out: cancel the whole line, which the May 2026 accounting officer assessment does treat as a formal counterfactual, offset by the £33 billion to £58 billion cancellation cost, a magnitude obtained by dividing that cost by the future cost, where the cancellation cost itself has no comparable benchmark (NAO p.26, p.27, p.28); keep 360 km/h and automatic train operation, dropped after the assessment between January and May 2026 on the testing cost against the £1 billion to £2.5 billion saving (NAO p.25, p.26); dispose of all the Phase Two land, with the Secretary of State announcing in January 2026 that 2a would be held for a future north-south line and more than 350 compensation cases settled by HS2 Ltd (NAO p.13, p.47); pause construction and restart once the baseline is set, not adopted, since reset and construction continue in parallel and the Euston tunnels were still being bored in early 2026 (NAO p.18, figure 3, note 1; p.16); run the civils contracts on their original terms without renegotiation, replaced after the 2024 failure by the 2025 approach of settling historic changes first and negotiating future spend afterwards (NAO p.10, p.29); treat the £1.3 billion of lost benefits from the lower speed as grounds for rejection, with the loss written into the assessment and the conclusion unchanged (NAO p.26); keep Phase Two alive on paper and decide after Phase One, where the October 2023 cancellation was described as final and the new government reintroduced the 2b western leg bill in July 2024 for a different line (NAO p.17, figure 1, note 3).

Early signals: ① January 2020, the NAO progress report naming cost and governance risk, overridden by the project's cross-party support and a notice to proceed already issued. ② November 2021, the Integrated Rail Plan cutting the eastern leg of Phase Two, so that Leeds would no longer be reached directly by HS2 but through the East Midlands Parkway direction, filed as a scope optimization with the project continuing as a whole (NAO p.17, figure 1, note 1). ③ September 2023, the main civils cost forecast up £6 billion,

overridden as something a reset and a contract renegotiation could absorb, with the same contracts failing at renegotiation in 2024 (NAO p.29, p.10). ④ April 2024, the breach notice, filed as an organizational capability problem that left scope and schedule untouched, the breach standing until the reset is complete (NAO p.20, figure 3, note 2). ⑤ March 2025, the chief executive's assessment stating that construction had started too early, split across four workstreams and taken as input to the reset rather than as a trigger to stop (NAO p.18). That last sentence reads afterwards like an exit condition; at the time it was read as a diagnosis for the reset. What the five share is a translation. Every one became a way to continue better, and none became a question of whether to continue. The third is the cleanest: £6 billion is HS2 Ltd's own number, one month before Phase Two was cancelled, and it was used to support continuation. Laid side by side, the five compress one standard of judgement into five forms: what counts as under control.

The point of escalation: May 2026, when the Permanent Secretary signed the accounting officer assessment that wrote the cancellation cost into the continuation case's benefit-cost ratio as a deduction (NAO p.26, p.27). Three numbers hold the step in place. The cost of undoing Phase One was about £11 billion when estimated in 2023 and £33 billion to £58 billion in 2026, comparable to completing it (NAO p.27). Spending rose over the same period from £24.6 billion (end September 2023, 2019 prices) to £40.9 billion (end September 2025, nominal) (NAO p.27). The NAO states plainly that this cancellation estimate has no comparable benchmark (NAO p. 28). The escalation point sits here: the first assessment in which the price of exit was put on a level with the price of completion. The date is May 2026; the action was publishing the revised estimate and the lower-speed redesign; the approvers were the Permanent Secretary, through the accounting officer assessment, and the Secretary of State, through the January instruction on speed. Every top-up after this is argued from an exit price that keeps rising,

rather than from new benefit. The three numbers point the same way: the more that has gone in, the dearer exit becomes, the more rational continuation looks, and every one of the numbers was measured by the side arguing for continuation. The report also notes that the fourfold rise in the cancellation cost is a significant factor in the rising benefit-cost ratio (NAO p.28).

Failure chain: technical layer: the scope and quantities of the main civils were added to repeatedly after work started, and 66 per cent of the cost increase came from something other than inflation, namely underestimation 32 per cent, inefficient delivery 23 per cent and scope change 11 per cent, where scope change means necessary works that the original scope had missed (NAO p.23, figure 4, note 3). Organizational layer: governance was judged not fit for purpose by the Stewart Review, whose 28 governance recommendations were more than half implemented by February 2026 with the rest in progress; before that, a decision had to pass HS2 Ltd internal assurance, the department's approval points, IPA review and Cabinet Office and Treasury approval, a chain up to nine months long that rarely changed anything substantive (NAO p. 11, p.36). The April 2024 breach notice and the 2025 organizational assessment point at the same thing, that the body was not built to manage work at this volume (NAO p.18, p.20). In February 2026, 45 of 124 external review recommendations had been implemented (NAO p.21). Incentive layer: the argument for continuing strengthens as construction advances, and the price of exit rises as construction advances, with both measured by the same people. Reset money is booked to the continuation account; cancellation money exists as an estimate with no benchmark, and the NAO says so (NAO p.18, p.27, p.28). The 2023 cancellation was made by one government, and in July 2024 the next one reintroduced the 2b western leg bill (NAO p.17). The break sits between the organizational and incentive layers: the people entitled to reprice exit are the people delivering the continuation, so no seat at the review

table holds an interest on the exit side, and while the accounts are updated every quarter the seats never change.

Minimum counterfactual: in October 2023, when Phase Two was cancelled, add one line to the Permanent Secretary' s accounting officer assessment: the benefit-cost ratio recomputed from zero today, set beside the continuation option. One action, inside the month after the decision to continue, and no luck required. The department showed in May 2026 that the number can be produced, and produced 0.3 to 0.4 (NAO p.27). The same demand model and the same cost estimates were in hand at the time; what was missing was one more pass with sunk spending taken out of the denomin - ator. The line costs a page. It would not have stopped the project by itself. It would have put 0.3 to 0.4 into the parliamentary record two and a half years earlier. The difference those two and a half years make is where the record sits: the 2023 figure would have been in the same document as the Phase Two cancellation, without waiting for the auditor to recompute it in 2026. Where the worry is overreach, the line can stand as a footnote that changes no con - clusion, with the date and the signatory untouched. The line is the zero-based re-ask in written form.

Outcome and sources: the project continues, with potential cost up to £102.7 billion and full opening pushed into the 2040s (NAO p. 22, p.23). Primary source: Comptroller and Auditor General, High Speed Two reset, Session 2026-27, HC 52, National Audit Office, 29 June 2026 (signed 22 June 2026, Commons printing order 25 June, NAO p.1, p.3). The report runs to 56 pages and carries 8 recom - mendations, on re-evaluating the reset timetable, capability gaps, cost and commercial management, train formation and network integration, governance adjusting by stage, and benefits strategy (NAO p.15). The previous HS2 report was published in July 2024, when the funding envelope was still £44.6 billion (2019 prices); between the two reports sit two scope changes and one change of

government (NAO p.8, p.22). The audit scope covers the planning, assurance and implementation of the reset, and excludes integration with the existing network after completion (NAO p.8). Page numbers in this card are PDF pages; the link is in Appendix B under po-004.

The HS2 numbers deserve one pass with this chapter's scalpel. The ratio between £46.8 billion already spent and a potential ceiling of £102.7 billion is criterion ① (visible investment) in its national form: at that size the visible commitment makes the word stop hard to pronounce in any parliament, since the cost of cancelling is counted in tens of billions, and escalation becomes the default across successive governments. The structural value of the NAO report is that it has written the decision chain of each reset into a public record, and that record can be laid on the table at any future escalation review. The auditor's role in governing escalation is visible from this. It cannot block a decision, since democratic authority sits with the government, but it can make the ledger of each decision permanently readable, and readability is a deterrent, because the next government faces a recorded history one chapter longer.

Case 2 (Aviation): After the Runway Change

Case ID: av-025.

When and where: 27 December 2025, 1130 UTC (British winter time, the same locally); Talgarth Airfield, Powys, Wales, the northwest runway and the field adjoining it to the west. The airfield elevation is 970 feet and the northwest runway is 360 metres long. This was the first tow of the day, before gliding operations began (AAIB p.1, p.2, p.7).

What and who: towing operations at the Black Mountains Gliding Club. The tug was a Eurofox 2K, registration G-CMAX, with a Rotax

915 iS engine, club-owned and used only for towing gliders (AAIB p.1, p.5); the ASK13 two-seat glider on the rope carried two licensed gliding instructors (AAIB p.2). Responsibility by post: the duty instructor, who assessed the day's flying conditions; the tug pilot, commander of the combination until release. Guidance comes from the British Gliding Association (BGA); the investigator was the AAIB field investigation team.

What they were doing: flying started the way the club usually starts. The morning briefing was run by the duty instructor and covered the weather and the day's threats; the site briefing pack and video were mandatory viewing, and the airfield sits on the western slope of the Black Mountains, with ground shaped like an upturned saucer, smaller than most gliding sites and prone to ridge rotor turbulence (AAIB p.7). The airfield has no official weather source and a short runway, and conditions in the takeoff area are sensitive to rain; the club assembles the day's conditions from hillwalking forecasts, the Met Office aviation pages, two webcams and the clubhouse weather station (AAIB p.6, p.7). The plan was to take off to the east, into wind. During setup the takeoff area was found wet and soft and unusable (AAIB p.1). The duty instructor and the tug pilot conferred and switched to the northwest runway, accepting a 90° crosswind (AAIB p.1–2). The glider was moved to the new takeoff area. At the same time G-CMAX took off alone from the northwest runway for the club's customary 10 to 15 minute conditions check, whose purpose was to look at the rotor turbulence (AAIB p.2). The check flight came back on a reverse approach: the crosswind was at right angles to the runway, so neither approach direction had an into-wind component, and on final the aircraft was shaken hard enough that a member commented on it to another member at the time (AAIB p.2, p.10). The aircraft went around, approached again and landed. It taxied to the waiting ASK13 and the tow rope was attached; the combination accelerated

slowly on the wet soft grass, with the launch point already as close to the boundary fence as it could be (AAIB p.2).

The decision point:

① Morning of 27 December 2025, during setup: change the takeoff runway from the east to the northwest and accept a 90° crosswind, rather than stop flying for the day. Setting: a conversation in the takeoff area between the duty instructor and the tug pilot, recorded in the report as a discussion with no written record; the premise was that the original runway was unusable. Approved by: the duty instructor, as the person assessing the day's flying operations, and the tug pilot (AAIB p.1–2).

② The same morning, the check flight: take off alone, enter the airspace the tow would later climb through, and meet severe turbulence there. Setting: the club's customary conditions check, 10 to 15 minutes; approved by the tug pilot as commander (AAIB p.2). The report records no written conclusion from the check flight, only its purpose, to confirm rotor turbulence near the site before gliding began (AAIB p.2).

③ The same morning, after the check flight landed: attach the rope to the ASK13 and begin the ground run. Setting: the takeoff area; approved by the tug pilot. BGA guidance states that until release he is commander of the combination and, after consulting the person in charge, has a duty to stop the tow if he does not wish to continue, without waiting for the club to call a halt (AAIB p.9, p.2). Two instructor pilots were on the field, both with more than 20 years of flying (AAIB p.2, p.8). Past this step there is no ground option: the grass runway is 360 metres, and the ground run had already begun near the boundary fence (AAIB p.2, p.7).

④ 1630:37 UTC: throttle reduced to 21 per cent for about two seconds, then back to 100 per cent (AAIB p.4). The record shows power coming off before it went on, and the report considers that

the first may have been an attempt to land back on the field and the second an attempt to clear the trees and the single-storey building ahead (AAIB p.10). Approved by: the pilot. Those two seconds of reduced power, and the full power after them, are the only explicit record in the flight of a decision about whether to land (AAIB p.4, p.10).

What was visible then:

The morning briefing was run by the duty instructor and covered the weather and the day's threats (AAIB p.7). The clubhouse weather station read 12 to 16 knots with gusts of 23, wind from east-northeast, across the runway in use, air temperature 4°C (AAIB p.6). The station records every 5 minutes and the day's trace can be read back, with speed swinging widely and direction steady (AAIB p.6, p.10). Larkhill, about 2 nautical miles from the accident site, recorded 14 knots from the west with gusts to 28, and cloud rising from two oktas to four between 1500 and 1600 (AAIB p.13). The windsock was visible at the start of the takeoff run (AAIB p.6). Flight manual: maximum takeoff crosswind 15 knots, maximum tailwind 4 knots (AAIB p.5). BGA guidance: tugs are prone to loss of control in attitude, light tugs more so; the cumulative factors include turbulence in the takeoff area, rough ground, a significant crosswind and a short tow rope, and three or more of them should prompt a decision not to launch (AAIB p.8, p.9). Performance data: towing the ASK13 needs 190 metres on asphalt, 270 metres on grass with the upslope correction, and 340 metres with 70 metres added for obstacle clearance, and those figures only get worse on wet soft ground (AAIB p.6). The turbulence on the check flight's approach had been seen at first hand (AAIB p.2, p.10).

What was learned afterwards: airframe and engine were serviceable (AAIB p.4, p.10); the wind after liftoff may have exceeded 15 knots (AAIB p.10–11); the left wing drop came from crosswind and turbulence together (AAIB p.11); pulling to clear the obstacle cost

airspeed and led into the stall (AAIB p.10). None of it took shape until the data were read and the wreckage examined after the flight.

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: after liftoff the tug rolled left about 60° without command, the glider released and landed safely on the field, and the tug turned left about 60° into the lee, where the tailwind component grew; the mainwheel briefly touched the ground with trees and a single-storey building ahead; the aircraft lifted off again, climbing almost vertically as seen from the launch point, cleared the trees and the hangar, then the left wing dropped and it went into a field beyond the boundary. The left wingtip struck the top of the fence first, the nose hit the ground, and the fuselage came to rest having turned through 270°; both wings were displaced about 45°, the cockpit was badly damaged, one blade of the four-blade propeller had broken off about a quarter of its length from the hub, and the sole pilot was seriously injured (AAIB p.1, p.2, p.3). AAIB conclusion: the takeoff was made towards a turbulent area in gusty crosswind conditions, and the wing drop after liftoff led to loss of control (AAIB p.11). Back to the three questions. The most common answer to the first is to wait another ten minutes and watch the wind; in this case that action buys little margin, ten minutes on 360 metres of grass, and the real cost is admitting the day is not flyable. The item most easily missed in the second is the turbulence the

check flight brought back, which looks like a routine test flight and is in fact a measurement taken ten minutes earlier on the same track. The third lands before the rope goes on: compare the gust value on the station with the 15 knot limit, and if it is over, end towing for the day.

Options ruled out: stop flying for the day, waiting for the wind to drop or coming back another day, not adopted, since the report records the day's gliding conditions as favourable with two gliders and their people already on site, so stopping meant the two gliders and their crews waiting through a morning of not flying, and the forecast thermals were to last until 1800 (AAIB p.11, p.13); launch in a gap between gusts, where the station records every 5 minutes and the webcams show wide swings in speed, and the decision was still made on the judgement that the surface wind was below the station reading (AAIB p.10); go back to the east runway or use a third one, where the east runway was wet and soft and no other runway enters the record (AAIB p.1, p.7); shorten the tow rope, use a heavier tug or reduce the glider's weight, all adjustable items on the BGA factor list that the report does not record as discussed that day (AAIB p.8–9); exercise the commander's authority to call enough, and cancel towing for the day, which the BGA explicitly grants and which was not used (AAIB p.9); put the launch off until the afternoon when thermals weakened, not adopted, since the day was recorded as one of favourable gliding conditions (AAIB p.11, p.13); hand the judgement to the club collectively, where the report records no collective review at all and both the conditions assessment and the launch decision sat with one person (AAIB p.6, p.9).

Early signals: ① the takeoff area wet and soft and the east runway unusable, handled as a change of runway rather than a change of day (AAIB p.1); ② the station reading 12 to 16 knots gusting 23 against a 15 knot limit, overridden by the possibility that the sur-

face wind along the runway was lower (AAIB p.6, p.10); ③ the check flight shaken hard on final, with a member commenting at the time, overridden by the aircraft having landed safely, the report's own words being that it was shaken enough for one member to comment to another (AAIB p.2, p.10); ④ 360 metres of grass against 340 metres of obstacle clearance requirement, a margin of 20 metres, overridden because the launch point was already as close to the boundary fence as it could be (AAIB p.2, p.6, p.7); ⑤ 1 hour in the previous 90 days and 0 hours in the previous 28, against 54 hours on type, overridden by a licence held for 45 years and more than 12,000 tows (AAIB p.1, p.8). The report adds a sentence of its own: operations had succeeded at this site in similar conditions many times before, which may have supported a view that today would be no different (AAIB p.10). The hardest of the five is ②, since the limit is printed in the manual and the reading was on the clubhouse screen, with one comparison between them. The reason it was overridden is in the report: the surface wind along the runway may be lower than the station reading, and the surface wind could not be measured (AAIB p.10–11). The one thing overridden was the criterion itself. The softest of the five is ①, which was already a concession, recorded as an arrangement of the site. Next softest is ③, where one landing was taken as proof that this track was acceptable.

The point of escalation: 27 December 2025, one or two minutes before the accident time of 1130 UTC, at the moment the rope went onto the ASK13 and the combination began its ground run on 360 metres of grass. Action: launch. Approved by: the tug pilot, commander of the combination. In the ten minutes before, he had been shaken by turbulence in that same airspace and had landed (AAIB p.2, p.10). The investment here is counted in hours: the day's journey, the rigging, the towing and one check flight, plus two gliders, two instructors and the club's expectation of an operation. The price of exit was not flying that day, and the person

who had to say it was the only person entitled to say it. Past that step every remaining action is in the air, and only the record can reconstruct it. The report goes one step into this: the day's gliding conditions were favourable, and the goal of towing the glider may itself have kept the pilot from stopping to re-examine the danger from the crosswind and the turbulence (AAIB p.11). Date, action and approver are all present, and the money column is empty: at personal scale the top-up is counted in hours and in trips. The reasons for going were more than one. The wind might be lighter, the aircraft had just flown, the people were on the field, the glider was on the rope. Each stands on its own, and only together do they make a launch.

Failure chain: technical layer: a 90° crosswind means no into-wind component on the takeoff run, wet soft grass lengthens the ground run, obstacle clearance needs 340 metres against a 360 metre run-way, a margin of 20 metres; liftoff leads straight into ridge rotor turbulence, and the pull to clear the obstacle after the wing drop costs airspeed and leads to the stall (AAIB p.6, p.7, p.10, p.11); the engine record rules out power as a factor (AAIB p.4, p.10). Organ-izational layer: the site has no official weather source and condi-tions are judged by the duty instructor on local experience; the daily briefing and the mandatory site briefing were both in place, and neither carried a quantitative launch gate; the BGA's three-or-more-factors advice carries no releasing authority, and the same guidance leaves the enough-is-enough call to the person on the spot (AAIB p.9); club oversight of towing rested on one person, who both assessed the conditions and made the launch (AAIB p.6, p.7). Incentive layer: gliding conditions were favourable, people and equipment were in place, and towing is the club's core activity; the cost of stopping was paid that day and the probability of loss of control did not show that day; nobody else held a formal veto; and the improvements the report records stop at information and self-questioning, with no hard limit introduced (AAIB p.9, p.11). All

three layers point at one gap: the criterion was handed to experience and the veto to the person executing. The break sits between the organizational and incentive layers, since the authority to stop was written into the guidance and the cost of using it was paid on the spot by the person holding it.

Minimum counterfactual: after the check flight lands and before the rope goes on, compare the gust value from the station reading, 23 knots, with the 15 knot crosswind limit in the manual, and end towing for the day if it is over. One action, doable within ten minutes, with the reading on the club website and the clubhouse screen that day and the limit printed in the manual (AAIB p.5, p.6). It needs no better meteorology, only the use of a reference value as a limit. It needs no different person either, since the one doing it is still the one entitled to stop towing. If the club writes that line at the end of the morning briefing, tomorrow's action depends neither on memory nor on who happens to be there. The step does not touch the incentive layer: the cost of stopping still falls on the person on the day, and only the trigger moves from judgement to a reading. The surface wind could not be measured then (AAIB p.11), so judgement had to rest on experience, and until that changes the same launch will come again.

Outcome and sources: the sole pilot was seriously injured, the aircraft was damaged, and the two people in the glider were unhurt (AAIB p.1, p.2). After the accident the club improved its weather reporting, providing a live two-minute mean wind and ten-minute maximum gust, broken down by headwind, tailwind and crosswind for each of the three runways, and promoted an internal question of whether to fly rather than whether one can (AAIB p.11). Primary source: AAIB, Bulletin 09-26, AAIB-31440, Eurofox 2K G-CMAX, Talgarth Airfield, 27 December 2025, published 20 August 2026 (AAIB p.11). The report is a field investigation bulletin with no safety recommendations, and the safety actions

listed were completed by the club (AAIB p.11); the aircraft had flown 482 hours and made 3,379 landings, with airframe and powerplant records normal (AAIB p.5, p.10). Dates and times in this card come from the report, and page numbers are PDF pages; the link is in Appendix B under av-025.

Case two (av-025) at personal scale needs a footnote of its own: the day's flying was already scheduled, with two gliders, two instructors and one check flight on the field, and the launch was decided on top of what had already gone into the day. At personal scale the investment is not money; it is the journey, the rigging, the towing and the flight already flown. The price of exit is paid the same day, which is no flying today, and the probability of losing control does not show itself that day. The personal-scale defense has to move to the ground before it matters, as a decision rule set before takeoff: a reading over the limit ends towing for the day, no exceptions. In the air there is no room to negotiate, so moving the rule forward shifts the ground of escalation from the air to the ground, from a moment's judgement to a sheet of paper.

One execution detail on that sheet belongs in the text. The exception clause of a personal rule has to be harder to invoke than the rule itself. A glider version written as "unless the weather suddenly improves" turns the exception into an exit taken every time, since every time there is a reason. A harder form requires a second signature from whoever else is on duty that day, so that an exception on the ground borrows another pair of eyes. The general form: the exception procedure of a personal commitment involves a second person, and its cost is set deliberately above the cost of compliance, so that complying is cheaper than excepting.

Cheap compliance has a social design layer as well. Make the personal rule public, with club members and crews knowing each other's rules, and the published rule acquires peer custody: the person flying with you knows your rule and can remind you. A

network of custody upgrades individual discipline into group discipline, and the line about everyone here having that rule, heard in gliding clubs and in fishing communities, survives fatigue better than any individual willpower. The organizational version of group discipline is Device 4 in this chapter, since an organization that praises the person who stops a project is issuing peer custody to every member's personal rule.

Case 3 (Maritime): Sleep Debt and the Trip

ma-003. The prawn trawler Crystal Stream (LH 147), 6 June 2025, Loch Fyne, Scotland. The MAIB report (Report 11/2026, pages read for this book) records that the skipper, alone on watch, fell asleep and the vessel grounded, a constructive total loss, with the skipper slightly injured (the casualty table was checked and reads minor injuries); sleep debt had been accumulating through the week before. The economics of fishing make every trip hard to abort: fuel, the boat's schedule and the market window for the catch. The sunk and expected costs of a trip together make continuing the default, and continuing presupposes the skipper still on watch, even if that means working on overdrawn sleep.

Stop at the decision point: the morning before the accident trip left port. On the table: a week's fatigue account, the trip's fixed outlay, and what aborting would do to the schedule and to income. Outcome: grounding, constructive total loss. Primary source: MAIB, Report 11/2026, 2026; link at Appendix B, ma-003.

Case three (ma-003) has an economic detail worth writing out. In the market structure for catch, the cost of aborting a trip is amplified: the landing window is short and the price swings, so losing a day can lose the season's price. The structural inducement to escalate, a real opportunity cost, stacks on the psychological one, sunk cost. The distinction matters for defense, since psychological inducement is treated with the four devices and structural in-

ducement with institutional design: reserve capacity inside a rotation system, a pool of relief skippers, the scheduling defense from Chapter 15 case six. Leave the structure alone and the psychological line stands naked in front of a real opportunity cost every day. Change the structure and the psychological line has ground under it.

Case 4 (Fintech): More Money for an Unvalidated Model (condensed)

st-001. The legal tech startup Atrium, about 200 employees and about 450 clients. The Failory Cemetery archive, quoting the founder Justin Kan's public statements, read for this book: the company closed after raising $75.5 million. The founder's retrospective is a textbook sample of criterion ②: adding more money to a situation of lack of product-market fit only means spending the money faster. The structure of the funding rounds and the founder's reputational investment together made the fuel, since every round arriving on the account reinforced the reasoning that one more push would do it, while the core problem of product-market fit was never solved. The decision point is the board meeting that approved the next expansion with product-market fit unvalidated. Minimum counterfactual: tie each expansion tranche to a product-market fit validation milestone (the same lesson as Chapter 24 case seven). Primary source: Failory Cemetery (quoting Justin Kan's public statements), 2020; the link is in Appendix B under st-001 (evidence grade E).

The four cases sit on three scales of this chapter's spectrum: national (po-004, a £46.8 billion state project), organizational (st-001, a $75.5 million company) and personal (av-025, a launch after a check flight; ma-003, one week of sleep). That the three scales share one chain of criteria is the corner this chapter most wants to push the reader into. Escalation is a disease of decisions. You have a small-scale version of it running now.

The Evidence Gap

Primary evidence on escalation is thin by nature. Institutions rarely publish the decision chain of why they did not exit, which is another form of Chapter 2's survivorship bias: exited projects vanish and escalated ones leave reports. Wartime and large-policy escalation cases are scattered across archives and retrospective inquiries, and within this book's collection scope none has a verifiable full text, so the gap is marked as it stands. Directions for further collection are in, legislative renewal inquiries and long-tail loss studies from the insurance industry.

27.4 What Shows Up Earliest

Six weak signals, each with the reason that usually overrides it.

1. "Just a little longer" and "we have come this far" appear more often in meetings. Reason given: "morale".
2. The assumptions in a re-forecast are made more optimistic by one side, and the person who adjusts them does not sign. Reason given: "a model update".
3. Somebody writes an exit option and nobody puts it on the agenda. Reason given: "the timing is not right".
4. Milestones are redefined: the standard for reaching the original target is replaced by a standard that can be reached. Reason given: "a pragmatic adjustment".
5. People who argue against renewal are moved to another project. Reason given: "workforce optimization".
6. The project's outward narrative switches from the language of return to the language of persistence ("strategic value", "we cannot leave it half done"). Reason given: "one message". The switch in language comes before the budget request, and it is the joint between Chapter 13 and this chapter.

Now put the six signals against the four cases. Signal 1, the rising frequency of "just a little longer", belongs to case two: the same airspace had just shaken him into a landing, and the rope went on again. How many times the sentence was said inside his head can only be inferred, and the shape of the judgement chain is written in the launch record. Signal 2, assumptions made optimistic by one side, belongs to case one: every re-justification inside each reset was an occasion for retuning the return calculation on the remaining section. Signal 3, an exit option nobody tables, belongs to case one: both reset agendas were agendas for finishing the job, and a full cancellation was never formally laid beside them. Signal 4, redefined milestones, belongs to case four: product-market fit never arrived, and the expansion milestones have been through three generations. Signal 5, the dissenter moved, takes the form in a public project of the dissenting department's budget being rearranged. Signal 6, the language switch, is the standard prose of case one's reset press releases, and at national scale the switch from the language of return to the language of persistence is carried out in legislative language.

Reorder the six by the cost of evidence. Signals 4 and 6 are cheapest, read from the wording of documents and the text of milestones. Signals 1 and 2 are middling, read from minutes and from the version history of the model. Signals 3 and 5 are dearest, needing agenda records and the trail of personnel moves. Start with signal 4: pull three years of a project's milestone text and read the versions side by side, and the loosening of the achievement standard is plain. Every loosening should carry a signed reason, and one without a signature is handled under criterion ② of this chapter.

27.5 How to Defend

Device 1: exit conditions up front. The approval document must contain an exit clause: which signals trigger an exit review (quantified), who adjudicates (named), and a statement that at the review the money already spent no longer counts as an argument (stated in the text). The clause is agreed at signing, when all parties are calmest, and it is the approval-stage version of Chapter 8's Device 3. On wording: the trigger signal must be externally observable, in schedule, in external assessment, in a third-party milestone, and never "when management considers it necessary".

Device 2: an external adjudicator. The person who adjudicates an exit review must have no history with the project, having neither signed its approval nor spoken for it, and the appointment covers the project's whole expected life. His only agenda is to answer the zero-based re-ask, not to balance the parties.

Device 3: the zero-based re-ask. The first item on every review agenda is fixed: "if nothing had been invested as of today, would this project be started?" Answer that before discussing reasons to continue. The HS2-style reset review lacks exactly this step, since every minute of the reset discussion goes to allocating how the money already put in gets finished.

Device 4: a reward for termination. Put the person who stopped their own project into the same recognition and promotion channel as the people who succeed. Without it, exit clauses decay into text automatically, because the career interest of the person executing the clause runs against it (Chapter 19's alarm logic applies here too: whoever executes an exit has to be protected first).

The four devices applied to this chapter's cases. Device 1 on case one: a rail project's approval clause would name trigger signals, a ceiling on full cost per kilometre and the external audit conclusion at an opening milestone, with the trigger obliging a review, and the

review carrying the zero-based re-ask. Device 2 on case four: the board's product-market fit review brings in an outside member who took no part in the funding narrative, with the zero-based re-ask as the only agenda. Device 3 on case three: translate the question into a morning rule on a fishing boat, whether this trip would leave port if the fuel and the schedule had not yet been committed, one question asked in a moment. Device 4 in an organization: put the record of a stop into the standard material for promotion review, so that the person who stopped and the person who succeeded are ranked on the same sheet.

The failure modes of the four. An exit clause dies when its trigger is reinterpreted, with management declaring the signal inapplicable once the quantified threshold is met; the patch is to bind the adjudication of the signal to an external data source, a third-party milestone or an audit conclusion, so that the reading is not the project's to make. An external adjudicator dies when the list of unbound people runs out, since in a large organization nobody is unbound from every project; the patch is borrowing across organizations, independent directors, outside experts, the counterpart post in a sister body, and the cost of borrowing is far below the cost of being bound. The zero-based re-ask dies of formality, with item one completed and item two arguing from sunk cost as before; the patch is a structured minute, where the answer to question one has to appear and any appearance of the phrase already invested among the reasons to continue sends question one back for a second answer. The termination reward dies of being spoken rather than written, praised in the room and ignored in the appraisal; the patch is the stop record entering the personnel file as standard material, bound to Device 4's clause as written. All four patches share one check sentence: has this device ever actually changed a project's fate.

Installation order matters as well. Device 3 first, the zero-based re-ask, which costs nothing and works immediately; Device 1 next, exit clauses, applied to any new project; Device 2 next, external adjudicators, appointed annually; Device 4 last, the termination reward, the largest piece of institutional engineering. The order follows the depth of the intervention: change the question at each review, then change the project's birth certificate, then change who sits at the table, then change the organization's career incentives. An organization that reverses the order, calling for the reward first while its project documents carry no exit clause at all, gets one award ceremony and zero exits.

27.6 Boundaries with Adjacent Mechanisms

Against the escalation ladder (Chapter 8): Chapter 8 handles the process, which rung you are standing on; this chapter handles the decision point, why you are still climbing. Test in that order, grade with Chapter 8 first, then treat with the exit machinery here.

Against narrative before evidence (Chapter 13): escalation's outward expression is usually carried by narrative ("strategic value"), and narrative also preheats escalation, since announcing an expectation first makes exit a broken promise. Separate by object: managing expectations belongs to Chapter 13, accounting for the investment stays here.

Against normalization of deviance (Chapter 11): the cost overruns and delays of an escalating project get normalized one at a time ("big projects are all like this", which is the NAO's own wording on Sizewell), and normalization is escalation's lubricant. On large projects the ledger of Chapter 11 is overrun and delay themselves.

27.7 Five Self-Checks

1. On which page of your project's approval document is the exit clause? If there is none, is there still time to write one today?
2. At which meeting did you last argue for a budget increase with "we have put in so much already"? Did the project turn around after it?
3. Who last adjusted the assumptions in your project's re-forecast? How many adjustments went one way and how many the other?
4. What post does the last person in your organization to stop their own project hold now?
5. Zero-based re-ask: if nothing had been invested as of today, would you start the project you are holding? Put the answer beside your reasons to continue for one night, and read both in the morning.

Chapter 28 Horizon Mismatch; Short-Termism

> What this chapter covers: how a structure with the benefit in the current period and the cost years later mass-produces failure; the criteria, two field sites, the handover of the bill, and making the forward liability visible. When to turn back: when setting a term of office and its assessment cycle, when making a commitment that spans years, or when assessing work whose payoff arrives in the next term. Estimated reading time: 50 minutes.

Horizon mismatch needs no ill will from anyone. Put an ordinary person into a term shorter than the consequence cycle, give him a ledger that records only the current period, and let the handover carry no unfulfilled obligations. His choices will lean toward the present systematically, and each lean is rational and defensible at the time. The disease is structural: term, ledger and handover together guarantee that the cross-period cost arrives after the decision-maker's period of responsibility has closed. The evidentiary difficulty is real. First-hand deep cases here are the hardest to collect, since decision chains are scattered across audit and budget documents from several administrations, so the chapter is built from two skeletal field sites and an honest account of the gap.

28.1 What It Is

Definition: horizon mismatch describes a decision whose benefits and costs fall on different time scales, while the decision-maker's term of office, assessment cycle or accounting basis covers only the benefit side.

The criteria (all three present, and the structure stands):

① The benefit lands in the current period, the cost three years out or later; ② The person making the decision serves a term shorter than the cost cycle; ③ Nobody is accountable for the cross-period result: no one signs for receipt, and no handover ledger exists (the interface with Chapter 30).

One intuition about units comes before the criteria. Terms are counted in years, infrastructure in decades, ecology and population in generations. The failed conversion between the three is the meta-structure of horizon mismatch: the literature on term politics works the same ground (electoral cycles against governance cycles), and corporate governance runs a gentler version (average executive tenure well short of the payback period on strategic investment). Nobody is at fault for the failed conversion. It is the original parameter of institutional design, and the preventive work of this chapter is a patch on that parameter: extend the period of responsibility past the term, and pull the cost from outside the ledger into it.

Feasibility has cross-institutional evidence behind it. Pension and sovereign fund governance, with fiduciary duty reaching decades ahead. Generational balance clauses in environmental law, some constitutions writing a duty toward future generations into the text. The trust structure of perpetual funds, where trustees change and the trust does not. Together they prove that a period of responsibility longer than the term can be built, and the common parts are three: a fiduciary structure (duty written to the post, not the person), statutory funding (money locked into a dedicated account when the commitment is made), and a public ledger (progress against each commitment published annually). Those three parts are the public-administration prototype of this chapter' s Devices 1 to 3, and a reader designing cross-period accountability can copy them directly.

28.2 How It Grows

Three generating conditions. Condition one: the assessment cycle is set by the term. Political terms, executive terms, the subscription and redemption cycle of a fund manager, all shorter than the cost cycles of infrastructure, education, ecology and maintenance. The gap is not an anomaly; it is the normal state of these fields. Condition two: accounting and statistical bases record the current period only. Forward liabilities sit outside the current statements (technical debt, deferred maintenance, demographic structure), and a liability nobody can see does not constrain this period's decision. The matching principle has little purchase in front of a long-cycle asset. Condition three: handover is where the bill is dropped. The outgoing holder does not hand over a schedule of bills, and the incoming one does not know what debt he has received (Chapter 30's rupture, drawn along a time axis). Every handover resets unfulfilled obligations to zero.

The accounting of that reset deserves one more layer. Forward liabilities stay off the statements for two reasons, basis (Chapter 2's condition) and measurement: the amount is hard to estimate, since deferred maintenance depends on future material and labour prices, and technical debt on the future shape of the business. Measurement is genuinely hard, and its corollary gets misused: from cannot estimate well, to do not record. Accounting has a mature answer to poor estimation, interval estimates with sensitivity disclosure, and the answer was never transplanted onto forward liabilities because nobody was required to transplant it. Device 2 asks for that transplant. Figures reported as intervals, sensitivity disclosed every year, error corrected against earlier disclosures: from cannot estimate to corrected annually, a forward liability turns from an impossible task into an ordinary estimation problem.

The organizational side of the reset has a finer mechanism, the direction of handover. An exit handover is usually led by the out - going holder (what to hand over, how, in what depth), and the in - coming one starts from a weaker information position. Unfulfilled obligations are exactly what the outgoing holder has least incentive to write clearly, since writing them down is a public admission of shortfall. The answer is to transfer ownership of the document to the incoming holder: he demands documents against a list, and the outgoing holder completes against it. Device 1’ s receipt clause is the documentary form of that transfer. One execution detail fol - lows. The template has to exist in advance, because a list impro - vised on the day of receipt will be missing items, and its main - tainer is the post itself rather than whoever holds it in a given term.

28.3 Field Sites

Case 1 (Public): More Than a Decade of Scope Swings

po-007. Grade A material: a value for money audit by the UK Na - tional Audit Office (NAO), 48 pages, published 11 March 2026 (HC 1734), signed by the Comptroller and Auditor General on 3 March. Page numbers in this card are PDF pages.

Case ID: po-007.

When and where: June 2014 to January 2026, the span of four documents; the report was published on 11 March 2026. The loca - tion is London: the chain of spending reviews and business cases between the Department for Transport (DfT) and HM Treasury. The three scope changes are on paper in the Integrated Rail Plan for the North and Midlands of November 2021, Network North of October 2023, and the Northern Growth Strategy: Case for Change (CP 1485) of January 2026.

What and who: DfT is sponsor, holding the total budget and delivery oversight (NAO p.17). Delivery splits between two arm's-length bodies: Network Rail upgrades the existing lines east of the Pennines (phase 1), and HS2 Ltd holds the scope and parliamentary bill for the new Liverpool to Manchester line (phase 2) (NAO p. 17). Transport for the North is the statutory adviser and, before November 2021, co-client with DfT (NAO p.19); the housing and communities department holds the local funds that may support growth benefits (NAO p.17). Ministers decide the scope, and the report writes "the government decided" throughout without naming a signatory. DfT answers to Parliament as the accounting entity, and no single post at programme level carries the signature.

What they were doing: through 2025 and into January 2026, DfT had four live workstreams. A revised business case, expected in summer 2026, to carry the strategic case, the route to be taken and indicative costs (NAO p.26). Three phases advancing at uneven depths: phase 1 has the clearest scope, with electrification and upgrade of the Leeds to Bradford, Sheffield to Leeds and Leeds to York lines, redevelopment of the Leeds and Sheffield stations and a new station at Bradford, detailed design not yet started and Network Rail expecting to begin within two years; phase 2, the Liverpool to Manchester line, is unfixed on many sections and its parliamentary authorisation will not complete until around 2030; phase 3 has no delivery body (NAO pp.31, 32). Development funding of £1.1 billion across four financial years, 2026-27 to 2029-30, at cash prices, to take the scope to the point where costs can be estimated (NAO p.24), with the £45 billion ceiling resting on the cost range of the early options and on a long-term affordability assessment (NAO p.31). Governance under review: as of February 2026 the structure was still being reviewed (NAO p.27); after the January announcement, collaboration agreements were signed with the combined authorities and a delivery committee was planned, advisory to ministers only (NAO p.29), alongside a cross-programme

team that handles interfaces and takes no decisions (NAO p.34). Cumulative spend to the end of March 2026 was £410 million at cash prices, mostly design (NAO p.25).

The decision point:

One, November 2021, the Integrated Rail Plan for the North and Midlands. Three options on the table, cost range £17.2 billion to £46 billion at 2019 prices, all premised on the planned HS2 Manchester line (NAO p.17). Transport for the North's statutory advice was the largest. DfT chose the smallest, at £17.2 billion, and the report records that the others were unaffordable and integrated poorly with the Transpennine upgrade (NAO pp.17, 20). The choice was irreversible: the smallest scope went into the plan, and two years of detailed design followed it (NAO p.20).

Two, October 2023, Network North. HS2 Phase 2 was cancelled, the Millington to Manchester section this programme had been bor - rowing was transferred in whole, and Sheffield, Hull and Bradford joined the scope. The two together lifted the estimate from £17.2 billion to £30.6 billion at 2019 prices, up £13.4 billion, and in the same document the benefit-cost ratio fell from about 0.8 to 0.4 (NAO p.23). Scope, ledger and appraisal were rewritten on one sheet of paper.

Three, September 2024, DfT commissioned a study of alternative routes and stations between Liverpool and Manchester, including Manchester Airport, with an indicative cost ranking (NAO p.23). That turned "do we keep the former HS2 section" into a question a minister could answer in 2025; the report records that the section is not necessary to meet the Liverpool to Manchester objectives (NAO p.23).

Four, January 2026, the Northern Growth Strategy. Three things settled at once: the former HS2 section is retained as the interface to a future north-south line; delivery splits into three phases, phase

1 in the 2030s and phases 2 and 3 in the mid-2040s (NAO p.32); a spending ceiling of £45 billion at 2025 prices is set, with the formal cost estimate deferred until the scope is defined further (NAO p. 24). A fourth thing the document wrote itself: the ceiling excludes money already spent before the end of March 2026 (NAO p.25).

What was visible then:

- The three Transport for the North options of 2019 and the statutory advice attached to them, which was the largest (NAO p. 20).
- The text of the November 2021 plan: the £17.2 billion option and the judgement that the others were unaffordable, side by side in one document (NAO pp.17, 20).
- Network North of October 2023: the transfer of the Manchester section and the addition of three cities (NAO p.20). Over the next seven months, October 2023 to July 2024, DfT folded the changes into the existing design, with the emphasis on upgrades originally due for early delivery, Bradford station among them (NAO p.20).
- The same October 2023 document issued two things bound together: the cancellation of HS2 Phase 2 and the expansion of this programme's scope (NAO p.21). The benefit lost to the cancellation and the cost added by the expansion were booked on one sheet, and a reader cannot separate them.
- A decision delayed through 2025, which left local authorities without a route update for the entire year (NAO p.24).
- The July 2024 King's Speech commitment, including the revived and repurposed Crewe to Manchester bill (NAO p.21), and the ministerial scope review that began the same month and ran to January 2026 (NAO p.21).
- The alternatives study commissioned in September 2024, its publication deferred through 2025: it had been due in spring

2025 alongside the spending review and the ten-year infrastructure strategy (NAO p.23).

- Transport for the North's written concerns to the sponsor board in September 2024, about the number of open items and the time DfT took over them (NAO p.27), and the January 2026 finding by the Government Internal Audit Agency that the DfT programme team had not applied the lessons already identified consistently (NAO p.36).
- The year-by-year spending table: 2015-16 to 2025-26, between £1.02 million and £81.24 million a year, £409.5 million cumulative (NAO p.25).

Known only afterwards: the report's conclusion sentence about "more than a decade" (NAO p.13); the combined authorities' account in interview that involvement was ineffective and detailed analysis once shared was no longer visible (NAO p.24); and where the line went after March 2026.

You are sitting in this position, with the material above in hand.

> **Stop here. Decide before you turn the page.**
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> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: by March 2026 the account read as follows. The formal cost estimate was pending, with nothing but a £45 billion ceiling at 2025 prices counted from 2026-27 (NAO pp.22, 24). Outside the ceiling sat £410 million of design spend already incurred (NAO p.25). The first benefits land with phase 1 in the 2030s, the main benefit section in the mid-2040s (NAO p.32). On governance,

as of January 2026 no formal structure brought the central and local funders together (NAO p.29), the structure was still under re‑view in February (NAO p.27), and the planned delivery committee advises ministers without deciding (NAO p.29). DfT identified 11 lesson themes and 24 actions itself (NAO p.35). The units of the two scales are plain here: the current period runs on financial years and cash spend, the cross-period on through-years and journey times, and between them there is no conversion table.

The three questions, read back. The commonest answer to the first is "settle the scope once and for all". In this case that action costs this term a signature for a 2040s benefit, and the report shows the previous two settlements held for 23 months and 27 months re‑spectively (the book's calculation from the dates). The most commonly missed item under the second question is the year-by-year spending table: it looks like financial data, and it is the only thing that lays out, year by year, how every term spent money and no term left a scope behind. The checkpoint under the third be‑longs on the day the October 2023 document was published: check whether the new scope text carries an item-by-item column against the previous version, marked carried over, revised or dropped, with the write-off value of the dropped items.

Options ruled out:

One, the largest scope at £46 billion. Rejected in November 2021, with the reasons in the plan: unaffordable, and integrated poorly with the existing network, the Transpennine upgrade above all (NAO p.17).

Two, dropping the former HS2 section and building a low-cost Liverpool to Manchester line only. Formally studied in September 2024, with alternative routes, station options, a Manchester Airport station and a cost ranking (NAO p.23); rejected in January 2026 because the former HS2 section is the only interface that can con‑

nect a future north-south line to the west side of the Pennines (NAO p.23).

Three, publishing the plan on schedule in spring 2025. Delayed (NAO p.23); the report records DfT's account that the ministerial scope decision was outstanding and what could be said publicly was therefore limited (NAO p.24).

Four, giving the formal cost estimate in January 2026 as well. Not given, on the ground that the scope had to be defined in more detail first (NAO p.24).

Five, giving the local combined authorities decision rights on the delivery committee. Not given: the committee is attended by the combined authority mayors and by ministers, and it advises only (NAO p.29). The January 2026 announcement wrote this into the institutional design, and the combined authorities' formal role ends at advice and collaboration agreements.

Early signals:

① August 2015: a blueprint promised £13 billion across rail, strategic roads and local transport (NAO p.20). That figure is not on the same order of magnitude as the £17.2 billion, £30.6 billion and £45 billion that followed. Why it was passed over: it was a package of investment commitments, not this programme's budget.

② 2019: the three options spanned £17.2 billion to £46 billion, a spread of 2.7 times (NAO p.20). Why it was passed over: a wide range is normal in an early estimate.

③ October 2023: the change raised the estimated cost by 78% and halved the benefit-cost ratio (NAO p.23). Why it was passed over: on the books it was recorded as an expansion of scope.

④ The spending curve followed political milestones, not engineering progress: £30.59 million in 2022-23, back up to £81.24 mil-

lion in 2024-25 (NAO p.25). Why it was passed over: annual spend moves with the design stage, which is normal.

⑤ November 2021 and October 2023 were 23 months apart, and local authorities were asked to redo their spatial and local transport plans between the two documents (NAO p.24). Why it was passed over: the line the report quotes is "just two years after the November 2021 plan".

⑥ Transport for the North's written concerns of September 2024, and the January 2026 internal audit finding that lessons had not been applied consistently (NAO pp.27, 36). Why it was passed over: a governance review was under way, and the lessons were listed as 11 themes and 24 actions.

The point of escalation: October 2023, when the estimated cost of the HS2 Manchester section was transferred into this programme whole while Sheffield, Hull and Bradford were added to the scope (NAO pp.20, 23). What came across was the Millington to Manchester section, including Manchester Airport station and Piccadilly station (NAO pp.17, 22). After that step, cancelling the former HS2 section amounted to cancelling the future north-south line: the September 2024 alternatives study wrote the consequence down plainly, and the January 2026 announcement retained the section on that basis (NAO p.23). The July 2024 King's Speech revived the bill, and the route entered the procedure of legal text (NAO pp.21, 31). The table on page 22 sets the scope of the three announcements side by side, item by item, so which sections were added and which removed is readable at a glance (NAO p.22). That table is the physical evidence of this step. Between "the scope can change" and "this section cannot change" sits one document that transferred a cost. The signatory is the government, and the report names no one.

Failure chain:

Technical layer: the engineering itself did not fail. The break sits between the engineering breakdown and the commitment. In the three-phase plan, phase 2's parliamentary authorisation will not complete until around 2030, and the breakdown assumes the preceding section's scope will not move (NAO p.31). The interface is specific down to a platform: in November 2021 DfT widened the scope and funding of the Transpennine upgrade to include elements this programme would later use, among them a possible extra platform at Huddersfield and a freight passing loop (NAO p. 34).

Organizational layer: ministers decide the scope; DfT and two arm's-length bodies deliver; the benefits have to be realised by the housing and communities department, local government and the combined authorities (NAO p.17). As of January 2026 the formal structure that would bring these parties together did not exist (NAO p.29), the delivery committee does not decide (NAO p.29), the sponsor board's effectiveness was challenged in writing in September 2024 (NAO p.27), and the team managing the interfaces explicitly holds no decision rights (NAO p.34).

Incentive layer: each term's books run on financial years, and the spending table is laid out by year (NAO p.25); the commitment runs on the whole programme, and the price base switched from 2019 to 2025 (NAO p.22); the £45 billion ceiling counts from 2026-27 and leaves the £410 million of earlier design spend outside it (NAO p. 25). Three sets of units that do not convert: annual spend cannot show a cross-term commitment, a change of price base makes the years incomparable, and a moved start date puts earlier money outside the allowance.

The junction breaks between the organizational and incentive layers. The cost of a change has no one to sign for it, and the organizational layer lacks exactly the structure that could record a signature.

Minimum counterfactual: move the start of the £45 billion ceiling from 2026-27 back to 2015-16, and count both the money already spent and the investment written off by each scope change against that ceiling. One accounting action, inside the same January 2026 decision, requiring no luck: the year-by-year spend data sits in DfT's finance system, and the table on page 25 is its product. With the change, a change of the October 2023 kind would eat into the ceiling in the current period, and scope change would carry a price for the first time. The same ceiling would move written-off design fees from a historical footnote onto a line of the decision table. It does not settle who signs; it settles how much one change costs, a question nobody had answered. The ceiling is an instrument that already exists, and moving its start date needs no legislation, only a different date in the same decision.

Outcome and sources: NAO, Northern Powerhouse Rail, HC 1734, Session 2024-2026, 11 March 2026, 48 pages, ISBN 978-1-78604-664-2; signed by the Comptroller and Auditor General, Gareth Davies, on 3 March 2026. Audit basis: fieldwork from October 2025 to February 2026, 23 interviews, of which 7 with DfT and one each with Network Rail and HS2 Ltd (NAO p.42); document review covering the agendas and minutes of the DfT executive committee, investment committee, programme board and sponsor board (NAO p.44). The report does not cover the wider growth plan for the north or the progress of the Transpennine upgrade itself (NAO p.8). Its classification is a value for money audit, and its conclusions land on whether value for money was obtained, not on who was responsible (NAO p.4). Link in Appendix B, po-007. Every figure in this card comes from that report, and the page numbers are PDF pages.

Every scope change in Case 1 can also be filed term by term against this chapter's criteria: which term made it, the reason given, the previous term's investment written off, the new position of the

delivery date. Four columns joined together make a chronicle of horizon mismatch, and the value of the chronicle is its symmetry. Each term's change had an adequate reason inside its own period (strategic adjustment, fiscal constraint, reordered priorities), and read in sequence they show one machine turning: every term changes the scope, and every change pushes the date back. The fuel is at criterion ③, since the cost of a change has no signatory. The previous term's investment is written off with nobody answerable for it on the books, the delivery date moves with nobody signing for the new position. Adding a column for the signatory, with "none" entered, is an action an audit can take immediately. On the day that column comes back empty in run after run, the mechanism is written in ink.

Evidence in ink has a multinational reference to borrow: the UK's national infrastructure commitments mechanism, where commitments are published as a register on a five-year cycle with a delivery status against each one, and the statutory funding of long-term projects in several countries, where the budget source is locked when the commitment is made. Both turn delivery from political will into a registration obligation. The register makes each term's commitments comparable and lets an unfulfilled commitment be named. Naming is not punishment, which needs procedure; its function is to keep a slipped delivery date in public view. A date may slip, but it may not slip out of sight.

Case 2 (Manufacturing): This Year's Sales and a Stove Years Later

mn-002. Grade B material: a US Consumer Product Safety Commission (CPSC) recall announcement (recall number 27-051, 29 October 2026) and the manufacturer's recall notice issued the same day. Both are announcement texts, not investigation reports: no design review records, no validation test protocol, no launch

decision meeting and no signatories. Where the announcement is silent on an internal decision, this card writes "not in the public record" and fills nothing in.

Case ID: mn-002.

When and where: manufacturing window 14 June 2021 to 20 March 2023 (Ariston recall notice); sales window May 2021 to July 2026, through independent HVAC contractors and wholesale distributors across the United States (CPSC announcement); recall announced 29 October 2026. The failure site is the boiler room or utility room of a user's home, and the site of a licensed technician's service call.

What and who: Ariston Canada Inc. (formerly NTI Boilers Inc., Canada) is the manufacturer, and Ariston Thermo USA LLC (Providence, Rhode Island) the importer (CPSC announcement). The regulators are the US CPSC and Health Canada. After-sales service runs through independent licensed technicians, and the goods move through wholesale distributors. The announcement presses the notice obligation onto four separate roles: owners, selling installers, wholesalers and installing contractors (Ariston recall notice). The job titles and holders of the product safety and compliance functions are not in the announcement.

What they were doing: in 2026 the company was running three things at once. A safety notice in February 2026 on the FTG series, requiring an upgrade kit on every unit manufactured before 8 January 2026 (manufacturer's recall page). Sales of the affected batches had not stopped: the CPSC records sales from May 2021 through to July 2026, at $3,200 to $7,700 a unit (CPSC announcement). A recall of 29 October 2026, issued with repair parts moving into the channel: a new controller, part 6501575600 for FTVN and 6501575700 for TFTN, free to contractors, labour reimbursed under standard service practice, parts arriving about three days after contact (Ariston recall notice). The action required of the channel

is two rounds of notice two weeks apart; the action required of wholesalers is to count and quarantine stock and report the list to the manufacturer (Ariston recall notice). On the manufacturer's own figure, the remedy means one visit to each of 6,329 units. All three workstreams share one unit of measure: output by unit, shipment by unit, repair by unit. Nothing is counted in operating cycles or in years installed.

The decision point:

One, 14 June 2021, the start of the manufacturing window. It is the manufacturing start date the announcement gives, and the only carrier of a "design frozen" moment this card can anchor to (Ariston recall notice). The dates, attendees and signed documents of the design review are not in the public record, and this card does not fill them. Ten models are affected: FTVN150, FTVN150C, FTVN199, FTVN199C, TFTN150 and TFTN199 from NTI, and EFTU-150WBN, 150WCN, 199WBN and 199WCN from HTP. The CPSC gives the serial numbers in two ranges, the first beginning in a date-like form and the second carrying a 3260 prefix; how the two ranges map onto the manufacturing window is not explained (CPSC announcement). The start points of the two texts do not line up either: sales records begin in May 2021, manufacturing records on 14 June 2021 (CPSC announcement, Ariston recall notice). The sales ledger and the manufacturing ledger are not on the same basis.

Two, 20 March 2023, the last batch left the factory (Ariston recall notice). For the next 40 months that stock kept selling in the chan‑nel, into July 2026 (CPSC announcement). Manufacturing stopped; release did not.

Three, 29 October 2026, the recall announcement (number 27-051). The remedy was settled the same day: replace the controller and revise the software, with the condensate tray replaced only where a technician's visual inspection finds it "appears dam‑

aged" (Ariston recall notice). This is the only decision in the case with a date, a text and a signatory, Ariston Canada Inc. issuing jointly with the CPSC and Health Canada. The economics of the remedy were fixed that day too: free per unit, labour reimbursed per visit, parts in about three days, cost following the installed base rather than the output (Ariston recall notice).

Four, 29 October 2026, the second instruction in the same notice: owners must not clear the lockout themselves, and operation may be restored only after assessment by a licensed technician (Ariston recall notice). That writes the field habit of "reset means the heat is back on" into a prohibition for the first time. The same notice leaves a way out: where the unit genuinely has to be used, fit smoke and carbon monoxide alarms first (Ariston recall notice). The sentence concedes that some units will keep running until they are repaired.

What was visible then:

- Serial numbers and manufacturing records from June 2021: the CPSC gives two serial ranges, Ariston gives start and end dates and a total of "about 6,329" units. The two records cover each other, and their start points differ (CPSC announcement, Ariston recall notice).
- Sales records from May 2021: channel type, price range and coverage (CPSC announcement).
- The FTG notice of February 2026, a whole-batch notice drawing its retrospective boundary at "manufactured before 8 January 2026" (manufacturer's recall page). The format is established practice at this company. It has nothing to do with the condensate tray defect; the item shows only that the retrospective mechanism was running.

- The 34 condensate tray rupture reports accumulated before the recall, 3 with the front cover detaching and 1 with a technician's jaw fractured (CPSC announcement).
- The mechanism as the announcement describes it: on delayed ignition the tray cracks, unburned gas accumulates in the sealed chamber, and on ignition the front cover detaches suddenly (CPSC announcement). The units are unsafe before repair and must be taken out of service (Ariston recall notice).
- The post-repair procedure, also in the announcement: replace the parts, inspect visually, run a combustion test with a flue gas analyser, and put a label in a conspicuous place so the next technician can see the unit has been repaired (Ariston recall notice).

Known only afterwards: the date, location and model behind each of the 34 reports; the material, thickness and supplier of the tray; the protocol and duration of design validation and ageing tests; the meetings, signatories and internal assessments behind the launch decision. None of the four is in the public record, and this card does not write them.

You are sitting in this position, with the material above in hand.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: 4,726 units in the United States and 2,261 in Canada (CPSC announcement); the manufacturer's own figure is "about 6,329" (Ariston recall notice). The two do not agree, and

this card records both without forcing them to reconcile. The known count is 34 rupture reports, 3 front cover detachments and 1 fractured jaw (CPSC announcement); on the manufacturer's figure, 34 reports come to about 0.54% of 6,329 units (the book's calculation from the two announcements). The remedy is a free controller replacement, a technician's visual check of the tray, a combustion test and a label, with parts arriving about three days after contact (Ariston recall notice). The units of the two scales on this batch are the current period in units and unit price, the cross-period in ignition cycles and years installed, with no conversion table between them either.

The three questions, read back. The commonest answer to the first is "lengthen the validation period". In this case that action costs the following: the manufacturing window was only 21 months and the release window 62, so validation set by the manufacturing period cannot reach the fourth or fifth year after installation. The most commonly missed item under the second question is the in-struction not to clear the lockout. It reads as an operating tip, and it is the only sentence in the case that records a precursor. The checkpoint under the third belongs at the moment the controller software was finalised, and what it checks is whether reset counts were being recorded.

Options ruled out: the announcement does not say what was con-sidered at the time or on what grounds anything was rejected. The four items below are the boundaries of the option space the text allows one to infer, each with its basis noted, no reasons added. They fall on the product fix, the level of notice, the party executing and the sales cut-off.

One, replacing the condensate tray itself. Not adopted as a general measure: the kit is a controller and software, and the tray is re-placed only where it looks damaged (Ariston recall notice).

Two, a safety notice only, with no recall. The same manufacturer used just such an upgrade kit notice for the FTG series in February 2026 (manufacturer's recall page); this time it used a recall, with "stop using immediately" (CPSC announcement).

Three, letting owners reset the unit or repair it themselves. Prohibited in plain terms: no clearing the lockout, no handling it themselves (Ariston recall notice). The cost is that every loss of heat becomes a service call, with the waiting time falling on the household.

Four, selling off the existing stock and then dealing with it. Sales stopped in July 2026, and federal law prohibits the sale of a recalled product (CPSC announcement, Ariston recall notice).

Early signals:

① Delayed ignition trips the lockout. The announcement goes out of its way to say "do not clear any lockout", which indicates that resetting is a common action in the field (Ariston recall notice). A lockout is a state the service procedure can handle on site; reporting it as a fault requires a different protocol. Why it was passed over: a lockout is the most common boiler service call, and a reset is standard practice.

② The tray cracks only after delayed ignition, and a single event leaves a crack in a plastic part visible only with the unit opened up (CPSC announcement). Why it was passed over: handled as a component replacement, inside the warranty terms.

③ Three front cover detachments. That is no longer a component-level fault, since the cover flies off and injury outside the boiler room is plausible. Why it was passed over: 3 events across an installed base of several thousand is about 0.05% (the book's calculation from the announcement figures), and the ratio reads as a one-off.

④ The same manufacturer's safety notice channel had been used three times before, with the retrospective boundary always written as a manufacturing date: the TFT seal in July 2018, the TRX burner door studs in March 2025, the FTG upgrade kit in February 2026 (manufacturer's recall page). The channel works. What is missing is not a channel but a metric that counts resets.

The point of escalation: 29 October 2026, the moment "new controller plus revised software" was chosen as the remedy. Before that, the risk sat in the strength of the tray across 6,329 units. After it, the risk sits in one execution condition, that every unit has been visited once. The announcement can guarantee only two rounds of channel notice and the wholesalers' stock quarantine (Ariston recall notice); reaching the end user still runs through a contractor, and all four roles act on their own initiative. After the recall the units do not move, so the risk scatters across thousands of homes at their installation addresses (CPSC announcement). The announcement also leaves a flexibility clause: some units will keep running until they are repaired (Ariston recall notice). The clause exists to make the recall executable, and its cost is that the risk deadline moves from the recall date to the day the last unit is visited, a day in no document. This step has a date, a text and a signatory, and no way back.

Failure chain:

Technical layer: on delayed ignition the unburned gas accumulates in the sealed chamber, the pressure releases through the front cover when it ignites, and the tray cracks first along that stress path. The remedy changes the ignition control logic and not the tray's strength: between the number of delayed ignitions the controller allows and the number the tray can take, there is no calibrated figure (CPSC announcement, Ariston recall notice). The judgement is by eye, since a technician's visual inspection decides whether the tray is damaged and the announcement sets no

quantified criterion (Ariston recall notice). Confirmation after the repair is also a one-off: a combustion test with an analyser and a label, with no further sampling (Ariston recall notice).

Organizational layer: manufacture in Canada, import into the United States, service through an independent contractor network, and a defect information path that runs contractor to wholesaler to manufacturer to regulator. To reach the end user the recall has two rounds of channel notice and nothing else; the requirement on installers is at least two rounds, two weeks apart (Ariston recall notice).

Incentive layer: the current period is measured in units and unit price, at $3,200 to $7,700 a unit; the cost in ignition cycles and years installed. The warranty counts years, fatigue counts cycles, and the two rulers have different sources. The recall's accounts count units and service visits, and they open only after the recall date (CPSC announcement).

The junction breaks between the technical and incentive layers. Because the reset count was never recorded, per-unit accounting looks entirely sound, and the 34 reports are the shape that blank accumulated into.

Minimum counterfactual: before the first production run, make "lockout reset count" a factory-recorded field in the controller, shipped with the unit. One action, taken before June 2021, requiring no luck: the controller already handles the delayed ignition and lockout decision, and the 2026 remedy changes exactly that software (Ariston recall notice). With the record in place, a cumulative reset count reaching a threshold would bring the tray into view without waiting for 34 reports. The same record makes "which batch fails first" into comparable data, and nobody can make that comparison today. The cost is one data field and one software addition; the return is turning 34 reports from a count into a rate. It does not change the strength of the tray, only the date on which it

is found. This card does not claim the action was considered at the time.

Outcome and sources: CPSC, NTI and HTP Brand Gas Boilers Re - called Due to Risk of Serious Injury or Death from Fire, Carbon Monoxide (CO) Poisoning and Impact Hazards, recall number 27-051, 29 October 2026; Ariston Canada Inc. (formerly NTI Boil - ers), recall notice of the same date, ntiboilers.com/safety-recalls/ ftvn-tftn-product-recall. Link in Appendix B, mn-002. The two texts were published on the same day and serve as checks on each other: one for the regulatory record, one for execution detail. Grade B: both are announcements without the evidence chain of an invest - igation report, so the factual statements here stay inside those two texts, and the card describes no internal decision process because no verifiable text exists. Where the two disagree (4,726 plus 2,261 against about 6,329) the figures are recorded side by side without conversion, and no media retelling was used.

The material for Case 2 stops at the announcement, and the pro - tocol and duration of accelerated ageing tests are not in the public record; this passage covers only the general method. An acceler - ated test compresses time by amplifying stress (high temperature, high humidity, high cycling), and the conversion depends on an ageing model that extrapolates the relation between stress and life. The model's assumptions are the gap between test and reality. The management answer to that gap is settled practice in reliability engineering: keep a real-time tracking sample alongside the ac - celerated test, a batch of early product held for long-term obser - vation, with measured life backfilling the model's parameters. The cost of the sample is storage and follow-up; the return is a conversion factor for the next generation that stands on measure - ment. From Case 2's angle, the 34 rupture reports were them - selves a late backfill, paid for with a user's stove and a technician's jaw.

Backfilling has a public version in consumer products, in the relation between the CPSC's incident report database (the public report stream at saferproducts.gov) and the trigger for a recall. The stream aggregates use across millions of households into a statistical signal (the six hundred power bank reports in Chapter 16's fourth case were its product), and the timeliness of the signal sets the lag before a recall: a defect with few, scattered reports, like the 34 here, waits longer. Consumer products therefore have a public route out of horizon mismatch. The aggregation speed of defect reports is the error-discovery speed of the whole product line, and the firm-side counterpart is to monitor the stream and aggregate by its own model numbers (Chapter 20's two-way flow of the industry library, on its third appearance here).

The Evidence Gap

First-hand deep cases in horizon mismatch run into two difficulties. In political and infrastructure cases the decision chain is scattered through audit and budget documents from several administrations, and the cost of collection and reassembly is high. In corporate cases the forward cost (technical debt, deferred maintenance) is almost never written up as a postmortem, because writing it up hands the successor a bill, which is exactly the self-concealment this mechanism relies on. The A-grade deep cases in this book's library are currently mostly public audits, and the directions for further collection are renewal reviews by legislatures and research on long-tail losses in insurance.

Beside the gap, four extrapolations, labelled mechanism extrapolation rather than case card. Extrapolation one, software technical debt: every expedient implementation borrows time from the future, since the benefit of shipping lands at once and the maintenance cost accrues like quarterly interest; the interest is paid by everyone who takes the code over, the principal by a rewrite years

later. The check under criterion ③ is a technical debt ledger (what is owed, at what interest, due when) with a receipt at handover. Extrapolation two, infrastructure maintenance: roads and pipe networks are designed in decades and maintained on annual budgets, and deferred maintenance shows up year by year in the surface and in burst mains (Chapter 29's home ground), while on the books it stays a discretionary line. The check is a deferred maintenance ledger and statutory funding. Extrapolation three, pensions and population: the generational mismatch between contributors and recipients is the century-scale version, with this period's contributions supporting this period's pensions and the generational cost paid by a population structure decades away; the check is actuarial disclosure, together with the political difficulty of a contract between generations. Extrapolation four, the asset management industry: the quarterly cycle of subscriptions and redemptions is mismatched against the multi-year cycle of value investing, and managers are pushed into trading to the rhythm of flows, setting the return on current assets under management against the long-term return. The check is a longer lock-up and a longer performance reconciliation period. The four cover four time scales, from code to civilization, and share one chain of criteria; converting them into real cards is also on the collection list.

The common solution across the extrapolations has a cross-scale abstraction worth stating. The terminal form of governing horizon mismatch is to turn the future cost into today's cash flow (statutory funding, depreciation and amortization, provisions), so that the future line appears in this period's cost table. Cash flow is the only language both markets and politics hear, accounting and budgeting being the common inputs of both systems, and visibility (Device 2) is the forward post of that conversion: first the number, then the money, then the liability. The order cannot be scrambled.

Visibility without a number is a slogan, visibility without money is a ledger, visibility with money but no liability is a bad debt.

28.4 Earliest Signals

Six weak signals, each with the reason it usually gets passed over.

1. The budget table has no line for future cost, or the line exists in name only. Reason given: "the distant future cannot be estimated."
2. No table compares the decision-maker's term against the consequence cycle. Reason given: "who would make such a table."
3. Key milestones cluster before the term boundary, and every benefit milestone sits after it. Reason given: "that is how the plan is staged."
4. A key post changes hands in mid-project, and the handover contains no unfulfilled obligations. Reason given: "the handover was done to procedure."
5. "The next five-year plan will deal with it" becomes the closing sentence of this plan. Reason given: "we are phasing delivery."
6. Maintenance and capability-building spend is classed as discretionary and finishes last in the queue year after year. Reason given: "protect the priorities first." (Chapter 29 picks up here.)

Put the six back against this chapter's cases and extrapolations. Signal one, the missing future cost line, matches Case 1: across the many scope changes on the rail programme, written-off investment was recorded in history and never on the current decision table. Signal two, the absent term comparison, matches everything: a table setting term against consequence cycle is the one table this

chapter should standardise, and the absence of that table is criterion ② admitting itself. Signal three, asymmetric milestones, matches Case 2: the launch milestone sits inside the assessment period, the defect exposure milestone outside the warranty. Signal four, a handover without obligations, matches Case 1: the handover between administrations carried no statement of what was being taken on from the previous term's commitments. Signal five, the next five-year plan, matches extrapolation one: every entry in the technical debt ledger is annotated "to be repaid at the next refactor". Signal six, maintenance last, matches extrapolation two: the discretion in a discretionary line is exactly the exit horizon mismatch reserves for itself.

Reorder the six by cost of evidence. Signals one and six are cheapest, needing only the budget table. Signals three and four are middling, needing the plan and the handover documents. Signals two and five are most expensive, needing comparison across terms and textual analysis. The starting recommendation is signal one: open the budget table and look for a column on future cost. Its absence can be confirmed without any permission, and the act of confirming it is the first image of this mechanism inside your own organization.

One snapshot mechanism on the time axis deserves a rule of its own: an interim bill at the midpoint of the term, when the forward liabilities of the current term are made visible and published once. It gives the term a reconciliation point, so the hidden cost of the current period is exposed once in the middle rather than saved as a surprise for the successor, and its institutional prototype is the mid-year budget review in the public sector. The interim bill and Device 1's handover list are the same list at two moments: half at the midpoint, all at exit. The difference between the two, the liability added in the second half of the term, is the trace left by the

final-year sprint, and the trace is itself a soft constraint on sprint - ing.

28.5 Prevention

Device 1: a bill handover system. When a key post changes hands, the handover is required to contain a list of unfulfilled obligations: what was promised, what is owed, when the cost falls due, where the money is. The list is signed for by the incoming holder, and gaps are completed by the outgoing one, which puts Chapter 30's handover onto a time axis. The force of the list depends on how seriously the signature is taken: signing means the incoming holder confirms the list is true, and a false list can be pursued.

Device 2: make the forward liability visible. The forward cost of technical debt, deferred maintenance, and ageing population and assets is converted each year into a note to the current statements and published. The figure can be disputed; its absence cannot be. Visibility does not buy precision. It buys the change from nothing to something, which gives every postponement a line that can be cited.

Device 3: a cross-period signatory. Create a post accountable for the result five years out (a long-term value officer, a steward of legacy assets), with every assessment metric defined outside the term and the protection of the post written into the charter. The post answers criterion ② directly: the term is shorter than the consequence cycle, and the period of responsibility can still be longer than the term.

Device 4: a milestone symmetry review. At plan approval, check the distribution of milestones: where the gap between a decision milestone and a benefit milestone exceeds the term, require an account of the mechanism that will carry delivery across the gap (a

dedicated body, statutory funding, a treaty commitment), and where no account can be given, shorten the commitment.

The four devices against this chapter's cases and extrapolations. Device 1 on Case 1: make the handover between administrations carry a schedule of commitments assumed, with every commitment of the previous term marked carried over, revised or dropped, and a signed reason for each drop. Device 2 on extrapolation two: convert deferred maintenance into an interval figure in the notes to the annual budget, and publish the curve of the backlog year by year. Device 3 on extrapolation three: place the signatory for actuarial disclosure outside the current budget department, with protection of the post in the institution's charter. Device 4 on Case 2: bring the exposure period into the decision table at the product freeze, aligning the launch schedule with the exposure cycle and calibrating the warranty terms to it.

The failure modes of the four devices. The bill handover dies for want of competition over the completeness of the list, since the outgoing holder submits a template of empty lines and the incoming one signs it. The patch is a two-source list: the incoming holder builds his own from his own enquiry, the two are read together, and the differences are the points of suspicion. Visibility of the forward liability dies of drift in the basis, with the method changed every year and the history made incomparable. The patch is a frozen basis with a change register (the same as Device 4 in Chapter 12). The cross-period signatory dies of the ornamental post, where the holder attends ceremonies and never vetoes. The patch is a statutory veto list, naming which decisions require that signature, written into the charter. The milestone symmetry review dies of rhetorical justification, where the account of the cross-period mechanism fills up with vision language. The patch is hard fields: the establishment number of the dedicated body, the account code of the statutory fund, the clause number of the treaty

text. If none of the three can be filled in, shorten the commitment. All four patches share one check-up question: at the most recent cross-period commitment, who was in the room on the due date.

The order of installation matters too: Device 2 first (visibility, since the number is the foundation of everything after it), then Device 1 (handover, hanging the list on an existing process), then Device 4 (milestone review, applied to new plans at once), and Device 3 last (the cross-period post, the largest piece of institutional engineering). The order runs opposite to Chapter 27, which starts with the zero-based question, and the reason lies in where each mechanism is cut open: escalation's lesion is in the argument for a decision, mismatch's in the measurement of one. Change the question first in one case, change the ledger first in this one. Neither operation should wait, since the cost of mismatch grows every quarter, so the real answer is to schedule both first steps in the next quarter: one quarter to install the zero-based question and the forward liability note, at almost no cost. The review question is narrower than feasibility: who will be in the room on the due date.

28.6 Boundaries with Adjacent Mechanisms

With the invisibility of maintenance (Chapter 29): horizon mismatch explains why maintenance gets cut, since the cost of the cut lands outside the term, and the invisibility of maintenance explains why cutting it does not hurt, since the cut cannot be seen. The two chapters use different faces of the same cases: the delay at the Palace of Westminster (po-005) belongs to Chapter 29, and the multi-term structure behind the delay belongs here.

With sunk cost and escalation (Chapter 27): escalation pushes the cost further onto the successor, since the reset project accumulates a fresh set of bills, and horizon mismatch guarantees that the successor can clean up without ever holding the predecessor to ac-

count. The criterion: if the decision-maker has to be present in some form to bear the consequence, the structure corrects itself. What fails to correct is this chapter's territory.

With misaligned incentives (Chapter 17): a term shorter than the consequence cycle is misaligned incentives in the time dimension. Chapter 17's accounting for externalities, seen across periods, takes the shape of this chapter's bill handover and forward liability disclosure.

28.7 Five Self-Checks

1. Of the decisions your organization is making now, which year is the latest in which a cost falls due? Where will the person making the decision be when it does?
2. How much of your term remains? Of the things you are pushing forward, how many pay off after the term ends?
3. At the last handover of a post, did you receive a list of unful - filled obligations? Did you ever hand one over?
4. In your organization's budget, which line holds the money that must be spent five years from now? If there is no such line, whose surprise is that money turning into?
5. Would you be willing to come back in five years and give your successor an account of what you are signing off now? List the items you would not.

Chapter 29 The Invisibility of Maintenance

> What this chapter covers: why maintenance is always the first thing cut, and why the arrears arrive all at once years later; four sites, the backlog booked as a liability, and a wall around the maintenance budget. When to turn back: budget-cut discussions, approvals to run equipment in a known-defective state, and the weeks after some "sudden" failure. Estimated reading time: 50 minutes.

Maintenance is the one class of work whose success is that nothing happens. A gear replaced and a gear left alone turn the same ma - chine just as smoothly today; a seal repaired and a seal ignored leave the same wall just as dry today. Its output is the absence of events, and absence is the only cost line whose value cannot be verified in the period that paid for it. So when budgets tighten, maintenance is cut first. Nobody has to be incompetent for that to happen. It is the only line whose removal leaves no trace in the current quarter. Interest on the deferral compounds with the age of the equipment, and the bill is delivered years later in a single instalment, in the form of a breakdown, a shutdown, a collapse, or an explosion. The public and the people who decide call that an accident.

Put the delivery mechanism more coldly. A maintenance backlog is the one variety of organizational liability that comes with its own detonator. A financial default is procedural, handled through col - lection, litigation, restructuring. A maintenance default is physic - al. It accumulates inside the material and is redeemed under stress, and the redemption point is chosen by physics rather than by a creditor. Two consequences follow, and both belong on the page. The timing is not negotiable: no extension, no restructuring,

wall thickness at zero is at zero. The damage cannot be apportioned: a collapse does not select who is inside the building. Together they argue that a maintenance backlog should outrank a financial debt of the same size in governance priority. Financial debt can be carried along. Physical debt only presses down. A balance sheet cannot see the difference, which is what Device 2 in this chapter is for: writing the difference into the statement.

The invisibility itself has parts, and the parts differ by role, because each role is blind for its own reason. The user cannot see the waterproofing under the decorative layer. The manager cannot see the backlog scattered across the blank cells of a thousand work orders and asset registers. The auditor cannot see it, because a financial audit reads expenditure and not condition. The regulator cannot see it, because inspection samples cover a fraction of the estate. Stack the four and the backlog acquires the most complete concealment an organization can offer: no role sees the whole of it, and every role believes its own layer is fine. The general tool that breaks the concealment is direct measurement of condition, the measured corrosion depth, wall thickness, insulation resistance. Measurement turns the invisible into a number, and a number breaks all four invisibilities at once. Device 3 in this chapter, the condition trigger, is that breaking made into procedure.

29.1 What It Is

Definition: the invisibility of maintenance means that time and probability sit between what maintenance produces (availability, service life, safety) and what is spent on it, so that no appraisal period can measure the elasticity. Deferred maintenance is therefore a hidden liability shared by every kind of organization: governments, companies, households.

Three trigger criteria. Any one of them appearing means the mechanism has started:

① Maintenance done well shows no value, and the organization has no mechanism for booking "nothing went wrong" anywhere (the interface to Device 1 of Chapter 20);

② When budgets tighten, the maintenance line is cut first and restored last;

③ The consequences of the backlog arrive concentrated, as sudden failures, and the public and the decision-makers call them accidents.

29.2 How It Grows

Three generating conditions.

Condition one: no output metric. Time and probability sit between what maintenance produces and what it costs, so elasticity inside an appraisal period is zero. With no output ledger (Chapter 20), maintenance is scored as producing nothing in every budget round.

Condition two: external invisibility. The interior state of equipment, corrosion depth, remaining lubrication, insulation ageing, seal ageing, is invisible to users, visitors and auditors alike. Anything that shows poorly is weak in resource allocation, and maintenance is the least visible of all asset work.

Condition three: the decision-maker's term is shorter than the backlog's maturity (Chapter 28's time condition). The bill for cutting maintenance always lands on the next person, and the next person's appraisal does not cover it either.

29.3 The Mechanism in Four Domains

Case 1 (Aviation): The Debt Loaded into the Cargo Hold

Case ID: av-035.

When and where: 14:13:42 on 11 May 1996 (US Eastern Daylight Time). ValuJet Flight 592 lifted off from Miami International Air‑port (MIA) and about ten minutes later went into the Florida Ever‑glades; the main impact crater lay roughly 17 miles northwest of MIA (NTSB AAR-97/06 p.15, p.18). The work site is a month earlier: SabreTech's Miami repair station, and ValuJet's cargo area at MIA.

What and who: ValuJet Airlines (a 14 CFR Part 121 carrier); Sab‑reTech, Inc. (a 14 CFR Part 145 domestic repair station, certificate RD3R811L, first issued 3 December 1969, operating under the SabreTech name from 1 February 1996, p.75). In the accident month the station had 265 permanent employees and 351 contract workers, 57 percent contract (p.76); of the 587 people who booked hours against these three MD-80s, 442 came from six labour con‑tractors (p.76). The positions at issue: the mechanic who signed the MECH SIGNOFF block of work card 0069; the SabreTech inspector who signed the Final Inspection block of the non-routine card; the logistics supervisor; the shipping clerk; the receiving clerk; and the three ValuJet technical representatives assigned to the station. Aircraft type DC-9-32, registration N904VJ.

What they were doing: On 31 January 1996 ValuJet bought two MD-82s (N802VV, N803VV), and on 1 February a third, an MD-83 (N830VV). All three were ferried to the SabreTech Miami station for modification and repair (p.18). One task was to check whether the cabin chemical oxygen generators had passed twelve years from the date of manufacture (p.18). The check found that every gener‑ator on N830VV expired in 1998 or later, while most of those on

N802VV and N803VV were already out of date (p.20). ValuJet therefore ordered all generators on the two aircraft replaced, under engineering order E.O. 15-3520-B0820 and routine work card #0069, "Chemical Oxygen Generator, Removal/Installation" (p. 20). The card was taken from the one McDonnell Douglas issued for the same task, with the content unchanged; ValuJet had not used it before (p.20 n.13). Work started in mid-March (p.27). The recorded removals on N802VV came to 72 generators, 67 three-mask and 5 two-mask, most built in 1979 and 1980; N803VV has no serial-by-serial record (p.28). The contract set delivery of N802VV for 24 April and N803VV for 30 April, with $2,500 deducted for each day late (p.27 n.25). A station memorandum of 26 April put all maintenance personnel, management included, on seven-day weeks until the three MD-80s were delivered, because of the workload (p. 30 n.29). Sign-off on the cards came on 30 April for N803VV and 5 May for N802VV (p.28). Both aircraft were then added to ValuJet's operations specifications and left: N803VV was added and departed the same day, 1 May, and N802VV was added on 7 May and departed on 9 May (p.28 n.26). On the day the aircraft left, five boxes were still on the floor of the receiving area.

The decision point: The first is the morning of 4 May 1996, when a supervisor handed the mechanic the task of signing card 0069; he signed the N802VV card on 5 May (p.28, p.30). He said the caps were not discussed at the signing, and that his attention was "entirely on the airworthiness of an aircraft with new generators installed" (p.30). The card carried no inspector sign-off block, because the task was not on ValuJet's list of required inspections (p. 26). His own count of removals was 30 to 35; the recorded figure for N802VV is 72. According to the station's quality control director, 72 people booked about 910 hours against the work covered by card 0069, the station had no consistent shift handover briefing procedure, and no system tracked which specific tasks each shift performed (p.28, p.29).

The second is the signature in the Final Inspection block of the non-routine card, in the same period. The SabreTech inspector said he knew at the time of signing that the generators needed caps, had raised it with the shop lead, and was told that the SabreTech supervisor and the ValuJet technical representative both knew and "would handle it in the warehouse". He signed after receiving that answer (pp. 30–31).

The third is 7 or 8 May 1996, when the logistics supervisor came to the receiving hold area and ordered the floor cleared. The reason was a prospective customer audit, after another customer had written up the area's housekeeping as deficient; he said he did not know what was in the boxes, and he gave no instruction for disposing of them (p.32).

The fourth is 8 May 1996, when the shipping clerk asked the logistics supervisor whether he should seal the boxes and get them ready for Atlanta. The answer was "Okay, that sounds good to me." (pp. 32–33). The clerk redistributed the generators among five boxes, filled the top of each with 2 to 3 inches of bubble wrap, sealed them, and affixed blank SabreTech address labels and ValuJet COMAT labels reading "aircraft parts" (p.33). The next day he asked the receiving clerk to make out an air waybill and handed him a note asking for "Oxygen Canisters - Empty". The clerk abbreviated Oxygen to Oxy, put Empty in quotation marks, and dated the document 5/10/96 (p.33). On 11 May a truck delivered the shipment to ValuJet's ramp, and ValuJet employees signed for it (pp. 33–34).

What was visible then: ① Work card 0069, a seven-step process, step 2 reading "if the generator has not been expended, install the shipping cap on the firing pin" (p.24). The card omits step 8 of the McDonnell Douglas manual, "store or dispose of the oxygen generator" (p.25). It carries no hazardous materials marking and no requirement for an inspector's signature (p.26). ② Section 2.D of

the McDonnell Douglas MD-80 maintenance manual 35-22-01: no generator may be disposed of before it has been fired and fully expended, and disposal requires securing it in a noncombustible fixture, pulling the release pin, and waiting for the shell to cool (p. 25). ③ The green "Repairable" tag on the generator itself (Form M021), with "Reason for Removal" reading "out dated", "out of date", "expired" (p.27). ④ The four tag definitions on page 44 of section five of SabreTech's inspection procedures manual: green means awaiting repair, red means scrap awaiting disposal, must be segregated, must carry an inspector's signature (p.77). ⑤ Around 10 April 1996, a ValuJet technical representative saw generators on a parts rack without caps, warned the mechanic that they would catch fire, and suggested disposing of them "with the rest of SabreTech's hazardous waste". He then went to the inspector, the supervisor and the project manager; the boxes were moved more than three weeks after that first approach, and he did not know where they went (p.31). ⑥ The five cartons on the receiving floor on 8 May, and one fact about them: no written procedure at the time required anyone delivering items to that area to state the contents or whether they were hazardous (p.32). Information learned afterwards: of roughly 144 generators removed, only about 6 were reported as expended, and the other 138 or so have no disposal record (p.28); the SabreTech Miami station never trained its employees in hazardous materials recognition or in air transport, and ValuJet's policy of carrying no hazardous materials was never passed to the contractor (p.79, p.136); four other MD-80 operators each had a practice, Alaska requiring on-site firing and disposal from 1992 and American from 1994, Delta and TWA fitting caps and shipping the generators back to base as hazardous materials by COMAT (p.106).

You are in the seat now, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 14:10:03 the cockpit voice recorder picks up an unidentified sound and the captain asks "What was that?"; at 14:10:22, "we need to go back to Miami"; at 14:11:12 a flight attendant shouts "completely on fire"; at 14:12:45 calls from the ground go unanswered; at 14:13:42 the aircraft hits the ground (pp. 16–17). All 110 people aboard died: 2 pilots, 3 flight attendants, 105 passengers (p.15). The aircraft was destroyed, valued at $4 million; salvage and cleanup cost more than $10 million (p.34). Back to the three questions. The commonest answer to the first is "add a disposal step to the work card", and that step costs nothing, because the McDonnell Douglas card already had step 8 and it was dropped in transcription. The item most easily missed under the second is the pair of quotation marks around Empty: the receiving clerk added them, they look like a spelling habit, and they wrote "unexpended" as "empty". The checkpoint under the third belongs at the moment the maintenance site hands material over to the freight staging area, where no written handover procedure existed (p.32).

Options ruled out: Firing the generators on site: Alaska and American already did so before the accident, SabreTech was doing this task for the first time, the caps were not in stock, and under the contract they were "peculiar expendables" to be supplied by ValuJet or separately agreed (p.29 n.28, p.106). Fitting caps and shipping the generators back to base as hazardous materials: the Delta and TWA practice, which requires the carrier to be author-

ized to accept hazardous materials, while ValuJet's policy was to carry none (p.106, p.80). Handing the shipment to Dangerous Goods Consultants for packing, labelling and documentation: SabreTech held a contract with the company and the shipping clerk could call it directly, but this shipment never entered that flow (p. 80, p.33). Waiting for ValuJet's Atlanta warehouse to decide on disposal: a technical representative called on 9 May to coordinate and was told to wait until 13 May to settle who would do the in‑ventory (p.32). Using the contract's written authorization clause: the service agreement stated that equipment removed and not re‑installed was retained by SabreTech and could be disposed of only with the customer representative's written authorization, and that if no authorization came within thirty days SabreTech had to notify the customer in writing of its intention to dispose of the items (p.32 n.36). That path was written in the same contract. Nobody started it.

Early signals: ① On 10 April 1996 the technical representative saw the uncapped generators and reported them up the chain, to the inspector, the supervisor and the project manager, all three of whom later denied having been approached; the boxes were moved more than three weeks later (p.31). What overrode it: the matter "had already been reported", and each position assumed the next step sat with someone else. ② No caps in stock: a supervisor told the mechanic the company had none (p.29), and another su‑pervisor said nobody had raised the need with him (p.29). What overrode it: this was the first time the task had been done, and under the contract the caps were not SabreTech's stock respons‑ibility. ③ Life-limited parts should have carried a red tag under the manual, meaning scrap, segregated, inspector's signature, and what they actually carried was a green tag, meaning awaiting re‑pair (p.77, p.27); the shipping clerk read green as meaning empty (p.33). What overrode it: the tag read differently at different posi‑tions, and no crosswalk table existed. ④ On 22 and 24 December

1995, two fires broke out in packages of expired generators at Delta's Atlanta maintenance station; one was found to have its safety clip not fully inserted, and Delta moved to on-site disposal on 26 January 1996 as a result (p.107 and n.114). What overrode it: the two fires were not reportable to the FAA, and they were not reported. ⑤ On 10 August 1986 a DC-10 at O'Hare was destroyed by a fire from generators packed on the backs of COMAT seatbacks in the cargo hold; the FAA issued a notice on the 13th of that month, the notice lapsed a year later, and no periodic follow-up replaced it (pp. 107–108). What overrode it: a one-time notice. ⑥ On 10 February 1981 the Safety Board recommended, on the same class of cargo hold, that the FAA reassess Class D certification and either require detection and suppression or require a liner able to contain such a fire (A-81-012) (p.41). Fifteen years later the forward hold of this aircraft was still Class D.

The point of escalation: The oral approval of 8 May 1996. Sealing the boxes and labelling them with a COMAT tag reading "aircraft parts" turned five boxes from maintenance residue awaiting disposal into company material: the identity changed from "to be disposed of" to "to be delivered", and the action from "dismantle and destroy" to "transport and deliver". From that moment every downstream position, the receiving clerk, the truck driver, the ramp lead, the hold loader, the first officer, handled company material only. Before it, every node still held an interface for reversal: the card could have had step 8 added, the inspector could have declined to sign, the boxes could have stayed on the rack. After the sealing, one reversal interface remained, the commodity description on the air waybill.

Failure chain: Technical layer: the generator went without its shipping cap, the retaining pin shifted in handling and the generator fired; the shell's operating temperature runs 450 to 500 degrees Fahrenheit (p.21); the DC-9 forward cargo hold is Class D

under 14 CFR 25.857(d), a class that requires neither smoke detection nor fixed fire suppression (p.40). Organizational layer: the work card was missing the disposal step (p.25); an item outside the required inspection list had no inspector signature (p.26); a life-limited part carried the wrong tag (p.27, p.77); the receiving area had no written handover procedure (p.32); contractor staff had no hazardous materials training (p.79, p.136); ValuJet technical representatives were contractually barred from supervising or directing the contractor's employees (p.85). Incentive layer: the delivery deadlines and the $2,500 daily penalty (p.27 n.25), the seven-day weeks (p.30 n.29), a customer audit imminent after the area's housekeeping had already been written up (p.32); and the hold had no cargo restraint, so the boxes pressed on the tyres (p.34). One relation between the layers is worth recording: every link in the technical layer is written in a manual, and every link in the organizational layer is written in nobody's job description. The break at the junction sits between the organizational and incentive layers: the floor was cleared to pass an audit, the boxes were sealed to clear the floor, and the missing disposal process was filled that day by the act of shipping.

Minimum counterfactual: On 9 May 1996 the shipping clerk writes not "Oxy Cannisters - Empty" but "Unexpended chemical oxygen generators, oxidizing solid, Class 5.1", and hands the shipment to the contracted dangerous goods agent for documentation. One action, one position, doable that day, no spares needed, no additional approval (p.80 records that the clerk had authority to contact the agent directly), and nobody required to admit a previous error. The downstream consequence is certain: ValuJet's policy was to carry no hazardous materials, so a Class 5.1 oxidizer on the paperwork would have kept this shipment off any aircraft at any receiving dock (p.80).

Outcome and sources: 110 people died (p.15). The report finds SabreTech's failure to prepare, package and identify the unexpended chemical oxygen generators correctly to be a cause of the accident (p.133), and ValuJet's inadequate oversight of SabreTech to be a cause as well (p.137); SabreTech surrendered its repair station certificate on 15 January 1997 (p.96). Primary source: NTSB, Aircraft Accident Report NTSB/AAR-97/06, adopted 1997-08-19, link in Appendix B av-035. Page numbers here are PDF pages (printed page equals PDF page minus 14).

The arrears chain in Case 1 carries one more reading, at the level of industry structure. Outsourced maintenance is the normal form in aviation, especially among low-cost carriers. The efficiency gain is immediate, in labour and in parts. The boundary of quality responsibility stays blurred, with the contractor's disposal process, the carrier's acceptance check and the regulator's classification all three incomplete. The revisions the industry made after ValuJet, hazardous materials classification for maintenance residues and contractor qualification and oversight, were in substance an attempt to restore visibility to the outsourced chain: maintenance can be subcontracted, and responsibility cannot be subcontracted with it. Generalized across industries, wherever maintenance is outsourced, in property, equipment or IT, the acceptance checklist must include disposal items, what happens to the old parts and how the residue is stored. A disposal backlog kills as reliably as a repair backlog.

The blank in the record in Case 2 has a manufacturing-process detail worth enlarging. The opening of the door plug for riveting rework was a legitimate temporary state inside the flow. The backlog is created by the closing of that state going unrecorded: no work record on reinstallation, no trigger for a re-check. Legitimate temporary states are everywhere in manufacturing, openings left for access during servicing, fasteners left loose for a trial fit. Their

shared hole is that the closing action has no checkpoint. The fix is at process level: a temporary opening must be registered, opening paired with closing, and any location whose closing is unregistered enters the pre-delivery re-check list automatically. Paired registration turns every legitimate temporary state into a trackable object, and the backlog, an opening never closed, acquires a ledger.

The palace in Case 3 offers a governance comparison worth writing down. The British national heritage system never stopped assessing the building's condition, and the annual reports all describe the ageing. The backlog is a failure to get scheduled. That failure is half Chapter 18 (diffusion of responsibility) and half Chapter 28 (horizon mismatch); this chapter adds a third half, losing the priority competition. In a budget ranking, maintenance competes with growth spending, new projects get ribbon-cuttings, and maintenance gets no ceremony. The competition comes out the same way in most years. The levers that change the result are Device 2 in this chapter, booking the backlog as a liability so the figure is nailed into each year's budget document, and a narrative lever, rewriting maintenance from a cost into the preservation of value, since keeping a heritage building's value is itself an output. Both are slow variables, and the year a slow variable starts sets the year the backlog curve turns. Start one year late and the curve climbs for one more year.

The progressive collapse in Case 4 comes with an engineering point that belongs in this chapter. What makes progressive collapse frightening is the concealment of load transfer: one connection fails, the load quietly reroutes, the overload accumulates in the next member, and nothing shows on the outside. A maintenance backlog runs the same pattern. The interest is never settled, because there is no interim payment event; it moves inside the structure, from this waterproofing to that rebar, from this cover layer to that concrete, and the ledger of the transfers is published

only on the day of collapse. The one tool that breaks the transfers is direct condition measurement, Device 3's condition trigger. Measurement is a snapshot of the internal transfers, and the frequency of snapshots sets how long a transfer can hide.

Case 2 (Aviation): Assembly Without a Reinstallation Record

av-030. Alaska Airlines 1282, a Boeing 737-9 (N704AL), 5 January 2024, Portland. NTSB investigation report (AIR-25-04; the PDF was downloaded and read for this book): during the climb the left mid-exit door plug separated, the cabin depressurised rapidly, and the aircraft returned safely; 7 passengers were lightly injured, 169 were uninjured, and the airframe was substantially damaged (the report states on page 10 that the four retention bolts "were not located"). The report's safety issue points directly at the state of work on that door during final assembly at the Renton factory: the plug had been opened for riveting rework, and on reinstallation the positioning bolts were missing with no work record. The form the backlog takes here is a blank in the record: one opening of a critical area triggered no human re-check anywhere along the release chain.

Stop at the decision point: the release node after the door was closed and before the aircraft passed into delivery service. On the table: incomplete work-order records, and inspections ticking items off an existing checklist, a checklist with no line reading "reinstallation after opening a critical area must be re-verified". 〔Outcome〕 Fleet-wide grounding for inspection, and multiple revisions to the manufacturing process and to regulatory oversight. Primary source: NTSB, AIR-25-04, 2025, link in Appendix B av-030.

Case 3 (Public): The Palace, Thirteen Years of Repair Without Work Begun

po-005. The Restoration and Renewal programme for the Palace of Westminster. NAO 2026 update report (pages read for this book): the two Houses established the programme in 2013 and decided in 2018 to decant for the works, and at publication the programme was still in the early stages of the first scheme, seven years and capped at £3 billion. The building's maintenance backlog accumulates in units of a decade: every year's decision to wait was, in its own year, the money-saving decision, and the building's ageing curve does not pause for a budget cycle. World Heritage status means the bill for the backlog is written into every annual report, and the start of work still recedes. A backlog visible to this degree that still cannot be moved is the strongest form of this mechanism: nobody is unaware of it, and knowing does not get it onto the schedule.

Stop at the decision point: the approval moments, budget cycle after budget cycle, at which "the major works will not start this year". On the table: the fiscal room available in that year, the building's condition survey, and the political weight of the scheme. 〔Outcome〕 The backlog keeps accumulating and the scheme remains in its early stages. Primary source: NAO, Restoration and Renewal 2026 Update, 2026, link in Appendix B po-005.

Case 4 (Construction): A Forty-Year-Old Building's Progressive Collapse

Case ID: ce-009.

When and where: About 1:30 a.m. on 24 June 2021 (US Eastern Daylight Time), Champlain Towers South condominium, Surfside, Florida. The failure began three weeks earlier: in early June 2021, two slab-column connections between garage columns and the

pool deck failed in punching shear, one after the other (NIST technical findings, released 2026-06-22).

What and who: Champlain Towers South Condominium Association, Inc., the 136-unit owners' association and its board; Morabito Consultants, the structural engineer retained by the association, which submitted a structural field survey report on 8 October 2018; the Town of Surfside building department and its Building Official; the NIST National Construction Safety Team (NCST). Responsibility sits on three positions: the association board, for repair decisions and apportioning the cost; the retained structural engineer, for inspection and reporting; the building official, for review and clearance. The building was twelve storeys plus a penthouse, 136 units, three joined sections, reinforced concrete flat plate without beams, completed in 1981 (NIST news release, 2026-06-22).

What they were doing: Maintenance work continued throughout, and it still leaked. About fifteen years after completion, around 1996, on evidence of cracking, leakage and rebar corrosion, the association carried out a major repair of the pool deck: most of the original quarry tile was removed and some left in place, a waterproof membrane was laid over the existing sloped concrete topping, and a sand bed and new pavers went on top (NIST technical findings transcript). NIST's wording is that the added pavers and sand put load on a system that was already structurally inadequate. NIST also found that rebar corrosion had been visible more than twenty-five years before the collapse, and had progressed far enough by 1996 and 1997 to require structural repair and waterproofing; problems continued after that repair, with leakage and rust staining still present. A row of large planters stood on the pool deck, planted with palms, and was removed after the 2017 hurricane. In 2018 the association retained a structural engineer to begin the 40-year recertification, several years ahead of the 40-year

date; the report of 8 October went to the association treasurer (Miami-Dade grand jury report p.20), and the repair estimates it listed came to more than $14 million (p.21). In April 2021 the board president wrote to the owners: because most of the concrete and waterproofing work lay below ground, "almost the entire ground level" would have to be opened up, including the pool deck, the whole entrance driveway, the ground-level parking, the north-side contractor parking, and the planter landscaping. The letter said the visible damage in the garage and elsewhere had become signi‑ficantly worse since the initial inspection, and it explained the mechanism: when concrete spalling is visible, the rebar holding it together is rusting and losing section, and the deterioration accel‑erates (p.21). By April 2021, more than 29 months after the report was received, the association had not started any repair work (p. 21).

The decision point: The first is the board's regular meeting of 15 November 2018. The minutes record that the "40-year certifica‑tion" and the "structural engineer's report" had been dis‑cussed and had been reviewed by the then Surfside building offi‑cial, and they relay that official's finding that the required data had been collected and that the building appeared to be in very good condition (grand jury report p.20, quoting the minutes). That meeting produced no repair project and no vote on a source of funds. The grand jury report records one checkable link: structural repair requires a building permit, and on that basis it notes that the absence of any repair work should not have been invisible to the building official at the permit stage (p.21).

The second is the president's letter to owners in April 2021. It set the scope as opening the entire ground level and the deterioration as accelerating, but gave no start date and no schedule for appor‑tioning the cost (p.21). The letter also stated the trajectory, that the observed concrete damage would multiply over the coming years

(p.21). It was written before bids were opened, and not one sentence in it carried a date.

The third is early June 2021. A long horizontal crack opened in the masonry planter wall on the pool deck, with severe damage and displacement where the planter box met the wall; witnesses photographed it, and the same planter in a photograph from fourteen months earlier shows none of that damage (NIST technical findings transcript). In the same period the door from the pool deck to the street-level garage jammed and could not be used. Three things, on one deck, in one stretch of weeks, triggered no structural inspection.

The fourth is the board meeting that had been called to open the repair bids. Per the grand jury report, the building came down thirteen days after that scheduled date (p.21). Thirty-two months had passed since the October 2018 report.

What was visible then: ① The structural field survey report of 8 October 2018, the repair scope it listed, and the estimate above $14 million (pp. 20–21). ② The board minutes of 15 November 2018, including the relayed "very good condition" (p.20). ③ The president's letter to owners of April 2021 (p.21). ④ Visible rust, leakage and concrete spalling in the garage and on the pool deck: NIST records that after the 1996 and 1997 structural repair and waterproofing the problems continued, and leakage and rust staining did not stop (NIST transcript). ⑤ The early June 2021 photographs of the horizontal crack in the planter wall and the displaced planter box, alongside a clean photograph of the same place at the same angle from fourteen months earlier. ⑥ The jammed door. ⑦ A letter that translated spalling into rusting rebar: the April 2021 letter told every owner what to look for, and after it the rust in the garage was no longer invisible to a non-engineer (p.21). Information learned afterwards: the first punching shear failure came in early June and the second in mid-June, and for the two or three

weeks after that load rerouted across the deck (NIST news release); the collapse began at the pool deck, not at the tower; corrosion was measured across hundreds of rebar samples, with moderate to severe corrosion concentrated under the planters and where chloride-bearing water had wetted the concrete (NIST transcript).

You are in the seat now, with the material above in front of you.

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At about 1:30 a.m. on 24 June 2021 the middle and east sections of the building collapsed, only the west section remained standing, and 98 people died (NIST news release; grand jury report p.4). NIST excluded several possible triggers: vibration from construction at a neighbouring site, foundation failure and sinkhole settlement, hurricane and storm surge, impact loads such as a dropped load or an explosion, and overload from the roofing work under way at the time of collapse (NIST news release). Back to the three questions. The commonest answer to the first is "call the engineer when the planters cracked", an action whose cost is one site visit, when the association already had a structural engineer of record. The item most easily missed under the second is the major repair of around 1996 itself, which looks like evidence that maintenance was done and in fact added load at the connections. The checkpoint under the third belongs on abnormal displacement of non-structural components: a jammed door, a long crack in a planter wall, a displaced planter box, three at once on the same

deck for about three weeks, with no procedure at the time treating them as a structural signal.

Options ruled out: Replacing the pool deck waterproofing together with the sloped topping: what was chosen around 1996 was a membrane over the old topping, then sand, then pavers (NIST transcript). Starting repairs as soon as the report arrived: bids above $14 million apportioned across 136 units, and twenty-nine months passed after the report without work starting (grand jury report p.21). Inspecting earlier: the association had already done the inspection ahead of time, and that link was not the missing one (p.21). A structural emergency inspection of the planter damage in early June 2021: there were photographs, witnesses and a jammed door, and no procedure defined abnormal displacement of a non-structural component as a trigger for structural inspection (NIST transcript). Opening the bids and signing the contract at the pace then in force: even signed in mid-June, the work was months away and would have come after the June failure (grand jury report p. 21).

Early signals: ① Leakage and rust staining continued after the 1996 and 1997 repair (NIST transcript): treated as a cosmetic prob - lem, patched and repainted each time. ② The same repair added a sand bed and new pavers, raising the dead load: recorded as an improvement. ③ The report of 8 October 2018 and the building official' s relayed "very good condition" (p.20): the report entered the recertification flow, and that flow ends at submission, not at construction. ④ "The deterioration is accelerating" in the April 2021 letter (p.21): written into an explanatory letter to own - ers, and into no engineering deadline. ⑤ The horizontal crack in the planter wall and the displaced planter box in early June 2021 (NIST transcript): entered the repair channel for planters and doors, and not the structural one; the photograph from fourteen months earlier, at the same angle, was clean. ⑥ The row of large

planters on the pool deck is not on the original drawings: NIST compared the drawings with aerial photography and confirmed the planters are absent from them; they held palms for most of the building's life, the palms were removed after the 2017 hurricane, and the planters stayed. The addition never entered any record of structural change, and its load stayed on the slab.

The point of escalation: The pool deck repair of around 1996. The action was to remove part of the tile, lay a membrane over the old sloped topping, then sand, then new pavers, and the association initiated it: NIST's words are that evidence of cracking, leakage and rebar corrosion prompted the association to carry out a major repair of the pool deck. That step did two things at once, adding dead load to connections already short of capacity and turning waterproofing into a layer stacked on a layer. The construction detail stays in the record too: only most of the original tile was removed, some stayed in place, and the membrane went over the old sloped topping. The repair stopped at the surface. For the next twenty-five years every leak and every rust stain happened on top of that assembly, and each repaint reinforced it.

Failure chain: Technical layer: the punching shear margin at the slab-column connections was close to zero as built (NIST news release); load was added around 1996; rebar corroded in a chloride environment and connection capacity fell with it (NIST transcript: moderate corrosion alone can significantly reduce the capacity of a connection); in early June 2021 came the K-13.1 punching shear failure, with the slab held by membrane action and the adjacent connections, load rerouted, and a second failure in mid-June; on 24 June the south edge of the pool deck pulled away from the supporting wall, the north edge broke away from the face of the mid-rise tower, two tower connections were pulled out, the middle section collapsed progressively, and the east section followed (NIST news release). Organizational layer: the 40-year recertification is a

submission regime, the report is submitted, the official comments orally, and the repair need not become a project (grand jury report pp. 20–21); the vote on the repair budget and the determination of structural safety sat in two separate flows; abnormal displacement of a non-structural component appeared in no inspection trigger (NIST transcript). Incentive layer: $14 million across 136 units; and doing the inspection early lengthened the gap between report and construction, October 2018 to June 2021, thirty-two months (p.21). The break at the junction sits between the organizational and in - centive layers: the report moved the state from unknown to known, but "known" had no corresponding action and no date, and the calendar of the funding vote was not the calendar of physical fail - ure.

Minimum counterfactual: In early June 2021, within forty-eight hours of the horizontal crack, the displaced planter box and the jammed door, have the structural engineer of record inspect the slab-column connections at the pool deck and put the conclusion in writing. One action, doable at the time: the engineer was of re - cord, the contract relationship existed, no new budget approval was needed (grand jury report pp. 20–21); triggering it took one phone call, and nobody had to first admit the building was dan - gerous. What the inspection looks at is specific: deflection of the slab around the columns, diagonal cracking in the connection zone, and rust-staining cracks or water marks on the soffit, which are exactly the appearances NIST later identified as the punching shear failure (NIST transcript). The most likely result of that in - spection was closing the pool deck and adding temporary shoring, spending the remaining three weeks on unloading.

Outcome and sources: 98 people died (NIST news release). NIST's technical findings were released on 22 June 2026 and the final report is not yet published; every NIST citation here rests on the release page and the official transcript, without page numbers.

The timeline of the association's documents cites the Miami-Dade grand jury report (December 2021, a 43-page PDF), with page numbers as PDF pages (printed page equals PDF page minus 3). Primary source: NIST, Technical Findings on the 2021 Partial Collapse of Champlain Towers South, 2026-06-22, link in Appendix B ce-009.

Case 5 (Chemical): The Dust Collector Nobody Cleaned (condensed)

ch-008. The polishing shop of Zhongrong Metal Products Co., Ltd., Kunshan, Jiangsu, 2 August 2014. The State Council "8·2" accident investigation team's report (published on the Ministry of Emergency Management website, full text read for this book): the dust removal system had not been cleaned as required for a long period; aluminium dust in collection bin No. 1, damp and oxidising, heated and ignited, and the workshop detonated as a whole; 97 people died, 163 were injured, and direct economic loss came to 351 million yuan. Accumulated dust in a dust removal system is the most typical maintenance object there is, since one cleaning costs hours, and the backlog here took the form of each successive relaxation of the cleaning interval. The lower explosive limit of aluminium dust means that once the accumulation passes a threshold it is an explosive; the interest on this backlog is measured in kilograms and the bill is settled by a detonation. The decision point sits in the scheduling documents that relaxed the cleaning cycle, one after another. Minimum counterfactual: write the cleaning frequency into process discipline and pair it with a measured dust depth. Primary source: State Council "8·2" accident investigation team, 2014, link in Appendix B ch-008.

Case 6 (Aviation): The Locking Tab Nobody Could See (condensed)

av-012. Helicopter external-load logging operations, January 2023, United States. NTSB report (2024, read for this book): after a gov - ernor lag adjustment the previous day, work continued with the locking tab not clamped; a ground check could not reveal that state, and during the next day' s operation power dropped ab - ruptly, the helicopter rolled inverted, the pilot was uninjured, and the airframe was substantially damaged. The invisibility of main - tenance takes a physical form here: whether the tab is clamped is buried in the mating faces of the component, and no one standing and looking, including the person who did the work, can confirm it. Only a targeted check action can, by touch or with a gauge. Minimum counterfactual: add an explicit check step at the close of maintenance on that location, the action written as a line item, from the same definition of a check action as Chapter 22. Primary source: NTSB, 2024, link in Appendix B av-012.

Case 7 (Drugs and Devices): The Two Years That Arrived Late (condensed)

ph-004. Enoxaparin sodium injection, storage and transport tem - perature excursion. FDA enforcement report (the API record was read for this book): the excursion occurred in early 2022 and was reported only in January 2024, and the recall covered 2 units while distribution was nationwide. A deviation in maintenance and storage, cold chain drift, is invisible when it happens, and a re - porting lag of two years turned the invisible into the normal. The recalled quantity is tiny, and the value of the case is the lag itself: for two years the same storage conditions could have recurred with nobody the wiser. Minimum counterfactual: continuous temper - ature recording in storage and transport, with a real-time alarm on excursion, so a deviation is reported when it happens rather than

discovered in a review. Primary source: FDA, 2024, link in Appendix B ph-004.

Case 8 (Nuclear): The Register Gap the Stocktake Found (condensed)

nu-003. The laboratory of the Saint-Étienne School of Mines, France, September 2026. ASNR event notification (read for this book): a stocktake found 2 natural uranium samples, 40 milligrams in total, missing, and an unregistered americium-241 sealed source; the event was rated level 1 on the INES scale. The gaps in the register, an unregistered source and samples of unknown whereabouts, were invisible to everyone before the stocktake: the data was not ugly, nobody had counted. ASNR inspection had already required an annual physical inventory, and the exposure of the gaps is itself proof that the frequency of the inventory had not been implemented. Minimum counterfactual: run the annual physical inventory as the inspection required and keep the record. Primary source: ASNR, 2026, link in Appendix B nu-003.

29.4 What Shows Earliest

Six weak signals, each with the reason that usually overrides it.

1. Maintenance as a share of the budget falls year by year, while the number of assets and their age stay flat or rise. Override: "efficiency gains."
2. Maintenance shifts from planned work to "fix it when it breaks". Override: "condition-based maintenance is more economical." It varies by equipment, and it is often the opening of a backlog.
3. The size of the backlog is quantified for the first time in a report, and for many institutions the first quantification is itself late. Override: "getting a clear picture of what we hold."

4. Approvals to run with a known defect begin citing earlier approvals to run with a known defect. Override: "nothing happened last time." (Chapter 11.)
5. A maintenance crew is pulled off to support a production or business peak, and does not come back. Override: "the wider interest."
6. In the post-event report, the failed item "had exceeded its design life". Override: "service beyond life is normal."

Put the six signals back against this chapter's eight cases. Signal one, the share falling year by year, maps to Cases 3 and 5: both the palace's annual maintenance and the shop's cleaning frequency have a downward curve that can be drawn. Signal two, planned work turning into repair on failure, maps to Case 6: the locking tab check had no place in the maintenance plan, and the state was found only when it broke, and by someone else. Signal three, the first quantification, maps to Case 4: the structural assessment at Surfside had already put the repair figure onto the owners' agenda, and the first quantification is the eve of the general offensive. Signal four, running with a known defect by precedent, maps to the extract fan in ch-001, Chapter 11's case. Signal five, crews pulled away, maps to Case 5: relaxation of a cleaning frequency usually travels under the name of supporting production. Signal six, service beyond life as normal, maps to the pipework in Case 5: the 1998 pipe on the 2024 bill is that sentence.

Classify the eight cases by what carries the backlog. Material state carries Cases 4, 5, 6 and 8, the concrete, the dust, the mating faces, the register. Process record carries Cases 2 and 7, the reinstallation record and the reporting lag. A budget line carries Case 3, the annual postponement. The three carriers need three different detection tools: material by direct condition measurement (Device 3), record by paired registration and latency monitoring, since a backlog is an overdue record, and budget line by liability booking

and a wall (Devices 2 and 4). Run the self-check across the three: which class holds most of your organization's backlog? The answer points at your first purchase.

29.5 How to Defend

Device 1: an auditable maintenance ledger. Maintenance moves from soft spending to a hard ledger, with what was done, when, against which standard, and at what cost entered line by line and open to inspection, the asset version of the trace-keeping requirement in Chapter 38. The ledger changes the game by itself: when maintenance is cut, what gets cut is an entry with a number and a history, not an anonymous percentage. The recommended auditor of the ledger is the insurer, since a backlog drives its payouts directly and it is the only reviewer with an economic motive built in.

Device 2: the backlog booked as a liability. Deferred maintenance is converted at the industry's standard rates into a liability figure, entered in the notes to the balance sheet, and published annually, which is mature practice in public asset management. The backlog turns from "work not done" into "money owed", and the semantics of the budget discussion turn with it, from "should we spend this" into "when do we repay this".

Device 3: a dual trigger of age and condition. Maintenance triggers upgrade from "calendar due" to "calendar or condition, whichever comes first", and spending on condition monitoring, vibration, thermography, corrosion survey, is part of the maintenance budget rather than an optional extra. The structural inspections at Surfside, the residue disposal at ValuJet, the reinstallation record on 1282: all three cases were short of exactly this layer, the condition trigger.

Device 4: a wall around the maintenance budget. The maintenance line gets a floor, set as a proportion of asset replacement value, and

cuts below the floor require the consent of an external independ‑ ent party, an auditor or an insurer. The wall's purpose is not to make maintenance uncuttable. It is to make every cut confront a party with a stake in the answer: the question the insurer will ask, who pays for this backlog later, is precisely the question missing from the budget meeting.

Four devices, applied to this chapter's cases. Device 1 on Case 6: explicit check entries at the close of maintenance go into the ledger, what was checked, who did it, what the gauge read. Device 2 on Case 3: the palace's repair backlog converted annually into a note in the public accounts. Device 3 on Case 6: the locking tab is inspected on a fixed interval, and a borescope finding sets an im‑ mediate re-check, so whichever comes first triggers the work. Device 4 on Case 4: a structural maintenance reserve inside the owners' fee, drawable only with the joint signature of an independent engineering party and the insurer.

The failure modes of the four devices. A maintenance ledger dies when the ledger and the labour hours come apart, with booked hours estimated and actual hours unreconciled; the patch is an interface between the ledger and the scheduling system, so main‑ tenance hours are written from the roster as measured. Liability booking dies when the conversion factor loses value, left ten years without an update so the figure no longer means anything; the patch is to tie the factor to replacement cost and refresh it each year against a construction cost index. The dual trigger dies in a coverage gap in condition monitoring, with critical equipment monitored and the long tail running on the calendar, while the backlog concentrates in the long tail; the patch is sampled condi‑ tion rounds across long-tail equipment, a number of units each year, with the results used to recalibrate the calendar intervals. The wall dies when its height is guessed, a floor percentage with no basis behind it; the patch is to derive the floor from the booked li‑

ability, so the size of the backlog sets the height of the wall and the wall follows the debt. The four patches share one check sentence: when was the backlog's converted figure last calibrated against measurement?

One position in the organization of maintenance deserves a recommendation of its own: an advocate for maintenance. Maintenance has no seat of its own at the budget meeting, since the maintenance manager usually also runs production or operations, and the knife that cuts maintenance is often in the maintenance manager's hand. The form can be light. The maintenance crew elects one of its number to attend budget review, with a voice and no vote, and once a year that person gives the governing body a maintenance condition report, the backlog curve, the catches record, the long-tail equipment list. The cost is two meetings. What it changes is the seating at the table: a proposal to cut maintenance has to be read aloud in front of maintenance.

29.6 Boundaries with Adjacent Mechanisms

Against the slack of success (Chapter 20): slack cuts defensive spending, backups and drills, while this chapter cuts sustaining spending, maintenance and overhaul. The two ledgers share a structure and differ in line items, and building them together is the recommendation.

Against horizon mismatch (Chapter 28): the decision structure of cutting maintenance, with the cost landing outside the term of office, is horizon mismatch, and Westminster and Surfside are claimed by both chapters. The criterion: to analyse why the cut was possible, use this chapter; to analyse why the person who cut bears no consequence, use Chapter 28.

Against margin erosion (Chapter 25): a maintenance backlog eats the condition reserve, with corrosion eating wall thickness, while

margin erosion eats the design reserve, the redundancy consumed in a single cut. Record the two ledgers on the same page, since a system's remaining life and its remaining margin are one conversation.

29.7 Five Self-Checks

1. What is your organization's maintenance backlog, converted into money? When was that figure first quantified?
2. Has the ratio of maintenance budget to total asset value risen over the last five years?
3. In your ledger, when is the most recent entry recording that maintenance caught a problem? If you cannot answer, the ledger has not been built.
4. Is any equipment running with a known defect right now? Which precedent did its approval cite?
5. If the maintenance budget were cut by thirty percent tomorrow, how long would it take before the consequence showed in the numbers? That delay is the mechanism's entire timetable.

Chapter 30 Handover Rupture; Organizational Forgetting

> What this chapter covers: where knowledge falls between one pair of hands and the next; Polanyi's tacit knowledge, the criteria, four sites, the three-part handover template, and the shadow period. When to turn back: staff changes, project transfers, shift handovers, supplier changes. Estimated reading time: 50 minutes.

Michael Polanyi's epistemological proposition, that we know more than we can tell, is cashed out in an organization every time a person is replaced, a shift changes over, or a supplier is swapped. What can be written into a handover file is one layer of knowledge: parameters, states, conclusions. Judgement, feel, and most of the answer to "why was it done this way" live in people. When the person goes, the paper is promoted to the whole, and everything under the paper is lost.

What makes handover rupture distinctive is that it is painless. At the moment of rupture there are no symptoms. Every handover was completed, every form was signed, and the knowledge fell at the instant the departing person turned around.

30.1 What It Is

Definition: handover rupture means critical knowledge is lost along the transfer chain. It has two forms: lateral distortion at shift and step handovers, where information does not reach the next party at the same level; and longitudinal loss at personnel change, organizational forgetting, where judgement and lessons leave with the person.

Three trigger criteria. Any one of them means the mechanism has started:

① Nobody can answer "why is it done this way" for a critical process, or the answer exists only inside one head;

② Handover transfers documents and states, not judgement and lessons (the field for "what has gone wrong here before" is missing);

③ After the long-serving person leaves, answers to the same question begin to diverge by version.

30.2 How It Grows

Three generating conditions.

Condition one: documents carry conclusions only. A procedure says what to do, not why, and not what went wrong that year. The omission is usually made for schedule reasons and because the point seems obvious, and the obvious is exactly the part most easily lost.

Condition two: people move faster than documents are updated. Knowledge is leaving and the paper is not being topped up. Turnover rate multiplied by the tacit share of the knowledge is the annual loss.

Condition three: project-based organization. Teams assemble for a job and disperse when it is finished, and cross-project memory has no institutionalized container. Every project end is a miniature organizational dissolution.

The distinction from Chapter 16 (attenuation up the hierarchy) sits here: attenuation is a message deforming as it travels vertically, while rupture is knowledge falling as it travels laterally and across

time. Both share a root, a transfer bandwidth compressed until only conclusions fit through it.

Polanyi's proposition has a second half worth adding: we can tell more than we can write. Speech (apprenticeship, mentoring, the pre-shift briefing) carries more tacit knowledge than documents, because conversation is two-way, the listener can ask and the speaker can watch a face and adjust, while a document is one-way, the writer guessing at the reader's questions and dropping whatever the guess misses. That layered distinction gives handover design a direct rule: documents carry what can be written, conversation carries what can be said, drills carry what can be done. Three layers, three carriers, and mixing any two of them leaks (using documents where conversation belongs, using conversation where a drill belongs). Device 1 in this chapter, the template, governs the document layer; the shadow period, Device 2, governs the conversation and drill layers. The division of labour now has an epistemological basis.

One public finding from the research history of organizational forgetting is worth citing: an organization's memory of its accidents decays over years (the share of staff who lived through the event, the usage rate of review material, the overlap between drill scenarios and past accidents, all measurable decay indicators). Aviation has a mature paradigm against that decay: recurrent training scenarios are required to include adaptations of historical accidents, so every crew's annual recurrent training contains one scenario drawn from a real accident. That is institutionalized forgetting-resistance, the organization training itself continuously on its own accidents. Transplanting the paradigm to other industries has only two thresholds: a review library (no material, no adaptation) and mandatory recurrent training (no mandate, only goodwill). Those two thresholds are the relay between Device 2 in this chapter and the protective mechanisms of Chapter 19: material

comes from a review culture, the mandate comes from teeth in the safety culture.

30.3 The Mechanism in Four Domains

Case 1 (Aviation): Two Understandings Before the Stop Line

Case ID: av-037.

When and where: 17:32 to 17:47:27 on 2 January 2024 (Japan time). Tokyo International Airport (Haneda), on Runway C (34R) north of its intersection with taxiway C5; Tokyo Aerodrome Control (the tower); aircraft A taxied from stand N957 via taxiways H and C to C5.

What and who: Haneda Air Base, 3rd Regional Coast Guard Headquarters, Japan Coast Guard: DHC-8-315, JA722A (aircraft A), 6 on board, the captain as PF and the first officer as PM. Ministry of Land, Infrastructure, Transport and Tourism, Tokyo Aerodrome Control: the positions on duty were aerodrome control east (Tower East), ground control east (Ground East), and the aerodrome co‑ordination position (TC); Tokyo Terminal Control, the Haneda de‑parture coordination position (DF) and the Haneda search co‑ordination position south (AFS). Japan Airlines flight 516, A350-941, JA13XJ (aircraft B), 379 on board. Positions are given as the report names them; the report gives no names.

What they were doing: The Noto Peninsula earthquake struck on 1 January, and the 3rd Regional Headquarters had to move disaster support supplies from its own building to Niigata Airport; the formal instruction reached Haneda base at 14:55 on 2 January (p.4). The aircraft the base had allotted was G-V, and specialist officer A and captain A agreed to change it to JA722A (p.4). The supplies ar‑rived late, at 15:55, and loading took 25 minutes (p.4). At 16:32 the aircraft was towed to N957, and on the way the APU generator

dropped twice the moment it was connected; the maintenance technician judged it a GCU fault, and while he was away at the hangar for a part he was told Niigata could lend a ground power unit, so the plan became starting on a GPU at the destination (p.5). That chain of handling pushed the departure slot from 16:45 to 17:25, renegotiated by chief pilot A with the flight operations of - ficer (p.5). Aircraft A began to taxi at 17:32 (p.6). One item was still open on board: no GPU had been arranged at Komatsu, and the legs beyond Niigata were still undetermined at takeoff (p.5).

The control side was changing shifts in the same period. The team on duty numbered 10, one shift from 12:00 to 21:15, briefed at 14:05 and taking over the operations room from the previous shift at 14:15 (p.11). Tower East had already been through three positions that day: aerodrome control west at 15:05, ground control east at 15:50, aerodrome control south at 16:35, with about 20 minutes of rest between each, and from 17:20 he was at aerodrome control east (p.11). At 17:42 he saw aircraft A on taxiway H, and the flight plan had already told him this was a supply transport, not a search and rescue task of the priority kind (p.12). At 17:43:02 aircraft B checked in, and at 17:43:26 aircraft C (p.13). At 17:44:36 he told DF one decision: put aircraft A after B landed and before D landed, on the reasoning that B and D were about 7 nautical miles apart, A was slow and C was fast, and sending the slow one first would save the interval for the fast one (p.13).

The decision point:

① The aircraft change on the morning of 2 January. From G-V to JA722A, with specialist officer A reporting to the chief operations officer A at the 3rd Regional Headquarters that it was executable (p.4). The carrier was the base allocation and the flight plan filed afterwards: movement began at 17:25, 6 on board, destination Niigata, purpose of flight written as transport of disaster support supplies (p.6). The change moved two crew who had been standing

by on G-V onto JA722A, and put this aircraft into the evening peak departure sequence; captain A said in his oral account that he had stated the purpose when filing, and later read that as the reason the order had been moved up (p.7).

② The takeoff order at 17:44:36. A inserted between B and D, carried by DF's out-of-scope transmission and one verbal decision, with nothing in writing (p.13). In the wake of the same decision, DF said C's takeoff clearance would be issued once A had gone (p.14). Once the order was set that way, every later action was executing it.

③ The instruction at 17:45:14. Tower East transmitted "JA722A, Tokyo TWR, good evening. No.1, taxi to holding point C5"; the carrier is the control frequency recording, transcribed word for word in Attachment 1 (p.152). It came four seconds after A's first call on the new frequency, and used the format of the example sentences in the ATC Procedure Standards (p.46): the first half has a procedure behind it, the added No.1 has none. Inside one string of speech, half has a source and half does not. On the recording, at 17:45:18 the first officer read back "To holding point C5, JA722A. No.1, Thank you", and between 17:45:21 and 17:45:22 the captain repeated two words only, "No.1" and "C5" (p.152).

④ The stop bar lights were out of service during renewal of the control unit. The report states that Runway C has stop bar lights and that they were out of service because of renewal work on the control unit (p.53). Start and end dates, the approving document, and whether substitutes were in place are not stated in the report. The same page records the normal use: in service, departures on Runway C may enter the runway only via four taxiways, C1, C2, C13 and C14, and the controller manually switches off that taxiway's lights each time an entry clearance is issued (p.53). While the lights were out, that one-taxiway-at-a-time release went with them.

What was visible then: ① At 17:43:02 aircraft B checked in. At 17:44:56 the tower issued B a landing clearance for 34R, and at that

moment A was still on another controller's frequency, having been transferred at 17:44:54 and checking in on the new frequency only at 17:45:10 (p.152). So the material on A's table runs as follows: Ground East's taxi instruction and the readback "Continue C holding point" (p.152); the 17:45:14 instruction carrying No.1; at 17:45:23 a voice in the cockpit saying "no problem"; at 17:45:25 the captain ordering the before takeoff checklist (p.8, p.152); at 17:45:41 the first officer calling "Anti-Collision, WHITE" and turning the beacon white (p.152); at 17:46:05 the gust lock coming off (p.152); at 17:46:13 the aircraft crossing the stop position marking at C5 (p.9); at 17:46:26 the Haneda communications operator calling in about the Komatsu GPU (p.9); at 17:46:46 the aircraft stopping on the 34R centreline, 560 metres from the southeast end of Runway C (p.9); from 17:47:03 to 17:47:26 no voice in the cockpit at all (p.153).

On the control side the table held 5 aircraft of his own, plus two departures on taxiway C heading for Runway 05 (p.128), and an airport surface surveillance display. After D checked in at 17:45:55, Tower East counted the five: A into C5, B cleared to land, C holding at C1, D slowing, E queued behind the two going to 05. All five matched what he expected (p.14). Information learned afterwards: at 17:46:20 the runway occupancy monitoring support function had already raised an attention alert, and it persisted until 17:47:28 (p.51); the function makes no sound, the tower had no handling rule, no training and no document explaining its principle, and all three controllers involved later could not recall anything lighting up (p.51); the flight monitoring position in principle watches traffic on Runway A only, not Runway C (second progress report p.4); captain A had flown 0 hours 00 minutes on the type in the previous 30 days (p.25).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 17:47:27 aircraft B landed on 34R and struck aircraft A standing on the runway. A was severely damaged and caught fire: 6 on board, the captain seriously injured, the other 5 dead. All 379 on B evacuated, 1 passenger seriously injured, 4 lightly injured, and 12 examined by a doctor for feeling unwell (p.1, p.2). Back to the three questions. The commonest answer to the first is "have the crew read back and confirm", and that action costs nothing here, because the readback was completed at 17:45:18, word for word correct. The item most easily missed under the second is the landing clearance at 17:44:56: it is short, it is ordinary, and it was issued in the 32 seconds when A was not on the frequency. The checkpoint under the third falls at the moment of frequency transfer, where the transfer would carry "clearances already issued on this frequency", which arrives earlier than one more look at the runway. The report reaches no causal conclusion: the second progress report still lists the items above as under analysis (second progress report p.3, p.5), the final report has not been published, and this card therefore stops at the point where two understandings part before the stop line.

Options ruled out: Having C take off before B, rejected: C's taxi speed before C1 was below expectation, and the wake turbulence behind a long-haul international departure would affect D's landing (p.13). Putting A behind C in the sequence, rejected: it would miss the window between B and D, and A is slow, so waiting behind C means waiting longer (p.13). Letting A enter the runway

from C1, rejected: C5 is the intersection nearest its position, and only via C5 could it take off immediately after B landed (p.14). Holding the stop position physically with the stop bar lights, unavailable: out of service for renewal work (p.53). Relying on the occupancy alert as a backstop, unavailable: no handling rule, no training, no sound (p.51). A go-around for B at the last moment, not done: the tower issued no go-around instruction, and the report lists it as an action the tower did not take (p.127). Passing the presence of a landing aircraft on 34R to A as traffic information, not done: the recording has the tower telling B "there is a departing aircraft", and that direction runs one way (p.13).

Early signals: ① Stop bar lights out of service during renewal (p. 53), a safety layer offline with nobody watching. What overrode it: the lights are lit only when visibility is below 600 metres or when the controller judges it necessary (p.53), and the visibility observed at 17:30 on 2 January was above 10 kilometres (p.52). ② A March 2008 review meeting on preventing runway incursions produced the occupancy monitoring function, in service at Tokyo Aerodrome Control from fiscal 2010 (p.49, p.50); the function had been live for more than ten years with none of the three, a handling rule, a training course, an explanation of principle (p.51). What overrode it: it was treated as a visual aid only, with the view out of the window taking precedence. ③ The same function arrived at Chitose Aerodrome Control in 2013 already carrying a three-level audio alert (p.50 n.34), and Haneda had none. What overrode it: each authority runs a different system (p.50 n.32). ④ On 10 June 2023 a Thai Airways wingtip struck an EVA Air tail on Runway A at Haneda, and around 17:46 Tower East was recalling it (p.15). The case was closed and filed when handled, and never became a review of the visual division of labour on Runway C. ⑤ The ATC Procedure Standards wrote standard wording and example sentences for "taxi to the runway holding position" (p.46), and wrote

no wording at all for departure sequence (p.46). No document marks the gap, and no one treats it as a defect.

The point of escalation: 17:45:25, captain A saying "yes, then, Before Takeoff Checklist"; the first officer begins reading at 17:45:30, calls "Anti-Collision, WHITE" at 17:45:41, and reports "Checked" at 17:46:05 while taking off the gust lock (p.152). All three actions carry second-level timestamps on the voice recording. The before takeoff checklist belongs procedurally to the items after "runway entry clearance obtained" (p.8, p.126), and once it is read the aircraft's configuration moves toward flight: beacon from red to white, control lock off. At 17:46:13 the aircraft crosses the C5 stop position marking; at 17:46:46 it stops on the centreline, the captain says "I have Control" and the first officer answers "You have Yoke" (p.153). After that, at 17:46:55 the cabin crew member asks "Captain, is now a good time?", the first officer answers "ah, later would be better", and the captain answers "yes, a little later"; from 17:47:03 to 17:47:26 the cockpit is silent (p.153). From the first word of the checklist to the collision is 2 minutes 2 seconds, and nothing in that stretch pulls it back. In the same stretch the tower's attention is on B's landing and D's deceleration (p.128).

Failure chain: Technical layer: the stop bar lights were out, so the physical barrier was absent (p.53); the occupancy function turns the runway yellow on a screen and makes no sound, and it is eligible to report only from 48 seconds before an arriving aircraft passes the entry end (p.51), with the first actual report coming 7 seconds after A crossed the stop position marking (p.128), so its entire work is to display the error once the error has happened. Organizational layer: the transfer passed the track and not the landing clearance already issued, and the 17:44:56 clearance fell in the 32 seconds before A entered the frequency (p.152); the standards give wording for taxi and none for sequence (p.46); the mon-

itoring function sat in place for over ten years with the handling rule and the training missing, its alert displayed on 14 screens while all three controllers could not recall it (p.50, p.51); the flight monitoring position's remit excludes Runway C (second progress report p.4); Tower East, from seeing A turn into C5 to the collision, said not one word to another controller about A's movement (p. 127). Incentive layer: 17:47 sits in the period before the day's traffic peak (p.47), the departure slot had already slipped from 16:45 to 17:25 (p.5), Niigata's operating hours end at 21:30 (p.4), and the tower had to fit a slow aircraft into a 7 nautical mile gap (p. 13). The break at the junction sits between wording and rule: one document prescribes how to bring an aircraft to the holding point, and the other thing, how to tell it that it is number one, has no document at all.

Minimum counterfactual: At 17:45:14, split "No.1" out of the taxi instruction and issue it as its own transmission (taxi to holding point C5 first, then you are number one for departure). One action, doable at the time, violating no procedure, precisely because the standards prescribe no wording for departure sequence (p.46), so splitting it or not both sit inside what the rules allow. The cost is one extra transmission, about three seconds. The gain is that "stop at C5" and "you are first" no longer travel inside one string of syllables. The report lists No.1 separately as one contrib‑uting factor (p.127), and records that the captain's readback covered two words only, "No.1" and "C5" (p.152). What it does not stop is two things: the stop bar lights out of service, and the landing clearance missed at frequency transfer. Each has its own entry point, and neither is in this sentence.

Outcome and sources: JTSB published the first progress report on 25 December 2024 and the second on 25 December 2025; the final report is not yet published, and this card rests on the two progress reports and the control communications record in Attachment 1.

The second progress report adds three items for further analysis: that Haneda base had not introduced a sterile cockpit rule, that the first officer ran the before takeoff checklist after obtaining entry intent, and that the crew did not confirm to each other that no other aircraft was on the final approach path (second progress report p.3). Page numbers are the footer page numbers of the reports; in the first progress report the PDF page is 8 higher than the footer. Primary sources: JTSB, Air Accident Investigation Progress Report (1st and 2nd), links in Appendix B av-037.

The cross-agency handover in Case 1 deserves one more organizational reading. Civil aviation and the coast guard each have mature internal handover systems, with their own radio procedures and their own instruction vocabulary, and the rupture appeared at the interface between the agencies, where the two vocabularies have no institutional translator. Interface governance has precedents in aviation and at sea (joint exercises, standardized phraseology templates on shared frequencies), and the precedents share one feature: the interface is made an object of training rather than an object of assumption. Assuming the other party understands is the most expensive assumption available. A question any organization doing cross-agency work can put to itself: when was your interface with the partner agency last jointly exercised? An organization whose answer is counted in years has an interface shaped like Case 1.

The three-station transfer in Case 2 has a detail worth taking apart. The error passed the check at every station, because each station checked its own process and not the truth of what arrived from upstream. Trust between stations is a design choice: trust saves time, since nothing is checked twice, and discounted audit costs staff. The lesson here is that trust should be graded. Identity data is high risk and the receiving station re-verifies it independently; status data is low risk and can be accepted as given. The basis for

grading is the consequence of an error propagating, an identity error amplifying down the whole chain, so the fields with large propagation consequences are the ones worth re-verifying. Limiting mandatory re-verification to a handful of high-risk fields is what keeps the audit cost bearable.

The state handover in Case 3 has a general industrial solution worth writing out: a third eye on clearance verification. A line clearance at a strength change is self-certified by the shift, which clears, signs and confirms its own work, and the blind spot of self-certification is familiarity, since clearing the same line every day blurs the eye to residue. A third eye, QC or the neighbouring shift cross-verifying the clearance, gives self-certification an adversary. The cost of cross-verification is scheduling complexity; its benefit is written in the recall quantity, and 33,280 bottles of mixed pack is the price tag the self-certification model carried. Any state handover, a clearance, a changeover, a reset, can ask itself whether this step has an adversarial second pair of eyes. An eye that is missing is a blank left by the rupture.

The label rupture in Case 4 has an institutional fix worth spreading: a single trigger point for change. When a formula change, a label change and a purchasing change each run their own process, the coupling between them (change the formula and the label must change) is held by human memory. A single trigger point means any change to a formula element automatically generates a label review work order, a system-level link that does not depend on memory, and no change takes effect until that work order closes. The cost is one process rebuild, and it closes this class of rupture at the root. Listing the coupling points that depend on memory and systematizing them one by one is the roadmap for governing handover rupture in manufacturing.

Case 2 (Healthcare): A Wrong Identity Passed Through Three Stations

Case ID: me-006.

When and where: The report keeps a relative diary and gives no absolute dates: pre-admission assessment on day 0, admission on day 2, the first emergency call around 20:00 on day 14, the second around 08:45 on day 19, the third around 15:00 on day 20, surgery from 08:30 on day 21. Places: a care home in England; the ambulance trust's emergency operations centre (EOC); the majors reception desk in an acute trust's emergency department; radiology; the surgical assessment unit and the operating theatre; pathology. Report published 22 March 2022.

What and who: The care home (resident a 93-year-old woman, called Mrs E in the report, with dementia, able to walk with a frame, prone to falls); the ambulance trust (the 999 call handling desk, the EOC tactical commander, the crew, the electronic patient report form ePRF); the acute trust (the emergency department majors flow co-ordinator and majors receptionist, emergency nurses, radiology, the surgical assessment unit, the operating theatre, pathology, the medical admissions team); HSIB local integrated investigation pilot 2. Responsibility is counted by position; the report gives no institutional or personal names.

What they were doing: Mrs E was transferred into the care home by arrangement with the acute trust, for dementia and safeguarding reasons (p.9). Because of COVID restrictions, care home staff and acute trust staff completed the pre-admission assessment by telephone, and the date of birth was recorded wrongly at that step: day and year right, month wrong (p.9). She was admitted two days later, and the acute trust's discharge summary arrived at the home carrying the correct date of birth, to be used in building the

personal care plan. The wrong assessment was never withdrawn and stayed in the same binder as the other paper (p.9).

The first fall came around 20:00 on day 14 and the nurse on duty called 999. This time the handler recorded the name spelling and the date of birth correctly, the crew's handheld device found a match in PDS, and the NHS number was written automatically into the ePRF (p.10). The second fall came around 08:45 on day 19, call volume was high, the ambulance trust was operating its emergency rules apply procedure, and the handler did not have to take a full name for the crew (p.10). The date of birth the handler recorded came from the pre-admission assessment and was wrong; the surname spelling was not requested, and two letters were wrong in the middle of the name (p.10). The crew attended, assessed and decided to convey. The handheld PDS search failed, and what appeared on the ePRF was "processing or communication error" (p.10). The care home prepared the documents to travel with her: the medication chart, the precautions and concerns form, and the ReSPECT form, on which the surname spelling, date of birth, address, GP, NHS number and next of kin were all correct (p.11).

The emergency side was busy in the same period. The majors reception desk had a dedicated receptionist only between 14:00 and 22:00, and the flow co-ordinator covered the rest of the day; observation records show that over about two hours in the morning the flow co-ordinator spent 5 to 10 minutes at the desk (p.27). When an ambulance could not get anyone at the desk, the crew waited 5 minutes and then booked in somewhere else (p.27).

The decision point:

① The telephone assessment on day 0 and the filing on day 2. The date of birth was wrong by one month and written into the pre-admission assessment; two days later the correct discharge summary went into the same binder and did not replace it (p.9). Two paper documents carried it, the responsible positions were care

home staff and acute trust staff, and the report states the assessment was completed by both on the telephone. From then on, every 999 call read the date of birth off that page (p.10, p.12).

② Emergency rules apply on day 19. The report describes it as a dynamic decision taken with the EOC tactical commander, switched on in bands according to call volume, and reviewed on risk against benefit (p.21). An EOC internal record carried it. The same page states that there is no evidence the ambulance trust had any obligation to tell other healthcare providers about this kind of risk assessment (p.21).

③ Booking in around 10:00 on day 19. The acute trust had no formal booking-in procedure, only a flow chart limited to how to save the ePRF into the hospital system (p.26). Booking in was done verbally between the crew and the flow co-ordinator, with the date of birth asked first and taken from the ePRF (p.11). The hospital system, the emergency system and the NHS Spine all had no record of Mrs E, so the system created a new record and a new admission number using the wrong date of birth and surname (p.11). On day 20 the person handling it was the dedicated receptionist (p.25), and what came across was the same new record.

④ The wristband on day 19. The emergency nurse applied the addressograph sticker printed with the wrong details to a white band (p.11). The acute trust's PPI policy states in terms that "a hospital addressograph sticker is not to be attached to a Patient Identification Band" (p.29), and requires the band to carry surname, first name, date of birth, NHS number and admission number (p.29); Mrs E's band had no NHS number (p.29).

What was visible then: ① At the 999 call the nurse on duty held the full set of paper with the correct details, NHS number included (p.11, p.19). ② The crew held the same set (p.11). ③ PDS results on the ePRF appear in three forms: day 14 "PDS found a match", day 19 "PDS trace failed due to processing or communication error",

day 20 “PDS trace was invoked and a response was received. A match was not found” (p.22 table 1). ④ On the day 19 ePRF the GP and next of kin details were correct and had not been entered by the handler (p.23). ⑤ At the desk the date of birth was asked first and the surname second, both fields taken from the ePRF (p.11). ⑥ The band sticker carried the wrong spelling, the wrong date of birth, the new admission number, the emergency episode number, the correct care home address and a barcode, with the NHS number field empty (p.29). ⑦ On radiology’s CRIS the day 19 entry reads “confirmed by Mrs E herself” and the day 20 entry reads “confirmed by the care home escort”, while the escort said she did not know Mrs E’s identity details and remembered it being done between the radiographer and Mrs E (p.38). ⑧ The surgical checklist was signed, confirming that band and notes agreed (p.38). ⑨ Pathology treated the first blood tube as a new patient (p.40); the second, crossmatch tube carried the same wrong details, so the two tubes matched each other (p.40). Information learned afterwards: PDS returns no candidates if a single field is inaccurate (p. 24); the trust recorded 192 events of this kind between October 2020 and October 2021 (p.7); Mrs E suffered no substantive harm from the wrong identity and received no transfusion (p.41).

> **Stop here. Decide before you turn the page.**
>
> 1. What would you do? Write the action, not “pay more attention”.
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: A day 20 X-ray confirmed a fractured neck of femur, she was transferred to the surgical assessment unit, and a second set of inpatient paper notes was opened under the wrong details (p.13). At 08:30 on day 21, because the patient could not

sign, "form 4" was signed, noting no known allergies, and the surgical checklist confirmed band and notes agreed (p.13). The second pre-operative blood tube went to pathology, and the pathology system could not link it to the first automatically, be‑cause the link runs on the NHS number alone (p.40). Pathology found a patient with a slightly different surname spelling, a date of birth one month out, the same care home, the same GP and an NHS number, and reported it to theatre; by then Mrs E had left theatre and was in recovery (p.41). Theatre staff telephoned the GP and the care home, the correct details were confirmed, and the two sets of notes were merged (p.41). The report makes 5 local safety recommendations (R/2022/170 to 174) and 2 local safety observa‑tions (O/2022/143, 144), and records one national safety risk: where the NHS number is not used as the primary identifier, patient identification is at risk (p.6). Back to the three questions. The commonest answer to the first is "re-check identity on arrival", and the cost of that action here is not money. It is that nobody was assigned to do it, since the desk had someone in place for 5 to 10 minutes in two hours (p.27) and no line in the procedure says whose step it is (p.26). The item most easily missed under the second is the empty NHS number field on the band, because a blank looks like a data tidiness issue and not like a signal. The checkpoint under the third falls at the moment the system finds no such patient and is about to create a record, where the recorded action was to create one directly, without asking the care home and without asking the escort standing there (p.28).

Options ruled out: Asking the care home or the escort to verify before booking in, not done: the care home's name was on the ePRF and the escort was on site before the new record was created (p.28). Checking the ePRF against the paper in the vehicle, which crews usually do and did not do here: the report gives two possible reasons, that the home is very close to the hospital and that clinical care took priority (p.23), and states it could not interview the

paramedic to confirm (p.23). Having pathology link the two tubes manually by the typed admission number, which needs a manual override, not done: the report says the link runs on the NHS number alone and stops at that step when the number is missing (p.40). Pausing surgery until crossmatch closed, a policy other acute trusts hold and this one did not (p.41). Obtaining a number through the NHS App or a similar route, which the report notes was possible and the call flow at the time did not require (p.19). The emergency department telling downstream at booking that PDS had not matched, not done: that state entered no field (p.23). Writing the band by hand instead of applying the addressograph sticker, which is what the policy says (p.29), not followed on the floor.

Early signals: ① Day 14 PDS matched and the NHS number wrote itself into the ePRF (p.10), so the whole chain looked sound; the one success and the two failures went through the same steps without a procedure, and a success does not add a check to the next attempt (p.26). ② Day 19 the ePRF read "processing or communication error", and the report calls such disconnections "not an uncommon event" (p.22). What overrode it: the device was under a procurement and replacement review (p.23); and the crew had the correct paper in the vehicle (p.11). An unreliable device and correct documents in hand were true at the same time, and no step connected them. ③ The trust logged 192 events related to correct patient identification in 12 months (p.7). What overrode it: identification was treated as an industry-wide problem (p.7, citing WHO 2007). ④ NPSA/2008/SPN001 in 2008, reissued in 2009 to the chief executives of organizations in England and Wales, required the NHS number as the national identifier (p.19, p.20). What overrode it: it was an alert and not a mandate, and the 2014 NHS Number survey was still saying the number had to move from good practice to core practice (p.19). ⑤ Two documents with different dates of birth, one right and one wrong, lay in the same care home

binder (p.9). What overrode it: a complete file was taken for a reliable file.

The point of escalation: Around 10:00 on day 19, when three systems had no record of Mrs E and creating a record was the only action that let the process continue; once created, every later station took it as true (p.11). The new admission number and the sticker on the band carried it. One other person was present, the care home escort. The report says she did not know there had been any difficulty in booking in, that she believed she could have helped, if only by pointing at the documents that came with the patient, and that her recollection is that from entering the hospital to leaving that day she never saw that set of documents again (p. 28). In the following 24 hours radiology, the surgical assessment unit and theatre all checked, and what they checked was the new record against its own copies: radiology wrote "confirmed by herself" on CRIS, and theatre signed a checklist confirming that band and notes agreed (p.38). This is the step between nobody can be found and no way back. Of everyone who checked across the three stations, none asked why a 93-year-old from a local care home had no previous notes (p.38); by day 21 the question came from pathology (p.40).

Failure chain: Technical layer: PDS returns no candidates when any field is inaccurate (p.24); the crew's search had also hit a communication error (p.23); the pathology system links two tubes automatically only by NHS number, and without it a manual override is needed (p.40). Organizational layer: the emergency department had no formal booking-in procedure, only a flow chart for saving the ePRF (p.26); the PPI policy forbids putting the sticker on the band and it was done both times (p.29); the policy requires the NHS number on the band and the field was empty (p.29), while under DSCN 04/2009 the four core identifiers should be surname, first name, date of birth and verified NHS number (p.29); the policy

pushes confirmation onto the care home escort, who may never have met the resident (p.36); the report's own conclusion is that a correctly detailed band applied on admission is the only line of defence against a wrong identity following the patient between departments (p.39); radiology guidance says most identification events originate in the emergency department (p.37) and emergency practice did not change because of it; a 93-year-old from a local care home with no previous notes raised no question (p.38). Incentive layer: the EOC traded risk against benefit while 999 calls queued, and the benefit was counted as one more question against one fewer call answered (p.21); the desk had someone in place 5 to 10 minutes in two hours (p.27); a crew waits 5 minutes and books in elsewhere (p.27). The break at the junction sits between two documents: the risk was accepted, reviewed and approved inside an internal assessment, and the receiving organization never received it (p.21).

Minimum counterfactual: Before the crew left on day 19, correct the surname spelling and the date of birth on the ePRF once, using the paper travelling with the patient. One action, the documents were in hand (p.11), the correction is a single comparison, and the report puts it as a recommendation to the ambulance trust (R/2022/171, p.5). The report also states that crews locally already checked the ePRF against the travelling documents (p.23); in other words the action needs no new capability, equipment or authority, only a change from custom to requirement. It does not change how busy the department was and does not depend on anyone being especially careful. It asks only that an error does not travel while correct documents are in the room. What it does not stop is the next three stations: nobody at the desk is assigned to check, the sticker still goes on the band, pathology still links by NHS number alone. Put the other way, adding a check in the emergency department without touching this station would still have the new record created at 10:00 on day 19 with the same wrong details.

Outcome and sources: Mrs E suffered no substantive harm from the wrong identity, surgery needed no transfusion, and identity was confirmed after the operation by telephone with the GP and the care home (p.41). The report's safety recommendations all point at standardization and process, and the authors write that such recommendations are "unlikely to eliminate incidents" (p. 41); they also write that the investigation could not interview the paramedic, so why the documents were not checked has no answer (p.23). This report is one of four in the HSIB local integrated investigation pilot (p.2), and the five recommendations point at the care home, the ambulance trust and the acute trust (p.5, p.6). Primary source: HSIB, Local integrated investigation pilot 2: Incorrect patient details on handover, published 22 March 2022, PDF on the HSSIB site, link in Appendix B me-006. Page numbers are PDF pages.

Case 3 (Drugs and Devices): Two Strengths in One Bottle

ph-003. Mixed Amphetamine Salts Tablets 5mg / Teva Pharmaceuticals. FDA recall record (D-1285-2020, API read for this book): some bottles contained tablets of a different strength, and 33,280 bottles were recalled (Class II). The packaging line has a handover interface between a batch changeover and a strength changeover: clearance confirmation, material reconciliation, first-piece inspection. The exact location of the rupture is defined by the investigation; its type is textbook, in that information about the previous state, what was left in the line, was not fully passed to the operation of the next state. A line's handover does not happen between people, it happens between states, which makes it less visible than a shift handover: no briefing can remind anyone, and only records can speak.

Stop at the decision point: the handover moment between strength changeover production, line clearance and first-piece confirma-

tion. On the table: the ticks on the clearance record, the visual similarity of the two strengths, the scheduling pressure. 〔Out - come〕 Class II recall. Primary source: FDA, D-1285-2020, 2020, link in Appendix B ph-003.

Case 4 (Agriculture and Food): The Allergen That Never Reached the Label

ag-009. The 19th Hole Snack Mix (Better Business...Better Foods, Inc.). FDA recall record (event 96758 / F-0801-2025, API read for this book): the product contained undeclared almond and sesame, and 18,000 units were recalled (Class II). A rupture sits between the formula change information and the label update flow: allergen declaration depends on the formulation room and the label room speaking the same language, and any change notice that fails to arrive turns the rupture into product on a shelf. The consequence of a wrong allergen declaration is counted on the most sensitive population: a consumer allergic to almond or sesame does not need to misread the ingredient list, only for the label to be out of date.

Stop at the decision point: the moment the formula adjustment is signed off and the label print file circulates. On the table: the new formula’s ingredient list, stock of the old labels, the distribution list for the change notice. 〔Outcome〕 Class II recall. Primary source: FDA, F-0801-2025, 2025, link in Appendix B ag-009.

Case 5 (Construction and Transport): The Stop the Radio Never Delivered (condensed)

ce-004. A crane pushing a wagon struck and injured two track workers, March 2025, United Kingdom. RAIB report (2026, pages read for this book): the radio stop instruction did not reach the crane driver, the crane had no forward view, the wagon carried no lights, 1 person was seriously injured and 1 slightly injured. The

rupture takes a channel form here: the link between sending and receiving, the frequency, the blind spots, the confirmation process, was never closed, the sender believed the message had arrived, the receiver never heard it, and no mandatory handshake sat in between. Minimum counterfactual: radio instructions in mobile work carry a mandatory read-back, with no read-back treated as not delivered. Primary source: RAIB, 2026, link in Appendix B ce-004.

Case 6 (Construction and Transport): Where the Rule Book Did Not Reach (condensed)

ce-005. A wrong points position in a reversing move caused a de - railment, June 2025, United Kingdom. RAIB report (2026, pages read for this book): the reversing scenario was not covered by the rule book, the signaller worked from experience where there was no rule to follow and made an error, the 4 crew on the train were uninjured, and the train and infrastructure were damaged. The non-transferability of experience is exposed in the uncovered scenario: routine scenarios travel by rule, exception scenarios travel by experience, and experience lost, or never passed on, makes exception scenarios the worst ground for rupture. Minimum counterfactual: write a rule for each exception handover scenario (turn personal experience into a work instruction), and update the list of uncovered scenarios annually. Primary source: RAIB, 2026, link in Appendix B ce-005.

Case 7 (Aviation): The Two Numbers Missing from the Joint Briefing (condensed)

av-016. A glider was towed over speed and the rope released, May 2024, Japan. JTSB report (2025, read for this book): the joint brief - ing before takeoff did not confirm the tow speed limit or the emergency release altitude, the rope released over speed during the tow, the forced landing at low altitude struck trees, the pilot

was slightly injured, and the glider was destroyed. Tug and glider are a temporary pairing of two sets of people, towing across clubs, and the joint briefing is the only window where the two sets of experience hand over; the two numbers that window missed, the speed limit and the release altitude, are exactly the parameters that save lives in an emergency. Minimum counterfactual: joint briefings for cross-crew operations always include a double confirmation of emergency parameters, each reading them to the other and getting a read-back. Primary source: JTSB, 2025, link in Appendix B av-016.

Case 8 (Nuclear): Two Tagging Errors in Two Weeks (condensed)

nu-001. Gravelines nuclear power plant, unit 3, major outage. ASNR event notification (October 2026, notification read for this book): an electrical tagging error during the outage left a standby diesel generator without power beyond the 24-hour limit (INES level 1), and the same plant repeated it within two weeks (nu-002, equipment unavailable for 7 hours 45 minutes). Tagging and restoration during an outage is the highest-density handover site there is, with hundreds of tags and multiple crews rotating, and two similar errors in a short interval show the handover interface was not corrected after the first: the lesson from the error went through a handover rupture of its own, in that the fall the previous shift took never entered the next shift's briefing. Minimum counterfactual: tagging errors become a fixed review item in the daily outage briefing. Primary source: ASNR, 2026, link in Appendix B nu-001.

30.4 What Shows Earliest

Six weak signals, each with the reason that usually overrides it.

1. A newcomer asks "why", and the answer is "it has always been done this way". Override: "getting up to speed comes first."
2. The handover checklist carries states and parameters only, with no "what has gone wrong here before". Override: "that is what the review is for."
3. The same question draws different answers from different long-serving staff, and each calls theirs the standard practice. Override: "everyone has their own experience."
4. A key post's handover period is under a week, or there is none. Override: "the work cannot wait."
5. After a supplier or partner change, the wording of quality complaints changes and the number does not. Override: "running-in period."
6. The phrase "it has not been the same since X left" circulates in the organization. Override: "hero culture." (Put the other way: hero culture is a symptom of organizational rupture, not a cause.)

Put the six signals back against this chapter's eight cases. Signal one, asking why and being told it has always been done this way, maps to the simulated scenario: in the place with no rule to follow (Case 6), the why was never asked out loud. Signal two, a checklist without what went wrong, maps to Case 8: the outage briefing carried no record of the previous shift's fall. Signal three, several versions of one answer, maps to Case 1: where the same aircraft should go next had one version at the tower, one at the flight monitoring position and one with the captain, and the standards wrote wording for taxi and none for departure sequence. Signal four, no handover period, maps to Case 7: the temporary towing

pairing had not even a formal handover period, only the briefing window. Signal five, wording changing after a supplier change, maps to Case 3: the running-in period of a strength changeover was priced in a recall. Signal six, not the same since X left, is the organizational signature of the simulated scenario: the more scenarios without rules, the more often the phrase is heard.

Reorder the six by the cost of evidence. Signals one, two and four are cheapest (interview a newcomer, read a template, check a roster); signal five is middling (compare complaint texts); signals three and six are dearest (they need cross-person checking and exit interviews). The starting recommendation is signal one: next week, find someone within three months of joining and ask them to list the three whys they did not dare ask. That list is the first map of the rupture, and every point on the map is an object for a redemption plan (Device 4).

30.5 How to Defend

Device 1: the three-part handover template. A handover document carries three fixed parts: status (current parameters and open items), why (the reason for key settings and how they evolved), and what has gone wrong here before (incidents and near misses in this step, each with its review number). Parts two and three are where rupture concentrates, and a document without them is not signed for. Format point: parts two and three carry a minimum line count, and a blank fails, because "there is no why" and "nothing has ever gone wrong" are themselves claims that need verifying.

Device 2: the shadow period. A key post handover gets at least two weeks with two people on the job, the incoming person operating and the outgoing person supervising without doing the work, since rupture is cheapest to expose during the shadow. The acceptance standard is not "handover completed" but "the incoming per-

son completed N full tasks independently". Where two people are impossible, a single-crewed fishing boat or a one-person depart - ment, a written question list replaces it: the incoming person asks, the outgoing person answers in writing, and the record of the exchange is the handover document.

Device 3: continuity of decision records. When the three lines of the Chapter 7 decision record (point of no return, confirmation seat, what was visible then) are archived entry by entry, those re - cords become the core of the handover pack: a new holder reading ten decision records beats ten pages of state description. Decision records turn handover from "I tell, you listen" into "you look at the desk as it was", with the lowest distortion of the three.

Device 4: a redemption plan for the why. For any process that only one person knows, set a redemption deadline: N months before that person leaves, the judgement is written down and verified by a second person executing it. Verification means the second person completes one full task from the text alone; where they stall is the break point, and each stall is written up. The schedule is set at hiring, because starting redemption when the resignation arrives is usually too late.

Four devices, applied to this chapter's cases. Device 1 on Case 2: the pre-hospital handover form gains a what-has-gone-wrong sec - tion, covering this patient's previous identity events and the known risks in this handover. Device 2 on Case 7: the shadow period for towing is the double confirmation in the joint briefing, plus listening to each other on the first tow. Device 3 on Case 1: radio traffic at the cross-agency interface is archived on the three decision-record lines (instruction, confirmation, what was visible), so the review reads the desk directly. Device 4 on Case 6: the sig - naller's exception-scenario experience is redeemed into text one line at a time against the uncovered-scenario list, and verification is a second person running the scenario from the text alone.

The failure modes of the four devices. The three-part template dies in padding of parts two and three, with the why written as "per procedure" and the incident record written as "nothing abnormal"; the patch is padding detection, cross-reading someone else's handover pack, and returning any pack in which "per procedure" appears. The shadow period dies when the shadow becomes a helper, the incoming person fetching and carrying for two weeks with an independent-task count of zero; the patch is a counter of independent tasks, with N completions as the sign-off metric. Decision records die in lagging archiving, the record sitting on a personal laptop and not to be found in the pack; the patch is system-level archiving, in the same repository as the work orders. The redemption plan dies in loose verification, the second person reading and not executing; the patch is execution verification, one full task from the text alone, with any stall returned for rewriting. The four patches share one check sentence: in the most recent handover, did the incoming person independently find a rupture. A handover process that has never found one is a formality.

One dimension of handover frequency deserves a line of its own: the handover at an abnormal moment. Routine handover has a template (state, parameters, open items), and handover at an abnormal moment, mid fault handling, mid temporary change, mid drill, is the easiest to break, because both parties are busy and the template does not get filled. The simplest form is one structured call: what state we are in, what I am doing, what you do first when you take over, what must not be touched. Four elements said out loud and the minimum handover is complete. The template goes into the emergency plan, on the same clause as the crisis command provision in Chapter 18, and it is drilled, because the muscle for abnormal handover has to grow in drills.

30.6 Boundaries with Adjacent Mechanisms

Against information attenuation up the hierarchy (Chapter 16): attenuation is vertical deformation, rupture is lateral and longitudinal falling. Chapter 16 repairs the channel (side-by-side reading, direct copies); this chapter repairs the container (templates, shadow periods).

Against the invisibility of maintenance (Chapter 29): the loss of tacit knowledge is the hidden form of maintenance, since the upkeep of judgement (reviews, mentoring, document updates) also produces nothing visible and is also postponed. The ledger principle of Chapter 29 applies to knowledge assets: a knowledge ledger has to be kept too.

Against why the truth-teller goes first (Chapter 19): the commonest silence in a ruptured organization is newcomers no longer asking why, either because no answer comes or because the question embarrasses the old hands. Restoring a culture of asking is the precondition for Device 4 in this chapter.

30.7 Five Self-Checks

1. How many things on your post can you list that only you know? Do those entries have a redemption arrangement in your exit plan?
2. In the handover material you received most recently, was there a "what has gone wrong here before" section?
3. In your team, how many versions of the answer exist for one operational question? Have those versions ever been reconciled?
4. How long is the spoken part of your shift handover? How much of it enters no record at all?

5. If the most senior person in your team left tomorrow, which processes would have to be reinvented? Did you ask them for that list before they left?

Chapter 31 Regulatory Lag and Capture | Stigler

> What this chapter covers: why rules always arrive half a step late, and how a regulator can end up speaking for the regulated; Stigler's capture theory, the criteria, four field sites, and the defence of a diversified information supply. When to turn back to it: when you are assessing a sector's regulatory environment, forecasting the risk window of a new business model, or when your compliance team understands the business's real holes better than the business team does. Reading time: about 50 minutes.

Regulatory lag and capture is the only pair of mechanisms in this book where the invisible hand makes a fully frontal appearance. The market's hand steers the regulator's attention, its personnel and its information pipes toward the interests of the regulated, and none of it needs a bribe or a conspiracy. George Stigler's 1971 paper, The Theory of Economic Regulation, Bell Journal of Economics and Management Science 2(1): 3-21 (the reprint record held by Crossref has been checked), makes the core claim: regulation is usually requested by the industry and operated for it, and regulation benefits the incumbents. Fifty years on, the platform economy, artificial intelligence and gene technology have merely moved his argument onto new ledgers. This chapter treats lag (rules slower than technology) and capture (the regulator sliding toward the regulated) together, because each causes the other: the lag period is the seedbed of capture, and capture lengthens the lag.

The extension of Stigler's argument needs one main line drawn through it: capture theory has moved from interest capture to cognitive capture, in which the regulator's way of thinking is assimilated by the industry, so that even with personnel and interests

kept separate, the frame through which a problem is seen is already the industry's frame. Cognitive capture is harder to govern than interest capture because it leaves no transaction that could be prohibited. Nobody rotated through the door, yet everyone's whiteboard carries the same set of industry terms. Detection therefore runs only on indirect indicators: the similarity between the wording of regulatory reports and industry white papers, the density of dissent recorded inside the regulator, the share of independent research the regulator cites. All three are measurement ideas discussed in the literature, and this book hands them to the reader for one use, self-check: has your organization's risk and compliance function undergone cognitive capture, is the language of risk now entirely in tune with the language of business? Capture of an internal relationship sits closer than capture of a regulatory one. It lives in your own building.

One institutional design direction for cognitive capture deserves to be written down: diversify the regulator's information diet (a deepening of Device 1). The extreme form is a technical body fully independent of the industry, a national-laboratory type of institution that serves no company and lives on independent testing. The product of such a body, independent test data and methodology, is the culture medium the regulator grows on; without it the regulator's technical judgment can only grow out of the regulated party's white papers. Neutral bodies are traditional in nuclear power and aviation, in the form of national laboratories and government test centres, and are generally absent in the new sectors (platforms, algorithms, genes). An absent neutral body is the deepest pit in a lag period. If your sector has a policy discussion about founding one, this chapter's position is to support it: it is the most expensive and the most fundamental of all the defences.

31.1 What It Is

Definition: regulatory lag means rules updating more slowly than technology and business models change, so that risks inside the window sit unclaimed. Regulatory capture means a deep fusion of regulator and regulated across personnel, information and interest, so that the regulator's objective function slides toward the regulated party's.

Three criteria start it. Any one is enough.

1. The regulator's main information source is the regulated party: data reported by them, technology explained by them, people supplied by them.
2. The time between the rule's last update and the change in the business exceeds one technology generation.
3. The expected cost of a penalty is below the expected gain from the violation, so the fine becomes a membership fee.

31.2 How It Grows

Three generative conditions.

Condition one is industry concentration. The regulated parties are few and large, and the regulator faces a full-time, professional opposing team while its own staff turn over and its budget arrives through an approval process. Asymmetry in specialist knowledge is the technical precondition of capture. Cognitive capture arrives before interest capture.

Condition two is insufficient regulatory resources. What the regulator cannot understand it can only have explained by the regulated party, and it believes what it is told. Data are submitted by the regulated, definitions are set by the regulated, technical training is

provided by the regulated. The single information pipe was settled the day the budget passed.

Condition three is the revolving door. People move in both directions between regulator and regulated, which puts tomorrow’s employer into today’s case file. A revolving door does not necessarily corrupt. It necessarily affects how sharp the wording is.

31.3 The Shape in Four Domains

Case One (Aviation): Thirty Days, Two Years, Four Documents

av-038.

When and where: 21 March 2022, 14:16 to 14:21:40 Beijing time; a valley running south-east to north-west near Molang Village, Langnan Town, Teng County, Wuzhou, Guangxi Zhuang Autonomous Region. The main impact point was a water-filled crater about 45 square metres in area and 2.7 metres deep, at 23°19′25.52″ N, 111°06′44.30″ E. The issuing office for the bulletin of 2022-04-20 was the Aviation Safety Office.

What and who: the technical investigation team organized by the Aviation Safety Office of the Civil Aviation Administration of China (CAAC), under the Regulation on the Technical Investigation of Civil Aircraft Events (CCAR-395) and Annex 13 to the Convention on International Civil Aviation; China Eastern Yunnan Airlines as operator; Boeing as design and manufacturing party; the US National Transportation Safety Board (NTSB) carrying out the recorder readout, the work performed by NTSB personnel with CAAC personnel present to supervise and record (NTSB report p. 4). Positions involved: head of the investigation team, authorised representative of the State of Design, safety director of the operator, airworthiness inspector of the regional administration. The book has

not obtained the team membership list or the appointment docu ‑ ments.

What they were doing: at 13:16 the aircraft took off from runway 21 at Kunming Changshui Airport; at 13:27 it reached cruise alti ‑ tude 8,900 metres; at 14:17 it entered the Guangzhou control area along airway A599; 14:16 was the last normal air-ground commu ‑ nication (CAAC bulletin 2022-04-20, paras 3 and 5). At 14:20:55 Guangzhou area control radar raised a "deviation from assigned altitude" alert. The aircraft left cruise altitude, the controller called the crew, and no reply came. At 14:21:40 the last radar re ‑ cord showed standard pressure altitude 3,380 metres, ground speed 1,010 km/h and heading 117 degrees; the signal then disap ‑ peared (same bulletin, para 3). Records before departure were clean: the last A check (31A) and C check (3C) both inside main ‑ tenance programme limits; that day's pre-flight and transit re ‑ lease with no fault report and no deferred defect; loading within requirements; no cargo or mail declared as dangerous goods; flight crew and cabin crew holding valid licences and certificates; flight, duty and rest times meeting the rules; the pre-flight medical ex ‑ amination passed that day (2022-04-20 bulletin, para 5; 2024-03-20 bulletin, para 6). The site survey followed: wreckage fragments were concentrated on and under the ground between bearings 0 and 150 degrees from the impact point; the trailing edge of the right winglet was found about 12 kilometres from the main impact point; vegetation in the forest showed signs of burning (2022-04-20 bulletin, para 4).

The decision point:

One. On 23 March 2022 the crash-survivable memory unit of the cockpit voice recorder (CVR) was recovered, and on 27 March the same component of the flight data recorder (FDR). On 28 March CAAC and NTSB personnel formed a recorder data recovery group (NTSB report pp. 4, 5, 14). The legal basis was written into the bul ‑

letin: CCAR-395 and Annex 13 (2022-04-20 bulletin, para 2). CAAC had already tried to download the CVR in Beijing using a substitute chassis; decompression threw an error and produced no intelligible audio (p. 5).

Two. On 20 April 2022, day 30 of the accident, CAAC published a bulletin on the preliminary report. It states its basis plainly: Annex 13 requires the State conducting the investigation to send a preliminary report to ICAO and to the States participating in the investigation within 30 days, with content "usually the factual information obtained so far, not including analysis of the causes and conclusions" (2022-04-20 bulletin, para 2). The same bulletin described the recorders as "severely damaged by impact, with data recovery and analysis still in progress" (para 5). That day is the boundary of "when did they know". Control knew within 55 seconds that the aircraft was down. On day 30 the investigators did not know why.

Three. On 1 July 2022 the NTSB issued the Cockpit Voice and Flight Data Recorder Combined Download Report. All four audio channels (observer, first officer, captain, cabin) were rated Excellent. The first three ran about 120 minutes each and the cabin channel about 180 minutes, with the accident segment recorded in full (NTSB p. 13, Table 1). The recovery group had examined the CVR board microscopically, repaired pins and replaced connectors, and the final download ran without error (pp. 11-13).

Four. 20 March 2023 and 20 March 2024, two anniversary bulletins. The first called the accident "very complex, extremely rare", and recorded that over the year more than 100 pieces of significant wreckage had been tested and that verification work had been run on flight simulators and on a real aircraft (2023-03-20 bulletin, paras 2 and 3). The second itemised crew licences and medicals, airworthiness certificates, maintenance personnel qualifications, pre-flight faults, airport support, air traffic control qualifications

and equipment, weather, loading and dangerous goods, concluded that no anomaly had been found in any of them, and closed with one line about continuing experimental verification and cause analysis (2024-03-20 bulletin, paras 6 and 7).

What was visible then: the seat is March 2024, the Aviation Safety Office of CAAC, the person drafting that year's anniversary bulletin. On the desk: the exclusion list from the 2022-04-20 preliminary report, eight items, each with its basis (para 5); the statement in the same bulletin that the preliminary report had gone to ICAO and the participating States under Annex 13 (para 2), which means this list is all that other States' operators received (para 5); the download report of 2022-07-01, four channels with quality ratings and durations, ten minutes to read, not one word about cause (NTSB p. 13); the draft of the 2023-03-20 bulletin and the questions that followed its publication; the test records and submission lists for more than 100 pieces of wreckage (2023-03-20 bulletin, para 2); the plans and results of the simulator and real-aircraft verification; operating data from the same type continuing in normal service at home and worldwide; and two sets of questions that have to be answered, from the families and from international civil aviation. One further item needs a decision at this desk: whether to keep the draft sentence "repair the damaged recorders and analyse the recovered data". The NTSB download report had recorded all four channels as Excellent on 1 July 2022 (NTSB p. 13), and by this day that is 20 months. Information that became known only afterwards: the specific content of the accident-segment recorder data (released in April 2026 through the US Freedom of Information Act process, and this book quotes no parameter from it); the sequence of control inputs before ground impact; confirmation of any single point of failure.

> **Stop here. Decide before you turn the page.**

> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: all 123 passengers and 9 crew on board died, 132 people (2022-04-20 bulletin, para 1). The official documents published since number three: the preliminary report bulletin of 2022-04-20 and the progress bulletins of 2023-03-20 and 2024-03-20. There is no cause conclusion, and therefore no targeted airworthiness directive based directly on this accident, and no grounding or restriction of the 737-800 fleet. At the 2025 and 2026 anniversaries no comparable bulletin from CAAC was seen (Deutsche Welle citing Reuters, 2026-03-22, Grade E). The investigation is in its fifth year. Back to the three questions. The most common answer to the first is "wait another year until the experiments finish". The cost of that action in this case is one anniversary briefing and one meeting with the families each year, and the price is that the new material available for each bulletin shrinks year on year. The item most easily missed under the second is the download report of 2022-07-01. It looks like a technical annex, and it is the only document in the case that can strike "the data could not be read" off the list of explanations. The checkpoint for the third belongs at the bulletin that follows the data readout: if a bulletin cannot say what it adds to the one before, it is only a mark of time passing.

Options ruled out: grounding the 737-800 (the global fleet runs into the thousands, a grounding without technical basis is not feasible and has no precedent); issuing a targeted airworthiness directive (an AD requires first establishing an "unsafe condition", and no cause means no finding); publishing the recorder

content (Annex 13 constrains disclosure of investigation records, and criminal and family litigation are running in parallel); writing a cause into the preliminary report (Annex 13 states expressly that a preliminary report contains no cause analysis, a sentence the bulletin itself quotes); transferring the investigation to the State of Design (Annex 13 puts the organizing duty with the State of Occurrence, with the State of Design sending an authorised representative); closing the case as "pilot error" (the 2024-03-20 bulletin excluded anomalies in licences, medicals, duty and rest, and training records item by item, and the existing documents do not support that conclusion); an independent third-party review (Annex 13 provides no such procedure and there is no precedent).

Early signals: (1) From the alert at 14:20:55 to loss of signal at 14:21:40, 45 seconds, the aircraft fell from 8,900 metres to 3,380 metres at a ground speed of 1,010 km/h on heading 117 (2022-04-20 bulletin, para 3). That profile does not resemble a gradual loss of control, and it was pressed flat by "wait for the wreckage analysis". (2) Between 14:16 and 14:20:55, 4 minutes 55 seconds, no abnormal call and no distress call (same paragraph). It was pressed flat by "the communications were normal". (3) The main crater is 45 square metres and 2.7 metres deep, the right winglet trailing edge came down 12 kilometres away, and the forest burned (para 4). That distribution points to high speed, a steep angle, and parts leaving the aircraft first. It was read past by the bulletin's sentence that the aircraft broke up after hitting the ground. (4) Both recorders were severely damaged, every download attempt in Beijing with a substitute chassis failed, and the memory boards had to go to the NTSB laboratory (NTSB pp. 5, 15). It was pressed flat by "send it out and it will be read", when the readout took three months. (5) More than 100 pieces of wreckage were tested and still produced no single point of failure (2023-03-20 bulletin, para 2). The phrase "complex, rare" absorbed it, and for two

years afterwards no public document mentioned the sentence "there is still no failure point" itself.

The point of escalation: 1 July 2022. The data could be read, the accident segment was complete, and all four channels were rated Excellent (NTSB p. 13). For the investigating side this was the day of highest availability: a 33-page report, four channels readable, the accident segment complete, after which any claim that technical conditions do not permit can only point at analysis and no longer at recovery. The report's structure carries the same message. It concerns removing boards, repairing pins, replacing connectors, correlating time and rating quality; there are two appendices, an audio quality rating table and a table of verified parameters, and not a page discusses a possible cause (pp. 4-32). For the industry nothing happened that day: the fleet kept flying and the bulletins continued. There was no press conference and no Chinese-language material mentioned the report. It went into the file and stayed there for nearly four years, until it surfaced through the US Freedom of Information Act process in April 2026 (media record, Grade E).

Failure chain: technical layer. A rapid descent from cruise at 8,900 metres to 3,380 metres in 45 seconds at a ground speed of 1,010 km/h. The debris distribution and the winglet trailing edge 12 kilometres away point to an in-flight breakup. Crew, maintenance, fuel, loading, air traffic control and weather were excluded item by item (2024-03-20 bulletin, para 6). No single point of failure has been confirmed, and that layer is still blank.

Organizational layer. The organizer of the investigation is also the industry regulator of the operator. CCAR-395 places the duty to organize the investigation with CAAC, and the bulletin's wording is drafted and approved by agencies in the same chain. The technical investigation team was formed "together with the relevant departments" (2022-04-20 bulletin, para 2), so member units and

the regulated entity sit inside one administrative sequence. Annex 13 mandates only the 30-day preliminary report; the timing of the final report is held by the investigating State. Without a cause conclusion the airworthiness authority cannot write an "unsafe condition" into a directive, and the fleet's response can go no further than generic notices and operator self-checks.

Incentive layer. The 737-800 is a global fleet, the cost of a grounding runs into the tens of billions of dollars, and a grounding without technical basis is not feasible. The longer the investigation runs, the shorter the industry's memory, the more new topics arrive in the same period, and the lower the marginal return on pressing the question. In a period with no cause, "wait for the report" is the least trouble for every party.

The junction breaks between the organizational and incentive layers. For the regulator, "no conclusion" is a tolerable steady state. For the industry it is an exposure with no maturity date.

Minimum counterfactual: when the preliminary report was published on 20 April 2022, publish an investigation timetable alongside it: the date of the next public bulletin, the categories to be published (wreckage, recorder data, simulation verification), and a commitment that if there is still no cause by then, two lists will be published, items excluded and items under examination. One action, needing no new evidence, no prior finding of cause, and no one changing a conclusion. A timetable promises no conclusion, only nodes. Once the nodes are public, the reason for any delay has to be public too, and the length of the window shifts partly from "scientific progress" back to "governance commitment".

Outcome and sources: three official bulletins (CAAC 2022-04-20, 2023-03-20, 2024-03-20, HTML full text, official release without page numbers; this card numbers the paragraphs of the official text, and a paragraph-order cross-reference is in the evidence file); NTSB, Cockpit Voice and Flight Data Recorder Combined Download Re-

port, report number DCA22WA102, 1 July 2022, 33 pages, released through the Freedom of Information Act process after April 2026, page numbers as PDF pages. The passage on the absence of new bulletins at the 2025 and 2026 anniversaries rests on media records (Grade E); the book has not obtained the final accident report, nor any internal document of the investigation. Link at Appendix B, av-038.

The no-conclusion state deserves one more reading, from the side of the global safety system. The force of an international civil aviation special telegraph, in which a State's regulator sends safety advice to its own operators after a major accident, depends on a cause conclusion. In a period without a cause, what goes out can only be generic notices about stronger training and procedure checks, and the incremental constraint such notices place on operators of the same configuration is close to zero. The no-cause period is therefore not only the waiting of the State concerned. It is a public window for the whole industry, and its length is set by the scientific progress of the investigation, not by any governance intention. This book passes no comment on how fast any individual case moves. It writes down the structure: a prolonged investigation is itself an industry-level risk exposure, and the hedges against that exposure (tighter generic inspection, voluntary self-checks by operators of the same configuration) fall inside the operator's own discretion during the window. On how reliable discretion is when there is no external compulsion, see Chapter 11.

Case Two (Chemical): Forty to Sixty Tons, Five Regulatory Layers, Zero Storage Inspections

ch-002.

When and where: 17 April 2013, smoke and fire reported at 19:29, detonation at 19:51; West, McLennan County, Texas, United States, at the West Fertilizer Company fertiliser warehouse and one side of

the rail spur. The warehouse was built in 1961 and in production by 1962; in a 1972 photograph the nearest house stood about 265 feet from the plant boundary; the US Geological Survey recorded the explosion as magnitude 2.1 (CSB p. 20, p. 84).

What and who: West Fertilizer Company (WFC; privately held; fertiliser blending, retail and distribution; the same site held the grain silos of Adair Grain, another family company). Five regulatory layers: the US Department of Labor Occupational Safety and Health Administration (OSHA); the Environmental Protection Agency (EPA); the Department of Homeland Security Chemical Facility Anti-Terrorism Standards (CFATS) programme; the Texas Office of the State Chemist (OTSC) and the Texas Commission on Environmental Quality (TCEQ); the McLennan County Local Emergency Planning Committee (LEPC) and the West Volunteer Fire Department (WVFD). Positions involved: the company manager, the fire chief and assistant chief, the county LEPC coordinator, the state SERC duty line. The report does not state a company safety officer position.

What they were doing: 40 to 60 tons of fertiliser-grade ammonium nitrate (FGAN) sat in wooden bins inside the warehouse, and one railcar on the spur was not fully unloaded, about 100 tons (p. 19). The same site held 17 tons of anhydrous ammonia, with 34,000 pounds actually present in April 2013 (p. 19, p. 180). The company blended to customer formula, and blended material counted under EPCRA as "work in progress" rather than "finished goods for sale" (p. 206). When the warehouse was built in 1961 the surrounding land was open and the town had no zoning. The town then grew toward the warehouse: a park less than 150 feet away, an apartment building at 450 feet, a nursing home at 500 feet, a middle school at 552 feet (p. 20, p. 16). The 2012 Tier II form listed FGAN for the first time, submitted to WVFD, the county LEPC and the state health services department (p. 204). On 12 February 2013

pallets and branches were burned on site and the neighbouring elementary school was evacuated for about 30 minutes; the school had no advance notice, and 911 dispatch did not confirm that this was a notified burn (p. 203). Fire broke out on the evening of 17 April. The cause remains undetermined, and law enforcement kept three possibilities open: an electrical wiring fault, a short circuit in an electric golf cart, arson (p. 18). The day before, a farmer had bought a bag of FGAN and ammonium sulphate blend, and that bag became the only sample in later testing that represented the material inside the bin before the blast (p. 80). The fire department sent two pumpers, two small attack vehicles and a tanker. The usable hydrant stood by the high school, 1,600 feet away, and after laying 1,000 feet of four-inch hose the crews were still about 700 feet short (p. 31).

The decision point:

One. 2004, change of ownership. An EPA inspector came on site and issued a citation for failure to resubmit the risk management plan (RMP) registration; the company then hired its insurer to write the anhydrous ammonia RMP (p. 174). On 21 June 2006 TCEQ issued a violation referral for anhydrous ammonia without an air permit (p. 174, note 440). Both visits concerned ammonia. The report also records that the owner "did make corrections" after being notified of non-compliance (p. 174). The RMP that followed covered anhydrous ammonia, not ammonium nitrate; the report does not state the inspection date, the inspector's name or the checklist, nor whether the inspection reached the warehouse side.

Two. 2008, when DHS extended the Top-Screen submission period for agricultural facilities serving farmers by 73 FR 1640, expressly not applicable to chemical distribution facilities (p. 178). WFC never submitted. Against the CFATS threshold of 2,000 pounds, it held at most 120,000 pounds on site, sixty times the threshold. The

CSB's conclusion sentence reads: until the explosion in 2013, DHS did not know that this facility stored FGAN (p. 178).

Three. The 2012 Tier II submission. The form carried the quantity range for FGAN, the storage method and the on-site location. The county emergency response plan had last been formally revised in 2010, and Annex Q required every EPCRA-regulated facility in the county to be listed; WFC was not on the list (p. 203-204). The written judgment of the county LEPC was that WFC's ammonia and FGAN "appeared" to qualify for the agricultural exemption, and that judgment was confirmed by the state SERC (p. 205). WFC also appears on no attendance roster of any county LEPC meeting across a span of more than 21 years (p. 203).

Four. 17 April 2013, 19:29 to 19:51, twenty-two minutes. The 19:29 alarm came from a member of the public; a patrol officer arriving saw flames and called dispatch to upgrade the call to a structure fire (p. 30). The assistant chief, on arrival, advised officers to begin evacuating nearby residents and radioed for a ladder truck to stand by at the apartment building; none was available. Full evacuation was still being assessed when the explosion came, with flames then pouring from the roof skylight and the north-east door to a height of 40 to 50 feet (p. 32). In those twenty-two minutes not one of the five regulatory layers was present. The only thing that could arrive was a volunteer fire department with no pre-plan.

What was visible then: the seat is March 2013, coordinator of the McLennan County LEPC. On the desk: the 2012 Tier II form, with annual maximum and daily average ranges for FGAN, the storage method and the on-site location, submitted including to the county fire department (p. 198, p. 204); the county plan in its 2010 revision, body plus 22 annexes, Annex Q being the hazardous materials and oil spill response annex, whose regulated facility list omits WFC (p. 203); the fact that Tier II electronic submissions are received through the state health services department and the county's

copy went down the same channel (p. 199); the LEPC attendance rosters, with WFC absent from every one from the committee's founding in 1992 to 2013 (p. 203); the school evacuation record of 12 February 2013 and the school's demand that future burns be notified in advance (p. 203-204); the state SERC position on the agricultural exemption, on which the county concluded that WFC's ammonia and FGAN "appeared" to qualify (p. 205); the fire department's training records, which show anhydrous ammonia drills and no FGAN drill (p. 122). What was not on the desk is equally clear: an impact-range calculation for this facility, a drill record for this facility, a federal inspection report dated after 1985 (p. 171, p. 244). The Tier II form itself omits three things: the material of the bins, the spacing inside the building, and how far a blast would reach. Its fields are name, quantity range, storage method, location, and whether the location is confidential (p. 198). Information that became known only afterwards: the bins were wooden, with no sprinklers and no detection; the roof was combustible asphalt shingle; the seed room stood next to the bins (p. 244); blended material was not eligible for the agricultural exemption, and an EPA letter to the Fertilizer Institute dated 3 September 2010 stated that "blending must be reported" (p. 206); the Top-Screen was never submitted; the 1971 provision 1910.109(i) had always applied to facilities storing more than 1,000 pounds of AN (p. 169).

> **Stop here. Decide before you turn the page.**
>
> You are in this seat, with the material above on the desk.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: 15 people died, 12 of them fire and rescue personnel and 3 members of the public; 252 people were injured directly by the explosion, and the report counts more than 260 injured (p. 16, p. 33). More than 350 homes were damaged and 150 destroyed; a middle school, a high school, a 22-unit apartment building and a 145-bed nursing home were demolished (p. 16, p. 243); more than 150 buildings around the facility were damaged. Insurance-related losses came to about 230 million dollars, and federal disaster relief exceeded 16 million dollars (p. 90). WFC held a general liability limit of 1 million dollars and property cover of about 4.45 million dollars, and the company went bankrupt afterwards (p. 89, p. 16). On 9 October 2013 OSHA issued 24 citations, 9 of them serious violations of 1910.109(i): inadequate ventilation, non-fire-resistant construction, non-compliant stacking and bin height (p. 173); the case was settled without an admission (p. 174). The CSB entered the site on 18 April 2013, and for the first four weeks the scene was controlled by the Bureau of Alcohol, Tobacco, Firearms and Explosives and the State Fire Marshal's Office as a criminal case, which delayed evidence collection and interviews (p. 18). The report was unanimously approved at a public meeting in Waco on 28 January 2016; of its 19 recommendations 3 remain open, one of them requiring fertiliser-grade ammonium nitrate to be listed as a regulated substance under the federal Risk Management Program, still marked open (CSB project page). Back to the three questions. The most common answer to the first is "add the facility to Annex Q and schedule a joint inspection". The cost of that action here is one site visit and one item on a quarterly meeting agenda, and the price is that nobody has the authority to make it happen. The item most easily missed under the second is the 2012 Tier II form. It looks like an annual return, and it is the only document that states both how many tons and where they sit. The checkpoint for the third belongs in the week the information ar-

rives: within 30 days of the form arriving, did the facility enter Annex Q.

Options ruled out: on 31 January 1994 the EPA proposed listing commercial high explosives under RMP, with a threshold set at 3.0 psi lethal overpressure (59 FR 4487, see p. 190, note 530); on 6 January 1998 the category was removed after a settlement with the Institute of Makers of Explosives (63 FR 640, same note). In the late 1990s OSHA considered listing FGAN under PSM, and it did not happen for "resource constraints and other priorities" (p. 180). The 2008 DHS extension went to farms only, not to distributors (p. 178). Until Texas amended its commercial fertiliser rules in June 2014, the Office of the State Chemist held sales registration authority only, with no authority to inspect storage areas (p. 217). The listing criteria for county Annex Q covered only facilities holding extremely hazardous substances, and FGAN was not on that list (p. 197); adding it to the EHS list would require changing the federal list, which is not a county decision. OSHA itself had two provisions it could not cite: in 1910.109(i), the two clauses on approval of large-scale storage and on continued use of existing buildings reserve authority to the "authority having jurisdiction" (p. 175). HB 417, introduced in 2015 to give the insurance commissioner rulemaking authority, was still in committee in December 2015 (p. 218). The industry's voluntary guidance, self-assessment every three years, arrived only after the event, in February 2014 (p. 219).

Early signals: (1) OSHA last entered the site in 1985 and cited anhydrous ammonia, respiratory protection and recordkeeping, not ammonium nitrate; the CSB found no 1910.109(i) citation against the facility at all, and no inspection record for the following 28 years (p. 171). It was pressed flat by "it was inspected last time and fixed". (2) Across the United States 1910.109(i) was cited 19 times between 1974 and 1999 and zero times between 1999 and 2013; no

facility with the same industry code was cited under that provision in the same period, and the GAO concluded in 2014 that before West there had been exactly one citation in the whole country for ammonium nitrate storage at a fertiliser facility (p. 173). It was pressed flat by "nobody fines it, so it does not apply". (3) The annual survey forms of the insurer Triangle between 2006 and 2009 recorded ammonium nitrate on site, handled it as a security question, ticked "yes" to compliance with state storage and transport rules, and the state had no such rule at the time; the same forms examined grain dust and anhydrous ammonia closely against federal standards (p. 89). It was pressed flat by "the form passed". (4) On 12 February 2013 the school was evacuated for 30 minutes, 911 dispatch did not confirm a notified burn, and the school district afterwards asked only for advance notice (p. 203). It was pressed flat by "it was only a burn". (5) The 2012 Tier II form had already reported FGAN to the fire department, and the report's own words are that "no mechanism ensured that a drill or inspection would follow" (p. 122). It was pressed flat by "reported means managed".

The point of escalation: 2010, when Triangle declined to renew, on the grounds that WFC had repeatedly failed to act on its survey recommendations, replacing corroded wiring among them; WFC moved to U.S. Fire, with general liability of 1 million dollars (p. 89, p. 244). The CSB asked U.S. Fire for its site inspection records for the policy period and did not receive them; its outside counsel told CSB investigators that a policy of 1 million dollars "did not require much site activity" (p. 90). The report does not state the policy signing date or the signatory. The 2012 Tier II form carried one further column, asking whether the owner elected not to disclose the location to the public (p. 198): the design of the form itself presumes that reporting is not publishing. From the change of insurer in 2010 to April 2013 there was no external party holding both site access and a motive to find fault. OSHA had stopped, the state

had no storage inspection authority, and the insurer had changed. The 2012 form added knowing. It did not add looking.

Failure chain: technical layer. Wooden bins, no automatic sprinklers, no fire detection, a combustible roof and combustibles hard against the pile; radiant heat from the fire warmed the surface of the pile, and soot, molten asphalt and PVC melting from the conveyor above contaminated the ammonium nitrate into an explosive mixture (p. 244). This layer is not in dispute.

Organizational layer. Five layers each covered a segment and each left a gap. OSHA's standard stopped in 1971, and from the mid-1970s the agency lost the authority to bring in new consensus standards (p. 169, note 403); the standard allowed wooden bins "treated to resist impregnation" without defining impregnation, and required sprinklers only for bagged product above 2,500 tons (p. 167); the PSM list omitted FGAN, a reactivity class 3 material (p. 180); the 50 per cent test in the retail exemption let anhydrous ammonia out of scope as well (p. 182). The EPA's RMP list held no FGAN, and WFC's 2011 hazard review listed toxic release, equipment failure and earthquake, with no fire and explosion (p. 188). The EPCRA agricultural exemption was read by county and state as a blanket exemption (p. 205). CFATS has a threshold and a list, and where a facility does not report, DHS does not know (p. 178). The state handled sales registration and not storage (p. 217). The county plan's annex did not list the facility (p. 204). The fire department had no pre-plan, for the stated reason that "there was no such regulatory requirement" (p. 244). No federal agency regulates municipal fire departments (p. 122).

Incentive layer. From 1999 to 2013 enforcement citations for this class of violation were zero, so the expected cost of a violation was close to zero (p. 173). The insurance market stopped pricing the risk in 2010, and the remaining underwriter sized its site investment to the premium (p. 90). The town's housing spread toward

the facility, with the nearest apartment at 450 feet, the nursing home at 500 feet and the elementary school at 552 feet (p. 16), and no party in that expansion had an incentive to set a separation distance.

The junction breaks between the organizational and incentive layers. All five layers collected information, and none converted information into an obligation to inspect the site. Without that obligation, no layer's budget had to answer for how many tons stood inside that building.

Minimum counterfactual: one line added to the checklist of the 2004 EPA site inspection: is ammonium nitrate stored on this site, in what quantity, in what packaging, and in what building material. The inspector was already on site, and one line needs no legislation, no new budget, and no one changing their view. An answer of 40 to 60 tons in wooden bins with no sprinklers would have put the facility inside the scope of RMP or PSM, or inside the 30-foot combustible separation requirement that Texas only wrote into its rules in 2014: nine years earlier. One action, nine years before the event, at the cost of one question.

Outcome and sources: the primary source is the final report of the US Chemical Safety Board, West Fertilizer Company Fire and Explosion, Report 2013-02-I-TX, approved by the board on 28 January 2016, 267 pages as a PDF; page numbers here are PDF pages, equal to the printed page plus 3 (printed p. 201 is card p. 204). Relevant chapters: Chapter 5 on insurance (p. 85 ff), Chapter 7 on emergency response (p. 107 ff), Chapter 8 on regulatory analysis (p. 167 ff), Chapter 10 on key findings (p. 243 ff), Chapter 11 on recommendations (p. 245 ff). The status of the recommendations (19 in total, 3 open, one of them the listing of FGAN as a regulated substance under the Risk Management Program) comes from the CSB project page. Link at Appendix B, ch-002.

Case two's gap carries one more footnote from the sociology of fire. The concentration of volunteer firefighter casualties is the dark note of this subject: the absence of separation distances hits small communities hardest, since facilities in large cities get professional inspection and facilities in townships rely on volunteers. Gaps are distributed with strong geographic inequality, because regulatory density falls as urbanisation thins, and the bill for a gap is therefore posted mostly to small communities. That is the common distribution of every gap-type regulatory failure. Repair priority should run against the urban gradient, low-density places first, and the repair tool is a listed cross-layer joint check: turning this chapter's test, do the three layers together cover it, into an annual cross-layer joint verification checklist.

Filling that checklist has one technical point. Fill it by object, not by jurisdiction: the rows are this facility, this business line, this product; the columns are the regulatory layers; each cell records whether that layer covers the object and under which provision. A list filled by object exposes the empty cells directly, and an object covered by no layer is visible at a glance. A traditional list filled by jurisdiction only demonstrates each agency's diligence. Once the empty cells are exposed, the handling is the registration of Device 2 in this chapter: empty-cell objects go on the window-period list, with a self-reporting duty and a bond attached. The long-run fix for an empty cell is legislative gap-filling, assigning the ownerless object explicitly to one layer. Joint check, registration and assignment form the annual cycle of gap governance, chaired by the joint meeting of the layers, where what counts is the consolidation of the object lists; the meeting is only the occasion.

Case Three (Finance): 2.56 Million Fractional Share Trades

fi-011. Raymond James & Associates. The FINRA penalty of August 2026 (the monthly summary PDF has been checked by this book):

failure to report about 2.56 million fractional share trades settled in principal, fined 125,000 dollars. Fractional share trading grew out of odd-lot investing and automatic dividend reinvestment plans (DRIP). The business grew outside the trade classifications of the rule, and these trades were handled as bookkeeping entries. The rule was written in an era when fractional shares did not exist or did not reach scale, and the scaling of the business turned one classification error into a reporting gap in the millions. The ratio between the fine, 125,000 dollars, and the size of the gap, 2.56 million trades, is the arithmetic display of criterion 3, the fine as membership fee.

Stop at the decision point: the memorandum written when the fractional share business went live, in which internal compliance determined its reporting status. On the desk: the literal text of the rule, the shape of the new business, and the peer practice, that others do it too. **Outcome:** the violation was established, and the rule catch-up for the new business began after the penalty. Primary source: FINRA, Disciplinary and Other FINRA Actions - October 2026, 2026, link in Appendix B, fi-011.

Case three's fractional shares carry a lesson in taxonomy worth writing down. Rule classification always lags, so at the moment a business goes live its classification decision is the last line of defence, and the business side's compliance memorandum is the only gate during the lag period. The quality of that memorandum sets the strength of the gate. In this case the memorandum classified fractional shares as a bookkeeping matter, which was a misclassification; the misclassification was reinforced by peer practice, since others reported the same way, and peer practice then became the evidence for the memorandum, which closed the loop of judgment on itself. Only one question breaks such a loop: if this classification is wrong, what is the worst consequence. The answer in this case, a regulatory penalty for the reporting gap, could have

pushed the memorandum toward a regulatory consultation, and asking the regulator is an order of magnitude cheaper than guessing the regulator. The either-or lesson, in one sentence for the compliance side of every new business: with a classification you are unsure of, ask and do not guess. Whether the regulator answers is the regulator's business. Whether you ask is your compliance duty.

Case Four (Maritime): Convention Replaced the Assessment

ma-012. The pot-fishing vessel Amadeus (TH7), December 2023, UK waters. MAIB report (Report 13/2026, pages checked by this book): a deckhand leaned outboard during pot hauling, went into the water and died; the investigation points to an operation that required leaning outboard in an unstable posture with no fall protection, and to an industry practice that had long stood in for a formal risk assessment. Regulatory density on small fishing vessels is low by nature: where the rule cannot reach, convention is the law, and what convention eats is exactly the space a risk assessment was meant to cover. (The death and the report number have been checked against the pages; the detail of the convention is in the body of the report.)

Stop at the decision point: the moment this method of working was first carried on without protection and accepted by the rest of the fleet. On the desk: the efficiency of the method, a record of never having had an accident, and the impossibility of a regulator being present. **Outcome:** one person died. Primary source: MAIB, Report 13/2026, 2026, link in Appendix B, ma-012.

Case four's substitution of convention for assessment has one structural footnote from the fishing industry. On a small vessel the crew are the owners, working self-employed, and the cost of regulatory reach is very high, with boarding inspections covering a single-digit percentage of the fleet, so the self-copying of conven-

tion has no external calibration at all: what a new crew member learns from an old one has never been checked against a written standard. The workable lever for governing convention therefore sits in industry organization rather than in inspection: the operating guidance of a fishermen's association, and the operational requirements carried in insurance rates, since insurers board vessels more often than regulators do and a rate lever reaches far more operators than a fine. The insurance lever deserves its own line in this chapter. Insurers are among the few external calibrators whose interest runs alongside the regulator's in loss reduction, and in a sector where insurance rates carry the safety standard, the self-copying of convention gains one more external check.

The theme of external calibration carries the chapter into the signal list of section 4. Put the six signals against this chapter's cases and its reasoning. Signal one, manuals drafted by the leading firms, matches the gap in case two: each layer's inspection manual is written to its own jurisdiction, and no one drafts the joint manual written from the object's side. Signal two, a new business three years old with no rule, matches cases three and six: fractional shares and crypto assets are both windows on the three-year scale. Signal three, fines that get costed, matches case three: 125,000 against 2.56 million trades is the arithmetic an operator runs. Signal four, departure destinations concentrated in one industry, is the main signal of cognitive capture, and the three indirect indicators in section 1 are its operationalisation. Signal five, compliance findings that do not travel upward, matches case three: the classification doubt in the internal memorandum never reached the level of a regulatory consultation. Signal six, a self-regulatory code with no cost of breach, matches case four: industry convention is exactly self-regulation at zero cost.

Reorder the six signals by the cost of the evidence. Signals three and two are cheapest, since they need only the timestamps on fines

and on rules. Signals one and six sit in the middle, needing the drafting list and the self-regulatory text. Signals four and five are most expensive, needing personnel trajectories and a trace of internal information flow. The starting recommendation is signal two: list the new business models in your sector, annotate each with the year of the corresponding rule, and take the list of windows older than three years straight to the governance layer. That list is the entire stock of lag in your sector.

Case 5 (Drugs and Devices): One Month Shipping Without Clearance (condensed)

ph-005. Adept Medical's ear ventilation T-tube (9mm). FDA device recall (Z-0966-2025, API record checked by this book): distributed without 510(k) clearance, 6,280 units recalled (2,500 in the United States, 3,780 outside it), distributed to Canada, Australia and New Zealand. The regulatory position on the device, its classification and its clearance pathway, was itself clear. What failed was the internal release gate: the decision chain that shipped before clearance arrived had no blocking position. Regulatory lag gives way here to a collapse inside regulatory execution, in which the rule was in place and the gate had no latch fitted. Minimum counterfactual: a hard validation of clearance status in the release system, so that no shipping order can be generated without it. Primary source: FDA, 2025, link in Appendix B, ph-005.

Case 6 (Fintech): The Banking Channel Is the Regulation (condensed)

st-002. An Indian cryptocurrency exchange (peak monthly trading volume 246 million dollars, December 2017). Failory Cemetery archive quoting the CEO Rahul Raj's post-mortem verbatim, checked by this book: it kept operating for fourteen months after its banking channel was frozen and then shut down. Regulatory uncertainty in crypto, meaning blank and shifting rules, travelled

through the banking compliance channel and arrived as real pressure: the regulator said nothing, and the financial infrastructure spoke first. The case shows the true position of the regulated during a lag period. An absent rule is not freedom; it is uncertainty in the form of access to infrastructure, and the price that uncertainty sets on innovation, the cost of fourteen months of struggle, is carried by the company alone. Minimum counterfactual: a business continuity plan for a period of regulatory uncertainty, with the cost of diversifying channels calculated in advance. Primary source: Failory Cemetery (quoting Rahul Raj's postmortem), 2019, link in Appendix B, st-002 (evidence grade E).

31.4 What Shows Up Earliest

Six weak signals, each with the reason usually used to press it flat.

1. The regulator's technical lectures, data platform and inspection manuals are drafted with the participation of the leading regulated firms. Reason: "They understand it best."
2. A new business has existed for more than three years with no matching rule and no explicit exemption. Reason: "Let it run a while."
3. The ratio of fine to violation gain is being costed openly by practitioners. Reason: "It is only a cost."
4. Where regulatory staff go when they leave is concentrated in the regulated industry. Reason: "Talent is a market."
5. The compliance team knows the business's real holes better than the business team, and that knowledge does not travel upward. Reason: "That is exactly what compliance is for."
6. The trade association's self-regulatory code has become a substitute for the missing rule, and breaching it costs nothing. Reason: "Industry self-governance."

31.5 How to Prevent It

Device 1: diversify the information source. Force the regulator's and the risk function's information inputs to exceed the regulated party's own reporting: independent sampling, third-party data, complaint and public sentiment channels, cross-region comparison. A single information source is the technical base on which capture stands. Plurality does not prevent capture; it preserves a floor, and while the floor holds, everything else still has a chance.

Device 2: register the window. A new business registers a list of blank points in the rules at launch, and during the blank period it carries a higher self-reporting duty and a bond. This rewrites the lag period from "nobody is watching" into "conditional release", and the condition is the deposit plus the reporting frequency. Had the fractional share business in fi-011 gone live with a window registration, the classification problem in 2.56 million trades would have surfaced in the first year rather than at the millionth trade.

Device 3: link the penalty to the gain. Set the penalty base against the gain from the violation, as a multiple of illegal proceeds, so that the membership fee arithmetic stops working; for repeat offenders, restrict the business rather than adding a linear penalty. The deterrent force of a penalty does not depend on its absolute size but on its ratio to the gain. Below a ratio of 1, the fine is the sector's cost of entry.

Device 4: a cooling-off period on the revolving door. Staff leaving key regulatory posts for the regulated industry serve a cooling-off period, during which paid communication on matters concerning that agency is prohibited, and the same applies in reverse. A cooling-off period cannot prevent capture of the mind, but it cuts the coarsest interest pipe. Cognitive capture, the "believes what it is

told" of criterion 2, has to be diluted by diversifying the information source.

Set the four devices against this chapter's cases. Device 1 on case two: the object list for the cross-layer joint check is verified with the participation of a third party outside the trade association, an academic group or an insurer. Device 2 on case three: the regis - tration list for a fractional-share-type new business is submitted at launch, and the window period carries a monthly flash report on the structure of trading. Device 3 on case three: the penalty base for a reporting violation becomes the number of trades or the number of clients involved, linked to scale, so that the fine is no longer decoupled from it. Device 4 keeps the personnel policy as written and adds symmetric execution of the reverse clause, with the regulated party's waiting obligation when hiring a former regulator written into the contract as well.

The four devices also have four ways of dying. Diversifying the in - formation source dies when the definitions converge across the multiple sources, with third-party data still collected under defin - itions the regulated party wrote; the patch is independent author - ity over definitions, so that at least one source holds the right to define them. Window registration dies when the registration deadline drifts out of control, a three-year window becoming ten and the registration turning into a permanent licence; the patch is a sunset clause, with registration valid for one year and renewal requiring a fresh assessment whose report is published. Linking penalty to gain dies in the accounting contest, with the determin - ation of illegal proceeds dragged into a numbers game; the patch is to objectify the penalty base on business volume measures such as trades, clients or data volume, which bypass the determination of proceeds. The cooling-off period dies when the list is marginalised, with the definition of a key post narrowed until most leavers fall outside it; the patch is to reverse the burden on the list, with the

regulator deciding whether a post is key and the regulated party holding no right to apply for exemption. All four patches share one health-check question: when was this device's parameter, its definitions, its deadline, its penalty base or its list, last revised independently by the regulator.

The enterprise-side mirror of these defences belongs in the same section, because most of this chapter is written for regulators and the regulated party needs its own set of moves for surviving a lag period in a captured environment. Four of them. Treat the regulatory window as a risk exposure, accruing the risk reserve for window-period business on the assumption that no regulator will catch the fall. Register voluntarily, which is the enterprise side of Device 2, since a window registration buys a predictable expectation from the regulator and costs less than a penalty afterwards. Keep a direct channel for compliance, so that a compliance finding reaches the governance layer with a record, which is the cure for signal five. Cool the revolving door yourself, setting an internal cooling-off period for key posts when hiring former regulatory staff even where the law does not require it, since this is the cheapest reputational insurance a firm can buy in a captured environment. The common logic of the four: when the regulatory environment fails, the structural moves on the enterprise side are the only part still under its own control.

31.6 Boundaries with Adjacent Mechanisms

Against compliant and unsafe (Chapter 32): the organizational form of capture is "every inspection passed", seen in the certification philosophy around TWA800 and in the gap inspections of this chapter, and Chapter 32 supplies the micro-mechanism, in which nobody owns the space between the compliance items. The relation between the two chapters: Chapter 31 is the climate, Chapter 32 is the weather.

Against misaligned incentives (Chapter 17): capture is misaligned incentives in its regulatory form, where the regulator's own measures (budget, headcount, calm) point away from its real objective (risk control). The version of Chapter 17's overall-result signatory, in this setting, is the independent parliamentary scrutiny and the press.

Against extrapolating past the validated domain (Chapter 23): operating inside a lag period is in essence a large-scale extrapolation that nobody declared, with the business extending the applicability of an old rule to a new form. The counterpart of Chapter 23's declaration form in a regulatory setting is window registration.

31.7 The Five Self-Checks

1. In what year was your sector's rule last updated, and what changed in the business over the same period? List three differences.
2. Through which channel did the largest hole your compliance team has found reach the regulator or the governance layer, and how long did that take?
3. Has anyone costed the ratio of fine to violation gain in your sector? What did it come out at?
4. Where did the regulatory officials you know go when they left? What does that flow mean for your information strategy?
5. If your business entered a regulatory blank tomorrow, would you register voluntarily? What does your answer depend on?

(This chapter: 12,058 characters.)

Chapter 32 Compliant and Unsafe (Hollnagel: Safety-I and Safety-II)

> What this chapter covers: why a system that passes every item can still be lethal; Hollnagel's split between Safety-I and Safety-II, three sites, and the scenario inspection that follows from it. When to come back to it: after every check passed and the accident happened anyway; whenever someone proposes adding one more compliance requirement. Reading time: about 50 minutes.

A compliance system makes a promise it never states out loud: meet every item and you are safe. Erik Hollnagel calls this mode of inspection, which looks for failure and assigns blame, Safety-I, and he names its blind spot. It can prove the absence of a violation. It cannot prove safety. Safety is never the sum of clauses; it is how the system performs as a whole in real situations. The seams between clauses, the new situations no clause covers, the way a clause gets carried out on a wrong premise: all three sit outside Safety-I's field of view. The three sites in this chapter show what that blind spot costs, three times over. One is a physical gap you can compute. One is a wrong classification. One is a consent mechanism that went on violating the rules after two rounds of remediation.

32.1 What It Is

Definition: compliant and unsafe describes the structure in which an inspection system passes item by item while an accident still happens. Rules are written clause by clause, inspections are run clause by clause, every individual pass is real, nobody owns the space between clauses, and the compliance record gradually takes the place of safety judgment as the proof of safety (Hollnagel,

Safety-I and Safety-II: Safety Thinking in Action, Ashgate, 2014; the Crossref record has been checked).

Three trigger criteria. Any one of them means the mechanism is already running:

① Every inspection item passes and the accident happens anyway; ② Nobody owns the gaps between compliant items: no position signs for "is the whole thing safe"; ③ The compliance record starts being used as proof of safety ("we meet every standard" becomes the final answer to a challenge about risk).

32.2 How It Grows

Three generating conditions.

Condition one is the historical path by which rules are written. Regulations and standards grow naturally in clause form. A clause is a unit you can inspect, and inspection can only see units. Condition two is the cost structure of checking. Item-by-item checking can be standardized, outsourced and documented; a whole-system assessment cannot, and so it loses ground in every budget round. Condition three is self-protection in the assignment of responsibility. For the inspector, "each item compliant" is the safest conclusion available; for the inspected, "each item meets standard" is the cheapest defence available. Supply and demand push the inspection system toward clauses together, and the end point of that push is criterion ③, where the compliance certificate replaces the safety judgment outright.

Hollnagel's distinction deserves a second layer, because Safety-II is often misread as a soft cultural slogan. His original argument is an engineering one. Safety-I, the search for failure, is mathematically condemned to a blind spot. Failure is a low-probability event, so inspecting for failure means inspecting for the absence of a

sparse sample. Success is the normal state, and the normal state is where the system's real way of working is stored. So the operational definition of Safety-II is observation first. Treat the ordinary, successful way of working as the object of study. Which workarounds, which deviations, which on-the-spot combinations let the system run every day. Where the workarounds cluster is where the map of the gaps between rules lies. Safety-I can never draw that map, because inspecting for failure only ever marks failure points. This is the theoretical base of Device 3 in this chapter, sampling: sampling success is not for reward, it is for the map.

One more correction, in the other direction. Safety-II does not replace Safety-I; compliance inspection certainly continues. It adds a layer of whole-system judgment on top. The relation is the one between a medical check and a life. The check answers whether there is disease; the life answers whether the body is well. A person who only has check-ups and never looks at how they live can go down with every number in range, which is the whole meaning of the three cases in this chapter. The combined system has a ready name, borrowed from the sentence pattern of Chapter 11: a clause ledger plus a scenario ledger. The formats of both ledgers are delivered in section 5.

32.3 Field Sites

Case 1 (Aviation): A Valid Certificate and a Flammable Tank

Case ID: av-036.

When and where: 17 July 1996, 20:19 US Eastern Daylight Time, takeoff from runway 22R at New York's Kennedy International Airport; 20:31, the aircraft broke up in the air over the Atlantic about 8 miles south of East Moriches on Long Island and went into the sea, at an altitude of about 13,800 feet when the explosion oc-

curred. Before departure it stood at gate 27, Terminal 5, at Kennedy (AAR-00/03 p.19, p.22, p.88).

What and who: Trans World Airlines, Inc. (the operator, running a scheduled international service from Kennedy to Paris Charles de Gaulle under 14 CFR Part 121); the Boeing Company (the aircraft was a 747-131, part of the 747-100 series, and Boeing held the type certificate); the Federal Aviation Administration (the authority that writes airworthiness standards and issues type certificates). 230 people were aboard: 4 flight crew (captain, first officer and check captain, flight engineer, flight engineer and check captain), 14 flight attendants, 212 passengers (p.19, p.22).

What they were doing: The aircraft had left Athens at 05:37 that day as TW881 and landed at Kennedy at 16:31, at gate 27 by 16:38. TW800 had been scheduled to depart at 19:00 and was delayed by a ground service vehicle that broke down and blocked the stand, and by a question about a passenger and a bag that matched (p.19). From 16:38 to 20:19 the aircraft sat at the gate with the auxiliary power unit and two of its three air conditioning packs running, about two and a half hours, which was standard TWA ground practice (p.19, p.34 n.33). The work done in that window was turnaround support: fuelling, loading, cabin cleaning, crew change. The centre wing tank had been full at the Athens depar - ture and the transatlantic leg had burned it down to about 300 pounds; it arrived at Kennedy with that much in it, no fuel was added to it during refuelling, and after 16:38 nothing further was done to that tank. Total fuel after refuelling was 181,800 pounds (pp. 60–61). In later tests the report describes the quantity as "50 gallons, similar to TW800" (p.148 n.249), against a tank capacity of 12,890 gallons (p.159 n.287). On the 747-100 the centre wing tank sits inside the centre section structural box of the wing, directly above the bay holding the three air conditioning packs, and the hot air those packs discharge passes through louvers in the bay bulk -

head into that bay (p.30, pp. 34–35). For those two and a half hours there was no insulation between the heat source and the tank, 300 pounds of fuel in the tank, and vapor in the rest of it.

The decision point: Three, and in each the signatory can be recorded only as a body and a document; the report gives no individual names.

① 1969: the FAA installed a nitrogen inerting system on a DC-9 in a flight test. The conclusion was that the system "provided adequate inerting of the fuel tank", but that "the cost of the system at the time outweighed the benefit" (p.215 n.395, quoting the Federal Register notice of proposed rulemaking of 3 April 1997, 62 FR 16013). The same notice cites safety recommendation A-71-59 of 1971: in an accident at New Haven, 27 of the 28 occupants survived the impact and died in the fuel fire and explosion that followed it. The recommendation was classified on 13 November 1985 as "closed, acceptable action".

② 8 March 1979: the FAA issued airworthiness directive AD 79-05-04, requiring Teflon sleeving on wiring inside the main fuel tanks of some 747s and repeated inspection. It was triggered by reports of fuel pump wiring chafing through its insulation inside an aluminum conduit and arcing against the conduit hard enough to burn a hole in it. A fleet survey after the directive found many aircraft carrying long runs of wiring "missing up to 80% of its total insulation thickness", and AD 79-06-02 of 19 March extended the inspection to other aircraft (p.230). The remedy was inspect and sleeve.

③ The initial certification of the 747-100. In NPRM 99-18 of 29 October 1999 the FAA looked back: at certification, "the degree of fuel tank contamination, and the significance of certain failure modes of the fuel tank system, were not considered to the extent that recent events and service experience have shown to be necessary". The compliance showing at the time was that the fuel

quantity indicating system wiring, even in contact with the highest voltage on the aircraft, would not arc under contamination-free conditions (p.237). The proof began from no contamination.

What was visible then: The crew had the flight manual. Its note on the centre wing tank temperature limit said that running the air conditioning packs could raise the tank temperature by 10°F to 20°F; flight tests afterwards measured 40°F to 60°F (p.220). The cockpit had no centre wing tank temperature indication, so the crew could not read the temperature in the tank at all (p.221). The operator had a continuing airworthiness maintenance programme accepted by the FAA, with tank inspection done as a zonal visual check. NPRM 99-18 records how tank system inspection on transport aircraft had historically been assembled: operational checks, functional checks after maintenance, component overhaul done to avoid delay, and zonal visual checks merged with other maintenance. "Dedicated maintenance instructions to find and correct conditions that reduce fail-safe capability were considered unnecessary, because the existing fail-safe capability was assumed not to degrade in service" (p.237). The airworthiness authority had §25.981, two sentences long: a maximum temperature must be established below the lowest expected fuel autoignition temperature with a safety margin, and no temperature at any place in the tank where fuel ignition could occur may exceed that temperature under all possible operating conditions, failures and malfunctions of any component (p.195). Both halves prove something about temperature. What was learned afterwards, the flammability of the vapor in the tank and the paths by which ignition energy reached it, came from the wreckage recovery, fuel sampling and laboratory combustion testing done after July 1996. The investigation concluded that the most likely ignition source was a short circuit outside the tank that sent excessive voltage into it along the fuel quantity indicating system wiring, and that neither the energy re-

lease mechanism nor the point of entry could be determined (p. 326).

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 20:31:12 on 17 July 1996 the cockpit voice recorder stopped, the last radar return was recorded in the same second, the aircraft broke up at about 13,800 feet, the wreckage fell into the sea about 8 miles south of East Moriches, and all 230 people aboard died (p.19, p.88). The report was adopted on 23 August 2000 (p.3). Back to the three questions. The most common answer to the first is "include flammable vapor in the certification checks". In this case that action has a price on the record: inerting or foam in the tank, at least $20 billion over ten years; targeted ventilation, $3.5 billion over ten years; raising the fuel flash point to 140°F, $15 billion in the United States and $60 billion elsewhere, with a possible aviation fuel shortage (p.234). All of those answers had been put forward, and each time the cost was booked. The item most often missed under the second question is the scope of §25.981. The words "fuel ignition" appear in the clause and read as though ignition were already governed; what the clause governs is a maximum temperature, and whether vapor is present is not in the text. The checkpoint under the third belongs on the conclusion page of the 1969 inerting evaluation: write "technically feasible, cost not justified" as a decision with a review date attached, rather than as a final ruling.

Options ruled out: Five, and the report records a cost or a reason for each. ① Raise the fuel flash point to 140°F: $15 billion in the United States and $60 billion elsewhere over ten years of fuel manufacture, plus engine redesign and recertification, and the working group judged the change "beyond the current experience base" (p.234). ② Inerting or foam filling: at least $20 billion over ten years, and retrofitting the in-service fleet "very difficult" (p. 234). ③ Targeted ventilation: $3.5 billion over ten years, feasible only for the in-service fleet, while a new design can simply move the heat source (p.234). ④ Restrict ground operation of the air conditioning packs and use ground cooling carts instead: the FAA's initial assessment of 3 December 1997 replied that this would have "little effect on the overall exposure" (p.216); a re-calculated version on 17 May 2000 had the exposure falling from about 35% to about 25% (p.219); Boeing's service letter of 5 May 2000 only "recommended" the carts where they were "available and practicable", and it never became mandatory. ⑤ Fit temperature probes in the tank with a cockpit display (A-96-177): the FAA judged a tank temperature red line tied to flammability impractical on the existing configuration, and noted that "adding another set of electrical components and wiring inside the tank would introduce another possible ignition source" (p.221). What the five share: each was converted into a ten-year total, and each total went onto the same table as the expected value of an accident.

Early signals: Five. ① Since 1959 the FAA's list carries at least 26 explosions or fires in the fuel tanks of military and civil transport aircraft, TW800 among them (p.197). Why it was filed away: the mechanisms differed from case to case, and there was no common clause to hang them on. ② 9 May 1976: a 747-131 of the Iranian Air Force suffered a left wing tank explosion on approach to Madrid, all 17 people aboard died, and the wreckage showed the explosion had begun in the left wing tank near a fuel valve (p.197). Why it was filed away: lightning and a JP-4 fuel mix, meaning an external

energy source. ③ The fleet survey after AD 79-05-04 of 8 March 1979, with insulation missing up to 80% of total thickness in many places (p.230). What was done: inspect and sleeve. Vapor was not touched. ④ A Boeing study of centre wing tank heating done for the US Air Force (document D226-20582-1) concluded that "the primary heat load appears to come from the high temperature of the air conditioning equipment bay" and recommended insulation between the packs and the tank (p.159). Why it was filed away: it was done for the military E-4B and never entered civil certification or operating documents, and Boeing staff retrieved it only late in the investigation (p.159 n.286). ⑤ The same study mentions an aircraft of a Japanese airline, departing Hawaii, losing centre wing tank fuel feed during climb (p.159). Why it was filed away: it read as a fuel pump performance issue, and the temperature limit in the manual existed to prevent fuel freezing and to guarantee feed on a hot day, not to prevent the vapor from becoming flammable (p. 220). All five went the same way. Not one of them could be hung on the checklist of the day.

The point of escalation: 18 February 1997, when the FAA answered in writing the four recommendations A-96-174 to -177 that the Safety Board had issued two months earlier. The reply stated that the current airworthiness standards "assume that the fuel vapor in the tank is flammable, and that the design requirement is to eliminate potential ignition sources in the tank". Controlling the flammability of the vapor "would require a major change in design concepts", and holding fuel temperature below the flammable range was "a major task with significant economic impact" (pp. 214–215). On 3 April 1997 the notice went out for comment: 340 commenters, 976 pages. Public comment concentrated on the need to act; industry comment concentrated on technical soundness and cost (p.216). On 23 July 1998 the FAA's Fuel Tank Harmonization Working Group concluded that bringing the whole fleet down to the flammability level of the wing tanks was a

worthwhile goal, but that "retrofitting the in-service fleet is not currently feasible" (p.218). On 21 September 1999 the Safety Board recorded that two and a half years after the recommendations went out, "no design changes to the aircraft have been made", and classified A-96-174 and -175 as "open, unacceptable response" (pp. 217–218). No hearing took place between these four documents; the whole exchange was in writing. This is the step from a known gap to a point of no return. The framework wrote its own premise into the reply, and from then on the gap was part of the premise.

Failure chain: Technical layer: the centre wing tank sits directly above the bay holding the three air conditioning packs, and it held only 300 pounds of fuel. Two and a half hours of ground heating pushed the fuel temperature into the 104°F to 122°F band, which at the pressure of 13,800 feet is already above the lower flammability limit of Jet A; in the laboratory the fuel ignited at 96.4°F (p.150). The fuel quantity indicating system wiring carried the energy of a short circuit outside the tank into it, the vapor ignited, and the aircraft broke up. Organizational layer: § 25.981 governs the relation between temperature and autoignition temperature and says nothing about whether vapor is present (p.195); the compliance showing at initial certification started from "no contamination" (p.237); the maintenance programme assumed that the fail-safe capability of the original design would not degrade in service (p.237); failure mode and effects analysis and fault tree analysis were used as proof that ignition sources had been eliminated, and the investigation states plainly that they "should not be used as the sole means" (p.325, conclusion 19). The report also records that the condition of this aircraft's wiring system "was not unusual for an airplane of this age" (p.326, conclusion 23). Incentive layer: the three paths to lower flammability cost between $3.5 billion and $20 billion over ten years, and the cost-benefit table converted "do nothing for ten years" into "equivalent to three

TW800s" and still judged that acceptable; this is the position the working group's consumer representative quoted word for word in a letter of dissent (p.235 n.414). The break at the junction sits between the organizational and the incentive layers. Because the rules wrote flammable vapor in as a premise, no spending to reduce flammability could be booked as correcting a noncompliance. It could only be booked as a new cost.

Minimum counterfactual: 1969, in §25.981, the FAA writes that the fuel tanks of new transport aircraft types must reduce the flammability of the vapor, and does not judge inerting cost-ineffective. One action, on the conclusion page of that DC-9 test, and no luck required: the system had already been shown on the same DC-9 to "provide adequate inerting", and the installation and its operating data sat in the same document (p.215 n.395). The action needed no new experiment, no fleet survey, and no one's goodwill. What was missing was the term of the judgment "cost outweighs benefit": written as a requirement for new types, rather than as a one-off retrofit of the entire in-service fleet. Run it the other way and the chances in 1979, 1988 and 1996 each added to the ignition source list without any of them reaching flammability; every extension of the list extended the gap with it. Had the clause existed in 1969, certifying the 747-100 would have required answering why vapor is allowed to be there, and the space above those 300 pounds of fuel on 17 July 1996 would not have been flammable.

Outcome and sources: 230 people died, the aircraft was destroyed, and its insured value was $11 million. The report was adopted on 23 August 2000 and found the probable cause to be an explosion caused by ignition of flammable vapor in the centre wing tank, with the design and certification concept of preventing tank explosions solely by eliminating all ignition sources listed as a contributing factor (p.326). Conclusion 20 states that this concept "is

fundamentally flawed, because experience has shown that all possible ignition sources cannot be predicted and reliably elimin - ated" (p.325). The report also issues 5 new recommendations, A-00-105 to -108, and records the disposition of each of the four 1996 recommendations (pp. 327–328). By the time the report was adopted, measures to reduce flammability were still at the study and comment stage, and the 1999 notice of proposed rulemaking covered new designs only (p.241). Primary source: NTSB, Aircraft Accident Report NTSB/AAR-00/03, adopted August 23, 2000, link at Appendix B av-036.

Case 2 (Chemical and Process): A Dust Workshop Whose Classification Passed

Case ID: ch-008.

When and where: 07:34 on 2 August 2014, at 189 Nanhe Road, Kunshan Economic and Technological Development Zone, Kun - shan, Suzhou, Jiangsu Province: polishing workshop No. 2, that is Building 4, of Kunshan Zhongrong Metal Products Co., Ltd. The workshop had a floor area of 2,145 square metres, 44.24 metres north to south and 24.24 metres east to west, a two-storey rein - forced concrete frame with a storey height of 4.5 metres and 3 bays on each floor (report I(2)1).

What and who: Kunshan Zhongrong Metal Products Co., Ltd (wholly owned by Taiwanese capital, registered capital US$8.8 million, designed annual capacity 500,000 pieces, main business revenue in 2013 RMB 165 million); the designer of Building 4, the Huai' an Architectural Design Institute of Jiangsu Province; the builder, the Lianshui County Construction and Installation Com - pany of Jiangsu Province; the general contractor for the dust re - moval system, Kunshan Lingzheng Electromechanical Environ - mental Protection Equipment Co., Ltd; the drawing review body, the development zone office of Liyue Drawing Review Company;

the fire design review and acceptance authority, the Kunshan Public Security Fire Brigade; the completion acceptance filing authority, the quality supervision station of the Kunshan Housing and Urban-Rural Development Bureau (report I(1), III(2)).

What they were doing: The workshop was running its daily output of wheel hub grinding and polishing. It was designed for 32 production lines, 16 on each floor, 12 stations per line, 384 stations in all; in fact 13 lines ran on the first floor and 16 on the second, 348 stations, with all grinding and polishing done by hand using handheld electric grinders. Each floor's 1,072.5 square metres carried 16 lines, each 13-metre line carried 12 stations, some adjacent lines had workers standing back to back less than 1 metre apart, and hubs were stacked in the aisles. The layout and the choice of equipment were done by the general manager from his own experience, with no formal design (report I(2)2, III(2)1(1)). Dust removal was 8 systems built in March 2006, one system shared by every 4 lines and 48 stations, using mechanically shaken bag filters; after a modification in 2012 all 8 outdoor discharge ducts were connected together and discharged through a single main duct. Neither the dust equipment and collection ducts nor the hand tool sockets and their distribution boxes had any grounding (I(2)3). The workshop had no dust cleaning regime: the first direct cause found in the report is that "the dust removal system in the accident workshop had not been cleaned as required for a long period, and aluminum dust had accumulated", and the management cause states that "the enterprise did not clean the dust in time as required, leaving large amounts of aluminum dust in the dust ducts and in the work area" (III(1), III(2)1(3)). The workshop had 250 people on its books before the accident, working 07:00 to 19:00; 265 were on site when it happened (I(2)4, I(3)).

The decision point: Four, and in each the signatory can be recorded only as a post and a document; the report records every individual by post and by the dates that post was held.

① June 2004, when construction of phase three began and the fire hazard category of Building 4 was set at the design stage as Class E. The designer's basis was the use of the building as a "metal products grinding workshop", as supplied by Zhongrong. The report finds that the designer, "without seriously understanding the hazards of the various metal dusts", took one statement of intended use supplied by Zhongrong and improperly classified the workshop as Class E (III(2)4).

② The drawing review, which took place before February 2005 (the report dates it by the tenure of the deputy director in charge, April 2000 to February 2005), and whose conclusion was Class E. The report finds that the building "should have been approved as Class B but was approved as Class E", and that "there were errors in the review procedure" (III(2)3(4), IV(2)35).

③ The fire design review and acceptance. The building went into use in 2005, and the review and acceptance by the Kunshan Public Security Fire Brigade "failed to find and correct, in accordance with the Code for Fire Protection Design of Buildings (GBJ16-87, 2001 revision), the erroneous fire hazard classification made by the design department, and the simplified review and acceptance procedures were not strict" (III(2)3(2)).

④ The completion acceptance filing. The quality supervision station of the Kunshan Housing and Urban-Rural Development Bureau "filed in violation of the regulations, the preconditions for filing not being met", and the filing materials were missing the environmental approval document (III(2)3(4), IV(2)32). All four shared one input: Class E. Once that input was in force, every later inspection, every approval and every form took it as a premise.

What was visible then: Design, drawing review and fire all held the same drawing set and the same statement of intended use. The enterprise held the design parameters of the dust system: each hood 500 millimetres by 200 millimetres, the hub centre 500 millimetres from the hood mouth, 600 cubic metres per hour per hood, 28,800 cubic metres per hour per system, and a mean branch duct velocity of 20.8 metres per second (III(2)1(3)). The enterprise also held standards that were already in force. GB17269-2003, the safety regulation for dust explosion protection in aluminum and magnesium powder processing, requires a mean branch velocity of 23 metres per second, which for this layout means 31,850 cubic metres per hour, a shortfall in the original design of 9.6%. GB/T15605-2008, the guide for dust explosion venting, requires venting devices. GB50058-1992, the code for electrical installations in explosive and fire hazardous atmospheres, requires explosion protection and grounding (III(2)1(2), 1(3), 1(5)). The regulator held a dust monitoring report issued by Jiangsu Laibo Environmental Testing Technology Co., Ltd, and the report finds that the company "did not, as GBZ159-2004 requires, sample dust concentration at the polishing positions under normal production conditions before issuing the monitoring report" (III(2)4). What was learned afterwards, the moisture-driven exothermic path in the aluminum powder inside a collection bin, the possible part played by a thermite reaction, and the way the absence of venting and the shared main discharge duct turned one explosion in a bin into an explosion of the whole workshop, came from post-accident site inspection, simulation and bench testing (III(1)).

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".

> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: At 07:34 the No. 1 dust collector exploded, the shock wave travelled along the dust ducts into the workshop, and the aluminum dust it raised exploded in series. 47 people died on the spot, 28 died in hospital the same day despite treatment, and 185 were injured (II(1)). Within the 30-day reporting period the count was 97 dead, 163 injured and RMB 351 million of direct economic loss; the report separately records 49 further deaths after the reporting period (opening section). Back to the three questions. The most common answer to the first is "correct the category and redo the whole review". In this case that action costs four retrofits: recompute the venting area on the first floor to Class B, convert the evacuation stairs to enclosed stairwells, make all the electrical installations explosion-proof, and add vents to the dust system and redo the air volumes. The item most often missed under the second question is the monitoring report. It has an issuing body, a conclusion and a signature; the one thing missing is the sentence "sampled under normal production conditions". Because it is formally complete, it is the least gap-like document in the entire set. The checkpoint under the third belongs in the conclusion column of the drawing review: one check, whether the process behind the drawing still matches the category. Change the process and leave the category alone, and every later inspection will pass.

Options ruled out: The report leaves no record of an options comparison at all, and that absence is itself testimony. What can be seen in the documents is six options already written into the rules at the time, each replaced by a cheaper practice. ① Approve the building as Class B: replaced by deciding on the strength of the intended-use statement the enterprise supplied and nothing else

(III(2)4). ② Fit venting devices to the dust system (GB/T15605-2008): none were fitted, and the report finds that "because there was no venting device, the hot gases and combustion products of the explosion were discharged instantly through the dust ducts and out of every hood" (III(1)). ③ Explosion protection and grounding for the electrical installation (GB50058-1992): not done, "none of it explosion-proof", and the cable and wiring installation was non-compliant (III(2)1(5)). ④ Design the branch ducts at 23 metres per second: actual 20.8 metres per second, a shortfall in the original design of 9.6% (III(2)1(3)). ⑤ Give the dust system to a firm qualified to design and install it: it was awarded as a whole to an unqualified firm, which took design, fabrication, installation and the later modification together (III(2)1(2)). ⑥ Test dust concentration under normal production: not done, and the report was issued anyway (III(2)4). The six replacements all point the same way: work that needed professional judgment was exchanged for a form that could be signed.

Early signals: Five. ① The Kunshan development zone had "seen several metal dust deflagration accidents" before this one, and local supervision did not draw the lesson and organize a full survey (III(2)2(1)). How it was pushed aside: the zone's accidents were never converted into an inspection item for this enterprise. ② In 2004 workshops 3 and 4 were built without approval and without environmental impact procedures, and the environmental inspection squadron "did not find and correct this" (IV(2)31). How it was pushed aside: built first, paperwork afterwards. ③ In 2007 the environmental facilities for two new production lines, and in 2012 the technical modification of the dust system in Building 4, neither went through completion acceptance (IV(2)28, 29). How it was pushed aside: modification was scheduled to production, acceptance was scheduled behind it. ④ In 2013 the Jiangsu provincial fire service ran a campaign on fire safety in illegal buildings, and the Kunshan Public Security Fire Brigade "implemented it poorly and

did not identify the problems at Zhongrong" (III(2)3(2)). How it was pushed aside: the campaign worked from a roster, and this plant was not on the roster. ⑤ The Work Safety Committee Office of the State Council set up a special programme on safety in aluminum and magnesium processing enterprises. The Kunshan safety supervision bureau did not verify the township statistics, "so that Zhongrong was not entered on the list of aluminum and magnesium processing enterprises and did not carry out the special programme as required", and the second safety supervision division of the Suzhou bureau "did not forward the superior's work plan for the special programme" (III(2)3(1), IV(2)16). How it was pushed aside: the document stopped at municipal level. The five have the same shape. Every campaign really happened, and every one of them walked past this building.

The point of escalation: 2012, when the dust system in Building 4 was modified and all 8 outdoor discharge ducts were connected into one main duct (I(2)3). The work went to the unqualified Kunshan Lingzheng; neither the ducts nor the collectors were given vents, no grounding was installed to drain static, and no environmental completion acceptance was organized afterwards (III(2)1(2), III(2)3(3)). The motive is not stated in the report; what can be verified is the arrangement after the modification and the firm that carried it out. Before this step, the 8 systems were 8 units that could each relieve pressure on its own. After it, an explosion in any collection bin had a duct path to all 348 stations. The report describes the consequence this way: the shock wave travelled along the dust ducts into the workshop, the dust it raised exploded in series, and "because there was no venting device, the hot gases and combustion products of the explosion were discharged instantly through the dust ducts and out of every hood" (III(1)). The date of the modification is recorded only to the year; the report gives no month or day.

Failure chain: Technical layer: the collection bin of the No. 1 dust collector was corroded through, and about 20 kilograms of aluminum powder had accumulated in it. Rain had fallen continuously for the two days before, with a mean temperature of 31°C, a maximum of 34°C and humidity peaking at 97%; moist air entered through the hole in the bin floor, the aluminum powder absorbed the moisture and oxidized exothermically, and the iron oxide produced by the corrosion could support a thermite reaction once the earlier heat had triggered it, raising the temperature to the ignition temperature of the dust cloud. At 07:10 the dust extraction fan started, creating negative pressure and a dust cloud above the bin; at 07:34 it ignited (III(1)). Organizational layer: the fire hazard category of Building 4 was set as Class E in 2004, so the whole set of requirements that follows from a category, venting area, stair form, electrical explosion protection class and vents on the dust system, never entered that workshop's inspection list (III(2)1(1)); the enterprise was not on the aluminum and magnesium processing list, so the special programme did not cover it (III(2)3(1)); repeated safety inspections "all failed to find the major hazard that dust at this company had long exceeded limits and could cause an explosion" (III(2)3(1)). Incentive layer: the single shift from 07:00 to 19:00, the 348 stations, the back-to-back spacing under 1 metre and the hubs stacked in the aisles are all the shape of capacity; the dust system modification went to an unqualified firm and was not accepted afterwards (III(2)1(1), 1(2)). The break at the junction sits between the organizational and the incentive layers. One classification decision lowered the compliance cost and the safety level at the same time, and both effects are recorded on the same event: the review passed.

Minimum counterfactual: At the design review in 2004, approve Building 4 as Class B. One action, in the conclusion column of the drawing review, and no luck required: that polishing aluminum wheel hubs produces combustible aluminum dust was not new

knowledge in 2004 and needed no experiment at this plant, and the category table of the fire code was in the reviewer's hands; the report's own wording is that the building "should have been approved as Class B". Change the category and four requirements trigger at once, the venting area on the first floor, the enclosed stairwell, explosion-proof electrical installations and vents on the dust system, and all four fall before the building is occupied, without stopping production for a minute. The cost of this step can be checked: the category table, the drawings and the review record were all on the desk in 2004, and the change happens on paper. In the other direction, after the building went into use in 2005 the same change means stopping lines, altering structure and redoing fire acceptance. None of the four happened in the nine years before the accident.

Outcome and sources: 97 dead and 163 injured within the 30-day reporting period, RMB 351 million of direct economic loss; judicial measures were taken against 18 people, disciplinary action within the Party and the administration was recommended against 35, the Jiangsu Provincial People's Government was to be criticized in a circular and required to report to the State Council, and Zhongrong was fined at the upper limit (IV). The accident was classified as a production safety responsibility accident (III(3)); part five of the report sets out the rectification requirements. Primary source: the State Council investigation team for the "8·2" particularly serious explosion at Kunshan Zhongrong Metal Products Co., Ltd, "Investigation Report on the '8·2' Particularly Serious Explosion Accident at Kunshan Zhongrong Metal Products Co., Ltd, Kunshan, Suzhou, Jiangsu Province", published on the website of the Ministry of Emergency Management on 30 December 2014, link at Appendix B ch-008.

The two rounds of remediation in case three still owe a layer of psychological reading. At the end of the second round the team

was certain it met the requirements, since every remediation item carried a closure record, and the source of that certainty is the self-reinforcement of compliance records: the internal version of criterion ③, in which the remediation record becomes proof of safety. What breaks self-reinforcement is an outside view, here the users' true consent rate, meaning not that users clicked agree but that they really knew and were really free. Measuring true consent has an established method in user research, the comprehension test, which has users restate what they agreed to; the method costs far less than the €325 million of tuition, and it is absent because the compliance list never carried that item. The list asks whether a consent mechanism exists, not whether consent actually happened. Adding one line to the list costs one user study, and that line is where Safety-II lands in data compliance.

The common point of the three cases can be made precise here. Their safety gaps all grew in the blind spot of the question form: TWA800 did not ask about vapor, Kunshan did not ask how far the category was from reality, and the CNIL case did not ask whether consent was real. That blind spot is structural, because the question list comes from past accidents, and a new class of gap is never on the list before the next accident. The only remedy is Device 2 in this chapter, the scenario inspection: the scenario is new, so the questions follow the scenario. What scenario inspection really does is keep the question form fluid. Clauses are static, scenarios are dynamic, and only a dynamic question lights up a new gap between static clauses.

Case 3 (Software): Still Non-Compliant After Remediation

sw-007. Google LLC and Google Ireland. The CNIL decision (pages read for this book): September 2025, a total fine of €325 million for inserting advertisements between emails in Gmail without consent, and for cookies set when accounts were created without val-

id, freely given, informed consent (Google LLC €200 million, Google Ireland €125 million). The decision records a more telling sequence: two rounds of remediation had already been carried out, and the authority found the violations still in place. The compliance process was running in full: remediate, re-inspect, remediate again. The whole question, whether users really consented and did so informed and free, did not change with item-by-item remediation. Remediation changed how clauses were implemented. What was violated was the overall design of the consent mechanism.

Stop at the decision point: the moment the second remediation plan was signed off and the team confirmed that it met the requirements. On the table: the item-by-item closure of the remediation list, the true consent rate from user research, and the compliance status of each of those two sets of information. 〔Outcome〕 A €325 million fine, and the overall design of the consent mechanism named. Primary source: CNIL, 2025, link at Appendix B sw-007.

Note on the Evidence Gap

The fourth case slot in this chapter is reserved for the same shape in agriculture and food (systemic outbreaks under compliance systems such as HACCP; the compliance history of the outbreak cases already held in CDC MMWR needs retrospective verification) and will be filled once collected; the three existing cases already cover aviation, chemical and process, and data compliance. The reserved slot is filled for now by a case from the same library: ag-003, case four of Chapter 13, revisited here with a different reading and replaceable by a deeper HACCP-specific case once collection is complete.

Case 4 (Agriculture and Food): Cheese Made to a Compliant Process (condensed)

ag-003. Whatcom Blue, a raw milk cheese from Twin Sisters Creamery. FDA enforcement report (event 97880 / H-0193-2026, API record checked for this book): E. coli O103:H2 detected, Class I recall of 3,438.2 pounds. Making raw milk cheese violates no current process tradition, the compliance narrative of the aging process is complete, and safety rests on a process assumption, that aging inactivates the pathogen. The state laboratory's finished-product test broke the assumption. Compliant and unsafe takes this form here: compliance is about the process, how the cheese is made, and safety is about the result, what is in the cheese; every item of the process passes and the test of the result is absent. Minimum counterfactual: put finished-product testing into the release standard, which puts the safety question onto the compliance list. Primary source: FDA, 2026, link at Appendix B ag-003.

Case 5 (Drugs and Devices): Voids in a Seal (condensed)

ph-010. Voids in the Tyvek pouch seal of a surgical pack. FDA recall (December 2025, API record checked for this book): the seal voids put the sterility barrier in doubt, and 14,789 units were recalled (8,651 in the United States, 6,138 outside it). The compliance checks on the packaging (seal parameters, material specification, process records) can all pass; a void is a defect at the level of the result, since a passing parameter does not mean every seal is intact. Sterile barrier integrity is a safety attribute and a parameter is a compliance attribute, and the difference between them is this chapter's definition. Minimum counterfactual: add sampling destructive testing of seal integrity to batch release, trading the destruction of the test for confirmation of the barrier. Primary source: FDA, 2026, link at Appendix B ph-010.

32.4 What Shows Earliest

Six weak signals, each with the reason usually used to push it aside.

1. The safety meeting's whole agenda is "is there a violation", and no item asks "where could it go wrong". Reason: "compliance is the floor."
2. The findings of an inspection are all single-clause items, with no systemic finding that crosses clauses. Reason: "inspections are designed by clause."
3. "Everything meets standard" is used to answer a challenge that has nothing to do with the clauses. Reason: "let the data speak."
4. The approval route for exceptions and waivers runs more smoothly than the remediation route. Reason: "waivers have a dedicated channel."
5. A risk raised at the front line has no inspection item to answer to, so it never enters the inspection result. Reason: "no clause, no check."
6. The inspection clauses added after each accident accumulate year on year; inspection hours double and the accident rate does not move. Reason: "keep strengthening."

Put the six back against the five cases of this chapter and each one lands somewhere. Signal 1, meetings that only discuss violations, lands on case three: the agenda of every remediation meeting was clause closure, and the authenticity of consent never reached the agenda. Signal 2, no finding that crosses clauses, lands on case one: every finding of the certification inspection was a single-item pass, and the fact of flammable vapor as a whole had no clause to hang on. Signal 3, a compliance answer to a challenge, lands on all the cases: it is criterion ③ in the field. Signal 4, a smoother waiver channel, lands on case four: the waiver for a process tradition, raw

milk without pasteurization, ran through a cultural channel and not a testing one. Signal 5, front-line risk with no item to hang on, lands on case two: the maintenance arrears of the dust system had no line on the Class E checklist. Signal 6, clauses accumulating while the accident rate holds, is the long-run portrait of a Safety-I system: the clauses grow, and the gaps grow with them.

Reorder the six by the cost of the evidence. Signals 1 and 3 are cheapest, since they need only the minutes and the reply texts. Signals 2 and 4 are middling, needing the finding structure of the inspection reports. Signals 5 and 6 are the most expensive, needing statistics that cross clauses and a historical comparison. The place to start is signal 3. The next time "everything meets standard" is offered as an answer about safety, ask one follow-up: do the com‑pliant items, taken together, cover the scenario in front of us today. The follow-up costs one sentence, and it leaves a challenged mark on every run of criterion ③ in the organization.

32.5 How to Defend

Device 1, the extra question after the compliance check. On the conclusion page of every compliance inspection, append one fixed question: "Taken together, are these compliant items safe?" The answer has to come from a person who signs for the whole (Chapter 18, the single accountable person), and it has to cite at least one cross-clause integrated analysis, a scenario walkthrough or a joint failure analysis. One question does not close the gap. It puts the gap into the text.

Device 2, the scenario inspection. Once a year, switch the inspec‑tion system from clause mode to scenario mode: pick three real accident scenarios, from your own library or the industry's, walk every relevant clause and the site along the scenario, and record the stretches the clauses do not reach. Those stretches are the map

boundary of the system. The misclassified category at Kunshan, the flammable vapor on TWA800 and the overall design of consent at the CNIL are all places on that boundary.

Device 3, Safety-II sampling. At regular intervals, collect how things go right. This is observation rather than violation hunting: a record of the difference between the actual practice of an ordinary successful day and the rule, which is Hollnagel's method in operational form. Most of these differences are harmless and a minority sit exactly on the gap. They can only be seen from success, because failure exposes the failure point and success exposes the whole way of working.

Four devices against this chapter's cases. Device 1, the extra question, on case one: the question on the certification conclusion page writes down the physical fact of vapor flammability and where the residual risk sits. Device 2, the scenario inspection, on case two: walk the dust explosion scenario through every clause that applies to the workshop, and record the list of stretches the Class E standard does not reach. Device 3, Safety-II sampling, on case three: sample how users actually complete the consent flow, how many screens they skip and how many points they misread, and the place where the workarounds cluster is the gap map of the consent mechanism. Device 4, the use prohibition, on all the cases: write the ban on compliance answers into the spokesperson's handbook and the rules of internal meetings.

Each of the four devices has a way of dying. The extra question dies in watered answers, the word "safe" filling the blank; the patch is a citation requirement, so that an answer must cite the number of the scenario walkthrough and an unnumbered answer is returned automatically. The scenario inspection dies in selection bias, the same three old scenarios every year, where the new gap cannot be found; the patch is rotation with outside input, half the scenarios taken from the industry's latest accidents. Safety-II sampling dies

in sampling drift, where sampling becomes an award ceremony that records good deeds and not workarounds; the patch is anonymization, so that a record carries the place and the practice and not a name. The use prohibition dies in rephrasing, the compliance answer returning as "we went through all the necessary reviews"; the patch is a semantic version of the ban, so that answering a question about overall safety with the completeness of inspection is prohibited whatever the wording. All four patches share one check sentence: when did the scenario inspection last find a new stretch the clauses do not reach. A scenario inspection that has never found one has turned the scenarios back into clauses.

The formats of the two ledgers are delivered here. The clause ledger stays as it is: item, compliance status, inspection date, governing clause. The scenario ledger adds four columns: scenario (the name of a real accident or an industry accident scenario), coverage judgment (the stretches covered and the breakpoints left after the clause chain is walked through the scenario), breakpoint disposition (whether the breakpoint went into a clause revision or was listed as a known boundary), review date (the annual re-walk of the scenario). The two ledgers have different readers, the clause ledger for the inspector and the scenario ledger for the designer and the governance layer, and the scenario ledger's slot on the quarterly agenda should sit before the compliance report. Look at the gaps first and the compliance second, and the language of the meeting changes with the order.

Device 4, the prohibition on using compliance as proof. Answering a challenge about safety with "everything meets standard" is prohibited; the answer has to point to a scenario-based integrated analysis. This prohibition changes a language habit, and until the habit changes, criterion ③ is already running in every organization.

32.6 Boundaries with Adjacent Mechanisms

Against regulatory lag and capture (Chapter 31). Lag and capture supply the compliance system with its stale rules, which is the climate; this chapter runs inside the old rules, which is the weather. The longer the old rules run, the louder Chapter 32's "everything meets standard" sounds.

Against the illusion of redundancy (Chapter 22). The illusion is redundancy's same-source error, counting the number and not the kind; compliant and unsafe is the clause-ization of inspection, counting clauses and not scenarios. Both share one audit habit: counting the countable and ignoring the uncountable.

Against normalization of deviance (Chapter 11). The form normalization takes inside an inspection system is criterion ③, answering a challenge about risk with the compliance record. Chapter 11 watches the register; this chapter watches the language of meetings.

32.7 Five Self-Checks

1. At the last accident in your field, did every compliance check pass beforehand? What share of the causal chain did those passing checks cover?
2. How many items on your checklist were added ten years ago? When did each of them last stop a problem?
3. When was your organization's compliance certificate last used to answer a challenge about safety, and on what occasion? Who asked about the scenario as a whole?
4. If next year's budget halved the compliance inspection and doubled the scenario walkthroughs, would the visibility of risk in your organization rise or fall?

5. Take one scenario you own, a fire, an outage, a contamination, and walk it along the accident chain through all your clauses: how long is the written list of stretches the clauses do not reach?

Chinese draft: 11,392 characters.

Chapter 33: Cascading Failure | Contagion

> What this chapter covers: how a failure at one point amplifies along dependency chains into a wide area; the criteria, four field sites, and a method for managing a critical dependency as if it were your own single point. When to turn back: supplier consol - idation, cloud migration, a just-in-time squeeze on inventory, and the moment before your risk register is updated for the last time. Reading time: 50 minutes.

The amplification mechanism sits inside one line of business ad - vice: focus on what you are good at and outsource the rest. Each half of that sentence holds on its own. Put together, they are the entire condition of this chapter. Outsourcing creates a dependency. Focus removes the backup. And the dependency graph is never a required field on the decision form. Then one point fails (an in - gredient plant, a piece of automation code, a component's pack - aging), and the failure runs down the dependency chain into every downstream that focused on its own strength. Each downstream's continuity measures fail at the same moment, because all of them lean on the same upstream. Scale makes this worse. The more concentrated the dependency, the larger the radius of the single point, and the larger the radius, the more of the industry's risk register needs rewriting. Registers are rarely rewritten.

33.1 What It Is

Definition: cascading failure is failure that propagates along de - pendency relations and amplifies at each step. A single node (a supplier, a cloud region, an ingredient plant, a first-tier compon - ent) develops a problem, concentration carries it into many

downstreams, and the continuity measures downstream all fail together because they depend on the same upstream. The division of labour with Chapter 21, hidden coupling, runs like this: that chapter deals with lines nobody knows exist, this one deals with lines that exist and are already concentrated. Concentration is what translates a local event into a system event.

The arithmetic of that translation is worth stating first. The radius of a concentration grows linearly with the number of downstreams: one upstream, N downstreams, radius N. Secondary damage grows faster than linearly, since each of the N downstreams has downstreams of its own, and the radius multiplies at every layer. The consequence is blunt for anyone who has to sign the decision. Only two levers control the radius. Reduce the number of downstreams hanging on one point, which means diversifying upstream. Or cut the propagation layer, which means giving downstreams an alternative path. Both are expensive. Diversification gives up the volume discount. An alternative path is idle capacity. The price of either lever has to appear in a decision table before anyone can compute with it, which is the arithmetic under Device 2 in this chapter, single-point pricing. Most memos on centralisation decisions carry one column, efficiency. The work of this chapter is to get the second column onto the same page: expected loss from a supply interruption.

One further layer belongs on the translation. A concentrated upstream failure turns a technical event into a public event: an industry-wide shortage, a recall spanning several countries, an outage across a whole network. Public events are handled by a different logic from technical ones, since they bring in regulators, press, and recall coordination across legal jurisdictions. That leaves a third hidden column in the cost of centralisation, the premium on handling a public event, meaning the coordination cost of a multi-jurisdiction recall and the opportunity cost of a regulatory inquiry.

A decision table with all three columns (efficiency, interruption loss, public-event premium) is the minimum a centralisation de - cision deserves. Most organisations have the first column only. This chapter fills the table out to three.

Three criteria start the mechanism. Any one of them is enough.

1. Concentrated dependency: the number of suppliers, paths or regions for a critical input sits below the threshold you set yourself.
2. No isolation: downstreams share one upstream and hold no switchover path to another.
3. The third party’s risk is not on your own risk register: “the supplier’s problem is not my risk”.

33.2 How It Grows

Three generative conditions.

Condition one is the homogenisation of efficiency. Just-in-time, centralised purchasing, multi-cloud collapsed to a single cloud, regional warehouses merged: each of these is a winner on current-period cost, and their combined effect is concentrated dependency. No individual decision is wrong. Their sum is.

Condition two is the homogenisation of the downstream. Every firm in the industry makes the same “optimal” choice, so the whole industry shares one single point, and a failure at that point is promoted from a company event to an industry event.

Condition three is a risk table that stops at the wall. Supplier management and risk management run as two systems that never read or write each other, so the risk of your supplier’s supplier’s supplier has never entered your field of view.

33.3 Four Domains, Four Shapes

Case One (Software): One Thousand One Hundred Prefixes Withdrawn by Mistake

sw-003. Cloudflare's BYOIP (Bring Your Own IP) service, 20 February 2026. The official postmortem, read and checked for this book: a defect in the API query issued by an automated cleanup subtask, in which a pending_delete parameter sent empty by the client was read by the server as "all", caused about 1,100 BYOIP prefixes (roughly a quarter of that service) to be withdrawn from BGP announcement. The fault lasted 6 hours 7 minutes. What it hit was the thousand-plus customers who had entrusted IP assets to one shared piece of automation. A parameter whose meaning was ambiguous, carried through a shared automation base, became the simultaneous failure of a thousand customers.

The decision point is the moment before the cleanup subtask went live, with the semantics of an empty parameter still undefined and untested. On the desk: the API's default behaviour, the test coverage of the automation pipeline, and one assumption, that nobody would ever pass this parameter empty.

> **Stop here. Decide before you turn the page.**
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What happened: 6 hours 7 minutes of fault. Recovery ran in two stages, a rollback that restored 800 prefixes, and the remaining

~300 whose service configuration had been deleted from the edge, restored by engineers by hand at 23:03 because self-service could not reach them. Primary source: Cloudflare, 2026. Link at Appendix B, sw-003.

The recovery structure in Case One deserves a second look through this chapter's lens. The rollback restored 800. The remaining 300 came back because customers acted for themselves, and they could act because the automation base no longer worked for those prefixes at all. That split says dependency has two depths. A shallow dependency returns on its own when the upstream rolls back. A deep dependency needs the customer's own hand. The recovery radius is layered exactly as the failure radius is, and most continuity plans write only the shallow layer. Register granularity therefore has to carry the recovery path: which dependencies heal with an upstream rollback, and which need your own action. That information comes from only one place, a drill, and the layered map is one of the outputs of the interruption drill under Device 3. Its value shows up at night, during a real incident, when someone has to know which customers to telephone and which ones not to bother.

Case Two (Agriculture): One Bag of Stabilizer and Ovens in Many States

Case ID: ag-008.

When and where: initiation 3 May 2024 (recall_initiation_date, identical across all 36 records); FDA classification 6 June 2024 (center_classification_date); enforcement report published 12 June 2024 (report_date); termination 4 September 2024 (termination_date). Site address: Danisco USA Inc., 4 New Century Pkwy, New Century, Kansas 66031-1144. Distribution field: 24 states in the United States (AZ, CA, CO, FL, IA, ID, IL, IN, KS, KY, MI, MN, MO,

NE, NJ, NY, OH, OK, PA, TN, TX, UT, VA, WI), plus Canada and Mexico.

What and who: Danisco USA Inc., one plant, New Century, Kansas. Under event_id 94565 sit 36 food recall records, numbered F-1305-2024 through F-1340-2024 in an unbroken run, all Class I, all initiated on the same day, all classified on the same day, all ter‑minated on the same day. The product field does not describe one product. It describes 36 formulations: 33 stabilizer and emulsifier systems, 2 preservatives (Germantown Gard, Germantown Gard (A)), and 1 sweetener blend (Reese, SG FR Sweetener Blend CM50). The brand field carries Grindsted, Germantown, Cremodan, Reese, Avicel-plus and CRE-90, along with unbranded GSD, ASC and GS series numbers. Packaging comes in three forms: 20, 22.68 and 25 kg bags, 907.2 kg and 2270 lb totes, and 1000 kg bulk bags. The classifying body is the FDA Center for Food Safety and Applied Nutrition (CFSAN). Responsibility is recorded by post: food safety and quality management at the plant own the release of finished goods; formulation and sales own the customer mapping. The records name no individual.

What they were doing: the shape of the production line can be checked item by item. One plant, 36 formulations in production at the same time, shipping at the same time. Three packaging forms running side by side say the plant held a small-bag line as well as tote and bulk lines, and that the product left the plant into some‑body else’ s line: bakery, dairy, beverage, ice cream, seasoning. The quantity field orders a sequence from 50.00 kg to 61,871.04 kg: 50.00 (F-1331, CRE-90), 75.00 (F-1322, Cremodan SL-64-A), 500.00 (F-1332), 816.48 (F-1329), 884.52 (F-1319), 884.52 (F-1334), 989.00 (F-1327), 1,814.40 (F-1312), 1,814.40 (F-1313), 1,814.40 (F-1310), 2,721.60 (F-1337), 3,000 (F-1330), 3,628.80 (F-1323), 3,674.16 (F-1308), 4,490.64 (F-1340), 4,513.32 (F-1325), 5,443.20 (F-1307), 7,030.80 (F-1309), 7,257.60 (F-1333), 10,297.00 (F-1326), 11,000.00

(F-1328), 13,608 (F-1317), 14,991.48 (F-1324), 18,144.00 (F-1336), 19,028.52 (F-1305), 21,000.00 (F-1339), 31,865.40 (F-1318), 33,600.00 (F-1335), 35,448.84 (F-1306), 36,265.32 (F-1316), 36,288 (F-1315), 36,288 (F-1320), 36,741.60 (F-1314), 45,360.00 (F-1338), 61,826.80 (F-1311), 61,871.04 (F-1321). The total is 575,026.84 kg, which at the 22.68 kg standard bag works out to about 25,354 bags.

The batch field is more informative still. Across the 36 records, code_info lists 688 non-duplicated batch numbers, every one of them beginning 1204, with the tail running from 1204520956 all the way to 1204633668. This is the heaviest single item of shape in the case. 688 batches judged suspect at once says the suspect object is not one kettle or one day but a long stretch of production. The record gives batch numbers only, never the production date attached to a batch number, and this book does not read the prefix as a date.

The decision point: the grade of the material decides that this section can reach the document and go no further. There is no person to sign for it. Four moments, listed by the record that carries each one, each with a note of what the record is missing.

One. Initiation on 3 May 2024. The voluntary_mandated field reads "Voluntary: Firm initiated". 36 records generated on one day means the judgement covered all 36 formulations inside a single day rather than assessing them one at a time. The date the decision was taken, the test results it rested on, the signatory: not stated in the record.

Two. The choice of notification channel. The initial_firm_notification field reads "Two or more of the following: Email, Fax, Letter, Press Release, Telephone, Visit". Six channels, any two or more. Which two were used, what day they went out, which customers received them: not stated. This choice is the most consequential in the whole case. Whether a downstream bakery heard before its finished product shipped depended on whether that column chose the telephone or chose the letter. B2B ingredient

notification carries a structural problem of its own: the receiving party is purchasing and the warehouse, the using party is formulation and production, and a notice delivered to the first is not a notice delivered to the second. The record says "two or more" and does not say whose desk it reached.

Three. Classification on 6 June 2024. The classifying body is the FDA, not the firm. The conclusion is Class I: reasonable probability that use of the product will cause serious adverse health consequences or death. The classification date sits 34 days after initiation. The basis for the judgement (positive samples, serotype, exposure assessment) is not stated, and the reason field holds six English words: Potential Salmonella Contamination. This is worth setting against the device records. A device record's reason field writes out the full spectrum of consequences (possibly no injury, local infection, rarely life-threatening). A food record gives a six-word conclusion and no layering of consequences. One enforcement database, two centres, two conventions of writing, and the food entry hands the reader less.

Four. Termination on 4 September 2024. 124 days from initiation. The criteria for termination (recovery rate, verified destruction, downstream confirmation) are not stated. Three fields appear with the same value on the same day across 36 records, which says these 36 were handled as one thing. The action of handling them together is in the record. The reason is not.

What was visible then: on 3 May 2024 the plant held four classes of material that the record can confirm.

One. The production and shipping ledger for 688 batches. Which batch, how many kilograms, to which customer in which state: the plant had it, the public record does not. Two. The composition tables for 36 formulations. Which components in a given formulation came out of the same blending step is a question only the formulation table can answer, and the record does not state it. Three.

Environmental monitoring records. In a dry-blend facility a Salmonella positive usually appears in environmental sampling before it appears in finished product. Sampling dates, locations, results: all not stated. Four. The customer distribution list. 24 states plus two countries is the plant's shipping record, and it is also the source of the distribution_pattern field in the record.

Information that arrived only afterwards: which process step the contamination occurred at, which input batch carried it in, the serotype of the positive sample, whether there were cases and how many. None of the four is in the record. This gap is not an omission by this book. It is the boundary of the material. A recall record is an enforcement ledger, not an investigation file.

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What happened: all 36 Class I, all status Terminated. Total affected quantity, summing the product_quantity field across the 36 records, is 575,026.84 kg (this book's addition, not a figure the record states directly). Largest single record 61,871.04 kg (F-1321, bulk tote); smallest 50.00 kg (F-1331). Distribution 24 states plus Canada and Mexico. 124 days from initiation on 3 May to termination on 4 September. The records state no case count, no deaths, no recovery rate, no destruction tonnage. Back to the three questions. The most common answer to the first is to work down the downstream customer list, notifying each customer and stopping shipments, and the time cost of that route is exactly the 124 days, with

the further condition that it depends on the downstream cooperating. The item most easily missed under the second is the packaging column: totes and bulk (the 36,288, 61,871.04 and 61,826.80 kg records) mean the product went into industrial vats rather than onto a kitchen counter, and once it is in a vat there is no batch left to separate. The third question lands on environmental monitoring rather than finished-goods testing: finished-goods testing is a sample, environmental monitoring is a trend, and a trend can tell you which day it started.

Options ruled out: the five below are actionable alternatives read back from the shape of the record. The record states no discussion of any of them and no reason for rejecting any.

One. Recall only the high-risk uses. Stabilizer bound for ready-to-eat dairy recalled first, stabilizer for bakery products that get reheated deferred or released. In the record all 36 formulations go on the same day, at the same class, with a scope covering every use. Two. Recall only the positive batches. Split on test results and release the rest. The record gives each formulation a run of batch numbers, 688 in total, a scope set by formulation and every batch in it, not by positive result. Three. Compress the recall into one record. The FDA split it into 36 by formulation, in an unbroken numbering run, each one independent. Merging them would take 35 checking entries away from the downstream. Four. Stop shipments first and announce the recall once the investigation concludes. The record runs the other way: initiation on 3 May, classification on 6 June, the recall initiated 34 days before the classification, on the firm's own judgement rather than on a regulatory finding. Five. Handle by packaging form. Totes and bulk (into industrial vats, no traceability by bag) recalled first, bagged product (onto a kitchen counter, checkable by bag) deferred. The record makes no such distinction: the 50.00 kg bag (F-1331) and the 61,871.04 kg bulk tote (F-1321) are both Class I, both initiated the

same day, both terminated the same day. The five share one property: every scope pulled out to the coarsest setting, with no intermediate layer left in place.

Early signals: (1) One plant supplying 36 formulations across the full range from bakery to dairy to beverage: the reuse surface of the single point is countable line by line in the product_description field. (2) 688 batches suspect at once, the tail running from 120452 to 120463: the suspect interval is long, which says the judgement was not extrapolated back from one kettle. (3) Quantities spanning three orders of magnitude, 50.00 kg to 61,871.04 kg: one cause landing on both a pilot batch and an industrial tote, with packaging form providing no isolation. (4) Distribution across 24 states plus two countries, while the name Danisco never appears on the end product: a B2B ingredient supplier's failure radius is visible only in the distribution_pattern field, and the consumer never sees it. (5) 124 days from initiation to termination: a Class I B2B ingredient recall is recovered through downstream cooperation, and that length says the resistance to recovery sits downstream, not upstream. One more item can only be recorded as "not seen": searching the openFDA food enforcement database by recalling_firm returns 36 records for Danisco, all from this event, with nothing earlier. The database's coverage is not complete, so this reads as "not in this database" and not as "never happened". It also says that anyone trying to predict this firm from its recall history would have received no warning at all beforehand.

The point of escalation: 6 June 2024, when the classification landed as Class I. That step rewrote "voluntary recall after the firm's own checks" into "regulator's judgement of a reasonable probability of serious harm". The date is in the record. The basis of the judgement is not. The cost of the escalation falls downstream: every bakery, dairy and beverage plant in 24 states and two countries that used these 36 formulations is pulled into

one recall coordination, and the formulation sheet in their hands says "stabilizer", not which Danisco batch. Class I carries a second consequence at the finished-goods end. The cakes, yoghurt and ice cream the downstream made are already on shelves, and an ingredient recall has to travel upward into finished-product recalls, whose trigger conditions, deadlines and announcement language appear in none of the 36 records. The upstream 575 tonnes closed out inside 124 days. The downstream finished-product recalls are not on this case's account.

Failure chain: technical layer. Salmonella entered a dry-blend product (process step, input and date all unrecorded), and the fin - ished goods left the plant still carrying it. Organizational layer. One plant supplied 36 formulations, 24 states and two countries; the recall was cut into 36 records by formulation, each record giving batch numbers and no batch-to-customer mapping, so the down - stream has to match 36 records back to its own formulations. In - centive layer. Downstream gatekeeping relies on the certificate of analysis travelling with the goods, the certificate is a sample, and sampling cannot catch a distributed contamination; switching supplier would mean re-running formulation stability trials, which costs more than accepting the sampling risk. All three layers break at once. The boundary break sits between the organizational and the technical layer: contamination happens at the level of process step and batch, the recall is issued at the level of formulation (SKU), and the layer between them, batch to customer, is entirely absent from the record. What the downstream receives is a 36-line list, not a table it can match against its own stock. The cost of that gap can be estimated. Of the 688 batches, only a few lines probably entered any one downstream's stock, and without the mapping those few lines are handled at the full 36-formulation scope.

Minimum counterfactual: on 3 May 2024, at initiation, publish alongside the code_info of each of the 36 records a batch-to-cus -

tomer table: which batch, how many kilograms, to which customer in which state. One action, taken on the day of initiation, requiring no luck, since the data sits in the plant's shipping system already. The effect is computable. The downstream need not pull a whole formulation; it can isolate by batch, and of 688 batches only a few lines entered any one stock. Without that table the downstream has one option left, treat the entire formulation as suspect. The same data is a shipping document in the plant's hands and 688 numbers in the public record. The distance between those two forms is the length of this failure chain.

Outcome and sources: 36 records, all Class I, totalling 575,026.84 kg (this book's addition), 24 states plus Canada and Mexico, terminated 4 September 2024. The eight largest are 61,871.04 (F-1321), 61,826.80 (F-1311), 45,360.00 (F-1338), 36,741.60 (F-1314), 36,288 (F-1315), 36,288 (F-1320), 36,265.32 (F-1316) and 35,448.84 (F-1306), which together come to 350,089.60 kg, 60.9 per cent of the total; the smallest is 50.00 kg, and the ratio between them is 1,237. One cause, landing on a 50 kg pilot batch and a 61 tonne bulk tote, handled on identical terms. No further record appears under the same event_id, so these 36 are the whole case. The handling channel for the two export countries (Canada's counterpart agency, Mexico's) is not in the record, which gives one phrase, "Canada, and Mexico". Primary source: FDA food recall enforcement report, event_id 94565, recall_number F-1305-2024 through F-1340-2024, openFDA food/enforcement endpoint, data updated 21 October 2026; link at Appendix B, ag-008.

Case Two carries one detail worth taking separately, in recall coordination across three countries. Coordination across jurisdictions (the FDA, its Canadian and Mexican counterparts, the local regulators of each downstream bakery) is the bulk of the public-event premium from Section 33.1. The cost rises with the number of jurisdictions, since each one brings its own notification lan-

guage, its own deadlines, its own reporting format, while contamination does not recognise borders; ingredient movement across North America is a natural consequence of logistics. A B2B ingredient supplier's compliance toolkit should therefore hold a multi-jurisdiction recall plan: jurisdiction tags on the downstream list, template versions in several languages, a contact table for local agencies. Upkeep costs a few hours a year. Its value on the day of a real recall is counted in days.

Case Three (Aviation): One Tyre and Two Hydraulic Systems

av-007. An Atlas Air Boeing 747-400F (N404KZ), 12 August 2024, Narita. The JTSB report (aircraft accident investigation report, 2025, PDF downloaded and checked for this book): a tyre was damaged during taxi, both hydraulic systems failed after takeoff, and the aircraft returned; the 7 people on board were uninjured (死傷者はいなかった, wording checked against the original), the airframe took moderate damage, and the runway was closed for about 7 hours. Tyre, wheel well structure and hydraulic line form one physical dependency chain: damage at a single point travelled along a structural path into two independent hydraulic systems, and the redundancy of the hydraulics (the subject of Chapter 22) failed in the face of physical propagation.

The decision point is the moment during taxi when abnormal vibration was felt and the decision was taken to keep taxiing rather than stop for an inspection. On the desk: the attribution of the vibration (the crew felt it "like running over something" but heard no abnormal noise, wording checked, report p. 2), the procedural cost of interrupting the taxi, and the configuration of tyre-state visibility from the flight deck (detail in the body of the JTSB report).

What happened: a return and landing, and the airport closed for about 7 hours. Primary source: JTSB, AA2025-6-2-N404KZ, 2025. Link at Appendix B, av-007.

The physical dependency chain in Case Three has an engineering coda. The physical path through tyre, wheel well and hydraulic line is visible on the drawings; it is drawn on the layout. The reason a visible path still becomes a failure chain is that path risk is assessed within system boundaries. The hydraulics are assessed as hydraulics, the tyre as a tyre, and cross-system physical proximity, meaning what a fragment can strike, sits on no single system's assessment sheet. Cross-system proximity is shared territory between Chapter 21 and this chapter: the mapping chapter draws the line, this chapter assesses what the line can carry, meaning how large a failure it will transmit. The joint output is a physical proximity list, the groups of components that share a bay, a structure or a route, and every row of that list is a candidate path for an av-007 style failure.

Case Four (Drugs and Devices): A Component Maker's Wrinkle, a Whole Kit Recalled

Case ID: ph-008.

When and where: initiation 12 June 2026 (recall_initiation_date); FDA classification 31 August 2026 (center_classification_date); enforcement report published 9 September 2026 (report_date). Address of the recalling firm: 3 Lakes Dr, Northfield, Illinois 60093-2753, United States. The distribution field reads US Nationwide distribution. The manufacturing address of the component maker is not in the record, which gives the recalling firm's address only.

What and who: Medline Industries, LP, kit assembler and initiating firm, Northfield, Illinois. This record, Z-3057-2026, covers two

models: BLOOD CULTURE TRAY PEDIATRIC (model DYNDH1193) and BLOOD CULTURE KIT 2 W/CONNECT (model DYNDH1487A). The components named inside the kits are the BD ChloraPrep Clear 1 mL and FREPP Clear 1.5 mL applicators. Under the same event (event_id 99334) sit 16 device recall records, all initiated by Medline, all initiated on the same day. The classifying body is the FDA Center for Devices and Radiological Health (CDRH). Responsibility is recorded by post: quality and regulatory affairs at the assembler own the sterility claim of the kit; the component maker owns the packaging seal of the applicator. The records name no individual.

What they were doing: these 16 records describe a production shape that can be checked item by item. Initiated on one day, under one block of reason text, covering 16 kits whose clinical uses have nothing to do with each other: blood collection and blood culture (Z-3051, Z-3057, Z-3063), intravenous cannulation (Z-3056), arterial cannulation (Z-3055), pre-operative preparation and ophthalmology (Z-3058, Z-3061), caesarean section and delivery (Z-3059, Z-3060), interventional and orthopaedic (Z-3062), colorectal (Z-3052), breast ultrasound (Z-3053), tracheostomy and anaesthesia (Z-3048), kits listed by catalogue number with no product name (Z-3049, Z-3050), and emergency trays mixed with blood collection (Z-3054). The quantity field forms an extremely steep stair: 104 kits (Z-3048), 176 (Z-3052), 300 (Z-3053), 352 (Z-3060), 1,840 (Z-3055), 2,080 (Z-3061), 3,650 (Z-3057), 3,752 (Z-3059), 4,636 (Z-3049), 13,032 (Z-3058), 17,750 (Z-3051), 18,290 (Z-3062), 22,736 (Z-3056), 58,037 (Z-3050), 434,595 (Z-3054), 996,692 (Z-3063). The span is a factor of 9,583. What can put 104 and 996,692 inside one block of reason text is a component reused again and again: the 1 mL and 1.5 mL applicators entered every category of kit from rare procedures to routine blood collection.

One earlier shape is in the record. On 12 June 2025 the same Medline firm recalled 15,700 units of the same component (Z-2155-2025, event_id 97154), with a reason field stating that the components "were recalled by BD due to potential to exhibit an open seal on its packaging", distribution limited to California, Florida, Pennsylvania and Texas, and the distribution field explicitly reading "No international distribution". The 2026 defect is described differently: wrinkles in the paper lidding may extend into the seal area. The two descriptions are not the same sentence, and the record does not say whether they involve the same process step, the same line, the same sealing machine. What can be established is the same component, the same brand, 12 months apart, and both times the assembler converting a component problem into a kit recall.

The decision point: the grade of the material decides that this section can reach the document and go no further. There is no person to sign for it. Four irreversible moments, listed by the record that carries each one, each with a note of what the record is missing.

One. Fixing the component in the kit catalogue. The carrier is the product description field itself: all 16 records write "BD ChloraPrep Clear - 1 mL" and "FREPP Clear 1.5 mL" under the Medline kit's name, with model numbers and millilitres stated. This is the outcome of a fixing decision, and it says the kit's sterility claim was built on these two specifications and never changed specification afterwards. The review date, the review record and the signatory for that fixing: not stated. The weight of this item is that the design freedom of a kit sits entirely at the moment of component selection, and afterwards the component's packaging process becomes the ceiling on the kit's sterility claim.

Two. Batch release. The carrier is the code_info field: 12 batch numbers for this record, 6 for each model (DYNDH1193: 24ABA808,

24BBV222, 24GBT333, 24IBU725, 23KBX578, 23LBP536; DYNDH1487A: 25BMG311, 24AME318, 24BMI775, 24CMJ550, 24IMG215, 23JME381). 12 batches named at once means all 12 passed the assembler's release criteria at the time of release. What the criteria were, which component certificate they rested on, who signed: not stated. Of everything on this card, this is the only check that passed. The check was not skipped. It was performed, it passed, and the kit failed anyway.

Three. Initiation on 12 June 2026. The voluntary_mandated field reads "Voluntary: Firm initiated", and initial_firm_notification reads "Letter". 16 records generated together on one day says the scope of this recall was the catalogue, not a model-by-model assessment. The decision was taken before 12 June; the date it was taken and the signatory are not stated.

Four. Classification on 31 August 2026. The classifying body is the FDA, not the firm. The conclusion is Class II: use of the product "may cause temporary or medically reversible adverse health consequences", with a low probability of serious harm. The reason field supplies the spectrum of consequences itself: possibly no injury, local infection, less commonly a need for medical attention, antibacterial treatment, an intervention or hospital admission, and in a very few cases critically ill or immunosuppressed patients may be at risk to life. This classification is the price struck in the case. A quantity of 1,578,022 kits bought a Class II, not a Class I.

What was visible then: before 12 June 2026 the assembler held five classes of material that the record can confirm.

One. The kit structure list. The product description field states the component's brand, name and specification (1 mL, 1.5 mL), which says the component is a named item on the kit's bill of materials and not part of a bulk mixture. Two. Its own batch ledger. The 12 batch numbers in this record, and the batch numbers across all 16, are the assembler's own batch record numbers,

and which component batch went into which kit batch can be matched only from the assembler's material issue records. Three. The public record of 12 June 2025 (Z-2155-2025). Same component, same brand, 15,700 units, four states, and a reason field reading "open seal", available to anyone who looks in the FDA enforcement database. Four. The component maker's inspection documents travelling with the goods. The kit's sterility claim rests on them; the record states neither their content, nor the sampling plan, nor the acceptance criteria. Five. Three of its own recalls between 2023 and 2024 caused by components inside its kits: 233,638 kits on 15 May 2023 for Turkuaz ultrasound gel inside the kits failing to meet sterility specification (Z-2321-2023); 86,699 units on 15 November 2023 for Nurse Assist saline inside the kits, classified Class I (Z-1117-2024); and 8 April 2024 for plastic syringes inside the kits (Z-2768-2024, quantity field empty). Three times, the same sentence pattern: the component goes wrong, the kit is recalled.

Information that arrived only afterwards: the physical mechanism, a wrinkle in the paper lidding extending towards the seal area; the specific batch range of the affected component (the 2026 records give kit batch numbers and no component batch numbers); and whether there were infections, how many, in which state. None of the three is in the record.

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What happened: Class II, status Ongoing (as at the 21 October 2026 data update read for this book). This record, Z-3057-2026, covers 3,650 kits; the 16 records under the event, summed on the product_quantity field, come to 1,578,022 kits (this book's addition, not a figure the record states directly), 432 times this record's own count. Distribution nationwide. The records state no casualties, no infection count, no recall completion rate, no destruction quantity. Back to the three questions. The most common answer to the first is to trace the component batch and recall only the kits that received the problem batch, but the 16 records are split by kit model and not by component batch, which says the key did not exist at the time. The cost of that one action is a single column of data; what it would have saved is the order-of-magnitude gap between 1,578,022 and 3,650. The item most easily missed under the second is the 15,700 unit record of June 2025, same component, same brand, which reads like last batch's business. The third question lands on batch release: look in the release records for a component batch number, and its absence is the answer.

Options ruled out: the four below are actionable alternatives read back from the shape of the record. The record states no discussion of any of them and no reason for rejecting any.

One. Isolate by component batch. Recall only the kits that received applicators from the problem batch, release the rest. The record does the opposite: 16 records split by kit model, each listing every affected kit batch for that model, with no component batch number anywhere in them. Two. Handle by clinical risk. Blood culture and interventional kits recalled first, ophthalmology, pre-operative and obstetric deferred. In the record the 2,080 kit ophthalmology set (Z-3061) and the 352 kit caesarean set (Z-3060) go on the same day, at a scope covering every clinical use. Three. Merge the 16 records into one. The FDA's enforcement records are split by kit model, 16 independent numbers, and merging them would coarsen

the traceability granularity and take 15 checking entries away from the downstream hospital. Four. Let the component maker recall alone and the assembler do nothing. That route does not close structurally: the recall strikes the kit's sterility claim, and once the kit is opened in use what the clinician sees is Medline's bag, not BD's box. A fifth shape deserves a note. Write the recall as "a potential sterility gap in the component" and leave the kit on the shelf. In the record all 16 pull the whole kit, and not one of them handles the component alone.

Early signals: (1) Z-2155-2025 on 12 June 2025: same component, open seal, 15,700 units, four states. Checkable at the time, in the public record. It was buried as "that was the last batch". (2) The component has two specifications only, and was built into 16 kits with entirely different clinical uses: the wider the reuse, the harder the single point. This one becomes visible only when 16 records appear on the same day; beforehand it is a single line on a bill of materials. (3) The quantity distribution spans a factor of 9,583 (104 to 996,692), which says the component entered both rare proced‑ures and routine blood collection: the failure radius is set by the largest record, not by the average. (4) The applicator's sterile barrier has one interface only, paper lidding against seal. That in‑terface is made by the component maker, and the assembler can verify it but cannot change it. The record states no second source for the component. (5) The same assembler recalled once in May 2023, once in November 2023 and once in April 2024 for compon‑ents inside its kits (Turkuaz ultrasound gel, 233,638 kits; Nurse Assist saline, 86,699 units, Class I; plastic syringes, quantity field empty), and again in June 2025 for the applicator. Four times, one sentence pattern: the component goes wrong first. The pattern had to repeat a fourth time before anyone treated it as a structural problem.

The point of escalation: 12 June 2026, when 16 records were initiated on one day. That step rewrote "a component in some kits has a problem" into "the sterility claim of the entire kit catalogue cannot be defended". Who approved it is not in the record. The act of approving is: 16 records on one day, one block of reason text, one "Letter" notification method, one classification date. The price of the escalation appears at the same time, a Class II and not a Class I: the quantity reached a million kits and the probability judgement on health consequences did not keep pace with the quantity. One coincidence is visible in the fields. The 2025 recall (Z-2155-2025) was also initiated on 12 June. Two years, the same calendar day. The record states no link between that date and any internal milestone (annual quality review, supplier requalification), and this book draws no inference, recording it for later checking.

Failure chain: technical layer. The wrinkle in the paper lidding extended into the seal area, the applicator's sterile barrier held a gap, and the kit's single sterility claim failed with it. Organizational layer. One component specification was reused across 16 kit groups, the recall was cut into 16 records by kit model, and component batch numbers are entirely absent from the record. Incentive layer. The economics of a kit come from not making the component yourself; the fewer the components and the wider the reuse, the broader the catalogue and the lower the quoted price. All three layers break at once. The boundary break sits between the organizational and the technical layer: validation of the component's sterilisation belongs to the component maker, the sterility claim of the kit belongs to the assembler, and the two validation regimes share no batch key. The 2026 record holds 12 kit batch numbers and not one component batch number. That absent column is the interface of the failure chain. It also explains why the recall could only be cut by model: without a key there is no finer cut. The same defect reached four states and 15,700 units in

2025, and the whole country and 1,578,022 kits in 2026. What changed in a year is the component's distribution footprint, not the defect.

Minimum counterfactual: add one column to the kit release record, the batch number of the component loaded, entered on the same line as the kit batch number. One action, taken at any release before 12 June 2026, requiring no luck, since the data already sits on the assembler's goods-receipt documents and material issue ledger; what is missing is writing it into the release record. With that column, a notice from the component maker is answered by a lookup down the column, and the recall scope shrinks from 16 models to the batches that actually received the problem component. The record supplies its own evidence for the counterfactual: no component batch number appears in any of the 16 records, which says the column did not exist at the time. Cost: maintaining one column of batch numbers, one further data entry.

Outcome and sources: Class II, Ongoing, 3,650 kits in this record and 1,578,022 kits across the 16 records (this book's addition), 432 times as many. Later corroboration, all dated after this case: on 14 July 2026 AVID Medical initiated Z-3109-2026 for the same component (98 kits, distribution in Nebraska only); on 20 July Medline initiated Z-3073-2026 (222,653 units); on 22 July AVID initiated Z-3194-2026 (288 kits); on 23 July Medical Action Industries initiated Z-3181-2026 (6,420 kits). One component, four assemblers within six weeks. The four are independent recalling firms, and the only common item in the records is the component's name. The defect wording is not consistent across them (Z-3073 and Z-3181 read open or incomplete seal, Z-3194 adds mould contamination), and this book counts them on "same component" and not on "same defect". Primary source: FDA device recall enforcement report, recall_number Z-3057-2026, event_id 99334, openFDA

device/enforcement endpoint, data updated 21 October 2026; link at Appendix B, ph-008.

Case Four has a mirror image in brand responsibility worth looking at. The gain of private label is not having to build a line (asset light). The cost is that naming rights over a defect pass to the brand: the recall notice carries the brand's name, and the factory's name appears in a footnote. Naming and responsibility bind together in public perception, since consumers know only the name on the recall notice, so a brand's risk register has to manage the state of its private-label suppliers as its own state: the supplier's inspection records, change notifications and recall history all enter the brand's register, which is Chapter 29's maintenance ledger applied directly at the supplier dimension. The more common the private-label model, the more the brand's supplier ledger matters. It does not appear on the brand's balance sheet. It appears on the brand's recall notice.

The four cases share one structure, and it can be closed into a single statement. A cascade is at bottom a mismatch between the granularity of responsibility and the granularity of dependency. Dependency runs fine, down to component and parameter; responsibility runs coarse, up at company and brand. Failure never enters at the wrong granularity, since it happens at the finest layer, while responsibility is always struck at the wrong one: a recall strikes the brand, an outage strikes the platform. Governance follows from that. Sink responsibility to the depth of the dependency: register whose part, define semantics at parameter level, put a clinical signature at catalogue level. The cost of sinking is amortised over years. The cost of not sinking is paid in one instalment.

Case Five (Agriculture and Food): Another Name Inside the Kit (condensed)

ag-005. A pesto pasta salad kit. The FDA enforcement report (H-0090-2026, API record read and checked for this book): Listeria risk inside the kit, a Class I recall of 3,252 cases; the separately sealed components inside the kit were recalled in the same batch. The cascade logic is isomorphic with Case Four (a component's defect disappears inside the kit's name), and the incremental information here is the independent seal: components each packaged separately, touching nothing. Recalling the separately sealed components in the same batch says the recall scope has already moved from the contamination path to the kit as a whole. Tightening the scope is progress in the recall system. It is also the mark of a widening cascade radius, since the commercial loss on the whole kit exceeds the actual risk of the defective component. Minimum counterfactual: assess kit recall scope component by component, recalling the defective component and mitigating the sound ones. Primary source: FDA, 2026. Link at Appendix B, ag-005.

Case Six (Healthcare): The Safety Needle That Left the Catalogue (condensed)

me-002. The NHS in the United Kingdom, U-500 high-concentration insulin administration. The HSIB report (2022, archived snapshot read and checked for this book): a dedicated safety needle disappeared from the supply catalogue after a contract change, nurses were left drawing the concentrated solution with standard U-100 syringes, and two patients had hypoglycaemic episodes; 11 comparable insulin administration events occurred nationally between 2017 and 2022. One line of change in the supply chain (a catalogue withdrawal) rewrote clinical practice directly, since the substitute tool was never clinically assessed, and the path of that change (purchasing contract to nurses' bench) bypassed

every assessment gate that should have stood in it. Cascading failure is present at both ends: upstream a supplier's contract decision, a single point; downstream the administration practice of many hospitals, exposed simultaneously to one catalogue change. Minimum counterfactual: make clinical assessment a precondition of any supply change affecting a clinical tool, by adding a clinical signature position to the change notification chain. Primary source: HSIB, 2022. Link at Appendix B, me-002 (archived link, grade A–).

33.4 What Shows Up Earliest

Six weak signals, each with the reason normally used to bury it.

1. The risk register has no supplier column, or suppliers are registered by purchasing category rather than by dependency criticality. The reason: "supplier management is purchasing's job."
2. Your critical supplier's supplier, do you know who it is? Answering no means two layers of dependency is a black box.
3. Has the isolation and degradation plan been drilled? No drill means the cascade takes everything.
4. Has anyone estimated the exit cost from a single-source supplier? Not estimated means the exit capability is evaporating.
5. A third party's incident report has never triggered a dependency check of your own. The reason: "they use a different model from ours."
6. The "multi-active" cloud regions, data centres or logistics routes are in fact one physical region or one backbone link. The reason: "it looks separate on the topology diagram."

Put the six signals back against this chapter's cases and two new ones. Signal one (no supplier column) matches Case Four: the risk

of the private-label component has no row in the brand's register. Signal two (two layers is a black box) matches Case Two: the ingredient's ingredient, meaning where the stabilizer's own raw material comes from, sits two layers outside the bakery's field of view. Signal three (no drill, total loss) matches Case One: no BYOIP customer had ever drilled an alternative path against the shape of a mistaken prefix withdrawal. Signal four (exit cost never estimated) matches Case Six: the exit cost of the safety needle, a clinical risk, was never in the purchasing assessment. Signal five (a third party's incident triggers no check) matches the lesson of the two new cases: a recall in your own industry is a free dependency checklist. Signal six (multi-active in one region) is this chapter's physical criterion: the two hydraulic systems in Case Three are the sample of a topology that is separate on paper and co-resident in fact.

The six sort again by cost of evidence. Signals one and five are cheapest, since they need only reading the register's structure and the response log to incident bulletins. Signals two and four are mid-priced, asking the supply chain about two layers and checking whether an exit assessment exists. Signals three and six are dearest, needing drills and physical verification. The starting recommendation is signal five: subscribe to industry incident bulletins (regulatory recall databases, safety notices), run every entry past your own dependencies, a question that costs ten minutes each, and mark into the register whatever the check confirms. The compounding pays out two years later, when the supplier column of the register has verifiable depth for the first time.

33.5 How to Prevent It

Device 1: the critical dependency register. Manage a critical dependency as your own single point: register the supplier, the alternative path, the switchover time and the last verification, item

by item. Granularity reaches down to "whose part, inside which of your products". Below that level the register is decoration (the lesson of ph-008).

Device 2: single-point pricing. For the most concentrated dependency, write the loss from an interruption scenario into the purchasing decision: the "discount" on a single source has to be compared on the same table as the expected interruption loss. Most memos on a single-sourcing decision have no such column.

Device 3: the isolation drill. At least once a year, an "upstream disappears" drill: the critical input cut for 48 hours, switchover to the alternative path per plan. What it measures is not the ideal switchover time; it is the time taken to discover what is missing.

Device 4: joint review of third-party events. When an upstream incident happens in your industry, complete a dependency check of your own within 72 hours (do we use the same firm, the same region, the same version) and put the conclusion into the register. A third party's incident is free intelligence, and declining it is waste.

Set the four devices against this chapter's cases. Device 1 on Case Four: the kit's component list down to the item, whose disinfectant, whose culture bottle, whose bag, with the alternative possibility registered for each. Device 2 on Case Two: the single-source decision on the stabilizer ingredient gains the expected interruption loss column, with the coordination cost of a multi-state recall inside it. Device 3 on Case One: a BYOIP customer, or a simulated customer on the platform side, runs one switchover drill against a mistaken prefix withdrawal. Device 4 on Case Six: an industry bulletin about a catalogue-change event, a needle vanishing or a packaging change, checked against your own catalogue within 72 hours.

The four devices also have four ways of dying. The critical dependency register dies of a granularity gamble, registering to the supplier and not to the part, betting that the failure will not be in that part; the patch is tiered granularity, class A critical parts registered to the part, class B to the supplier, with the tier set by failure radius. Single-point pricing dies of arithmetic laziness over expected loss, counting the direct interruption only and not the second-order loss from recall and reputation; the patch is a three-layer loss model, interruption, recall, public event, which is the three columns of Section 33.1. The isolation drill dies of a script that is too friendly, 48 hours, a weekday, notice in advance, when a real interruption is not friendly; the patch is a randomised script, with time, duration and break point drawn by lot, the same method as true rehearsal in Chapter 22. The joint review dies of being a one-off, checked and filed while the next incident starts the scramble over again; the patch is routine registration of check results, each check updating the matching register row, so the register is the sediment of the checks. The four patches share one health-check question: when was this register last updated by a real event.

Governance of cascading failure has one further view that deserves its own line, at industry level: common resilience. Homogeneous downstreams (condition two) mean industry-level single points, ingredients, equipment and software shared across a whole sector, and an industry-level single point needs an industry-level plan, whether shared stockpiles, mutual backup agreements, or a shortage mutual-aid arrangement run by a trade body. The economics of common resilience, meaning the sector sharing the cost of redundancy, coexist with the prisoner's dilemma of who pays to build it first. The force that usually breaks the deadlock is a regulator, in the form of mandatory industry stockpiles on the strategic-materials model, or a large customer, whose supplier diversification requirements protect the whole sector as a side effect.

Early examples of common resilience can be found in every industry, spare-part pools in aviation and mutual support arrangements in banking among them. Generalising those examples is the first item on this chapter's list of recommendations to industry bodies.

33.6 Boundaries with Neighbouring Mechanisms

Against hidden coupling (Chapter 21): coupling is about the invisibility of a connection (the line exists and nobody drew it); cascade is about the propagation of concentration (the line can be drawn and has no isolation). The dependency graph of Chapter 21, taken down to the supply chain, is this chapter's register.

Against the illusion of redundancy (Chapter 22): a cascade often has an illusion in front of it, where "multiple suppliers" turn out to be two brands out of one contract manufacturer. Chapter 22's common-cause list, projected onto the supplier dimension, is this chapter's single-point pricing.

Against crossing a scale threshold (Chapter 24): scale amplifies the concentration of dependency and the radius of propagation; the dependency consolidation of a growth phase, which is Chapter 24's efficiency logic, is exactly where this chapter's concentration comes from. In a growing firm the two chapters have to be read together.

33.7 The Five Self-Checks

1. Write down the top five critical dependencies of your product or service. Has a second supplier for each one been qualified?
2. In your risk register, how deep is the deepest layer at which "the supplier's supplier" appears? If the answer is more than three layers, has anyone actually verified it?

3. At the last upstream incident in your industry, did you organise a dependency check? How long did it take?
4. Has the process you sit in ever run a 48-hour interruption drill? What was the first missing item it found?
5. Your "multi-active" architecture, multiple supply sources, multiple routes: are they physically separate, and who verified that?

(This chapter: 11,385 characters.)

Chapter 34: Path Lock-in (David; Arthur)

> What this chapter covers: how an early accidental choice grows into a structure you cannot leave; the classic argument from David and Arthur, the criteria, two field sites with the evidence gap stated, and how to run a from-zero derivation. When to turn back: at a selection review, in a standards fight, and any time "we can't afford to switch" has become the premise of the discussion. Reading time: 45 minutes.

Path lock-in is the only mechanism in this book that produces no accident. There are no casualties, no penalty and no investigation report. The cost is the sum of every alternative that did not happen. Paul David used the century-long history of the QWERTY keyboard to show how a layout designed to stop typebars jamming went on to rule electronic keyboards a hundred years after jamming itself had disappeared (David, "Clio and the Economics of QWERTY", American Economic Review 75(2): 332–337, 1985). Brian Arthur proved the generality of the pattern with a mathematical model: increasing returns let the early leader's advantage compound with time (Arthur, "Competing Technologies, Increasing Returns, and Lock-In by Historical Events", Economic Journal 99(394): 116–131, 1989, Arthur's bibliographic record verified in Crossref). Lock-in is not the same as error. The locked path usually performs well. What has gone is the capacity to re-evaluate it.

The QWERTY story is worth telling in full, because it shows every part of a lock. Typewriter mechanics in the 1870s imposed a physical constraint: adjacent typebars jammed when struck in fast succession, and the Sholes layout spread the common letters apart on purpose. The constraint disappeared decades later as the

mechanism improved. By then the layout had grown its own lock-ing parts. A typist's muscle memory was an investment in human capital. The typing curriculum was an institutional investment. The keyboard maker's tooling was a fixed investment. The electronic keyboard inherited the entire compatibility requirement. The parts list maps onto this chapter's generating conditions one for one: learning cost, complementary investment, network effects, institutional fit. The lesson sits at the end of the story. No single part was a mistake, and the lock needed nobody to make an error. It needed time and inertia. That is where this chapter differs in temperament from the rest of the book. Other mechanisms have a perpetrator. Lock-in has only a calendar.

The taxonomy of network effects is worth completing, because naming the type is the first step in judging how hard a lock will be to break. Direct effects: the more users, the more useful, as in communication protocols and social platforms. Indirect effects: the more complements, the more useful, as in application ecosystems and accessory markets. Data effects: the more use, the more accurate, as in algorithms and maps. Social effects: using it is the career-safe default, as in a job seeker's skill choice or a firm's me-too procurement. Difficulty rises across the four. A direct effect yields to a standard converter, an indirect effect to a compatibility layer, a data effect to migration tools. The social effect is hardest, because what it holds is the risk calculation in people's heads rather than the system. Device 4, the alternative archive, records them differently for that reason. The first three take technical parameters and price. The fourth takes the adopter list, meaning who uses what, and the adopter list is the only measure of a social effect.

34.1 What It Is

Definition: path lock-in means an early choice, often accidental, that reinforces itself over time through network effects, learning costs, complementary investment and institutional fit, until the feasible space for alternatives contracts and closes.

Three start criteria. Any one of them means the mechanism is already running.

① Alternatives no longer appear in the selection discussion. They were not rejected. Nobody raised them.

② Migration cost only rises and is never re-estimated. Every "we can't afford to switch" cites a figure from years ago.

③ "We have always used this" has been promoted from a statement of fact to a statement of reasons.

The three differ in how loudly they announce themselves. Criterion ① is silent, since a missing agenda item is invisible by construc‑tion. Criterion ② is written, and leaves a trail of estimate dates. Criterion ③ is spoken, usually in a meeting where nobody writes it down. Of the three, ② is the only one that can be audited from a filing cabinet.

34.2 How It Grows

Three generating conditions.

Condition one: increasing returns. The more it is used, the better it works, through ecosystem, plug-ins, the talent pool and compatible parts, so the leader's advantage compounds with time. This is Arthur's core argument.

Condition two: complementary investment hardens. Training, tools, contracts and downstream interfaces built around the early

choice each act as a hidden tax on migration. Individually none is large. Added together they are a cost wall.

Condition three: the evaluation capacity atrophies. Years without a selection exercise let the muscle for it waste, meaning the market scan and the alternative file. When the capability is finally needed, the organization can no longer assess, and leans further toward not switching.

The three feed each other. What gets locked is the organization's assessment function, not the system.

The atrophy has a ledger of its own, and the ledger explains why nobody notices. The true cost of lock-in is opportunity cost, the return the alternative would have produced, and opportunity cost is not booked under any accounting standard, because accounting records what happened. An unbooked opportunity cost leaves the locked balance sheet permanently presentable: no default, no impairment, no penalty. A presentable statement then argues for staying, since a clean sheet offers no grounds for treatment. The deeper purpose of Device 2, migration cost re-estimated, is to write opportunity cost down as a number for the first time. The re-estimate covers what migration would cost and also what staying forgoes each year, drawn from the annual improvement list of the alternatives, with Device 4's file as the data source. Set the one-off migration cost beside the annual opportunity cost of staying and the decision is complete for the first time. A complete decision does not have to choose migration, and lock-in often remains the better answer. What a complete decision ends is the state of no decision. Most organizations did not choose to stay. They never did the arithmetic.

There is a follow-on action for organizations that have done it: book the returns of lock-in as well, which is where the rationality argument in 34.6 lands. Those returns stay out of the accounts too, the stability dividend of a settled ecosystem, the efficiency of deep

optimization, the inspection-free cost of compatibility. Write them into an annual list of what staying saved this year, set that list beside the opportunity cost list, and lock-in becomes an account with income and expenditure instead of an atmosphere. The difference shows on the review date. An account with books goes through an income statement once a year: benefit still there, renew; benefit gone, discuss migration. An account without books survives on feeling, and the life of the feeling is the life of the organization.

34.3 Field Sites

Case 1 (Public): A Railway Built for 360 and Opened at 320

Case ID: po-004.

When and where: October 2023 to June 2026. London, at the Department for Transport, and Birmingham, at HS2 Limited; the sites are Old Oak Common, Curzon Street and Euston, with tunnelling, viaduct and earthworks on the route between London and Birmingham. The report records decisions to the month, not to the day.

What and who: the Department for Transport (DfT), the project's funder and sponsor; HS2 Limited, a non-departmental public body of the DfT, responsible for design, construction and delivery of the railway into use (p.16); Euston Delivery Company Limited, set up by the DfT in April 2026 to take over the London terminus and the related works (p.12). Phase One civils are delivered by four joint ventures, each holding a section (p.29). Responsibility is recorded by post: the Secretary of State for Transport, the Permanent Secretary at the DfT, the chief executive of HS2 Limited.

What they were doing: from January 2025 the DfT and HS2 Limited were running the third reset, with four workstreams in parallel (cost and schedule; supply chain and commercial; organizational capability; governance) and a planned close in spring 2027. The

reset carries a budget of £153 million, of which £101 million had been spent by the end of March 2026 (p.18). Of the 124 recommendations accepted from external reviews, 45 were complete by February 2026, 36 per cent, with the rest scheduled to close with the reset (p.21). Construction did not pause. The Colne Valley viaduct was finished, and tunnelling towards Euston began in early 2026 (p.18). The cost estimate was rebuilt from the ground: the project split into 21 sub-projects, each checked for scope, schedule, cost and risk, then run through scenario tests to produce a range (p.25). Contracts were in negotiation. Talks with the four civils joint ventures, begun in spring 2024, had failed; in April 2025 the approach changed to settling historic accounts first and amending future terms after that, with a target of autumn 2026, about six months later than planned (pp.29–30). In January 2026 the Secretary of State asked HS2 Limited to assess what a lower speed would save (p.25). In March 2026 the DfT appointed a delivery agent for the disposal of Phase Two land: 560 properties, of which about 350 fall under the Crichel Down rules and must first be offered back to former owners (p.47). At Euston, £3.8 billion had been spent, of which £1.3 billion on land and compensation and £1.5 billion on tunnels and the approach lines, with £4.1 billion more required under the current design (p.44). On Phase Two, land and property had cost £643 million, covering 1,256 properties across 18 square kilometres, and several purchases were still being completed in March 2026 (p.46).

The decision point:

① October 2023: the government cancels the whole of Phase Two, the 2a and 2b western and eastern legs together (p.17, figure 1, note 2). The report says only "the government" and records neither a signed document nor a signatory. The cut turned Phase One from the first section of a line to the north into a spur that ends at Birmingham: trains must descend to the existing West Coast Main

Line north of Birmingham (p.31), and that line's platform dimensions were fixed long before HS2 existed. The report's section records the cancellation alone, with no review that day of the train contract, the platform design or signalling compatibility; the next record on train formation comes two and a half years later (p.31).

② May 2026: the DfT agrees three scope changes. Maximum speed falls from 360 to 320 km/h, automatic train operation is deleted, and signalling drops to a version already in use elsewhere (pp.25–26). In the same month the Permanent Secretary publishes the Accounting Officer Assessment, with a benefit-cost ratio of 1.5 to 6.4 (p.26). The report names neither the meeting that took the scope decision nor the signatory. The shape of the decision needs to be read closely. The civils were completed to the original specification: tunnel cross-section, track spacing and structures all built as designed for 360 km/h. Everything deleted sits at the system level (p.25). What could be deleted was deleted, and what could not be deleted was kept. The report gives one further reason worth quoting: testing 360 km/h itself takes time and money, and the test might prove the speed infeasible (p.26). Speed therefore moved from an engineering question to an operating parameter. The decision carries a real benefit too, in a lighter test burden and an opening at least a year earlier (p.26). The benefit is genuine, and it lands entirely in the layer that could be deleted.

③ 2026: the final decision on train formation and fleet size is deferred to 2028 (p.31). This is not the absence of a decision. It moves the decision to a year when the platforms have already been poured. Civils at Curzon Street and Old Oak Common are advancing, and their costs have doubled, from £600 million to £1.3 billion and from £2 billion to £4.3 billion (p.31). The report's reason for the deferral is practical: decide the formation in 2028 and the supply chain can still adjust (p.31). The platform is not in the supply chain. It is in the civils.

④ April 2026: Euston is separated from the main line and handed to a newly created delivery company, which the DfT calls a "single accountable body" (p.12). At that point the company was not yet formally operating and its chief executive was still acting (p.12). The interface between station and line turned from coordination inside one organization into a contract between two.

What was visible then: before May 2026, the following was on the table. The 2020 cost estimate of £35 billion to £45 billion at 2019 prices (p.22). The NAO report of July 2024 and the Stewart Review of June 2025, the latter finding the governance structure not fit for purpose (p.35). The DfT's April 2024 notice to HS2 Limited that it was in breach of its delivery obligations (p.20, figure 3, note 2). The four civils joint venture contracts and their change orders, with cost verification backlogs averaging 21 months and covering close to £5 billion (p.30). The train contract signed in 2021, with formation ordered as two 200-metre units (p.31). The recorded dimensions of the existing West Coast Main Line platforms (p.31). The list of 1,256 properties and 18 square kilometres of land already bought (p.46). The location map of 550 boreholes awaiting backfill (p.48). And the new estimate from May 2026 of £87.7 billion to £102.7 billion, against which HS2 Limited says it is aiming for £93.2 billion (p.22). The process that produced that estimate was on the table as well: 21 sub-projects verified one by one, spot-checked by an independent expert panel that included the Infrastructure and Projects Authority, with the DfT itself writing that the figures still carry high uncertainty (p.25).

What was learned afterwards: cancelling Phase One would cost £33 billion to £58 billion (p.27), the same order of magnitude as completing it and more than four times the 2023 estimate (p.28); had today's cost been known in 2020, the benefit-cost ratio would be 0.3 to 0.4 (p.27); full opening falls between May 2040 and December 2043 (p.23); the new trains are slower than the incumbent Pen-

dolino on some sections (p.31); a major decision passes five serial approval gates and can take up to nine months (p.36).

> **Stop here. Decide before you turn the page.**
>
> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: Phase One continues. The cost range is £87.7 billion to £102.7 billion, with a target of £93.2 billion; £46.8 billion had been spent by the end of March 2026, of which £2.6 billion went on the cancelled Phase Two (p.22). Passenger service from Old Oak Common to Birmingham begins between May 2036 and October 2039; the full Euston line opens between May 2040 and December 2043 (p.23). The lower speed and the deleted automatic operation are expected to save £1 billion to £2.5 billion, at a cost of about £1.3 billion in long-term benefits (p.26). Back to the three questions. The most common answer to the first is "stop and recalculate" or "re-tender". In this case, stopping is priced at £33 billion to £58 billion (p.27), and the report records no assessment of re-tendering at all, only the sentence that the remaining time on the contracts is reducing and the opportunity to avoid risk and save cost is receding quickly (p.30). The cost of re-tendering is a tender document nobody has written. The price of stopping includes site restoration and termination compensation (p.26, p.27). The item most easily missed in the second is the 2021 train contract: signed before Phase Two was cancelled, formation ordered as two 200-metre units, and the platforms it will stop at broke ground three years later (p.31). It lives in the contract file, not on the construc-

tion programme, so every review sees progress and never sees it. The checkpoint in the third lands on the week of October 2023: when Phase Two was cancelled, did anyone change the train contract at the same time.

Options ruled out: cancel the whole project, which was costed, with the restoration and termination fees folded into the numerator as avoided cost, pushing the benefit-cost ratio up to 1.5 to 6.4 (p.26, p.28), and rejected on price. Run the line at 360 km/h to the end, which requires testing whether trains and track can run safely at that speed, a test that costs time and money and may return a negative (p.26). Change the trains to single 200-metre units, which gives less capacity than the incumbent West Coast Main Line trains and cuts project benefits with it (p.31). Re-tender the civils, for which the report records no assessment and leaves only the sentence about contract time running out (p.30). Keep Euston inside the main-line reset, which was not done: it was separated into an independent delivery body in April 2026, with no assessment of a combined option recorded (p.12). Replace signalling and train control together, where the deletion of automatic operation left signalling at a version used elsewhere, and the report records no in-house or new-version option (p.26).

Early signals: ① the 2020 estimate of £35 billion to £45 billion against a 2024 funding envelope of £44.6 billion, with both the DfT and HS2 Limited accepting that delivery inside the envelope was not possible (p.22); the number at approval and the judgement at execution parted company in year four. Overridden because the reset was under way. ② October 2023, Phase Two cancelled and the 2021 train contract left as it was, formation unchanged (p.31); the contract fixes car length, platform length follows from it, and the platforms had not started. Overridden because the contract was signed and the platforms not yet built. ③ spring 2024, the renegotiation fails: HS2 Limited did not understand costs already in-

curred, had no reliable total cost estimate, and the transfer of risk to the joint ventures was not accepted (p.29). Overridden because it was handled as a commercial problem. ④ a cost verification back‑log averaging 21 months and close to £5 billion, with single con‑tract verification periods running from 6 to 60 months (p.30). Overridden because accounts can be caught up later. ⑤ both sta‑tions doubling in cost, which the report attributes to immature design, underestimated complexity, site constraints and re-plan‑ning (p.31). Overridden because it read as a delivery efficiency problem.

The point of escalation: the start of civils. HS2 Limited's own March 2025 assessment uses the words "construction works hav‑ing commenced too soon", and the same passage records that the contracts "have not delivered the expected performance" (p.18). The report names neither the start date nor the approval docu‑ment. From the day work began, alignment, tunnel cross-section, track spacing, longitudinal gradient and platform length left the drawing and entered concrete. Those five can afterwards only be accommodated, never re-chosen, and the space of alternatives contracts from how the whole network connects to how this one work site gets modified. Chapter 27 handles every later decision to continue. This chapter handles the start itself. The money it spent was not the largest item in the project. What it spent was the option set.

Failure chain: technical layer: the civils were completed to the 360 km/h specification and are operated at 320; automatic operation was deleted and signalling dropped to a version used elsewhere (pp.25–26); the platforms are designed for two 200-metre units, the existing line's platforms cannot take them, and a single unit car‑ries less than the incumbent trains (p.31); the new trains are slower than the Pendolino on some sections (p.31). All four point the same way: the changeable part was made smaller and the unchangeable

part was finished. The report's phrase for the civils is that they were completed to the original specification supporting 360 km/h trains (p.25), so the cross-section, track spacing and structural margin reserved for that speed stay in the concrete and go unused in service. Organizational layer: there was no reliable total cost estimate in 2024, so the renegotiation had no floor price (p.29); the verification backlog ran 21 months and close to £5 billion (p.30); the independent panel recommended filling the systems integration capability gap early, because that function holds every interface in the project (p.33); responsibility is split between the DfT, HS2 Limited and the new delivery company, and the station-to-line interface became a contract (p.12); major decisions pass five serial approval gates and can take up to nine months (p.36). Incentive layer: the price of cancelling is the same order as the price of finishing, so continuing is the cheaper side (p.28); the remaining term on the civils contracts keeps shrinking and the bargaining window is closing (p.30); exit also carries land restoration, with part of the 1,256 properties to be disposed of and 550 boreholes to be backfilled between December 2027 and March 2028 (pp.46–48). The break sits between the organizational and the incentive layers: the organizational layer lost the capacity to reprice, so nobody was left who could change the price structure in the incentive layer.

Minimum counterfactual: in the week of October 2023, when the cancellation of Phase Two was announced, send a change notice to the 2021 train contract, changing "two 200-metre units" to "formation to be determined, platforms designed in parallel for single and double units". One action, taken before platform construction starts at the two stations, costing one contract variation and two sets of platform drawings. It changes nothing about the cancellation of Phase Two and requires nobody to forecast demand. It takes the formation decision back out of the concrete. Per the report, the formation decision is made in 2028 while the platform design is fixed in 2026 (p.31). The window for that change

notice is a few months: once the platforms are poured, parallel design collapses into a choice between two.

Outcome and sources: Phase One continues, with a cost ceiling of £102.7 billion; Phase Two is cancelled, leaving 1,256 properties and 550 boreholes awaiting backfill on the books. One NAO recommendation faces this chapter's interface directly: when the trains are specified, they should be placed in the future operation of the whole network, with seat capacity assured (p.15). Primary source: Comptroller and Auditor General, High Speed Two reset, Session 2026-27, HC 52, National Audit Office, published 29 June 2026 (signed 22 June 2026), 56 pages, ISBN 978-1-78604-684-0; page numbers cited here are PDF pages. The link is in Appendix B under po-004.

Case one's lock has a further anatomy in the time dimension. The window shrinks with each project stage: at route selection the space of alternatives is network-wide, at land acquisition it contracts to the corridor, after the start of works to the work site, and after opening to operations alone. The engineering reason for the shrinkage is the materialization of input, since money turned into concrete is no longer money. The governance inference follows: in a large project the substitution assessment has to be moved ahead of materialization, and the from-zero derivation at feasibility is the only cheap window, because asking the from-zero question after opening gets a political answer. Front-loaded window management is therefore the core technique of large-project governance. Every stage review should label the substitution space of that stage, whether network, corridor or work site, and the width of that space is the length gauge for the review's agenda: at a stage where only the work site can still move, the review should no longer be discussing the route. The label costs one line in the review template. What it prevents is the mistimed debate, since arguing about the

route at a stage when only a work site can be adjusted consumes all the patience an organization has.

The product history in case two (mn-006) carries a consumer-goods detail worth writing out. Two locks hold case two in place. The user’s lock is the spare-part buying habit and the memory of how a new carafe fits the pot. The firm’s lock is tooling, suppliers and brand consistency. Each reinforces the other, and fifty-two years of two-way reinforcement turned the object of a material substitution assessment from one material into an ecosystem: changing the material means changing the spare-part supply chain, which means changing what users believe the product is. Consumer goods usually break a lock by running a second line rather than by substituting: old and new material side by side, and the market votes with its feet. A parallel line costs double inventory across two supply chains, and its obstacle is channel objection, since retailers will not clear shelf space for a trial line. The true cost of running both, trial-line inventory and shelf cost against the expected loss from the hidden risk, almost never appears on an internal state‑ment, because the two ledgers belong to procurement and to brand and no single sheet puts them together. Device 2’s pricing table is used across departments here: putting the two ledgers on one sheet is the step that comes before deciding whether to change the material.

Case three (av-029) gains a regulatory-document view. The two pages of a safety bulletin are the regulator’s standard action on a design inheritance, since it cannot mandate a modification and can only alert the industry, and the effect of the alert depends on op‑erators’ filing habits, meaning whether an old bulletin gets pulled when a new aircraft of the same configuration is certified. Civil aviation has mature carriers for that retrieval, in the mandatory search of airworthiness directives and service bulletins. The ultr‑alight field, with low regulatory density and dispersed operators,

lacks the equivalent, so transmission of the design inheritance degrades to person to person, old hands teaching new ones, and person-to-person transmission is Chapter 30's rupture curve. Design inheritance in a low-density industry can only be governed by industry self-organization: club safety and technical sessions, type owners' groups. The community is the first safety net in a low-density industry, and its activity level is the gauge of that industry's real safety level.

With the lessons of three cases laid side by side, this chapter can offer a quick identification check, the six signals in compressed form: a sentence about not being able to afford it, quoting an old number; an agenda item that has gone missing, the alternative dropped from the agenda; a phrase that has been promoted, always having used this now serving as a reason; a note of irritation, the from-zero derivation dismissed as boring; a silent archive, with no alternative file; a batch of characterized defects, defects retold as custom. The use of the six is a pocket audit. At the next selection meeting, the next procurement review, the next technical discus - sion, run the six sentences past your ear, and two hits are enough to start this chapter's annual derivation. The threshold is zero, since listening is free, and that is the minimum viable defence this chapter can offer.

Case 2 (Manufacturing): Fifty-two Years and a Plastic Spoon

Case ID: mn-006.

When and where: 1974 to 29 October 2026. The United States market, with the importer Bodum USA, Inc. of New York; manu - facture in Portugal and Thailand. The recall notice was published by the U.S. Consumer Product Safety Commission (CPSC) on 29 October 2026, number 27-055.

What and who: Bodum USA, Inc., the importer of record, responsible for sales in the United States and for this recall; the CPSC as publisher of the notice; and the holders of 5.9 million French press coffee makers as the recalled population. The notice does not name the brand's design or engineering department, does not name the glass supplier for the carafe, and does not say whether the two production sites are company plants or contractors.

What they were doing: a product line that had run for fifty-two years was in production and on sale. The notice gives the sales window as 1974 to October 2026 and lists the channels in one run: Target, Walmart and home goods stores nationwide, plus Bodum.com, Amazon and other online sellers, at about $15 to $65 (CPSC, Sold At). That list is a fifty-two-year total and the notice does not break it down by year. There are eight models: Bistro, Brazil, Caffettiera, Chambord, Crema, Java, Jesper, Kenya. Capacities run to two only, 8-cup and 12-cup, and the common feature is the replaceable glass carafe (CPSC, Description). The notice records three inscriptions: the brand name on the frame, the brand name on the inner lid, and a warning etched on the glass carafe itself (ibid.). One carafe interface covers the whole $15 to $65 price band (an inference: the notice gives no pricing basis). The company issued a statement the same day, describing itself as family-owned for 72 years, with a formula of high quality at an affordable price, and calling the recall voluntary (Bodum statement, as quoted by Daily Coffee News). On the risk side the notice records nothing at all: no design change record, no material assessment, no structural revision, no failure analysis.

The decision point:

① 1974: the product enters the market and the carafe appears as a replaceable part. The freeze date, the review record, the reason for the material choice and the signatories are not in the notice. What can be verified is the result: fifty-two years later the eight models

on sale still carry the words replaceable glass carafe. The consequence of that naming is inferable. The carafe became a consumable, a consumable breaking is a spare-part sale on the books rather than a product failure, and the locus of responsibility for failure moved from the factory to the kitchen. The replaceable part has a second function: switching cost here is paid per event rather than once. Every new carafe is another vote for the design, and no single vote is large enough to trigger a re-selection. One more point on the counting. Breakage happens in kitchens, so the count depends on users reporting it; 70 is the number of reports received, not the number of incidents, and the notice gives neither a reporting basis nor a denominator.

② The range extends to eight models. The notice gives no launch year for each model. What can be confirmed is the method: eight names attached to one carafe interface, with two capacities only and variation in body shape, frame and inner lid, never in the interface. A fixed interface shares tooling and spare parts, and the price is that coverage of a single-point defect equals the whole product line, with 5.9 million as one number for eight models together. The recalled population is written by interface too, "models with replaceable glass carafes", so the unit of identification is the carafe and not the model.

③ 29 October 2026: the remedy is finalized. The CPSC notice's Remedy field carries one word, Repair. The remedy consists of two objects, a plastic stirring spoon and a warning label (CPSC, Remedy). The company statement says that products shipped from then on will include the spoon and the label (original: "Going forward, these products will be sold with the plastic stirring spoon and an additional warning label"). The notice records no participants in the remedy review, no price comparison of material options, and no reason why the carafe or the machine was not changed. A fifty-two-year material question reached the table on that day, and the

answer was two objects priced in cents. The remedy does not nar - row the recall: 5.9 million units, not one fewer. What it changes is how those 5.9 million get used. It has one further property. It is a permanent product change rather than a one-off recall action, and the change lands on the accessory rather than on the carafe.

What was visible then: before the remedy was fixed, the following was on the table. Seventy reports of breakage in use, 48 of them with burns (CPSC, Incidents/Injuries), all occurring during use, with no record of breakage while the pot stood idle. The hazard description puts a metal spoon into the causal chain: the carafe can be damaged or cracked by metal spoons and other implements and may then break suddenly during use (CPSC, Hazard). The warning text was already etched on the carafe (CPSC, Description), and etching happens as the carafe is formed, so those words are already in the tooling and any change of material or wording means re-etching. Eight models share one interface (ibid.). The sales window runs 1974 to 2026 at $15 to $65 (CPSC, Sold At). Two manufacturing sites (CPSC, Manufactured In) mean two lines must produce interchangeable carafes, so the tolerances have to be constrained by a single drawing (an inference: the notice gives no drawing). The notice gives no per-model sales figures, so a unit cost for the recall cannot be calculated. What the label has to warn about is written specifically: do not use a carafe that is cracked or damaged, and do not stir with a metal spoon (CPSC, Remedy). The boundary of use for this product was rewritten in this recall, and the carrier was a sticker. What was learned afterwards: the time distribution of the 70 reports, the severity grading of the 48 burns, the per-model sales split, the aftermarket volume of replacement carafes, none of which appears in the notice. In this case that column is empty, empty enough that it has to be said out loud.

> **Stop here. Decide before you turn the page.**

> You are in that seat now, with the materials above in front of you.
>
> 1. What would you do? Write the action, not "pay more attention".
> 2. Which item here are you most likely to miss, and why does it look unimportant?
> 3. If you may set one checkpoint, when is it and what does it check?

What happened: 5.9 million units recalled, number 27-055. Seventy breakages, 48 of them with burns, seven in ten (this book's arithmetic on the notice's fields). The injury type is single: hot liquid escaping at the moment of breakage. Two more divisions on the same fields give about 113,000 units a year across fifty-two years and 5.9 million, and about 1.2 reports per 100,000 units on 70 reports against 5.9 million (ibid.). The recall covers stock sold between 1974 and October 2026, so fifty-two years of product are handled by a single notice, and the scope fits in one sentence because the whole line has one interface. Reviewing the three questions: the most common answer to the first is "recall and replace the carafe", which is not among this case's options, since the two objects were a spoon and a label, and the cost of that action cannot be calculated from the notice because it was never done. The item most easily missed in the second is the warning already on the carafe, which proves that reminding the user has been the chosen route for decades while reports still accumulated to 70. The checkpoint in the third belongs to the crossing of material and method of use rather than to the user's hands. Ask whether the carafe can take a metal spoon, rather than whether users will stir with one.

Options ruled out: change the material, to borosilicate or tempered glass, which the notice does not assess at all, and whose inferable cost is recutting the tooling and rebuilding the spare-part line at both production sites in step. Change the carafe, meaning

recall every carafe, which the remedy does not include, and the logistics and breakage of 5.9 million glass carafes would be a new account on its own. Refund or replace the machine, where the Remedy field says Repair and the company statement lists only a spoon and a label, and whole-unit recovery would be out of proportion to a $15 to $65 price point (an inference). Change the structure, thickening the wall or altering the rim, which the notice does not record, and altering the rim would mean altering the frame and the inner lid, which means altering eight models. Remove the metal spoon from the use case, which was adopted, and it is the only one that was.

Early signals: ① the warning text etched on the carafe (CPSC, Description). The damage mechanism was already known, and the treatment was to put it into the tooling, so every unit leaves the plant carrying the same reminder. Overridden because writing it down was taken as warning. ② the hazard description treating a metal spoon as a normal use action (CPSC, Hazard). Stirring is a necessary step with a French press, and implement and vessel have always been paired this way on this line. Overridden because a usage problem was not treated as a product problem. ③ eight models sharing one interface, so coverage of a single-point defect equals the whole line. Overridden because interface uniformity was a cost advantage. ④ the fifty-two-year sales window (CPSC, Sold At). Any year could have put the material on the table, and no year left a record. Overridden because selling well was taken as validation. ⑤ 70 reports, 48 of them burns, with one injury type repeating. Overridden as low incidence, in the company's own words: the rate of reported incidents involving carafe breakage during use is extremely low.

The point of escalation: products shipped after 29 October 2026 include the plastic spoon and the warning label (Bodum statement). The spoon is this case's escalation point. From that day the

combination of glass carafe and metal spoon moves in the file from pending assessment to mitigated, and the whole line acquires closure eligibility. The cost is one injection moulding run and one label print. What it buys is a treatment record that can be cited, and anyone opening the page afterwards sees that remedial action was taken. The spoon has a second effect: shipping the remedy with the product writes the recall into the standard configuration, so every later shipment repeats the sentence that this problem has been dealt with. The word the company attaches to the spoon is Bodum-designed (original: "a free Bodum-designed plastic stirring spoon"), which makes it a new part in the product line rather than an emergency item.

Failure chain: technical layer: the glass carafe cracks under point load, the crack propagates in use until the carafe breaks, and hot liquid escapes. Glass is brittle, the crack sits in the wall where the user cannot see it, and invisible damage leaves care in use as the only enforceable line. The hazard wording uses "including", which makes the metal spoon one known path of damage and leaves the others unlisted. Of three variables, glass, point load and heat, the remedy touched only the middle one, and touched it by changing the tool. Material grade, wall thickness and thermal tolerance are not in the notice. Organizational layer: eight models share one interface, so the defect covers the line horizontally; the constraints on use are etched on the product with enforcement handed to the user; the recall's category is Repair, putting material and structure outside the remedy; the unit of identification is the interface, so any material change would need a validation covering eight models at once, and the notice records no such validation system; the notice records no design review and no failure analysis. Incentive layer: 5.9 million units in the field, $15 to $65 price points, two production sites and fifty-two years of channel relationships. Changing the material means recutting tooling and rebuilding the spare-part line; shipping a spoon means one

moulding run. The break sits between the organizational and in - centive layers: the cost structure of the remedy determines that it can only land on behaviour, never on material.

Minimum counterfactual: before the remedy was fixed on 29 Oc - tober 2026, put carafe resistance to point load from a metal spoon onto the remedy options list, quoted in the same table as shipping a spoon. One action, costing one point-load test and one supplier quotation. It commits to no material change and asks only that both options sit in one table. It depends on no after-the-fact in - formation either, since the test uses the very action already written into the hazard description. The effect is on the agenda: the table gains a column that wants a number, and whether or not the column can be filled, its presence turns shipping a spoon from the only option into one option. The notice records no such item, and nothing in the verifiable content of the remedy shows a trace of one.

Outcome and sources: on 29 October 2026 the CPSC announced voluntary recall 27-055: 5.9 million units, 70 breakages, 48 burns, remedy repair, being one plastic stirring spoon and one warning label; manufactured in Portugal and Thailand; importer Bodum USA, Inc. of New York. The notice does not say whether the recall runs in batches, whether it covers every carafe in the field, or how replacement carafes will be supplied. This document is a recall notice, not an investigation report: no failure analysis, no test data, no internal documents. This chapter can write only what the no - tice' s format holds and marks everything else as not stated. Primary source: U.S. Consumer Product Safety Commission, "Bodum USA Recalls 5.9 Million French Press Coffee Makers with Replaceable Glass Carafes Due to Risk of Serious Burn Injury", Recall No. 27-055, 29 October 2026. The company statement is quoted in full from Daily Coffee News, 29 October 2026, a second -

ary source used only for the company's own words. The link is in Appendix B under mn-006.

Case 3 (Aviation): The Design Inheritance of a Two-Stroke Engine

av-029. Ultralight aircraft fitted with ROTAX two-stroke engines, spark plugs facing down. The Spanish aviation safety agency (AESA) safety bulletin BIS 01/24, published 8 October 2024 and read as a PDF for this book, reports that the lubrication system on this engine type can be installed incorrectly, leading to multiple engine failures. The bulletin runs to two pages and carries no individual casualty figures (no public quantification). What it records is a design inheritance: the spark-plug-down two-stroke configuration has been passed down through generations of types in the ultra‑light field, the lubrication system's installation sensitivity travels with the configuration from generation to generation, and each generation of users faces only its own aircraft.

Stop at the decision point: the moment at each certification or modification approval of a new aircraft of the same configuration when the prior notices on lubrication system installation errors are or are not retrieved. On the table: the certification requirement items, the archive of prior notices (AESA issued a consolidated bulletin only in 2024), and the individual confidence that "my in‑stallation is fine". Outcome: AESA issues the bulletin to alert the industry, the bulletin carries no binding force, and individual handling depends on the operator. Primary source: AESA, BIS 01/24, 2024; link at Appendix B, av-029.

The Evidence Gap

There is a forensic difficulty specific to deep cases of lock-in: lock-in usually produces no accident report, since its cost is opportunity

cost and not injury, and this book's strong cases therefore cluster in public infrastructure and long-lived consumer goods. Primary documents on technical standards fights, such as the historical battles over electricity distribution and videotape formats, sit mostly in the archives of the competing parties. The gap is marked as it stands; directions for further collection are dispute archives held by standards bodies and switching-cost arguments made in antitrust review.

Beside the gap, this chapter offers four derivations, marked as mechanism derivations and not case cards. Derivation one, a software stack: framework and cloud choices take root in team skills, the CI pipeline and legacy code, and the migration cost curve rises faster than linearly with the age of the codebase; the check form for criterion ② is a re-estimate of migration cost, priced on paper every two years. Derivation two, a medical device ecosystem: a hospital's equipment choice grows compatibility roots in consumable specifications, the training system, the maintenance contract and the electronic record interface, and a single-brand ward ecosystem makes changing brand equal to rebuilding the ward; the check form for criterion ① is the alternatives item in the procurement review, since several bidders does not equal an alternatives assessment, the bids being for models inside the same ecosystem. Derivation three, a city and its transport: expanding the road network reinforces car dependence, through induced demand, since wider roads produce more driving, and the feasible space for transit and for walking and cycling contracts with it; the check form for criterion ③ is the annual curve of the active travel share in the transport budget. Derivation four, an organizational process: ERP and process shape each other, with the process set up for the software and the software written for the process, so changing ERP means changing the process, which means changing organizational habit; the check form for criterion ② is the why paragraph in the process document, Chapter 30's Device 1 third

paragraph, where a process that cannot say why it exists is usually a process shaped by software. The four derivations run from a codebase to a city and share one chain of criteria; their promotion into case cards is on the collection list.

The general solution across the four is a design principle that works at every scale: build a translation layer into the contract and the architecture of a new selection, with data formats, interface protocols and skill requirements all written so that they can be translated, and changing supplier costs whatever the translation layer costs. The translation layer is the valve on the depth of the lock. A lock without one is welded in; with one, it is plug-connec - ted. Its price is a complexity premium in the design, and its return appears on the quotation for every future replacement. The only moment to build it in is at selection, since adding it later means rewriting, which ties it to Device 3, exit terms up front, in the same signing: the translation layer handles the technical side and the exit terms the commercial side, one soft and one hard, and both of them live in the drawer on signing day.

34.4 What Shows Up Earliest

Six weak signals, each with the reason that usually overrides it.

1. No alternatives item on the agenda of the selection meeting. Reason given: “there is nothing to choose, so there is nothing to discuss.”
2. A claim about migration cost that quotes a figure from years ago. Reason given: “we assessed it at the time.”
3. A new project defaults to the old selection and nobody asks for a case. Reason given: “consistency.”
4. Attrition among suppliers or among the technical community, in people and in compatible parts, begins to show and nothing is done. Reason given: “market fluctuation.”

5. The organization shows irritation at the hypothetical "if we started from zero". Reason given: "a pointless question."
6. Defects in the locked object are systematically redescribed as characteristics or as industry practice. Reason given: "being practical."

Now put the six back against this chapter's cases and derivations. Signal one, the disappearing alternative, belongs to case one: by HS2's review seasons, a network-wide alternative was on nobody's agenda. Signal two, the old number, belongs to derivation one: the migration quote is older than the codebase. Signal three, the default, belongs to case two: fifty-two years of carafe material and never a serious quotation. Signal four, unattended attrition, belongs to derivation four: the loss of ERP-ecosystem people and compatible parts is a late signal of a lock. Signal five, irritation at the from-zero question, is the standing condition of case three's design inheritance in industry discussion, in the form of "my installation is fine". Signal six, defect as characteristic, belongs to case two: across fifty-two years of a fragile carafe, care with glass was the product's personality rather than its defect.

Reorder the six by the cost of evidence. Signals one and three are cheapest, read from meeting agendas and from the defaults in a project charter. Signals two and six are middling, read from the age of an estimate and from the wording of a text. Signals four and five are dearest, needing staff turnover data and observation of meetings. The recommended start is signal two: collect every claim in the organization that something cannot be afforded to switch, a single all-hands transcript will do, and mark the year of the estimate each one cites. Any claim citing an estimate more than five years old goes into Device 2's re-estimation queue. The length of that queue is the map of the lock's stock, and every coordinate on

the map deserves a twenty-minute comparison (the action in 34.7, question five).

34.5 How to Defend

Device 1: the from-zero derivation. Once a year, take the core selections and run them on paper: if we started from zero today, what would we choose, and what would the difference cost. The derivation commits to nothing. It exists to keep the assessment function from atrophying, and its output is an annual difference list, whose rate of growth is the rate at which the lock is deepening, or at which the case for staying is improving.

Device 2: migration cost re-estimated. Re-estimate migration cost every two years and record it: which costs have amortized, which technologies have made migration easier. Most claims that something cannot be afforded turn, after re-estimation, into claims that it can be afforded but would be a nuisance, and the space for a decision sits between nuisance and impossibility.

Device 3: exit and compatibility terms up front. Write a ceiling on exit cost and an openness requirement for data and interfaces into the contract and the architecture of every new selection. The depth of a lock is priced on signing day, and half a page of terms beforehand beats a feasibility study ten years later.

Device 4: an alternative archive. Keep a living file of the outside world for each key selection, one page per alternative technology per year, covering maturity, price and adopters. The reader of that file is not the decision meeting. It is your future self. What stands against lock-in is the knowledge that replacement exists, rather than replacement itself.

The four devices on this chapter's cases and derivations. Device 1 on case one: at the feasibility stage of a large project, run the net-

work-wide from-zero derivation as a fixed step, with the difference list against the actual scheme entering the public parliamentary record. Device 2 on derivation one: refresh the technology migration quote every two years, and set the refreshed quote against the opportunity cost list, which is Device 4's data. Device 3 on case two: write interface openness into the spare-part ecosystem contracts, as a compatibility requirement for third-party carafes, and give the brand's tooling design a tolerance for material change. Device 4 on case three: have the community archive of the design inheritance carry the safety record of alternative configurations, one page a year on a lubrication system alternative.

The failure modes of the four. The from-zero derivation dies of hostility, since it gets read as an attack on an existing decision and the person running it gets labelled the naysayer; the patch is authorization and indemnity, making the derivation a routine annual action with the author's identity protected or rotated, from the same source as the protection in Chapter 19. Migration re-estimation dies of distorted quotation, since a supplier quotes a migration price designed to avoid the job; the patch is multiple quotations and staged pricing, quoting a migration of one tenth first to find the real unit price. Exit terms die of unmeasured terms, where a ceiling on exit cost is written and the way to calculate the ceiling is not; the patch is a measurement annex, listing the components of exit cost and how each is priced, attached to the contract, the same pattern as the clause template in Chapter 27. The alternative archive dies of being one person's, maintained by a single pair of hands and a single body of knowledge, so the file dies when the person leaves; the patch is a rota, with a different author each year, and the rotation of authorship is what makes several people familiar with the file. The four patches share one check sentence: when was this device's output last cited in a decision.

The psychology of the derivation carries one more obstacle that deserves separate handling. A from-zero derivation often confirms the status quo: run the exercise, and the answer is that we would still choose this. The confirmation makes the derivation look superfluous, and the appearance has to be taken apart. A confirming derivation is not wasted work, because it turns continuation from a default into a conclusion, and continuation renewed every year is a different thing from continuation by habit a decade old. The difference between a conclusion and a habit sits in the accountability structure. A habit has no signatory, since nobody knows on what day it was decided. A conclusion has one, and this year's derivation report carries a name. Signed continuation and unsigned continuation are treated very differently in an accident review: the first is a tested choice and the second is an unclaimed inheritance. The annual cost of the derivation is one meeting and one sheet of paper, and what it buys is the signature on every core selection the organization holds. That is the best value among this chapter's four devices.

34.6 Boundaries with Adjacent Mechanisms

Against sunk cost and escalation (Chapter 27): lock-in explains the structure, why switching is not possible, and escalation explains the behaviour, why more keeps going in. HS2 occupies both chapters. Chapter 27 handles the decision to continue at each reset, and this chapter handles the fact that continuing had long since stopped being a choice between options.

Against the slack of success (Chapter 20): slack shrinks defensive spending and lock-in shrinks assessment spending, and both are by-products of an absence of bad news. Chapter 20's output ledger and this chapter's alternative archive are two accounts in one system of record.

On the rationality of path dependence, one concession has to be made plainly: lock-in has returns, in ecosystem stability, in the efficiency of deep optimization, in the free-of-inspection cost of compatibility, and question group 34 in Chapter 37 asks specifically for those returns to be booked. The defences in this chapter promise no release from the lock. They promise to keep the option to leave visible.

34.7 Five Self-Checks

1. When did an alternative last appear on the agenda for your organization's most important selection?
2. Which year's figure does your "we can't afford to switch" estimate come from, and has it ever been re-estimated?
3. How much of your own skill set is bound to one platform or one technology? Can you tell the part that is an asset from the part that is a lock?
4. Has a from-zero derivation, once a year and on paper, ever been run in your organization? If not, who is preventing it?
5. Of the things you use because they are fine, which one have you not seriously compared in five years? Twenty minutes today is worth whatever the answer turns out to be.

Aviation | av

> Five columns on this page: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn. Case IDs point to Appendix B.

1 Record Quality

Aviation keeps the most complete failure record of any domain in the book; all four devices of Chapter 3 are present. What follows is where the primary material sits and how far it can be checked. The retrieval mechanics are in.

United States, NTSB. Every accident carries an identifier, DCA21-FA085 for instance. Formal reports in the AAR, AIR and AAB series are published as full-text PDFs, and the CAROL database covers the whole set of accident and incident records. One note from testing the channels: the NTSB's HTML pages carry anti-scraping limits, while the direct PDF links and the data interfaces answer normally. That is what decides the right path for automated collection. Locate the number in the database, then pull the PDF from the direct link.

United Kingdom, AAIB. Formal reports carrying AAR numbers, and general investigation reports, sit in full on gov.uk, in HTML and PDF together, with nothing in the way of collection. The list pages run newest first and cover every published report.

Japan, JTSB. Two tracks, the final report and the 経過報告 (progress report), full text in Japanese, hosted under the Ministry of Land, Infrastructure, Transport and Tourism domain. The www.jtsb.go.jp domain does not resolve; the working entry is the mlit.go.jp path. The progress-report institution deserves attention. On a major accident, instalments come out before the final report exists, and the

Haneda collision (av-037) was still at the progress-report stage almost two years after the event.

France BEA, Germany BFU, Italy ANSV, Spain AESA. Full text in French, German, Italian and Spanish PDFs. The depth is there. The language is where the cost lands.

CAAC. Reports and progress bulletins come out inside the ICAO Annex 13 frame, the MU5735 bulletins being one example (av-038). The official channel is usable, and how much of the text is published moves with the tier of the case.

Overall: the world's highest density of Grade A material, and the non-English material is as deep as the English. Below the reporting threshold, voluntary systems (NASA ASRS among them) fill part of the gap, and the skew in a voluntary sample has to be handled the way Chapter 2 handles it.

2 Feedback Delay

Aviation shows two extremes of delay at once.

Design and management decisions return their feedback in years. Lengthen a maintenance interval, issue an airworthiness directive, rewrite a training syllabus, and the consequences need years, sometimes hundreds of thousands of flight hours, before they ap‑pear. The lubrication-interval extension on Alaska Airlines 261 (av-034) ran from approval to consequence on a scale of years. The 737-9 door plug (av-030) took about eight months from the final-assembly work to the event, and from the standpoint of the design certification system the delay is counted in decades. So the people deciding at the organizational and incentive layers almost always meet the consequences of their own decisions after their own ten‑ure has ended. That is the soil Chapter 28, horizon mismatch, grows in.

Crew decisions return their feedback in seconds. A takeoff, an approach, a go-around: the effect cashes in immediately. Because the delay is so short at this layer, the evidence chain for crew error is the most complete and the most reproducible in the book. An NTSB brief final report, five pages or so, will usually lay out the whole operational chain.

Between the two ends sits a middle band. The maintenance front line runs in weeks. The company safety management system runs in quarters. A reader borrowing aviation practice should fix which band they sit in before borrowing anything. Borrow interval management and the validation has to be planned over years. Borrow the checklist and it can be validated over weeks.

3 Coupling

Coupling inside one airframe is pressed down by design to a very low level: redundant systems, fail-safe design, segregation requirements. Seventy years of institutional spending produced that. Hidden coupling in Perrow's sense is controlled at the level of the single aircraft.

Coupling across aircraft is where aviation's real risk sits, and it takes the form of common-cause failure. Three diffusion paths. The same fleet: the worldwide grounding after the 787 battery events (av-002), the grounding of every 737-9 for inspection after the door plug left its frame (av-030). The same maintenance and management system: the interval extension on Alaska 261 applied across the fleet (av-034). The same training and operating culture: the automation dependence on Asiana 214 carried features common to an airline and to a region's training culture (av-031). Isolation instruments exist and are used as institutions, among them grounding orders, airworthiness directives and survey-style fleet inspections. Diffusion is usually cut off inside days.

The conclusion to carry into other industries is not that coupling here is low. It is high coupling already tamed by institutions, and taming costs what the devices of Chapter 3 cost. An industry whose coupling looks low without a matching institutional spend has probably not seen its coupling yet (Chapter 21).

4 The Three Deadliest Mechanisms

Practical drift (Chapter 15). Aviation carries the densest procedure set in the world, which makes the fork between formal process and actual practice the most instructive one to study. Samples: an air ambulance crew looking down in flight to consult the record book (av-020, the behaviour chain complete); getting out of an R22 with the engine running becoming routine (av-010, where the manufacturer's safety notice had stated the risk years before); a 737 whose takeoff performance was computed for the full runway while the takeoff ran from an intersection, with nobody stepping in (av-028). Every generating condition for practical drift is present here. The rules are written by people who do not fly, the workload is heavy, and a violation almost never carries a consequence.

Margin erosion (Chapter 25). Samples: approved extensions of the horizontal stabiliser lubrication and inspection intervals (av-034, where the cut had a document, an approver, and delivered itself flight by flight); a 747 freighter cargo door left unsealed in rain, letting water reach the electronics bay (av-026, no recheck required of the fairing and the seals). Aviation keeps a ledger tradition for margins. The ledger is held by the operator, and regulatory approval is the legal channel through which a margin gets cut.

Automation and skill decay (Chapter 26). Samples: the autothrottle inadvertently shut off during Asiana 214's visual approach (av-031); a helicopter cleared to a helipad that landed on a runway instead (av-017, no reconfirmation before final approach).

Automation dependence combined with compressed training hours is the failure path this industry has validated most often. The industry's own countermeasures, manual-flying quotas and scenario-based simulator training, are worth taking as a complete set by anyone borrowing from this page.

5 What Outsiders Most Easily Mislearn

Mislearning one: taking the checklist for a culture. An aviation checklist works because three supports sit under it. The person signing has the authority to stop the line. Findings are recorded, and the record is not used to assign blame. Missed checks get caught by a data system, FDM, flight data monitoring. Remove the three and the checklist decays into a signature campaign (Chapter 32, compliant and unsafe). Transplanting the form without the system is the commonest cross-industry mis-transplant there is.

Mislearning two: taking one case's lesson for the mechanism. The safety recommendations at the end of a report address specific conditions; lift one out (say, "improve communication") and you have taken the prescription without seeing the patient. The right use is the mechanism field on the case card. Classify first, meaning decide which chapter the case hangs on, then read Section 5 of that chapter for the countermeasures.

Mislearning three: mistaking aviation's transparency for the industry norm. Aviation can publish full reports because a disaster paid for the institution (Chapter 4). Arguing inside your own organization that aviation publishes, so why can't we, points in the right direction. Do not overestimate how easy the publishing is. Behind every page of a public report stands a statutory power to investigate.

(First draft of this page: 2,325 characters.)

6 Three Questions from Readers Outside the Field

What use is an aviation case to me, when my industry has no airworthiness rules? The use sits in the mechanism, not the reg - ulation. Normalization of deviance (Chapter 11) and margin erosion (Chapter 25) are isomorphic across industries; the regulation is only the parameter.

Aviation’ s accident rate is already low, so is there anything left to learn? The low rate is an institutional output, produced by the four devices of Chapter 3. What you learn is the mechanics of the institution, and copying the clauses is not the same exercise.

Isn’ t this too technical, all that airworthiness vocabulary? The twelve fields of the case card have already flattened the vocabulary into a sequence of actions. The one-sentence event description in field two is written for a reader outside the field, and the technical detail stays in the body text for verification.

The common answer to all three: aviation is the best specimen library of mechanisms in the book because its records are the best (Chapter 3), and for no other reason.

7 Mechanism Distribution at a Glance

The book holds 40 cases in aviation. Distribution by mechanism, largest first:

- Hidden coupling (10): av-001, av-002, av-004, av-007, av-015, av-023, av-026, av-027, av-036, av-040
- The invisibility of maintenance (10): av-001, av-006, av-012, av-013, av-019, av-026, av-029, av-030, av-034, av-035
- Extrapolating past the validated domain (8): av-005, av-014, av-020, av-042, av-025, av-033, av-039, av-040

- Where intuitive expertise breaks down (8): av-008, av-009, av-011, av-014, av-016, av-022, av-023, av-024
- Normalization of deviance (6): av-004, av-005, av-006, av-010, av-028, av-029
- Regulatory lag and capture (5): av-002, av-030, av-034, av-035, av-038
- Practical drift (5): av-003, av-020, av-030, av-032, av-033
- Margin erosion (5): av-003, av-013, av-015, av-025, av-039
- The illusion of redundancy (5): av-007, av-022, av-027, av-028, av-037
- Handover rupture (4): av-009, av-016, av-031, av-037
- Automation and skill decay (4): av-017, av-018, av-031, av-032
- Information attenuation up the hierarchy (1): av-017
- Misaligned incentives (1): av-019
- Sunk cost and escalation (1): av-025
- Compliant and unsafe (1): av-036

How to use the list: when a new accident bulletin from this domain arrives, guess the mechanism against this table before reading. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the domain.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the new Grade A and Grade B reports in aviation and check them against the five columns here and against the distribution in Section 7. A mechanism that has not appeared before gets a new line in Section 4. A shift in frequency gets an updated count in Section 7. A new Grade A case goes into the library under the field standard of Chapter 5.

The trigger, the owner and the review date go into the user's calendar (Chapter 39). Like any ledger, a domain page that is not updated is dead.

9 The Starter Kit

Three items for a reader new to aviation. Entry case: av-001, the parked 787 whose emergency locator transmitter lithium battery caught fire and damaged the aft fuselage (AAIB, 2015, link in Appendix B). Read that one first for the house style and granularity of a report in this domain. Entry data source: the primary libraries listed in the aviation part of. Find the entry points, bookmark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

10 Reservations

Three limits apply to everything judged on this page. The book's case library samples toward events with rich public records, and the record skew is the subject of Chapter 2. The frequency ordering of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quotation. All three limits are the book's general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in Aviation

Read aviation cases through the three-layer model of Chapter 9 and one accent stands out. Breakpoints at the organizational and in - centive layers rarely appear alone; they almost always arrive to - gether with a missing signal at the technical layer. The reason is the density of the four recording devices (Chapter 3). A technical signal nearly always leaves a trace, and the failure sits in whether the organization translated it, which is breakpoint one in Chapter 9. The typical incentive-layer form is the tension between a busi - ness model paid per departure and the redundancy that safety needs (Chapter 17, av-032), and that line is tightest in the regional carrier system. For a reviewer: on the nine-cell walkthrough of an aviation case, the incentive cell most often needs to be traced back to the day the commercial model was signed, which is the longest retrospective reach of any intervention.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
av-001	Chapter 21 Hidden Coupling	Parked 787: emergency locator transmitter lithium battery fire damages the aft fuselage
av-002	Chapter 31 Regulatory Lag and Capture	787 on stand: auxiliary power unit lithium battery fire and smoke
av-003	Chapter 15 Practical Drift	S-92 helicopter accident on the Plymouth hospital site
av-004	Chapter 21 Hidden Coupling	Sightseeing aircraft on approach strikes an undeclared kite line, pitot fails

Case ID	Mechanism chapter	One line
av-005	Chapter 23 Extrapolating Past the Validated Domain	R44 low-G manoeuvre: hub strikes the mast, aircraft breaks up in flight
av-006	Chapter 29 The Invisibility of Maintenance	777 right engine fan blade fatigue fracture, nacelle burns, aircraft returns
av-007	Chapter 21 Hidden Coupling	747 freighter tyre fails in the taxi, dual hydraulic failure after takeoff, return
av-008	Chapter 14 Where Intuitive Expertise Breaks Down	Student solo applies brakes on the landing roll, overruns into a fence
av-009	Chapter 14 Where Intuitive Expertise Breaks Down	Helicopter blade strikes a fuel-shed roof in the taxi, tail boom separates
av-010	Chapter 11 Normalization of Deviance	Pilot leaves with the engine running, helicopter goes out of control and rolls over
av-011	Chapter 14 Where Intuitive Expertise Breaks Down	Light aircraft bounces on landing, swings, right wing strikes the ground
av-012	Chapter 29 The Invisibility of Maintenance	Logging helicopter loses power on the external load and ends up inverted
av-013	Chapter 25 Margin Erosion	Training flight loses engine power twice, ditches at sea, aircraft destroyed
av-014	Chapter 23 Extrapolating Past the Validated Domain	VFR light aircraft enters cloud, spatial disorientation, uncontrolled descent
av-015	Chapter 25 Margin Erosion	Sightseeing aircraft approaches too low, strikes power lines, ground fire

Case ID	Mechanism chapter	One line
av-016	Chapter 30 Handover Rupture	Glider towed too fast releases, low forced landing into trees
av-017	Chapter 26 Automation and Skill Decay	Helicopter cleared for the helipad lands on the runway
av-018	Chapter 26 Automation and Skill Decay	Sightseeing aircraft goes around and comes down beyond the airport boundary
av-019	Chapter 17 Misaligned Incentives	GA helicopter with a hydraulic failure ferried without authority, crashes after takeoff
av-020	Chapter 15 Practical Drift	Air ambulance crew looks down at the record book, hits trees, forced landing
av-021	Chapter 6 What Was Visible Then (walkthrough)	Helicopter takes off at night in fog, comes down and burns
av-042	Chapter 23 Extrapolating Past the Validated Domain	Hunter T7 display manoeuvre fails, aircraft hits a road, 11 people on the ground killed
av-022	Chapter 22 The Illusion of Redundancy	Helicopter and floatplane collide at about 1,000 feet
av-023	Chapter 14 Where Intuitive Expertise Breaks Down	Air ambulance loses control after the hover and dives into a road
av-024	Chapter 14 Where Intuitive Expertise Breaks Down	Sightseeing helicopter hard-lands and rolls over on approach
av-025	Chapter 23 Extrapolating Past the Validated Domain	Glider tug goes out of control after takeoff and hits the ground

Case ID	Mechanism chapter	One line
av-026	Chapter 21 Hidden Coupling	747 freighter L1 door left open in rain, water ingress, false airspeed in flight
av-027	Chapter 21 Hidden Coupling	767 freighter aileron control jams in the climb
av-028	Chapter 11 Normalization of Deviance	737 performance computed for the full runway, takeoff made from an intersection
av-029	Chapter 29 The Invisibility of Maintenance	ROTAX two-stroke oil system assembled wrong, a series of failures follows
av-030	Chapter 29 The Invisibility of Maintenance	737-9 door plug separates in the climb, rapid depressurisation, return
av-031	Chapter 26 Automation and Skill Decay	Visual approach drops below the glidepath and hits the seawall
av-032	Chapter 15 Practical Drift	Night approach: stall warning, wrong control input, stall into houses
av-033	Chapter 15 Practical Drift	Rudder over-controlled after takeoff, vertical tail separates, crash in a built-up area
av-034	Chapter 29 The Invisibility of Maintenance	Horizontal stabiliser jackscrew nut wears through, pitch control lost, sea impact
av-035	Chapter 29 The Invisibility of Maintenance	Cargo hold oxygen generator fire, aircraft comes down in a swamp
av-036	Chapter 32 Compliant and Unsafe	Centre wing tank explodes, 747 breaks up in flight and falls into the sea
av-037	Chapter 30 Handover Rupture	Haneda runway: JAL A350 and a Japan Coast Guard aircraft collide and burn

Case ID	Mechanism chapter	One line
av-038	Chapter 31 Regulatory Lag and Capture	737-800 in cruise enters a steep descent and crashes
av-039	Chapter 23 Extrapolating Past the Validated Domain	Aerobatic display hits a road, 11 people on the ground killed
av-040	Chapter 21 Hidden Coupling	Helicopter lifts from the centre circle of a stadium, loses control, ground fire

How to use the table: on a news item or a bulletin from this domain, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

The last word for aviation's governing layer. The four devices are thick enough already. What this domain has left to improve sits at the execution end rather than the recording end: tracking whether safety recommendations actually land (the implementation-rate logic of Chapter 19), and the real effect of training (the manual-flying quota of Chapter 26). Those two are where the next stage of spending pays. The value of these cases does not lie in telling aviation how to work, since aviation is more professional at this than the book is. It lies in letting other industries take away the mechanics: the four devices of Chapter 3, the protection logic of Chapter 19, the freshness regime of Chapter 26. Aviation's position in this book is that of a template rather than a specimen.

Aviation holds the most cases of any domain, 41 in the av series. That reflects where aviation sits in the library: the main ground on which the mechanisms are validated, with all twenty-four mechanisms having at least one aviation sample. A reader who wants to

learn the whole mechanism set from a single domain should pick aviation for coverage. The same coverage has a side effect worth naming. It can leave the impression that failure happens only in industries with a formal investigation system. Read Chapter 2 (calibrating for record skew) after this page, or aviation's transparency will be mistaken for the normal state of the world.

16 Closing Notes on the Aviation Page

This page is the largest of the domain pages because its public material is the thickest. Formal reports from NTSB, AAIB, JTSB, BEA and BFU form a band of evidence that compares across borders, and one mechanism can be evidenced repeatedly, in different chapters of different reports. One warning goes with the material. Thick material breeds aviation exceptionalism, the belief that this industry prevents failure because of resources no one else has. Aviation's actual distinction is a single thing: the institutional density of mandatory reporting and independent investigation (first among the five precedent domains of Chapter 3). Institutional density can be borrowed by other domains. The thickness of the reports is only its output. When you meet any device on this page, ask first where it snags in another industry. Most of the time the answer is institutional density, and that answer can be changed.

Chemical & Process | ch

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

The U.S. Chemical Safety and Hazard Investigation Board (CSB) is the primary benchmark in this domain. It investigates and does not prosecute, publishes reports in full, tracks its recommenda - tions, and attaches an animated reconstruction to each case page (retrieval route in). The UK HSE regime covers occupational and process safety. On the Chinese side, the Ministry of Emergency Management’ s section for reports on especially serious accidents carries the full text of State Council investigation team reports (Qingdao 11·22, Kunshan 8·2 and others), while accidents beneath that grade usually appear as bulletins rather than full reports. The Chinese part of this book’ s chemical sample is layered accord - ingly, with ch-007 and ch-008 as Grade A full texts. Overall: material on major accidents is rich, material on small and medium acci - dents depends on company records, and the public sample leans toward the heavy tail.

2 Feedback Delay

Three layers coexist. At the process layer, loss of control is meas - ured in seconds and minutes (runaway reaction, leaking vapour cloud) and feedback arrives almost at once. At the equipment layer, degradation (corrosion, fatigue, seal ageing) runs in months and years. At the management layer, arrears in maintenance, in classi - fication and in regulatory coverage run in decades, and they settle in a single payment usually shaped like a major accident (the dust

workshop of ch-008, the regulatory gap of ch-002). A chemical de - cision maker faces the whole spectrum of delays at once, and budgets and appraisals naturally favour the fast-feedback layer.

3 Coupling

Coupling inside the plant is high. Pipelines, utilities and tankage interleave in one physical space, so a small fault can amplify along a process route; the relief line in ch-005 became the channel through which an explosion travelled. Coupling across the bound - ary rises with the distance to the community: the evacuation radi - us of a toxic release connects a plant decision straight to public emergency response. Single units are physically segregated by design, and shared utilities (power, nitrogen, steam, the effluent system) are the classic gathering place for dark lines.

4 The Three Deadliest Mechanisms

Hidden coupling (Chapter 21): a relief system tied back into the process (ch-005), and the dark lines that run through shared utilit - ies. Margin erosion (Chapter 25): reactivity testing that never happened, relief capacity sized too small, a process safety man - agement system absent altogether (ch-006). Compliant and unsafe (Chapter 32): after a building’ s hazardous-area classification was entered wrong, every inspection round still passed (ch-008); inside a regulatory gap, several layers of inspection each covered one segment and nobody covered the seams (ch-002). Next in fre - quency: normalization of deviance (Chapter 11, the 32 alarms in ch-001) and information attenuation up the hierarchy (Chapter 16, the reporting chain in ch-007).

5 What Outsiders Most Easily Mislearn

Mislearning one: taking a hazardous chemicals permit for safety management. A permit covers the moment of admission; process safety management covers the whole of operation. The cost difference between the two is a factor of ten.

Mislearning two: taking the CSB animation for the complete truth of an event. The animation is a communication tool. The evidence chain lives in the body of the report and in the case file.

Mislearning three: learning only from the countermeasures to big accidents. Small and medium releases and near misses carry most of the learning available in this domain, and they are exactly what gets recorded least (Chapter 2, record skew). Learning from major accidents alone is training everyday decisions on a tail distribu - tion.

6 Three Questions from Readers Outside the Field

Is a chemical plant explosion really that far from me? The com - mercial equivalents of these mechanisms sit close by. A dust workshop maps onto dust in woodworking and food processing, a runaway reaction onto any exothermic process, a gap in storage and transport onto any warehouse compliance regime. Mechanism has no boundary.

The CSB only covers the United States, so is there anything to learn from Chinese chemical cases? Yes. ch-007 and ch-008 are both Chinese Grade A full-text reports, and a State Council invest - igation team report is complete in structure with its list of ac - countability public. The Chinese chemical record is more usable than most readers assume.

Isn’ t safety spending in chemicals just money burnt? Safety spending is another form of insurance premium. The cost of not

spending settles as an accident; the direct economic loss in ch-008 was 351 million yuan. The arithmetic is in Section 2 of this chapter and in Chapter 29.

7 Mechanism Distribution at a Glance

The book holds 9 cases in chemical and process industries. Distribution by mechanism, largest first:

- The invisibility of maintenance (4): ch-001, ch-005, ch-008, ch-009
- Compliant and unsafe (3): ch-002, ch-004, ch-008
- Practical drift (2): ch-001, ch-007
- Regulatory lag and capture (2): ch-002, ch-009
- Extrapolating past the validated domain (2): ch-003, ch-006
- The illusion of redundancy (2): ch-003, ch-004
- Hidden coupling (2): ch-005, ch-007
- Normalization of deviance (1): ch-001
- Margin erosion (1): ch-006
- Information attenuation up the hierarchy (1): ch-007
- Cascading failure (1): ch-008

How to use the list: when a new accident bulletin from this domain arrives, guess the mechanism against this table before reading. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the domain.

8 How This Page Gets Updated

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9 The Starter Kit

Three items for a reader new to chemical and process industries. Entry case: ch-001, methyl mercaptan leaking into a building with 4 workers killed (U.S. Chemical Safety and Hazard Investigation Board (CSB), 2019, link in Appendix B). Read that one first for the house style and granularity of a report in this domain. Entry data source: the primary libraries listed in the chemical and process industries part of. Find the entry points, bookmark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

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of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quotation. All three limits are the book's general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in This Domain

The accent in chemical cases is that the physical layer is the loudest. Process upsets, dust explosions and vapour clouds present a technical failure at physical scale, in tonnes and megawatts. Two review traps follow. The drama of the physical layer absorbs all the attention, so breakpoints at the organizational and incentive layers get mentioned in passing; the misclassified building in ch-008 and the absent process safety management system in ch-006 both sit at the organizational or incentive layer. Public and press attention then goes to whose operating error it was, while the value the CSB adds is precisely in pushing attribution up to structure. For a reviewer: on the nine-cell walkthrough of a chemical case, the evidence in the incentive row has to reach back to the process selection and contract structure of the plant's construction phase. The retrospective reach is the longest on the page, and the return is the largest.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
ch-001	Chapter 11 Normalization of Deviance	Methyl mercaptan leaks into a building, 4 workers killed

Case ID	Mechanism chapter	One line
ch-002	Chapter 31 Regulatory Lag and Capture	Ammonium nitrate store explodes, 12 volunteer firefighters and 2 members of the public killed
ch-003	Chapter 23 Extrapolating Past the Validated Domain	Hurricane flooding cuts power, organic peroxide self-heats, a chain of explosions follows
ch-004	Chapter 22 The Illusion of Redundancy	Combustible gas flows back into the cracker furnace, explosion, 95 million dollars of property damage
ch-005	Chapter 29 The Invisibility of Maintenance	Work light left behind in maintenance punctures a rupture disk, explosion and ethylene oxide release
ch-006	Chapter 23 Extrapolating Past the Validated Domain	Reactor runs away and decomposes, explosion, vessel fragments thrown off site
ch-007	Chapter 16 Information Attenuation up the Hierarchy	Pipeline leak sends crude into a storm drain, ignition during the response, explosion
ch-008	Chapter 29 The Invisibility of Maintenance	Aluminium dust collection system explodes, the whole workshop goes up with it
ch-009	Chapter 31 Regulatory Lag and Capture	Low-pressure vessel leaks, vapour cloud explodes

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

The lessons of this domain compress into one sentence: the phys - ical world does not negotiate over its margin. Reactivity testing cannot be skipped, relief design cannot be sized on an estimate, a misclassified building has to be corrected on purpose. Each of those is an entry in the margin ledger of Chapter 25, and chemical margins are the most physical in the book. No software rolls them back and no announcement recalls them. After the signature and the startup, the reactor tied to the grid and the tank roof sealed, the only move left is to have worked through the decision table of Chapter 7 before the irreversible point. A chemical reader who prints that table and tapes it up in the control room has taken the most concrete deliverable this chapter offers.

Nine cases is a small number, and the density of lessons in them is among the highest in the book: 97 dead, 62 dead, 14 dead, 4 dead, 2 dead. Chemical Grade A cases almost all carry fatalities. That is not evidence that chemistry is more dangerous. It is the filing threshold of the domain filtering out the light accidents (Chapter 2). The lessons of a small chemical accident settle inside the com - pany, where no outsider sees them, and that is the shape of the re - cord skew here. What you are reading is the part of the iceberg above water. For the volume below it, consider that the 5 recom - mendations of ch-006 were rated unacceptable: a company can refuse even a CSB recommendation, so the internal lessons of a small accident travel outward even less.

14 How the Mechanisms Vary in This Domain

The chemical variation on the invisibility of maintenance. Main - tenance arrears in this domain come with a physical amplifier built in. Read the general criteria in the local language: no output measure for maintenance reads as “the turnaround window yields

to the production schedule" (the exhaust fan in ch-001 went unrepaired for months, on the grounds that a shutdown loss counted against the appraisal); budget pressure reads as "the utility refurbishment slips down the schedule" (the dust collection system in ch-008). One criterion is specific to this domain: verification of what a turnaround delivered. The work light left in the explosion-proof area (ch-005) sits outside the general weak-signal list, and the local list needs an added line on whether the closeout of a turnaround has a record of leftover items. The point of the variation: maintenance failure in chemicals does not decay. A year of arrears is paid in tonnes.

The chemical variation on extrapolating past the validated domain. The direction of extrapolation is fixed here: from rated conditions toward boundary conditions. The lesson of ch-003 is that a grid outage was extrapolated into "the backup supply will hold", and a flood level into "the cold store will not be inundated"; every hop of that extrapolation was written into the emergency plan, and no one was responsible for the hop distance. The lesson of ch-006 runs the other way: reaction calorimetry from a small-scale test was extrapolated to a production reactor, and relief capacity was designed on the old data. The reading discipline for this domain: wherever a plan or a design document contains "shall", "as customary" or "with reference to a similar unit", there is an extrapolation point. Register each one (the first device of Chapter 23).

The chemical variation on normalization of deviance. Alarms are the main ground on which normalization happens here. The 32 alarms in ch-001 were acknowledged one by one over several hours, and an acknowledgement costs two seconds on the DCS. The convenience of acknowledging an alarm is itself an accelerator of normalization in this domain. How to read it: sort the alarm list into three classes by response type (an operator must attend, the

process needs adjusting, information only), and a rising share in the information-only class is a weak signal. Reading that needs no specialist knowledge, only a DCS report. The line connects to the three-state register of the first device in Chapter 11: alarm entries get three states and no more, and the fourth state, acknowledged but not actioned, is forbidden.

15 Cross-Reading Suggestions

The highest return comes from reading this page against three others. Against the nuclear page: the two domains share three kinds of site, the tag, the alarm and the overhaul, and the inde - pendent verification device on the nuclear page (the lesson of nu-002) transfers directly into chemical isolation management for turnarounds. Against the manufacturing page: the recall ma - chinery of the mn series and chemical process safety share the language of batch traceability, and the discussion of mandatory standards on the manufacturing page (a variation on Chapter 32, compliant and unsafe) and the misclassification in ch-008 stand as footnotes to one another. Against the public policy page: read ch-007 (Qingdao) alongside the large-project chapter of the po series to see how the mechanisms of emergency decision making stack up on the seam between government and company. All three pairings point at one sentence: chemical failures rarely stay inside chemicals.

Nuclear & Energy | nu

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn. The case library for this domain has not been collected yet, so this page sets the frame first: the institutional picture and the mechanism judgements rest on the institutional analysis of Chapter 3. The empty case slots show up as a wholly empty column in Appendix A and will be filled once the collection is done.

1 Record Quality

Institutional depth here is the greatest of any industry, and the public face is split in two. Mandatory reporting and international sharing: the International Atomic Energy Agency (IAEA) and the OECD Nuclear Energy Agency run the Incident Reporting System (IRS) together, and operating events from member states enter a shared pool; the U.S. NRC requires licensees to submit operating event reports (10 CFR Part 50.73). The other half is closed: the reports and assessments of the World Association of Nuclear Operators (WANO) are proprietary, circulating inside the industry and invisible from outside. What the open world can get is each regulator's own documents, NRC inspection reports and event reports, French ASN notices, with the depth of detail set by the country in question; retrieval routes and the direction for filling the gaps are in.

2 Feedback Delay

The longest in any industry. Design defects and material ageing return their feedback over decades. Operating events return theirs in months, filtered through several layers of review. Organizational

and safety-culture consequences take longer still: the lessons of Chernobyl and Fukushima entered operating practice on a scale of decades. The one short-feedback layer is training and the simulat - or. This domain has almost no fast feedback, and all of its learning depends on institutional memory (Chapter 30).

3 Coupling

Perrow classes nuclear power as the archetype of a system that is high on both interactive complexity and tight coupling: the inter - actions between subsystems exceed anyone’ s full understanding, and failures travel faster than any intervention. Seventy years of defence amount to taming that coupling inside multiple physical barriers and defence in depth. Nuclear is therefore the other ex - ample of high coupling held down by institutions, standing along - side aviation but with different instruments: aviation through in - vestigation, nuclear through physical barriers and redundancy. Coupling between plants is low, since sites are independent. Coupling across generations is high: designs carry over, standards carry over, and lesson sharing depends on international systems.

4 The Three Deadliest Mechanisms (from Institutional Analysis, Case Slots Pending)

The illusion of redundancy (Chapter 22): common-cause failure of multiple defences is the standard shape of a nuclear accident, where common cause means an external event or a systemic design or cultural defect, and the shared-source failure of standby power and emergency systems is the most studied lesson in the field. Handover rupture (Chapter 30): passing operating experience between generations depends on closed industry systems (WANO, INPO), and institutionalizing organizational memory is the lifeline of this domain. Regulatory lag and capture (Chapter 31): licence

extensions for design life and the certification of new reactor types keep a standing tension between the rules and the technology. This column will be revised against actual cases once the collection is done.

5 What Outsiders Most Easily Mislearn

Mislearning one: reading WANO's closed doors as an absence of failure records. The internal record density here may be the highest in the world; it is simply not published. Concluding that nuclear offers nothing to learn is a double misjudgement.

Mislearning two: taking the INES scale for a linear measure of severity. It is logarithmic, and it tracks international public impact more than loss.

Mislearning three: taking the visible layers of defence in depth for the actual line of defence. The common-cause check of Chapter 22 is a life-and-death question in a nuclear setting, and treating it as a compliance exercise is the error.

6 Three Questions from Readers Outside the Field

What use is a nuclear case to an industry that is not nuclear? Nuclear is the archetype of defence in depth and conservative de - cision making. Its redundancy design, its graded event regime and its mechanism for sharing operating experience are all reference frames for other industries (Chapter 3).

The safety record is so good, is there anything left to learn? Pre - cisely because the recording threshold is so low, with even an INES level 0 entered, this domain holds the thickest near-miss library anywhere. The learning value of near misses is the subject of Chapter 2: a low threshold produces earlier lessons rather than more records.

What do the event levels mean? INES runs 0 to 7 on a severity scale, 0 having no safety significance and 7 a major release. Every nuclear case in this book sits between 1 and 3, and the lower levels carry the higher density of lessons: they are numerous, fully de - tailed, and most of them came out of luck rather than defence.

7 Mechanism Distribution at a Glance

The book holds 4 cases in nuclear and energy. Distribution by mechanism, largest first:

- Practical drift (2): nu-001, nu-004
- Handover rupture (2): nu-001, nu-002
- Normalization of deviance (2): nu-002, nu-003
- The illusion of redundancy (1): nu-002
- The invisibility of maintenance (1): nu-003
- Diffusion of responsibility (1): nu-004

How to use the list: when a new event report from this domain ar - rives, guess the mechanism against this table before reading. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the do - main.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the new Grade A and Grade B reports in nuclear and energy and check them against the five columns here and against the distribution in Sec - tion 7. A mechanism that has not appeared before gets a new line in Section 4. A shift in frequency gets an updated count in Section 7. A new Grade A case goes into the library under the field standard

of Chapter 5. The trigger, the owner and the review date go into the user’s calendar (Chapter 39). Like any ledger, a domain page that is not updated is dead.

9 The Starter Kit

Three items for a reader new to nuclear and energy. Entry case: nu-001, a wrong tag during an outage that took a standby diesel out of service past the 24-hour limit (ASNR, the French nuclear safety and radiation protection authority, 2026, link in Appendix B). Read that one first for the house style and granularity of a report in this domain. Entry data source: the primary libraries listed in the nuc - lear and energy part of. Find the entry points, bookmark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

10 Reservations

Three limits apply to everything judged on this page. The book’s case library samples toward events with rich public records, and the record skew is the subject of Chapter 2. The frequency ordering of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quota - tion. All three limits are the book’s general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in This Domain

The accent here is that the organizational layer is the thickest. Procedures, tags, watchkeeping, review levels: the density of organizational defence is the highest of any industry, and the INPO and WANO assessment systems are a complete model of organizational audit. Two consequences follow. A break at the organizational layer alone is very unlikely in a nuclear case, since that layer almost always catches it, so real failure almost always needs the incentive layer to cooperate: a budget cut leaves the organizational gate unrepaired for years, and appraisal pressure makes conservative decision making give way. For a reviewer: on the three-layer walkthrough of a nuclear case, the technical breakpoint usually enters only at the last moment, and before that there is a long rope running through the organizational and incentive layers. The far end of the rope is often a budget meeting ten years back.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
nu-001	Chapter 15 Practical Drift	Wrong tag during an outage takes a standby diesel out of service past the 24-hour limit
nu-002	Chapter 11 Normalization of Deviance	Tagging error defeats the compensatory measures, equipment unavailable for 7 hours 45 minutes
nu-003	Chapter 11 Normalization of Deviance	Laboratory loses natural uranium samples, an unregistered americium-241 sealed source is found

Case ID	Mechanism chapter	One line
nu-004	Chapter 15 Practical Drift	NRC proposes a fine against an industrial radiography company for four safety violations

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

One line for the nuclear reader: this industry has already installed the four devices to their limit (Chapter 3), and the real value of its cases for this book is the view of what residual risk looks like once defence density has hit the ceiling. The residual no longer comes from a missing record or an absent independent investigation, since both are there. It comes from the slow loosening of the or - ganizational and incentive layers (Chapter 25 and Chapter 28). The nuclear case tells the reader that an institution is a starting point, and that keeping it running is the work. That is also why this book treats nuclear cases with the most caution, with an evidence grade and an archival source marked on every one.

Four cases make this one of the thinnest domains in the book, and the thinness is itself a result of success: operating events run into the tens of thousands, major accidents into single digits. The INES grading lets a reader calibrate severity precisely. Level 1, which covers nu-001, nu-002 and nu-003, marks an operating anomaly; level 3 and above is an accident. The value of low-level cases was set out in Chapter 2: they are near misses turned into text. The re - cording regime of this industry converts what other industries leave unrecorded into a written record, and that is this domain’ s

particular contribution to the book. Few cases do not mean few lessons. Lesson density is lessons divided by cases, and nuclear has the highest lesson density of any domain here.

14 How the Mechanisms Vary in This Domain

The nuclear variation on practical drift. Drift here happens almost entirely at the interface between paperwork and action, and the tag is the largest such interface. The two tagging errors in nu-001 and nu-002 share one shape: the logic on the tag sheet was right, the physical match was wrong, the wrong valve or the wrong cabinet was tagged, and the compensatory measures were recorded on paper with nobody going to the site to verify them. How to read it: whether the verification action was independent, a second pair of eyes, is the core detection point for drift in this domain. Independent verification is a written requirement of nuclear safety culture; what varies in practice is the density with which it is executed. The general criterion, that the rule is written so that nobody can follow it, barely holds here, since nuclear has a surplus of rules. The local criterion for drift has to be rewritten as: the rule set is complete and the verification step is omitted by custom.

The nuclear variation on normalization of deviance. Normalization happens mainly on temporary exemptions for unavailable equipment. The technical specifications of a unit set hard limits on how long key equipment may be unavailable (the 24 hours in nu-001), and going past the limit requires a shutdown or a power reduction, so recovery just inside the limit becomes standard practice: the item comes back at hour 23 and 50 minutes, and the next cycle runs the same way. How to read it: take the distribution of recovery times against the limit, meaning the share of recoveries that land between hour 20 and hour 24. A share that is improbably high says the limit is being used as a target rather than a boundary, which is goal displacement (Chapter 12) mirrored into nuclear. The three-

state register needs a fourth column here: how the item was restored, really or on paper.

The nuclear variation on the invisibility of maintenance. The maintenance ledger here is the thickest in any industry; the thin one is the verification ledger. The lesson of nu-003, samples lost and an unregistered sealed source, shows another kind of maintenance: the ledger for sources and samples is kept by hand, the entry depends on an individual researcher's diligence, and the institution has no stock-count mechanism. How to read it: the frequency of a stock reconciliation audit, meaning the three-way check that the ledger has it, the site has it, and nothing is on site that the ledger lacks, is the right indicator for invisible maintenance in this domain. An audit that reads repair records and never reconciles stock is thick and hollow.

15 Cross-Reading Suggestions

The first priority is reading this page against the chemical page: tagging, alarms and overhauls are the same three kinds of site in both, and the three variations in Section 14 of the chemical page can each be converted back into nuclear terms. Against the aerospace page: read the discussion of review culture in mi-001, where the flight rationale goes into the ledger, alongside the operating experience feedback system of nuclear (the WGOF tradition at OECD/NEA), and the different institutional densities with which two high-reliability industries handle precedent evidence become visible. Against the healthcare page: read the piped oxygen supply in me-011 against the utility redundancy here, and the gap between the two industries in the verification density of life-support systems shows up plainly. Healthcare can borrow the whole vocabulary of independent verification from this page.

Healthcare | me

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

The latecomer that built the institution properly is the United Kingdom. The Health Services Safety Investigations Body (HSSIB, formerly HSIB) runs its investigations on the aviation pattern, and its reports carry the event sequence, the safety recommendations, and a national figure to anchor them, such as "roughly 192 wrong-site surgeries nationwide each year". Every me case in this book comes out of that archive. The original site was unreachable from the network used to collect material for this book, so the reports were read through Internet Archive snapshots, and records which snapshot stands behind which case.

The American material is a step removed. AHRQ's Patient Safety Network (PSNet) and Web&M&M publish case commentaries, secondary curation written for teaching, strong on framing and weak as evidence. On the device side the FDA's MAUDE database holds reports in volume, with single-entry quality uneven.

China's health commission bulletins are available only intermittently. The workable substitute is the judgment in a medical damage liability dispute, which supplies a dated sequence, expert opinion, and a finding on causation, at the price of a legal frame wrapped around a clinical event (Chapter 3).

The verdict: full-text investigation is best in the UK, and its narrative depth is unusual because the patient's identity is protected while the whole chain stays legible. The sample per case is small,

which is why the national figure carries as much weight as the report itself.

2 Feedback Delay

Three layers, and they run at wildly different speeds. At the medication and surgical layer feedback is counted in minutes: a wrong site or a wrong concentration shows on the table or by the end of the day. At the process layer it is counted in weeks and months: follow-up that does not happen, results that nobody claims, with the twelve-day reporting delay of me-005 as the type specimen. At the system layer it is counted in years: how far the reconciliation routine has drifted, what the identification-event rate is doing.

What separates healthcare from the rest of the book is the spread inside a single shift. One nurse's day holds second-scale and year-scale risk at the same time, and a reflex trained at one scale misfires at the other.

3 Coupling

Inside a hospital, coupling is moderate. Processes are templated closely enough that a defect replicates isomorphically from ward to ward, and physical separation is comparatively generous, so a bad process does not take the building down with it. Across the industry, coupling is high. A process validated at one hospital gets adopted nationally and the defects travel with it: the medication-error shapes of me-001 through me-003 recur across HSIB's national figures instead of staying local. A national protocol is a single point with national reach. The colour coding of injectable medicines is the standing example; when the coding fails, it fails everywhere at once.

4 The Three Deadliest Mechanisms

Handover rupture (Chapter 30). Healthcare carries the densest handover surface in the book: shift to shift, ward to theatre, hospital to community, road to resuscitation room. me-006 carries an identity error from the prehospital crew into a duplicate record in the emergency department. me-005 leaves an abnormal chest film with no owner for twelve days. Every one of those surfaces is a candidate rupture, and the number of surfaces, not the diligence of the staff, sets the exposure.

Compliant and unsafe (Chapter 32). The paperwork is done while the act is not. me-001's Stop Before You Block was signed off as the block went into the wrong leg. me-002 walked around the concentration coding system with an ordinary choice of syringe.

The illusion of redundancy (Chapter 22). me-003 took out two defences in one event: the purple syringe that was meant to mark the route, and the second pair of eyes that was meant to catch it.

Next in frequency, diffusion of responsibility (Chapter 18). me-008's incompatible implant came to light afterwards through a registry.

5 What Outsiders Most Easily Mislearn

Mislearning one: treating individual accountability as system improvement. Healthcare's review culture has been occupied for decades by the question of who was at fault, and the institutional innovation at HSSIB is the separation of investigation from blame. That structure is the transferable asset. No particular checklist is.

Mislearning two: imagining that a hospital's strict processes are strict in execution. A checklist in use and a check performed are two different states, and the gap between them is where several cases in this book live (Chapter 32).

Mislearning three: undervaluing the national figure. A single hos - pital' s case teaches little on its own; the real contribution of an HSSIB report is a national frequency to divide by. Any lesson learned at one site should be read against that base before it is generalised.

6 Three Questions from Readers Outside the Field

Are the healthcare cases only for clinicians? No, and the reason is structural. Healthcare is the prototype site for decisions taken un - der time pressure, on incomplete information, with a lethal con - sequence attached, and the mechanisms transfer whole. A triage priority conflict and a medication check that breaks at a handover are shapes a trading desk or a control room meets as well.

Will the hierarchy make the cases hard to follow? The twelve fields have already flattened clinical language into actions. A block in - jected into the wrong leg, a chest report that arrives twelve days late: the sequence is readable by anyone. Terminology stays in the source report for those who want it.

Is the blame culture in medicine especially heavy? It varies widely by jurisdiction, and the second-victim problem, the injury done to the clinician who made the error, is a public topic in medicine earlier than anywhere else. The repair work on blame culture started here first, and its lessons travel (Chapter 19).

7 Mechanism Distribution at a Glance

The book holds 16 cases in healthcare. Distribution by mechanism, largest first:

- handover rupture (6): me-003, me-005, me-006, me-010, me-012, me-016

- the illusion of redundancy (5): me-001, me-003, me-004, me-009, me-012
- compliant and unsafe (3): me-001, me-002, me-014
- information attenuation up the hierarchy (3): me-005, me-007, me-010
- hidden coupling (3): me-006, me-008, me-014
- normalization of deviance (2): me-001, me-004
- diffusion of responsibility (2): me-007, me-008
- where intuitive expertise breaks down (2): me-013, me-015
- cascading failure (1): me-002
- practical drift (1): me-009
- crossing a scale threshold (1): me-011
- the invisibility of maintenance (1): me-011
- regulatory lag and capture (1): me-013
- narrative before evidence (1): me-015
- goal displacement; Goodhart’s Law (1): me-016

How to use the list: on any new incident notice from this domain, guess the mechanism from the list before reading the report. The hit rate is your familiarity with the domain’s accent. Then read and check. The places you guessed wrong mark the thickness of industry skin still between you and the field.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the Grade A and Grade B reports added to this domain, walk them against the five columns above and the table in Section 7, and revise. A mechanism that appears for the first time gets a line in Section 4. A shift in frequency updates the counts in Section 7. A new Grade A case goes into the library against the field standard of Chapter 5.

The trigger, the owner, and the review date belong in the user’s calendar (Chapter 39). A domain page that is not updated is dead in the same way a ledger is: the information does not rot, it simply stops describing the field.

9 The Starter Kit

Three items for a reader new to this domain, about one working day in total.

The entry case is me-001, a nerve block for right ankle surgery injected into the left leg, investigated by HSIB (the predecessor of HSSIB) in 2018, with the link in Appendix B. Read it first for the format and the grain of the reports in this field.

The entry source is the first-hand library listed under Healthcare in. Find the entrance, bookmark it, visit once a quarter.

The entry mechanism is the first line of the Section 7 table. Read the mechanism chapter beside the case from this domain and take the pair twice, once for the criterion in the chapter and once for the parameters here. When the three items are done you own a first coordinate system for the field.

10 Reservations

Three limits apply to every judgement on this page. The book’s case library samples toward events with rich public records, which is the record skew of Chapter 2. The frequency ranking in Section 7 describes the book’s library, not the true distribution of the domain. Institutions and technology both move, so this page has a shelf life measured in years, and a citation should carry the year it was read.

All three are the book’s general disclaimer, and the threshold check of Chapter 1 applies to this book as much as to anything it examines.

11 How the Three Layers Sound in This Domain

The accent here is breakpoint one, the place where a technical signal never gets translated by the organizational layer, and it is the most crowded of the three. A clinical signal, whether a symptom, a laboratory value, or an imaging finding, reaches an organizational action such as a consultation, a referral, or a follow-up only through several translations: triage, handover, report routing. Each translation is a high-incidence breakpoint, and me-005’s delayed chest film and me-016’s broken discharge information both sit at one. The incentive layer takes the form of bed occupancy and turnover targets, with rewards going to the teams that move pa - tients fastest, and that tension acts directly on handover quality.

For the reviewer: in a healthcare nine-cell table the organizational layer usually holds the densest evidence, because the record sys - tem timestamps itself, while the technical layer often surfaces only in post-mortem pathology. Getting the timeline right is the whole of the walkthrough.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
me-001	Chapter 32 Compliant and Unsafe	Nerve block for right ankle surgery injected into the left leg
me-002	Chapter 32 Compliant and Unsafe	U-500 insulin drawn up in a U-100 syringe, overdose

Case ID	Mechanism chapter	One line
me-003	Chapter 22 The Illusion of Redundancy	Oral midazolam transferred into an intravenous syringe and injected
me-004	Chapter 11 Normalization of Deviance	Postpartum packing left in place five days, unnoticed
me-005	Chapter 30 Handover Rupture	Abnormal chest film never passed on, cancer diagnosis delayed, patient died
me-006	Chapter 30 Handover Rupture	Prehospital identity error builds a duplicate record in the emergency department
me-007	Chapter 18 Diffusion of Responsibility	MRI cancelled and never rebooked, diagnosis delayed
me-008	Chapter 18 Diffusion of Responsibility	Incompatible implant fitted during hip replacement
me-009	Chapter 15 Practical Drift	Cardiac arrest patient confused with the neighbouring bed, resuscitation delayed
me-010	Chapter 16 Information Attenuation up the Hierarchy	Admission information for a patient with learning difficulties arrives late, condition deteriorates
me-011	Chapter 24 Crossing a Scale Threshold	Hospital piped oxygen supply falls short, the trust declares a major incident
me-012	Chapter 30 Handover Rupture	Cauda equina syndrome missed over repeated attendances, decompression delayed again after diagnosis

Case ID	Mechanism chapter	One line
me-013	Chapter 14 Where Intuitive Expertise Breaks Down	Pulmonary embolism decisions in the emergency department without systematic support
me-014	Chapter 32 Compliant and Unsafe	Morphine oral solution: concentration on the label read as the dose, elderly patient died
me-015	Chapter 14 Where Intuitive Expertise Breaks Down	Aortic dissection recognised late, patient died in transit
me-016	Chapter 30 Handover Rupture	Three-month-old infant deteriorates within four hours of discharge, readmitted, died

How to use the table: on a news item or a bulletin from this domain, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

The last word to the governance layer in healthcare. The HSSIB reports are among the strongest accident analysis texts produced anywhere, and what they share is a method: no blame, and a system diagram instead of a verdict. The largest lesson in this book's healthcare cases belongs to system designers. me-005's ownership of follow-up, me-013's decision support, me-015's triage differential: in each case the defence lands on a structure, not on a person. That is the HSSIB method, and it is the spirit of the Systems Engineering Initiative for Patient Safety. A clinical reader who carries the structural analysis back to his own department has taken the chapter's delivery.

The cases on this page come from independent HSIB investigation, and their common quality is a full reconstruction plus recommendations pitched at system level. The method, with its no-blame frame, its systems analysis, and its publicly tracked recommendations, is the most mature institutional experiment in safety that medicine has run, and these cases show what kind of knowledge product it yields. An HSIB report reads like a postmortem written with the nine-cell table; the three-layer analysis of Chapter 9 is its standard structure. For a medical student the practical use is to read these reports as an advanced form of case teaching. What is learned is systems thinking, and systems thinking is the safety net under clinical skill: the denser the net, the safer the patient.

14 How the Mechanisms Vary in This Domain

Handover rupture at maximum density. One inpatient admission passes through a dozen handovers (prehospital, emergency, ward, imaging, theatre, nursing shift), and every one of them reorganises the information once. me-005 and me-006 stand for the two lesions. In the first, a result has no owner and the loop never closes. In the second, the key is wrong and the record duplicates. The local measure is closure rate rather than record thickness: of the result-type messages sent out, how many were explicitly received and acted on within the window. Count it by department, and put Chapter 30's device wherever the rate falls below the threshold. Rupture in healthcare almost never presents as not knowing. It presents as assuming the other side knew.

Normalization with a moral shell. "For the patient" is the most common licence for compressing a check in this domain. In me-011, where the piped oxygen supply fell short, the emergency itself exempted the verification steps. In me-016, where a three-month-old deteriorated after discharge, discharge criteria applied in bulk absorbed one infant's abnormality into the mean. The

local instrument is a register with two columns, emergency exemption and convention exemption, counting every check skipped under the heading of urgency or of routine. The monthly totals are a thermometer for normalization. Exemptions cannot be abolished, since resuscitation is what it is, but each one must carry a name and a time. Anonymous exemption is the incubator.

Truth-telling against a steeper gradient than any firm's. me-015 and me-016 both contain a junior doubt that was never escalated, and the authority structure of a medical team is what kept it where it was. The only reliable indicator of health here is the dissent record of departmental meetings: who raised a different view on which case, and what happened to it afterwards. An anonymous survey cannot measure a culture. Chapter 19's list of alarms has its healthcare counterpart in one question: is the difficult-case conference actually discussing the case, or endorsing it?

15 Cross-Reading Suggestions

Healthcare against maritime is the most surprising pair in the book. Both run small teams, high pressure, and explicit rank, and the bridge and the theatre turn out to share one problem in attention management: ma-002's alarm masked by music reads beside me-001's checklist degraded to a signature.

Against Drugs and Devices: me-014's misread morphine concentration and the recalls in the ph series meet at the same design question, since the error space of a device label and the error space of a drug label are one space. Chapter 26 deserves conservative handling in this domain; me-013 shows what an absent decision support costs, and it does not follow that decision support is a cure.

Against nuclear and energy: see the cross-reading section on that page.

Drugs and Devices | ph

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

This domain runs on mandatory reporting plus a public database (Chapter 3), and the American first-hand channel is the most complete of the three. Recall and enforcement reports can be pulled programmatically through the openFDA interface, across the drug, device, and food databases, and every ph case in this book was checked against those API records. The device adverse event database (MAUDE) varies in quality entry by entry and makes up for it in volume. An FDA safety communication that carries an investigation timeline can be graded strong.

Europe routes through the EMA and the national agencies. China's NMPA notices, recalls, and annual adverse reaction reports are the standard outlet, and the detail per entry is thinner.

The verdict: rich in signals and thin in review. What this domain records is the event. Why it happened is mostly left out, which is Section 3.5's pattern of signal detection standing in for accident investigation.

2 Feedback Delay

The structural feature of the field is that delays are long. Clinical trial feedback is counted in years. Post-market risk feedback is counted in years too, since an adverse reaction signal needs epidemiological verification between accumulation and confirmation.

Feedback on device fatigue and ageing tracks intensity of use rather than the calendar.

The distribution and use end moves faster: recalls are executed, adverse events are reported. The reporting rate itself is disturbed by how aware the reporter is and how hard the causal link is to judge. The common illusion among decision makers is to read “no signal yet” as “no risk yet”.

3 Coupling

Global distribution means simultaneous global exposure. A device enters several countries through a reprocessing protocol (ph-007). A packaging defect in one component enters hospitals across the United States inside a kit (ph-008).

Regulatory coupling comes next. One licensing decision, or the absence of one (ph-005), is exported along the distribution network to every market that recognises the licence or has no corresponding review of its own.

Production coupling concentrates at sterility assurance. A sterility failure is batch-level and crosses company boundaries (the kit case of Chapter 33).

4 The Three Deadliest Mechanisms

Extrapolating past the validated domain (Chapter 23). A single-use device is reprocessed and reused (ph-007). An unlicensed model reaches distribution (ph-005). A device built for one concentration scale is used on another, which is the same failure as me-002 on the clinical side.

Cascading failure (Chapter 33). The kit model carries a component defect across company lines (ph-008).

Regulatory lag and capture (Chapter 31). Marketing without a li‑cence, reporting that lags by years (two years in ph-004), and new product forms landing outside the old classification.

Next in frequency, normalization of deviance (Chapter 11). The missing sterility assurance in ph-001 stays invisible until the consequence arrives.

5 What Outsiders Most Easily Mislearn

Mislearning one: reading the recall database as a safety ranking. Reporting culture, adjudication criteria, and product volumes are not standardised across firms or countries, so a comparison of recall counts means nothing.

Mislearning two: taking the absence of adverse event reports as evidence of safety. In a field where the reporting rate sits far below the true rate, silence carries no information.

Mislearning three: reading the Class designation as the size of the loss. Class describes an assumption about the severity of the health hazard, not the scale of the damage done. Small volume spread wide and large volume held narrow are different arithmetic en‑tirely: the two units of ph-004 and the 5.9 million pieces of mn-006 sit at the two ends of the same page of the manual.

6 Three Questions from Readers Outside the Field

What does a drug recall have to do with me, when I do not make drugs? Recalls are the most complete textbook of supply chain failure available: one defect travelling from component to finished product, which is ph-008’s kit model. Any industry with a supply chain stands on the same propagation path.

What can a database like MAUDE tell me? The density of risk, not its size. A report count is not an incidence count, and the distinc - tion between signal and evidence in Chapter 3 is compulsory reading before the database is used.

How does marketing without a licence (ph-005) happen? The shape of a gate that fails is the same in every industry: the licence status was never hard-checked inside the system, and the design of the release system is Chapter 31. Drug and device cases display gate failure more clearly than any other material in the book.

7 Mechanism Distribution at a Glance

The book holds 10 cases in drugs and devices. Distribution by mechanism, largest first:

- compliant and unsafe (6): ph-002, ph-004, ph-005, ph-007, ph-009, ph-010
- normalization of deviance (3): ph-001, ph-003, ph-006
- the invisibility of maintenance (2): ph-001, ph-004
- cascading failure (2): ph-008, ph-010
- handover rupture (1): ph-003
- regulatory lag and capture (1): ph-005
- goal displacement; Goodhart’ s Law (1): ph-006
- extrapolating past the validated domain (1): ph-007
- hidden coupling (1): ph-008
- margin erosion (1): ph-009

How to use the list: on any new incident notice from this domain, guess the mechanism from the list before reading the report. The hit rate is your familiarity with the domain’ s accent. Then read and check. The places you guessed wrong mark the thickness of industry skin still between you and the field.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the Grade A and Grade B reports added to this domain, walk them against the five columns above and the table in Section 7, and revise. A mech - anism that appears for the first time gets a line in Section 4. A shift in frequency updates the counts in Section 7. A new Grade A case goes into the library against the field standard of Chapter 5.

The trigger, the owner, and the review date belong in the user’ s calendar (Chapter 39). A domain page that is not updated is dead in the same way a ledger is: the information does not rot, it simply stops describing the field.

9 The Starter Kit

Three items for a reader new to this domain, about one working day in total.

The entry case is ph-001, semaglutide injection recalled for insuf - ficient sterility assurance, FDA, 2025, with the link in Appendix B. Read it first for the format and the grain of the reports in this field.

The entry source is the first-hand library listed under Drugs and Devices in. Find the entrance, bookmark it, visit once a quarter.

The entry mechanism is the first line of the Section 7 table. Read the mechanism chapter beside the case from this domain and take the pair twice, once for the criterion in the chapter and once for the parameters here. When the three items are done you own a first coordinate system for the field.

10 Reservations

Three limits apply to every judgement on this page. The book’ s case library samples toward events with rich public records, which

is the record skew of Chapter 2. The frequency ranking in Section 7 describes the book's library, not the true distribution of the domain. Institutions and technology both move, so this page has a shelf life measured in years, and a citation should carry the year it was read.

All three are the book's general disclaimer, and the threshold check of Chapter 1 applies to this book as much as to anything it examines.

11 How the Three Layers Sound in This Domain

The accent in drugs and devices is that the incentive-layer breakpoint is the one that pays first. Design choice and supply contract are fixed before the product reaches the market, and the failure is repaid years later in the form of a recall: ph-007's reprocessing protocol stretches the original manufacturer's boundary outward, and ph-008's kit model lets a component defect travel between firms. The lookback from an incentive breakpoint here is the longest in the book, running from the recall date back to the date the contract was signed, a distance counted in years, with the evidence scattered between contract files and test records.

For the reviewer: in a nine-cell walkthrough of a drug or device case, the incentive cell should be read for the contract clauses as closely as for the appraisal scheme, and specifically for the weight given to price against the weight given to quality. The lesson of ma-001 applies in this industry as it does in any other.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
ph-001	Chapter 11 Normalization of Deviance	Semaglutide injection recalled for insufficient sterility assurance
ph-002	Chapter 32 Compliant and Unsafe	Camphor massage ointment recalled for CGMP violations
ph-003	Chapter 30 Handover Rupture	Mixed amphetamine salt tablets: wrong strengths packed in one bottle, recalled
ph-004	Chapter 29 The Invisibility of Maintenance	Enoxaparin injection stored and shipped outside its temperature range
ph-005	Chapter 31 Regulatory Lag and Capture	Airway T-piece marketed without 510(k) clearance
ph-006	Chapter 12 Goal Displacement; Goodhart’s Law	Gonorrhoea culture medium fails to detect some strains
ph-007	Chapter 23 Extrapolating Past the Validated Domain	Reprocessed trocars may crack and separate in reuse
ph-008	Chapter 33 Cascading Failure	Sterile packaging breached inside a blood culture kit, barrier lost
ph-009	Chapter 25 Margin Erosion	Oropharyngeal airway may be partly or completely blocked
ph-010	Chapter 33 Cascading Failure	Voids in the Tyvek seal of surgical packs

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

One line for readers inside the industry. The failure chain in drugs and devices runs from a component supplier's packaging line all the way into a patient's vein, and a break at any link ends at the patient: ph-008's kit, ph-005's unlicensed distribution, ph-004's cold chain drift. Every link sits in the register of Chapter 33. The component supplier's name and the post that owns it belong in your risk register, and a supplier who is not on the register is a risk that is not on the register. The last step for a reader here is to open the supplier list and run it through Chapter 33's criteria. That step is available today.

The cases on this page come from the FDA recall and enforcement database, and every API record has been checked for this book. The structured format gives this domain an unusually complete evidence chain, since recall number, product code, distribution scope, and remedy are each a field, which makes these the finest-grained case records in the book. The case count is moderate, and the lessons run the full length from component to shelf: ph-008's kit model is the most complete display of supply chain failure the book contains, and one case read through one supply chain beats ten cases each seen in part. There is a dividend for readers here, because the FDA publishes an open API and a monitoring script is a weekend's work; Chapter 39's automation advice has its most direct technical entry at this point.

14 How the Mechanisms Vary in This Domain

Compliant and unsafe, the drug variation. Almost all ten recalls in the ph series came out of routine compliance inspection and mar - ket surveillance, which means the recall is the system working. The peculiar form here is a timing mismatch: the moment the compliance certificate covers and the period in which the risk is

live do not coincide. In ph-001 the problem sits inside production batches while the inspection happens off site. In ph-010, where the Tyvek seal carries voids, the problem sits in drift of the packaging line parameters while periodic inspection samples by batch.

The local reading: the sampling density of compliance against the coverage needed for process parameter drift is the measure of whether compliance feeling equals safety in this field. A jurisdiction with many recalls is not a jurisdiction with high risk, since recall visibility is the positive form of the record skew in Chapter 2, and this point runs against outside intuition.

Extrapolation, the drug variation. In this domain extrapolation happens on the storage and transport chain. ph-004, where enoxaparin went outside its temperature range, carries the lesson: the conditions covered by the registration studies and the conditions of the real logistics network are two different distributions, the duration of exposure at the excursion point decides the quality outcome, and most quality systems handle an excursion as a binary event rather than accumulating a duration distribution. The reading: a temperature chain report that gives only the number of excursions, without duration and batch linkage, makes the extrapolation risk invisible. Chapter 23's Device 1 has its counterpart here in a table comparing registered conditions against actual logistics conditions, updated once a year.

Maintenance invisibility, the device variation. In devices it shows up as ageing of the reprocessing validation. ph-007's cracked trocars carry the lesson: the reprocessing protocol for a reused device is validated against a design life, and the gap between actual reuse counts and the validated count is tracked by nobody. The reading: set the reprocessing count ledger against the validated count ceiling, since use beyond the ceiling is a weak signal peculiar to this field. Read beside the consumer recalls of the mn series, where mn-006's forty-year-old product cracked, and the general

property becomes visible: a validation has a shelf life. Devices count uses, consumer goods count years.

15 Cross-Reading Suggestions

Drugs and devices against healthcare is the first pairing to take, because the user end and the manufacturing end are two halves of one chain; me-014 and ph-009 meet at the design of a label.

Against manufacturing: the recall regimes of the mn and ph series share the language of CPSC and FDA, and the standards discussion in section 14 of the manufacturing page applies here without modification.

Against software: batch record traceability read beside deployment rollback shows that the two industries put their irreversible point in different places, drugs at the moment of injection and software at the moment of the push, while the management of the valida - tion window before that point is one problem in both.

Construction and Infrastructure | ce

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

The sample for independent investigation of transport infrastructure is the UK's Rail Accident Investigation Branch (RAIB). Reports sit in full on the gov.uk platform, and a formal report carries the maintenance and decision sequence, not just the physics, which is why the ce cases in this book rest mainly on RAIB material. The American benchmark for building failure is NIST's post-disaster investigation, with the Surfside condominium as the reference case and a technical conclusion that took five years to reach. Construction-phase safety in the UK falls to HSE.

On the Chinese side, the Ministry of Housing and Urban-Rural Development and the Ministry of Transport issue mainly bulletins, and the availability of an investigation report for a major bridge or building accident varies from case to case.

The verdict: transport infrastructure has the most thoroughly institutionalised record of the three. Building and municipal work records its events after the fact, in assessment reports, and the record of maintenance decisions during operation is thin. That thinness is Chapter 29 appearing in a building context.

2 Feedback Delay

This domain has the widest spread in the book. At the structural layer, feedback from concrete carbonation, rebar corrosion, and bolt fatigue is counted in decades, as in the fatigue fracture of

ce-002's baseplate bolts. At the construction layer it is counted in hours: a mistake in a lift or a pour shows on the spot.

Operating decisions, such as the platform inspection in ce-007 or the occupation of the recess in ce-003, give their feedback in years, and the normal state in between is no accident at all. The public usually calls the resulting event sudden. Every investigation report in this field spends its pages refuting that word.

3 Coupling

Coupling to the public is higher here than in any other domain, because the asset is public. A railway, a bridge, a level crossing: when one fails it acts directly on people who never chose to be exposed (the train collision in ce-001, the pedestrians at the crossing in ce-008, the collapse of the condominium in ce-009).

Cross-system coupling is dense as well. Civil works, track, signalling, power supply, and communications share one constrained space, and ce-003, where communications cable filled the refuge recess, is coupling in its physical form. Isolation measures exist, in the shape of single-line block closure and structural redundancy. They fail through accumulation, as maintenance and occupation both draw the isolation down (Chapter 25).

4 The Three Deadliest Mechanisms

The invisibility of maintenance (Chapter 29). Surfside's structural debt (ce-009), the bolt failure at Audenshaw that nobody recorded (ce-002), the rulebook at Denbigh Hall that did not cover the scenario in front of it (ce-005).

The illusion of redundancy (Chapter 22). At Talerddig both sanding systems failed, automatic and manual together (ce-001). At Ealing

Broadway the door obstruction detection and the anti-drag measure came up empty in the same event (ce-006).

Diffusion of responsibility (Chapter 18). The level crossing warning sits on an interface between asset owner and operator (ce-008), and facilities held in common by several parties have governance gaps built into their ownership.

Next in frequency, practical drift (Chapter 15). Inspection degrades into a signature in ce-003 and ce-007.

5 What Outsiders Most Easily Mislearn

Mislearning one: reading accident rate as a measure of construc - tion quality. Most infrastructure failure happens in the operating period, which is more than nine tenths of the life cycle, so con - struction quality is the opening chapter of the story. Learning in - frastructure through a construction lens means reading only the title page.

Mislearning two: taking "once in a hundred years" as evidence that protection is adequate. That phrase and the margin arithmetic of Chapter 25 combine into the scenario Chapter 22 exists to de - scribe.

Mislearning three: underestimating the accumulation of permitted occupation. A recess, a space under a bridge, a safety clearance: each temporary occupation looks harmless alone, and together they are the full cause of an event like ce-003. The public sees only the last one.

6 Three Questions from Readers Outside the Field

What use is a building accident to a reader outside construction? Buildings are the prototype of multi-party governance. One build -

ing involves owner, managing agent, designer, contractor, supervisor, and regulator, and the accountability interfaces of Chapter 18 are drawn most completely in building cases. Any industry that coordinates across organisations can read itself into them.

Is a collapse like Surfside a low-probability event? A collapse is the settlement of a debt. Decades of deferred maintenance are cleared in one tide, so what looks like low probability is high-certainty accumulation (Chapter 25).

What has a bridge or a tunnel to do with the management of an office block? The mechanism is the same: structural margin eaten away, inspection reduced to attendance, repairs shelved. Only the scale differs, and the invisibility of maintenance in Chapter 29 holds for any owner of a physical asset.

7 Mechanism Distribution at a Glance

The book holds 9 cases in construction and infrastructure. Distribution by mechanism, largest first:

- the illusion of redundancy (4): ce-001, ce-003, ce-006, ce-008
- compliant and unsafe (4): ce-002, ce-004, ce-007, ce-009
- practical drift (4): ce-003, ce-005, ce-006, ce-007
- the invisibility of maintenance (3): ce-001, ce-002, ce-009
- information attenuation up the hierarchy (2): ce-002, ce-006
- handover rupture (2): ce-004, ce-005
- extrapolating past the validated domain (1): ce-001
- diffusion of responsibility (1): ce-008
- cascading failure (1): ce-009

How to use the list: on any new incident notice from this domain, guess the mechanism from the list before reading the report. The hit rate is your familiarity with the domain's accent. Then read

and check. The places you guessed wrong mark the thickness of industry skin still between you and the field.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the Grade A and Grade B reports added to this domain, walk them against the five columns above and the table in Section 7, and revise. A mech - anism that appears for the first time gets a line in Section 4. A shift in frequency updates the counts in Section 7. A new Grade A case goes into the library against the field standard of Chapter 5.

The trigger, the owner, and the review date belong in the user' s calendar (Chapter 39). A domain page that is not updated is dead in the same way a ledger is: the information does not rot, it simply stops describing the field.

9 The Starter Kit

Three items for a reader new to this domain, about one working day in total.

The entry case is ce-001, two passenger trains on wet rail, braking short, a head-on collision, investigated by RAIB in 2026, with the link in Appendix B. Read it first for the format and the grain of the reports in this field.

The entry source is the first-hand library listed under Construction and Infrastructure in. Find the entrance, bookmark it, visit once a quarter.

The entry mechanism is the first line of the Section 7 table. Read the mechanism chapter beside the case from this domain and take the pair twice, once for the criterion in the chapter and once for

the parameters here. When the three items are done you own a first coordinate system for the field.

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Three limits apply to every judgement on this page. The book's case library samples toward events with rich public records, which is the record skew of Chapter 2. The frequency ranking in Section 7 describes the book's library, not the true distribution of the domain. Institutions and technology both move, so this page has a shelf life measured in years, and a citation should carry the year it was read.

All three are the book's general disclaimer, and the threshold check of Chapter 1 applies to this book as much as to anything it examines.

11 How the Three Layers Sound in This Domain

The accent here is the deepest interweaving of the incentive layer with the timeline. Design margin, maintenance debt, and deferred repair all grow inside the incentive layer on a scale of decades: the structural debt at Surfside ran for several decades, and po-005's scheme was put off for thirteen years. The technical breakpoints of a building, concrete spalling and a fractured bolt, are the terminal outlet of that debt. Trace to the root and the physical evidence turns out to be the endpoint, while the ledger is the path.

For the reviewer: in a building case's nine-cell walkthrough the incentive row almost always contains a multi-owner governance interface (Chapter 18). Finding no single person to hold accountable is not a dereliction of duty; it is the true shape of the governance structure, and the repair of that structure is Chapter 18's subject.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
ce-001	Chapter 22 The Illusion of Redundancy	Two passenger trains on wet rail, braking short, head-on collision
ce-002	Chapter 29 The Invisibility of Maintenance	Baseplate bolt fatigue fracture derails a freight train
ce-003	Chapter 15 Practical Drift	Platform refuge recess filled with cable, passenger who fell had nowhere to go
ce-004	Chapter 30 Handover Rupture	Crane pushing a wagon injures two track workers
ce-005	Chapter 30 Handover Rupture	No rule for the reverse shunting scenario, wrong point position, derailment
ce-006	Chapter 22 The Illusion of Redundancy	Train door closes on a passenger’ s hand and drags him
ce-007	Chapter 15 Practical Drift	Mobile ramp missing for four years, train departs without it
ce-008	Chapter 22 The Illusion of Redundancy	Tram strikes two pedestrians at a level crossing
ce-009	Chapter 33 Cascading Failure	Garage column to pool deck connection fails, progressive collapse of a seaside condominium

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

The hardest lesson this domain hands a reader is the irreversibility of margin. Concrete cannot be un-poured and a pile cannot be re-driven, so the irreversible conclusion of Chapter 7's decision table holds permanently here. There is one exit from that: do the checking before the irreversible point. Surfside's structural re‑pair assessment and the sign-off on the Westminster scheme both waited; every deferral pushed the irreversible point backwards and the consequence forwards. The self-check for a reader in this field is to pull out the structural inspection report on his own project and read it again in the margin ledger format of Chapter 25. Look for the line that says no repair needed. That line is where the reader should stop.

The cases here span unusually long periods. Surfside's debt ran for decades, the Westminster scheme for thirteen years, and nobody knows how long the recess at Ickenham had been filling up. On that time scale Chapter 25, margin erosion, and Chapter 28, horizon mismatch, run in their most primitive form: no software can be restarted and no contract terminated at speed, and the curing period of concrete is the cooling-off period of governance, in which every decision collects interest for more than ten years. The lookback in these cases is the longest of any domain in the book, and the incentive-layer evidence in a nine-cell walkthrough often reaches back beyond ten years, which asks the reviewer for the skills of an archival historian. Chapter 33's register, in this field, is the building archive.

14 How the Mechanisms Vary in This Domain

Maintenance invisibility, the infrastructure variation. The asset has a century of life, and the debt is settled across generations, which makes Chapter 29's horizon mismatch reach the largest scale in

the book. In ce-002 the fatigue fracture of a bolt at Audenshaw entered no maintenance ledger and no report, the assurance system did not notice it for years, and the indicator stayed at zero while the target got worse.

The reading: the first question about maintenance visibility here is the overlap between the list of items the ledger covers and the list of structurally critical parts, meaning whether the critical parts are on the ledger at all. The second question is the ratio of inspection interval to component life, since an interval longer than the life means there is no inspection. Surfside (ce-009) pushes the variation to its limit: the gap between the repairs named in the structural assessment and the repairs actually done was accumulated by nobody until physics settled it.

Hidden coupling, the infrastructure variation. The carrier of coupling here is space, and ce-003, the platform recess filled with cable, is the cleanest teaching case in the book: every occupation went through a compliance approval, and the safety function was spent in the course of being compliant.

The reading: dual-register recording of space, one register engineering and one safety function, is Chapter 21's Device 1 in its infrastructure form. The same volume of space holds one state on the engineering drawing and another in the safety ledger, and a conflict between the two is the weak signal. Service pipes under a bridge, tunnel clearances, platform recesses, and escape routes are the four classes to register first.

Practical drift, the infrastructure variation. ce-005, the reverse shunt with no rule to follow, shows the distinctive form: in a non-standard scenario there is no procedure to drift from. Forward work has a method, while reverse, temporary, and non-standard work is improvised on the spot, and the failures in this domain concentrate exactly there. The reading: the prior-authorisation regime for non-standard work, meaning whether anyone approved

it and on what information, is the substitute question for drift in this field. ce-006 and ce-007 show the other half: in standard scen - arios the strength of execution decays with time, and the four-year absence of the ramp was carried by the fact that nothing had ever gone wrong.

15 Cross-Reading Suggestions

Construction and public policy is the first pairing, since the HS2 programme, Westminster, and Northern Powerhouse Rail in the po series are all infrastructure projects. The mechanism of operating an asset lives in the ce series and the mechanism of deciding about one lives in the po series, and only the two pages together give a full life cycle.

Against maritime: ce-009’s progressive collapse and the pipework ageing of ma-004 share the language of silent accumulation, corrosion and fatigue.

Against software: the occupation approval here read beside the permission changes of sw-002 shows that the cumulative effect of legitimate actions is a source of failure common to physical and digital networks.

Maritime and Transport | ma

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily learn wrong.

1 Record quality

The UK Marine Accident Investigation Branch is the global sample. Reports sit in full text on the gov.uk platform in two tracks, formal reports and general investigations, and all 13 ma cases in this book come from its 2026 output. The investigation runs independent of the maritime judiciary (Chapter 3). The IMO framework sets the principles, but jurisdiction fragments along flag lines: the same accident under a different flag can end with no public text at all (Section 3.4). The notification channel of China's Maritime Safety Administration is of limited availability [: pending verification]. Overall: small and medium vessels under flags of convenience are the dark zone of the record, while the small fishing boats inside MAIB jurisdiction form the densest sample of small-accident records anywhere. Nine of the 13 cases here are fishing vessels and small workboats. That ratio is the MAIB's value in one line. It turned the failures of unimportant people into searchable knowledge.

2 Feedback delay

Operational feedback runs in seconds and minutes (a watchkeeper dozing off, a grounding, a collision), and the shape matches aviation. Maintenance and fatigue feedback runs in weeks and months: the sleep debt in ma-013 built up over a single week, and the sea water piping in ma-004 went aboard in 1998. Institutional

feedback, meaning flag-state regulation and the survey system, runs in years. Fishing has a structure of its own. Voyage economics make "carry on" the default (Chapter 27), so the delay gets absorbed as a matter of course.

3 Coupling

Inside one vessel, coupling is moderate. Propulsion, power, windlass and firefighting form physical chains, and ma-001 is the textbook form: loss of electrical power took the anchor with it. Between vessels, coupling climbs steeply in close-quarters work (the tanker and the fishing vessel in ma-005, the collision inside a traffic separation scheme in ma-008). Coupling to the outside world changes with the water. Coastal and traffic areas wire a single-vessel accident straight into the public order of the waterway through vessel traffic services, and the edge of VTS authority is a regular scene for diffusion of responsibility (ma-008).

4 The three deadliest mechanisms

Practical drift (Chapter 15): the watch alarm buried under music (ma-002), a master alone on watch and asleep (ma-003), a new procedure executed off its own script (ma-007). The invisibility of maintenance (Chapter 29): sea water piping installed in 1998 (ma-004), the non-OEM bearing that left the anchor useless after a power failure (ma-001), an inspection gap on a lifting chain (ma-013). Margin erosion (Chapter 25): freeing ports cut smaller year after year (ma-009), working posture hardened into habit (ma-012). Next in frequency come normalization of deviance (the give-way assumption in ma-005) and handover rupture (the VTS authority boundary in ma-008).

5 What outsiders most easily learn wrong

Wrong lesson one: build the picture of the industry from large ships and cruise liners. Most hulls in the world merchant and fishing fleets have no modern bridge and no redundancy, and half the lessons in MAIB reports come from vessels under 25 metres. Wrong lesson two: read flag-state registration as a regulatory guarantee (the fragmentation in Chapter 3). Wrong lesson three: discount small accidents as report material. Capsize and man-overboard cases from fishing boats are the best teaching material available on single-person, single-point, low-cost defence. The de - fences are cheap (a watch alarm, an immersion suit, a second an - chor), the failures are cheap and fatal, and the sense of proportion across industries is all inside them.

6 Three questions from an outsider

Question one: what do fishing and small-boat cases offer deep-sea shipping? The mechanism does not pick a hull size. Fatigue on watch, custom standing in for assessment, emergency equipment failing at the moment it is needed: the small-boat detail is the most complete, MAIB investigates a small accident with the seriousness it gives a large one, and a small case is a transparent specimen of a large mechanism. Question two: why do maritime reports always mention the flag state? Jurisdiction fragmentation is the institu - tional feature of this field, and the flag decides who investigates. Chapter 3 describes the fragmentation and carries the retrieval countermeasures. Question three: does any of this carry to road freight? Watchkeeping, handover, deferred maintenance, cascad - ing failure, and the non-OEM bearing chain of ma-001 hold in any mode of transport. Distance and fatigue are the trade’ s common enemies.

7 Mechanism distribution at a glance

This domain holds 13 cases, distributed as follows, most frequent first:

- Margin erosion (7): ma-003, ma-004, ma-005, ma-008, ma-009, ma-011, ma-013
- Practical drift (6): ma-002, ma-003, ma-007, ma-009, ma-011, ma-012
- Invisibility of maintenance (4): ma-001, ma-004, ma-006, ma-013
- Illusion of redundancy (3): ma-002, ma-006, ma-009
- Cascading failure (1): ma-001
- Normalization of deviance (1): ma-005
- Information attenuation (1): ma-007
- Diffusion of responsibility (1): ma-008
- Hidden coupling (1): ma-010
- Compliant and unsafe (1): ma-010
- Where intuitive expertise breaks down (1): ma-010
- Regulatory lag and capture (1): ma-012

How to use the list: when a new bulletin lands, guess the mechanism against this table before reading the report. Your hit rate is your familiarity with the field's accent. Then read the text and check. Where you guessed wrong is the industry skin still sitting between you and the domain.

8 How this page is updated

A domain page is a live document. Once a year, take the new A and B grade reports in Maritime and Transport and check them against the five columns and the list in section 7. A mechanism that appears for the first time gets a line in section 4. A shift in frequency

gets a new count in section 7. A new A-grade case enters the library under the field standards of Chapter 5. Trigger, owner and review date go into the user's calendar (Chapter 39). A domain page, like the ledger, is dead the moment nobody updates it.

9 The starter kit

Three items for a reader new to Maritime and Transport. Entry case: ma-001 (a non-OEM bearing wore early and the generator was destroyed by fire; MAIB, 2026; link at Appendix B). Read it first for the house style and the grain size of a report in this field. Entry data source: the primary collection listed under maritime and transport in. Find the entrance, bookmark it, visit once a quarter. Entry mechanism: start with the top row of the section 7 list and read that chapter against a case from here. One mechanism gets two passes, once in the mechanism chapter for the criteria, once in this domain for the local parameters. The three items take about one working day, and at the end of it you own a first coordinate system for the field.

10 Reservations

Three limits sit on the judgments above. The case library samples toward events with rich public records, and the skew is described in Chapter 2. The frequency order reflects this book's library, not the true distribution in the field. Institutions and technology move, so this page has a shelf life measured in years, and any citation of it should carry the year. Those three are the book's general dis‑claimer, and the threshold check of Chapter 1 applies to this book as much as to any other artifact.

11 The three layers in the maritime accent

The maritime accent is heaviest at the organizational layer. On fishing boats, inland craft and small vessels, operating procedure exists largely as custom rather than text: outboard work with no assessment in ma-012, a watch with no reserve in ma-003. Custom standing in for text makes practical drift in Chapter 15 and the es - calation ladder of Chapter 8 almost impossible to separate here, because the end point of drift is custom, and custom is itself the first step of the ladder. For the reviewer: in the nine-box walk - through of a maritime case, run the custom archaeology in the or - ganizational box first, asking which generation of vessel the cus - tom started on. The answer points at Chapter 30, handover rup - ture. Custom cannot be traced back for the plain reason that it was never written down.

12 Case-to-chapter lookup

Case ID	Mechanism chapter	One line
ma-001	Chapter 29 The Invisibility of Maintenance	Non-OEM bearing wears early, generator destroyed by fire
ma-002	Chapter 15 Practical Drift	Watchkeeper asleep, grounding, watch alarm masked by music
ma-003	Chapter 15 Practical Drift	Master alone on watch, asleep, grounding
ma-004	Chapter 29 The Invisibility of Maintenance	Ageing sea water piping fails, flooding, sinking
ma-005	Chapter 11 Normalization of Deviance	Neither vessel acts under the collision rules, tanker sinks the fishing boat
ma-006	Chapter 29 The Invisibility of Maintenance	Unsupported flushing pump suction pipe floods and sinks the vessel

Case ID	Mechanism chapter	One line
ma-007	Chapter 15 Practical Drift	Seafarer crushed to death by a trailer during ro-ro vehicle loading
ma-008	Chapter 18 Diffusion of Responsibility	Two cargo ships collide in the North Sea traffic separation scheme, 5 dead
ma-009	Chapter 15 Practical Drift	Trawler runs before a following sea, water trapped on deck, capsize
ma-010	Chapter 21 Hidden Coupling	Rescue vessel: successive entries into an oxygen-deficient anchor chain locker, 3 dead one after another
ma-011	Chapter 15 Practical Drift	Fishing vessel setting crab pots, two men overboard
ma-012	Chapter 15 Practical Drift	Crab boat hauling pots, crew member leans over the side, overboard and dead
ma-013	Chapter 29 The Invisibility of Maintenance	Lifting chain breaks, deck crew member struck by the towing block

How to use the lookup: guess the mechanism from the section 7 distribution, then come here to find the chapter that carries the criteria and the countermeasures. The table does not replace the reading. It is a map, not the journey.

13 Closing note

A last piece of advice for maritime readers. The density of cases proves a commonplace: a small boat does not mean a small lesson. MAIB investigated each of these sinkings with the same care, Re - port 8/2026 through Report 17/2026, every one public in full, and

that seriousness about small accidents is the field's institutional wealth. Small accidents in other industries still get a line in the news and nothing more. The largest lesson the maritime cases hold for a reader outside the field is that the risk of custom is no fishing peculiarity. Any industry leaning on oral tradition rather than written standards sits somewhere on ma-012's extension line. Does your operation have ma-012-style outboard work, a high-risk act nobody has written down? The answer is in the blank space of your work instruction.

One more feature of this page is the hull span, from ro-ro ships of tens of thousands of tonnes down to fishing boats of tens of tonnes. A wide hull span carries a wide mechanism span, and the same water produces failures from a ro-ro's radio link dropping to a trawler's freeing ports cut smaller. Reading across the span trains you to strip the equipment parameters and look only at structure. The result of that training: hand any maritime bulletin, whatever the size of the ship, and within a minute you can place it among this book's mechanisms. The method is simple. Cover the vessel name and the tonnage, read only the process description, and repeat ten times. Recognition then runs without deliberate effort.

14 How the mechanisms vary here

Normalization of deviance, maritime variation: the fatigue and dozing cases (ma-002, ma-003) compress the scene to its minimum, one person, deep night, no second pair of eyes. Read it this way: the watch alarm is this field's own device, and its effectiveness has to be verified under worst-case audibility. The lesson of ma-002 is that passing a monthly test does not mean audible beneath music. Single-handed watchkeeping passes on flag-state rules, and the gap between compliance and safety is filled with physical exhaustion, which makes maritime the small-organization template of Chapter 32, compliant and unsafe. One further reading rule: if the

handling of a dozing event stops at "educate the person", normalization has already started. The person is processed and the structure is left as it was.

The invisibility of maintenance, maritime variation: ma-001 (a non-OEM bearing wearing early), ma-004 (ageing sea water piping) and ma-006 (an unsupported suction pipe) make a maintenance spectrum of spare-part quality, component life and installation workmanship. The maintenance economics of a small owner is the structural constraint of the field: maintenance spending is far more sensitive to freight rates there than at a large operator. Read it this way: the comparison between what class survey covers and where failures actually land is the right measure of maintenance visibility, and the share of failure points outside survey coverage is the blind-spot rate of the system. ma-013 (a lifting chain parting) pushes the blind spot into lifting gear accessories. Survey covers primary structure, deck rigging depends on crew self-inspection, and the failure landed precisely in the fatigue band of self-inspection.

The illusion of redundancy, maritime variation: ma-010, three dead in succession in a rescue vessel's oxygen-deficient anchor chain locker, is the book's heaviest case of the rescuer becoming the victim. Confined-space management failed on an emergency response vessel, and the urgency of a rescue exempted the gas testing procedure. Read it this way: the collapse in enforcement of the confined-space entry permit in rescue scenes is the variant of the illusion of redundancy, where the procedural layer of redundancy is absent in the one scene that needs it most.

15 Reading in pairs

Read the maritime page against the healthcare page (small teams, high pressure, steep hierarchy; see the paired-reading section

there). Against manufacturing: ma-001's non-OEM part reads well beside the supply-chain quality cases in the mn series, where the rule that the source of a part decides the life of the part holds in both. Against construction, see the paired-reading section on that page. Against aviation: av-019 (a hydraulic failure and an out-of-rule ferry flight) reads against the sail-while-sick structure of the ma series, since the ferry segment of aviation and the fishing segment of maritime are both low points in their own regulatory density. Put the low-point cases side by side and the correlation between regulatory density and failure density shows up as one of the most stable empirical regularities in the book.

Software and Internet Services | sw

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily learn wrong.

1 Record quality

The world's only public postmortem ecosystem that no statute requires (the exception in Chapter 4). Vendor postmortems, the kind Cloudflare and GitHub publish along with monthly availability reports, give a timeline and a narrative of decisions; that is D-grade material, and the depth is substantial. Regulatory penalty documents from CNIL, the FTC and the ICO cover the data compliance side at C grade. The 9 sw cases here are made up of vendor postmortems (6) and regulatory penalties (3). The gap is security incidents. Intrusions and breaches almost never get a public postmortem, because the postmortem text doubles as an attack manual and legal liability has no ceiling; the public postmortem library has also been shrinking in recent years (Section 4.2). Retrieval channels are in, where blog scraping that times out should be retried with curl.

2 Feedback delay

The shortest in any industry. Change feedback arrives in seconds and minutes, since a canary release exposes the fault on a percentage of traffic first. Incident feedback arrives in minutes, through monitoring and alerting. Postmortem feedback arrives in weeks. That speed is the material basis of software's postmortem culture: feedback fast enough to carry a high-frequency learning loop. Inside the field the useful distinction is control plane against

data plane. Data plane failures report back fast; control plane fail - ures (dashboards, permissions, configuration) report back just as fast but are more easily underestimated, and the retry storm in sw-004 happened in the control plane.

3 Coupling

Coupling at global infrastructure scale. A single vendor' s edge network carries distribution and security for traffic across the in - ternet, as the three global events sw-001 to sw-003 show, and a shared substrate turns one point of semantic ambiguity into sim - ultaneous failure for thousands of customers. Automated coupling adds a second layer: changes propagate by themselves (the feature file in sw-002 spreading with the deployment) and client retries amplify a fault (sw-004). Isolation measures are real (regionaliza - tion, canaries, rollback), yet common-mode failure keeps penet - rating them, because the same configuration system, the same permission model or the same cleanup job sits underneath. The conclusion on coupling matches aviation' s. High coupling has been pushed down by engineering, and the pushing down is a continuous process that carries a cost.

4 The three deadliest mechanisms

Hidden coupling (Chapter 21): a database permission change doubles the feature file (sw-002), a retry storm (sw-004), the empty-parameter semantics of a BYOIP cleanup (sw-003). Crossing a scale threshold (Chapter 24): record load breaks capacity that was cal - ibrated on historical peaks (sw-005), bulk abuse of a registration system (the mirrored form of fi-004). Cascading failure (Chapter 33): downstream synchronization failing because the automation was shared. Next in frequency come misaligned incentives (the

structural conflict in the data compliance cases sw-007 and sw-008) and practical drift (operations and documentation diverging).

5 What outsiders most easily learn wrong

Wrong lesson one: read postmortem culture as proof that the industry is safe. Postmortems mostly cover availability incidents that fell ill and recovered, not security incidents and not commercial failure, which is the boundary drawn in Chapter 4. Wrong lesson two: copy the word blameless without the machinery behind it. A blameless postmortem with no anonymization process and no separation from performance review is blameless rhetoric, the name-reality gap of Section 4.2. Wrong lesson three: treat fast rollback as a universal answer. Software can roll back because deployment is atomic and state can be versioned. Writing rollback into another industry's emergency plan without asking whether our state can be versioned is the commonest error in cross-industry transplant, and Chapter 4 states the four conditions that decide it.

6 Three questions from an outsider

Question one: what use is software's postmortem practice outside software? Software is the only living sample of a public postmortem culture, and Chapter 4 takes it apart. The blameless process, the game day exercise and the status page are all parts that travel, and Chapter 4 gives the conditions under which transplanting them works. Question two: are cloud platform outages far from my business? If your business depends on any cloud service or SaaS product, you are the downstream party in the cases on this page (Chapter 33, cascading failure). Do those providers appear on your supplier list? Question three: how can a software fault be isomorphic to a chemical explosion? The three layers of the failure

chain do not care about industry. A bug in the technical layer, a missing review in the organizational layer, launch pressure in the incentive layer: when all three break at once the shape is the same, which is why Chapter 9 puts the two industries inside one model.

7 Mechanism distribution at a glance

This domain holds 9 cases, distributed as follows, most frequent first:

- Hidden coupling (6): sw-001, sw-002, sw-003, sw-004, sw-006, sw-008
- Cascading failure (3): sw-003, sw-004, sw-005
- Margin erosion (2): sw-001, sw-009
- Crossing a scale threshold (2): sw-002, sw-005
- Misaligned incentives (2): sw-007, sw-008
- Extrapolating past the validated domain (1): sw-001
- Illusion of redundancy (1): sw-006
- Compliant and unsafe (1): sw-007
- Invisibility of maintenance (1): sw-009

How to use the list: when a new incident write-up lands, guess the mechanism against this table before reading it. Your hit rate is your familiarity with the field's accent. Then read the text and check. Where you guessed wrong is the industry skin still sitting between you and the domain.

8 How this page is updated

A domain page is a live document. Once a year, take the new A and B grade reports in Software and Internet Services and check them against the five columns and the list in section 7. A mechanism that appears for the first time gets a line in section 4. A shift in fre -

quency gets a new count in section 7. A new A-grade case enters the library under the field standards of Chapter 5. Trigger, owner and review date go into the user's calendar (Chapter 39). A do - main page, like the ledger, is dead the moment nobody updates it.

9 The starter kit

Three items for a reader new to Software and Internet Services. Entry case: sw-001 (a WAF regular expression backtracks cata - strophically, exhausts CPU, global outage; Cloudflare, 2019; link at Appendix B). Read it first for the house style and the grain size of a write-up in this field. Entry data source: the primary collection listed under software and internet services in. Find the entrance, bookmark it, visit once a quarter. Entry mechanism: start with the top row of the section 7 list and read that chapter against a case from here. One mechanism gets two passes, once in the mechan - ism chapter for the criteria, once in this domain for the local parameters. The three items take about one working day, and at the end of it you own a first coordinate system for the field.

10 Reservations

Three limits sit on the judgments above. The case library samples toward events with rich public records, and the skew is described in Chapter 2. The frequency order reflects this book's library, not the true distribution in the field. Institutions and technology move, so this page has a shelf life measured in years, and any citation of it should carry the year. Those three are the book's general dis - claimer, and the threshold check of Chapter 1 applies to this book as much as to any other artifact.

11 The three layers in the software accent

The software accent is that the technical break is the youngest of the three. Code and configuration breakpoints travel from deploy to exposure in hours and days (the empty parameter in sw-003, the regular expression in sw-001), and the breakpoints in the organizational and incentive layers are just as young: the design of a monitoring blind spot, the approval exemption granted under launch pressure. The entire history of the industry is a few decades. The practical consequence for the nine-box walkthrough is that one table shows a full set of young breakpoints at every layer. Elsewhere a decade of evolution is needed to lay them down; software does it in a single release. The walkthrough is therefore efficient, and the lesson density is correspondingly high.

12 Case-to-chapter lookup

Case ID	Mechanism chapter	One line
sw-001	Chapter 21 Hidden Coupling	WAF regular expression backtracks catastrophically, exhausts CPU, global outage
sw-002	Chapter 21 Hidden Coupling	Database permission change pushes the feature file past its limit, core traffic interrupted
sw-003	Chapter 21 Hidden Coupling	BYOIP cleanup defect withdraws about 1,100 prefixes by mistake, outage over 6 hours
sw-004	Chapter 21 Hidden Coupling	Retry storm from a dashboard overwhelms the tenant service

Case ID	Mechanism chapter	One line
sw-005	Chapter 24 Crossing a Scale Threshold	Record load breaks an under-provisioned component, authentication fails in cascade
sw-006	Chapter 22 The Illusion of Redundancy	Background cleanup job drags down the primary database, order creation widely blocked
sw-007	Chapter 17 Misaligned Incentives	CNIL fines Google €325 million: ads inserted between emails, invalid cookie consent
sw-008	Chapter 21 Hidden Coupling	CNIL fines FNAC DARTY €7 million: app sent browsing data without consent
sw-009	Chapter 25 Margin Erosion	CNIL fines France Travail €5 million: no MFA, weak passwords, data leak

How to use the lookup: guess the mechanism from the section 7 distribution, then come here to find the chapter that carries the criteria and the countermeasures. The table does not replace the reading. It is a map, not the journey.

13 Closing note

A last piece of advice for software readers. The public postmortem culture is the industry's gift to everyone else (Chapter 4), but the gift's shelf life depends on conditions inside the industry, and the warning at the end of Chapter 4 is that culture has no legislative moat. Two self-checks. When was your organization's last public postmortem? If you cannot answer, the public culture is shrinking where you work. Does your blameless commitment have any isol - ation from the IT system, in the sense of Chapter 8? Software has

already proved this can grow. Keep the four conditions of Chapter 4 in place and it will not be a flash in the pan.

The cases on this page share a feature no other industry has: the party writes its own postmortem. Cloudflare's and GitHub's texts are published in full, with technical detail reaching down to API parameters and individual lines of code, which would be unthinkable elsewhere and is itself one of the objects Chapter 4 studies. The reading advice is to read them in series. One company, four cases, sw-001 through sw-004: read four in a row and a single organization's failure fingerprint appears. The root cause differs every time and the propagation path repeats its pattern. Recognizing that pattern is the advanced training for a software reader, and what it produces is sensitivity to the fingerprint of your own organization. One more observation: every case here sits at a top platform. Outages at small and mid-size software companies get no public postmortem, so survivorship skew runs in this field too.

14 How the mechanisms vary here

Hidden coupling, software variation: the coupling cases share one signature, in that whoever executes the change does not know what the change touched. The lesson in sw-002 (a database permission change pushing the feature file past its limit) and sw-003 (a cleanup defect withdrawing prefixes by mistake) both sit in the coverage of the dependency graph: the dependencies of the permission system, the routing system and the cleanup job were never modelled. Read it this way: the reach rate of a change, meaning the ratio of the impact surface declared beforehand to the one found afterwards, is Chapter 21's criterion expressed as a number here. Software coupling is more hidden than chemical coupling in one respect: dependencies grow by themselves as versions drift (library upgrades, configuration merges), so a manual inventory decays faster than in any other industry. Automatic generation of the de-

pendency graph is therefore a floor requirement, not an optimiza - tion.

Crossing a scale threshold, software variation: sw-005 (record load breaking an under-provisioned component) and sw-004 (a retry storm overwhelming the tenant service) show the two scales of software, load magnitude and feedback speed. Feedback here is measured in seconds, the fastest in the book, while capacity de - cisions are validated on a quarterly cycle through load testing. That mismatch of cycles is the structure of scale failure in software. Read it this way: component coverage between the load test envir - onment and production, meaning how many components have never met real pressure. The lesson of sw-005 is that the weak component sat outside that coverage. The retry storm adds a layer specific to software: a defence (retry) becomes an amplifier under abnormal conditions, so any automatic retry needs a circuit breaker and jitter. That rule is the field's correction to the concept of redundancy in Chapter 22.

Regulatory lag and capture, software variation: sw-007 through sw-009, the three CNIL cases, show the enforcement style of data regulation. Fines are set as a share of turnover, remediation orders come as a system, and violations are found through complaints and audits. Read it this way: the ratio of compliance cost to violation benefit is badly asymmetric here, since the marginal return on using data is high and the chance of being caught is low, and the escalation of fines in the GDPR era is regulation catching up. Chapter 31's criterion, that rules update more slowly than the business, is measured in generations in this field, as the ratio of platform iteration cycle to legislative cycle. The sense that regula - tion is dense and still late is stronger for readers here than in any other field.

15 Reading in pairs

Read the software page against the public policy page, where po-003's dashboard rating and sw-004's dashboard retry make a mutual irony: in both, the metric layer has become the new failure surface. Against finance: the sw and fi enforcement cases share a time structure of long behavioural period and short discovery period, and fi-004's shell advisers beside sw-007's cookie consent shows compliance formalized in the same shape in two industries. Against manufacturing: software change management beside the manufacturing change order shows how far apart the two industries sit in the maturity of one device, since the change order is a more complete instrument than the software change approval.

Finance | fi

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily learn wrong.

1 Record quality

Enforcement documents are the gold mine. The SEC publishes litigation releases case by case, numbered and citable (5 fi cases here); FINRA's monthly disciplinary summary PDF covers the middle and long tail of brokerage violations; the CSRC publishes administrative penalty decisions in full (4 fi cases here, with body pages that can be scraped while the list page is loaded dynamically, all of it recorded in). UK FCA final notices were unreachable from this book's network environment [pending verification]. The structural gap is that enforcement documents record what got caught. The build-up of systemic risk, meaning leverage, concentration and liquidity mismatch, has almost no full text before it breaks: the material for a 2008-class event is a commission report written afterwards, not a supervisory document written before. Overall verdict: micro-level violations are recorded in great thickness, and macro-level risk depends on post-crisis ratification.

2 Feedback delay

Two clocks. On the market clock, price and liquidity feedback arrives in seconds, which produces the illusion of the high-frequency stop-loss; most risk does not report back on the market clock at all. On the regulatory clock, violation feedback arrives in years (the conduct period in fi-001 ran 2019–2021 with the penalty in 2026; the bulk fraud in fi-004 ran 2025–2026 up to indictment), and risk

feedback arrives in cycles, since credit losses are invisible during a boom. The career incentives of a financial decision-maker are perfectly mismatched to the regulatory clock: the bonus is paid on the market clock and the penalty lands on the regulatory clock.

3 Coupling

The highest of any domain. Institutions are wired to each other through interbank liabilities, clearing and collateral networks, so a single default travels along the clearing chain. On the asset side they are wired through common holdings, which makes the diversification that rises and falls together a pseudo-diversification. On the information side, market sentiment transmits infection instantly. The isolation devices of finance, meaning margin, clearing houses and deposit insurance, essentially convert coupling from disorderly into orderly. Coupling itself never goes away; it is institutionalized into a settleable form. The supplier line of a risk register corresponds here to the counterparty line, and the checklist in Chapter 33 is a life-and-death document in this field.

4 The three deadliest mechanisms

Misaligned incentives (Chapter 17): the tension inside audit independence, where two firms failed in succession on the same subject (fi-008, fi-009), and the structural conflict between sales and risk control. Goal displacement (Chapter 12): the beautification of loan loss metrics (fi-001), and a risk rating that departs from actual risk, 24 of 53 (the GAO case in Chapter 12, the public edition of financial governance). Regulatory lag and capture (Chapter 31): fractional share trading falling outside an old classification (fi-011), bulk registration fraud (fi-004), the compliance gap around an AI financing narrative (fi-002). Next in frequency come narrative be-

fore evidence (the offering frauds in fi-002 and fi-005) and diffusion of responsibility (the four-signature chain in fi-007).

5 What outsiders most easily learn wrong

Wrong lesson one: read penalty documents as moral evidence. They record violations that can be proved, not the whole of an or - ganizational culture, and using them to judge culture carries the same sample bias as the survivorship skew in Chapter 2. Wrong lesson two: rank severity by the size of the fine. The $125,000 in fi-013 and the $17.78 million in fi-005 do not measure the same di - mension. Wrong lesson three: take diversification for safety. Di - versification that synchronizes across peers is a single exposure under stress (Chapter 33), and most of the simultaneous failure of diversified portfolios in financial history carries a common factor’ s name.

6 Three questions from an outsider

Question one: what use are financial institution cases to an ordin - ary investor? A due diligence checklist grows straight out of them: auditor independence, cross-checking against the rating, the signs of a backlog of alerts. The five-step audit in Chapter 12 is a ready tool for investors. Question two: the fines look light, $125,000 against 2.56 million trades, so is breaking the rules a good trade? That is criterion 3 shown as arithmetic, where the fine becomes a membership fee (Chapter 31). A light penalty is not grounds for feeling lucky. It is evidence that supervision failed. Question three: is the compliance vocabulary too hard? The action sequences in these cases, cancel the order, release it, report it, are ordinary commercial actions, and the terms live in the source document titles. The mechanism layer is readable by anyone.

7 Mechanism distribution at a glance

This domain holds 13 cases, distributed as follows, most frequent first:

- Misaligned incentives (6): fi-002, fi-003, fi-005, fi-006, fi-008, fi-009
- Narrative before evidence (5): fi-001, fi-002, fi-003, fi-005, fi-006
- Goal displacement (2): fi-001, fi-010
- Regulatory lag and capture (2): fi-004, fi-011
- Where intuitive expertise breaks down (2): fi-008, fi-012
- Practical drift (2): fi-009, fi-010
- Crossing a scale threshold (1): fi-004
- Diffusion of responsibility (1): fi-007
- Information attenuation (1): fi-007
- Hidden coupling (1): fi-011
- Extrapolating past the validated domain (1): fi-012
- Normalization of deviance (1): fi-013
- Automation and skill decay (1): fi-013

How to use the list: when a new enforcement action lands, guess the mechanism against this table before reading it. Your hit rate is your familiarity with the field's accent. Then read the text and check. Where you guessed wrong is the industry skin still sitting between you and the domain.

8 How this page is updated

A domain page is a live document. Once a year, take the new A and B grade reports in Finance and check them against the five columns and the list in section 7. A mechanism that appears for the first time gets a line in section 4. A shift in frequency gets a new

count in section 7. A new A-grade case enters the library under the field standards of Chapter 5. Trigger, owner and review date go into the user's calendar (Chapter 39). A domain page, like the ledger, is dead the moment nobody updates it.

9 The starter kit

Three items for a reader new to Finance. Entry case: fi-001 (thousands of delinquent timeshare loans rescinded and loan loss metrics dressed up; SEC, 2026; link at Appendix B). Read it first for the house style and the grain size of a document in this field. Entry data source: the primary collection listed under finance in. Find the entrance, bookmark it, visit once a quarter. Entry mechanism: start with the top row of the section 7 list and read that chapter against a case from here. One mechanism gets two passes, once in the mechanism chapter for the criteria, once in this domain for the local parameters. The three items take about one working day, and at the end of it you own a first coordinate system for the field.

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Three limits sit on the judgments above. The case library samples toward events with rich public records, and the skew is described in Chapter 2. The frequency order reflects this book's library, not the true distribution in the field. Institutions and technology move, so this page has a shelf life measured in years, and any citation of it should carry the year. Those three are the book's general disclaimer, and the threshold check of Chapter 1 applies to this book as much as to any other artifact.

11 The three layers in the finance accent

The finance accent is that the incentive-layer breakpoint is the clearest of the three. Nearly every penalty decision describes one: the provisioning pressure in fi-001, the offering narrative in fi-005, the arbitrage of a rule gap in fi-011. The decision is itself the official confirmation of an incentive-layer breakpoint, which is why evid - ence grades in this field run high by nature, since C-grade docu - ments are produced by a process that runs automatically. For the reviewer: in the nine-box walkthrough, the incentive row has the most complete evidence, and the decision can be quoted directly. The bottleneck is usually the technical layer instead, because the detail of the trading system gets compressed when the decision is written.

12 Case-to-chapter lookup

Case ID	Mechanism chapter	One line
fi-001	Chapter 12 Goal Displacement	Thousands of delinquent timeshare loans rescinded, loan loss metrics dressed up
fi-002	Chapter 13 Narrative before Evidence	AI startup overstates financial prospects and model feasibility in a crowdfunding offering
fi-003	Chapter 13 Narrative before Evidence	Investment adviser falsifies ADV office information and uses another person’s CRD number
fi-004	Chapter 24 Crossing a Scale Threshold	38 shell adviser entities file false ADVs in bulk
fi-005	Chapter 13 Narrative before Evidence	Cannabis company securities offering fraud, court finds liability on summary judgment

Case ID	Mechanism chapter	One line
fi-006	Chapter 13 Narrative before Evidence	Offering documents and annual report fabricate material content, fraudulent issuance
fi-007	Chapter 18 Diffusion of Responsibility	Four responsible persons held liable for fraudulent issuance and a false annual report
fi-008	Chapter 17 Misaligned Incentives	Accounting firm fails its duty of diligence on an annual report audit and issues a report containing false statements
fi-009	Chapter 15 Practical Drift	A second accounting firm fails the audit of the same subject
fi-010	Chapter 12 Goal Displacement	Option order origin codes recorded inaccurately in the books over a long period
fi-011	Chapter 21 Hidden Coupling	About 2.56 million fractional share trades not reported as required
fi-012	Chapter 23 Extrapolating Past the Validated Domain	Tender offer over-filed by 150,000 shares, net long calculation failed to subtract short options
fi-013	Chapter 11 Normalization of Deviance	Anti-money-laundering alerts with no structured disposition, many closed within seconds

How to use the lookup: guess the mechanism from the section 7 distribution, then come here to find the chapter that carries the criteria and the countermeasures. The table does not replace the reading. It is a map, not the journey.

13 Closing note

A last piece of advice for finance readers. The lesson of these cases condenses to one word, basis. The accounting basis in fi-001, the classification basis in fi-011, the conversion basis in fi-012, the audit basis in fi-008: financial failure almost never happens at the trading layer. It happens at the definition layer, and the definition layer is the battlefield of Chapter 12. The self-check is to open your own metric definition document, walk it line by line with the basis-change register format from Chapter 12, and ask whether each change of basis is recorded. A basis change with no record is a potential fi-001.

These cases carry an evidential advantage no other domain has. A penalty decision is a statutory document: SEC, FINRA, CNIL and CSRC decisions all pass through legal process, sit at evidence grade C, and their facts have been tested. The cases on this page there - fore rank above the D and E grade cases of other industries as a matter of course. There is also a reading technique. The findings of fact are the section worth close reading, because they record the regulator's list of established facts rather than legal argument, and the basis of that list is the regulator's own visibility boundary (Chapter 31). The case count is moderate and the amounts span widely, from €80,000 to €325 million; the way to read that span is in Chapter 31 as well. What matters is the ratio of penalty to gain. The absolute figure is only the yearly dues on a membership.

14 How the mechanisms vary here

Goal displacement, finance variation: fi-001, the timeshare loan rescissions, is the archetype case of Chapter 12, and the variation specific to finance is that the metric coincides with the regulatory report. One number both assesses people inside the firm and gets disclosed to the regulator, so once displacement happens the dis -

tortion enters two loops at once. Read it this way: the disclosure obligation list for basis changes, meaning which metrics require an explanation to investors when their basis changes, is the core compliance asset of this field, and the failure point in fi-001 was the missing disclosure. A self-check for the reader: does the metric you are responsible for have a basis sensitivity note, saying which accounting judgments move the number and by how much.

Narrative before evidence, finance variation: fi-002 (an AI startup overstating model feasibility) and fi-005 (a cannabis company fraud) show the narrative pressure of a fundraising scene, where the funding window, the competitive story and the founder's reputation all push toward an optimistic description of prospects. Read it this way: the ratio of narrative strength in an offering document to the support of the underlying evidence is the right object of due diligence, which is Chapter 37's question used in diligence. fi-003 and fi-004, a forged ADV and shell adviser entities, show the industrialized form of narrative: once false filings can be produced programmatically, supervisory verification density has to be matched to the production rate, and that is the point of the SEC's bulk cases.

Misaligned incentives, finance variation: fi-008 and fi-009, two accounting firms failing on the same subject one after the other, are the structural form of the mechanism inside professional service intermediaries. The audit fee is paid by the audited party, so the economic foundation of independence tilts by construction. Read it this way: audit opinion shopping, measured by the frequency and stated reasons for changing auditors, and the quality curve across an auditor's tenure, are the two correct observation points on the incentive structure here. fi-010 through fi-013, filing errors and alerts closed in seconds, show the incentive layer of the compliance post: the cost of a compliance action is immediate and the probability of regulatory discovery is discounted. The SEC's

monthly penalty summary, the main source for the fi series, is ex - actly the institutional device that raises that discovery probability.

15 Reading in pairs

Read the finance page against the software page (see the paired-reading section there). Against public policy: po-008, the central budget execution audit, reads beside the fi enforcement cases, where audit works as the same device on the state's books and on the market's books. Against startups: the deaths in the st series read beside the offering frauds in fi, where the health of the nar - rative market along the funding chain depends on the density of due diligence and disclosure. Read those two domain pages to - gether and they amount to a financing risk manual.

Startups and Products | st

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

The worst record in the book, and structurally so: the extreme form of the survival skew described in Chapter 2. First-hand material is mostly the founder's own postmortem (all four st cases here were collected and verified through Failory Cemetery, and all four carry the founder's own words). The sample is self-selected, the attribution self-reported, and the writing sits under emotional and legal constraint at once (non-disclosure agreements, refund clauses). Bankruptcy and liquidation filings give a factual skeleton with no narrative attached. Media retellings do not count as evidence, and the Chapter 19 rule applies to them too. The gap has a shape: the fastest and largest deaths, the cash-burnout type, are the least postmortemed. Read this domain's failure material with the three quality checks of Chapter 2 running the whole way.

2 Feedback Delay

The cash-flow clock is measured in months and it is not linear: when financing tightens, every company's runway shortens at the same time (st-004's Xinja attributed its shutdown directly to "an increasingly difficult funding environment"). Product-market fit is measured in quarters, but the growth narrative defers the reading (st-001's Atrium kept expanding while fit had not arrived). The regulatory clock, in crypto and fintech, runs in years and is exogenous (st-002's Koinex, where a freeze on banking channels carried regulatory uncertainty straight into the business). What

separates this domain from the others is that the distribution of its feedback delays is set by variables the company does not control.

3 Coupling

Coupling between individual companies is low: one death does not directly infect the next. Coupling inside the ecosystem is high. The funding environment, channel policies and platform rules are an upstream shared by the whole industry, the form Chapter 33 de - scribes as dependency concentrated on the VC cycle. For platform startups coupling rises with success. A company dependent on one acquisition channel, one cloud vendor, one ecosystem buys that dependence cheap while it is small, and discovers at scale that it has bought its own jugular (Chapter 33).

4 The Three Deadliest Mechanisms

Sunk cost and escalation (Chapter 27): money kept going in after product-market fit had failed to arrive (st-001, where the founder’ s own sentence is the most honest line of postmortem in this book). Narrative before evidence (Chapter 13): the growth story ran ahead of the product (st-003). Horizon mismatch (Chapter 28): the cost cycle of the acquisition model set against the cycle of the funding environment (st-004). Next in frequency: regulatory lag and capture (st-002’ s banking channels) and crossing a scale threshold (organisation and product failing on two fronts during expansion).

5 What Outsiders Most Easily Mislearn

Wrong lesson one: taking the attribution in a postmortem for the truth. A founder’ s account passes three filters, self-report bias, confidentiality clauses, and the needs of the story being told. Read

a postmortem for the information environment of the time; do not read it for the cause of death. Wrong lesson two: treating a surviv - or’ s lesson list as the complete set (Section 2.2: what is missing is the deadliest death, and cash exhaustion, co-founder rupture and a regulatory veto have the lowest rates of appearance in public postmortems). Wrong lesson three: taking one case for a method. “Raise densely, validate quickly” appears in the sample as both success and st-001-style failure, and one case cannot give you the prior probability. The book returns to this point repeatedly. In this domain it is loudest.

6 Three Questions from Readers Outside the Field

Why is the evidence grade in startup cases so low? Public documents on startup death are thin by nature, and the survival skew of Chapter 4 is heaviest here. Every startup case in this book is marked E and used only to illustrate a mechanism. Marking the grade honestly is the protection the book offers its startup readers.

Is the method for a product failure the same as for a safety ac - cident? The analysis of decision points and points of no return applies without change (Chapter 7: shipping, expanding, betting on a new market are all points of no return). narrative before evidence in Chapter 13 is at its most typical in a funding round. The method has no boundary.

Will studying failure make a founder more afraid to start? The aim is to make the cost of failure predictable. A founder who knows how each road ends tends to test faster than one who does not. Chapter 40 deals with the failures that should be encouraged. Fear comes from the unknown, and the unknown is what a case library removes.

7 Mechanism Distribution at a Glance

The book holds 4 cases in startups and products. Distribution by mechanism, largest first:

- Crossing a scale threshold (1 case): st-001
- Sunk cost and escalation (1 case): st-001
- Regulatory lag and capture (1 case): st-002
- Cascading failure (1 case): st-002
- Narrative before evidence (1 case): st-003
- Misaligned incentives (1 case): st-003
- The slack of success (1 case): st-004
- Horizon mismatch (1 case): st-004

How to use the list: when a new failure report from this domain arrives, guess the mechanism against this table before reading. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the domain.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the new Grade A and Grade B reports in startups and products and check them against the five columns here and against the distribution in Sec - tion 7. A mechanism that has not appeared before gets a new line in Section 4. A shift in frequency gets an updated count in Section 7. A new Grade A case goes into the library under the field standard of Chapter 5. The trigger, the owner and the review date go into the user’ s calendar (Chapter 39). Like any ledger, a domain page that is not updated is dead.

9 The Starter Kit

Three items for a reader new to startups and products. Entry case: st-001 (a legal tech startup that raised $75.5 million and then closed; Failory Cemetery, citing founder Justin Kan's public statement, 2020, link in Appendix B). Read that one first for the house style and granularity of a report in this domain. Entry data source: the primary libraries listed in the startups and products part of. Find the entry points, bookmark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

10 Reservations

Three limits apply to everything judged on this page. The book's case library samples toward events with rich public records, and the record skew is the subject of Chapter 2. The frequency ordering of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quota - tion. All three limits are the book's general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in Startups

The three-layer accent in startups is that all three layers are young at once and each accelerates the others. The founder's intuition (Chapter 14) becomes a product decision directly, in the technical

layer. The product story becomes a funding promise directly (Chapter 13). The funding arrives and becomes an expansion budget directly (Chapter 27). The coupling distance between layers, ten years in a large company, compresses to eighteen months in a startup, and the life cycles of st-001 and st-003 are samples of that compression ratio. For the reviewer: in a startup case the time column of the nine-cell walkthrough comes out shockingly short, and the break in the incentive layer and the break in the technical layer can fall in the same month. A nine-cell table spanning a few months reads like an incident flash. The density of the lesson does not thin because the page is short.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
st-001	Chapter 24 Crossing a Scale Threshold	Legal tech startup raised $75.5 million, then closed
st-002	Chapter 31 Regulatory Lag and Capture	Indian crypto exchange closed after banking channels froze
st-003	Chapter 13 Narrative before Evidence	Accounting software company raised $103 million, then closed
st-004	Chapter 20 The Slack of Success	Australian neobank returned its licence and closed

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

A last word for founders: a low evidence grade is no reason to dis - count a case. Because the survival skew is heaviest in this domain, one startup failure honestly marked E is worth more than ten suc - cess books put together. The self-check for a startup reader is to run the decision table of Chapter 7 over your next point of no re - turn: shipping, signing the lease, hiring the tenth person. The output of that check is not a stop order. It is an awareness of where the door closes behind you, and a founder who knows what they are doing gets up fastest after a fall. Chapter 40, on the failures worth encouraging, is waiting.

The cases on this page carry the lowest evidence grade in the book, E, and they come from startup graveyards such as Failory Cemetery: a CEO’s words quoted, the page assembled by media. A low grade is no reason to leave them out. A grade A report on a startup failure does not exist, because a company that has fallen does not write its own investigation report. This is the survival skew of Chapter 4. The value of an E grade startup case is the first-handness of the CEO’s words. Assembled by media they may be, but the tone and the logic survive, and the speaker’s frame of understanding is part of the lesson. Treat the E mark as an alarm. Where the evidence is thin, complete it with your own judgement, and the completing is the startup education.

14 How the Mechanisms Vary in This Domain

The startup variation of sunk cost and escalation: the three fund - ing deaths in the st volume (Atrium, ScaleFactor, Xinja, with Koinex outside them as an environment type) share one escalation signature, in which the size of the raise becomes the collateral of the promise. Atrium’s $75 million and ScaleFactor’s $103 mil - lion turned into irreversible organisational commitments the mo -

ment the money landed: expansion, pricing, headcount. The cost of a strategic turn is amplified by the size of the raise. How to read the variation: the first major pivot after a raise is more hostage to sunk cost than any decision made before it, and the test is the fre‑quency with which "already invested" appears as an argument in the pivot meeting's materials. The external adjudicator of Chapter 27 has a startup counterpart, the board, and the board's independent authority over a pivot is the most valuable governance device in this domain.

The startup variation of narrative before evidence: narrative pres‑sure here comes with a timetable, the funding round, and the mo‑ment the narrative closes is predictable, the final draft of the pitch deck. The lessons in the st volume cluster around the speed at which narrative and product evidence come apart. The gap between ScaleFactor's promise of automated bookkeeping and the degree of automation actually in the product was invisible in‑side the funding narrative. How to read the variation: if the power to define the core metric (what counts as "automated", what counts as an "active user") sits entirely with the side telling the story, the evidence system has already failed. The Chapter 13 device has a startup counterpart: write the definition of the core metric into an annex of the financing documents, so the definition itself becomes a disclosure.

The startup variation of horizon mismatch: Koinex is the template for an environment mismatch, where the ratio between the product cycle (weeks) and the regulatory cycle (months to years) deteriorates sharply and the premise of the business model is pulled out from under it. How to read the variation: list every de‑pendency on regulation and platform policy in the business plan (banking rails, app stores, cloud terms), and assess the single-point exposure and substitutability of each one beforehand. The st volume shows that default optimism about these dependencies is

an industry-wide habit. Xinja, which returned its licence, is the reverse case: it identified that the expected value of continuing was negative and exited, the cleanest stop-loss in the book's startup volume.

15 Cross-Reading Suggestions

Read this page against finance first, as the two ends of the finan - cing chain (see the cross-reading section on the finance page). Against software: read the product deaths of the st volume beside the operational deaths of the sw volume. Startups die of product-market mismatch, mature companies die of scale and coupling, and the two mechanism sets barely intersect; read together they show how organisational maturity maps onto failure type. Against public policy: read Koinex beside the regulatory decisions in the po volume. A single firm's survival hangs on a public policy timetable, and policy uncertainty is itself part of a startup's risk exposure.

Public Policy and Administration | po

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn.

1 Record Quality

The public reports of audit institutions are this domain's distinctive asset. The US GAO and the UK NAO track the failure of government projects audit by audit, year by year and period by period, so one project accumulates several reports across several years (HS2 and the VA electronic health record each have multiple records in this book's library). The annual audit work report of China's National Audit Office is submitted to the National People's Congress in the name of the State Council and published as a PDF (po-008). The value takes a different form from other investigations: what is recorded is not a single accident but the year-on-year divergence between plan and execution, a public edition of feedback delay. Royal commissions and commissions of inquiry, the deep reviews of major administrative failure, are scattered across countries. Assessment: strong on tracking, weak on narrative. Read the domain by project, whole series at a time, aligned by year.

2 Feedback Delay

The longest in the book. Policy feedback is measured in decades (po-007's rail plan went more than ten years without being delivered), project governance in years (GAO's annual comparison), fiscal execution in years (the annual audit report). The institutional consequence: the decision-maker's term (Chapter 28) is naturally shorter than the cycle of the policy's consequences, and inter-

temporal debt (maintenance arrears, ageing populations and assets, committed obligations) has no line item in the budget. po-005's restoration arrears at Westminster are one form of it.

3 Coupling

Coupling inside a project is low to moderate: it can be phased, a section can be stopped. Coupling between policies is high. The same object is acted on by successive governments, several departments and several layers of governance, and the cost of a scope change is absorbed by whoever comes next (po-007). Cross-domain coupling becomes visible in a crisis. The cost of a public project's failure, service interruption and fiscal loss, is borne by an unspecified public, which makes public failure a political event by construction, and makes the pressure of narrative before evidence heaviest here.

4 The Three Deadliest Mechanisms

Sunk cost and escalation (Chapter 27): HS2's two resets and the cancellation of phase 2, with £46.8 billion already spent serving as the argument for continuing (po-004). Goal displacement (Chapter 12): an IT dashboard whose risk ratings diverged from actual risk (po-003), and a conceptual schedule handed to the oversight body (po-002). Diffusion of responsibility (Chapter 18): thirteen years without an agreed plan for the restoration, with authority split (po-005), and scope swinging across governments with nobody answerable for delivery (po-007). Next in frequency: the invisibility of maintenance (po-005), and information attenuation up the hierarchy (po-008's early-collection incentive and its information structure).

5 What Outsiders Most Easily Mislearn

Wrong lesson one: reading a single audit report as event news. The value of a GAO or NAO report lies in the series. Five years of reports on one project, read together, are what reveal metric drift and ac - cumulating arrears; the latest report alone is a lab result with no medical history behind it. Wrong lesson two: attributing project failure to "government inefficiency". The attributions in audit reports are structural: governance structure, definitions, incent - ives. Mechanisms that are isomorphic across the public and private sectors (Chapters 12, 17, 27) simply keep a different set of books in the public domain. Wrong lesson three: underestimating the de - terrent value of public audit. Periodic tracking plus published im - plementation rates for recommendations (Chapter 19) is one of the few workable intertemporal accountability devices the public domain has.

6 Three Questions from Readers Outside the Field

What use are public policy cases to a corporate manager? Hori - zon mismatch, diffusion of responsibility and path lock-in are all isomorphic inside large corporate projects. HS2 across several governments and a multi-vendor IT programme running for years are the same failure chain. Read Chapters 24, 27 and 28 together.

Are audit office and NAO reports worth reading outside the public sector? Yes. The reports are complete in structure, graded in evidence and numbered in recommendation, which makes them the best models available of what a competent failure report looks like (Chapter 3).

Accountability here is visible. Where is the organizational learning? Public accountability is step one. Step two is tracking whether the recommendations land. Of po-002's 18 recommend -

ations, 4 were implemented, and tracking that rate is itself a learning tool (the implementation-rate logic of Chapter 19).

7 Mechanism Distribution at a Glance

The book holds 8 cases in public policy and administration. Distribution by mechanism, largest first:

- Diffusion of responsibility (4 cases): po-002, po-003, po-005, po-007
- The invisibility of maintenance (2 cases): po-001, po-005
- Information attenuation up the hierarchy (2 cases): po-002, po-008
- Practical drift (1 case): po-001
- Crossing a scale threshold (1 case): po-001
- Goal displacement (1 case): po-003
- Sunk cost and escalation (1 case): po-004
- Path lock-in (1 case): po-004
- Extrapolating past the validated domain (1 case): po-006
- Narrative before evidence (1 case): po-006
- Horizon mismatch (1 case): po-007
- Misaligned incentives (1 case): po-008

How to use the list: when a new audit report from this domain ar - rives, guess the mechanism against this table before reading. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the do - main.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the new Grade A and Grade B reports in public policy and administration and check them against the five columns here and against the distribution in Section 7. A mechanism that has not appeared before gets a new line in Section 4. A shift in frequency gets an updated count in Section 7. A new Grade A case goes into the library under the field standard of Chapter 5. The trigger, the owner and the review date go into the user's calendar (Chapter 39). Like any ledger, a domain page that is not updated is dead.

9 The Starter Kit

Three items for a reader new to public policy and administration. Entry case: po-001 (air traffic control modernisation with no full-life-cycle cost estimate, and 138 systems whose schedules do not line up; GAO, 2026, link in Appendix B). Read that one first for the house style and granularity of a report in this domain. Entry data source: the primary libraries listed in the public policy and administration part of. Find the entry points, bookmark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

10 Reservations

Three limits apply to everything judged on this page. The book's case library samples toward events with rich public records, and the record skew is the subject of Chapter 2. The frequency ordering

of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quotation. All three limits are the book's general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in Public Policy

The three-layer accent in public policy is that the break in the incentive layer keeps time with elections and changes of government. po-007's scope changes, po-004's reset and escalation and po-005's shelved plan all beat to a change of government or a budget cycle. Chapter 28's term mismatch reaches its purest form here. A company at least inherits a continuous KPI; a public project's KPI resets to zero when the government changes. For the reviewer: in the nine-cell walkthrough, the incentive-layer cell should record the election year and the decision year. Their overlap is the time seed of the failure, and a project with heavy overlap carries horizon mismatch in its design from the start.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
po-001	Chapter 29 The Invisibility of Maintenance	ATC modernisation with no full-life-cycle cost estimate; 138 systems with unaligned schedules
po-002	Chapter 16 Information Attenuation up the Hierarchy	EHR rollout paused after poor reviews, then restarted; 4 of 18 recommendations implemented

Case ID	Mechanism chapter	One line
po-003	Chapter 12 Goal Displacement	Federal IT dashboard risk ratings left unrevised, diverging from actual risk
po-004	Chapter 27 Sunk Cost and Escalation	High-speed rail project reset twice, phase 2 cancelled
po-005	Chapter 29 The Invisibility of Maintenance	Restoration programme set up 13 years earlier, no construction plan agreed
po-006	Chapter 23 Extrapolating Past the Validated Domain	Nuclear project sold as risk-reducing through a new model, though no comparable project has escaped staged overruns
po-007	Chapter 28 Horizon Mismatch	Regional rail plan, scope adjusted repeatedly over more than ten years
po-008	Chapter 17 Misaligned Incentives	Central budget execution audit found tax collected early and in excess

How to use the table: on a news item or a bulletin from this do - main, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

A last word for policy readers: what these cases ask for is a mech - anism designed to outlive a change of government. po-005’ s stat - utory deadline, po-007’ s trail of recorded changes, and po-004’ s zero-base re-ask are three designs in that family, each immune to an election. The vaccine costs one page of legislation or one clause in a charter. Its absence cost thirteen years of shelving and tens of

billions of pounds of overrun. The self-check for a public project reader is to pull out the approval document and find which page the exit clause is on. If it is not page one, the accent in Section 11 is the agenda for your next budget meeting.

The cases on this page come from the NAO and the GAO, the supreme audit institutions of two countries, and their reports are highly standardised: findings numbered one by one, recommendations tracked one by one, evidence itemised. Public audit reports are the most disciplined evidence format among the book's sources, and Chapter 3 sets out their institutional value. One feature is peculiar to this page: what gets audited is usually an institution rather than a company. HS2 is not a company, and the VA EHRM is not a person. An institutional case has to be walked across governments, agencies and budget cycles, so the nine-cell table of Chapter 9 expands into a timeline matrix, one nine-cell per government, with each change of government acting as a line break. Bring a timeline. An institutional case without one does not read.

14 How the Mechanisms Vary in This Domain

The public policy variation of horizon mismatch: the big projects of the po volume (HS2, Westminster, Northern Powerhouse Rail) push Chapter 28 to the largest scale in the book. The decision-maker's election cycle runs in years, the project's consequence cycle runs in generations, and the mismatch multiplier is far beyond anything in a company. How to read the variation: does a full-life-cycle cost estimate exist in the approval document (po-001's lesson, 138 systems and no full-life-cycle estimate), and who triggers a reset (po-004's two resets). The device to argue for is a cross-term project contract, with scope, budget band and decision gates written into legislation or a cross-party agreement. Its weak point is politics itself, since the next government need not honour

the last one's contract, so a policy project's register should be designed on the assumption that the contract will be torn up, with exit clauses to match.

The public policy variation of sunk cost and escalation: po-006, Sizewell C, shows the distinctive defence of policy escalation. The argument that a new model reduces risk is itself the narrative vehicle for escalation, while the fact that no comparable project has escaped staged overruns is put in a footnote. How to read the variation: the distribution of fact and narrative across the approval document, whether the decisive fact sits in the body or in a foot‑note, is a fast way to identify policy escalation. The external adju‑dicator of Chapter 27 becomes parliament and the audit office here. po-008 shows the annual audit acting as a continuous ex‑ternal adjudication of the national ledger, and the rate at which its recommendations are implemented is the core indicator of a system's capacity to correct itself.

The public policy variation of the invisibility of maintenance: po-005, thirteen years without a settled plan for the Palace of Westminster, is the template at political scale. The beneficiaries of the restoration are future users, who have no vote. The people who bear the cost are the current fiscal year, and they hold every veto. Here the invisibility is produced by the political structure itself; no manager has to be careless for it to happen. How to read the vari‑ation: a compulsion to report condition, meaning periodic, public and budget-linked condition assessment, is the only visibility device that works in this domain. po-002, with 4 of 18 recom‑mendations implemented, shows that recommendation tracking already has a mature form here, in the GAO's recommendation ledger, and that publishing the implementation rate is itself the device.

15 Cross-Reading Suggestions

Read this page against construction and infrastructure first (see the cross-reading section of that page). Against finance: read po-008 beside the enforcement cases of the fi volume, as the two ends of the audit device. Against startups: read the resets of policy projects beside the stop-losses of the st volume. Admitting failure costs an order of magnitude more in politics than in business, because a public admission hands ammunition to the opposition, and that extra cost is the structural reason policy projects escalate harder.

Military | mi

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, and what outsiders most easily mislearn. Public first-hand material is scarce in this domain. The case library holds one case (mi-001, a space mission provisionally filed here) and the judgements of the framework are marked as awaiting collection.

1 Record Quality

The thinnest public record in the book, and thin for institutional reasons: military accident investigations are mostly classified, and what is published comes as accident notices, declassified archives and select committee reports. The usable public samples are these. Independent investigation of space missions, where the Columbia Accident Investigation Board report is published in full and mi-001 is drawn from it, since NASA is a civil-military system and the investigation regime for crewed spaceflight is isomorphic with aviation's. Annual safety reports from national defence safety bodies, where the US military's safety centres publish statistics and lesson summaries while the deep case studies are mostly restricted. Historical declassified archives, with a time lag measured in decades. Collection directions are listed in (declassified archive collections and defence safety publications). Assessment: the structural judgement is trustworthy, in that the investigative institutions exist; the case material is still to be gathered.

2 Feedback Delay

Training and equipment operation give feedback on a mission cycle. Equipment acquisition gives feedback in decades, a heavier

horizon mismatch than public infrastructure. Organisational cul - ture gives feedback across generations. Crewed systems give feed - back immediately and irreversibly, with the points of no return of Chapter 7 operating in their harshest form, which is why military aviation and spaceflight produced the world's strictest checklists and the Crew Resource Management regime. Institutional density runs in proportion to the irreversibility of the consequence.

3 Coupling

Coupling inside a mission system is high and peculiar. In combat and in training, communications, identification, the chain of command and the rules of engagement couple under pressure and on incomplete information. The 1994 Black Hawk shootdown is the prototype case for Snook's theory of practical drift (Chapter 15), where the divergences inside each link add up at an accidental in - tersection. Coupling across units is constrained by establishment and doctrine, and the interface in joint operations is a high-incidence break point (Chapter 30).

4 The Three Deadliest Mechanisms (from CAIB and the Public Literature, Case Slots Pending)

Normalization of deviance (Chapter 11): CAIB's review of the "in-family" classification of foam shedding (mi-001) is itself a specimen of normalization inside a military-grade organisation, and mission standards that move a step at a time through success are a disease of high-risk organisations generally. Diffusion of re - sponsibility (Chapter 18): the separation of the command chain from the safety oversight chain, and committee-style mission re - view. Information attenuation up the hierarchy (Chapter 16): the filtering of upward information under mission pressure recurs

throughout published military investigation reports. To be revised once the collection is done.

5 What Outsiders Most Easily Mislearn

Wrong lesson one: taking the availability of declassified files for the whole truth. What can be declassified has passed a double filter of time and classification, the military form of the record skew in Chapter 2, and statistical inference drawn from declassified material needs unusual care about its sample. Wrong lesson two: imagining military discipline as a vaccine against drift. Snook's finding runs the other way: drift inside a disciplined organisation takes the form of each link doing the right thing by its own logic, and discipline does not produce alignment by itself. Wrong lesson three: using a space case as an aviation case. The institutional isomorphism of CAIB and the NTSB, independent investigation, full publication, tracked recommendations, is worth learning, but the one-shot mission structure of spaceflight means its feedback cannot be compared directly with the repetitive operations of civil aviation.

6 Three Questions from Readers Outside the Field

Are military cases open to outside readers? What is public is the redacted version of an investigation report. CAIB was a congressionally mandated independent investigation and is published in full. The organisational and incentive-layer lessons transfer completely. Technical detail is abstract where classification applies, which does not impede learning the mechanism.

Does a rank culture distort the lessons? Rank culture is the hardest testbed available for Chapter 19, on why the truth-teller goes first. CAIB's organisational analysis of Columbia reads most powerfully against exactly that background: a blameless review

was still possible inside a strong rank culture, and the comparison drawn from the software industry shows this is not a cultural des - tiny.

Where should a reader moving from military to civilian work start? With mi-001, Columbia. The report is a complete sample that reads outside the industry, and its organisational analysis is isomorphic with failure in any large organisation.

7 Mechanism Distribution at a Glance

The book holds 1 case in military. Distribution by mechanism, largest first:

- Normalization of deviance (1 case): mi-001
- Narrative before evidence (1 case): mi-001

How to use the list: when a new investigation report from this do - main arrives, guess the mechanism against this table before read - ing. Your hit rate is your familiarity with the accent of failure in this domain. Then read the report and check the guess. Where you guessed wrong is the layer of industry skin still between you and the domain.

8 How This Page Gets Updated

A domain page is a live document. Once a year, take the new Grade A and Grade B reports in military and check them against the five columns here and against the distribution in Section 7. A mech - anism that has not appeared before gets a new line in Section 4. A shift in frequency gets an updated count in Section 7. A new Grade A case goes into the library under the field standard of Chapter 5. The trigger, the owner and the review date go into the user's cal - endar (Chapter 39). Like any ledger, a domain page that is not updated is dead.

9 The Starter Kit

Three items for a reader new to military. Entry case: mi-001 (ex - ternal tank foam struck the left wing and the vehicle broke up on reentry; Columbia Accident Investigation Board (CAIB), 2003, link in Appendix B). Read that one first for the house style and granu - larity of a report in this domain. Entry data source: the primary libraries listed in the military part of. Find the entry points, book - mark them, visit once a quarter. Entry mechanism: start from the first row of the table in Section 7 and read the mechanism chapter against a case in this domain. Read one mechanism twice, once in the chapter for the criteria and once here for the parameters. Working through the three items takes about one working day and leaves you with a first coordinate system for the domain.

10 Reservations

Three limits apply to everything judged on this page. The book’ s case library samples toward events with rich public records, and the record skew is the subject of Chapter 2. The frequency ordering of mechanisms reflects the distribution inside this library, not the true distribution in the domain. Institutions and technology in the domain keep moving, so this page has a shelf life measured in years, and a reader quoting it should state the year of the quota - tion. All three limits are the book’ s general disclaimer, and the threshold check of Chapter 1 applies to the book itself as well.

11 How the Three Layers Sound in the Military

The three-layer accent in military cases is the highest rank density in the organisational layer. Every link in the chain of command is a translation, and Chapter 16’ s attenuation is institutionalised in military systems as the briefing. More translations do not mean

information must decay, because the briefing format is a mature answer to translation loss: read-back, confirmation, standard phraseology. The civilian portability of the format is discussed in Chapters 16 and 30. For the reviewer: the technical-layer cell in a military case is often abstract because of classification, and abstract does not mean absent. Evidence in the organisational and incentive layers still carries the walk-through. Write "classified" in the cell, following the blank-cell discipline of Chapter 9.

12 Case to Mechanism Chapter, Quick Reference

Case ID	Mechanism chapter	One line
mi-001	Chapter 11 Normalization of Deviance	External tank foam struck the left wing; the vehicle broke up on reentry

How to use the table: on a news item or a bulletin from this domain, guess the mechanism from the distribution in Section 7, then use this table to reach the criteria and countermeasures in the chapter. The table does not replace reading. It is a map, not the journey.

13 Closing Address

A last word for readers of military cases: the public versions (CAIB, the military-related parts of JTSB) are the densest samples of organisational failure available. Rank culture, information pressure, cross-agency coordination, the whole set of difficulties is present. For a non-military reader their value is that of a mirror: how strong is your organisation's rank culture, how often has cross-agency coordination dropped the line, and does mi-001's organisational lesson have a counterpart where you work. The mechanism mapping here (Chapters 11, 13, 18, 19) is cross-industry public prop-

erty, and classification blocks technical detail without blocking the structural lesson.

This page holds one case, mi-001, Columbia, and the CAIB report, but that one case carries the weight of a small library. The report runs to ten volumes, and volume one alone is over 500 pages; the book cites pp. 39 and 122–129 of it. CAIB's organisational analysis chapters are the deepest writing on normalization of deviance anywhere in the book, and easier to read than Vaughan's original because they are verified after the fact. There is a second reason this page is thin. How public a military investigation is depends on who authorised it: a congressionally mandated investigation such as CAIB is published in full, while an investigation kept inside the military, such as av-019, the Xinjiang Yaxin general aviation report, was never published in full and could not enter the main text. The selection standard is therefore a double threshold, an accident large enough and an investigation open enough. What survives the double filter is the densest evidence in the book, and mi-001's weight was stated at the start of this page.

14 How the Mechanisms Vary in This Domain

The military variation of normalization of deviance: mi-001 is one of the origin cases of the book's work on normalization, and the distinctive form here is the flight rationale regime, in which risks that cannot be ruled out are written into the flight release document one at a time until the release document itself becomes the ledger of normalization. How to read the variation: the age distribution of "accepted risk" entries in the release documents, meaning how old the oldest entry is, is the domain's normalization thermometer. The older the entry, the more the risk assessment rests on past releases rather than current evidence. CAIB tabulated the flight-by-flight history of foam shedding (Chapter 11

cites p.124), which is precisely this thermometer made into an ob - ject.

The military variation of narrative before evidence: the review culture around mi-001, where a foam strike was filed as a main - tenance problem rather than a safety problem, shows the power of narrative classification. Which category a phenomenon is put into decides which processing pipeline it enters and what level of re - source it can command. How to read the variation: does the risk classification standard provide a channel and a record for reclas - sification requests, who approves a downgraded classification, and on what evidence. The density of the paper trail on classification changes is the domain's measure of narrative immunity. The neighbouring av-017, a helicopter cleared for the helipad that landed on the runway, adds a second blind spot from a militarised organisation, academy-owned and under quasi-military manage - ment: the clearance travelled on a single-channel radio, and a clearance with no check is no clearance.

The military variation of diffusion of responsibility: a multi-layer command structure is a natural hot spot for Chapter 18. In mi-001's lessons, jurisdiction over the foam problem circulated for years among the project office, the space shuttle programme and the contractor, and each layer could name a specific reason why the responsibility was not its own. How to read the variation: a single-claimant rule for cross-layer risk entries, where each entry carries one and only one name at or above a stated rank, is the targeted device for this domain. An entry that circulates for ten years with the same claimant is a healthy signal. A claimant that changes automatically as the organisation chart changes is a symptom.

15 Cross-Reading Suggestions

Read this page against aviation first (mi-001 and the av volume share the language of airworthiness and release, and aviation's mandatory reporting system is the clearest proof of the institutional domestication this page discusses). Against public policy: read mi-001's project governance beside the big projects of the po volume, where the governance defects of very large technical projects are isomorphic on both sides of the civil-military line. Against nuclear: the safety culture literatures of the two industries reference each other, with CAIB and the Chernobyl and Fukushima investigation reports citing one another, and readers of this page should take the independent verification variation in Section 14 of the nuclear page with them.

Manufacturing and Supply Chain | mn

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, what outsiders most easily mislearn.

1 Record Quality

Recall notices carry this domain. In the United States, CPSC covers consumer products and NHTSA covers motor vehicles, both publishing case by case. All 7 mn cases in this book come from CPSC pages; NHTSA returned 403 in this book's network environment, which records. Notices from the Defect Product Recall Center under China's State Administration for Market Regulation are only partly reachable. Quality incident investigations run deeper, automotive consent orders and defect investigation reports among them, but availability varies case by case, and the NHTSA consent orders were never obtained here.

The overall verdict: notices are plentiful and thin. A reason, a quantity, a count of incident reports. Everything that would explain why the design failed is missing, so mechanism attribution falls to the reader. That is the job the mechanism index does in this domain.

2 Feedback Delay

Three clocks run at once, at very different speeds.

Harm to the user moves in months and years: the product reaches the user, the defect shows itself, the report is filed, the regulator learns. mn-001's 600 incident reports are the shape of that chain on record. Quality on the factory floor moves in hours and batches: first-article inspection, sampling, line checks. Design defects move

in product generations, and mn-006's French press had been selling for 50 years before the accounting came due.

Channel volume stretches the middle clock out of shape. At a fixed defect rate, exposures grow nonlinearly with units shipped (Chapter 24), and the delay between shipment and harm deforms with it.

3 Coupling

Channel coupling rises with volume. One e-commerce channel carries a single batch into hundreds of thousands of households (mn-003). Supply chain coupling is the second axis (Chapter 33): a component maker's defect travels through contract assembly into the brand's product, as with mn-002's Ariston boiler, and structurally the same way through the drug kit in ph-008.

Coupling across products is low, since most consumer goods are independent of one another. The dark thread is the invisible fam‑ily: one contract factory serving several brands. Recall notices never carry the factory's name, so a defect spanning several brands appears as several unrelated notices. Tracing has to work from batch numbers and process signatures.

4 The Three Deadliest Mechanisms

Crossing a scale threshold (Chapter 24): 338,000 units recalled within a year of launch (mn-003); 5.9 million glass carafes accumulated over 50 years (mn-006).

Misaligned incentives (Chapter 17): the mismatch between the cost of listing speed and the cost of compliance checking (mn-004, mn-007).

Margin erosion (Chapter 25): material and structural margins nar - row with each round of cost optimization, and the defect stays in - visible until thermal runaway, tipping or shatter (mn-001).

Next in frequency: hidden coupling and cascading failure, both present in mn-002’ s cracked condensate tray and its delayed ig - nition.

5 What Outsiders Most Easily Mislearn

Misreading one: treating a recall as evidence of bad quality. A re - call is a signal that monitoring and enforcement are working. In a market where the recall regime is thin, the same products appear in no database at all, which is the record skew of Chapter 2.

Misreading two: taking the recall count for the size of the risk. 338,000 heaters against 17 ignitions. Order-of-magnitude thinking needs usage hours in the denominator, and the notice never supplies them.

Misreading three: watching the assembler only. The root cause usually sits two or three tiers down, at mn-002’ s manufacturer Ariston or at the BD components inside the drug kit. Chapter 33’ s dependency inventory has to reach the process and material layer to be worth anything here.

6 Three Questions from Outside the Industry

Question one: what does manufacturing recall practice offer a service business? The full chain, detect, assess, report, recall, trace, is structurally the same as fault management in services. Services have recalls too: the apology notice, the compensation scheme, the root-cause fix. The mn cases’ problems with recall lag and granularity have a counterpart in every industry.

Question two: how is a cascading supply chain failure prevented? Chapter 33's four devices, register, price, drill, debrief, are general parts. mn-001's six hundred reports are the bill for a register that did not exist.

Question three: why do violations continue when mandatory standards exist, STURDY and PPPA among them? Standards govern design and channels govern listing. The coverage rate of a standard and the inspection rate at listing are two separate variables, and Chapter 17's incentive mismatch concentrates in small-batch import channels.

7 Mechanism Distribution at a Glance

This book holds 12 cases in manufacturing and supply chain. Distribution by mechanism, most frequent first:

- Margin erosion (7): mn-001, mn-003, mn-004, mn-005, mn-008, mn-010, mn-012
- Hidden coupling (5): mn-001, mn-002, mn-008, mn-010, mn-011
- Misaligned incentives (3): mn-004, mn-007, mn-009
- Cascading failure (2): mn-002, mn-012
- Crossing a scale threshold (2): mn-003, mn-006
- The invisibility of maintenance (1): mn-006
- Regulatory lag and capture (1): mn-009
- Extrapolating past the validated domain (1): mn-011

How to use the shorthand: on any new incident notice in this domain, guess the mechanism against this table before reading. Your hit rate is your familiarity with the domain's accent. Then read the source and check. Where the guess was wrong is the layer of industry skin still between you and the domain.

8 How This Page Is Updated

A domain page is a living document. Once a year, check the new A- and B-grade reports in manufacturing and supply chain against the five columns and the table in section 7. A new mechanism gets a line in section 4. A change in frequency gets new counts in section 7. A new A-grade case goes into the case base through the field standard in Chapter 5. The trigger, the owner and the review date go into the user’ s calendar (Chapter 39). Like a ledger, a domain page that is not updated is dead.

9 Starter Kit

Three items for a reader new to the domain.

Starter case: mn-001, a power bank’ s lithium battery in thermal runaway, recalled after 600 incident reports (CPSC, 2026, link at Appendix B). Read it first for the form and granularity of reporting in this domain.

Starter source: the first-party libraries listed under manufacturing and supply chain in. Find the entry point, bookmark it, visit once a quarter.

Starter mechanism: take the top row of the section 7 table. Read the mechanism chapter beside a case from this domain, so the mechanism is read twice, once for its criteria and once for its parameters here.

One working day for all three, and you have your first coordinate system for the domain.

10 Reservations

Three limits apply to every judgement on this page.

The case base over-samples events with rich public records, which is the record skew of Chapter 2. The frequency ordering reflects the distribution of this book's cases, not the true distribution in the industry. Institutions and technology change, so the page's shelf life is counted in years, and a reader citing it should state the year of citation. All three are the book's general disclaimer, and Chapter 1's threshold check applies to this book as much as to anyone else's system.

11 The Three Layers in a Manufacturing Accent

The manufacturing accent sits in the incentive layer, where the breakpoints are the most fragmented. Compliance cost on small imports, sampling coverage on platforms, margin pressure at contract factories. In the mn cases these breakpoints never appear as one large decision. They are diffused through the channel as tens of thousands of small pricing decisions.

Fragmented breakpoints cannot be repaired one at a time. Product-by-product management does not scale, because the products cannot be handled one at a time. Only the structural methods of Chapter 17 work at this layer: payment terms, compliance spot checks, performance bonds. Manage the rules and the products take care of themselves.

12 Mechanism Chapter Quick Reference

Case ID	Mechanism chapter	One line
mn-001	Chapter 21 Hidden Coupling	Power bank lithium battery in thermal runaway, recalled after 600 incident reports

Case ID	Mechanism chapter	One line
mn-002	Chapter 21 Hidden Coupling	Boiler condensate tray cracked and trapped gas, delayed ignition blew the casing open
mn-003	Chapter 25 Margin Erosion	Space heater overheated and caught fire, recalled at scale within a year of launch
mn-004	Chapter 25 Margin Erosion	Fabric-front drawer chest unstable and prone to tipping, in breach of the STURDY mandatory standard
mn-005	Chapter 25 Margin Erosion	Nursing pillow could obstruct an infant's breathing, in breach of a mandatory standard
mn-006	Chapter 24 Crossing a Scale Threshold	French press glass carafe shattered in use, 48 scald injuries
mn-007	Chapter 17 Misaligned Incentives	Mattress failed the mandatory flammability standard
mn-008	Chapter 21 Hidden Coupling	Child's sensory swing hook detached, 29 reports and 14 injuries
mn-009	Chapter 17 Misaligned Incentives	Minoxidil recalled for lacking child-resistant packaging
mn-010	Chapter 21 Hidden Coupling	Commercial mower fuel tank seeped, 106 reports including 1 fire
mn-011	Chapter 21 Hidden Coupling	All-terrain vehicle exhaust trapped debris and caught fire, 18 reports

Case ID	Mechanism chapter	One line
mn-012	Chapter 33 Cascading Failure	Fan heater motor mount fractured and caught fire, 7 reports and 2 injuries

How to use the table: on news or a notice from this domain, guess the mechanism from section 7, then use this table to reach the criteria and the countermeasures in the mechanism chapter. The table does not replace the reading. It is a map, not the journey.

13 Closing Note

One sentence holds the manufacturing lesson: channel efficiency is the leverage on risk. Faster listing, wider reach, larger batches amplify exposure speed and recall difficulty together. mn-003's recall of 338,800 units is that leverage demonstrated in physical units.

The self-check for a manufacturing reader is Chapter 24's fifth question. If scale doubled tomorrow, what breaks first? If the answer is compliance review in the channel, the resilience warning light is already lit. Chapter 33's key dependency inventory is the most direct tool this page hands over.

A second property of these cases is worth carrying. They come from CPSC's recall library, and their common profile is no injury or light injury at high volume. A recall is interception as well as penalty: the machinery takes the product off the shelf before it injures anyone, which is the largest institutional difference from the domains in this book that investigate after an accident.

The reading trick is the timeline. How long from launch to recall? mn-003 one year, mn-006 half a century, mn-012 two years. The gap measures exposure speed, and exposure speed is set by how hard the product is used and how visible the defect is when it starts (Chapter 25). Fast exposure forces a recall inside a year; slow ex-

posure settles after fifty. A reader can draw the same curve for their own product: cumulative hours of use, or user contacts, since launch. The inflection point is the expected date the defect sur - faces.

14 How the Mechanisms Bend Here

Extrapolation in manufacturing happens in the use domain. The product passes under laboratory conditions and fails in real hands. mn-006 is the sample: the carafe shattered four decades after launch, with 48 scalds. Extrapolation runs on two axes, duty (tem - perature shock, mechanical impact) and time (material aging). Consumer goods fail mostly on the time axis, because validation covers the first year while the design life is set at ten.

The reading device is a table with validation conditions in one column and use conditions in the other: laboratory drop height against the real distribution of drop heights, accelerated aging duration against design life. The gaps in the table are the extra - polation points. mn-002 crosses both axes: the cracked tray is aging, the delayed ignition is a combination of duty conditions that no single test reproduced.

Compliant and unsafe takes a specific shape here. mn-004, the unstable drawer chest, and mn-005, the nursing pillow, both breach mandatory standards, STURDY among them. The two cases show how that regime is enforced. Standards control admission before sale, recalls control the product after sale, and the seam between them is the period during sale, when stock already in the channel belongs to neither mechanism. The diagnostic question is the join between a standard update and the recall of existing stock: after the new standard takes effect, are the old-standard units still on the shelf? That answer gives the real coverage of the regime. The standard’ s text does not. mn-007 adds the cross-border layer.

Standards differ by jurisdiction, so compliant is a property relative to a jurisdiction, and a change of shipping route changes a product's compliance identity without touching the product.

Cascading failure runs up the bill of materials. Most mn recalls are triggered by CPSC notice or by market surveillance, and the deeper cascade sits upstream, where one part's defect spreads along the BOM into several finished-product lines. mn-010 (mower fuel tank, 106 reports) and mn-011 (ATV exhaust debris, 18 reports) teach the reporting radius: the user tells the retailer, the retailer tells the brand, the brand tells the regulator, and the signal attenuates at every hop (Chapter 16). The measure to keep is hop count and latency from the user's report to the design desk. That number is the honest measure of the learning loop in this industry.

15 Cross-Reading

Pair this page with drugs and devices, where the same recall institution speaks two regulatory dialects (see that page's cross-reading section). Pair it with software, reading the engineering change order against software change management (see that page's cross-reading section). Pair it with agriculture and food: ag-008, a stabilizer carrying bacteria into food plants across several states, reads against BOM diffusion here and shows ingredient diffusion in food and component diffusion in manufacturing as two raw-material forms of one cascading mechanism.

Agriculture and Food | ag

> Five columns: record quality, feedback delay, coupling, the three deadliest mechanisms, what outsiders most easily mislearn.

1 Record Quality

Two kinds of record carry this domain, and they differ in depth.

Outbreak investigations give the depth. CDC’ s MMWR reports carry a full timeline, with dates for first case, alert, activation, recall and closure. ag-001 and ag-002 were both read in full for this book and grade strong. FDA enforcement reports give the breadth, and the openFDA interface makes them retrievable programmat - ically; all 8 ag recalls here were checked through an exact event_id query. FSIS, which covers meat, poultry and egg, was unreachable in this book’ s network environment and is marked unverified in. On the Chinese side, sampling notices from the State Administra - tion for Market Regulation are only partly available.

The verdict splits the same way. Outbreak investigations are of high quality, since epidemiological training makes the timeline hard to fault. Recall notices are thin: a reason in one clause, a quantity in one field, and nothing about mechanism. Attribution has to be re - built from the timeline, which is what a reader with the mechanism chapters in hand can do.

2 Feedback Delay

Delay runs at three depths.

An acute outbreak moves in days and weeks. ag-001 went from first case on 21 September to recall on 24 October, 37 days for the whole chain, and that counts as a fast one. Chronic harm from heavy

metals or accumulated pesticide residue moves in years and arrives dispersed through epidemiological statistics. A process assumption moves in generations: the aging assumption in ag-003 held until state testing broke it.

Food has a structural feature the other domains lack. The delay is set jointly by the consumer's willingness to report and the clinician's ability to recognise. Many outbreaks' recorded first case is weeks later than the true first case, and no part of that gap is under the producer's control.

3 Coupling

Ingredient chain coupling (Chapter 33): one bag of business-to-business ingredient reached bakeries in 23 states (ag-008), and no finished product carried the supplier's name. Retail chain coupling: the retail end follows an upstream recall on its own schedule (ag-004's Whole Foods bulk sliced), and bulk bins and private label break traceability in the last metre before the customer.

Coupling across categories is low, but shared origin and a shared distribution centre make separate products share one microbial ecology. That is Chapter 21's dark thread in food form: shared water, shared people, shared cold chain.

4 The Three Deadliest Mechanisms

Extrapolating past the validated domain (Chapter 23): commercial raw milk sales rest on process and testing assumptions, and the outbreak ran to 171 cases (ag-001); 60-day aging was treated as a safety barrier (ag-003).

Compliant and unsafe (Chapter 32): legally sold raw milk was still the outbreak's source. Compliance paperwork cannot answer a question about microbial ecology.

Cascading failure (Chapter 33): B2B ingredient diffusion (ag-008), and retail follow-up carrying responsibility downstream (ag-004).

Next in frequency: handover rupture (ag-009, where a formula change and the label came apart, and an undeclared allergen is the sharpest form a handover break takes).

5 What Outsiders Most Easily Mislearn

Misreading one: reading a recall notice as an isolated event. Same-source recalls come in chains, ag-003's factory recall followed by ag-004's retail recall, and one notice read alone hides the shape of the dependency behind it.

Misreading two: treating "natural" or "traditional process" as a safety property. Raw milk and aged cheese both show that a process assumption has to be tested against the finished product. That sentence is the whole of Chapter 23 in a food setting.

Misreading three: imagining the risk boundary from the tolerance of a healthy adult. Food harm falls on children, the elderly and the immunocompromised: 70 per cent of ag-001's 171 cases were under 18. The population dimension belongs on the first line of any extrapolation checklist.

6 Three Questions from Outside the Industry

Question one: what does a food recall have to do with me, when I run no plant? Food is the most complete sample available of a failure chain running from raw material to shelf. ag-008's ingredient crossing 23 states, ag-009's label break: any product with a labelling duty sits on the same chain.

Question two: is a small workshop's accident worth studying? A small workshop shows a process assumption failing in its purest

form, with no corporate compliance department standing in front of it. The assumption sits directly on the shelf. ag-003 and ag-001 are both textbook in this respect.

Question three: what does the range in recall granularity mean, 8 boxes in one notice and 5.9 million units in another? Granularity reflects traceability, and traceability is a direct function of the supply chain ledger. Chapter 33's key dependency inventory in the food industry is the batch ledger: the finer the ledger, the more precise the recall and the smaller the commercial loss.

7 Mechanism Distribution at a Glance

This book holds 10 cases in agriculture and food. Distribution by mechanism, most frequent first:

- Cascading failure (5): ag-004, ag-005, ag-006, ag-007, ag-008
- Compliant and unsafe (2): ag-001, ag-003
- Extrapolating past the validated domain (2): ag-001, ag-003
- Hidden coupling (2): ag-002, ag-008
- Practical drift (2): ag-009, ag-010
- Diffusion of responsibility (1): ag-004
- The invisibility of maintenance (1): ag-005
- Handover rupture (1): ag-009

How to use the shorthand: on any new incident notice in this do‑main, guess the mechanism against this table before reading. Your hit rate is your familiarity with the domain's accent. Then read the source and check. Where the guess was wrong is the layer of industry skin still between you and the domain.

8 How This Page Is Updated

A domain page is a living document. Once a year, check the new A- and B-grade reports in agriculture and food against the five columns and the table in section 7. A new mechanism gets a line in section 4. A change in frequency gets new counts in section 7. A new A-grade case goes into the case base through the field standard in Chapter 5. The trigger, the owner and the review date go into the user’ s calendar (Chapter 39). Like a ledger, a domain page that is not updated is dead.

9 Starter Kit

Three items for a reader new to the domain.

Starter case: ag-001, commercial raw milk causing a Salmonella outbreak of 171 cases (CDC, MMWR, 2025, link at Appendix B). Read it first for the form and granularity of reporting in this do - main.

Starter source: the first-party libraries listed under agriculture and food in. Find the entry point, bookmark it, visit once a quarter.

Starter mechanism: take the top row of the section 7 table. Read the mechanism chapter beside a case from this domain, so the mechanism is read twice, once for its criteria and once for its parameters here.

One working day for all three, and you have your first coordinate system for the domain.

10 Reservations

Three limits apply to every judgement on this page.

The case base over-samples events with rich public records, which is the record skew of Chapter 2. The frequency ordering reflects the distribution of this book's cases, not the true distribution in the industry. Institutions and technology change, so the page's shelf life is counted in years, and a reader citing it should state the year of citation. All three are the book's general disclaimer, and Chapter 1's threshold check applies to this book as much as to anyone else's system.

11 The Three Layers in a Food Accent

The accent here is that the technical layer's breakpoints come with a biological clock attached. Pathogen incubation, allergen distribution, shelf-life decay: the physical clock of failure answers to nobody in the organisation. The defensive window is the hardest of any domain in this book, with recall deadlines counted in hours. ag-004 followed up in two days; ag-001 ran the whole chain in thirty-three days.

That hardness forces real-time information systems. Chapter 16's attenuation and Chapter 30's handover breaks meet the lowest tolerance in food, where the deadline is measured in hours. The industry did not choose this constraint. The pathogen's clock imposed it.

12 Mechanism Chapter Quick Reference

Case ID	Mechanism chapter	One line
ag-001	Chapter 32 Compliant and Unsafe	Commercial raw milk caused a Salmonella outbreak of 171 cases
ag-002	Chapter 21 Hidden Coupling	Organic walnuts contaminated with E. coli O157, 13 cases

Case ID	Mechanism chapter	One line
ag-003	Chapter 23 Extrapolating Past the Validated Domain	Raw milk blue cheese tested positive for STEC, Class I recall
ag-004	Chapter 33 Cascading Failure	Retail end followed with a recall of the same-source STEC cheese
ag-005	Chapter 33 Cascading Failure	Pesto pasta salad kit, Listeria, Class I recall
ag-006	Chapter 33 Cascading Failure	Ready-to-eat cucumber slices, Listeria, Class I recall
ag-007	Chapter 33 Cascading Failure	Caramel popcorn, potential Salmonella, nationwide Class I recall
ag-008	Chapter 21 Hidden Coupling	Stabilizer ingredient carrying bacteria reached food plants in several states
ag-009	Chapter 30 Handover Rupture	Snack mix with undeclared almond and sesame
ag-010	Chapter 15 Practical Drift	Lemon cookie recalled for a colour labelling error

How to use the table: on news or a notice from this domain, guess the mechanism from section 7, then use this table to reach the criteria and the countermeasures in the mechanism chapter. The table does not replace the reading. It is a map, not the journey.

13 Closing Note

One sentence holds the food lesson: the organism’s clock is the final consumer. Your information system can be slow, your han‑dover can break, your ledger can have gaps. A pathogen’s incub‑ation period will not wait. ag-001’s thirty-three-day chain, ag-004’s two-day retail follow-up, ag-008’s spread across states:

the information system has to outrun the clock, and that is the one constraint in this industry nobody can negotiate. Readers from slower domains should take the point as a piece of luck.

The self-check is Chapter 33's key dependency inventory applied to your batch traceability. Every hop from raw material to shelf: can you follow it?

These cases come from FDA and CDC recall libraries and outbreak investigations, with timelines precise to the day. First illness, report, test, recall, resumption of production, each with a date. That precision gives food the highest fidelity of causal reconstruction in this book; a reader can follow a pathogen day by day from farm to shelf to table to emergency room.

There is a structural feature worth naming. Recall granularity varies more here than anywhere else. ag-006 recalled 8 boxes, ag-007's quantity field is empty, ag-008's recall crossed three countries. Those differences are differences in traceability (Chapter 33), and they show up in the ledger before they show up on the shelf. A food reader has one extra tool: FDA's public API, which like MAUDE on the drugs page can be scripted for monitoring.'s search manual has the entry points.

14 How the Mechanisms Bend Here

Cascading failure has its cleanest sample in this book in ag-008, where a stabilizer carrying bacteria reached food plants in several states. Contamination of a single ingredient spread along recipes into several finished-product plants that had never met and did not know they shared a risk source. Two measures explain the blast radius. The first is shared-source concentration: how many finished lines draw on one upstream. The second is the recall's trace radius, the number of hops and the hours needed to walk from a

finished product back to an ingredient. That radius decides whether the affected area can be ringed in time.

ag-001 shows the opposite shape. Raw milk went from one farm straight to consumers, an extremely short chain traced very fast. A short-chain product's risk structure matches a small shipowner's maintenance economics: the source is the endpoint, with no buffer layer to absorb anything.

Hidden coupling in food travels on shared recipe and shared line. ag-003 and ag-004, the blue cheese factory recall and the retail sliced recall, are one risk source recalled twice at two channel levels. A retailer does not automatically receive the factory recall, so the retail follow-up needs a trigger of its own, and that missing notification is the coupling blind spot. The measure is the completeness of the notification path, factory to distributor to retail to the discount channel. That path is food infrastructure in the same sense as the cold chain, and a register of notification breaks is Chapter 21's dependency inventory translated into food.

Regulatory lag reads differently here. Raw milk is legal in some states and not others, so a product can be lawful and dangerous at the same time, and the gap is settled by CDC's outbreak investigation rather than by the statute. The measure is the separation between legality and risk: the share of legally sold food appearing in outbreak data. Against that, the recall classification system is the positive sample, with its 48-hour response requirement for a Class I recall. Response density by class is higher here than in most industries, and readers from other domain pages can borrow it.

15 Cross-Reading

Pair this page with manufacturing, where ingredient diffusion and component diffusion turn out to be one cascading mechanism in two raw materials (see that page's cross-reading section). Pair it

with healthcare: the outbreak investigations here use CDC’s case-control method, and read against me’s diagnostic delay they show recognition delay as one problem in two settings, namely how fast a signal climbs from a single case to a pattern. Pair it with public policy: raw milk legislation against po’s policy cycles shows the density of food safety rules swinging between legality and risk, a clean sample of a policy process shaped by values.

Chapter 35 The Situating Table

> What this chapter covers: a self-assessment table with six parameters. Fill it in and it points to the two or three failure mechanisms most likely running in your situation. This is the door that turns the book's twenty-four mechanism chapters into personal kit. When to come back to it: before taking over a new project, reviewing a new system, expanding the organization or changing the appraisal scheme; and at the first step of the thirty-minute walkthrough in Chapter 36. Reading time: about 35 minutes, including the time to fill in one table.

The door has a reverse meaning too. If you skip this chapter and go straight to a mechanism chapter in Part III because some case pulled you in, come back afterwards and fill the table once. Then ask whether the mechanism that attracted you is the main dish of your situation. Many readers are drawn in by a case's drama. The table's job is to make drama give way to relevance. What your situation needs defended matters more than which story moved you.

Only time can run the final test of relevance. Whether the mechanism your table hit eventually showed symptoms in your organization is a fact you learn six to twelve months later, and that revisit is how the table's accuracy gets calibrated. Calibration does not change the content of the mechanism chapters. It changes the precision of the map from parameters to mechanisms. The map has versions. Your own accumulated calibration data is the custom edition, and the custom edition beats the generic one, because your organization is not an average organization.

Average organization deserves one last pass, since it is what the table is measured against. The twenty-four mechanisms were distilled from cases in dozens of industries. Distillation pulls out the

common structure. Common means it applies to the largest num - ber of organizations, and the largest number means exact for none of them. The table is a bridge to your own organization: this book supplies the generic span, your calibration data casts the custom one, and the custom map is the warning system your organization actually needs. Building the custom edition begins with recording calibration data. That is how the chapter’ s deliverable turns from a table into a system.

35.1 Why Six Parameters

Checking twenty-four mechanisms every time is not possible. The table compresses your situation into six parameters, three levels each, and the combination points at a small number of chapters. The cost of that compression is missed judgment.

The table promises two things and nothing else: it puts the most likely mechanisms at the front of the queue, and it excludes the ones that clearly do not apply.

A third promise sits alongside them. The table is a reading route. The two or three hit chapters are required reading at 45 to 60 minutes each, so three chapters cost about three hours and the first deep pass fits in one afternoon. The time budget is therefore computable: half an hour to fill the table, half an hour to screen, three hours of close reading. One working day buys a targeted diagnosis of organizational failure, and it is the best-value day in the book.

There is a cross-chapter link worth naming. The candidate chapters you end with are your starting reading list, and the com - pletion mark is not finishing them. It is a line written in the margin of the table: which level would I now fill differently. How often the levels move is a direct readout of learning. If the levels before and after reading are identical, one of two things is true. Your judgment

of the situation was already accurate, or the chapter's criteria did not cut through your own narrative. Chapter 6's exchange test separates the two: hand the chapter to someone else and see whether their levels move.

A reader may already be forming a question the chapter does not raise. Could this be a tool that computes the mapping automatically? Technically yes, since the mapping is a lookup. The paper version has one reason behind it. Filling the table is the thinking. A tool returns an answer in a minute and what it saves is exactly the thinking; fifteen minutes by hand buys one pass of all six parameters through your head. Time saved and judgment trained do not arrive together. Use paper the first time, and the tool once the table is familiar.

What you pay for skipping it is worth counting as well. Your organization runs, the mechanisms run, and you do not know which ones. Chapter 16 already priced not knowing: the cost arrives late. The free version of the table is running the six parameters through your head, five minutes. An organization that will not spend even that has mechanisms running at full speed with nobody at the dashboard.

The table does not promise to be exhaustive, and the practical reason is that mechanisms manufacture each other's conditions. The normalization of Chapter 11 feeds the slack of Chapter 20. The misalignment of Chapter 17 drives the displacement of Chapter 12. A mechanism set behaves like a network rather than a list. What the table gives you is the entry mechanism, the one to check first. From there you follow the neighbours it pulls in, and section 6 of each mechanism chapter is the neighbour list. Entry plus neighbours usually covers the trunk of a situation.

Selection criteria for the six: each must be assessable by you alone, in half an hour, with no external audit. Any parameter you cannot answer with confidence gets the most dangerous level. The table

exists to find risk, and an optimistic fill only deceives the person writing it.

Cross-filling is the antidote to optimism. Have a colleague from an unrelated part of the business fill your table, and read the gap between the two versions as your blind spot. The logic matches the newcomer audit in Chapter 33 and the exchange test in Chapter 30. The output is not a verdict on whose version is right. It is a list of differences, and that list is the best available opening for the next conversation between the two of you.

The six are not independent. Parameter 3, incentive, correlates strongly with parameter 4, time: an organization that rewards the short term usually also runs on short time pressure. Parameter 1, coupling, grows together with parameter 5, scale. Correlation means some combinations are naturally more probable than oth - ers. High coupling with large scale and systemic misalignment turns up far more often than low coupling with a single point and aligned incentives. That is the real distribution of organizations, and the table is designed to the distribution rather than to ortho - gonality.

Provenance sets how much weight the table can carry, so it is worth one pass. The parameters went through three rounds. Round one built a candidate list of nineteen: the six in use plus thirteen others, among them team culture, leadership style, industry cycle, technology maturity, staff turnover and outsourcing ratio. Round two cut against three criteria. Self-assessable: the person filling the table must give a level in half an hour without an external audit, which removed turnover and outsourcing ratio, since the accurate numbers sit with HR and procurement and an individual can only guess. Stable mapping to at least three mechanism chapters: this removed industry cycle, whose correspondence to the twenty-four mechanisms is unstable in every direction, with the same point in the cycle producing completely different exposure in different or -

ganizations. A determinable boundary between the three levels: this removed team culture and leadership style, which are con‑tinuous spectra that every person defines differently, so level judgment would collapse into a personality quiz. Round three back-tested against verified cases, refilling every situation that has a full card in this book and checking whether the mapping pointed at the mechanism that actually ran. Candidates with a poor hit rate went out. Six survived.

The same screening states the table’s nature. The six do not cover every failure factor. They are the whole intersection of self-assess‑able, mappable and determinable. Factors outside the intersection keep operating. They are simply not this table’s business.

A twenty-parameter version is the obvious next question, and the answer is a trade between filling cost and judgment quality. Triple the parameters and you triple the time, while correlation means the new parameters add far less information than a linear estimate would suggest. The design target is a first screen in fifteen minutes that yields an executable reading route, and that target sets the table’s width. Finer granularity has its place, later in the chain: coarse positioning first, then a detailed check of the criteria in whichever chapters you hit. Granularity gets finer downstream.

35.2 The Six Parameters: Definitions and Three Levels

Parameter 1: coupling. How fast and how far a component failure propagates through your system.

- Low: the failure can be isolated physically or by process, and the affected surface can be counted (one production line, one customer).

- Medium: the failure spreads inside your department or along one business chain, and isolation takes a manual action (stopping a queue, switching a data centre).
- High: the failure crosses systems, crosses departments or goes straight to the outside, and the propagation path has no list. Chapter 21's test: nobody can answer "who does this outage affect".

Parameter 2: redundancy reality. How far your backup would survive a real switchover.

- High: redundancy is heterogeneous (different vendors, different principles, different people), and a switchover has been rehearsed under real pressure and succeeded.
- Medium: redundancy exists but shares its source with the primary system (same vendor, same drawings, same crew), or switchover has only been rehearsed on plan.
- Low: redundancy exists on paper and has never been verified, or the critical step was a single point to begin with.

Parameter 3: incentive direction. The angle between the appraisal and reward system and the exposure to risk.

- Aligned: the people who create the risk carry the consequence, and the consequence enters their appraisal.
- Partly misaligned: local metrics pull against overall risk, but a hedging role exists with real authority (risk control, safety supervision).
- Systemically misaligned: each layer's rational choice sums to a loss for the whole, and the person who finds the problem is the person who benefits from it (Chapter 17's test).

Parameter 4: time pressure. The window between a decision and its consequence.

- Short: the window runs in days, and the gain from skipping verification is visible now (a release date, a delivery date, a shift).
- Medium: the window runs in quarters, and compressing the process pays this period while the cost lands inside the year.
- Long: the window runs in years, and the decision-maker's tenure will most likely end before the consequence matures (the soil of Chapter 28).

Parameter 5: scale. The number of people coordinating, the number of nodes, the length of the dependency chain.

- Single point: one team, and every open question can be settled in one meeting.
- Multi-team: cross-team interface documents are needed to state who depends on whom.
- Network: outsourcing, franchising, multiple suppliers, a platform ecosystem, so a layer of dependency sits outside the organizational boundary (the territory of Chapters 24 and 33).

Parameter 6: regulatory density. The frequency, depth and penalty weight of external inspection.

- None or sparse: the outside barely looks at your failures, and the binding constraints are the market and yourself.
- Dense: frequent inspection, item-by-item checking, real penalties.
- Dense but captured: inspection is frequent, but the rules update more slowly than the business, the information source is single, and the penalty sits below the gain from violation (Chapter 31's test).

One trap sits on the regulatory density parameter: a high level does not mean safe. A dense inspection regime is where Compliant and Unsafe (Chapter 32) breaks out most often, because the proof of passing each item takes the place of the question whether the whole is safe. Be honest about your industry when you fill this in. Density on paper does not count.

There is a candidate fourth level for this parameter: the trust in - ertia that follows capture, where the regulated party has lived alongside the regulator long enough to form a culture in which the rule is the whole of the obligation, and compliance reads as safety. It stays out of the table because the regulated party cannot judge it from inside. An organization at that level usually does not know it, and detecting the level needs an outside view and the multiple in - formation sources of Chapter 31. A reminder for the reader: ten years or more in one industry without a single substantive regu - latory challenge may mean you are already at the fourth level.

Every parameter comes with an evidence list. Check the evidence before fixing the level, and let the evidence set the level.

Coupling: the impact list from your last cross-system failure, and whether the affected surface was countable; the age of your de - pendency documentation; and an oral test. Ask three senior people at random "if this component fails, who is affected" and see whether the answers agree. An organization where they do not agree moves its coupling self-assessment up one level.

Redundancy reality: the record of the last real switchover, with date and result; the vendor overlap between backup and primary; the difference list between the rehearsal script and the production environment. No real switchover in three years caps the level at medium.

Incentive direction: where risk items sit in the current appraisal form and what weight they carry; the last instance where someone

paid an appraisal price for blocking a risk; any documented veto by risk control or a safety post against a production decision. If you cannot find a veto instance, treat the parameter as prefilled at systemically misaligned.

Time pressure: this quarter’ s delivery commitments; the last three decisions to compress verification or review, with the reasons given; an estimate of the consequence cycle, meaning how long a problem typically takes to surface. Write the decision cycle and the consequence cycle side by side and the window’ s length becomes obvious.

Scale: how many interface documents exist and when they were last updated; the outsourcing and supplier list; how many teams one change has to be announced to. A notification list above ten people gets treated as prefilled at network.

Regulatory density: the number and depth of external inspections over the past year, item-by-item checking or listening to a briefing; the ratio of penalty to gain from violation in the cases you know; instances where a rule update lagged a business change. Three pieces of evidence in hand and the level is clear.

The evidence lists share one property: everything comes from documents and records you already hold, so filling the table starts no new work. Failing to assemble the evidence is itself a finding. Whichever parameter has the thinnest evidence in your organiza ‑ tion is the darkest area in your knowledge of it, and a dark area gets filled at the most dangerous level.

35.3 Combination Rules: From Parameter Set to Mechanism

The table below is the core of the chapter. Each mechanism carries its trigger combination. Abbreviations: C = coupling, R = redund ‑

ancy reality, I = incentive direction, T = time pressure, S = scale, G = regulatory density. The reading method: fill in the six parameters, then go down the rows. Wherever your combination falls inside a row's trigger region, that mechanism joins the candidate list. Candidates usually number 3 to 6. Section 35.4 narrows them to 2 or 3.

Ch.	Mechanism	Trigger combination
11	Normalization of Deviance	T = long (delayed consequence) + a deviation of the same class has occurred ⩾2 times with no consequence
12	Goal Displacement; Goodhart's Law	I = systemically misaligned + the layer above sees only the numbers
13	Narrative before Evidence	decision already publicly committed + time pressure short (too late to reopen the analysis)
14	Where Intuitive Expertise Breaks Down	S = network (entering an unfamiliar environment) + T = long (delayed feedback)
15	Practical Drift	T = short (no immediate consequence for breaking the rule) + rules do not match the actual workload
16	Information Attenuation up the Hierarchy	S = multi-team or larger + reporting carries appraisal weight (I ≠ aligned)
17	Misaligned Incentives	I = systemically misaligned (the strongest single-parameter trigger in this table)
18	Diffusion of Responsibility	S = multi-team + a multi-party signature structure

Ch.	Mechanism	Trigger combination
19	Why the Truth-Teller Goes First	I = systemically misaligned + performance decided unilaterally by the layer above
20	The Slack of Success	T = long + a long record of no accidents
21	Hidden Coupling	C = high + the dependency list cannot be answered (history of mergers, integrations, legacy systems)
22	The Illusion of Redundancy	R = medium or low + backup from the same source
23	Extrapolating Past the Validated Domain	a pilot-to-rollout or cross-industry transplant is under way
24	Crossing a Scale Threshold	S = climbing from multi-team toward network + fast growth
25	Margin Erosion	T = medium or long + cuts made ⩾2 times in a row with no cumulative ledger
26	Automation and Skill Decay	high automation + falling hours of manual-operation training
27	Sunk Cost and Escalation	sunk investment publicly visible + exit equals admitting error (I ≠ aligned)
28	Horizon Mismatch	T = long + the decision-maker’s tenure is shorter than the consequence cycle
29	The Invisibility of Maintenance	T = long + maintenance has no output metric + budget pressure

Ch.	Mechanism	Trigger combination
30	Handover Rupture	S = multi-team + fast staff turnover + documents that record conclusions only
31	Regulatory Lag and Capture	G = dense but captured
32	Compliant and Unsafe	G = dense + safety meetings discuss only whether a rule was broken
33	Cascading Failure	C = high + S = network + concentrated dependence (outsourcing, a single supplier)
34	Path Lock-in	system age ⩾5 years + alternatives no longer appear in selection discussions

Three reminders for reading the table. I is the heaviest parameter: systemic misalignment lights four rows at once, 12, 17, 19 and 27, and if you fill that level your candidate list stands on those four chapters. The pair C and R at high with medium or low lights 21, 22 and 33 directly, and that is the main axis of software and infra‑structure situations. Triggers of the "under way" kind, the rollout action in row 23 and the scale climb in row 24, behave differently from state triggers: they require a refill while the action is running. At the second region you roll out to, the parameters may already have moved.

35.4 Narrowing from Six Candidates to Two or Three

Three filters, run in order.

Filter one: evidence. Ask of each candidate whether your situation holds the specific evidence named by that mechanism's entry criteria. Chapter 15's practical drift, for instance, is tested by

“formal process and actual practice have diverged over time” . Where you cannot produce one concrete example, the mechanism drops to an observation item and leaves the main list.

Filter two: level (Chapter 9). Which layer can you move? A line engineer can move the technical layer and the organizational layer, so a mechanism triggered mainly at the incentive layer (12, 17, 27, 28) is a finding that has to be reported. Written as a traffic report that travels upstream, it has a route; treated as something to defend against locally, it has none.

Filter three: time window. Consequences landing inside your tenure stay on the main list. Consequences beyond it move to the ledger with a named recipient. This is not avoidance. A risk nobody picks up is handled only once it is written into a handover (Chapter 30).

After the three filters, the last step converts the main list into an action list. Each mechanism on it maps to section 5 of its chapter, How to Prevent It, where the devices are listed one by one: which device, who signs, at what rhythm. The action list format sits in the template of Chapter 38. Table to action list to template, Chapters 35 to 38 form a complete chain from finding to landing. This chapter is the first link; the chain closes in the rehearsal meeting of Chapter 39.

Two filled instances make the use concrete.

Instance one: the payments team of a SaaS company, fourteen people, just signed its second large customer and is about to take on a third. Six parameters. Coupling medium, because a payment failure spreads along the order chain but manual traffic switching can isolate it inside the business chain. Redundancy reality medium, because the standby payment channel comes from the same vendor as the primary and the switchover was rehearsed once, at launch. Incentive direction partly misaligned, because release ca-

dence counts in the team's appraisal while the risk control post holds a real veto over changes that cause financial loss. Time pressure short, because a quarterly revenue commitment makes the gain from compressing the verification cycle visible at once. Scale multi-team, because payments depend on risk control, reconciliation and the merchant interface team. Regulatory density sparse, because the industry has no strong regulator and the binding constraints come from customer contracts. Against the table in 35.3, four rows hit: 16 (multi-team plus reporting that carries appraisal weight), 22 (medium redundancy from the same source), 24 (climbing from multi-team toward network), and the edge of 12, where the evidence that the layer above sees only numbers is thin, so it drops to an observation item. The three filters: evidence keeps 16 and 22 (last quarter a financial-loss report took three weeks to reach the team, which is the concrete example for 16; the standby channel has never switched for real, which is the example for 22), while 24 stays because the action is under way, with a note on when to refill. Level puts all three inside the technical and organizational layers the team lead can move. The time window transfers nothing. Main list: 16, 22, 24. Action list, three items: read the prevention section of the three chapters; schedule one real switchover rehearsal for the standby channel and record the result; put "delay in information reaching us" on the agenda of the next quarterly report as an explicit item.

Instance two: a surgical nursing unit in a hospital, the sheet filled by the head nurse. Coupling high, because any checking error acts directly on the patient and there is nothing to isolate it. Redundancy reality medium, because the second-line duty draws on the same pool as the shift and the roster lists them as each other's backup. Incentive direction partly misaligned, because operating-room turnover counts toward performance while the nursing department holds independent inspection authority over safety checks. Time pressure medium, because complications show up

over weeks and months. Scale multi-team, because one operation is run jointly by surgery, anaesthesia and nursing. Regulatory density dense. Three rows hit: 16, 18 (a multi-party signature structure), 32 (dense regulation, and the department's safety meeting agenda is dominated by violation notices). Filters: evidence keeps 18 and 32 (the three-party check runs with the signature as the completion mark and the content check left to individual diligence, so 18 holds; six consecutive months of minutes discuss violation events and not one near miss, so 32 holds). Level demotes 16 to a reporting item, because the cross-department reporting structure is not something a head nurse can move. The time window transfers nothing. Main list: 18, 32.

Set the two instances side by side and one thing shows. Same six parameters, and the software team's main list lands on information and redundancy while the nursing unit's lands on structure and compliance. The difference in combination decides which chapters get read. The table does not hand you the answer. It takes you off random reading.

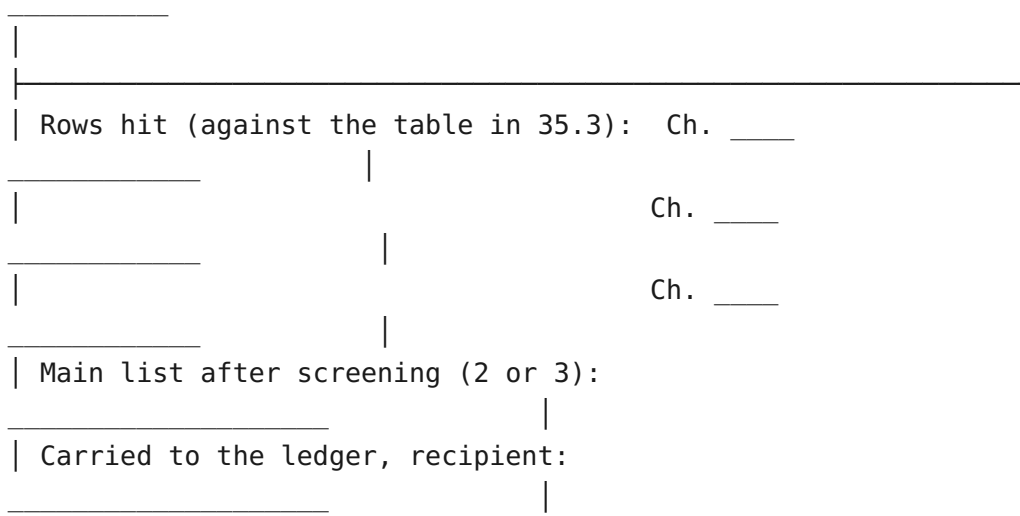
The table also works as a ninety-minute meeting. Everyone fills their own copy beforehand, fifteen minutes. In the room each parameter is called out with its basis read aloud, about sixty minutes for six parameters. Before people leave, one item only: the two parameters with the widest disagreement become actions, fifteen minutes. The discipline comes from the rehearsal meeting in Chapter 39. Argue with the basis, never with the person.

Of the three filters, the level filter is the one most often skipped, so here is a small worked case. A mid-level engineer at a platform company fills the table and the main list comes back with Chapter 28, horizon mismatch: the platform's exit clauses are missing, the consequence cycle clearly outlasts the tenure of the decision-makers, and the criteria hold one by one. The level filter then asks its question, and the engineer cannot move the exit clauses, which

belong to the financing structure and the board. Chapter 28's destination is a report, written in the shape of a finding: the criteria one by one, the evidence each points to, the facts verified, and the action recommended at the decision level, in a form the layer above can read. All of Chapter 16's techniques for the upward channel apply. Copy yourself on one baseline as well: a known structural risk that has been written down counts as doing your duty, and one that has not been written down is complicity. Here the level filter's real function shows. It converts "what I cannot defend against" into "what I can report", which is why the main list only ever holds locks you actually have keys to.

35.5 Printable Version

```
┌──────────────────────────────────────────────────────────────────────────────
│ Situating Table   date: _______   filled by: _______
object: _______    │
│ 1  Coupling: low / medium / high            basis:
__________          │
│ 2  Redundancy reality: high / medium / low  basis:
__________          │
│ 3  Incentive: aligned / partly / systemically mis-
aligned              │
│    basis:
__________
│
│ 4  Time pressure: short / medium / long     basis:
__________          │
│ 5  Scale: single point / multi-team / network
basis: _________         │
│ 6  Regulatory density: none or sparse / dense /
dense but captured     │
│    basis:
```



Filling discipline: every level needs a basis, one sentence is enough, and a level without a basis gets filled at the most dangerous setting. Chapter 36 takes over from here: fifteen minutes in the library, ten minutes writing the conclusion, and the output is "which three deaths I most resemble".

35.6 The Six Parameters at Organizational Level

The personal version is complete on its own. The organizational version turns it into a quarterly rolling instrument. Each quarter, every owner of a key project or system fills one sheet, and the sheets aggregate upward to the governance layer, which reads trend rather than single sheets. Is coupling rising or falling? Is redundancy reality improving or degrading? Has incentive direction worsened? A parameter's quarterly trend carries far more than a single reading. A single reading locates; a trend warns.

The second organizational use is the distribution chart: a bar chart for each parameter by level across all projects, counting how many

projects sit at high coupling, how many at low redundancy, how many at systemic misalignment. The extreme quadrant, the count of projects where high coupling, low redundancy and systemic misalignment hit together, is the organization's heaviest risk carrier. That carrier list sets the topic priority for the rehearsal meeting in Chapter 39.

The third use is entry positioning for new projects. At approval, the project owner fills one sheet as an attachment, standing beside the exit clauses and the list of irreversible points as the three documents of approval. Entry positioning hangs the warning on the project from day one, instead of waiting for the first anomaly to open the book.

Calibration data only accumulates value in a format. The recommended minimum is one line per record: date, object, six levels, chapters hit, revisit column. The revisit column gets two fields six months after the fill. Whether the object showed symptoms, meaning what happened, or an explicit statement that nothing did. Whether the hit held, where a symptom matching the mechanism of the hit chapter counts as a hit and a mismatch is recorded as a deviation with the mechanism that actually ran. Version control for the map follows from this. The table in 35.3 carries a version number, governance reviews hits and deviations once a quarter, and a row is revised only after three or more deviations accumulate, with the calibration source attached to the revision. The rule has one purpose: to make the table more accurate with use instead of leaving it at factory precision. Three years of calibration data carries a by-product. Your organization's own mechanism bias will show, the same two or three chapters hit repeatedly by similar objects, and that bias is the organization's failure fingerprint. The topic list for the rehearsal meeting in Chapter 39 is drawn straight from it.

Three typical fingerprints recur in high-frequency situations, listed for quick comparison.

Fingerprint A: an internet product in growth. Coupling medium to high, redundancy medium, incentive partly misaligned, time short, scale climbing, regulation sparse. Hits 16, 22, 24 and 33 at high frequency. The theme is expansion turning yesterday's redundancy into today's single point: during a scale climb, the backup that was enough last year does not cover this year's added dependency surface.

Fingerprint B: a heavily regulated operating unit (hospital, nuclear power, aviation maintenance). Coupling high, redundancy medium, incentive partly misaligned, time medium to long, scale multi-team, regulation dense or dense but captured. Hits 18 and 32 at high frequency, adding 31 where the capture evidence is sufficient. The theme is the feeling of compliance taking the place of the safety question, with the fact of passing every inspection read as evidence that the whole is safe.

Fingerprint C: long-cycle infrastructure (buildings, pipe networks, platform foundations). Coupling high, redundancy low to medium, incentive often systemically misaligned, time long, scale network, regulation dense with rules that update slowly. Hits 21, 25, 28 and 29 at high frequency. The theme is invisible maintenance meeting horizon mismatch: the consequence lands in the next person's tenure while the cut happens in this period's budget.

Fingerprints save time. A reader whose situation closely resembles one can read that fingerprint's chapters in the order of their hit frequency, then come back and complete the evidence for all six parameters as a check. A fingerprint is a starting point; evidence is the landing place. Where the two conflict, trust the evidence.

A rarer result is also real: six parameters filled, rows compared, no row hit. An empty result has three explanations, in order of prob-

ability. The likeliest is optimistic filling, a variant of error one, and the fix is to go back to the evidence lists with attention on incentive and redundancy, the two places where self-image pulls the level. The next is a genuinely clean situation: low coupling, real redundancy, aligned incentives, a sane tempo, controllable scale, effective regulation. Such combinations exist, mostly in small stable teams, and the table's job is done, with an annual refill. The one that deserves most suspicion is the third: every parameter green and the felt sense uneasy. Return to the line at the head of this chapter, where the numbers look good and the feeling is wrong. Unease usually comes from a dimension that has not been parameterized yet, the state of a key person, a process awkwardness nobody can name. Write it down as a candidate parameter for the next fill. An empty result is information, and the order for handling it: doubt the table first, doubt yourself second, trust the result last.

35.7 The Limits of the Table

No table can be exhaustive. Four limits are known and stated here to prevent misuse.

Limit one: the six parameters compress, and compression loses information. Dimensions outside parameters one to six (team culture, leadership style, an external shock) are not in the table. Lost information does not become unimportant; it falls outside this table's remit.

Limit two: three levels is coarse, and the world is continuous. The coarse grain leaves room for subjective judgment at the boundary, and the counter is the basis field. A level with a written basis at least lets a reviewer judge your judgment.

Limit three: the trigger regions overlap, and one combination hitting several rows is normal. Several hits does not mean all of them

are running. The table gives candidates, not a diagnosis, and dia - gnosis goes back to each mechanism chapter and checks its criteria one by one.

Limit four: the table does not replace intuition. It is a tool, and the judgment of the person holding it is the core. Chapter 14's warn - ing about the failure zone applies to the table as well: where the environment is irregular or feedback is delayed, your locating in - tuition may itself be off, and in that case the table deserves more weight than the intuition does.

The common conclusion from all four: the table is a first screen, not a final review. Its value is compressing twenty-four chapters into two or three, which gives attention a starting point. What lies past those chapters, going back to the chapter, checking the cri - teria, finding the evidence, is not the table's work. It is yours.

35.8 Five Common Filling Errors

Error one: optimistic levels. The sign is an empty basis field, or a basis that states a wish ("redundancy: high, we are planning a re - mote disaster-recovery site"). The design already carries a counter, since a level with no basis gets the most dangerous set - ting. What actually works is cross-filling: two fillers' difference in levels is more conspicuous than any reminder.

Error two: filling the target state. The organization is mid-reform and the filler writes the post-reform picture. Keep two sheets apart. A present-state sheet locates mechanisms; a target-state sheet ar - gues the reform; the difference between their levels is exactly what the reform is buying. Mixed into one sheet, every warning misses, because you compared an organization that does not exist yet against the trigger regions.

Error three: filling once. The sheet gets treated as a one-off diagnosis and filed. Mechanisms run dynamically and parameters move: expansion raises scale, layoffs lower redundancy, a new appraisal scheme worsens incentives. Put the table on a rhythm, quarterly routine plus event triggers, where a new appraisal scheme, an organizational expansion or a serious accident reported in your industry each earns a refill. The event-triggered sheet usually carries more information than several quarterly ones, because the event has just changed your situation.

Error four: taking a hit for a diagnosis. Main list of two or three chapters in hand, the reader skips the criteria and starts installing prevention. The table gives candidates. Diagnosis goes back to section 1 of each chapter and checks the entry criteria one by one, keeping what has concrete evidence and demoting what does not. Prevention installed past the criteria is a placebo.

Error five: filling it and not showing it. The sheet goes into a drawer because the levels look bad. The table's value to one person is finding risk, and its value to the organization is trend and distribution. Both depend on being seen. A sheet nobody sees is a sheet not filled: its whole information content has improved the filler's intuition, and Chapter 14 has just said under what conditions intuition fails.

Same table, different roles, different centre of gravity. Four common roles are worth separating.

Line engineer: what you can move is the technical layer and the organizational layer. Incentive direction will most likely come out partly or systemically misaligned, and by the level filter that is not a level you can change; it is material for a report. The engineer's version of the table therefore produces two columns: what can be defended (mechanisms in the technical and organizational layers, with the prevention section attached) and what can be reported (mechanisms in the incentive layer, written as a report with con-

crete evidence). Keep the columns apart and the table does not turn into a complaint list.

Middle manager: you are the direct operator of four of the six parameters (incentive, time, scale, regulation), and your filling accuracy sets the quality of the organizational roll-up. The error you are most likely to commit is error two, filling the target state, because you are usually the person executing the reform. The counter is to have your direct reports cross-fill your sheet. Their level for incentive direction sits closer to the truth than your own.

Executive and board member: your seat suits two parameters, scale and regulation, whose evidence is on your desk. The other four should come from the middle-management roll-up rather than your own pen. The core output of the executive version is the dis - tribution chart (section 35.6) and the list of extreme-quadrant car - riers. A single sheet is of limited use to an executive; the distribution is your view.

External role, audit, investment due diligence or regulation: you are filling in the situation of the subject, and the self-assessment standard is discounted because you do not hold their ledger. Work by evidence availability, dropping a level where evidence is miss - ing. Where you cannot get evidence, assume nothing optimistic, start at medium and mark it "not visible from outside". The greatest value of an external fill is comparison: set your sheet against the subject's own, parameter by parameter, and the widest gap becomes the first item on the due-diligence question list.

The shared discipline for all four roles is one line. Fill the para - meters you have evidence for, and cite the other parameters from whoever filled them. The higher the role, the easier it is to fill the line's levels from imagination, and the table distorts upward along the power gradient the way information attenuates upward through the hierarchy (Chapter 16).

A concrete schedule for landing it.

Week one: each core project lead fills a present-state sheet, one cross-fill happens, and the pair produces a main list and a draft action list. The week does not aim at an organizational roll-up; it aims at every sheet having a basis field.

Quarter one: two things inside three months. Turn the "read the chapter" item on the action list into one two-hour collective close reading, with a presenter for each hit chapter and a written answer afterwards from each participant, which level they now fill differently. Run one revisit cycle as well, recording the state of the previous round's objects in the revisit column, even if the first round's conclusion is "no symptoms, hit holds".

Year one: after four quarters, governance should hold four things. A map with a version number. At least three rounds of calibration data. One distribution chart. At least one rehearsal meeting scheduled off the distribution chart (Chapter 39). With all four, the table has turned from a sheet of paper into a system. Missing any one, it is still paper.

The most common break point in the schedule is the second quarter, when the first round's novelty is gone and nobody fills the revisit column. The counter is to write the revisit into the standing agenda of the quarterly meeting, ten minutes, never skipped. Agenda status is the only reliable defence against forgetting, and it shares a base principle with the invisibility of maintenance in Chapter 29: anything without a routine carrier disappears into busyness.

Five Self-Checks

1. Fill the table in 35.5 in five minutes. Which incentive direction level do you get, and would you show the basis to your team?

2. Of the six parameters, which do you answer with least confidence, and what kind of information are you missing that leaves it uncertain?
3. In your situation, which rows of the 35.3 table hit, and of those chapters, which have you never read?
4. Has your organization run a rollout into a new environment in the past year? Did anyone list what had changed relative to the pilot?
5. Which level did you give time pressure? Think of your last decision to compress a verification process: does the reason given then still hold?

35.9 Where the Table Sits in the Book's Toolchain

The table is the first link, and four things hang off it downstream. The thirty-minute walkthrough of Chapter 36 takes the table's output as its input: the first half of the walkthrough list, which three deaths, comes straight off the main list. A walkthrough without the table picks topics from memory, memory picks wherever the last accident happened, and that place is not necessarily where the mechanism is most active. Chapter 37's review questions are chosen by hit mechanism. Running the questions for all twenty-four chapters is unrealistic, so the design questions belonging to the hit chapters get asked first, and the question list carries chapter numbers for selection by number. Chapter 38's postmortem template uses mechanism names as archive tags: every postmortem writes "mechanism: Chapter X" in its conclusion field, archived postmortems aggregate by mechanism, and the aggregate feeds calibration of the 35.3 map, which closes the chain into a loop. Chapter 39's rehearsal topics come from the extreme quadrant of the distribution chart, since the quarterly organizational roll-up of section 35.6 produces the heavy-risk carrier list and the rehearsal schedules off it, one carrier per session. The

mark of a full circuit: a real hazard found in a rehearsal, archived through a postmortem, becomes one calibration record in the map. Read the four together and the table’s role is clear. It turns “read a thick book” into “read two or three chapters and do three con‑crete things”. The rest of the toolchain each holds one segment, and together they are an organization’s mechanism warning and rehearsal system.

Chapter 36 Thirty Minutes Before the Decision

> What this chapter covers: the full run from the position table to a conclusion page reading "these are the three deaths I most resemble", in a time box of 5 + 15 + 10 minutes, with a walkthrough of a synthetic example end to end. When to turn back: before any decision that touches a point of no return (Chapter 7); and when you want other people to use this procedure and need to hand them a minimum version. Estimated reading time: 30 minutes.

36.1 The Procedure at a Glance

The thirty-minute procedure is the executable form of the position table (Chapter 35) and the master gate for all the mechanism material in this book. It does one thing. Before a point of no return, it converts "which kinds of death do I most resemble" from diffuse anxiety into three specific mechanism names and three corresponding checking actions.

The compressed version for emergencies is worth giving up front as well. Each of the three time boxes shrinks to a third: two minutes filling the table, with only two parameters, incentives and time; five minutes in the library, checking only the first criterion of the first mechanism that hits; three minutes writing the conclusion, with only the likeliest death and one action. The ten-minute version produces one death and one action instead of three, and it is a hundred times better than zero minutes. Its precondition is a real emergency, a decision window measured in hours. If your window is measured in days, run the full thirty. A sense of emergency is not an emergency: often it is the amplifier from Chapter 13, narrative before evidence. The more you feel there is no time, the more you need the procedure.

One numerical coincidence in the design is worth naming. Five plus fifteen plus ten is thirty, and thirty minutes is roughly the limit at which attention stays at high quality on a single task. Beyond thirty, fatigue degrades the judgment you bring to the library search. Short of thirty, the table and the search lack depth. Thirty minutes is the optimum between cognitive load and judgment quality. Nobody designed the number that way; it was noticed afterwards, when the finished procedure was looked at as a whole and turned out to sit on the sweet spot of cognitive science. That is why it should not be compressed to fifteen, where quality fails, nor stretched to sixty, where fatigue cancels the gain. Thirty minutes is thirty minutes: physics, not a preference.

The three time boxes:

Stage	Time	Action	Output
Fill the table	5 minutes	Six-parameter positioning (Chapter 35, table 35.5)	6 grades with evidence
Search the library	15 minutes	Against the hit mechanism chapters: criteria, weak signals, comparison cases	Candidates narrowed to 2–3 mechanisms
Write the conclusion	10 minutes	Three-paragraph conclusion page	One page: three deaths, three checking actions, check person

The threshold for using the procedure: the decision touches a point of no return (signature, go-live, announcement, external commitment), or the commitment is large enough that withdrawal would draw on second-tier resources. Ordinary decisions below that threshold do not need the full run. Copy the format of the conclusion page instead, three deaths plus checking actions, and use it in a five-minute spoken version.

Timing has its own rules. Run the procedure before the decision point, and leave at least one overnight gap (Chapter 7, the cooling-off period). Run it after the meeting and the output is postmortem material. Run it before and the output is armour.

There is a deeper reason for that difference. The person who runs the procedure before the meeting is still inside the information fog, and the discipline of Chapter 6 comes naturally: you have no outcome to lean on, so judgment is forced to start from what was visible. Run it afterwards and all three disciplines fail. You already know the outcome, and creeping determinism paves every fork in the road into a highway. So the thirty-minute procedure has one legal timing: before the decision. What happens afterwards is a postmortem (Chapter 38), not a thirty-minute procedure. The two have different purposes, and armour and teaching material should not be packed in the same box.

Ten minutes of preparation come before the clock starts. They do not count toward the thirty, but without them the procedure idles. Four items.

Materials: the numbers the decision turns on, budget, schedule, dependencies, and the record of comparable past decisions, on one page at most. The thirty minutes will not digest a fifty-page feas - ibility report. It needs the conclusions and numbers you have already digested. If materials are missing, the gap is handled under the Chapter 35 evidence checklist: whichever class of number is missing, that parameter gets the most dangerous grade.

Room: a table you can write on and a stretch of time nobody in - terrupts. The three time boxes do not tolerate handling messages at the same time. Phone silent, and the thirty minutes marked on the calendar as non-displaceable. Displacement is the most common way this procedure dies. It looks postponable at any moment, and after the third postponement the decision point has arrived.

Role: confirm the candidate check person, against the standard in 36.4. Asking in advance works better than asking afterwards, be‑cause advance notice leaves the other person room to decline or to name someone better. Asked afterwards, only obligation remains.

Entrance: write one sentence answering “which point of no return does this decision touch”, using the Chapter 7 test: signature, go-live, announcement, external commitment, or withdrawal drawing on second-tier resources. The sentence goes at the top of the table. A decision whose point of no return you cannot name usually fails the threshold; take the spoken version and leave it there.

Two patches for the compressed version. The first: compression applies to time boxes only, and the check person is never dropped. The ten-minute output of one death and one action still carries a name and a time; a compressed page without a checker is self-comfort. The second: when a decision run in compressed form came out fine, give it a full thirty minutes later. Walk the judgment again with time to spare, and put the difference into the team’s card library, meaning the mechanisms the full run caught that the compressed run missed. Compression is an emergency posture. Routine compression is routine myopia.

36.2 The First Five Minutes: Filling the Table

Fill the six grades on the printable version at Chapter 35, 35.5. The discipline has one line: every grade carries evidence, and a grade with no evidence is filled at the most dangerous setting. A para‑meter you cannot write evidence for inside five minutes is exactly where your information is darkest, and that fact is worth recording. Section 36.6 comes back to it.

The common error is turning the table into a statement of position. For “incentives: aligned”, the evidence cannot be “our team has a good atmosphere”. It has to point at a specific appraisal clause,

and the test of that clause is whether the role that creates the risk bears the consequence. No clause, and the starting grade is partial misalignment.

36.3 The Middle Fifteen Minutes: Searching the Library

Take the 3–6 mechanisms, by chapter number, that the 35.3 table hit. Run three fixed actions on each, at a budget of 2–3 minutes per mechanism. The actions can be that fast because every chapter in Part Three has the same skeleton, in the same positions (writing standard, section 10).

Action one: check the activation criteria (section 1 of the chapter). Each mechanism chapter gives three criteria of the form "present means already started". Ask of each one: is there identifiable evidence of this in my situation? All three absent, and the mechanism leaves the main list.

Action two: scan the weak-signal list (section 4). Each chapter carries 6–10 weak signals, and each signal comes with the reason it was pressed down at the time. Any hit gets marked. Two hits or more, and the mechanism moves to the head of the main list.

Action three: read one comparison case (section 3 plus Appendix B). Read only the case card's decision-point paragraph and its minimum counterfactual paragraph, not the whole account. The purpose is to give the candidate mechanism a concrete face, so that when you discard it you are not discarding something you simply could not picture happening.

At fifteen minutes you should be holding a main list of 2–3 mechanisms, each with criterion evidence, 1–3 watch items where there is suspicion but no evidence, and the names of the mechanisms you moved off. Removal leaves a trace as well. Three months on, at

the postmortem, the judgment "this one did not look like us" is itself up for examination.

36.4 The Last Ten Minutes: Writing the Conclusion Page

Three fixed paragraphs, one page.

Paragraph one: the three deaths I most resemble. One line each: mechanism name with chapter number, one sentence describing the situation, and the criterion evidence that hit. A worked example: "Practical drift (Chapter 15): the spot-check procedure went live three months ago and runs at 2% while the rule says 5%, and nobody has the authority to change the difference (criterion 1 hit)."

Paragraph two: one checking action per death. The action must be completable before the decision point and it must leave an artifact. Example: "Before go-live, pull the actual execution rate of spot checks against the written value; if the gap exceeds half, the switch does not ship." One action per death, the same discipline as the minimum counterfactual. Three actions equal zero actions.

Paragraph three: check person and time. State who verifies paragraph two's output, and at what hour.

For the artifact, use a sticky note or a message rather than a formal document. The ceremony of a formal document is what makes people postpone writing the page. The note has three lines: three deaths, three actions, one check person with a time. It goes on the desk or is pinned in the project tool. A visible conclusion page is ten times more useful than one filed in a folder, and the Chapter 30 register works on the same principle: visibility is a tool's oxygen. A conclusion page without a check person is a diary, not a tool.

The check person is a key design element of this thirty-minute choreography and deserves its own explanation. The role does three things. It is a time lock: the action you wrote has to appear in the checker's calendar, or it stays words. It supplies an outside view: the checker sits outside your decision chain, so he reads your artifact without your bias. It is an escalation channel: if he finds the action undone or the result abnormal, he has the right to report it upward, a miniature of the stop authority in Chapter 18. The selection standard: no stake in the decision, time to look, and knows what abnormal looks like. People who meet all three are easier to find than you expect. A quality person on another team, a PM on the neighbouring project, or your own manager, as long as he will spend five minutes.

Fifteen minutes of library search only runs if the stock has been arranged beforehand. Three speed tricks.

Trick one: criterion cards. Make one card for each mechanism chapter you have hit, three activation criteria on the front and eight weak signals on the back. The source is a transcription of sections 1 and 4, and making the card is itself a close reading. After a dozen of your frequently used mechanisms, the first two actions become flipping cards instead of flipping the book.

Trick two: pre-marking. In the two or three chapters you use most, highlight the comparison case's decision-point paragraph and its minimum counterfactual paragraph. Action three then changes from finding the paragraph to reading the mark.

Trick three: a fixed format for the trace. Record every search on the same line: mechanism, criterion entries hit, weak-signal entries hit, reason for removal. A fixed format is what lets a postmortem three months later compare the judgment then against the facts now.

36.5 Walkthrough of a Synthetic Example

The following scene is a teaching composite, not a real case, and exists to show how the three time boxes join.

Scene: a payments team decides to replace manual line-by-line review of refunds with automatic approval plus 5% spot checks. Decision point: the act of flipping the configuration switch. Point of no return: automatic approval takes effect and users begin receiving refunds nobody reviewed.

Filling the table, 5 minutes. Coupling: medium, since a wrong refund hits a single transaction and the support queue, and a payment can be stopped by hand but volume would drown the staff. Realness of redundancy: low, since the automatic rule's backstop is a manual console on the same code, with no independent check line. Incentives: partly misaligned, since support hours are this group's appraisal item while the loss is carried by risk. Time pressure: short, since it must ship before the promotion. Scale: multi-team, with payments, risk and support involved. Regulatory density: dense, since payments is subject to regulatory examination and rule changes must leave a trace. Every grade has identifiable evidence.

Searching the library, 15 minutes. The 35.3 table hits four rows: 15, practical drift, time short plus rule against workload; 11, normalization of deviance, a spot-check ratio that is nominal and a consequence that arrives late; 12, goal displacement, the hours metric outweighing the loss; 26, automation and skill decay, manual review skill going unused. Action one, criteria: Chapter 12 criterion 3, "somebody has been penalized for pointing this out", has no evidence yet, so the mechanism drops to a watch item. Chapter 15 criterion 1, "the same deviation twice or more", hits, since last month brought two cases of "spot-check ratio temporarily lowered during peak" with no change process. Chapter 11 cri-

terion 2 hits, since the per-transaction loss is small and has always been explained as contained. Action two, weak signals: Chapter 15 hits two, training teaching something different from the manual, and new hires learning from the old hands. Action three, a case: the practical-drift comparison case confirms the shape, a fork that came from a rule written and not followed.

Writing the conclusion page, 10 minutes. The three deaths are practical drift (15), normalization of deviance (11), and automation and skill decay (26, promoted from watch item, on the argument that manual review skill decays as it goes unused and six months from now nobody may be able to run the rollback switch by hand). Actions: (1) before go-live, pull the actual spot-check execution rate for the last three months and compare it with the written value; if the gap exceeds half, do not ship; (2) add an independent check line beside the automatic rule, keeping refunds above a threshold on manual review, drawn from a heterogeneous source; (3) keep one manual review drill a month and write proficiency into the shift handover. Check person: the risk interface verifies (1) at the go-live review, and the duty manager verifies (3) monthly.

The most valuable part of this page is action 2. It comes from none of the three mechanism texts. It surfaced structurally once the three mechanisms had been located, as the obvious repair: put back the independent redundancy that is missing. What the pro - cedure produces is three concrete actions with named checkers, and the vocabulary is a side effect.

Each of the three paragraphs also has a sample for a different class of decision. Copy the format.

Infrastructure sample, data-centre migration. Death one, hidden coupling (Chapter 21): the migration list carries no dependency on the old backup system, and the criterion “nobody can state the full blast radius” hits. Action: run a full dependency scan before migration and compare it against the oral interviews. Death two,

the illusion of redundancy (Chapter 22): the standby hall and the primary hall sit with the same carrier, and the criterion "same-source backup" hits. Action: run the switchover drill once under conditions where the standby hall is actually carrying production traffic. Death three, handover rupture (Chapter 30): the migration window crosses a quarter boundary and the on-call roster changes over it. Action: the handover page of the migration manual gets a signed cross-check before the changeover. Check person: the platform lead verifies all three at the migration review.

Hiring sample, a key post. Death one, where intuitive expertise breaks down (Chapter 14): the interviewer's judgment comes from a highly regular interview environment. Action: add a work-sample test built on the same structure as the job's real tasks. Death two, narrative before evidence (Chapter 13): the requisition is already announced externally and the candidate profile was publicly committed at announcement. Action: hand the evidence list behind the hiring standard to a third-party colleague for review before the final round. Check person: the HR interface verifies before the offer goes out.

Contract sample, a supplier agreement. Death one, horizon mismatch (Chapter 28): the term runs five years while the counterparty's financial health moves on a quarterly cycle. Action: add an annual financial-health review clause and an exit clause. Death two, crossing a scale threshold (Chapter 24): the single supplier's share will pass four in ten. Action: prequalification of a second supplier starts in the same period as signing. Check person: legal verifies before the seal is applied.

What the three samples share: every death points at a criterion, every action has an artifact and a time of verification, and the whole thing fits on one page.

With the walkthrough finished, the three boxes deserve a commentary each, because the example hides design intentions that are easy to read past.

On the filling box. The heaviest grade among the six is realness of redundancy at low. Look at how the evidence is written: the backstop is a manual console on the same code, with no independent check line. That one sentence does two jobs. It evidences the grade, and it already implies the repair, an independent check line from a heterogeneous source. Good evidence carries its own repair clue. Weak evidence, "the backstop feels shaky", is a record of mood.

On the library box. Chapter 12 was demoted to a watch item because criterion 3 had no evidence. The move is worth enlarging. A mechanism with incomplete criteria can still be a real mechanism, and watch item means "current evidence does not support it", not "it is not happening". Watch items belong in the margin of the conclusion page; this example did not write them there, but it kept the trace. Searching has always included exclusion. Knowing which mechanism lacks evidence is worth, defensively, as much as knowing which one is confirmed.

On the writing box. Action 3, the monthly manual drill, serves death three, skill decay, and its verification rhythm is monthly, decoupled from the decision itself. That shows the page can hold two classes of action: one-off actions before the decision point, such as pulling the execution data, and periodic actions after it, such as the monthly drill. Both need a checker, and only the check time differs. The example wrote both because an automation decision naturally produces periodic after-effects, which is the theme of Chapter 26.

The method has a source. Research on prospective hindsight shows that when a task is framed as "imagine the event has already happened, then work back to its causes", people list significantly more causes, and better ones, than when they are asked directly

what could go wrong (Mitchell, Russo & Pennington, Journal of Behavioral Decision Making 2(1): 25–38, 1989; Crossref record checked). The writing section of the thirty-minute procedure borrows that effect directly: "which three deaths do I most resemble" is a hindsight question. Gary Klein developed it into the premortem, and Chapter 39 of this book is his ninety-minute version. The thirty-minute version here compresses the effect onto a decision at personal scale, and the price of the compression is a single viewpoint. One person cannot run past his own group's blind spots, which is why 36.4 insists on an external check person.

36.6bis The Cross-Check List for the Conclusion Page

Five minutes of cross-check after the page is written, the same instrument as cross-filling: ask a colleague with no connection to the decision to go through it. Five items, one at a time.

1. Does every death point at a criterion entry, Chapter X, criterion 1, 2 or 3? A death you cannot point at is a story. Go back to the library.
2. Does every action have an artifact? The artifact can be data, a record, a drill result. It cannot be "awareness", "atmosphere" or "attention".
3. Can every action be given a date on a calendar? An action you cannot date has no owner.
4. Does the check person meet all three standards: no stake, has time, recognises abnormal? Miss any one and change the person.
5. Do the three deaths cover different layers (the three layers of Chapter 9)? Three deaths all on one layer, all technical for instance, mean the search missed organizational and incentive candidates. Go back and search again.

Passing means all five. Two failures mean the page gets rewritten. Four failures mean the procedure itself has gone out of shape, and you go back to the 36.2 preparation list and start over. The checker signs initials in the box; the cross-check stays under ten minutes, because above that nobody uses it.

The page also needs lifecycle management. Once the actions have been verified, archive the note in a public place, a fixed column in the project tool, tagged twice: decision object and mechanisms hit. At the end of each quarter, pull the archive statistics, three num - bers only: usage (is anyone using it at all), the rate of actions veri - fied on time, and the cross-check item that fails most often. Each number maps to one improvement action. Low usage sends you back to step one of the rollout. A low verification rate sends you to the check person’s time commitment. The frequent failure sends you to the cross-check list itself. Statistics stop there, and a dashboard would be over-engineering.

36.6 Three Ways to Fail

Failure one: filling the table with the evidence skipped. A grade with no evidence is decoration, and the whole procedure fails with it. The test is easy. Show your table to a colleague with no connec - tion to the decision and ask which grade he does not believe.

Failure two: the library search turning into browsing cases. Stop - ping to read the stories blows the fifteen minutes. Read only the decision-point paragraph and the minimum counterfactual. The purpose is screening mechanisms, a case is a calibrator, and one calibrator is enough.

The calibrator deserves one more step of explanation. Case calib - ration exists to prevent matching mechanisms on paper. Without a comparison case, people tend to match any situation to the mech - anism they know best. The case anchors the match to concrete

detail from the scene, and anchored matching is more accurate. That is not theory; it is the common sense of information retrieval, where a query carrying an example beats a query of keywords.

Failure three: the page reading "heighten attention, keep tracking". An eight-word conclusion equals zero actions. Test: does paragraph two contain a checking action you can date on a calendar? If not, rewrite it.

A warning that runs the other way. The procedure can also do harm. Exploratory decisions (Part Six, Chapter 40) are not for this procedure, because in a situation that needs trial and error, rehearsing death kills the attempt. Judge whether your decision is exploratory or operational with the three criteria in Chapter 40, and only then decide whether to spend the thirty minutes.

One successful usage is left to write. The first run of the thirty-minute procedure must have someone alongside. What a first-timer needs most is a person who has been through it, and that person has exactly one job: when you state your evidence, ask how you know. That single question doubles the honesty of the table. The companion needs no formal appointment; any colleague who has read Chapter 35 will do, and a reading group, a training session or a morning stand-up can manufacture companions in batches. Someone alongside the first time, alone the third, teaching others by the fifth. After those three steps the thirty minutes stops being a tool and becomes a habit.

36.7 Five Common Questions

Question one: the decision maker has no thirty minutes. Delegate the run. An assistant or the team fills the table and searches the library from the materials, and the decision maker joins for the last ten minutes only. The delegated version loses honesty in the table, since an assistant writing your evidence will soften it a grade. The

counter is one extra question from the check person at the page: which of the six grades do you disagree with. Disagreements get refilled.

Question two: a superior overrules the action. An overrule is one of the procedure's normal outputs: the procedure put the risk on the table, and the table is where the decision chain decides. The page then gains one line: the action overruled, the reason for the overrule, and a signature reading "risk acknowledged". That line turns an overrule from a default into a trace, and a trace is itself a defence. The diffusion of responsibility from Chapter 18 is held off here by one sticky note.

Question three: several decisions run in parallel. Order them by the timing of their points of no return, and run the nearest one first. Parallel decisions share the same library stock, the criterion cards and the pre-marking, so the second thirty minutes usually runs shorter than the first. Never compress several decisions into one thirty minutes. The time budget of the three boxes is designed for one decision.

Question four: a remote team. Turn the boxes into a shared-document relay. The table and the search are done asynchronously, and the conclusion page is written face to face in a fifteen-minute video call. The remote version loses on action three, where the screen weakens the case's sense of place. The counter: one person reads the case aloud and everyone else only listens, since reading aloud forces slow reading and slow reading restores the sense of place.

The common ground under the four: every stage may change shape, and two things may never be dropped, the evidence column and the check person. The first keeps the table true, the second keeps the action happening. Delegation, relay and asynchrony are changes of carrier, not changes of procedure.

Question five: does this procedure make an organization timid. Teams that have run a large number of conclusion pages report one complaint: having to think of three deaths every time, is this teaching people not to decide. The answer sits in the output. The page produces three actions and a signature, and once the actions are done the decision proceeds as it would have. The procedure blocks two things only. It blocks the case where the actions will not be done and the decision is pushed through anyway, and blocking that is its job. It blocks exploratory decisions that wandered into it, which is the reverse warning in 36.6. Caution and timidity are told apart by output: caution produces actions, timidity produces delay. The procedure recognises actions only. One measurement note while we are here: in teams that run conclusion pages, the number of decisions does not fall. What falls is the number of firefights afterwards and the frequency of "we never thought of that" in postmortems. Those two are what the procedure actually buys.

36.8 Division of Labour with the Chapter 39 Premortem

The thirty-minute procedure and the ninety-minute premortem (Chapter 39) are two specifications of one family, split along three axes. Scale: the thirty-minute version serves a single decision, the premortem serves a project or a class of carrier (the extreme quadrant of the parameter map in 35.6). People: the thirty-minute version can be run alone, the premortem needs at least five, since diversity of viewpoint is its fuel and a solo premortem is a longer self-conversation. Timing: the thirty-minute version sits against the decision point, days before it; the premortem sits against project milestones, kickoff, architecture freeze, a month before launch. The outputs differ too: a three-line note against a hazard list with owners attached. The usual order is premortem first, thirty minutes later. Hazards found at the premortem enter the pro-

ject's checkpoints, and as a checkpoint approaches, each individual decision near it runs its own thirty minutes. With both levels running, an organization's protection at points of no return has scan precision at the single decision and coverage breadth at the project level.

36.9 Three Steps to Roll It Out

The thirty-minute procedure lives easily in one person and dies easily in an organization. Rollout takes three steps.

Step one: a pilot group. Pick a team where decisions are dense: product, operations, post-investment. Two or three people run it ten times. The pilot tracks two numbers only: how many runs, and which of the five cross-check items fails most. After ten runs, hold one postmortem and turn the mechanisms this team hits most into a team criterion card, a team library of its own, with a higher hit rate than the generic one.

Step two: make the template library and the card library public. The pilot's criterion cards, sample pages and cross-check list go into the organization's shared template library, placed inside tools everyone already passes through: the project kickoff process, a fixed step in the release process. No standalone knowledge base: nobody passes through a knowledge base, and only what you pass through gets used.

Step three: a quarterly sample. Governance pulls five archived pages each quarter and looks at two things, the rate of actions verified and the completeness of criterion citation. The result does not enter anyone's individual appraisal, because the moment it does, filling the table becomes a statement of position and the error 36.2 warned about returns. It has two uses only: adding to the card library and fixing the procedure. Rollout fails in a highly consistent shape. Step one has nobody alongside, and a lonely tool

does not survive its third use. Step two puts the templates in a knowledge base nobody passes. Step three turns the sample into an appraisal, and an appraisal forces acting. Follow the three steps and dodge those three pits, and after a year the conclusion page turns up in the organization's everyday vocabulary. At that point the tool has grown into the organization.

How the page interfaces with the governance processes already in place decides whether it is one more sheet of paper laid on top or part of the skeleton. Three interface points.

Interface one: the review meeting. If the organization holds kickoff reviews, architecture reviews or change reviews, the page's three paragraphs become the fixed first page of the review pack, with the rest of the pack unchanged. That first page turns a review from hearing a report into confronting risk: reviewers read the three deaths first and judge whether a worse one was missed, and a missed one goes onto the first page before anyone discusses approval. No charter change is needed, only one more sheet.

Interface two: the approval chain. The check person's signature box and the approval chain's sign-off box are different instruments and should not be merged. Sign-off is an act of authority, agree or not. Checking is an act of fact: did the action happen, does the artifact exist. Merged, the checker becomes the Nth signer and the factual content thins out, which is the Chapter 18 mechanism replicated at tool level. The correct merge is one required field in the approval chain, named "conclusion page archive number". No number, no passage.

Interface three: the postmortem. The postmortem of Chapter 38 looks back at the judgment made at the decision point, and the conclusion page is the snapshot of that judgment. The first move of the postmortem is to pull the archived page and check it line by line against what happened: how many of the three deaths occurred, how many of the three actions held, did the checker actu-

ally check at the time. The page gives the postmortem a primary record to compare against, and the ceiling on postmortem quality rises with it.

Common to all three: a new tool does not build its own stove. It makes itself one page and one field inside a process that already runs. Grafting outlives building.

A sample quarter with both specifications in use. At the start of the month, one project-level premortem, ninety minutes, with the hazard list going into the checkpoints. Mid-month, each decision near a checkpoint runs its thirty minutes: the spot-check switch, the canary percentage, the wording of an announcement, one note each. At month end, ten minutes with the month's notes and the premortem list side by side, marking where the two agree on mechanisms. A mechanism hit by both is the main theme in that project, and its criterion card belongs in the front row of the team library. The calendar demonstrates one thing. The two specifications do not compete for time; they compete for coverage. The premortem sweeps area, the thirty minutes sweep point, and coverage is complete only when the two line up.

Five Self-Checks

1. On the irreversible decision nearest to hand, which of the three time boxes can you not actually run? Is the obstacle time or authority?
2. The last external commitment you signed off without the overnight gap: looking back, was it worth it?
3. Can the actions in paragraph two of your conclusion page be dated on a calendar? Which one cannot?
4. Who on your team would make a good check person? Does that person have the authority to stop something?

5. What does action 2 in the synthetic example, adding an independent check line, correspond to in your own system? Does it exist today?

The check-person role has an organization-level version worth giving larger readers: a roster. The checker should not always be the same person, because a fixed checker turns the role into a formality. The logic matches the real drill of personnel redundancy in Chapter 22, and each rotation refreshes the checker's understanding as well. The rotation period follows decision frequency, monthly for dense teams and quarterly for sparse ones. The roster hangs in the team's scheduling system, on the same layer as the on-call roster. Treating checking as a kind of duty says by itself how much weight it carries, and duty cannot be skipped.

An upgrade path follows the five questions. Once the individual version is fluent, build the team version. What changes is that the library stage becomes a collective activity: each member claims one mechanism and checks its criteria, and five minutes later the group reports. Collective search hits more criteria, because different people associate different evidence with the same criterion, and the fifteen minutes usually turn up 1–2 more effective criterion hits than the solo version. The team version is more manual and suits decisions with high irreversibility. Low-risk decisions stay on the individual version.

A last paragraph for the reader still deciding whether to use any of this. The entire cost is one afternoon-tea slot and one page of sticky note; the entire output is three concrete checking actions. Cost against output, this is the best ratio in the book, cheaper than any ledger, drill or audit, because it needs thirty minutes before a decision and no project, no budget, no cross-department coordination. It is a tool one person can start alone. If you finish the book and want to carry away one thing you can use tomorrow, carry this chapter.

Chapter 37 Design Review Question List

> What this chapter covers: 130 review questions in groups, a general set plus the twenty-four mechanisms, each one carrying a note on what it means when you cannot answer. Use it with the locator table in Chapter 35: pick the groups your situation hits, then take the questions one at a time. When to come back to it: design reviews, decision meetings, due-diligence interviews, and any time you are setting an exam for a team below you. Reading time: about 50 minutes.

The question list is a toolbox. It comes off the shelf when needed and goes back, and nobody reads it through in one sitting. There are two occasions for taking it out. The first is a design review: ten minutes before the meeting, the questions for the mechanisms in play are copied into the agenda, and the review changes from a presentation with an audience into a pass down a list. The second is a self-check, one person walking their own system through the questions one at a time. The output of a self-check is the list of questions that could not be answered, and that list is the finding. A question nobody in the design ever considered is itself a gap.

Every question is worded for an answer out of knowledge: can you answer this, as distinct from have you done this. The first form points at knowledge, the second at process. Process on file with knowledge missing is the most common failure shape in this book, and it is practical drift (Chapter 15) seen through a review question.

The list has a second use, in training. Hand a newcomer the questions and send them to interview the veterans. The newcomer learns the structure and logic of the system; the veterans get asked things they never considered. Both sides gain. The barrier is as low

as a barrier gets, because nothing has to be taught to anybody. Hand over the list. The list teaches by itself.

Three rules, and they fit on three lines. Rule one: questions are used in groups. Take the mechanism groups the locator table hits, add the general group, and stop at twenty to thirty questions for one review; reading the whole table is an editing act, not a review act. Rule two: the arrow after each question gives the meaning of a failure to answer, and that meaning goes into the minutes, where it does more work than a follow-up question. Rule three: answers must land on dated evidence, and a verbal assurance counts as no answer.

Where the library came from is worth stating once. Not one of the 130 questions came out of theoretical reasoning. All of them were traced back from the decision-point field of this book's case cards: what material lay on the table at that decision point, and which question nobody asked out loud. That unasked question was then rewritten into a form you can put to yourself today. On the Alaska 261 card, the question lying next to the decision point was "whose inspection record is the extension based on" (the information source was the operator alone), and it became the prototype for question 3 of the Chapter 11 group. On the Travel + Leisure card it was "who independently reviewed the change in the counting rule", which became question 7 of the general group. On the Grafton card it was "once the alarm sounds, who has to stop what they are doing", which is question 2 of the Chapter 16 group. This provenance tells you what kind of document you are holding: an index to the decision points of a case collection, with a site behind every question that paid real tuition. It also explains the wording. Almost every organization in those cases had processes and had documents. What they lacked was somebody at the decision point asking the question and demanding an answer.

The relation to the trigger criteria needs drawing as well. A mechanism chapter's criteria answer whether the mechanism is running. A question answers whether your organization holds any knowledge about that mechanism. A criterion hit requires evidence. A question answered with "I do not know" requires only the admission. In a self-check the two tools divide the work: questions screen for black holes in knowledge, criteria confirm whether the mechanism is actually operating. Questions first, criteria second, because screening a black hole costs far less than collecting evidence, and the cheaper instrument belongs in front.

The General Ten (run this group first in any review)

1. At which second does this action become irreversible? → No answer means the decision-maker has not yet met the irreversible point (Chapter 7).
2. Who holds the last second-confirmation slot before that point, and what does that person check? → No answer means the defence is nominal.
3. Which of the items the decision rests on carry a timestamp? → None of them means the decision stands on retelling (Chapter 6).
4. If every dollar committed were written off tomorrow, who loses most? → No answer means nobody has costed the exit (Chapter 27).
5. Who holds the unilateral right to stop this, and in which document is it written? → No answer means the right does not exist (Chapter 18).
6. Was the last anomaly of this class attributed in the register? → No answer means the anomaly was explained away (Chapter 8, rung one).

7. Who independently reviewed the last change in the counting rule? → No answer means the ruler is being swapped (Chapter 12; Chapter 8, rung five).
8. When was the critical dependency list (third parties, suppliers, single systems) last updated? → No list means the dependencies are invisible (Chapters 21 and 33).
9. What happened, professionally, to the last person who stopped a project? → No answer means stopping is a career risk (Chapters 19 and 27).
10. Where does the person who pays for today's decision in three years sit now? → No answer means horizon mismatch is running (Chapter 28).

Normalization of Deviance (Chapter 11)

1. How many times has a deviation of this class occurred, and on what basis was each occurrence judged to have no consequence? → No register means normalization is already running.
2. In "there is still enough margin", is the margin a number or an adjective? → An adjective means the standard is softening.
3. When was the inspection standard last relaxed, and who approved it? → No answer means the relaxation was done in the dark.
4. How was the last question a newcomer raised about the old practice handled? → "It has always been done this way" means criterion ③ is hit.
5. Which release action can be completed without writing down why it is acceptable this time? → If one exists, release leaves no trace.

Goal Displacement; Goodhart’s Law (Chapter 12)

1. How is this metric being optimized right now? Name three actual workarounds. → Cannot name three means you have not been on the floor.
2. How long is the footnote to the metric’s definition, and who has the authority to interpret it? → A swelling footnote means the definition is drifting.
3. What happened afterwards to the last person who said the metric was distorted? → If they paid a price, criterion ③ is hit.
4. Which observation channel that cannot be scored is still running, and who reads it? → None means the organization has only the ruler left.
5. Has anyone written the list of differences between this metric and the goal it stands for? → Never written means the substitution runs unmonitored.

Narrative before Evidence (Chapter 13)

1. How many independent explanatory paths did the current conclusion pass through before it was fixed? → One means there is no competing hypothesis.
2. Whose job description includes “responsible for overturning this”? → Nobody means refutation has no post.
3. Is the hypothesis written in a form that can be overturned, with the evidence that would make you drop it named? → Cannot write that means it cannot be falsified.
4. How much evidence must an objection produce before it counts? → A higher bar than the proponent faces means the burden is asymmetric.

5. When did the early conclusion first enter a formal document? → Before the data was complete means the narrative closed its books early.

Where Intuitive Expertise Breaks Down (Chapter 14)

1. Does the experience this judgment rests on still hold in the current environment of rules, technology and people? → No mandatory check means running exposed.
2. How often do the rules of this environment change? → Fast change plus a practice built on experience puts you in the failure zone.
3. How long is the feedback loop on this class of decision, and who can afford to wait for it? → Longer than six months means no learning loop.
4. When did someone last answer data with "I have done this for twenty years"? → If it happened, criterion ① is hit.
5. In a new field, is the first move analogy or measurement? → Analogy means intuition is out of bounds.

Practical Drift (Chapter 15)

1. When was the diff between the manual and actual practice last run? → Never means drift is invisible.
2. Are newcomers trained from the manual or from a senior hand? → The senior hand means drift is passing down the generations.
3. Which rule can nobody in fact follow? → No answer means nobody dares admit it.
4. What is the immediate consequence of breaking that proced - ure? → None means drift has a place to grow.

5. Who approves the informed violation, and under what name? → Oral custom means a violation with no owner.

Information Attenuation up the Hierarchy (Chapter 16)

1. Where do the front-line version and the decision layer's version differ, item by item? → No answer means attenuation has passed the midpoint.
2. What channel carries information around the hierarchy, and who is responsible for reading it? → No channel means you hear retellings only.
3. Is the ratio of adjectives to numbers in reporting moving which way? → More adjectives means packaging in progress.
4. How many layers does bad news cross from the front line to the decision, and what does each layer delete? → If you cannot say what is deleted, every layer deletes.
5. What happened to the last person who reported over their manager's head? → If they were dealt with, the channel is dead.

Misaligned Incentives (Chapter 17)

1. Have the performance schemes of each layer been read side by side? Which clause is the angle between them? → Never read side by side means the misalignment is invisible.
2. Whose account carries the externality of this layer's decision? → Nobody's means the cost is discharged for free.
3. Who is responsible for the overall result, and does that person hold real authority or only the blame seat? → No authority means decoration.

4. Between the decision-maker's tenure and the cycle of the consequence, which is longer? → The shorter tenure means rational neglect.
5. How dense are "per the contract" and "per the rules" in the minutes? → High density means behaviour is taking cover in the contract.

Diffusion of Responsibility (Chapter 18)

1. In whose contract or job description does the stop authority sit, and at which clause? → No clause means stopping depends on courage.
2. How many people are on the signature chain, and what exactly does each one sign? → If you cannot say what each one signs, the signatures are idling.
3. What is the organization's first question after an accident? → "Who signed" in place of "where did it break" confirms a culture of diffusion.
4. How many layers of subcontracting, and who holds inspection rights at each layer? → If you cannot count the layers, it is a black box.
5. For a resolution the committee passed, who executes, who supervises, who closes it out? → All three in one seat means nobody owns the resolution.

Why the Truth-Teller Goes First (Chapter 19)

1. What is the three-year usage record of the internal reporting or dissent channel? → Zero use means the problems are present and nobody says so.
2. Where is the person who last raised a sharp question now? → Gone or marginalized means criterion ② is hit.

3. Who signs the performance review unilaterally? → The direct manager alone means the alarm is wired to the manager's mood.
4. Which questions have become bound to which leader's reputation? → Cannot list them means the binding runs deep.
5. How many of the organization's alarms (independent oversight, rotating audit, external assessment) still sound? → All silent means the only alarm left is the accident.

The Slack of Success (Chapter 20)

1. Which inputs have shown no visible output value for a long time? Is there a list? → No list means they are being read as surplus.
2. Has "this has never caused a problem" been used as grounds for a cut? → If it has, criterion ② is hit.
3. After the last cut to defensive spending, how long until an accident appeared? → Within two years means criterion ③ is hit.
4. In the finance ledger, does "nothing happened" count as output or as no output? → No output means it will be cut inside the budget cycle.
5. Who was last rewarded for "nothing happened"? → Nobody means this work runs on self-amusement.

Hidden Coupling (Chapter 21)

1. When was the real dependency graph last drawn, by whom, and has anyone checked it? → No graph means the coupling is invisible.
2. Which systems share something "unimportant" (power, accounts, database tables, domains, people)? → Cannot list three means the sharing has never been examined.

3. Where is the post-mortem of the last cause-unknown outage filed? → Not filed means the clues are evaporating.
4. Can "who does this outage affect" be answered automatically? → If not, the blast radius is guessed.
5. Who holds the list of acquisitions, outsourced pieces and legacy connections? → No owner marks the deepest coupling.

The Illusion of Redundancy (Chapter 22)

1. What is the list of items the backup shares with the primary system (same vendor, same drawings, same crew)? → No list means common origin has never been faced.
2. When was failover last rehearsed at peak load? → Never means the rehearsal is a performance.
3. Has the redundancy been verified on its own, by turning the primary off? → Afraid to turn it off means the redundancy is unproven.
4. Has anyone walked through the scenario where both systems fail at once? → Never considered means common-cause failure runs exposed.
5. Does the integrator's contract carry a redundancy liability clause? → Not written means you bought the redundancy on the drawing.

Extrapolating Past the Validated Domain (Chapter 23)

1. Which page of the scale-up plan holds the section "what changed from the pilot"? → No such section means the extrapolation was never declared.

2. Does the difference list say, item by item, why each difference does not affect the conclusion? → Hand-waved means the differences were never digested.
3. Has "we validated this at A" been used as grounds for skipping checks? → If so, criterion ③ is hit.
4. Who listed the differences in the new environment, and when? → Nobody did means differences are found by accident.
5. Is the valid boundary of the extrapolation written into the final conclusion? → Not written means the next extrapolation follows.

Crossing a Scale Threshold (Chapter 24)

1. Where is the re-validation record for the core assumptions at the last order-of-magnitude jump? → No record means the assumptions run exposed.
2. Is coordination cost measured, in meetings, management layers and interface documents? → Unmeasured means the nonlinearity is invisible.
3. Which practices had already changed character at twice the scale? → No answer means the change is unknowable.
4. Why was the extra management layer added? → For communication means a threshold signal.
5. Is there a record of veterans saying "it was not like this before"? → No record means the organization is losing its control group.

Margin Erosion (Chapter 25)

1. Where is the design margin register, and who maintains it? → No register means the cuts cannot accumulate.

2. Has anyone read the list of past cuts as a whole? → Per-event reading only means nobody holds the total.
3. What is the quantified basis for "still enough"? → No number means criterion ③ is hit.
4. Who would stop the fifth cut? → Nobody means a straight slide.
5. Can the original design value of the margin still be found? → If it cannot, the reference point is gone.

Automation and Skill Decay (Chapter 26)

1. What is the mandatory frequency of manual operation, and when was it last recorded? → No frequency means the skill is decaying on its own.
2. What share of training is takeover scenarios? → A low share with high automation means the takeover lands at the hardest moment.
3. When did "the system handles it" become a reason not to practise? → If it has, criterion ③ is hit.
4. The manual assumes actions people know; how do you verify they still know them? → No verification means the assumption is expiring.
5. Is the threshold for leaving automatic mode written down, and rehearsed? → Neither means the takeover is improvised.

Sunk Cost and Escalation (Chapter 27)

1. Which section of the project charter holds the exit condition, and who adjudicates it? → Not written means the only exit is a coup.
2. When did "we have already put in this much" last appear as a main argument, and in which meeting? → Frequent appearance means escalation is running.

3. Who is the external adjudicator, and do they have a stake in the project? → None means you judge yourself.
4. Who last adjusted the re-forecast assumptions in the optimistic direction? → One-sided adjustment means the model serves the conclusion.
5. What happened, career-wise, to the last person who argued for exit? → Moved aside means exit has a price.

Horizon Mismatch (Chapter 28)

1. Does the budget have a line for future cost? → If not, the far cost never enters the account.
2. Has anyone drawn the decision-maker's tenure and the con - sequence cycle on one chart? → Never drawn means the mismatch is never faced.
3. Who signs off on the cross-period result? → Nobody does means a commons.
4. How does a long-dated liability show up in the current-period statement? → If it does not, the ledger is forgetting on people's behalf.
5. Does the bill list the next incumbent will inherit exist? → If it does not, handover is where blame gets passed.

The Invisibility of Maintenance (Chapter 29)

1. Where is the three-year curve of maintenance as a share of budget? → Falling year on year with a constant asset count means the criterion is hit.
2. Is the maintenance register auditable, in what was done, when, and against which standard? → Not auditable means soft spending.

3. In the maintenance ledger, does "nothing happened" count as output? → If it does not, you lose every budget review.
4. Who signed the last decision to cut maintenance? → A signer who did not know the consequence means delayed settlement.
5. When was the maintenance standard last updated against the vendor manual? → A version cycle behind means the standard is expiring.

Handover Rupture (Chapter 30)

1. Does documentation exist for why key processes are done this way? → If only the how exists, the judgment is being lost.
2. Does the handover list carry a column for what has gone wrong here before? → If it does not, the lessons leave with the person.
3. After the veteran leaves, what channel carries questions beginning with why? → No channel means the rupture is a fact.
4. Can you list the places where the documentation states conclusions only? → If you cannot, the blind spots are unknown.
5. What answer did the last newcomer get when asking why? → "It has always been like this" means criterion ③ is hit.

Regulatory Lag and Capture (Chapter 31)

1. What share of the regulator's information comes from the regulated? → A dominant share means the structural condition for capture is present.
2. What is the lag between the last rule update and the change in technology? → A generation means lag is now normal.
3. Has anyone computed the ratio of penalty to profit from violation? → Below 1 means the fine is a membership fee.

4. What is the record of people moving between regulator and regulated? → Frequent movement both ways means the revolving door turns.
5. Who drafted the key clause? → The regulated party means drafting power has leaked out.

Compliant and Unsafe (Chapter 32)

1. Who owns the gaps between inspection items? → Nobody means the gaps are free.
2. After a compliance check, has anyone asked whether the compliant items taken together are safe? → Never asked means nobody owns the whole.
3. Are safety meeting agendas built on violations or on risk? → Violations only means a Safety-I loop.
4. Has a compliance certificate ever stood in for a safety judgment? → If so, criterion ③ is hit.
5. Is the list of exceptions and waivers getting longer or shorter? → Longer means the rules are failing while the paper stays clean.

Cascading Failure (Chapter 33)

1. Does the risk register carry a line for third-party dependency? → If not, someone else's risk is not risk.
2. Do you know who your key supplier's supplier is? → Not knowing means two layers down is a black box.
3. Has the isolation and degradation plan been rehearsed? → No rehearsal means a cascade takes everything.
4. Has anyone estimated the exit cost of a single-source supplier? → Not estimated means exit capacity is evaporating.

5. Has a third party' s accident ever entered your own post-mortem? → Never means the learning radius stops at your wall.

Path Lock-in (Chapter 34)

1. When was the last exercise on what you would choose if you started from zero today? → Never means locked in without knowing it.
2. How much has the current estimate of migration cost risen against three years ago? → Rising without review means the lock is deepening.
3. When did an alternative last appear in a selection review? → Long ago means the evaluation has become ritual.
4. Does anyone book the benefits of the lock-in, in compatibility, network effect and scale? → Booking the pain only means the exercise will be distorted.
5. Is the trigger condition for breaking the lock written down? → No trigger means the lock has no keyhole.

How to Run the Review Meeting

The pass has a fixed method, in four steps.

Step one, grouping, ten minutes, done by the chair beforehand. Take the mechanism groups the locator table hits, add the general group, land on 20 to 30 questions, and copy them into the agenda. Under each question leave three lines: answer, evidence (date plus source), verdict (answerable, black, refused).

Step two, the pass, each respondent time-limited. Questions are read out one at a time and the designated respondent answers. The answer must carry a date and a source, and an adjective counts as no answer. Ninety seconds per question: no dated evidence inside

ninety seconds is recorded as black, with no debate, no explana - tion, no filling of the gap. The count of black questions is the main output of the review.

Step three, disposition of the black questions, fifteen minutes. Each black question gets three things settled on the spot: what evidence is missing, who supplies it by which date, who reads it once sup - plied. A black question that cannot be disposed of this way, be - cause the evidence does not exist and nobody can produce it, is upgraded to a design gap and goes into the action column of the conclusion page (Chapter 36).

Step four, minutes, five minutes. One page: the black question list plus the disposition commitments. The reviewer of the minutes is chosen by the same standard as the conclusion page in Chapter 36.

The part of this method most often dropped is the time limit and the discipline of the black call. Without a time limit the review slides back into a briefing. Let explanations in and adjectives swamp the numbers again. Ninety seconds and the hardness of the black mark are the whole secret of turning the list from a document into a tool.

Here is a redacted specimen of the minutes, three black questions in their real format, with the details invented.

> Q8 (dependency list update): black. Missing: the automatic de - pendency list does not cover the three legacy-connection systems. Supply: architecture group to produce the legacy connection list within two weeks. Review: brought by default to the next review. Q9 (career outcome of the person who stopped it): black. Missing: no record. Disposition: no evidence to be supplied; upgraded to a design issue, with a stop-authority clause written into the project charter. Q7 (review of the definition change): answerable. Evid - ence: definition register, 2026-09-14, independent review signature by the risk interface owner.

Three entries, three dispositions: one that can be evidenced (a name and a date), one that cannot (upgraded to a design issue), one green (source given). Green questions go into the minutes too, for a reason that surfaces six months later. In a post-mortem you need to know which pieces of knowledge were lit at the time, and a lamp that was lit is also a coordinate for reconstructing what nobody knew then.

Due Diligence and the Outside View

In an outsider's hands the list is a different instrument. Investment due diligence, an audit entrance, an acquisition assessment, supplier qualification: each can put the questions and count the distribution of black. Two things differ from the internal self-check.

An outsider has no obligation to smooth things over, so black is black and the count comes out cleaner. Keep raw counts: how many of the general ten went black, how many went black in each hit mechanism group, and whether the black questions could be assigned an evidence owner on the spot. Those three numbers draw a profile of the organization's knowledge stock that is uglier and harder to fake than any qualitative paragraph in a due-diligence report.

The outsider also adds three questions the internal self-check does not use. When was your last internal self-check, and does the black list still exist? An organization that has never self-checked will come back near-total black from outside. After the last external review, how many black questions were evidenced? The completion rate is a direct readout of learning capacity. Is there any question considered awkward to ask inside? If there is, list them: the awkward questions are a map of the relationships.

The output is a two-page list: distribution of black, evidence commitments, date of the next review. Writing the review date onto the output is the last step that makes the method take effect.

Maintaining the Question Library

The library is not a finished document. Three rules keep it alive.

Growth comes from post-mortems. After each one (Chapter 38), ask whether this accident contained a question that, asked at the decision point, would have changed the direction. If it did, rewrite it as a question, add it to the library, and tag the source case and the date. The library then grows on the organization's tuition, and the tuition is not wasted.

Retire and de-rate. A question green for three consecutive reviews, with everyone answering and evidence complete, is demoted to a spot-check question. Green for six consecutive reviews, it leaves the standing set and joins the re-green file. The library stays sharp this way; a green question left in the standing set only dilutes the density of the pass.

Keep a version number and a change log. Every addition, deletion or edit gets one line: date, question number, action, source. The log turns "how long since your list was updated" into a question with an answer, and that question is the first of the three external ones.

The owner belongs in quality or safety, the same administrator who holds the deviation register (Chapter 11, Device 1) and the post-mortem archive (Chapter 38). Three registers under one keeper, reconciled once a quarter, and the cross-reading is where the insight comes from.

Quick Recipes

Three recipes for the common occasions, to save the time spent choosing groups.

Recipe A, new system launch review: the full general ten plus groups 21 (Hidden Coupling), 22 (The Illusion of Redundancy), 26 (Automation and Skill Decay) and 30 (Handover Rupture), 30 questions. A launch replaces yesterday's set of coupling relationships wholesale, so the weight falls on dependency and takeover.

Recipe B, organizational expansion or acquisition review: the full general ten plus groups 16 (Information Attenuation), 18 (Diffusion of Responsibility), 24 (Crossing a Scale Threshold) and 33 (Cascading Failure). Expansion lengthens the reporting chain, lengthens the signature chain and widens the dependency surface, and these four groups sit on exactly those three things.

Recipe C, budget cut review: the full general ten plus groups 20 (The Slack of Success), 25 (Margin Erosion), 29 (The Invisibility of Maintenance) and 28 (Horizon Mismatch). A cut rests on somebody's assumption that these inputs produce nothing, and these four groups check that assumption's books line by line.

A recipe is a starting point. If the locator table hits a mechanism outside the recipe, add that group. Recipes are revised every six months against post-mortem data, and the mechanisms that actually occurred in this organization always take priority in the next revision.

Three Ways a Pass Fails

Failure one: the pass becomes a performance. Respondents receive the questions in advance and prepare answers, and the review turns into a re-reading of the briefing. Test it by swapping the re-

spondent on two questions without notice; if the person swapped in cannot answer, the pass is acting. Countermeasure: name respondents on the spot, with no advance notice.

Failure two: question fatigue. After a year of use every question has a template answer, the pass runs shorter, black grows rarer, and the organization acquires the feeling of rapid progress. Test it by rewriting one question, same meaning with all the wording changed, and watch whether black returns. A return means the earlier green was memorization. Countermeasure: the de-rating and the rewording rotation in rule two.

Failure three: black questions never settle. Black marks pile up, everyone nods at the disposition step, and nobody supplies the evidence afterwards. Test it by spot-checking the previous round's black questions, to see whether the promised evidence exists and carries a date. Countermeasure: the disposition commitments go straight into the action column of the conclusion page, sharing its reviewer, so the question list rides the checking mechanism Chapter 36 already built instead of raising a second one.

The common antidote to all three is one thing: keep the list in contact with reality, by swapping people, rewording, and spot-checking commitments. Like any register, a question library that stops touching reality degrades into a printed document.

Five of the general questions carry the opening of a session, so they get their own notes; the texture of the first few minutes sets the quality of the whole meeting.

Q1, which second is irreversible: force the specific. "Launch day" is not an answer; "the second the switchover takes effect" is. The function of this question is to drag the meeting from the conceptual layer to the physical layer, and every question after it benefits from the pull. A failure to answer usually means the decision-

maker has not broken the decision down to the action layer, and at that point stopping to take the decision apart is worth more than continuing the pass.

Q3, which items carry timestamps: listen for the age of the evid - ence. If every timestamped answer predates the last change in the environment, the answer amounts to nothing. Follow with one question: what has changed in the environment since this material was produced? If it changed and the material was not updated, black.

Q5, where the stop authority is written: ask for the original text, not the retelling. A retold stop authority evaporates at the first conflict; a written one does not. "There is a consensus on this" counts as black, because a consensus that never enters a document does not exist. That is the oral version of the Chapter 18 criterion.

Q8, when the dependency list was updated: let the system answer for itself. The best answer is a list generated automatically, with the generation time at the link; second best is a manual list with an update date. A manual list untouched for more than a quarter counts as black.

Q10, where the person who pays in three years sits: this one is a thermometer, measuring the horizon awareness in the room. A cold answer, whether nobody can answer or the whole room laughs, is itself the heaviest line in the minutes. A cold room is not a deduction. After the cold, turn the question into one action on the conclusion page, a one-page handover forecast written for whoever takes over in three years, and the thermometer has become construction.

Training Use: A Three-Week Plan

The raw material for the plan is the list and nothing else.

Week one: the newcomer takes the general ten and interviews three veterans, half an hour each, recording only the answers and the hesitation inside them. Output: a three-way comparison across the ten. What the newcomer learns is the real distribution of knowledge in this organization, which questions have stock underneath and which are empty.

Week two: the newcomer takes two mechanism groups, the two most relevant to the post, about ten questions, and has to find one real example in this organization for each or confirm there is none. Questions with no example are tagged organizational blank and confirmed with a mentor. Output: a mechanism-group list with local footnotes. What the newcomer learns is how to hang an abstract mechanism on a concrete site.

Week three: the newcomer sits in on a real review with one task, counting the black questions and the disposition commitments in that session. Afterwards, 300 words of observation for the chair. What the newcomer learns is the method: the time limit, the black mark, the disposition. From week four the newcomer can time any review, and after three more rounds can serve as respondent.

Total cost, three half-days of interviews and one session sat in. Output, one usable review participant. Traditional training gives newcomers knowledge; this plan gives them questions. In failure prevention, questions are the scarcer resource (the knowledge-in - flation argument in Chapter 4), and scarce training resources belong on the scarcer side.

Links to the Tools in Chapters 38 and 39

The upstream and downstream relations are worth setting out. Upstream sits the locator table in Chapter 35, which decides which groups you take. Get that step wrong and the whole pass idles, so

outside recipes A, B and C, a hit mechanism group always outranks recipe habit.

Two downstream. The first: black questions feed the evidence layer of the Chapter 38 post-mortem. What a post-mortem lacks most is a snapshot of knowledge at the decision point, and the black list is the negative of that snapshot, the record of what nobody knew then. With the black list pulled up, reconstructing what was not known at the time has a basis, and hindsight contamination is blocked one layer further. The second: the pass is the twin of the Chapter 39 rehearsal. A rehearsal works backwards from a failure that already happened; the pass works forwards from the question of whether you can answer. Both probe the same dark region of knowledge, and the two outputs can check each other. If every hazard the rehearsal turned up can be ruled out by a green answer to some question, the rehearsal is going through the motions. If the pass comes back all green and the rehearsal finds three real haz - ards, the green answers are watered. Compare the two outputs side by side once a year, and the list has a two-way reference for its calibration.

Remote and Cross-Team Adaptation

Three points for remote reviews. Keep cameras on during the an - swering step, because the black call depends on raw information in faces and hesitations, and a text-only channel translates "cannot answer" into "will reply later", costing a grade of accuracy. Project the minutes and the black list live in a shared document so everyone watches the black count grow; visibility is what holds the discipline of the pass together (the meeting version of the sticky-note principle in Chapter 36). For teams across time zones, split the pass in two: the question and answer block sits in the overlap, with the time discipline unchanged, and the disposition block runs asynchronously, each black question claimed in the document with

the claim timestamped. A joint review across teams adds one step: each side runs the general ten alone before the joint session, and comparing the two black distributions exposes the knowledge gap better than a joint pass. Two teams' different answers to the same question are a risk portrait of the interface zone, the Chapter 33 territory projected onto questions.

Self-check runs on a different rhythm from a meeting: one person, one list. Three rhythms for three depths.

Monthly quick check, fifteen minutes: the general ten only, answered aloud, with the questions you cannot answer written down. The point is familiarity; questions atrophy like muscle without use.

Quarterly self-check, ninety minutes: the groups the locator table hits plus the general group, answered in writing, each answer with a date and a source, producing a black list. That black list goes straight into the action column of the Chapter 36 conclusion page, or joins the team review's black pool. No separate pile.

Annual comparison, half a day: lay the four quarterly black lists side by side and read three trajectories. Black in all four quarters is a structural blind spot, upgraded to a design issue. Black turning green is evidence of learning, written into the annual review. Green turning black is recent decay, handled first. This comparison carries the most information of anything in self-check: one black list tells you where the gaps are, four tell you which way the organization is drifting.

The most common break point is month three. The first two months are fresh; in the third a project squeezes the check out. The countermeasure is the same one the locator table's implementation schedule uses: put it on the agenda, first Monday of a fixed month, no exceptions.

One hidden return is worth writing down. A person's answer history against the list is their own growth record in failure knowledge. Turn back twelve months in the annual comparison and most people find their own optimism, and that finding is the self version of the Chapter 6 swap test.

One Question in Three Role Versions

Different levels need different wording for the same suspicion. Take question 5 of the general group, the stop authority, as the example; rewrite the rest by the same method.

Engineer's version: in the part of the work I hold, when I think something is wrong, what is the shortest path to stopping it, and who will listen? If the answer contains telling my manager and my manager telling his manager, the path is longer than the danger radius of the step, which is the field form of the Chapter 18 criterion.

Manager's version: how many hops does the stop signal travel from the person closest to the danger in my team to me, and does it decay at each hop? The question forces the manager to count his own reporting chain, and Chapters 16 and 18 merge here.

Executive's version: when was the last instance of a stop coming from below in this organization, and what happened to that person afterwards? The age of the instance and the fate of the person, taken together, are the whole evidence for whether the stop authority is real.

All three have the same structure: the suspicion does not change, and the addressee and the shape of the answer follow the power position. One rule governs the rewrite. The role version must not blunt the suspicion. Softening "who will listen" into "everyone needs to be aware of this" kills the question on the rewriting desk.

The Cost and the Return, Roughly

A rough account for readers still hesitating. On the cost side, one review pass including grouping and minutes runs about two hours; the quarterly self-check ninety minutes; criteria cards and list maintenance about an hour a quarter. That totals roughly forty work-hours a year for a ten-person team, one person's week.

On the return side, two references. The direct loss of any grade-A case in this book runs from a $950,000 fine to eleven deaths, and the reader can convert the proportion of that class the list might have blocked into their own organization's units. The account stops there; more arithmetic would be slick. The list still fails to stop a great many accidents. It stops the class where nobody ever asked, and that class appears often enough in post-mortem records to justify the week.

There is an intangible side too. The pass changes how meetings talk. Adjectives retire on their own in front of a demand for dated evidence, and the change spills into meetings where no question list is present. Spillover cannot be booked, but teams that have used the list report it: after one pass, "it should be fine" buys a good deal less in that team.

Backing One Group, Question by Question

The Normalization of Deviance group serves as the demonstration: five questions, each with a case from this book standing behind it, so you can see where each question came from.

Q1, how many times and on what basis each was judged con‑sequence-free, stands on Challenger and Columbia: the evaluation history of O-ring erosion at Marshall, in the transcription layer of CAIB and of the House investigation materials, and the per-flight no-disaster table for foam shedding (CAIB p.124). What the two

cases share is that a history of no consequence was taken as proof of no risk. The question puts that substitution back on the table and runs it again.

Q2, margin as number or adjective, stands on a language phenomenon shared across several chapters: from "the margin is still sufficient" to "still sufficient" to "sufficient", the evaporation history of three words. The function of the question is to make the organization write the number back. No number and no source counts as no margin (the review version of Device 2 in Chapter 11).

Q3, who approved the relaxation, stands on av-034: the extension of lubrication and end-gap inspection intervals was applied for by the operator and approved by the FAA, on a single information source. The question presses exactly on the information make-up of the approval chain, where an operator's application plus an operator's data means nobody checked.

Q4, how a newcomer's doubt was handled, stands on the review room in mi-001 and on the veteran narrative phenomenon that several cases share. A newcomer's question is a free blind-spot survey, and the way it is handled, answered or restated as custom, measures the organization's cognitive openness directly.

Q5, which release can be completed without a written reason, stands on nu-002, where nobody verified after the placard went up. The question forces out the list of waived releases. If the list exists, it is a map of the gaps. If it does not, the waivers are being granted in the dark. Both results need handling.

The demonstration can be repeated on any group. Find a case in this book for each question, searchable by mechanism in the case index of Appendix B, then find a local instance to land it on. The book's case proves the question is worth asking; your instance proves whether you are asking it in time. Put the two side by side and the question is actually installed.

The Closing Question

1. Of all the questions above, how many can you answer with dated evidence inside three minutes? → Fewer than five means the organization's stock of knowledge about failure is thinner than its memory of success. The cause sits in the system rather than in you (Chapter 4), but the window to change it is in your hands (Chapters 38 and 39).

The closing question deserves a second read. It presses the weight of all 130 into one number: how many answers you can produce with dated evidence in three minutes. Organizationally that num - ber measures the thickness of the knowledge stock. Personally it is more direct, because it is what you carry out of the room. Every output of the pass eventually converts into movement in this number, and comparing the number quarter to quarter is more honest than comparing any briefing's adjectives. The list closes here. The 130 questions are the tackle; the closing question is the scale; a catch only counts once it has been weighed. Put the scale on the platform every quarter, on the procedure already written in the self-check section. The first reading is usually unflattering, and that is normal, because the list begins by admitting the dark area is larger than you thought.

(The Chinese text of this chapter runs to 10,001 characters.)

Chapter 38 Writing a Postmortem That Lasts

> What this chapter covers: how to turn the case card into a post-mortem template; six rewrites set against their originals, paragraph by paragraph; and the record-keeping regime that keeps a postmortem citable three years later. When to turn back: when you have to write a postmortem, review someone else's, or discover that nobody can follow the one you wrote three years ago. Estimated reading time: 35 minutes.

A postmortem has one test of usability, and the test is a matter of time. Can a new colleague, three years from now, answer three questions from this document alone: what happened, why it happened, what to do if it happens again. A document that answers all three is a live postmortem. One that cannot is a dead file, and dead files outnumber live postmortems in most organizations by a wide margin. The cause is fixed at the moment of writing. They were not designed for the reader three years out. The template and the discipline in this chapter serve that single target.

A postmortem passes through three stages, each with its own reader and its own use. Stage one, the first week after it is written: the reader is the team itself, the use is immediate learning, and the critical item is the replay of the decision point (Chapter 6's two-column discipline). Stage two, three months to a year out: the reader is someone running a similar project, the use is prevention, and the critical item is whether the minimum counterfactual can be executed (Chapter 10's five-step method). Stage three, more than a year out: the reader is the person who inherits the work and the investigator who arrives later, the use is historical truth, and the critical item is the integrity of the numbered evidence chain (Chapter 6's procedural defense). One document, three different

requirements. A good postmortem meets all three at once, which means the template has to carry all three by design.

The twelve fields of the case card in Chapter 5 become the section structure of a postmortem directly: ID, one-line summary, subject, when and where, domain and mechanism, what was visible then, options ruled out, decision point, point of no return, scale of consequences, minimum counterfactual, evidence grade and sources. An organization may add or drop fields to suit its own work. The twelve-field skeleton is better left uncut. Cut into it and its postmortems stop being comparable with those of other organizations, and comparability is the precondition for knowledge moving at industry level (Chapter 1).

The other common way to kill a postmortem is to convert it into a blame document. Once a postmortem becomes the basis for punishment, the next one carries no truth (Chapter 19's alarm logic). Chapter 4 has a working model for the institutional design that prevents this, taken from the software case: a blameless principle plus data separation between the review system and the personnel system. What the separation changes is the IT boundary, two systems with two permission groups; it does not delete personnel records. Technical isolation holds where a cultural promise does not (Chapter 19).

A postmortem serves three publics, and the three do not read the same language. Decide who is reading before the first sentence.

Public one is the team. They want the decision point replayed and the next actions named, and the language is concrete: work order numbers, dates, names. Public two is the people who do similar work elsewhere. They were not in the room, they do not know the cast, and what they need is the mechanism in transferable form, in structural language: under which conditions, which mechanism starts. Public three is the investigator or successor three years out. They want the evidence chain, indexed by number, source and

material list. Most postmortems serve public one alone, which is why three years later only public one remembers the document, and public one has usually been reassigned.

Each public has a matching writing discipline. For public one, the factual section: two columns, a source on every line. For public two, the mechanism section: the mechanism field plus a paragraph describing the form the mechanism took on site, written the way Section 3 of a mechanism chapter is written. For public three, the evidence section: material list and grade labels, per Chapter 5. The three-part structure of the template in Section 38.2 gives each public one part, which is not a coincidence. The twelve fields were always an event record, a teaching text and an evidence file at the same time.

One more layer of purpose. The knowledge a postmortem produces comes in two kinds: about this event (this project's pothole) and about the mechanism (the general structure under these conditions). The first goes into the project ledger, the second into the organization's case library. The mechanism field is the interface where one turns into the other. Write the event, then abstract its mechanism into one transferable sentence; when the entry moves into the library the particulars can be de-identified and the mechanism kept. Library entries use the Chapter 5 card format, so an internal postmortem and this book's external cases speak one language and exchange without loss.

38.1 The Two Fates of a Postmortem

A postmortem that dies the moment it is filed is the normal case: submitted, archived, never opened again. The ones still cited three years later are a minority, and they share three features. The information environment at decision time can be rebuilt, because timestamped material survives. The conclusion traces to one con-

crete action, because a minimum counterfactual exists. Someone owns the follow-up, because a verifier and a date are written down. This chapter turns the three features into a template and a regime.

The causes of early death are just as clear. The fact layer carries no timestamps, so nothing can be checked. The conclusion layer is made of adjectives, so nothing can be executed. The output layer has no owner, so nothing lands. Three ways to die, three sections of this chapter.

38.2 The Twelve-Field Postmortem Template

The twelve fields of the Chapter 5 case card are the postmortem template. Write them in three parts.

Opening section, fields 1 to 5: ID, one-line summary, subject, when and where, domain and mechanism. The section pins the event down and prevents drift in the agreed account. The mechanism field may not be fillable with confidence at postmortem time; a first draft may carry a "suspected" tag and be settled at the monthly revision.

Process section, fields 6 to 9: what was visible then, options ruled out, the decision point, the point of no return. This is the body of the document, written in two columns (Chapter 6), the left column holding what was visible then and the right column holding what was reconstructed afterward. Here sits the one advantage an internal postmortem has over an external case card: you hold the internal material, so you can write the meetings and the signatures in finer grain than any investigation report. That is the whole advantage. Do not spend it on adjectives.

Output section, fields 10 to 12: scale of consequences, minimum counterfactual, evidence grade and sources. The output section of a postmortem adds three items, which makes fifteen fields: the

improvement action (one per mode of death, written the way the conclusion page of Chapter 36 specifies), the verifier and the date, and the ledger number (entered into the anomaly ledger, Chapter 8, Device 1).

The file header carries one fixed line of metadata: postmortem date, writer, and the material list, a dated list of files. The material list is the collateral behind the document's credibility, since it is how a reader judges what you actually had in hand. Group the list by evidence grade, A group first, and give each file its export date. Export date and content date are two different timestamps and both are kept: the export date proves that what you reviewed was the version as it stood, not a version updated afterward. The list ends with a gap section, and the honesty of the gap section is the litmus test of the whole document's honesty.

Then the twelve fields one at a time, with the filling rule and the common error for each. The rules are written to the question of who will use this line, and how, three years from now.

Field one, the ID. Issue a number that is never reused; year plus sequence plus subject code is a workable format. The common error is renumbering, where a ledger migration changes the numbers and breaks every old citation. An ID is the document's identity card. Once issued it stays.

Field two, the one-line summary. One sentence, confined to the mechanism, cause words forbidden. "Methyl mercaptan vapour accumulated during tank drainage; four people died" passes. "Accident caused by inadequate management" does not. Cause words belong in the mechanism field. The one sentence pins facts only.

Field three, the subject. Name an organizational unit and a role that can be pointed at, not a department nickname. A reader three

years out does not know who "the old platform crew" were. They know "Platform Infrastructure Department, Storage Group".

Field four, when and where. Minutes where the record supports minutes, and a plain statement of the precision where it does not. Ban "recently" and "in the earlier period". Write the time zone and the shift: a night-shift event and a day-shift event are often different mechanisms.

Field five, domain and mechanism. The mechanism name from this book's vocabulary plus the chapter number. Where two mechanisms are plausible, hang both as suspected and note the deadline for settling. The common error is a result label such as "operator error". That is a conclusion. The field wants a process: which two of Chapter 9's three layers broke, and how.

Field six, what was visible then. Every line with a source and a timestamp, left column only. The common error is hindsight leaking in: the writer knows the ending, and the pen slides toward "they should have seen it". Test: cover the right column and read the left. If it reads as wilful disregard, hindsight has entered the text.

Field seven, options ruled out. At least one option, with the reason for exclusion and the moment of exclusion. The common error is writing up the option that only the ending proved best as "what should have been chosen". Judge the exclusion on the information visible then (Chapter 6).

Field eight, the decision point. Write it at the grain of "what was on the table". The common error is writing a calendar day, as in "the decision of 12 September". A decision point is where materials met options, not a square on a calendar.

Field nine, the point of no return. The triggering act and the moment it took effect. The common error is merging it with the decision point. In most events the two sit hours to months apart, and

the interval between them is exactly the working window of the defenses.

Field ten, scale of consequences. Numbers plus basis, with the estimation method footnoted. The common error is direct loss alone. Delay cost, trust cost and opportunity cost go in as far as they can be estimated; where they cannot, say that they cannot.

Field eleven, the minimum counterfactual. One action, placed before the decision point, with the conditions it depends on noted. The common error is three actions. Three actions is an improvement list, not a counterfactual.

Field twelve, evidence grade and sources. Grade every load-bearing fact, A to E on the Chapter 5 scale. An internal postmortem holds A-grade material as a matter of course, the original records, which is the surplus it has over an external investigation. Grading makes the surplus visible.

38.3 Before and After: Six Rewrites

The passages below are teaching composites. Their sentence patterns are taken from the common forms of real postmortems (no specific organization). Each is rewritten paragraph by paragraph, with a note after the rewrite on what changed and why.

Original one: "At around 14:00 on 12 September 2025, a quality problem occurred in our company's Project X, causing a certain amount of economic loss."

The problem: the time is vague ("around 14:00" is a fact and acceptable; "a certain amount of economic loss" carries zero information). Rewrite: "2025-09-12 14:07 (time recorded on work order X-3312): hollowing and cracking found in the masonry wall on floor 5 of Building 3, Project X, across 42 walls (quality inspection record Q-258, 2025-09-12). Economic loss stated separately

under the basis given below." What changed: the vague quantifier became a number with a source; where the basis has not yet produced a figure, the text says it is stated separately rather than parking "a certain" in the slot.

Original two: "Analysis shows the main cause was insufficient safety and quality awareness among the construction crew, together with schedule pressure in the earlier period and weak links in management."

The problem: four evaluations (insufficient awareness, schedule pressure, weak links, main) and zero actions. The rewrite has two columns. Visible then: the crew meeting minutes of 28 August record that masonry work on Building 3 was assigned to a newly transferred crew, with no mock-up demonstration arranged; item 3 of the weekly quality report of 3 September had already flagged mortar joint fill below standard. Reconstructed afterward: tear-down inspection confirmed mortar strength below specification (laboratory report L-77). What changed: evaluations became dated evidence; "schedule pressure" landed on a specific scheduling change; what was visible then is separated from what was learned after.

Original three: "The project department attached great importance to the matter and organised rectification at the first opportunity."

The problem: no action, no subject, no date. Rewrite: "12 September 16:30, the project manager signed rectification order R-91; by 15 September, 31 of the 42 walls had been reworked (supervision log, 15 September); the remaining 11 wait on replacement of delivered material and are scheduled for 22 September (owner: [name], calendar milestone 9-22)." What changed: the whole action sequence landed on work orders, logs, signatures and dates.

Original four: "Next, we will strengthen training, improve procedures, tighten supervision, draw lessons across the board, and make sure similar problems never occur again."

The problem: four all-purpose counterfactuals (Section 10.3), which total zero. Rewrite: the minimum counterfactual is that after the flag in item 3 of the 3 September weekly quality report, a re-inspection within five working days as the procedure required (the clause reads "re-inspection shall be carried out"; no re-inspection record was found) would have caught the defect before it spread across the walls. Two improvement actions follow: (1) the re-inspection deadline for flagged weekly-report items goes into the supervisor's weekly checklist (verifier: the chief supervisor's representative, at every weekly inspection); (2) mock-up demonstration for an incoming crew goes into the conditions for starting work (verifier: the project manager, at every start of work). What changed: one minimum counterfactual, changing one action only; two verification actions that can be marked on a calendar; "never occur again" deleted and replaced by ledger tracking.

The four rewrites share one property. None of them adds a fact the original did not contain. They swap adjectives for sources, evaluations for actions, and four directions for one counterfactual. Almost all of a postmortem's information already sits in the raw material. Writing is the act of putting it into a shape that can be checked.

Two more pairs, covering different kinds of event, on the same principle.

A fifth pair (an IT incident). Original: "In the early hours the system jittered; late monitoring alerts allowed the impact to widen; service has been restored." Rewrite: "2025-11-08 02:14 (gateway log, NTP-synchronised), payment gateway connection pool exhausted; order failure rate 63% between 02:14 and 02:41 (sampled from the gateway access log, sampling basis in Annex A). The first

automatic alert fired at 02:29 on the failure-rate threshold, before human engagement (duty log 02:33); in the four minutes from alert to engagement, the escalation path on the duty roster pointed to a person who had already left the shift (roster v3, issued 11-07)." What changed: jitter became a failure rate with a sampling basis; the vague complaint about late alerting was broken into a timeline that exposed the real break, an escalation path pointing at someone off duty (Chapter 30, handover rupture).

A sixth pair (a near miss). Original: "A near-injury in the workshop; fortunately nobody was hurt; education was given on site." Rewrite: "2025-10-21 10:52 (workshop CCTV timestamp), line 3 packing station: the palletising robot started without waiting for the confirmation signal and emergency-stopped 0.8 metres from the operator (PLC log). Two earlier unconfirmed starts of the same model are on record (maintenance ledger 9-30, 10-07), both closed as sensor false alarms, neither triggering an interlock check." What changed: "fortunately nobody was hurt" became a distance and a stop record; "education" became a ledger sweep that surfaced the third occurrence of the same deviation (Chapter 11's criterion: the same deviation twice already). The value of a near-miss review lies almost entirely in the sweep. A near miss is the mechanism's free sample, and skipping the sweep voids the sample.

The last two pairs make one further point. Every fact used in the rewrite already existed in a system the organization was already running. What the postmortem lacked was never data. It was the act of putting data onto a timeline.

38.4 The Record System: The Postmortem Starts at the Decision

Whether a postmortem is usable three years later was decided at the moment of decision, by what was left behind. Four minimum record-keeping standards.

Standard one: the three-line decision record (Chapter 7). The point of no return, the second-to-last confirmation position, and the information visible at the time. The three lines go into a fixed field of the decision document or the meeting minutes.

Standard two: the minimum standard for minutes. Date, attendance, the order of agenda items, the conclusion and the dissent for each item, and the time of adjournment. Minutes with no dissent column cannot produce "options ruled out". The postmortem then reconstructs them from memory, and memory is grade D evidence.

Standard three: the anomaly ledger, entered and attributed as events occur (Chapter 8, Device 1). The ledger is the postmortem's primitive accumulation. Half the material in a postmortem written three years later comes from the ledger, and none of it from anyone's memory.

Standard four: the register of basis changes (Chapter 8, Device 2). Every adjustment to a measurement basis, a baseline or a standard leaves its own record. The condition a postmortem fears most is "why was it calculated that way at the time" with nobody able to answer. The register is the answer bank.

None of the four is expensive. Each is a fixed field in a document that already exists. In Chapter 4's terms, this is how an organization installs its own devices without waiting for legislation.

38.5bis Running the Review Meeting

Writing happens after the meeting, and a badly run meeting poisons the material before anyone writes. Five rules for the meeting itself.

Rule one: lock the materials before the meeting opens. Circulate the material list two days ahead, logs, minutes and ledger extracts included, and accept no oral fact in the room that is not on the list. This blocks the memory contest, in which the loudest voice's version becomes the record.

Rule two: project the two columns. The room's screen carries two columns throughout, the left one holding what was visible then sorted by timestamp, the right one left empty. Every contribution answers one question first: does this go in the left column or the right. Left requires a source. Right gets tagged "after the fact". Chapter 6's writing discipline moves forward into the meeting, at the cost of one repeated question.

Rule three: the decision-point replay. Advance the material to the moment before the decision and stop there. Ask those who were present to answer aloud, under the rules of Section 36.1 and on the left column alone: what would you do. Only then is the ending revealed. This is the one place in the meeting where "if we had it to do again" is allowed, and it is the cheapest classroom a new member will get.

Rule four: the procedural implementation of no-blame. The chair holds two iron rules. Questions address structures. Questions addressed to persons are handed to a separate channel; where the matter is the handling of a specific individual, it goes down the personnel line after the meeting, and the minutes carry four words, "handled on the personnel line". The technical base of those rules is the system separation described at this chapter's opening: two systems, two permission groups. With that in place

the chair's rules have institutional backing and do not depend on the chair's personal authority.

Rule five: collect three things before the room empties. A first draft of the mechanism field (suspicion allowed), candidates for the minimum counterfactual (two allowed, one settled after the meeting), and an owner for every improvement action. With those three, the writer has a skeleton. A meeting that ends with none of them was a memorial service.

38.4bis The Boundary of Blame

No-blame does not mean no responsibility, and the boundary has to be drawn precisely or the institution will be overturned after the first serious case. The line runs along two sides, intent and capacity. Negligence, process gaps, knowledge gaps, distorted incentives: all of these go into the postmortem, and the accounting targets structure. Deliberate violation, falsified records, working under the influence of alcohol: these go down the personnel line, and the postmortem records the factual chain only, without judging the person. The line is not drawn by the review meeting. Existing instruments draw it (safety regulations, professional codes), and the meeting's job is to write the factual chain to the precision those instruments need for a finding. The boundary has to be explained publicly when the institution goes live: what the postmortem protects is honest error, and the scope of that protection is written as a negative list and posted on the front page of the review system. The list itself defends in both directions at once, against the postmortem turning into a trial and against anyone hiding behind no-blame to shed all responsibility.

The four standards need an implementation path, and the path is to leave the institution alone and change the template. Standard one, the three-line decision record, embeds as a required field in

the approval flow: the approval form gains a section headed "declaration of the point of no return", three lines, and the flow does not proceed while it is blank. Standard two, the dissent column, embeds in the minutes template: two fixed lines at the end, "Objections:" and "No objections, confirmed by:". A team whose objection column is permanently empty deserves more suspicion than a team whose column has content. Standard three, the anomaly ledger, embeds in the shift and quality handover sheet: one line, "anomaly entry number this shift", with "none" written where there is none. Standard four, the register of basis changes, embeds in the change process for reports and dashboards: any submission that changes a basis must link to a register entry. All four sit on paperwork that already exists. No permission changes, no new systems, no change to headcount, and a week to roll out. Once rolled out, the supply of raw material for postmortems has a pipeline, and the quality ceiling of every future postmortem was set three years earlier by those four fields.

The evidence grades have a second use inside an internal postmortem: grading the material also exposes the organization's recording habits. Where a postmortem's A-grade share (system logs, original ledgers) is low and its C-grade share (after-the-fact accounts) is high, that finding belongs in the metadata. It says the decision moment left too little behind, and the postmortem rests on recollection. Several postmortems in a row with that shape move the object of improvement up a level, to the record system itself. The grade table therefore doubles as a health check on the record system, which is how the Chapter 5 table gets used inside an organization.

38.5 The Quality Check

A postmortem passes ten questions before it is delivered, one line each.

1. Does every statement in the factual section carry a source?
2. Are there separate columns for "visible then" and "reconstructed afterward"?
3. Is the subject of the conclusion section a structure or a person? (Chapter 9: if the subject is a person, send it back.)
4. Is at least one ruled-out option written down?
5. Is the minimum counterfactual a single action?
6. Do the improvement actions map one to one onto the modes of death?
7. Does every action have a verifier?
8. Does the verifier know? (It counts only if they have been asked.)
9. Is the mechanism field tagged with the right name from the vocabulary?
10. Is the metadata's material list complete?

Pass all ten and the postmortem enters the ledger. Fail any and it goes back for rewriting. This is the same table as the thirty-minute procedure in Chapter 36, put to a second use.

A postmortem can also catch the disease of Chapter 13. The first postmortem establishes a narrative, the second cites the first, and three years later the narrative has become fact. Three defenses.

Defense one: the mechanism section must carry a rival-mechanism paragraph. Beside the settled mechanism, list the alternatives that were considered and the reasons they were dropped, written in the manner of Chapter 13's comparison hypotheses. The point is a record of the reasoning rather than a tone of balance: a later reader

can re-examine whether the exclusion still holds and, when new evidence appears, knows which direction to reopen.

Defense two: citations of outside conclusions grade down. When a postmortem cites the conclusion of another postmortem, the cited content counts as grade D, a first-party account, whatever grade the original's factual section carried. This is compatible with the Chapter 5 grade table: grade belongs to the form of the material, and retelling never upgrades it. The narrative errors in public events work exactly this way, the wording of the first report fixed by every retelling that follows.

Defense three: a reopening clause, two years after the postmortem is settled. The keeper sweeps, at the two-year mark, the mechanisms still marked suspected, the key conclusions that were downgraded on citation, and the places marked "material gap". New evidence, a log newly released, a new investigation report, a participant willing to talk after leaving the job, triggers a partial reopening. The output of a reopening is an errata page, never a silent rewrite of the original. The errata regime preserves the document's historical truth: the reader can always see the judgment made at the time and the correction that came later, and both are knowledge.

The three defenses share one principle. A postmortem is writing about the past, and writing collapses toward narrative under its own weight. Evidence chains, rival paragraphs and errata pages hold the collapse back on the writer's behalf. The writer's private vigilance is not a mechanism.

38.3bis Two Variants: The Near Miss and the Success

The template covers two special objects with two variants.

The near-miss review. It is built like the incident review, with two relaxations and one tightening. Relaxation one: the decision-point material may be incomplete, provided the gaps are stated, since a near miss naturally leaves a thinner record than an accident. Relaxation two: the mechanism may stay suspected longer, since near-miss signals are weak and a forced settlement files the case wrong. The tightening: the ledger sweep is mandatory, as in the sixth rewrite, where prior occurrences of the same class must be traced back. Most of a near-miss review's value lies outside the event itself, in the third confirmation it supplies for its siblings. The intake bar is deliberately low. An incident review passes the ten questions; a near miss passes three (facts sourced, mechanism tagged as suspected, sweep done). A low bar buys volume, and volume is what turns scattered near misses into a pattern.

The success review. A conspicuous project win also deserves a postmortem, and its object is which defenses actually fired. The method mirrors the incident review: the decision-point replay is unchanged, and the conclusion section names which device stopped which path (the deviation ledger entry that caught something under Chapter 11, the live drill that avoided something under Chapter 22). The most valuable output is the other half of the counterfactual. Nothing went wrong: was that luck, or was one specific device doing its job? Only a review can separate the two. An organization that cannot separate luck from defense will copy the luck and cut the defense, which is exactly how the slack of Chapter 20 begins. The success review keeps the twelve fields, with the mechanism field reading "mechanism intercepted" and field eleven rewritten as "minimum defense, verified". Two or three a year gives the defense system a record of its own wins, and at budget review (Chapter 20's question 4) that record is the written evidence that not having an incident counts as output.

38.6 Conversion to Case Cards

Writing the postmortem is the first step. Its second life decides how long the knowledge lasts. Conversion into a case card is the main channel: compress the fifteen-field postmortem into a twelve-field case card on the Chapter 5 format and file it in the organization's case library. Three rules govern the trade-offs.

Rule one: de-identify the particulars, keep the structure true. Names become roles ("the duty engineer"); organizational units may survive down to department level; commercially sensitive magnitudes become bands ("losses in the millions"). The floor for de-identification is that the mechanism structure stays undistorted. The intervals in the timeline, the material composition of the decision point and the ruled-out options are the three things most easily smoothed while redacting, and a smoothed case is a dead case. Keep a restoration key, an ID cross-table, in a permission group separate from the library, for internal investigators to restore the identity.

Rule two: re-grade the evidence. A-grade material in the postmortem stays A on the card; oral accounts drop to D on the card. Conversion is a re-grading process, and the pass also corrects the grades the postmortem got wrong.

Rule three: settle the mechanism field. A mechanism left suspected in the postmortem is settled at conversion against the fixed criteria, checked one by one against the starting criteria in the mechanism chapter. A mechanism whose criteria lack evidence stays marked "suspected", with the missing evidence named. A library's search quality rests entirely on the discipline of this field. A case filed under the wrong mechanism is a lost case.

The library's rhythm: quarterly intake (three to five cards chosen from the quarter's postmortems for their second-life value), an annual compilation (aggregated by mechanism, producing the or-

ganization's own mechanism distribution), and one reflux line, where the library's cards feed the calibration of Chapter 35's mapping table and the additions to Chapter 37's question library every six months. Postmortem, case card, question library, mapping table: four tools on one data flow, and the postmortem is the source at the head of it.

38.7 Governing the Review Archive

Once postmortems accumulate they need governance, and governance has four duties only.

Duty one: the registry. Every postmortem enters one master ledger (ID, subject, mechanism, date, action-verification status). The registry prevents duplicated labor: postmortems of similar events cite each other instead of each rewriting the background.

Duty two: aggregation. Once a year, aggregate by mechanism: which mechanism draws the most postmortems, which has the lowest action-verification rate, which is cited most often three years on. The output is a one-page portrait of the organization's failure patterns, and it feeds the topic list for Chapter 39's rehearsal directly.

Duty three: the three-year test. Each quarter, draw three postmortems at random that have passed their third anniversary, and ask someone who was not involved at the time to answer the opening three questions from the document alone (what happened, why, what to do differently). The share fully answered is the library's survival rate, tracked year by year. This test is the opening standard of this chapter made into an institution, at a cost of one afternoon a year.

Duty four: retirement. A postmortem whose actions are all verified and whose mechanism has entered the case library moves to read-

only archive. Read-only is not a grave: the three-year test draws from exactly there.

Registry, aggregation, testing, retirement: about one day's work a quarter in total, and it belongs under the same keeper as Chapter 37's question library.

One last governance point concerns incentives. Postmortem outputs enter no one's appraisal denominator, and postmortem counts are never ranked. The only KPI in postmortem governance that can be published is the survival rate of the three-year test. It is the one measure that reflects quality without putting pressure on the writer, because the pressure lands three years later, when the writer is no longer in the post. That is what keeps the return of blame-ification at bay.

38.8 The Writing Budget

One common cause of death is "no time to write". Here is a realistic budget and a division of labor. Small events (no injury, loss under ten thousand yuan): four hours. Medium events (one review meeting, draft within a week): two working days. Large events (cross-department, formally investigated): follow the investigation's rhythm, with the postmortem due within two weeks of the investigation report being settled. Staffing: one writer, preferably someone who sat at the decision but did not own it; one materials clerk, who pulls logs and ledger extracts out of the systems; one chair, who runs the review meeting and executes the ten questions. Three people writing together does not work: a postmortem with three voices reads, three years later, like three people each wrote a section.

The last writing discipline is about honesty. Where you cannot proceed, because material is missing, memories conflict or the mechanism will not settle, write "missing: X here". Fluent spec-

ulation filling a hole is forbidden. A reader three years out can forgive a gap. No reader can forgive a hole patched so well it cannot be seen.

38.9 The Mirror at Chapter 39

Postmortem and rehearsal are a pair of mirrored tools, and each is clearer against the other. On the time axis, the postmortem follows the event and the rehearsal precedes it; one processes tuition already paid, the other tries to avoid paying it. On the information axis, the postmortem has an ending to lean on and dense material; the rehearsal has no ending, runs entirely on inference, and its output is rough by nature. On the psychological axis, the post‑mortem has to defend against hindsight, since someone who knows the ending replays the judgment distorted, while the re‑hearsal borrows hindsight by pretending the ending is known so that criticism rises. One coin, two faces: Chapter 6's discipline is a defense when used in a postmortem and a weapon when used in a rehearsal. On the output axis, the postmortem produces case cards and improvement actions, the rehearsal produces a hazard list and a division of dispositions, and the two outputs should cite each other in the ledger. Where a hazard listed at a rehearsal later becomes a real accident, the postmortem must cite that list and check whether the defense failed or the list was never executed. On the frequency axis, postmortems are driven by events and re‑hearsals by the calendar, quarterly; calendar-driven tools decay more slowly, event-driven tools depend on a supply of events, and the two complement each other. The sign that a pair has matured in an organization: a postmortem of an accident cites a hazard line from a rehearsal two years earlier, and that rehearsal's disposi‑tion column records why the line was not acted on. At that point the tuition loop closes: what the rehearsal could not recover, the

postmortem collects, and what the postmortem cannot prevent, the next rehearsal plays out in advance.

38.10 Reading Four of the Ten Questions

Four of the ten questions let the most through, so each gets a reading rule.

Question 3, whether the subject of the conclusion is a structure or a person. Read the grammatical subject. "The construction crew's safety awareness was insufficient" has a person as its subject, and goes back. "Mock-up demonstration was not written into the start-of-work conditions" has an institutional act as its subject, and passes. The gray zone ("the duty manager did not confirm against the roster") is judged on one test: could a competent person in that seat have done the same? If yes, keep the sentence; if no, it is a shot at a person.

Question 6, whether the improvement actions map one to one onto the modes of death. Play connect-the-dots. Draw a line from each mode of death in the conclusion section to an action in the output section. Two modes of death sharing one action (usually an all-purpose action of the "more training" kind), or a mode of death with no line at all (found, but nothing is going to be done about it), sends the document back. The exercise takes five minutes, and it catches the commonest half-finished postmortem on the spot, the one whose actions are disconnected from its conclusions.

Question 8, whether the verifier knows. Read the form of the evidence. What counts is the verifier's own confirmation, a message reply, an accepted calendar entry, a signature. The writer's "they should know" does not count. This question exists to catch the courtesy claim, agreement in the room and no knowledge afterward.

Question 10, whether the material list is complete. Judge the hon - esty of the gap section, not the length of the list. A short list is not the problem. A hidden gap is: a log that should exist and was not exported, minutes that should exist and were lost. State them in the gap section at the end of the list, and only then is the list com - plete. The gap section is what lets a reader three years later tell which silences belong to missing material and which belong to missing writing.

38.11 Calibration Against the Book's Cases

The cases in this book have one more use in the review room: cal - ibration. When the meeting reaches the mechanism section, pull one case card of the same mechanism (Appendix B indexes them by mechanism) and lay it beside your own. Compare how much material stood at your decision point against how much stood at theirs. Check whether your ruled-out options column contains the option theirs contained. Set your scale of consequences beside theirs and count the orders of magnitude between them. The pur - pose is to stop the mechanism section from sliding back into local exceptionalism ("our situation is different"). Laid side by side, the tables at two organizations' decision points look alarmingly alike, and what differs is usually only the industry's vocabulary. Rotate the calibration case monthly. After six months the room begins to recognize mechanisms on sight, which is the second path to internalizing them, alongside the question pass.

38.12 The Banned-Words List

The decay of postmortem writing starts in the vocabulary, which is why the vocabulary gets a list. Seven banned words or phrases, each with its replacement.

One: “insufficient awareness” or “not enough attention paid”. An inner state cannot be checked. Name the specific act that was missing (“no mock-up demonstration was arranged”). Two: “further strengthen” or “continuously improve”. A verb with no artifact behind it. Replace with an action, a verifier and a date. Three: “certain”, “serious” or “major”, modifying a loss with no basis. Replace with a number and an estimation method; where no number can be computed, write “basis stated separately”. Four: “at the first opportunity”. A time adjective. Replace with a timestamp to the minute. Five: “similar problems”. A vague scope. Replace with the ledger search string that defines the class (“unconfirmed starts, same model, line 3”). Six: “deep reflection” or “draw lessons across the board”. An emotion verb. Replace with the specific sweep it supposedly triggers (“all interlocks of the same model inspected, list in the annex”). Seven: “eliminate” or “never again”. An unverifiable promise. Replace with a ledgered acceptance clause (“closed after two consecutive inspection cycles with no entry of the same kind”).

Enforcement can be done by machine. Run the seven phrases through full-text search before the ten questions and rewrite every hit. The list will miss variants (“effectively enhance” among them), so a human read follows the machine, and the human has one instruction only: at any sentence describing an inner state, an emotion or a promise, stop and ask for evidence. The list and the ten questions together make up the document’s language discipline, and the language discipline is the ground floor. Template and process stand on top of it. Revise the list once a year with the governance cycle: each year’s additions come from that year’s black questions and from the evasive phrasings that recurred in that year’s rejected postmortems, and the list’s growth is the history of the organization’s language discipline.

38.13 Freeze Before Judging

The boundary of blame has one implementation detail left, the sequencing of the two lines. The personnel finding and the writing of the postmortem may run in parallel, but the factual section must be frozen in one version before the personnel finding is made. When the personnel process needs the factual chain, it cites the frozen version, so that the pressure of the finding cannot work backward onto the facts. A hash or a paper signature will do; the form does not matter, the sequence does. Facts frozen first, findings after, and any correction to the postmortem goes through an errata page. The sequencing protects both directions: the person under investigation against a record slowly rewritten, and the writer against the transfer of blame for a verdict ("the version you wrote is what destroyed him"). Boundary plus sequence is what makes the blameless principle stand. It is a procedural promise, and a procedural promise's credibility comes entirely from the record of the procedure being kept. The first exception resets that record to zero.

Five Self-Checks

1. Take your organization's most recent postmortem and check it paragraph by paragraph against the four rewrites in Section 38.3. How many of its paragraphs are written the way Original One is written?
2. How many files in your postmortem's material list carry timestamps? If the answer is none, the problem began on the day of the decision.
3. Where is the dissent record from your most recent postmortem? Minutes with no dissent column: how much of them do you believe?

4. Of the four record-keeping standards in Section 38.4, which can your organization put in place immediately? The one it cannot, who is blocking it?
5. Can the postmortem you wrote three years ago still answer "why was it decided this way at the time"? For the one that cannot, which record-keeping standard was missing?

The chapter closes here. The whole of this practice compresses back into the three questions it opened with: what happened (the sources and timestamps of the factual section), why it happened (the three-layer structure and the rival-mechanism paragraph of the mechanism section), and what to do if it happens again (one minimum counterfactual plus two actions with named verifiers). Each question has a reader waiting three years out, and a writer who keeps that reader in mind produces a postmortem that does not die. At the level of the organization only one measure is worth carrying: the survival rate of the three-year test. The demand is narrow: a stranger three years later can understand the document and act on it, which is a low bar. It does ask that the writing give up performance at the moment of writing, because performance does not last three years and evidence does. A reader who has come this far in the book carries twenty-four mechanism chapters, thirty-eight devices and one hundred and thirty questions, and all of it becomes an organizational asset only by landing in one concrete postmortem after another. The postmortem is the book's exchange counter: mechanisms convert into actions here, actions into ledger entries, ledger entries into the starting point of someone else three years later. That word, starting point, is this chapter's whole ambition. A postmortem does not try to make the future's decisions in advance. It tries to set the person who comes later one step higher than you stood.

If the answers to the five questions cluster on the last one, what your organization lacks is record-keeping (Section 38.4), and the

embedded fields come before any work on writing. If they cluster on the first three, what it lacks is writing discipline (the rewrites in Section 38.3), and the remedy is to take the most recent postmortem and rewrite it paragraph by paragraph as an exercise. Either gap can be closed within a quarter, which is why this tool has the steepest improvement curve in the book. Money spent this month shows this month, and once it shows, the ledger, the case library, the question library and the mapping table all have a common source feeding them.

Chapter 39 The Ninety-Minute Dry Run

> What this chapter covers: a meeting script for rehearsing failure before the fact. Who chairs, how the opening goes, what gets asked, how it is recorded, how it closes, who follows up; with a one-page agenda. When to turn back: when a project is approaching an irreversible point, or a team needs a kickoff or a go-live review. Drop the meeting in and use it. Estimated reading time: 35 minutes.

The premortem's method comes from Gary Klein's prospective hindsight: the team assumes the failure has already happened and reasons backwards to the causes. In the experiments, groups told to assume the failure had occurred listed about thirty percent more risk causes than groups doing ordinary planning. The reason is social. Assuming it happened removes the pressure, because nobody has to be the first to say that something bad might happen. Saying that it already did gives everyone permission.

The ninety minutes divide into six blocks, on the time boundaries of the agenda table in Section 39.3. Minutes 0 to 10, opening: the chair reads out the decision and the irreversible point (the result of Chapter 7's decision table), announces the rules, and sets the frame as problems only, no debate, every entry recorded without a name, no rebuttal. Chapter 19's silence rule starts running here. Minutes 10 to 25, independent writing: each member writes causes of death on sticky notes, alone, without discussion and without looking at anyone else's. Independent writing exists to defeat anchoring and conformity. Three entries minimum, more is better. Minutes 25 to 55, reading out: each person reads their deaths aloud, recorded only, no rebuttal, exhausted before any grouping. Minutes 55 to 70, grouping and collision: the individual deaths are

hung on candidate mechanisms (Chapter 9's three layers, technical, organizational and incentive), and anything new is added on the spot. Minutes 70 to 85, claiming actions: each selected death gets one defensive action and one detection indicator, with an owner and a check date claimed in the room. Minutes 85 to 90, closing: the chair reads the one-pager, dissent is itemised, the meeting ends.

The division of labour with Chapter 36's thirty-minute process runs like this. The thirty minutes is a personal tool, run by one person before a decision. The premortem is a team tool, run by a group before the irreversible point. Neither substitutes for the other. The thirty minutes covers the blind spots of individual judgement, the premortem covers the blind spots of collective judgement, and the two work best in series: the individual runs the thirty minutes and produces the conclusion page, the conclusion page goes in as input to the premortem, and the premortem's output comes back to update the individual page. The series takes about two hours, and a high-irreversibility decision is worth the two hours.

Three failure forms are known. Formality: the notes get written and filed, and nobody follows the actions. The counter is to put actions into work orders (Chapter 10's work-order conversion). Power performance: the senior people speak first and the rest follow. The counter is strict enforcement of the independent writing block, with nobody allowed to open his mouth for the twenty minutes. Complaint session: the room blames the external environment and the decisions above it. The counter is that the chair follows every external attribution with one question, what can we do inside these conditions, which pulls the attribution back into the range of action.

Three tests decide whether a decision deserves a premortem. All three must pass, or the meeting is devalued by being held too often.

Test one: an irreversible point genuinely exists (Chapter 7's decision table has been passed), and the cost of withdrawal is higher than the cost of the meeting. A reversible decision can use the personal thirty minutes in Chapter 36. A reversible decision does not need a rehearsal.

Test two: at least three of the participants hold information that does not overlap. A room with homogeneous information cannot produce blind spots. The deaths written will be highly repetitive, and a repeated joint list is worth less than one person's list. The organizer judges overlap before the meeting, quickly: three roles, three kinds of experience, or at least one outside view.

Test three: the political temperature of the decision allows bad news to exist. If the project has already been publicly endorsed from the top (Chapter 13's narrative has closed the books), deaths raised at the premortem get translated automatically into "we know, but it is already decided". Fix the temperature first, or split the meeting into the anonymous written version in Section 39.7. Hold it anyway, and what comes out is form one, the credit-claiming meeting, in premortem dress.

What the three tests share is a view of the premortem as a scarce resource. Its output quality is set by conditions that exist before anyone enters the room. Most organizations do not hold too few meetings. They hold too many. Rehearse everything and you have rehearsed nothing. Two to four a year, each with its conditions fully met, beats one a month with the conditions compromised.

One more point belongs to scheduling. Hold the meeting early enough before the irreversible point to leave a remediation window, because defensive actions need execution time. Held right up

against the point, actions get claimed and cannot be done, and the premortem degenerates into an exoneration procedure: we warned. The recommended window is two to six weeks before the irreversible point.

39.1 Positioning: Before, Not After

The premortem and the postmortem are a pair. The postmortem comes after the outcome; the premortem comes before the irreversible point. The psychology comes from the premortem method Klein proposed: the whole room assumes the project has already failed, then each person writes down why. Once the death is assumed, criticism stops being doom-saying and becomes autopsy. (Klein, 'Performing a Project Premortem', Harvard Business Review, September 2007; reprinted in IEEE Engineering Management Review, 2008; Crossref record checked.)

Ninety minutes comes from a single constraint: the meeting has to fit inside one ordinary meeting, with no extra resources. A premortem that runs past ninety minutes never gets held a second time, which is the same as never holding it. The output of the whole session is one page: a list of three to five modes of death, one defensive action per mode, one verifier and one date per action. That page has the same shape as Chapter 36's conclusion page. The premortem is the conclusion page executed as a meeting.

39.2 Preparation (organizer, about 60 minutes, not counted in the ninety)

Three items make up the materials pack: the conclusion page from Chapter 35's positioning table, with two or three candidate deaths already screened (if it was never done, it has to be done before the meeting, since the premortem cannot start from zero); the three

most recent same-class entries from the anomaly ledger; the current version of the dependency graph or dependency list. The pack goes out 24 hours ahead, merged with the cooling-off period in Chapter 7.

The chair comes from outside the project under review. Rank is not the qualification. The authority to keep the meeting on script is. The project lead's role is respondent, not chair, and chairing your own premortem makes the defendant the judge.

The room: everyone on the project plus one external reviewer, from a sibling team, a downstream user or risk. Five to ten people. Above ten, split into two sessions.

Six steps carry the script. The detail below is written for the chair, step by step.

Opening, minutes 0 to 10. After the three fixed sentences, one more action: put the three pack items on the wall (the conclusion page, the three ledger entries, the dependency graph) and say plainly that deaths have to hang on these three materials, and that anything conjured from thin air will not be taken. That sentence blocks half the emotional entries. The difference between a death at a premortem and a confession in a counselling room is the material anchor.

Silent writing, minutes 10 to 25. The chair's only work in this block is the clock and the silence. Two practical details. Sticky notes come in three colours, one for each layer, technical, organizational and incentive, with the meaning declared before anyone writes. Forcing the colour choice is the first layer of attribution training. The notes of the highest-ranking person are collected last, laid face down on the table, and read last, which is a physical defence against anchoring. If someone starts discussing early, the chair points at the wall clock and says nothing. Keeping silence works better than correcting it.

Round-robin reading, minutes 25 to 55. The discipline is record only, no rebuttal, and the hard part is the order of reading and writing. Record with the left hand and do not judge with the right: after each item, one question, which layer it belongs to and which mechanism chapter it resembles. An item that cannot answer goes to the edge of the board, into a pending zone, and is not forced onto a chapter. Ten to twenty items in thirty minutes is normal density. Past thirty, someone has been writing sub-items instead of deaths, and they get merged in the room.

Classification, minutes 55 to 70. The tagging discipline is chapter number before intuition. Every death gets a specific chapter number. The ones that will not fit are marked off-vocabulary and kept on file, because off-vocabulary deaths are the raw material for revising Chapter 35's mapping table, and they are never thrown away. The ten general questions get sampled in this block, and only against the deaths with votes. Votes come from a quiet tally: after one round of reading, each person puts dot stickers on the three deaths he agrees with most, thirty seconds for the whole room.

Action claim, minutes 70 to 85. Every selected death passes three columns on the spot: action, check date, claimant. The trap is the collective claim, the one that says our group will watch it. The chair has to force out a single name, and an action with no name becomes an escalation item there and then. Escalation items carry a recipient and a deadline as well, and an item with two blank columns does not go into the minutes.

Close, minutes 85 to 90. The chair reads the one page aloud, then asks one last question: how will this page be verified in three months. The answer goes into the footer, with the review date and the comparison method. An output with no verification plan is, by Chapter 38's standard, a dead document.

Three details on preparing the pack. The first item may reuse an organizational conclusion instead of a newly filled page, as long as

the date it was filled is marked; a page older than a year gets re‑filled before the meeting. The second item is chosen by similar mechanism, not similar project: a first launch with no precedent in this project takes same-mechanism records from other projects, since same class means the same class of death, not the same line of business. The third item is the one that most often fails. Most organizations have no dependency graph sitting ready, and the substitutes go in order of preference: an auto-generated depend‑ency list, an annex to an architecture review, and only then a hand-drawn sketch. A project with none of the three has produced the premortem’s finding number one before the meeting started, that its dependencies are invisible, and that finding is written up as one known death in the pack.

One distribution rule. Send conclusions, not the raw library. A thirty-page log gets read by nobody, which makes sending it the same as sending nothing. The conclusion page stays inside three pages, the raw library goes out as a link, and whoever wants depth fetches it.

39.3 The Script

Time	Step	Action
0-10	Opening	The chair reads the script and the rules, and announces the assumption that the project has already failed
10-25	Independent writing	Everyone silent, each writes “how we died”, three entries minimum per person

Time	Step	Action
25-55	Round-robin	Each person reads his deaths in turn, record the item, not the argument, exhaust first and classify later
55-70	Classification	Hang individual deaths on candidate mechanisms, new ones added in the room (pointing back to Part Three chapter numbers)
70-85	Action claim	One defensive action per selected death, claimant and date claimed in the room
85-90	Close	The chair reads the one-pager, objections itemized, meeting ends

The opening words are the chair's, and there are three of them, fixed. "This meeting assumes that three months from now the project has failed. We do one thing: name the causes of death." "Everything said is recorded and nothing is rebutted. Rebuttal starts after minute seventy." "At the end of every death I will ask what its earliest visible signal was." The second sentence is the floor the rest of the meeting stands on. The number one cause of a failed premortem is rebuttal arriving early: the first speaker finishes, gets explained away, and everyone behind him closes his mouth.

The independent write is the key component of Klein's method: write before speaking, so that the senior person's account cannot anchor the room. This is Chapter 13's narrative before evidence, in meeting form. The silence rule runs to the end of the block: the chair does not speak, and the highest-ranking person hands in his pen last.

Classification uses the mapping table in Section 35.3 and the question sets in Chapter 37. The chair opens with a sample from the ten general questions, which turns Section 36.2's library lookup into an oral examination, and any answer that hits a question goes into the evidence column of the one page on the spot.

Action claiming follows Chapter 36: one action per death; the action has to be markable as a date on a calendar; the claimant needs the authority to execute or the authority to escalate. A death with no action becomes an escalation item, with the recipient and the deadline written in. Only two things may leave a premortem, an action with a claimant and a report with a recipient.

One worked sheet follows, a teaching composite rather than a real project. The format can be copied.

> **Premortem one-pager** | Subject: warehouse automation, phase two go-live | Irreversible point: 11-20, 08:00, the cutover switch | Chair: L.W., Quality (not a project member)
>
> Death one: practical drift (Chapter 15). The emergency rollback procedure is written in five steps; execution in practice follows the three-step drill version. Defensive action: redo the rollback drill under 5% real traffic. Check date: 11-13. Claimant: K.Z., platform team.
>
> Death two: hidden coupling (Chapter 21). The legacy warehouse system's reconciliation job still connects directly to the new database, and after cutover the reconciliation windows overlap and lock the table. Defensive action: complete the historical integration list with a dependency scan, and delay cutover until it clears. Check date: 11-15. Claimant: D.M., architecture team.
>
> Death three: handover rupture (Chapter 30). Cutover day falls the day after a shift change, and nobody on the duty roster has been through a previous cutover in full. Defensive action: two-person duty on cutover day plus a cutover manual handover page. Check date: 11-18. Claimant: S.H., operations team.

> Escalation item: incentives point the wrong way (partly misaligned, the Chapter 17 group). The go-live milestone bonus structure makes the gain from compressing testing immediately visible. Recipient: general manager of the business unit. Deadline: written reply by 11-12.
>
> Footer: review date 12-20, comparison by checking the three deaths one by one against the cutover log (owner L.W.). Objection items, two: see minutes.

What the sheet shows is the interlock between columns. Every death carries a chapter number, every action a date, every escalation item a recipient. The whole page is under three hundred words and takes ten minutes to write, and its entire value is paid out on 20 December, when it is read back: how many of the three deaths hit, how many of the three actions blocked something. That review is the premortem's second meeting.

39.4 Records and the Close

Minutes follow the minimum standard in Section 38.4 of Chapter 38, with four columns added for this meeting: the death list with mechanism chapter numbers, the defensive actions, the verifier and date, and the objection items. Objection items are the line between a premortem and an ordinary review. An opinion rejected in the room is recorded with the reason for the rejection, and three months later those entries are the first batch of evidence for the postmortem.

The one-pager goes to everyone within 24 hours of the meeting, copied to both sides' superiors. A premortem that is not copied upward is a tea party: defensive actions need resources and calendar space, and neither arrives by a route that bypasses the level above. The copy also carries a design intent that has to be explained to the people receiving it. A superior who receives the page

has exactly two correct moves: assign resources to the actions, or state a position on a particular death and leave a record. Demanding an item-by-item verdict is too much. Expecting silent acceptance is also wrong. The footer keeps one line for a reply, filled with agreement to the actions and the schedule, or a comment on individual items, five words being enough. The reply line is what makes the copy a two-way commitment: the level below discloses, the level above catches. A one-pager whose reply line is blank twice in a row goes straight to governance before the next meeting, reported by the chair. A failed copy is measured, not guessed by feel.

39.5 After the Meeting

Follow-up has two items. Defensive actions enter the anomaly ledger with a 30-day review point, because the reason an action did not run is itself a diagnosis: no resources, no authority, or an action that was written as a wish. And the deaths identified at the premortem enter the comparison column of the postmortem template: at project close or at failure, the team checks item by item what was rehearsed, what happened, and why the rehearsal did not stop it. That comparison is the only feedback loop this institution has for improving itself. Skip it and the premortem degrades into a rite.

The three failure forms can be detected from the feel of a single meeting, and the chair can carry three questions.

Question one: in the first five minutes, did the highest-ranking person speak? A word before the silent writing block means form one, the credit-claiming meeting, at better than even odds, because he has already set the frame of the agenda. The fix belongs to the next meeting: the chair talks to him beforehand and asks him to hand in first and speak last.

Question two: what share of the death entries name a person? Past twenty percent carrying specific names, form one has crossed with blame and the meeting has left autopsy logic. The fix is on the spot: the chair restates the recording rule, and entries with names move to oral clarification and stay out of the minutes.

Question three: how many minutes did the claim block take? Under ten, the meeting was form three, going through the motions: thirty minutes reading deaths, ten minutes claiming in a hurry, and the output has had its teeth pulled. The fix is to set the next timetable backwards, locking minute 70 as the moment claiming must start, and cutting the death list to three rather than letting claims run out of time.

One indicator runs across meetings: the hit rate of the death lists, meaning the share of deaths that actually happened at the three-month review. The healthy band is twenty to forty percent. Zero means the premortem is writing safe words. Above half means the problem at this project level is severe enough that no rehearsal was needed to see it, and the meeting should escalate to governance. A team that lands outside the band three sessions running has a problem in the composition of the room, meaning homogeneous information, rather than in the process, and should change the people under test two of Section 39.1 before holding it again.

The psychological boundary deserves stating, so the tool is not overrated. It solves one problem precisely: social pressure that suppresses bad news, which was Klein's core experimental finding, and both the independent writing and the already-dead framing are designed around that suppression. Three classes of problem it does not solve, and the organization needs other instruments for each.

Patch one, knowledge blind spots. A problem nobody in the room can write cannot be rehearsed: if no one knows a coupling exists, it will not appear on the list. Only an outside view and materials can

light that area, which is why the three-item pack is not optional. The premortem's value equals the knowledge boundary of the participants plus the boundary of the materials, and meeting technique can approach that ceiling but not break it.

Patch two, incentive-layer deaths. Participants go soft on their own incentive structure, since writing that our appraisal scheme caused the compressed testing is writing the exam question for one's own boss, and incentive entries come out systematically thin. The remedy is the escalation channel the chair keeps open in Section 39.3, plus one direct question during classification: one level up, is this an incentive problem? Asking by name is more reliable than waiting for volunteers.

Patch three, insufficient diversity in the room. Five people from the same role, the same tenure and the same background produce five similar lists, and independence buys nothing. Diversity here means information trajectory, not demographics: someone who worked in another industry, someone transferred in from downstream, someone less than six months in the post, each of them carries a path nobody else in the room knows. The organizer selects on tra - jectory, and where the room cannot supply it, the external reviewer slot in Section 39.2 fills it.

What the three patches share: the premortem is an amplifier, and what it amplifies is the knowledge and the viewpoints carried into the room. Its output ceiling is fixed before the meeting, by materi - als and by people. A well-run meeting cannot rescue a compromised preparation.

39.8 Where Chairs Come From

Chair supply is the bottleneck on institutionalizing the premortem. Everyone has sat in meetings. People who will make the project

lead keep quiet for twenty-five minutes on script have to be grown. Three steps.

Step one, observation. A new chair sits in on two sessions with a counting task: how many times a rebuttal arrived early, how many minutes the claim block used, how many entries carried names. The task is designed to make him see the gap between script and room, and the gap list becomes his first textbook.

Step two, co-chairing. In real meetings the new chair gets an experienced co-chair. The split: the chair holds the process, the co-chair records items and keeps the clock. When the chair slips, letting a rebuttal through or missing the closing question, the co-chair does one thing during the meeting, which is pass a note, and one thing after, a ten-minute review. Chair errors get reviewed within forty-eight hours. Left longer, they become convention.

Step three, rotation and rating. Chairs rotate through the organization on the same logic as Chapter 36's verifier rotation. After each meeting the participants score three dimensions: process held, rebuttals intercepted, claims and close. A chair with stable scores across three rounds enters the organization's chair pool, and the pool stays at five or more. An organization with one good chair loses its premortems the day that person transfers.

The selection advice runs against intuition: the best chairs are often technical leads from other teams, not full-time safety or quality staff. A full-time functionary chairing tends to turn the meeting into an inspection; a peer technical lead keeps it a discussion among peers. Both kinds belong in the pool, at a suggested ratio of two to three.

Venue and equipment need practical specifications, because a small failure here drags down a large block.

Venue: a wall you can stick things to is a requirement, whether a whiteboard, glass, or a large sheet of paper taped up. Death entries

have to hang, physically or digitally. Notes lying flat on a table have no public visibility and classification cannot run. Remote sessions use a shared whiteboard, with the rule unchanged: once an item is posted, deletion leaves a trace, struck through rather than removed, which prevents disputes about the record after the meeting.

Materials: three colours of sticky notes, each colour enough for fifteen per person; dot stickers for voting, three per person; a timer at the chair’s hand, page turned when it rings. Check the day before. A chair who leaves the room to find sticky notes comes back to a meeting that has changed flavour.

Seating: no head of table, the chair sits at the side. The respondent, meaning the project lead, is placed deliberately furthest from the wall, so the path of reading items and hanging them crosses his field of view. The arrangement reminds the room physically that this meeting is about the project and also about him, and neither of those needs to be handled gently.

Remote sessions add three rules: cameras on, since reading faces is part of the work; whiteboard write access opened to everyone before the meeting, so each person posts his own in the silent block; and stricter timing, because online meetings lose time discipline by default, with the chair announcing the clock once per block.

Three challenges come up, and the chair should have an answer ready for each.

“This death cannot happen, we already have a defence.” Do not argue. Record it with a note attached, respondent says a defence exists, and test the evidence for that defence in the classification block, using question two of the ten general questions, the second-to-last confirmation point.

“This meeting is talking the project down.” Go back to the second opening sentence, death assumed and nothing rebutted,

and add one line: a death not named today will have to be named in three months, with interest.

"I am the project lead, I say how this meeting runs." The chair should be holding an authorisation obtained before the meeting, the chairing authority in Section 39.2, and now uses it. Authority is not debated in the room. The line is: we run on script, and objections go to my superior after the meeting.

39.9 Data Flow and the Toolchain

Four destinations carry the premortem's data inside the book's toolchain, listed here so the data does not die in the minutes.

Destination one, defensive actions into the anomaly ledger, as in Section 39.5, with the 30-day review point watched by the steward. Destination two, the chapter-number distribution of the death list into the calibration data of Chapter 35's mapping table: when premortems on one class of situation keep hitting the same two or three chapters, that is the failure fingerprint of the class, and Section 35.6 describes the organizational use. Destination three, off-vocabulary deaths and recurring variants into Chapter 37's supplement channel for the question bank, under rule one, that increments come from postmortems, with the premortem as the second increment source. A death that will not fit the vocabulary, if it is real, is worth a new question, and the question bank's boundary expands with each premortem. Destination four, the three-month hit comparison into the comparison column of Chapter 38's postmortem: the premortem list is the first thing a postmortem cites.

All four destinations sit under one steward, the same one who holds the question bank and the ledger, reconciled once a quarter. Completeness has a simple test: pull a random one-pager from a year ago and try to find, within ten minutes, the execution status of

its actions, its review conclusion and its citation in a postmortem. If all three come up, the toolchain is alive. If not, the premortem was a meeting, however lively, and not a system.

39.7 Three Variants

Micro, thirty minutes, three to five people. For medium irreversible points, an architecture change or an important release. Compression: silent writing and round-robin reading merge into write and post, post and read; classification is cut, with the micro version tagging mechanisms directly and running no collision; the claim block keeps its full fifteen minutes. Output is half a page: three deaths, three actions, three check dates. The difference from Chapter 36's thirty-minute process is the number of people. There is no case where three to five people each run the personal process and then combine the pages to better effect than a micro session. If they can sit in one room, hold the meeting.

Twin, two ninety-minute sessions a week apart. For mergers, outsourcing switches, platform migrations, situations with a wide dependency surface. The first session runs standard and produces the death list. The week between is the organizer's, filling in materials against that list: dependency scans, supplier financial checks, historical incident searches. The second session is a death hearing, not a repeat: each death passes through the new evidence and is confirmed, upgraded or withdrawn, with a reason required for withdrawal. The logic is investigation. The first session produces hypotheses, the interval collects evidence, the second produces conclusions. A large situation rehearsed in one sitting produces an output that is half true by necessity. Split it in two with one pass for evidence and the quality doubles.

Anonymous written version, one week, asynchronous. For situations with high political temperature or teams that cannot meet

across time zones. The organizer sends an anonymous form, three deaths, each tagged to a layer. After the deadline the chair classifies and tags chapter numbers independently, and the output goes out as a premortem report, deaths unsigned, classification reasons stated. What is lost is the collision block, meaning the associations that verbal exchange sparks. The remedy is a second round: after the report goes out, one round of anonymous additions is opened, for entries written after reading other people's deaths, and it usually recovers two or three more. The anonymous version produces less than a same-room version in quality, and far more than a same-room version where nobody dares say the true thing. When the temperature is not ready, anonymity is the only form that collects real deaths.

The three variants share one page format, one follow-up rule with actions into the ledger and a 30-day review, and one hit-rate statistic. Variants upgrade. A micro session whose deaths tag to more than three chapter numbers upgrades to standard; a standard session that cannot check its dependency surface upgrades to twin. The upgrade test is a single number: if more than thirty percent of the last output's deaths could not claim an action, the situation is larger than the meeting format, and the format upgrades.

The 30-day review point has a fifteen-minute micro format of its own. It is not a self-check email. It is three people standing at the ledger for fifteen minutes: was the action executed, checked against the output; if not, was the cause resources, authority, or an action written as a wish, one of the three, written into the ledger; and a wishful action gets rewritten into an executable one, following item two of Chapter 36's cross-check list. The commonest finding at thirty days is the third: actions claimed at the premortem turn out to have been wishes when the time comes to land them. That finding feeds the next claim block, where the chair spends two minutes on historical cases of wishful actions, and the quality

of claiming improves visibly. One task remains: the first pass at the hit grade, happened, partly or not, which the three-month formal grading revises later.

39.6 The Three Failure Forms

Form one, the credit-claiming meeting. Symptoms: the death list is all "customers too demanding" and "the market changed", and the defensive actions are all "strengthen communication". The disease is a chair who is not independent and rebuttal that arrives early. The fix is a new chair and a reread of the second opening sentence.

Form two, the anxiety meeting. Symptoms: the death list runs without end and not one entry claims an action. The disease is treating the premortem as confession. The fix is the rule that an action which cannot be claimed becomes an escalation item, which forces convergence, and the meeting ends at ninety minutes.

Form three, going through the motions. Symptoms: the one-pager is identical to the last one. The disease is that the premortem has no interface with the ledger or the postmortem, meaning the two items in Section 39.5 were never executed. The fix is to put the hit rate of the previous premortem's deaths on the first page of this meeting.

The premortem is not a one-off rite but a machine that needs maintenance, and its maintenance record is the comparison of successive one-pagers. Without the comparison, the third session slides to form three by necessity.

Appendix: A One-Page Agenda

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Premortem · Agenda (90 minutes)
Subject: ____   Irreversible point: ____   Chair: ____
```

```
(not a project member)
Sent 24 hours before: positioning table conclusion
page / last 3 ledger entries / dependency graph
0-10    Opening: the three script sentences, read
aloud
10-25   Silent writing of deaths (at least 3 per per-
son)
25-55   Round-robin reading, record only, no rebuttal
55-70   Tag mechanisms (Section 35.3 table) + sample
from the ten general questions
70-85   Action claim (one per death: action / check
date / claimant)
85-90   Close: read the one-pager; itemise objections
24 hours after: one-pager to all staff and both
superiors; actions into the ledger
```

Voting and priority need their own note, because this step decides the quality of the one page. The default rule is three dot stickers per person, voted on the deaths each person agrees with most, all three allowed on one death. After the ranking the chair does one thing: check the distribution across layers. If the top three all sit in one layer, usually the technical layer, because technical deaths are the easiest to write and the safest to name, he forces one from each of the other two layers into the top three. This is a targeted cor‑rection of the room’s comfort zone, since organizational and in‑centive deaths are hard to write and will be absent without force. The two forced nominations are marked on the page as the chair’s nomination and carry the same weight as the voted entries. The list stays at three to five entries, and beyond five the merging is done by action, since one action that defends several deaths is written once. Chapter 36 gave the principle: three actions equal zero actions, and five actions equal two and a half.

Hit-rate counting is specified down to the operation. On the review date, the one in the footer, the review owner pulls the actual cutover or close-out records and grades each death on the list in three bands: happened, meaning the process described actually appeared; partly happened, meaning a variant of the mechanism appeared with different details; did not happen. The bands are counted separately and the hit rate counts only happened. The grades go into the annotation column of the one page, which is not rewritten, only annotated in a different pen colour, and the annotated page is filed into the ledger. In the annual tally two numbers stand side by side: the organization-wide hit rate, and the count of happened entries that an action blocked. The first measures the premortem's eye and the second measures the defence's execution, and reading them separately is what keeps either from contaminating the other.

An annual calendar for the organization, as a model. Quarterly routine: one standard session, its topic taken from the extreme-quadrant carriers on Chapter 35's parameter distribution chart (Section 35.6), plus however many micro sessions the quarter produces, decided by each team with the organization counting pages only. The quarterly standard session sits in the second week of the quarter's first month, clear of quarter-end delivery pressure, with time to prepare the pack.

Event-triggered: after a peer's major accident bulletin, add a session for projects carrying a similar mechanism on site, with the third pack item replaced by the bulletin itself. The value of the trigger is a psychological window. An external accident makes bad news fully legitimate for a while, and deaths that cannot be said in ordinary weeks can be said in that window. The window expires: not held within two weeks of the bulletin, and the temperature falls back.

Milestone-fixed: one month before go-live, one month before a closing, one month before an outsourcing switch, three points written into the project template as an institution, by adding one line to the initiation checklist, premortem scheduled. The point of the template is to move the meeting from remembered to unmissable. Institutional memory does not run on people's memory.

Annual summary: hit rates across all pages, chair pool ratings, distribution of variants used, one page to governance. The most important line in that report is the absence count, listing the large irreversible points that have no premortem record. The absence list needs reading more than the hit list does, since the risk of a session that never happened is uncovered in full.

The calendar's total cost can be counted. Four standard sessions a year at six hours each including preparation, plus micro sessions and triggered ones, comes to about eighty person-hours a year for an organization of a hundred people. Against the organizational cost of one real accident, which is the order-of-magnitude table in Chapter 1, the arithmetic does not need doing a second time.

39.10 The Close

The ninety minutes, accounted for at the end: the premortem spends one meeting's time to buy the chance of converting "we never thought of it" into "we thought of it and did not stop it", before the irreversible point arrives. The gap between those two states is an order of magnitude, because the first only becomes visible after the point is passed, and the second still has an action with a name on it. The book's tools close the loop at this chapter: Chapter 35 positioning, which says what to defend against; Chapter 36 the personal walkthrough, which says how one person defends; Chapter 37 the question pass, which measures the cognitive stock; Chapter 38 the postmortem, which books the tuition already paid;

this chapter's premortem, which saves the tuition not yet paid. The five share one discipline: translate vague unease into statements with sources and actions with dates. The premortem is the only meeting among the five, and the only one that needs other people to cooperate, which is why it is the easiest to criticize as a fuss and why it has the lowest cost per session and the highest output per session of the set. The reason to hold one never needed more evidence, since Chapter 7's list of irreversible points is the whole of it: every real irreversible point added to that list deserves one more page.

Three points on using the agenda. The agenda goes out with the pack 24 hours ahead, and participants arrive holding a filled conclusion page from the positioning table, because a participant who skipped the personal thirty minutes will spend the first ten minutes of silent writing trying to recall project details. The time boundaries in the agenda are hard: the chair's timer answers to the agenda and not to the mood, an overrunning block cuts content in the room, cutting items and keeping claims, and no block is extended. The post-meeting action on the last line, the one-pager out within 24 hours, is done by the chair personally, since a delegated send quietly shortens the copy list.

One suggestion for the first session: start with a small and real micro version, on a medium irreversible point a team actually faces, and ten minutes after the meeting have everyone score it on three dimensions, freedom to speak, quality of entries, hardness of claims, then fix the low scores before the next one. The first meeting aims at walking the full process, not at meeting the standard. An organization that has walked it once faces half the resistance at the second session, because the doubt in the room has changed from what use is this meeting to what about the one we failed to stop last time. The second doubt is the healthy one: it means the output has entered the mental ledger.

Five objections follow, which the sponsor will meet in full, usually the person who read this book and wants to bring the process in.

"The project team knows its own risks best, so an outside chair is formalism." The team knows the part of the risk that has already been discussed. The premortem wants the part that has not been discussed, and independent writing and an outside chair are both doors onto that part.

"There is not that much to rehearse, our project is small." Small projects take the micro version, thirty minutes. The test is withdrawal cost, and plenty of small projects carry a withdrawal cost that has nothing to do with size.

"We can write it down, but nobody will act, the actions never land." A death that cannot claim an action becomes an escalation item, and escalation items have recipients and deadlines. If escalation also draws no response, that fact deserves escalation on its own, by Chapter 19's test, and the premortem has merely measured a deafness the organization already had.

"Meeting time costs too much." A wrong comparison will always look expensive. The comparison is the historical accident cost of the same class of irreversible point, and Section 38.7's organizational profile holds the numbers ready.

"A premortem makes people nervous." Look at which kind does: the hazard sealed in the corridor after the meeting, or the hazard written on a page with a claimant's name. Nerves come from unease that has nowhere to go, and the page is where it goes.

One ordering suggestion for the sponsor, after the five answers. The first step in promoting premortems has always been a team willing to try and one real decision, not a policy document. A policy brought to a meeting with no sample behind it draws an objection list much longer than this chapter's. With one fully walked micro sample in hand, a one-pager, three deaths, three ac-

tions and a review conclusion thirty days later, the policy discussion changes from whether to hold these meetings to how to hold them well, and once the question changes most of the resistance goes with it. The book's five usage chapters follow the same order. The positioning table and the thirty-minute process run without approval. The question pass borrows an existing review meeting. The postmortem rewrite starts with the next postmortem and costs no approval. The premortem comes last, because it needs people and authority. Moving from tools that need no approval to tools that need approval is itself a ladder of rising resistance, climbed in order, and every step of that ladder has a section of this chapter as a handrail. An organization that has climbed all five no longer holds Part Three's twenty-four chapters as a book. It has its own ledger, its own card library, its own hit rate, and an external book has grown into an internal system.

The Five Self-Checks

1. Of the three items in the premortem pack for your nearest irreversible decision, how many do you hold right now?
2. If that meeting were held tomorrow, who chairs it, and would he dare make the project lead keep quiet for twenty-five minutes?
3. At the last review meeting you attended, were the objections recorded or explained away?
4. The last premortem one-pager: did it go to both sides' superiors?
5. If you have organized premortems, where is the comparison table of successive one-pagers? With no comparison, which session will you start backfilling from?

Chapter 40 The Failures Worth Encouraging

> What this chapter covers: three checkable criteria that separate exploratory failure from preventable failure, and the two symmetrical diseases that follow a badly drawn line, dressing preventable failure up as trial and error, or punishing exploratory failure as dereliction. When to turn back: when you are setting the accountability standard for failure; when you hear "we need to be tolerant of failure", or "zero tolerance". Estimated reading time: 30 minutes.

Every page of this book is about preventing failure. Not one of them says failure is forbidden. This chapter draws the boundary of what should be encouraged: which failures deserve encouragement, which deserve sanction, and where the line sits.

An encouraged failure has to pass three criteria. One: the cost stays inside a budget, and the exploration budget is money that gets spent until it runs out, not a hole with no bottom. Two: the knowledge produced can be coded, and what was learned can be written down rather than only felt. Three: the failure did not touch a point of no return (Chapter 7: once you have signed, gone live, announced, the action has stopped being exploration). All three hit, and the failure deserves encouragement. The form of that encouragement is a public postmortem, recognition for the team, and the lesson entered in the knowledge base.

A failure worth sanctioning also has three criteria. One: a known rule was broken (the rule exists, you knew it, you broke it, which makes the act a violation and not an exploration). Two: it was concealed after the fact (worse than the failure itself, by the logic of Chapter 19). Three: the same failure repeated (Chapter 11: when the same class of anomaly appears a second time and draws the

same explanation again, the first step of the ladder is already in motion).

Between the three encouraging criteria and the three sanctioning ones lies a grey zone: cost over budget but knowledge of value, a point of no return touched but a new opportunity found by accident. Discretion over the grey zone belongs to the governance layer, and the decision goes into the case library, so the next grey-zone event has a precedent to consult. The grey zone exists because the line between encouragement and sanction is not a physical line. Organizations draw it in each instance, and each drawing leaves a trace. The traces are the organization's boundary history.

Before the criteria are used, three worked examples (pedagogical composites, not cases from the library) are worth filling in, because the sharpness of a criterion only shows against a filled form.

Case A. A team trials a new caching strategy in production without declaring it, saves budget, and on the third day a cache avalanche takes the service down for two hours. Criterion one, information increment: "cache stampede will avalanche" is already cited in the operations manual, so the information is pseudo-information, and the failure is preventable. Criterion two, cost constraint: no paper existed beforehand, undeclared, preventable. Criterion three, path declaration: no document, preventable. All three point at preventable, whatever the team lead calls it. Handling: ledger, postmortem, structural accountability.

Case B. A team declares a pilot on a new material, with a maximum loss of 500,000 written on the paper and three indicators named as stop-loss. Eight months later the indicators have not been met, the project closes, and the loss comes to 470,000. Information increment: the failure threshold of this material under this working condition is a new proposition, so this is exploration. Cost constraint: the actual loss sits inside the paper. Path declaration: the failure happened on the declared path. All three hit, exploratory.

Handling: public postmortem, information into the library, recognition for the team.

Case C. A team declares a pilot with a paper ceiling of 300,000, then goes past the ceiling to 800,000 mid-course to catch a market window, and the direction later proves sound. Criterion one: the direction was sound, so the information is real. Criterion two: 1.6 times past the paper, a miss. Criterion three: over the limit while staying on the path, a miss. Two misses, so it is handled as preventable, but the postmortem records one separate line: the direction judgment was correct. Whether the direction was right and whether the process was lawful are two accounts, and an organization that adds them together learns neither the direction nor the discipline. The taste of the grey zone appears here, and 40.6 comes back to it.

What the three examples teach in common: every document used to check a criterion is a pre-event document, which is what stops the urge to reword the story afterwards. The declaration holds the same position in this system that preregistration holds in an experiment. It turns exploration from a retrospective figure of speech into a status held in advance.

40.1 One Boundary, Two Diseases

Failure is not one category. It is two.

A failed clinical trial, a closed product line, a technology route disproved: these are prices an organization pays to buy information. Shipping with a known defect, skipping verification, ignoring an alarm: these are the consequences of a known rule being broken. The information in the first kind did not exist before the failure. The information in the second kind was written into the rules before the failure happened. The two need opposite governance. Exploratory failure wants a budget, a declaration, and a

reward for the information it produced. Preventable failure wants a ledger, a postmortem, and accountability traced back to structure. An organization that runs both through one standard will develop one of two diseases, and often both at once.

Disease one: preventable failure dressed up as exploration. A launch that skipped verification becomes "moving fast and testing cheaply". A gamble that ignored the alarm becomes "the courage to commit". An operation with no record becomes "agile". The dressing keeps preventable failure out of the ledger, so the same death recurs, and each recurrence arrives with the sentence "we learned a great deal".

Disease two: exploratory failure punished as dereliction. A team that tried and failed is held to account, nobody declares the next attempt, and exploration moves underground. Failure continues. It simply stops being recorded and stops being discussed. The organization's visible failure count falls while its information flow stops.

Both diseases point at the same boundary. The test is not the size of the failure, and not the attitude of the person who owned it. It is three objective conditions that anyone can check.

40.2 The Three Criteria

Criterion one: information increment. Did this failure produce a piece of information that did not exist before it?

The information from exploratory failure takes the form "this road does not go through": the drug does nothing at the target, the market will not pay this price, this architecture cannot hold this scale. The "information" from preventable failure is pseudo-information: weak mortar will hollow out, an alarm that has rung for two hours is an instruction to go and look, low tyre pressure will

burst a tyre. Those rules existed before the failure, in a standard, a manual, or an industry lesson. The check: write the cause of the failure in one sentence and ask whether that sentence is a citation of some existing document. A citation means preventable. A new proposition means exploration.

Criterion two: cost constraint. Was the cost of the failure bounded in advance?

Exploratory failure is information bought under a capped bet. A phase-one trial of a few hundred subjects, one prototype, one pilot city: the cost appears in a budget and the ceiling can be calculated. The cost of preventable failure has no ceiling. Its size is set by luck, and a run of quiet years only means its turn has not come. The check: ask whether a piece of paper existed before the event stating the maximum loss. If it existed and the actual loss fell inside it, exploration. If there was no paper, or the loss ran far past it, preventable.

Criterion three: path declaration. When the failure happened, was it on a path declared beforehand?

Declaration is what turns an attempt into an organizational decision instead of a personal adventure. The plan, the boundary, the stop-loss point and the approver all sit in a document. A failure inside the declared path gets exploratory governance. A failure on a departure from that path, a skipped step, an exceeded limit, a private modification, gets preventable governance, however much the intention resembled innovation. The check: take out the pre-event declaration and see whether the failing action appears on it. If no document exists, the default is preventable, because nothing can show the attempt was controlled.

How the criteria are used: all three hit, exploratory. Any one miss, preventable. A grey zone, two hits and one weak, may be marked

“pending classification”, but it must be entered in the ledger. Being unclassified is allowed. Being unrecorded is not.

The point of the criteria is to prevent the after-the-fact explanation. The urge to relabel a preventable failure as exploration never goes away, and since every criterion points at a pre-event document, the relabelling has nothing to work with.

Now the exploratory trio, item by item.

Portfolio management. The path limit starts from an operational formula: the number of paths open at one time does not exceed what the organization can honestly postmortem at the same time. For most teams the real number is three to five. The limit is checked monthly. When the limit is full and someone wants a new path, the only legal move is to close an old one, and running over the limit is not permitted. What the limit prevents is not waste. It prevents the dilution of attention, because an organization splitting attention five ways, a fifth each, has five paths running naked.

Loss budget. Three tiers. A small note, where a single path’s loss cannot exceed a tenth of the team’s quarterly budget, is approved by the team. A medium note, between a tenth and a half, is countersigned one level up. A large note, above a half, is approved by the governance layer. The tiers make the cost of deciding proportional to the cost of being wrong: small notes open fast and close fast, large notes open slowly and close under scrutiny. The mandatory review triggered by an overrun has its own detail: additions freeze on the day the line is crossed, a review conclusion is due within 48 hours, and the review asks two questions only. Is the reason to continue new evidence or old hope (Chapter 13), and who is delaying the stop-loss.

Quick funeral. Once the speed of falsification becomes a metric, the reverse game starts: stop-loss lines set low so that paths can be closed quickly and look decisive. The declarant sets the stop-loss

line, the budget approver reviews it, and the two rights are kept apart. One ritual is worth keeping at the funeral: the closed path gets a hundred-character epitaph (what was learned, under what conditions this road reopens) and the epitaph goes into the know - ledge base. The epitaph is the final product of exploratory failure. A failure that can produce a hundred characters has completed the information purchase. A failure that cannot produce them has not even closed its books.

The preventable trio is already laid out elsewhere (ledger in Chapter 11, postmortem in Chapter 38, structural accountability in Chapter 9). One item is unique to this chapter: keep the two ledgers apart. Put exploratory entries and anomaly entries in one book and two years later nobody can say which failure was money spent on information and which was a fee paid for nothing. A shared book is the breeding ground for prettification, and grey-zone entries are the easiest of all to paint over as exploration once they sit in a mixed book. Separate books, separate colours, separate field definitions for the mechanism column. This is infrastructure-level separation, and it works on the same principle as the separation of the postmortem system from the personnel system in Chapter 38: where a cultural promise will not hold, a permission group will.

Criterion one needs one more layer, because pseudo-information has an in-house variant. A rule sits in an industry standard or a vendor manual, and this organization never internalized it. Nobody read it, or someone read it and it never entered a proced - ure. The cause of the failure is still "a citation of an existing doc - ument", so the failure is handled as preventable. That lenient reading attracts objections ("we genuinely did not know"), and there are two reasons to hold it. The criterion system exists to protect the structure of the pre-event paper, and accepting "a rule nobody read does not count" would reward ignorance, making illiteracy a passport into exploration. And information unknown to

this organization was available at low cost, one search, so the claim of information increment does not stand. The humane side of the lenient reading sits in Chapter 9: structural accountability runs as usual (why did this rule never enter a procedure, which is a matter for the ledger and the postmortem), while the treatment of indi - viduals goes through the fair-culture substitution test. Leniency and mercy are two different things. The criteria settle the account. The culture sets the penalty.

40.3 Governance of the Two Classes

The exploratory trio: portfolio management (a cap on the number of paths open at once, read as a portfolio rather than case by case), loss budget (each path states its maximum loss in advance, and a breach triggers mandatory review), quick funeral (the speed of falsification is the metric, one month to disprove beating a year to confirm, and delayed falsification is the largest cost in exploratory governance).

The preventable trio: ledger (every event entered, tagged with a mechanism chapter number), postmortem (the Chapter 38 tem - plate), structural accountability (ask about process and incentives, not only about people). The two systems share one infrastructure: records, ledger, independent review (Section 38.4). They differ in standard and consequence.

Here is the minimum design of a declaration system, so that cri - terion three can actually be applied. One page, five required fields.

Field one: hypothesis. One sentence stating what proposition this attempt tests (“this adhesive holds under working condition XXX”). An attempt whose proposition cannot be written does not enter the process. An action that cannot say what it tests has no information increment to offer and falls outside criterion one by construction.

Field two: boundary. In what scope, under what conditions, for how long. This field is the paper against which criterion three checks the failing action.

Field three: maximum loss. Money, person-hours, and the largest impact on existing business, in numbers. This sheet is the whole of criterion two.

Field four: stop-loss point. Which signal means stop. The signal must be observable (an indicator, a date, an event), and "stop when it feels wrong" is not a stop-loss point.

Field five: information outlet. Where the learning gets written and who uses it. This field moves the codability requirement of criterion one to the front: decide where the learning goes before the fact, and afterwards there will be more than a feeling.

Approval has a time discipline. Small note, 48 hours. Medium, one week. Large, two weeks. Past the deadline, the default is refusal, because an approval process that dies of delay is worth exactly as much as no process. Both decisions are written in two lines. An approval gives a reason that can be tested later, since the approver is also declaring a judgment. A refusal has to name a specific defect in one of the five fields ("the boundary is too wide", "the stop-loss is not observable"). Refusals of the "not for now" kind are not allowed, because a refusal with no information pushes the declarant back into an undeclared attempt.

A declared attempt enters the exploratory ledger, and a failure goes through the exploratory governance of 40.3. Changing the boundary or the stop-loss mid-course requires a change declaration: edit the paper, leave the trace. Verbal changes are prohibited, because a paper amended by mouth leaves criterion three with nothing to check afterwards.

Declaration rates do not rise on exhortation. They rise on three structural changes.

Change one: make the win rate visible. How many refused declarations came back on a different path and succeeded, and how many epitaphs from closed paths were later cited by other projects. Publish both numbers. Declarants have to see the system produce value before they keep paying the cost of a page.

Change two: make refusal cost something. An approver who rejects a declaration signs the ledger, and the annual statistics report each approver's refusal rate alongside the number of times work went around the process and then had an incident. Approval becomes a recorded judgment, and a careless refuser becomes visible too.

Change three: decouple the declaration window from the performance cycle. The flood of declarations before a review and the drought after it show that declaring has been filed as a performance act. Decouple by keeping the declaration ledger out of the appraisal numerator and denominator, the same rule the post-mortem runs on, and let appraisal look only at the portfolio's final output.

The three changes share one logic. The enemy of the declaration rate was never laziness. It is a negative expected return on declaring. Fix visibility, liability and timing, and the return turns positive, and behaviour walks over on its own.

40.5bis Organizational Detectors

Both diseases have detectors at the organizational scale, and they find the condition earlier than case-by-case diagnosis does.

Disease one has three. Detector A: the ratio of exploratory entries to anomaly entries in the two ledgers. Where exploratory entries dominate to an implausible degree, with more than eight in ten failures described as exploration, the criteria are being used as rhetoric rather than as a check. Detector B: the frequency of "we

learned a great deal" set against the completion rate of the mechanism field's chapter number. Said often, and no chapter number to show, is evidence of prettification. Detector C: the annual inventory of large undeclared losses. Every one it turns up is direct evidence of a case that slipped past the dressing.

Disease two has three as well. Detector A: the annual curve of declaration count. Flat or falling while business volume rises means declaring is dying. Detector B: tracking behaviour after a refusal. What share of refused teams came back on a different path, which is healthy, and what share simply went ahead, which is the disease presenting. Detector C: one question in internal recruitment interviews, "would you say in advance here that you were about to attempt something that might fail". The pause before the answer is the reading.

Both sets are read the same way. Ignore the culture as stated. Look at the shape of the ledger and the curve of behaviour. Both diseases are early on paper and late in personnel files.

Four interfaces with the rest of the book, so that this chapter's criteria do not contradict the twenty-odd chapters before it.

Interface one, with Chapter 20 (the slack of success). What slack consumes is defensive investment: inspection, maintenance, redundancy. This chapter's exploration budget is informational investment. Both kinds of money look useless while nothing goes wrong, and both get cut easily, but the reasons for cutting differ and the diseases they prevent are unrelated. Book them as separate lines. Protect defensive investment with the Chapter 20 questions and the exploration budget with this chapter's portfolio management. Book them together and whoever wants to cut will always find a line that looks useless.

Interface two, with Chapter 27 (sunk cost and escalation). Additional spending on an exploration and additional spending on an

escalation are indistinguishable in the cash flow. The distinction is the pre-event paper. An addition inside the declared boundary, with the stop-loss point not yet triggered, is the normal rhythm of exploration. An addition after the stop-loss point has triggered is escalation, and the external adjudicator of Chapter 27 takes over. Criterion three's paper does its second job here: the first job settled the character of the failure, exploration or violation, and the second settles the rhythm, continue or stop.

Interface three, with Chapter 13 (narrative before evidence). The declaration is the antonym of a narrative: it forces the proposition to be fixed before the data arrives, and fixed in falsifiable form. The quality of a team's declarations can stand as a proxy for that team's immunity to Chapter 13. A declaration with a vague proposition is almost certain to slide into narrative reconstruction after it fails.

Interface four, with Chapter 36 (the reverse reminder from the thirty-minute procedure). Chapter 36 said that exploratory decisions are not served by the death rehearsal, because a rehearsal looks for preventable ways to die and an exploratory decision has no preventable death to find. That gives the two chapters a precise division of labour: operational decisions use the rehearsal to find death, exploratory decisions use the declaration to buy life. One instrument manages dying and one manages living, and an organization needs both. An organization with only one installed is either over-cautious or running naked.

The common thread through all four: these criteria are not a loose sheet. They close the loop of the book's mechanism system on the side that permits failure. The other half of that loop, preventable failure, occupies twenty-odd chapters. This chapter supplies the missing side, and only then is the system drawn complete.

The accounting detail behind criterion two deserves its own passage, because "how much do we write as the maximum loss" is

the field most easily gamed in any declaration system. The motive to write it small is obvious: a lower paper ceiling is easier to approve and lighter to answer for once it is breached. Three countermeasures.

Countermeasure one: define maximum loss as inclusive of spillover. Direct spend plus the largest impact on existing business (downtime, recall, compensation), verified by the budget approver, not calculated by the declarant. Estimate the spillover with the cascading-failure approach of Chapter 33: if this path fails, who receives it along the dependencies.

Countermeasure two: decouple the paper ceiling from the penalty. A breach is handled under the ledger logic of Chapter 11 (entry, postmortem, structural accountability) rather than fined by the size of the overrun. Once the overrun amount is tied to the penalty, declarants learn a reverse skill and write the paper enormously high to stay under it, which destroys the budget. Accountability always attaches to "no paper" and "concealed", and the amount itself is only a number in the postmortem.

Countermeasure three: audit the loss definition after the fact. Each quarter, pull two closed paths and check whether the calculation of actual loss matches the definition used on the paper. A paper that counted spillover and a postmortem that counted only direct spend is a silent overrun. Definition audit guards against bookkeeping compliance, where the ceiling was never breached because the definition went on a diet first.

The principle under all three: criterion two protects the structure of the capped bet. How large the bet is remains a matter for both parties to agree, and any unilateral change after agreement, writing it smaller or counting it smaller, hollows the structure out.

40.10 Three Misreadings

Misreading one: encouraging failure means raising the failure rate. The objective function of the criteria was never the number of failures. It is the unit price of information: the same piece of information bought as cheaply as possible. Portfolio management and the quick funeral press the unit price down, and the failure rate falls behind it, because once pseudo-exploration leaves, the remaining failures are all money spent on information. An organization that reads this chapter as "fail more often" has installed the shell of the policy without the core of the criteria.

Misreading two: a tolerant organization needs no accountability. The other half of the criteria, the three preventable ones, is printed on the same sheet as the three encouraging ones. The more exploration you encourage, the harder the preventable standard has to be nailed down, or violators will spend the exploration allowance (Section 40.1 explains why the two diseases usually arrive together). The fair-culture spectrum in high-hazard industries, graded between pure no-blame and full accountability, is structurally the same as these criteria: the grade is set by intent and by the substitution test, whether another person would have done the same. This chapter moves the material for that judgment forward into a pre-event paper, so the judgment returns to a document instead of a memory.

Misreading three: a declaration system kills inspiration. The improvised attempt born of inspiration is in fact the declaration system's best customer. One page, five fields, and an inspiration that can fill them is worth pursuing, while an "inspiration" that cannot state a proposition is usually an impulse. The system does not filter out inspiration. It filters out impulses wearing inspiration's clothes. Real gold is occasionally among the filtered impulses, and that is the insurance premium of the system, an order of magnitude cheaper than the tail cost of undeclared attempts.

The common antidote to all three: go back to the criteria. The criteria do not reward failure and do not punish failure. They separate two accounts, money spent on information and fees paid for nothing. An organization that keeps the accounts apart can afford to spend from both.

40.9 A Compliance and Legal Addendum

In heavily regulated industries the declaration needs one more layer: a mapping against the rules. Three additions.

Addition one: put a "rule mapping" field ahead of field five. It lists the current rule clauses this attempt touches, and the expected compliance position (inside current rules, exemption required, not covered). Mapping turns legal from the last veto into the person who draws the boundary in advance. The regulatory lag of Chapter 31 meets the demand for exploration here: the ground the rules do not cover is the lawful space for exploration, and it is the reason regulatory sandboxes exist.

Addition two: file the exemption in step. Where an exemption is needed, the exemption application goes in with the exploration declaration, rather than being sorted out after a failure. An exemption obtained afterwards looks no different to a regulator than a violation (the compliance logic of Chapter 32), while filing both together moves the boundary dispute ahead of the loss.

Addition three: design the transparency strategy toward the regulator in advance, deciding what range and what wording will be disclosed, which precedents and epitaphs the regulator may see. Do not decide this during an inspection. Chapter 31 described how a single information source produces capture, and a regulated party voluntarily supplying structured exploration information is one of the few available levers for changing the composition of that

source: when the regulator can see where your exploration boundary runs, a rule update has somewhere to land.

The floor under this section is one sentence. Every freedom this chapter grants rests on a pre-event paper, and in a regulated in - dustry the paper carries one more layer than elsewhere. Freedom costs more there. The structure of the criteria applies unchanged.

40.8 The Morphology of Encouragement

For an exploratory failure that hits all three criteria, the word "encouragement" has to become something concrete. Three forms, each with its taboo.

Form one: the postmortem changes specification. An exploratory failure enters the Chapter 38 template with the mechanism field tagged "exploratory path disproved". The session does not hunt for structural defects. It extracts the proposition and the evidence: which later judgments this falsification changed. Taboo: letting preventable-accountability questions into an exploratory postmortem. Wrong book, wrong questions.

Form two: courtesy toward the information. The epitaph enters the knowledge base, the proposition enters the case library, and the falsification feeds the position table and the question library, so a failed path leaves the question library one question larger. What receives the courtesy is the information, not the feelings: a citation can omit the word "brave" and must include "the proposition is falsified, evidence number X". Taboo: turning the person who ran the exploratory failure into a tragic hero. Hero narrative loads the next attempt with a performance obligation, and people suppress stop-loss signals to live up to it. The calmer the wording of the courtesy, the more durable the system.

Form three: renewal of the budget. A failure does not shrink that team's exploration budget, because the budget is assessed on portfolio performance rather than on a single path. Taboo: letting renewal become automatic. For a team whose last three epitaphs produced no codable information, renewal comes with conditions, a change of proposition domain or a change of the person holding the proposition. The whole design target of the encouragement machinery is one thing: the next person dares to declare, and must declare by the rules. Calibrate the three forms to that target and encouragement will not deform.

Portfolio management has one internal structure still to explain: the relation between paths. Inside the limit, a hedged pair is allowed, two paths with opposite propositions open at once, each with an independent stop-loss. The value of a hedge is that its information output runs in both directions: whichever side is falsified, the opposite proposition strengthens. The hedge has one rule. The two paths may not share a data source for their stop-loss signals. A shared stop-loss is the common-cause failure of Chapter 22 replayed in an exploration setting: the data source goes down and both stop-losses go blind at once. Beyond the portfolio view, the paths also need a read on crowding out. Once a star path absorbs the attention, the stop-loss signals on the others get read late, because attention is a resource. Add one question to the monthly portfolio check: in the past month, how many times was each path's ledger opened. The spread in those counts is the reading on crowding out.

40.4 Counterexample: Three Marks of Prettification

Mark one: the story contains no criteria. When you hear "that trial was valuable", ask three questions. Where is the pre-event declaration, what maximum loss was written, and what was the information increment. Three blanks and the prettification is

confirmed. A real explorer is boring before the fact (documents, limits, stop-loss points) and talks about value afterwards. The prettifier runs in reverse: nothing before, all meaning after.

Mark two: the cost only becomes known afterwards. The cost of exploratory failure is capped in advance. The cost of a prettified failure is startling afterwards. Whenever "trial and error" appears in the same breath as "far over budget", handle it as preventable.

Mark three: the same death appears a second time. Exploratory failure is the purchase of information, and one piece of information should not be bought twice. A pattern that recurs means the first round's information never reached the ledger, or that what happened was not information but dereliction.

There is a fourth mark, and it lives in the language. Prettified accounts are dense with unverifiable sentences: "the whole industry is like this", "the direction was right, execution was the problem". Whether the direction was right is exactly what the proposition and the stop-loss in the declaration were meant to answer. Whether execution was the problem is exactly what the ledger and the postmortem were meant to answer. Each sentence has a paper it belongs on. Saying them out loud while avoiding the paper is an admission that the paper holds nothing in your favour. When you hear either sentence, ask which version of the declaration carries the proposition. The absence of an answer is the fourth mark.

40.5 Counterexample Two: What Punishing Exploration Costs

An organization that punishes exploration loses more than a few attempts. It loses the declaration system. Once a person who ran an exploratory failure is publicly held to account, rational practitioners stop declaring, because declaring leaves evidence, and shift to

undeclared attempts: success becomes performance and failure becomes an incident. The organization's real exposure is unchanged and its visibility falls to zero. This is the form that information attenuation up the hierarchy (Chapter 16) takes on innovation topics. The governance test follows: look at an organization's declaration rate rather than its failure rate. A low failure rate with a low declaration rate is usually not safety. It is silence.

40.6 The Grey Zone and the Precedent Library

The grey zone deserves full operational detail, because it is the ground both diseases fight over. The prettifier wants to read the grey zone as exploration. The fault-finder wants to read it as dereliction. The precedent library is the governing form for the grey zone.

Six columns per entry: a de-identified summary of the event; the result of the three criterion checks, which hit and which missed; the point in dispute, with each side's reading; the governance layer's ruling and its reasons; the after-the-fact verification, meaning a look two years later at whether this class of behaviour became more or less common after the ruling; and the precedent number. The fifth column is what distinguishes a precedent library from an ordinary case library. The consequence of a ruling is tracked, and the ruling itself is submitted for testing.

Three operating rules. Rule one: the ruling is made by the governance layer, but the right to open a case sits with the front line. Any team whose failure was judged preventable may apply for grey-zone review, and the review is free, does not overturn the original finding, and only adds a precedent. Rule two: precedents carry retrieval value and no force of law. When the next grey zone appears, precedents inform the discussion and the ruling is made fresh, because the criterion check has to rest on that event's own

pre-event papers. Rule three: the library is reviewed once a year, in both directions. Check whether any precedent now rewards a disguised violation, which means the ruling was too loose, and whether any precedent punishes real exploration, which means the ruling was too strict.

A forecast of scale: a functioning organization produces roughly one to three grey-zone events a year. The library grows slowly, and each entry that appears is a developing agent for the organization's values. What counts as exploration here and what counts as dereliction is answered more honestly in a precedent than on a culture poster.

40.7 The Criteria at Every Level

The three criteria are not confined to the project level. They run upward and downward, changing shape a little at each level.

The individual career level. Personal exploratory failure, a change of field, learning a new skill, testing a side business, takes the same three criteria: a budget (living costs and time set aside), a declaration (candour with family or a partner), a stop-loss point. The individual version catches the same diseases. A preventable repeated mistake gets dressed as "I am experimenting" (the third fall into the same pit), or necessary exploration gets punished as "not serious work", where criterion three means declaring to yourself.

The industry level. The regulatory sandbox is a declaration system at industry scale: a limit (caps on users and on money), a declaration (the plan for entering the sandbox is published), a stop-loss (the exit clause). The sandbox shows the three criteria are executable at that scale too. Industry-level diseases have their analogues: dressing up consumers as free test subjects ("innovation always has someone paying for it"), and the fault-finding that bans every new business form ("no precedent means it is a violation"). Both

refuse the dull device of declaring in advance, and both want the freedom to explain afterwards.

Cross-level consistency has one test question. Take a single matter and ask "where is the paper" at four levels: individual, team, organization, industry. If the answers thin out as you go up, the declaration system has failed at the top, and nobody up there dares ask for paper on exploration. This is the horizon mismatch of Chapter 28 projected onto exploration: the consequences of senior exploration already take longer to arrive, and without paper the bet has no boundary at all.

One recommendation on sequence, the same shape as the other usage chapters. Start with the one-page declaration template. Try it in a team that genuinely has an exploration need, where one path reaching closure inside a quarter makes a complete sample. Run the criteria three times for real, the way Cases A, B and C are run, and only then discuss the organization's policy standard. The wrong order looks like this: policy first, so the organization paints the slogan ("we are tolerant of failure"), the slogan lands with no criteria and no template to catch it, and six months later the slogan is fuel for disease one. Template first, criteria with it, policy last. This is the same rollout ladder as earlier chapters: cost rises from zero, resistance rises from zero, and you climb in order.

Closing. This is the only chapter in the book whose main business is saying what not to defend against. The twenty-four mechanisms and every device work to shrink an organization's exposure. This chapter opens one controlled hole in that exposure: paper sets its size, declaration sets its position, and the epitaph collects its output. An organization without the hole relieves the pressure somewhere else, since undeclared attempts are water seeping through a crack in the foundation. An organization with an uncontrolled hole takes preventable failure in through it. The whole job of the criteria is to make the hole a valve: three criteria open it, three cri-

teria close it, and the grey zone goes to precedent. Put together, the book's position on failure is one line. For the unknown, buy. For the known, defend. Between buying and defending, the paper is explicit.

Those four words, the paper is explicit, are also the whole deliverable of this chapter: one template, two sets of criteria, an empty precedent register, and the blank cells waiting for your organization's first declaration. On the day the first cell is filled, the book's position on failure moves off the page and into your ledger, which is a more solid milestone than any reaction you might have to finishing a chapter. The criteria stop being three lines to be read and become a form in use. A form speaks for itself, and more accurately than this chapter does.

If your answers to the five questions cluster on the third one, no loss budget, then your organization's exploration is currently running with no paper at all, and the first sheet is the starting point. If they cluster on the second, no declaration channel, fix the channel before discussing tolerance, because an organization with no channel can only be tolerant of buried mines. Both are jobs you can start within a week: the template sits one section before 40.5bis, the channel hangs off the existing approval flow, and the budget starts at one percent of next quarter's project budget. A low starting point does no harm. The virtue of a criterion system is that it holds for small organizations too: a three-person team can still cap its exploration at three paths, declare on one page, and enter epitaphs in a register. The value of a system is in its structure, and structure does not discount for scale.

Five Self-Checks

1. Take the last failure your organization publicly praised for "daring to try". Run the three criteria on it. How many does it pass?
2. Take the last failure your organization held someone to account for. Was there a declaration before it? Having no declaration channel is itself the finding.
3. Does your team have the concept of a loss budget? Of the attempts running now, how many have a sheet stating a maximum loss?
4. How long did it take the last attempt to be falsified? Is that period getting longer or shorter?
5. Which process does a failure declaration travel through in your organization? If the answer is "there is no process", who is running undeclared attempts right now?

Chapter 41 Normal Accidents and Black Swans

> What this chapter covers: what Perrow's normal accident theory and Taleb's black swan theory each explain and each fail to explain, and where this book's mechanism layer sits between the two. When to turn back: when someone ends a discussion with "that is an inherent property of the system", or "that was a black swan nobody could have foreseen". Estimated reading time: 30 minutes.

This chapter deals with the boundary of the book's own method. Perrow's normal accident theory says the accidents of some systems are structural necessities. Black swan theory says some events cannot be predicted. If both are right, what use are these tools?

The answer comes in two layers.

Layer one: this book never promises to eliminate failure. It promises to move failure from invisible to visible. Visible is not the same as controllable, and visible failure still happens. What visibility buys is traceability, comprehension, and a response plan prepared in advance. Invisible failure is the expensive kind, because you do not know it has arrived. Perrow's structural necessity refers to the interaction of the technical layer with the organizational layer. He did not claim that misalignment in the incentive layer is also necessary, and the incentive layer is where this book has the most room to act: change the appraisal, change the contract, change who signs. Those are within human reach.

Layer two: Taleb's black swan is about the limits of cognition. Unpredictable events do exist. Most organizational failure, though, is not a black swan but a grey rhino, and the record skew of Chapter 2 is what makes a grey rhino look like a black swan. Not

knowing is not the same as unknowable. The grey rhino has already been written up as a report in someone else's industry, and your ignorance is a failure that circulates rather than a limit of cognition.

The two conclusions combine into one practical instruction: sort risk into three classes. Class one is preventable and measurable, and it takes the twenty-four chapters of Part Three; most organizational failure lives here. Class two is measurable but not preventable, structural risk; it takes emergency preparation and redundancy design, and the point of Chapter 22 is that measuring it at least lets you prepare. Class three is neither measurable nor preventable, the true black swan; it takes organizational resilience, meaning the intertemporal responsibility of Chapter 28, the knowledge pack of Chapter 30, and the supply chain resilience of Chapter 33. Resilience is not defence against a named risk. It keeps the organization from shattering when an unnamed one arrives. The sorting is itself the first act of prevention. Most organizations talk about classes two and three as one thing, and the result is that neither class gets a countermeasure of its own.

The genealogy of both theories is worth setting out, so the reader knows what is in hand. Perrow's book appeared four years after Three Mile Island, and its immediate motive was dissatisfaction with the nuclear industry's safety narrative of the time, that small probability plus multiple defences equals safe enough. He pressed organization theory, engineering and historical material into a two-dimensional frame, interactive complexity and tight coupling, and used it to triage systems. Over the following forty years that frame produced a whole literature of criticism and inheritance. The most systematic opposition came from high-reliability organization research, represented by Weick and Sutcliffe's Managing the Unexpected. Working on aircraft carrier decks, in air traffic control and among wildfire crews, they found tightly coupled organiza-

tions with long records of low accident rates, and argued that reliability comes from specific organizational practices: a preoccupation with failure, a reluctance to simplify, and deference to expertise. The dispute still reads sharply today, structural determinism against practised malleability. The mechanism layer stands between the two, on the practice side. It accepts that structure frames probability, which is Perrow's point, and argues that practice moves probability, which is the HRO point. Every mechanism chapter is a specific lever for moving it.

Taleb's black swan is the middle volume of a trilogy. Fooled by Randomness covers survivorship bias, The Black Swan covers the limits of cognition, and Antifragile covers benefiting from volatility. Read him with his target in view: the risk management industry built on normal assumptions. His weapons are distribution mathematics and epistemology, not organization theory. Read Taleb as organization theory and the conclusion is "management is useless", which his text does not support. His advice to practitioners, redundancy, low leverage, exposure control, the barbell, is entirely operational governance action. The real disagreement with the mechanism layer is smaller than it looks. It comes down to one empirical question: in the distribution of organizational failure, what share belongs to the tail. This book's 158 cases answer a small share. That answer is empirical rather than doctrinal, and the share will move with industry, era and regulatory environment.

One more discipline for reading both: cite the year and the edition. Perrow has a 1984 first edition and a 1999 expanded edition, which added Y2K and near-midair cases. The Black Swan has a 2007 first edition and a 2010 second edition, which added the grey swan discussion. Second-hand accounts routinely blend the two editions' positions. When checking a claim, check which edition it comes from.

41.1 Perrow: the Accident Is a Property of the System

Charles Perrow's Normal Accidents (1984) makes a cold proposition. When a system has interactive complexity, meaning interactions among components that exceed anyone's understanding, and tight coupling, meaning failure propagates faster than any intervention window, then accidents are not the product of management error. They are a necessary output of the system's structure.

Three Mile Island is the prototype: several independent small failures combined in ways the designers had not imagined, operators made reasonable moves on the strength of wrong information, and the accident happened anyway.

What the theory explains. It explains why, in some systems, severe accidents keep happening no matter how many people are replaced or how many procedures are added: the structure and the coupling sit there, and human effort is being applied at the wrong layer. It also explains why the "operator error" conclusion in an accident report is so suspect. In an interactively complex system, the operator is facing a situation that cannot be understood from inside, which is the extreme form of the boundary break in Chapter 9.

What it fails to explain. It underestimates the domesticating power of institutions. Aviation is a high-technology coupled system, and by Perrow's criteria it belongs in the accident-prone zone, yet seventy years of mandatory reporting, independent investigation and tracked safety recommendations pushed its accident rate to a level the theory does not predict. The theory turned "possible" into "inevitable". Interactive complexity plus tight coupling raises the probability significantly, and probability is not fate. It also says almost nothing about organizational learning: two tightly coupled systems, one with mandatory post-event reporting and one

without, are two different worlds several decades later. That variable has no place in Perrow’s frame.

For the frame to become a daily tool, the two abstract dimensions have to be turned into questions someone can answer. Here is a two-axis diagnostic.

Axis one, interactive complexity, has three questions. Does a component interaction list exist, and if it does, when was it last updated. Can any single role describe a full failure propagation path, tested by asking the most senior person to draw one on a whiteboard, where an incomplete drawing means complexity is over the line. Does the record contain a postmortem whose attribution points at a combination nobody had imagined. Two bad answers out of three and complexity counts high.

Axis two, tight coupling, has three questions of its own. Between a failure and the spread of its impact, is there a manual intervention window measured in minutes. Do buffers exist (inventory, margin, queue), and if they do, when were they last measured. Are the sequences reversible, meaning can steps be reordered and can you hold. Two bad answers and coupling counts tight.

The two axes give four quadrants, and each has a different management centre of gravity.

Low complexity with loose coupling is the comfort zone of this book’s twenty-four chapters, and the mechanism layer covers it completely.

High complexity with loose coupling puts the weight on dependency visibility (Chapter 21) and documentation: move the complexity onto drawings.

Low complexity with tight coupling puts the weight on buffers and redundancy (Chapters 22 and 25): buy the intervention window more life.

High complexity with tight coupling is Perrow's normal accident zone. The mechanism layer admits its ceiling here, and management moves to three things: carve the system into isolatable partitions to slow propagation, impose mandatory anomaly reporting with independent investigation, the aviation model, and replace confidence in prevention with emergency drills. The existence of this fourth quadrant is Perrow's real legacy to the governance layer. It makes "should this system exist at all, should it shrink, should we exit" a legitimate management agenda rather than the opening line of a failure.

Run the self-assessment once a year and file it. Drift between quadrants, low complexity sliding toward high, is itself an early warning of Chapters 21 and 24. The output is two letter grades and a year for each key system on the system inventory, a table of five lines or fewer that the governance layer can read at a glance.

The returns on the black swan label, and the mechanism it engages elsewhere in the book, are worth taking apart, because the label is the after-the-fact form of narrative before evidence. It pays in three layers.

The first layer is immunity (Chapter 18). An unpredictable failure has no responsible party, only victims, and the diffusion of responsibility releases at the instant the label lands.

The second layer is budget (Chapter 20). An unpredictable risk cannot be budgeted against, so cuts to defensive investment become rational under the label. There is no point defending against what you cannot name.

The third layer is time (Chapter 28). The consequences of extreme events stretch out by nature, rebuild, postmortem, litigation, and the unpredictability narrative pushes responsibility past the tenure of everyone involved.

Stack those three and the label costs almost nothing to hold and pays well. That is the structural reason it is in demand out of all proportion to its supply in the post-accident discourse market. The mechanism layer's response has never had anything to do with rhetorical quality. It checks the retrieval results for the objects. A market where the label is over-demanded clears by raising the verification cost on the supply side.

Perrow's frame and this book's case library are compatible on a point the reader can check. The twenty-four mechanisms all work at the organizational and incentive layers, while Perrow's claim of necessity covers the interaction of the technical and organizational layers. The two do not fight on the same ground: technical interaction can come close to necessity, and the organizational and incentive responses, reporting systems, independent investigation, signature structure, remain variables with a great deal of freedom. Aviation's seventy years is the chronicle of those freedoms being used. That expands the judgement at the end of 41.1. The best reading of Perrow is a probability theory of the technical layer, and the mechanism layer continues with a probability theory of the organizational layer. Two volumes make a complete risk book. This book chose to write the second, because the room to act there is empirically far larger: rewriting an appraisal scheme costs nine orders of magnitude less than rebuilding a reactor.

41.2 Taleb: Extreme Events Dominate

The Black Swan (2007) handles a different face of the problem. The extreme events that actually shape history are close to unpredictable before they happen, and people can always weave an explanation for them afterwards. What it explains: why risk management built on historical data fails systematically, since the sample contains no tail; why redundancy and low leverage are among the few strategies that still work under extreme conditions; why an ex-

pert's confidence in a forecast has little relation to the quality of the forecast.

What it fails to explain. It cannot account for attributable failure chains. In the great majority of this book's cases, a postmortem can name at least one piece of information visible at the time, one option that was ruled out, one minimum counterfactual. For those events, "unpredictable" is rhetoric rather than fact. Packing an attributable failure into a black swan narrative amounts to issuing a disclaimer for dereliction. It also underestimates the returns on misusing the label. Claiming unpredictability is the standard post-accident move of narrative before evidence (Chapter 13) and the amulet of diffusion of responsibility (Chapter 18).

Taleb deserves one fair point in his favour. His theory addresses the tail of a probability distribution, while most of this book's cases are failures in the body of the distribution. The two theories treat different diseases. The black swan covers "nobody thought this could happen". The mechanism layer covers "it happens, and the chain can be taken apart". Discuss the two pathologies in one breath and neither comes out clear.

The criterion can be operationalized one layer deeper. The question "which link, if it had been different, would have dropped the probability significantly" has a ready handle in investigation reports, in their causes and recommendations sections. Use the handle in reverse: read each recommendation against the failure chain, and any recommendation that points at a node is evidence that the node could be changed, since an agency would not spend a recommendation on a structure it considered unalterable. That reverse reading yields a practical corollary. The number and granularity of a report's recommendations roughly measure how much room the mechanism layer has in that case. Three Mile Island's report still lists recommendations in columns, even though Perrow cites the case as his sample of necessity, which shows that

even the archetype of inevitability left large changeable nodes. That is about the most honest footnote available on the relation between structure and necessity: necessity never swallows every node, and each node dismantled moves the structure an inch.

41.3 Where the Book Sits: the Mechanism Layer

The book occupies one layer between the two theories, the mechanism layer. The upward question, whether this class of system should exist, is Perrow's territory. The downward philosophy, what posture to take under radical uncertainty, is Taleb's. The mechanism layer handles the large decomposable middle, where a chain exists, information was once visible, and structure can still be changed.

The mechanism layer's self-limitation has to be honest: it does not guarantee that every failure can be taken apart. The test is this. In a postmortem, ask which link on the chain, if the information or the structure had been different, would have dropped the probability of the event significantly. A case that can name one link goes to the mechanism layer. A case where no link can be named gets marked "true unknown" and enters the appendix matrix as a record of humility, not the body of the book. In all the writing so far, no body case has reached the far end of that test, and that fact is itself an estimate of how often "black swan" is misused.

41.4 Three Practical Implications

Implication one: the pricing of redundancy. Perrow says redundancy in interactively complex systems can create new interactions, and rescue systems have introduced new failure surfaces before now. Taleb says redundancy is the only insurance under extreme conditions. The mechanism layer's compromise is Chapter 22: redundancy must be diverse and must be actually exercised. Re-

dundancy that fails both tests is held under Perrow's warning. Redundancy that passes both is held under Taleb's advice.

Implication two: the relation between stress testing and rehearsal. For failures in the body of the distribution, the premortem of Chapter 39 is enough. For tail events, a stress test asks a different question: what combination kills me, rather than how will this one die. Ask both questions, and ask them separately. Push tail questions into a premortem and you get a table of anxiety. Push mechanism questions into a stress test and you get a table of talk.

Three specifications keep a stress test from becoming talk. The combination is written down and held by the facilitator until the session, because a test whose combination is known in advance measures the participants' acting. The pass criteria are fixed beforehand: which level the response returns to within how many minutes, which loss line is capped. And a failed test is logged in the same ledger as a real incident, since a failed test is the cheapest tuition available and unlogged it is tuition paid for nothing. All three specifications have the same enemy, the performance test. The feeling of safety that follows a test costs more than not testing at all, because it burns budget and vigilance together. The fastest detector of a performance test comes at the tea break afterwards, when you ask one participant what they were afraid of during the exercise. A specific threat means it was a test. "It was fine" means it was a performance.

Implication three: accountability that resists narrative. Allocation of responsibility has to be written into pre-event documents, on the logic of Chapter 40's third criterion, because in the post-accident contest over the microphone, "systemic, inevitable, unpredictable" is always the stronger side's rhetoric. Neither Perrow nor Taleb set out to supply ammunition for disclaimers, and it gets used that way regardless. The mechanism layer's response is

to pull the discussion back to checkable objects: which document, which alarm, which signature.

41.5 Forensics for a Grey Rhino

Deciding whether a "black swan" is really a grey rhino takes an executable procedure of four steps.

Step one: search other libraries. Take the mechanism keywords, not the event name, and search the incident databases of other industries and other countries. Nuclear events go to IRS, aviation to NTSB and its overseas counterparts, chemical to CSB, maritime to MAIB. If the mechanism has appeared in another industry, your industry's "unpredictable" is immediately downgraded to "not searched". The search costs an hour, and that hour is the best-value spending in the whole procedure.

Step two: check internal priors. Look through the ledger, the anomaly reports and the internal discussions for a record of a similar warning, even one that survives as a joke or a complaint. Chapter 2's record skew works in reverse here: a record that reads as a joke afterwards was information beforehand. If a prior exists, the label drops to "measurable but not preventable" or lower.

Step three: interview experts ante hoc. Did anyone in the industry publish a close warning before the event, in a conference, a paper, a trade newsletter. If someone did and the warnings are retrievable, the "nobody foresaw this" rhetoric is bankrupt. Guard against the reverse trap at this step: a prophet can always be found afterwards, and finding one does not make the prophecy actionable. What counts is that a warning existed and that this organization sat inside its circulation radius. A warning ignored inside the radius is the hard evidence of a grey rhino.

Step four: run a minimum counterfactual stress test. Invent one by force: if a specific action had been taken beforehand, would the probability or the magnitude have changed significantly. Even a strained yes gives the mechanism layer a foothold. Only when all four steps come back empty does the case go into the true-unknown register.

The output is a label change form: what the label changes from, which evidence from which of the four steps, who verified it, and the date. The form itself enters the ledger. It settles accounts with the rhetoric and it reminds the next person who reaches for the label.

Fukushima sits in most nuclear practitioners' memory. The National Diet of Japan's independent commission (NAIIC, 2012) called the accident "a manmade disaster" that "could and should have been foreseen and prevented", and studies of tsunami height together with internal calculations both existed before 2011. A failure that the world's media processed as a natural and unforeseeable disaster changes label to grey rhino in front of four steps of forensics. It is the best teaching sample for the form.

41.6 A Humility Procedure for True Unknowns

The self-limitation in 41.3 needs a procedure someone can actually walk. A posture is not enough. Cases where the criteria come back empty and the label is "true unknown" go through three steps.

Step one: register. The entry goes into the humility zone of the appendix matrix, in five columns: the event, why all four forensic steps came back empty, the response cost at the time, which design decisions changed as a result, and the review date. The third column matters most. True unknown does not mean zero cost, and

registering the cost is what calibrates the organization's budget for tail risk.

Step two: replace prevention with resilience. The governance resource for that risk moves from the prevention budget to the response budget: redundancy, isolation, insurance, drills. The move leaves a trace in the form of a budget line change, which guards against the habit of wanting both, since a tail prevention budget usually produces a feeling of safety rather than prevention.

Step three: review on a schedule. True unknown is not a lifetime label. Industry knowledge grows through new incidents elsewhere, new research, new monitoring, and every two years the humility zone goes back through the four steps. Any new evidence and the entry migrates out into the mechanism layer. Migration is itself a registration event. How many entries an organization redeems from "unknowable" each year is a real measurement of its cognitive boundary.

The common meaning of the three steps: humility is not a disclaimer. Humility is a posture with a ledger. Claiming unknowability is free; registering it costs money, in cost registration, budget migration and a review obligation. Charge that fee and the count of true-unknown entries returns to the scarcity it deserves.

The triage itself can be made executable, with an owner and a rhythm.

The sequence: a risk item, arriving from the risk register, the ledger, the deaths named in a premortem, or the migration candidates in the humility zone, first passes the mechanism layer's four questions from 41.3. Can the chain be taken apart, was the information visible at the time, can the structure be changed, can a minimum counterfactual be written. One yes sends it to class one, preventable and measurable, tagged with a mechanism chapter and run through the twenty-four-chapter tools. No yes at all, but a

hit somewhere in the four forensic steps, sends it to class two, measurable but not preventable, with emergency preparation attached: a response plan, a drill date, a redundancy line. No yes and empty forensics sends it to class three and the humility procedure. The output is one class label and one of three disposition paths, one label per cell, and double labels are banned. A double label is the first symptom of triage failure: an item described as both preventable and survivable spreads resources across both and does neither.

The owner: the risk steward makes the first call, the same steward who keeps the ledger and the question library, and governance reviews a ten percent sample each quarter. The sample exists to check label inflation. Class one stuffed into class three is a disclaimer move, and class three stuffed into class one is overconfidence. Disputed items take the same channel as the grey-zone precedent library in Chapter 40, 40.6, and consistency grows on its own as precedents accumulate.

The portfolio view: once triage is done, count a share for each class, and watch two quarterly trends. A steadily falling share for class one means the mechanism layer is losing ground and the tools are going unused. A steadily rising share for class three means the unknown zone is expanding, either because the business moved into a new field or because triage has given up. Each class carries its own budget line, prevention, emergency, resilience, and moves between lines leave traces. Governance reads the portfolio view once a month.

This triage replaces the chapter's theoretical vocabulary, inevitability and unforeseeability, with operational label words. At that point the theory has done its job: it supplied the coordinate system, and every step after the label is governance action.

Three comparison groups show why the body cases hold up under pressure from both theories.

Group one: events outsiders call chance. The jackscrew failure on Alaska 261 (av-034) drew the phrase “metal fatigue is unforeseeable” in post-accident public discussion, and the report’s executive summary dismantles it in a line, since extending the lubrication and inspection intervals had compressed the window in which the crack was visible (NTSB AAR-02/01). The chain can be taken apart, the information was visible, the structure could have been changed by not extending the interval. All three questions pass, and the case is class one. The same cleanup applies to most of the library. What outside rhetoric calls chance has a name and a page number in the report.

Group two: the fourth-quadrant cases. Three Mile Island style events, mi-001 and its class, sit in the zone of high complexity and tight coupling. Notice, though, how they enter the body of the book. The body always discusses the nodes that can be taken apart, the review that wrote the flight rationale into the ledger, the per-flight foam table, and never claims that the whole event was preventable. The honest mechanism-layer treatment of a fourth-quadrant case is to take it apart as far as it goes and hand the remainder to resilience, which 41.3 states in writing. A reader comparing the two will notice that the book’s humility is local, conceding that certain nodes will not come apart, while the rhetorical humility is global, claiming the whole event was unknowable. An evidence river runs between them.

Group three: positive tail samples. A few library cases carry a tail lesson as their main point, low-probability high-magnitude combinations, and their governance conclusions all land in classes two and three: redundancy design, isolation partitions, emergency drills. The mechanism layer’s contribution to those cases is to take the tail apart, naming which combination and which chain it depends on. A tail taken apart can at least be rehearsed. Rehearsal

cannot make it preventable. It does stop the response from being improvised.

The common reading of the three groups: boundary questions about theory turn, at case level, into an act of assignment, this node to the mechanism, that node to resilience. The finer the assignment, the harder it is for either theory to serve as a shield for ending the conversation.

The operational detail behind the anti-narrative stance in 41.4 deserves expansion, because it is where this chapter's position merges with the declaration system of Chapter 40. Three cards get played in the contest over the microphone after an accident. The inevitability card: the system is like this, Perrow said so. The unknowability card: black swan, Taleb said so. The wholeness card: it was a local problem, fix it and we are fine. All three bypass checkable objects. The response is not a lesson in debate technique. It pins the discussion physically to three classes of object: pre-event documents (declaration, response plan, approval chain), pre-event signals (ledger, alarms, inspection records), and pre-event structure (signatures, intervals, the margins as they stood).

A three-column table does the pinning. The left column carries the narrative assertion, "unforeseeable". The middle column carries the retrieval result for the matching object, "ledger entry 2023-04; inspection interval extension approved 2021". The right column carries the label change, per 41.5. Paste the table into the investigation minutes and the cost of moving the rhetoric becomes uneconomic. The freedom to tell a story remains, but every story now faces a column of retrieval results.

The inevitability card is handled a little differently. Sometimes it is true, in the fourth quadrant. The right column then carries a triage result and resilience entries rather than a rebuttal. Claiming necessity is cheaper than contesting it, and once it is claimed the governance agenda shifts from whose fault this was to the partition

plan and the drill calendar. The narrative fight turns into construction.

One line on where finance and insurance sit across the three classes. Class one needs no insurance, since mechanism tools are cheaper and create no moral hazard. Class two can transfer magnitude, using emergency preparation plus a policy. Class three cannot be priced by the insurance market, because tail pricing depends on historical samples, which is exactly what the black swan definition denies, so class three risk capital is self-retained as a buffer under 41.7. Each class gets its own instrument, and mixing them, buying insurance on class one for convenience, raises cost and slackens defence at the same time.

41.8 Objections and Responses

The position in this chapter will meet four objections. Each gets a response.

Objection one: the mechanism layer suffers from survivorship bias. You only postmortem the failures that happened, and the ones you prevented are invisible. Response: the prevented are indeed invisible, which is why Chapter 38 sets up the success postmortem, for the minimum defence that was validated, and why Chapter 39's hit-rate statistics test rehearsed deaths against reality. The method is self-aware about this. It does not claim to see everything. It claims to see the portion that was recorded, and it marks how clearly it sees with evidence grades. The survivorship accusation lands on the induction enthusiast who derives general laws from cases. The mechanism layer's method is criterion first, where the chain must come apart before a case enters the body, and a criterion can be falsified.

Objection two: four steps of forensics are not feasible for a small organization with no search capacity and no expert network. Re-

sponse: restate the cost structure in terms of the resources actually available. Cross-library search is free at the level of public investigation reports, which every national agency publishes. The internal priors check draws on your own ledger. Expert interviews can be replaced by trade press archives. What is genuinely infeasible is claiming a black swan without doing forensics, and that is unaffordable at any scale.

Objection three: mechanizing every failure will kill legitimate criticism of systemic defects, and some systems should not exist. Response: that mixes layers. The mechanism layer handles how to kill fewer people inside a system that exists. Perrow's original question, whether the system should exist, belongs to the governance layer, and the two questions run in parallel. Because the mechanism layer takes failure apart in detail, the governance layer gets the material for a system-level verdict. The fourth quadrant of the 41.1 table lists shrinking and exiting as legitimate agenda items.

Objection four: you say no body case has turned out to be a true unknown. Is the criterion too loose, since anything can be taken apart a little. Response: readers can test the tightness themselves. Pull a case card from any chapter and try to break its minimum counterfactual by finding a path it does not defend. A successful break should be marked for return, which is exactly the design of the appendix's humility zone. It sits empty waiting for readers to put back the cases that would not come apart.

41.7 The Four Pieces of Resilience

The countermeasure for class three is resilience, and resilience that is not broken into pieces becomes another piece of rhetoric. Four pieces, each with its own ledger.

Piece one: buffers. Time buffer, the spare days on the critical path. Capacity buffer, the headroom at peak. Financial buffer, months of

cash. The buffer ledger keeps three columns: design value, current value, date last drawn on. The difference from margin erosion in Chapter 25 is bookkeeping. Margin is design headroom, an engineering quantity. Buffer is operating reserve, a management quantity. Both share the same enemy, the temporary borrow that is never returned, and the third column is the borrowing tracker.

Piece two: diverse redundancy. Structurally identical to Chapter 22, diverse and actually exercised. The resilience addition is the degradation ladder for switching: full speed, half speed, read only, safe shutdown, each level with its trigger condition and its recovery action. A system with no ladder has two speeds under extreme conditions, full and stopped, and the cost of stopped is routinely underestimated.

Piece three: intertemporal responsibility. Chapter 28's device means one thing here. The maintenance of response capacity needs a carrier that outlives a tenure, a drill calendar, an on-call roster, an annual review of the response plan, and it may not depend on the memory of any one cohort. The evidence that the carrier exists is a calendar and attendance sheets, not a document.

Piece four: the knowledge pack. Chapter 30's pack means, in a resilience setting, drawings in the dark: what a responder still has to read when the normal systems are down, whether paper plans, offline backups, or the oral history of long-serving staff. The annual test is a real darkness drill. Shut the normal information systems for two hours and see whether the response holds on what is in the pack. The result is almost always ugly, and the ugliness is the output, since the missing items produce a list on the spot.

The check runs quarterly, aligned with the triage, and the output is a four-line scorecard. Resilience now has a physical examination, and the rhetoric has turned into a metric. One use of the scorecard gets missed: the four scores are read aloud at the governance meeting, with the well-maintained and the badly maintained teams

in the same room. Publicity sets the psychological cost of borrowing a buffer, so that approval turns from signing a line into explaining in public, and the defence stands in that rise and fall. Four scores plus the three class shares fill the governance layer's quarterly page. That page is this chapter's whole coordinate system folded into a management interface, and the weight of the theory ends up on the desk as a few numbers.

A note on where the term grey rhino comes from, for retrieval purposes. It entered the risk literature in the writing of Michele Wucker around 2013, meaning a high-probability, high-impact threat that is looked away from. It is kin to Taleb's grey swan discussion and to the experimental tradition in cognitive psychology around knowing something and acting as if you did not. Word discipline follows: grey rhino stresses visibility and inaction, black swan stresses distribution and the limits of cognition, and the two words have long contaminated each other in public discussion. The four-step forensics of 41.5 is the scalpel that redraws the line between them.

Five Self-Checks

1. In your industry, which system would Perrow place in the accident-prone zone? What is the standard of accident recording there?
2. The last time you heard "that is a black swan", did the post-mortem produce a minimum counterfactual? If it did, what was the label then used for?
3. Does your organization's risk management ask "how will this one die" or "what combination kills me"?
4. Does your organization's most tightly coupled system, if it has one, carry aviation-style mandatory reporting and independent

postmortems? If not, is Perrow's prediction working better or worse for you?

5. Write the minimum counterfactual for your organization's most recent failure. If you cannot, does it belong to the mechanism layer or to the true unknowns?

After the five questions, one note on using the chapter. This is the most bookish chapter in the six parts, and most of what it gives are coordinates rather than tools. Coordinates pay off the next time you hear "inherent property" or "unforeseeable", in whether you can pull the discussion back to the three-column table inside thirty seconds. Pull it back and the chapter has earned its fifty minutes. Fail to pull it back and the other side is holding power rather than theory, and that fight is fought in Chapters 16 and 19: send the objects out, through the skip-level channel or the external search results, so the signal goes around the direct opponent of the rhetoric.

Closing. Perrow and Taleb are this book's two most important neighbours. They drew the terrain outside the boundary, which is what lets the mechanism layer draw the streets inside it with confidence. Normal accident theory reminds the mechanism layer that not every chain can be taken apart, and that the street ends in the fourth quadrant at the governance agenda. Black swan theory reminds it that the street map has a cognitive boundary and the humility zone needs blank pages kept in it. Both reminders are now inside procedures, the quadrant self-assessment, the four-step forensics, the three-step humility procedure and the three-column table. Theory has stopped being the end of a conversation. It has become labels, ledgers and budget lines, and the line numbers move every quarter, reminding everyone that argument at the boundary deserves respect and the streets inside it need someone sweeping every day. The sweeper has a form in hand, numbers on the form, and sources behind the numbers. That is what this

chapter hopes the reader carries out. It is how the book treats every large theory: respect it, then take it apart and file the pieces.

Beyond the five questions, a two-hour exercise. Fold the diagnostic at the end of 41.1, the label change form of 41.5 and the resilience scorecard of 41.7 onto one sheet. Column one lists the organiza - tion’ s first five key systems, each with a quadrant. Column two registers the existing claims of black swandom, each run through four steps. Column three scores the four resilience pieces. Two hours, with materials already at hand, and your use of the chapter turns from reading into triage, which is the entrance to the 41.6 flow. The entrance runs on your sheet. The exit is on the governance budget table.

Chapter 42 When This Does Not Work

> What this chapter covers: the three ceilings on the book's method (its own admission about record skew, the residue of hindsight attribution, and failure that cannot be prevented), plus four ways the tools get pointed in the wrong direction. When to turn back: when you notice the mechanism vocabulary being used as a charge sheet, when defences have piled up until they fight each other, or when a premortem has turned into a form. Estimated reading time: 30 minutes.

The book ends where it started, honestly: with the conditions under which its tools stop working. Listing conditions is not modesty. It gives the reader an accurate expectation, and knowing a tool's boundary matters more than knowing its function. Confidence that runs past the boundary is narrative before evidence (Chapter 13) at the level of method.

Condition one: no governance authority. Chapter 17 said that misaligned incentives can only be fixed by someone who holds governance authority. If you sit in an organization where authority is locked, the best scheme in the world will not clear approval. This chapter cannot help you, and this book cannot help anyone change what they have no authority to change. Two moves remain: use the tools to the full inside your own radius, and record what you find outside it and wait for a window.

Condition two: a culture in refusal. Chapter 19 covers it. If your organization is punishing people who speak plainly, then every ledger, checklist and signature this book recommends is a target painted on your back. Your priority is not building a system. It is protecting yourself: write the knowledge pack of Chapter 30 for yourself, and use the external channels of Chapter 19, regulator reporting, industry circles. Keep the hills green.

Condition three: extreme uncertainty. This is Chapter 41's true black swan. Every tool in the book rests on patterns from the past, and genuine novelty has no pattern to look up. The only thing left is resilience: redundancy, diversity, fast adaptation. None of that is the core of this book, but honesty requires saying that here your tools are finished.

Not working does not mean the reading was wasted. Someone who knows the boundary has one more layer of judgment than someone who does not, and judging which condition you are standing in is the first capability the book teaches, positioning (Chapter 35). A reader still using the tools after the boundary chapters is a reader who understood them. Such a person does not force the tools where they do not belong, and does not flinch from using them where they do.

The position of this chapter inside the whole should be clear. In the six-part structure it sits at the centre of Part Six. Chapter 40 draws the boundary of permitted failure. Chapter 41 draws the boundary of theoretical explanation. This chapter draws the boundary of the method itself. The three boundary chapters together form the book's pressure relief valve. A book that only takes air in will burst, and a tool book that teaches tools without teaching boundaries manufactures the very confidence it claims to prevent. A failure that happens with tools in hand and no boundary in hand is harder to postmortem than a failure without tools, because it arrives with complete paperwork saying "we followed the chapter". What the reader should carry out of this chapter is a brake and a dotted line on a map: beyond the dotted line, the book no longer counts.

The first ceiling gets one estimation method, for sizing the unrecorded zone. Three steps.

Step one, adjacency scoring. Score your own industry against the five precedent domains (Chapter 3) item by item: regulatory dens-

ity, investigation regime, reporting culture, litigation intensity, industry age. The mean of the five is your adjacency.

Step two, the discount rate. Each step down in adjacency deepens the coverage discount the mechanism library suffers in your industry. A rough starting rule: for every step down across the five grades, cut your confidence in direct citation by twenty percent.

Step three, local stocking. In the mechanism domain where the discount runs deepest, build local cases from a small sample in your own ledger, three cases to start, in the Chapter 5 card format. Reading three local cases beside the book's cases is more honest than an analogy stretched across a large discount.

The output is a mechanism coverage map for your industry: which chapters can be read directly, which need a discount, which need local samples. Hang it on the team wiki and cite it in the reading route you hand a new hire in their first week.

The second ceiling gets operating rules, because the mechanism label is an after-the-fact coordinate system and single tagging's arbitrariness needs diluting. Three rules.

Rule one: the mechanism field in a body postmortem may carry two mechanisms, a primary and a secondary, and the secondary must hit at least one criterion.

Rule two: cross-author sampling. Each quarter, pull two postmortems and have someone else who has read the mechanism chapters tag them independently. The disagreement rate is a health metric for postmortem quality. Too high means the criteria are idling. Too low means the tagging is going through the motions. A healthy rate runs between twenty and forty percent.

Rule three: disagreement is handled by co-writing rather than by adjudication. Both primary mechanisms enter the ledger entry, and six months later you look at which one explains the improve-

ment actions that followed. Double tagging turns a precision problem in the coordinate system into a knowledge problem, since the multi-layer structure of the event shows up in the tension between the two labels.

The third ceiling gets a ledger format for a decision not to prevent, five columns: the object and the risk, the estimation basis for expected loss, the prevention cost and operating friction, the reason for the decision, and the review date. The fifth column is the heart of the format, because a decision not to prevent still gets a review: if the loss did happen, was the arithmetic right at the time. Once the format is in use, the organization has a chain of evidence for rational inaction for the first time. When audit or investigation meets a decision not to prevent, it meets an account that was calculated and recorded, and the difference between that and negligence sits on one page.

42.1 The Three Ceilings

Ceiling one: record skew is the foundation defect. The cases in this book come from domains that have investigation regimes, so the sample passed through the triple filter of Chapter 2. Two consequences have to be stated plainly.

One: the mechanism library covers only recorded failure. When you carry it across industries you must estimate the size of the unrecorded zone yourself. The further your industry sits from the five precedent domains of Chapter 3, the lower the coverage, and the bigger the discount you must apply when arguing from an adjacent industry.

Two: "we have never heard of that mechanism in our industry" carries two possible explanations. It does not exist, or it is not recorded. The book leaves the first judgment to you and takes re-

sponsibility only for reminding you that the second explanation is always in the room.

Ceiling two: the residue of hindsight attribution is never zero. The discipline of Chapter 6 pushes the retrospective view down. It does not reach zero. The mechanism label is itself a retrospective act: the mechanism field on a case card was assigned by an author who already knew the outcome, and a different author might assign a different primary mechanism to the same event. That is not a de- fect being covered up. It is a statement of the tool's precision. The mechanism is a coordinate system for organizational knowledge, not a physical property of the event. Cases where two authors as- sign different primary mechanisms, Vaughan's organizational attribution sitting beside a committee's physical attribution (Chapter 2), show the coordinate system doing its job: the dis- agreement exposes the multi-layer structure of the event (Chapter 9) rather than a failure of the coordinates.

Ceiling three: some failures cannot be prevented, and some are not worth preventing. The first kind is governed by Chapter 41's cri- teria: a chain with no nameable intervention point gets marked true unknown and enters the matrix. The second kind gets dis- cussed less and needs adding here. Prevention has a cost, and prevention whose cost exceeds expected loss plus operating fric- tion is a negative-value investment. For an operation running a million cycles a year, adding a two-person check against a small loss that arrives once a decade may not balance. The book does not supply the calculation table, because expected values differ by in- dustry, but it supplies the discipline: a decision not to prevent also enters the ledger, stating the comparison between expected loss and prevention cost. A decision to abandon prevention and a de- cision to prevent are the same kind of decision, and both leave a trace.

The relation among the three ceilings matters too. They reinforce each other rather than standing as three parallel walls. Record skew (ceiling one) feeds the residue of hindsight attribution (ceiling two), since a filtered sample makes tagging more arbitrary, and together they compress the reliability of the prevention judgment, which is ceiling three: nobody knows how many items on the "not worth preventing" list were actually worth preventing and merely invisible to the record. The common upstream of all three is one fact. Knowledge of failure is scarce by nature, and any method built on scarce knowledge carries the damp of its foundation. The book's entire handling of that foundation comes down to three words that recur: local stocking, evidence grades, ledger traces. None of them cures scarcity. All of them make scarcity visible, and visible scarcity can be priced, and priced scarcity drives stocking, and stocking is the only baton passed between three generations of users.

The fix for the second way the tools go wrong needs its own detail: the exit tax on over-engineered defence. Give a new device a mandatory accounting rule. Each device added must be logged together with the old device or process step it displaces: whose time it consumes, whose check it routes around. A device that cannot produce that entry is frozen for a quarter before it ships. The exit tax guards against installing without uninstalling, and a device inventory needs pruning the way a codebase does. At the annual audit, ask which device did not fire once last year and was never cited, and what the maintenance cost of a silent device on the process is. Keep a small ceremony for retiring one: three lines of retirement note (the path it guarded, what replaced it, who watches that path now), filed into the precedent library. Installing leaves a trace and uninstalling leaves a trace, which is the full meaning of defence budget and risk budget being one book.

42.2 Four Ways the Tools Go Wrong

Way one: the mechanism as a noun. The twenty-four mechanisms get used as a charge sheet, and the postmortem turns into a la - belling contest: attach a word, end the discussion. The vocabulary was designed as a retrieval coordinate, taking you from a situation to a chapter. It was never designed as the endpoint of attribution. Identifier: once the mechanism is attached, the meeting produces no action and no check person. The fix returns to Chapter 36. The mechanism is a way station. The destination is an action.

Way two: defence over-engineering. Readers who finish Part Three tend to catch one disease: every mechanism gets a defence, a checklist on top of a checklist, a node on top of an approval chain. Defence is itself part of the system, and it creates new coupling (Chapter 21) and new compliant-but-unsafe conditions (Chapter 32), while every skipped defence node becomes a fresh drift point (Chapter 15). The correct practice is the discipline of Chapter 36: one action per death, with the ledger and the postmortem as the backstop. Defence budget and risk budget are one book.

Way three: the postmortem as ritual. Chapter 39 listed the third failure form of a premortem, and postmortems catch the same disease: template complete, fields complete, nobody reads it again. The diagnostic is a single question. In the past twelve months, was any minimum counterfactual cited into a process or into training. If none was, the postmortem is already ritual. The fix is not to stop holding them. It is to connect the output to two rigid interfaces: improvement actions into the ledger (Chapter 8, Device 1), and deaths into the comparison column of the next premortem (the session tracking in Chapter 39).

Way four: the position table lying to itself. Chapter 35’ s table gets used to prove “we have already thought about it” , with all six grades filled at the middle grade and the evidence column full of

boilerplate. The table’s value is entirely the truth of the evidence column, and the person filling it should expect their manager to sample that evidence, which is the detector for the first failure use in Chapter 36. The deeper self-deception treats the walkthrough itself as proof of diligence. The thirty-minute procedure raises the quality of judgment. Responsibility for judgment stays with the judge, and no tool signs a release on your behalf.

The common pathology behind the four deserves naming: the tool replacing judgment. All four share one deep move. The user hands the responsibility for judgment over to the tool, and under pressure a tool will always grow a pathology: the vocabulary becomes a charge sheet, the checklist becomes a mountain of checklists, the postmortem becomes a rite, the position table becomes an amulet. The psychological mechanism matches Chapter 12. A tool that can be assessed, recorded and displayed gets optimized into its proxy action, and the optimization runs until proxy and purpose come apart. Three devices against this are already buried in this chapter, in principles one to three, and they gather into one line of practice. The tool makes your question accurate. The answer always grows at the junction of evidence and judgment. Once the hand filling forms cannot stop, come back and read this chapter again.

Under the three ceilings there is still a large remaining space, and it deserves a positive statement. The sweet spot of these tools is far wider than the boundary. The sweet spot has two criteria: the failure chain lives inside human institutions, meaning there are people, processes, appraisals and records, and you hold authority inside a radius, even a radius as small as your own checklist. Meet both and the twenty-four mechanisms, the five usage tools and the three boundary criteria all sit within working distance.

Readers outside the sweet spot, the ones without authority, the ones inside a culture of refusal, the ones facing a true unknown, have their march routes in the first three paragraphs of this

chapter: use the radius fully, leave traces outside the window, keep the hills green. None of the three routes is surrender. Using the radius fully builds the showroom, and when the window opens others can see it. Leaving traces is ammunition stored for your future self. Protecting yourself keeps one alarm in the organization's signalling system that still rings. The one thing the boundary chapters refuse to hand over is a reason for despair.

Each of the three conditions gets a manual, one page, all of it executable.

Manual for condition one, no authority.

Radius inventory: list everything you can actually move, your own checklist, your own team's agenda, your own signature authority, and confirm each one. Most people underestimate their radius.

Showroom construction: pick the most painful mechanism inside the radius and fit it out completely, ledger, postmortem, question review, to the point where you have data to show.

External record: findings outside the radius get written up in the upward format of Chapter 16, one a month, written even when no reply comes, because continuity of record is itself credit when a window opens.

Window monitoring: three kinds of window exist, a change of personnel in the governance layer, the aftermath of a major incident announcement, and the arrival of an external audit. The showroom and the reports go in together at that moment.

The core of this manual is patience and compounding. A missing governance authority will not come back because you are in a hurry. A showroom will.

Manual for condition two, a culture in refusal.

The priority order is cold and worth repeating. Safety first: the bottom-line actions for personal and professional safety, not one of

them conceded. Self-protection second: the knowledge pack written for yourself, the judgments you handled, the evidence, the dissents, all dated, kept in a private copy. Channels third: external channels built while you do not need them, industry circles, regulatory routes, legal advice.

The one piece worth expanding is the courage not to sign. Refusing to sign a record you know to be wrong is the most expensive and the most valuable action available in a culture of refusal, and its protection comes from the knowledge pack prepared in advance: with a complete chain of evidence, the refusal has standing.

The last line of the manual is written for a future window. Cultures do change, and the person who left true statements in the ledger is the veteran the new culture needs most.

Manual for condition three, extreme uncertainty.

Tool switch: prevention tools off, the four pieces of resilience fully on, buffers, diverse redundancy, intertemporal responsibility, knowledge pack, each scored on the Chapter 41 scorecard.

Exposure control: split the exposure by affordable loss. Fix how much you can lose before discussing how to try.

Adaptation loop: compress the feedback cycle. Weekly post-mortems, double the rhythm of the quick funeral, since the one certainty under uncertainty is the value of iteration speed.

Psychological budget: set a tuition cap for the worst case, stop at the line, and write the line down on paper.

The common principle: where certainty is low, the only structural advantages left are learning speed and survivability, and both run on the four items above.

The shared ground of the three manuals: the boundary chapters do not turn into philosophy. They always give the next action. Philosophy is elsewhere. The ledger is here.

Two refinements to the fixes for ways one and three. For the mechanism-as-noun problem, add an action column to the white‑board in the postmortem room, and make it the facilitator's rule that every mechanism label is followed by one question: which device in section 5 of that chapter does this point to. A label that cannot name a device does not reach the conclusion section. That rule pulls the vocabulary back from the endpoint to the way sta‑tion, since the only lawful exit from a label is a device. For the ritual problem, install a simple counter for the adoption rate: the number of minimum counterfactuals cited into a process or a training over twelve months, which the ledger can produce auto‑matically, read aloud once a quarter at the postmortem meeting. A postmortem process that reads zero for two consecutive quarters goes into remediation under Chapter 38's governance rules, working through the rigid interface list of 38.5 item by item. Both refinements share one point: fixes do not restate attitudes, they give the process a countable exit.

Principle four joins the first three: keep a record of your own de‑viations. Every time you use these tools and deviate, a field skipped, a stage compressed, a discipline forgotten, it earns its own line, reviewed quarterly. The record has two uses. It is the differ‑ence list between your version and the standard version, and the differences hold the particularity of your situation, which is exactly what the standard does not cover and you must stock locally. It is also raw data for improving the tool: forty-two devices deform when they reach your site, and the shape of the deformation is design input for the next version. Deviation is not shameful. Invisible deviation is.

A short set of questions users actually ask.

Too many tools, which one first. Install in ascending order of res‑istance: the postmortem rewrite, which needs no approval; the three-state ledger, which is one table; the question review, which

borrows one meeting; the position table, which goes on the quarterly agenda; the premortem, which needs people and authority.

Six months in and no effect, what now. Check three counts: ledger entries, postmortem adoption rate, premortem hit rate. All three climbing means it is working, since effects show up in the records first and in the accident rate later, and that order cannot be reversed. All three flat and you should go back to 42.2 and look for the four diseases.

How does this coexist with our ISO or Six Sigma programme. These tools govern the seams of those systems, between checklist items and at process interfaces, and they do not replace them. Chapter 32 covered the arrangement: the compliance system governs items, the mechanism tools govern between items.

What is the biggest risk in the book. Being shelved after reading, which is the fate of any reading material, or being installed and deformed, which is the normal state of any device. Both are normal, and the deviation record plus the quarterly review is how you live alongside both.

Will this make me unpopular. Yes, in the short term, with a fair probability. Ledgers, questions and check persons all demand payment from inertia and from muddling through, and the person demanding payment usually collects rolled eyes before the first instalment. Three protections exist: the knowledge pack in advance, so your chain of evidence is always complete; allies outside the matter, the facilitator pool of Chapter 39 and the check-person candidates of Chapter 36; and patience afterwards, the compounding of the Chapter 40 showroom. With all three in place, popularity stops depending on your personality and depends on whose records hold up when the next accident arrives. People whose records hold up have always been the most welcome people in any organization over the long run. The loop is slow and the

direction is right: those who walk it lose for three years, and after the third the road widens, because every organization, once it hurts, goes looking for the person holding a ledger, and a ledger never betrays its holder. It remembers every loss you saved the organization, and every warning you gave that went ignored.

After the map and before closing the book, one last note on use. The density of the tools across the book is deliberately uneven, and rereading has a different method from first reading. A first reading seeks completeness, meaning what is there. A rereading seeks the gap, meaning what you lack.

The gap method: take three real documents from your organization, a postmortem, a checklist, an appraisal form, walk the table of contents, and ask of each chapter whether the device casts a shadow in your document. Chapters with a shadow get skipped. Chapters without one become next quarter's reading list. This compresses a thousand-page book into a personal gap list, and the length of that list is your distance from the book. The distance is good news, since it can be measured and shortened.

Merge the rereading rhythm with the audit rhythm of principle three. On the day of the annual audit, reread the contents by the gap method and produce next year's gap list. Count once a year and the tools do not rust, and a rusted tool behaves exactly like an uninstalled one when a failure arrives.

42.5 The Map of the Book

Before the cover closes, the map gets drawn once more so the reader can see the road taken.

Part One, the meta-questions, answers why failure deserves systematic study: record skew makes the knowledge of failure scarce (Chapter 2), knowledge inflation lets success narratives crowd out

memory (Chapter 4), and failure is isomorphic across industries (Chapters 1 and 3). Part Two, the anatomy, supplies the language for the whole toolset: the twelve fields of the case card (Chapter 5), the three disciplines (Chapter 6), points of no return and the cooling-off period (Chapter 7), the escalation ladder (Chapter 8), the three-layer failure chain (Chapter 9), the minimum counterfactual (Chapter 10). Part Three, the twenty-four mechanisms, is the main hall: three criteria, four domain cases, eight weak signals and four devices in every chapter. Part Four, the fourteen domain pages, puts the mechanisms back into the accent of each industry. Part Five, the five usage chapters, turns the book into a toolbox: positioning, walkthrough, questions, postmortem, premortem. Part Six, the three boundary chapters, draws the boundary of permitted failure, of theory, and of method.

The six parts share a single thread: translate a vague unease into a statement with a source and an action with a date. Every device, every table and every question in the book is a variant of that translation rule. If the reader may carry away one sentence, carry this one. Unease is data, translation is method, and the ledger is home.

Two cross-route combinations are worth adding, because real readers often hold compound identities.

Combination A, manager and engineer at once. Walk your main route, then borrow the other route's first action once. A middle manager on route two also runs route one's checklist diff; an engineer on route one also runs route two's upward comparison. The largest risk of a double identity is seeing failure only through the lens of your own post: the cross-route first action costs an hour and buys the view from the other bank.

Combination B, a veteran who runs a team plus a newcomer. Each walks their own route, the veteran the governance line, the newcomer the front line, and two months later they exchange answers

to the five self-checks. The output is a first draft of the team's mechanism map, since the reefs the veteran knows and the shallows the newcomer sees combine into one drawing for the first time when the two answer sheets lie side by side. Chapter 33's newcomer audit and Chapter 6's exchange test hold their last reunion here.

Both combinations rest on the same principle as the rest of the book: a failure map is never complete, and one more lens removes one more blank. Combining adds no reading volume, only order, at near-zero cost.

42.4 Five Readers, Five Routes

The book has more than one kind of reader. Five routes, each with a starting chapter and a first action.

Route one, the front-line engineer. Start with Chapter 15 (practical drift) and Chapter 25 (margin erosion), the two chapters whose phenomena sit closest to the line. First action: diff your own checklist against the manual, the single-person version of Device 1 in Chapter 15, and produce three differences. Then continue with Chapter 36 (the personal walkthrough) and Chapter 11 (daily use of the three-state ledger).

Route two, the middle manager. Start with Chapter 16 (information attenuation) and Chapter 30 (handover rupture), since the middle is the epicentre of both. First action: read three of your upward reports beside the raw material that went down, and mark what each layer deleted. Then continue with Chapter 35 (position table), Chapter 37 (question review) and Chapter 38 (postmortem rewrite), three tools that cover the meetings a middle manager sits in every week.

Route three, the executive and the governance layer. Start with Chapter 17 (misaligned incentives) and Chapter 28 (horizon mismatch), since the levers in both sit only in senior hands. First action: map the clauses of your current appraisal form against Chapter 17's criteria, clause by clause. Then continue with Chapter 39 (resources and facilitation authority for the premortem), Chapter 41 (three-class triage and the resilience scorecard) and Chapter 20 (protection of defensive investment).

Route four, the regulator and the auditor. Start with Chapter 31 (regulatory lag and capture) and Chapter 32 (compliant and unsafe), the back of the mirror. First action: map your institution's checklist against Chapter 32's question set, asking who owns the gaps between items. Then continue with Chapter 37 (external use of the question library) and Chapter 2 (record skew, so you can read how the material you receive was filtered).

Route five, the student and the researcher. Start with Chapter 2 (record skew) and Chapter 5 (case card methodology) to establish the evidence discipline first. First action: take any case card and do an independent evidence-grade review, scoring item by item against the Chapter 5 grade table. Then continue with any three chapters of Part Three read closely, plus Chapter 6 (the three disciplines), since the first lesson of research is rebuilding the scene rather than citing the conclusion.

All five routes end in the same place: the ledger. Whichever identity you enter by, the tools come to rest in your organization's record system. Finish the book with no record improved and it was reading material. One record changed because of it and it is a device.

42.3 The User’s Principles

Principle one: tools serve judgment, and judgment cannot be out-sourced. Every table, checklist and script in this book exists to or-ganize attention you already have onto the right question. None of them answers for you. The most valuable voice in a review is still the person with specific evidence saying “I disagree”, not the person with the most complete form.

Principle two: keep one question outside the form. Each time you finish using the tools, keep one doubt that no field covered. Write it down and carry it into the next use. The boundary of a tool, which is this whole chapter, is patrolled by these questions without co-ordinates.

Principle three: audit the method itself. This method also fails, in the form of blind spots in the mechanism library, side effects of defences, and ritualized postmortems. Set a rhythm, once a year for instance, and look back over how your organization has used the tools: which deaths the premortems named and how many hit, how many postmortem counterfactuals were adopted, whether any mechanism the position table screened out was later confirmed. The method needs a ledger of its own, which is the last and most self-proving piece of advice in the book.

The common spirit of the three principles collects into one word: operationalized humility. In most management writing, humility is a tone. In this chapter it is a record format. Principle one writes final responsibility for judgment back onto the person, in the format “the tool does not sign a release”. Principle two gives questions without coordinates a fixed place to live, in the format of one uncovered doubt per line. Principle three opens a ledger sub-ject for the method’s own failures, in the format of three review numbers at the annual audit. Tone blows away. Formats stay. Three years on, whether a user understood the book is measured more

reliably by whether those three formats sit in the toolbox than by any reflection the user offers about it. This is also the graduation standard the six parts share: Part Three tests whether you can find the mechanism, Part Five whether you can install the device, and this chapter whether you can say where the device does not reach.

42.6 A Last Word: to the Reader Three Years From Now

This section is written for you, opening the book three years from now, wherever you are by then.

In three years you will probably live through one real failure. It may be your project, your organization, or someone else's battlefield where you only handled the closing. In the week it happens, the book's four hundred thousand characters will shrink to three things. You will reach for a timeline, and the discipline of Chapter 6 will block part of the hindsight self-torture. You will want to hang a mechanism name on it, and the vocabulary of Part Three will give you a coordinate so you know which cloud your rain came from. You will want to write something down, and the template of Chapter 38 will tell you that every line needs a source and a date, your own pain included. Three things done, and you hold a post-event document more honest than most. That is not consolation. It is the whole design target: to let the knowledge of failure outlive the people it happened to.

You will also watch others refuse this. Someone will use one of the three conditions from the opening of this chapter to explain why no device is going to be installed. You will notice that half of the explanation is true, since the boundary is real, and half is rhetoric, since the boundary is being used to avoid responsibility inside the radius. Your discrimination comes from this chapter: only someone who knows the boundary can hear it being abused. No

argument is needed at that point. Open your ledger. Data is the only way this method argues, and the way it never loses.

You will also find that the book needs updating. It was written in a particular era, its cases stop at a particular moment, and the industry is walking a road it has not seen. If you already hold a ledger, a precedent library and a deviation record, you are one of the authors of its next edition. The words "local stocking", which recur throughout the book, mean you.

The last line is for the moment you close the cover. Of everything this book teaches, the cheapest and the most expensive item is "write it down at the time". Cheap, because a pen is enough. Expensive, because it means stopping, in an organization busy looking forward, for something nobody has yet asked you to look at. That minute of stopping is the minute the four hundred thousand characters were trying to buy.

42.7 Where the Three Boundary Chapters Converge

The division of labour among the three chapters of Part Six gets checked once here, so the reader does not carry the wrong one away.

Chapter 40's boundary is the policy boundary: two standards for two classes of failure, and one valve. Its user is the person who sets rules. Chapter 41's boundary is the theoretical boundary: the reach of two large theories and a four-step forensic procedure. Its user is the person who makes judgments. This chapter's boundary is the method boundary: three ceilings and four pathologies. Its user is the person who picks the tools up every day.

The three answer one question together, where does this stop, and the answer has three parts. In policy, it should not govern the exploration that deserves encouragement (Chapter 40). In theory, it

cannot reach the true unknown (Chapter 41). In method, it cannot hold a user who has abandoned judgment (Chapter 42). The wide ground outside those three segments is its working area. The equipment list for that area is already complete in Part Five, and the most important discipline is already complete in this chapter: tools serve judgment, judgment cannot be outsourced, and the ledger records everything, including the tool's own failures.

The three boundary chapters also share a private motive worth admitting. Their final purpose is to let the reader, on closing the book, dare to say no to it. A book that teaches you to question everything while forbidding anyone to question it becomes a new dogma. The boundary chapters are the book's own insurance against that: when a reader audits the method under principle three of 42.3, every criterion for the audit was supplied by the book itself. That design is the last device in the book and the least conspicuous. It produces nothing in normal times. It speaks only when the book is being misused.

After the five questions, a three-minute closing action. Write three lines. Line one, the first action you will carry away from the five routes, in one sentence from the route chapter. Line two, the name of your organization's ledger system, or the number you intend to give it, and if there is no ledger yet, write "starting today". Line three, where you will put this book, physically or digitally, so that your self three years from now can reach it with one hand, since a working manual belongs on the desk and not on the top shelf.

With those three lines written, the book's mission is handed over to your desk. The author's part ends there. The user's part starts at your next shift.

The last line of the book belongs at the end of this section, as the close. May your ledger stay green, your rehearsals stay true, and your postmortems keep being cited. May you defend what should be defended and afford what you want to buy. May failure in your

hands turn from cost into capital. That is what these tools are for, and it is also what you still hold when they do not work: an honest recorder has a place in any institution.

Five Self-Checks

1. When was the last time your organization's mechanism vocabulary, if it is in use, was used as a charge? Did that discussion produce an action?
2. Inventory the defences in your part of the system. Did any de - fence added in the past year make a process more roundabout or easier to skip?
3. In the past twelve months of postmortems, how many minim - um counterfactuals were adopted into a process? If none, who are the postmortems for?
4. The last time you filled a position table, would you dare circu - late the evidence column for all six grades to your team?
5. At what rhythm will you audit this method itself, and on what date does the first audit fall?

Appendix A: Mechanism × Domain Matrix

> Cells hold case IDs, and only cases with a first-hand full text are entered. A blank cell is material still to be collected. Appendix C gives the reading method.

Mechanism	Aviation	Chemical & Process	Maritime & Transport	Software & Internet Services	Finance	Healthcare	Drugs & Devices	Construction & Infrastructure	Public Policy & Administration	Military	Manufacturing & Supply Chain	Agriculture & Food	Startups & Products	Nuclear & Energy
Normalization of Deviance (Chapter 11)	av-004, av-005, av-006, av-010, av-028, av-029	ch-001	ma-005		fi-013	me-001, me-004	ph-001, ph-003, ph-006			mi-001				nu-002, nu-003
Goal Displacement; Goodhart's Law (Chapter 12)					fi-001, fi-010	me-016	ph-006		po-003					
Narrative before Evidence (Chapter 13)					fi-001, fi-002, fi-003, fi-005, fi-006	me-015			po-006	mi-001			st-003	
Where Intuitive Expertise Breaks Down (Chapter 14)	av-008, av-009, av-011, av-014, av-016, av-022, av-023, av-024		ma-010		fi-008, fi-012	me-013, me-015								
Practical Drift (Chapter 15)	av-003, av-020, av-030, av-032, av-033	ch-001, ch-007	ma-002, ma-003, ma-007, ma-009, ma-011, ma-012		fi-009, fi-010	me-009		ce-003, ce-005, ce-006, ce-007	po-001			ag-009, ag-010		nu-001, nu-004
Information Attenuation up the Hierarchy (Chapter 16)	av-017	ch-007	ma-007		fi-007	me-005, me-007, me-010		ce-002, ce-006	po-002, po-008					

Mechanism	Aviation	Chemical & Process	Maritime & Transport	Software & Internet Services	Finance	Healthcare	Drugs & Devices	Construction & Infrastructure	Public Policy & Administration	Military	Manufacturing & Supply Chain	Agriculture & Food	Startups & Products	Nuclear & Energy
Misaligned Incentives (Chapter 17)	av-019			sw-007, sw-008	fi-002, fi-003, fi-005, fi-006, fi-008, fi-009				po-008		mn-004, mn-007, mn-009		st-003	
Diffusion of Responsibility (Chapter 18)			ma-008		fi-007	me-007, me-008		ce-008	po-002, po-003, po-005, po-007			ag-004		nu-004
Why the Truth-Teller Goes First (Chapter 19)														
The Slack of Success (Chapter 20)													st-004	
Hidden Coupling (Chapter 21)	av-001, av-002, av-004, av-007, av-015, av-023, av-026, av-027, av-036, av-040	ch-005, ch-007	ma-010	sw-001, sw-002, sw-003, sw-004, sw-006, sw-008	fi-011	me-006, me-008, me-014	ph-008				mn-001, mn-002, mn-008, mn-010, mn-011	ag-002, ag-008		
The Illusion of Redundancy (Chapter 22)	av-007, av-022, av-027, av-028, av-037	ch-003, ch-004	ma-002, ma-006, ma-009	sw-006		me-001, me-003, me-004, me-009, me-012		ce-001, ce-003, ce-006, ce-008						nu-002

Mechanism	Aviation	Chemical & Process	Maritime & Transport	Software & Internet Services	Finance	Healthcare	Drugs & Devices	Construction & Infrastructure	Public Policy & Administration	Military	Manufacturing & Supply Chain	Agriculture & Food	Startups & Products	Nuclear & Energy
Extrapolating Past the Validated Domain (Chapter 23)	av-005, av-014, av-020, av-042, av-025, av-033, av-039, av-040	ch-003, ch-006		sw-001	fi-012		ph-007	ce-001	po-006		mn-011	ag-001, ag-003		
Crossing a Scale Threshold (Chapter 24)				sw-002, sw-005	fi-004	me-011			po-001		mn-003, mn-006		st-001	
Margin Erosion (Chapter 25)	av-003, av-013, av-015, av-025, av-039	ch-006	ma-003, ma-004, ma-005, ma-008, ma-009, ma-011, ma-013	sw-001, sw-009			ph-009				mn-001, mn-003, mn-004, mn-005, mn-008, mn-010, mn-012			
Automation and Skill Decay (Chapter 26)	av-017, av-018, av-031, av-032				fi-013									
Sunk Cost and Escalation (Chapter 27)	av-025								po-004				st-001	
Horizon Mismatch (Chapter 28)									po-007				st-004	

Mechanism	Aviation	Chemical & Process	Maritime & Transport	Software & Internet Services	Finance	Healthcare	Drugs & Devices	Construction & Infrastructure	Public Policy & Administration	Military	Manufacturing & Supply Chain	Agriculture & Food	Startups & Products	Nuclear & Energy
The Invisibility of Maintenance (Chapter 29)	av-001, av-006, av-012, av-013, av-019, av-026, av-029, av-030, av-034, av-035	ch-001, ch-005, ch-008, ch-009	ma-001, ma-004, ma-006, ma-013	sw-009		me-011	ph-001, ph-004	ce-001, ce-002, ce-009	po-001, po-005		mn-006	ag-005		nu-003
Handover Rupture (Chapter 30)	av-009, av-016, av-031, av-037					me-003, me-005, me-006, me-010, me-012, me-016	ph-003	ce-004, ce-005				ag-009		nu-001, nu-002
Regulatory Lag and Capture (Chapter 31)	av-002, av-030, av-034, av-035, av-038	ch-002, ch-009	ma-012		fi-004, fi-011	me-013	ph-005				mn-009		st-002	
Compliant and Unsafe (Chapter 32)	av-036	ch-002, ch-004, ch-008	ma-010	sw-007		me-001, me-002, me-014	ph-002, ph-004, ph-005, ph-007, ph-009, ph-010	ce-002, ce-004, ce-007, ce-009				ag-001, ag-003		
Cascading Failure (Chapter 33)		ch-008	ma-001	sw-003, sw-004, sw-005		me-002	ph-008, ph-010	ce-009			mn-002, mn-012	ag-004, ag-005, ag-006, ag-007, ag-008	st-002	
Path Lock-in (Chapter 34)									po-004					

Counts: 24 mechanisms by 14 domains gives 336 cells, 138 filled and 198 blank. The collection holds 159 cases, and more than one case to a cell is the norm, since a single case can witness several mechanisms at once.

Three ways to read it. Read a row: one mechanism's evidence across the domains, and the wider the spread, the more general the mechanism, with Hidden Coupling, The Invisibility of Maintenance and Practical Drift the three widest rows. Read a column: the mechanism density of one domain, and where a column is mostly blank the domain is either subject to recording bias (Chapter 2) or short of this book's sampling, so a reader testing a local sample should fill that column first. Read for blanks: a blank does not mean the mechanism is absent from that domain. It means the book holds no first-hand full text to put in the cell, and the reading method in Appendix C outranks the table.

One last lesson in reading a table. The blanks in the matrix deserve more care than the IDs inside the cells, and every reading method in Appendix C applies to a blank. Before reading a blank, ask whether it is recording bias, a sampling gap, or a genuine absence of the mechanism. Three kinds of blank, three responses, and once the check is done the blanks have become the reader's own collection list.

Appendix B: Case Index

> ID format: `domain code-serial`. Evidence grades are defined in Appendix C (A = full text of an independent investigation or audit report, B = official bulletin or recall notice, C = judgment or penalty document, D = the party's own public postmortem). Archived links are graded half a step down under C.4 (A–). `Chapter` gives the mechanism chapter in which the case appears in the main text. Every link below was checked against its document in October and November 2026.

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ag-001	Compliant and Unsafe	Agriculture & Food	2023-09	US	B	CDC（MMWR） | Outbreak of Salmonella Typhimurium	23
ag-002	Hidden Coupling	Agriculture & Food	2024-02	US	B	CDC（MMWR） | Notes from the Field: Outbreak of	Reserve
ag-003	Extrapolating Past the Validated Domain	Agriculture & Food	2025-10	US	B	FDA | FDA Enforcement Report event 97880	13
ag-004	Cascading Failure	Agriculture & Food	2025-10	US	B	FDA | FDA Enforcement Report event 97931	Reserve
ag-005	Cascading Failure	Agriculture & Food	2025-09	US	B	FDA | FDA Enforcement Report event 97713	Reserve
ag-006	Cascading Failure	Agriculture & Food	2025-05	US	B	FDA | FDA Enforcement Report event 97011	Reserve
ag-007	Cascading Failure	Agriculture & Food	2024-05	US	B	FDA | FDA Enforcement Report event 94571	Reserve
ag-008	Hidden Coupling	Agriculture & Food	2024-05	US	B	FDA | FDA Enforcement Report event 94565	33

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ag-009	Handover Rupture	Agriculture & Food	2025-04	US	B	FDA | FDA Enforcement Report event 96758	30
ag-010	Practical Drift	Agriculture & Food	2025-06	US	B	FDA | FDA Enforcement Report H-0088-2025	Reserve
av-001	Hidden Coupling	Aviation	2013-07	GB	A	AAIB | Aircraft Accident Report 2/2015 –	Reserve
av-002	Regulatory Lag and Capture	Aviation	2013-01	US	A	NTSB | Auxiliary Power Unit Battery Fire,	20
av-003	Practical Drift	Aviation	2022-03	GB	A	AAIB | Aircraft Accident Report AAR 2/202	25
av-004	Hidden Coupling	Aviation	2023-09	FR	A	BEA | Accident survenu au Robin DR300-14	Reserve
av-005	Extrapolating Past the Validated Domain	Aviation	2021-10	DE	A	BFU | Untersuchungsbericht BFU21-0949-3X	Reserve
av-006	The Invisibility of Maintenance	Aviation	2021-02	US	A	NTSB | Aviation Investigation Final Repor	11

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
av-007	Hidden Coupling	Aviation	2024-08	JP	A	JTSB｜航空事故調査報告書，ボーイング式７４７－４００Ｆ型Ｎ４０４ＫＺ 離陸	33
av-008	Where Intuitive Expertise Breaks Down	Aviation	2024-06	JP	A	JTSB｜航空事故調査報告書，パイパー式ＰＡ－２８－１５１型ＪＡ３７１２ 着陸	Reserve
av-009	Where Intuitive Expertise Breaks Down	Aviation	2021-05	IT	A	ANSV｜RELAZIONE D’INCHIESTA – INCIDENTE	Reserve
av-010	Normalization of Deviance	Aviation	2018-06	US	A	NTSB｜Aviation Investigation Final Repor	Reserve
av-011	Where Intuitive Expertise Breaks Down	Aviation	2025-07	US	A	NTSB｜Aviation Investigation Final Repor	14
av-012	The Invisibility of Maintenance	Aviation	2023-01	US	A	NTSB｜Aviation Investigation Final Repor	Reserve
av-013	Margin Erosion	Aviation	2023-07	FR	A	BEA｜Accident du CESSNA 177 « CARDINAL	Reserve
av-014	Extrapolating Past the Validated Domain	Aviation	2021-02	DE	A	BFU｜Untersuchungsbericht BFU21-0041-3X	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
av-015	Margin Erosion	Aviation	2019-11	IT	A	ANSV｜Relazione d’inchiesta – INCIDENTE	Reserve
av-016	Handover Rupture	Aviation	2024-05	JP	A	JTSB｜航空事故調査報告書，アレキサンダー・シュライハー式ＡＳＫ１３型ＪＡ２	Reserve
av-017	Automation and Skill Decay	Aviation	2024-08	JP	A	JTSB｜航空重大インシデント調査報告書，ユーロコプター式ＥＣ１３５Ｐ２＋型Ｊ	19
av-018	Automation and Skill Decay	Aviation	2023-07	IT	A	ANSV｜RELAZIONI D’INCHIESTA – INCIDENTE	Reserve
av-019	Misaligned Incentives	Aviation	2025-09	CN	B	CAAC｜Investigation and handling report on the Yaxin General Aviation helicopter accident published	Reserve
av-020	Practical Drift	Aviation	2023-03	US	A	NTSB｜Aviation Investigation Final Repor	15
av-021	Sunk Cost and Escalation	Aviation	2023-03	US	A	NTSB｜Aviation Investigation Final Repor	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
av-022	The Illusion of Redundancy	Aviation	2023-09	US	A	NTSB	Aviation Investigation Final Repor	22
av-023	Where Intuitive Expertise Breaks Down	Aviation	2023-04	US	A	NTSB	Aviation Investigation Final Repor	Reserve
av-024	Where Intuitive Expertise Breaks Down	Aviation	2022-08	US	A	NTSB	Aviation Investigation Final Repor	Reserve
av-025	Extrapolating Past the Validated Domain	Aviation	2025-12	GB	A	AAIB	AAIB investigation to Eurofox 2K,	23, 27
av-026	Hidden Coupling	Aviation	2024-02	GB	A	AAIB	AAIB investigation to Boeing 747-4	Reserve
av-027	Hidden Coupling	Aviation	2025-10	GB	A	AAIB	AAIB investigation to Boeing 767-3	Reserve
av-028	Normalization of Deviance	Aviation	2025-04	GB	A	AAIB	AAIB investigation to Boeing 737-8	Reserve
av-029	The Invisibility of Maintenance	Aviation	2024-10	ES	B	AESA	Boletín Informativo de Seguridad B	34
av-030	The Invisibility of Maintenance	Aviation	2024-01	US	A	NTSB	In-Flight Separation of Left Mid E	29

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
av-031	Automation and Skill Decay	Aviation	2013-07	US	A	NTSB｜Descent Below Visual Glidepath and	26
av-032	Practical Drift	Aviation	2009-02	US	A	NTSB｜Loss of Control on Approach, Colga	17
av-033	Practical Drift	Aviation	2001-11	US	A	NTSB｜In-Flight Separation of Vertical S	15
av-034	The Invisibility of Maintenance	Aviation	2000-01	US	A	NTSB｜Loss of Control and Impact with Pa	11
av-035	The Invisibility of Maintenance	Aviation	1996-05	US	A	NTSB｜In-Flight Fire and Impact with Ter	29
av-036	Compliant and Unsafe	Aviation	1996-07	US	A	NTSB｜In-flight Breakup Over the Atlanti	32
av-037	Handover Rupture	Aviation	2024-01	JP	A	JTSB｜海上保安庁所属ボンバルディア式ＤＨＣ－８－３１５型ＪＡ７２２Ａ及び日	30

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
av-038	Regulatory Lag and Capture	Aviation	2022-03	CN	B	CAAC	Notice on the progress of the investigation into the China Eastern MU5735 accident of 21 March	31
av-039	Extrapolating Past the Validated Domain	Aviation	2015-08	GB	A	AAIB	Aircraft Accident Report AAR 1/201	Reserve
av-040	Hidden Coupling	Aviation	2018-10	GB	A	AAIB	Aircraft Accident Report AAR 1/202	21
av-042	Extrapolating Past the Validated Domain	Aviation	2015-08	GB	A	AAIB	Aircraft Accident Report 1/2017, Hawker Hunter T7 G-BXFI, Shoreham	23
ce-001	The Illusion of Redundancy	Construction & Infrastructure	2024-10	GB	A	RAIB	Report 08/2026: Collision between	22
ce-002	The Invisibility of Maintenance	Construction & Infrastructure	2024-09	GB	A	RAIB	Report 10/2025: Derailment of a fr	12
ce-003	Practical Drift	Construction & Infrastructure	2025-03	GB	A	RAIB	Report 02/2026: Fatal accident at	15

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ce-004	Handover Rupture	Construction & Infrastructure	2025-03	GB	A	RAIB | Report 03/2026: Two track workers	Reserve
ce-005	Handover Rupture	Construction & Infrastructure	2025-06	GB	A	RAIB | Report 04/2026: Derailment at Denb	Reserve
ce-006	The Illusion of Redundancy	Construction & Infrastructure	2024-11	GB	A	RAIB | Report 01/2026: Passenger trapped	Reserve
ce-007	Practical Drift	Construction & Infrastructure	2025-07	GB	A	RAIB | Report 09/2026: Dangerous occurren	20
ce-008	The Illusion of Redundancy	Construction & Infrastructure	2025-06	GB	A	RAIB | Report 10/2026: Collision between	18
ce-009	Cascading Failure	Construction & Infrastructure	2021-06	US	A	NIST | Champlain Towers South Collapse In	29
ch-001	Normalization of Deviance	Chemical & Process	2014-11	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Toxic Chemical Release at the DuPo	11

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ch-002	Regulatory Lag and Capture	Chemical & Process	2013-04	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | West Fertilizer Explosion and Fire	31
ch-003	Extrapolating Past the Validated Domain	Chemical & Process	2017-08	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Arkema Inc. Chemical Plant Fire	22
ch-004	The Illusion of Redundancy	Chemical & Process	2025-06	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Furnace Explosion and Fire at Shel	14
ch-005	The Invisibility of Maintenance	Chemical & Process	2023-07	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Explosions, Fires, and Toxic Ethyl	21

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ch-006	Extrapolating Past the Validated Domain	Chemical & Process	2024-11	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Final Report: Fatal Runaway Reacti	25
ch-007	Information Attenuation up the Hierarchy	Chemical & Process	2013-11	CN	A	State Council 11·22 accident investigation team, published on the Ministry of Emergency Management site | Investigation report on the particularly serious 11·22 leak and explosion at the Sinopec Donghuang crude oil pipeline, Qingdao, Shandong	16

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ch-008	The Invisibility of Maintenance	Chemical & Process	2014-08	CN	A	State Council 8·2 accident investigation team, published on the Ministry of Emergency Management site | Investigation report on the particularly serious 8·2 explosion at Zhongrong Metal Products, Kunshan, Suzhou, Jiangsu	32
ch-009	Regulatory Lag and Capture	Chemical & Process	2021-04	US	A	U.S. Chemical Safety and Hazard Investigation Board (CSB) | Yenkin-Majestic Resin Plant Vapor	19
fi-001	Goal Displacement; Goodhart’s Law	Finance	2026-09	US	C	SEC | Travel + Leisure Co. (Litigation R	12
fi-002	Narrative before Evidence	Finance	2026-08	US	C	SEC | GenesisAI Corp. and Archil Cheishv	13

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
fi-003	Narrative before Evidence	Finance	2026-08	US	C	SEC | Ichcoin Tech Corp. (Litigation Rel	Reserve
fi-004	Crossing a Scale Threshold	Finance	2026-08	US	C	SEC | False Forms ADV Filings (Litigatio	24
fi-005	Narrative before Evidence	Finance	2026-07	US	C	SEC | American Patriot Brands, Inc., et	Reserve
fi-006	Narrative before Evidence	Finance	2026-10	CN	C	CSRC | Administrative penalty decision (〔2026〕 No. 57)	Reserve
fi-007	Diffusion of Responsibility	Finance	2026-10	CN	C	CSRC | Administrative penalty decision (〔2026〕 No. 59)	18
fi-008	Misaligned Incentives	Finance	2026-10	CN	C	CSRC | Administrative penalty decision (〔2026〕 No. 56)	17
fi-009	Practical Drift	Finance	2026-09	CN	C	CSRC | Administrative penalty decision (〔2026〕 No. 54)	Reserve
fi-010	Goal Displacement; Goodhart’s Law	Finance	2026-08	US	C	FINRA | Disciplinary and Other FINRA Actio	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
fi-011	Hidden Coupling	Finance	2026-08	US	C	FINRA	Disciplinary and Other FINRA Actio	31
fi-012	Extrapolating Past the Validated Domain	Finance	2026-08	US	C	FINRA	Disciplinary and Other FINRA Actio	14
fi-013	Normalization of Deviance	Finance	2026-08	US	C	FINRA	Disciplinary and Other FINRA Actio	11
ma-001	The Invisibility of Maintenance	Maritime & Transport	2025-01	GB	A	MAIB	Catastrophic engine failure and su	21
ma-002	Practical Drift	Maritime & Transport	2025-10	GB	A	MAIB	Grounding and subsequent loss of t	15
ma-003	Practical Drift	Maritime & Transport	2025-06	GB	A	MAIB	Grounding and subsequent loss of t	15
ma-004	The Invisibility of Maintenance	Maritime & Transport	2024-03	GB	A	MAIB	Flooding and foundering of the ste	25
ma-005	Normalization of Deviance	Maritime & Transport	2024-04	GB	A	MAIB	Collision between the crude oil ta	Reserve
ma-006	The Invisibility of Maintenance	Maritime & Transport	2024-02	GB	A	MAIB	Foundering of the fishing vessel F	20

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ma-007	Practical Drift	Maritime & Transport	2024-07	GB	A	MAIB	Fatal accident to a crew member on	16
ma-008	Diffusion of Responsibility	Maritime & Transport	2023-10	GB	A	MAIB	Collision between the bulk carrier	18
ma-009	Practical Drift	Maritime & Transport	2014-11	GB	A	MAIB	Capsize and sinking of stern trawl	25
ma-010	Hidden Coupling	Maritime & Transport	2007-09	GB	A	MAIB	Entry to an enclosed space on emer	14
ma-011	Practical Drift	Maritime & Transport	2024-05	GB	A	MAIB	Double man overboard from the fish	Reserve
ma-012	Practical Drift	Maritime & Transport	2023-12	GB	A	MAIB	Fall overboard from the crab potti	31
ma-013	The Invisibility of Maintenance	Maritime & Transport	2023-10	GB	A	MAIB	Fatal injury to a deckhand followi	25
me-001	Compliant and Unsafe	Healthcare	2018-09	GB	A	HSIB, predecessor of HSSIB	Administering a wrong site nerve b	15

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
me-002	Compliant and Unsafe	Healthcare	2019	GB	A	HSIB, predecessor of HSSIB | Administering high-strength insuli	23
me-003	The Illusion of Redundancy	Healthcare	2019	GB	A	HSIB, predecessor of HSSIB | Inadvertent administration of an o	22
me-004	Normalization of Deviance	Healthcare	2019	GB	A	HSIB, predecessor of HSSIB | Detection of retained vaginal swab	20
me-005	Handover Rupture	Healthcare	2017-07	GB	A	HSIB, predecessor of HSSIB | Failures in communication or follo	16
me-006	Handover Rupture	Healthcare	2021	GB	A	HSIB, predecessor of HSSIB | Local integrated investigation pil	30
me-007	Diffusion of Responsibility	Healthcare	2022	GB	A	HSIB, predecessor of HSSIB | Clinical investigation booking sys	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
me-008	Diffusion of Responsibility	Healthcare	2017	GB	A	HSIB, predecessor of HSSIB | Implantation of wrong prostheses d	18
me-009	Practical Drift	Healthcare	2022	GB	A	HSIB, predecessor of HSSIB | Identification of critical patient	Reserve
me-010	Information Attenuation up the Hierarchy	Healthcare	2022	GB	A	HSIB, predecessor of HSSIB | Caring for adults with learning di	Reserve
me-011	Crossing a Scale Threshold	Healthcare	2021-06	GB	A	HSIB, predecessor of HSSIB | Oxygen issues during the COVID-19	24
me-012	Handover Rupture	Healthcare	2021-08	GB	A	HSIB, predecessor of HSSIB | Timely detection and treatment of	Reserve
me-013	Where Intuitive Expertise Breaks Down	Healthcare	2022-03	GB	A	HSIB, predecessor of HSSIB | Clinical decision making: diagnosi	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
me-014	Compliant and Unsafe	Healthcare	2022-04	GB	A	HSIB, predecessor of HSSIB | Unintentional overdose of morphine	Reserve
me-015	Where Intuitive Expertise Breaks Down	Healthcare	2020-01	GB	A	HSIB, predecessor of HSSIB | Delayed recognition of acute aorti	14
me-016	Handover Rupture	Healthcare	2021-12	GB	A	HSIB, predecessor of HSSIB | Recognition of the acutely ill inf	Reserve
mi-001	Normalization of Deviance	Military	2003-01	US	A	Columbia Accident Investigation Board (CAIB) | Columbia Accident Investigation Bo	11
mn-001	Hidden Coupling	Manufacturing & Supply Chain	2026-10	US	B	CPSC | HAIARA Power Banks Recalled Due to	Reserve
mn-002	Hidden Coupling	Manufacturing & Supply Chain	2026-10	US	B	CPSC | NTI and HTP Brand Gas Boilers Reca	28
mn-003	Margin Erosion	Manufacturing & Supply Chain	2026-10	US	B	CPSC | DREO Recalls DR-HSH015 Space Heate	24

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
mn-004	Margin Erosion	Manufacturing & Supply Chain	2026-10	US	B	CPSC | SOOWERY Recalls 8-Drawer Fabric Dr	17
mn-005	Margin Erosion	Manufacturing & Supply Chain	2026-10	US	B	CPSC | PILLANI Recalls Nursing Pillows (V	Reserve
mn-006	Crossing a Scale Threshold	Manufacturing & Supply Chain	2026-10	US	B	CPSC | Bodum USA Recalls 5.9 Million Fren	34
mn-007	Misaligned Incentives	Manufacturing & Supply Chain	2026-10	US	B	CPSC | Longjili Trading Recalls Pamag Pie	Reserve
mn-008	Hidden Coupling	Manufacturing & Supply Chain	2026-10	US	B	CPSC | Harkla Recalls Children’s Sensory	Reserve
mn-009	Misaligned Incentives	Manufacturing & Supply Chain	2026-10	US	B	CPSC | RootStim Recalls Minoxidil Bottles	Reserve
mn-010	Hidden Coupling	Manufacturing & Supply Chain	2013-04	US	B	CPSC | MTD Products Recalls Cub Cadet Com	Reserve
mn-011	Hidden Coupling	Manufacturing & Supply Chain	2013-01	US	B	CPSC | BRP Recalls Can-Am Side-By-Side Ve	23
mn-012	Cascading Failure	Manufacturing & Supply Chain	2014-01	US	B	CPSC | Sears and Kmart Recall Kenmore Osc	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
nu-001	Practical Drift	Nuclear & Energy	2026-10	FR	B	ASNR, French nuclear safety and radiation protection authority | Indisponibilité d’ un groupe électr	Reserve
nu-002	Normalization of Deviance	Nuclear & Energy	2026-10	FR	B	ASNR, French nuclear safety and radiation protection authority | Non-respect d’ une mesure compensat	Reserve
nu-003	Normalization of Deviance	Nuclear & Energy	2026-09	FR	B	ASNR, French nuclear safety and radiation protection authority | Perte et découverte de sources rad	Reserve
nu-004	Practical Drift	Nuclear & Energy	2026-10	US	C	NRC, US Nuclear Regulatory Commission | NRC Proposes $36,000 Civil Penalty	Reserve

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ph-001	Normalization of Deviance	Drugs & Devices	2025-10	US	B	FDA	openFDA drug enforcement record D-	Reserve
ph-002	Compliant and Unsafe	Drugs & Devices	2024-06	US	B	FDA	openFDA drug enforcement record D-	Reserve
ph-003	Handover Rupture	Drugs & Devices	2020-05	US	B	FDA	openFDA drug enforcement record D-	30
ph-004	The Invisibility of Maintenance	Drugs & Devices	2022-01	US	B	FDA	openFDA drug enforcement record D-	29
ph-005	Regulatory Lag and Capture	Drugs & Devices	2024-11	US	B	FDA	openFDA device enforcement record	13
ph-006	Goal Displacement; Goodhart’s Law	Drugs & Devices	2026-06	US	B	FDA	openFDA device enforcement record	Reserve
ph-007	Extrapolating Past the Validated Domain	Drugs & Devices	2026-08	US	B	FDA	openFDA device enforcement record	23
ph-008	Cascading Failure	Drugs & Devices	2026-06	US	B	FDA	openFDA device enforcement record	33

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
ph-009	Margin Erosion	Drugs & Devices	2026-06	US	B	FDA	openFDA device enforcement record	Reserve
ph-010	Cascading Failure	Drugs & Devices	2025-12	US	B	FDA	openFDA device enforcement record	Reserve
po-001	The Invisibility of Maintenance	Public Policy & Administration	2026-09	US	A	GAO	Air Traffic Control Systems (GAO-2	24
po-002	Information Attenuation up the Hierarchy	Public Policy & Administration	2026-09	US	A	GAO	VA Electronic Health Record Modern	16
po-003	Goal Displacement; Goodhart’s Law	Public Policy & Administration	2026-10	US	A	GAO	IT Dashboard: Selected Agencies’ I	12
po-004	Sunk Cost and Escalation	Public Policy & Administration	2026-06	GB	A	NAO	High Speed Two reset	27, 34
po-005	The Invisibility of Maintenance	Public Policy & Administration	2026-06	GB	A	NAO	Restoration and Renewal of the Pal	18
po-006	Extrapolating Past the Validated Domain	Public Policy & Administration	2026-05	GB	A	NAO	Sizewell C (HC 33, 2026-27)	13
po-007	Horizon Mismatch	Public Policy & Administration	2026-03	GB	A	NAO	Northern Powerhouse Rail	28

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
po-008	Misaligned Incentives	Public Policy & Administration	2026-06	CN	A	National Audit Office | State Council report on the audit of central budget execution and other fiscal revenue and expenditure for 2025	17
st-001	Crossing a Scale Threshold	Startups & Products	2020-03	US	D	Failory Cemetery, quoting founder Justin Kan in public remarks | Atrium, Failory Startup Cemetery	27
st-002	Regulatory Lag and Capture	Startups & Products	2019	IN	D	Failory Cemetery, quoting CEO Rahul Raj's retrospective verbatim | Koinex, Failory Startup Cemetery	Reserve
st-003	Narrative before Evidence	Startups & Products	2020-06	US	D	Failory Cemetery, quoting CEO Kurt Rathmann | ScaleFactor, Failory Startup Cemetery	13

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
st-004	The Slack of Success	Startups & Products	2020-12	AU	D	Failory Cemetery, quoting CEO Eric Wilson’s email to customers | Xinja, Failory Startup Cemetery	28
sw-001	Hidden Coupling	Software & Internet Services	2019-07	US	D	Cloudflare | Details of the Cloudflare outage o	Reserve
sw-002	Hidden Coupling	Software & Internet Services	2025-11	US	D	Cloudflare | Cloudflare outage on November 18,	21
sw-003	Hidden Coupling	Software & Internet Services	2026-02	US	D	Cloudflare | Cloudflare outage on February 20,	21
sw-004	Hidden Coupling	Software & Internet Services	2025-09	US	D	Cloudflare | A deep dive into Cloudflare’s Sept	26
sw-005	Crossing a Scale Threshold	Software & Internet Services	2026-08	US	D	GitHub | The August 17 outage, and the work	24
sw-006	The Illusion of Redundancy	Software & Internet Services	2026-09	US	D	GitHub | GitHub availability report: Septem	12

ID	Primary mechanism	Domain	Event (year-month)	Country	Grade	Primary source	Chapter
sw-007	Misaligned Incentives	Software & Internet Services	2025-09	FR	C	CNIL | Cookies and advertisements inserte	17
sw-008	Hidden Coupling	Software & Internet Services	2026-10	FR	C	CNIL | Cookies and mobile application: 7	Reserve
sw-009	Margin Erosion	Software & Internet Services	2026-01	FR	C	CNIL | Closure of the order issued agains	Reserve

159 entries in total (aviation 41, healthcare 16, finance 13, maritime & transport 13, manufacturing & supply chain 12, agriculture & food 10, drugs & devices 10, construction & infrastructure 9, chemical & process 9, software & internet services 9, public policy & administration 8, nuclear & energy 4, startups & products 4, military 1); 88 are used as cases in the mechanism chapters, and the rest are held in reserve for the matrix.

Three notes on use. One, this index and the Appendix A matrix carry identical numbering, and a mechanism chapter citing a case uses the same number; where two disagree, check this index before doubting the chapter. Two, the one-line description of a case is the plainest form there is: who, where, what happened, in one line. The full fields (time, place, subject, source, link, evidence grade) sit in the case card itself. Three, the chapter column records where the case appears in the main text, and some cases appear in more than one mechanism chapter from different angles; a reader working case by case should start with the first chapter the ID sits in, then follow the column.

Appendix C: Evidence Grades and the Reconstruction Method

> This appendix is the operating manual for the grade system of Chapter 10: the five definitions with their tests, how to read the matrix, the labelling convention for reconstruction, how links are archived, and the working sheet for the four-part number record.

C.1 Five grades, definitions and tests

Grade	Definition	Test	What it can support in the body
A	Full text of an independent investigating body's report	The body is independent of the party investigated and of the penalty function; the full text, not a summary, is public; it carries a factual part and an analytical part, or an equivalent structure. A process report, such as the JTSB's 経過報告, is graded A−	The decision sequence, the options ruled out, the minimum counterfactual, the escalation point
B	Official bulletins, communiqués, regulatory announcements	Officially signed, one party's statement, no evidentiary section	The magnitude of the consequence, conclusive findings. Entry into the body requires the "reconstructed" label

Grade	Definition	Test	What it can support in the body
C	Judgments, penalty decisions, arbitration documents	The facts have been through adversarial testing, but the selection of facts serves the issue in dispute	Amounts, responsible parties, the timeline of acts. Technical process outside the dispute is labelled “reconstructed”
D	The party’s own public postmortem	Published voluntarily by a company or an individual; a postmortem carrying a timeline is the strong form, an announcement the weak one	The shape of internal organizational process. Supports a finding of fact only after cross-verification against A, B or C
E	Media reporting	Any retold text	Lead generation only. Never a source of fact for the body

The rules for movement between grades: a core fact corroborated by B and C together can support a body case labelled “reconstructed”; an A– case does not support the escalation point, which turns on detail at the level of the signatory; a case whose links have died and which has no archive is downgraded to pending verification; quoted text that does not match its original is removed. the link check and the number-to-source check carry this out.

C.2 Reading the matrix

The mechanism × domain matrix in Appendix A takes three readings.

Start with the blanks, which carry two meanings. A blank marked “to collect” means the cell has not been searched yet. A blank

marked “no record” means it has been searched and no qualifying primary material was found. The two are entirely different: the first is a gap in this book, the second may be a structural silence in the domain, which is Chapter 2’s evidence of recording skew.

Row density comes second. A row, meaning one mechanism, with strong cases in several domains says the mechanism holds across industries. A row with cases in only one domain is first suspected of thin collection coverage, and only after that of the mechanism being industry-specific.

Column density comes last. A column, meaning one domain, with many cases usually reflects a strong recording institution (aviation, chemical processing) rather than a high accident rate. A column that is nearly blank calls first for a check on whether that domain’s primary library is reachable (Chapter 3’s device checklist), and only then for a conclusion.

C.3 Labelling a reconstruction

Cases on medium evidence, mostly grades B, C and D, are still worth writing, on one condition: the inference is labelled explicitly. The convention: a factual sentence in the body states the fact and cites the source in parentheses; an inference opens with “Reconstructed:” and closes with its basis in parentheses. An example:

“Minutes of the 14 March review meeting show the risk listed as agenda item 7 (factual sentence; grade C: meeting minutes). Reconstructed: from the agenda order and the meeting’s total length, the item got less than 5 minutes and no resolution entry was formed (inference; basis: the time allocation on page 2 of the minutes).”

Three disciplines. An inference must be re-checkable by the reader along the basis it cites. An inference with no re-check path is not written. Inference may not prop up inference, since two “recon - structed” sentences still add up to one inference.

C.4 Links, archives and the soft 404

Every case’s primary link has to be opened by hand and read to its content: WebFetch returning body text, or curl returning HTTP 200 with a file type, a page count and a subject that match. Two common traps need testing separately. The soft 404 is an HTTP 200 that returns an error page or a placeholder, and the test is any one of three: an abnormally small file, text containing “404/Forbid - den/Not Found”, or a subject that does not match expectation. The anti-scraping placeholder is a status 200 whose content is a CAPTCHA or a login wall.

When the original link is dead, an Internet Archive (web.archive.org) snapshot is allowed on three conditions: the body notes “original link dead, archive link”; the snapshot date is later than the publication of the cited content; the grade drops half a step from the original’s (A to A−, B to B−).

C.5 Working sheet for the four-part number record

Every number in the body gets its four parts registered at the material end. The template:

```
Number:
Issuing body:
Document and number:
Date of publication:
Sample and definition:
```

```
Link:
Chapter and section where used:
```

When a part is missing, go back to the source and fill it first. Where it cannot be filled, delete the number and rewrite the mechanism narration. The difference audit takes this sheet as its authority: the set of numbers in the body is a subset of the set of numbers in the material cards, and a non-empty difference sends the chapter back.

Three notes on use. The grade evaluates the form of the material, not its truth: a speculative passage inside an A-grade report is still speculation and is cited as speculation, and a dated, specific fact inside an E-grade report may be accepted, though any key number must be traced back to a higher grade. The rules of downgrade and upgrade are not symmetric: retelling does not upgrade, since a conclusion taken from second-hand material is graded as second-hand, while a new investigation can upgrade, since a fact confirmed by an independent investigation is graded by the new material. The table also serves as the template for a reader's own case library: internal postmortem material grades the same way (system logs A, meeting minutes B, participants' oral accounts D), so an organization's library and this book's library can cite each other directly.

Appendix D: A Genealogy of Mechanisms

> The twenty-four mechanisms are not independent entries standing side by side. They are twenty-four projections on the surface of a few tensions underneath. This appendix records the family relations among them: which ones usually travel in pairs, which one disguises itself as which, and what a combination means when it shows up. It cross-links with Section 6 of each mechanism chapter.

D.1 Three underlying tensions

Tension one: knowing against admitting, in the information layer. The organization knows the truth, or could know it, but the cost of admitting it deforms the truth in the archive, in the reporting and in the narrative. Family members: Normalization of Deviance (11, distortion at the archival layer), Narrative before Evidence (13, distortion at the expressive layer), Information Attenuation up the Hierarchy (16, distortion at the transmission layer), Why the Truth-Teller Goes First (19, silencing at the source), Goal Displacement; Goodhart's Law (12, distortion at the measurement layer). All five share one physical process: the truth is distorted layer by layer inside the organization's information structure. Reading a combination: when more than three of the five appear together, the organization's sensing system has failed as a whole, no single-point repair will help, and an independent sensing channel has to be brought in from outside (the devices of Chapters 16 and 19).

Tension two: now against later, in the time layer. Present gain is traded for future cost, and the structure permits it. Family members: Horizon Mismatch (28, the tenure structure), The Invisibility of Maintenance (29, the asset side), Margin Erosion (25, the design

side), The Slack of Success (20, the will side). All four run on one licence: someone later pays. Reading a combination: in an organ - ization where all four appear, the risk state is older than any in - dicator shows, since indicators reflect the current period while the debt sits buried on the time axis (Surfside is the specimen holding all four: ce-009).

Tension three: one against all, in the structural layer. Between local rationality and collective loss sits an externality that cannot be priced. Family members: Misaligned Incentives (17, the direction of interest), Diffusion of Responsibility (18, where responsibility is lodged), Cascading Failure (33, transmission through dependency), Crossing a Scale Threshold (24, the limit of coordination). Reading a combination: in an organization running both 17 and 18, wrong direction with nobody accountable, failure is only a matter of time; in an industry running both 33 and 24, dependency concentrated and scale expanding, what is being waited for is simultaneous failure (the 38 shell consultants of Chapter 24, the thousand cloud customers of Chapter 33).

D.2 Common pairs

Pair	Relation	Where it appears in this book
15 Practical Drift × 11 Normalization of Deviance	drift supplies the raw material, normalization completes the filing	av-020 (the logbook check in flight), ma-002 (an alarm buried under music)
31 Regulatory Lag and Capture × 32 Compliant and Unsafe	the old rule is the climate, item-by-item compliance is the weather	av-036 (certification philosophy), ch-008 (a compliant check on a misclassification)
25 Margin Erosion × 29 The Invisibility of Maintenance	arrears on both sides of the margin, design and condition	ce-009 (Surfside), ma-004 (the seawater pipe)

Pair	Relation	Where it appears in this book
27 Sunk Cost and Escalation × 34 Path Lock-in	lock-in explains why the switch cannot be made, escalation explains why it continues	po-004 (HS2)
21 Hidden Coupling × 33 Cascading Failure	the hidden line exists, the dependency is concentrated	sw-002 (permissions × feature files), ph-008 (kit × component)
12 Goal Displacement × 17 Misaligned Incentives	a misaligned interest finds its own ruler	fi-001 (the loan-loss indicator), po-003 (the risk rating)
13 Narrative before Evidence × 27 Sunk Cost and Escalation	an expectation stated early makes exit a broken promise	fi-005 (issuance fraud), po-006 (the Sizewell replication narrative)
18 Diffusion of Responsibility × 16 Information Attenuation	nobody claims it across, nobody carries it up	ma-008 (VTS authority), po-002 (the conceptual schedule)
22 The Illusion of Redundancy × 26 Automation and Skill Decay	a paper defence replaces both the real defence and the real practice	ce-001 (double sanding), me-003 (the two-person check)
23 Extrapolating Past the Validated Domain × 14 Where Intuitive Expertise Breaks Down	the conclusion crosses the boundary together with the person judging it	ph-007 (reprocessed trocars), me-002 (U-500 drawn in a U-100 syringe)

D.3 Mutual disguise, and how to tell them apart

Disguise one: 17 dressed as 12. An appraisal problem gets dis - cussed in the shape of an indicator design problem: change the indicator, leave the incentive alone. The test: does behaviour go back to the old pattern once the indicator changes? If it goes back, 17 is running (Chapter 12, Section 6).

Disguise two: 25 dressed as 29. Eating the margin, which is cutting redundancy, and arrears in maintenance, which is deferring ser -

vice, get lumped together as aging. The test: is the object a design value (25) or a condition value (29)? At Surfside both layers were present.

Disguise three: 19 dressed as 16. When no news comes up the channel, separate first whether nobody spoke (19) or somebody spoke and it was diluted (16). The diagnostic question: the last time someone spoke sharply, what became of that person?

Disguise four: 24 dressed as 22. Simultaneous failure exposed by growth gets handled as common-cause failure, and the answer is another backup. The test: is the trigger external (22) or the scale itself (24)? GitHub's record load (24) against the Arkema flood (22) is the textbook contrast.

Disguise five: 30 dressed as 18. Nobody being accountable after a broken handover looks like diffusion, when the knowledge never arrived at all. The test: does the receiving side know it is account - able? If not, 30. If it knows and assumes someone else will handle it, 18.

D.4 Combination warnings

Three combinations, each of which on appearing calls for handling to be escalated at once, from a ledger walkthrough to a dedicated review.

Combination one, the sensing-failure package: three of 11, 13, 16 and 19 present. The organization is already systematically deaf, and the sensing gets repaired before anything else is discussed.

Combination two, the time-bomb package: two of 25, 28 and 29 present. The debt sits on the time axis and has no signatory, which is the reading of Surfside and of Westminster.

Combination three, the systemic-exposure package: two of 21, 24 and 33 present. The shared platform of an expansion phase is ac -

cumulating the conditions for simultaneous failure, and sw-005 in Chapter 24 and sw-003 in Chapter 33 come from the same company, which is not a coincidence.

Three notes on use. The genealogy is a map for reading, not a pedigree for theory: two chapters sitting next to each other does not mean two mechanisms share an origin, only that their tests are easily confused, and the boundary discussion in Section 6 is the formal text that separates them. The three-generation division, upstream disciplinary concept, mechanism chapter, field form, matches three citation needs: cite upstream in a paper, use the mechanism chapter in management, copy the field form into a ledger, and each reader takes the layer they need. The genealogy grows, since Section 6's boundary discussions are revised with each version, so the table gets checked against the body text once a year and new adjacencies are added as they appear; the table is built as a living document.